

Wiley Finance Series

The Bank Credit Analysis Handbook

A Guide for Analysts, Bankers and Investors

SECOND EDITION

JONATHAN GOLIN
PHILIPPE DELHAISE

WILEY

The Bank Credit Analysis Handbook

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Preface to the New Edition

In early 1997, Jonathan Golin applied for a position of bank credit analyst with Thomson BankWatch. He had limited experience in financial analysis, let alone bank financial analysis, but Philippe Delhaise, then president of BankWatch's Asia division, had long held the view that outstanding brains, good analytical skills, a passion for details, and a degree of latent skepticism were the best assets of a brilliant bank financial analyst. He immediately hired Jonathan.

Jonathan joined a team of very talented senior analysts, among them Andrew Seiz, Damien Wood, Tony Watson, Paul Grela, and Mark Jones. Philippe and the Thomson BankWatch Asia team produced, as early as 1994 and 1995, forewarning reports on the weaknesses of Asia's banking systems that led to the Asian crisis of 1997.

After the crisis erupted, Philippe made countless presentations on all continents, and he conducted, with some of his senior analysts, a number of seminars on the Asian crisis. This led to a contract with John Wiley & Sons for Philippe to produce a book on the 1997 crisis that was very well received, and which we hope the reader will forgive us for quoting occasionally.

When in 1999 John Wiley & Sons started looking for a writer who could put together a comprehensive bank credit analysis handbook, Philippe had neither the time nor the courage to embark on such a voyage, but he encouraged Jonathan to take the plunge with the support of unlimited access to Philippe's notes and experience, something Jonathan gave him credit for in the first edition of the *Bank Credit Analysis Handbook*, published in 2001.

Meanwhile, Thomson BankWatch—at one point renamed Thomson Financial BankWatch—merged with Fitch in 2000, but Philippe and Jonathan quit prior to the merger. Philippe carried on teaching finance and conducting seminars on bank risk management in a number of countries. Recently, in Hong Kong, Philippe cofounded CTRisks Rating, a new rating agency using advanced techniques in the analysis of risk. Jonathan moved to London, where he founded two companies devoted to bank and company risk analysis.

During the 2000s, the risk profile of most banks changed dramatically. Many changes took place in the manner banks had to manage and report their own risks, and in the way such risks shaped a bank's own credit risk, as seen from the outside. Jonathan's book needed an overhaul rather than a cosmetic update. This is how eventually Jonathan and Philippe joined forces to present this new, expanded edition to our readers.

In the preface of the first edition, Jonathan thanked Darren Stubing for his substantial contribution to several chapters, and most likely some of Darren's original input still pervades this new version of the book. The same applies to texts contributed by Andrew Seiz in the first edition, and there is no doubt that research done by the Thomson BankWatch Asia team, together with some of their New York-based

colleagues, permeates the analytical line adopted both in Jonathan's first edition and in the present new edition of *The Bank Credit Analysis Handbook*. The only direct outside contribution to this edition is coming from Richard Lumley in the chapter on risk management. We are thankful to all direct and indirect contributors.

DRAMATIC CHANGES

The crisis that started in 2007 is still on at the time of writing. Banks and financial systems should share the blame with profligate politicians, outdated socioeconomic models, and a shift of the world's center of gravity toward newcomers.

However deep the resentment against banking and finance—often fanned by otherwise entertaining political slogans¹—banks are here to stay.

Banks remain a major conduit for the transformation of savings into productive investments. It is particularly so in emerging countries where capital markets are still not sufficiently developed and where savers have limited access to direct credit risk opportunities. Even in advanced economies, access to market risk often involves dealing with banks whose contribution as intermediaries is sometimes—and often justifiably—questionable.

More than most other financial intermediaries, banks do carry substantial credit and market risks. They act as shock absorbers by removing from their depositor's shoulders—and charging, alas, hefty fees for the service—some of that burden.

As we shall point out in this book, weak banks actually rarely fail—they often merge or get nationalized—or at least their problems rarely translate into losses for depositors² or creditors. Major disasters do occur, though, and we should not dismiss the view that the mere possibility of such an occurrence is enough for state ownership or state control of banks to gain respect in spite of the huge inefficiencies such models introduce. At the very least, banks should be submitted, within reason, to better regulatory control.

Banks, however, cannot survive unless they take risks. The trick for them is to manage those risks without destroying shareholder value—the fatter the better, from a creditworthiness point of view—and without endangering depositors and creditors.

STRUCTURE OF THE BOOK

This book explores the tools available to external analysts who wish to find out for themselves whether and to what extent a bank or a group of banks is creditworthy.

It is a jungle out there. A wide range of theoretical research is available. Extreme opinions exist on most topics, making it difficult to reach a consensus on a middle ground where depositors, creditors, and regulators should confine the banking systems' risk analysis.

Our book is a modest attempt at balancing the wealth of research and opinions within a useful handbook for analysts, regulators, risk assessment offices, and finance students.

Dividing bank credit analysis in separate chapters was a headache. Asset quality has an impact on earnings and on capital adequacy, liquidity on asset quality and earnings, management skills on asset quality, earnings on capital, accounting rules on earnings and capital—all on convoluted Möbius strips.

The first three chapters explore the notions associated with the credit decision, with the tools used in creditworthiness analysis and generally with the business of banking, more specifically with those activities that expose banks to risk.

Chapters 4 and 5 explore the earnings—or income, or profit and loss—statement and the balance sheet of a bank, together with the increasingly important off-balance sheet. Those documents are the first documents an analyst will be confronted with. Except for the reader already familiar with bank financial statements, those chapters are essential to understand how the various activities of the bank find their way into the final published documents that disclose—and sometimes conceal or disguise—the facts, figures, and ratios that should shape the analyst’s opinion on the bank’s creditworthiness.

The two accounting chapters pave the way for the introduction, in separate chapters, of the five basic elements of CAMEL, the mainstream model for assessing a bank’s performance and financial condition. Each of those five chapters relates back, in some way, to the two accounting chapters.

Chapter 6 discusses earnings and profitability, with their many indicators. Chapter 7 is the most important as it attempts to describe how the analyst can assess the asset quality of a bank, and how the bank monitors its assets and deals with nonperforming loans and with its exposure to other impaired assets or transactions.

Management and corporate governance are covered in Chapter 8, where the analyst will, among other things, learn how to appraise a bank’s overall management skills, which, in spite of tighter external regulations, remain a critical factor.

Chapter 9 is about capital and its various definitions and indicators. This is where a first round of comments touches on the Basel Accords, because the earlier versions of those accords focused almost exclusively on capital adequacy.

Liquidity, which is in Chapter 10, has become a major issue in the wake of the 2007–2012 crisis. It is also a very difficult parameter to analyze. No single indicator is able to describe a bank’s liquidity position, to the point where even the proposed liquidity requirements under Basel III do not bring much light to the debate.

Chapter 11 is about country and sovereign risks, which used to be relevant only to emerging markets but came to the fore during the 2011–2012 debt crisis in Europe. Globalization and the free circulation of funds around most of the world have now pushed the analysis of country and sovereign risk way beyond the traditional ratios describing such basic factors as inflation or balance of payments. Bank creditworthiness is more than ever influenced by macroeconomic factors.

Risk management is analyzed in Chapter 12, together with the second part of our exploration of the Basel Accords, to which we added a section on ratings. Risk management is no doubt the topic that saw the most changes over the past few years.

The banking regulatory regime is explored in Chapter 13, with its structural and prudential regulations as well as its impact on systemic issues.

The regional and worldwide crises of the past 20 years have generated considerable research on the causes of, and remedies to bank crises, financial crises, debt crises, sovereign crises, and their various combinations. Those crises are described and explained in Chapter 14, while Chapter 15—our last chapter—is devoted to the resolution of banking crises specifically.

We decided against offering a glossary of financial terms, as the book is already heavy and, in this day and age, the reader will no doubt find excellent glossaries on the Internet.

In our attempt to render the reader's task easier by dividing the book into 15 chapters, we created the need for many cross-references to other chapters. We believed that the reader would have neither the courage nor the need to swallow many chapters in one sitting, and we wanted, as much as possible, our chapter on, say, asset quality to cover most or all of what the reader would want to know when reading that chapter in isolation. Inevitably, as a result, there is a—small—degree of duplication here or there.

We would like to beg our readers' forgiveness for offering many examples from Asia. Both authors are thoroughly familiar with banking systems in that region—which admittedly is no justification in itself—while, more importantly, Asia is by far the largest financial market outside of the EU and the United States. In addition, whatever the definition of an emerging market, Asia without Japan arguably harbors the biggest emerging market banking system in the world, a fertile ground for dubious banking practices.

Considerable research is available on banking systems, banking crises, and other topics relevant to the bank credit analyst. As a matter of fact, so much information and so many opinions are offered that the analyst would need to invest a year of her life just to get acquainted with the existing literature on bank creditworthiness. Our modest ambition was to distill academic research into something palatable, to pepper our findings with information gathered over our many years of experience in bank credit analysis, and to offer our reader a useful reference handbook.

London and Port Arthur
September 2012

NOTES

1. Malaysia's Prime Minister Mahathir produced an interesting opinion in a speech on September 20, 1997 reported by the Manila Standard newspaper on September 22, 1997: "Currency trading is unnecessary, unproductive and immoral. . . . It should be illegal." As reported in French by *Le Parisien* newspaper on January 12, 2012, socialist François Hollande said on that day in a meeting during his campaign for the French presidency "Dans cette bataille qui s'engage, mon véritable adversaire n'a pas de nom, pas de visage, pas de parti, mais il gouverne; cet adversaire c'est le monde de la finance," which freely translates as: "In the battle that is starting, my true opponent has no name, no face, no party, but it reigns; this opponent is the world of finance."
2. Especially so where deposit insurance schemes exist.

The Bank Credit Analysis Handbook

The Credit Decision

CREDIT. Trust given or received; expectation of future payment for property transferred, or of fulfillment or promises given; mercantile reputation entitling one to be trusted;—applied to individuals, corporations, communities, or nations; as, to buy goods on credit.

—Webster’s Unabridged Dictionary, 1913 Edition

A bank lives on credit. Till it is trusted it is nothing; and when it ceases to be trusted, it returns to nothing.

—Walter Bagehot¹

People should be more concerned with the return of their principal than the return on their principal.

—Jim Rogers²

The word *credit* derives from the ancient Latin *credere*, which means “to entrust” or “to believe.”³ Through the intervening centuries, the meaning of the term remains close to the original; lenders, or *creditors*, extend funds—or “credit”—based upon the belief that the borrower can be *entrusted* to repay the sum advanced, together with interest, according to the terms agreed. This conviction necessarily rests upon two fundamental principles; namely, the creditor’s confidence that:

1. The borrower is, and will be, *willing to repay* the funds advanced
2. The borrower has, and will have, the *capacity to repay* those funds

The first premise generally relies upon the creditor’s knowledge of the borrower (or the borrower’s reputation), while the second is typically based upon the creditor’s understanding of the borrower’s financial condition, or a similar analysis performed by a trusted party.⁴

DEFINITION OF CREDIT

Consequently, a broad, if not all-encompassing, definition of credit is the realistic belief or expectation, upon which a lender is willing to act, that funds advanced will

be repaid in full in accordance with the agreement made between the party lending the funds and the party borrowing the funds.⁵ Correspondingly, *credit risk* is the possibility that events, as they unfold, will contravene this belief.

SOME OTHER DEFINITIONS OF CREDIT

Credit [is] nothing but the expectation of money, within some limited time.

—John Locke

Credit is at the heart of not just banking but business itself. Every kind of transaction except, maybe, cash on delivery—from billion-dollar issues of securities to getting paid next week for work done today—involves a credit judgment. . . . Credit . . . is like love or power; it cannot ultimately be measured because it is a matter of risk, trust, and an assessment of how flawed human beings and their institutions will perform.

—R. Taggart Murphy⁶

Creditworthy or Not

Put another way, a sensible individual with money to spare (i.e., savings or capital) will not provide credit on a commercial basis⁷—that is, will not make a loan—unless she believes that the borrower has both the requisite willingness and capacity to repay the funds advanced. As suggested, for a creditor to form such a belief rationally, she must be satisfied that the following two questions can be answered in the affirmative:

1. Will the prospective borrower be *willing*, so long as the obligation exists, to repay it?
2. Will the prospective borrower be *able* to repay the obligation when required under its terms?

Traditional credit analysis recognizes that these questions will rarely be amenable to definitive yes/no answers. Instead, they call for a judgment of probability. Therefore, in practice, the credit analyst has traditionally sought to answer the question:

What is the likelihood that a borrower will perform its financial obligations in accordance with their terms?

All other things being equal, the closer the probability is to 100 percent, the less likely it is that the creditor will sustain a loss and, accordingly, the lower the credit risk. In the same manner, to the extent that the probability is below 100 percent, the greater the risk of loss, and the higher the credit risk.

CASE STUDY: PREMODERN CREDIT ANALYSIS

The date: The last years of the nineteenth century

The place: A small provincial bank in rural England—let us call the institution Wessex Bank—located in the market town of Westport

Simon Brown, a manager of Wessex Bank, is contemplating a loan to John Smith, a newly arrived merchant who has recently established a bicycle shop in the town's main square. Smith's business has only been established a year or so, but trade has been brisk, judging by the increasing number of two-wheelers that can be seen on Westport's streets and in the surrounding countryside.

Yesterday, Smith called on Brown at his office, and made an application for a loan. The merchant's accounts, Brown noted, showed a burgeoning business, but one in need of capital to fund inventory expansion, especially in preparation for spring and summer, when prospective customers flock to the shop. While some of Smith's suppliers provide trade credit, sharply increasing demand for cycles and limited supply have caused them to tighten their own credit terms. Smith projected, not entirely unreasonably, thought Brown, that he could increase his turnover by 30 percent if he could acquire more stock and promise customers quick delivery.

When asked by Brown, Smith said he would be willing to pledge his assets, including the shop's inventory, as collateral to secure the loan. But Brown, as befits his reputation as a prudent banker, remained skeptical. Those newfangled machines were, in his view, dangerous vehicles and very likely a passing fad.

During the interview, Smith mentioned in passing that he was related on his father's side to Squire Roberts, a prosperous local landowner well known to Brown and a longstanding customer of Wessex Bank. Just that morning, Brown had seen the old gentleman at the post office, and, to his surprise, Roberts struck up a conversation about the weather and the state of the timber trade, and mentioned that he had heard his nephew had called on Brown recently. Before Brown had time to register the news that Roberts was Smith's uncle, Roberts volunteered that he was willing to vouch for Smith's character—"a fine lad"—and, moreover, added that he was willing to guarantee the loan.

Brown decided to have another look at Smith's loan application. Rubbing his chin, he reasoned to himself that the morning's news presented another situation entirely. Not only was Smith not the stranger he was before, but he was also a potentially good customer. With confirmation of his character from Roberts, Brown was on his way to persuading himself that the bank was probably adequately protected. Roberts's indication that he would guarantee the loan removed any remaining doubts. Should Smith default, the bank could hold the well-off Roberts liable for the obligation. Through the prospective substitution of Robert's creditworthiness for that of Smith's the bank's credit risk was considerably reduced. The last twinge of anxiety having been removed, Brown decided to approve the loan to Smith.

Credit Risk

Credit risk and the concomitant need for the estimation of that risk surface in many business contexts. It emerges, for example, when one party performs services for another and then sends a bill for the services rendered for payment. It also arises in connection with the settlement of transactions—where one party has advanced payment to the other and awaits receipt of the items purchased or where one party has advanced the items purchased and awaits payment. Indeed, most enterprises that buy and sell products or services, that is practically all businesses, incur varying degrees of credit risk. Only in respect to the simultaneous exchange of goods for cash can it be said that credit risk is essentially absent.

While nonfinancial enterprises, particularly small merchants, can eliminate credit risk by engaging only in *cash and carry* transactions, it is common for vendors to offer credit to buyers to facilitate a particular sale, or merely because the same terms are offered by their competitors. Suppliers, for example, may offer *trade credit* to purchasers, allowing some reasonable period of time, say 30 days, to settle an invoice. Risks arising from trade credit form a transition zone between settlement risk and the creation of a more fundamental financial obligation.

It is evident that as opposed to trade credit, as well as settlement risk that emerges during the consummation of a sale or transfer, fundamental financial obligation arises where sellers offer explicit financing terms to prospective buyers. This type of credit extension is particularly common in connection with purchases of big ticket items by consumers or businesses. As an illustration, automobile manufacturers frequently offer customers attractive finance terms as an incentive. Similarly, a manufacturer of electrical generating equipment may offer financing terms to facilitate the sale of the machinery to a power utility company. Such credit risk is essentially indistinguishable from that created by a bank loan.

In contrast to nonfinancial firms, which can choose to operate on a cash-only basis, banks by definition cannot avoid credit risk. The acceptance of credit risk is inherent to their operation since the very *raison d'être* of banks is the supply of credit through the advance of cash and the corresponding creation of financial obligations. Success in banking is attained not by avoiding risk but by effectively selecting and managing risk. In order to better manage risk, it follows that banks must be able to estimate the credit risk to which they are exposed as accurately as possible. This explains why banks almost invariably have a much greater need for credit analysis than do nonfinancial enterprises, for which, again by definition, the shouldering of credit risk exposure is peripheral to their main business activity.

Credit Analysis

For purposes of practical analysis, credit risk may be defined as the risk of monetary loss arising from any of the following four circumstances:

1. The default of a counterparty on a fundamental financial obligation
2. An increased probability of default on a fundamental financial obligation
3. A higher than expected loss severity arising from either a lower than expected recovery or a higher than expected exposure at the time of default
4. The default of a counterparty with respect to the payment of funds for goods or services that have already been advanced (settlement risk)

The variables most directly affecting relative credit risk include the following four:

1. The capacity and willingness of the obligor (borrower, counterparty, issuer, etc.) to meet its obligations
2. The external environment (operating conditions, country risk, business climate, etc.) insofar as it affects the probability of default, loss severity, or exposure at default
3. The characteristics of the relevant credit instrument (product, facility, issue, debt security, loan, etc.)
4. The quality and sufficiency of any credit risk mitigants (collateral, guarantees, credit enhancements, etc.) utilized

Credit risk is also influenced by the length of time over which exposure exists. At the portfolio level, correlations among particular assets together with the level of concentration of particular assets are the key concerns.

Components of Credit Risk

At the level of practical analysis, the process of credit risk evaluation can be viewed as formulating answers to a series of questions with respect to each of these four variables. The following questions are intended to be suggestive of the line of inquiry that might be pursued.

The Obligor's Capacity and Willingness to Repay

- What is the capacity of the obligor to service its financial obligations?
- How likely will it be to fulfill that obligation through maturity?
- What is the type of obligor and usual credit risk characteristics associated with its business niche?
- What is the impact of the obligor's corporate structure, critical ownership, or other relationships and policy obligations upon its credit profile?

The External Conditions

- How do country risk (sovereign risk) and operation conditions, including systemic risk, impinge upon the credit risk to which the obligee is exposed?
- What cyclical or secular changes are likely to affect the level of that risk? The obligation (product): What are its credit characteristics?

The Attributes of Obligation from Which Credit Risk Arises

- What are the inherent risk characteristics of that obligation? Aside from general legal risk in the relevant jurisdiction, is the obligation subject to any legal risk specific to the product?
- What is the tenor (maturity) of the product?
- Is the obligation secured; that is, are credit risk mitigants embedded in the product?
- What priority (e.g., senior, subordinated, unsecured) is assigned to the creditor (obligee)?
- How do specific covenants and terms benefit each party thereby increasing or decreasing the credit risk to which the obligee is exposed? For example, are there

any call provisions allowing the obligor to repay the obligation early; does the obligee have any right to convert the obligation to another form of security?

- What is the currency in which the obligation is denominated?
- Is there any associated contingent/derivative risk to which either party is subject?

The Credit Risk Mitigants

- Are any credit risk mitigants—such as collateral—utilized in the existing obligation or contemplated transaction? If so, how do they impact credit risk?
- If there is a secondary obligor, what is its credit risk?
- Has an evaluation of the strength of the credit risk mitigation been undertaken?

In this book, our primary focus will be on the obligor bank and the environment in which it operates, with consideration of the credit characteristics of specific financial products and accompanying credit risk mitigants relegated to a secondary position. The reasons are twofold. One, evaluation of the first two elements form the core of bank credit analysis. This is invariably undertaken before adjustments are made to take account of the impact of the credit characteristics of particular financial products or methods used to modify those characteristics. Two, to do justice to the myriad of different types of financial products, not to speak of credit risk mitigation techniques, requires a book in itself and the volume of material to be covered with regard to the obligor and the operating environment is greater than a single volume.

Credit Risk Mitigation

While the foregoing query concerning the likelihood that a borrower will perform its financial obligations is simple, its simplicity belies the intrinsic difficulties in arriving at a satisfactory, accurate, and reliable answer. The issue is not just the underlying probability of default, but the degree of uncertainty associated with forecasting this probability. Such uncertainty has long led lenders to seek security in the form of collateral or guarantees, both to mitigate credit risk and, in practice, to circumvent the need to analyze it altogether.

Collateral—Assets That Function to Secure a Loan

Collateral refers to assets that are either deposited with a lender, conditionally assigned to the lender pending full repayment of the funds borrowed, or more generally to assets with respect to which the lender has the right to obtain title and possession in full or partial satisfaction of the corresponding financial obligation. Thus, the lender who receives collateral and complies with the applicable legal requirements becomes a *secured creditor*, possessing specified legal rights to designated assets in case the borrower is unable to repay its obligation with cash or with other current assets.⁸ If the borrower defaults, the lender may be able to seize the collateral through foreclosure⁹ and sell it to satisfy outstanding obligations. Both secured and unsecured creditors may force the delinquent borrower into bankruptcy. The secured creditor, however, benefits from the right to sell the collateral without necessarily initiating bankruptcy proceedings, and stands in a better position than unsecured creditors once such proceedings have commenced.¹⁰

It is evident that, since collateral may generally be sold on the default of the borrower (the *obligor*), it provides security to the lender (the *obligee*). The prospective loss of collateral also gives the obligor an incentive to repay its obligation. In this way, the use of collateral tends to lower the probability of default, and, more significantly, reduce the severity of the creditor's loss in the event of default, by providing the creditor with full or partial recompense for the loss that would otherwise be incurred. Overall, collateral tends to reduce, or mitigate, the credit risk to which the lender is exposed, and it is therefore classified as a *credit risk mitigant*.

COLLATERAL AND OTHER CREDIT RISK MITIGANTS

Credit risk mitigants are devices such as collateral, pledges, insurance, or guarantees that may be used to reduce the credit risk exposure to which a lender or creditor would otherwise be subject. The purpose of credit risk mitigants is partially or totally to ameliorate a borrower's lack of intrinsic creditworthiness and thereby reduce the credit risk to the lender, or to justify advancing a larger sum than otherwise would be contemplated. For instance, a lender may require a guarantee where the borrower is comparatively new or lacks detailed financial statements but the guarantor is a well-established enterprise rated by the major external agencies. In the past, these mechanisms were frequently used to reduce or eliminate the need for the credit analysis of a prospective borrower by substituting conservatively valued collateral or the creditworthiness of an acceptable guarantor for the primary borrower.

In modern financial markets, collateral and guarantees, rather than being substitutes for inadequate stand-alone creditworthiness, may actually be a requisite and integral element of the contemplated transaction. Their essential function is unchanged, but instead of remedying a deficiency, they are used to increase creditworthiness to give the transaction certain predetermined credit characteristics. In these circumstances, rather than eliminating the need for credit analysis, consideration of credit risk mitigants supplements, and sometimes complicates it. Real-life credit analysis consequently requires an integrated approach to the credit decision, and typically requires some degree of analysis of both the primary borrower and of the impact of any applicable credit risk mitigants.

Since the amount advanced is known, and because collateral can normally be appraised with some degree of accuracy—often through reference to the market value of comparable goods or assets—the credit decision is considerably simplified. By obviating the need to consider the issues of the borrower's willingness and capacity, the question—*What is the likelihood that a borrower will perform its financial obligations in accordance with their terms?*—can be replaced with one more easily answered, namely: “Will the collateral provided by the prospective borrower be sufficient to secure repayment?”¹¹

As Roger Hale, the author of an excellent introduction to credit analysis, succinctly puts it: “If a pawnbroker lends money against a gold watch, he does not need credit analysis. He needs instead to know the value of the watch.”¹²

Guarantees

A *guarantee* is the promise by a third party to accept liability for the debts of another in the event that the primary obligor defaults, and is another kind of credit risk mitigant. Unlike collateral, the use of a guarantee does not eliminate the need for credit analysis, but simplifies it by making the guarantor instead of the borrower the object of scrutiny.

Typically, the guarantor will be an entity that either possesses greater creditworthiness than the primary obligor, or has a comparable level of creditworthiness but is easier to analyze. Often, there will be some relationship between the guarantor and the party on whose behalf the guarantee is provided. For example, a father may guarantee a finance company’s loan to his son¹³ for the purchase of a car. Likewise, a parent company may guarantee a subsidiary’s loan from a bank to fund the purchase of new premises.

Where a guarantee is provided, the questions posed with reference to the prospective borrower must be asked again in respect of the prospective guarantor: “Will the prospective guarantor be both willing to repay the obligation and have the capacity to repay it?” These questions are summarized in Exhibit 1.1.

EXHIBIT 1.1 Key Credit Questions

		Binary (Yes/No)	Probability
Willingness to pay	Primary Subject of Analysis (e.g., borrower)	Will the prospective borrower be willing to repay the funds?	What is the likelihood that a borrower will perform its financial obligations in accordance with their terms?
Capacity to pay		Will the prospective borrower be able to repay the funds?	
Collateral	Secondary Subject of Analysis (Credit risk mitigants)	Will the collateral provided by the prospective borrower or the guarantees given by a third party be sufficient to secure repayment?	What is the likelihood that the collateral provided by the prospective borrower or the guarantees given by a third party will be sufficient to secure repayment?
Guarantees		Will the prospective guarantor be willing to repay the obligation as well as have the capacity to repay it?	What is the likelihood that the prospective guarantor will be willing to repay the obligation as well as have the capacity to repay it?

Significance of Credit Risk Mitigants

In view of the benefits of using collateral and guarantees to avoid the sometimes thorny task of performing an effective financial analysis,¹⁴ banks and other institutional lenders traditionally have placed primary emphasis on these credit risk mitigants, and other comparable mechanisms such as *joint and several liability*¹⁵ when allocating credit.¹⁶ For this reason, *secured lending*, which refers to the use of credit risk mitigants to secure a financial obligation as discussed, remains a favored method of providing financing.

In countries where financial disclosure is poor or the requisite analytical skills are lacking, credit risk mitigants circumvent some of the difficulties involved in performing an effective credit evaluation. In developed markets, more sophisticated approaches to secured lending such as *repo finance* and *securities lending*¹⁷ have also grown increasingly popular. In these markets, however, the use of credit risk mitigants is often driven by the desire to facilitate investment transactions or to structure credit risks to meet the needs of the parties to the transaction rather than to avoid the process of credit analysis.

With the evolution of financial systems, credit analysis has become increasingly important and more refined. For the moment, though, our focus is upon credit evaluation in its more basic and customary form.

WILLINGNESS TO PAY

Willingness to pay is, of course, a subjective attribute that can be ascertained to a degree from the borrower's reputation and apparent character. Assuming free will,¹⁸ it is also essentially unknowable in advance, even perhaps to the borrower. From the perspective of the lender or credit analyst, the evaluation is therefore necessarily a *qualitative* one that takes into account information gleaned from a variety of sources, including, where possible, face-to-face meetings that are a customary part of the process of *due diligence*.¹⁹

The old-fashioned provincial banker who was familiar with local business conditions and prospective borrowers, like the fictional character described earlier, had less need for formal credit analysis. Instead, the intuitive judgment that came from an in-depth knowledge of a community and its members was an invaluable attribute in the banking industry. The traditional banker knew with whom he was dealing (or thought he did), either locally with his customers or at a distance with *correspondent banks*²⁰ that he trusted. Walter Bagehot, the nineteenth-century British economic commentator put it well:

*A banker who lives in the district, who has always lived there, whose whole mind is a history of the district and its changes, is easily able to lend money there. But a manager deputed by a central establishment does so with difficulty. The worst people will come to him and ask for loans. His ignorance is a mark for all the shrewd and crafty people thereabouts.*²¹

In general, modern credit analysis still takes account of willingness to pay, and in doing so maintains an unbroken link with its past. It is still up to one or more

individuals to decide whether to extend or to repay a debt, and manuals on banking and credit analysis as a rule make some mention of the importance of taking account of a prospective borrower's character.²²

Indicators of Willingness

Willingness to pay, though real, is difficult to assess. Ultimately, judgments about this attribute, and the criteria on which they are based, are highly subjective in nature.

Character and Reputation

First-hand awareness of a prospective borrower's character affords at least a stepping-stone on which to base a credit decision. Where direct familiarity is lacking, a sense of the borrower's reputation provides an alternative footing upon which to ascertain the obligor's disposition to make good on a promise. Reliance on reputation can be perilous, however, since a dependence upon second-hand information can easily descend into so-called *name lending*.²³ Name lending can be defined as the practice of lending to customers based on their perceived status within the business community instead of on the basis of facts and sound conclusions derived from a rigorous analysis of the prospective borrowers' actual capacity to service additional debt.

Credit Record

Although far more data is available today than a century ago, assessing a borrower's integrity and commitment to perform an obligation still requires making unverifiable, even intuitive, judgments. Rather than put a foot wrong into a miasma of imponderables, creditors have long taken a degree of comfort not only in collateral and guarantees, but also in a borrower's verifiable history of meeting its obligations.

As compared with the prospective borrower who remains an unknown quantity, a track record of borrowing funds and repaying them suggests that the same pattern of repayment will continue in the future.²⁴ If available, a borrower's payment record, provided for example through a credit bureau, can be an invaluable resource for a creditor. Of course, while the past provides some reassurance of future willingness to pay, here as elsewhere, it cannot be extrapolated into the future with certainty in any individual case.²⁵

Creditors' Rights and the Legal System

While the ability to make the requisite intuitive judgments concerning willingness to pay probably comes more easily to some than to others, and no doubt may be honed with experience, perhaps fortunately it has become less important in the credit decision-making process.²⁶ The concept of a *moral obligation*²⁷ to repay a debt—which perhaps in the past arguably bolstered the will of the faltering borrower to perform his obligation in full—has been to a large extent displaced in contemporary commerce by legal rather than ethical norms.

It is logical to rank capacity to pay as more important than willingness, since willingness alone is of little value where capacity is absent. Capacity without willingness, however, can be overcome to a large degree through an effective legal system.²⁸ The stronger and more effectual the legal infrastructure, the better able a creditor is to enforce a judgment against a borrower.²⁹ Prompt court decisions backed by the threat of the seizure of possessions or other means through the arm of

the state will tend to predispose the nonperforming debtor to fulfill its obligations. A borrower who *can* pay but *will not*, is only able to maintain such a position in a legal regime that is ineffective or corrupt, or very strongly favors debtors over creditors.

So as legal systems have developed—along with the evolution of financial analytical techniques and data collection and distribution systems—the attribute of *willingness* to repay has been increasingly overshadowed in importance by the attribute of *capacity* to repay. It follows that the more a legal system exhibits creditor-friendly characteristics—combined with the other critical attributes of integrity, efficiency, and judges' understanding of commercial requirements—the less the lender needs to rely upon the borrower's willingness to pay, and the more important the capacity to repay becomes. The development of capable legal systems has therefore increased the importance of financial analysis and as a prerequisite to it, financial disclosure. Overall, the evolution of more robust and efficient legal systems has provided a net benefit to creditors.³⁰

Willingness to pay, however, remains a more critical criterion in less-developed markets, where the quality of the legal framework may be lacking. In these instances, the efficacy of the legal system in protecting creditors' rights also emerges as an important criterion in the analytical process.³¹

CREDIT ANALYSIS IN EMERGING MARKETS: THE IMPORTANCE OF THE LEGAL SYSTEM

Weak legal and regulatory infrastructure and concomitant doubts concerning the fair and timely enforcement of creditors' rights mean that credit analysis in so-called *emerging markets*³² is often more subjective than in developed markets. Due consideration must be given in these jurisdictions not only to a prospective borrower's willingness to pay, but equally to the quality of the legal system. Since, as a practical matter, willingness to pay is inextricably linked to the variables that may affect the lender's ability to coerce payment through legal redress, it is useful to consider, as part of the analytical process, the overall effectiveness and creditor-friendliness of a country's legal infrastructure. Like the evaluation of an individual borrower's willingness to pay, an evaluation of the quality of a legal system and the strength of a creditor's rights is a highly qualitative endeavor.

Despite the not inconspicuous inadequacies in the legal frameworks of the countries in which they extend credit, bankers during periods of economic expansion have time and again paid insufficient attention to prospective problems they might confront when a boom turns to bust. Banks have faced criticism for placing an undue reliance upon expectations of government support or, where the government itself is vulnerable to difficulties, upon the International Monetary Fund (IMF). Believing that the IMF would stand ready to provide aid to the governments concerned and thereby indirectly to the borrowers and to their creditors, it has been asserted that banks have engaged in imprudent lending. Insofar as such reliance has occurred, it has arguably been accompanied by a degree of obliviousness on the part of creditors to the difficulties involved in enforcing their rights through legal action.³³

While the quality of a country's legal system is a real and significant attribute, measuring it is no simple task. Traditionally, *sovereign risk ratings* functioned as a proxy for, among other things, the legal risk associated with particular geographic markets. Countries with low sovereign ratings were often implicitly assumed to be subject to a greater degree of legal risk, and vice versa.

In the past 15 years, however, surveys have been conducted in an attempt systematically to grade, if not measure, comparative legal risk. Although by and large these studies have been initiated for purposes other than credit analysis—to assess a country's investment climate, for instance—they would seem to have some application to the evaluation of credit risk. The table in Exhibit 1.2 shows the scores under such an index of rule of law. Some banks have used one or more similar surveys, sometimes together with other criteria, to generate internal creditors' rights ratings for the jurisdictions in which they operate or in which they contemplate credit exposure.

EXHIBIT 1.2 Rule of Law Index: Selected Countries 2010

Country	Score	Country	Score
Finland	1.97	Bermuda	1.16
Sweden	1.95	Guam	1.16
Norway	1.93	Estonia	1.15
Denmark	1.88	Portugal	1.04
New Zealand	1.86	Barbados	1.04
Luxembourg	1.82	Slovenia	1.02
Netherlands	1.81	Tuvalu	1.02
Austria	1.80	Taiwan, China	1.01
Canada	1.79	Korea, South	0.99
Switzerland	1.78	Antigua and Barbuda	0.98
Australia	1.77	Czech Republic	0.95
United Kingdom	1.77	Monaco	0.90
Ireland	1.76	San Marino	0.90
Greenland	1.72	Martinique	0.89
Singapore	1.69	Netherlands Antilles	0.89
Iceland	1.69	Reunion	0.89
Germany	1.63	Virgin Islands (U.S.)	0.89
Liechtenstein	1.62	Cayman Islands	0.89
United States	1.58	Israel	0.88
Hong Kong Sar, China	1.56	Qatar	0.87
France	1.52	St. Vincent and the Grenadines	0.86
Malta	1.48	Mauritius	0.84
Anguilla	1.42	St. Lucia	0.82
Aruba	1.42	Latvia	0.82
Belgium	1.40	Brunei	0.80
Japan	1.31	Hungary	0.78
Chile	1.29	Puerto Rico	0.77
Andorra	1.23	Lithuania	0.76
Spain	1.19	Palau	0.74
Cyprus	1.19	Uruguay	0.72
French Guiana	1.18	St. Kitts And Nevis	0.71
American Samoa	1.16	Macao Sar, China	0.71

Country	Score	Country	Score
Dominica	0.69	Rwanda	−0.31
Poland	0.69	Maldives	−0.33
Bahamas	0.68	Colombia	−0.33
Oman	0.67	China	−0.35
Botswana	0.66	Bosnia–Herzegovina	−0.36
Samoa	0.65	Belize	−0.36
Greece	0.62	Serbia	−0.39
Slovakia	0.58	Moldova	−0.40
Kuwait	0.54	Uganda	−0.40
Malaysia	0.51	Senegal	−0.41
Costa Rica	0.50	Mongolia	−0.43
Bahrain	0.45	Albania	−0.44
Cape Verde	0.42	Mali	−0.46
Nauru	0.41	Armenia	−0.47
United Arab Emirates	0.39	Guyana	−0.48
Italy	0.38	Vietnam	−0.48
Vanuatu	0.25	Zambia	−0.49
Namibia	0.23	Swaziland	−0.50
Jordan	0.22	Jamaica	−0.50
Croatia	0.19	Mozambique	−0.50
Saudi Arabia	0.16	Tanzania	−0.51
Grenada	0.11	Gambia	−0.51
Tunisia	0.11	Gabon	−0.51
Bhutan	0.11	Syria	−0.54
Turkey	0.10	Philippines	−0.54
South Africa	0.10	Cuba	−0.55
Tonga	0.09	Mexico	−0.56
Kiribati	0.07	Niger	−0.57
Romania	0.05	Argentina	−0.58
Seychelles	0.02	Peru	−0.61
Brazil	0.00	Kazakhstan	−0.62
Montenegro	−0.02	Indonesia	−0.63
India	−0.06	Kosovo	−0.64
Ghana	−0.07	Lebanon	−0.66
Micronesia	−0.08	São Tomé and Príncipe	−0.69
Bulgaria	−0.08	Solomon Islands	−0.70
Sri Lanka	−0.09	Djibouti	−0.71
Suriname	−0.09	Niue	−0.72
Egypt	−0.11	Benin	−0.73
Panama	−0.13	Ethiopia	−0.76
Malawi	−0.14	Algeria	−0.76
Morocco	−0.19	Bangladesh	−0.77
Thailand	−0.20	Russia	−0.78
West Bank Gaza	−0.20	Pakistan	−0.79
Georgia	−0.21	Ukraine	−0.80
Burkina Faso	−0.21	Dominican Republic	−0.81
Trinidad and Tobago	−0.22	Nicaragua	−0.83
Marshall Islands	−0.27	Madagascar	−0.84
Macedonia	−0.29	Honduras	−0.87
Lesotho	−0.30	El Salvador	−0.87

(Continued)

EXHIBIT 1.2 (Continued)

Country	Score	Country	Score
Mauritania	-0.88	Nigeria	-1.21
Azerbaijan	-0.88	Timor-Leste	-1.21
Laos	-0.90	Burundi	-1.21
Cook Islands	-0.90	Côte d'Ivoire	-1.22
Iran	-0.90	Angola	-1.24
Fiji	-0.90	Equatorial Guinea	-1.26
Paraguay	-0.92	Eritrea	-1.29
Togo	-0.92	Kyrgyzstan	-1.29
Papua New Guinea	-0.93	Korea, North	-1.30
Sierra Leone	-0.94	Central African Republic	-1.30
Libya	-0.98	Sudan	-1.32
Liberia	-1.01	Guinea-Bissau	-1.35
Kenya	-1.01	Haiti	-1.35
Nepal	-1.02	Uzbekistan	-1.37
Guatemala	-1.04	Turkmenistan	-1.46
Cameroon	-1.04	Chad	-1.50
Belarus	-1.05	Myanmar	-1.50
Yemen	-1.05	Guinea	-1.51
Comoros	-1.06	Congo, Democratic Republic	-1.61
Bolivia	-1.06	Iraq	-1.62
Cambodia	-1.09	Venezuela	-1.64
Congo	-1.13	Zimbabwe	-1.80
Ecuador	-1.17	Afghanistan	-1.90
Tajikistan	-1.20	Somalia	-2.43

Source: World Bank.

It is almost invariably the case that the costs of legal services are an important variable to be considered in any decision regarding the recovery of money owed. A robust legal system is not necessarily a cost-effective one, since the expenses required to enforce a creditor's rights are rarely insignificant. While a modicum of efficiency may exist, the costs of legal actions, including the time spent pursuing them, may well exceed the benefits. It therefore may not pay to take legal action against a delinquent borrower. This is particularly the case for comparatively small advances. As a result, even where creditors' rights are strictly enforced, willingness to pay ought never to be entirely ignored as an element of credit analysis.

EVALUATING THE CAPACITY TO REPAY: SCIENCE OR ART?

Compared with *willingness* to pay, the evaluation of *capacity* to pay lends itself more readily to *quantitative* measurement. So the application of financial analysis will generally go far in revealing whether the borrower will have the ability to fulfill outstanding obligations as they come due. Evaluating the capability of an entity to perform its financial obligations through a close examination of numerical data derived from its most recent and past financial statements forms the core of *credit analysis*.

The Limitations of Quantitative Methods

While an essential element of credit evaluation, the use of financial analysis for this purpose is subject to serious limitations. These include:

- The historical character of financial data.
- The difficulty of making reasonably accurate financial projections based upon such data.
- The inevitable gap between financial reporting and financial reality.

Historical Character of Financial Data

The first and most obvious limitation is that financial statements are invariably historical in scope, covering as they do past fiscal reporting periods, and are therefore never entirely up to date. Because the past cannot be extrapolated into the future with any certainty, except perhaps in cases of clear *insolvency* and *illiquidity*, the estimation of capacity remains just that: an estimate.

Even if financial reports are comparatively recent, or forward looking, the preceding difficulty is not surmounted. Accurate financial forecasting is notoriously problematic, and, no matter how sophisticated, financial projections are highly vulnerable to errors and distortion. Small differences at the outset can engender an enormous range of values over time.

Financial Reporting Is Not Financial Reality

Perhaps the most significant limitation arises from the fact that financial reporting is an inevitably imperfect attempt to map an underlying economic reality into a usable but highly abbreviated condensed report. As with attempts to map a large spherical surface onto a flat projection, some degree of deformation is unavoidable, while the very process of distilling raw data into a work product small enough to be usable requires that some data be selected and other data be omitted. In short, not only do financial statements intrinsically suffer from some degree of distortion and omission, these deficiencies are also apt to be aggravated in practice.

First, the rules of financial accounting and reporting are shaped by people and institutions having differing perspectives and interests. Influences resulting from that divergence are apt to aggravate these innate deficiencies. The rules themselves are almost always the product of compromises by committee that are, at heart, political in nature.

Second, the difficulty of making rules to cover every conceivable situation means that, in practice, companies are frequently afforded a great deal of discretion in determining how various accounting items are treated. At best, such leeway may only potentially result in inaccurate comparisons; at worst, this necessary flexibility in interpretation and classification may be used to further deception or fraud.

Finally, even the most accurate financial statements must be interpreted. In this context too, differing vantage points, experience, and analytical skill levels may result in a range of conclusions from the same data. For all these reasons, it should be apparent that even the seemingly objective evaluation of financial capacity retains a significant qualitative, and therefore subjective, component. While acknowledging both its limitations and subjective element, financial analysis remains at the core of

effective credit analysis. The associated techniques serve as essential and invaluable tools for drawing conclusions about a company's creditworthiness, and the credit risk associated with its obligations.

It is, nevertheless, crucial not to place too much faith in the quantitative methods of financial analysis in credit risk assessment, nor to believe that quantitative data or conclusions drawn from such data necessarily represent an objective truth. No matter how sophisticated, when applied for the purpose of the evaluation of credit risk, these techniques must remain imperfect tools that seek to predict an unknowable future.

Quantitative and Qualitative Elements

Given these shortcomings, the softer more qualitative aspects of the analytical process should not be given short shrift. Notably, an evaluation of management—including its competence, motivation, and incentives—as well as the plausibility and coherence of its strategy remains an important element of credit analysis of both *nonfinancial and financial companies*.³⁴ Indeed, as suggested in the previous subsection, not only is credit analysis both qualitative and quantitative in nature, but nearly all of its nominally quantitative aspects also have a significant qualitative element.

While evaluation of willingness to pay and assessment of management competence obviously involve subjective judgments, so too, to a larger degree than is often recognized, do the presentation and analysis of a firm's financial results. Credit analysis is as much art as it is science, and its successful application relies as much on judgment as it does on mathematics. The best credit analysis is a synthesis of quantitative measures and qualitative judgments. For reasons that will soon become apparent, this is particularly so in regard to financial institution credit analysis.

QUANTITATIVE METHODS IN CREDIT ANALYSIS

These remain imperfect tools that aim to predict an unknowable future. Nearly all of the nominally quantitative techniques also have a significant qualitative element. To reach optimal effectiveness, credit analysis must therefore combine the effective use of quantitative tools with sound qualitative judgments.

Credit Analysis versus Credit Risk Modeling

At this stage, it should be noted that there is an important distinction to be drawn between credit risk analysis, on the one hand, and credit risk modeling and credit risk management, on the other. The process of performing a counterparty credit analysis is quite different from that involved in modeling bank credit risk or in managing credit risk at the enterprise level. Consider, for example, the concept of rating migration risk.

Rating migration risk, while an important factor in modeling and evaluating portfolios of debt securities, is not, however, of concern to the credit analyst

performing an evaluation of the kind upon which its rating has been based. It is important to recognize this distinction and to emphasize that the aim of the credit analyst is not to model credit risk, but instead to perform the evaluation that provides one of the requisite inputs to credit risk models. Needless to say, it is also one of the requisite inputs to the overall risk management of a banking organization.

RATING MIGRATION RISK

Credit risk is defined as the risk of loss arising from default. Of all the credit analyst roles, rating agency analysts probably adhere most closely to that definition in performing their work. Rating agencies are in the financial information business. They do not trade in financial assets. The function of rating analysts is therefore purely to evaluate, through the assignment of rating grades, the relative credit risk of subject exposures. Traditionally, agency ratings assigned to a given issuer represent, in the aggregate, some combination of probability of default and loss-given-default.

However, the fixed income analyst and, on occasion, the counterparty credit analyst, may be concerned with a superficially different form of credit risk that, ironically, can be attributed in part to the rating agencies themselves. The fixed income analyst, especially, is worried not only about the expected loss arising from default, but also about the risk that a company's bonds, or other debt instruments, may be downgraded by an external rating agency. Although rating agencies ostensibly merely provide an opinion as to the degree of default risk, the very act of providing such assessments tends to have an impact on the market.

For example, the downgrade of an issuer's bond rating by one or more external agencies will often result in those bonds having a lower value in the market, even though the actual financial condition of the company and the risk of default may not have altered between the day before the downgrade was announced and the day after. For this reason, this type of credit risk is sometimes distinguished from the credit risk engendered by the possibility of default. It is called downgrade risk, or, more technically, rating migration risk, meaning the risk that the rating of an obligor will change with an adverse effect on the holder of the obligor's securities.

At first glance, downgrade risk might be attributed to the role rating agencies play in the market as arbiters of credit risk. But even if no credit rating agencies existed, a risk akin to rating migration risk would exist: the risk of a change in credit quality. Through the flow of information in the market, any significant change in the credit quality of an issuer or counterparty should ultimately manifest in a change in its credit risk assessment made by market participants. At the same time, the changes in perceived creditworthiness would be reflected in market prices implying a change in risk premium commensurate with the price change. Nevertheless, the risk of a decline in credit quality is at the end of the day only of concern insofar as it increases the risk of default. It can therefore be viewed simply as a different manifestation of default risk rather than constituting a discrete form of risk. Nevertheless, rating

migration is used in some credit risk models, as it usefully functions as a proxy for changes in the probability of default over time.

Ideally, downgrade risk should be equivalent to the risk of a decline in credit quality. In practice, however, there will inevitably be gaps between the rating assigned to a credit exposure and its actual quality as the latter improves or declines incrementally over time. What distinguishes the risk of a decline in credit quality from default risk as conventionally perceived is its impact on securities pricing. A decline in credit quality will almost always be reflected in a widening of spreads above the risk-free rate and a decline in the price of the debt securities affected by the decline. Since price risk by definition constitutes market risk, separating the market risk element from the credit risk element in debt pricing is no easy task.

CATEGORIES OF CREDIT ANALYSIS

Until now, we have been looking at the credit decision generally, without reference to the category of borrower. While *capacity* means having access to the necessary funds to repay a given financial obligation, in practice the evaluation of capacity is undertaken with a view to both the type of borrower and the nature of the financial obligation contemplated. Here the focus is on the category of borrower.

Very broadly speaking, credit analysis can be divided into four areas according to borrower type. The four principal categories of borrowers are consumers, nonfinancial companies (corporates), financial companies—of which the most common are banks—and government and government-related entities. The four corresponding areas of credit analysis are listed together with a brief description:

1. Consumer credit analysis is the evaluation of the creditworthiness of individual consumers.
2. Corporate credit analysis is the evaluation of nonfinancial companies such as manufacturers, and nonfinancial service providers.
3. Financial institution credit analysis is the evaluation of financial companies including banks and nonbank financial institutions (NBFIs), such as insurance companies and investment funds.
4. Sovereign/municipal credit analysis is the evaluation of the credit risk associated with the financial obligations of nations, subnational governments, and public authorities, as well as the impact of such risks on obligations of nonstate entities operating in specific jurisdictions.

While each of these areas of credit assessment shares similarities, there are also significant differences. To analogize to the medical field, *surgeons* might include orthopedic surgeons, heart surgeons, neurosurgeons, and so on. But you would not necessarily go to an orthopedic surgeon for heart surgery or a heart surgeon for brain surgery. Although the primary subject of this book is the credit analysis of banks, in describing the context in which this specialist activity takes place, it is worth taking a broad look at the entire field.

To begin, let us consider how one might go about evaluating the capacity of an individual to repay his debts, and then briefly consider the same process in reference to both nonfinancial (i.e., corporate)³⁵ and financial companies.

Individual Credit Analysis

In the case of individuals, common sense tells us that their wealth, often measured as *net worth*,³⁶ would almost certainly be an important measure of capacity to repay a financial obligation. Similarly, the amount of incoming cash at an individual's disposal—either in the form of salary or returns from investments—is plainly a significant attribute as well. Since for most individuals, earnings and cash flow are generally equivalent, *income*³⁷ and net worth provide the fundamental criteria for measuring their capacity to meet financial obligations.

CONSUMER CREDIT ANALYSIS

The comparatively small amounts at risk to individual consumers, broad similarities in the relative structure of their financial statements, the large number of transactions involved, and accompanying availability of data allow consumer credit analysis to be substantially automated through the use of credit-scoring models.

As is our hypothetical Chloe, below, most individuals are employed by businesses or other enterprises, earn a salary and possibly bonuses or commissions, and typically own assets of a similar type, such as a house, a car, and household furnishings. With some exceptions, cash flow as represented by the individual's salary tends to be fairly regular, as are household expenses. Moreover, unsecured³⁸ credit exposure to individuals by creditors is generally for relatively small amounts. Unsurprisingly, default by consumers is very often the result of loss of income through unemployment or unexpectedly high expenses, as may occur through sudden and severe illness in the absence of health insurance.

Because the credit analysis of individuals is usually fairly simple in nature, it is amenable to automation and the use of statistical tools to correlate risk to a fairly limited number of variables. Moreover, because the amounts advanced are comparatively small, it is generally not cost-effective to perform a full credit evaluation encompassing a detailed analysis of financial details and a due diligence interview of the individual concerned. Instead, scoring models that take account of various household characteristics such as salary, duration of employment, amount of debt, and so on, are typically used, particularly with respect to unsecured debt (e.g., credit card obligations). Substantial credit exposure by creditors to individual consumers will ordinarily be in the form of secured borrowing, such as *mortgage lending* to fund a house purchase or auto finance to fund a car purchase. In these situations, scoring methodologies are also employed, but may be coupled with a modest amount of manual input and review.

CASE STUDY: INDIVIDUAL CREDIT ANALYSIS

Net worth means an individual’s surplus of assets over debts. Consider, for example, a hypothetical 33-year-old woman named Chloe Williams, who owns a small house on the outskirts of a medium-sized city, let us call it Oakport, worth \$140,000.³⁹ There is a remaining mortgage of \$100,000 on the house, and Chloe has \$10,000 in savings, solely in the form of bank deposits and mutual funds, and no other debts (see Exhibit 1.3).

Leaving aside the value of Chloe’s personal property—clothes, jewelry, stereo, computer, motor scooter, for instance—she would have a net worth of \$50,000. Chloe’s salary is \$36,000 per annum after tax. Since her salary is paid in equal and regular installments in arrears (at the end of the relevant period) on the fifteenth and the last day of each month, we can equate her after-tax income with cash flow. Leaving aside nominal interest and dividend income, her total monthly cash flow (see Exhibit 1.4) would be \$3,000 per month.

EXHIBIT 1.3 Chloe's Net Worth

Chloe's Assets	\$	Chloe's Obligations and Equity	\$	Remarks:
Chloe's House				2-bedroom house at 128 Bayview Drive, Current market value: \$140,000
Portion of house value STILL owned to bank	100,000	Liabilities— mortgage owed to bank (Chloe's mortgage on her house: financial obligation to bank)	100,000	A single major liability—the funds she owes to the bank, which is an obligation secured by her house
Portion of house value NOT owned by bank—relatively illiquid	40,000	Home equity— unrealized if she sells the house	40,000	]—Chloe's Net Worth = \$50,000
Cash and Securities		Equity in		
Owens in full without margin loans—liquid assets	10,000	securities— unrealized unless she sells them	10,000	
	<u>150,000</u>		<u>150,000</u>	

Notes: Value of some assets (house, securities) depends on their market value. Traditionally a business would value them at their fair value or cost.

EXHIBIT 1.4 Chloe's Cash Flow

	Annually \$	Monthly \$
Chloe's after-tax income	36,000	3,000
Less: Salary applied to living costs and mortgage payment	(26,000)	(2,167)
Net cash flow available to service debt	10,000	833

Net Cash Flow

Net cash flow⁴⁰ is what remains after taking account of Chloe's other outgoings: utilities, groceries, mortgage payments, and so on.

To analyze Chloe's capacity to repay additional indebtedness, it is therefore reasonable to consider her net worth and income, together with her net cash flow, her track record in meeting obligations, and her level of job security, among other things. That Chloe has an impeccable credit record, has been with her company, an established Fortune 500 corporation for six years, with a steadily increasing salary and significant net worth would typically be viewed by a bank manager as *credit positive*.⁴¹

Evaluating the Financial Condition of Nonfinancial Companies

The process of evaluating the capacity of a firm to meet its financial obligations is similar to that used to assess the capacity of an individual to repay his debts. Generally speaking, however, a business enterprise is more difficult to analyze than an individual. Not only do enterprises vary hugely in the character of their assets, the regularity of their income stream and the degree to which they are subject to demands for cash, but also the financial structure of firms is almost always more complex than it is for individuals. In addition, the interaction of each of the preceding attributes complicates matters. Finally, the amount of funds at stake is usually significantly higher—and not infrequently far higher—for companies than it is for consumers. As a consequence, the credit analysis of nonfinancial companies tends to be more detailed and more hands-on than consumer credit analysis.

It is both customary and helpful to divide the credit analysis of organizations according to the attributes to be analyzed. As a paradigm, consider the *corporate credit analyst* evaluating credit exposure to a nonfinancial firm, whether in the form of financial obligations in the form of bonds issued by multinational firms or bread-and-butter loans to be made to an industrial or service enterprise. As a rule, the analyst will be particularly concerned with the following criteria and this will be reflected in the written report that sets forth the conclusions reached:

- The company's *liquidity*
- Its *cash flow* together with
- Its near-term *earnings capacity* and *profitability*
- Its *solvency* or *capital* position

Each of these attributes is relevant also to the analysis of financial companies.

Evaluating Financial Companies

The elements of credit analysis applicable to banks and other financial companies share many similarities to those applied to nonfinancial enterprises. The attributes of liquidity, solvency, and historical performance mentioned are all highly relevant to financial institutions. As with corporate credit analysis, the quality of management, the state of the economy, and the industry environment are also vital factors in evaluating financial company creditworthiness.

Yet, as the business of financial companies differs in fundamental respects from that of nonfinancial businesses, so too does their analysis. These differences have a substantial impact on how the performance and condition of the former are evaluated. Similarly, how various financial characteristics of banks are measured and the weight given to various categories of their performance contrast in many respects with the manner in which corporates are analyzed.

Suffice it to say for the moment that the key areas that a credit analyst will focus on in evaluating a bank include the following:

- *Earnings capacity*—that is, the bank's performance over time, particularly its ability to generate *operating income* and *net income* on a sustained basis and thereby overcome any difficulties it may confront.
- *Liquidity*—that is, the bank's access to cash or cash equivalents to meet current obligations.
- *Capital adequacy* (a term frequently used in the context of financial institutions that is essentially equivalent to *solvency*)—that is, the cushion that the bank's capital affords it against its liabilities to depositors and the bank's creditors.
- *Asset quality*—that is, the likelihood that the loans the bank has extended to its customers will be repaid, taking into account the value and enforceability of collateral provided by them.

Even in this brief list, differences between the key criteria applied to corporate credit analysis and those important in the credit analysis of financial and nonfinancial companies are apparent. They are:

- The importance of asset quality.
- The omission of cash flow as a key indicator.

We will discuss these differences, as well as the reasons for them, in succeeding chapters.

As with nonfinancial companies, qualitative analysis plays a substantial role, even a more important role, in financial institution credit analysis.

Finally, it should be noted that there is a great deal of diversity among the entities that comprise the financial sector. In this book, we focus almost exclusively on banks. They are the most important category of financial institutions and also probably the most numerous. Banking organizations, so defined, nonetheless embrace a wide range of institutions, and the category embraces a number of subcategories, including commercial banks, specialized, wholesale banks, trust banks, development banks, and so on. The number of categories present within a particular country's financial sector depends upon the structure of the industry and the applicable laws governing

it. Equally, the terminology used to refer to the different categories of banking institutions is no less diverse, with the relevant statutory definitions for each type varying to a greater or lesser extent from jurisdiction to jurisdiction.

Aside from banks, the remainder of the financial sector is composed of a variety of other types of entities including insurance companies, securities brokerages, and asset managers. Collectively, these entities are referred to as nonbank financial institutions, or as NBFIs. As with the banking industry, the specific composition of the NBFI sector in a particular jurisdiction is influenced by applicable laws, regulations, and government policy.

In these pages, we will focus almost exclusively on commercial banks. An in-depth discussion of the credit analysis of NBFIs is really the subject for another book.

A QUANTITATIVE MEASUREMENT OF CREDIT RISK

So far, our inquiry into the meaning of credit has remained within the confines of tradition. Credit risk has been defined as the likelihood that a borrower will perform a financial obligation according to its terms; or conversely, the probability that it will default on that commitment. The probability that a borrower will default on its obligation to the lender generally equates to the probability that the lender will suffer a loss. As so defined, credit risk and default risk are essentially synonymous. While this has long been a serviceable definition of creditworthiness, developments in the financial services industry and changes in regulation of the sector over the past decade have compelled market participants to revisit the concept.

Probability of Default

If we think more about the relationship between credit risk and default risk, it becomes apparent that such probability of default (PD), while highly relevant to the question what constitutes a “good credit”⁴² and what identifies a bad one, is *not* the creditor’s *only*, or in some cases even her central concern. Indeed, a default could occur, but should a borrower through its earnest efforts rectify matters promptly—making good on the late payment through the remittance of interest or penalty charges—and resume performance without further breach of the lending agreement, the lender would be made whole and suffer little harm. Certainly, nonpayment for a brief period could cause the lender severe consequential liquidity problems, should it have been relying upon payment to satisfy its own financial obligations, but otherwise the tangible harm would be negligible. Putting aside for a moment the impact of default on a lender’s own liquidity, if mere default by a borrower alone is not what truly concerns a creditor, about what then is it really worried?

Loss Given Default

In addition to the probability of default, the creditor is, or arguably should be, equally concerned with the severity of the default that might be incurred. It is perhaps easier to comprehend retrospectively.

Was it a brief, albeit material default, like that described in the preceding paragraph, that was immediately corrected so that the creditor obtained all the expected benefits of the transaction?

Or was it the type of default in which payment ceases and no further revenue is ever seen by the creditor, resulting in a substantial loss as a result of the transaction?

Clearly, all other things being equal, it is the expectation of the latter that most worries the lender.

Both the probability of default and the severity of the loss resulting in the event of default—each of which is conventionally expressed in percentage terms—are crucial in ascertaining the tangible expected loss to the creditor, not to speak of the creditor's justifiable level of apprehension. The loss given default (LGD) encapsulates the likely percentage impact, under default, on the creditor's exposure.

Exposure at Default

The third variable that must be considered is exposure at default (EAD). EAD may be expressed either in percentage of the nominal amount of the loan (or the limit on a line of a credit) or in absolute terms.

Expected Loss

The three variables—PD, LGD, and EAD—when multiplied, give us expected loss for a given time horizon.⁴³

It is apparent that all three variables are quite easy to calculate after the fact. Examining its entire portfolio over a one-year period, a bank may determine that the PD, adjusted for the size of the exposure, was 5 percent, its historical LGD was 70 percent, and EAD was 80 percent of the potential exposure. Leaving out asset correlations within the loan portfolio and other complexities, expected loss (EL) is simply the product of PD, LGD, and EAD.

EL and its constituents are, however, much more difficult to estimate in advance, although past experience may provide some guidance.

The Time Horizon

All the foregoing factors are time dependent. The longer the *tenor*⁴⁴ of the loan, the more likely it is that a default will occur. EAD and LGD will also change with time, the former increasing as the loan is fully drawn, and decreasing as it is gradually repaid. Similarly, LGD can change over time, depending upon the specific terms of the loan. The nature of the change depends upon the specific terms and structure of the obligation.

Application of the Concept

To summarize, expected loss is fundamentally dependent upon four variables, with the period often assumed to be one year for the purposes of comparison and analysis. On a portfolio basis, a fifth variable, correlation between credit exposures within a credit portfolio, will also affect *expected loss*.

The PD/LGD/EAD concept just described is extremely valuable as a way to understand and model credit risk. It will be developed at length in subsequent chapters.

Major Bank Failure Is Relatively Rare

While bank credit analysis resembles corporate credit analysis in many respects, it differs in several important ways. The most crucial difference is that, *broadly speaking*, modern banks, in sharp contrast to nonfinancial firms, do not *fail* in normal times. That may seem like a shocking statement. It is an exaggeration, but one that has more truth in it than might first appear, considering that, quite often, weak banks are conveniently merged into other—supposedly stronger—banks. Most bank analysts, if you press them hard enough, will acknowledge the declaration as generally valid, when applied to the more prominent and internationally active institutions that are the subject of the vast majority of credit analyses.

Granted, the present time, in the midst of a substantial financial crisis, does not qualify as normal time. In each of 2009 and 2010, roughly 2 percent of U.S. banks failed, and in 2011, so did roughly 1.2 percent of them. The rate of failure between 1935 and 1940 was about 0.5 percent per year, and it remained below 0.1 percent per year in the 20 years after World War II. Between 2001 and 2008, only 50 banks failed in the United States—half of them in 2008 alone, but that left the overall ratio of that period below 0.1 percent per year.

In the United States alone, other data show that the volume of failures of publicly traded companies numbered in the thousands, with total business bankruptcies in the millions. To be sure, the universe of banks is much smaller than that of nonfinancial companies, but other data confirms that bank collapses are substantially less probable than those of nonfinancial enterprises. This is, of course, not to say that banks *never* fail (recall the foregoing qualification, “broadly speaking”). It is evident the economic history of the past several centuries is littered with the invisible detritus of many long-forgotten banks.

Small local and provincial banks, as well as—mostly in emerging markets—sometimes larger institutions are routinely closed by regulators, or merged or liquidated, or taken over by other healthier institutions, without creating systemic waves.

The proportion of larger banks going into trouble has dramatically increased in the past few years, particularly in the UK and in the United States, but also in Europe. The notion of *too big to fail* has always been accepted in the context of each separate market. In November 2011, that notion was extended to include a systemic risk of contagion, with the publication by the Financial Stability Board (FSB) of a list of 29 “systemically important financial institutions” which would be required to hold “additional loss absorption capacity tailored to the impact of their [possible] default.” There are of course many more “too big to fail” financial institutions around the world.

The notion of “too small to fail” also exists since it is often cheaper and more expedient—not to mention less embarrassing—for governments to arrange the quiet absorption of a small bank in trouble.

A wide danger zone remains in between those two extremes.

A thorough discussion about default can be found in later chapters of this book.

Bank Insolvency Is Not Bank Failure

The proposition that banks do not fail is, it must be emphasized, an overstatement meant to illustrate a general rule. There is no intent to convey the notion that banks

do not become insolvent, for especially with regard to banks (as opposed to ordinary corporations), insolvency and failure are two distinct events. In fact, bank insolvency is far more common, even in the twenty-first century, than many readers are likely to suspect.⁴⁵ Equally, insolvent banks can keep going on and on like a notorious advertising icon so long as they have a source of liquidity, such as a central bank as a *lender of last resort*. What is meant is that the bankruptcy or collapse of a *major* commercial banking institution that actually results in a significant loss to depositors or creditors is an extremely rare event.⁴⁶ Or at least it did remain so until the crisis that started in 2008. For the vast majority of institutions that a bank credit analyst is likely to review, a failure is highly improbable. But because banks are so highly leveraged, these risks, and perhaps more importantly, the risks that episodes of distress that fall short of failure and may potentially cause harm to investors and counterparties, are of such magnitude that they cannot be ignored.

Why Bother Performing a Credit Evaluation?

If major bank failures are so rare, why bother performing a credit evaluation? There are several reasons.

- First, evaluating the default risk of an exposure to a particular institution enables the counterparty credit analyst working for a bank to place the risk on a rating scale, which helps in pricing that risk and allocating bank capital.
- Second, even though the risk of default is low, the possibility is a worrisome one to those with credit exposure to such an institution. Consequently, entities with such exposure, including nonfinancial and nonbank financial organizations, as well as investors, both institutional and individuals, have an interest in avoiding default-prone institutions.⁴⁷
- Third, it is not only outright failure that is of concern, but also events short of default can cause harm to counterparties and investors.
- Fourth, globalization has increased the risk of systemic contagion. As a result, the risk on a bank has become a twice-remote risk—or in fact a risk compounded many times—on that bank's own risk on other financial institutions with their own risk profile.

Default as a Benchmark

That bank defaults are rare—barring systemic crises—in today's financial industry does not detract from the conceptual usefulness of the possibility of default in delineating a continuum of risk.⁴⁸ The analyst's role is to place the bank under review somewhere on that continuum, taking account of where the subject institution stands in terms of financial strength and the potential for external support. The heaven of pure creditworthiness⁴⁹ and the hell of bankruptcy⁵⁰ define two poles, somewhere between which a credit evaluation will place the institution in terms of estimated risk of loss. In terms of credit ratings, these poles are roughly demarcated by an AAA rating at one end, and a default rating at the other.

In other words, the potential for failure, if not much more than a remote possibility in most markets, nonetheless allows us to create a sensible definition of credit risk and a spectrum of expected loss probabilities in the form of credit ratings.

In turn, these ratings facilitate the external pricing of bank debt and, internally, the allocation of bank capital.

Pricing of Bank Debt

From a debt investor's perspective, the assessment of bank credit risk distilled into an internal or external rating facilitates the determination of an investment's value, that is, the relationship of risk and reward, and concomitantly its pricing. For example, if hypothetical Dahlia Bank and Fuchsia Bank, both based in the same country, each issue five-year subordinated floating-rate notes paying a semiannual coupon of 6 percent, which is the better investment? Without additional information, they are equally desirable. However, if Dahlia Bank is a weaker credit than Fuchsia Bank, then all other things being equal, Fuchsia Bank is likely to be the better investment since it offers the same rate of return for less risk to the investor.

These evaluations of bank obligations and the information they convey to market participants function to underpin the development and maintenance of efficient money and capital markets. The relationship between credit risk and the return that investors will require to compensate for increasing risk levels can be depicted in a *rating yield curve*. The diagram in Exhibit 1.5 illustrates a portion of such a curve. Credit risk, as reflected in assigned credit ratings, is shown on the horizontal axis, while the risk premium, described as basis points above the risk-free rate demanded by investors, is shown on the vertical axis. Observe that as of the time captured, for a financial instrument having a rating of BBB, corresponding roughly to a default probability within one year of between 0.2 percent and 0.4 percent,⁵¹ investors required a premium of about 200 basis points (bps) or 2 percent yield above the risk-free rate.

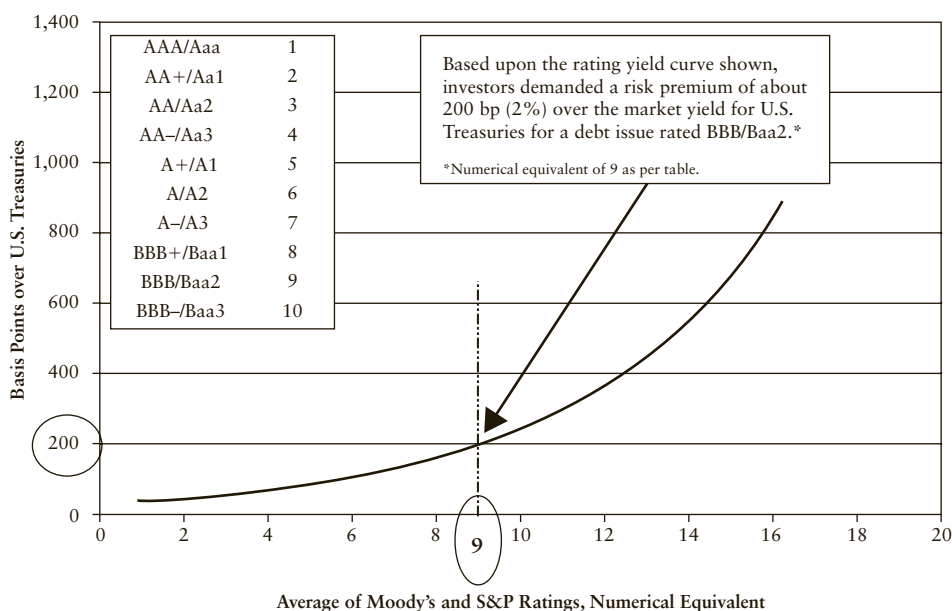


EXHIBIT 1.5 External Ratings and the Rating Yield Curve

Allocation of Bank Capital

From the counterparty credit analyst's perspective, the same assessment process enables the institution for which she works to better allocate its risks and its capital in a manner that is both prudent and compliant with relevant regulatory prescriptions.⁵² For internal risk management purposes, bank analysis facilitates the setting of exposure limits with regard to the advance of funds, exposure to derivatives, and settlement.

Events Short of Default

The default of an entity to which one has credit exposure is obviously something to be avoided. The risk of default is also useful conceptually to define a spectrum of default risk. But what about events short of default? How do they figure in the calculus of the creditor or investor? As alluded to, a bank does not have to fail for it to cause damage to a counterparty or creditor. A *technical default*, not to speak of a more material one, can have critical consequences. If a company's treasurer, for example, loses access to funds on deposit with a bank even temporarily, this loss of access can have serious knock-on effects, even if all the funds at stake are ultimately repaid.⁵³ Likewise, if one bank is relying upon another's creditworthiness as part of a larger transaction, the first bank's default, again even if only technical or temporary, can be a grave matter for its counterparty, potentially harming its own credit rating and reputation in the market. In all of the foregoing cases, irrespective of the likelihood of outright bank failure, bank credit analysis provides the means to avoid fragile banks, as well as the tools to steer clear of failure-prone institutions in markets where bank collapses are not so uncommon.

Banks Are Different

Banks are different in that they are highly regulated and their assessment is intrinsically highly qualitative. In many respects, however, bank credit analysis and corporate credit analysis are more alike than they are dissimilar. Yet there are vital differences in their respective natures that call for separate approaches to their evaluation both in respect to the qualitative and quantitative aspects of credit analysis. Some of these differences relate to the structure of a bank's financial statements as compared with a nonfinancial entity. Others have to do with the role of banks within a jurisdiction's financial system and their impact on the local economy.

The banking sector is among the most tightly regulated of all industries worldwide. This fact alone means that the scope, character, and effectiveness of the regulatory apparatus will inevitably affect the performance and financial condition of institutions that come within its ambit. Consequently, consideration of the adverse and beneficial consequences of existing regulations, and proposed or promulgated changes, will necessarily assume a higher profile than is normally the case in connection with nonfinancial companies.

The reason that banks are heavily regulated is in large part attributable to the preeminent role they play within the financial markets in which they operate. As crucial components within a national payment system and the extension of credit within a region or country, their actions and health inevitably have a major impact

on the climate of the surrounding economy. Consequently, banks are more important than their apparent size, measured in terms of revenue generation and employment levels, would suggest. It should not be surprising therefore that governments around the world take a keen interest in the health of the banks operating within their borders, and, supervise them to a far greater degree than they do nonfinancial enterprises. In contrast, nonfinancial firms, with a few exceptions, are lightly regulated in most jurisdictions, and governments generally take a hands-off policy toward their activities.

In most contemporary market-driven economies, if an ordinary company fails, it is of no great concern. This is not so in the case of banks. Because they depend on depositor confidence for their survival, and since governments neither want to confront irate depositors, nor more critically, contend with a significant number of banks unable to function as payment and credit conduits, deposit-taking institutions are rarely left to fend for themselves and go bust without a passing thought. Even where *deposit insurance* exists and depositors remain pacified, the failure of a single critical financial institution may be plausibly viewed by policymakers as likely to have a detrimental impact on the health of the regional or national financial system. Moreover, the costs of repairing a banking crisis typically far outweigh the costs of taking prudent measures to prevent one. Governments therefore actively monitor, regulate, and—in light of the importance of banks to their respective economies—ultimately function as lenders of last resort through the national central bank, or an equivalent agency.

Owing to the privileged position that banks commonly enjoy, their credit analysis must give due consideration to an institution's role within the relevant financial system. Its position will affect the analyst's assessment concerning the probability, and degree, of support that may be offered by the state—whether explicitly or more commonly implicitly—in the case the bank experiences financial distress. Making such assessments not only calls for consideration of applicable laws and regulations, but also relevant institutional structures and policies, both historic and prospective. Moreover, the analysis must consider policies that, in an effort on the part of government to reduce *moral hazard*,⁵⁴ may be quite opaque. In sum, this aspect of the analytical process necessarily requires keen judgment, and is in consequence principally qualitative in character.⁵⁵

NOTES

1. Walter Bagehot, *Lombard Street: A Description of the Money Market* (1873), hereafter *Lombard Street*. Bagehot (pronounced “badget” to rhyme with “gadget”) was a nineteenth-century British journalist, trained in the law, who wrote extensively about economic and financial matters. An early editor of *The Economist*, Bagehot's *Lombard Street* was a landmark financial treatise published four years before his death in 1877.
2. Various attributions; see for example Global-Investor.com; *500 of the Most Witty, Acerbic and Erudite Things Ever Said About Money* (Harriman House, 2002). Author of *Adventure Capitalist* and *Investment Biker*, Jim Rogers is best known as one of the world's foremost investors. As co-founder of the Quantum fund with George Soros in 1970, Rogers's extraordinary success as an investor enabled him to retire at the age of 37. He remains in the public eye, however, through his books and commentary in the financial media.

3. See, for example, “credit. . . . Etymology: Middle French, from Old Italian *credito*, from Latin *creditum*, something entrusted to another, loan, from neuter of *creditus*, past participle of *credere*, to believe, entrust.” *Merriam-Webster Online Dictionary*, www.m-w.com. *Webster’s Revised Unabridged Dictionary* (1913) defines the term to mean: “trust given or received; expectation of future payment for property transferred, or of fulfillment or promises given; mercantile reputation entitling one to be trusted; applied to individuals, corporations, communities, or nations; as, to buy goods on credit.” www.dictionary.net/credit. Walter Bagehot, whose quoted remarks led this chapter, gave the meaning of the term as follows: “Credit means that a certain confidence is given, and a certain trust reposed. Is that trust justified? And is that confidence wise? These are the cardinal questions. To put it more simply, credit is a set of promises to pay; will those promises be kept?” (Bagehot, *Lombard Street*).
4. This is assuming, of course, that the financial condition of the borrower has been honestly and openly represented to the creditor through the borrower’s financial statements. The relevance of the assumption remains important, as the discussions concerning financial quality later in the book illustrate.
5. As put by John Locke, the seventeenth-century British philosopher, “credit [is] nothing but the expectation of money within some limited time . . . money must be had, or credit will fail.”—vol. 4 of *The Works of John Locke in Nine Volumes*, 12th ed. (London: Rivington, 1824), www.econlib.org. Credit exposure (exposure to credit risk) can also arise indirectly as a result of a transaction that does not have the character of loan, such as in the settlement of a securities transaction. The resultant settlement risk is a subset of credit risk. Aside from settlement risk, however, credit risk implies the existence of a financial obligation, either present or prospective.
6. R. Taggart Murphy, *The Real Price of Japanese Money* (London: Weidenfeld & Nicolson, 1996), 49. Murphy’s book was published in the United States as *The Weight of the Yen: How Denial Imperils America’s Future and Ruins an Alliance* (New York: W. W. Norton, 1996). Although Taggart’s book is primarily concerned with the U.S.–Japan trade relationship as it evolved during the post–World War II period, Chapter 2 of the text, entitled “The Credit Decision,” provides an instructive sketch of the function of credit assessment in the commercial banking industry.
7. The phrase “on a commercial basis” is used here to mean in an “arm’s-length” business dealing with the object of making a commercial profit, in contrast to a transaction entered into because of friendship, family ties, or dedication to a cause, or as a result of any other noncommercial motivation.
8. There are four basic types of collateral: (1) real or personal property (including inventory, trade goods, and intangible property); (2) negotiable instruments (including securities); (3) other financial collateral (i.e., other financial assets); and (4) floating charges on business assets. *Current assets* refers to assets readily convertible to cash. These are also known as *liquid assets*.
9. *Foreclosure* is a legal procedure to enforce a creditor’s rights with respect to collateral pledged by a delinquent borrower that enables the creditor to legally retain or to sell the collateral in full or partial satisfaction of the debt.
10. *Bankruptcy* is the legal status of being insolvent or unable to pay debts. Bankruptcy proceedings are legal proceedings in which a bankruptcy court or similar tribunal takes over the assets of the debtor and appoints a receiver or trustee to administer them. Unsecured creditors may also be able to initiate bankruptcy proceedings, but are less sure of compensation than the secured creditor.
11. A related question is whether the legal framework is sufficiently robust to enable the creditor to enforce his rights against the borrower. Where creditors’ rights are weak or difficult to enforce, this consideration becomes part of the credit decision-making process.

12. Roger H. Hale, *Credit Analysis: A Complete Guide* (New York: John Wiley & Son's 1983, 1989). The traditional reliance on collateral has given rise to the term "pawnshop mentality" to refer to bankers who are incapable or unwilling to perform credit analysis of their customers and lend primarily on the value of collateral pledged. See for example Szu-yin Ho and Jih-chu Lee, *The Political Economy of Local Banking in Taiwan* (Taipei, Taiwan: NPF Research Report, National Policy Foundation, December 10, 2001). "Because of the pawnshop mentality and practices in banking institutions, the SMEs are not considered good customers for loans," www.npf.org.tw/English/Publication/FM/FM-R-090-069.htm. SME is an acronym for *small- and medium-size enterprise*. Murphy, referring to banking practice in Japan in the 1980s, observed that "Japanese banks rarely extend domestic loans of any but the shortest maturity without collateral" (*Real Price*, 49).
13. Typically, in this situation, the loan would be advanced by an auto manufacturer's finance subsidiary.
14. *Financial analysis* is the process of arriving at conclusions concerning an entity's financial condition or performance through the examination of its financial statements, such as its balance sheet and income statement. Financial analysis encompasses a wide range of activities that may be employed for internal management purposes (e.g., to determine which business lines are most profitable) or for external evaluation purposes (e.g., equity or credit analysis).
15. *Joint and several liability* is a legal concept under which each of the parties to an obligation is liable to the full extent of the amount outstanding. In other words, where there are multiple obligors, the obligee (creditor) is entitled to demand full repayment of the entire outstanding obligation from any and all of the obligors (borrowers).
16. As well as having an impact on whether to advance funds, the use of collateral, guarantees, and other credit risk mitigants may also serve to increase the amount of funds the lender is willing to put at risk.
17. *Repo finance* refers to the use of repurchase agreements and reverse repurchase agreements to facilitate mainly short-term collateralized borrowings and advances. Securities lending transactions are similar to repo transactions. Both are explained in more detail in Chapter 3.
18. *Free will* has been defined as the power of making choices unconstrained by external agencies, wordnet.princeton.edu/perl/webwn. If so defined—that is, meaning having the freedom to choose a course of action in the moment—it is by definition not predetermined, and therefore the actions of an entity having free will cannot be 100 percent predictable. This is not to say, however, that predictive—if not determinative—factors cannot be identified, and that the probability of various scenarios unfolding cannot be estimated, especially when the number of transactions involved is large. Indeed, much of credit risk evaluation is underpinned by implicit or explicit statistical expectations based on the occurrence of a large number of transactions.
19. *Due diligence* means the review of accounts, documentation, and related written materials, together with interviews with an entity's principals and key personnel, for the purpose of supporting a professional assessment concerning the entity. A due diligence investigation is typically performed in connection with a prospective transaction. So a law firm would likely perform a due diligence investigation before rendering of a legal opinion concerning an anticipated transaction. So, too, will a rating agency undertake a due diligence review before assigning a rating to an issuer.
20. A *correspondent bank* is a bank that has a relationship with a foreign banking institution for which it performs services in the correspondent bank's home market. Since few, if any banks, can feasibly maintain branches in all countries of the world, correspondent banking relationships enable institutions without branches or offices in a given jurisdiction to act

on a global basis on behalf of such institutions' clients. Typical services provided by a correspondent bank for a foreign institution include check clearing, funds transfers, and the settlement of transactions, acting as a deposit or collection agent for the foreign bank, and participating in documentary letter of credit transactions.

21. *Lombard Street*, note 2 supra, quoted in Martin Mayer, *The Bankers: The Next Generation* (Penguin, 1996), 10. The quotation comes from Chapter 3 of the book, entitled "How Lombard Street Came to Exist, and Why It Assumed Its Present Form," and the passage discusses the evolution of commercial banks from institutions reliant on note-issuance to those dependent upon the acceptance of deposits. Note that a leading textbook on bank management also pays homage to the axiom that bankers understand their own geographic franchise best and "are more apt to misjudge the quality of loans originating outside [it]. . . . [and] loan officers will be less alert to the economic deterioration of communities outside their trade areas." George H. Hempel, Donald G. Simonson, and A. Coleman, *Bank Management: Text and Cases*, 4th ed. (hereafter *Bank Management*) (New York: John Wiley & Sons, 1994), 377.
22. For example, *Bank Management*, note 21 supra, observes that there is a consensus among bankers that "the paramount factor in a successful loan is the honesty and goodwill of the borrower," and rates a borrower's character as one of the four fundamental credit criteria to be considered, together with the purpose of the funds, and the primary and secondary sources of loan repayment. In a specialist book focused on emerging markets, *character* is one of five "Cs" of credit, along with capacity, capital, collateral, and conditions. Waymond A. Grier, *The Asiamoney Guide to Credit Analysis in Emerging Markets* (Hong Kong: Asia Law & Practice, 1995), 11.
23. Ironically, *name lending* is also called "character lending." Distinct from this phenomenon is *related-party lending*, which means advancing funds to a family member of a bank owner or officer, or to another with whom the owner or officer has a personal or business relationship separate from those arising from his or her capacity as a shareholder or as an employee of the bank.
24. It should be borne in mind that whether relying on first-hand knowledge, reputation, or the borrower's credit history, the analytical distinction between *willingness* to pay and *capacity* to pay is easily blurred. In discussing character, *Bank Management* distinguishes among fraudulent intent, moral failings, and other deficiencies, such as lack of intelligence or management skills, some of which might just as easily come under the heading of *management assessment*. Ultimately, the creditor is only concerned whether the borrower is good for the funds "entrusted" to him, and as a practical matter there is little to be gained for this purpose in attempting to parse between how much this belief rests on willingness to pay and how much it rests upon capacity to pay.
25. Where, however, there exist a large number of recorded transactions, stronger correlations may be drawn between the borrower's track record and future behavior, allowing the probability of default to be better predicted. This, of course, increases a bank's ability to manage risk, as compared with other entities that engage in comparatively few credit transactions and provides banks with an essential competitive advantage in this regard.
26. The value of such experience comes not only from the bank officer's having reviewed a greater variety of credit exposures, but also from having gone through an entire business cycle and seen each of its phases and corresponding conditions. As with the credit analysis of large corporate entities, in the credit analysis of (rather than by) financial institutions, the criterion of character tends to be given somewhat less emphasis than is the case in respect to individuals and small businesses. As various financial scandals have shown, however, this reduced emphasis on character is not necessarily justified.
27. The term *moral obligation* is used here to distinguish it from a legally enforceable obligation. A legal obligation may also represent a moral obligation, but a moral obligation does not necessarily give rise to a legal one.

28. While full recovery may nonetheless still be impossible, depending upon the borrower's access to funds and the worth of the collateral securing the loan, partial recovery is generally more likely when creditors' rights are strong. The effectiveness of a legal system encompasses many facets, including the cost and time required to obtain legal redress, the consistency and fairness of legal decisions, and the ability to enforce judicial decisions rendered. It may be added that the development of local credit reporting systems also may affect a borrower's willingness to fulfill financial obligations since a borrower may wish to avoid the detrimental effects associated with being tagged as a less than prime credit.
29. In this book, the terms and phrases such as *legal efficiency* and *quality of the legal infrastructure* are used more or less synonymously to refer to the ability of a legal system to enforce property rights and creditors' rights fairly and reliably, as well as in a reasonably timely and cost-effective manner. In a scholarly context, these terms are also used to refer to the ability of legal institutions to reduce "idiosyncratic risk" to entrepreneurs and to prevent the exploitation of outside investors from insiders by protecting their property rights in respect of invested funds. See for example Luc Laeven, "The Quality of the Legal System, Firm Ownership, and Firm Size," presentation at the Stanford Center for International Development, Stanford University, Palo Alto, California, November 11, 2004.
30. It is apparent that there is almost always a risk that a credit transaction favoring the creditor may not be enforced. This risk is often subsumed under the broader rubric of *legal risk*. Legal risk may be defined generally as a category of operational risk or event risk that may arise from a variety of causes, which insofar as it affects credit risk becomes a proper concern of the credit analyst. Types of legal risk include (1) an adverse change in law or regulation; (2) the risk of being a defendant in time-consuming or costly litigation; (3) the risk of an arbitrary, discriminatory, or unexpected adverse legal or regulatory decision; (4) the risk that the bank's rights as creditor will not be effectively enforced; (5) the risk of ineffective bank supervision; or (6) the risk of penalties or adverse consequences incurred as a result of inadvertent errors in documentation. Note that these subcategories are not necessarily discrete, and may overlap with each other or with other risk classifications.
31. Regrettably for lenders in emerging markets, effective protection of creditors' rights is not the norm. As was seen in the aftermath of the Asian financial crisis during 1997–1998, the legal systems in some countries were demonstrably lacking in this regard. Reforms that have been implemented, such as the revised bankruptcy law enacted in Thailand in 1999, have gone some distance toward remedying these deficiencies. The efficacy of new statutes is, however, dependent upon a host of factors, including the attitudes, expertise, and experience of all participants in the judicial process. In Thailand and Indonesia, as well as in other comparable jurisdictions where legal reforms have been implemented, it can be expected that it will take some years before changes are thoroughly manifested at the day-to-day level. Similarly, the debt moratorium and emergency laws enacted in Argentina in 2001 and 2002 curtailed creditors' rights in a substantial way. Incidentally, in June 2010 in Iceland, not exactly an emerging market, the Supreme Court ruled that some loans indexed to foreign currency rates were illegal, shifting the currency losses from borrowers to lenders. Similar decisions may yet be taken in Hungary or in Greece.
32. Coined in 1981 by Antoine W. van Agtmael, an employee of the International Finance Corporation, an affiliate of the World Bank, the term *emerging market* is broadly synonymous with the terms *less-developed country* (LDC) or *developing country*, but generally has a more positive connotation suggesting that the country is taking steps to reform its economy and increase growth with aspirations of joining the world's developed nations (i.e., those characterized by high levels of per capita income among various relevant indicia). Leading emerging markets at present include, among others, the following

countries: China, India, Malaysia, Indonesia, Turkey, Mexico, Brazil, Chile, Thailand, Russia, Poland, the Czech Republic, Egypt, and South Africa. Somewhat more developed countries, such as South Korea, are sometimes referred to as *NICs*, or *newly industrialized countries*. Somewhat less-developed countries are sometimes referred to anecdotally as *subemerging markets*, a term that is somewhat pejorative in character.

33. This is an illustration of the problem of *moral hazard* that is discussed later in the book.
34. The term *financial company* is used here to contrast financial intermediaries with non-financial enterprises. Not to be confused with the term *finance company*, *financial company* refers to banks as well as to nonbank financial intermediaries, abbreviated NBFIs. The concept of intermediation is explored further in Chapter 3.
35. While banks and other financial institutions are usually organized as corporations, the term *corporate* is frequently used both as an adjective and as a noun to generically refer to *nonfinancial enterprises*, such as manufacturers, wholesalers, and retailers, electrical utilities and service providers, owned by mainly private investors as opposed to those principally owned or controlled by governments or their agencies. The latter would usually fall under the rubric of state-owned enterprises.
36. *Net worth* may be defined as being equal to assets less liabilities, and is generally synonymous with the following terms which are often used to describe the same concept in relation to companies: equity capital; total equity; net assets; owners' equity; stockholders' equity; shareholders' funds; net asset value; and net tangible assets (net assets less intangible assets such as goodwill). More generally, it may be observed that financial terms are often associated with a plethora of synonyms, while, at the same time, fundamental terms such as *capital* or *nonperforming loans* may have distinctly different meanings depending upon the circumstances.
37. *Income* is an accounting concept distinct from *cash flow* in that it seeks to match past and future cash flows with the transaction that generated them, rather than classifying such movements strictly on the basis of when—that is, in which financial reporting period—they occurred. The differences between income and cash flow are discussed in Chapter 6.
38. *Unsecured* means without security such as collateral or guarantees.
39. Note that for an individual, net worth is frequently calculated taking account of the market value of key assets such as real property, in contrast to company credit analysis, which, with some exceptions, will value assets at their historical cost.
40. *Net cash flow* can be defined as cash received less cash paid out for a given financial reporting period.
41. *Credit positive* means tending to strengthen an entity's perceived creditworthiness, while *credit negative* suggests the opposite. For example, "[The firm's] recent disposal of the fiber-optic network is slightly credit positive." Ivan Lee et al., *Asia-Pacific-Fixed Income, Asian Debt Perspective-Outlook for 2002* (Hong Kong: Salomon Smith Barney, January–February 2002), 24.
42. A *good credit* means, of course, a good credit risk; that is, a credit risk where the risk of loss is so minimally low as to be acceptable.
43. For simplicity's sake, variables such as the period and correlations within a loan portfolio are omitted in this introductory discussion.
44. *Tenor* means the term or time to maturity of a credit instrument.
45. Moreover, not all episodes of bank distress reach the newspapers, as problems are remedied by regulators behind the scenes.
46. In short, while historically a sizable number of banks have closed their doors, and bank defaults and failures are not unknown even today. The types of institutions that do suffer bank collapses are almost always local or provincial, with few, if any, international counterparty relationships.
47. Retail depositors will also be concerned with a bank's possible default, although their deposits are often insured by governmental or industry entities to some extent.

48. As discussed, credit risk is largely measured in terms of the probability of default, and loss severity (loss given default).
49. In other words, 0 percent risk of loss, that is, a risk-free investment.
50. The worst-case scenario refers to an institution in default, subject to liquidation proceedings, in which 100 percent of principal and interest are unrecoverable.
51. Such default risk ranges are associated with ratings. In the past, each rating agency defined ratings in vague terms rather than as probabilities of default—other than as ranges of historical observations—and each rating agency would have its own definition. The advent of Basel II has now forced them to map each rating level to a range of probabilities of default.
52. A discussion of the mechanics of economic capital allocation is unnecessary for the purposes of this chapter, but it will of course be developed later in the book.
53. A decline in the credit quality of the bank has prompted the analyst to seek a reduction in limits for the exposure to the bank.
54. Industrial and service companies are thus far less likely to benefit from a government bailout, although this likelihood depends on the political-economic system. Even in highly capitalist countries, there are exceptions where the firm is very large, strategically important, or politically influential. The bailout of automaker Chrysler Corporation in the United States, a company that was later acquired by Daimler-Benz, was a notable illustration. More recently, the 2008 crisis prompted substantial government intervention in Europe and in the United States. In more *dirigiste* economies, state bailouts are less rare. Still, even in these economies, most corporates have no real lender of last resort, and must remain solvent and liquid on pain of fatal collapse. Perhaps, in consequence, their quantitative performance and solvency indicia tend to be scrutinized more severely than those of their financial institution counterparts.
55. Aside from their relevance in such extreme circumstances, because of the degree to which bank performance is affected by government regulation and supervision, the same considerations are important in the ongoing analysis of a bank's financial condition.

The Credit Analyst

Though the principles of the banking trade appear somewhat abstruse, the practice is capable of being reduced to strict rules. To depart upon any occasion from those rules, in consequence of some flattering speculation of extraordinary gain, is almost always extremely dangerous, and frequently fatal, to the banking company which attempts it.

—Adam Smith, *Wealth of Nations*

If you warn 100 men of possible forthcoming bad news, 80 will immediately dislike you. And if you are so unfortunate to be right, the other 20 will as well.

—Anthony Gaubis¹

What is a credit analyst?

What are the various types of credit analyst?

How do their roles differ?

Where do bank credit analysts fit in?

These are the questions this chapter seeks to answer.

Although all credit analysts undertake work that involves some similarity in its larger objectives, the specifics of each analytical role may vary a great deal. In the previous chapter, it was suggested that the approach to credit evaluation is contingent upon the type of entity being evaluated. In addition, the scope and nature of the evaluation will depend upon the functional role occupied by the analyst. Hence, while the same core questions underpin the analytical process, the time and resources available to perform the credit risk evaluation will vary.

This chapter seeks to survey the various subfields of credit analysis, as well as the different roles that can be undertaken within each corner of the field. The aim is to provide a practical overview of the field and to define the subject of our inquiry that is explicated in the chapters that follow. To this end, credit analysis and credit analysts are classified in three different ways: by function, by the type of entity analyzed (as referenced above), and by the category of employer. Under this approach, some repetition is unavoidable. It is hoped, however, that by the end of the chapter the reader will have a good understanding of what credit analysis is, and where bank credit analysis fits into the larger picture.

THE UNIVERSE OF CREDIT ANALYSTS

Common sense tells us that the job of the credit analyst is to assess credit risk. Used without further modification, this encompasses a wide range of functions running the gamut from the evaluation of small business loan applications to rating corporate customers at a global bank. Consider, for example, the following four job descriptions below, each for a “credit analyst,” drawn from actual advertised positions. While each listing is nominally for a credit analyst, the positions differ substantially in their content, scope of responsibility, and compensation.

Job Description 1: Credit Analyst

Manage pipeline of loans. Review loans and customer documentation to ensure they meet requirements and to determine loan status (approve/decline), including data entry. Review property appraisals and relevant documentation and perform basic mortgage calculations to validate score-based approval. Refer loan applications over \$750,000 to higher level.

Consumer Credit

The first position deals with retail consumer credit, primarily mortgage lending. From the phrasing of the ad, we can see that the position is a relatively junior one with limited authority. The emphasis is less on detailed fundamental analysis and more on ensuring that documentation is in order, and that the loan applicant meets predetermined scoring criteria.

Job Description 2: Credit Analyst

Review and analyze scoring analytics, interfacing as necessary with the risk management group. Develop, test, implement, and maintain a variety of origination and collection scorecards. Review modifications to existing systems. Prepare reports to support risk decisions.

Credit Modeling

The second position also concerns consumer credit, but is not so involved with the analysis of individual exposures as the first. Instead of reviewing applications, this job involves the review and development of more refined consumer credit scoring systems. Both this and the first position are rather far afield from bank and financial institution credit analysis, which is the focus of this book.

Job Description 3: Credit Analyst

Global investment bank seeks an experienced credit analyst to have responsibility for the analysis and credit rating of the bank’s international corporate clients. The role will also include involvement in credit modeling and participation in credit committee presentations.

Corporate Credit

The third advertisement is for a corporate credit analyst, since the scope of the position extends only to corporate entities, as opposed to financial institutions or sovereign credits. It also includes some duties with regard to the development of credit risk models.

Job Description 4: Credit Analyst

Monitor exposures to counterparties, which comprise primarily banks, brokers, insurance companies, and hedge funds. Prepare counterparty credit reviews, approve credit limits, and develop and update credit policies and procedures. The credit review process includes detailed capital structure and financial statement analysis as well as qualitative assessments of both the counterparty and the sector in which it operates.

Counterparty Credit

Only the final listing specifically addresses the type of work that is the main subject of this book. It calls for a counterparty credit analyst to analyze banks and other financial institutions, while, like the third advertisement, it also embraces some supplementary responsibilities. From these job descriptions, which by no means take in all the main analytical roles, it is apparent that the field of credit analysis is wide and varied.

THE COUNTERPARTY CREDIT ANALYST

Those credit analysts who evaluate the creditworthiness of financial intermediaries are known generally as bank and financial institution analysts. Within this broad category, the field can broadly be divided into two areas: (1) the analysis of banks and (2) the analysis of nonbank financial institutions (NBFIs) such as insurance firms or asset managers. When credit analysts are employed by a financial institution to analyze another financial institution, their evaluations are usually performed with a view to a prospective bilateral transaction between their employer and its opposite number as a counterparty. The credit risk arising from this type of transaction is termed *counterparty credit risk*, and those who evaluate such risk on behalf of prospective transaction participants are *counterparty credit analysts*.

As part of the evaluation process, *counterparty credit analysts* ordinarily assign the counterparty an internal rating. In contrast to the rating agency analyst who assigns an external rating, the counterparty credit analyst may be called upon to make recommendations concerning:

- Prudent limits in respect of particular credit risk exposures
- The approval or disapproval of a particular credit application
- Appropriate changes to the amount of exposure, tenor, collateral, and guarantees or to contractual provisions governing the transaction

The majority of credit analysts are employed by financial intermediaries participating in the money and capital markets. Others work for nonfinancial corporations and for organizations that provide market-support functions, such as rating agencies and government regulators. While all credit analysts evaluate credit risk, the analytical role embraces a broad range of situations and activities—as the preceding descriptions make apparent. The credit officer at a small rural bank may have to decide whether a loan should be extended to a retail hardware store owner whose establishment the officer visits regularly. At the other end of the spectrum, the head of credit at a multinational bank may hold responsibility for setting country risk limits and for determining the credit lines to be extended to specific banks and corporations in that country, as well as having on-the-spot authority for approving or rejecting specific transactions.

The evaluations undertaken by credit analysts at rating agencies such as Moody's Investors Service, Standard & Poor's, and Fitch Ratings in assigning unbiased rating grades to debt issued by governments, corporations, financial institutions, and other entities, represent another corner of the field. *Sell-side* and *buy-side* fixed-income analysts, who work respectively for investment banks or banking divisions, or for hedge funds, and proprietary trading units, are concerned not only with credit risk but with also with the relative value of debt instruments in comparison with issues in the same class and their corresponding desirability as investments. Finally, bank examiners and supervisors are also engaged largely in credit analysis as they evaluate the soundness of individual institutions as part of their supervisory function.

Classification by Functional Objective

Within the universe of credit analysis, practitioners can also be differentiated in terms of their functional role. Most are employed primarily to evaluate credit risk as part of a larger risk management function performed by all manner of financial institutions, as well as by many nonfinancial companies. A smaller group are fixed-income analysts who are employed to help choose investments in debt securities or to find investment opportunities. This distinction between risk management and investment selection has a significant impact on the analyst's objectives and on his or her day-to-day work.

Risk Management versus Investment Selection

The role of most credit analysts is to facilitate risk management, in the broadest sense of the term, whether at the level of the individual firm or at the level of national policy. The sort of risk management with which a credit analyst is concerned is, of course, credit risk management. Credit risk management forms part of the broader enterprise risk management function that includes the oversight and control of *market risk*, *liquidity risk*, and *operational risk*. Within the private sector, the main responsibility of credit analysts operating in a risk management capacity is to research prospective customers and counterparties, to prepare credit reports for internal use, to make recommendations concerning transactions and risk limits, and to generally facilitate the risk management of the organization as a whole.

Within the public sector, bank examiners are at heart credit analysts. They are employed by agencies that regulate financial institutions. As part of their supervisory

function, they undertake independent reviews of specific institutions, typically from a credit perspective. The usual aim of these regulatory agencies is to maintain a sound financial system while seeking both to encourage investment and foster economic growth and to facilitate the creation of deep and liquid financial markets.

Rating agency analysts, although not directly involved in risk management, evaluate issuers, counterparties, and debt issues from a similar perspective. Their mission is to provide unbiased analysis as the basis upon which to assign ratings to issuers or counterparties, as well as to specific debt issues or classes of debt issues, when required. These ratings are, in turn, used to facilitate both risk management and investment selection. In addition to the rating agencies, a number of other sources exist, offering various degrees of independence, relevance, and analytical depth.

Those credit analysts involved in investment selection represent a smaller portion of the field. Most credit analysts that perform this function can be classified as fixed-income analysts. Investment selection, of course, refers to the identification of potential investments (or those to avoid), and the making of recommendations or business decisions concerning how to allocate funds available for investment, from which the investor expects to make a return over time. In the investment context, credit risk is of particular importance in respect to fixed-income securities and other debt instruments; hence, its assessment comprises the larger part of the fixed-income analyst's work. It is normally less of a concern in respect to equity securities since the inherent trade-off accepted by equity investors is that a significant upside potential implies greater credit risk. Nonetheless, equity analysts implicitly take account of credit concerns and do, from time to time, address those concerns explicitly in investment reports. For both the equity analyst and the fixed-income analyst, the main objective is to reach a conclusion as to whether a particular investment will generate the expected return and whether it is more apt to exceed expectations or fall short of them.

In analyzing a fixed-income security, the risk of default is always an underlying concern, if not the analyst's chief focus. In most instances, the analyst's principal concern is the potential deterioration in an investment's credit quality and the corresponding risk of a decline in its price. All other things being equal, increased credit risk will cause a given debt security's price to fall. (See the box further on.) In contrast to the counterparty credit analyst, the fixed-income analyst is concerned not only with the credit risk of a contemplated transaction, but also with the investment's *relative value*. In brief, relative value refers to relative desirability of a particular debt security vis-à-vis securities in the same asset class and having the same assigned rating and other fundamental characteristics. It represents a key input in the fixed-income analyst's recommendation concerning a security—for example, whether to buy, sell, or hold it as an investment.

It should be noted that often within a financial institution, the functions of risk management and investment selection are largely separate domains. Their separation may be reinforced through the establishment of a so-called Chinese Wall constructed at the behest of regulators to limit the flow of certain types of information to prevent the unfair exploitation of inside information by customers or traders. Such barriers and the divergent objectives of credit analysts employed in a risk management capacity vis-à-vis those employed in an investment selection tend to discourage collaboration between the two types of staff.

Primary Research versus Secondary Research

Although in respect to the evaluation of credit risk, the basic elements of each of the previously mentioned analytical roles are similar, the amount of time and resources available to an analyst to assess the relevant credit risk depends very much upon the nature of the position. Credit analysis, when undertaken by one or more individuals, requires time and resources. Accordingly, the cost of analysis is higher when it is performed with human input, as opposed to being processed using a credit scoring mechanism. The more primary research required, the higher the cost. Notwithstanding the benefits of conducting a comprehensive credit risk review, it may not be cost-effective to perform an in-depth assessment in all situations where credit risk arises. The cost factor explains why the “analysis” of small standardized transactions is frequently automated, by the use of a *credit risk model* incorporated into a computer software application. While this book does touch upon the automated modeling systems that underpin *credit scoring*, it is primarily concerned with *analyst-driven credit research*. Such research and the evaluation of credit risk based on that research takes into account both quantitative and qualitative criteria, and considers both microeconomic (bank-specific) variables as well as the macroenvironment, including political, macroeconomic, and industry/systemic factors. This type of evaluation process may also be termed *fundamental credit analysis*.²

The same cost rationale constrains counterparty credit analysts as well, albeit to a lesser extent. The analysis of banks and other financial institutions, for reasons to be discussed, is not wholly amenable to quantification and thus cannot be fully automated. Nevertheless, the time and resources to perform credit reviews is limited, given that a counterparty credit analyst employed by a financial institution may well be responsible for an entire continent or region—for example, Asia—and his or her brief may extend to a hundred or more banks. Obviously, such an individual will not be able to visit every bank within his or her purview, nor spend several days analyzing a single institution. Although, ideally, counterparty credit analysts will conduct an independent review of the bank’s financial statements, and may, in some cases, periodically call or visit the subject bank, the greater part of the counterparty credit analyst’s work will tend to be taken up by researching the ratings produced by third parties, taking into account recent developments and utilizing other available sources of information to arrive at a synthesis of the institution’s credit story. Naturally, the conclusions reached will incorporate the analyst’s own assessments. The form of the resulting credit report will vary from institution to institution. Since the analysis and the recommendations made are intended purely for internal purposes, the reports that contain them will be briefer than those produced by rating agencies and considerably briefer than those produced by sell-side analysts at investment banks. Ordinarily, the entire report will rarely exceed two or three pages, and will simply include a short executive summary followed by a page or two of supporting text.

The credit analyst employed by a rating agency is, as a rule, under much less severe limitations. One reason is that nowadays most ratings are *solicited*, meaning that they are paid for by the party being rated or issuing the instrument that is to be assigned a rating. Unlike in-house corporate and counterparty analysts who often rely upon the assessments made by the rating agencies, it is the rating agency analysts themselves who are expected to produce a comprehensive and in-depth credit evaluation. By undertaking the intensive primary research that forms the foundation of the rating

assignment, rating agency analysts provide the value-added service to their subscribers that allows the latter to complete credit reviews in an expeditious manner. In addition to examining the bank's financials, rating agency analysts almost invariably visit the bank in question to form an independent conclusion as to its creditworthiness. Bank visits and accompanying due diligence investigations are fairly time-consuming, taking at least the better part of a day, and sometimes significantly longer. Additional time is needed to prepare the final report and have it approved by the agency's rating committee. To ensure that enough resources are allotted to produce a high-quality credit evaluation, each rating agency analyst typically covers a fairly small number of institutions, often in a small number of countries.³

A Special Case: The Structured Finance Credit Analyst

Finally, another type of credit analyst whose province does not easily fit within the preceding categories and whose functions are generally outside the scope of this book must be mentioned: the structured finance credit analyst.⁴ Structured finance refers to the advance of funds secured by certain defined assets or cash flows. The credit analysis of structured products is often complex because the resulting credit risk depends primarily on the manner in which such assets and cash flows are assembled, and in particular upon the forecasting of the probability of various contingencies that affect the ownership of such assets and the amount and timing of the associated cash flows to create a transaction framework.

Although, in principle, structured finance methods resemble ordinary secured lending backed by collateral, they are often considerably more complex, and the additional security is typically provided in a considerably more sophisticated manner, either by means of the transfer of assets to a special purpose vehicle (SPV) or synthetically, through, for example, the transfer of credit risk using credit derivatives. Instead of being based solely on the intrinsic creditworthiness of the issuer or borrower, structured products analysis takes account of a large variety of other criteria. Note that in the wake of the global credit crisis of 2007–2010, demand for structured products fell significantly. It can be expected, however, that at some stage demand may resume, albeit not to the same extent as previously nor for the breadth of complex instruments as existed before the crisis.

By Type of Entity Analyzed

Credit analysis is usually categorized into four fields. These correspond to the four basic types of credit exposures, namely:

1. Consumer
2. Corporate
3. Financial institution
4. Sovereign/municipal (subnational)

As illustrated in the first two job descriptions provided at the beginning of this chapter, the consumer credit analyst only rarely engages in intensive examination of an individual's financial condition. Because, as suggested, case-by-case intensive analysis of individuals for personal lending purposes is seldom cost-effective, most consumer credit

analysis is highly mechanized through the use of scoring models and similar techniques. As a result, unless concerned with modeling, systems development, or collateral appraisal, consumer credit roles often tend to be broadly clerical in nature. Hence, for our purposes, the main areas of credit analysis can be simplified to three as follows:

1. Corporate
2. Financial institution
3. Sovereign/municipal

Each of these fields—corporate credit analysis, sovereign credit analysis, and financial institution credit analysis—is a specialty in its own right. This book focuses on the analysis of financial institutions, and, more particularly, on the analysis of banks. Both corporate credit analysis and sovereign/municipal analysis are, however, discussed as a passing knowledge of each field is useful background for the bank credit analyst.

Note that, in practice, there is often a degree of overlap among the categories. Within a particular institution a single analyst may, for example, be responsible for both financial institutions and corporates. In a similar fashion, the analysis of sub-sovereign entities such as municipalities or public sector agencies may be grouped as a separate category from sovereign analysis or combined with corporate or financial institution analysis.

Finally, in the realm of both counterparty credit analysis and fixed-income analysis, in respect to the three categories of credit analysis discussed below, a distinction can be made between (1) the generic credit evaluation of an issuer of debt securities, without reference to the securities issued; and (2) a credit evaluation of the securities themselves. As a rule, an analysis of the former is a prerequisite to conducting an analysis of the latter.

Corporate Credit Analysts

Corporate credit analysts evaluate the credit risk of nonfinancial companies, such as industrial enterprises, trading firms, and service providers, generally for purposes of either lending to such organizations, holding their securities, or providing goods or services to them that give rise to credit risk. Since banks primarily lend not to other banks but to nonfinancial organizations, the preponderance of credit analysis performed within banks is corporate in nature. Compared to the analysis of financial institutions, where ongoing and long-standing counterparty relationships with other financial institutions involving multiple transactions are customary, corporate credit analysis tends not only to be more specialized by industry, but also more oriented toward specific transactions as opposed to the establishment of continuing relationships.

The largest of the three main areas in which analyst-driven research is performed, corporate credit analysis is also the most diverse, ranging considerably in terms of the industrial and service sectors, products, scale, and the geographical regions of the firms that are the targets of evaluation. While the core principles of corporate credit analysis remain largely the same across nonfinancial sectors, specific industry knowledge is often an important part of the corporate credit analyst's skill set. It follows that, while corporate credit analysis itself is an area of practice, within the field as a whole analysts frequently concentrate on particular industry sectors such as

retailing, oil and gas, utilities, or media, applying sector-specific metrics to aid in their assessment of credit risk.

Such specialization within the realm of corporate credit risk evaluation is most apparent in fixed-income analysis and at the rating agencies, since both almost always involve more intensive and primary research than is required for risk management within financial institutions.⁵ As an illustration, a U.S.-based global investment bank would approach the division of responsibility among corporate credit analysts with each analyst taking responsibility for just one or two sectors, while some analysts would have a regional brief. In this example, corporates are broadly classified as falling into one of the following sectors:

- Transportation and vehicle manufacture
- Paper and forest products
- Natural resources (excluding forest products)
- Chemicals
- Energy
- Property
- Telecom/media
- Utilities
- Sovereigns

In the same way that the sector being analyzed influences the analytical approach, so too may the scale of the business affect the analytical methodology. The analytical tools and metrics applied to small businesses—which are increasingly the target of bank business lending as large enterprises gain access to the capital markets—may differ from those applied to publicly listed multinational enterprises. Compared to large listed organizations about which there is much publicly available information, more field and primary research may be needed in respect to small and medium-size enterprises as well as more intensive scrutiny of the owners and managers. Lastly, since cash flow analysis is especially critical in evaluating corporate credit risk and the analyst is likely to assess the creditworthiness of firms in more than one industry, accounting skills perhaps take on somewhat greater importance in the corporate credit realm than in respect to financial institutions.

Bank and Financial Institution Analysts

Another category of credit analysis looks at banks and other financial institutions, and its corresponding objective is to assess the creditworthiness of financial intermediaries. In contrast to corporate credit analysis, this function will be only infrequently performed for the purpose of making conventional lending decisions. Instead, the analysis of a particular bank is generally undertaken either in contemplation of entering into one or more usually multiple bilateral transactions with the bank as a counterparty. As noted previously, banks and other financial institutions can also be assessed with reference to and as part of an analysis of debt instruments or securities issued by such institutions.

Counterparty Credit

As noted previously, the term *counterparty* refers to a financial institution's opposite number in a bilateral financial contract. The credit risk that arises from such transactions

is often called *counterparty (credit) risk*, and bank and financial institution analysts whose role is to evaluate the credit risk associated with the transaction are frequently called *counterparty credit analysts*. Whatever their title, the focus of counterparty credit analysts is on the potential credit risks that result from financial transactions, including *settlement risk*, which is discussed later.

Counterparty credit analysts may also have responsibility for setting exposure limits to individual institutions or countries, or participate in the process of making a decision as to whether to extend credit or not. Because the vast majority of financial transactions involve banks or other financial institutions on at least one side of the deal, counterparty credit analysts are generally employed mainly by such organizations. For the same reason, banks and other financial institutions are the principal targets of this type of credit analysis.

Product Knowledge

To both counterparty and corporate credit analysts employed by a financial institution, the type of exposure anticipated, whether a conventional plain vanilla transaction or one more complex is contemplated, will often affect the analysis undertaken and the conclusions reached. The reason for this, which was discussed in the preceding chapter, is that the type of product from which the prospective credit risk arises can have a substantial impact on the severity of any loss incurred. Equally, the length of time over which the credit exposure will extend—that is, the tenor of the exposure—is an important credit consideration.

The vast majority of counterparty transactions involve the following product categories:

- Financing or obtaining funding directly through the interbank market on a senior unsecured basis
- Financing or obtaining funding through repurchase (repo)/reverse repurchase (reverse repo) transactions
- Financing or obtaining funding through the lending or borrowing of securities
- Factoring, forfeiting, and similar types of receivables finance
- Holding or trading of debt securities of banks and other financial companies for trading or investment purposes
- Foreign exchange (FX or forex) dealing, including the purchase or sale of FX options and forwards
- Arranging or participating in other derivative transactions including, for instance, interest rate swaps, foreign-exchange swaps, and credit derivatives
- Holding or participating in securitizations or structured finance that gives rise to counterparty credit risk
- Correspondent banking services, including trade finance effected through documentary letters of credit
- Custodial and settlement services

Sovereign/Municipal Credit Analysts

A third category of credit analysis concerns the assessment of sovereign and country risk. Governments throughout the world borrow funds through the issue of fixed-income securities in local and international markets. Sovereign risk analysts are

therefore employed to assess the risk of default on such obligations. (Technically, governments do not go bankrupt, although they may default on their obligations.) Sovereign analysis is, however, relevant not just to profiling the risk associated with government debt issues. It also provides the context for evaluating credit risk in respect to other exposures. Hence, sovereign analysts appraise the broader risks arising from cross-border transactions as well as from transactions directly with a nation, its subnational units (e.g., provinces and cities), or governmental agencies.⁶

Sovereign analysts make use of tools that are analogous to those utilized by analysts assessing the risk of corporate entities, but that take into account the peculiar characteristics of governments. Instead of looking at company financials, sovereign analysts examine, among other metrics, macroeconomic indicators to gauge whether a government will have the wherewithal to repay its financial obligations to local and international creditors. Most sovereign risk analysts therefore have a strong background in economics. Sovereign risk also takes account of political risk, so it is also necessary for the analyst to have an understanding of the political dynamics of the country that is under review.

The Relationship between Sovereign Risk and Bank Credit Risk

Sovereign risk and bank credit risk are closely linked, and each affects the other. In brief, the strength of a nation's financial system affects its sovereign risk and vice versa. For this reason, the level of country or sovereign risk associated with a particular market is a significant input in the credit analysis of banks located in that market. Although sovereign risk analysis is a distinct field from bank credit analysis, the bank credit analyst should have at least a passing familiarity with sovereign risk analysis (and vice versa). As part of the process of forming a view about the impact of the local operating environment on a particular banking industry, many bank analysts engage in a modicum of sovereign risk analysis while also relying upon the sovereign risk ratings and accompanying analyses published by the rating agencies or from internal divisions responsible for in-house assessments of sovereign risk.

Sovereign risk has two distinct but related aspects.

1. One is the evaluation of a sovereign entity as a debt issuer as well as the evaluation of specific securities issued by a sovereign nation, or by subnational entities within that nation.
2. The other is the evaluation of the operating environment within a country insofar as it affects the banking system.

Although the process of evaluating each aspect of sovereign, or country, risk is similar in both situations, they represent two discrete facets of credit risk analysis. The analysis of sovereign debt issues may seem to be outside the scope of this book, but, to the extent that such instruments are now found in the books of a growing number of banks, and even of small domestic banks, this will be of further concern to us.⁷ In contrast, the analysis of sovereign risk itself, as part of an evaluation of a bank's operating environment, is of critical importance to the overall evaluation of the institution's credit risk profile.

While sovereign risk is in itself relevant to the analytical process, bank credit analysts are particularly interested in the systemic risk associated with a given

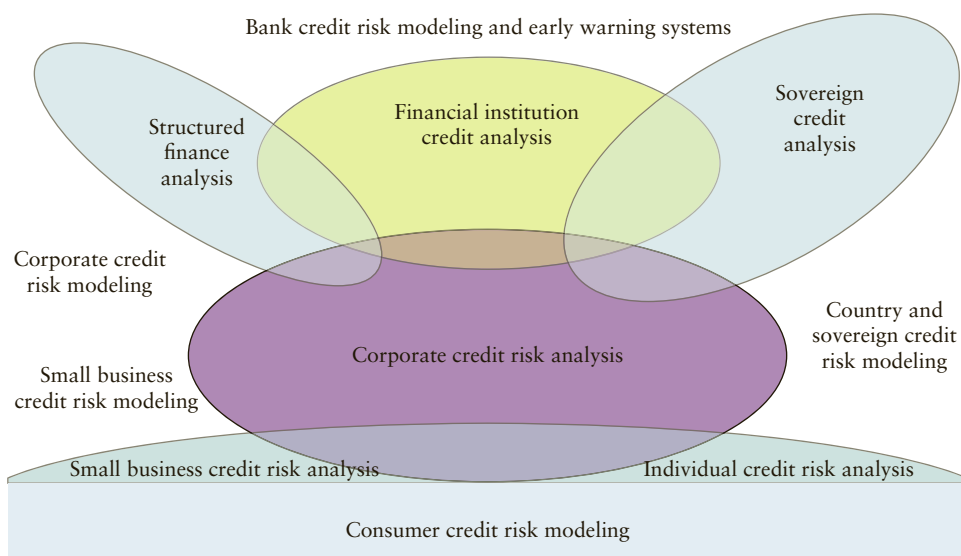


EXHIBIT 2.1 The Universe of Credit Analysis by Subject of Evaluation

banking industry. Systemic risk, which is closely related to sovereign risk and arguably a subset of it, refers to the degree to which a banking system is vulnerable to collapse, and conversely, to the strength and stability (or conversely the fragility) of the banking sector as a whole. Systemic risk is largely synonymous with the risk of a banking crisis, a phenomenon characterized in part by the roughly contemporaneous collapse or rescue prior to collapse of multiple banks within a single jurisdiction. Sovereign risk and systemic risk are discussed in greater detail in later chapters, together with the systemic risk and the aftermath of its realization. Exhibit 2.1 depicts the universe of credit analysis in a graphic format.

Classification by Employer

Another way to understand the work that credit analysts perform is to look at the types of organizations that employ them. A bank credit analyst, for instance, generally works in one of four primary types of organizations, which closely correspond to the functional roles described above. They are:

1. Banks and related financial institutions
2. Institutional investors, including pension funds and insurance firms
3. Rating agencies
4. Government agencies

The first two categories are not entirely discrete. As discussed in the box on the next page, banks and other financial institutions such as insurance companies may simultaneously function as issuers, lenders, and institutional investors, while other organizations that have investment as their primary function may offer additional services more typically associated with banks and may also include a significant risk management group.

Banks, NBFIs, and Institutional Investors

Banks constitute the largest single category of financial institutions, and are the largest employer of credit analysts. Aside from banks, nonbank financial institutions (NBFIs) are also significant users of these skill sets as well as being themselves objects of analysis. A major subcategory of NBFIs is comprised of investment management organizations. As discrete organizations, mutual funds, unit trusts, and hedge funds fall within this grouping.

BUY-SIDE/SELL-SIDE

Institutional investors, and the investment analysts employed by them, are collectively referred to as the *buy-side*. Intermediaries that attempt to sell or make markets in various securities, together with the analysts who work for them, constitute the *sell side*. There is some overlap, and it is possible for a financial institution to be on the buy side and the sell side at the same time.

For example, a bank may sell securities to customers while also trading or investing on a proprietary basis. Similarly, while insurance firms are nominally in the business of risk management, they are also major institutional investors. The premiums they collect need to be invested on a medium- or long-term basis to fund anticipated payouts to policyholders, and, as a result, they are important institutional investors. As with banks, credit analysts employed by investment management organizations generally work in either a risk management capacity or an investment selection capacity.

Rating Agencies

Rating agency analysts are credit analysts who work for rating agencies to evaluate the creditworthiness of banks, corporations, and governments. The three major global agencies are Moody's Investor Services, Standard & Poor's Rating Services, and Fitch Ratings. In addition, local rating agencies in various countries, sometimes affiliated with the big three, may play an important role in connection with domestic debt markets.

The three-step purpose of a rating agency analyst performing a credit evaluation for the first time will be to:

1. Undertake an overall assessment of the credit risk associated with the issuer
2. Evaluate the features of any securities being issued in respect to their impact on credit risk
3. Make a recommendation concerning an appropriate credit rating to be assigned to each

The credit rating will be used by risk managers and investors to determine whether the exposure or investment is attractive, as well as at what price it might be worth accepting.

THE RATING AGENCY ANALYST

Credit analysts are employed by rating agencies to perform risk assessments that are distilled into ratings represented by rating symbols. Each symbol, through its letter or number designation, is intended to classify the rated institution as a strong, average, or weak credit risk, and various gradations in between. The assignment of a rating to the bank will typically be supported by an analytical profile, which represents the fruit of considerable primary research on the part of the analytical team.

Government Agencies

Governments function both as policy makers and regulators on the one hand, and as market participants on the other, issuing debt or investing through government-owned organizations. Government bank and insurance examiners are essentially credit analysts who function in a regulatory capacity, assessing the riskiness of a bank or insurance company to determine the institution's soundness and its eligibility to continue to do business. With regard to governments or their agencies that act as market participants, the scope and legal status of such wholly or partly state-owned entities vary considerably from country to country. Generally, credit analysts within these institutions function in a similar manner to their counterparts at privately owned enterprises.

THE RATING ADVISOR

One role that makes use of credit analytical skills but does not easily fit into the classifications in this chapter is that of the rating advisor. Usually a former rating agency analyst, the rating advisor is normally employed by investment banks to provide guidance to prospective new issuers in the debt markets. As a rule, the rating advisor will make an independent analysis of a prospective issuer to gauge the rating likely to be assigned by one or more of the major agencies, and then counsel the enterprise on how to address its likely concerns. The rating advisor's guidance will include advice on how to make a presentation to the rating agency analysts and how to respond to their questions. His or her job is as a behind-the-scenes advocate, seeking to obtain the best rating possible for the prospective issuer and working to see that it is given the benefit of the doubt when there is uncertainty as to whether a higher rating is justified over a lower one.

Organization of the Credit Risk Function within Banks

In looking more closely at those credit analysts who perform a risk management function on behalf of market participants, it may be useful to examine some common approaches to the organization of this function within institutions. There are several

typical approaches. The usual one is to divide the functions between corporate, financial institution (FI), and sovereign credit risk. The specific topic of structured credit analysis might also constitute a separate team within the FI group.

ROLE OF THE BANK CREDIT ANALYST: SCOPE AND RESPONSIBILITIES

Having surveyed the various types of credit analysts, we now examine the principal roles of the bank credit analyst in more detail. In the previous section, we saw that within the category of bank credit analyst are a number of different subtypes. In this section, our focus is on the counterparty credit analyst and the fixed-income analyst.

The Counterparty Credit Analyst

The counterparty credit analyst is concerned with evaluating banks and other financial intermediaries as part of his or her own organization's larger risk management function. The need for the evaluation of credit risk exposure to banks is an especially important one.

The Rationale for Counterparty Credit Analysis

Undertaking credit risk exposure to other financial institutions is integral to banking. Banks take on credit risk exposure in respect to other banks in a number of different circumstances including trade finance and foreign exchange transactions. With regard to trade finance, banks ordinarily seek to cultivate correspondent banking relationships globally in order to build up their capacity to offer their importing and exporting customers trade finance services. Depending upon the structure of the banking system within a particular country, the proportion of banks that are internationally active may represent a small or large percentage of the significant commercial banks.

Such banks will have correspondent banking relationships with hundreds of financial institutions worldwide, and in large countries, local and provincial banks will have similar relationships with their counterparts in other geographic regions. Hence, unlike a bank's corporate borrowers, which in most cases are likely to be based in the same region as the bank's head offices (unless the bank has significant overseas operations), its exposure to other financial institutions in the form of exposure to correspondent banks may extend halfway around the world.

In addition to the need of many banks to have international relationships with other banks around the world, under normal market conditions, banks frequently lend to and borrow from other banks. Such interbank lending serves to maintain a market for liquid and loanable funds among participating banks to meet their liquidity needs. Overall, exposure to other banks and other financial institutions is likely to be a substantial proportion of the bank's overall credit risk (although not so large in net terms as it is in nominal or notional terms). Despite the need for banks to maintain ongoing transactional relationships with other financial institutions, their characteristic high leverage, among other traits, makes them not insignificant credit risks. Elevated leverage on both sides of a transaction makes each counterparty extremely sensitive to risk, and protecting against such risk constitutes a significant cost of doing business.

Exacerbating the vulnerability of banks to credit risk with respect to other banks is the potential for a collapse of a single, comparatively small bank to have repercussions far out of proportion to its size. The adverse effects can affect the business climate and economy of a whole nation or region, while the failure of a major bank or multiple banks can be catastrophic, potentially resulting in a collapse of the local banking system.⁸ Although the total collapse of an internationally active bank is fairly rare in normal times, the events of 2008–2012 show how real a possibility they are.

Credit Analyst versus Credit Officer

Like the fixed-income analyst, the counterparty credit analyst's research efforts are undertaken with the objective of reaching conclusions and recommendations that will influence business decisions. In the case of a counterparty credit risk evaluation, this often takes the form, as previously suggested, of a recommendation that a particular internal rating be assigned to the institution just analyzed. The context for such a recommendation may be an annual review or a specific proposal for business dealings with the subject institution (through the establishment of country limits or credit lines).

The scope of analytical responsibility varies from bank to bank. At some institutions, the roles are entirely separate. The credit analyst's responsibility may be limited to analyzing a set of counterparties, as well as particular transactions, and preparing analytical reports, but might not extend to making credit decisions, or undertaking the related work of recommending credit limits and making presentations to the credit committee. Instead, this function might be the sole responsibility of the credit officer. Normally, under such a structure, the credit officer has first gained experience as a credit analyst and has acquired the intensive product knowledge enabling him or her to rapidly gauge the risk associated with a specific transaction.

At other banks, these roles may be more closely integrated. The credit officer may also perform relevant credit analyses or reviews, and prepare applications for new credit limits or for annual reviews of existing limits. Irrespective of how the functions are defined at particular organizations, the term *credit officer* implies a greater degree of executive authority than that associated with the term *analyst*, which tends to connote an advisory rather than an executive function. Within financial institution teams, there may be a further functional separation between the role of the analyst and the role of the credit officer.

At the executive level, the principal objectives of the counterparty credit risk team are:

- To implement the institution's credit risk management policy with respect to financial counterparties by subjecting them to a periodic internal credit review, and with the aim of establishing prudent credit limits with respect to each counterparty.
- To evaluate applications for proposed transactions, recommending approval, disapproval, or modification of such applications, and seeing the process through to its final disposition.

As a practical matter, the relevant decision-making responsibility customarily extends to:

- *Authorizing the allocation of credit limits* within a financial institution's group or among various product lines.

- *The approval of credit risk mitigants* including guarantees, collateral, and relevant contractual provisions, such as break clauses.
- *The approval of excesses over permitted credit limits*, or the making of exceptions to customary credit policy.
- *Coordination with the bank's legal department* concerning documentation of transactions in order to optimize protection for the bank within market conventions.

Depending on the organizational structure, credit officers may liaise with other departments within the bank such as those that are charged with monitoring limit violations, margin collateral, and market risk. In addition, credit officers may have responsibility for reviewing and recommending changes in bank credit policies.⁹

The advent of Basel II and Basel III has meant that credit analysts must provide their skill also to a new, gigantic, range of tasks for the benefit of risk management departments. This has introduced a new set of parameters in how the credit analyst approaches his or her duties.

Product Knowledge

Credit analysis cannot be divorced from its purpose or context. Certainly, the overall and ultimate objective of the credit risk management framework within which counterparty credit analysis takes place is to optimize—within regulatory constraints and internal parameters—return on risk-adjusted capital. The myriad of financial products that a bank offers to its customers, however, together with the various trading and investment positions it takes in the operation of its business, engender a multitude of specific credit exposures. The more common of these were enumerated in the preceding section concerning bank and financial institution analysts. A more systematic list is provided in Exhibit 2.2.¹⁰

Although counterparties are likely to first be graded without regard to a particular transaction, a full estimation of credit risk requires that specific transactions also be rated with reference to the type of obligation incurred.¹¹ In comparison to the expansive categories of obligations evaluated by rating agencies, a bank goes beyond a “rating exercise” to make decisions concerning specific limits on exposure and the approval or disapproval of proposed transactions, together with required modifications if not approved in full. Such decisions cannot prudently be made without

EXHIBIT 2.2 Selected Financial Products

Simple Financial Products	More Complex Financial Products
Term loans	Mortgage-backed securities
Documentary letters of credit	Asset-backed securities
Money market investments/obligations	Credit default swaps
Investments in bonds/bond issues	Structured investment facilities
Spot transactions	Structured liquidity facilities
Interest rate swaps	Weather derivatives

product knowledge,¹² which refers to the in-depth understanding of the characteristics of a broad range of financial products. These characteristics include:

- The impact of the proposed transaction on the borrower's financials
- The features of the obligation or product and its risk attributes
- The amount and type of credit risk mitigation
- Any covenant agreed to by the borrower

There, credit analysis of the counterparty as a discrete entity merely represents only one input in the decision-making process. An equally essential part of the analytical process is the understanding of the risks associated with particular products or transactions. The counterparty credit analyst provides the initial credit evaluation supporting a recommendation concerning decisions on these points. Depending upon how responsibilities are divided, it may be up to the credit analyst or the credit officer to supply the product knowledge against which appropriate credit judgments can be made. These decisions—as to whether an individual transaction will be entered into, and under what terms—are, of course, made with reference to internal policies and procedures.

AN ADVERSARIAL ROLE?

Whether supporting proprietary trading operations or customer business, the relationship between the credit officer and the front office is rarely adversarial. While the credit officer usually has the right to reject a proposed transaction, he or she must generally remain cognizant of the fact that banking is a risk-taking business and that it is profits from such risk taking that pay the bills.

Consequently, the credit officer is generally expected to be receptive to concerns of the business and to look for ways to meet changing business needs while protecting the bank against undue credit risk exposure. The most successful banks are those that welcome and manage risk, and that price and hedge that risk appropriately, and the most effective credit officers are those that understand that a balance that must be maintained between profits and prudence.

The Fixed-Income Analyst

Credit analysts, as we have seen, may function not only as risk analysts, assessing and managing risk, but also as investment analysts assisting in the selection of investments. A relatively small proportion of these fixed-income analysts cover financial institutions. Such analysts may specialize in banks, or banks may just comprise a portion of their portfolio. Like equity analysts, fixed-income analysts make recommendations on whether to buy, sell, or hold a fixed-income security such as a bond. That is, they must ascertain the relative value of the security. Is it undervalued and, therefore, a good buy, or overvalued and consequently best to sell?

THE FIXED-INCOME ANALYST

The fixed-income analyst's goal is to help his or her institution make money by making appropriate recommendations to traders and to clients. As part of this objective, the fixed-income analyst seeks to determine the value of any debt securities issued by the bank, taking account of market perceptions, pricing, and the issue's present and prospective creditworthiness. This analysis is used to make recommendations to traders or investors to help them decide whether to buy, sell, or hold a given security.

Approaches to Fixed-Income Analysis

Fixed-income analysis can be divided into fundamental analysis and technical analysis. Fundamental analysis explores many of the same issues that are undertaken when engaging in credit analysis for risk management purposes; that is, default risk. But the definition of credit risk applied may differ to a degree from that utilized by the counterparty credit analyst or the corporate credit analyst. Technical analysis looks at market timing issues, which are affected by the risk appetite of institutional investors and market perception, as well as pricing patterns. Investor appetite is often strongly influenced by headline events, such as political crises, foreign exchange rates, and rating actions, such as upgrades or downgrades, by credit rating agencies. Most fixed-income analysts consider both fundamental and technical factors in making an investment recommendation.

Impact of the Rating Agencies

The impact of rating agencies on fixed-income analysis is twofold.

First, by providing independent credit assessments of bond issues and of sovereign risk, rating agencies facilitate the establishment of benchmark yield curves, and thereby strengthen markets and enhance liquidity. At the same time, assigned ratings tend to foster a market consensus on a particular issuer and thereby provide a basis for the determination of the relative value of the issuer's securities.¹³

Second, the perception of the likelihood of a rating action, both as an indicator of fundamentals and irrespective of them, may be factored into an investor's calculus. Ratings therefore play a critical role in fixed-income analysis in regard not only to the fundamentals they reveal, but also the probability of rating actions being taken and the timing of such actions. See box entitled "Rating Migration Risk" in the previous chapter.

Whether designed for a bank's customers or the bank itself, fixed-income analysis requires a good understanding of:

- The elements that affect creditworthiness
- How the issue and the issuer are perceived by the market
- Market movements and dynamics
- How rating agencies operate

Often fixed-income analysts have had prior experience working as rating agency analysts.

A Final Note: Credit Analysis versus Equity Analysis

Much of the published analysis available on banks is produced by equity analysts for stock investors rather than by credit analysts. The reason is that bank stocks are often of greater interest to the larger investment community than bank debt securities, which, at least in the past, were not as widespread globally as equity securities.

The focus of equity analysis, it must be acknowledged, is often antithetical to the aims of credit research. Equity analysis concentrates on determining whether a prospective investor should invest in the shares of a particular firm. The core questions that equity analysis seeks to answer are:

- Which course of action will best profit an investor: to buy, sell, or hold the securities of the subject company?
- What is the appropriate value of the company's securities, based on the best possible assessment of its present and future earnings?

Bank equity analysts, therefore, almost exclusively confine their analysis to publicly listed financial institutions (i.e., banks listed on a stock exchange), although they might also analyze a bank that is about to list or a government-owned bank that is about to be privatized.

A principal indicator with which equity analysts are concerned in determining an appropriate valuation is return on shareholders' equity (ROE), a number that reflects the equity investor's return on investment. Since ROE is closely correlated with leverage, higher profitability does not necessarily imply higher credit quality; instead, as common sense would dictate, risk often correlates positively with return. In contrast to the equity analyst, the credit analyst tends to give greater weight to a variety of financial ratios that reflect a bank's asset quality, capital strength, and liquidity. Together, such indicators reflect the institution's overall soundness and ability to ride out harsh business conditions rather than merely its ability to generate short-term profits.

Another salient difference between credit analysis and equity analysis concerns the extent to which financial projections are utilized. Equity analysts normally base their share price valuations on financial projections. (Such projections are, of course, derived from the historical data.) In contrast, historical financial data is the principal, if not sole, focus of credit analysts.¹⁴

Despite this critical difference in approach between the equity analyst and the credit analyst, neither equity nor credit analysts are necessarily oblivious to credit or valuation concerns. Since shareholders are theoretically in the first loss position should a bank fail, it is understandable, and indeed crucial, that equity analysts pay some attention to credit risk. Indeed, credit considerations come to the fore during times of economic stress. The Asian crisis of 1997–1998 highlighted the need for analysts in the region to take into account a company's financial strength and external support, as well as its profitability. Following the crisis, as Lehman Brothers' analyst Robert Zielinski noted:

In the past, most of the focus of an analyst's research was on the earnings line of the income statement. The analyst projected sales based on industry growth, profit margins, and net income. The objective was to come up with a

*reasonable figure for EPS growth, which was the main determinant of stock valuation. . . . Today, the analyst places most of his emphasis on the balance sheet. Indeed the most sought-after equity analysts in the job market are those who have experience working for credit rating agencies such as Moody's.*¹⁵

In a similar vein, a banking institution with a high proportion of bad loans and correspondingly high credit costs will probably not be the first choice for an equity analyst's buy list.

ARE EQUITY INVESTORS SUBJECT TO CREDIT RISK?

In discussing the influence of credit analysis on equity analysis and vice versa, an interesting question arises as to whether equity investors are exposed to credit risk. Are they?

Although the purchase of ordinary shares will often be subject to settlement risk, a form of credit risk, if that risk is put aside, then the answer, formally speaking, is "no." Credit risk presumes the existence of a definite financial obligation between the creditor and the party to whom the credit is exposed to the risk of loss through the possibility of default. In other words, for credit risk to exist there must be a corresponding financial obligation, either present or prospective, between the issuer and the investor.

A common shareholder of an equity security is subject to a risk of loss but as a rule there is no financial obligation to redeem the investor's shares. His or her investment is perpetual, and no firm claim exists upon the firm's assets; the claim is only upon the excess of assets over obligations to creditors at the time of liquidation. Similarly, there is no right on the part of the ordinary shareholder to dividends. Hence an equity shareholder is not subject to credit risk, though he or she is subject to market risk—the risk that the value of the investor's shares will drop to zero.

In view of the weak position of shareholders vis-à-vis creditors in the event of a bank's failure, an equity investor is likely to be sensitive to the possibility of the value of his or her investment being wiped out completely. Insofar as the prospect of not just a zero percentage return but the loss of the entire investment is linked to the credit profile of the bank, which it indeed is, equity investors in practice are likely to perceive credit risk even if as shareholders they have no formal right to redeem their investment and must wait in line behind creditors in the event of liquidation. In any event, whether or not credit risk formally exists in relation to equity investors, there is no reason to think that credit assessment techniques should not be of benefit to equity investors in certain instances. In the same way that a fall in the credit quality of a debt security generally results in a lower market price for the debt security, a decline in the credit quality of an enterprise that issues both debt and equity will tend to register downward pressure on the prices of both its debt and equity securities—all other things, of course, remaining equal. Hence, it would be entirely appropriate for equity analysts to employ the techniques of credit analysis to evaluate their prospects in such situations.

Likewise, equity market conditions and performance of a particular bank stock may, on occasion, be of interest to the bank credit analyst. For instance, dramatic falls in a bank's stock price as well as the existence long-term adverse trends are worth noting as they may, but not necessarily, suggest potential credit-related problems. Similarly, the credit analyst should have some sense of the bank's reputation in the equity markets as that may have some effect on the institution's capacity to raise new capital if required.¹⁶ This, in turn, will influence the perceived capital strength and liquidity of the institution.

APPROACHES TO EQUITY ANALYSIS

Equity analysis can be divided into two broad approaches: fundamental analysis and technical analysis. Fundamental analysis examines the factors affecting a company's earnings, including the company's strategy, comparative advantages, financial structure, and market and competitive conditions. It attempts to ascertain whether the firm's shares are undervalued or overvalued with respect to the firm's present and projected future earnings. Thus, the core of the equity analyst's work revolves around the constructing of financial projections upon which the analyst's estimated valuations are based. Making projections is largely about making assumptions. Assumptions inevitably embody an element of subjectivity, and small differences in assumptions can result in large differences in the resulting calculations of expected future stock prices. Regardless of how estimated future prices are calculated, the resulting figures will determine in large part whether a recommendation to buy, sell, or hold is made.

Technical analysis looks at patterns or, more accurately, perceived patterns, in share price movements to attempt to predict future movements. To the technical analyst, these patterns express common archetypes of market psychology, and technical analysis emphasizes the timing of the decision to buy or sell. Most equity analysts, whether covering banks or other companies, employ fundamental rather than technical analysis as their primary tool, although technical factors will often be given some consideration. As opposed to technical analysis, fundamental analysis is relatively unconcerned with market timing issues. Instead, it presupposes a generally efficient market amid which temporary inefficiencies may arise, enabling investors to find bargains. A corollary belief is that the market will ultimately recognize the true value of such bargains causing share prices to rise to a level that better corresponds to that true value.

CREDIT ANALYSIS: TOOLS AND METHODS

As with any field, credit analysis utilizes various tools and resources, employs recognized methods and approaches, and generates customary types of work products. In the usual course of events, the analyst will:

- *Gather information* concerning a subject entity and industry from a range of sources.

- Distill the data into a *consistent format*.
- *Compare* the financial and other data with similar entities (peers), and to past performance.
- *Reach conclusions* (and possibly make recommendations) that are ordinarily expressed in writing as credit reports or credit profiles.

Although credit analysis in its various permutations has the same paramount goal—to come to a determination as to the magnitude of risk engendered by a credit exposure, or conversely the creditworthiness of an entity—the analytical approach used will differ according to the circumstances.¹⁷ Hence, the combination of tools, methods, and the resulting work product will differ according to the nature of the analyst's role.

Qualitative and Quantitative Aspects

Credit analysis, as suggested in Chapter 1, is both a *qualitative* and a *quantitative* endeavor, involving a review of the company's past performance, its present condition, and its future prospects. Aside from the purely mechanical credit scoring exercise, it is practically impossible to undertake an entirely objective credit analysis that considers only quantitative criteria. Similarly, a solely qualitative evaluation performed without quantitative indicia to support it is arguably more vulnerable to inconsistency, human prejudice, and errors of judgment. In practice, the two aspects of analysis are inextricably linked.

Quantitative Elements

The *quantitative* element of the credit assessment process involves the comparison of financial indicators and ratios—for example, percentage rates of net profit growth or, in the case of a bank, its risk-weighted capital adequacy ratios. The juxtaposition of such indicators allows the analyst to compare a company's performance and financial condition over time, and with similar companies in its industry.¹⁸ In short, the quantitative aspect of credit analysis is underpinned by *ratio analysis*.

RATIO ANALYSIS

Ratio analysis refers to the use of financial ratios, such as return on equity, to measure various aspects of an enterprise's financial attributes for the purpose of respectively identifying rankings relative to other entities of a similar character and discerning trends in the subject institution's financial performance or condition. Ratios are simply fractions or multiples in which the numerator and denominator each represent some relevant attribute of the firm or its performance. The most useful financial ratios are those in which the relationship between such attributes is such that the ratio created becomes in itself an important measure of financial performance or condition. To return to the initial example, return on equity, that is, net income divided by shareholders' equity, shows the relationship between funds placed at risk by the shareholders and the returns generated from such funds, and for this reason has emerged as a standard measure of a firm's profitability.

Qualitative Elements

Not all aspects of a company's financial performance and condition can be reduced to numbers. The *qualitative* element of credit analysis concerns those attributes that affect the probability of default, but which cannot be directly reduced to numbers. Consequently, the evaluation of such attributes must be primarily a matter of judgment. For example, the competence of management is relevant to a firm's future performance. It is management, of course, that determines a firm's performance targets, plans how to reach these objectives while effectively managing the company's risks, and that is ultimately responsible for a company's success or failure. Ignoring such qualitative criteria handicaps the analyst in arriving at the most accurate estimation of credit risk. Certainly, management competence should be considered in the process of evaluating the firm's creditworthiness. Taking it into account, however, is very much a qualitative exercise.

FINANCIAL QUALITY: BRIDGING THE QUANTITATIVE–QUALITATIVE DIVIDE

If corporate credit analysts benefit from an ability to rely more on quantitative analysis, and if financial institution credit analysts benefit from being able to focus on a comparatively homogeneous sector, one area where both face a new challenge is in the analysis of *financial quality*. By financial quality analysis is meant financial evaluation that goes beyond reported numbers to look at the quality of those numbers and the items they are measuring. Consider one aspect of financial quality to which attention has long been given: *asset quality*. To a bank credit analyst, an evaluation of asset quality, the assessment of a bank's loan book, is a critical and traditional part of the analytical process. To a corporate credit analyst, asset quality usually means the value of a firm's inventory, or to a lesser extent, its fixed assets. Financial quality encompasses other financial attributes including earnings quality—how real is the income reported?—and capital quality. If assets are dubious, then by definition, so is the corresponding equity. The all-important matter of liquidity quality in banks has shown its relevance in the long financial crisis that started in 2007, in particular during the subprime crisis of 2008 and the European debt crisis of 2011. These matters are discussed in greater depth both in this and in subsequent chapters.

The qualitative and quantitative aspects of credit analysis are summarized in Exhibit 2.3.

Intermingling of the Qualitative and Quantitative

Certain elements of credit analysis are inherently more qualitative in nature while others are more quantitative (see Exhibit 2.4), although almost always some of both can be found. As previously suggested, the evaluation of a borrower's stand-alone capacity to service debt is, in general, predominately quantitative in nature. Evaluation of its willingness, however, is predominately qualitative in character.

EXHIBIT 2.3 Qualitative versus Quantitative Credit Analysis

Quantitative		Qualitative	
The drawing of inferences from numerical data. Largely equivalent to ratio analysis. Nominally objective.		The drawing of inferences from criteria not necessarily in numerical form. Nominally subjective.	
Requires criteria to be reducible to figures. More amenable to statistical techniques and automation.		Relies heavily on analyst's perceptions, experience, judgment, reasoning, and intuition.	
Pros	Cons	Pros	Cons
Good starting point for analytical process.	Ignores assumptions and choices that underpin the figures.	Holistic approach that does not ignore what cannot be easily quantified.	Making the relevant distinctions may be difficult—more labor-intensive than quantitative analysis.
Permits use of various quantitative techniques.	Numbers may often only approximate economic reality leading to erroneous conclusions.	Takes account of human judgment—does it pass the sniff test?	Works best when analyst is highly skilled and experienced so requires more training and judgment.
Shows correlations explicitly.	Ratios may not be answering the relevant questions.	Potentially allows financial vulnerabilities and ill-timed strategies to be identified as early as possible.	May encourage inconsistency in ratings owing to differing individual views of the importance of different factors.
Facilitates consistency in evaluation.	Not all elements of credit analysis can be reduced to numbers.		

EXHIBIT 2.4 Quantitative-Qualitative Matrix

Element	Method of Evaluation		Emphasized Evaluation Mode	Mainly Affects
Obligor	Capacity	Financial analysis	Quantitative	PD
	Willingness	Reputation, track record	Qualitative	
Conditions		Country/systemic risk analysis	Mix	All
Obligation characteristics		Product analysis	Qualitative	
Collateral (credit risk mitigants)		Appraisal (for collateral) and characteristics of obligation (if a financial collateral); capacity and willingness (for guarantor), etc.	Mix	LGD and EAD

Indeed, practically all facets of credit analysis simultaneously include both quantitative and qualitative elements. A bank's *loan book*, for instance, can be evaluated quantitatively in terms of *nonperforming loan ratios*, but a review of the character of a bank's credit culture and the efficacy of its credit review procedures is, as with an evaluation of management, largely a qualitative exercise. In the same way, those essentially qualitative elements of credit analysis, such as economic and industry conditions, are often amenable, to a greater or lesser degree, to quantitative measurement through statistics such as GDP growth rates or levels of nonperforming loans. Conceptually, any qualitative analysis can reach a degree of quantification, if only through the use of a scoring approach. The qualitative component of quantitative indicators may not always be as obvious. (See the box entitled "The Hidden Qualitative Aspects of Quantitative Measures.") Other considerations, such as the degree of ability of the central bank to supervise banks under its authority, are more subjective in nature, but nevertheless comparative surveys on bank regulations or bank failures may function as rough quantitative proxies for this attribute.

THE HIDDEN QUALITATIVE ASPECTS OF QUANTITATIVE MEASURES

Even seemingly quantitative indicators often have a qualitative aspect. In particular, financial ratios are considerably more malleable than may be first assumed. Accounting and regulatory standards, and the scope and detail of disclosure, vary considerably around the world, with the more highly industrialized countries typically maintaining stricter standards. For this reason, analysis of banks in emerging markets frequently requires a greater component of qualitative assessment, since the superficially precise financial disclosure is not always to be trusted. We will return to accounting issues affecting bank analysis in later chapters.

Macro and Micro Analysis

The process of bank analysis cannot be done in isolation. Instead, the analyst must be aware of the risk environment of the markets in which the bank is situated and in which it is operating, as well as the economic and business conditions in the financial sector as a whole. In this context, sovereign and systemic concerns must also be taken into account, as must the legal and regulatory environment, and the quality of bank supervision.

At the same time, although the foregoing macro-level criteria will influence a bank's credit risk profile, to be of practical use the credit risk of a particular institution must be gauged relative to its previous results and to similar entities. In other words, to rank a bank's comparative credit risk, the analyst needs to judge the bank he or she is appraising with reference both to the bank's own *historical performance* and to its *peers*, while taking account of operating conditions affecting players within and outside of the financial industry.¹⁹ Exhibit 2.5 summarizes the principal micro- and macro-level criteria to be considered in the analytical process.

EXHIBIT 2.5 Micro Level versus Macro Level

Micro Level Criteria		Macro Level Criteria	
Found at the individual bank level and in relation to close peers, for example, financial performance, financial condition (liquidity, capital, etc.) management competence		Found in the operating environment, e.g., market/sectoral trends, sovereign/systemic/legal and regulatory risk, economic and business conditions, industry conditions, government support	
Quantitative	Qualitative	Quantitative	Qualitative
Comparing a bank's earnings and profitability with its peers; observing changes in bank's capital strength over time	Evaluation of bank management, its reputation and business strategy	Establishing correlations between financial variables such as increasing sector loan growth and NPL ratios	Reviewing systemic risk and impact of changes in the business environment—e.g., from new legislation
Projecting future changes in financial attributes	Judging the quality of reported results	Noting changes in industry profitability over time; forecasting future changes	Assessing the likelihood of government intervention (support)

An Iterative Process

This said, when looking at a market for the first time, the question arises: *Analyze the individual banks first, or the banking system as a whole?* The analyst confronts something akin to a chicken-and-egg problem. Since individual banks must be viewed in context—that is, in relation to other institutions within the industry, particularly to those similarly situated—the relevant banking system requires an analyst's early attention.

But the system or sector as a whole cannot be fully understood without knowledge about the problems and prospects of specific banks. For example, key ratios such as average loan growth may not be available until a large proportion of the individual banks have been analyzed and the data entered into a system spreadsheet.²⁰

To analyze an unfamiliar banking industry, it might be helpful to begin with initial research into the structure of the system as a whole, the characteristics of the industry, and the quality of regulation. In this way, a background understanding of the level and nature of sovereign and country risk may be obtained. This could be followed by a review of the major commercial banks, to provide a foundation and a benchmark for a review of smaller and likely second- and third-tier institutions. Finally, with a more detailed and comprehensive understanding of the sector, the analyst might return to a more macro perspective, preparing a review of the country's entire banking sector, highlighting the impact of key players, and differentiating the various categories of institutions operating within the industry.

Peer Analysis

It is evident that a comprehensive bank credit analysis incorporates both quantitative and qualitative reviews of the subject bank, and comparing it against its peers and

with the bank's historical *performance*.²¹ The comparison with peers is called *peer analysis*, and the comparison with historical performance is called *trend analysis*. The comparison with peers is undertaken to establish how a bank rates in terms of financial condition and overall creditworthiness among comparable institutions in the banking system.

WHAT IS A PEER?

The term *peer* is often used in bank credit analysis to refer to an entity of a similar size and character to the entity being examined. It is essentially synonymous with the term "competitor." A peer group might vary in size from 3 institutions to 50 or more. In most cases, however, the number in the group will be between 3 and 15.

Usually, but not always, the institutions will be based in the same jurisdiction. When evaluating mid-sized commercial banks, for example, the peer banks will in most cases be banks of similar size based in the same country. When evaluating institutions having a global reach, the largest investment banks for example, institutions of similar size but based in different countries might be selected.

Finally, when the relevant market includes more than one country, such as Europe, banks of a similar nature, such as Spanish *cajas* and German *Spar-kassen* (both forms of regional savings banks) may be compared on a transnational basis. When matching up entities in jurisdictions that have different regulatory regimes, however, care must be taken that such variances—in loan classification, for example—are taken into account.

Resources and Trade-Offs

While time available and the depth of any accompanying written analysis may vary, the analyst's principal tools remain the same. See Exhibit 2.6. It is evident that the volume of resources applied to each type of bank credit analysis will differ according to the analyst's situation and aims, as well as availability.

Limited Resources

At one end of the spectrum is the bank rating analyst, upon whom the counterparty credit analyst may rely, who will be engaged in the production of independent research based on intensive primary and field research. In addition to examining the bank's annual reports and financial statements, he or she will typically visit the bank, submit a questionnaire to management, and perform a due diligence investigation. In some markets, the United States, for example, the rating agency analyst is permitted access to nonpublic information that is not available to investment and counterparty credit analysts.

At the spectrum's other end is the counterparty credit analyst assisting in the process of establishing credit limits to particular institutions. Owing to time and resource limitations, he or she is likely to rely largely on secondary source material,

EXHIBIT 2.6 Basic Source Materials for Bank Credit Analysis

Material	Contents	Remarks
Annual reports	Income statement, balance sheet, and supplementary financial statements. These are generally, but not in all cases, available on the web. If not, they are usually available by request.	Accompanying web-based analyst/investor presentations and press releases may also be a useful source of information. Financial data for a minimum of three years is recommended.
Interim financial statements	Interim financials are often limited to an unaudited balance sheet and income statement.	In some jurisdictions, interim statements will only be provided in a condensed or rudimentary form with considerably less detail than in the annual statements.
Financial data sources	A variety of electronic database and other search data services may be part of the bank credit analyst's kit. Some key ones include Bankers' Almanac, Bloomberg, and Bankscope. Bankscope, in particular, is widely used by bank credit analysts. It provides re-spread data and ratios drawn from bank end-year and interim financial statements. In addition, a range of informational databases, statistical data sources, credit modeling and pricing tools are also available from various vendors.*	Although it is always good practice to consult the original financial statements, proprietary data services such as Bankscope are widely employed. Financial data services provide the advantage of consistency in presentation but may not always be available in a timely fashion for all institutions required to be evaluated. In addition, regulatory agencies in various markets may provide data useful to the analyst. In the United States, the Securities and Exchange Commission's EDGAR database is one, while the bank database maintained by the Federal Reserve Bank is another.
News services	News articles concerning acquisitions, capital raising, changes in management, and regulatory developments are important to consider in the analysis. Among the most well-known providers of proprietary news databases are Bloomberg, Factiva, and LexisNexis.	Newspaper and magazine clippings can be helpful but are time-consuming to collect; proprietary data services such as <i>Factiva</i> function as electronic clipping services can collect reams of news articles very quickly. Where there is no access to such services, much of the same information can be obtained free of charge from the web.

(Continued)

EXHIBIT 2.6 (Continued)

Material	Contents	Remarks
Rating agency reports and other third-party research	Reports from regulatory authorities, rating agencies, and investment banks. Reports from the major rating agencies, Moody's, S&P, and Fitch Ratings, are invaluable sources of information to counterparty credit analysts and fixed-income analysts.	Counterparty credit analysts will necessarily rely to a great extent on rating agency reports when preparing their own reviews. Fixed-income analysts will engage in their own primary research but compare their own findings with those of the agencies in seeking investment opportunities.
Prospectuses and offering circulars	Prospectuses and other information prepared for the benefit of prospective investors may include more detailed company and market data than provided in the annual report.	Documents prepared for investors often, as a matter of law or regulation, must enumerate potential risks to which the investment is subject. This can be quite helpful to bank credit analysts. In many jurisdictions, however, prospectuses are not easily accessible or may not add much new data.
Notes from the bank visit and third parties	For rating agency analysts, the bank visit is likely to be supplemented by a questionnaire submitted by the agency and completed by the bank. Fixed-income analysts ordinarily will frequently make bank visits. Counterparty credit analysts are likely to make such visits only occasionally.	Banks often prepare a packet of information for rating agency analysts reviewing or assigning a rating. In addition to information formally obtained in the course of a bank visit, the analyst may also seek to obtain informal views about the bank from various sources.

*Stock and bond prices available from sources such as Bloomberg, which is also a major financial news provider, may also be used for analytical purposes.

such as reports from rating agencies.²² Visits to institutions will usually be relatively brief and limited to those markets about which the analyst has the greatest concern.²³ In general, the rating agency analyst will engage in primary research to a greater extent while the counterparty credit analyst will depend more heavily on secondary research sources.

Primary Research

Fundamental to any bank credit analysis are the bank's annual *financial statements*, preferably *audited*²⁴ and preferably available for the past several years—three to five years is the norm—accompanied by relevant *annual reports*, and any more recent *interim* statements. Other resources may include regulatory filings, *prospectuses*, *offering circulars*,²⁵ and other internal or public documents.

As suggested above, thorough primary research would encompass making a visit to the bank in question, preferably to meet with senior management to gain a better understanding of the bank's operating methods, strategy, and the competence of its management and staff.

THE BANK VISIT

Bank visits for *due diligence purposes* are most frequently made by rating agency analysts, for which they are a matter of course with regard to the assignment of a rating. Fixed-income analysts, as well as equity analysts, will also frequently visit with bank management, although often such meetings will take place collectively at analysts' meetings conducted by management. These usually coincide with the release of periodic financial statements. Because the evaluation by an agency or fixed-income analyst can have a large impact on the ability of the bank to raise financing, it is generally easier for these analysts to gain access to senior managers than for the counterparty analyst.

Counterparty credit analysts tend to make bank visits less frequently. There are two principal reasons. First, in view of the larger universe of banks that counterparty credit analysts generally cover, they will usually have comparatively little time available to make bank visits. Second, senior bank officers cannot afford to be continually meeting with the hundreds of correspondent banks and other institutions with which they have a relationship. Unless the transaction is an especially important one to the counterparty, the analyst may be relegated to less senior staff, whose role it is to manage correspondent and counterparty banking relationships.

The bank visit is practically a prerequisite for the rating agency analyst. Where such a due diligence visit is made, the rating agency analyst will almost invariably submit written questions or a questionnaire to the bank, and visit management. Indeed, best practice is for a team of at least two analysts to make a formal visit to the bank, with the visit lasting the better part of a day or more. The exception to this procedure comes in the case of *unsolicited ratings*,²⁶ which are prepared by the rating agency analyst on the basis of information publicly available. Even for such ratings, the agency analyst may nevertheless visit the institution and have an informal discussion with bank staff. For the bank rating analyst, however, such visits will normally be made whenever possible.²⁷

For the counterparty credit analyst, the decision whether to attempt to make a bank visit will naturally be contingent upon the resources available in terms of time and budget, the importance of the relationship with the entity to be analyzed, and the degree of market consensus on the entity's financial condition, as well as the likelihood that significant information will be gleaned from such a visit. As would be expected, managers at the bank to be evaluated must themselves be receptive to a visit.

REQUISITE DATA FOR THE BANK CREDIT ANALYSIS

The items needed to perform a bank credit analysis will depend upon the nature of the assignment undertaken, but in general the following resources should be reviewed:

- The annual report, including the auditor's report, the financial statements and supplementary information, as well as interim financial statements
- Financial data services and news services
- Rating agencies, data from regulators, and other research sources
- Notes from primary and field research

The Annual Report

Although the annual report may be full of glossy photos and what appears to be corporate propaganda, it should not be ignored. Much can be learned from it about the culture of the bank, how the bank views business and economic conditions, and management's strategy. As the annual report is prepared for the bank's shareholders and prospective equity investors, its thrust will be on putting the bank's operating performance in the best possible light. Bearing this in mind, an understanding of the management's side of the story can nevertheless provide a useful counterpoint to a more critical examination of bank performance.

READING BETWEEN THE LINES

When reading a company's annual report, the analyst should ask himself or herself, "What points are being glossed over?" If liquidity appears to be a weak point, how does the company treat this issue? Is the concern addressed, or is it mentioned only in passing? What other scenarios might unfold besides management's rosy view of their firm's future?

In addition to the intangible impressions it may engender, the bank's annual report will sometimes supply information on particular aspects of the bank's operations not available in the financial statements. Not infrequently it may contain a wealth of mundane factual information, such as the institution's history and the number of branches and employees, as well as useful industry and economic data. Similarly, significant information relating to the regulatory environment, such as changes in accounting rules or banking laws, can frequently be found in the annual report.

The Auditor's Report or Statement

The analyst should turn to the auditor's report at the start of the analysis to determine whether or not the auditor of the bank's accounts provided it with a clean or *unqualified opinion*.²⁸ The auditor's report will normally appear just prior to the financial statements. In essence, a *clean opinion* communicates that the auditor does

not disagree with the financial statements presented by management. It does not mean that the auditor might not have presented the financial information differently, choosing a different accounting approach or disclosing additional data. In other words, an unqualified opinion means that the financial statements as presented meet at least the minimum acceptable standards of presentation.²⁹

Content and Meaning of the Auditor's Opinion

The auditor's opinions vary somewhat in length and content depending upon the jurisdiction in which the audit was performed and the standards applied. Much of the content will be boilerplate language used as a standard format, and designed primarily to shield the auditor from any legal liability.³⁰ What is important is to watch out for any language that is *out of the ordinary*. A typical unqualified auditor's report will contain phrases more or less equivalent to those in Exhibit 2.7.³¹

EXHIBIT 2.7 The Auditor's Opinion: An Unofficial Translation Guide

Boilerplate	What this means:
The auditors have audited specified financial statements of a certain date.	"Do not blame us, the auditors, for anything that occurred or became apparent after that date."
Financial statements are the responsibility of the management of the company.	"We can only base our opinion on data provided by the company. If the data is inaccurate or fraudulent, blame company management, not us."
The financial statements have been prepared in accordance with generally accepted local accounting standards and are free from material misstatement.	"The financial disclosure provided meets minimally acceptable local accounting standards or relevant regulations governing such disclosure. We have not detected any egregious errors or inaccuracies that are likely to have a major impact on any conclusion you may draw about the company for investment purposes."
The audit involved examining evidence supporting the statements on a test basis, which provide a reasonable basis for the auditor's opinion.	"We have not scrutinized every single item of financial data or even most of them. This would cost a small fortune and take an exceedingly long time. Instead, as is deemed customary and reasonable in our profession, we have tested some data for discrepancies that might indicate material error or fraud."
In the opinion of the auditors, the financial statements present that financial position fairly in all material respects as of the date of the audit.	"The financial statements might not be perfect, but they present a reasonable picture of the company's financial condition, subject to the present standards set forth in law and generally followed in the industry, notwithstanding that higher standards might better serve investors."

As a reading of the table will make clear, a clean auditor's report does not ensure against fraud or misrepresentation by the company audited. The fairly standardized language of the auditor's report, although varying from country to country, has evolved to emphasize the limitations in what should be drawn from it. Although audits could arguably be more thorough, the expense involved in checking most or all of the source data that make up a company's financial statements other than on a test basis has been reasonably asserted to be prohibitive.

Qualified Opinions

A *qualified opinion*, that is, one in which the auditors limit or qualify in some way their opinion that the financial statements provide a fair representation of the bank's financial condition, can be discerned in cases where additional items other than those mentioned above are added. A qualified opinion is easily identifiable by the presence of the word *except* in the auditor's statement or report. It is typically found in the concluding paragraph which usually starts with "In our opinion."

Typical situations in which an opinion will be qualified by the auditors include the following:

- The existence of unusual conditions or an event that may have a material impact on the bank's business
- The existence of material related-party transactions
- A change in accounting methods
- A specific aspect of the financial reports that is deemed by the auditor to be out of line with best practice
- Substantial doubt about the bank's ability to continue as a going concern

Of course, the last type of qualification is the most grave and will justifiably give rise to concern on the part of the analyst. Not all qualifications are so serious and should be considered bearing in mind what else is known about the bank's condition and prospects, as well as the prevailing business environment. An extremely rare phenomenon is the adverse opinion, in which the auditors set forth their opinion that the financial statements do not provide a fair picture of the bank's financial condition.

Philippine National Bank, as an example, attracted a serious qualification from its auditors in 2004 in a situation where a specific aspect of the financial reports was deemed to be irregular.

CASE STUDY: THE AUDITOR'S OPINION: THE CASE OF PHILIPPINE NATIONAL BANK

Consider the case of Philippine National Bank (PNB), one of the banks in the Philippines—but which is not the central bank of the country—that were hardest hit by the Asian financial crisis of 1997. The auditor, SGV & Co., said in the last paragraph of the 2004 annual report: "In our opinion, except for the effects on the 2004 financial statements of the matters discussed in the third paragraph, the financial statements referred to above present fairly in all

material respects the financial position of the Group and the parent company as of December 31, 2004 and 2003, and the results of their operations and their cash flows for each of the three years in the period ended December 31, 2004, in conformity with accounting principles generally accepted in the Philippines.”

In the third paragraph, the auditor described a transaction involving PNB’s sale of nonperforming assets to a special-purpose vehicle. The losses from the sale of the transaction were deferred over a 10-year period in accordance with regulatory accounting principles prescribed by the Philippine central bank³² for banks and other financial institutions availing of certain incentives established under the law. But SGV & Co. noted that had such losses been charged against current operations, as required by generally accepted accounting principles, investment securities holdings, deferred charges, and capital funds as of December 31, 2004, would have decreased by P1.9 billion, P1.1 billion, and P3.0 billion, respectively, and net income in 2004 would have decreased by P3.0 billion. This would have been taken against a posted net income of about P0.35 billion in 2004.

In his report on the 2010 accounts, the auditor still had to qualify his opinion as the reporting of the transaction did not comply with the rules of the Philippine GAAP for banks. This is not, of course, to say that the bank was doing anything illegal or was attempting to conceal the transaction.

Although most auditors’ opinions are unqualified and therefore generally do not provide any useful information about the bank, a qualified opinion is a red flag even if it is phrased in diplomatic language, and even if the bank can hide behind the leniency of some regulation. The irregularities noted should be closely scrutinized for their impact on financial reporting.

Change in Auditors

Like a poor credit rating, a *qualified opinion* is not something a company’s management wants to see. As it is management who generally selects and pays the auditing firm, it is sometimes not unreasonably perceived that when a company changes auditors it is the result of a disagreement about the presentation of financial statements or because the particular accounting firm is unwilling to provide a clean opinion. This is certainly not always the case, and the reasons for a change in auditor may be entirely different. In some countries, a change in auditors after a period of several years is mandatory, as a means of preventing too cozy a relationship developing between the auditor and the audited company. Nonetheless, changes in auditors should be noted by the analyst for possible further inquiry.

Who Is the Auditor?

Finally, mention should be made of the organizations that perform audits. The accounting profession has consolidated globally into a few major firms.³³ In some countries, however, independent local firms may have most or all domestic banks as their clients. While privately owned banks are usually audited by independent accounting

firms, government banks sometimes are not. Special government audit units may perform the audit, or they may not be audited at all.

Although there may be no significant difference in quality vis-à-vis a less well-known firm, an audit by one of the major international accounting firms may be perceived as affording a certain imprimatur on a bank's financial statements. The critical issue from the analyst's perspective should not be the name of the firm, but whether the auditor has the expertise to scrutinize the enterprise in question. Bank accounting, for instance, differs in some key respects from corporate accounting, and a modicum of comfort can be drawn from an audit performed by a firm that has experience with financial institutions. Another point to keep in mind is that some local auditors might carry an internationally renowned brand name under arrangements that do not include following all technical and ethical rules in place within that international audit firm.

The Financial Statements: Annual and Interim

The essential prerequisite to performing a credit analysis of a bank, or indeed any company or separate financial entity, is access to its financial statements, either in original form or prespread into a format suitable for analytical purposes.³⁴ Without such financial data, quantitative analysis will be practically impossible.

There are three primary financial statements:

1. The balance sheet—to include off-balance-sheet items
2. The income statement
3. The statement of cash flows

Of these, the balance sheet and the income statement are by far the most important to the analysis of banks, and are discussed in more detail in subsequent chapters.

In respect to nonfinancial companies, the statement of cash flows is often considered to be the most important. The cash flow statement is not particularly helpful, though, in bank credit analysis.

A fourth financial statement, *the statement of changes in capital funds*, is useful in both financial company and nonfinancial company credit analysis. When available, it is particularly helpful in bank credit analysis, as it clearly shows changes in the capital levels reported by the institution.³⁵

FINANCIALS

Sometimes referred to as *financials*, *financial statements* are a form of published accounts that show a company's *financial condition* and performance. There are three principal financial statements:

1. The *balance sheet* (also called *the statement of condition*)
2. The *income statement* (also called the *profit and loss statement* or *P&L*)
3. The *statement of cash flows* (*cash flow statement*)

Other financial statements may be published of which perhaps the most common is the *statement of change in capital funds (the statement of changes in shareholders' equity)*. To facilitate the analytical process, the reports published by companies will usually be modified, or *re-spread* to present the financial data in a consistent manner across the sector so that like items can be compared with like items.

Audited financial statements will ordinarily be found in a company's *annual report* together with the *auditor's report*, *supplementary footnotes*, and a report from management, the last of which may take a variety of forms (e.g., Chairman's Letter to Shareholders). Depending upon the jurisdiction and applicable rules, a company may issue interim financial statements semiannually or quarterly. Of course, for internal or regulatory purposes, special financial statements, normally unaudited, may be prepared.

Timeliness of Financial Reporting

The more timely the financial statements, the more useful they are in painting a picture of an institution's current financial condition. Unfortunately, not all banks issue their annual reports as soon as might be preferred. At best, publication of annual reports will follow within one to two months following the end of the financial year. On occasion, extraordinary circumstances such as the need to restate³⁶ figures as the result of a regulatory action may delay publication. This is usually not a positive sign.

In other circumstances, the reasons for late publication are more innocuous. The bank may still be in the process of translating the report into another language, or there may be delays in printing. In such cases, depending upon the purpose and urgency of the review, the analyst might attempt to obtain preliminary or unaudited figures directly from management.

As a rule of thumb, the less developed the market, the longer the delay in the publication of financial reports tends to be. The most dilatory in reporting financial data tend to be banks in emerging markets that are either government-owned institutions or are not publicly listed on a stock exchange. In extreme cases, an interval of up to two years may pass following the end of fiscal year before audited or official financial data are available for state-owned banks. It is evident that in such cases, the value of the reports will almost certainly be very limited.³⁷ Still, some data is better than none and can at least provide the basis for a less than laudatory report.

SPREADING THE FINANCIALS

While the evaluation of the creditworthiness of a bank is, as suggested, both a quantitative and a qualitative endeavor, an examination of the quantitative financial attributes of the institution is usually the first stage in forming a view concerning its overall credit quality. In this section, therefore a bit more emphasis is placed on preparing to engage in the quantitative part of the analytical process. The initial step in this first stage, unsurprisingly, is to examine a bank's financial statements.

Making Financial Statements Comparable

At the outset, we confront an obstacle that is not unique to the banking industry. In any given market, it is rarely the case that all banks, or all nonfinancial firms, for that matter, will adhere to any single standard format of account presentation. Instead, there is a great deal of variation in how banks present their accounts. While the key elements of published bank accounts tend to be arranged in a similar fashion, the proverbial devil lies in the details. Though guided by regulatory requirements—which have improved in recent times—and the advice of their auditing firm, each bank management will present its results according to its own preferences.

Classification of particular line items and the level of disclosure will differ, as will the lucidity of the accompanying notes and explanatory material. Some financial statements provide extremely detailed data; others are more cursory. Likewise, some provide a great deal of useful information in the financial statements themselves; others relegate data of interest to the accounting notes. Finally, some can be clearly understood, and others can be aggravatingly opaque. Were an analyst to proceed to examine a set of banks solely with reference to the financial statements released by management together with their annual reports, the exercise would be fraught with difficulties. Owing to variations in disclosure, presentation, and classification, he or she would be constantly turning pages back and forth, checking definitions and adjusting for differences in disclosure or categorization. Obtaining a clear picture of how each bank stacks up against others in the sector could be more frustrating than necessary.

To simplify the analytical process and reduce the risk of error, it is common practice to arrange the financials of the banks to be analyzed on a spreadsheet in a simplified and consistent manner, so that like items can be more easily compared with like items. This process is called *spreading the financials*. The analyst's initial task therefore is to spread the financials to present the bank's accounts consistently to facilitate their comparison. With the key financial data presented consistently, the analyst's next steps are to derive those ratios that will provide the best indication of the financial performance and condition of the bank and to collect the facts and information to be used to render the requisite qualitative judgments as part of the analytical process.

DIY or External Provider

Spreading the data may be done in any number of ways. On the one hand, the process may be highly automated through links to internal or proprietary databases. In the case of banks, the leading product in the field is Bankscope, which is ubiquitous in counterparty credit departments around the world, although there are, of course, other sources. On the other hand, the analyst may need to convert the financial data independently provided by each bank into the format decided upon. The format itself may be prespecified, or it may be left up to the analyst's discretion. Naturally, even when a format is specified, the analyst may use his or her own working calculations to supplement the formal procedure.

An advantage of spreading one's own financials is that by completing the process the analyst has already learned quite a bit about the bank, and is well on the road to preparing a report. Spreading financials requires an understanding of how the bank has characterized various items on the accounts, and in the process of making adjustments to fit the standardized spreadsheet, the analyst will glean a great deal of insight into the

nature of the bank's activities and the performance of its business. Another advantage is that external data providers are not infallible and may either as a result of policy or error characterize particular items in a nonoptimal manner, and occasionally in a manner that can give a misleading impression. Where the analysis is of both a critical and intensive nature, rather than one that is routine, there is really no substitute for preparing one's own spreads. Finally, where the requisite data are not available from a third-party provider, spreading one's own financials will be the only alternative.

The primary disadvantage to spreading financials independently is that it is often highly time-consuming and tedious work. In some cases, the accounts may be in a foreign language, further complicating matters. Moreover, the analyst's workload, particularly the time allotted for each evaluation, may make spreading the data from scratch impractical. Having all the data available through an external provider in a standardized format obviously speeds up the review process and enables the analyst to get on with making comparisons rather than spending a great deal of time re-entering data and deciphering items that may be irrelevant to the final review. With regard to an analytical team as a whole, the use of an external provider (or an internally maintained database) is likely to encourage a greater level of consistency in how the data are spread, absent close supervision of analysts, since the format used by the data provider will be the same for each.

One has to recognize, though, that external providers are faced with the same problem as an analyst who performs his own data spreading: they sometimes have to take a view as to how some items should be split into separate subitems, or to the contrary as to how several items should be combined into a single one. While banks in advanced economies have little freedom in the way they publish their figures, the situation could be different in emerging markets. As a result, good analysts might use external providers' data as a basis, but would always revisit the matter through a direct study of raw data published by the bank.

One Approach to Spreading

Assuming no formal procedure is in place, it is a fairly simple matter to prepare a spreadsheet for the key data to be entered. Those skilled in manipulating Microsoft Excel and similar programs, of course, can build customized spreadsheets that are highly automated and include built-in analytical tools. Exhibit 2.8 contains a very simplified version of a uniform spreadsheet that might be used for bank analysis. In this condensed version, we have only shown one year of financial data, and have also included formulas for the reader's reference, but comparisons across years are easily derived. While we have not yet discussed the particular financial attributes that the bank credit analyst seeks to evaluate, the illustration may perhaps serve nevertheless as a reference point for some of the items we will be discussing in the chapters that follow.

It is important to note that a number of indicators cannot directly be derived from such a financial spreadsheet, as some figures are simply not disclosed by financial institutions, or when disclosed are not sufficiently transparent.

At the upper left of the spreadsheet, descriptive information is provided concerning the institution, which in this case is a hypothetical name, "Anybank." The left side of the spreadsheet shows a condensed income statement for Anybank, while the right side contains a condensed balance sheet. The far left-hand column shows how each line item in the income statement is derived and the column to the right of it

EXHIBIT 2.8 Simplified Uniform Spreadsheet for Bank Analysis

Bank Name:	ANYBANK				
Location:	ANYTOWN, ANYLAND				
Period:	Fiscal Year Ending: 12/31/ (Consolidated)				
		200X	Currency	ANYUNIT, millions	200X
Definitions	INCOME STATEMENT		Definitions	BALANCE SHEET—ASSETS	
A	Interest Income	5,000	a	Cash and Near Cash	2400
B	Interest Expense	1000	b	Interbank Assets	1800
C=A−B	Net Interest Income	5000	c	Government Securities	3600
D=E+F	Noninterest Income	3000	d	Marketable Securities	3000
E	Fees and Commission Income	2000	e	Unquoted Securities	200
F	FX and Trading Accounts	1000	f	Total Loans and Advances	171400
G=C+D	Operating Income	8000	g	Due from Holding and Subsidiaries	0
H=I+J+K	Noninterest Expense (Operating Expense)	4000	h=sum(b:g)	Total Earning Assets	180000
I	Compensation and Fringe Benefits	2500	I	Average Earning Assets	177500
J	Occupancy Expenses	500	j	Subsidiaries and Affiliates	0
K	Other Expenses	1000	k	Fixed Assets	20000
L=G−H	Preprovision Income (PPI)	4000	l	Other Assets	0
M	Loan Loss Provision (LLPs)	500	m	Intangible Assets	0
N=L−M	Net Operating Income after LLPs (NOPAP)	3500	n=a+h+j+k+l	Total Assets	200000
O	Nonoperating Items	0		BALANCE SHEET—LIABILITIES and CAPITAL	
P=N−O	Pretax Profit (excluding ⁱⁱ nonoperating items)	3500	o	Interbank Deposits	4600
Q	Tax	1200	p	Customer Deposits—Demand	110000
R=P−Q	Net Income before Minority Interest	2300	q	Customer Deposits—Svgs and Time	55000

S	Minority Interest	200	r=p+q	Total Customer Deposits	165000
T	Preferred Dividends	100	s	Due to Holding and Subsidiaries	0
U=R-S-T	Net Income Attributable to Common Shareholders	2000	t	Other Liabilities	15400
V	Common Dividends	800	u=sum(o:t)	Total Liabilities	185000
W=U-V	Retained Earnings	1200	v	Subordinated Debt and Loan Capital	0
			w	Minority Interest in Subsidiaries	0
	RATIOS			Shareholders' Equity	15000
	PROFITABILITY			CAPITAL	
X=R/aa	Return on Assets	1.16%	y	Tier 1 Capital	15000
Y=R/z	Return on Equity	15.86%	z	Average Equity	14500
Z=C/I	Net Interest Margin	na	aa	Average Assets	197500
ZA=H/G	Efficiency Ratio		ab	Contingent Accounts	0
ZB=H/aa	Cost Margin		ac	Risk-Adjusted Capital	15000
ZC	Effective Tax Rate, Reported		ad	Risk-Weighted Assets	180000
ZD=R/ad	Return on Risk-Weighted Assets		ae	Tier 2 Capital	0
	CAPITAL ADEQUACY			LOAN BOOK—ASSET QUALITY ITEMS	
ba=z/aa	Equity/Assets (Avg)	7.34%	af	Net Charge-Offs (NCOs)	200
bb=z/I	Equity/Earning Assets (Avg)	8.45%	ag	LLRs	1800
bc=x/f	Equity/Loans	8.75%	ah	Nonperforming Loans	1000
	<i>BIS Risk-Adjusted Capital</i>		ai	90-Day Past Due and Accruing (i.e., not counted as official NPL) ⁱⁱⁱ	20
	<i>Ratios:</i>				
bd=y/ad	Tier I	8.06%			
be=ae/ad	Tier II	0.00%	aj	Restructured Loans	50
bf=ac/ad	BIS Total	8.06%	ak	Other Real Estate Owned (OREO) ^{iv}	100
bg=n/x	Leverage (times)	13.3	al=sum(ah:ak)	Nonperforming Assets	1170

(Continued)

EXHIBIT 2.8 (Continued)

Bank Name:	ANYBANK				
Location:	ANYTOWN, ANYLAND				
Period:	Fiscal Year Ending:				
	12/31/ (Consolidated)	200X	Currency	ANYUNIT, millions	200X
	LIQUIDITY			ASSET QUALITY RATIOS	
bh=f/n	Loans/Assets	85.70%	am=l/f	LLPs/NCOs	250.0%
bi=t/r	Loans/(Customer) Deposits	103.88%	an=ag/f	Loan Loss Reserves/Gross Loans	1.1%
bj=t/(r+o)	Loans/Total Deposits	101.06%	ao=am/(f+ak)	NPAs/Loans + Other Real Estate	
				Owned	0.7%
bj=b/n	Interbank Assets/Assets	0.90%	ap=am/f	NPAs/Loans	0.7%
bk=b/o	Interbank Assets/Interbank		aq=ag/ah	Loan Loss Reserves/Nonperforming	
	Deposits	39.13%		Loans	180.0%
bl=(a+b+c+d)/n	Quasi-Liquid Assets Ratio	5.40%	ar=am/x	NPAs/Total Equity	7.8%
bm=(a+b+c)/n	Liquid Assets Ratio	2.10%	as=ag/f	LLRs/Loans	1.1%

i. Depending upon the definitions used, some cash/near cash items might be deemed to be "earning assets."

ii. In contrast to usual reporting practice, for analytical purposes nonoperating items may be subtracted before calculating pretax profit or net income.

iii. For reasons of space, special mention items (that is, problematic assets that do not meet aging criteria) are not shown. In some circumstances special mention items would not be considered official or technical NPLs and might be broken out as a separate line item. This line item hence could be regarded as official NPLs (meeting aging criteria) but nevertheless accruing (by reason of the bank's judgment) minus special mention items (nonofficial NPLs deemed by the bank to be problematic).

iv. This line refers to foreclosed assets.

v. Analytical definitions of liquid assets would likely subject category "c" to a liquidity check taking account of local conditions and any formal restrictions on negotiability.

describes the line item. For example, interest income is designated as “A,” interest expense as “B,” and net interest income “C” is therefore defined as A–B. The third column from the left shows amounts for each item. The right side of the spreadsheet showing assets, liabilities, and equity is analogous to the income statement on the left, with the item defined in the left-hand column (fourth from the left margin of the spreadsheet), the item description in the middle column, and the amount in the far right-hand column.

ADDITIONAL RESOURCES

The Bank Website

The advent of the Internet has made the bank credit analyst’s job easier.³⁸ No longer is it always necessary to request a bank’s annual report and wait weeks for its arrival. Annual reports, financial statements, news releases, and a great deal of background information on the bank and its franchise can be obtained from the web. Moreover, the depth, interactivity, and overall quality and style of the site will say something about the bank as well as its online strategy. In addition, many banks will post webcasts of their results discussion with analysts together with the accompanying presentation.

News, the Internet, and Securities Pricing Data

Annual reports are just about out-of-date the day they are published. Much can happen between the end of the financial year and the publication of the annual report, and the analyst should run a check to see if any material developments have occurred. An examination of the bank’s website can be very helpful here, but alternatively, a web search or the use of proprietary electronic data services such as *Bloomberg*, *Factiva*, or *LexisNexis* can be extremely valuable in turning up changes in the bank’s status, news of mergers or acquisitions, changes in capital structure, new regulations, or recent developments in the bank’s operations.

Bond pricing will, of course, be a primary concern of the fixed-income analyst, but counterparty credit and rating agency analysts can make constructive use of both bond and equity price data when the bank is publicly listed or is an issuer in the debt markets. Anomalous changes in the prices of the bank’s securities can herald potential risks. The market will be the first to pick up news affecting the price of the bank’s securities, and in this sense it can function as a kind of early-warning device to in-house and agency analysts.³⁹ Real-time securities data in emerging markets, as provided by Bloomberg, for example, can be costly, but then again the web with the emergence of search engines like Google has leveled the playing field making much of the same or similar business and financial news easy to access.

Prospectuses and Regulatory Filings

Prospectuses and offering circulars intended for prospective investors are published to enable them to better evaluate a potential investment. Generally speaking, their format and content is restricted by regulation to compel securities issuers to present the benefits of the investment in a highly conservative manner and to highlight possible risks. In view of this latter objective, these documents can be, but are not always, rich sources of information about a bank. Their value depends a great deal on the market

and upon the type of securities issue for which the prospectus was prepared. Prospectuses for equity and international debt issues may add substantial data beyond what was included in the latest annual report. In contrast, prospectuses for bank loan syndications and local bond issues may provide relatively little helpful information. Prospectuses, however, are not always easily accessible.⁴⁰ Ongoing regulatory filings made with the securities regulator have a similar function, although they are intended to benefit buyers in the secondary market as well as the purchasers of new issues.

Secondary Analysis: Reports by Rating Agencies, Regulators, and Investment Banks

The use of secondary research will depend on the type of bank credit report being prepared. Rating agency analysts will often review official reports from central banks and government regulators, but, like fixed-income analysts, will avoid the use of competitor publications. Bank counterparty credit analysts, however, will rely to a greater extent on secondary research sources and less on primary sources. Their credit reviews are not intended for external publication, and the views of the rating agencies are not usually ignored. Investment reports prepared by equity analysts, although they take a different perspective from bank credit reports, can nevertheless be useful in helping to form a view concerning a bank. Since these reports are ordinarily prepared for their investor-customers, very recent ones may not be easy to obtain. Such reports may be purchased, sometimes on an embargoed basis, from services such as those offered by Thomson Reuters.⁴¹

CAMEL IN A NUTSHELL

Once all information, including an appropriate spreadsheet of the financials over several years, is available to them, bank credit analysts almost universally employ the CAMEL system or a variant when evaluating bank credit risk. Although originally developed by U.S. bank supervisors in the late 1970s as a tool for bank examination,⁴² it has been widely adopted by all rating agencies and counterparty analysts. Even many equity analysts draw on the CAMEL system to help them in making recommendations concerning the valuation of bank stocks. It is the approach we reluctantly explain in this book.⁴³

What is the CAMEL system? CAMEL is an oversimplification that does not catch all it should, and that does not give proper weight to the various elements. CAMEL is simply an acronym that stands for the five most important attributes of bank financial health. The five elements are:

- C for Capital
- A for Asset Quality
- M for Management
- E for Earnings
- L for Liquidity

All but the assessment of the quality of “management” are amenable to ratio analysis, but it must be emphasized that “liquidity” is very difficult to quantify. The

meaning of these facets of financial ratio analysis and the ratios used to measure them will be discussed in detail in the chapters that follow.

Since the term CAMEL was coined, banks have ventured into a number of types of transactions that no longer fit into those five categories. Many such transactions are recorded off the balance sheet or, if they are recorded on the balance sheet, are prompted by asset/liability management needs, making the term *asset quality* too restrictive. But a camel would no longer be a camel without its “a” or with too many additional letters.

Although sometimes termed a model, the CAMEL system is really more of a checklist of the attributes of a bank that are viewed as critical in evaluating its financial performance and condition. Nevertheless, it can provide the basis for more systematic approaches to evaluating bank creditworthiness.

As used by bank regulators in the United States, the CAMEL system functions as a scoring model. Institutions are assigned a score between “1” (best) and “5” (worst) by bank examiners for each letter in the acronym. Scores on each attribute are aggregated to form composite scores, and scores of 3 or higher are viewed as unsatisfactory and draw regulatory scrutiny.

CAMEL, CAMELS, AND CAMELOT

The acronym CAMEL can also function as a mnemonic as illustrated in the list on the opposite page. With no disrespect intended to the animal, the two humps on the camel that provide reserves of nourishment can be thought of as signifying C for capital and L for liquidity, both of which provide a bank with the reserve buffer necessary to absorb economic shocks. The animal’s front legs pull it forward as do a bank’s earnings, so long as they are not hindered by asset quality problems coming from behind. Finally, the camel’s head and eyes, which scan the desert horizon for the next oasis or dust storm, could stand for bank management. It is management’s job to ensure the institution’s survival by obtaining the necessary sustenance while avoiding the perils that may befall it, particularly in turbulent time. Note that while the elements of the acronym are examined sequentially in this chapter, a more functional approach is taken in succeeding chapters, beginning with earnings, which is followed by a discussion of asset quality, capital, management, and liquidity.

In addition to the acronym CAMEL, another widespread variant, CAMELS adds an “S” for sensitivity to market risk. This was officially adopted by the Uniform Financial Institutions Rating System (UFIRS; see quotation later) in 1997.

Under the UFIRS, the regulatory agencies evaluate and rate a bank’s financial condition, operational controls, and compliance in six areas. These areas are Capital, Asset quality, Management, Earnings, Liquidity, and Sensitivity to market risk. Each of these components is viewed separately and together to provide a summary picture of a bank’s financial soundness.

The CAMEL model is also used in equity analysis. A variant termed CAMELOT, developed by Roy Ramos at investment bank Goldman Sachs (GS), added an “O” and “T” to the basic CAMEL root to represent evaluation of the bank’s operating environment and assessment of transparency and disclosure.

NOTES

1. Anthony Gaubis (1902–1989) was an investment analyst and advisor who published a stock market newsletter, *Business and Investment Timing*. Obituary abstract, *New York Times*, October 11, 1989.
2. Until now, we have looked at credit analysis in a general way. With this chapter, we begin to concentrate on the credit analysis of financial institutions generally and on banks specifically, a topic that shall occupy most of the remainder of the book.
3. It is worth noting here that between the two extremes just discussed—the due diligence undertaken by rating agency analysts and the automated scoring of credit risk using a quantitative model—hybrid systems are employed in the counterparty credit context, encouraged also by the requirements of Basel II and Basel III. Under a hybrid approach, some rating inputs are generated using quantitative data supplied from internal sources or an outside provider, while more qualitative scorings are entered by the analyst. They represent an attempt to reduce costs and enhance the consistency of ratings, while seeking to incorporate analyst judgments where quantitative models are unwieldy.
4. Although basic structured finance transactions such as mortgage-backed securities and similar securitizations are discussed briefly in some of the pages that follow, the analysis of such transactions remains wholly outside the scope of this book.
5. There are, of course, exceptions, as when a bank is contemplating advancing a very large loan or engaging in another type of transaction of comparable magnitude.
6. A distinction is sometimes made between sovereign risk analysis and country risk analysis. The term *country analysis* tends to connote a greater emphasis on the impact of a nation's political, legal, and economic regime, together with potential changes in that regime, upon debt issuers within its borders, as well as those affecting the risks of foreign direct investment in the country. The difference between country risk and sovereign risk has become largely semantic, however, and in practice the terms are often used interchangeably. The distinction between sovereign risk analysis and country risk analysis, insofar as one can be made, is discussed further in Chapter 11.
7. This distinction is, nevertheless, reprised in Chapter 11 dealing with the subject of sovereign risk.
8. This is discussed further in Chapters 14 and 15 of this book.
9. Note that in regard to credit officers covering financial institutions, some banks make a distinction between trading-floor credit officers, who have responsibility for credit decisions involving investment and trading operations, and those responsible for approving simple trade finance transactions, correspondent banking, and routine cash management activity, such as interbank borrowing and lending.
10. Although for illustrative purposes we have frequently used the example of simple loan transactions, the credit exposures to which a bank may be subject are extremely diverse. They run the gamut from such basic term loans to highly sophisticated derivatives transactions and structured finance transactions. Each has its own risk characteristics, and risk-conscious financial institutions will ordinarily have credit policies in place governing their exposure to various types of transactions.
11. Similarly, external rating agencies will ordinarily evaluate a counterparty in a general manner, while debt securities will be assigned ratings based on features of the sort just mentioned. Issuer ratings may be made with reference to a specific debt issue or to a class of generic issues. Whether the issue rating is affected by the issuer rating assigned will depend upon the characteristics of the issue. In the recent past, the ratings of many structured products were decoupled from the ratings of the issuer or originator.
12. It is not unusual for considerations beyond the estimated credit risk to affect the decision as to whether to approve or reject an application for a loan or other service or product subjecting the bank to credit risk. Such considerations could include the bank's historical

- or desired future relationship with the customer, including the prospect of other business and similar considerations.
13. Fixed-income analysts tend to integrate fundamental and technical analysis to a greater degree than do equity analysts. (Equity analysis is discussed in the following subsection.) Both equity and fixed-income analysis vary in respect of the audience to which the analytical reports are targeted. In the case of investment banks and brokerages, fixed-income analysis may be intended for clients, often institutional investors or asset managers, who will use the research to make their own trading decisions. Alternatively, fixed-income analysis may be intended primarily for a firm's own traders, who will use the research internally.
 14. This was not because credit risk analysis was not forward-looking, but because traditionally financial projections were perceived as too unreliable. While accuracy in financial projection remains notoriously difficult to achieve, it would not be surprising if the use of financial projections to a limited degree, for the purpose of identifying potential unfolding scenarios, for example, could become more commonplace in the credit review context.
 15. Robert Zielinski, Lehman Brothers, "New Research Techniques for the New Asia," December 14, 1998.
 16. Likewise, the bank's share price and recent or long-term price trends or volatility may very well have an impact on its ability to raise capital, or access liquid funds. For this reason, some credit analysts keep a weather eye on a bank's share price as a proxy for impending difficulties that may manifest in credit problems. More broadly, the usefulness of equity research, its techniques, or the fundamental data upon which it is based, depends on the bank analyst's role, the comparative availability of data for the institution that is the subject of analysis, and, naturally, upon whether the analyst has access to such material.
 17. Recall that in the Chapter 1, we observed that the category of borrower would influence the method of evaluating the associated credit risk.
 18. In most cases, credit analysis relies on historical financial data. In some situations, however, quantitative projections of future financial performance may be made.
 19. Peers and peer groups are discussed elsewhere in the book.
 20. The issue of where to start will be of lesser concern when third-party data providers are used, but there may be occasions when such data is unavailable on a timely basis and sector benchmarks must be established independently.
 21. Although less relevant than it was in the past, the CAMEL model of analysis, introduced later in this chapter, provides a generally accepted framework for analyzing the creditworthiness of banks. CAMEL is an acronym for Capital, Asset quality, Management, Earnings, and Liquidity.
 22. Rating agencies will be discussed in Chapter 12.
 23. In effect, by relying on external rating agencies, the counterparty analyst is outsourcing part of the credit function. Regulatory considerations may also come into play. Reference to the opinion of independent agencies may be required to satisfy rules governing the extension of credit or the making of investments at the organization at which the analyst is employed. Conversely, however, where the employing bank utilizes an internal rating-based approach to allocating capital, the use of an internal rating system that is independent of external agency rating assignments may be an element of regulatory compliance. The role of rating agencies in bank analysis is discussed further in later chapters.
 24. The adjective *preferably* is inserted here only because in some emerging markets, recent audited financial statements may be impossible to obtain, particularly in regard to state-owned institutions; and a trade-off surfaces between taking account only of audited data that is out of date and considering unaudited data that is current. Where government support is the primary basis for the creditworthiness of the entity, the use of unaudited data may shed additional light on the organization's performance. However, organizational credit policies may prohibit the use of unaudited data, and as a general rule

- unaudited data should not be used. Moreover, the absence of audited data in itself may fairly be regarded as an unfavorable credit indicator.
25. Offering circulars and prospectuses are documents prepared in advance of a securities offering to inform prospective investors concerning the terms of the offering. Information concerning the issuer, its activities, and notable risks is typically included in these documents. Often, the format and content of such documents will be governed by regulation.
 26. Unsolicited ratings are assigned by an agency on its own initiative, either on a gratis basis or without a formal agreement between the agency and the issuer or counterparty. When the rating industry began in the United States in the early twentieth century, all ratings were unsolicited. The rating agencies relied on subscription revenue from investors to support their operations and generate a profit. As the industry became established, however, an external rating effectively became a prerequisite to a successful debt issue. The rating agencies were then in a position to charge issuers for a rating. Such paid-for ratings are called solicited ratings, and have become the norm among the major global rating agencies. Nevertheless, unsolicited ratings still exist, particularly outside of major markets or in respect of companies that do not issue significant debt, but which are of interest to investors and counterparties. Publicly available information provides the basis for such ratings.
 27. Note that unless the analyst is working in the capacity of bank examiner, there is normally no right of access to bank management. When performing a solicited (i.e., paid) rating, rating agencies will ordinarily enter into an agreement that ensures analyst access to management. Otherwise, such visits are made on a courtesy basis only, and in exceptional cases the analyst may be unable to meet in person with management. In-house analysts, in particular, may have difficulty or may be shunted off to the investor relations or correspondent bank relations staff who often will be able to add little to the data provided in the bank's annual report.
 28. "Unqualified" means that the auditor has attached no additional conditions to its opinion, that is, the opinion is without further qualification.
 29. John A. Tracy, *How to Read a Financial Report: for Managers, Entrepreneurs, Lenders, Lawyers and Investors*, 5th ed. (New York: John Wiley & Sons, 1999).
 30. In other words, the language is largely intended to provide a defense against litigation that would seek to hold the auditor liable for any fraud or misrepresentation subsequently discovered in the financial statements.
 31. The vernacular translations supplied for each statement are the interpretations of the authors and, needless to say, have no official standing.
 32. In spite of its name, Philippine National Bank is not the central bank. The central bank is Central Bank of the Philippines or Bangko Sentral ng Pilipinas.
 33. Note that the large global accounting firms often operate through local affiliates that often have names different from their global affiliates.
 34. The three key financial statements are discussed in more detail in subsequent chapters. It is assumed that readers will have some degree of understanding of accounting principles, if not an accounting background.
 35. The statement of changes in capital funds is sometimes reported in combination with one of the other statements, and at other times it is reported separately. In some markets, it may be omitted entirely.
 36. Restated financial statements (restatements), also called restatement of prior-period financial statements, are adjusted and republished to correct material errors in prior financial statements or to revise previous financial statements to reflect subsequent changes in an entity's accounting or reporting standards.
 37. Where the subject institution is wholly owned by the national government, its credit risk may at times be found to be essentially equivalent to sovereign risk. In this situation, the standalone financial strength of the bank would be a secondary consideration and the delay in financial reporting less critical than it otherwise would be.

38. The pervasive publication of annual reports and financial statements on the World Wide Web has been a great boon to credit analysts generally and considerably reduced problems in obtaining financial data in a timely manner. Consolidation in the financial services industry together with the ubiquity of third-party data providers—most notably Bankscope mentioned earlier—has facilitated the bank analyst's work by reducing the time spent on collecting and entering data. Nevertheless, there are some institutions that do not make their financial reports freely available through these channels. In such cases, there may be no alternative but to contact the bank directly and request that they be posted, e-mailed, or faxed as the case may be.
39. One should not read too much in such signals, as will be discussed later in the book.
40. Some prospectuses may be available from online data providers on a subscription basis.
41. Selected reports also may be available to Bloomberg or Factiva subscribers.
42. Under the Uniform Financial Institutions Rating System adopted in the United States in 1979, the CAMELS system was formally adopted as the most comprehensive and uniform approach to assessing the soundness of banks, although as a formalized methodological approach appears to date back to the practices of bank examiners in the early twentieth century.
43. It should be emphasized, however, that the financial services industry is rapidly evolving as banks engage in new activities. Refinements and alternative models to bank credit assessment methodologies, therefore, cannot be ignored.

The Business of Banking

Banks are still preeminent in the financial system . . . [and] are vital to economic activity, because they reallocate money, or credit, from savers, who have a temporary surplus of it, to borrowers, who can make better use of it . . . [and by] collaborating to clear payments, they help individuals and firms fulfill transactions.

—*The Economist*¹

Banking is a business like any other—a raw material (in this case money) goes into the process we call banking, and hopefully, a profit . . . comes out . . . the other end.

—Howard Palmer²

A bank is a place that will lend you money if you can prove that you don't need it.

—Bob Hope, comedian and actor (1903–2003)³

Banking, at its core, is a simple business. Banks transform the savings of many individual depositors into credit finance. One of several types of *financial intermediaries*—a category that also includes *insurance companies*, *securities underwriters and brokers*, and *investment managers*, a bank, as do these institutions,⁴ functions as a conduit between those with funds to deposit or invest and those in need of funds. Classically, in banking, intermediation occurs when a bank accepts deposits from those with surplus funds (depositors) and lends the same funds to entities in need of funds (borrowers).⁵ That is, traditionally, banks borrow money from depositors to lend to those who borrow money from it. Although banks perform other functions besides taking deposits and advancing loans, this core intermediation function remains at the heart of commercial banking.⁶

BANKS AS INTERMEDIARIES

Banks intermediate between savers and borrowers by taking deposits from those with surplus funds (depositors) and lending those funds to entities in need of funds.⁷

How Banks Add Value

Through their deposit-taking and lending operations, banks add value for their customers by offering terms that are respectively attractive to both depositors and borrowers. Banks are able to add value to both depositors and borrowers through their ability to:

- Package savings vehicles and advance funds in a manner that prospective depositors and borrowers would find difficult to undertake on their own.
- Spread their risks over many transactions, and in the process mitigate risk.
- Gain an information advantage not only from their participation in a volume of transactions, but also through their access to the financial accounts of borrowers, actual and prospective, and depositors.

Depositors benefit from banks accepting their deposits as a consequence of being able to:

- Place excess funds with the bank in the amount and at the time suitable to their individual needs.
- Earn interest on all or a portion of those funds deposited.
- Withdraw the funds with few or no restrictions as they see fit.
- Use the bank's payment services (e.g., checking accounts) to transfer funds to third parties.

Borrowers generally benefit from:

- A ready source of finance on terms that, at least to a degree, meet their particular needs.
- A degree of certainty concerning the cost of finance and the term over which it will be available.
- A degree of confidentiality (which is typically lost when the borrower issues paper in the money and capital markets).
- Ancillary services offered by banks.

In exchange for advancing funds to borrowers, the bank earns for itself a spread (or margin) between what it pays on funds borrowed from savers and what it earns on the funds it lends to customers.⁸ At the macroeconomic level, banks are generally viewed as providing a net benefit as a result of their contribution to economic growth. They do this by facilitating the transfer of savers' excess funds to borrowers in need of financing to grow their businesses, and by providing a secure means to effect payment between parties. Banks also contribute to economic growth in other ways, as discussed later in the book.

A crucial contribution of banks is that they carry the risk that borrowers may not be able to reimburse their loans. Taking away that risk from depositors is arguably the best way to allocate savings into investments, as retail depositors—and even more sophisticated depositors—neither have the time, the skills, nor the inclination to gauge the risk of lending directly to borrowers. This is a bank's intermediation function, for which it should be rewarded through interest rate spreads between lending and borrowing rates.

Banks as Deposit-Taking Institutions

What makes a bank different from other financial intermediaries is being permitted to take deposits from the public.⁹ By obtaining funding in this manner, and effecting the first step of the described intermediation function, the bank is in a position to complete the process by lending to borrowers.

Customer deposits are usually divided into three categories for purposes of reporting and analysis as follows:

1. Demand deposits
2. Time deposits
3. Savings deposits

INTERMEDIATION AND DISINTERMEDIATION

Financial intermediation refers to the process in which an entity acts as a middleman (intermediary) between depositors and borrowers enabling the channeling of funds from those with excess savings—which constitutes capital—to those in need of such capital (see Exhibit 3.1). A bank is one type of financial intermediary—and probably the most ubiquitous—but not the only type.¹⁰

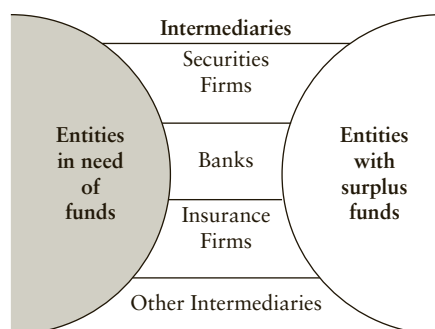


EXHIBIT 3.1 Intermediation

Intermediation

To better understand the intermediary function that banks perform, imagine a typical saver and a typical borrower.

The saver might be an individual who saves a portion of his or her income for retirement or for the purchase of an asset, such as a house, while he or she—in his or her role as a consumer—spends the remainder.

The borrower might be a firm that wishes to take advantage of the financial leverage on borrowed funds to expand its operations. Absent access

to efficient capital markets, the firm would be able to finance such an expansion only through the retained earnings of its business (internal capital generation) or from the contribution of additional capital from its owners, or from the sale of their ownership rights in the form of new shares. In a hypothetical best-case scenario, the reason for the company's desire to expand may be that it has a product or service in such great demand that the only obstacle holding back its ability to expand its sales (and profits) tremendously is a lack of capital.

Unless it has the ability to tap the resources of savers, the firm's growth and, correspondingly, general economic growth would be impeded. Limited in its access to outside capital, the pace of expansion on the part of both the firm and the economy in which it operates would tend to be painfully slow, and market opportunities could be lost.

Savers would also suffer. With opportunities to lend or invest their savings constrained by the absence of financial intermediators, they might find it difficult to earn a return on their nest eggs without incurring excessive credit risk. Without the capacity to generate returns on their capital, savers' store of wealth would be subject to the risk of loss of value over time. They might even have to pay for the safekeeping of their savings, which would otherwise be vulnerable to theft.

In this bleak environment for both savers and prospective borrowers, the bank offers advantages to both. To savers, its prospective depositors, it offers two key benefits: safety and the ability to earn interest on funds deposited with the bank.¹¹ Prospective borrowers gain access to a new source of credit, made available through savers' deposits with the bank.

Although savers want the opportunity to generate a return on their savings, and borrowers need finance, the desires of prospective savers and borrowers are otherwise at odds. In addition to the desire to earn a return either as interest, in dividends, or through appreciation, savers generally want their funds to be secure and to be easily accessible (i.e., *liquid*). All other things being equal, for savers, the greater the return earned the better. In contrast, borrowers favor maintaining flexibility of the use of the funds borrowed and the certainty of retaining the funds for a fixed period, while being able to repay the loan when it is most convenient to their situation. Without these benefits, borrowers may face reduced liquidity or lose investment opportunities. Obviously, all other things being equal, borrowers prefer the interest rate to be as low as possible.

The bank earns its keep by meeting the respective and conflicting preferences of savers and borrowers. In its capacity as an intermediary, it repackages savers' deposits into loans to borrowers under terms comparatively attractive to both groups. Other types of financial intermediaries, including investment banks, securities houses, and insurance firms, perform an analogous role in different ways.¹²

Investment banks match entities with surplus funds (investors) with entities in need of funds that wish to issue securities (issuers) in exchange for those funds. Investment banks take a fee from the issuer and commissions from investors in the process.

Securities houses perform a similar matching role in the secondary market, benefiting from commissions from buyers and sellers as well as, in their role as market makers, from a bid–offer spread.¹³

Insurance firms receive premiums from their policyholders and invest those funds in securities, which form part of a pool of assets out of which proceeds to insured parties are paid when certain contractual conditions or contingencies occur.

Disintermediation

Disintermediation is the opposite phenomenon to intermediation. Intermediation occurs when banks are accepted as safe repositories of savers' funds and as a reliable source of finance for firms. Disintermediation occurs when the role of banks in intermediating between those with excess capital and those in need of capital is diminished. The immediate cause is almost always competition from other institutions. One example of disintermediation occurs when companies that in the past were only able to obtain finance through banks in the manner just described are able to raise funds directly in the money and capital markets through the issue of securities. Another occurs when savers are able to obtain attractive returns on their savings through money-market accounts and similar products, and consequently have alternatives to banks.

The development of various institutional alternatives to bank loans as a source of funding and to bank deposits as savings vehicles inevitably results in a degree of disintermediation. In other words, increased intermediation by nonbanks may result in decreased intermediation by banks. As financial systems have evolved, and local equity and debt markets have become established, or local firms have gained access to global capital, the phenomenon of disintermediation has become a trend in many markets. Its causes include deregulation, both at the national and international level. At the national level, deregulation has enabled the creation of new finance and investment vehicles that compete with banks for both deposits and lending business. At the international level, deregulation has fostered the freer flow of capital across borders and an increased role for institutional investors in global markets.

The differences among them mainly concern the degree of access the customer retains to funds placed on deposit and the rate of interest payable on those funds. While deposit interest rates are dictated by the overall yield curve of the relevant currency as well as by competitive constraints, the more access the depositor is willing to forego, the greater the rate of interest that can normally be obtained.

Demand Deposits

Demand deposits represent deposits that may be withdrawn at will without notice by the depositor to the bank. Depositors have traditionally been able to write checks

against demand deposits; consequently, they are commonly referred to in some markets as *checking accounts*. Another widely used term for a demand deposit account is a *current account*. In exchange for their convenience, demand deposits traditionally either do not pay any interest to the depositor or the interest paid is minimal.

Time Deposits

A time deposit is a form of deposit in which the depositor commits to leave funds with a bank for a specified period at a specified rate of interest. A *certificate of deposit* or *CD* is a form of time deposit. In practice, banks often permit time deposits to be withdrawn upon a designated period of notice being given or subject to a penalty payment.

Saving Deposits

Saving deposits are a hybrid between a time deposit, where a designated sum of money is deposited with the bank under specified terms, and the demand deposit, which can be withdrawn without notice. All funds deposited into a *savings account* represent saving deposits. It is not necessary for the depositor to commit a designated sum or forego access to the funds for a significant period. Instead, the savings account pays a minimum rate of interest on all funds deposited, and these funds can be withdrawn instantaneously or after a relatively short notice period. In practice, banks often waive the notice requirement, although they may place restrictions on savings accounts, limiting the number of withdrawals, and as a rule do not permit checking privileges on funds so deposited. In some countries, banks reward a savings account with additional interest on amounts kept in the account during a specified period, for example one full year.

How Banks Earn Money

Using the funds that depositors have placed with it, the bank provides credit to customers in the form of loans, *revolving lines of credit*, *overdraft facilities*, and similar mechanisms, among its other activities.¹⁴ As observed in the previous section, banks earn income on the difference between their funding costs (the interest rate paid to depositors and others who advance funds to the bank) and their yield on loans (the interest rate charged to borrowers). This difference is the bank's spread.

Lending activity can be broadly divided into wholesale and retail lending.

- *Wholesale lending* means the provision of bank credit to large enterprises.
- *Retail lending* means the provision of bank credit to small and medium-sized enterprises (SMEs), as well as to individuals.

In addition to generating interest income from *earning assets*,¹⁵ banks may also obtain revenue from sources other than *interest income*. The most common sources of noninterest income are gains from trading activities, and fees and commissions paid by customers.

The composition and specific sources of bank revenue vary greatly from one market to another depending upon its state of development and the strength of the

underlying legal infrastructure. In more developed markets, so-called nontraditional sources of bank revenue—from credit cards, transaction processing, asset management services, and investment banking,¹⁶ for example—have come to comprise an increasingly large proportion of total bank revenue. As an example, the diagram in Exhibit 3.2 shows the sources of revenue for large U.S. banks circa 2003 divided into traditional and nontraditional sources, and indicates that a substantial proportion of total bank revenue comes from the latter category. As discussed later in this book, banks globally have since then increased their reliance on nontraditional sources of revenue. That strategy certainly played a role in the financial crisis that started in 2007.

SOURCES OF BANK REVENUE: INTEREST INCOME VERSUS NONINTEREST INCOME

Conventionally, a bank's revenue is divided between that earned from interest-generating assets and that generated from other sources. The first category is commonly called interest income or interest revenue; the second is called noninterest income or noninterest revenue.

Loans and similar advances of funds, together with holdings of debt securities, represent financial assets to a bank that generate interest revenue. For this reason they are called earning assets. Many of the same assets—debt securities most commonly but also loans—may be traded to realize a gain (or loss). This trading activity falls within the category of noninterest revenue. Another type of noninterest revenue is fees and commissions that may be generated through the brokering of financial assets, or from the provision of ancillary services to bank customers.

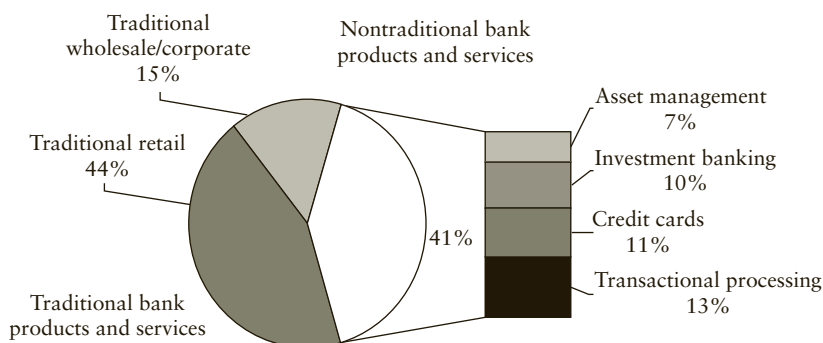


EXHIBIT 3.2 Sources of Bank Revenue, U.S. Large-Cap Banks Circa 2003

Source: Adapted from U.S. Bancorp: Piper Jaffrey, Andrew B. Collins, Steven M. Truong, and R. Neal Kohl, *Equity Research, Large Commercial Banks, The "60 Second" Bank Stock Primer* (Minneapolis: U.S. Bancorp Piper Jaffray, April 2003), 30 (hereafter *60 Second Primer*).

Note, however, that even in this highly evolved financial market, the bulk of bank revenue on average still comes from traditional banking products that, for the most part, represent forms of loans held on the bank's balance sheet. They include retail-lending products, including primarily mortgage loans, as well as wholesale and corporate lending products.

So-called nontraditional products, which were once the province of specialist institutions, comprise the remaining sources of bank revenue. Broadly speaking, these can be classified into four areas:

1. Asset management
2. Investment banking
3. Credit cards
4. Transactional processing

These categories are discussed later in this chapter.

The Organization of the Bank

Banks are structured in many different ways depending upon the size of the institution, the scope of its business activities, and the breadth of its geographic operations. A small community bank will obviously have a simpler organizational structure than a global financial institution such as Citicorp or Deutsche Bank. Management preferences, regulatory requirements, business trends and fashions, and not infrequently internal politics influence the organization of banks. In large institutions, organizational restructuring is practically an ongoing activity.

Notwithstanding the great variety of business structures, most banking organizations can be divided into at least three principal areas: operations, management, and support.

1. *Operations.* Actual banking operations embrace a bank's business operations, including funding, treasury activities, lending to those in need of finance, and providing other banking products.
2. *Management.* As does any organization, banks invariably have a management group, which is responsible for setting and executing strategy.
3. *Support.* Finally, banks require a support group to provide the infrastructure through which banking operations are performed. This includes legal and audit functions, compliance, risk management, human resources, information technology, and other related activities.

Exhibit 3.3 illustrates a hypothetical structure of a major bank operating within a single country. In the associated diagram, management sits at the top of the organizational chart overseeing banking operations. It is supported by a risk management group and a second support group encompassing legal services and other support functions. Risk management has now become a major tool for even the most traditional banks, not only to guide the bank's management in their commercial decisions but also to comply with increasingly complex regulations regarding capital adequacy, as will be discussed at length later in the book.

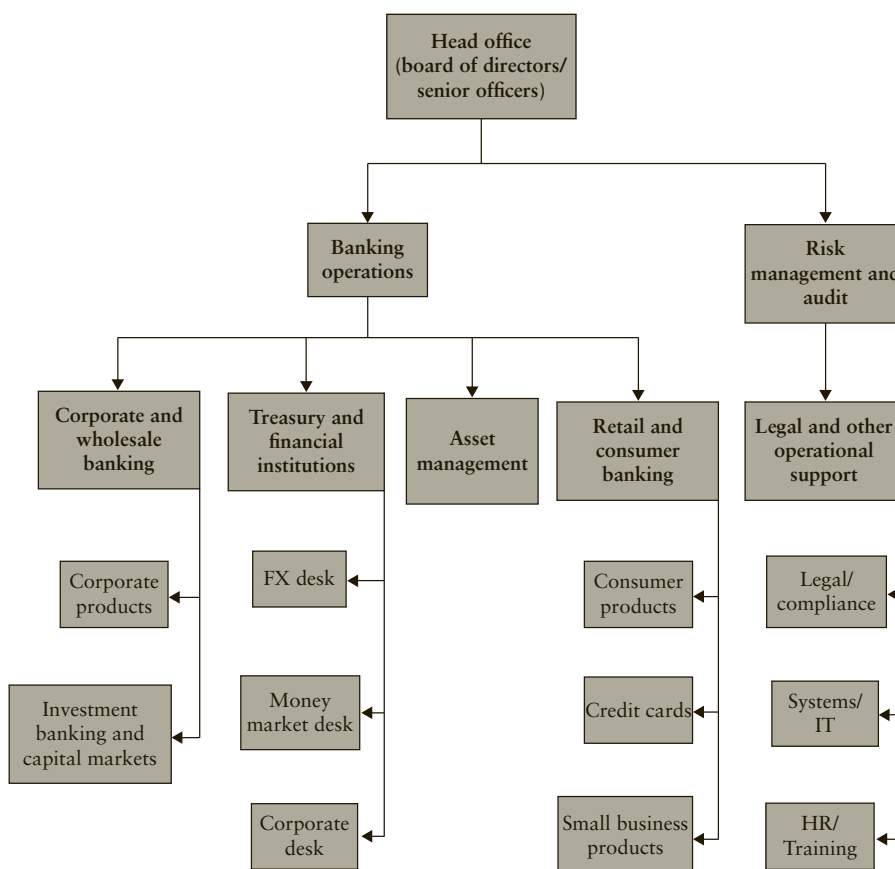


EXHIBIT 3.3 Sample Organizational Chart

More specifically, banking operations are divided into four areas:

1. *Wholesale and corporate banking*, responsible for providing financial services to large enterprises
2. *Retail and consumer banking*, responsible for providing financial services to individuals and small businesses
3. *Treasury*, responsible for managing the bank's funding and liquidity needs
4. *Asset management*, responsible for investing funds on behalf of customers.

In this simplified diagram, the executive committee, which oversees management, is omitted, as are some of the more specialized functions and those lower down the organizational hierarchy.

BANKS AS LENDERS

Historically, for the commercial bank as it evolved from the early nineteenth to late twentieth centuries, lending—in the broad sense of the term—was the institution's

main business. A bank accepted deposits from individuals and firms with a view to advancing a portion of those same funds to its customers (ordinarily commercial or noncommercial organizations rather than individual consumers). So long as interest earned from borrowers exceeded interest paid to depositors, the bank could earn a profit. Traditional loans, as shown in Exhibit 3.2, comprise the bulk of the average bank's financial assets. They are also on average the riskiest assets it holds, if one ignores derivatives, which, of course, are not assets in a strict sense. Nevertheless, by accepting the risks implicit in lending, a bank is able to earn substantial revenues and through its high leverage often lucrative returns. Much of the risk borne by the bank is credit risk, and the quality of a bank's loan assets in turn affects its own credit quality. Besides making loans directly to borrowers, to maintain liquidity banks typically also hold investment securities and other interest-earning assets.¹⁷

Categories of Lending

Much of the credit advanced by banks is in the form of loans, broadly defined.¹⁸ In general, *credit products* can be categorized by common parameters including the following:

- *Tenor*: short term vs. long term
- *Collateral*: secured vs. unsecured
- *Currency*: local vs. foreign
- *Legal obligation*: committed vs. uncommitted

Beyond the variables just enumerated, further distinctions can be drawn. *Credit facilities* offered by a bank can be classified by product type, variant structures within each type, and by the customer base to which they are marketed. For instance, within the broad class of loan products is the term loan, in which funds are advanced at the beginning of the loan term, and repaid with interest either at maturity, or periodically. Likewise, a bank may concentrate on lending to consumers, small businesses, or large industrial enterprises. Sometimes, product distinctions parallel customer segmentation. Consumers have no need of letters of credit, while large industrial enterprises will not be the main market for mortgages for owner-occupied residential housing.

By Financing Vehicle/Product Type

Loan facilities encompass recognizably distinct types of instruments, some of which are so distinct as to be regarded as discrete business areas. *Syndicated lending* and *Securitization*, for example, may be viewed as a separate lending franchise from ordinary bank lending.

By Customer Franchise

Besides the distinctions just enumerated, which generally span the entire universe of bank products, credit facilities can be divided by:

- Industry sector (e.g., agriculture, heavy industry)
- Broad product categories—such as *loans*, *receivables finance*, *documentary trade finance*
- Customer base to which they are marketed, such as consumers, small business, or blue-chip industrials¹⁹

LOANS VERSUS OTHER CREDIT PRODUCTS AND FACILITIES

Loans

A loan can be defined as a legally binding agreement between a lender and a borrower under which the lender advances a certain sum of money to the borrower for a certain period and the borrower promises to repay the sum advanced—the principal—with interest as provided in the loan agreement.

Other Credit Products

However, credit can also be provided to a customer through other legal mechanisms—for example, a financial lease. The mechanisms and structures that form the universe of products through which a financial institution provides credit to customers are frequently referred to collectively as credit facilities or, more broadly, as credit products—often shortened to facilities or products.²⁰

Facilities

A facility can therefore be defined as an arrangement made by a financial institution with a customer to provide a given type of credit support. The term facility embraces a broader group of credit products than loans.

Many bank products and almost all facilities can be grouped according to the following parameters:

Short Term versus Long Term

Short term: In the context of traditional commercial banking practice, *short term* means having a term of less than one year; *long term* means one year or longer. (Note that this terminology differs from that employed with respect to debt securities issued in the money and capital markets.)

Secured versus Unsecured

The term *secured* means that the risks to the bank are at least nominally mitigated through the provision of a guarantee by an ostensibly creditworthy third party; or that assets have been pledged to the bank or that the bank otherwise has a legal claim to designated assets in the event of borrower default.

Local Currency versus Foreign Currency

Local currency facilities are those in which the repayment obligation is in the domestic currency of the locality in which the facility was made.

Committed versus Uncommitted

Facilities can be classified as committed—meaning that the bank has entered into a binding legal agreement to provide the facility—or as uncommitted.

Traditional lending products include those listed in the tables in Exhibits 3.5, 3.6, and 3.7.

Each of the foregoing groups can be further subdivided, and each of these resultant subcategories may be regarded as a particular market niche. To illustrate, *consumer lending*, which broadly speaking may include the making of personal loans, residential mortgages, credit cards issuance, and auto and other vehicle finance loans, can be further categorized according to the financial characteristics of various consumer classes. Some institutions specialize in credit card issuance or in mortgage lending, for instance, and within these respective categories, to the *prime* market or to the *subprime* market and various gradations thereof.

To simplify the many distinctions that can be drawn, it is helpful for basic analytical purposes to initially recognize four fundamental categories of lending:

1. *Commercial and industrial lending*, meaning loans to business for working capital or longer-term finance purposes²¹
2. *Commercial real estate lending* to property developers and managers
3. *Consumer lending*, including mortgage lending to enable individuals and families to buy their own dwellings
4. *Lending to financial institutions*, which is normally provided through the interbank or money market and typically comes under the rubric of treasury operations.

Of course, the actual categories commonly in use will vary from market to market, as will the composition of the typical bank loan book. As an illustration, Exhibit 3.4 shows the breakdown of lending in Brazil in 2010.

Note that loans to financial institutions in general are not provided in the example. In recent years, as especially apparent in the debt crisis in Europe, banks have increased their balance sheet—on both sides—with huge amounts of interbank borrowing and lending, and huge amounts of lending to sovereign borrowers. The topic of loan portfolios is discussed in greater detail in the chapter devoted to asset quality.

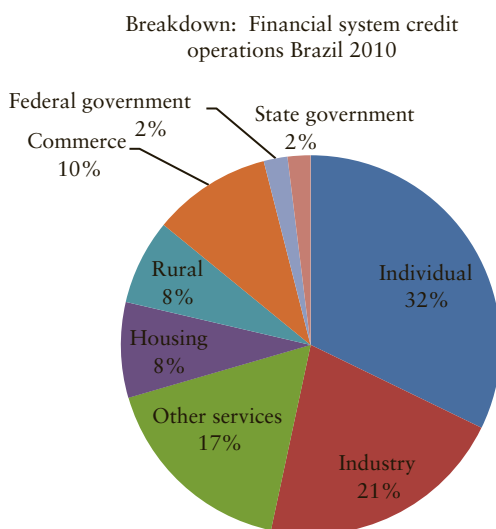


EXHIBIT 3.4 Breakdown of Loans and Advances in Brazil 2010

Loans and the Lending Process

The category of loans embraces among other products:

- *Term loans* represent the simplest type of loan, being for a fixed term. It may provide for payment of the principal amount advanced at loan maturity or in amortizing installments throughout the term of the loan. Likewise, interest may be payable at maturity, in a balloon or bullet payment,²² or in periodic installments.
- *Rollover loans* are essentially term loans that renew upon maturity according to predetermined criteria.
- *Call loans*, in contrast to rollover loans and term loans, may be terminated before maturity by either party, but notably by the lender.
- *Short-term advances* usually are working capital loans, sometimes part of a larger arrangement, with a maturity not exceeding one year.
- *Bridge loans* are a type of short-term advance that is advanced in contemplation of longer-term funding being put in place.

The salient characteristics of these loans are described in greater detail in Exhibit 3.5.

EXHIBIT 3.5 Selected Lending Products

Product Type	Definition	Typical Terms and Usage	Risks
Term loan	Normally extended for a period of one year or more, a term loan constitutes an agreement by the borrower to repay to the bank the principal sum advanced at the end of the loan term, together with interest, at either a fixed or variable rate, in the form of a balloon (bullet) payment or on a periodic (e.g., monthly, quarterly) basis. An amortizing term loan means that the principal balance is gradually reduced during the term of the loan.	Term loans are primarily used for long-term (i.e., more than one year) financing requirements such as for capital expenditures. Usually from 1- to 5-year maturity, although 7 to 10 years or longer is not unknown. Loans with bullet repayment terms usually require a higher standard of creditworthiness on the part of the borrower, and should be appropriate to the customer's situation.	Loss of principal advanced and anticipated interest payments. Loss exposure will decline over time on an amortizing loan, whereas exposure risk for loan with bullet terms in respect of principal and interest will remain constant until maturity.
Rollover loan	A rollover loan is a special category of term loan, equivalent to a revolving line of credit. The interest rate of the rollover loan is fixed for	Like term loans, the maturity of rollover loans may be as long as 7 to 10 years, but they may also be used for short-term credit	As with term loans, the exposure risk will vary according to the amount of principal and interest outstanding in accord

(Continued)

EXHIBIT 3.5 (Continued)

Product Type	Definition	Typical Terms and Usage	Risks
	a limited period of time, usually ranging from one month to one year. Preagreed extensions of the loan (rollovers) are periodically set prior to the next succeeding term based on predefined criteria, e.g., an interbank interest rate such as LIBOR plus a risk premium appropriate to the borrower's credit standing.	provisions in response to a customer's working capital needs.	with the loan provisions.
Short-term advance	A short-term advance is simply a loan extended for a period of between one day and one year whereby a single bullet payment is made at maturity. A short-term advance may constitute a part of a larger credit arrangement such as a revolving line of credit.	Customarily, short-term advances are either for a round number of months or days and used for working capital finance or other short-term needs.	Since short-term advances employ bullet repayment terms, exposure will be equal to the full amount of unpaid principal and interest.
Call loan	A call loan is a loan extended for an indefinite period, which may be terminated at any time by either party upon notice according to the loan terms.	Call loans are generally used for temporary financing needs of an uncertain duration. Typically, notice of termination of a call loan may be given every day before 12.00 A.M., in which case final repayment must be made within two working days.	Since call loans have no fixed duration, exposure risk excludes anticipated future interest income beyond the notice period.
Bridge loan	A bridge loan is a type of short-term advance utilized to cover an anticipated gap between the time payment arising from a given transaction is confirmed and the actual receipt of the proceeds pursuant to the settlement of the transaction.	Since the duration of the period between consummation of the transaction and settlement can be estimated with some high degree of probability, bridge loans are usually of a fixed term, often less than one month.	Same as other short-term advances of fixed maturity.

Lines of Credit

A line of credit—that is, a commitment by a financial institution to advance funds up to a designated maximum during a given period²³—is similar to a rollover loan in that it offers the borrower greater flexibility than the conventional fixed-term loan. In contrast to a fixed-term loan, which is characteristically very specific concerning the amount of funds to be lent and the schedule for repayment, a line of credit allows for a variable amount to be drawn in a given period and similarly permits some leeway in the repayment schedule. Facilities that fall under the broad category of lines of credit include:

- Overdraft facilities
- Revolving credit facilities
- Standby facilities or standby rollover facilities
- Multipurpose facilities
- Specialized facilities (e.g., note issuance facilities liquidity support facilities)

Credit lines come in a variety of flavors as described in Exhibit 3.6.

EXHIBIT 3.6 Selected Facilities

Product Type	Definition	Typical Terms and Usage	Risks
Overdraft facility	An overdraft facility is a credit arrangement under which a bank advances funds to a borrower in varying amounts up to a specified limit, and the latter's repayment obligations are, within predefined parameters, flexible. That is, within specified limits, the date and amount repaid within any given financial period is not fixed but may vary according to the customer's needs. An overdraft facility may be a discrete product, or part of another credit product.	Overdraft facilities are most commonly used to provide working capital to companies through the cycle of manufacturing or providing goods or services and converting them into collected cash. The limits on overdraft facilities are customarily reset on an annual or similar periodic basis. Note that of the facilities described in this table, the overdraft facility generally affords the greatest flexibility in drawdown and in repayment.	Unlike many loan products, the credit exposure in respect of overdraft facilities and other similar lines of credit incorporates a maximum or nominal exposure figure equal to the amount of credit advance permitted under the facility and the expected exposure.
Revolving credit facility	A revolving credit facility is quite similar to a rollover loan, described in the	Like term loans, the maturity of rollover loans may be as long as	

(Continued)

EXHIBIT 3.6 (Continued)

Product Type	Definition	Typical Terms and Usage	Risks
Standby facility/ Standby rollover facility	previous table, the main difference being that revolving credit facilities do not have maturities of greater than five years. A standby facility is similar to an overdraft facility but differs in that its tenor may be longer than is customary with the latter in order to meet the particular needs for which it is obtained. Exposure for a standby facility is incurred both during the period in which it may be exercised and during the period in which, if drawn upon, funds must be repaid. In addition the facility may be provided on a multicurrency basis. Repayment terms, however, are usually fixed rather than flexible. A standby rollover facility is a standby facility structured with a view to being extended on a rollover basis.	7–10 years, but they may also be used for short-term credit provisions in response to a customer's working capital needs. A standby facility is typically used for backup purposes to provide a customer with assurance that credit will be available for a period of time, often in order to ensure the customer's ability to consummate a major transaction such as a large investment or acquisition.	
Multipurpose facility	A multipurpose facility is a variant facility that may incorporate characteristics of overdraft facilities, revolving credit facilities, and standby facilities.	Multipurpose facilities are used to provide both working capital and longer-term finance to customers.	
Specialized facilities	These include note-issuance facilities and liquidity-support facilities. Note-issuance facilities are provided in connection with a note-issuance program whereby a bank or a group of banks underwrites a borrower's issuance of short- or medium-term notes in the money or capital markets. Liquidity-support facilities are used to provide liquidity in connection with securitizations such as the origination by a customer of mortgage-backed securities (MBS).		

Asset-Based and Receivables Finance

Besides loans, commercial banks customarily offer a variety of types of asset-based or receivables financing. These facilities are commonly provided to merchants and to manufacturers to finance domestic and overseas trade, and therefore may be categorized as types of export financing or commodity financing. Asset-based and receivables finance instruments are summarized in the table in Exhibit 3.7.

EXHIBIT 3.7 Asset and Receivables Finance Instruments

Product Type	Definition	Typical Terms and Usage
Acceptance borrowing (bankers' acceptances)	<i>Acceptance borrowing</i> refers to the use of bankers' acceptances, also known as bills of exchange (technically speaking, the instrument can be classified as a <i>negotiable time draft</i>), to finance the import or export of goods. The instrument is called an <i>acceptance</i> because by its terms it obligates the bank that accepts it to pay its face value to the bearer of the instrument at maturity.	One of the oldest forms of finance, bankers' acceptances (bills of exchange) facilitate trade by allowing the merchant to substitute a recognized bank's creditworthiness for its own. This rendered the instruments negotiable (marketable), and historically bills of exchange functioned as a form of money to consummate transactions in lieu of coin or currency. Modern bankers' acceptances ordinarily have a maximum tenor of six months.
Commodity financing	<i>Commodity financing</i> refers to the financing of a designated quantity of commodities such as agricultural products, mineral products, petroleum products, and so on in advance of sale. Hence, they are ordinarily short-term in character.	The title documents (e.g., warehouse receipt, bill of lading) to the commodities to be sold function as collateral for the loan. As a rule, the commodities are required to be stored at a bonded warehouse, in which case a warehouse receipt serves as title to the goods.
Export financing	Export financing encompasses a number of products, including loans directly to exporters, for further extension to importers, or directly to the importers or their banks.	Export loans are a form of credit that is primarily used to finance the export of capital goods where it is contemplated that the overseas purchaser will not pay the entire cost of the goods at delivery, but will finance the balance of the purchase over some period of time, typically more than one year. To mitigate risks, <i>export credit insurance</i> provided by <i>export credit agencies</i> , is purchased.

Some of the products employed for this purpose, such as documentary letters of credit, represent *contingent* risks to the bank rather than outright *on-balance-sheet* risk and are characteristically more fee-driven than interest-income driven. Letters of credit are among the most popular, and are discussed later in the chapter under the heading of “Trade Finance and the Letter of Credit.” Others such as bankers’ acceptances (bills of exchange) are also used for financing trade, but in addition constitute money-market instruments, and therefore are linked with investment banking and related dealing activities undertaken by banks in the money and capital markets.

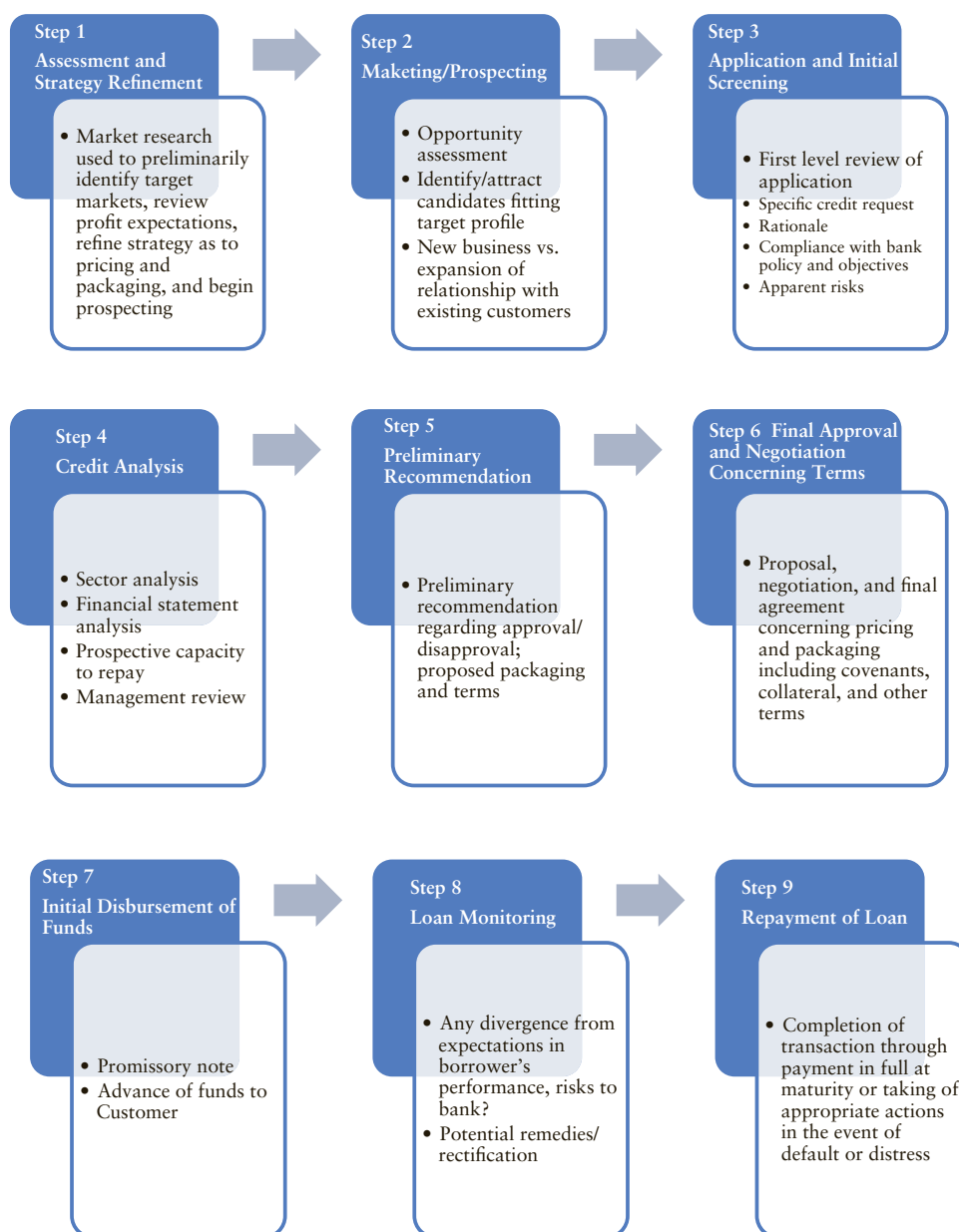
The Lending Process

Operating a successful lending business is a more complex activity than it might first seem. To make the activity profitable, funds must be sourced at a reasonable cost to lend to financially sound borrowers. Sourcing funding frequently means attracting new depositors or attracting new deposits from existing depositors. Making sound loans depends heavily on the identification of creditworthy loan applicants. Both activities involve appropriate pricing on the one hand and cost-effective marketing on the other. With regard to lending, the prospective customer must be approached or attracted to the institution. To this end, marketing is used to convey the appropriate image on the part of the bank and to educate the prospective customer concerning the benefits arising from a relationship with the bank. Naturally, pricing is another essential component. Terms governing the loan agreement need to be both acceptable to the customer and to the bank. The major steps in the lending process²⁴ are shown in Exhibit 3.8.²⁵

The process begins with market research, the refinement of lending strategy and the formulation of tactics to attract the type of customers the bank seeks. Naturally, for this purpose (and also to appeal to depositors and facilitate inexpensive and stable funding), some sort of distribution network must be in place. Traditionally, this meant the development of a branch network to bring the bank close to the customer, or as happened in the United States, the development of a highly localized banking system, which discouraged the creation of national branch networks by the largest institutions. A distribution network was also critical to developing a strong lending business, because many customers prefer to deal with an institution that is locally accessible. The World Wide Web has brought with it e-banking, but not all types of lending activities are amenable to web distribution, and banks that are purely web-based have found it more difficult than some expected to establish strong deposit networks.

Assuming a suitable distribution infrastructure has been constructed, the first operational step is to market the bank’s lending and other financial services to attract desired loan applicants. With an application having been submitted, an initial review is performed to determine whether it broadly fits within the bank’s guidelines, whether the candidate is of a caliber sought by the bank, and whether risk levels seem acceptable. This is followed by the process of packaging the loan to a particular applicant, and the initiation of negotiations concerning pricing and loan terms, including those governing collateral and covenants.

In the next phase of the lending process, credit analysis is undertaken, and a recommendation made.²⁶ If affirmative, the proposal is considered by the appropriate credit committee. If approved, the agreement with the customer can then be consummated. It is not uncommon for the credit committee to require some modification to the

**EXHIBIT 3.8** The Lending Process

terms, particularly where large sums are at stake, and if a final agreement is reached, this phase of the process concludes, the agreement is formalized and funds advanced.

The final phase of the process involves monitoring the customer and taking appropriate action in case of default or the emergence of new risks. Credit control staff assess and monitor collateral, while the bank's legal department keeps associated lending documentation on file and, in case of borrower distress, works out

problematic loans. Finally, with the maturity of the loan and its repayment, the transaction concludes.

Syndicated Lending

Syndicated lending is the business of arranging large loans and then selling off participatory interests in the loan to other financial institutions. Correspondingly, a syndicated loan is one in which the funds are advanced by two or more lenders. In practice, the number of banks participating in a *loan syndicate* will ordinarily be at least 4 to 7, and on larger deals subscribers may exceed 30. Despite the participation of multiple lenders, there is a single loan agreement that entitles each creditor to designated rights of action against a delinquent borrower. The process of syndicating a loan is summarized in the diagram in Exhibit 3.9.

In most loan syndications, there will be a more senior group of banks that actively collaborate with the arranging bank, also called the lead manager, in pricing the deal and setting up further distribution to other subscribers. One of these so-called lead banks will be appointed as agent to oversee the loan following the closing of the transaction and to administer the subsequent relationship between the customer and syndicate participants. The arranging bank typically benefits from a substantial fee as compensation for originating the transaction, while fees may also be earned for co-arranging the loan, acting as agent, and underwriting the transaction by agreeing to provide liquidity according to predetermined terms.

Generally speaking, the more senior members of the syndicate, being those that tend to have a more active role in the syndication process, generally receive greater remuneration than its junior members. The latter, being not actively involved in arranging and marketing the deal usually function merely as lenders. In some circumstances, however, these ordinary subscribers may receive a participation fee merely for entering the syndicate. Finally, insofar as part of the loan remains undrawn by the borrower, syndicate members may earn additional fees in lieu of interest and commitment fees to compensate for the cost of committing regulatory capital and of liquidity management.

Rationale and Similarities with Public Debt Issues

The rationale for loan syndication is that the financing needs of a particular borrower may be so large as to make the risks to an individual lender prohibitive, especially

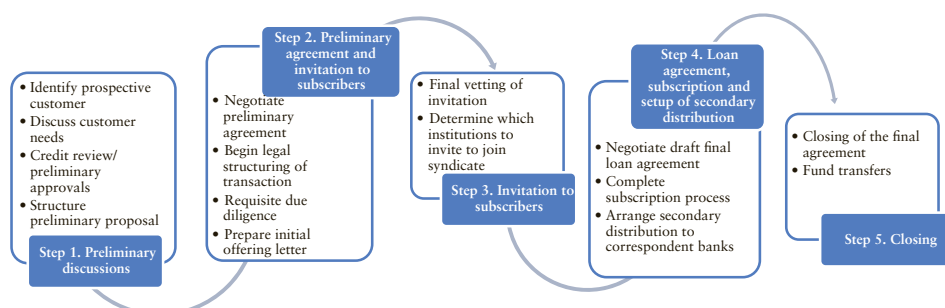


EXHIBIT 3.9 Process of Loan Syndication

with regard to concentration risk. As with the issuance of notes or bonds in the debt capital markets—such issuance is the primary alternative to loan syndications—the purpose of syndicated loan financing is frequently to obtain medium-term credit to finance or refinance capital expenditures (capex).

In addition, the business of arranging syndicated loans resembles the *underwriting* of debt securities not only in terms of the size of the financing, but also in that the activity is significantly fee-driven. Indeed, the size of a syndicated loan is frequently comparable to that of a typical corporate bond issue, although the maximum maturity tends to be somewhat shorter at 3 to 5 years versus 7 to 10 years for corporate bond issues.

Differences from Public Debt Issues

The key difference between syndicated loan financing and public debt financing is that funding is provided privately through banks and other financial institutions instead of through a public securities issue in the capital markets. (It should be noted, however, that a secondary market has developed in syndicated loan participations in some markets, particularly in the U.S. private placement market.) Its private character affords syndicated lending some advantages over the issue of notes or bonds in the public capital markets.

The key advantages offered by syndicated loans include:

- Greater speed and flexibility in the financing process and structure
- Reduced public disclosure requirements
- The absence of a need for a public debt rating²⁷

More specifically, syndicated loans can be structured more along the lines of a bank lending facility, rather than in terms that find favor with the bulk of fixed-income investors. For example, syndicated loans can incorporate a revolving line of credit that can be used flexibly by the customer. As a form of private financing, disclosure and compliance requirements tend to be less onerous than is the case in the public debt markets, and syndicated loans are frequently faster to arrange than comparably sized bond issues. Finally, while an agency credit rating is practically mandatory in the public debt markets, this is not necessarily the case in the syndicated-loan market, which again can be an advantage both in terms of cost and speed.

Trends in Loan Syndication

The nature of syndicated lending has altered over time in key markets. Where in the 1980s, syndicated loans were a significant source of finance to sovereign borrowers, as well as to major commercial enterprises, many such former users of syndicated loan finance can now access the debt capital markets directly. The result has been a change in the character of syndicated lending in some markets toward uses such as to finance mergers and acquisitions or leveraged buyouts, in which the principal borrowers desire a high degree of confidentiality, which would not be possible by means of a bond issue. This has caused the market to expand enormously in some markets worldwide, for example in Australia with over 150 deals in 2011.

LEASE FINANCE

Finance leases although having a different legal form from loans nevertheless have a quite similar function. Instead of a business customer borrowing cash to finance the purchase of, for instance, new vehicles for its employees' use, it may instead enter into a financial leasing arrangement in which it leases the vehicles from the bank, or a bank agency, making periodic payments to the lessor in the same way that it would make periodic interest and principal payments on an amortizing loan. At the end of the lease period, rather than repaying the loan in full, having effectively done the equivalent by making its lease payments, it may exercise the option to take full title to the vehicles.

Securitization

Securitization began in the 1970s, grew explosively during the 1980s and 1990s, and in the process this form of financing has transformed the traditional lending function. This is particularly so in the more developed regions with highly evolved capital markets. This section elaborates upon how the process works.

THE DEVELOPMENT OF SECURITIZATION AND ITS IMPACT ON BANK LENDING

The growth of securitization over the past three decades has given new life to lending activities in many markets. Through securitization, loans—traditionally illiquid assets—are aggregated into pools and repackaged into new relatively liquid securities that are sold to investors. In the process, they are removed from the balance sheet of the originating bank. This has enabled banks to more rapidly realize the value of the originated loan while offloading the accompanying credit risk to investors. Analogously, similar mechanisms have been evolving to enable the securitization of a wide variety of financial flows to offer highly sophisticated financing products to customers.

The development of securitization essentially began in the early 1970s in the United States. A U.S. governmental agency, the Government National Mortgage Association (GNMA or Ginnie Mae), pioneered the purchase and sale of pools of standardized mortgage loans. With the agency's guarantee, the creation of mortgage-backed securities—then termed single-class mortgage pass-throughs—became feasible. Other quasi-governmental agencies including the Federal Home Loan Mortgage Corporation (FHLMC or Freddie Mac) and the Federal National Mortgage Association (FNMA or Fannie Mae) got into the act in the next decade. These new securities were backed by the full faith and credit of the respective agencies, which were either government agencies (Ginnie Mae) or quasi-government agencies (Fannie Mae and Freddie Mac). U.S. government backing, either explicit in the case of Ginnie Mae or perceived in the case of

Freddie Mac and Fannie Mae, enabled these securities to attract the AAA rating that made them attractive to institutional and other investors looking for a higher-yielding substitute to U.S. government paper.²⁸ The 2008 crisis of course brought some changes to that risk perception.

With the acceptance of agency-backed mortgage-backed securities by investors, the market soon evolved attracting nonagency issuers and spawning other products. So-called private label issues used a variety of credit enhancements to attract sufficiently high investment-grade ratings. These included external credit risk mitigants such as guarantees or insurance provided by entities other than the government, and internal enhancements effected through the manner in which the transaction was structured, including overcollateralization. The use of multiclass issues, in which several classes of investments were created, each with different risk and reward characteristics, facilitated this process since they could be marketed to a broader population of investors.²⁹ The mix of maturity, credit risk, and cash flow characteristics of each class of securities would appeal to different investor groups. The development of collateralized mortgage obligations, together with asset-backed securities (ABS) that were collateralized by assets other than residential real estate, such as credit cards, auto loans, government-sponsored student loans, and financial leases—underpinned a tremendous expansion in the market for securitized instruments.

What Is Securitization?

Securitization can be defined as the aggregation, or pooling, of financial assets that individually are comparatively illiquid into highly liquid marketable securities. Securitized instruments come in many varieties having a myriad of risk, reward, and maturity characteristics. As a rule, however, the underlying assets that form the pool from which the securitized instruments are created are both revenue-generating and relatively homogenous, and when pooled have comparatively predictable risk and return characteristics.

The first securitized instruments relied upon pools of mortgage securities both to generate interest revenue to make the coupon payments on the repackaged security and to provide collateral backing to investors. Subsequently, many other types of assets have formed the basis of securitization. In addition to mortgage-backed securities or MBSs, the universe of securitizations encompasses ABS and collateralized mortgage obligations (CMOs) among others. The securities issued through securitizations can be simultaneously categorized both as debt securities and derivative securities.

In form, these obligations resemble debt securities, such as bonds, because they typically generate a coupon payment subject to a predetermined formula. However, the characteristics of each class of securities are structured using the pool of underlying assets, and the value of the security is based both on the characteristics of the underlying pool and the structure of the particular class relative to other classes. Equally significantly, as are other derivative instruments, securities are almost

invariably structured so as to remove the credit risk from the entity—often a bank—originating the transaction—off balance sheet to a bankruptcy-remote special purpose vehicle (SPV). Unsurprisingly, as a consequence, both the credit analysis and valuation of these multiclass securities is considerably more complicated than in respect to a plain vanilla corporate bond issue.

DEFINITION OF SECURITIZATION

Securitization is a method of funding revenue-generating assets that is effected by aggregating the corresponding receivables produced by such assets (which individually are comparatively illiquid) into large asset pools, and then repackaging both the cash flow generated by them and the associated collateral into single or multiclass issues. Owing in part to the *law of large* numbers, the resulting securities possess relatively predictable risk, return, and maturity characteristics (in comparison to the individual underlying assets) and are therefore relatively liquid, and consequently marketable to investors.

In a sense, the process of securitization creates a whole that is greater than the sum of its parts, since each individual underlying assets is comparatively less liquid and more volatile than a corresponding portion of the aggregated asset pool. Likewise, per unit of principal, the securities that can be carved out of that pool are worth more to a prospective purchaser than the individual underlying assets. From another perspective, securitization can be viewed as a form of off-balance-sheet receivables financing that is effected through the transfer of the underlying assets off the balance sheet of the originator to a bankruptcy-remote SPV, which in turn issues the securitized instruments.

Rationale for Securitization

The *raison d'être* for securitization depends in large part upon the regulatory environment in which it is undertaken. Generally speaking, it enables financial institutions to transform the relatively illiquid underlying assets on their balance sheets (which they have originated or acquired) into marketable and liquid securities. In so doing, they are afforded the ability to sell the underlying assets en masse for an immediate gain. Another key advantage of securitization is that it enables financial institutions to raise funds at a lower cost than through a straight debt issue since credit enhancements allow for the rating of the securitized issue higher than the issuer itself.³⁰ A further benefit of securitization is that it arguably enhances bank liquidity by facilitating funding diversification.³¹ In addition, under the Basel Accords, securitization—especially for residential mortgages—reduced the cost of capital for many banks.³² Last, of course, banks can generate significant fee income by providing their securitization expertise to their corporate clients.

For investors, securitized instruments have the advantage of offering a combination of attractive risk, return, and maturity characteristics, often generating comparatively higher yields than similarly rated conventional instruments. There is

no free lunch, however, and in exchange for the higher yields, investors may take on greater risks than those associated with conventional corporate bond issues, such as prepayment risk, call risk, and extension risk.³³ The particular risk and return characteristics of the repackaged security will depend on both the characteristics of the underlying assets and how the particular issue is structured.

For example, the risk, return, and maturity attributes of mortgage loans vary depending upon among other things:

- The type of property (i.e., residential or commercial, single-family detached houses or flats)
- The purpose of the loan (e.g., owner occupied, buy to let, property development)
- The class of underlying borrower (e.g., prime vs. subprime)
- Lending, appraisal, and borrower eligibility criteria
- The loan-to-value (LTV) ratio³⁴
- Underlying instrument characteristics (e.g., maturity, fixed vs. variable interest rate, prepayment penalties)
- Relevant law and regulation that may affect some of the foregoing attributes as well as remedies in the event of default

Process of Securitization

To effect a securitization, the bank aggregates the financial assets to be securitized into a pool, and these are sold to an SPV, which acquires the legal rights previously held by the originating bank. As the purchaser of this pool of assets, the SPV steps into the originating bank's shoes, and becomes entitled to interest payments, principal repayment, and the right to the collateral associated with the assets in the event of borrower default.

Typically, at least two classes of debt securities are issued, together with an equity component. By contract, each class of MBS is structured so that in effect the rights to revenue and collateral cover from the acquired assets are allocated to the corresponding classes of investors to create different risk and reward characteristics. The holders of the equity portion of the issue absorb the first loss arising from defaults on the underlying mortgages, while the junior note holders represent the second line of defense for the senior note holders. This last group, unsurprisingly, are provided the greatest degree of insulation from a shortfall in revenue.

The example of MBS, which comprised the first securitizations, and remain a large part of the market, can be used to illustrate the process. Therefore, to create MBS, the underlying individual mortgage loans are aggregated and sold to an SPV in the manner just described. The diagram in Exhibit 3.10 shows the process of securitization with reference to a simple two-class MBS with a residual equity component retained by the originator.

Credit Cards

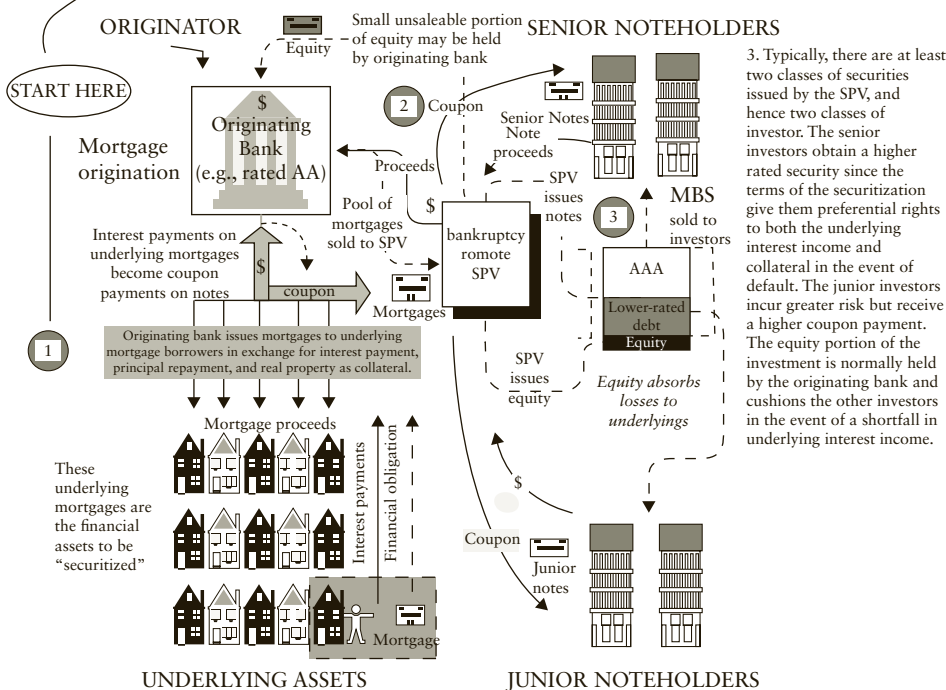
Most people are familiar with credit cards, charge cards, and their more recent siblings, debit cards.³⁵ In contrast to a credit card, a charge card requires the cardholder to pay the outstanding balance in full each month. Although charge cards actually preceded the development of credit cards, credit card use has far

Presecuritization

1. Origination of mortgages by originating bank.

Securitization

2. Securitization begins with the aggregation of a pool of mortgages which is sold to an SPV. The SPV in turn issues securities to investors which pay them interest generated by the underlying mortgage borrowers. The collateral securing these mortgages also provides security to the investors. Upon the sale of the underlying assets to the SPV, the originating bank generally receives immediate payment, which is funded by the sale of the mortgage-backed securities to investors.

**EXHIBIT 3.10** The Process of Securitization Simplified

outstripped that of charge cards. Debit cards, which evolved after credit cards, are linked directly to the cardholder's bank account, and therefore represent a form of electronic cash.

Sources of Revenue

Credit cards offer consumers a line of unsecured credit that resembles in some respects an overdraft facility (discussed in the previous section) but owing to the wide acceptance of cards by merchants is considerably more convenient to use.³⁶ A crucial advantage of the credit card to the holder is the convenience it affords. The user need not carry large amounts of cash, nor be inconvenienced with frequent retrievals of cash. At the same time, the line of credit provided by the card enables the user to ride out temporary shortages of cash, and finance larger transactions over an extended period. From card users, banks earn interest on unpaid balances as well as from annual fees, application fees, and penalty fees including late fees and over-limit fees.³⁷

DEFINITION OF A CREDIT CARD

A credit card is a line of unsecured credit evidenced to merchants by means of a plastic card that contains identifying data about the facility that has been established with the card issuer.³⁸

The advantage to the merchant is threefold. First, although banks collect fees from merchants for payment processing, acceptance of credit cards effectively offers a form of receivables finance, because the merchant is paid promptly for customers' purchases less a percentage retained by the former. Second, while the fees cut into merchants' margins, the increased volume of custom that accepting cards for purchases brought usually more than compensated for the fees paid. Historically, credit cards first gained ground with merchants that sold high-markup goods and services, and those that catered to expense account customers who preferred to use credit cards and be reimbursed within the payment period by their companies, rather than from their own pockets. Finally, credit cards also relieve merchants from the onerous task of collecting funds from defaulting customers.

In summary, sources of revenue from credit cards include the following:

- Interest on cardholders' outstanding balances fees
- Fees collected from the merchant who accepts the card in payment for a transaction
- Fees collected from cardholders for annual use and for penalties

Most but not all fees fall into the category of yield-related income, so that they constitute interest income, even though they are nominally called fees.

It is evident that the credit card represents a hybrid financial product that while superficially an interest-earning product also generates fee revenue directly or indirectly through transaction processing. So credit cards operations encompass both a lending and a fee-generating business. Although banks often generate lofty rates of interest on unpaid balances, marketing expenses and credit losses are also relatively high. Nevertheless, the credit card industry is generally viewed as a fairly profitable sector within the banking industry.

Growth of the Credit Card Industry

Since the introduction of the first bank credit card, the credit card industry has expanded tremendously, both in developed markets and globally. Interest rate deregulation, the development of sophisticated credit information databases, and the evolution of securitization techniques³⁹ in the 1980s provided a huge impetus to card issuance. Receivables skyrocketed during the 1990s and 2000s. The number of cards in circulation⁴⁰ exploded, partly thanks to new markets such as Brazil, and especially China (see Exhibit 3.11). The North American credit market is now relatively mature, however, and growth in credit card issuance is much more rapid in other countries. Credit card distribution is greatest in developed countries and those large

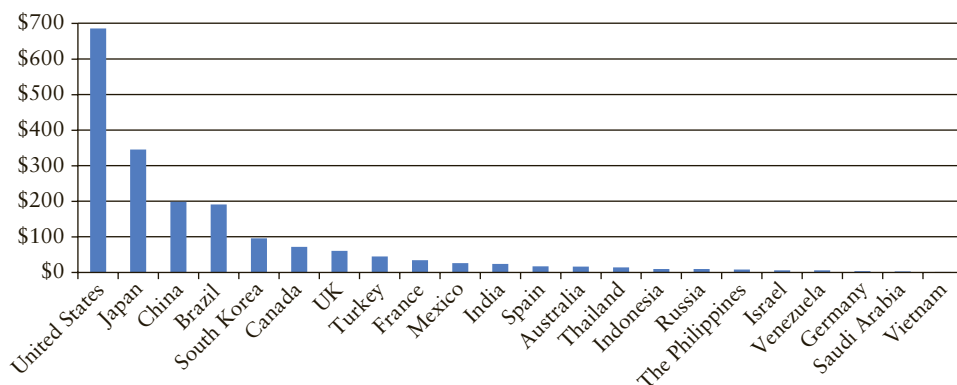


EXHIBIT 3.11 Global Bank Credit Cards in Circulation in 2009 (US\$ million)

emerging markets experiencing rapid growth of the middle class. Receivables have grown substantially in the 2000s, although the financial crisis put an end to that growth after 2007.

Despite this nominal distinction, from an economic and an analytical perspective, most so-called fee income arguably represents a form of interest revenue. In comparing results and analyzing the profitability of credit card operations among various card issuers, it is important that consistent classifications be used. Otherwise, results will be distorted. Historically, U.S. credit card issuers do not always report net interest income consistently. For example, back in FY2000, of six major issuers, three defined net interest income as being limited to spread income, two defined it as spread income plus late fees, and one defined it as spread income plus late fees plus other nominally noninterest items.⁴¹ A detailed discussion of these matters is outside the scope of this book.

In most cases, bank credit analysts will not need to evaluate a financial institution's credit card operations in depth, unless of course credit card issuance is a very large part of the organization's business. When it is necessary, however, caution should be exercised, as there can be considerable variation in how revenue and expense items are classified.

The credit card business could be very profitable, but the temptation exists for banks to venture beyond safe limits, usually by extending credit card facilities to fragile cardholders, not dissimilar to subprime mortgage borrowers. Banks gamble on credit card holders getting wealthier relatively quickly, a frequent occurrence in dynamic emerging markets. This is fraught with dangers, as was the case in Korea in 2003, after the number of credit cards doubled and the credit card assets quadrupled in the previous three years.

Therefore, even a bank's seemingly innocent exposure to credit card assets could very well become relevant to the external analyst.

A detailed discussion of these matters is, however, outside the scope of this book.

Credit Card Terminology

The credit card industry uses some very specific terminology, some of which resembles terms used in other areas of banking. They are not detailed in this book,

but readily available from many sources, for example from the websites of large banks. The analyst would easily figure out the potential credit risks associated with each of them.

DEFINITION OF A DEBIT CARD

Like an ATM card, a debit card is linked to the cardholder's bank account and also allows the cardholder to make purchases with other parties in the same manner as credit cards or charge cards, the difference being that the funds are withdrawn directly from his or her account. Therefore, there is no credit risk to the card issuer. Often ATM cards double as debit cards. Some debit cards use the cardholder's signature, rather than being authorized electronically at the point of sale. These are also known as deferred debit cards or as offline debit cards.

For the external analyst's purpose, the notion of *card receivable* is important. It is the revenue generated from individual cardholders encompassing finance charges, including penalty fees, over-the-limit fees, and other similar fees. Depending upon the defining body, it may also include merchant fee revenue. Charge-off ratio (for credit cards) is obtained by dividing credit card receivables written off in a given period as percentage of total card receivables registered at the end of the period.

BANKS AS FINANCIAL SERVICE PROVIDERS

Although lending and other forms of credit extension have long been the mainstay of the banking industry, this activity is often highly competitive, relatively risky, and labor intensive. As a consequence of its highly competitive nature, as well as the impact of disintermediation, margins on lending have declined in many markets. The adverse impact on margins arguably has been exacerbated both by the increased availability of data relevant to credit risk assessment, which has reduced the competitive advantage particular institutions may have once enjoyed, and by capital regulations that have increasingly required banks to account for the risks associated with lending. On the whole, these secular developments have encouraged banks to look to alternative sources of revenue to what is sometimes termed a bank's *warehouse activities* or *spread business*. This section therefore looks at banks' non-spread business, or *nonwarehouse activities*, that is, activities other than lending-related business, from which revenue may be generated.

Overview: The Impact of Disintermediation

As large companies have gained better access to the money and capital markets, becoming less reliant on bank finance in the process, they no longer need banks to act as financial intermediaries as much as they did in the past. Instead of borrowing from banks, and generating correspondingly large financial assets for banks, they can issue

their own paper more cheaply. This benefits those banks that can offer investment banking services at the expense of those that cannot. In markets where such asset-side disintermediation has become pervasive, traditional corporate lending has generally become less profitable than previously. Similarly, as discussed, disintermediation occurs on the liability side of the balance sheet when consumers obtain access to savings vehicles other than savings accounts, such as annuities, money-market accounts, or brokerage accounts. Where individuals are offered a variety of alternatives, a substantial number are likely to be attracted to products offering different characteristics from traditional bank deposit accounts.

THE BASEL ACCORDS AND DISINTERMEDIATION

A contributing factor to increasing disintermediation in the banking industry in the 1990s and 2000s was the implementation of the first Basel Accord in 1988, which increased the amount of capital required to be allocated to many types of loans. This had the effect of requiring lending activity to generate higher returns to justify the amount of capital allocated. Partly as a result, in some markets the spread that banks could earn on lending to those large enterprises able to access these markets was too tight to be attractive, while the risks implicit in lending to small and medium-sized enterprises could not be covered and provide an adequate margin at any reasonable rate of interest. Rivalry among banks for the remaining prospective borrowers who offered an attractive risk/reward proposition quickly resulted in squeezed margins in the remaining profitable lending niches.

The overall result is that, as disintermediation has taken hold in the banking sector, the proportion of operating revenue generated by sources other than earning assets has risen, while the proportion of funding contributed by customer deposits has declined. From another perspective, disintermediation results in the conversion of spread revenue into fee and commission revenue. Among the 50 largest U.S. banks, revenue from fees, commissions, and trading activity on average increased from 20 percent of operating revenue in 1990 to almost 40 percent in 2000. Correspondingly, the proportion of funding obtained from sources other than deposits increased from 15 percent in 1990 to more than 40 percent 10 years later.⁴²

The trend extended to the 2000s when it also spread to Europe and other markets. The frenetic quest for easy revenue is largely to blame for the distrust that established itself within the interbank markets as early as 2005 and 2006, leading—although this was not the only cause of the problems—to the disastrous crisis of 2007.

Treasury

In addition to holding financial assets in the form of loans and similar instruments, most banks hold securities, primarily debt securities, for investment and trading purposes. They also lend to other banks in the interbank market. These and related

activities comprise a bank's treasury operations. Correspondingly, a bank's securities portfolio is usually managed within its treasury department. Foreign exchange dealing, which is explored in greater depth in a later section of this chapter, also customarily comes within the scope and responsibility of treasury operations. Generally speaking, the purpose of treasury operations is both to supplement bank funding through borrowings in the money and capital markets and to provide a cushion of easily liquefiable financial assets. With the advent of regulations imposing a more proactive management of risk, the traditional asset/liability management activities—so far mostly meant to balance cash flows over multiple periods—expanded into mitigating a broader range of risks. More specifically, banks engage in trading and investment activity in the money and capital markets for the following reasons:

- To maintain a requisite degree of liquidity
- To establish an ongoing presence in the markets to diversify its sources of funding
- To manage interest rate risk
- To earn a modest return on excess cash

TREASURY FUNCTIONS AND STRATEGIES

Treasury activities fulfill several functions for a bank—liquidity enhancement, asset-liability management, risk management, and profit supplementation. In the last context, treasury activities come within the realm of proprietary trading, which refers to trading done on a bank's own account—that is, by putting its own capital at risk—rather than trading made on behalf of customers where the bank collects an investment management fee but ordinarily does not share in profits. Banks are not primarily investment managers, and consequently treasury operations rarely are core sources of operating profits for banks.

Although trading activity is a valuable supplemental source of revenue and in some circumstances can be a substantial contributor to the bottom line, if a bank becomes dependent on trading revenue to generate positive results, this can be a source of concern for the credit analyst. Such a scenario suggests that the bank's spread business is weak or that its risk profile is less conservative than ordinarily associated with commercial banking. A bank that is deriving the bulk of its profits from proprietary trading is using a different business model from that ordinarily associated with deposit-taking institutions. This is because trading gains, over the medium term, invariably represent a more erratic (i.e., volatile) source of revenue than spread income.

During periods of strong loan demand, banks often reduce their securities holdings, only to increase their liquid asset holdings again during periods of weak demand for credit. This has the benefit of allowing them to capture profit opportunities from lending quickly when the economy begins to turn around

and also provides added liquidity during periods of economic stress. Likewise, banks seek to benefit from changes in interest rates associated with the credit cycle.

- When interest rates are high, banks usually try to maximize and lock in yields for longer maturities, so that they will benefit from these investments when interest rates ultimately fall.
- When interest rates are low, however, they will often focus on short-term investments to minimize their exposure to interest rate risk and to take advantage of higher rates when the opportunity comes to roll over the funds invested short-term.

Banks can also shape their securities holdings to hedge against particular types of obligations to reduce or eliminate interest rate risk.⁴³

Overview of Treasury Operations

Treasury operations, like those of the bank as a whole, are often divided into front-office, middle-office, and back-office functions. Although nearly all banks do some trading on their own account, some institutions engage in a substantial amount of proprietary trading activity. This business may be conducted within general treasury operations, along with liquidity and cash management functions. Alternatively, it may be handled by a separate proprietary trading unit, the purpose of which is to generate profits in the same fashion as an investment management fund.

Organization of Treasury Operations: Front Office

In the context of treasury operations, the front office is responsible for executing the bank's trading, investment, and hedging strategies. It is characteristically divided by function, conventionally into the following groups:

- Money-market desk
- Corporate desk
- Foreign exchange (FX) desk

The money-market desk deals in short-term money-market instruments.⁴⁴ The corporate desk deals primarily with longer-term instruments, including corporate bonds. The FX desk deals with foreign currency transactions, which are discussed later in the chapter.

Organization of Treasury Operations: Back Office and Middle Office

The back office is responsible for processing, checking, settling, and confirming transactions, monitoring transactions and collateral, handling funds transfers, maintaining the IT infrastructure, and communications infrastructure support.

As part of its functions, the back office is responsible for supplying the requisite data to the middle office, which oversees compliance, treasury, and related risk management oversight and reporting, and supervising the treasury operation as a whole. The middle office therefore collaborates with the front office to manage the bank's cash (liquidity) position and its foreign currency position. It supervises compliance with internal trading limits. It also liaises with the back office to ensure that the bank complies with statutory and regulatory reporting requirements imposed by government agencies.

Trading Book versus Banking Book

Where the bank trades on its own account⁴⁵—that is, where the object is to profit from realized gains—this revenue is normally attributed to the bank's treasury activity, separate from the stream of revenue generated from periodic interest payments. However, since interest income may, as with treasury bills or commercial paper, be realized through the sale or redemption of the securities in question, the line between interest income and noninterest income can be indistinct. More importantly, accounting treatment varies among regions or countries and it also depends upon the period for which securities are intended to be held. A somewhat arbitrary distinction nevertheless must be made between assets held for trading purposes, that is, to realize a gain on sale, and those that may be held until maturity and therefore will be used primarily for their interest-generating capacity. The distinction drawn is between a bank's *trading book* and its *banking book*, and a time threshold is often considered to determine whether a securities holding forms part of the former or the latter.

The distinction between assets held for trading or held to maturity is not always very clear. In the context of substantial variations in the value of such assets—for example, the sovereign debt of some European countries in 2011 and 2012—some accounting rules may help hide disastrous situations.

TRADING BOOK VERSUS BANKING BOOK

A bank's *trading book* can be defined as its positions in financial instruments that are held with the intention of reselling them in the short term to take advantage of changes in prices and interest rates. The trading book also encompasses positions held for hedging purposes. All other positions—that is, those intended to be held over a longer time horizon—comprise the *banking book*.

Instruments Used in Treasury Operations

In developed countries, the vast majority of money- and capital-market securities purchased and sold by the bank and corresponding instruments sold by it are of investment-grade quality, being typically government debt issues or those of blue-chip companies, which are readily marketable in the secondary market. In addition,

interbank loans can be effected through an established interbank market such as the market for federal funds in the United States or informally between banks as counterparties through interbank deposits and reverse repo and repo transactions (discussed in a separate section). Overall, money-market instruments, discussed in the following subsection generally comprise the bulk of bank security holdings. As well as conventional money-market instruments, highly liquid capital market securities—that is, those with a maturity greater than one year—may be used in treasury operations. Those of greatest interest to banks include investment-grade issues of:

- Government debt
- Corporate debt
- Highly rated issues of MBS and ABS

Since medium- and long-term securities are as a rule highly liquid, they function in a similar manner to money-market instruments for trading purposes, owing to their typically high degree of liquidity. If held to maturity or for the longer term, they may be useful in asset liability management to hedge interest rate risk on longer-term lending exposures.

The Money Market

The money market is the primary playing ground in which banks undertake their treasury activities. Bank holdings of long-term securities generally represent a considerably smaller proportion of their total securities holdings. The principal instruments traded in the money markets—which exist in various shapes and denominations in most countries—are described in the succeeding pages. They are as follows:

- Interbank funds
- Negotiable CDs and eurodollar time deposits
- Treasury bills (T-bills)
- Commercial paper
- Repurchase (repo) and reverse repo and securities lending

THE MONEY MARKET

The money market is the market for short-term debt instruments, including federal funds, repos and reverse repos, T-bills, interbank placements, negotiable CDs, bankers' acceptances, and commercial paper, together with similar instruments. Most money-market instruments have a maturity of one year or less, although there are some exceptions. The participants in the money market include financial institutions, including banks that wish to borrow or to lend, government and corporate issuers that wish to borrow funds, traders and investors in money-market instruments, and brokers and dealers who arrange the purchase and sale of money-market instruments for a commission.

Interbank Funds

Interbank funds refer to bank deposits placed with other banks. A bank that places a deposit is effectively lending to the bank that receives the deposit, and the bank accepting the deposit is effectively borrowing from its counterpart. More generally, the *interbank market* refers to the informal market that functions both internationally and domestically among financial institutions primarily for short-term financing. Strictly speaking, the interbank market refers to the market among banks for funding, often overnight, but generally for a term of less than six months. However, the term can be used more broadly to encompass both funds made available to banks directly or indirectly through a country's central bank, for example, in the United States, the Fed's discount window⁴⁶ and *federal funds*⁴⁷ as well as funds placed on deposit with (i.e., lent to) other banks. The rate at which financing can be obtained in international money centers is often published, for example, LIBOR, the London Interbank Offered Rate.

Negotiable CDs

A CD is a fixed-deposit receipt issued by a bank, which evidences funds deposited with the bank for a specified period of time (time deposit) that usually pays a fixed rate of interest, either periodically until maturity or at maturity.⁴⁸

Negotiable CDs are large-denomination CDs that are highly marketable and therefore resemble debt securities. Like bonds, their price fluctuates with prevailing interest rates, rising as rates fall and declining as rates rise. There are variant types of CDs that exist both in respect to the U.S. banking system, the global banking system, and banking systems in particular countries. In the United States, a negotiable CD is defined to be a large-denomination CD (\$100,000 or more), ordinarily having a term of one year or less, that is transferable (negotiable) in the secondary market.⁴⁹ *Eurodollar* CDs are similar to U.S. domestic negotiable CDs in that they are large denomination and issued in U.S. dollars.⁵⁰ The difference is that eurodollar CDs, also called *euro-dollar time deposits*, are issued outside the United States, primarily in London.

T-Bills

T-bills are short-term government securities. T-bills do not pay interest to the holder periodically, but instead are sold through a competitive auction at a discount. That is, the difference between the amount at which the security is purchased and the principal amount payable at maturity represents interest to the holder. Similar instruments may be issued by subnational governmental units in other countries. In the United States, these are referred to as municipal notes.

Commercial Paper

Commercial paper is a form of short-term unsecured debt issued in the shape of an unsecured negotiable promissory note, which like T-bills is sold at a discount to its face value. It can be viewed as the corporate equivalent of a T-bill. In the United States, commercial paper has a maturity of 2-to-270 days as a consequence of Securities and Exchange Commission rules that exempt shorter-dated issues from registration.⁵¹ Typical maturities are 50 days or less. Finance companies are major issuers of commercial paper.

COMMERCIAL PAPER

Commercial paper is a money-market instrument that is analogous to a T-bill and municipal note. It is, however, issued by commercial firms, including industrial corporations and banks, rather than by government.

Repo Finance

Repo and reverse repo transactions, together with security lending, are forms of short-term borrowing collateralized by high-grade securities. Repo comprises a transaction in which a party with securities in its possession (ordinarily highly rated investment-grade fixed-income securities) sells the securities to a buyer in exchange for cash, subject to an obligation to repurchase the securities from the seller at a premium at a specified time. The period of time is typically short term, but repos of long duration, for example, one year or longer, are not unknown.

Substance vs. Form

Although the transaction takes the form of a sale and repurchase of securities, economically a repo constitutes collateralized institutional borrowing. That is, transacting a repo is equivalent to lending cash with the simultaneous “purchase” of securities as collateral to the counterparty that has advanced the funds. In the same vein, the difference between the sale price and the repurchase price (which will invariably be higher than the sale price) is the economic equivalent of an interest payment by the seller/repurchaser (the cash “borrower”) to the buyer/reseller (the cash “lender”) of the securities. When one party transacts a repo, the other party is simultaneously transacting a reverse repo. In other words, both terms refer to an identical transaction. Therefore, in a reverse repo, the “seller” of the securities effectively borrows cash from the “buyer,” again with the securities as collateral for the term of the transaction.

Repo vs. Reverse Repo

Each term by its use customarily identifies which is the cash borrower and which is the cash lender. So in respect of Bank ABC: in a repo—that is, Bank ABC transacts a repo—Bank ABC is the economic borrower of cash; in a reverse repo—that is, Bank ABC transacts a reverse repo—Bank ABC is the economic lender of cash. While technically a repo to one party is a reverse repo to the other (two terms being mirror images of the same transaction), confusingly in common parlance, a repo may refer to the borrowing of funds by a dealer, and a reverse repo to the borrowing of funds by a nondealer. Irrespective of terminology, the collateral under a repo/reverse repo agreement is typically marked to market daily, and the seller of the securities (i.e., the borrower) must make up any shortfalls in collateral.

Tri-Party Repo

Tri-party repo is a transaction in which the cash provider and cash taker enter into a three-way agreement with a custodian or clearing agent to facilitate settlement and

daily margining. Eligible collateral is specified in the supporting documentation and is selected within these parameters by the cash taker. All transactions are settled on a delivery versus payment (DVP) basis. The economics and credit risks of a tri-party repo are the same as a bilateral repo/reverse repo transaction but for any risks attributable to the activities of the custodian or clearing agent. Tri-party repo facilitates efficient use of bond portfolios in the provision of collateral and the ability for timely substitution of collateral when required. The transactions involve an exchange of title resulting in the “lender” of the cash obtaining beneficial ownership of the collateral (securities) and the “borrower” becoming beneficial owner of the cash. During the life of the transaction, the “lender” retains economic ownership of the securities except for voting rights.

REPO DEFINED

A *repo* is a transaction in which one party will sell and subsequently repurchase securities, ordinarily highly rated investment-grade fixed-income securities, at a specified date and price. A reverse repo is the mirror image of a repo. In a reverse repo, the counterparty will buy and subsequently resell securities at a specified date and price. Economically, the designated “profit” built into the sale and repurchase is equivalent to interest. The seller of the securities is the borrower, while the buyer and reseller of the securities is the lender.

Securities Lending

Repos and securities lending transactions are very similar, and structured in the same way. The differences are as follows:

- Repo transactions ordinarily involve the selling and repurchase of highly rated investment-grade debt securities, while securities lending involves the selling and repurchase of equity securities.
- The economic motivation of repo transactions is the desire on the part of the seller of the collateral to obtain liquid funds, often to finance other transactions, while the motivation for securities lending is the need for one counterparty to cover short positions or to hedge trades.
- Repo transactions can be and often are initiated by any institutional holder of high-grade securities, while securities lending is typically only engaged in by large investment or commercial banks that are able to intermediate between conventional asset managers such as insurance companies and long-only funds on the one side and hedge funds or other proprietary trading entities on the other; securities lending to such intermediaries is frequently a natural outgrowth of their need to make a market in particular securities.

Trade Finance and the Letter of Credit

Even before lending became the major business of commercial banks, one of their primary functions was to finance trade, including both domestic trade and trade

between merchants exporting and importing goods over long distances. For some banks, trade finance and related transaction processing continue to be a major source of revenue. Some trade finance products take the form of loans. Others such as a bankers' acceptance function as money-market instruments. Still others, such as the documentary letter of credit, discussed later, function as contingent risk products—that is, the bank provides a form of backup credit that remains unused so long as the seller and purchaser perform their respective obligations. In sum, banks earn income from foreign trade transactions by:

- Providing credit to importers and exporters
- Guaranteeing funds to exporters through, for example, the issue of letters of credit
- Processing trade transactions.⁵²

Loans do not require further elaboration at the moment, and having discussed bills of exchange, we now turn to the letter of credit. Until the advent of sophisticated global electronic data communications systems, the documentary letter of credit was, together with the bill of exchange, one of the most important mechanisms used to finance foreign trade transactions. This instrument is described further in Exhibits 3.12 to 3.14 that accompany the discussion. Although letters of credit remain an important instrument for the finance of trading, electronic systems have been developed in recent years, some by individual banks and others by third-party organizations.

Documentary Letters of Credit

A documentary letter of credit, abbreviated L/C, is an instrument used to facilitate the financing of international trade transactions. It performs this function by supplementing whatever credit a prospective exporter and importer are willing to grant each other (absent the intervention of a third party) by involving a reputable bank as an intermediary.

In brief, the essence of the L/C transaction is as follows. By issuing an L/C, the bank:

- Promises the seller that he will be paid upon the shipment of the designated goods.
- Promises the buyer that payment will only be made so long as the seller has complied with the terms and conditions specified in the L/C.

An L/C therefore represents a formal commitment by the bank issuing the letter of credit (the issuing bank) to pay a third-party beneficiary (ordinarily the exporter or seller of the goods) up to the amount prescribed (in a designated currency) and within a prescribed time against the delivery of the goods as evidenced by presentation of prescribed documents conforming to those described in the applicant's instructions. The issue of an L/C is initiated at the request of its customer (also referred to as the applicant, who is ordinarily the importer or buyer of goods from overseas), although it is the exporter or seller who may demand the use of this form of payment as a condition of sale.

EXHIBIT 3.12 Types of L/Cs

Type of Letter of Credit	Distinguishing Characteristics
Documentary	Standard type of L/C used in trade transactions described above. Usually irrevocable, but may be confirmed or unconfirmed.
Back-to-back	In a back-to-back L/C, a first L/C functions as collateral for the issue of a second L/C. This is used where there is a chain of middlemen between the supplier and the ultimate purchaser.
Deferred	The buyer is granted a specified period of time by the issuing bank to pay for the goods after delivery. The deferred L/C therefore functions as a financing mechanism enabling the buyer to arrange to sell the goods and obtain funds to pay the original seller.
Red clause	Under a red-clause L/C, the confirming bank is authorized to advance funds to the seller-beneficiary prior to presentation of the commercial documents. Normally, such an advance payment is made against a certificate that confirms the beneficiary's promise to dispatch the goods and to present the documents to the bank required by the letter of credit. The red-clause L/C can be advantageous to a seller (exporter) since there is an assurance of being able to obtain payment before it ships the designated goods. Like a deferred L/C, the red clause L/C can facilitate the finance and production of the goods to be dispatched.
Revolving	The issuing bank provides the applicant with a revolving line of credit linked to ongoing L/C transactions. This device is used where a trader is engaging in frequent trades of a similar character, or where the goods are to be shipped in installments or at periodic intervals.
Transferable	While L/Cs are generally negotiable, the transferable L/C specifies that the seller may transfer the funds received (or direct those funds) to a designated party. The transferable L/C is often used where another party has played a critical role in facilitating an import-export agreement, as where that party has financed the goods to be supplied by the seller. A transferable L/C often gives the middleman specific rights in connection with the transaction.

Source: Credit Research Foundation, *Understanding and Using Letters of Credit*, www.crfonline.org/orc/cro/cro-9-2.html.

Characteristics of Documentary L/Cs

There are variations on the basic L/C structure. These include the following:

Confirmed versus Unconfirmed L/Cs

Confirmed L/Cs are those that have been guaranteed (confirmed) by another bank (usually located in the same country as the seller) in addition to the issuing bank. Unconfirmed L/Cs have been guaranteed only by the issuing bank. Confirmed L/Cs provide greater comfort to the seller or exporter since he or she is likely to be familiar with the local institution and will be able to pursue a claim against it with greater ease than against a bank based in another country.

Irrevocable vs. Revocable L/Cs

Irrevocable L/Cs may not be changed after issue unless both buyer and seller agree. A revocable L/C may be changed by the buyer after issue, but because this provides less comfort to the seller, it defeats much of the purpose of the L/C. Therefore, most documentary L/Cs are irrevocable in character.

Payable on Demand versus Payable on Maturity

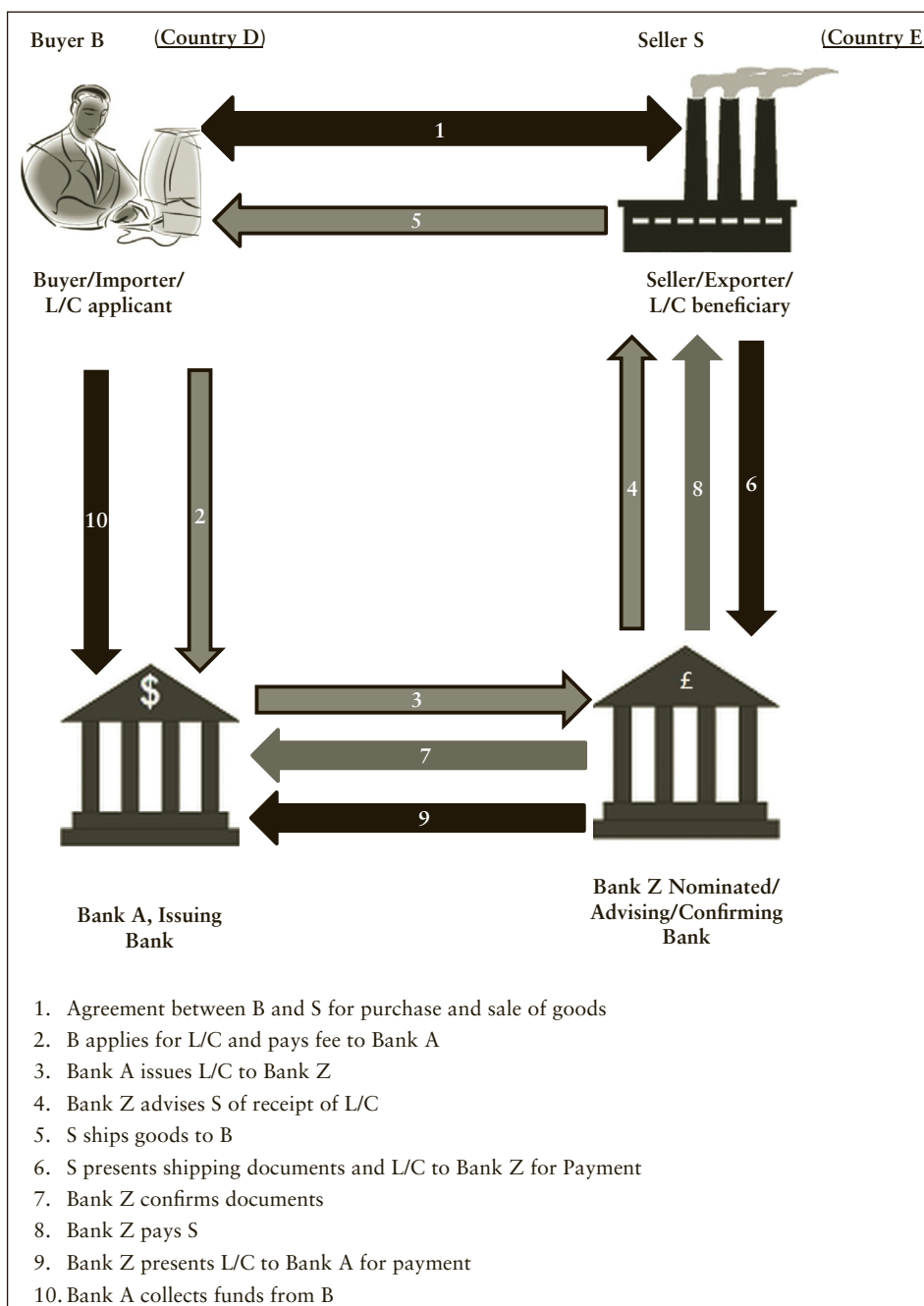
L/Cs may be subject to immediate payment upon presentation of proper documentation (sight drafts) or payable at a later time (time drafts or date drafts), for example, 30 days from acceptance. L/Cs are widely used in international trade because they afford protection to both buyers and sellers. Sellers are protected against shipping costly goods to a deadbeat buyer in a country where legal redress could be prohibitively expensive. Buyers similarly have protection against remitting funds and receiving nothing in return. While the terminology used in referring to the parties to and benefiting from the L/C can be somewhat arcane, the transaction itself is normally quite simple. In effect, the bank in issuing a documentary L/C allows the buyer or importer to substitute the bank's creditworthiness for that of the buyer, thereby giving comfort to the overseas seller or exporter that it will receive payment on the delivery of conforming goods. An L/C therefore functions as a contingent payment guarantee. The transaction associated with the issue of a confirmed L/C is illustrated graphically in Exhibit 3.13.

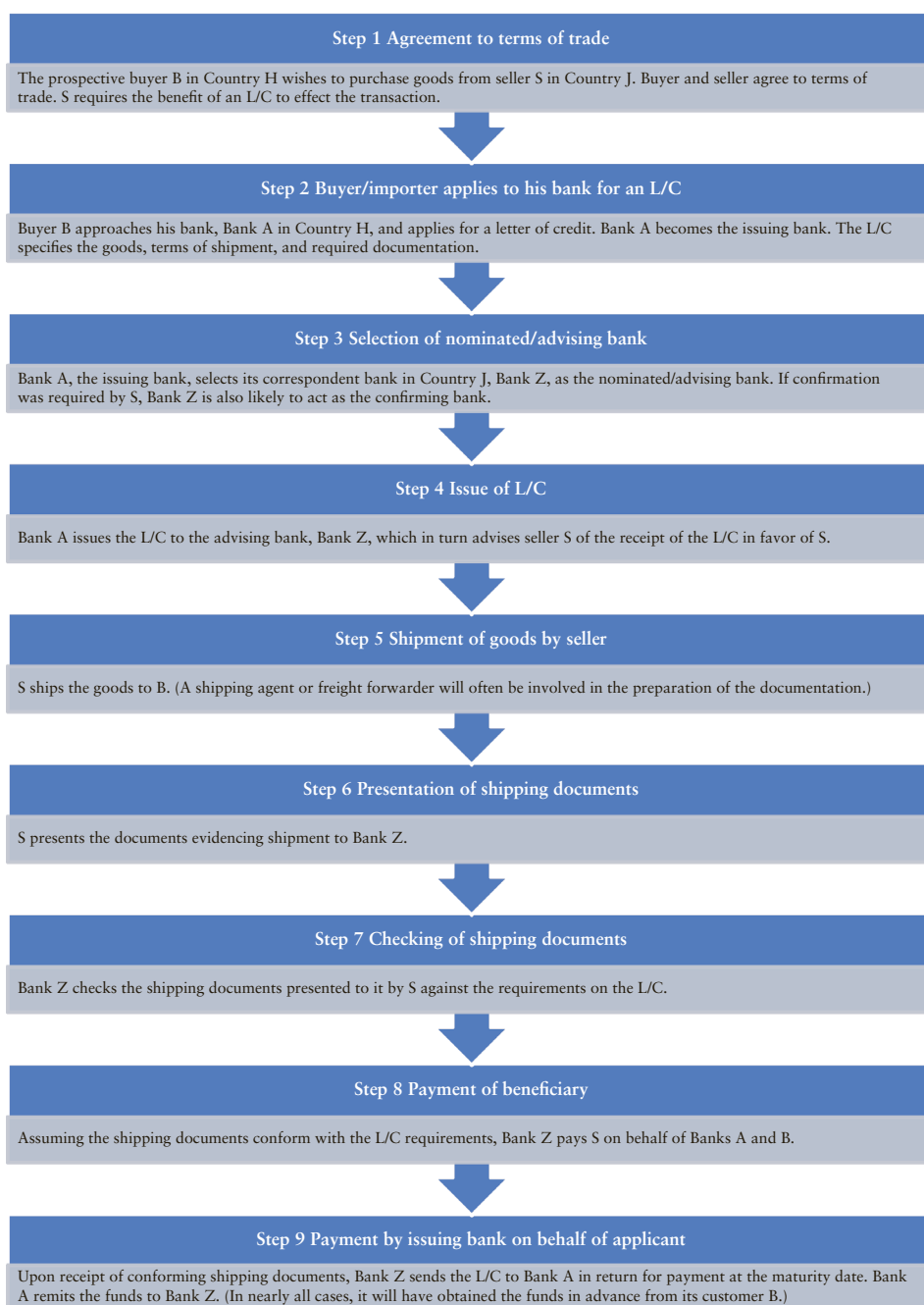
While there may be variations in procedures, the steps, enumerated in Exhibit 3.14, generally are as follows:

- Step 1: Buyer (importer) and seller (exporter) agree to terms, which require an L/C.
- Step 2: Buyer (importer) applies to his bank for an L/C.
- Steps 3 and 4: Buyer's bank nominates its correspondent bank acceptable to the seller (exporter) in his or her country to confirm the transaction. Buyer's bank issues an L/C for which buyer pays a fee. The correspondent bank advises seller on the receipt of the L/C.
- Steps 5 and 6: The seller ships the goods to the buyer and presents shipping documents to the correspondent bank.
- Steps 7, 8, and 9. Correspondent bank pays seller, collects funds from buyer's bank, which in turn collects the funds from buyer (step 10).

Since an L/C essentially functions as a guarantee of payment by the buyer of the underlying goods—the issuing bank advances principal and pays any interest on behalf of the applicant—such costs are passed through to him or her. The fees that the bank earns are analogous to insurance premiums. Unsurprisingly, the fees will typically be calculated as a percentage of the total amount at risk. Regulations promulgated by the International Chamber of Commerce (ICC) govern L/Cs.⁵³ While the documentary L/C described is perhaps the most common form of L/C and that used most frequently in trade transactions, there are other instruments that are referred to by the term *letter of credit*.

As Exhibit 3.12 shows, variant types of L/Cs fulfill different customer needs. Most of these are used for international trade purposes; mention should be made, however, of the standby L/C described below, which functions as a performance guarantee and is not generally used in trade transactions.

**EXHIBIT 3.13** How an L/C Works

**EXHIBIT 3.14** Confirmed Irrevocable L/C Transaction

STANDBY L/C

A standby L/C is a type of performance guarantee. Under a standby L/C, the issuer undertakes to pay to the beneficiary an agreed sum in the event the applicant defaults on a specified obligation. As a rule, the beneficiary must present a statement, subject to the terms of the L/C, that a material event of default has occurred. Although it incorporates the phrase “letter of credit,” it is quite different from a documentary L/C. It is not a trade finance product but may be employed in a wide variety of situations. Standby L/Cs have been compared to bank demand guarantees and may be used in situations where a bid bond, performance bond, or other form of payment guarantee would otherwise be used.

Foreign Exchange Dealing and an Introduction to Derivatives

Banks have long played an important role in facilitating foreign exchange (forex) transactions. When a bank’s customers have business dealings with parties in countries other than their own, as discussed in the preceding subsection, the resulting commercial transactions may need to be consummated in a currency other than the customers’ local medium of exchange.

Foreign Exchange Transactions

The business of providing foreign currency to a bank’s customers is generally handled by the forex desk of the institution’s treasury.

For quite a long time now, the major volumes transacted in forex have ceased being generated solely—or almost solely—by commercial transactions. The banks still enjoy good spreads on forex transactions (see Exhibit 3.15) brought to them by captive customers, but both the wish to make additional profits in proprietary trading and the need to manage assets and liabilities under modern risk management techniques now provide the banks with other sources of potential profits or losses.

Forex transactions are of two basic types: spot transactions and hedged transactions. In a spot forex transaction, one currency is immediately sold for another.

ILLUSTRATION OF A SPOT TRANSACTION

Suppose Sussex Cycles, a hypothetical manufacturer of bicycles based in the UK, exports its products to a distributor in the United States and receives export proceeds of \$1 million for a shipment of 2,500 bikes. It is likely that Sussex Cycles will wish to convert the funds received into the local currency, British pounds sterling. To do so, it may enter into a spot contract with its bank to sell the \$1 million in exchange for pounds at the prevailing exchange rate, for example, \$1.60 per pound sterling or £625,000.

EXHIBIT 3.15 Basic Forex Products

Product Type	Definition	Typical Terms and Usage	Risks
FX-spot contract	A foreign exchange (FX) spot contract is a binding agreement executed between two counterparties to exchange a specified amount of one currency for a specified amount of another (at a <i>countervalue</i> based on an agreed rate of exchange). <i>Spot</i> means “for essentially immediate delivery”; in practice the transaction conventionally must be executed within two working days ($T + 2$).	The purpose of an FX-spot contract is usually one of the following: To obtain foreign currency (i.e., foreign exchange) to facilitate a transaction, directly or indirectly on behalf of a customer or To facilitate a trade on the bank’s own account. As part a bank’s foreign exchange dealing activities or other proprietary trading activities.	Excluding market risks arising out of a bank’s trading on its own account, the risks associated with an FX-spot transaction are generally confined to settlement risk, the risk exposure being equal to the nominal amount of the contract.
FX-forward contract	An FX-forward contract is a binding agreement executed between two counterparties to exchange a specified amount of one currency at a specified foreign exchange rate for the corresponding <i>countervalue</i> at a specified future date. <i>Future</i> in this context means any date more than three working days from the date of the transaction.	An FX-forward contract is a type of derivative often used to hedge foreign exchange risk. Buyers of FX-forward contracts include companies doing business internationally, as well as banks and other financial institutions, while sellers of FX-forward contracts are mainly financial institutions. The vast majority of FX-contracts mature within 2 years, although FX-forward contracts of up to 10 years or more are used occasionally.	FX-forward contracts are subject to market risk, which is calculated through the use of statistical models. Also, FX-swap contracts are subjected to default risk, where the parties owing greater amounts can default.
FX-swap contract	An FX-swap contract is a binding agreement to make a currency exchange between two foreign parties. They exchange currencies for a certain length of time and agree to reverse the	An FX-swap contract also is a type of derivative often used to hedge foreign exchange risk. End users of FX-forward contracts are usually companies doing	Other than market risk, which is calculated through the use of statistical models, FX-swap contracts are also subjected to default risk, since the swap market is

Product Type	Definition	Typical Terms and Usage	Risks
	transaction at a later date. An FX-swap contract is just a long position in a forward contract on a foreign currency and a short position in a forward contract on the same currency with a different expiration.	business internationally, banks, and other financial institutions. The swap market is made up of dealers, which are banks or other financial institutions. These dealers make market in swaps, quoting bid and ask prices and rates, thereby offering to either side of a swap transaction.	almost exclusively an over-the-counter market. Default is possible whenever a payment is due.
FX-future contract	An FX-future is a also a binding agreement to exchange one currency for another at a specified date in the future at a price (exchange rate) that is fixed on the purchase date, but it is traded in future exchange and guaranteed by a clearinghouse so that it is free from default risk.	A type of derivative like the FX forward, the FX option contract is often used to hedge foreign exchange risk. It also is utilized to support various trading strategies.	FX-future contracts are subject to market risk, which is calculated through the use of statistical models.
FX options	An FX option grants the option holder the right but does not oblige him to buy or sell a specified sum of foreign currency at a specified price during the option period.	A type of derivative like the FX forward, the FX option contract is often used to hedge foreign exchange risk. It also is utilized to support various trading strategies.	FX options are subject to market risk. The Black-Scholes formula was first created to enable the accurate pricing of options.

In a hedged forex transaction, which can be effected through forwards, options, and other derivative instruments, an agreement is undertaken to convert one currency for another at a predetermined rate.

Introduction to Derivatives

Derivatives—short for derivative instruments, derivative assets, or derivative securities—can be defined as financial instruments that derive from an underlying asset or pool of assets, the value of the instrument being subject to changes in the

value of the underlying assets or changes in interest rates, exchange rates, and other similar publicly available data, according to some predetermined calculation. Often derivative contracts are bilateral contracts between two counterparties, one of which will almost always be a financial institution. Functionally, derivatives act to alter either the timing or currency of cash flows, or both.

Derivatives Defined

Derivatives are financial contracts that create instruments or securities, the value of which is derived from underlying assets or pools of assets. Typically, the value of the derivative will be affected, often disproportionately, by marginal changes in interest rates, exchange rates, prices of securities, financial or commodity indices, or other publicly available data. Common types of derivative instruments include:

- Futures
- Forwards
- Options
- Swaps

More complex derivatives can be created by using these basic types of derivative structures as building blocks.⁵⁴ Credit derivatives have come to play an increasing role in creating what are called *synthetic securitizations*. While derivatives are probably best used for hedging purposes, that is, laying off (reducing) risk, they can also be traded or used as an element of trading strategies. They may also be used to structure various types of synthetic securities, such as collateralized bond obligations, to create any combination of risk and reward characteristics. Derivatives can be highly complex, posing pitfalls for the unwary, and comparatively unsophisticated institutions have allegedly been victimized in particular transactions. The high leverage that derivatives afford, the high notional amounts (that do not take account of netting, that is, where offsetting positions are not taken into account), and the potential for systemic risk resulting from their abuse have given rise to concerns, which have been aggravated by headline cases in which companies or other entities have suffered great losses (e.g., Orange County, Lehman Brothers' sponsored retail bonds, Dexia's structured loans). Such cases have given derivatives a perhaps not quite deserved notoriety. A bank that engages in significant derivatives activity should have suitable risk control mechanisms in place and be able to explain them coherently to the analyst.

As the foregoing definition suggests, derivatives are not limited to the forex context. But since this category of financial instruments evolved with commodities trading and forex dealing, it is appropriate to introduce the concept in this section, and to provide some background on derivative products. Derivatives can be classified in several ways, including:

- Traditional (e.g., forex forwards, commodity forwards) vs. modern (interest rate swaps)
- Exchange traded (i.e., listed and traded on a market exchange) vs. over-the-counter (customized instruments with varying degrees of marketability)
- The character of the underlying assets, including interest rate-based, currency-based, commodity-based, index-based, and specialized (e.g., weather derivatives)

- Plain vanilla (e.g., ordinary interest rate swaps, forex swaps) vs. complex (e.g., structured investment vehicles)

Note that in the realm of forex, forwards, options, and swap transactions are the most commonly used mechanisms.

Although derivatives originated hundreds of years ago, they have grown tremendously in popularity for a number of reasons. Notable factors in the derivatives boom include the increased sophistication and deregulation of financial systems and their underlying legal frameworks, the increased volatility in markets worldwide, the globalization of the financial markets, and the benefits that derivatives can provide in reducing risk, particularly market risk, through hedging strategies and by providing an indicator of market perception of future risk. In particular, the standardization of derivative contract terms under uniform contracts promoted by trade organizations such as the International Swaps and Derivatives Association (ISDA), and their consequent sale through market exchanges have reduced the need to customize transactions and has also diminished legal uncertainty about enforceability, reducing their risk and facilitating their purchase and sale.

Forwards

A forward is a contract between a buyer and a seller to transfer a commodity or financial asset on a fixed delivery date at a fixed price. In contrast to options and futures, neither the buyer nor seller of a forward ordinarily has the right to opt out of the agreed-upon transaction.

FORWARD DEFINED

A forward is a binding exchange-traded contract between a buyer and a seller for the delivery of a specified number of uniform lots of goods at a specified price at a specified date.

Forwards cannot be counted upon to be standardized contracts, nor are they as a rule exchange traded. They can be understood as over-the-counter future contracts. Since they are not traded on an exchange, there is no mandatory margin requirement, but correspondingly, the parties must bear credit or counterparty risk. From an accounting perspective, forwards, like other derivative instruments, are contingent and off-balance-sheet items whose value depends upon whether the fixed price at the time of the delivery date is above or below the market price at that time. Forwards are therefore subject to market risk as well as to default risk. A future is a forward bought or sold through a third-party futures exchange. The exchange reduces risk since the contracts used are standardized, thereby affording liquidity, and the imposition of margin and collateral requirements settled daily by the exchange practically eliminates default risk.

Futures

A future, also called a future contract, is a standardized contract for the delivery of a uniform lot of items, usually a commodity or financial asset, to be delivered by the seller to the buyer at a specified price on a specified date, normally within one year. While, like options, futures contracts are considered to be a derivative instrument, unlike with options, both parties are obliged to consummate the transaction. However, most future contracts do not result in the physical delivery of the financial asset that was the subject of the contract, but instead are closed out through the financial exchange in which they are traded.

FUTURE CONTRACT

A future contract is similar to a forward contract except that it is standardized and therefore tradable through an exchange.

The traditional purpose of futures is to enable the parties to the transaction to hedge risks, such as the risk that prices of a commodity will rise or fall or interest rate risk (in the case of financial futures), but like all derivatives speculators and dealers buy and sell futures in the hope of making profits, often short term. So they are commonly used in commodities markets (commodity futures contracts) to hedge the purchase or sale of commodities such as wheat, corn, and pork bellies, while financial futures (financial futures contracts) serve a similar function with regard to foreign currency, for example. To protect the sellers against default risk, that is, counterparty risk, the prospective purchaser is obliged by an intermediary exchange to make a margin deposit at the time of entering into the contract, and such deposits may be required to be increased if market conditions or the position of the purchaser changes.

FINANCIAL FUTURES DEFINED

Financial futures contracts are used to purchase financial assets such as a specified basket of U.S. Treasury bonds or assets linked to financial indexes such as the Nikkei 225. Currency futures are futures contracts linked to currency exchange rates.

Options

An option is a financial contract that grants, ordinarily in exchange for a premium paid, the purchaser or holder the right (but does not impose an obligation) to buy (a call option) or sell (a put option) an asset at a predefined fixed price, known as the strike price, for delivery in the future as specified in the contract.

CALL OPTION AND PUT OPTION DEFINED

A call option grants the holder of the option the right (but not an obligation) to buy the designated asset at the designated price and at the time designated in the agreement.

A put option grants the holder of the option the right (but not an obligation) to sell the designated asset at the designated price and at the time designated in the agreement.

The rational holder of a call option will exercise if the market price of the asset rises above the price fixed in the contract. Likewise, the rational holder of a put option will exercise it if the market price falls below the price fixed in the contract.

Sometimes, conventional securities will include embedded options. For example, a corporate bond may include a call provision that allows the issuer to compel redemption of the issue under predefined circumstances.⁵⁵ Likewise, companies may sometimes issue convertible bonds, which under certain conditions may be exchanged for equity shares in the issuer.

Swaps

A swap is a type of derivative transaction in which two counterparties agree to swap cash flows generated by designated assets to obtain certain perceived benefits. Swaps of this nature are almost invariably in the form of either an interest rate swap or a currency swap or both. A generic term for such swaps is *asset swaps*.

Interest Rate Swaps

An interest rate swap is often motivated by one party's desire to protect itself against the risk of higher interest rates. For example, an industrial company may only be able to borrow funds from a bank at a floating rate of interest, but would prefer the certainty of a fixed rate of interest. An interest rate swap allows such a company to swap its floating-rate obligation for a fixed-rate obligation. A swap may be arranged—this is often done by a bank as an intermediary in exchange for a fee—with another enterprise that can obtain fixed-rate financing cost-effectively and in exchange for a premium paid or received by the counterparty may be willing to carry the risk—financially positive or negative—of making the fixed-rate interest payments.

The risks that each party will bear will depend upon the movement of interest rates. The firm financing on a fixed-rate basis will be better off if prevailing rates increase, and would be worse off if prevailing rates declined, since its floating-rate financing costs would decrease in a falling interest rate environment. However, it may prefer the stability of fixed rates to any potential windfall benefit from falling interest rates. The counterparty accepting a floating-rate obligation for a fixed-rate one may believe that interest rates will fall, and prefer to take the gamble that they might move in an unexpected direction.

INTEREST RATE SWAP DEFINED

An interest rate swap is a transaction that obliges two or more counterparties to exchange interest rate obligations. An ordinary interest rate swap involves obligations in the same currency. No principal payments are ordinarily exchanged, however, either initially or at maturity.

Example: UVW, Inc., a large creditworthy U.S. company, issues a five-year \$10 million bond paying a fixed rate. XYZ Ltd., a second-tier British company issues a five-year \$10 million bond paying a floating rate. UVW's CFO prefers floating-rate funding, fearing a fall in interest rates. XYZ Ltd. desires the stability of fixed-rate funding. Through a bank, which acts as an intermediary and collects a fee for its services, they swap their obligations to pay interest, making adjustments for the rates in a mutually agreeable fashion through the bank.

Currency Swaps

If the currencies are not the same, the swap is called a currency swap, or sometimes an interest-rate currency swap.

CURRENCY SWAP DEFINED

A *currency swap* is a type of swap transaction between two or more counterparties for the purpose of exchanging interest payments under a debt obligation or other payments in different currencies, coupled with the transfer of principal payments at maturity (and occasionally upon entering the swap transaction).

Variant Swaps

While the preceding types of transactions represent the most common types of interest rate and currency swaps, swaps may also be structured in particular ways to meet specific needs. Special types of swaps include the following:

- A cancelable interest rate swap is an interest rate swap that is cancelable at one or more times during the life of the deal without penalty.
- A constant maturity swap is an interest rate swap most often used as a building block to construct investment products. This type of swap is one in which either one payment stream is indexed to a swap rate or where the two payment streams are floating and one is indexed to a long-dated swap rate.

Not all swaps refer to the exchange of cash flows. So-called total-return swaps and other types of swaps represent forms of more complex derivatives including credit derivatives discussed in the following section.

Credit Derivatives Generally

Instead of transferring and thereby hedging a counterparty against market risk—that is, the adverse movement of interest rates, forex rates, asset prices, and so on—credit derivatives can be used to provide a form of default insurance in the same way as does a guarantee or a political risk insurance policy provided by an export credit agency.

Types of credit derivatives include:

- Credit default swaps (CDSs)
- Total-return swaps (TRSs)
- Credit-linked notes

As a rule, credit derivatives normally are over-the-counter derivatives, rather than exchange-traded instruments. Nevertheless, they tend to be standardized to a degree through the incorporation of the 2003 ISDA Credit Derivatives default definitions (and ulterior supplements) into the contract.

CREDIT DERIVATIVE DEFINED

A class of derivative in which credit risk is transferred from one party to another, rather than market risk as is the case with most other derivatives.

CDSs

Under a CDS, two counterparties agree that one party will pay the other periodically a fixed premium payment, often specified as a percentage of a reference asset (e.g., 30 basis points) until the expiration of the agreement, while the other party need make no payment whatsoever unless a credit default, as defined in the agreement, occurs. Upon this default, the party liable for making the payment transfers payment to the other party, and the agreement terminates. Settlement can be in the form of a cash transfer or the transfer of a specified asset.

CDS DEFINED

A CDS is a contract between two parties in which one counterparty agrees to bear the risk of default (a “credit event”) with respect to a specified reference asset, and the other party pays the first counterparty a premium in exchange for the default protection. If a default occurs as defined in the contract, the second counterparty collects a specified payment from the first. A CDS is a contract where neither of the parties has to hold a stake in the underlying asset. Under the 2003 ISDA standardized contract, default is defined with respect to the underlying asset as the:

- Bankruptcy of the issuer or counterparty
- Failure to pay by the issuer or counterparty
- Restructuring event as defined in the contract

TRS

The TRS may be classified as a form of credit derivative because it is designed to protect one counterparty from loss of value of the underlying asset, whether resulting from a credit event or from changes in market conditions. The total return received by this party is defined as a predetermined fixed or floating income stream. The name derives from the fact that one party agrees to pay the other the total return of the reference assets (from inception of the swap to its maturity) to the counterparty, where the return is defined to be the sum of net price movements, interest and dividend payments (received or accrued), together with any dividend payments received or accrued, while the second party agrees to pay the first party specified fixed or floating payments based on the same notional amount.

TRSs are typically intended to be financing transactions, so they have a similar credit profile to a reverse repo, with the key difference being that there is no legal obligation for either the party or the counterparty to transfer title through the sale or repurchase of the assets; instead, the assets that generate the return, the reference assets, remain with the original holder. This structure enables one counterparty (the investor or total-return receiver) to obtain the benefits of holding as security, that is, cash flow from dividends or interest payments and any appreciation in the market value of the security without actually taking title to the security, while the other counterparty (the holder of the reference asset or total return payor) is insured against any decline in the market value of the reference asset.

In other words, during the life of the transaction, the total-return payor pays to the total-return receiver the total interest or dividend payments generated by the reference asset, together with, at the conclusion of the transaction, any appreciation in the market value of the reference asset. The countervailing obligation of the investor or total-return receiver is to pay a predetermined fixed or floating rate of interest to the total-return payor and to compensate the total-return payor for any decline in the market value of the reference asset from its initial market value. Economically, the investor or total-return receiver is in effect obtaining financing from the total-return payor. The total-return payor obtains a predetermined interest rate plus compensation for any decline in the market value of the reference asset, while the investor or total-return receiver receives all the income from the reference asset and any price appreciation in existence at the transaction's conclusion. TRSs are often used by hedge funds as a financing tool to obtain the benefits of taking a position in a security, without taking the security on its balance sheet, while banks and other holders of securities use TRSs to obtain additional fee revenue.

Investment Banking and Securities Brokerage

Investment banking is a type of banking business that encompasses the provision, for a fee, of advisory and financing services to large enterprises, particularly the underwriting of securities issuance in the money and capital markets and related services. Historically, in many markets, investment banking was undertaken by specialist institutions, while in those markets in which the universal banking model prevailed, large commercial banks also provided investment banking services. These include:

- Advising on financing issues generally, particularly of large enterprises
- Advising on mergers, acquisitions, and other forms of business combinations, and providing funding to facilitate such activities
- Underwriting and distribution of securities (including brokering and making markets in securities)
- Proprietary trading and managing the assets of institutions and high-net-worth individuals

Investment Banks and Forex Products

More complex and sophisticated forex products are outside the scope of this book, but the names of a few are listed in Exhibit 3.16 for illustrative purposes. While commercial banks usually earn the bulk of their income from interest-generating activities, investment banks generally obtain the bulk of their revenue from fees for the services listed and from trading gains. Another difference, at least in their historical functions, is that commercial banks tend to concentrate on arranging short-term working capital finance, while investment banks focus on long-term finance through the issue of stocks and bonds. Over the past several decades, the boundary between investment-banking and commercial-banking activities has become less distinct, and commercial banks increasingly engage in activities once reserved to specialist institutions. There is some movement in the other direction as well, as investment banks have merged with other entities to broaden the scope of the services they provide.

EXHIBIT 3.16 Selected Complex Hedge-Type Forex Products

Product Name	Intended Use—Hedge Against
Call-Premium Stepper	Rise in value of the underlying currency without requirement of initial premium payment
Put-Premium Stepper	Fall in value of the underlying currency without requirement of initial premium payment
Knock-Out Call Option	Rise in value of the underlying currency option with lower premium than ordinary call (but option terminates if spot price reaches “knock-out level”)
Knock-Out Put Option	Fall in value of the underlying currency option with lower premium than ordinary put (but option terminates if spot price reaches “knock-out level”)
Range Forward (Cylinder or Risk Reversal)	Defined unfavorable spot rate movements
Participating Forward	Designated unfavorable spot rate movements but with opportunity to benefit from favorable spot rate movements (variants: American Conditional Forward, European Conditional Forward, American Forward Adjuster, European Forward Adjuster)
Bonus Forward (Variant: Chaos Forward)	Rise or fall in interest rates under more-advantageous terms within a designated range and less-advantageous terms outside that range
Nondeliverable Forward	Risk in a nonconvertible currency with settlement in U.S. Dollars

Source: Rabobank.

Types of Investment Banks

The investment-banking industry can be divided between the major global players, such as Morgan Stanley and Goldman Sachs, and specialized and regional players focused on investment-banking activities. A distinction can also be drawn between the former, which engage almost exclusively in investment-banking activities, and integrated financial groups such as Deutsche Bank, UBS, and Citigroup, which encompass both large commercial-banking and investment-banking operations. Aside from the major global players, smaller investment banks operate locally in many countries, handling local-currency transactions or deals involving smaller firms. Similarly, commercial banks with investment-banking arms may act in a similar capacity. Some investment banks specialize in particular industries. For instance, Keefe Bruyette and Woods developed as a boutique investment bank targeting mid-sized U.S. commercial banks.

INVESTMENT BANKING AND MERCHANT BANKING

An investment bank is a financial intermediary that provides a variety of financing and advisory services, especially to organizations that wish to raise finance in the capital and money markets. The characteristics of investment banks that most distinguish them from commercial banks are that they rely much less on deposit funding and interest generation from earning assets. Instead, they derive the bulk of their revenue from fee-generating activity, most notably the underwriting of debt and equity securities, and trading gains. The principal activities of investment banks include the following:

- Underwriting of securities issues
- Making markets in securities (dealing)
- Securities brokerage
- Proprietary trading⁵⁶
- Asset management
- Mergers and acquisition advice and financing
- Private equity and venture capital investment

Investment banks were traditionally structured as partnerships, and depended to a much greater extent on placing their own capital at risk, rather than engaging in highly leveraged and comparatively low-margin lending operations such as retail deposits. Today, however, most major investment banks are publicly traded corporations, and may be part of a larger financial group, which also provides commercial-banking functions. The boundaries between different types of financial institutions have become increasingly blurred with deregulation. A bank that engages in both investment banking and various types of commercial banking, as well as providing other financial services, is sometimes known as a universal bank.

Investment banks have become increasingly important as capital markets have developed worldwide, and as traditional and legal barriers between

commercial and investment banking have broken down. The traditional British term for an investment bank is a *merchant bank*. Merchant banks also engaged in what is today termed *private equity*, that is, the strategic investment in companies, a function that investment banks also may engage in to some degree, depending upon the jurisdiction. In contemporary usage, merchant banking refers to “private equity investment by financial institutions in the unregistered securities of either privately or publicly held companies.” Such investments typically consist either of common stock or of debt instruments that are convertible into equity. In the United States, the bank subsidiaries are major investors in the private equity market.

Underwriting

Perhaps the most important function of investment banks is what they may be best known for—underwriting the issue of debt and equity securities in the capital markets. Investment banks are also active in the money markets particularly on behalf of corporates. Investment banks work not only with industrial and service corporations, but also with state and multinational entities, as well as other financial institutions to arrange financing. The origination of such securities takes place in the primary markets, while the subsequent purchase and sale of “second-hand” securities takes place in the secondary market.

A strong secondary market is a prerequisite to a strong primary market. Unless investors have some assurance that they will be able to sell any securities purchased directly from the issuer, they will be loath to put their capital at risk for the long periods implicit in all equity and many debt issues. To a much greater extent than the purchaser of an automobile, for example, the purchaser of a corporate bond, even if his or her intention is to hold the security until maturity, almost invariably buys with a view to the instrument’s marketability. As a consequence, investment banks have sought to develop a strong presence in the secondary market where they act as market makers and brokers, ensuring a liquid market for the stocks and bonds they originate on behalf of their customers. The brokering function is discussed in the next section.

UNDERWRITING

The underwriting function includes the following elements:

- Origination—obtaining the underwriting assignment, designing the structure and terms of the securities issue—and winning the mandate.
- Structuring and pricing of the securities issue and overcoming the commercial and regulatory hurdles to bring it to market.
- Distribution of the securities (the marketing and sale of the securities to investors, ordinarily with the assistance of other intermediaries).

Types of Underwriting

There are several common approaches to underwriting, which differ in the amount of risk the underwriter is willing to incur to place the securities to be issued. There are also variations in required or customary practice from country to country and market to market, depending upon the nature of the issue and which laws and regulations apply. Exploration of these matters in depth is beyond the scope of this book. Some of the key concepts, however, are set below, along with the basic underwriting process and some key terminology.

The Underwriting Process

The process of underwriting securities to be issued in the money and capital markets is not dissimilar to the process of arranging a syndicated loan, as discussed. A key difference is that the number of securities to be distributed is generally much larger than the subscription shares in respect of a syndicated loan, and regulatory requirements are often much more onerous.⁵⁷

Phases of Underwriting

From the investment bank's vantage point, the underwriting process can be divided into three basic phases:

1. Premandate
2. Postmandate
3. Postissue

The first is the premandate phase, which involves identification of the opportunity and obtaining the mandate to act on the issuer's behalf. The process may be initiated by the investment bank or by the prospective issuer, and the competitiveness of the process will vary considerably. Investment banks also vie fiercely for key deals because they will affect their place on the "league tables," which denote those institutions that have managed the greatest volume of issues. For large government-related and corporate issues, the issuer may hold a beauty contest in an effort to obtain the most favorable terms, irrespective of whether a formal competitive bidding process is employed. This stage concludes with preliminary formation of the underwriting group (syndicate), negotiation of terms, and a final underwriting agreement.

The second stage is the postmandate stage, which begins with the decision by the issuer to employ a particular investment bank and ends with the closing of the transaction, the issue of the securities, and ultimately with the remittance of the issue proceeds to the issuer. This stage typically includes the following activities:

- Discussions with the issuer to tailor the transaction to meet the customer's needs.
- Formation of the underwriting group (syndicate).
- A due diligence investigation undertaken by the lead manager of the issuer.
- Hiring legal professionals to formalize the commercial terms agreed and to ensure compliance with regulatory requirements.
- Pricing of the issue and completion of arrangements to distribute the securities.
- Ratings advisory services to obtain the optimal credit rating.
- Issue of preliminary prospectus.

- Issue of final offering documents.
- Obtaining requisite regulatory permissions.
- Stabilization of the market, in which the underwriter seeks to effect a smooth distribution of the securities at a favorable price.

UNDERWRITING CONCEPTS

The underwriting process is most expansive in the case of an initial public offering of equity securities or in the case of a new debt issue. It is ordinarily more limited where equity shares have already been issued to the public and a “rights issue” of additional shares is contemplated, or where a debt issuance program has been established.⁵⁸ There are two basic types of underwriting.

1. *Negotiated underwriting.* A negotiated underwriting is one in which the terms of the issuance of the securities are negotiated between the investment bank and its customer, the issuer, rather than being let out to bid.
2. *Competitive underwriting.* A competitive underwriting is one in which the issuer selects an investment bank as its underwriter, and the terms of the issuance of the securities issue are negotiated between it and the investment bank.

Underwriting may be further classified by the structure of the underwriting agreement and the degree of risk that the underwriter incurs. The more common types are as follows:

Firm-commitment underwriting. The underwriter promises to purchase the entire issue of securities from the issuer and is responsible for distributing (i.e., selling) the shares to the public.

Best-efforts underwriting. The underwriter promises to purchase from the issuer only as much of the issue as it can sell.

Standby underwriting. This type of underwriting is commonly used in connection with rights issues where the issuer is uncertain whether it will be able to distribute all the shares to existing shareholders. Consequently, for a fee, the underwriter agrees to stand by to provide backup distribution capacity.

The final stage is the postissuance stage. During this phase, the investment bank will typically make a market in the securities and attempt to continue to bolster the image of the issuer and increase its reputation.

Securities Brokerage

In some markets, securities brokers were at one time largely separate from commercial banks and investment banks. There has, however, been a convergence

between the securities brokerage and investment banking industries and with a few exceptions, primarily in the retail brokerage arena, securities firms have, to a great extent, been absorbed within investment banks.

As a discrete business function, securities brokers generate revenue primarily from commissions on customer trades as well as the bid-offer spread and associated market-making activities. They also earn income from ancillary activities including market making, as well as through asset management activities, commissions generated on marketing investment funds, and margin finance. A more detailed discussion of the securities brokerage industry is beyond the scope of this book.

In this section, other fee-generating businesses that banks engage in are enumerated and explained briefly, primarily for reference purposes. A number of these—securities custody and settlement, credit card processing, trade-related services, payments and receivables processing, and cash management—can be categorized as transaction-processing services.

Securities Custody and Clearing

Securities custody and clearing services are administrative services in respect to securities that are provided by a bank to issuers and investors.

When provided to retail investors, securities custody and clearing services encompass at a basic level:

- Receiving, delivering, or holding securities for safekeeping on behalf of investors.
- Administering bond interest coupons and stock dividends.
- Placing unused funds in interest-earning accounts.
- Settlement of security purchases and sales.
- Processing of corporate actions in connection with securities holdings.
- Processing tax payments related to securities holding.
- Providing record-keeping services and reporting to the investor concerning his or her securities holdings.

For institutional investors, securities custody and clearing services may also include:

- The domestic and global settlement of securities transactions
- Administering securities on a multicurrency basis
- Tax reclamation
- Processing the redemption of securities
- Proxy voting
- Providing performance analytics

When provided to issuers, such services may embrace acting as a stock transfer agent for equity securities. The transfer agent administers shareholder records and corporate actions, handles notification of shareholder meetings, and provides related ancillary services on behalf of the issuer to shareholders. The corresponding role for fixed-income securities is to function as a corporate trustee and paying agent for debt issues. Other services provided to issuers include the processing of depository receipts, such as American depository receipts (ADRs), which are U.S. securities (which trade in the United States in U.S. dollars) but represent shares in a non-U.S. company.

Banks that provide securities custody and clearing services generally earn a fee or commission for providing them, although sometimes they are provided as part of a package to retail investors that includes securities-brokerage services, asset management, or trust-banking services. Investment funds, which function both as investors, on behalf of their own investors, and as issuers with regard to fund vehicles, are major users of institutional securities custody and clearing services. This is regarded as a low-risk business that is comparatively profitable, particularly among the relatively small number of institutions that have carved out a niche serving institutional investors and issuers. A relatively small number of major banks dominate the market for global custody and clearing, while in the United States, a fairly small number of players dominate the institutional business.

Asset Management and Trust Banking

Asset management is the business of investing other people's money, typically for a fee, which is usually based on a percentage of the total assets managed. Banks offer asset management services under a variety of umbrellas. Traditionally, banks provided such services under the rubric of trust-banking services to wealthy individuals and organizations as a part of helping these customers with estate planning. Over the past several decades, however, various developments have altered the asset management landscape, making it both more variegated and more competitive.

First, the fund management industry, which provides largely equivalent services through retail mutual fund products and similar vehicles, has grown enormously throughout the world. Banks may either create their own investment products to sell to customers or alternatively market the funds of independent companies. In many markets, banks compete with fund management firms or other nonbank financial institutions in the provision of these services. Consequently, asset management services are now available to a broad range of customers in many markets in the form of mutual funds or similar vehicles.⁵⁹

Second, the management of institutional pension funds, particularly as a result of the growth of private pension schemes over the past several decades—or other funds such as endowments, nonprofit foundations, and similar institutions—has become a prominent franchise within the overarching category of asset management. Under most such discretionary asset management schemes, the client and the bank will implement a plan providing that the funds under management will be invested according to defined guidelines.⁶⁰ Competition is also a feature of institutional money management. Over the same period, and for many of the same reasons, companies also have much more choice and data on the management of pension funds. While in the past it would have been common for these organizations to entrust the management of their financial assets to their main bank, nowadays there are many rivals for this business. Specialist investment managers, particularly those that have gained a strong reputation in the market, have attracted those with funds to invest, both individuals and institutions, away from the traditional bank trust department.

Finally, traditional trust banking—now more often than not referred to as private banking—has continued to develop as a franchise, although again banks face increasing competition in attracting this type of business. As a consequence, private banking has become more personalized and service oriented. Sometimes called *wealth management*, this area of banking is marketed to the high-net-worth

individual as being a cut above mere asset management. Not only does the bank offer its private banking clients special privileges such as “free” funds transfers and premium interest rates, but the client is designated a relationship manager to respond to all his or her banking needs, effectively offering an “integrated banking solution.” Like a concierge in an exclusive hotel, the account manager’s role is to ensure the client’s relationship with the service provider is a pleasant one that encourages customer loyalty.

Despite the fierce rivalry for customers, in an environment of disintermediation, asset management is an attractive franchise for banks. It adds almost no assets to the balance sheet, and therefore requires only modest amounts of capital. The 1990s were a period of substantial wealth creation in many countries, and there are a lot more wealthy individuals, families, and organizations in need of asset management services than in the past. In the United States, for instance, the number of households having a net worth of \$1 million or more nearly doubled between 1992 and 2000 from 2.8 million to 5.4 million. During the same period, total U.S. mutual fund assets increased, amid a rising market, from \$1.6 trillion in 1992 to reach a peak of \$7 trillion in 2000. The size of the market was such that banks benefited as well. The volume of assets in separate asset management accounts rose from \$478 billion to \$1.3 trillion over roughly the same period, while personal discretionary trust assets increased from \$528 billion to \$955 billion. The decline of the U.S. and many other stock markets between 2001 and 2003 resulted in considerable wealth destruction, as well as much dampened interest in these investments. Many of the factors that drove the 1990s boom in asset management are still in place in spite of the financial crisis that developed in 2008. However, and as markets recover, opportunities for banks to generate revenue through the provision of asset management services should remain strong. The volatility of the markets drove a good proportion of the affluent, who had flirted with a more eclectic and aggressive approach to investment in the late 1990s, back toward more traditional safe harbors of trust and personal banking.

TYPES OF RETAIL FUND PRODUCTS

Banks offer different types of mutual funds to their many customers. The different types of mutual funds include a pure bond fund, a pure equity fund, and a balanced fund (combination of bonds and equities). The income stream of a bond fund is more predictable and therefore would generally appeal to conservative investors. The income stream of an equity fund is much more volatile, but the rewards are higher on the average, hence, attracting investors with a greater risk appetite. Investors who are in the middle in terms of risk preferences will choose balanced funds.

Cash Management

Cash management refers to an array of services provided by banks that enable growing companies, particularly those operating internationally, to monitor and effect the collection, transfer, and remittance of funds on a local, regional, national,

or global basis. More concretely, these services encompass those offered by banks in relation to:

- The collection of receivables for the accounts of their (corporate) customers.
- Liquidity management of the funds collected.
- The disbursement of funds to the suppliers or other payees.

The aims of cash management services are primarily to:

- Optimize a firm's return on unused funds.
- Move funds across borders rapidly and on a cost-effective basis.
- Manage the disbursement and remittance of funds.
- Provide dynamically updated information concerning balances and the status of transactions.

In effect, cash management represents the packaging of discrete services to enable companies to realize cost efficiencies through volume discounts as well as to out-source their treasury operations to some extent. A package of cash management services might include a combination of the following:

- Check clearing and wire transfer services.
- Automated clearinghouse.
- Electronic funds transfer.
- Remittance-processing services (sometimes called lockbox services).
- Controlled disbursement services that provide daily reports of available surplus funds together with other sophisticated data reporting, account reconciliation, and collection services.
- Sweep accounts or demand deposit accounts that automatically sweep idle cash in a current account into a money-market account and sweep it back when needed.

In addition, cash management services may be provided on a multicurrency and international basis, providing similar services but affording global funds transfer and multicurrency transaction processing. Such services are usually only provided by global banks. Cash management services may be coupled with other transaction-processing services, such as the processing of payables to vendors, as well as with trade finance services.

WHO USES CASH MANAGEMENT SERVICES?

Consider a utility company in an emerging market that has a few or limited offices in the country but has millions of household and business customers. Collection of funds is likely to be a tremendous challenge for this company. The bank can make it easier for the company because of its much wider collection platform. This platform includes its branch collection, network collections, credit card acceptance, automated collection machines, lockbox, pickup services, incoming

transfers, and direct debits. Because of this more effective collection platform and process, collected funds will be higher and will be received faster. The company thus benefits in the following ways:

- The quicker receipt of cash can earn interest income.
- The company can choose to make payments to its suppliers sooner to take advantage of discounts (the typical supplier requires payment in 30 days, but if the buyer pays within 10 days, a discount of, say, 2 percent is given).
- Working capital requirements are reduced.

On the disbursement front, the process can again be made more efficient by the bank especially in the case of repetitive payments that have to be made by the company. Payment options include electronic funds transfer and direct credit. Cash management services provide dynamically updated information concerning balances and the status of transactions.

Insurance and Bancassurance

Depending upon the jurisdiction and whether permitted by regulation, banking groups may own insurance subsidiaries that provide insurance services, or may themselves act as agents for subsidiary or independent insurance companies. *Bancassurance* is a term often applied to the distribution of insurance products by banking organizations. Entry into the insurance business is sometimes seen as a way for banks to use their branch networks and benefit from cross-selling opportunities. The insurance industry is divided into two basic types of business: life insurance (called life assurance in the UK); and property and casualty insurance (general insurance in the UK). The former functions primarily to provide benefits to dependents upon the policyholder's death. However, life insurance is frequently combined with various types of investment vehicles in which the policyholder can benefit from the policy other than through his or her own demise. Property and casualty insurance aims mainly to insure property against loss or damage. Like banks, insurance companies provide their services under clearly defined contracts. Under a life insurance contract, there are usually three parties: the insurer, the policyholder, and the designated beneficiary. Under property and casualty insurance contracts, the policyholder and the insured are generally one and the same.

In addition to the preceding distinction, insurance companies divide their business activities between business lines and personal lines. Personal lines include life, disability, and health insurance products, as well as pensions and annuity products. The analysis of insurance firms is beyond the scope of this book.

NOTES

1. "The Business of Banking," *The Economist*, October 30, 1999.
2. Howard Palmer, *Bank Risk Analysis in Emerging Markets* (London: Euromoney Publications, 1998), 1.

3. Various attributions. Born Leslie Townes Hope near London, England, the entertainer achieved great commercial success in the United States beginning in the 1930s.
4. Nonbank financial intermediaries are often referred to as nonbank financial institutions or NBFIs.
5. See the quote from *The Economist* news magazine that leads this chapter. The process of intermediation (and disintermediation) is discussed in more detail later.
6. This chapter explores the business of banking to provide a foundational understanding for the process of evaluating the credit risks to which exposure to banks engenders.
7. See the quote from *The Economist* news magazine that leads this chapter.
8. Although not a broker in the traditional sense, a bank that makes loans is in a sense a broker of liquid funds, making cash provided by depositors and other creditors available to the borrower. Historically, banks' primary function was to provide short-term, that is, working capital, loans to businesses, primarily on a seasonal basis. Long-term finance was mainly provided by the owner's capital or obtained through the capital markets in the form of outside equity investment or medium- and long-term debt.
9. Indeed, one definition of a bank is an institution that is allowed to accept deposits from the public in return for providing financial services. Among them is the channeling of that money into lending activities. So the terms *deposit-taking institution* and *depository institution* are synonyms for a bank, broadly defined. Of course, there exist institutions denominated as "banks" that fund themselves from sources other than retail deposits. However, the generalization holds insofar as the classic contemporary commercial bank is the model institution that is the subject of this book.
10. Note that, depending upon the context, the term *intermediation* may be used to refer to financial intermediation generally or to bank intermediation particularly.
11. That is, it offers borrowers access to cash in exchange for which they are willing to pay "rent," that is, pay interest on the funds advanced to them as compensation for their use of those funds over a particular period.
12. In contrast to the manner in which commercial banks function as financial intermediaries, investment banks and securities houses perform their role by matching prospective buyers of a company's stocks and bonds (i.e., investors) with the companies that sell the securities (i.e., issuers), while insurance firms channel premiums from insured parties into a pool of assets comprised largely of securities. Still other financial intermediaries include asset management firms that accept investments from investors. In exchange for a management fee, investors' funds are used to purchase securities and other financial assets. Income and gains from the sale of those assets are transferred to the firm's investors when those investments are redeemed. Note that increased intermediation by nonbank financial institutions such as securities firms and insurance firms may result in rising disintermediation vis-à-vis banks.
13. More specifically, investment banks *underwrite* the issuance of securities and distribute the stocks and bonds issued to the purchasers in the primary market, while securities brokers perform the same function in respect of the market for *second-hand* securities, that is, the secondary market. For instance, if a technology firm wishes to issue new stock to raise capital, it approaches an investment bank, which handles the issuance and sells the stock through brokers to various buyers in the market. The brokers then continue to intermediate the purchase and sale of the shares in the secondary market matching sellers of such shares with buyers. Typically, the buyer and the seller each pay a commission to the broker. Where the broker acts as a *market maker*, maintaining a liquid market in certain securities, it will in addition benefit from a bid-offer spread, that is, the difference between the price for which the market maker is willing to buy the shares and the price at which he or she is willing to sell them. Note that the secondary market does not provide new capital to the issuer, but by making the shares liquid it supports the ability of the issuer to consummate the issue in the first place.

14. Again, lending comprises the fundamental part of a bank's intermediary function, in which it takes deposits and makes loans, acting as a go between by matching savers' funds with lenders' needs. Note that these credit instruments are further discussed later in the chapter.
15. Earning assets are assets, such as loans and other items, that generate a stream of interest revenue.
16. Note that while *investment banking* and *asset management* are not of recent origin, in many markets they have been customarily performed by specialist institutions rather than by commercial banks. In other markets where so-called *universal banking* prevails, these services have long been provided by major banking institutions.
17. Managing these securities holdings is done by a bank's treasury division, discussed later in the chapter. As has been noted, loans usually form the largest part, if not the majority, of a bank's financial assets. A bank that makes no loans is functioning more as an investment manager or as a proprietary trader than as what is normally thought of as a bank.
18. In banking parlance, however, the term *loan* frequently is used more narrowly to refer to a specific class of bank credit *products* or *facilities*.
19. Some of these groups may overlap depending upon how the categorization is effected. For instance, many forex products are simultaneously derivative products.
20. The term *bank products* embraces the widest scope of offerings to customers, including products that are not primarily used to extend credit, although the bank may incur some degree of credit risk in offering them.
21. A subcategory of this group, but sometimes divided into a separate category is *agricultural lending*, which comprises loans to farmers to purchase seed, fertilizer, equipment, and labor, to be repaid after the crop is harvested and sold.
22. The terms *balloon payment* and *bullet payment* are synonymous, referring to the return of the principal, together with any outstanding interest, in a lump sum at the loan's maturity.
23. Technically, a distinction can be made between a *nonrevolving* line of credit and a *revolving* line of credit. The former may include a specified repayment schedule and limit the number of times funds may be accessed. The revolving line of credit, however, typically does not impose a specified repayment schedule, but does mandate a minimum periodic payment to at least cover interest owing and to amortize (i.e., pay down) the principal balance. At the consumer level, credit card advances, cash reserve checking accounts, and overdraft accounts are examples of revolving lines of credit.
24. The lending and credit approval process is examined more closely in a subsequent chapter, which deals with the evaluation of asset quality.
25. Preliminary to the process of identifying desirable candidates for loans, the bank engages in market research and an assessment of its competitive advantages and strategic goals to ascertain target sectors and subsectors (step 1). It then implements a marketing strategy to existing and prospective customers (step 2). Upon finding eligible loan applicants seeking funding, a process of negotiation is initiated in respect of loan pricing and the terms of the credit advance (step 3). Upon consummation of a preliminary agreement, subject to credit review, a credit analysis is performed (step 4), although it may have been undertaken on a preliminary basis as part of the application process and as negotiations progressed. A recommendation is then made by the bank's credit officer or credit committee, and any necessary modifications to the terms of the transaction are made (step 5). Final terms are consequently concluded, and the loan agreement is formalized (step 6). Funds are subsequently disbursed (step 7), the loan is monitored until repaid (step 8), and finally the loan is repaid or if default occurs, collection and workout procedures are begun (step 9).
26. For lending to businesses, effective credit analysis is essential. For consumer lending, including credit card lending, a high investment in marketing and credit scoring technology is needed.

27. Sidestepping the need to submit to the scrutiny of the rating agencies contributes to a faster financing process and will tend to reduce out-of-pocket costs. Recent trends, however, under Basel rules, would lead to banks relying increasingly on external opinions for compliance basis, and such opinions carry a cost.
28. In these so-called agency transactions, credit risk was effectively borne by these government-linked institutions, and they were essentially risk free to investors.
29. At a minimum, a two-tranche structure created two classes of instruments, one having a senior priority of payment and other a subordinated priority. The subordinate class held what was sometimes called the *first-loss piece*, since it provided the senior class of investors with a first line of defense against losses arising from defaults in the underlying pool assets. That is, any such defaults would be first allocated to these investors while the senior investors remained unaffected. For this reason, the senior tranche would normally be entitled to a higher credit rating than the subordinate tranche. Various other mechanisms are also typically used to enhance the position of the senior class or classes of bondholders.
30. Conventional debt obligations, even where secured and therefore backed by the issuer's assets, can as a rule be rated no higher than the rating of the issuer itself. The process of securitization, however, functions to separate the assets backing the issue from assets securing the general creditors of the company. This is accomplished by transferring the assets backing the securitized issue into an SPV that is bankruptcy remote from the issuer. In addition, various credit enhancements may be used to reduce the credit risk to which the investors in the securitized instruments are subject. The result is to decouple the creditworthiness of the issuer from that of the securitized issue. Consequently, an issuer may benefit from the higher rating of the securitized issue, enabling it to raise funds more cheaply than if it were to issue an obligation on its own, notwithstanding that the assets securing the securitized obligation might be identical to those that would otherwise remain on the issuer's balance sheet.
31. For a financial institution, the use of securitization is particularly attractive in that it renders the process of lending essentially *self-financing*. That is, if Bank A wishes to make £100 million in loans, and hold those loans on its balance sheet until maturity, it must raise a corresponding amount in funding, avoiding too great a mismatch between the maturities of the loans advanced and the funds borrowed to finance those loans. The unpredictability of interest rates gives rise to a degree of market and liquidity risk. However, if the loans can be securitized, the bank need not bear those risks to the same extent and for the same period since they can be disposed of relatively rapidly and be taken off the institution's balance sheet.
32. In brief, under the first Basel Accord (now largely modified), for example, although commercial loans carried a higher risk weighting (100 percent) than residential loans and property loans, which required only half as much capital for the same volume of exposure, by removing loans from its balance sheet through the process of securitization, banks were almost invariably able to free up capital for other uses, notwithstanding that they might still retain a residual equity interest in the securitization that was subject to a 100 percent risk weighting. Moreover, since U.S. government agency-backed securitized issues attracted a risk weighting of just 20 percent weight, while Ginnie Mae securities required no capital backing whatsoever from a regulatory capital perspective, holding such agency issues for investment purposes could be more attractive than holding similar underlying loan assets.
33. *Prepayment risk* refers to the risk that the obligees under the underlying mortgages are more likely to prepay their mortgages early when interest rates fall to refinance at lower rates and to prepay the same loans later when interest rates rise. Prepayment risk is a form of call risk since if a high proportion of underlying mortgage borrowers prepay, the maturity of the MBS will decrease and may be "called" earlier than expected. Extension

risk is the opposite phenomenon, where in a rising interest rate environment, the term of the MBS will lengthen as underlying borrowers repay their mortgages more slowly than expected.

34. The loan-to-value ratio refers to the maximum amount lent to the underlying borrower divided by the appraised value of the assets securing the loan.
35. Credit cards represent, in effect, a revolving credit facility. Charge cards are akin to trade credit in that outstanding balances must be paid in full within a short period, usually one month, after being incurred. Debit cards provide for the electronic transfer of funds from the account of the payor to that of the payee either instantaneously or within a few days after the transaction.
36. While it is obvious that credit cards are at times misused as a substitute for other credit facilities, their primary attraction remains their convenience.
37. Some card issuers also benefit from charges imposed either directly or indirectly on transactions made in a foreign currency.
38. *Unsecured* means that there is no collateral backing the funds advanced by the bank or credit card company advancing credit to the cardholder. Note that consumers in their capacity as depositors are sometimes offered overdraft facilities, although the intended use of “overdraft protection” is often more as a backup facility rather than as a line of credit. Before the development of credit cards, obtaining credit for purchases was at the discretion of the merchant. So-called layaway plans under which the customer made regular deposits on a reserved purchase were one means by which durable goods were sold to customers unable to foot the entire expense immediately. Obtaining a personal loan for a major purchase was often a cumbersome process. Banks rarely extended unsecured personal loans. Most such credit was advanced by consumer finance companies for purchases of designated durable goods such as automobiles, home furniture, and television sets, and title to these goods was held by the bank as collateral. If the borrower defaulted, the lender could repossess the designated collateral securing the loan. Automobiles are still often financed this way, as, of course, are house purchases through mortgage loans.
39. The process of securitization was discussed in the preceding section. Securitization enables credit card issuers to pool their credit card receivables into ABS. It functions both to finance credit card advances to end users and to eliminate most of the credit risk borne by the card issuer. Many issuers continue to generate fees, however, through processing credit card transactions on behalf of the SPV to which the pooled receivables have been sold.
40. Note that although the terminology used in the industry is somewhat more complex, including designations such as acquiring bank, issuing financial institution merchant bank, Visa issuer, Visa merchant, and so on, to simplify the discussion, we refer to the three main parties to a credit card transaction simply as the card issuer (i.e., the bank), the cardholder (i.e., the consumer), and the merchant (i.e., the seller).
41. Similar types of variations exist with respect to a number of key items including the accrual of fee-based revenue, such as that generated from the imposition of late fees, and the reversal of such revenue when likely to be uncollectible; the amortization of marketing expenses; and, the recognition of losses. See John McDonald and Daniel Nickolich, “Credit Card Accounting Basics,” UBS Warburg, New York, May 25, 2001.
42. As might be expected, the proportion of operating income generated from sources other than interest-earning assets varied widely across the industry. Banks having a trust banking and securities custody focus generated more than 70 percent of their operating revenue from these sources, while more traditional commercial banking institutions generated less than 30 percent of operating revenue in this manner.
43. This may be done for any number of reasons, including eliminating certain risks arising from brokering transactions between two parties so as to profit only from the fee charged in arranging the transaction.

44. The term *money market* generally encompasses the interbank market.
45. Trading on the bank's own account is also referred to as *proprietary trading*.
46. According to the U.S. Office of Thrift Supervision, a regulatory agency responsible for supervising thrift banks (also called savings and loans), the discount window is "a figurative expression referring to the Federal Reserve facility for extending credit directly to eligible depository institutions (banks and thrift institutions with transaction accounts or nonpersonal time deposits)."
47. *Member banks* of the U.S. Federal Reserve system, that is, major commercial banks in the United States, that have funds in excess beyond that which they are required to maintain on reserve with the Federal Reserve Bank ("the Fed") may lend it to another bank overnight. Within the U.S. banking system, such funds are called *federal funds*.
48. Most CDs are fixed rate, but variable-rate CDs do exist.
49. Note that negotiable CDs differ from the small-denomination CDs often held by individuals both in terms of the principal amount and their "negotiability."
50. Eurodollars are U.S. dollars held overseas, usually in the form of bank deposit receipts, and they are not directly subject to the control of U.S. bank regulators.
51. In some countries, commercial paper may refer to longer-dated debt issues of one year or more.
52. Income from forex dealing is often the result of banks' involvement in trade finance transactions.
53. Documentary credits are subject to "Uniform Customs and Practice for Documentary Credit" (UCP), International Chamber of Commerce Publication No. 600, 2006 Revision.
54. Depending upon how they have been structured, securitized instruments—such as ABS—can be categorized as derivative instruments since the value of the security is dependent upon the value of underlying assets.
55. A call provision in the agreement between the issuer and bondholders gives the issuer the option to call the bond, that is, require early redemption. The converse of a callable bond is a puttable bond, that is, one that gives the bondholders the option under certain conditions to require the issuer to redeem the obligation before maturity. This is favorable to the bondholder and detrimental to the issuer.
56. Proprietary trading, or trading on the bank's own account, is similar to the treasury function performed by commercial banks, except that the emphasis is more on profiting through gains on trades rather than on maintaining adequate liquidity and managing the bank's asset-liability mix.
57. In the past, investment banks did not necessarily engage in securities brokering directly but worked with independent brokers to distribute securities they underwrote. Today, however, most investment banks have a strong presence in the brokerage industry, especially in respect to institutional investors. Independent securities brokers do exist, of course, but frequently concentrate on the consumer and retail brokerage market, rather than on acting as prime brokers for institutional investors.
58. A company that has already issued stock to the public may decide to issue additional shares through a rights issue. Typically, existing shareholders will have a preemptive right in such case to purchase the new shares before they are offered to others. Somewhat analogously, under an existing note issuance program, a debt issuer may issue new debt securities with less involvement on the part of the underwriter.
59. With the growth of discount brokerages and the widespread availability of financial data now available through the Internet, the opportunities for individuals to manage their own funds have also expanded tremendously. Mutual funds are often more attractive to individuals, especially those of a risk-averse persuasion, on the basis of the fact that his or her invested funds are professionally managed by the fund manager, who has a recognized degree of expertise as well as access to professional research, and the pooling of funds advanced by many individual investors allows effective diversification to take place.

The fund manager benefits from other economies of scale including lower transactions costs; for example, commission fees charged to institutional investors are much lower than those charged to retail investors. The ability of professional fund managers who actively manage their portfolios to beat the market has been questioned by academic theory, and as a result some believe that passively managed funds that track a recognized fund index such as the S&P 500 afford investors a safer alternative to actively managed funds. The fees charged by so-called index funds are ordinarily lower than those for actively managed funds since the process of allocating funds under investment is essentially mechanical in nature.

60. In relation to companies, the pension fund and employee benefit management that banks have provided to their corporate customers has in some cases evolved into more general institutional money management services.

Deconstructing the Bank Income Statement

To allege that making a provision against bad and doubtful losses was outside the ordinary activities of a bank and therefore not expected to recur . . . seemed to fly in the face of reality. Bad debts are of course a normal part of banking, and had been an even more regular feature of Midland's performance.

—Terry Smith¹

This book explores the creditworthiness of banks from several angles, many of them articulated around the income statement. Readers might be familiar with general accounting rules, and this chapter will introduce them more specifically to the bank's income statement. Inevitably, a number of concepts or definitions have to be discussed at this stage, while a more thorough analysis will be offered in later chapters, thereby introducing some degree of unavoidable duplication.

Evaluating a bank's earnings requires an understanding of how banks make money and how they differ from nonfinancial firms. This was the subject of Chapter 3, which discussed the business of banking and observed that a bank generates earnings from two primary sources: (1) assets that produce *interest income*;² and (2) *fees, commissions* (including net income from assuming risk, whether on or off balance sheet), and *trading gains* (or dealing gains). This characteristic affects the structure of the bank's income statement³ and accounts for some of the differences between how banks and nonfinancial companies report income. Principal differences between bank income statements and corporate income statements concern their respective:

- Streams of revenue⁴
- Terminology and manner of presentation
- Importance of allowance for bad debts⁵

ANATOMY OF A BANK INCOME STATEMENT: AN OVERVIEW

The bank *income statement* serves fundamentally the same purpose as does a corporate income statement—to report changes in the volume and composition of *net income* over a given period—and it is constructed in essentially the same way.

Expenses incurred are subtracted from *income* accrued to determine *net income* or *net profit*. A bank's *income statement* therefore, like that of its nonfinancial counterparts, measures the institution's *net profit* (or loss) by registering its *financial flows*,⁶ which are abstractions arising from the use of the *accrual method*⁷ of accounting.

As opposed to the static quality of the balance sheet, the income statement is *dynamic* in nature, showing financial flows within a given interval and encapsulating a company's ability to generate earnings over a period of time. In a sense, the income statement is a temporary account, being reset to zero at the start of each period—unlike the balance sheet, which is cumulative and extends over the life of the business.

THE INCOME STATEMENT

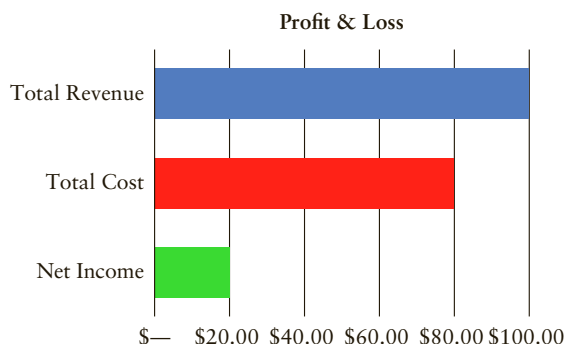
Also known as the *profit and loss account* or the P&L, this financial statement itemizes *total revenues* and *total expenses* for a particular accounting period to obtain *net income* for the period, which is the difference between the two. *Net income*, also called *net profit* or *net earnings*, is the so-called bottom line.

The income statement can also be viewed as an extension of the balance sheet in that it provides additional detail concerning changes in assets relative to liabilities and equity that have resulted from income and expense recognition between two balance sheet dates or over any particular interval of time. The remainder after deducting gross expenses from gross income represents *net income* earned—also known as the bottom line from its traditional position on the income statement. Retained earnings, that is, those not distributed in the form of dividends, become internally generated capital. Thus, in general, positive net income tends to increase equity capital, while negative net income—that is, losses—tends to reduce equity capital. In addition, since income statement accounts relate to a single financial reporting period, they are, in a sense, temporary accounts, as compared with balance sheet accounts, which are cumulative and continue throughout the life of the business.

The income statement reflects *profits*, income and profit being synonymous, and from net income, its constituents and other financial attributes, numerous ratios and indicators can be calculated for the purpose of comparing the performance of one enterprise with another. Exhibit 4.1 illustrates the income statement in graphical form.

The Cash Flow Statement

The *cash flow statement*, or *statement of cash flows*, records inflows and outflows on a cash basis, as opposed to the accrual basis used by the income statement. In contrast to the abstraction called *income* registered by the income statement, the cash

**EXHIBIT 4.1** The Income Statement

flow statement captures movement of cash in a given reporting period. The difference between cash inflow and cash outflow is net cash flow. If cash received exceeds cash paid out, net cash flow is positive; if cash received is less than cash paid out, net cash flow is negative.

Of immense importance in analyzing nonfinancial companies, the statement of cash flows is of little use in analyzing banks and financial companies owing to the fact that cash is a bank's stock in trade and movements of cash do not closely correspond to financial performance in the same way that they do in nonfinancial companies.

CASH FLOW STATEMENT

The cash flow statement registers the movement of cash into and out of a firm during a particular reporting period in order to measure net cash flow for the period. At the beginning of the next reporting period, all entries are reset to zero.

Income versus Cash Flow

It cannot be overemphasized that *income* is *not* equal to the movement of *cash*, and consequently the income statement neither indicates how much cash a company possesses, nor does it reflect the transfer of cash into and out of the firm's accounts. This movement is registered by the cash flow statement.

Note that cash may be defined to include *cash equivalents*, which are assets immediately liquefiable into cash without discount so physical delivery here means, effectively, that funds are available in the firm's accounts for disbursement (or can be readily obtained by the sale or pledge of marketable securities for immediate conversion into cash). Physical delivery here encompasses the electronic transfer of funds between accounts.

RELATIONSHIP BETWEEN THE BALANCE SHEET, THE INCOME STATEMENT, AND THE CASH FLOW STATEMENT

All the financial statements link together although the connections may not always be immediately apparent.

Income Statement versus Cash Flow Statement

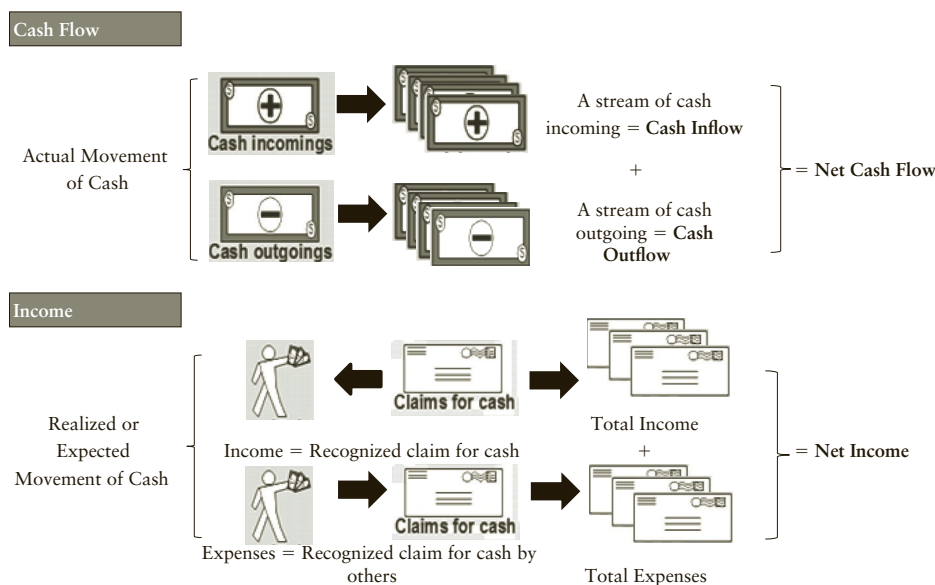
The relationship between the income statement and the cash flow statement derives from the fact that they each measure two different, albeit related, items. Cash flow is the movement of cash. Income, however, is an abstraction based on the accrual concept of accounting. Ultimately, the two converge, but usually cash flow will lag income since the time required for settlement and trade credit conventions result in payment often being received after the transaction date. In addition, not all incoming or outgoing cash flows are recognized as income or expenses. For example, capital expenditures are not recorded as expenses. Another example is depreciation, which is a noncash expense but which does not show up directly on the cash flow statement.

Balance Sheet versus Income Statement

The balance sheet is closely linked with the income statement. Recall that shareholders' equity on the balance sheet is comprised of paid-in capital, which includes surplus, and retained earnings. It is therefore directly affected by the amount of retained profits on the income statement. If the latter increases by \$1 million dollars, so will shareholders' equity. Moreover, as the discussion of the cash flow statement in the text that follows shows, cash flow can be derived from an examination of the balance sheet and income statement. For example, an increase in accounts payable for a given level of pretax income corresponds to a decrease in cash flow.

Balance Sheet versus Cash Flow Statement

Since the cash flow statement can be derived from the balance sheet and the income statement, it is evident that the balance sheet is also closely related to the cash flow statement. Recall that not all cash outlays are reflected in the income statement. As a general rule, however, cash outlays will have an impact on the balance sheet. An increase in an asset account generally signals a cash outflow, either immediately or later. For example, if a firm purchases new production machinery, this purchase will not show up on the income statement, other than the noncash charge for depreciation and amortization in the subsequent period. The purchase, however, will be mirrored on the balance sheet as the account for fixed assets increases to show the purchase price of the machinery. Of course, this could correspond to the conversion of one category of assets, namely cash, into another; or it could be the result of the borrowing of funds and a corresponding increase in liabilities.

**EXHIBIT 4.2** Cash Flow versus Income

Because income and cash flow are measured differently, it is not especially unusual for net income to be positive while net cash flow is negative. This phenomenon is illustrated in Exhibit 4.2. In the top half of the diagram net cash flow is negative if more cash is going out than is coming in. Similarly, recognized income, shown in the bottom half of the diagram, can be positive or negative. Recognized earnings for a period are those that translate in present cash inflows or are expected to result in future cash inflows. Expenses incurred for a period result in present cash outflows or are expected to result in future cash outflows. A positive result occurs if recognized earnings exceed expenses incurred.

Observe that the income statement is a register of realized and probable future cash flows or financial commitments. The statement of cash flows, by contrast, is a register of actual cash flows. In other words, the earning of revenue and the incurring of expenses that determine the net profit that the income statement depicts may bear only a vague relationship to the net flow of cash into and out of a company's coffers. The reason for this may be found in the *accrual* system of accounting discussed below.

No less for banks than for nonfinancial firms, income can be *overstated*, and losses masked where *financial inflows* are not matched properly to their corresponding *cash incomings*.⁸ With banks, this can notably occur where latent charges against income from *nonperforming loans*⁹ or from off-balance-sheet items are not yet *recognized*¹⁰ as such. Under the *accrual principle*, interest income continues to accrue on these unrecognized bad (i.e., nonperforming) assets and pass through the *income statement*, contributing to net income. The economic reality in such circumstances, however, may be that the matching cash payments have been delayed, perhaps never to be ultimately received as expected.¹¹ During

deteriorating economic conditions, such as those immediately following the Asian crisis of 1997, a situation of that nature is not uncommon. The income statement shown in Exhibit 4.3, drawn from the audited results of Dah Sing Financial Holdings (DSFH), a Hong Kong bank for the year ending December 31, 1998, illustrates several phenomena characteristic of a bank confronting deteriorating economic conditions.

Note that as economic conditions deteriorated and loan-loss provisions deducted from operating income increased to compensate for higher loan losses, both net interest income and net income fell. Specifically, with the Asian financial crisis at flood tide, DSFH's *net interest income* fell by 16.1 percent¹² year on year, *preprovision income* (operating profit before provisions) dropped 14.6 percent, *loan-loss provisions* almost quadrupled, and *net income* plunged sharply by 57.8 percent. Based on the P&L alone, the reasons for this decline cannot be identified with certainty. It may have been the result of reduced accrued interest income attributable to eroding asset quality or the impact of changing interest rates on the bank's asset-liability mix. Although no specific assertion is made with regard to DSFH, in a rapidly deteriorating business environment it may be surmised that, notwithstanding its substantial deterioration, the

	1998	1997	Variance
	HKS'000	HKS'000	%
Interest income	3,649,049	2,269,736	
Interest expense	(2,687,378)	(2,123,494)	
Net interest income	961,671	1,146,242	-16.1
Other operating income	458,930	375,361	22.3
Operating income	1,420,601	1,521,603	-6.6
Operating expenses	(717,133)	(698,134)	2.7
Operating profit before provisions	703,468	823,469	-14.6
Charge for bad and doubtful debts (node 1) ←	(255,172)	(66,819)	281.9
Operating profit after provisions ← NOPAP	448,296	756,650	-40.8
Profit on disposal of fixed assets and long-term investments	68,815	8,551	
Operating profit before exceptional items	517,111	765,201	-32.4
Exceptional items (node 2) ← Nonoperating items	(192,510)	(12,025)	
Profit on ordinary activities	324,601	765,201	-57.6
Share of net losses of associated companies	(2,223)	(1,551)	
Profit before taxation ← Pretax profit	322,378	763,650	-57.8
Taxation (node 3)	(50,356)	(88,404)	
Profit after taxation ← Net income	272,022	675,246	-59.7
Minority interests	(16,007)	(12,025)	
Profit attributable to shareholders	256,015	663,221	-61.4
Dividends	(145,565)	(273,968)	
Profit for the year retained	Below the line items 110,450	389,253	

EXHIBIT 4.3 Audited Results of Dah Sing Financial Holdings (DSFH), a Hong Kong Bank, December 31, 1998

Source: Company reports.

income statement conceivably may not have fully reflected the adverse impact of eroding asset quality.

Accrual Accounting

Advantages and Disadvantages of Accrual Accounting

Accrual-based accounting, unlike cash-based accounting, which is rarely used by commercial entities of any size, allows a firm to match its costs to its sales rather than to the timing vagaries of incoming cash collection and outgoing expense payments. By matching expenses with revenue, accrual accounting potentially functions as a superior and more accurate indicator of long-term financial health than does the statement of cash flows. The presence of healthy cash inflows is welcome, but whether a company possesses an ability to generate future cash flow is not determined by the actual cash payments and receipts recorded in the financial accounts.

This potential, however, is not always achieved, and as a result the accrual approach intrinsically suffers from a severe shortcoming from the vantage point of corporate credit analysis.

The decoupling of the timing of *actual* cash flows from that of *committed* cash flows, which affords accrual accounting its advantage, is also the cause of its principal deficiency. This deficiency is the inherent *mismatch* between the *recognition* of revenues and expenses on the one hand, and the actual *movement* of incoming and outgoing cash on the other. Since credit analysis is concerned with an *obligor's* capacity to make periodic cash payments as set forth under the financial obligation either actual or contemplated under review, this shortcoming of the income statement is a critical one.

ACCRUAL ACCOUNTING

While accrual accounting does match income and expenses arising out of the same accounting event, it does so at the price of permitting an inevitable divergence between the flow of cash and the flow of income.

The inevitable gap that therefore results between, on the one hand, the time when a unit of income is recognized and the time when the cash corresponding to that income is delivered—or, on the other hand, between the time an expense is incurred and the cash corresponding to that expense is paid out—has the following significance: The *income statement*, without adjustment, does *not* provide an accurate picture of the company's cash position. This is true irrespective of whether one flow occurs before or after the other.

Addressing the Deficiencies of Accrual Accounting

The income statement, on its face, is incapable of showing the amount of cash a company has generated, or which it has on hand, or whether or not the enterprise's

cash balance is increasing or decreasing. Since credit analysis is concerned with an *obligor's* capacity to make periodic cash payments as required under the financial obligation being contemplated, this shortcoming of the income statement is a critical one. This deficiency is remedied by the cash flow statement, which attempts to directly display the movement of cash into and out of the company's accounts and the change in its net cash position.

DIFFERENCE BETWEEN CASH FLOW AND INCOME IN THE LIGHT OF ACCRUAL ACCOUNTING

The essence of the difference between the statement of cash flows and the income statement derives from the fact that cash is real while income, as commonly reported using accrual-based accounting systems, is an abstraction, albeit a convenient one. Accrual accounting is governed by the matching principle, which seeks to recognize all revenues earned within a given financial period with the related expenses incurred, irrespective of the underlying movements of cash.

This principle allows revenues and expenses to be conceptually separated from the flows of cash generated by a business in any given period of time. For example, rather than depending upon receipt of cash from a customer, revenue from a sale is generally recognized as earned when the sale is materially consummated and there is a reasonable assurance that cash will ultimately be collected in a timely manner. Likewise, expenses are recognized as they are incurred, that is, when a commitment has been made to make payment, without regard to precise timing of the payment.

In a similar fashion, not all cash flows appear on the income statement and vice versa. Those regarded as capital expenditures, such as the purchase of machinery used to produce goods, do not appear directly on the income statement notwithstanding that an actual outgoing cash flow has occurred. Correspondingly, depreciation is a noncash item that appears on the income statement to account for the use of the machinery and a concomitant decline in its value.

Consequently, because revenue may be earned well before the corresponding cash is received, expenses may be incurred well in advance of the corresponding cash being remitted, and because not all cash flows correspond directly to income and expenses (and vice versa), the income statement has a flexible relationship to the movement of cash.

In the corporate credit context, the cash flow statement has another advantage over the income statement. Because cash flow is determined by the *physical delivery* of cash—it is in the bank and available for disbursement or it is not—the cash flow statement is much less subject to manipulation through management discretion than is the income statement. By reviewing the cash flow statement, creditors, and their

analyst proxies, can gain a better understanding of a company's liquidity—*liquidity* meaning its ability to access sufficient cash to meet its obligations on an ongoing basis. For these reasons, the cash flow statement is an invaluable tool for creditors and investors, and in respect to nonfinancial companies, the most important of the three financial statements for credit evaluation purposes. But, for reasons to be discussed, it is usually not an important financial report for the purposes of bank analysis.

Why the Cash Flow Statement Is Not Especially Useful for Bank Analysis

It might seem that the same method of cash flow analysis could be applied to a bank as it is commonly applied to a nonfinancial company. After all, as has been suggested, when a bank's loans become *nonperforming*, these assets may continue to accrue interest *without* generating a cash return. This will result in a similar gap between cash flows and financial flows that functions as a warning signal with respect to corporates.

For banks, however, the business of which is at bottom to use cash inflows (much of which is in the form of deposits and other funding) to generate more cash in the form of interest and fees, the statement of cash flows is, paradoxically, largely irrelevant to their credit analysis, particularly when performed by the external credit analyst. As is further explained later, the statement of cash flows for banks is simply not the reliable counterpoint to the income statement that it is for nonfinancial companies. Consequently, the *income statement*, rather than the *statement of cash flows*, is the best financial report available to measure a bank's performance from a credit perspective. This is not to say that the cash flow statement is irrelevant to the analytical process. Cash flow statements prepared weekly for internal management purposes are important tools in managing bank liquidity. For the analyst outside the bank, however, the annual or quarterly statement of cash flows is generally of little use in evaluating bank credit risk.

The Dual Role of Cash in Banks

The underlying reason for this phenomenon is that unlike for nonfinancial businesses that deal in ordinary goods and services, cash is a bank's *stock in trade*. This distinction has two important consequences that diminish the importance of strong and consistent cash inflow to a bank. The first is that while cash functions for a bank in the same way it does for other firms—that is, as a means to pay the rent, suppliers, staff salaries, and so on—a bank's cash *advances* (its loans and similar instruments) are also its primary *assets*. The second, discussed in the next subsection, concerns the differing liquidity requirements characteristic of banks and nonfinancial firms.¹³

For a bank, a high net cash flow must necessarily precede the cash inflow corresponding to the return of principal and interest generated by the bank's assets, particularly when a country's economy and its banks are developing rapidly. In other words, the more cash a bank advances to borrowers, the greater its volume of *earning assets*, and the higher its potential ultimate *income*.¹⁴ Indeed, a bank growing at a relatively brisk pace would be expected to have substantial net cash

outflow; that is, within a particular financial reporting period, advances of funds to borrowers may very well exceed those received in the form of deposits or other funding. Since high or rising net cash outflow is closely correlated with corresponding future flows of interest income, this phenomenon does not have the same negative impact that it has for corporates. The upshot is that, in contrast to *non-financial firms*, high or rising net cash outflows are not necessarily the signs of *financial distress* they would be for a nonfinancial firm, but instead may be an indicator of a healthy, growing bank.¹⁵

Somewhat paradoxically, while a period of prolonged moderate negative cash flow may be the sign of a healthy bank (since it characteristically means that the bank is growing its loan book rapidly in an expanding economy), a period of prolonged net positive cash flow may not be the salutary indicator it might seem to be at first glance. The latter scenario might reflect a scramble for new funding on the part of the institution or an inability to advance new loans at the same pace as occurred in the recent past—both credit-negative scenarios.

Differing Liquidity Requirements

The fact that cash is a bank's *stock in trade* has a second consequence, which makes cash flow analysis less useful in a credit evaluation of banks. Liquidity for a bank differs qualitatively from that of a typical nonfinancial enterprise. In brief, a bank does not need cash in the same way and for the same reason that a nonfinancial enterprise does.¹⁶ Since cash is its stock in trade, under normal circumstances it will tend to be awash in cash from depositors, even if most of its assets are either in the hands of its borrowers as loans or in the form of other *interest-bearing financial obligations*.¹⁷ Of course, as a matter of prudent banking, the bank will not lend all of its funds, but must maintain a *core reserve*, a significant portion of which must be in *liquid assets* or *quasi-liquid assets*. Still, absent unusual circumstances, a bank's liquidity needs will generally be modest.¹⁸ It should be noted that unusual circumstances have taken place with increasing frequency in recent times.

Nonfinancial companies, in contrast, require a regular stream of cash from sales not only to meet *operating expenses* and *current obligations*, but also to repay the *current portion* of long-term financing. They do not generally enjoy the benefits of receiving loans in the form of deposits placed with them, often interest free, and other funding in the ordinary course of business. Banks, however—so long as those who place funds with it are either paid a sufficient return, or receive other services justifying the maintaining of a deposit account with the institution—and so long as depositors and other creditors retain confidence in the institution—benefit from a recurrent inflowing stream of cash advances in addition to the funds they receive as revenue (i.e., interest paid by borrowers, fees charged to customers, and trading gains). What is more, for any particular bank, the stream of inflowing cash ordinarily can be increased by raising the interest rate paid to depositors, thereby attracting more advances from depositors and other creditors looking for a better return.

But therein lies the problem that banks confront, but nonfinancial companies generally do not, or at least not in the same extreme fashion. If, for whatever reason, a bank is unable to continue to attract deposits—or other financing such as *overnight* or

short-term money from other banks—it will face a serious predicament. In these circumstances, to meet its ongoing payment obligations it could only rely on the stream of cash it collects from revenue-generating activities, together with whatever liquid assets it holds. Since nearly all banks by the very nature of their business have a preponderance of *illiquid assets* to potentially *current liabilities*, their liquidity position is inherently unstable.¹⁹ In this set of circumstances, a bank's liquidity position would be severely weakened, and a *bank run*,²⁰ a phenomenon that constitutes the chief exception to a bank's usual ability to easily access needed cash, could arise.

What would cause this phenomenon to occur since, under normal business conditions, a particular bank need only gather funds from interbank markets or increase its deposit rate to attract a flow of new deposits? As suggested in the preceding paragraph, the most likely reason that depositors might be unwilling to make further deposits would be if they lose confidence in the soundness of the bank; that is, if they come to fear that the bank might fail, and believe that their deposits are at significant risk. Under these conditions, the marginal risk premium represented by a higher interest rate would, in the minds of depositors, be outweighed by the apparent risk of default—the loss of their entire deposit or advance. Should depositors lose confidence in a bank, not only would they cease depositing new funds in the institution, they would withdraw what funds they had on deposit, and place them elsewhere for safekeeping. In this scenario, the bank would confront a potentially catastrophic scenario: a bank run.²¹ A loss of confidence can all too easily and rapidly become self-fulfilling.

To sum up, banks require cash to pay operating expenses, as do corporates, but the overall cash requirements of banks differ substantially from those of their non-financial counterparts. They are vulnerable to a run on deposits because liquid funds represent a fraction of funds loaned and because the funding obligations supporting a bank's loans are subject to immediate withdrawal. This means that a bank's liquidity needs may change dramatically in a very brief period. By the time the cash outflow caused by a bank run was reflected on the statement of cash flows, it would be too late to benefit from analysis. For all the preceding reasons, the cash flow statement is not the valuable indicator that it is in corporate credit analysis, and, for this reason, is given comparatively little attention by bank credit analysts.²²

THE BANK RUN AND THE IMPORTANCE OF CONFIDENCE TO BANKS

Banks not only seek to have confidence in those to whom they lend funds, so too are they concerned with the confidence that those from whom they borrow, commercial creditors and retail borrowers alike, have in them. One commentator succinctly stated the bank's dilemma as follows: "The strongest bank . . . will last only so long as the people will have sufficient confidence in it to keep their money there."²³

A bank run is probably a bank manager's greatest fear. The Federal Reserve Bank of Minneapolis has defined the term as "a series of unexpected

cash withdrawals caused by a sudden decline in depositor confidence or fear that the bank will be closed by the chartering agency.” The need to maintain public and market confidence—and at the extreme to avoid the catastrophic loss of confidence that a bank run embodies—causes bank managers to seek to cultivate an image of propriety, which, at a fundamental level, must be acknowledged to be at odds with the intrinsic instability of the business of banking. Both the highly leveraged financial structure of banks, resulting from the maintaining of fractional reserves backing advances made to borrowers, and their concomitant vulnerability to sudden liquidity demands are at the root of this intrinsic instability.²⁴ An increasing reliance by many banks on interbank and other external funding has in recent years further exacerbated their instability.

The same root causes are responsible for an additional contributing factor to bank instability: the mismatch between the maturity of their assets (which are likely to be generally illiquid or comparatively longer term) and their liabilities (which are likely to be potentially immediate or comparatively shorter term).

Differences in Terminology and Income Statement Structure

The ways in which the individual *line items* (and the terminology employed)²⁵ on the bank’s *income statement* differ from those of its corporate counterparts can be seen in Exhibit 4.4.

Single Source versus Dual Sources of Revenue

In contrast to the usual nonfinancial company, which generally has a *single* source and category of operating revenue (i.e., sales revenue), a bank typically displays two distinct sources of *operating revenue*.²⁶

EXHIBIT 4.4 Bank Income Statement versus Corporate Income Statement (not to scale): Basic Comparison

Corporate Income Statement		Bank Income Statement	
Gross sales	X	Gross interest income	X
Less: Goods returns	(X)	Less: Interest expenses (i.e., funding expenses)	(X)
Discounts	(X)	Net interest income	NII
Net Sales	X	Add: Noninterest income	X
		Less: Operating expenses (excluding loan loss provision)	(X)
Less: COGS (excluding cost of debt finance)	(X)	Preprovision income	PPI

Corporate Income Statement		Bank Income Statement	
Gross Profit	X	Less: Loan loss provisions	(X)
Less: Operating expenses (Includes allowance for bad debts)	(X)	Net operating profit after provisions	NOPAP
Operating income	OI	Add: Nonoperating income	X
Add: Nonoperating income	X	Less: Nonoperating expenses	(X)
Less: Nonoperating expenses	(X)	Profit before tax	EBT
Profit before interest and tax	EBIT	Less: Taxation	(X)
Less: Cost of debt finance	(X)	Net income	NII
Profit before tax	EBT		
Less: Taxation	(T)		
Net income	NI		

Loan-Loss and Other Provisions

Another noticeable difference between the nonfinancial company's income statement and that of a bank concerns *loan-loss provisions*. For a nonfinancial firm, the category of *credit losses*, and the *allowance for bad debts* that accounts for them, is one of many *operating expenses*. Typically, for most nonfinancial firms, it is a comparatively inconsequential item. For a bank, however, credit losses on advances—and the *allowance* or *provisions* that account for them are of immense importance. The result is that loan-loss provisions typically have (and should have) a prominent place on a bank's income statement.²⁷ Consistent with the most prevalent usage, we employ the term *loan-loss provisions* to refer to the accounting line item for this allowance, although it goes by many names. Banks will also suffer losses or potential losses in respect of assets other than loans, as well as in respect of off-balance-sheet items. Accounting rules about provisions against such other losses are diverse and often adjusted—with some delay—to changes in financial instruments. Until now, we have examined the primary components of the major financial statements both generally and with specific reference to banks and other financial firms.

BANK INCOME STATEMENTS VERSUS NONFINANCIAL COMPANY FINANCIAL STATEMENTS: POINTS TO REMEMBER

- A nonfinancial company derives its income from sales; a financial company's income is derived from earning assets that generate interest income,

and from non–interest-income sources. Customarily, a nonfinancial firm’s most significant expense is the cost of goods sold; for a financial company, it is traditionally interest expenses.

- In the past, most commercial banks earned the bulk of their income from their *spread business*, and noninterest income was incidental. Today, however, noninterest income is often a large contributor to a bank’s operating income.
- While the structure of bank and non–financial-company income statements are similar, there are characteristic differences in presentation owing both to convention and to the different functions of the two categories of business. Some differences are mainly the result of differences in customary terminology.
- Interest expense is placed less prominently in a nonfinancial company compared with a bank. Similarly, loan-loss and similar provisions are of critical importance to a bank compared with a corporation’s bad debt provisions.
- Nonoperating items include losses and gains that are of an unusual and nonrecurrent character.
- Net income (attributable to shareholders) is either paid out in dividends or reinvested as internally generated capital.
- Selected alternative terms for noninterest income are:
 - Noninterest revenue
 - Other income
 - Other operating income
 - Other operating revenue

In the following section, the focus turns to the essential elements of a *bank’s* income statement, which are explored in depth, proceeding from *gross interest income*—the top line of the statement—down to *net income*, the bottom line.²⁸ Since many of the same concepts raised in early chapters are discussed in the remainder of the chapter, a degree of repetition is unavoidable.

A FURTHER DISSECTION

It bears repeating that specific financial statement line items can be referred to by a number of terms that mean the same thing. For example, the terms *interest expense*, *interest paid*, and the *cost of funding* are three of any number of synonyms that could be applied to the same line item. For some income statement line items the multitude of synonyms can be the source of confusion, particularly when any given term has more than one meaning.²⁹ The terms used in the following pages are in general widely used, but the selections made inevitably are somewhat arbitrary.

While the bank income statement can be viewed simply as a procession from *gross interest income* at its top to *net income* at the bottom line, for analytical

EXHIBIT 4.5 The Three Primary Building Blocks of the Bank Income Statement in Tabular Format

Line Items	Building Blocks	Calculation of Remainder Items
(Gross) Interest income Interest expense Net interest income	Building Block 1. Derivation of net interest income	Interest income – <u>Interest expense</u> = Net interest income +
Noninterest income Operating income Operating expenses (Net) operating income	Building Block 2. Derivation of preprovision income	<u>Noninterest income</u> = Operating income – <u>Operating expenses</u> = Preprovision income –
Loan loss provisions Preprovision income Nonoperating items Pretax profit Tax Net income	Building Block 3. Derivation of net income	<u>Loan loss provisions</u> <u>Preprovision income</u> – Nonoperating items = Pretax profit – <u>Tax</u> = <u>Net income</u>

purposes the statement can be divided into three fundamental elements. As can be seen in Exhibit 4.5, which provides a conceptual illustration:

- The first is the computation of *net interest income*.
- The second the computation of *operating income* and *preprovision income*.
- The third the computation of *net income*.

It is useful to think of the income statement in this way rather than merely as a succession of line items since its basic building blocks do not change, although presentation of the income statement may vary.

In brief, the first building block—the derivation of net interest income—concerns the profitability of a bank’s spread business—that is, earnings generated by lending, investment in debt securities, and other interest-producing activities. The second, the derivation of preprovision income, relates to the impact of the institution’s other businesses as well as its control over expenses. Finally, the last building block, the derivation of net income, takes account of the bank’s credit costs and tax liability as well as miscellaneous items.

INTEREST SPREAD AND INTEREST MARGIN

In banking, the interest spread is the difference between the rate of interest charged to borrowers and the rate of interest paid on the associated funding.

The interest spread is the difference between a bank’s *yield on earning assets* (i.e., the composite interest rate it earns on money lent or invested) and its *cost of funding* (i.e., the composite interest rate it pays to borrow funds from depositors, lenders, or investors).

The interest margin is the difference between a bank's *interest income from earning assets* (i.e., the total interest it earns on money lent or invested) and its *cost of funding* (i.e., the total interest it pays to borrow funds from depositors, lenders, or investors).

Having obtained a bird's-eye-view of a bank income statement, each of the three basic line item categories outlined in Exhibit 4.5 can be examined in greater detail providing a more refined picture of this account. To facilitate a fuller understanding of the income statement, it would be helpful to focus on a single example.

In Exhibit 4.6, the data from the published income statement of KeyCorp, a major bank holding company in the United States, have been placed into an ordinary spreadsheet. This institution has been chosen because the presentation is relatively simple and straightforward compared with that of some of its peers.

Under nearly all accounting regimes, management has long had a degree of discretion as to how financial results are displayed, and disclosure conventions vary by jurisdiction. For instance, specific categories employed to categorize interest income, as well as other items in a bank's financial statements, would vary from bank to bank and country to country. In recent years, though, accounting standards, such as U.S. Generally Accepted Accounting Principles (GAAP) and International Financial Reporting Standards (IFRS), whether they represent new sets or improvements over existing standards, have made it easier to compare banks across jurisdictions. Categorization may be made directly in the income statement, as is KeyCorp's practice (see Exhibit 4.6); alternatively, it may be provided in the accounting footnotes.

EXHIBIT 4.6 KeyCorp, Statement of Income, 2010

Consolidated Statements of Income			
Year ended December 31 (<i>dollars in millions, except per share amounts</i>)	2010	2009	2008
INTEREST INCOME			
Loans	\$2,653	\$3,194	\$3,732
Loans held for sale	17	29	76
Securities available for sale	644	460	404
Held-to-maturity securities	2	2	3
Trading account assets	37	47	56
Short-term investments	6	12	31
Other investments	49	51	51
Total interest income	3,408	3,795	4,353
INTEREST EXPENSE			
Deposits	671	1,119	1,468
Federal funds purchased and securities sold under repurchase agreements	6	5	57
Bank notes and other short-term borrowings	14	16	130

Consolidated Statements of Income			
Year ended December 31 (<i>dollars in millions, except per share amounts</i>)	2010	2009	2008
Long-term debt	206	275	382
Total interest expense	897	1,415	2,037
NET INTEREST INCOME	2,511	2,380	2,316
Provision for loan and lease losses	638	3,159	1,537
Net interest income (expense) after provision for loan and lease losses	1,873	(779)	779
NONINTEREST INCOME			
Trust and investment services income	444	459	509
Service charges on deposit accounts	301	330	365
Operating lease income	173	227	270
Letter of credit and loan fees	194	180	183
Corporate-owned life insurance income	137	114	117
Net securities gains (losses)	14	113	(2)
Electronic banking fees	117	105	103
Gains on leased equipment	20	99	40
Insurance income	64	68	65
Net gains (losses) from loan sales	76	(1)	(82)
Net gains (losses) from principal investing	66	(4)	(54)
Investment banking and capital markets income (loss)	145	(42)	68
Gain from sale/redemption of Visa Inc. shares	—	105	165
Gain related to exchange of common shares for capital securities	—	78	—
Other income	203	204	100
Total noninterest income	1,954	2,035	1,847
NONINTEREST EXPENSE			
Personnel	1,471	1,514	1,581
Net occupancy	270	259	259
Operating lease expense	142	195	224
Computer processing	185	192	187
Business services and professional fees	176	184	138
FDIC assessment	124	177	10
OREO expense, net	68	97	16
Equipment	100	96	92
Marketing	72	72	87
Provision (credit) for losses on lending-related commitments	(48)	67	(26)
Intangible asset impairment	—	241	469
Other expense	474	460	439
Total noninterest expense	3,034	3,554	3,476
INCOME (LOSS) FROM CONTINUING OPERATIONS BEFORE INCOME TAXES	793	(2,298)	(850)
Income taxes	186	(1,035)	437
INCOME (LOSS) FROM CONTINUING OPERATIONS	607	(1,263)	(1,287)
Income (loss) from discontinued operations, net of taxes of (\$14), (\$28), and (\$103)	(23)	(48)	(173)
NET INCOME (LOSS)	584	(1,311)	(1,460)

Source: Company reports.

Deriving Net Interest Income

Net interest income is a measure of how much profit a bank is generating from its “spread” business. Put another way, it represents a bank’s *gross profit* produced from its *interest-earning assets*. As an arithmetic expression, *net interest income*, as we have seen, is simply the difference between *interest income*—the *gross interest revenue* earned from *interest-earning assets*—and *interest expense*—the *cost of funding* such assets.

Gross Interest Income

Interest income, or *interest received*, refers to the funds earned from lending and related functions that fall under the category of economic interest. Revenue need not be formally termed *interest* to be classified as such. For instance, payments received as rent or lease payments from *financing leases*³⁰ as opposed to *operating leases* or *coupon payments*³¹ received on *debt securities*³² held are indistinguishable from interest on money lent in respect of their economic function. So while the largest portion of interest income generated by commercial banks will usually be the result of loans made to the institution’s customers, interest earned on finance leases, debt securities, and similar instruments may also form a substantial portion of this revenue stream.³³

FINANCING LEASES VERSUS OPERATING LEASES

As opposed to an ordinary operating lease, where the party leasing the property (the lessee) has no right to acquire the property leased by the owner (the lessor) at the end of the lease term and merely rents the property for the duration of the lease term, a financial (finance, financing) lease is a lease used to finance the acquisition of the lessor’s property by the lessee. So while under an operating lease, the owner of the property (the lessor) retains ownership of the property at the end of the lease, under a financial lease, the lessee acquires, or has the option to acquire, the leased property at the end of the lease term.³⁴ In a financing lease, the lease payments are analogous to the combined interest and principal repayments of an amortizing installment loan.

As with many of the other line items in bank accounting, *interest income* may be designated using a variety of terms. Whatever the terminology applied, this line item generally encompasses income generated from interest earned on loans advanced and securities held, both of which are assets to the bank. On a global basis and in most countries, interest income comprises the lion’s share of the revenue generated by banks, and net interest income constitutes the greater part of preprovision income.

Although banks traditionally earned the bulk of their interest income from lending to customers—for KeyCorp in 2010, as Exhibit 4.7 shows, interest earned from loans was about 78 percent of all interest income generated³⁵—interest earned from advancing funds to other banks and from holding debt securities contributes to this

revenue stream. Typically, the yields on interbank lending and securities holdings are lower than those earned on loans because of their perceived low credit risk and high liquidity.

It is common for banks to report interest income by *source*, such as interest income from commercial loans, interest income from consumer loans, interest income from interbank lending, interest income from short-term securities, and so on. For KeyCorp, interest income has been broken down into several categories such as interest from loans or from loans held for sale, which comprises the vast bulk of interest revenue and other interest income, or interest generated from securities. The last category encompasses interest generated by:

- Securities available for sale, meaning those with respect to which the holding period is indeterminate.
- Held-to-maturity investment securities, meaning those intended to be held to maturity, or for the long term.
- Trading account assets.
- Short-term investments, usually meaning *securities held for trading purposes*, typically having a holding period of 120 days or less (the definitional holding period will vary with the jurisdiction).
- Other investments (a catch-all category).³⁶

Overall, interest generated from debt securities, as a proportion of total interest income, has increased over time for KeyCorp. This can be clearly seen by comparing figures from 2008 to 2010 in Exhibit 4.7. Ten years earlier, that proportion was in the vicinity of 10 percent only.

KeyCorp's approach is a reasonable one. But, of course, it is just one way of classifying interest income. Further refinements are possible. Interest income generated by loans may be broken down further as between commercial, consumer, and real estate loans, as well as interbank lending, while that from securities may be split between taxable and tax-exempt securities. Additionally, the foregoing categories can be grouped according to whether their source is domestic or foreign. Common ways of grouping interest income are shown in Exhibit 4.8.

EXHIBIT 4.7 KeyCorp, Interest Income Breakdown

Year Ended December 31, U.S. dollars, in millions	% of Total Interest Income			
	2010	2010	2009	2008
Loans	77.85%	2,653	3,194	3,732
Loans held for sales	0.50%	17	29	76
Securities available for sale	18.90%	644	460	404
Held-to-maturity securities	0.06%	2	2	3
Trading account assets	1.09%	37	47	56
Short-term investments	0.18%	6	12	31
Other investments	1.44%	49	51	51
Total interest income	100.00%	3,408	3,795	4,353

Source: Company reports.

EXHIBIT 4.8 Common Groupings Used to Categorize Interest Income

	Interest Income from Loans	Interest Income from (Finance) Leases	Interest Income from Debt Securities	Interest Income from Other Earning Assets
By Sector	Consumer, commercial, real estate, financial institution	Not usually categorized by sector		
By Purpose	For example, mortgage lending, property management, property development		Investment vs. trading	
By Market/Location	Foreign vs. domestic; state by state			
By Tax Status	Taxable vs. tax exempt			

Interest Expense (Expense Item)

Interest expense usually constitutes one of the largest elements of a bank's total expenses, if not the largest. It represents the cost of funding the bank's assets; that is, the amount of interest the bank must pay to its depositors and creditors from whom it has borrowed funds to on-lend to bank customers or to purchase securities or other financial assets.

This line item also may be referred to on bank income statements as the *cost of funding*. It goes without saying that *interest income* to the lender refers to the very same funds that the borrower classifies as *interest expense*, and the two terms refer to the same income stream viewed from different vantage points.

For most banks, the larger proportion of interest expense is incurred on customer deposits. Some institutions, however, are more dependent upon *money-market funding*,³⁷ including the *interbank market*.³⁸ This category of funding is sometimes referred to as *purchased funds*, since unlike retail *customer deposits*,³⁹ it is effectively purchased through brokers who, for a fee, obtain the funds on the bank's behalf. The depositors are generally commercial entities that may have no other relationship with the bank and are therefore highly sensitive to the interest rate offered and the degree of credit risk associated with the institution.

PURCHASED FUNDS VERSUS CUSTOMER DEPOSITS

The term *purchased funds* ordinarily refers to bank funding *other than* retail customer deposits and a bank's long-term borrowing (usually accomplished through the issue of debt securities). Purchased funds, which are often obtained ("purchased") through brokers, may also be referred to as *wholesale funds*,

commercial funds, or *commercial deposits*. More colloquially, this category of funding may be called *hot funds*, *hot money*, or *confidence funding*. In very simple terms, purchased funds are actively sought by the financial institution, while customer deposits are obtained passively. Whatever the designation, this type of funding generally encompasses:

- Interbank borrowings
- Large certificates of deposit (generally uninsured)
- Other short-term borrowings (excluding securities issues)

For analytical purposes, however, it is important to note that short-term interbank borrowing itself should—whenever possible—be divided between interbank funds that are actively sought by the bank and interbank funds that are passively received by the bank.

So short-term bank funding can be divided into three basic categories:

1. Customer deposits (including industrial and commercial customers)
2. Passively obtained interbank deposits (not too dissimilar to commercial deposits)
3. Actively obtained interbank deposits (of a “purchased funds” nature)

The term *net purchased funds* sometimes means purchased funds net of money-market assets, but that would imply a perfect liquidity of such assets.

EXHIBIT 4.9 KeyCorp, Interest Expense Breakdown

Year Ended December 31, U.S. dollars in millions	% of Total Interest Expense 2010	2010	2009	2008
Deposits	74.80%	671	1,119	1,468
Federal funds purchased and securities sold under repurchase agreements	0.67%	6	5	57
Bank notes and other short-term borrowings	1.56%	14	16	130
Long-term debt	22.97%	206	275	382
Total interest expense	100.00%	897	3,424	2,037

Source: Company reports.

As with interest income, interest expense can be categorized by its source (e.g., customer deposits), and the classification will typically be provided either directly in the income statement, or indirectly in the accounting footnotes. KeyCorp provides this data on its income statement and categorizes interest expense as set out in Exhibit 4.9.

Most commercial banks obtain the bulk of their funding from *customer deposits*. For KeyCorp, this category of funding has been relatively stable over the past eight

years, in a range of between US\$57.6 billion to US\$65.6 billion. Other funding sources have—as they should—fluctuated much more, but they only represented between 25 and 30 percent of total funding between 2004 and 2008, to fall to only about 20 percent in the three years to 2011. Naturally the ratio between total funding costs from customers and total funding costs from other sources is bound to be influenced by factors other than just the volume sourced. Although almost all banks rely to some degree on sources other than deposits for funding, some institutions—*wholesale banks*,⁴⁰ for example—tend to rely to a great extent on such *commercial* sources of finance. This funding—referred to previously as *purchased funds*⁴¹—may be in the form of overnight or *interbank borrowing*,⁴² issues of debt in the money or capital markets, or other forms of medium- and long-term borrowing. In Exhibit 4.9, the latter three of the four interest expense line items represent *commercial funding*. While most of KeyCorp’s funding is from customer deposits, a significant portion is from commercial sources.

Net Interest Income

Net interest income is a critical figure that corresponds to the profits earned on the bank’s *spread business* and one of the few terms that, together with its variant net interest revenue, is almost universally used to designate this particular *line item*.

Subtracting *interest expense* from *interest income* leaves *net interest income* as the remainder.

Net interest income is more important than each of its constituents—that is, (gross) interest income and interest expense—since each will typically vary greatly with prevailing economic conditions and interest rates, while net interest income being the difference between them will tend to be relatively less sensitive to change.⁴³ Significant increases or decreases in net interest income are therefore of greater interest to the analyst than respective changes in interest income and interest expense alone. In general, moderate and continuous increases in net interest income, in line with asset growth, and in line or somewhat above average relative to the bank’s peers, are the ideal.⁴⁴ Excessive net interest income, however, may hide unacceptable risk taking.

ILLUSTRATION: NET INTEREST INCOME AND ITS CONSTITUENTS

The stability of net interest income compared to its constituents is reflected in KeyCorp’s income statement excerpted in Exhibit 4.10, which shows KeyCorp’s quarter-by-quarter (for years 2009 and 2010) interest income, interest expense, and net interest income. Figures shown in italics represent an increase over the immediately previous period. Net interest income over the eight quarters moved higher in absolute terms, but most importantly higher in relation to interest income. This was the result of a softening of interest expense, which was due to the fact that, in 2010, the bank relied less on expensive sources of funding such as CDs and fixed deposits. Generally, though, all three (interest income, interest expense, and net interest income) move in parallel.

EXHIBIT 4.10 KeyCorp, Interest Income and Expense, FY2009 and FY2010

	US\$, millions, by quarter							
	4Q10	3Q10	2Q10	1Q10	4Q09	3Q09	2Q09	1Q09
Interest income	811	844	861	892	933	977	945	977
Interest expense	182	204	244	267	303	388	376	388
Net interest income	629	640	617	625	630	589	569	589
Net interest income/ Gross interest income	77.56%	75.83%	71.66%	70.07%	67.52%	60.29%	60.21%	60.29%
Change of:								
Interest income	−4%	−2%	−3%	−4%	−5%	3%	−3%	
Interest expense	−11%	−16%	−9%	−12%	−22%	3%	−3%	
Net interest income	−2%	4%	−1%	−1%	7%	4%	−3%	

Source: Company reports.

Changes in net interest income are more important than changes in gross interest income or interest expense individually, and net interest income is regarded as a more valuable indicator than its constituent parts. There are two principal reasons for this.

First, net interest income is a vital indicator of the performance of a bank's spread business in the same way that net income is an indicator of a bank's overall performance. Since, historically at least, a bank's spread business—that is, its interest earnings on loans and securities—was a bank's core business activity, a strong flow of net interest income has long been viewed as a critical measure of a bank's financial health.⁴⁵

Second, *net interest income* values are more stable and therefore more important than those of its constituents. This stability reflects the fact that large changes in gross interest income and interest expense do not necessarily result in significant changes to net interest income. When prevailing interest rates rise, the cost of borrowing increases, but so does interest revenue—most of the time, anyway. Likewise, when prevailing interest rates fall, interest revenue declines, but so as a rule does the cost of funding. This normal parallel movement of lending rates and funding rates means that they tend to cancel each other out resulting in little or no change to net interest income.⁴⁶ Net interest income values are therefore less volatile

than interest income and interest expense values individually. Three points should be noted, though:

1. Higher interest rates often allow for wider spreads.
2. The shape of the yield curve has an obvious influence on the risk margins of financial institutions, as they tend to borrow short and lend long.
3. The repricing of interest is not simultaneous on all assets and liabilities.

NET INTEREST INCOME SIGNIFICANCE

Net interest income represents profits earned on the bank's interest-earning businesses, as opposed to business that generates fees and commissions or trading gains.

While on a sector-wide basis, funding rates and lending rates typically move in tandem as prevailing interest rates rise or fall, circumstances affecting a particular bank or group of banks can prevent changes in funding costs from offsetting interest revenue and vice versa. These circumstances include situations in which a bank has either locked in funding costs or interest revenue through long-term borrowings or long-term lending at a fixed rate.⁴⁷ For instance, a bank may have made a number of fixed-rate five-year loans when prevailing interest rates were low. Should prevailing interest rates rise, it must pay more on the money market for funding, and could also be forced to pay more to its retail depositors. So despite an increase in prevailing interest rates, its interest revenue would remain flat or increase slowly with new loans made at higher rates, while the rise in funding costs remains unstoppable. Net interest income in this situation would, of course, decline.

Naturally, interest paid or received may have increased or decreased as a result of substantial changes in the bank's volume of business—either organic or resulting from a merger. Whatever their sources, changes in interest income and changes in interest expenses may appear in various periods, and they do not always go in the same direction, resulting in a potential dramatic impact on net interest income.

Illustration: Net Interest Income on a Bank Income Statement

Exhibit 4.11 shows an excerpt of income statements from Mitsubishi UFJ Financial Group for the years 2009, 2010, and 2011.

The volumes of both interest income and interest expenses went down substantially between 2009 and 2010, but with a limited impact on net interest income, even though interest income fell considerably more (by about 51.6 percent) than interest expenses (by 29.2 percent). Between 2010 and 2011, though, interest expenses grew substantially (by 105.1 percent) while interest income remained almost flat (down by 7.5 percent).

Composition of Noninterest Income

Noninterest income takes in all *operating revenue* produced by a bank *other* than revenue produced from *interest-earning assets*.

The types of businesses other than spread business in which banks engage vary considerably from country to country, and within each market, different classes of banks may specialize in certain types of activity or be limited in the activities in which they can engage by regulation. The diversity in sources of noninterest income accounts to some degree for a corresponding diversity both in how banks report noninterest income and in how such income is categorized by analysts.⁴⁹

Noninterest income, also called *noninterest revenue* (or *other operating revenue* among various terms applied), is ordinarily made up of two subcategories of income:

1. *Fees and commissions* generated by a variety of a bank's business activities constitute a comparatively stable form of noninterest income.
2. *Trading income*, which includes *foreign exchange dealing profits* and *securities trading gains* is generally regarded as a relatively volatile form of noninterest income.⁵⁰

These two broad categories are useful for analytical purposes, even though terminology and classification differs from institution to institution and across markets. Those listed in Exhibit 4.13, an excerpt from Bank of America Corporation's 2010 income statement, are not atypical.

Bank of America Corporation and Subsidiaries

Consolidated Statement of Income

(Dollars in millions, except per share information)		Year Ended December 31		
		2010	2009	2008
Interest income	Like most commercial banks, the bulk of BofA's operating revenue stream drives from its spread business (net interest income). Noninterest income provides a substantial supplement to this core revenue source.	\$ 50,996	\$ 48,703	\$ 56,017
Loans and leases		11,667	12,947	13,146
Debt securities		1,832	2,894	3,313
Federal funds sold and securities borrowed or purchased		6,841	7,944	9,057
Trading account assets		4,161	5,428	4,151
Other interest income				
Total interest income		75,497	77,916	85,684
Interest expense	Most of BofA's noninterest income came from fee and commission associated with <i>investment and brokerage services, service charges, and credit cards</i> . Trading income, the more volatile component of non-interest income is represented here mainly in the categories of <i>equity investment gains</i> and <i>trading account profits</i> , although other categories may include some trading-type gains.	3,997	7,807	15,250
Deposits		3,699	5,512	12,362
Short-term borrowings		2,571	2,075	2,774
Trading account liabilities		13,707	15,413	9,938
Long-term debt				
Total interest expense		23,974	30,807	40,324
Net interest income		51,523	47,109	45,360
Noninterest income				
Card income		8,108	8,353	13,314
Service charges		9,390	11,038	10,316
Investment and brokerage services		11,622	11,919	4,972
Investment banking income		5,520	5,551	2,263
Equity investment income		5,260	10,014	539
Trading account profits (losses)		10,054	12,235	(5,911)
Mortgage banking income		2,734	8,791	4,087
Insurance income		2,066	2,760	1,833
Gains on sales of debt securities		2,526	4,723	1,124
Other income (loss)		2,384	(14)	(1,654)
Other-than-temporary impairment losses on available-for-sale debt securities:				
Total other-than-temporary impairment losses		(2,174)	(3,508)	(3,461)
Less: Portion of other-than-temporary impairment losses recognized in other comprehensive income		1,207	672	—
Net impairment losses recognized in earnings on available-for-sale debt securities		(967)	(2,836)	(3,461)
Total noninterest income		58,697	72,534	27,422

EXHIBIT 4.13 Bank of America Corporation, Income Statement, 2010, Excerpt (annotated)

Source: Company reports.

This bank divides noninterest income into several designated categories and one catch-all category. The main groupings are:

- Card income (i.e., income from credit cards)
- Service charges (e.g., account maintenance charges)
- Investment and brokerage fees (e.g., commissions on securities transactions)
- Investment-banking income (e.g., advisory fees and underwriting fees)
- Equity investment gains (e.g., venture capital gains)
- Trading-account profits (e.g., trading gains from debt securities holdings and foreign exchange)
- Mortgage-banking income (e.g., income from securitizing mortgage loans)
- Insurance
- Profits from the sale of debt securities

NONINTEREST INCOME DEFINED

Noninterest income—also called *other income*—encompasses all revenue *not* derived from *interest-earning assets* (*earning assets*). The principal sources of noninterest income are *fees and commissions* obtained by a bank in exchange for providing various and sundry services ranging from the mundane to the highly sophisticated. For example, within this category of revenue are fees for the rental of *safe deposit boxes*, fees for the supply of guarantees and *letters of credit* (L/Cs), and fees for providing *investment advisory services*. In addition to fee and commission income, banks may also supplement their revenue stream from profits earned on *foreign-exchange dealing* and other *trading activities*. The growth of trading in derivatives and generally in all types of instruments different from traditional earning assets has recently opened the door to substantial noninterest income to the point of reaching amounts that are sometimes larger than those associated with traditional commissions and fee income. Needless to point out, such trading income may prove very volatile, and even at times negative.

KeyCorp, as Exhibit 4.14 shows, has grouped *noninterest income* into several discrete categories and one catchall category. Seven of the categories correspond to sources of *fee and commission* income described, while one category relates to net securities gains, corresponding to *trading income* (discussed in the following section). It would be necessary to explore notes to the accounts to figure out whether gains or losses on loan sales and on principal investment should be strictly categorized as trading income. Other income might also include bits and pieces belonging to the two main categories. Note that most of KeyCorp's noninterest income derives from fees and commissions rather than trading income, which is regarded as more volatile than the fee and commission income. Overall we can see that noninterest income has been very stable over the past three years.⁵¹

EXHIBIT 4.14 KeyCorp, Noninterest Income

NONINTEREST INCOME, U.S. dollars in millions Year Ended December 31, 2010	F&C or TI	2010 % Noninterest Income	2010 \$millions	2009 \$millions	2008 \$millions
Trust and investment services income	F&C	22.7%	444	459	509
Service charges on deposit accounts	F&C	15.4%	301	330	365
Operating lease income		8.9%	173	227	270
Letter of credit and loan fees	F&C	9.9%	194	180	183
Corporate-owned life insurance income	F&C	7.0%	137	114	117
Net securities gains	TI	0.7%	14	113	(2)
Electronic banking fees	F&C	6.0%	117	105	103
Gains on leased equipment		1.0%	20	99	40
Insurance income	F&C	3.3%	64	68	65
Net gains(losses) from loan sales		3.9%	76	(1)	(82)
Net gains(losses) from principal investment		3.4%	66	(4)	(54)
Investment banking and capital markets income	F&C	7.4%	145	(42)	68
Gains from sale/redemption of Visa Inc. shares		0.0%	—	105	165
Gain related to exchange of common shares for capital securities		0.0%	—	78	—
Other income		10.4%	203	204	100
<i>Total noninterest income</i>		100.00%	1,954	2,035	1,847

Source: Company reports.

Fee and Commission Income

Fees and commissions are produced by a broad range of banking services, running the gamut from the provision of various types of deposit accounts that generate maintenance, check-clearing, and penalty fees, and other small but high-volume charges to high-level advisory services that may engender single fees in the millions of dollars. A range of sources of fee and commission income is illustrated in Exhibit 4.15.

The breakdown of noninterest income at KeyCorp is fairly typical and illustrative of the approach taken by many banks. Each component is described further in the following paragraphs.

Trust and Investment Services Income

The category of trust and investment services income includes fees earned for investment management on behalf of customers. Trust banking services include management of investments held in trust accounts (often inheritances) where the bank acts as a fiduciary to the trust beneficiaries. Investment banking, another category described below, of course covers a much wider range of services associated with the bank's role as an intermediary for customers' access to various markets. Another way to classify

EXHIBIT 4.15 Bank Business Activity Most Likely to Generate Non–Yield-Related Fee and Commission Income

Category of Activity	
Bookkeeping/Data Management Services	
■ Check verification ■ Courier services ■ Credit card issuance ■ Factoring ■ Insurance brokerage/agency ■ Investment advising ■ Issuance of small denomination debt ■ Lending and similar extensions of credit ■ Letters of credit/trade finance ■ Loan servicing ■ Management consulting ■ Mortgage banking	■ Mutual funds/unit trust business ■ Property management ■ Real estate appraisal ■ Real estate brokerage ■ Real estate syndication/development ■ Safe deposit boxes ■ Sale of travelers' checks and similar ■ Trading in coin and bullion ■ Trust/asset management services ■ Underwriting credit insurance ■ Underwriting life/health insurance ■ Underwriting property/casualty ■ Venture capital

this stream of revenue is as *fees and commissions from asset management services*. For KeyCorp, this type of business comprised the largest single component of the bank's noninterest income stream. For FY10, trust and investment services income amounted to US\$444 million, or over 20 percent of the total noninterest income stream of US\$1,954 million. All other things being equal, trust and investment services income tends to increase during bull markets and decline during bear markets.⁵²

Service Charges on Deposit and Credit Card Accounts

While banks incur *interest expense* on interest-earning deposits, they may also generate noninterest income in the form of account-service fees. For instance, many banks charge minimum maintenance fees for customer accounts, although frequently these are waived if a threshold deposit level is reached. In addition, other fees on both interest-earning and non–interest-earning accounts, including check-cashing fees, fall into this category. Credit card fees—beyond interest income—could be quite substantial. Although each fee charged is ordinarily relatively small, their high volume adds significantly to a bank's revenue stream. For example, fees charged to customers for maintaining business and consumer accounts represented the next-largest category of noninterest income at KeyCorp, amounting to about 15 percent of noninterest income.

Investment Banking and Capital Markets Income

The category of *investment banking and capital markets income* encompasses fees from:

- Corporate advisory work
- Securities underwriting, together with arranging corporate financing
- Securities broking
- Related activities on behalf of customers

This foregoing type of business is associated with investment banks in particular, although it will often be a significant source of revenue for larger commercial banks.

L/Cs and Loan Fees

This category of *fee income* includes fees in connection with a bank's providing L/Cs for trade finance and other purposes. It may also include certain loan-related fees, such as designated fees obtained for providing loans to customers or for servicing loans. For KeyCorp, this category of noninterest income is a steady and safe source. The treatment of loan fees varies depending upon whether the fees are classified as yield related or as non-yield related. If the former, they will be deemed interest income; if the latter, they will be considered as noninterest income.

INTEREST INCOME OR NONINTEREST INCOME?

Although in most cases the distinction between *interest income* and *noninterest income* is abundantly clear, occasionally questions arise as to how a particular item should be classified. For instance, banks often charge fees for arranging loans, or commitment fees for unused credit or for loan disbursements committed to in the future. Are such fees part of the fees and commissions category, and therefore to be deemed noninterest income, or are they simply part of the income expected to be generated by the loan?

The answer is: it depends. A strong argument can be made that fees charged for arranging loans are effectively part of the yield the asset is expected to generate, and therefore they should be grouped with interest income rather than with noninterest income within the fees and commissions category. Nevertheless, some regulators and banks classify certain fees—for example, *syndication fees* and *mortgage servicing fees*⁵³—as non-yield-related, and therefore as part of noninterest income for financial reporting purposes.⁵⁴ Consequently, when comparing banks, care should be taken to measure like with like.

Electronic Banking Fees and Corporate-Owned Life Insurance Income

A self-explanatory item, electronic banking fees charged for automated teller machines (ATMs) and online banking service fees constituted a small but significant and growing category of noninterest income for KeyCorp. Be aware that in some countries, by law or custom, ATM withdrawals are free to customers or those maintaining a relationship with the ATM provider. The volume of fees generated will vary accordingly.

Corporate-owned life insurance income is a form of incidental income and a separate line item corresponding to it is not widely seen on a global base, although it is not uncommon to find it on income statements in the United States.⁵⁵

Trading Income

Trading income refers to gains arising from:

- Foreign exchange (forex) dealing
- Securities trading and investment
- Related activities

Trading income or trading gains are profits earned on investing or trading activity on the bank's account—for example, through dealing in foreign exchange, in interest rate-sensitive instruments or in various types of derivative products. However, investment income, insofar as it derives from interest earned on securities held for investment, will usually be construed as part of interest income.

Classification of revenue varies, for example, with the nature of the holdings of securities and the source of income. Generally speaking, if deemed *trading securities*, both *realized* and *unrealized gains* and losses form part of net trading income and are reflected as income on the income statement.⁵⁶

Net Securities Gains and Net Gains from Loan Securitizations and Sales

In addition to *forex dealing income*, banks may post gains (or losses) from securities holdings. For most banks, securities holdings are—or should be—primarily, if not exclusively, in the form of high-grade debt instruments. Such revenue generated from securities held for trading purposes is classified as *trading income*. A related source of revenue usually deemed to be part of trading income results from gains generated from loan securitizations—that is, the packaging and sales of loans, such as residential mortgage loans by a financial institution—and the sales of such instruments. For KeyCorp this category of income registered a significant if still relatively low percentage of total noninterest income. In less sophisticated markets, net gains from loan securitizations and sales may be negligible.

As for many domestic U.S. banks, KeyCorp's *trading income* represented a small part of its total noninterest income. In common with its peers, the bank's revenue from forex dealings was relatively small and not deemed significant enough to be broken out separately from "securities trading gains" or "other income" on the face of the income statement. Its *net securities trading gains*, though grouped separately, were also quite small. For some internationally active banks, however, forex dealing income would be substantial and identified as a discrete category of non-interest income.

Operating Income

Added together, as illustrated in Exhibit 4.16, the two revenue streams—*net interest income* and *noninterest income*—comprise a bank's *operating income*. The adjective "operating" denotes that the revenue is *recurrent* in character, and is generated as a part of the enterprise's ordinary and ongoing activities.⁵⁷ As banks have ventured into substantial nonwarehouse activities, the two revenue streams—and in particular the noninterest income—have lost some of their recurrent character: They do recur, but their volume is less predictable.

EXHIBIT 4.16 Operating Income: A Bank Has a Dual Revenue Stream

Statement of Income in €millions	2010	2009	2008
Interest and similar income	28,779	26,953	54,549
Interest expense	13,196	14,494	42,096
Net interest income	15,583	12,459	12,453
Provision for credit losses	1,274	2,630	1,076
Net interest income after provision for credit losses	14,309	9,829	11,377
Commissions and fees income	10,669	8,911	9,741
Net gains (losses) on financial assets/liabilities at fair value through profit or loss	3,354	7,109	(9,992)
Net gains (losses) on financial assets available for sale	201	(403)	666
Net income (loss) from equity method investments	(2,004)	59	46
Other income (loss)	764	(183)	699
Total noninterest income	12,984	15,493	1,160
Compensation and benefits	12,671	11,310	9,606
General and administrative expenses	10,133	8,402	8,339
Policyholder benefits and claims	485	542	(252)
Impairment of intangible assets	29	(134)	585
Restructuring activities	—	—	—
Total noninterest expenses	23,318	20,120	18,278
Income (loss) before income taxes	3,975	5,202	(5,741)
Income tax expense (benefit)	1,645	244	(1,845)
Net income (loss)	2,330	4,958	(3,896)
Net income (loss) attributable to noncontrolling interests	20	(15)	(61)
Net income (loss) attributable to Deutsche Bank shareholders	2,310	4,973	(3,835)

The significance of operating income is that it measures an institution's gross *earnings capacity* before any deductions are made for operating expenses, loan-loss provisions, taxes, or other items, whether ordinary or unusual. By comparing operating income with preprovision income, the efficiency of a bank in managing its operating expenses can be discerned.

As discussed, Exhibit 4.16 shows a bank's operating income as coming from two sources: interest-generating business and non-interest-generating business. Again, the latter is derived from *fees and commissions*, as well as *trading gains* or *dealing gains*.

NET INTEREST INCOME AS OPERATING INCOME

Observe that the classification of (net) *interest income* as *operating income* is contrary to the treatment generally applied to nonfinancial firms, where interest income is normally judged to be *nonoperating* income. This is attributable to the earning of interest from financial assets representing banks' core source of revenue. For a nonfinancial firm, however, this activity is incidental to its primary activity of selling automobiles, manufacturing computers, providing consulting services, or whatever line of business it engages in.

EXHIBIT 4.17 Deutsche Bank, Consolidated Income Statement, 2010 Excerpt

Statement of Income in €millions	2010	2009	2008
Interest and similar income	28,779	26,953	54,549
Interest expense	13,196	14,494	42,096
Net interest income	15,583	12,459	12,453
Provision for credit losses	1,274	2,630	1,076
Net interest income after provision for credit losses	14,309	9,829	11,377
Commissions and fees income	10,669	8,911	9,741
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Policyholder benefits and claims	485	542	(252)
Impairment of intangible assets	29	(134)	585
Restructuring activities	—	—	—
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Income (loss) before income taxes	3,975	5,202	(5,741)
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Net income (loss)	2,330	4,958	(3,896)
Net income (loss) attributable to noncontrolling interests	20	(15)	(61)
Net income (loss) attributable to Deutsche Bank shareholders	2,310	4,973	(3,835)

Source: Company reports.

In Exhibit 4.17, which is an excerpt from the 2010 income statement of Deutsche Bank, a major German financial institution, a representative presentation of operating income and its components can be seen. Interest income (here referred to somewhat unusually as “interest and similar income”) and noninterest income added together comprise operating income. Noninterest income is broken down into its two major categories, commissions and fees income and gains or losses on financial assets with no clear indication as to whether such gains or losses arose from trading or holding those assets—of course details are available elsewhere in the published accounts.⁵⁸

Operating income, also commonly referred to as *operating profit* (among other terms), is another important line item that signifies a bank’s total revenue stream *net* of *interest expense*. Indeed, perhaps because of its importance, it is sometimes termed *total revenue*, although this usage can give rise to confusion as the same appellation is more commonly applied to a bank’s *gross* revenue stream without deduction of interest expense.⁵⁹

ALTERNATIVE TERMS FOR OPERATING INCOME

- Operating profit
- Income before operating expenses
- Total revenue

Noninterest Expenses (Operating Expenses)

Noninterest expense, as would be expected, generally embraces all operating expenses *other than* a bank's *funding costs*. That is, it includes all expenses other than interest expenses incurred on borrowed funds. The terms *operating expense* or *expenses, overhead*,⁶⁰ and *noninterest expense*, as with many bank accounting terms, are often used interchangeably.

Exhibit 4.18, excerpted from a Russian bank's website, shows the portion of an income statement referencing noninterest expenses. Bank of Moscow's income statement groups its noninterest expenses into several categories. As with most banks, compensation (salary and bonuses) is the largest single category of noninterest expense.⁶¹

A close look at *noninterest expenses* is an important part of the analytical process, since changes in these costs are a major variable affecting performance. It is also an item over which management has a substantial degree of control. Significant changes in *noninterest expenses* are of key interest to the analyst, as higher expenses normally mean lower profits and vice versa. In many ways, a bank's operating expenses resemble those of other service businesses. There is a key difference, however, in their manner of classification. Banks as a rule have *no* cost of goods sold or cost of services line item in their income statements. The closest analog—and a substantial one—to cost of goods sold is a bank's funding expense (i.e., interest expense). As a consequence, a bank's costs of operation are essentially the same as its noninterest expenses—that is, the costs incurred by a bank other than the interest expense it pays to fund its loans and other financial assets—and the terms are used synonymously. As do other service providers, a bank incurs costs merely by keeping its doors open. Before a bank can make any loans, or earn fees for services, it must have premises from which to

EXHIBIT 4.18 Noninterest Expenses: Bank of Moscow, Income Statement, 2010, Excerpt
Group of Joint Stock Commercial Bank—Bank of Moscow (Open Joint Stock Company)
Notes to the Consolidated Financial Statements for the Year Ended December 31, 2010
(in thousands of Russian Roubles)

24. General and Administrative Expenses	Note	2010	2009
Staff costs		9,492,443	7,430,049
Rent		2,049,986	2,045,801
Taxes other than income tax		1,340,812	1,083,060
Professional services (security, communications, and other)		1,120,713	1,220,347
Expenses related to premises and equipment		1,037,036	612,705
Depreciation and amortization	12	914,248	848,117
Administrative expenses		624,272	484,626
Advertising and marketing		372,286	328,068
Charity		61,885	88,032
Other		104,516	177,000
Total general and administrative expenses		17,118,197	14,317,805

Staff costs include remuneration to employees and insurance contributions.

Source: Company reports.

operate and staff to offer financial services to customers. The following categories of expenses comprise the largest part of the bank's *cost base*.⁶²

- *Compensation* expenses
- *Occupancy* expenses (i.e., rent on the bank's premises and fees to utilities suppliers)
- *Information technology* expenses
- *Marketing and advertising* expenses
- *Administrative* expenses and other related costs such as the purchase of office supplies

A catchall category of "other expenses" is also typically included. Banks will often provide additional disclosure concerning the source of expenses either directly in the income statement or through footnotes.

Again, these groupings are not atypical, and serve as a useful framework to discuss a bank's operating expenses in more detail as follows.

Compensation Expense

A bank's largest noninterest expense is usually salaries and other payments to staff as compensation for their employment (compensation expenses). KeyCorp is not unusual in this respect. Staff compensation at KeyCorp amounted to 48.5 percent of total noninterest expense for FY10. It can also be seen that this expense has been steadily decreasing in the past three years. Some analysts compare staff costs to overall expenses, but the volatility of such expenses takes away the significance of that type of comparison. Note that in contrast to some nonfinancial service companies, where certain compensation expenses will be considered to be an integral part of the cost of providing the services, all compensation costs to a bank are as a rule considered to be operating expenses.⁶³

Occupancy Expense

Like all businesses, a bank must have *premises* in which to operate. The costs of renting office space and related services such as utility charges may be a significant expense, particularly to an institution with a large branch network. Such outlays may be classified under the heading of "occupancy expense." The next largest category of noninterest expense after compensation, KeyCorp's occupancy expense usually amounts to about 7 or 8 percent of operating expenses.

IT Expenses

Banking is increasingly a highly data- and technology-intensive industry. Unsurprisingly, the costs of maintaining the necessary infrastructure to meet data- and information-processing requirements are substantial, but they have dropped over recent periods. For KeyCorp, which classified these items separately as computer processing and equipment, the costs in the aggregate amounts to about 5 percent of operating expenses.

Marketing and Advertising

As service businesses that both are highly regulated in respect to their activities and deal in the ultimate fungible commodity—money—banks often confront highly

competitive markets. So marketing and advertising expenses can be substantial. KeyCorp spent 2 percent of its operating expense budget on these services.

Administrative and General Office Expenses

Like almost all enterprises, general administrative and office expenses represent a cost of doing business. There is no distinct boundary between this category and several others that may be broken out separately. Professional services such as those supplied by accountants, lawyers, and consultants are not infrequently included within this category, although they may be grouped separately or included within “other expenses.” KeyCorp, however, has broken out professional services (i.e., legal, accounting, and consulting services) as a separate line item.

Other Expenses

As with nonfinancial service companies, a bank’s other operating expenses will include a variety of miscellaneous costs, including those for noncash charges such as depreciation. The category of “other expenses” often functions as a catchall grouping to accommodate these miscellaneous items, which individually may be too small to itemize separately.

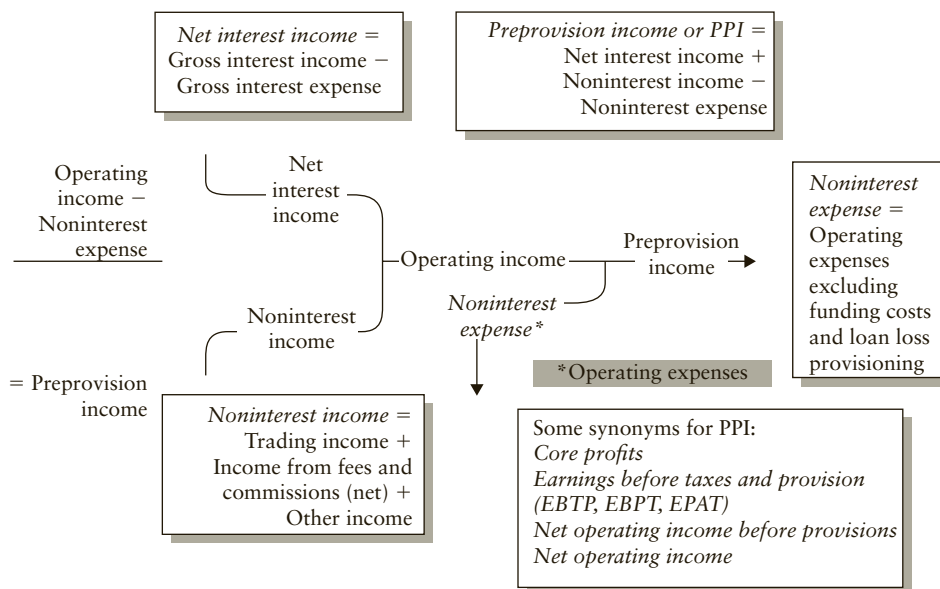
Preprovision Income (Net Operating Income, before Provisions)

When *noninterest expenses*, or *operating expenses*, are deducted from *operating income*, what is left is an important line item that is referred to by a variety of labels. For the sake of clarity, the primary term used in this book for this remainder item is *preprovision income* (or “PPI”) but there are other labels applied to it.

ALTERNATIVE TERMS FOR PREPROVISION INCOME

- Core profits
- Earnings before taxes and provisions (EBTP, EBPT, EPAT)
- Net operating income before provisions
- Net operating income
- Operating profit
- Operating profit before provisions
- Preprovision operating income
- Preprovision profit
- Underlying profit

Unless more particularly defined, each of these terms refers to precisely the same thing—that is, the sum of *net interest income* plus *noninterest income* (which can be negative) minus *noninterest expense*—and each has its proponents. A schematic of preprovision income is shown in Exhibit 4.19.

**EXHIBIT 4.19** Preprovision Income

As was implied earlier, the significance of PPI is that it represents a bank's earnings capacity adjusted for its operating cost base. In other words, it embodies the stream of income that, subject to deduction of credit costs, taxes, and nonoperating items, is available to restore, maintain, or improve a bank's financial health.

At this stage, it is relevant to remember that expected losses might conceivably be treated as recurrent expenses that should reduce the gross operating income well ahead of the provisions that would be needed on account of unexpected losses. Doing otherwise might result in credit losses that may mistakenly signal a weakness in the banks' management of its credit risk.

Deriving Net Income

After PPI is calculated, it only remains for *loan-loss provisions*, unusual or miscellaneous income or expense items (i.e., *nonoperating items*) and *taxes on income* to be taken into account to reach the bottom line: *net income*.⁶⁴ It is at this point, however, that the income statement *as reported* and the income statement *that is most desirable for analytical purposes* are likely to diverge. It is partly because of this divergence that the analyst will typically re-spread financial statements into a consistent format. (The other reason is to maintain consistency in financial data classification.)

Loan-Loss Provisions

Loan-loss provisions comprise the bank income statement line item that refers to a designated allowance for losses arising from the institution's lending activity. It corresponds to a corporation's *allowances for bad debts*, but is a much more significant expense item for banks. This is because the extension of credit, rather than

being an incidental part of a bank's business, usually forms the basis for a major part of its earning stream.

Loan-loss provisions superficially seem to be a noninterest expense like any other operating expense. But in a similar manner to depreciation, loan-loss provisions do not represent a cash expense. Instead, they are a noncash charge—an estimation of earnings foregone as a result of a borrower's failing to repay an advance in accordance with the terms prescribed. For this reason, loan-loss provisions are reasonably distinguished from other noninterest expenses.

Although loan-loss provisions are among the deductions made from operating income to compute net income (see Exhibit 4.20 at the conclusion of this section), this line item is one of the most critical on a bank's income statement. Loan-loss provisions reflect management's estimation, subject to regulatory guidelines, of the *loan losses* that the bank will experience within the relevant *fiscal period*.⁶⁵ They therefore function as a means to adjust for a bank's inevitable mistakes in extending credit—the *credit costs* a bank faces as a result of engaging in the business of lending. Although, in theory, banks ought to be able to predict their credit costs within a reasonable range, in practice they will often fail to gauge their credit costs accurately.

Loan-loss provisions are discussed in greater detail in Chapters 6 and 7, but for the moment it is important to be aware that disclosure and presentation of this expense item category probably varies more than any other line item on a bank's income statement. Some banks classify loan-loss provisions separately, highlighting this important item. Others show loan-loss provisions as merely another noninterest, or operating, expense, while still others subtract provisions from net interest income. DBS, Singapore's largest bank, follows this first approach, as shown in the excerpt from its recent income statements in Exhibit 4.20. KeyCorp follows the second procedure, deducting loan-loss provisions from net interest income to obtain another line item: *net interest income after provisions for loan and lease losses*. Each of these methods of presentation has its justifications. Some analysts prefer the first approach, while the second approach would provide a truer picture of net interest income if expected losses on riskier loans were to adjust downward some of the associated excess interest income.

EXHIBIT 4.20 DBS, Income Statements, 2008, 2009, and 2010, Excerpt

	2010 \$millions	2009 \$millions	2008 \$millions
Operating profit before provisions	3,113	3,929	3,109
Less: Loan-loss provisions	796	1,414	524
Operating profit	2,317	2,515	2,585

Source: Company reports.

In truth, loan-loss provisions should be viewed as a cost of engaging in the business of lending. Loan-loss provisions from this point of view represent a credit cost, so to speak—one not dissimilar to a bank's interest expense. Applying this logic, it is reasonable to deduct loan-loss provisions from net interest income.⁶⁶ From an analytical viewpoint, however, loan-loss provisions are important in their own right as an indicator of a bank's ability not only to manage its credit costs but also to avoid

problem assets. Additionally, they reflect a bank's ability to manage its funding costs, distinct from its ability to manage other expenses. Overall, it is probably best to analyze banks using both approaches separately. Meanwhile, in this book, however, we shall carry on dealing with loan-loss provisions separately from both interest expenses and noninterest expenses, so we define both *operating income* and *preprovision income* (sometimes called *net operating income*) as being exclusive of a deduction for loan-loss provisions. For the same reason, we make a distinction between *preprovision income* (also known as *net operating income*) and *pretax profit*, defining the former as exclusive of deductions for loan-loss provisions and the latter as inclusive of such a deduction.⁶⁷

As hinted above, conceptually, the traditional approach of considering a loan loss as a mistake has evolved in recent times. There should be nothing wrong for a bank to venture into riskier lending practices provided the higher interest margin associated with them more than compensates the bank for the higher risk. The loan loss is then simply viewed as an additional expense entered into with the view of generating more revenue. The insurance business operates on that model. Under that approach, loan losses—or *claims* in insurance parlance—should be divided into expected (or statistical) losses and unexpected losses, and their accounting treatments should be different. Strictly speaking, only the unexpected losses should be labeled “mistakes.”

Loan-loss provisions are both a financial flow item that registers on the income statement and—unlike most other expenses so enumerated—a *noncash* expense. A *noncash expense* (noncash charge) is an accounting entry that does not reflect an actual cash outflow. The charge against operating income that loan-loss provisions represent is analogous to charges for *depreciation*.⁶⁸

LOAN-LOSS PROVISIONS

While physical assets—for instance, a bank's computer system—are subject to the loss of value from wear and tear or obsolescence and therefore understandably attract a charge for depreciation, financial assets are vulnerable to a loss of value for other reasons. Loans and debt securities, for example, may lose value as a consequence of changes in prevailing interest rates and market prices. Another cause of a loss of value, which loan-loss provisions attempt to account for, is asset impairment, which occurs where the asset becomes non-performing, or becomes subject to higher credit risk.

Both loan-loss provisions and depreciation have a similar impact on the income statement—they diminish net income. Just as a physical asset with a short useful life engenders a large charge against income, so too does a loan asset highly vulnerable to default prompt the charge of substantial loan-loss provisions against it. But in the same way that depreciation charges can only attempt to approximate losses resulting from the consumption of the asset, and may not necessarily equal actual losses, so too do loan-loss provisions function merely to estimate credit costs while diverging considerably from the amount of actual loan losses.

Loan-loss provisions are a noncash charge against operating income made to account for expected or unexpected loan losses. They are customarily divided into two kinds:

1. General provisions, which are intended to account for loan losses not yet identified but anticipated as a result of historical experience.
2. Specific provisions, which are intended to account for loan losses attributable to identified problematic loans.⁶⁹

Other terms for loan-loss provisions⁷⁰ include:

- Allowance for loan losses
- Bad and doubtful debts
- Charge for bad and doubtful debts
- Credit costs
- Loan-loss provision expense
- Provisioning

Although the principle of setting aside an allowance for loan losses to account for *credit costs* is almost universally accepted by banks and bank regulators throughout the world, and although it is increasingly regulated, determining the appropriate amount of loan-loss provisioning beyond mandatory coverage can be difficult.⁷¹ There are several reasons banks often underestimate their credit costs and underprovision.

- First, during an economic boom, the resultant atmosphere of “irrational exuberance” blinds market participants to a developing *asset bubble*.
- Second, because provisions are deducted from operating income and therefore cut into the bottom line, management often has an incentive to keep provisioning costs to a minimum to prevent profits from being reduced and drawing the ire of shareholders and equity analysts.
- Third, tax regulations may discourage adequate provisioning to maximize bank profits and tax collection.

It is of course also possible that a bank may *overprovision* to reduce *tax liability*, or to *smooth* out profits. The use of loan-loss provisions to manage earnings is discussed later in the book.

Finally, it should be noted that *loan-loss provisions*, which are as a rule charged to a bank’s income statement, and *loan-loss reserves* on a bank’s balance sheet are entirely distinct (albeit related) categories.

- *Loan-loss provisions* are a financial *flow* item found on a bank’s income statement that corresponds to the loss estimation for a particular fiscal period.
- *Loan-loss reserves*—as opposed to such a charge attributable to a particular financial period—represent the *cumulative* sum of such estimated losses, less actual loan *write-offs* and *recoveries*.

In other words, loan-loss reserves are a *stock* item, while loan-loss provisions are a *flow* item.⁷²

LOAN-LOSS PROVISIONS VERSUS LOAN-LOSS RESERVES

Loan-loss provisions: The amount charged to the income statement for (specific and general) provisions against loan losses in a given reporting period. Loan-loss provisions are a flow item reset to zero at the beginning of a financial period.

Loan-loss reserves: The cumulative stock of loan-loss reserves at the end of a given reporting period after new loan-loss provisions have been added and write-offs of delinquent (nonperforming) loans have been subtracted. Loan-loss reserves are a stock item representing the cumulative amount registered at the beginning of a financial period adjusted for changes during that period.

Exhibit 4.21 illustrates the derivation of pretax profit, which is what is left after loan-loss provisions are deducted from preprovision income and nonoperating items are taken into account.

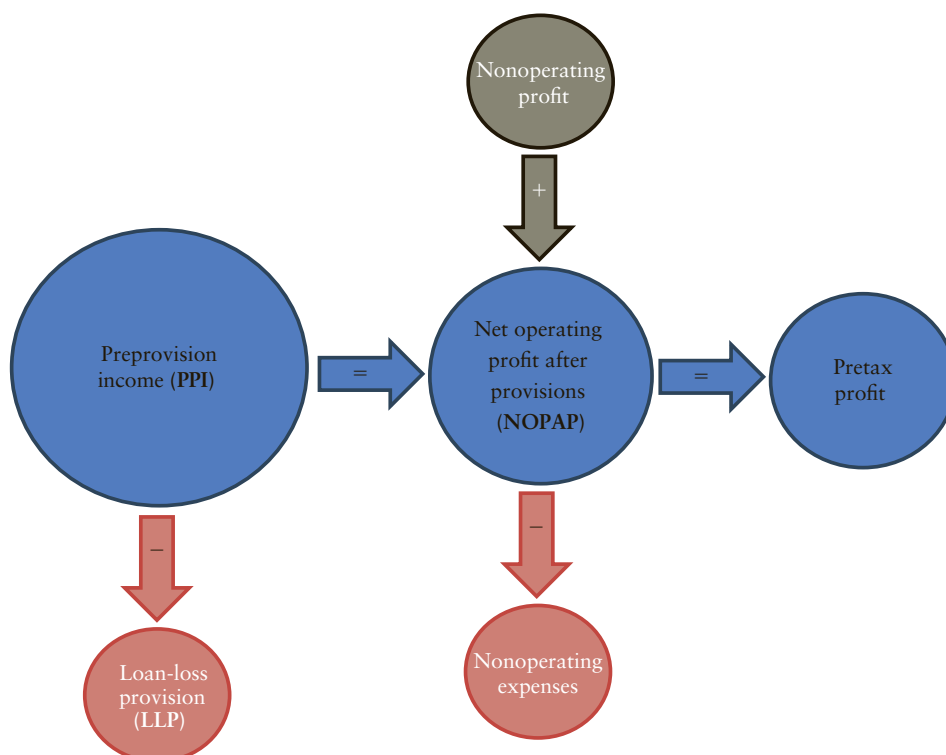


EXHIBIT 4.21 Pretax Profit

Net Operating Profit after Provisions

Deducting *loan-loss provisions* from *preprovision income* results in a remainder item that goes by a variety of names. As with the preceding category of *preprovision income*, there is no uniformly agreed-upon term for it. For the sake of clarity, we have termed this item *net operating profit after provisions*. There are alternative terms attached to the difference between *preprovision income* and *loan-loss provisions*.

ALTERNATIVE TERMS FOR NET OPERATING INCOME AFTER PROVISIONS

- Operating profit
- Operating profit after provisions
- Net operating income after provisions
- Pretax net operating income
- Net operating income

Nonoperating Items

Nonoperating items are income or expense items of an unusual character not associated with the enterprise's ordinary business activity. Customarily, these nonoperating items fall into the category of either *exceptional* items or *extraordinary* items.⁷³

Operating Items versus Nonoperating Items

By definition, *operating income* denotes an enterprise's *recurrent* income stream—that is, revenue generated by its main business activity. Similarly, *operating expenses* are those costs incurred as a result of the firm's main activity. Unsurprisingly, *nonoperating items* are gains and losses of a *nonrecurrent* nature. These income or expense items tend to be either *one-off* gains or losses, or income and expenses resulting from occasional or ancillary business activities. As a result, they have been traditionally separated from the main income stream for financial reporting purposes. *Exceptional* and *extraordinary* items, discussed later, are categories of nonoperating items.

Tax Treatment of Nonoperating Items

Because a company is liable for tax on all *net income* irrespective of whether it is *recurrent* or *nonrecurrent*, any *nonoperating* items, including *exceptional* or *extraordinary* items, typically remain *above the line* for computation of pretax income.⁷⁴ Although taxable income, they may be broken out for financial reporting purposes; that is, reported net income may be adjusted to take account of the distortions caused by nonoperating items. Even if not separated, *nonoperating items* are

appropriately distinguished for analytical purposes to show an enterprise's core earnings strength uncontaminated by atypical financial flows peripheral to the entity's core business activities. These adjustments, while sensible for the purposes of analysis, should be clearly distinguished from the organization's official reported net income.

Extraordinary or Exceptional Items

While the traditional designations of "extraordinary" and "exceptional" items are both nonrecurrent income or expenses, the traditional difference is as follows:

- *Extraordinary items* are those that arise outside of the bank's ordinary business (e.g., through the one-off sale of a subsidiary).
- *Exceptional items* are unusual gains or losses, income or expenses, arising in the course of the bank's ordinary business (e.g., a loan syndication fee as an arranger for a bank that rarely plays that role and has no intention of pursuing the loan syndication business).

From one year to the next, various atypical events may cause a bank's profits to fluctuate wildly. For example, if the institution sells a subsidiary and books a one-time gain, its net income may increase dramatically. Such nonoperating items, traditionally referred to as "extraordinary" or "exceptional" gains or losses (income or expenses) may make the institution's bottom line appear much better or worse than is justified by its ongoing business. For instance, absent the sale, it is conceivable that the bank's net income would have decreased. Local accounting rules differ in how these items are reported, and almost inevitably, a degree of discretion exists on management's part as to whether or not an item may be deemed a nonoperating item. Where comparatively strict accounting standards are enforced, nonoperating items will be clearly labeled as such. However, where accounting standards are lax, and often in respect to interim unaudited statements, closer scrutiny may be required as these categories can be used to obfuscate actual bank performance.⁷⁵

It is evident that there is a temptation to classify the unusual item in the manner that makes results look better:

- When an unusual event creates a *negative* result for a bank, there is an incentive to explain the adverse impact by noting that it is unlikely to be recurring.
- When the unusual event has a *positive* result, however, there is a disincentive to draw attention to the income being unlikely to be recurring.

In some situations, unusual items, although perhaps not contravening local accounting rules in being characterized as such, may be arguably viewed by the analyst as nonrecurrent, yet be properly deemed by the bank to be operating items under applicable rules. Because of the potential for abuse, under several accounting regimes, the use of nonoperating items has become subject to greater restrictions in the past 20 years. Some jurisdictions have restricted the use of the designation of extraordinary items to prevent abuse of this accounting category, and the classification is not used at all under international accounting standards (IAS).

ANALYTICAL PRESENTATION VERSUS FINANCIAL REPORTING PRESENTATION

For analytical purposes, it is important that the analyst spread (i.e., reformat) a bank's financials in a consistent manner and in a way that makes critical financial statement items most intelligible. Since financial reporting varies from country to country and banks often are allowed significant discretion in the presentation of their financial statements, the presentation of financial data for the purpose of credit analysis often differs from that used by the bank for official financial reporting purposes. Differences may be evident both in the sequence in which line items are presented and in the treatment of specific line items. Although differences in treatment can arise at any point in the income statement, the line item for loan-loss provisions is one that not infrequently is subject to a reordering or even readjustment by the analyst. For instance, KeyCorp shows loan-loss provisions as a deduction from net interest income rather than from PPI. (Compare DBS's presentation in Exhibit 4.19.) It should be possible, in fact preferable, from an analytical perspective, however, to analyze loan-loss provisions independently, whether they are deducted from PPI or from net interest income, as they are an important indicator of a bank's health, and since in doing so the respective effects of operating expenses and credit costs on core profits would be clearly revealed.

Pretax Profit

Pretax profit, also known as *pretax income* or *earnings before tax*, simply designates income that is subject to tax. Like preprovision income and net operating profit after provisions, *pretax income* is another indicator of core earnings capacity. There are alternative terms for *pretax profits*.

TERMINOLOGY FOR PRETAX PROFITS

- Earnings before taxes (EBT)
- Income before income taxes
- Net income before taxes
- Profit on ordinary activities
- Pretax income
- Profit before tax
- Profit before taxation

It is a short step from the computation of net interest income and noninterest income and expense to *pretax profit* (or *pretax income*), as can be seen in Exhibit 4.22.

EXHIBIT 4.22 KeyCorp, Pretax Profit, and Net Income

	2011	Change FY10 to FY11	2010	2009	2008	2008 and 2009 Combined
Income before income taxes [Pretax profit]	1,301	68.96%	770	-2,346	-1,134	-3,480
Income taxes	369	98.39%	186	-1,035	334	-701
Effective tax rate	28.36%		24.16%	44.12%	-29.45%	20.14%
Net income	932	59.59%	584	-1,311	-1,468	-2,779

Source: Company Reports.

Pretax profit is another important remainder item. Occasionally, the use of tax credits or additional tax liabilities may cause this figure to differ significantly from a normal percentage of net income, and these discrepancies should be noted. Another important function of this line item is to enable the calculation of tax-adjusted sensitivity analyses. In brief, these are what-if scenarios, which show the impact on the bottom line of differing accounting treatment of particular line items.

Exhibit 4.22 also illustrates that while *pretax income* and *net income* usually move in the same direction and at comparable rates, they can also move at significantly different paces. In some cases, they can move in opposite directions, and small percentage changes in certain line items can have seemingly a disproportionate impact on the bottom line. When, unfortunately, losses are recorded, the impact of taxes might get magnified.⁷⁶

ILLUSTRATION: IMPACT OF TAXATION

For example, Keycorp's pretax profit was negative in both financial year 2008 and financial year 2009, as was the case for many U.S. banks as a result of the subprime crisis. The bank paid income taxes in 2008 to the tune of US\$334 million even though the profits were negative. In 2009, the bank benefitted from a positive adjustment (a tax refund) of US\$1,035. An analysis of the profitability of the bank in 2008 and 2009 separately may lead to conclusions that are heavily influenced by the tax treatment in those two years. The net income of 2008 looks unnecessarily penalized by the income tax, while the tax relief in 2009 looks unnecessarily generous. By combining the years 2008 and 2009, it is possible to smooth out the impact of the tax treatment, and to show that the effective tax rebate rate at 20.14 percent (for 2008 and 2009 combined) remained in line with the effective tax rate usually achieved (24.16 percent in 2010, 28.36 percent in 2011, and in the vicinity of 25 percent to 30 percent in previous periods).

Taxation is ordinarily the only intervening item between *pretax profit* and *net income*. This line item only includes taxes imposed on the firm's income, whether at the national or local level, and does not generally include other types of taxes, such as tax on property, tax on sales, value-added tax, or miscellaneous fees for licenses or permits. The latter are usually deemed to be *operating expenses*.

Net Income

Net income, of course, is what remains after tax has been deducted from pretax profit and other adjustments have been made in respect of minority interest, income from affiliates, and other miscellaneous items. While vulnerable to manipulation through the discretionary treatment of individual constituent line items, the definition of net income is subject to little debate. It is certainly the one that is most frequently disclosed and is used in key ratios, such as *return on assets* and *return on equity*. For these reasons, it is probably the most important of the income remainder items, and certainly the one that receives the most attention in both the financial and popular press.

Net income for a bank, as for any enterprise, represents the *bottom line* of the income statement—that is, the *officially reported net income* figure. So the term *bottom line* has become a synonym for net income. As the final stage in a series of deductions from total income, net income is computed by deducting taxation from net income subject to taxation (taxable income). While it is a universal income statement item, it is given a variety of monikers.

The inflows of revenues, the outflows of expenses, and the derivation of remainder items are shown schematically in Exhibit 4.23, which represents a reconfigured composite of the flow diagrams in this section.

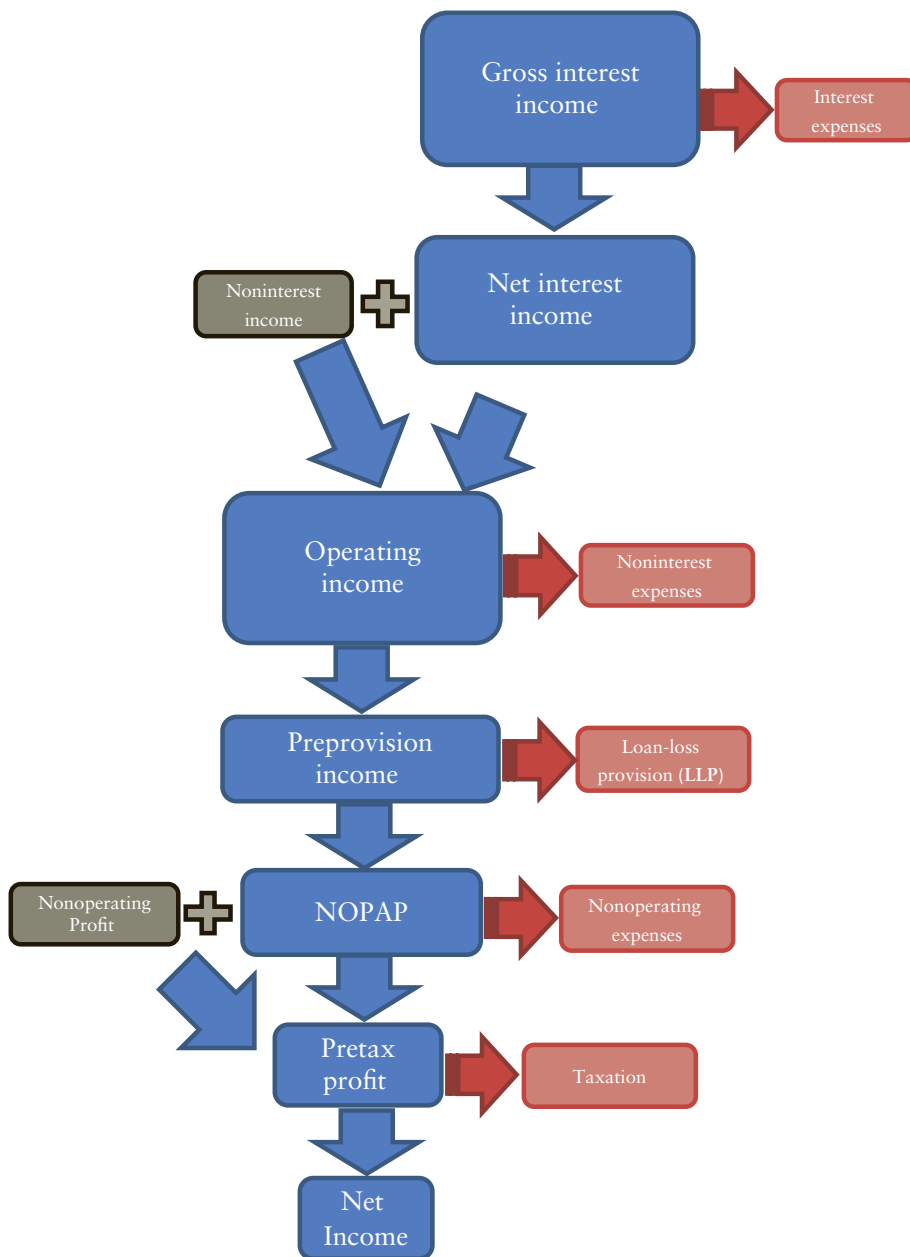
ALTERNATIVE TERMINOLOGY FOR NET INCOME

- Net profit
- Net earnings
- Earnings after tax
- Net profit attributable to shareholders
- Net profit attributable*
- Attributable profit*
- Profit after taxation
- Net income after tax

*Sometimes the terms marked with an asterisk are used almost interchangeably with net income. Technically, however, these terms refer to the portion of net income to which the ordinary (i.e., common) shareholders are entitled.

Minority Interest

Minority (minorities') *interest(s)* refers to the portion of a partly owned subsidiary's net income that is nominally owned by shareholders other than the bank's shareholders.⁷⁷ Income attributable to the minority shareholders is computed at the consolidated level.

**EXHIBIT 4.23** Composite Financial Flow Diagram

MINORITY INTEREST: AN ILLUSTRATION

Minority interest refers to the ownership interest in subsidiaries of a given company, where a given *subsidiary* is *not wholly owned* by the company, and therefore a minority ownership interest in the subsidiary is in the hands of

noncompany shareholders. To reflect that a portion of the income generated by the company's non-wholly owned subsidiaries does not belong to its shareholders, adjustments are made in consolidated financial statements. To illustrate how this works, assume that hypothetical Tulip Bank has a single non-wholly owned subsidiary, Subco, which is 80 percent owned by the bank and 20 percent by other shareholders. Subco generates 5 percent of Tulip Bank's consolidated net income, which was £20 million last year. In other words, Subco generated £1 million of total earnings and one-fifth (20 percent) of that earning stream was attributable to nonbank shareholders. At the consolidated level, the minority interest would be £200,000 or 1 percent of consolidated net income. This amount is subtracted to calculate net income attributable to Tulip Bank's ordinary shareholders, which in this case would be £19.8 million.

Rules governing the recognition of minority interest vary. Ordinarily, however, where the proportion of ownership in the subsidiary or affiliate reaches a qualifying level, an appropriate amount representing the interest of minority shareholders in the bank's earnings is deducted to obtain *net income attributable to shareholders*. Rules under U.S. GAAP reflect this approach.

TREATMENT OF MINORITY INTEREST ON THE INCOME STATEMENT UNDER U.S. GAAP

In the United States, not specific to the banking context, three possible scenarios are recognized for the treatment of minority interest. They depend upon the ownership interest of the reporting entity in the affiliate or subsidiary.

- Where the ownership interest is 20 percent or less, the *cost method* is generally used. In this case, the ownership interest is akin to a strategic investment. The reporting enterprise may not include within its own consolidated income any portion of the net income of the company in which it has an ownership interest other than dividends received as a result of its equity ownership.
- Where the ownership interest is more than 20 percent but less than 50 percent, the reporting enterprise may include within its consolidated income as designated revenue from affiliates its *proportional* share of its affiliate's net income, even if it receives no dividends from it.
- Where the ownership interest is 50 percent or more, the reporting enterprise shows 100 percent of the subsidiary's net income on its income statement, but then makes a *deduction* for the portion owned by minority shareholders. The deduction itself is commonly referred to as minority interest or interests.

Deutsche Bank

Financial Report 2010

02 – Consolidated Financial Statements

Consolidated Statement of Income

Consolidated Statement of Income

in €m.

Notes

2010

2009

2008

Interest and similar income

6

28,779

26,953

54,549

Interest expense

13,196

14,494

42,096

Net Interest income

15,583

12,459

12,453

Provision for credit losses

1,274

2,630

1,076

Net Interest Income after provision for credit losses

14,309

9,829

11,377

Commissions and fee income

10,669

8,911

9,741

Net gains (losses) on financial assets and liabilities

(Omitted Material)

Total interest expenses

23,318

20,120

18,278

Income (loss) before income taxes

3,975

5,202

(5,741)

Income tax expense (benefit)

34

1,645

244

(1,845)

Net income (loss)

2,330

4,958

(3,896)

Net income (loss) attributable to noncontrolling interests

20

(15)

(61)

Net income (loss) attributable to Deutsche Bank shareholders

2,310

4,973

(3,835)

Observe that in this income statement, net income and net income attributable to Deutsche Bank shareholders are nearly identical. The net income attributable to minority interest or preferred shareholders is negligible.

EXHIBIT 4.24 The Bottom Line: Deutsche Bank, Income Statement, 2010, Excerpt

Source: Company reports.

Note that in many cases there will be no *preferred shareholders*—further discussed below—nor any material minority or associates' interest, so that *net income* (without further qualification) will be practically equivalent to net profit attributable to shareholders. Observe that in Deutsche Bank's 2010 income statement excerpted in Exhibit 4.24, the line items for net income attributable to common shareholders, at EUR 2,310 million, and net income are almost identical.

Income from Affiliates (Associates)

Income from affiliates (income from associates) is the mirror image of minority interest. That is, it refers to income attributable to a minority shareholder of an entity in which the majority of the shares are owned by another entity. Reporting and consolidation treatment of this item vary.

Preferred Shareholders

Shareholders can be divided into common or ordinary shareholders and preferred or preference shareholders. Common shareholders possess a pure equity stake in the firm whose shares they hold. Preferred shareholders' interest resembles that of a creditor in that they possess some preferred right to periodic dividends. They are termed preferred shareholders because they rank higher than *common shareholders* (*ordinary shareholders*) both in the issue of dividends and in the event of the liquidation of the company or its insolvency (but below junior and senior debt holders and most, if not all other creditors).

Although *preferred shareholders* are nominally owners of the entity in which they hold shares, in reality their interest is a hybrid between equity and debt. The

degree to which it inclines toward one or the other depends upon the terms under which the preferred shares are issued. Unlike common shareholders, who possess only a residual *equity interest* in the company (equity being what, if anything, remains after the company's liabilities are subtracted from its assets), the right of preferred shareholders to this equity is generally—but not always—limited to the extent of their investment or some other liquidation value plus dividends owed. For this reason, equity analysts frequently set aside the portion of net income attributable to preferred shareholders since the common shareholders on whose behalf they evaluate a company are not entitled to this residual claim or income stream.

When evaluating bank credit risk, the relevance of *net income attributable to preferred shareholders* depends upon the purpose for which the analysis is undertaken. In either case, however—whether an equity analysis or a credit analysis is performed, the amount, on a percentage basis, is usually comparatively small and often not material to the review. Preferred shares are most relevant to bank credit analysis in the context of capital adequacy, which is discussed in Chapter 9.

PREFERRED SHARES DEFINED

Preferred shares are shares that in contrast to common shares give their holder a preferential right to a defined dividend ahead of common shareholders, provided sufficient net profits exist out of which to pay such dividends. Preferred shareholders also rank above common shareholders (but below bondholders and other creditors) in the event of the liquidation of the company and the subsequent distribution of its assets.⁷⁸ Another key difference between preferred shareholders and common shareholders is that normally the former have no voting rights, although, depending upon the terms of under which their shares have been granted, they may acquire voting rights in particular circumstances. Preferred shares are typically issued at par. Unlike common shares, par is the principal amount of the shares and usually more closely corresponds to the actual value of the shares than the par value of common shares corresponds to their actual value. In this and other ways, preferred shares resemble fixed-income securities such as bonds, and the pricing of preferred shares follows the same general principles as bond pricing. Preferred shares can have many of the same attributes as bonds, such as callability, convertibility, and being subject to *sinking fund requirements*.⁷⁹ Unlike most bonds, however, preferred shares are ordinarily perpetual, and have no fixed maturity, being issued for an indefinite length of time. Another difference is that the right of preferred shareholders to be paid dividends may vary with circumstances as specified in the agreement between the issue and issuer.

Net Income Attributable to Shareholders

Net income attributable to shareholders is the final bottom line, after the portion of a bank's earnings attributable to minority shareholders has been deducted.

Assuming for the present an equivalence between *net income* and *net income attributable to shareholders*, the income stream reaches the apparent terminus of its journey with the so-called bottom line. But is the journey, in fact, complete? What happens to the income that a bank, or any business, generates?

There are two courses available as illustrated in Exhibit 4.25.

The first is that *net income* may be channeled *outside* the entity and be distributed to shareholders as *dividends*.

Dividends are discretionary payments to common shareholders and, in the case of preferred (preference) dividends, obligatory payments to *preferred* (preference) *shareholders*. As was mentioned, depending upon the terms of the security instrument, preferred shareholders are normally entitled to a fixed dividend ahead of common shareholders, but behind creditors.

The second possibility is that net income may be channeled back into the entity and reinvested in it. In other words, *net income* may be either funneled back into the enterprise as *internally generated capital* or returned in the form of *dividends* to shareholders.

Taking away dividend payouts leaves *retained earnings*. Retained earnings represent profits plowed back into the bank's business. In other words, they represent a component of the bank's capital that has been *internally generated*, as opposed to the injection of new contributed funds.

In this way, the income statement and the balance sheet are directly linked. Note that in either case, the owners of the enterprise benefit. They either have cash in hand or shares in a better-capitalized company and, owing to its greater net worth, one that is of greater value.

Exhibit 4.26 summarizes the key elements of the bank income statement and the bank balance sheet, discussed in greater detail in the next chapter, side by side. It also

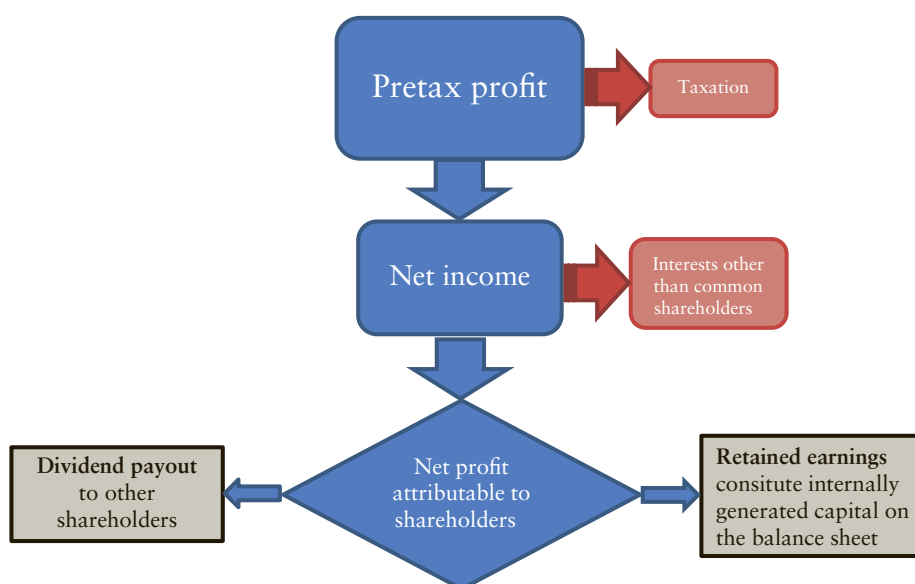


EXHIBIT 4.25 Net Income and below the Bottom Line

Balance Sheet		Income Statement	
Liquid assets	Deposits	Interest income	X
Marketable securities		Less: Interest expenses (i.e., funding expenses)	(X)
Loans	Other short-term funding	Net interest income	NII
		Add: Noninterest income	X
Other assets	Shareholders' equity	Less: Operating expenses (excluding loan loss provision)	(X)
		Preprovision income	PPI
		Less: Loan-loss provisions	(X)
		Net operating profit after provisions	NOPAP
		Add: Nonoperating income	X
		Less: Nonoperating expenses	(X)
		Profit before tax	EBT
		Less: Taxation	(X)
		Net income	NII
		Less: Minority interest	(MI)
		Net income attributable to common shareholders	NISH
		Dividend payout	(D)
		Retained earning	RE

EXHIBIT 4.26 Retained Income Is Internally Generated Capital

shows the link between the two, retained earnings adding to shareholders' equity. Note that the increase in shareholders' equity is accomplished indirectly. Since net income represents new assets, usually in the form of cash, upon which there is no outside claim, there is no increase in liabilities corresponding to the putative cash increase with the result that shareholders' equity rises.

INCOME STATEMENTS UNDER IAS

So far we have been discussing the accounting treatment of various bank financial statements without reference to any particular accounting regime. That will remain the case for the most part throughout the book since our objective is to consider the financial disclosure of various items from a general analytical perspective, rather than from the vantage point of compliance with particular accounting rules. Mention ought to be made, however, of the accounting framework known as international accounting standards, or IAS, which are bringing about a degree of global convergence in the financial reporting of banks. Widely adopted in Europe, IAS and U.S. GAAP represent the two leading accounting regimes used worldwide. Presentation of the income statement under IAS is quite similar to that used in U.S. GAAP and other commonly applied methods of financial reporting. This can be seen in the specimen income statement depicted in Exhibit 4.27.

Nevertheless there are some significant differences between the treatment of various line items under IAS compared with other accounting regimes, and from time to time we will mention specific differences. A detailed cross-comparison of the IAS or other bank accounting frameworks is, however, outside the scope of this book.

EXHIBIT 4.27 Specimen Income Statement under IAS

		Year Ended		
		Dec. 31, 200X + 1	Dec. 31, 200X	
A	Interest and similar income	7,750	6,937	
B	Interest expense and similar charges	−4,982	−4,365	
C	Net interest income	2,768	2,572	$C = A + B$
D	Fee and commission income	1,172	1,117	
E	Fee and commission expense	−51	−56	
F	Net fee and commission income	1,121	1,061	$F = D + E$
G	Dividend income	93	35	
H	Net trading income	287	269	
I	Gains less losses from investment securities	49	120	
J	Other operating income	139	105	
K	Impairment losses on loans and advances	−128	−146	
L	Operating expenses	−2,993	−2,737	
M	Operating profit	1,336	1,279	$M = C + F + G + H + I + J + K + L$
N	Share of profit of associates	10	10	
O	Profit before income tax	1,346	1,289	$O = M + N$
P	Income tax expense	−403	−401	
Q	Profit for the year	943	888	$Q = O + P$
	Attributable to:			
R	Equity holders of the Company	933	878	
S	Minority interest	10	10	
		943	888	

Source: PricewaterhouseCoopers.

NOTES

1. Terry Smith, *Accounting for Growth: Stripping the Camouflage from Company Accounts* (London: Century Business, 1992). The comment was made in respect of Midland Bank's attempt in 1987 to classify its provisions for lending to less developed countries as an extraordinary rather than as an exceptional item.
2. These assets are referred to as *earning assets*, short for *interest-earning assets*.
3. The income statement is also commonly known as the *profit and loss statement* or P&L.
4. A bank usually has two or more categories of operating revenue, while a nonfinancial company's operating revenue usually comes almost solely from sales.
5. Where for most nonfinancial companies the allowance for bad debts is a minor component of the income statement, its counterpart on a bank's income statement—often called loan-loss provisions—is of major importance.
6. The term *financial flows* is used here as a synonym for revenue and expenses and to distinguish these from cash flows.
7. The accrual method is based on the accrual principle, which stands for the following proposition: that income and expenses should be reported when *earned* or *incurred* (e.g., where an offer has been made and accepted and legal obligation has consequently

resulted), irrespective of the timing of *cash flows*—that is, whether the cash flows corresponding to the income or expense *item* have been received or paid in the same reporting period. The accrual principle is a corollary of the matching principle that requires that a firm match expenses incurred with the revenues that result.

8. Conversely, income may be understated if, for example, expenses are overstated or exaggerated. The latter situation can occur where a bank sets aside much greater loan loss provisions than is appropriate for its circumstances. Note that the term *cash incomings* is used here as a synonym for positive or inbound cash flows.
9. *Nonperforming* is bank jargon for *in default* or *delinquent*. So the term *nonperforming loan*, abbreviated NPL, generally refers to a loan that is *delinquent*, that is, no longer performing and generating interest according to the terms of the governing loan agreement. As discussed later in the book, the definition of an NPL is often highly specific in practice.
10. Recall that under the accrual principle (see note 7) an expense is not registered until it is *recognized*.
11. Suppose, for instance, that economic conditions began to deteriorate toward the end of a financial year, and the number of borrowers who fail to pay back their loans in accordance with the loan terms exceeds estimates. In this situation, it would not be uncommon to find that the bank has accrued, and will therefore record, a certain amount of *interest income* (also referred to as *interest revenue*), which may very well exceed the actual amount of cash interest payments ultimately received from borrowers. For this reason, *best practice* in *bank accounting* requires that interest income on nonperforming loans cease to be accrued and be placed into a *suspense account*. Furthermore, such interest already accrued should be *clawed back* (i.e., deducted) where such cash interest payments are not ultimately received.
12. While gross interest income jumped sharply as prevailing interest rates rose, interest expense rose even more rapidly causing the decline in net interest income.
13. This characteristic results from the nature of a bank as a highly leveraged financial intermediary. As opposed to a bank, a nonfinancial firm will have a low proportion of financial assets to nonfinancial assets on its balance sheet. When it comes to off-balance-sheet items, that difference may prove even more substantial.
14. To be sure, the same could be said of capital-intensive nonfinancial companies where machinery and raw materials must be purchased before products can be manufactured, sales made, and income earned. One difference between the two is that a bank generally begins earning interest immediately upon the advance of funds, while for a nonfinancial firm there is almost always a significant lag between the establishment of productive capacity and the receipt of income. Another is the difference in the liquidity requirements of the two types of enterprises, discussed later.
15. A bank's situation differs from a nonfinancial firm in yet another respect. Once established and past its early growth phase, it is the rare nonfinancial company that will be constantly in need of new outside funding. This need, however, is intrinsic to fractional-reserve commercial banking, which in many countries has an incessant appetite for new outside sources of funding to be used for the purpose of advancing those funds to others.
16. Of course, as does any commercial enterprise, a bank needs cash to pay its bills, for the use of premises, electricity, and so on. But unlike nonfinancial companies, what it trades is money that can be immediately exchanged for other goods and services, so its situation is significantly different from such firms.
17. Another term for *interest-bearing financial obligations* is *interest-bearing liabilities*. If an obligation bears interest, it is by definition a financial obligation.
18. Unusual circumstances may result from a run on the bank's deposits or exceptionally *volatile* market conditions. A *bank run* on deposits occurs when depositors, as a

consequence of their losing confidence in the soundness of a bank, attempt to withdraw their deposits en masse. Even if the loss of confidence is based solely on rumor and not reality, a bank run may be self-fulfilling and those depositors who are not convinced of the institution's lack of soundness initially are likely to be persuaded—as other depositors withdraw their funds—that the bank's loss of *liquid assets* will inevitably place it in a precarious position, and that, from a rational perspective, it is in their best interest to join the line for withdrawals. A bank run thus creates a vicious cycle, and rumors without foundation may be transformed into self-fulfilling prophecies. The Federal Reserve Bank of Minneapolis has defined a bank run as “a series of unexpected cash withdrawals caused by a sudden decline in depositor confidence or fear that the bank will be closed by the chartering agency . . . since [a bank's] cash reserve . . . is only a small fraction of its deposits, a large number of withdrawals in a short period of time can deplete available cash and force the bank to close and possibly go out of business” (Federal Reserve Bank of Minneapolis, Glossary, www.minneapolisfed.org/glossary.cfm).

19. This is inherent in the fractional-reserve nature of commercial banking, in which its loans and other advances almost always exceed liquid reserves.
20. See note 18 defining the term. Federal Reserve Bank of Minneapolis, Glossary, www.minneapolisfed.org/glossary.cfm.
21. Bank runs will be further discussed in Chapter 10 on liquidity.
22. This is not to say that there may not be situations in which a bank cash flow statement may not be useful.
23. Attributed to Carter G. Woodson, eminent Afro-American educator. www.brainyquote.com/quotes/authors/c/carter_g_woodson.html.
24. “[S]ince [a bank's] cash reserve . . . is only a small fraction of its deposits, a large number of withdrawals in a short period of time can deplete available cash and force the bank to close and possibly go out of business” (Federal Reserve Bank of Minneapolis, Glossary).
25. There are many variants to the terminology used but in most cases they refer to the same item, as defined by applicable accounting standards. To paraphrase Shakespeare, a rose by another name is still a rose. The actual language comes from the play *Romeo and Juliet*. See William Shakespeare, *Romeo and Juliet*, II, ii, 1–2. “[W]hat's in a name? That which we call a rose by any other name would smell as sweet.”
26. Naturally, a corporation's *nonoperating* revenues *could* come from interest income, although these sources will generally be a minor or incidental part of the company's business. Note that some businesses such as property development companies closely resemble financial institutions, which characteristically generate income through rental income (which is analogous to interest income) and investment gains (analogous to trading and investment gains). They are sometimes grouped together with financial institutions.
27. Not all institutions, to be sure, are always as forthright in making their credit costs as conspicuous as might be desired, and sometimes loan-loss provisions are either not disclosed or must be ferreted out with some difficulty.
28. Note that from this point in the book, comparisons with corporates become less frequent.
29. As an illustration, the term *net operating income* has several different meanings that refer to different line items. So the use of this term can create confusion unless it is precisely defined.
30. Another term for *financing leases* is *financial leases* or *finance leases*.
31. The term *coupon payment* (*coupon*) refers here to the periodic interest payment made by the issuer of a fixed-income security to bondholders. The coupon is typically payable semiannually as in the U.S. bond market, or annually as in the eurobond market.
32. *Debt securities* are *fixed-income* securities—that is, securities where payment to the holder is predetermined—such as a *bond*, *commercial paper*, or a *Treasury bill*. The term

debt instrument is largely synonymous with the term *debt security*, but the latter tends to connote nominally *marketable* or *negotiable* paper while the former embraces *non-marketable* and *nonnegotiable* instruments such as many conventional loans.

33. Collectively, revenue from all these sources is classified as interest income.
34. In the latter case, the lessee will usually receive credit against the purchase price for the lease payments made.
35. We are referring here specifically to accounting income—that is, income as reported—calculated on an accrual basis. Accrual accounting contrasts with cash-based accounting. The latter calculates net income purely on the basis of cash receipts and cash disbursements within a given financial period, and as a consequence bears a close similarity to the statement of cash flows from operations. Cash-based accounting, however, is relegated primarily to very small businesses that either lack or do not require the sophistication to report their financial results in the more conceptual manner that accrual accounting permits. Note that, in theory, there are a number of ways to define income: as *economic earnings* (net cash flow + change in asset value); *distributable earnings* (amount of earnings that can be paid out in dividends without altering a company's value); *permanent earnings* (a firm's market value multiplied by the required rate of return); and *sustainable income* (income sustainable at a given level of capital investment). See Gerald I. White, Ashwinpaul C. Sondhi, and Dov Fried, *The Analysis and Use of Financial Statements*, 2nd ed. (New York: John Wiley & Sons, 1998), 116. Unless otherwise specified, references to income in this text are to accounting income, meaning income as reported in a company's financial statements.
36. For KeyCorp that last category is only about 1.5 percent of gross interest income, not a material amount. Note that a further analysis of interest income by type of lending is not provided by the bank, although the balance sheet, which we will examine in the next chapter, provides more detail about the composition of the bank's loan book.
37. The *money market* is the market for short-term debt instruments, including *repos* and *reverse repos*, *treasury bills*, *interbank placements*, *negotiable certificates of deposit*, *bankers' acceptances*, and *commercial paper*.
38. The *interbank market* is the market among banks for *short-term funds*, often *overnight funds*.
39. *Customer deposits* are also referred to as *core deposits*—although core deposits might sometimes include stable or relatively stable deposits from other sources.
40. In contrast to *retail banking*, *wholesale banking* is mainly concerned with lending and providing other financial services to large or highly creditworthy organizations, including major corporations and government agencies, and to sound banks and institutional investors. Wholesale banks usually engage in a smaller number of high-value transactions, while the retail bank business model requires a high volume of relatively low-value transactions. As a rule, wholesale banks maintain smaller branch networks than their retail counterparts and correspondingly, given their reduced access to consumers, rely more upon institutional and commercial depositors and creditors for their funding.
41. Purchased funds are often described as being sourced from money markets, but they can as easily define funds that are actively sought by a financial institution, as opposed to being deposited with it.
42. What elsewhere might be classified under the rubric of “bank deposits” here falls within the category of “federal funds purchased,” U.S. terminology for interbank borrowing, which amounted to about 5 percent of total funding costs.
43. Net interest income is a crucial figure to any bank, not only because it is a more stable indicator than interest income or interest expense individually, but also because it ordinarily comprises a large portion of a bank's operating revenue and changes in net interest income are likely to have a more direct impact on the bank's bottom line. Moreover net

- interest income figures as highly in such key profitability measures as the net interest margin, a primary indicator of the profitability of a bank's spread business.
44. In some circumstances, gross interest income could actually increase while net interest income decreases. This scenario will occur when a bank's interest expenses rise at a sharply higher rate than its interest income. An institution that is excessively reliant on money-market funding and also locked into low-yielding term loans may find itself in this situation. We will return to this topic in a future chapter, but the lesson for the moment is that looking at interest income or interest expense alone tells us little; instead, net interest income is the more significant metric.
 45. With increasing disintermediation and emphasis on noninterest income, its importance is diminishing, but net interest income still remains a core performance metric.
 46. In other words, while interest income and interest expense may vary greatly depending upon prevailing interest rates and economic conditions, any advantage or disadvantage caused by a change in one is frequently cancelled out by the corresponding movement in the other. For instance, if gross interest income increases from \$80 million to \$120 million in a given financial period, but gross interest expense increases from \$30 million to \$70 million, there would be no change to net interest income, which would remain flat at \$50 million.
 47. Changes in the shape of the yield curve, discussed in a later chapter, can have a similar effect.
 48. *Spread income* refers generally to *net interest income*.
 49. *Noninterest income* may be defined as the sum of income from fiduciary activities; service charges on deposit accounts in domestic offices; investment banking, advisory, brokerage and underwriting fees; venture capital revenue; net servicing fees; net securitization income; income from derivatives; insurance commissions and fees; net gains (losses) on sales of loans and leases; net gains (losses) on sales of other real estate owned; net gains (losses) on sales of other assets (excluding securities); and other noninterest income. Many of those items can be measured on a net basis, with any losses taken against profits, and as a result they could be negative.
 50. A bank's trading income may be distinguished from its investment income. Investment income can be broadly defined as all income arising from debt securities holdings. It is comprised of two components: coupon interest and trading gains. The first is usually considered to be part of net interest income, while the second is usually considered to be part of noninterest income, specifically trading income. Actual treatment will depend upon local accounting rules and how the securities holdings are classified.
 51. Except for investment banking and L/C and loan fees—and ignoring very volatile gains or losses from loan sales and principal investment—all categories of noninterest income have either declined or displayed very modest growth at KeyCorp over the three years represented in the data in Exhibit 4.14. The analyst reviewing KeyCorp's figures for the first time is likely to wonder why these categories of income have increased while others have declined or remained relatively flat, with a stable level of noninterest income. A full analysis of the notes to the accounts is always needed when attempting to assess the value or the creditworthiness of a bank. The financial crisis that started in 2007 certainly had an impact on that source of income. Comparisons with figures for previous years and with peer banks would be interesting.
 52. Most of the time, no breakdown between the letter of credit fees and loan fees is provided directly in the income statement. Further examination of the notes to the financial statements and other financial disclosure is required to shed further light on this point.
 53. *Mortgage-servicing fees* are fees charged by a bank to administer and collect mortgage payments from mortgage holders.
 54. For example, the state of Florida in its instructions for semiannual income statement reporting mandates that "non-yield-related fees, such as mortgage servicing fees and

- syndication fees applicable to loans which are not assets of the [bank]" be included as "service charge income." State of Florida, Department of Financial Institutions, *Instructions, Semi-annual Statement of Income* [for commercial banks].
55. *Corporate-owned life insurance* is a form of self-insurance vehicle used by businesses to insure key employees. It has a number of tax advantages that have encouraged its use in the United States. Since in common with some types of life insurance policies, the product has both an insurance element and a savings element, accrued income may be generated by such a mechanism insofar as premiums exceed insurance costs creating interest or investment income associated with the insurance policy. See U.S. Government General Accounting Office, Report GAO-04-303, *Business-Owned Life Insurance: More Data Could Be Useful in Making Tax Policy*, May 13, 2004. Because of the long-term nature of corporate-owned life insurance policies, and the inability of the policy holding company to realize the income until the termination of the policy, such income is generally of low quality for analytical purposes.
 56. Any interest earned by the bank during the period it holds trading securities is generally considered to be *interest income* as opposed to *noninterest income*.
 57. Some make a distinction between *operating revenue* and *operating income*, calling what we term here *operating income*, "operating revenue." Applying this usage, a line item farther down the income statement is referred to either as *operating profit* or *operating income*. We prefer not to rely on such nuanced differences in terminology. Note that in the United States, regulators employ the expression *adjusted operating income* to refer to "the sum of net interest income on a *taxable equivalent basis* and noninterest income." The item "adjustments for tax equivalency," however, is often omitted by analysts at this level of the income statement.
 58. In the excerpt from the consolidated income statement for Deutsche Bank in 2010, "consolidated" means that the statement encompasses the bank, which comprises the largest part of the broader enterprise, as well as any affiliated subsidiaries (which may also publish separate accounts). Unless the banking operation forms a relatively minor part of a larger group, it is preferable to use consolidated statements for counterparty credit risk purposes.
 59. In other words, although the most common practice is to define *total income* as *gross interest income* plus *gross noninterest income*, some banks define it as *net interest income* plus *gross noninterest income*.
 60. *Overhead* may be defined as total noninterest expense including salaries and employee benefits, expense on premises and fixed assets (net of rental income), *amortization expense* of intangible assets, and other noninterest expense.
 61. A legal firm, for example, will typically bill clients on the basis of the law partner's or associate's hourly rate.
 62. The term *cost base* is sometimes used to refer to a bank's operating expense (noninterest expense) burden.
 63. As with all enterprises, net income forms the proverbial bottom line of the bank's income statement.
 64. As we approach the bottom of the income statement, we approach the harder reality—what the bank actually generates in *net profit (net income)*. Confusingly, some labels for the line item that is the subject of this subsection are apt to result in misunderstandings. For example, occasionally the term *operating profit* is used to refer to what is left after loan-loss provisions and noninterest expenses (operating expenses) have been subtracted from the sum of net interest income and noninterest income. If so applied, it would not be synonymous with the term *operating income*, which is almost universally used to designate the total of *net interest income* and *noninterest income*. Similarly, the term *net operating income* may also be ambiguous—exactly what is it *net* of? Although not in common parlance, we prefer to use the phrase *net operating profit after provisions* (or

NOPAP) to refer to this line item. It has the virtue of being self-explanatory while pointing to the terminology in wider use—that is, *operating profit*, *operating profit after provisions*, and *net operating income after provisions*. Although *net operating profit after provisions* is a bit cumbersome, it is not apt to be misconstrued with any other term. In addition, the acronym NOPAP, or “no pap” (*pap* meaning soft food or “worthless or oversimplified ideas”) may be useful as a mnemonic. Obviously, in practice, the analyst can apply whatever label is accepted practice at the institution for which he or she works, or choose his or her own. The purpose here is not to impose any standard usage, but merely to communicate our meaning clearly.

65. The term *fiscal period* is used synonymously in this book with *accounting period*, *financial period*, *financial reporting period*, and *reporting period*. The term *fiscal year* or *financial period* is abbreviated as FY. For example FY12 means the financial year corresponding to 2012, regardless of which month of that calendar year marks the end of the financial year.
66. In actuality, however, loan-loss provisions can only approximate credit costs, which usually (but not always) more closely correspond to net charge-offs. This may be affected by the applicable regulatory framework governing provisioning and charge-offs.
67. The reader should be aware that there are a variety of approaches to accounting for loan-loss provisions, and this is but one of them, albeit one that makes clear distinctions between each critical element of the income statement.
68. Recall that depreciation functions as a present (and periodic) *noncash* deduction for a past cash expenditure meant to account for the impairment of an asset's value over time. Recall also that neither a company's capex nor—quite obviously—a bank's advances to customers in the form of loans are recorded on the income statement as expenses. Depreciation functions as an indirect indication of capex expenditures on the income statement since it accounts for the consumption of the capital assets over several periods. Arguably, charges for the impairment of a physical asset could represent a better analogy. Also, when a physical asset suddenly loses value, as when a factory is decimated by fire, it is as a rule immediately written off, rather than being subject to an intermediate provisioning stage.
69. The respective categories of general provisions and specific provisions predate the Basel accords and do not dovetail closely with the concepts of economic capital and unexpected vs. expected loss. The two categories are explained further later in the book. From a risk management standpoint, loan-loss provisions are meant to account for probable loan losses—that is, expected loss as discussed in Chapter 1.
70. As with the other income statement classifications, a variety of phrases is used to designate this important line item. Aside from differences in terminology, it also is worth drawing attention to the distinction between loan-loss provisions and loan-loss reserves. The phrase “loan-loss reserves” is frequently used to refer to cumulative net loan-loss provisions reflected on a bank's balance sheet, as opposed to the charge made periodically to the income statement.
71. Regulations may in some situations discourage banks from being overly prudent in their provisioning policy.
72. Loan-loss accounting and the relationship between loan-loss provisions and loan-loss reserves are discussed in greater depth with the topic of asset quality in Chapter 7.
73. The difference between the two categories is discussed later in this section.
74. The exception is where they are specifically tax related, in which case they would usually be discussed in a footnote to the taxation line.
75. Determining whether an item is an extraordinary or exceptional item involves reference to applicable accounting guidelines, and in the end may be a judgment call.
76. Note that an increase in KeyCorp's effective tax rate has dampened net income growth in 2011.

77. This situation commonly arises where a bank—or any company, for that matter—owns more than 50 percent but less than 100 percent of the shares of another company. Treatment of minority interest is discussed in more detail in the second half of the chapter.
78. In this case, preferred shareholders are entitled to a predetermined liquidation value, while common shareholders only have a claim to what is left over after all creditors and preferred shareholders have been paid.
79. A *sinking fund* provision is an obligation on the part of the issuer, set forth in the *bond indenture* that constitutes the agreement between the issuer and investors, to retire some proportion of the total issue outstanding according to a specified schedule. The primary purpose of a sinking fund is to ensure that the issuer has the wherewithal to redeem the securities at maturity, that is, repay the investors their principal. By requiring early redemption, the amount of the balloon payment the issuer must make at maturity is lessened, thereby reducing the credit risk borne by the investor.

Deconstructing a Bank's Balance Sheet

[A] typical bank's balance sheet presents, on the face of it, an alarmingly precarious situation: its liabilities are mostly short term, but most of its assets are realizable only in the long term, and it is highly geared or "leveraged." Depositors and other creditors must be persuaded that the whole pack of cards will not come crashing down at a moment's notice.

—Fitch IBCA¹

This book explores the creditworthiness of banks from several angles, many of them articulated around the balance sheet and, increasingly, around off-balance-sheet items. Bank financial statements are very complex. Readers may be familiar with general accounting rules, and this chapter will introduce them more specifically to the bank's balance sheet—and its off-balance-sheet items. Inevitably, a number of concepts or definitions have to be discussed at this stage, while a more thorough analysis will be offered in later chapters, thereby introducing some degree of unavoidable duplication.

The balance sheet, also known as the *statement of condition* or *statement of resources* records *assets* on one side and *liabilities* and *equity* on the other. The two sides must balance, hence the term. Some analysts, however, prefer to separate the liability side into equity—which for a bank is usually just a small fraction of the total—and actual liabilities toward third parties, as can be schematically seen in Exhibit 5.1.

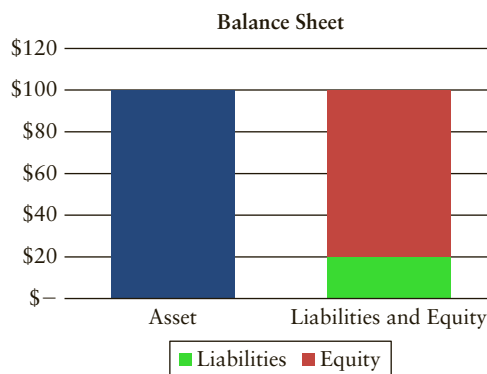


EXHIBIT 5.1 The Balance Sheet

The balance sheet merely captures a snapshot of the composition of a firm's assets and the claims upon it at one moment in time—usually December 31 or the end of a quarter. Since it is a *static* picture, it does not say much about the company's past performance or about its likely future. Hence, the balance sheet provides precious little information about the company's ability to service debt on an ongoing basis. Of course, by comparing the firm's balance sheets at various times, some inferences can be drawn about its *debt servicing capacity*.

As with the income statement, ratios and indicators can be derived from the balance sheet to measure liquidity, capital strength, and asset quality. From both statements, additional profitability and other financial indicators can be calculated.

BALANCE SHEET

Also known as a *statement of condition*, or a *statement of resources*, among other terms, the balance sheet is a snapshot showing the relationship between a firm's assets on the one hand, and its liabilities and therefore owners' equity on the other. The two sides must always equal each other; owners' equity will fluctuate to reflect any excess of assets over liabilities. If liabilities exceed assets, the firm is insolvent. The balance sheet offers an indication of a company's leverage, its level of liquidity, and its overall financial strength.

While the broad categories that comprise a bank's balance sheet are generally similar to those of industrial corporations, individual line items and, more significantly, the overall composition of the balance sheet will differ substantially from the nonfinancial firm. Exhibit 5.2, which shows a representative corporate and bank balance sheet side by side, provides an illustration.

KEY DIFFERENCES BETWEEN THE BALANCE SHEETS OF BANKS AND NONFINANCIAL COMPANIES

The main ways in which a bank's balance sheet differs from that of a nonfinancial enterprise are as follows:

- *Composition.* Most of a nonfinancial company's assets are composed of physical assets such as inventory or fixed assets. In contrast, most of a bank's assets are financial assets, such as loans and securities, as opposed to physical assets.
- *Leverage.* The structure of a bank's balance sheet, particularly the degree of leverage, or gearing, employed. This differs considerably from that of a nonfinancial company, being on average much higher than its corporate counterparts.
- *Mirror image.* A bank's balance sheet is in some respects the mirror image of that of a nonfinancial company. For example, if a nonfinancial company has funds on deposit at a bank, that deposit is an asset. To the bank, a customer deposit is a liability, an obligation that must be repaid.

Each of these differences is discussed further in the following pages.

Bank Balance Sheet		Corporate Balance Sheet	
Assets	Liabilities	Assets	Liabilities
Cash		Cash	
Marketable securities	Customer deposits	Marketable securities	Account payable
		Account receivable	Short-term debt
		Inventory	Medium- and long-term debt
Loans	Bank deposits (Interbank borrowing)	Other assets	
	Short-term debt	Fixed assets	Shareholders' equity
Other assets	Medium- and long-term debt		
Fixed assets		Intangible assets	
Intangible assets	Shareholders' equity		

EXHIBIT 5.2 Composition of a Bank Balance Sheet versus Corporate Balance Sheet**EXHIBIT 5.3** Apple Inc.

Apple Inc., \$ millions	2011	2010	2011, % total
Other current assets	38,843	35,117	33.38%
Inventories	776	1,051	0.67%
Accounts receivable—net	5,369	5,510	4.61%
Long-term marketable securities	55,618	25,391	47.79%
Property, plant, and equipment—net	7,777	4,768	6.68%
Goodwill	896	741	0.77%
Intangible assets	3,536	342	3.04%
Other assets	3,556	2,263	3.06%
Total assets	116,371	75,183	100.00%
Current liabilities	27,970	20,722	24.04%
Other liabilities	11,786	6,670	10.13%
Total liabilities	39,756	27,392	34.16%
Shareholders' equity	76,616	47,791	65.84%
Total liabilities/total equity	0.52	0.57	

Source: Company reports.

To illustrate the difference between a bank balance sheet and a corporate balance sheet, it is helpful to examine some real-world examples. In Exhibit 5.3 and Exhibit 5.4, respectively, are the balance sheets for Apple Inc.,² a leading U.S. company, and KeyCorp, a typical U.S. bank.³

EXHIBIT 5.4 KeyCorp, Consolidated Balance Sheet, 2010

Consolidated Balance Sheets		
December 31 (in \$ millions, except per share data)	2010	2009
ASSETS		
Cash and due from banks	\$ 278	\$ 471
Short-term investments	1,344	1,743
Trading account assets	985	1,209
Securities available for sale	21,933	16,641
Held-to-maturity securities (fair value: \$17 and \$24)	17	24
Other investments	1,358	1,488
Loans, net of unearned income of \$1,572 and \$1,770	50,107	58,770
Less: Allowance for loan and lease losses	1,604	2,534
Net loans	48,503	56,236
Loans held for sale	467	443
Premises and equipment	908	880
Operating lease assets	509	716
Goodwill	917	917
Other intangible assets	21	50
Corporate-owned life insurance	3,167	3,071
Derivative assets	1,006	1,094
Accrued income and other assets (including \$91 of consolidated LIHTC guaranteed funds VIEs)	3,876	4,096
Discontinued assets (including \$3,170 of consolidated education loan securitization trust VIEs at fair value)	6,554	4,208
Total assets	<u>\$91,843</u>	<u>\$93,287</u>
LIABILITIES		
Deposits in domestic offices:		
NOW and money market deposit accounts	\$27,066	\$24,341
Savings deposits	1,879	1,807
Certificates of deposit (\$100,000 or more)	5,862	10,954
Other time deposits	8,245	13,286
Total interest-bearing	43,052	50,388
Noninterest-bearing	16,653	14,415
Deposits in foreign office—interest bearing	905	768
Total deposits	60,610	65,571
Federal funds purchased and securities sold under repurchase agreements	2,045	1,742
Bank notes and other short-term borrowings	1,151	340
Derivative liabilities	1,142	1,012
Accrued expense and other liabilities	1,931	2,007
Long-term debt	10,592	11,558
Discontinued liabilities (including \$2,997 of consolidated education loan securitization trust VIEs at fair value)	2,998	124
Total liabilities	<u>80,469</u>	<u>82,354</u>

Consolidated Balance Sheets		
December 31 (in \$ millions, except per share data)	2010	2009
EQUITY		
Preferred stock, \$1 par value, authorized 25,000,000 shares:		
7.75% Noncumulative Perpetual Convertible Preferred Stock,		
Series A, \$100 liquidation preference; authorized 7,475,000		
shares; issued 2,904,839 and 2,904,839 shares	291	291
Fixed-Rate Cumulative Perpetual Preferred Stock, Series B,		
\$100,000 liquidation preference; authorized and issued 25,000		
shares	2,446	2,430
Common shares, \$1 par value; authorized 1,400,000,000 shares;		
issued 946,348,435 and 946,348,435 shares	946	946
Common stock warrant	87	87
Capital surplus	3,711	3,734
Retained earnings	5,557	5,158
Treasury stock, at cost (65,740,726 and 67,813,492 shares)	(1,904)	(1,980)
Accumulated other comprehensive income (loss)	(17)	(3)
Key shareholders' equity	11,117	10,663
Noncontrolling interests	257	270
Total equity	11,374	10,933
Total liabilities and equity	<u>\$91,843</u>	<u>\$93,287</u>

Source: Company reports.

Composition of Assets and Liabilities

Asset Composition

For any company, assets take in the firm's ownership rights in specific property or items having a measurable value. These items of value include primarily physical assets—such as buildings and equipment—and financial assets, which are generally composed either of cash or of legally enforceable claims against others for payment in cash at a given time.⁴ Another type is the intangible asset, such as goodwill. Assets may be grouped by type, purpose, maturity, and liquidity. That is, assets can be divided according to whether they are tangible or intangible, financial or physical, liquid or illiquid, held to maturity or available for sale, short term or long term, and so on.⁵

Assets: Illustrative Composition

Although the terms describing the categories differ somewhat, both Apple and KeyCorp hold some of the same types of assets. For instance, Apple's property, plant, and equipment classification corresponds roughly to KeyCorp's premises and equipment.

Asset Composition of Nonfinancial Companies

Nonfinancial companies hold physical assets in the form of inventory and fixed assets. Generally speaking, nonfinancial firms possess a higher proportion of physical to nonphysical assets than do financial institutions. Although a corporate may hold some

financial assets for cash management and strategic investment purposes, these assets would not normally be held mainly to generate interest income.⁶ For manufacturing firms, physical assets will usually comprise a large proportion of total assets.

As shown in Exhibit 5.3, Apple's inventory and fixed assets in FY10 comprised about 24 percent of its assets other than cash, cash equivalents, and securities. Apple is an industrial company with more physical assets than a commercial or service company. Nonfinancial service companies, indeed, bear a greater resemblance to their financial service counterparts in that they tend to have a comparatively small proportion of physical assets to total assets. However, unlike financial service companies, a nonfinancial service company's financial assets other than cash and trade-related accounts receivable are likely to be a small proportion of total assets.⁷ Apple is an exception owing to its substantial cash chest accumulated through retained earnings.

Asset Composition of Banks

As a financial institution, it should not be surprising that most of a bank's assets are in the form of financial assets that generate a return in the form of interest income.⁸ These (interest) earning assets are ordinarily composed primarily of loans, and to a lesser degree various forms of securities and other financial instruments.⁹ Nonearning assets include cash in the vault, which does not earn interest, as well as other non-interest-earning cash accounts, and physical assets.¹⁰ Like nonfinancial companies, banks will own a variety of physical assets. These would ordinarily include items such as premises, furnishings, and equipment—desks, filing cabinets, and computers, for instance—and foreclosed assets obtained from defaulting borrowers, often real estate. Since banks sell no newly manufactured physical assets—checkbooks provided to customers being one of the incidental exceptional items that proves the rule—inventory does not even rate a mention on the bank balance sheet.

For most banks, fixed assets will comprise a comparatively small proportion of total assets. Illustrating this point, it can be seen in Exhibit 5.4 that fixed assets classified by KeyCorp as premises and equipment comprised a relatively small proportion of total assets. In FY10, they amounted to a mere \$908 million out of almost \$92 billion in total assets, less than 1 percent. In contrast, Apple's corresponding line item—property, plant, and equipment, was \$4,768 million, a much more significant percentage of the total.

Liability Composition

Liabilities are claims against a company's assets, excluding equity, which represents the owners' residual claims. Put another way, liabilities are obligations owed by the firm to other parties. These obligations are typically grouped by tenor and type. Current liabilities are those that are short term in nature, typically including trade payables and short-term debt obligations (as well as the current portion of long-term obligations). Medium- and long-term debt obligations usually comprise the bulk of noncurrent liabilities (other than equity).

Liabilities: Illustrative Composition

The *liability side* of the balance sheet of a bank shows the funding for a bank's assets. These will be funded primarily by deposits and short-term financial obligations. A nonfinancial company will not fund itself through the taking of deposits—if it accepted deposits for this purpose, it would by definition be a financial institution.

Any deposits held by a nonfinancial enterprise will almost always be negligible in amount, and represent partial payment on sales rather than advances for the purpose of earning interest income. It is plain that loans and other securities holdings form the bulk of KeyCorp's assets and that the liabilities funding these assets are composed mainly of deposits and other borrowings. Apple's balance sheet, however, shows neither an entry for loans advanced nor one for deposits.

Gearing

In addition to dissimilarities in the composition of their balance sheets, banks and nonfinancial companies typically have much different overall structures, particularly in respect to gearing, or leverage.¹¹ These quasi-synonymous terms, as illustrated figuratively in Exhibit 5.4, measure the proportion of the owners' funds placed at risk—that is, equity—relative to the firm's total liabilities, and by extension to its total assets. Essentially, gearing refers to how much an enterprise is financed from borrowed funds as opposed to the funds that the owners themselves have put at risk.

In a highly leveraged or highly geared firm, a relatively small amount of equity capital is supporting a large amount of funding provided by those lending to the firm. High leverage magnifies returns on equity (ROE), but an enterprise that employs it beyond reason is, all other things being equal, more vulnerable to failure and would most likely not be able to keep borrowing.¹² A bank's balance sheet is almost always much more highly geared than that of its corporate counterpart. Observe that at FY10, as shown in Exhibits 5.3 and 5.4, the ratio of liabilities (not counting equities) to equity for Apple was about 174 percent (1.74 times), while the corresponding ratio for KeyCorp's was about 753 percent (7.53 times).

In contrast to a bank, a nonfinancial enterprise typically has ratios of debt financing and equity financing not too far from one to one, meaning that the two fundamental components of the liability side of the balance sheet, debt and equity, are close to equal as a proportion of the total. Indeed, to exhibit a very conservative capital structure, a nonfinancial firm would register a debt-to-equity ratio of less than one, meaning that shareholders' funds at risk are greater than the funds that creditors have placed at risk. Even with a gearing higher than one-to-one, of course, a given amount lent by creditors faces a lower risk than the same amount contributed by shareholders, since in case of a failure creditors' claims enjoy seniority over shareholders' claims.

At one extreme, some nonfinancial firms may have no debt other than trade payables; at the other, depending upon the industry, a debt-to-equity ratio of two or more would ordinarily be regarded as aggressive. As should be expected, banks as a rule employ levels of gearing considerably higher than the most highly leveraged corporations. Notwithstanding their usual reputation as conservative enterprises, it is not at all unusual for a bank's liabilities to be 10 times or more its equity. In fact that ratio has, for many banks, increased a lot in the past 20 years, until the 2008 crisis prompted some deleveraging. The diagram in Exhibit 5.5, not to scale, illustrates the differences in typical gearing.

Bank Balance Sheet as Mirror Image of Corporate Balance Sheet

By definition one party's financial obligation to another is the other's asset. Even a company's equity, which is a financial obligation for the company—albeit conditional on the company holding more assets than liabilities—represents an asset for its shareholders. Consequently, what is an asset to one entity should be a liability to one

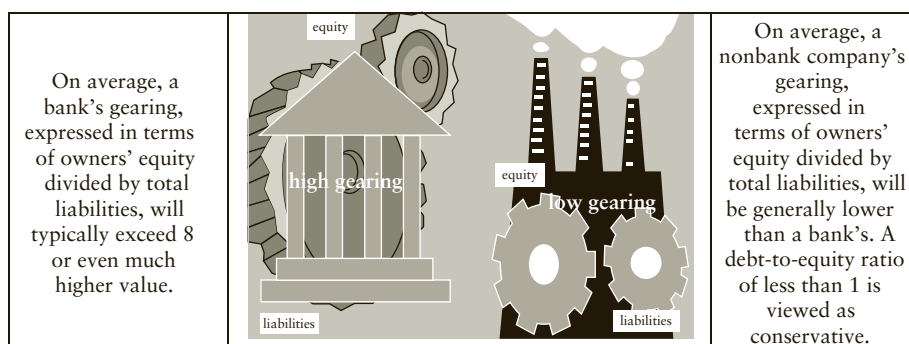


EXHIBIT 5.5 Differences in Typical Gearing

or more other parties. This can be clearly seen in financial assets, such as loans and bonds, which represent claims on another entity for the payment of interest or principal. In other words, financial assets when viewed from the opposite side of the transaction are financial obligations. For instance, a bond simultaneously constitutes both a financial asset and a financial obligation, or liability. Determining the characterization on the balance sheet depends upon the enterprise's status as borrower (issuer) or lender (investor, bondholder).

In light of the dual character of financial instruments just described, the balance sheet of a bank, by virtue of the institution's role as a financial intermediary, represents in some respects the reverse image of that of a nonfinancial firm. To a bank, the loans it has advanced to its customers, and the debt securities it has purchased in the market, are its assets, as they generate interest income on the principal advanced. But, of course, these very same financial instruments that are assets to the bank are liabilities to the borrowers and issuers who must return with interest the funds they have borrowed from the bank. Similarly, a bank's liabilities—such as the securities it has issued in the market and the deposits it has taken from its depositors—are assets to the customers who have deposited funds in accounts with the bank and investors who have invested in its securities. To the bank, however, they are obligations to the investors and depositors respectively that must be fulfilled.

To further illustrate this point, we can see in Exhibit 5.6 that the loan made by Bank DEF to customer ABC, Inc. is an asset to the bank, but a liability to the company. Likewise, to Ms. GHJ, a depositor with Bank DEF, her positive savings account balance with the bank is an asset to her, but to the bank this deposit represents a liability. By accepting Ms. GHJ's deposit and promising to return it to her on demand with interest, the bank has effectively borrowed funds from the depositor. Just as with any other creditor, it has an obligation to the depositor to honor that obligation and repay the funds according to the terms of the agreement entered into between the depositor and the bank.¹³

THE ESSENTIAL LINE ITEMS OF THE BANK BALANCE SHEET: ASSET SIDE

As with the presentation of the income statement, the manner in which a bank presents its balance sheet varies somewhat from country to country, and from bank to bank.

ABC, Inc. (Borrower)		DEF Bank		Consumer, GHJ (Depositor)	
Assets	Liabilities and Equity	Assets	Liabilities and Equity	Assets	Liabilities and Equity
Cash	Accounts payable	Cash	deposits	Cash in bank *	Consumer debt
Receivables		Securities		Savings deposits in bank	Mortgage
Inventory	Loan from bank	Loans to companies	Debt	Securities	
Fixed assets	Owners' equity	Fixed assets	Owners' equity	Equity in house	
					Net worth

N. B. Balance sheets not to scale

*The category described here would include immediately available funds such as those held in checking accounts (or current accounts) or in money-market accounts.

EXHIBIT 5.6 Bank Balance Sheet as Mirror Image of Corporate Balance Sheet

KeyCorp's balance sheet depicted in Exhibit 5.3 shows how a typical bank reporting under U.S. Generally Accepted Accounting Principles (GAAP) displays this data, while Exhibit 5.7 shows a specimen format for bank reporting under the Canadian GAAP format. It is evident that while the two balance sheets are broadly similar, there are also some differences.

EXHIBIT 5.7 An Illustrative Bank Balance Sheet: Specimen Canadian GAAP Format

Consolidated Balance Sheets		
As of October 31 (C\$ millions)	2011	2010 (1)
Assets		
Cash and due from banks	\$ 13,247	\$ 8,440
Interest-bearing deposits with banks	12,181	13,254
Securities (Note 3)		
Trading	145,274	144,925
Available-for-sale	34,284	38,594
	179,558	183,519
Assets purchased under reverse repurchase agreements and securities borrowed	84,947	72,698
Loans (Note 4 and 5)		
Retail	228,484	214,937
Wholesale	69,758	60,107
	298,242	275,044
Allowance for loan losses	(1,958)	(2,038)
	296,284	273,006
Other		
Customer's liability under acceptances	7,689	7,371
Derivatives (Note 7)	100,013	106,155
Premises and equipment, net (Note 8)	2,490	2,139
Goodwill (Note 10)	7,703	6,660
Other intangibles (Note 10)	2,115	1,710
Assets of discontinued operations (Note 11)	27,143	34,364
Other assets (Note 12)	18,332	16,890
	165,485	175,289
	\$ 751,702	\$ 726,206
Liabilities and shareholders' equity		
Deposits (Note 13)		
Personal	\$ 166,030	\$ 151,347
Business and government	258,494	239,233
Bank	19,657	23,981
	444,181	414,561

Consolidated Balance Sheets		
As of October 31 (C\$ millions)	2011	2010 (1)
Other		
Acceptances	7,689	7,371
Obligations related to securities sold short	44,284	46,597
Obligations related to assets sold under repurchase agreements and securities loaned	46,188	41,207
Derivatives (Note 7)	101,437	108,908
Insurance claims and policy benefit liabilities (Note 14)	6,875	6,273
Liabilities of discontinued operations (Note 11)	20,071	24,454
Other liabilities (Note 15)	29,580	28,220
	256,124	263,030
Subordinated debentures (Note 16)	7,749	6,681
Trust capital securities (Note 17)	—	727
Noncontrolling interest in subsidiaries (Note 19)	1,941	2,256
Shareholder's equity (Note 18)		
Preferred shares	4,813	4,813
Common shares (shares issued – 1,438,376,317 and 1,424,921,817)	14,017	13,378
Contributed surplus	212	236
Treasury shares – preferred (shares held – 6,341 and 86,400)	—	(2)
– common (shares held – (146,075) and 1,719,092)	8	(81)
Retained earnings	24,282	22,706
Accumulated other comprehensive loss	(1,625)	(2,099)
	41,707	38,951
	\$ 751,702	\$ 726,206

Source: Company accounts.

Not only does a bank balance sheet differ significantly from a corporate balance sheet, so too does the manner of presentation of bank balance sheets differ depending upon the applicable reporting guidelines followed. We will discuss these differences later as we examine each component of the bank balance sheet.

Asset Classification

Manner of Presentation

Assets are customarily listed on the bank's balance sheet in order of decreasing liquidity, which refers to the rapidity with which they can be liquefied or transformed into cash.¹⁴ For example, less-liquid assets such as houses and vehicles are less rapidly and efficiently converted into cash, contrasting with more liquid assets such as securities being more readily and efficiently converted. Cash, whether in the form of coin and currency, or in the form of funds in a checking account, is obviously the most liquid asset,¹⁵ while

nonstandardized long-term loans (together with fixed assets) are generally considered to be the least liquid. (Intangible assets such as goodwill are typically classified below fixed assets.) In between are readily negotiable assets such as long-term government securities (assuming that a liquid secondary market exists) and other high-grade debt securities issued by private companies or other organizations.¹⁶ These are sometimes termed quasi-liquid assets.¹⁷ In some jurisdictions, banks may hold equity securities. In some cases, these will be readily liquefiable, while in other situations, such as when they are held as a strategic investment by the bank, they may not be.¹⁸

The presentation of securities holdings on the bank's balance sheet will, as elsewhere, vary with the applicable accounting and reporting standards. Under international accounting standards (IAS), it is recommended that trading securities and investment securities be separately grouped. This approach can be seen on the specimen balance sheet in Exhibit 5.8, where there are separate line items for each, and investment securities are divided into held-to-maturity securities and available-for-sale securities.

Significance of Asset Liquidity

A bank's liquidity position, discussed in detail in Chapter 10, is an important part of its credit evaluation, and the classification of assets according to their degree of liquidity is therefore useful to the analyst. Broad categories of assets classified by liquidity are shown in Exhibit 5.9. In this table, there is some overlap between categories with respect to their liquidity, so the preceding classifications are merely indicative. While as a general rule securities tend to be more liquid than loans, this is not always the case. Some loans are highly liquid, while some securities are highly illiquid. The foregoing illustration, however, offers an indication of how major categories of bank assets rank in terms of liquidity, and provides a useful framework for a discussion of the asset side of a bank balance sheet.

EXHIBIT 5.8 An Illustrative Bank Balance Sheet: Specimen IAS Format

(All amounts expressed in millions of [local currency]) As of Dec. 31	2011	2010
ASSETS		
Cash and balances with central banks	6,080	4,315
Treasury bills and other eligible bills	1,485	771
Due from other banks	8,576	6,604
Trading securities	8,721	9,376
Derivative financial instruments	5,325	5,442
Loans and advances to customers	59,203	53,208
Investment securities—available-for-sale	4,006	1,201
Investment securities—held-to-maturity	3,999	1,009
Investments in associated undertakings	112	108
Goodwill and other intangible assets	122	204
Property, plant, and equipment	1,519	1,555
Deferred tax assets	273	255
Other assets	2,110	2,111
Total assets	101,531	86,159

(All amounts expressed in millions of [local currency]) As of Dec. 31	2011	2010
LIABILITIES		
Due to other banks	15,039	13,633
Other deposits	16,249	12,031
Derivative financial instruments and other trading liabilities	4,039	6,277
Due to customers	51,775	42,698
Debt securities in issue	1,766	1,232
Other borrowed funds	2,808	2,512
Other liabilities	2,871	2,224
Current taxes	101	173
Deferred tax liabilities	1,109	693
Retirement benefit obligations	237	221
Total liabilities	95,994	81,694
Minority interest	36	37
SHAREHOLDERS' EQUITY		
Ordinary shares	1,200	1,150
Share premium	857	818
Less: Treasury shares	-47	-52
Retained earnings	2,359	1,920
Other reserves	1,132	592
Total shareholders' equity	5,501	4,428
Total equity and liabilities	101,531	86,159

EXHIBIT 5.9 Bank Assets by Nominal Liquidity Rank

Liquidity Ranking	Line Item	Remarks
Most Liquid Assets		
Liquid assets	■ Cash	Since cash is the fundamental element to which all assets are liquefied, and therefore necessitates no further conversion, it is the most liquid asset. Highly creditworthy financial assets for which there exists a deep and liquid secondary, i.e., trading market, such that they may be converted to cash on short notice without significant discount may be regarded as equivalent to cash (cash equivalent), and may even be defined as cash for financial reporting purposes.
	■ Due from the central bank	
	■ Due from other commercial banks	
	■ Government securities	
Moderately Liquid Assets		
Quasi-liquid assets	■ Marketable securities	Highly marketable securities that are not deemed cash or cash equivalent might include listed or unlisted securities for which a deep and liquid secondary market exists, e.g., government-guaranteed obligations.
	■ Other government obligations	
	■ Deposits with other banks	

(Continued)

EXHIBIT 5.9 (Continued)

Liquidity Ranking	Line Item	Remarks
Less-Liquid Assets		
	■ Loans and advances ■ Other securities and financial obligations ■ Equity securities or direct investment that generally are not easily liquefiable	Financial assets not falling within the two categories above are considered less liquid.
Other and Fixed Assets		
	■ Premises and equipment ■ Goodwill, intangibles, and miscellaneous other assets	For purposes of liquidity classification, fixed assets may be regarded as illiquid, notwithstanding that they might be converted to cash within a reasonable time period subject to varying discounts. Some intangible assets, such as goodwill, however, may be completely illiquid.

Most-Liquid Assets: Cash and Cash Equivalent

The category of assets referred to as cash and cash equivalent encompasses a bank's most liquid assets. Broadly speaking, the terms *cash* and *cash equivalent* correspond to and are synonymous with the terms *liquid assets* and *quasi-liquid assets*.

LIQUID ASSETS

A liquid asset is one capable of being quickly converted to cash with negligible risk of loss. Such instruments generally include instruments such as government bills (e.g., treasury bills), investment-grade commercial paper issued by financial institutions or corporations, and similar instruments. Sometimes the term *quasi-liquid assets* may be used to refer to those assets that are not formally classified as *liquid assets* but have essentially the same characteristics. Broadly speaking, the terms *cash* and *cash equivalent* correspond to, and are synonymous with, the terms *liquid* and *quasi-liquid assets*.

Cash

Customarily, cash includes bullion, coin, and currency in the bank's vault, currency in automated teller machines (ATMs), as well as from the bank's vantage point, checks in the process of collection and similar items of a very liquid character.¹⁹ Under IAS, the definition of cash embraces not just currency and coin but also funds that are placed on deposit with other financial institutions provided they are permitted to be withdrawn without notice.²⁰ The financial crisis that started in 2008 showed that such a definition should perhaps be reviewed, as deposits with other

financial institutions have at times become difficult to recover. The presentation of securities holdings on the bank's balance sheet will, as elsewhere, vary with the applicable accounting and reporting standards. Under IAS, it is recommended that trading securities and investment securities be separately grouped. This approach can be seen on the specimen balance sheet in Exhibit 5.4, where there are separate line items for each, and investment securities are divided into held-to-maturity securities and available-for-sale securities.

Cash Equivalent

The term *cash equivalent* is sometimes used to distinguish between cash and other types of highly liquid assets similar to cash. Under IAS 7, cash equivalent consists of short-term, highly liquid investments that are readily convertible to known amounts of cash and are subject to an insignificant risk of changes in value.²¹ An illustration of the more liquid portion of the asset side²² of the balance sheet can be seen in Exhibit 5.10.

On this excerpt from KeyCorp's balance sheet, there are six line items shown: cash and due from banks;²³ short-term investments; trading account assets; securities available for sale; held-to-maturity investment securities; other investments. Many such assets fall within the broad category of cash and cash equivalent as defined,²⁴ but some of them belong to less-liquid asset categories, as discussed below.

Moderately Liquid Assets: The Securities Portfolio

Aside from loans and cash, most of a bank's assets are in the form of securities.²⁵ As noted, some highly liquid short-term securities may be functionally equivalent to cash, although they may be nominally classified with other securities for certain reporting purposes. In many markets, the vast majority of securities held by a bank will be both highly rated and marketable, so they will have a fairly high degree of liquidity.²⁶ It is, however sometimes the case that seemingly marketable securities are in reality not readily liquefiable or liquefiable only under benign market conditions.

EXHIBIT 5.10 KeyCorp: Cash, Cash Equivalent, and Securities, Data Excerpted from 2010 Balance Sheet

Year ended December 31 U.S. dollars in millions	2010	2009
...	...	...
Cash and due from banks	278	471
Short-term investments	1,344	1,743
Trading account assets	985	1,209
Securities available for sale	21,933	16,641
Held-to-maturity securities (fair value: \$17 and \$24)	17	24
Other investments	1,358	1,488
...	...	...
Total assets	91,843	93,287

Source: Company reports.

Classification of Securities by Purpose

Banks ordinarily group their securities holdings²⁷ into one of three categories for financial reporting purposes, and the major accounting regimes, including IAS and U.S. GAAP, treat each category in roughly the same manner. The classification is important because it affects how each type of security is valued on the balance sheet, and also how gains or losses are recognized for income statement or equity measurement purposes.

Trading Securities

Trading securities, as the term suggests, are securities that have been purchased for trading purposes, meaning with the objective of realizing a gain from short-term price fluctuations.²⁸ Securities in this category are valued at cost, including transaction costs, at the time of acquisition²⁹ and—most of the time—subsequently marked to market.³⁰

Investment Securities

There are two types of investment securities: held-to-maturity securities and available-for-sale securities.

Available-for-Sale Securities

Available-for-sale securities encompass debt or equity investments that the bank has the intent to hold indefinitely but that it may sell as a consequence of changing interest rates, exchange rates, or prices, or for liquidity management or asset-liability management purposes.

Held-to-Maturity Securities

Held-to-maturity securities are debt instruments having a fixed maturity and in respect to which the bank has the intent and the ability to hold the securities until maturity.

The difference between held-to-maturity securities and available-for-sale securities is based upon management's capacity and intention as indicated at the time of purchase.

Significance of Distinction between Held-to-Maturity and Available-for-Sale Securities

The distinction between the two classes of investment securities is significant because the revenue generated from each class of holdings and the value of each is measured differently. Held-to-maturity securities traditionally are valued at cost, subject to amortization and provision for impairment³¹ in their value.

In contrast, available-for-sale securities are marked to market (or revalued at fair value using an approved method). Like trading securities, which are also valued at fair value rather than cost, realized and unrealized gains or losses of held-to-maturity securities are recognized. But unlike trading securities, such gains or losses are not treated as positive or negative income and therefore do not pass through the income statement. Instead, they are applied directly against the equity component of the balance sheet, although in some jurisdictions such gains or losses may not affect certain capital ratios.

WHY BANKS HOLD SECURITIES

Banks hold securities for several reasons.

- *Cash/liquidity/asset-liability management.* The primary reason is for cash management purposes, which for a bank are largely equivalent to liquidity management or asset-liability management purposes—liquidity management comes within the larger category of asset liability management. Because of the intrinsic mismatch between a bank's comparatively long-term assets and its short-term liabilities, banks will use holdings of longer-term securities as one means to reduce this mismatch.
- *Collateral.* Another reason for holding securities is as financial collateral against loans or other advances. Repo finance, for example, makes use of debt securities as collateral to provide financing at comparatively low interest rates.
- *Proprietary trading and market making.* A further reason that a bank may hold securities is for proprietary trading purposes. Investment banks, or commercial banks that are originators of securitized assets, may hold securities as part of the underwriting process. Similarly, investment banks especially may wish to make a market in certain types of securities. In other words, to facilitate and maintain its role as a market participant in certain types of financial instruments, a bank may effectively be required to maintain some inventory of those securities.
- *Strategic investment.* Finally, a bank may hold equity securities for strategic investment purposes. For instance, it may hold shares in a subsidiary or affiliated company or in a joint venture with another bank or with a nonfinancial company.

Presentation of the Securities Portfolio: A Comparison of Two Banks

KeyCorp

Further detail on the KeyCorp's securities holdings seen in the previous section can be found in the notes to its financial statements. In note 7 to KeyCorp's 2010 financial statement, additional information concerning the composition of the bank's securities portfolio is provided, which is excerpted in Exhibit 5.11. In addition to classifying securities by purpose and type, KeyCorp provides for each category of securities data concerning:

- Amortized cost
- Gross unrealized gains
- Gross unrealized losses
- Fair value

Recall that, under the valuation rules discussed, only investment securities are valued at cost less amortization. Securities available for sale are valued at fair

EXHIBIT 5.11 Composition of KeyCorp's Securities Portfolio, 2010

December 31, in \$ millions	2010				2009			
	Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value
SECURITIES AVAILABLE FOR SALE								
U.S. Treasury, agencies and corporations	\$ 8	—	—	\$ 8	\$ 8	—	—	\$ 8
States and political subdivisions	170	\$ 2	—	172	81	\$ 2	—	83
Collateralized mortgage obligations	20,344	408	\$ 87	20,665	14,894	187	\$ 75	15,006
Other mortgage-backed securities	998	71	—	1,069	1,351	77	—	1,428
Other securities	15	4	—	19	100	17	1	116
Total securities available for sale	<u>\$ 21,535</u>	<u>\$ 485</u>	<u>\$ 87</u>	<u>\$ 21,933</u>	<u>\$ 16,434</u>	<u>\$ 283</u>	<u>\$ 76</u>	<u>\$ 16,641</u>
HELD-TO-MATURITY SECURITIES								
States and political subdivisions	\$ 1	—	—	\$ 1	\$ 3	—	—	\$ 3
Other securities	16	—	—	16	21	—	—	21
Total held-to-maturity securities	<u>\$ 17</u>	<u>—</u>	<u>—</u>	<u>\$ 17</u>	<u>\$ 24</u>	<u>—</u>	<u>—</u>	<u>\$ 24</u>

Note 7 to KeyCorp's financial statement 2010, SEC Filing 10-K.

Source: Company reports.

value—that is marked to market—and any gains or losses are reflected directly in shareholders' equity.

Note that KeyCorp shows no holdings of trading securities, and that its held-to-maturity securities were negligible.

Over 99 percent of KeyCorp's securities portfolio is allocated to securities available for sale. Of these, U.S. Treasury, municipal, and corporate debt securities account for about 1 percent of the total portfolio of about \$22 billion. The balance is largely taken up with securitized assets, namely collateralized mortgage obligations and other mortgage-backed securities (MBS). This large proportion of securitized assets reflects KeyCorp's status as a significant originator of such securities and the large and highly developed market for such assets in the United States. Collateralized mortgage obligations comprise \$20,665 billion, or 94 percent of the total, other MBS \$1,069 million or 5 percent of the total, with the rest about 1 percent of the total.³²

DSFH

Compare KeyCorp's balance sheet with that of Dah Sing Financial Holdings in Exhibit 5.12, which reports under Hong Kong GAAP. The data in this illustration is drawn from the face of DSFH's December 31, 2010, balance sheet.

A portion of 15 percent of DSFH's securities portfolio is in the form of trading securities, and 24 percent in the form of held-to-maturity securities, in contrast to KeyCorp, which held none or a negligible amount of them. Correspondingly, the available-for-sale portion, which for KeyCorp constituted 99 percent of its portfolio, only amounted to 47 percent of DSFH's portfolio.

Possible Reasons for Difference

The disparity in the composition of the securities portfolios reflects differences in the nature of each bank's business model, and the comparison is not meant as a judgment on those models.

- KeyCorp's securities holdings, it seems, mainly function to support the business of securitizing mortgage assets, which in turn facilitates and funds the bank's lending business. The minimal proportion of investment securities suggests that the bank is able to find reasonably good yields by allotting its funds to other assets, while the absence of trading securities indicates that the bank engages in negligible proprietary trading. The lack of trading securities also implies that the bank is able to manage its liquidity through other means.

EXHIBIT 5.12 Dah Sing Financial Corp., Balance Sheet, 2010, Line Items Relating to Securities Portfolio

	HK\$ Thousand	% of Total
Trading securities	5,568,876	15%
Financial assets designated at fair value through P&L	5,784,468	15%
Available-for-sale securities	17,858,053	47%
Held-to-maturity securities	9,114,454	24%
Total securities	38,325,851	100%

Source: Company reports.

- DSFH only categorizes the type of securities it holds by whether they are debt or equity securities, listed in Hong Kong, outside Hong Kong, or unlisted.³³ Roughly half of its securities portfolio is available for sale, a third of which is listed outside of Hong Kong, while another third is unlisted. Why does DSFH maintain such a large holding of investment securities and significant holding of trading securities? One possible reason is that lending opportunities in its home market are limited—although the primary loan-to-deposit ratio is above 75 percent—while the significant portion allotted to trading securities suggests that they are being used for liquidity management purposes or as a means to generate additional profits through trading.
- While this may not have a significant impact, the reporting is not done under the same accounting rules.

Less-Liquid Assets: Loans and Advances

Loans are nominally among the least liquid of a bank's earning assets in spite of what can be seen in some loan agreements. The reality, however, has changed to a degree with the advent of securitization and the more widespread standardization of loan terms than was the case in the past. KeyCorp, as an active originator of securitized assets backed by loans, holds a significant portion of its loans for sale, which it groups separately from its other loans. On a global basis, however, loans, which comprise the bulk of most banks' financial assets, nominally rank last among financial assets in terms of liquidity.

Presentation of the Loan Book: On the Face of the Balance Sheet

KeyCorp presents data about its loan assets in three line items as excerpted in Exhibit 5.13. In common with other U.S. banks, KeyCorp presents the sum value of its loans, recorded at their principal amount (i.e., "net of unearned income"). A separate line item shows the total allowance for loan losses (i.e., loan-loss reserves) set aside to account for expected credit losses. A third line shows total loans net of such reserves.

Under the IAS format, loans may be presented as a single line item on the face of the balance sheet, with all additional data about the composition of the loan portfolio in the notes to the accounts. Further information about loan-loss reserves is provided in the accounting notes. These are discussed later in this section.

EXHIBIT 5.13 KeyCorp, Consolidated Balance Sheet Excerpt, 2010

KeyCorp, Consolidated Balance Sheet, US\$ millions, December 31		
ASSETS	2010	2009
...	...	...
Loans, net of unearned income of \$1,572 and \$1,770	50,107	58,770
Less: Allowance for loan losses	1,604	2,534
Net loans	48,503	56,236

Source: Company reports.

Loan Portfolio Composition

In the same way that interest income on the income statement may be grouped by type of earning asset, as well as by location, industry sector, or tax status, similar classifications may be found on the balance sheet or in the accompanying footnotes. Common subcategories of loans are:

- Commercial and industrial loans
- Real estate loans
- Consumer loans
- Loans to depository institutions
- Agricultural loans
- Other loans and leases

The method of categorization depends upon applicable local rules and customary practice as well as management's discretion.

KB Financial Group

The composition of the loan portfolio of KB Financial Group, a large Korean banking group, which includes Kookmin Bank, drawn from its published data, is depicted in Exhibit 5.14. About 56 percent of KB financial group's loans are loans to households, of which 25 percent are to housing. Most of the rest goes to commercial borrowers.

Of the 31 percent of loans recorded as going to households for general purposes, some are in fact *small office home office* or SOHO loans, meaning loans to small enterprises run by households or individuals.

As this may not appear immediately from aggregate figures, the analyst should invest some time in researching the matter with the view of understanding the risk profile of the bank.

KeyCorp

The composition of KeyCorp's loan portfolio is shown in Exhibit 5.15. The bank's loan portfolio shows a portion of 31 percent for consumer loans, including housing loans, while, for example, 23 percent goes to commercial real estate loans.

Reasons for Differences

Each market is different, and each bank is bound to select a business model that suits its skills and its business environment.

The two banks may exhibit different shades of loan asset quality, and it is obvious that, however detailed financial statements or annual reports may be, they usually do not disclose sufficient information for a safe opinion to get established. Nevertheless, such reports offer considerably more information than in the past.

KeyCorp has always been very transparent, and Korea was for long the most advanced Asian country in terms of the volume—if not the accuracy—of its banks' financial disclosure, well ahead of Hong Kong, especially when Hong Kong was still under colonial rule.

We will discuss the significance of the composition of a bank's loan book in more detail in Chapter 7 concerning asset quality.

EXHIBIT 5.14 KB Financial Group, Loan Portfolio Composition, 2010

KB Financial Group Loan Receivable Portfolio	2010 Won million	2009 Won million	% Total 2010	% Korean Loan Book 2010
Commercial				
Working capital loans				
General-purpose loans	40,118,932	40,704,835	20.00%	23.25%
Notes discounted	501,816	604,481	0.25%	0.29%
Overdraft accounts	302,860	283,190	0.15%	0.18%
Trading notes	912,900	957,654	0.46%	0.53%
Others	7,421,062	8,519,544	3.70%	4.30%
Facilities loans				
General facilities loans	20,708,698	18,428,467	10.32%	12.00%
Others	1,627,460	1,684,730	0.81%	0.94%
Households				
General purpose loans	54,675,330	51,585,539	27.25%	31.68%
Housing loans	42,632,804	45,112,757	21.25%	24.71%
Remunerations on mutual installment savings	23,794	28,900	0.01%	0.01%
Others	549,170	471,568	0.27%	0.32%
Public Sector				
Public operation loans	2,548,970	2,872,236	1.27%	1.48%
Public facilities loan	69,759	161,439	0.03%	0.04%
Others				
Property formation loans	300	382	0.00%	0.00%
Others	471,655	347,676	0.24%	0.27%
Total Loans in Korean won	172,565,510	171,763,398	86.02%	100.00%
Loans in foreign currencies inside Korea	26,840,562	26,020,714	13.38%	
Total loans for use in Korea	199,406,072	197,784,112	99.40%	
<i>Loans in foreign currencies outside Korea</i>	<i>1,212,739</i>	<i>985,239</i>	<i>0.60%</i>	
Total loans receivable (Before provision of loan losses)	200,618,811	198,769,351	100.00%	

Source: Company reports.

Accounting for Loan Losses on the Balance Sheet

While information concerning the total amount of loan-loss reserves may sometimes be included on the face of the balance sheet, more often than not it is provided in the notes to the financial statements. The illustration in Exhibit 5.16 shows the status of KeyCorp's loan-loss reserves at the end of the financial year. The first line indicates total loan-loss reserves at the start of the year. From this amount, charge-offs of nonperforming loans³⁴ are deducted. Charge-offs represent those loan assets the value of which has been written down to zero. Since loan-loss reserves were established to account for any expected loan losses, it is sensible that these reserves absorb any actual loan losses in so far as possible.

EXHIBIT 5.15 KeyCorp, Loan Portfolio Breakdown, 2010

In US\$ Millions, December 31, 2010		% of Total
Commercial, financial, and agricultural	16,441	32.51%
Commercial real estate:		
<i>Commercial mortgage</i>	9,502	18.79%
<i>Construction</i>	2,106	4.16%
Total commercial real estate loans	11,608	22.95%
Commercial lease financing	6,471	12.80%
Total commercial loans	34,520	68.26%
Residential-prime loans:		
Real estate—residential mortgage	1,844	3.65%
Home equity		
<i>Key Community Bank</i>	9,514	18.81%
<i>Other</i>	666	1.32%
Total home equity loans	10,180	20.13%
Total home equity loans	12,024	23.78%
Consumer other—Key Community Bank	1,167	2.31%
Consumer other:		
<i>Marine</i>	2,234	4.42%
<i>Other</i>	162	0.32%
Total consumer other	2,396	4.74%
Total consumer loans	15,587	30.82%
Total loans held for sale	467	0.92%
Total Loans	50,574	100%

Source: Company reports.

EXHIBIT 5.16 Accounting for Loan-Loss Reserves: KeyCorp, 2010

Allowance for loan and lease losses			
In US\$ millions, Year ended December 31	2010	2009	2008
Balance at beginning of year	\$ 2,534	\$ 1,629	\$ 1,195
Charge-offs	(1,822)	(2,396)	(1,240)
Recoveries	252	139	109
Net loans charged off	(1,570)	(2,257)	(1,131)
Provision for loan losses	638	3,159	1,537
Allowance related to loans acquired, net	—	—	32
Foreign currency translation adjustment	2	3	(4)
Balance at end of year	\$1,604	\$2,534	\$1,629

Source: Company reports.

The act of setting aside loan-loss provisions against a loan asset, or writing off the asset, is often based on the likelihood of an ultimate outcome. In some cases, loans that have been provisioned against through the establishment of loan-loss provisions will display superior performance to that predicted or feared, and their actual value will exceed their net reported value after provisions. In other words, if a particular loan was written off completely, it may nevertheless have an actual value greater than zero. If partially provisioned against, the actual value may be in excess of nominal value less loan-loss provisions set aside. This occurs because unless the loan is written off as worthless and never resumes performance, the actual value of a particular loan will be established when, for example, it is sold to a collection agency, or collected against by the bank. Alternatively, the loan may resume performance, in which case its stated value will depend upon local accounting rules.

In each of these situations except the first, the bank will recover assets, or a portion of the assets, which it has already written off the balance sheet. The next line item, recoveries, adjusts for the receipt of such revenue. Since the amount of impairment has already been subtracted from the loan-loss reserve account, upon recovery the amount recovered must be added back to it. Note that charge-offs less recoveries equals net charge-offs, the next line item, which becomes the actual adjustment to the loan-loss reserve account. New loan-loss provisions charged to the income statement are added here to bring the total loan-loss reserves up to the levels deemed adequate by the bank. The remaining three line items are adjustment items that in most cases will not be particularly significant.

Other Financial Assets

The main categories of financial assets held by banks are those discussed: namely, cash, cash equivalents, securities, and loans. Other financial assets will either be classified in one of the foregoing groups or in a catchall category of “other assets,” some of which conceivably may be nonfinancial assets.

Fixed Assets and Other Assets

Fixed Assets

The category of fixed assets, of course, embraces the bank’s premises and related physical assets. As discussed in a previous section, this asset category will comprise a relatively minor part of the bank’s total balance sheet, in contrast to the balance sheets of many nonfinancial enterprises.

Goodwill, Intangibles, and Miscellaneous Other Assets

Goodwill, as discussed earlier, is created when the cost of an acquisition of a company exceeds its book value. Treatment of goodwill under U.S. GAAP changed with the implementation of SFAS 142,³⁵ and a subsequent update SFAS 157 dated September 2011. Under this standard, goodwill and certain other intangible assets are required to be subject to impairment testing, at least annually. Previous practice allowed acquiring companies to amortize goodwill over a period.

THE ESSENTIAL LINE ITEMS OF THE BANK BALANCE SHEET: LIABILITY SIDE

The assets on a bank's balance sheet are balanced by its liabilities and shareholders' equity. Recall that liabilities are simply claims upon the assets of the bank, or put another way, what the bank owes its creditors.³⁶ Shareholders' equity merely constitutes the remainder left over upon the subtraction of total liabilities from total assets. Together, liabilities and equity can be viewed as providing the funding for all of the bank's assets.³⁷

Liability Classification

In the same way that assets are listed on the balance sheet by liquidity ranking, which loosely corresponds to their maturity, liabilities are listed beginning with claims due currently, followed by short-term obligations, and concluding with long-term debt. As with nonfinancial companies, banks may borrow from other banks, raise funds in the money market, or issue medium- or long-term debt in the capital markets. They may also incur other liabilities³⁸ that are not normally deemed debt, including obligations to:

- Vendors for the provision of goods and services—trade payables
- Internal financial commitments to subsidiaries and affiliates
- Obligations to the government in the form of tax liabilities.

Banks and similar deposit-taking financial institutions are distinctive from nonfinancial companies, however, in that they are the only class of enterprises that finance a substantial portion of their assets through deposits. Nevertheless, from an accounting perspective, deposits are just another form of debt, and depositors are just another—albeit a rather special—class of creditors.

Funding liabilities can be broadly divided into customer deposits and purchased funds. Customer deposits represent retail deposits from individuals or businesses, while purchased funds are made up of commercial deposits, often in the form of large-denomination certificates of deposit, interbank borrowing, repo finance, commercial paper, or banknote issues, and other short-term borrowings. Interbank borrowing may be considered as not having been “purchased” if they were obtained passively, pretty much like customer deposits.

In Exhibit 5.17 showing KeyCorp's liabilities, the first seven line items refer to various types of deposits, with the eighth being a subtotal item.

Below deposits, we find two additional line items:

1. Federal funds purchased
2. Other liabilities (including long-term debt)

These take in various forms of short-term and long-term debt, other than deposits, and are discussed in more detail later. The IAS format is similar to U.S. GAAP but is structured slightly differently. More information concerning nonfunding liabilities is provided on the face of the balance sheet. The corresponding line items are shown in Exhibit 5.18.

EXHIBIT 5.17 KeyCorp, Balance Sheet, Liabilities, 2010**LIABILITIES**

	2010	2009
Deposits		
Deposits in domestic offices:		
[1] NOW and money market deposit accounts	\$27,066	\$24,341
[2] Savings deposits	1,879	1,807
[3] Certificates of deposit (\$100,000 or more)	5,862	10,954
[4] Other time deposits	8,245	13,286
[5] Total interest-bearing	43,052	50,388
[6] Non-interest-bearing	16,653	14,415
[7] Deposits in foreign office—interest-bearing	905	768
Total deposits	60,610	65,571
Purchased Funds		
Federal funds purchased and securities sold	2,045	1,742
Bank notes and other short-term borrowings	1,151	340
Other Liabilities		
Derivative liabilities	1,142	1,012
Accrued expense and other liabilities	1,931	2,007
Long-term debt	10,592	11,558
Discounted liabilities	2,998	124
Total liabilities	80,469	82,354

Source: SEC filing 10k.

EXHIBIT 5.18 Presentation of Bank Liabilities, IAS Format

Due to other banks
Other deposits
Derivative financial instruments and other trading liabilities
Due to customers
Debt securities in issue
Other liabilities
Current taxes
Deferred tax liabilities
Retirement benefit obligations
Total liabilities
Other borrowed funds

Customer Deposits

Customer deposits in the main refer to core deposits (also called retail deposits) as opposed to purchased funds (also called commercial deposits).³⁹ Most commercial banks obtain the greater part of their funding from such core deposits, and where this is the case, it is generally regarded as credit positive. The rationale is that in many markets customer deposits represent a relatively cheap source of finance and are also a more stable source of funding.⁴⁰

The reason for this stability is that retail depositors often remain loyal to their bank, notwithstanding that it may not offer the highest return on their savings. Other factors may make depositors reluctant to transfer deposits to another institution merely to benefit from an incremental difference in return. These potentially include:

- Satisfaction with the breadth and quality of services
- The perceived creditworthiness of the bank
- Transaction costs involved in changing banks or moving funds

U.S. GAAP

Looking at the presentation of KeyCorp (see Exhibit 5.17), which reports under U.S. GAAP, it can be seen that deposits are grouped together on the face of the balance sheet and that, as is commonplace, they collectively comprise the larger part of the bank's debt financing.⁴¹ Domestic deposits and foreign deposits are separated, with domestic deposits itemized by type, and by whether the deposits are interest bearing or non-interest bearing. Traditionally, current or checking accounts do not pay interest, and these probably constitute most of the \$16.7 billion in non-interest-bearing accounts held by KeyCorp. Today in many markets, however, the distinction is blurred, and some interest-bearing accounts allow checking privileges.⁴²

The remainder of the \$60.6 billion in deposits are interest bearing. These are divided into several types:

- Money-market accounts⁴³
- Traditional savings accounts
- Large-denomination certificates of deposit (CDs)
- Time deposits

It is evident that the foregoing line items can be characterized for the most part as comprising customer deposits, since interbank deposits in the U.S. market fall under the category labeled as federal funds purchased. However, large-denomination CDs would probably be best classified as purchased funds, rather than as customer deposits.

IAS

Under the IAS format, customer deposits are clearly listed under the heading "due to customers." In a footnote to the accounts, these are further categorized as to the type of depositor, including:

- Large corporate customers
- Small and medium-size enterprises
- Retail customers

Purchased Funds and Interbank Borrowing

As opposed to customer deposits, which are perceived to be somewhat less sensitive to short-term interest rate fluctuations, purchased funds are viewed as being relatively more interest-rate-sensitive. Purchased funding is often arranged through deposit brokers

acting on behalf of corporate or bank treasurers and noncommercial considerations are much less relevant, if they exist at all, than in respect to customer deposits.

Coming within the category of purchased funds are interbank deposits, wholesale deposits, and other similar types of commercial funding obtained through brokers and the money markets. This type of funding is viewed as more volatile than customer deposits, so brokered deposits in the form of interbank deposits are placed by bank treasurers looking to optimize their return on unused funds, or in the case of wholesale deposits by their corporate counterparts.

U.S. GAAP

Federal funds purchased is the term used in the United States to refer to borrowings in the interbank market established by the Federal Reserve Bank (the Fed). The terms *securities sold* refers to funds borrowed under repurchase agreements; that is, repo finance, a form of collateralized borrowing. Repo finance is ordinarily undertaken between two financial institutions. In the same manner, *federal funds sold* refers to interbank lending, and *securities purchased* refers to funds lent under repo agreements.

IAS

Under the IAS, the category of interbank deposits, together with other similar interbank funding, is found under the heading “due to other banks.” This line item includes both deposits from other banks, and items in the course of collection.

Short-Term Debt

Aside from customer deposits and interbank borrowing, banks may source short-term funding from the money markets. They typically accomplish this by issuing short-term obligations, such as commercial paper, or by issuing other forms of short-term securities, such as banknotes.

U.S. GAAP

KeyCorp lists several sources of funding that allow for the issuance of short-term debt, including:

- A bank note program
- A commercial paper program
- A euro note program.

They are quite similar in nature, the main difference being that the first two are issued in the U.S. market while the last is issued to non-U.S. investors.

IAS (Short-Term and Long-Term Debt)

An IAS bank balance sheet does not distinguish on its face between short-term and long-term funding. Instead, the focus is on distinguishing between the types of instrument used to raise funds. Therefore, a discussion of how IAS treats both long-term and short-term debt is included in this section. The relevant line items are as follows:

Other deposits: This category includes certificates of deposit and money market accounts issued to depositors.

Debt securities in issue: This category includes all debt securities irrespective of maturity ranging from short-term instruments to medium and long-term notes.

Other borrowed funds: This category includes *subordinated* or *hybrid* debt instruments such as subordinated debt, convertible bonds, and redeemable preference shares. This category embraces debt that generally qualifies for regulatory capital purposes.⁴⁴

Long-Term Liabilities

The category of long-term debt is fairly self-explanatory. Although usage varies globally, for balance-sheet purposes the adjective *long-term* is often employed simply to refer to liabilities having a term of more than one year. As is evident in the IAS presentation of bank debt holdings, long-term debt includes subordinated debt, a type of funding that may qualify as regulatory capital.

Other Liabilities

Other liabilities, as alluded to, include tax liabilities, pension obligations, funds owed to suppliers, and other miscellaneous items. These items usually play a minor role for funding purposes but may be significant for the purposes of credit analysis.⁴⁵

Shareholders' Equity

Shareholders' equity,⁴⁶ that is, equity capital, denotes the residual claims upon the assets of the bank, which, if not needed to satisfy outstanding claims of creditors or depositors, revert to the institution's shareholders. Shareholders' equity therefore represents the interest in the bank (specifically in its assets) on the part of its owners. To the bank's creditors, it is viewed as a cushion against losses.

Common Shares versus Preferred Shares

Shareholders are principally divided into two types: common or ordinary shareholders and preferred or preference shareholders.⁴⁷

Common Shares

Common shares are the simplest form of shares. They represent a pure residual equity interest in a firm's assets. Subject to local law, common shares, however, may be divided into various classes, with some classes enjoying only limited voting rights.

Preferred Shares

In actuality, preferred shares (preference shares) are a hybrid form of capital falling somewhere on the spectrum between debt and equity. Their debt-like character arises from the fact that there usually exists some obligation on the part of the issuing corporation to pay dividends to preferred shareholders—although this obligation may be constrained depending upon the enterprise's financial results. In contrast, there is no obligation to pay dividends to common shareholders. In addition,

preferred shareholders may be entitled to the return of the funds advanced at a specific time, although some preferred shares are, like common shares, perpetual in character, with earlier redemption either not permitted except on the dissolution of the corporation or permitted contingent upon the occurrence of specific circumstances or conditions. Consequently, depending on the structure of the preferred shares, their hybrid nature may bear a closer resemblance to equity or a closer resemblance to debt.

Components of Shareholders' Equity: Overview

Excluding minority interest, which represents at the consolidated level the ownership interest of nonbank shareholders in the institution, shareholders' equity is comprised of four primary components:

1. Share capital
2. Retained earnings
3. Equity reserves
4. Treasury shares

Note that the terminology used to refer to these elements of shareholders' equity varies depending upon the jurisdiction, applicable accounting rules, and customary usage. The illustrations in Exhibit 5.19 and Exhibit 5.20 show the typical treatment of shareholders' equity on the balance sheet. Except for minority interest being highlighted on the face of the IAS balance sheet, the two presentations are quite similar.

EXHIBIT 5.19 KeyCorp, Shareholders' Equity, as Shown on the Balance Sheet

Consolidated Balance Sheets		
December 31 (in millions, except per share data)	2010	2009
Extracted from KeyCorp's Annual Report 2010		
EQUITY		
Preferred stock, \$1 par value, authorized 25,000,000 shares:		
7.75% Noncumulative Perpetual Convertible Preferred Stock, Series A, \$100 liquidation preference; authorized 7,475,000 shares; issued 2,904,839 and 2,904,839 shares	291	291
Fixed-Rate Cumulative Perpetual Preferred Stock, Series B, \$100,000 liquidation preference; authorized and issued 25,000 shares	2,446	2,430
Common shares, \$1 par value; authorized 1,400,000,000 shares; issued 946,348,435 and 946,348,435 shares	946	946
Common stock warrant	87	87
Capital surplus	3,711	3,734
Retained earnings	5,557	5,158
Treasury stock, at cost (65,740,726 and 67,813,492 shares)	(1,904)	(1,980)
Accumulated other comprehensive income (loss)	(17)	(3)
Key shareholders' equity	11,117	10,663
Noncontrolling interests	257	270
Total equity	11,374	10,933

Source: Company reports.

EXHIBIT 5.20 Shareholders' Equity, IAS Format**SHAREHOLDERS' EQUITY**

Ordinary shares
Share premium
Less: treasury shares
Retained earnings
Other reserves
Minority interest
Total shareholders' equity

Share Capital

The term *share capital*, or *paid-in capital*, refers to funds contributed by shareholders in return for shares. It is equal to the product of the par value (or stated value) of the shares multiplied by the number of shares outstanding, plus the cumulative difference between that product and the amount actually paid in exchange for the shares. The latter amount constitutes the share premium, or surplus.

Paid-in capital is comprised of two subcomponents; namely *legal capital* or *stated capital*⁴⁸—the number of issued shares multiplied by the *par value* of the shares; and *surplus* (also called *paid-in surplus* or *capital surplus*) means the amount contributed for shares in excess of stated capital or legal capital.⁴⁹ In fact, the expression *paid-in* serves the purpose of distinguishing between authorized capital and the actual capital contributed as such by the shareholders. *Authorized capital*⁵⁰—also called nominal or registered capital—simply refers to the amount that the shareholders have agreed upon as the maximum capital that the company will be allowed to issue.

Retained Earnings

Retained earnings (*undivided profits*) sometimes called undivided profits, represent cumulative profits or the earnings (i.e., net income) of the bank that have not been distributed as dividends to shareholders, and therefore can be viewed as having been plowed back into the institution. Another way to look at retained earnings is as internally generated capital.

As with each element of shareholders' equity, retained earnings have no intrinsic substance of their own.⁵¹ Although real legal interests, retained earnings, like profits, exist as an abstraction rather than as a source from which cash can necessarily be obtained. As with other elements of shareholders' equity, their character as a residual item depends entirely upon the nature of the assets held by the bank.

Preferred stockholders are entitled to specified dividends that typically accrue if unpaid. However, they have limited ownership rights, and their shares, which are valued at par, may be perpetual or have a fixed duration. Since preferred stockholders are entitled to regular payments and do not as a rule obtain an ownership stake in the company, they represent a form of hybrid capital that combines elements of equity and debt.

Equity Reserves

Other capital account items include equity reserves and treasury shares. Not to be confused with loan-loss reserves, equity reserves are special reserves set aside from

revenues⁵² and used to provision against losses or impairment of assets from sources other than the bank's loan or securities portfolio.

Treasury Shares

Not permitted in all jurisdictions, treasury shares are shares that have been issued to investors and then reacquired by the company.⁵³ They are therefore owned by the corporation itself and not, at the time of designation, by any shareholders. They function as a pool of shares out of which shares may be distributed to existing or prospective shareholders, whether through the exercise of stock options, new issues or rights issues, or other similar allocation mechanisms. Since treasury shares are owned by the bank, they do not function to finance assets, and by definition, they do not form part of capital. Since their contribution to shareholders' equity is zero, the par value of treasury shares is deducted from the total number of shares outstanding to obtain actual shareholders' equity.

OFF-BALANCE-SHEET ITEMS AND DERIVATIVES

Overview

Assets and liabilities on the balance sheet of a bank, or any firm, are so reported because they represent existing resources, owned by the enterprise, or existing claims against those resources. Consequently, at any given moment—recall that the balance sheet is a snapshot of a bank's asset and liabilities at a point in time—their value is comparatively fixed and relatively easily measurable.

A loan, for example, although it is subject to credit risk, can be valued in several ways, including the manner in which typically it is reported: at its par or principal amount, less any repayments, interest income, and subject to provisions against loan losses. Correspondingly, the amount owed in connection with traditional on-balance-sheet financial obligations can be similarly measured.

Contingent Items

Some financial instruments, however, are contingent in nature. Depending on the circumstances, no asset may come into being and a potential obligation may never arise.

Documentary Letter of Credit

The documentary L/C—often simply called an L/C—is a letter of credit used for trade finance purposes. The bank upon the payment of a fee by its customer promises to honor its customer's obligation to pay an overseas vendor for goods purchased, subject to the shipment of the goods supported by specified documentation attesting to the shipment. The L/C is structured so that the primary obligation is on the part of the bank's customer, the purchaser of the goods. Provided the documentation is in conformity with the terms of the L/C, the vendor's bank pays the vendor for the goods and claims the funds from the purchaser's bank, which is always obliged to pay if the documents are in order, even if its customer fails to make payment. In other words, the liability of the buyer's bank is largely contingent upon the default of its customer. Meanwhile, however, the bank remains the owner of the goods on the strength of the documents, hence the term. Documentary L/Cs are very complex. Any

discrepancy in the document or any fraud may result in one of the players—buyer, buyer's bank, seller, or seller's bank—losing the value of the goods to another player.

Standby Letter of Credit or Guarantee

Another elementary example is that of a standby L/C or guarantee. The bank only becomes liable for the full amount of the guarantee should the bank's customer fail to meet the obligations guaranteed.

Because it is difficult to establish the bank's actual exposure in the case of such contingent liabilities, and in any event since there is no cash flow involved until the guarantee is triggered, traditionally they were not recorded on the balance sheet; hence the term off-balance-sheet items. In lieu of being placed on the balance sheet, these items would be disclosed in a footnote to the accounts.

Derivatives Defined

The term *derivatives*, or *derivative instruments*, refers to financial contracts whose value is derived from an underlying asset, pool of assets (i.e., an index), or an established benchmark. In consequence, the value of a derivative instrument is subject to changes through shifts in the value of the underlying assets or in interest rates, exchange rates, or other similar publicly available data, according to some pre-determined calculation.

Traditional Derivatives: Options, Forwards, and Futures

The most common types of derivatives are options, forwards, and futures, which have been in existence for centuries. Their simplicity, however, belies the difficulties in measuring their value, a problem that has long baffled financial theorists. A sound approach to pricing options, for example, dates back only to 1973 and the advent of the Black-Scholes formula.

OPTIONS, FORWARDS, AND FUTURES DEFINED

Option

What is it? An *option contract*, or *option*, is a financial contract that grants (ordinarily in exchange for a premium paid) the purchaser or holder the right—without any obligation—to buy (a *call option*) or sell (a *put option*) a financial asset at a fixed price for delivery in the future as specified in the contract. That is:

- A call option grants the holder the right (but not the obligation on the holder) to buy the asset, which the holder will exercise if the market price of the asset rises above the price fixed in the contract.
- A put option grants the holder the right (but does not impose an obligation on the holder) to sell the asset, which the holder will exercise if the market price falls below the price fixed in the contract.

Forward

What is it? A forward contract, or forward, is a contract between a buyer and a seller to transfer a commodity or financial asset on a fixed delivery date at a fixed price.

How does it differ from an option? In contrast to an option, neither the buyer nor seller of a forward ordinarily has the right to opt out of the agreed-upon transaction.

How does it differ from a future? A forward contract is similar to a future contract) except that it is not normally standardized and, equally importantly, not traded through an exchange. A forward can be viewed therefore as an over-the-counter future contract. Since forwards are not traded through an exchange, there is no mandatory margin requirement, but as a consequence both parties to the contract are subject to default risk.

Future

What is it? A future contract, or future, is a standardized contract for a uniform quantity of items, usually a commodity or a financial asset, to be delivered by the seller to the seller at a specified price on a specified date, normally within one year.

How does it differ from a forward? Unlike forwards, futures are standardized contracts bought or sold through a third-party futures exchange. Since the contracts used are standardized, which increases liquidity, and are sold through an exchange that imposes margin and collateral requirements that must be settled daily, default risk to the parties to the transaction is minimized.

How does it differ from an option? Unlike options, both parties are obliged to consummate the transaction. To protect the sellers against default risk, that is, counterparty risk, the prospective purchaser is obliged by an intermediary exchange to make a margin deposit at the time of entering into the contract, and such deposits may be required to be increased if market conditions or the position of the purchaser changes. See also financial futures; forwards.

Swaps

In addition to options, forwards, and futures, derivatives products encompass swaps. Broadly speaking, a swap is a type of derivative transaction in which two counterparties for example swap their obligations to pay interest, making adjustments for the rates in a mutually agreeable fashion. As a rule, swaps are arranged by an intermediary bank, which collects a fee for its services.

An interest rate swap is a type of swap transaction that obliges two or more counterparties to exchange interest rate obligations. An ordinary interest rate swap involves obligations in the same currency. No principal payments, however, are ordinarily exchanged either initially or at maturity. An interest rate swap is often motivated by one party's desire to protect itself against the risk of higher interest rates. For example, an industrial company may only be able to borrow funds from a

Off-Balance-Sheet Items: Illustrative Types								
Nonderivative	Derivative instruments, or significant use of derivative instruments							
	Simple					Structured	Complex/Hybrid	
Letters of credit	Futures	Equity	Interest-rate	Currency	Commodity	Tranched securitizations	“Whole business” securitizations	
	Forwards							
	Options							
Guarantees	Swaps					Credit default swaps		

EXHIBIT 5.21 The Universe of Derivatives and Off-Balance-Sheet Products

bank at a floating rate of interest, but would prefer the certainty of a fixed rate of interest. An interest rate swap allows such a company to swap its floating-rate obligation for a fixed-rate obligation.

A currency swap is a type of swap transaction between two or more counterparties for the purpose of exchanging interest payments under a debt obligation or other payments in different currencies, coupled with the transfer of principal payments at maturity (and occasionally upon entering the swap transaction). Because the principal amount is usually transferred at the termination of the swap, currency swaps are more risky than interest rate swaps, all other things being equal.

The Universe of Derivatives and Other Off-Balance-Sheet Products

Over the past 25 years, increasingly advanced and multifaceted types of derivative products have been introduced. Exhibit 5.21 illustrates the main categories of derivatives and other off-balance-sheet products. The universe of off-balance-sheet products can be classified into traditional contingent products such as L/Cs and guarantees, and derivative products ranging in complexity from plain vanilla options, futures, and forwards to more sophisticated structured and complex hybrid products. Structured products can be assembled to offer a multitude of combinations of risk and reward characteristics to appeal to a wide range of market participants.

Derivatives: How Risky?

We view derivatives as time-bombs. . . .

[The] derivatives genie is now well out of the bottle, and these instruments will almost certainly multiply in variety and number until some event makes their toxicity clear. . . .

—Warren Buffett⁵⁴

Since options, futures, and forwards, and other more sophisticated derivatives began to comprise a significant part of the business activities of major financial institutions two decades ago, questions within the financial industry and among regulators have arisen concerning the impact of their use, and how to account for them properly in financial statements. A number of financial scandals in the late 1980s and during the 1990s in which derivatives figured highlighted the high risks

involved in their use. The recent financial crisis, hitting the world in 2008, has demonstrated the huge difficulty in pricing derivatives.

Defenders and Critics

While acknowledging that derivatives can pose pitfalls for the unwary, their defenders emphasize the benefits of these instruments' use for the purpose of hedging risk, and thereby increasing the stability of institutions that effectively employ them. They would argue that much confusion surrounds the dangers of derivatives and the hazards have sometimes overshadowed the legitimate use of derivative for risk management purposes.

In contrast, those wary of derivatives focus on the total notional value of the derivatives market, the proliferation of the most high-risk derivative instruments, and the use (and abuse) of derivatives for trading and speculative purposes, especially by unsophisticated market participants. Although some who criticize the use of derivatives undoubtedly do so out of ignorance, critics also include those whose financial acumen is beyond question, with Warren Buffett as a remarkable example.

The reality may lie somewhere in between. As may many modern technologies, derivatives may be used for good or for ill, and in the wrong hands there is no denying that they pose dangers to the unwary. Yet the expansion of the derivatives market over the past two decades is in large part a result of the growing recognition of their benefits. Increasing standardization in derivative products and supporting contracts, stronger legal frameworks, and better understanding on the part of market participants, together with greater depth and liquidity in derivative markets, have reduced some of the drawbacks associated with their use.

While the risks arising from the use of derivatives may be more difficult to understand than those associated with conventional on-balance-sheet products, that fact, until 2008, did not necessarily mean that derivatives were on the whole more risky. Indeed, in the early 2000s, credit charge-offs on derivatives for U.S. banks were negligible in comparison to the credit charge-offs on commercial and industrial loans. Even during the subprime crisis, the—substantial—losses in derivative holdings usually did not exceed those affecting their mortgage portfolios. One should note, however, that the analysis of such findings is very complex, as there are shifts of a bank's risk exposure between its direct portfolios and its derivatives portfolios.

The risks often seem greater than they are not only because of the high-profile scandals that have grabbed headlines,⁵⁵ but also for other reasons. Notional amounts of derivatives transactions exaggerate the actual netted risk exposures because the actual exposure at any given time is—often but not always—the net amount owed by one counterparty to the other, a sum that is likely to be far lower than the notional amount. In addition, with respect to interest rate swaps, which comprise the vast majority of all derivatives exposure, as periodic payments are made and as the swap nears maturity, the range of potential adverse interest rate scenarios narrows.

Not without Risk

Nevertheless, there are reasons for which the hazards of derivatives exposure should not be treated cavalierly. As a rule, any given derivatives transaction represents a zero-sum game in which at any particular moment one counterparty faces a loss, that is, is out of the money, at least in relation to how it would have stood had it not entered into the transaction. That one of the parties was in fact covering an existing exposure does not alter the reasoning, as that exposure had earlier created another

zero-sum game. As one research report by a U.S. investment bank put it in relation to interest rate swaps, such transactions either “generate or consume net interest income.”⁵⁸ Moreover, most derivatives, especially derivatives other than options, futures, and forwards, are over-the-counter products⁵⁶ not listed on any exchanges. Being private agreements made between two counterparties, the pricing and valuation of derivatives products may be problematic, relying as it often must on models, rather than on transparent market prices for the derivative instrument.

Deficiencies in the models used to price derivatives is likely to be difficult for nonpractitioners—and many bank analysts—to evaluate, even where the structure and assumptions employed by the model are fully disclosed. In brief, most derivatives pricing models rely upon the following inputs:

- Price of the underlying asset
- The exercise price
- Volatility
- Interest rates
- Time to maturity

Some of these such as interest rates, may be extremely difficult to forecast with accuracy over the long term. Consequently, the longer the maturity of the instrument and the greater its complexity, the more model risk becomes a serious concern. Difficulties in pricing have a knock-on effect exacerbating problems in estimating proper reserves to be set aside by banks and in allocating adequate levels of capital. Weaknesses in disclosure, regulation, and financial reporting only add to the perils.

The upshot is that coming to a full understanding of how much derivatives contribute to a particular bank's revenues or are likely to affect its balance sheet upon the occurrence of particular contingencies is a task fraught with difficulty. Greater disclosure concerning model assumptions together with net replacement values, value at risk (VaR)⁵⁷ data, and add-on capital requirements is a partial solution. But in the absence of uniform accounting and disclosure standards, evaluating the impact of derivatives on a bank's financial condition remains in large part a qualitative exercise.

Use of Derivatives

Derivatives, as suggested, potentially afford benefits to various market participants, which have contributed to their increasing popularity over the past two decades. Fundamentally, to counterparties derivatives can be used for hedging and risk management, on the one hand, and for trading and speculation, on the other. As one investment bank research report stated, “derivatives can be used either in a hedging mode—to provide insurance against rate changes, usually at some net cost—or in a proactive mode, attempting to anticipate and benefit from rate changes, albeit at a cost if anticipated changes fail to materialize.”⁵⁸ To financial intermediaries that arrange derivative transactions, setting up and administering derivatives deals also provide a source of fee income.

Risk Management

Derivatives potentially enable market participants to better manage their risks. Interest rate swaps and currency swaps, and futures, forwards, and options (especially in the

commodity markets) provide the clearest examples of this use. Risk management⁵⁹ can be of two sorts.

Cash Flow Management

One type of risk management involves changing cash flows to hedge against certain types of risk. Interest rate swaps, for instance, are frequently used to change fixed-rate funding obligations to floating rate and vice versa, or an obligation in one currency to those of another.

THE USE OF DERIVATIVES FOR RISK MANAGEMENT: EXAMPLE OF THE INTEREST RATE SWAP

The plain vanilla interest rate swap illustrates how derivatives are commonly used for risk management, or hedging, purposes. Suppose an industrial company wishes to borrow funds to expand its productive capacity. The rapidly growing company is not a giant in its sector, and does not yet have an established track record in the debt capital markets. Consequently, it finds it difficult to borrow cheaply at a fixed rate, and can only obtain reasonably priced finance on a floating-rate basis. Because its expertise is in its particular industry and not in the financial markets, it wishes to avoid taking on interest rate risk.

Another company, a more well-established issuer in the financial markets than the first, is able to source funds in the capital markets at a fixed rate easily, but has no pressing need for funds. However, its sophisticated treasury department is looking for ways to increase the returns it generates on unused liquid assets. A bank approaches the two companies and suggests an interest rate swap structured to allow the first company to obtain reasonably priced fixed-rate finance while the second can generate some additional revenue. Meanwhile, the bank earns a fee for arranging and administering the swap transaction. Each party sees a benefit in the transaction. The first company hedges its interest rate risk, the second company obtains additional revenue, while accepting additional risk that it can effectively manage, and the bank generates income as an arranger.

It is by no means the case that industrial companies are the primary users of derivatives. Banks themselves not only arrange derivatives transactions, such as an interest rate swap in this example, but are also major users as counterparties of both interest rate and currency swaps to hedge against corresponding risks in their fixed-income and foreign exchange (forex) portfolios.

Insurance

A second type of risk management effected through the use of derivatives involves products that mimic insurance products. For example, an industrial purchaser of a certain product may be concerned that the price of an important raw material may rise substantially in the next year. However, there is an equally strong probability that the price will fall during the same period benefiting the manufacturer. A large price rise could be disastrous for the company given its present contractual

commitments, while a price decline would merely result in a welcome windfall. By purchasing an option that enables the company to buy the product at its present price, the company can insure itself against the catastrophic situation.

CREDIT DERIVATIVES

A comparatively new type of derivative product, credit derivatives are employed to break out the credit risk in a transaction and effectively sell it to a party willing to accept the risk of default in particular circumstances in exchange for payment. Although the market for credit derivatives has grown rapidly since the late 1990s, this category of financial instruments long remained a relatively small portion of the total derivatives market. As an innovative and highly sophisticated product, the production of credit derivatives, earlier the province of the world's largest banks, grew substantially in the 2000s until the subprime crisis put a sudden brake to their expansion. The subsequent debt crisis in Europe confirmed the reality that many more players than originally thought were involved in derivatives that they did not understand.

Credit derivatives come in two basic varieties: credit default swaps and collateralized debt obligations.

Credit Default Swaps

Credit default swaps (CDSs) comprise the bulk of credit derivatives. In a CDS, one counterparty, the purchasing party, pays a fee to a counterparty, which agrees to accept the risk of default in respect to defined assets for a period in exchange for the fee. In effect, the first counterparty is purchasing insurance, while the second is taking on role of an insurer. A CDS may be arranged to "provide insurance" against the default of a single asset or a pool of assets. It is not necessary for either party to hold the asset or assets underlying the transaction.

Collateralized Debt Obligations

Another type of credit derivatives is the collateralized debt obligation (CDO), which resembles in some respects the asset-based security (ABS). CDOs are often used to facilitate ABS transactions as conduits, by acting to insure against particular default scenarios with regard to the assets underlying the ABSs. In a CDO, pools of credit risk exposure are aggregated and sold to investors, who receive payments in exchange, which provide an attractive nominal return on investment.

Trading and Speculation

Derivatives can also be used to facilitate trading, investment, and speculative activity, either by hedging certain types of risk to enable traders or investors to take on only those risks they wish in the hope of realizing a gain, or the derivatives themselves may be the subject of the transaction. Similarly, investment banks that make a market in

certain types of derivative instruments may be able to realize gains in the process of buying and selling such securities.

Analysis of Off-Balance-Sheet Items

Conventional bank analysis is oriented toward the income statement and items reported on the balance sheet. As a Morgan Stanley equity research report put it, “[b]ank analysts are, in our experience, happiest discussing on-balance-sheet items . . . [while] anything at the margin of the balance sheet . . . induces indifference or confusion.”⁶⁰ In addition to the fact that they are not easily accommodated by accrual-based accounting systems—the reason they were placed off balance sheet in the first place—the cause of this disinclination toward off-balance-sheet analysis can be attributed to derivative products coming in many varieties and being often highly complex. Consequently, the risk exposures are neither differentiated nor measured as easily as they are with on-balance-sheet items. As a corollary to this, the reporting of derivatives holdings has generally displayed less transparency compared with on-balance-sheet items, which are fairly standardized and on the whole widely understood.

This said, as the market for derivatives expands, and their use by financial institutions increases, they cannot be ignored in the process of evaluating a bank’s creditworthiness. Nonetheless, the foundational elements of on-balance-sheet analysis must be first explored before this more complicated subject can be profitably examined. Although a detailed discussion of the use of derivatives by banks and their effects is outside the scope of this book, derivatives are considered from time to time in succeeding chapters as we explore the process of bank credit analysis.

The Statement of Changes in Capital Funds

The *statement of changes in capital funds* registers changes in shareholders’ equity. Because banks are highly leveraged, determining their capital position is critical to bank credit analysis, much more so than for nonfinancial companies.

STATEMENT OF CHANGES IN CAPITAL FUNDS

Also called the *statement of changes in shareholders’ equity*, *statement of shareholders’ equity*, and similar terms, this financial statement accounts for changes in owners’ equity. Such changes result primarily from the following events:

- Earnings retained by the company
- Distributions to shareholders in the form of dividends
- Capital infusion made by shareholders either through an initial public offering (IPO) or a subsequent rights offering

In addition to the commonly reported financial statements just discussed, banks may provide supplementary data under various labels. In some jurisdictions, the format of fundamental financial statements is strictly regulated. In these circumstances,

additional information not required to be disclosed, but which is believed by management to be useful to investors may be provided.

Regulatory filings of various kinds, such as those required by securities regulators may also provide useful material. These should be distinguished from regular filings that most banks are required to make with bank supervisors. As a rule, these are kept confidential and are not accessible by the public.

NOTES

1. IBCA, *Bank Capital: A Vale of Tiers* (London: October 1997). IBCA is now part of Fitch Ratings.
2. KeyCorp's balance sheet is more detailed than Apple's, but this is partly a difference in presentation, with Apple providing more detail in the notes to the financial statements, rather than on their face.
3. KeyCorp, used as an example in the previous chapter, is actually a bank holding company. Most of the large U.S. banking groups have as their parent firm a bank holding company, which owns the lead bank.
4. Trade receivables or accounts receivable may be thought of as a category of financial asset in that each refers to a legally enforceable claim for payment by one party against another.
5. An intangible asset is an asset comprised of intangible property, meaning property whose value inheres in what it represents rather than in its physical nature. Goodwill is a special type of intangible asset. The IAS considers treatment of general intangible assets in IAS 38, which covers all nonphysical (tangible) assets other than goodwill (which is addressed by IFRS 3), and certain other forms of intangible assets including those held for sale or arising from insurance contracts, deferred tax assets, and mineral rights. Although financial assets are in a *sense* intangible assets, these monetary assets are typically treated differently from other forms of intangible assets and consequently the term *intangible assets* used without qualification usually denotes intangible assets of a nonfinancial character. Under IAS, an intangible asset is defined as "an identifiable nonmonetary asset without physical substance" (www.iasplus.com/standard/ias38.htm). Examples of intangible assets include brands, copyrights, exploration rights, franchise rights, and licenses. Intangible assets may have a limited life, such as patents or other legally protected intellectual property, or an unlimited life, such as goodwill. Intangible assets traditionally have been amortized to reflect their diminishing value as their useful life dwindles, but this treatment is no longer applied to goodwill.
6. Nor by definition would a nonfinancial company hold a large portion of financial assets for trading or portfolio investment purposes. It is not uncommon, however, for nonfinancial companies to use nonbank financial subsidiaries for these purposes. Although one purpose of such companies is likely to be to facilitate sales by the parent company, financial subsidiaries may become profit centers in their own right.
7. The nonfinancial company's financial assets will ordinarily be composed primarily of securities held for treasury and cash management purposes, together with equity investments in subsidiaries or in connection with strategic investments in other firms.
8. While commercial banks and investment banks frequently generate most of their returns on financial assets in the form of interest income, other financial companies such as asset managers aim to generate such returns mainly through trading and investment gains or through fees and commissions.
9. Earning assets encompass "any interest-producing or dividend-producing asset," including net loans and receivables, that is, net of unearned discounts and the allowance for loan loss reserve, interbank loans, and reverse repo transactions. As will be discussed later, using

total assets as the denominator of an indicator may no longer be very relevant, but it could still typically be commented that “earning assets account for about 85 percent–95 percent of total assets.” From Chaiyapat Paitoon and Bradford Ti, *Bank Accounts Made Easy*, vol. 1, *Recognition of Interest Income* (Hong Kong: Salomon Smith Barney, Asia-Pacific, February 12, 2001), 11.

10. Therefore, also included in the category of nonearning assets will be foreclosed assets—when real property, this is referred to in the United States as other real estate owned—and similar collateral or repossessed property.
11. The first usage is primarily British, the latter mainly American.
12. The illustration is intended as an analogy only.
13. Likewise, to a bank, its investment in bonds is an asset but to the company that issued them, the bonds are a liability. In the same way, to a bank its equity investment in Company X is an asset. But on Company X’s books, the bank is an owner of the firm and its claim constitutes shareholders’ equity in the company.
14. Transaction costs and the degree of a discount on the value of the asset to facilitate a quick sale also affect its liquidity. Liquidity is discussed in detail in Chapter 10.
15. Needless to say, since cash is cash, and therefore it need not be converted into cash, it is perfectly liquid.
16. Negotiable assets is a synonym for marketable assets, as is commonly used to refer to “tradable holdings of shares, bonds, derivatives, certificates in mutual funds, and convertible securities” (Bank of Israel, *Definitions and Clarifications*, www.bankisrael.gov.il/deptdata/pik_mth/ex_debt/define.htm [February 18, 2006]). Both terms encompass what are sometimes referred to as *quasi-liquid assets*. See the following endnote.
17. There seems to be no generally recognized definition for the term *quasi-liquid assets*, and the distinction between those assets that are deemed liquid and those that are considered quasi-liquid does not seem to be a precise one. The major difference between the two categories would seem to be that the latter category takes in a broader range of marketable securities than the former, encompassing securities having both short and long maturities. The term *quasi-liquid assets* is used both in bank financial statements (see, e.g., Bank Sinopac, 1999 Annual Report) and analyses (see, e.g., E. Wong, Barclays Capital, Asia-Pacific Fixed Income Research, “Bank of East Asia,” Barclays Capital: Hong Kong, May 13, 1999, p. 13, where quasi-liquid assets are defined as “Liquid Assets + Deposits + Marketable Securities”) as well as in macroeconomic literature (see, e.g., A.P. Thirwall, “The Mobilisation of Savings for Growth and Development in Developing Countries,” Departamento de Programación Monetaria e Investigación Económica Documento de trabajo 2004/02, Santo Domingo, Dominican Republic, February 2003, www.bancentral.gov.do/trabajos_investigacion/paperNo05.pdf).
18. Bank assets are also sometimes categorized by risk level, or, as mentioned, divided into interest-earning assets and nonearning assets. Risk categories include relatively risk-free assets such as government bonds of highly rated nations, low-risk assets, and moderate- and high-risk assets. Needless to add, the European debt crisis of 2012 perfectly illustrates the point that a perfect definition of a relatively risk-free asset does not exist.
19. The U.S. Federal Deposit Insurance Corporation defines cash as “U.S. and foreign currency and coin on hand or in transit, clearings, and cash items.” *Cash item* is a technical term referring to “checks or other items in process of collection payable in cash upon presentation.” Risk Management Manual of Examination Policies, Section 3.4—Cash and Due from Banks.
20. IAS stands for International Accounting Standards. Although it is not necessarily critical from an analytical standpoint whether particular assets are eligible to be classified as cash equivalent for IAS purposes, or under other accounting regimes, determining the relative liquidity of a bank’s nonloan assets is, as noted, facilitates the analysis of its liquidity position under the CAMEL framework. To reiterate, the term *cash equivalent* is a functional asset category, and the term may not necessarily appear on the balance sheet.

Instead, line items may refer to specific types of securities. This can be seen in the balance sheet of KeyCorp's, which reports under U.S. GAAP.

21. Under IAS 7.7, to qualify as "short term" for the purpose of being classified as cash equivalent, the securities may have a maturity of no more than three months from the date of purchase. Additionally, such assets must be held for treasury, that is, cash management, purposes, rather than for investment or operating purposes. The particular designation applied to the securities such as available for sale or held to maturity securities, although suggestive, is not necessarily determinative in this regard.
22. Balance sheets were traditionally presented horizontally; hence the terms *asset side* and *liability side*. KeyCorp follows contemporary practice of using a vertical format.
23. Generally, "due from banks" refers to demand deposits placed with other banks, but in some jurisdictions it may encompass time deposits as well.
24. While it is evident that cash and due from banks fall into the category of cash, examination of the balance sheet and the accounting notes would be necessary to determine which portions of the three subsequent line items would qualify as cash equivalent under applicable accounting rules.
25. Most bank securities holdings will be in the form of investment-grade debt securities. Except for shares held as strategic investments, including those of subsidiaries and affiliates, banks rarely possess large holdings of equity securities. Note that the term *securities* is used here loosely to embrace marketable financial assets other than cash. As discussed, some types of securities fall into the category of cash, while others are classified as cash equivalent. Note also that certain types of loans, termed purchased loans and receivables under the IAS, may be treated in the same manner as securities for valuation and income recognition purposes.
26. The high creditworthiness of government securities and deep markets for them in many countries tend to make them especially liquid. Although not all very creditworthy securities are also highly liquid, there tends to be a positive correlation between the two attributes.
27. Under IAS, purchased loans and receivables are treated in the same manner as securities for this purpose.
28. Although for most banks the vast majority of trading securities will be composed of debt securities, the category can encompass equity securities, listed or unlisted. Trading securities may also include securities "sold" but pledged to be repurchased under repo finance agreements.
29. The time of acquisition means the trade date, which is the date that a commitment to buy or sell the asset has been communicated to the counterparty.
30. To mark to market means to revalue assets at their "fair value," which is deemed to be their recognized market value at the current market price as reflected in quoted or published bids.
31. *Impairment* means that the financial asset's preprovision book value is greater than the asset's present value (i.e., the present value of its expected future cash flows), where present value is calculated using as the discount rate the asset's original interest rate. Alternately, impairment may be estimated using the fair value method. To mark the asset to market using this approach, the applicable discount rate is the current market rate of interest for a comparable asset.
32. Other securities in the available-for-sale portfolio, according to note 6, were primarily marketable equity securities.
33. So, while DSFH conceivably may have held some securitized assets, this was not discernible from the face of the balance sheet. In all probability, given that the market for local securitized assets was not well developed in Hong Kong at the time, it is likely that such holdings were, at best, a small proportion of total assets.
34. Nonperforming loans generally include loans that are more than 90 days past due, or in respect to which interest income has ceased to accrue.

35. The statement is summarized at the Financial Accounting Standards Board's website, www.fasb.org/st/summary/stsum142.shtml.
36. Recall that as presented on the balance sheet, liabilities are what the bank—or any other enterprise—would have owed others, should it have ceased operations and gone into liquidation on the balance sheet date.
37. Although technically all liabilities provide financing for a bank's assets, the term *funding liabilities* ordinarily refers only those deposits and other short-term debt funding, and not to incidental trade liabilities or to long-term liabilities that are deemed to form part of regulatory capital. See generally FDIC DOS "Manual of Examination Policies: Funding Sources and Other Liabilities," Section 6.2, www.almprofessional.com/Articles/98fundin.htm. Most liabilities are interest bearing, but incidental liabilities generally are not, unless the corresponding accounts become delinquent.
38. Observe that the referenced balance sheet shows such other liabilities in a single line item, entitled, Accrued expenses and other liabilities (the line item just above that shown as Long-term debt), which functions as a catch-all category.
39. Commercial deposits encompass both interbank deposits (interbank borrowing) and other wholesale deposits.
40. While formally having the character of debt, deposits are normally grouped separately from nominal debt, which denotes obligations in the form of fixed income securities issued by the bank. Depending on the market, interbank deposits (interbank borrowing) may be grouped with deposits or with debt. Here the latter occurs.
41. From the analyst's perspective, what is significant is whether they are interest bearing or not rather than whether checking services are provided.
42. NOW accounts are a form of accounts available in the United States, which is similar to money-market account.
43. The boundary between purchased funds and short-term debt is not a bright line. *Short-term debt* tends to connote funding sourced through money-market instruments such as commercial paper issues as opposed to the placement of deposits bilaterally or through brokers.
44. The use of subordinated debt as capital is discussed in Chapter 9 on capital.
45. See Chapter 9 concerning the analysis of capital adequacy.
46. A broader term is *owners' equity*.
47. The former usage is American, the latter mainly British.
48. Confusingly, the term *paid-in capital* may be used to mean what we have referred to in the text as *legal capital* or *stated capital*, and as such is to be calculated by multiplying the number of shares outstanding times the par value of each share.
49. See Peter S. Rose, *Commercial Bank Management* (New York: McGraw-Hill, 2002), 136. A significant feature of paid-in capital, including stated capital surplus, is that it is not available for distribution to shareholders, except upon liquidation.
50. Each country has its own definition for *authorized capital*, and some do not even make use of that notion. The United Kingdom, for example, long used it, but abolished it in 2006.
51. While it may be obvious to some, it is worth stressing that the term retained earnings merely signifies a portion of the shareholders' interests in the bank, and does not necessarily correspond to or represent any a pool of liquid funds available to the enterprise. (The character and composition of a firm's assets is displayed in the breakdown of assets and not in the equity portion of the balance sheet.)
52. Equity reserves include, among other items, contingency reserves, reserves set aside to pay dividends on preferred shares or to retire preferred shares or senior debt. They do not include loan-loss reserves.
53. The reason for the existence of treasury shares is that the organization may wish to have shares on hand for an acquisition (against shares) or for distribution to employees as a

form of compensation. Consequently, they do not represent outside capital invested in the company. Treasury shares also include issued shares repurchased by the company. Not all jurisdictions permit treasury shares and many require shares repurchased from investors to be “retired.” See generally C. Horngren, *Introduction to Financial Accounting*, 2nd ed. (Englewood Cliffs, NJ: Prentice-Hall, 1984).

54. Letter to Berkshire Hathaway Shareholders, 2002, as quoted in W. de Winton et al., “European Banks: Industry Overview—Derivatives: What You Need to Know,” London: Morgan Stanley, April 7, 2003.
55. These scandals have often pointed more to weaknesses in management policies and controls as much as to perils in off-balance-sheet products.
56. Within the universe of derivatives, a distinction can be made between listed or commonly traded over-the-counter derivative products, and those that are highly customized and are either nonmarketable or marketable only with difficulty.
57. VaR stands for value-at-risk. This concept is discussed in a later chapter.
58. Diane B. Glossman et al., “Banking Basics and Beyond,” UBS Warburg, New York, February 19, 2002, 61.
59. Risk management will be developed further in Chapter 12.
60. W. de Winton et al., “European Banks: Industry Overview—Derivatives: What You Need to Know” (London: Morgan Stanley, April 7, 2003), 5.

Earnings and Profitability

Earnings power is a key determinant of the long-term success or failure of a financial institution. It measures the ability of a bank to create economic value and, by adding to its storehouse of resources, to preserve or improve risk protection for creditors.

—Moody's Investor Services, *Bank Financial Strength Ratings: Revised Methodology*, September 2006

The keys to banking are as follows: Charge high enough lending rates to cover default risk and yield a hefty profit. Leverage your capital and customer relationships by providing services . . . in areas where you have a comparative advantage. Maintain a strong enough balance sheet so that counterparty credit concerns don't impinge on your ability to do business. Avoid businesses where you have no competitive advantage.

—Walter Altherr¹

An examination of the composition of a bank's earnings is an appropriate place to start a thorough analysis since the makeup of the various earnings flows conveys information about both its risk and performance profile. Different sources of earnings imply different levels of risk. In addition, an overdependency on any single source of revenue tends to make a bank more vulnerable to its sudden diminution. For instance, a bank that is heavily reliant upon commercial and industrial lending to small and medium-size firms is likely to be at a disadvantage to rivals that can draw on a number of revenue sources. Owing to its accepting higher risks, it will likely enjoy higher returns than an institution that engages primarily in lending to blue-chip corporate customers and secured mortgage lending. But over an extended period, its profitability may very well compare unfavorably with institutions having more diversified revenue streams that afford them more stable earnings production through the business cycle.

Having spread the financials of the bank to be evaluated—or having obtained these data in a standardized format to facilitate comparison, a number of key ratios are examined to begin the analytical process. Of the four financial elements of the CAMEL model—capital, asset quality, earnings, and liquidity—an examination of earnings is a good place to start, notwithstanding that the acronym begins with *capital*. This is because while all of a bank's financial attributes are interconnected,

the generation of earnings is at the heart of these interrelationships. Earnings are closely linked to both capital and liquidity, which comprise two of the components of the model. Therefore, the rate at which a bank creates profits measured on a periodic basis in relation to other variables provides the most immediate and sensitive indicator of its financial health. Provided its other ratios are within regulatory norms, a bank, like any other firm, by generating adequate earnings, will be able—barring deficient management or exceptional circumstances—to survive, grow, and prosper.²

Analysis of expenses is also part of earnings and profitability evaluation since income is simply what is left over when expenses have been subtracted from revenue. Looking at the composition of various categories of costs, their sources, their trends, and their relationship to other financial attributes, such as total assets or total net income, is an indirect way of examining earnings and profitability. The only difference is that we are approaching the issue from the other side of the coin, so to speak. The relative burden that a bank's operating expenses comprise in relation to its assets and revenue, and concomitantly management's ability to control and optimize its cost base, are loosely referred to as its *efficiency*.

THE IMPORTANCE OF EARNINGS

Though profitability levels are traditionally the focus of equity analysts, and in the past were given less emphasis than balance sheet measures of capital and liquidity, the dynamic ability of a bank to bounce back from economic shocks is increasingly seen as a critical attribute of credit strength.

From this perspective, the capacity of a bank to generate capital internally, attract new capital through high core profitability and earn its way back to financial health are regarded as paramount. They are viewed as more important than an impressive balance sheet, especially if high levels of capital and liquid assets are coupled with weak earnings and lackluster profitability.

Profits versus Profitability

Earnings within the CAMEL acronym refer to both profits and profitability, but with an emphasis on the latter. Note that earnings and profitability are two distinct but related attributes.

1. Profits are measured in absolute amounts or in terms of percentage change from one financial period to another.
2. Profitability is measured using ratios such as return on equity (ROE: net income/shareholders' equity) or return on assets (ROA: net income/total assets).

Note that a bank can register steadily increasing earnings in absolute terms at the same time that its profitability is steadily decreasing either in absolute terms or relative to its peers. At a more sophisticated level, profitability analysis takes account of the risks incurred to generate earnings—so-called risk-adjusted profitability—as well as earnings quality.

Risk-Adjusted Profitability

Risk-adjusted profitability metrics simply take account of the level of risk exposure in generating profit. These techniques embody the proposition that all things being equal, it is preferable to incur a lower level of risk to produce the same return. They will be discussed later in the ratio analysis section.

Earnings Quality

As with capital quality, asset quality, and management quality, there exists such an attribute as earnings quality. *Earnings quality*, in brief, refers to the sustainability of earnings. In other words, high-quality earnings are those that reflect a company's earnings capacity over time and that are reported without the use of accounting gimmicks. Such gimmicks, otherwise known as cooking the books or creative accounting embrace a variety of techniques that have in common the objective of making earnings—and other parameters—ultimately look better than they would applying best-practice accounting guidelines.

Needless to add, earnings quality should also consider the type of earning ratio used in the analysis. Only figures that are fully adjusted for risk would properly describe the bank's earnings. If accurate and truthful information is gathered by management for such an analysis and if the information is disclosed, the bank's apparent earnings ratios would not need potential negative adjustments on account of opacity.

Intrinsic Earnings Quality

In this light, the evaluation of earnings quality can be separated into two principal aspects. On the one hand, where no aggressive accounting practices are employed, the sustainability of various streams of income and expense burdens is considered. These can be roughly rated according to their volatility.

As a simple example, interest income from secured lending is customarily viewed as having a more recurrent character, and therefore higher quality, than income generated from gains on proprietary trading. Similarly, within the stream of interest revenue, income generated from secured loans to owner-occupied homeowners will typically be viewed as being of a higher quality than secured loans to real-estate developers.³

Accounting Risk

The other fundamental aspect of earnings quality analysis concerns the aggressiveness of accounting practices used by management. This facet of earnings quality requires a good understanding of applicable accounting rules and how the principles they are meant to safeguard may be circumvented through discretionary manipulation of those rules. The term *accounting risk* is sometimes used to denote the risk that reported earnings may diverge from earnings fairly reported under best accounting practices. Note that accounting may be aggressive and thereby adversely affect earnings quality without necessarily violating accounting rules or related laws.

Earnings Quality as a Warning Sign

Looking more closely at the importance of earnings to a bank's financial health, it is evident that if they are of a good quality, ample earnings and a strong level of profitability afford a bank the capacity to:

- Earn a sufficient *return* for the bank's owners, which makes it easier for the institution to access new investment in the capital markets if necessary.
- Maintain adequate *solvency* to absorb economic shocks by building up *capital* internally through retained income.
- *Fund new investment* to respond to customer needs and competitive pressures, thereby enabling the bank to expand at a comparable pace with its rivals.

Finally, a strong bottom line allows a bank to earn its way out of problems should it experience distress and have to restore its financial health.

So, while favorable earnings indicators are credit-positive and an indication of a robust bank, adverse material changes in earnings and profitability indicators are correspondingly negative and potentially herald a deterioration in the institution's overall financial condition and creditworthiness. More specifically, they can function as warning signs of any or all of the following:

- That the institution's *funding costs* are increasing.
- That *loan pricing* is being squeezed by competitive pressures.
- That asset quality is deteriorating.
- That the bank is being hit by trading or investment losses.
- That management is less able to contain *operating expenses*.

A decline in earnings and profitability can also be a precursor of less obvious deterioration in a bank's capital strength, asset quality, or liquidity. If such challenges are not overcome, the consequence can be a decline in the bank's overall financial health. By examining the income statement, the relationship between various parts of a bank's revenue stream and of its expense burden can be discerned.⁴

Impact of the Bank's Business Model on Earnings

To comprehend why a bank registered average, below-average, or above-average ratio values, and what caused these values to change, it is essential both to understand the interrelationships between earnings and profitability indicators as well as the nature of the business activities in which the bank to be analyzed engaged. Relevant lines of inquiry include the following.

What Is the Bank's Geographic Franchise?

What is the geographic scope of its operations?

- Global?
- Regional?
- National?
- Local?

What Is Its Business Focus, That Is, Its Business Franchise?

In what business activities does it focus? In what sectors does it generate the bulk of its earnings or enjoy a competitive advantage?

- Commercial banking?
- Investment banking?
- Custody and clearing?
- Government policy lending?

Who Are the Bank's Customers?

What is the nature of the bank's customer (borrower) and depositor base? Does it primarily serve:

- Multinational enterprises?
- Governments or nonprivate organizations?
- Large domestic firms?
- Small and medium-sized enterprises?
- Consumers?

What Is the Bank's Position within Its Market?

- How big is the bank?
- Where does it stand within the applicable banking sector?
- Is it one of the top three banks in the market?
- Is it a middle-market bank or a local community-based institution?

Performance Terminology

Related or similar-sounding terms are employed in this book. Some key terms are used in this chapter. One distinction for instance is between *profits* and *profitability*. Note also that by *profitability* we mean not mere earnings, profits, revenue, or income (depending upon the terminology used) but how earnings compare with other critical inputs or variables. Measuring profitability embraces the following queries:

- What is the relationship between net income and the amount invested by the owners in an enterprise?
- How does revenue resulting from interest paid by borrowers less funding costs compare with the amount of principal advanced?
- How does net income before taxes compare with the total resources owned by the enterprise?

EVALUATING EARNINGS AND PROFITABILITY: AN OVERVIEW

Before delving into more detail concerning the indicators mentioned in the preceding section, it is vital to be aware of the objectives of earnings and profitability analysis. In general, financial analysis for purposes of credit assessment seeks to break down a firm's accounts into their component parts to identify those elements that have functioned to strengthen or weaken the enterprise's financial condition. The goal is to

A NOTE ON TERMINOLOGY

Earnings Analysis

By *earnings* is meant *revenue*, *income*, or *profit*. The four terms are generally synonymous, although the connotation of the last two is that there must be some revenue in existence from which deduction can be made to obtain profit or earnings. In any case, when we talk about earnings analysis, we mean understanding the source of earnings or income—their composition, what drives them, and what hinders them.

Profitability Analysis

Profitability analysis has a somewhat narrower meaning than earnings analysis. Profitability is not the same as profits, and an increase (or decrease) in profits does not necessarily mean an increase (or decrease) in profitability. Output must always be related to inputs, whether inputs are measured in the form of shareholder funds (equity) or the assets the bank holds, since it is profitability and not mere profits that adds value to an equity investor's shares and generally provide comfort to the creditor. Profits could be the same for two banks, but their level of profitability could differ substantially.

Performance Analysis

By *performance analysis* is meant all of the preceding, as well as consideration of how earnings and expenses relate to nonfinancial inputs. So *performance analysis* includes *earnings analysis* and *profitability analysis*, as well as such ratios as net income generated on average per bank branch, or per activity, or per total assets per employee, to use simple examples. Perhaps needless to say, all these terms blur at the edges, and despite the distinctions drawn are often in practice used interchangeably.

understand the reasons for the most recent financial results, as well as for any significant changes from prior periods or divergence from industry norms.

Recognizing Trade-Offs

All other things being equal, above-average earnings growth and profitability are desirable characteristics and, within limits, usually construed as credit positive. But profit does not come without risk, and above-normal profits imply higher risk, or conceivably deficiencies in disclosure.⁵ To evaluate a bank's earning capacity and its profitability, analysts therefore examine those economic, business, and financial criteria that drive and constrain profit generation and seek to ascertain what choices management has made in terms of potential risk and potential reward. The trade-offs between earnings and other financial characteristics should always be borne in mind.

Risk versus Reward

A notable one is the trade-off between earnings and profitability on the one hand, and asset quality on the other. It is comparatively easy over the short term for a bank to increase its profits by relaxing lending standards and increasing the volume of funds advanced to borrowers. Over the longer term, however, the probable outcome is apparent. By allowing its asset quality to decline, a bank can cause its profits to grow, at least nominally and temporarily. But it is likely to pay the price later as its loans go sour at an increasing rate.

The degree to which earnings and profitability are sought will also influence other CAMEL elements besides the credit risk that is embodied in asset quality. For example, a bank that has reduced its holdings of liquid assets to a minimum will, all other things being equal, generate greater profits than a bank with a high proportion of liquid assets to loans. The increased earnings and profitability, however, will be at the expense of liquidity. That is, the increased profitability resulting from such decisions will probably be accompanied by higher liquidity risk.

In sum, the goal of earnings analysis is to identify the causes of fluctuations in financial performance, either positive or negative. Because, subject to a moderate degree of variation, banking is a comparatively homogeneous business and most deposit-taking institutions make money in fundamentally the same way, it is not particularly difficult to isolate the most critical variables. With a modicum of training and experience, the basic dynamics of the relationships that influence bank performance can be discerned and applied.

Barriers to Recognizing Trade-Offs

Although the illustrations just provided in respect of asset quality and liquidity risk are comparatively obvious, it must be acknowledged that the interrelationships among the myriad elements within a multifaceted banking operation can be both subtle and highly complex. The intricacies of their interactions in elevating or eroding performance may not always be within the grasp of the outside analyst. Often the reason is that disclosure is insufficient to determine which parts of the bank's operations are performing particularly well or badly. For example, within the stream of noninterest revenue, some components of the stream may increase for various reasons while other components are eroded as a result of other, perhaps overlapping influences. The apparent lack of change may mask the formation of portentous trends beneath the surface. In fact, this is true for many figures and ratios in bank analysis. Despite such obstacles, absent inadequate or deceptive financial reporting, it is almost always possible to make a reasonable assessment of its earnings capacity or the contribution of earnings and profitability to the institution's overall creditworthiness.

Sustainable Earnings and Profitability

It is also important to be aware that the concern of credit analysis is almost never solely with nominal earnings but with sustainable earnings. In other words, when assessing earnings and profitability, more is not always better. Therefore, in

addition to the question of how much risk has been traded for potential reward, another critical query concerns the sustainability of the institution's earnings. Erratic earnings mean greater volatility and imply greater risk. Consequently, ascertaining whether the risks are imprudent will almost certainly require more than a superficial effort.

NOMINAL EARNINGS VERSUS SUSTAINABLE EARNINGS

To illustrate the concept of the risk-reward trade-off, suppose that hypothetical Hyacinth Bank registered a return on assets (ROA) figure substantially higher than its peer, Daisy Bank. If so, the superficially superior profitability may have been the result of Hyacinth's incurring greater risks than that of its rival. When Hyacinth Bank's true risk exposure is taken into account, either qualitatively or quantitatively through the use of more refined risk-adjusted indicators, a more negative conclusion might be reached. Likewise, if the Hyacinth's apparently better results were the consequence of the use of accounting gimmicks that overstated its earnings, then the assessment reached when the full facts have been considered may diverge from those based on a superficial analysis.

Naturally, using more sophisticated profitability ratios, such as those in which numerator and/or denominator are adjusted for risk, allows for a sounder analysis, but those ratios are often difficult to establish at first glance.

A final point. Do not make the mistake of attributing much significance to marginal changes in absolute numbers. As a rule of thumb, for raw figures, changes or differences of 2 percent or less per annum—for example, 1.5 percent earnings growth—in a single annual reporting period border on the immaterial and as a practical matter may be construed as stable growth. Changes in ratios are more sensitive. In the case of ratios such as ROA or net interest margin, the analogous threshold might be 2 basis points (bps). For example, two banks having respective ROAs of 1.32 percent and 1.34 percent will justifiably not be regarded as materially different in terms of this indicator.

The concept of “sustainable earnings and profitability” just mentioned is essentially the same as the concept of earnings quality introduced earlier, and the two terms are used here more or less interchangeably. As discussed, earnings quality takes account of the intrinsic risks associated with the earnings stream together with deficiencies in disclosure and reporting of financial items.

To evaluate earnings quality, it is essential first to understand the nature of the business activities in which the subject bank is engaged, as was discussed in the preceding section. Second, the analyst should note any unusual features of the bank's financial statements that may signify aggressive, or even deceptive, accounting practices. Thus, in addition to understanding the business model of a bank as discussed in the first section of this chapter, it is worth taking some time to obtain an overview of the bank's financial statements before engaging in detailed ratio analysis. To this end, the following questions may prove useful.

- Are there any unusual features in the bank's accounts—anything out of the ordinary?
- What distinguishes this bank's financial statements from those of its peers?
- How do the elements of the financial statements fit together? Are there any apparent discrepancies?
- Is the disclosure adequate to understand the accounts?
- Have there been any recent restatements of the financial accounts that will make time series comparisons problematic?
- Have there been any business combinations that will do the same or provide fertile ground for creative accounting?

The foregoing questions are intended to be merely suggestive.⁷ The nub of the matter is to determine in what ways, and to what extent, the earnings and profitability registered by the subject institution differ from its peers.

EARNINGS ANALYSIS

Having obtained a bird's-eye view of a bank's franchise and financials, the core task of examining its various ratios and indicators can be taken up. By juxtaposing and noting differences in these ratio values, the institution's most recent results can be used in comparisons to the historical and recent performance of its competitors and in relation to its own past performance.

Interest versus Noninterest Income

Net interest income, as a general rule, is viewed by analysts as a more stable source of income than noninterest income. As does any generalization, it has its exceptions. But because banks have traditionally earned the bulk of their revenue from interest on loans or holdings of debt securities, and flows of interest income tend to be relatively predictable, the net interest income stream is usually regarded as being more sustainable and of a higher quality of earnings than noninterest income. At the same time, both investors and analysts are aware that noninterest income provides a useful complement to interest income since dependence upon a single category of revenue (as alluded to) may prove disadvantageous to a bank both in respect to long-term profitability and its credit risk profile.

One disadvantage of reliance upon interest income—and a corresponding advantage of having a substantial stream of noninterest revenue to complement it—is that capital must be allocated against the earning assets required to generate interest revenue. Sources of noninterest income are not so closely linked to asset holdings and are, therefore, to a large degree increasingly and thankfully constrained by recent regulations, decoupled from capital allocation requirements. A bank with a good mix of interest and noninterest revenue will often show higher levels of profitability, as measured by ROA and return on capital, particularly on *economic* capital as defined for regulatory purposes, than one that is solely depending upon interest income.

A second disadvantage of dependence upon a single source of revenue, even if it is of high quality and recurrent in character, is that such reliance introduces other

risks, namely that of overconcentration. Although all banks seek to minimize *concentration risk* within their loan books, relying entirely on interest revenue unavoidably creates a degree of concentration. If a bank enjoys a dual stream of earnings, then should difficult economic conditions arise resulting in a problematic lending environment, there remains the possibility that a bank may continue to generate substantial compensating revenues from noninterest income. So some degree of diversification resulting from a balance of interest income and noninterest income streams is usually perceived as credit positive.

A Deeper Analysis of Earnings

A crucial aspect of the analysis of earnings and profitability of a bank is arriving at an appropriate definition of earnings. The bank's income statement will report net income and various other remainder income items, such as pretax profit, according to its interpretation of the relevant accounting standards and reporting guidelines. But the definitions embodied in these standards and guidelines may not be those most useful to the bank credit analyst.

Analytically Adjusted Earnings versus Reporting Earnings

As observed elsewhere, there may be more than one bottom line.⁸ For the analyst there may be even more bottom lines to uncover, as well as intermediate items—such as pretax profit—to adjust.

Reported Net Income

Nominal, that is published net income, while it has the virtue of being readily accessible, may not be the best number to use to evaluate earnings growth (discussed in the next section) or profitability. Earlier it was observed that nonoperating items, for example one-off gains, while they may contribute to pretax profit, should be deducted from that number to obtain net income used for analytical purposes. The reason is simply that nonoperating items are by definition nonrecurrent and are apt to distort the bank's essential profitability. While most bank accounting regimes will conspicuously break out such items from recurrent items, there are some markets where this is not the case.

Core Earnings

Perhaps more essential than net income for analytical purposes is what is sometimes referred to as *core earnings*. Recollect that preprovision income (PPI) is equal to operating income (which has been defined as net interest income plus noninterest income) minus operating expenses (excluding loan-loss provisions).⁹ Accordingly, while net income figures in both earnings and in profitability analysis as the numerator in a variety of ratios, so does PPI. Different analysts and organizations have different ways of defining core earnings, but nearly all consider some variant of this ratio as an important profitability indicator for credit analysis purposes.

Adjusting for Other Items

To fine-tune PPI even further, two additional items are sometimes subtracted:

1. Preferred dividends¹⁰
2. In some circumstances, consumer loan write-offs

The reason for deducting preferred dividends at a higher level of the income statement is that while these payments are nominally dividends they are functionally indistinguishable from interest expense, which is, of course, subtracted to obtain preprovision income.¹¹ The reason for deducting consumer net charge-offs is to take account of actual credit costs of this type of business with the view of obtaining a measure of the risk-adjusted income.

Evaluating Earnings Growth

With the issue of operating revenue composition dealt with, there is still one more preliminary matter to consider before plunging into fundamental ratio analysis. It is *earnings growth*. Earnings growth shows how the subject institution's net revenues have changed since the past fiscal period.

Examining Changes in Earnings and Other Income Statement Items

To begin, the bank's most recently reported net profit figure can be reviewed, and also compared with the figure for the previous period. The value in the previous period can be compared with the one registered before that, and so on. The same approach can be applied to adjusted net income, as well as to other items on the income statement, including:

- Net interest income
- Noninterest income
- Noninterest expenses
- PPI
- Loan-loss provisions
- Net operating profit after provisions
- Pretax profit

To commence the analysis of earnings, the following questions may be posed with regard to each of the preceding items:

- How does the current figure compare with the previously reported period and on a year-on-year basis?
- Has it increased? If so, by what percentage?
- Has there been a decrease, and if so by how much?
- How does the percentage change compare with historical performance over the past two or three years?
- Is this the continuation of a trend, or possibly the reversal or beginning of one?

- How does the figure compare with those values registered by the bank's peers? Have other banks been affected similarly? Is the change sector-wide, or is it specific to the bank?
- Depending upon whether the change is sector-wide or bank-specific, what might have caused the change?
- What figures appear as if they might be out of the ordinary?¹²

SCAN FOR ANOMALIES

Steady and moderate increases over an extended period form the ideal growth profile from a credit perspective. Excessively high rates of growth, flat or negative growth, erratic growth, and worst of all making losses darken the credit picture. When examining net profit growth or similar ratios, watch out for distortions that arise from changes off a low base. This can cause the growth rate to appear better than it actually is. By at least scanning a bank's earnings over the most recent three-to-five-year period, the analyst can see whether any such distortions are present.

Delta Ratios

Delta ratios measure changes in earnings and other key financial statement items. Analysts customarily express earnings growth in relation to annual or year-on-year growth, quarter-on-quarter or sequential growth, and sometimes half-on-half growth. The formula for calculating the change from one period to another is shown in Exhibit 6.1.

Amplified Percentage Changes Off a Low Base and Other Percentage Change Peculiarities

When evaluating changes in earnings, or any other line item for that matter, be aware that small absolute changes can translate into large percentage gains, which may exaggerate the impact of the change, particularly on the positive side. Always consider the absolute change in line items as well as patterns of change over the past three to five prior periods, as the percentage changes may exhibit some peculiarities.

While negative growth rates are usually not to be cheered, they are not necessarily grim news.

If a bank's earnings increased 10 percent to \$2.2 billion for the quarter, as compared with 2.0 billion the previous quarter, and then declined by 9.1 percent back to \$2.0 billion the following quarter, the negative growth would still result in substantial profits, and in this case the profits would remain unchanged if the two periods were combined. And yet, if profits were negative in a given period, for instance -\$200 million in a given quarter, this would almost always be seen as an unfavorable outcome.

FORMULA	YEAR 2				YEAR 1			
CELL	A	B	C	D	E	F	G	H
FINANCIAL PERIOD	4Q	3Q	2Q	1Q	4Q	3Q	2Q	1Q
VALUE	10	20	30	30	20	30	30	20
YEAR-ON-YEAR (YOY) GROWTH, 4Q02 VS. 4Q01	$\begin{aligned} &= (A/E) - 1 \\ &= (10/20) - 1 \\ &= -0.5 \text{ or } -50\% \end{aligned}$							
QUARTER-ON-QUARTER (QOQ) GROWTH, 3Q02 VS. 3Q01		$\begin{aligned} &= (B/C) - 1 \\ &= (20/30) - 1 \\ &= -0.33 \text{ or } 33\% \end{aligned}$						
HALF-ON-HALF (HOH) GROWTH, 2Q02 VS. 4Q01			$\begin{aligned} &= (C/E) - 1 \\ &= (30/20) - 1 \\ &= 0.5 \text{ or } +50\% \end{aligned}$					

EXHIBIT 6.1 Delta Ratios

BEWARE OF AMPLIFIED PERCENTAGE CHANGES OFF A LOW BASE: AN ILLUSTRATION

For example, as shown in Exhibit 6.2, in years 1 and 2, hypothetical Ames Bank records net profits of \$90 million and \$100 million, respectively.

EXHIBIT 6.2 Net Profits over an Eight-Year Period, Ames Bank, a Hypothetical Institution

\$m	Yr 1	2	3	4	5	6	7	8
Net profit	90	100	10	60	110	1	−3	−12
% change		11%	−90%	500%	83%	−99%	na	na

In Year 3, amid a recession, Ames Bank records a profit of just \$10 million. Rebounding from the slump in Year 4, it records a profit of \$60 million. The absolute change is \$50 million, but the percentage change is 500 percent. Year 4 was a good year, but the 500 percent growth makes it look better than it was. Profits were actually the third-lowest in five years, but the increase looks large because it is jumping off a low base. Had the recession in Year 3 resulted in just \$1 million in profit, the increase would have been 5000 percent for the same Year 4 earnings.

In Year 5, Ames Bank's profit is \$110 million. Growth in absolute terms was also \$50 million, but this time the percentage change was 83 percent.

In Year 6, another recession hits and Ames Bank records only \$1 million in profit. The absolute change this time is \$109 million, the largest yet, but the percentage change is only -99 percent. Negative changes short of going into losses cannot exceed 100 percent, so negative results short of into losses can never reach the spectacular numbers that can be registered for positive growth.

Finally, in Year 7 the bank makes its first loss of \$3 million against \$1 million in profits the year before. Losses cannot be expressed meaningfully as percentage change from a positive base, so these are labeled n/a for not applicable. As the bank's loss difference in fact was four times the previous year's profit, Year 7 profits could conceivably be expressed as -400 percent of the previous year's profits. It would be trickier to transpose the reasoning from a loss to a profit, or from loss to loss, hence the n/a label.

The relationship between positive growth rates and the negative growth rates necessary to return to a starting value are shown in Exhibit 6.3 in both tabular and graphical form.

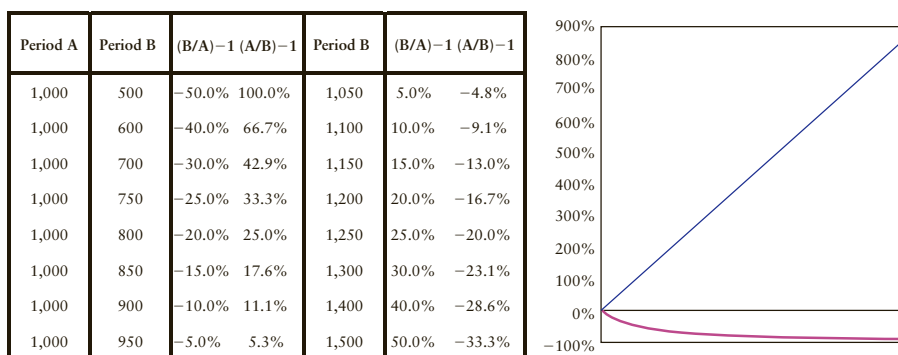


EXHIBIT 6.3 Positive Growth Rates and Corresponding Negative Growth Rates

PROFITABILITY RATIO ANALYSIS

For bank credit analysis purposes, the three major types of ratios—fundamental ratios, delta ratios, and common size ratios—are all utilized. Delta, or percentage change ratios, were already discussed earlier.¹³ We therefore turn to fundamental profitability ratios.

Overview of Fundamental Profitability Ratios

There are three principal types of fundamental ratios. They include:

1. Return-type ratios
2. Margin-type ratios
3. Cost efficiency-type (or cost-type) ratios

The most important examples of each category of ratio, together with their basic formulas, are shown in Exhibit 6.4.

Return-Type Ratios

Ratios in this group answer the question:

What return is generated on a particular input such as equity or assets?

Return on equity (ROE) and ROA are the two fundamental return-type ratios used in bank credit analysis.

1. ROE measures the return on the shareholders' equity (which may be different from his or her investment).
2. ROA shows how efficiently the enterprise is able to extract earnings from its assets.

The main difference between the two is that ROA effectively represents a core profitability measure that excludes consideration of leverage.

Margin-Type Ratios

Margin-type ratios show the profitability of a particular line of business, essentially asking the question:

How profitable on its own is this business line?

This category of ratios is similar in structure to return-type ratios, but differs in that rather than taking into account an institution's entire stream of operating income, margin-type ratios focus on a single component of the earnings stream corresponding to a type of activity.

As mentioned, all bank business activities fall under either one of two groups corresponding to a bank's two principal revenue streams: those that mainly generate interest revenue, and those that do not. Because it is easier to compute, it is not

EXHIBIT 6.4 Basic Performance Ratios

Return Ratios	Margin Ratios	Cost-Efficiency Ratios
Return on Assets: Net income/total assets	Net Interest Margin: Net interest income/ earning assets	Cost-Income Ratio: Noninterest expense/preprovision income
Return on Equity: Net income/shareholders' equity		Cost-Assets Ratio: Noninterest expense/total assets

surprising that the most common margin-type ratio for banks is that which measures, on an average—and therefore approximate—basis, its net interest income per unit of earning asset. This is the function of a bank's *net interest margin*, which is what a bank gets out of those assets. It is defined as net interest income divided by average earning assets.¹⁴

Cost-Efficiency Ratios

Finally, a third type of performance ratio looks at the cost side of the ledger. The two most common cost efficiency ratios are the cost-income ratio and the cost-asset ratio. Both measure a bank's operating expenses—usually defined as noninterest expense other than tax and loan loss provisions—as the numerator of a fraction in which either preprovision income or (average) assets function as the denominator.

1. The *cost-income ratio* reflects the degree to which a bank's overheads are adversely affecting its ability to generate profits.
2. The *cost-asset ratio*, also called *cost margin*, measures overheads as a percentage of total assets, a denominator that is less volatile than net income and therefore a more stable indicator of a bank's relative cost base.

Averaging the Denominator

In practice, there are variations of these basic ratios that are used to analyze a bank's performance. Nearly all of these variant indicators, however, are based on the five principal formulas. Although any of these five simple ratios are perfectly usable to gauge the profitability of the bank relative to its past performance and to other banks, in practice modifications are customary. By making these adjustments, these basic ratios can be adapted to answer the questions that are more truly relevant to the credit analyst's objective.

The most common adaptation is to average the denominator. This can be illustrated with reference to ROE. In brief, instead of using total equity as the denominator when calculating ROE, average equity can be used. Average equity is the sum of shareholders' equity in the current year and shareholders' equity in the previous year divided by two. The rationale for using average equity is that, in regard to annual results, end-year income reflects revenue generated throughout the 12-month period of the financial year, while shareholders' equity as reported on the balance sheet is that amount outstanding as of December 31, or whenever the institution's financial year ends. To calculate profitability, it is therefore more accurate to use an averaged denominator matching the income generated during the period with the amount of equity calculated at the midpoint of the reporting period as if equity had grown linearly over the year. It is accepted that using the average may not be totally correct, since equity does not grow linearly, but it is a better approximation than using equity either at the beginning or at the end of the year.

As might be expected, of course, the difference between averaged ROE and nonaveraged ROE increases the greater the change in equity from one year to the next. This can be illustrated with reference to the sample peer group as shown in Exhibit 6.5. Note that FirstMerit Bank registered a more than 50 percent increase in shareholders' equity between FY09 and FY10. As a consequence, the difference between ROE and return on average equity (ROAE) was significant at 1.5 percentage

EXHIBIT 6.5 ROE, Averaged versus Nonaveraged

Bank Name	Net Income US\$ m 2010	Return on Equity 2010	Return on Avg Equity 2010	Average Equity US\$ m 2010	Total Equity US\$ m 2010	Total Equity US\$ m 2009	Change in Equity	Diff in %: 2010 Equity and Average Equity	Difference: 2010 ROE and ROAE
American Express Centurion Bank	1,341	22.76%	25.64%	5,228	5,890	4,566	28.99%	12.66%	-2.88%
GE Capital Retail Bank	849	15.02%	15.98%	5,312	5,653	4,971	13.72%	6.42%	-0.96%
U.S. AgBank, FCB	511	11.06%	11.69%	4,377	4,625	4,128	12.04%	5.68%	-0.63%
Bank of America Oregon, National Association	345	27.74%	29.51%	1,168	1,242	1,093	13.67%	6.40%	-1.77%
Bank of America California, National Association	327	19.69%	18.89%	1,728	1,658	1,798	-7.79%	-4.05%	0.80%
BOKF NA	218	19.36%	19.53%	1,117	1,127	1,107	1.78%	0.88%	-0.17%
First Citizens Bank & Trust Company	208	13.52%	14.22%	1,461	1,536	1,385	10.92%	5.18%	-0.70%
East West Bank	169	7.49%	7.61%	2,228	2,263	2,192	3.26%	1.60%	-0.12%
City National Bank, N.A.	146	6.89%	7.21%	2,028	2,121	1,935	9.58%	4.57%	-0.31%
Silicon Valley Bank	129	12.00%	12.97%	994	1,075	914	17.56%	8.07%	-0.97%
FirstMerit Bank NA	106	7.48%	8.98%	1,184	1,421	947	50.13%	20.04%	-1.50%
Northern Trust Bank of Florida, N.A.	101	7.58%	8.54%	1,188	1,339	1,038	29.03%	12.67%	-0.96%
People's United Bank	98	2.34%	2.65%	3,703	4,194	3,213	30.53%	13.24%	-0.31%
BNY Mellon, National Association	89	3.40%	3.47%	2,558	2,612	2,505	4.23%	2.07%	-0.07%
UMB Bank NA	70	10.24%	10.75%	648	680	616	10.43%	4.96%	-0.51%
California Bank & Trust	59	5.01%	5.13%	1,147	1,174	1,120	4.82%	2.35%	-0.12%
Amegy Bank National Association	58	3.51%	3.57%	1,620	1,649	1,591	3.61%	1.77%	-0.06%
Associated Bank NA	23	0.76%	0.79%	2,915	3,011	2,819	6.79%	3.28%	-0.03%
MB Financial Bank, NA	21	1.56%	1.63%	1,293	1,345	1,240	8.43%	4.04%	-0.06%
Cathay Bank	15	0.98%	1.02%	1,451	1,507	1,395	8.01%	3.85%	-0.04%

Data Source: Bankscope.

Change in Assets (equity)		ROA and ROE: Nonaveraged Ratio Value versus Averaged Ratio Value		
Assets	☐	Where averaged assets are the ratio denominator and assets <i>grow</i> between the end of the previous period and the end of the current period	Averaged ROA will be <i>higher</i> than nonaveraged ROA	<
	☐	Where averaged assets are the ratio denominator and assets <i>contract</i> between the end of the previous period and the end of the current period	Averaged ROA will be <i>lower</i> than nonaveraged ROA	>
Equity	☐	Where averaged equity is the ratio denominator and equity <i>increases</i> between the end of the previous period and the end of the current period	Averaged ROE will be <i>higher</i> than nonaveraged ROA	<
	☐	Where averaged equity is the ratio denominator and equity <i>decreases</i> between the end of the previous period and the end of the current period	Averaged ROE will be <i>lower</i> than nonaveraged ROA	>

EXHIBIT 6.6 Impact of Averaging the Denominator

points, with ROAE 20 percent higher than ROE. Whereas a commonplace equity materially increases from year to year, nonaveraged ROE will be lower than averaged ROE; where equity decreases, the opposite will occur.

A similar adjustment is even more relevant to ROA, especially when total assets change substantially over a financial period.

The impact of averaging the denominator is shown in the matrix in Exhibit 6.6.

Other Ratio Adjustments

To fine-tune the ratio, further adjustments can be made to its numerator or denominator. To illustrate, in the case of ROE an equity analyst would probably be more interested in the return a bank is generating for the benefit of common shareholders, as opposed to preferred shareholders or shareholders of other companies. Also, to an equity analyst the market capitalization is of more relevance than the equity as per the financial statements. These alternative return-type ratios are more typically referred to by specific terms and abbreviations, but may sometimes be referred to by the more generic abbreviations, which at times is confusing. As an example, return on common equity and return on capital employed could both be abbreviated as ROCE, or just left as ROE. When reading research, it is therefore important to check the definitions. Let us now look at each of the ratio types in detail together with the key ratios in each group.

TERMINOLOGY WHEN THE RATIO DENOMINATOR IS AVERAGED: ROA VERSUS ROAA AND ROE VERSUS ROAE

When the denominator is averaged, the abbreviation for the ratio may or may not change.

Return on assets, is of course abbreviated as ROA. Return on average assets may be abbreviated as ROAA, or simply remains as ROA. Likewise,

return on average equity may be abbreviated as ROAE, or the designation ROE may be retained.

Unless otherwise stated, when ratios in which assets, loans, equity, or some other balance-sheet item form the denominator and an income statement item the numerator, in this book it is intended that the denominator shall be the averaged value over the current and previous period.¹⁵ The impact of averaging was shown in the matrix in Exhibit 6.6.

ROE

ROE versus ROA

ROE, together with ROA, are the two most elemental profitability indicators, neither of which is limited to the analysis of financial firms. They are properly discussed together because they are closely linked. As noted later in the chapter, the two are mathematically related, differing in value in proportion to the degree of leverage or gearing employed.

ROE VERSUS ROA

ROE and similar ratios measure net profits against shareholders' equity, while ROA and similar ratios measure net profits against total assets, which for a bank consist largely of financial assets such as loans. In general, ROA provides a more general measure of how effectively the bank's assets are being managed to generate revenues. In contrast, ROE is of particular interest to equity investors as it measures the return on their investment. Note that for any given level of ROA, ROE will vary according to the leverage employed—assuming a simple definition of leverage linking equity and assets. Nowadays, for reasons explained earlier in the book, it has become increasingly difficult to accept the concept that total assets represent an acceptable denominator for a ratio that measures how a bank manages its assets to generate revenue.

ROE Defined

ROE was discussed in a previous chapter, which introduced the CAMEL model and ratio analysis. There the simplest form of the ratio was described as the nominal net income divided by nominal shareholders' equity. The resulting value is normally stated in percentage terms as an indicator of shareholders' return on their investment. The calculation of simple ROE and average ROE are repeated in the formulae in a later section.

ROE and Credit Analysis

Although ROE is of particular importance to equity analysts, it is by no means irrelevant to credit analysts. Many commercial banks are owned entirely or in large part by

private shareholders, and consequently the ROE generated by a bank will not go unnoticed by its owners or in turn by bank management. To the extent that bank management is under pressure by its shareholders to generate ongoing profit growth and high ROE, the ROE figure registered by a bank may very well influence management's future strategy. Increasing the bank's leverage or using it to take greater risks, even within the constraints of capital adequacy ratios, is always tempting. The current ROE is therefore indirectly important in evaluating creditworthiness. It is obviously not always the case that a bank with a substantially above-average ROE is engaging in riskier practices—the enterprise may for example benefit from strong noninterest income flows. But extremely elevated ratios do bear further examination of why the bank is able to generate a return for investors so much better than its rivals.

EQUITY AND CREDIT ANALYST: DIFFERENT APPROACHES TO PROFITABILITY ANALYSIS

Because earnings and profitability analysis is important both to credit and to equity analysts, there is a large variety of approaches to evaluating this important financial characteristic. Equity and credit analysts, however, have different objectives in evaluating profitability and earnings.

Equity analysts, as would be expected, are primarily concerned with the assessment of these attributes from the perspective of the equity investor. Therefore, the focus is on valuing shares in the particular enterprise being analyzed, and as part of this process the returns generated for investors.

From a bank credit perspective, the importance of profits is not the value of the institution's shares, but instead its ability to meet its obligations to creditors, bolster capital, repair its balance sheet when assets are impaired, and in general their impact on solvency and credit risk. Therefore, credit analysts are comparatively little concerned with those earnings and profitability indicators that support an estimate of valuation, particularly quarterly earnings results. Instead, they are more concerned with those ratios that suggest a recurrent and sustainable earnings stream and thereby reflect the bank's ability to meet its commitments. Owing to the arithmetic relationship between ROE and ROA, a lofty ROE as well as affording greater profit to equity investors suggests the possibility of greater peril, either as a result of higher leverage or through holdings of high-risk financial assets.

Despite these differences, insofar as such ratios and the views of equity analysts and investors affect an enterprise's ability to attract finance in capital markets and thereby influence management's behavior, they are relevant to the credit analyst. What can be said without a doubt is that a bank that generates attractive ROE without incurring materially greater risks will have much less trouble attracting additional capital from equity investors than a competitor that registers only lackluster numbers.

ROE or ROAE

The formula for averaging equity to smooth out year-to-year changes in ROE is shown here.

$$\text{ROE (percent)} = \frac{\text{Net income}}{\text{Shareholders' equity}} \times 100$$

$$\text{ROAE (percent)} = \frac{\text{Net income}}{\text{Average equity}} \times 100$$

$$\text{Average equity} = ((\text{Shareholders' equity in Year 2}) - (\text{Shareholders' equity in Year 1})) / 2$$

Advantages and Disadvantages of ROE

While ROE has its deficiencies as an indicator, its greatest strength is that it does not differ hugely across industries, and in particular it allows some degree of comparability between financial firms and nonfinancial firms. Although it is difficult to make non-industry-specific generalizations about most ratios, ROE is one where a useful cross-sector rule of thumb can be applied.

The chief disadvantage of ROE as a credit metric is that the value it generates varies with leverage. If net income and average assets remain constant and only the amount of equity (and therefore gearing) vary, the same level of ROA will correspond to different levels of ROE depending upon the gearing employed. This is illustrated in Exhibit 6.7.

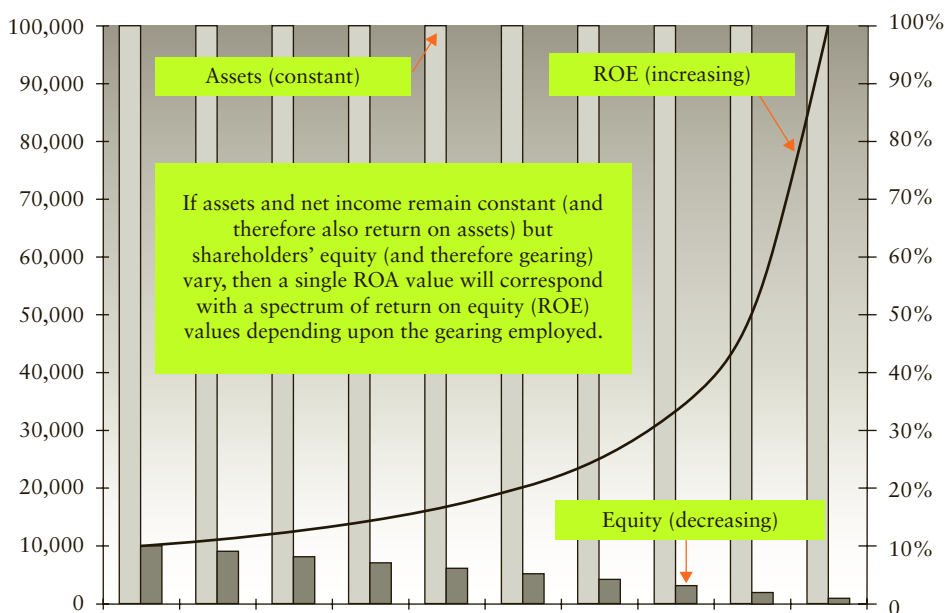


EXHIBIT 6.7 ROE versus ROA

IMPACT OF LEVERAGE: THE CREDIT ANALYST'S PERSPECTIVE

For equity shareholders, leverage is generally a benefit, since it tends to increase the returns on their funds at risk. However, the leverage that results from combining debt finance with equity finance, or gearing as it is sometimes called, is a two-edged sword. Leverage increases returns when an enterprise is doing well, but increases the risk of insolvency when conditions turn sour.

For the same reason, to debt investors and other creditors, high levels of leverage provide no advantage but instead, when taken outside of reasonable bounds, increase solvency risk. When the risk of insolvency is heightened, the risk is borne disproportionately by creditors, since creditors do not benefit from the possibility of high upside gains that shareholders enjoy. Therefore, high leverage is generally seen as credit negative.

ROE: Rule of Thumb

In many markets—but of course the general level of inflation and of interest rates in the local currency might dramatically change this view—a ROE of between 10 and 20 percent is considered acceptable performance under most circumstances. During good economic conditions, the average ROE for all companies in a given market might approach 20 percent and during a slump fall into the lower double digits. ROE of greater than 20 percent is generally considered high, while a ratio under 10 percent would often be deemed to be on the low side. As Bernard Baruch, the legendary financier reputedly said, “If you can’t make 10 percent on your money, drink it.”

The rule of thumb described here is, of course, too general. Inflation levels, yield curves, market perception of investment risk, all conspire to create background conditions that would call for different rules of thumb in different countries or for different currencies.

The diagram in Exhibit 6.8 illustrates a rule of thumb for evaluating the profitability of a bank in terms of ROE. Note that values below 0 percent are possible if the enterprise is making a loss, and it would of course be the same in respect of ROA.

The illustration in Exhibit 6.9 shows ROEs for the 10 Hong Kong banks with the highest ROE in 2010. Another 12 had a ROE below 10 percent, representing investments Mr. Baruch might have shunned, although their poor ROE might have more to

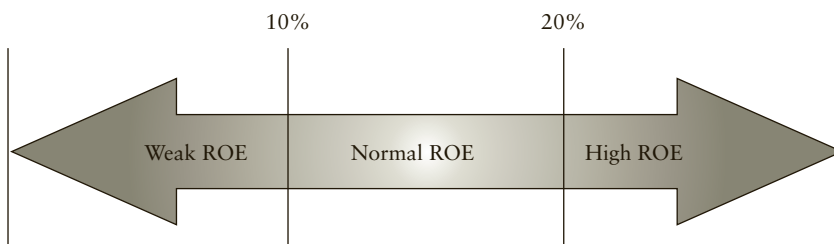


EXHIBIT 6.8 Rule of Thumb: Spectrum of ROE Values

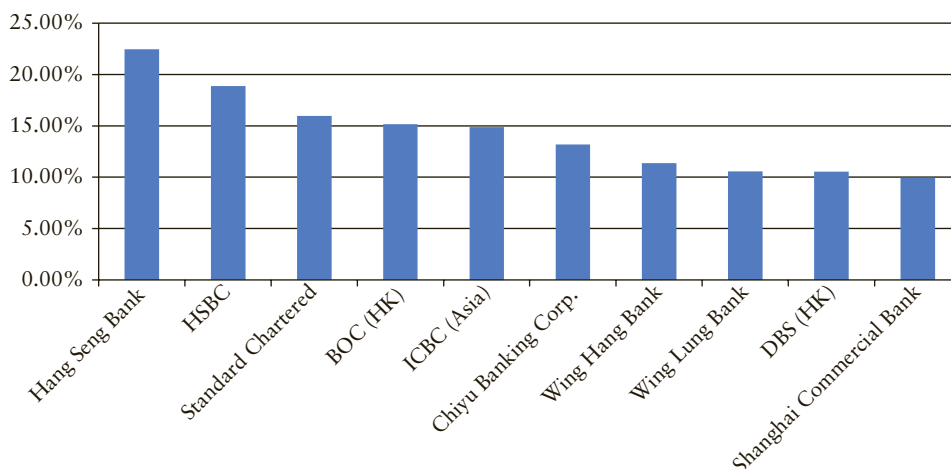


EXHIBIT 6.9 ROE, Selected Hong Kong Banks, 2010

Data Source: Banks' published financials.

do with the denominator than with the numerator and although equity is of course not equal to market capitalization. A commercial bank having an ROE consistently higher than 25 to 30 percent is quite unusual, and is either not engaged in conventional commercial banking or very likely taking on very high levels of risk. The most profitable of the group, Hang Seng Bank is a subsidiary of global giant HSBC.

Fine-Tuning ROE

For analytical purposes, ROE can be modified in several ways. The averaging of the denominator has already been discussed. Another adjustment is to deduct from net income the portion attributable to preferred shareholders and minority interests. Additionally, equity can be further reduced to tangible equity by subtracting intangible assets. From a corporate credit perspective in particular, the use of tangible assets in the ratio may be more useful for solvency evaluation. A number of adjustments may be used to create the variant ratios that measure ROE. Several of these are shown in the equations that follow, including return on common equity.

Note that where the numerator is reduced, the bank may register a somewhat higher number on the same earnings, while reducing the denominator will have the opposite effect, the result depending upon the proportional change in each.

$$\begin{aligned}
 &\text{Return on common} \\
 &\text{equity (ROCE)} \\
 &\text{(percent)} \\
 &(\text{= ROE attributable} \\
 &\text{to common} \\
 &\text{shareholders})
 \end{aligned}
 = \frac{\text{Net income} - \text{Portion of net income} \\ \text{attributable to minority shareholders}}{(\text{Average}) \text{ Shareholders' equity}} \times 100$$

$$\begin{aligned} \text{Return on tangible equity (ROE)} &= \frac{\text{Net income}}{\text{Shareholders' equity} - \text{Intangible assets (e.g., goodwill, deferred tax assets)}} \times 100 \\ \text{Return on tangible common equity (ROE)} &= \frac{\text{Net income} - \text{Portion of net income attributable to minority shareholders}}{\text{Shareholders' equity} - \text{Intangible assets (e.g., goodwill, deferred tax assets)}} \times 100 \end{aligned}$$

Fine-Tuning ROE to the Extreme

Measures such as risk-adjusted return on capital (RAROC), return on risk-adjusted capital (RORAC), and risk-adjusted return on risk-adjusted capital (RARORAC) are used to express profitability in risk-adjusted terms, but not necessarily only in the denominator.

We shall see later the need for an adjustment to the way a denominator such as total assets should be determined. A bank that registers a ROA of 2 percent would ordinarily be viewed as displaying better performance than one that exhibits an ROA of 1.5 percent. But if the bank with a 2 percent ROA was only able to reach this figure by accepting very high levels of risk, and the bank producing a return of 1.5 percent could do so by maintaining a conservative risk exposure, then it is possible that on a risk-adjusted basis, the institution with the nominally lower ROA could actually display greater profitability on a risk-adjusted basis than the one registering a higher nominal ROA.

The same approach should be applied to ROE. Banks should theoretically operate with equity funds at least equal to the *economic capital*¹⁶ determined by the regulators—whether or not inspired by Basel II and Basel III. Any excess equity, beyond the economic capital, is welcome by the regulators, but not mandatory. Since a bank with too much equity could just as well reduce its equity to the regulatory level, which is—or should be—the economic capital, the only ROE relevant to any comparative analysis is the return on economic capital rather than the return on equity. The term RORAC was coined to reflect that approach, with RAC standing for *risk-adjusted capital*, equivalent to the notion of economic capital.

The term RAROC, with RAR standing for *risk-adjusted return*, borrows from the insurance world. An insurance company would typically price the premium it charges on the basis that the premium should cover administrative costs, a profit margin, expected disbursements, and unexpected disbursements. Expected disbursements are part of the cost of running the business, and therefore the insurance company must make provisions in the shape of reserves to cover those disbursements. The provisions reduce the returns, hence the term *risk-adjusted return* for the return left after such provisions were taken.

The term RAROCAC takes in the two approaches within the same ratio.

RAROC, RORAC, AND RARORAC

RAROC (risk-adjusted return on capital) is a measure of return developed by Bankers Trust, a major U.S. bank, in the 1970s. It was an attempt to take into account the riskiness of the assets (typically with reference to the credit costs used to generate net income) by allocating equity and profits to specific business lines or profit centers. The concept could be defined broadly as allocated earnings divided by allocated capital. However, strictly speaking, RAROC should not address the matter of what the denominator ought to be. RAROC should instead do what its name implies: to adjust the numerator (the return) for expected losses, very much as an insurance company would set aside expected payments under its contracts.

RORAC (return on risk-adjusted capital), to the contrary, does not adjust the numerator, but it adjusts the denominator so that the capital is the economic capital adjusted for the risk involved in entering the transaction or set of transactions. RORAC became useful in the context of complex transactions where the actual risk exposure could differ widely from simple estimates.

It would make more sense to adjust both the numerator and the denominator so that the return (adjusted for expected losses) is compared to the economic capital it is using (adjusted for the transaction risk). In such a case, the RARORAC is used (risk-adjusted return on risk-adjusted capital). Naturally, the risk-adjusted capital for a transaction whose return is already penalized—as it is here—by a provision for expected losses would only have to be adjusted for unexpected losses. Only the RARORAC truly gives a correct comparative tool for risk/return analysis.

ROA

The other key basic return indicator is ROA. ROA measures the efficiency with which an enterprise is employing its assets. The formula for ROA and the formula for averaging assets to reduce distortion in the indicator (ROAA) are shown as follows.

$$\text{ROA (percent)} = \frac{\text{Net income}}{\text{Total assets}} \times 100$$

$$\text{ROAA (percent)} = \frac{\text{Net income}}{\text{Average assets}} \times 100$$

$$\text{Average assets} = \frac{(\text{Total assets in Year 2}) - (\text{Total assets in Year 1})}{2}$$

Significance of ROA

For bank credit analysis, ROA is actually more important than ROE since it excludes the impact of leverage. By disregarding leverage, ROA has some attributes of a core profitability indicator in that it measures earnings solely in relation to a bank's asset base, which, of course, is constituted mainly of earning assets and other financial assets, but disregards other items, in particular off-balance-sheet items, that generate income (or losses).

Though a crucial ratio for bank analysis, two important points should be kept in mind concerning ROA.

1. First, while it ideally should measure recurrent profitability, it is easy to neglect, in its calculation, to exclude one-off items such as gains from the disposal of subsidiary, thus distorting comparisons. Under most accounting systems, these should be separated on the income statement from the rest of the revenue stream, but not all published bank financials will in fact reflect this.
2. Second, when the bank has a high proportion of revenues that are not asset based, such as brokerage commissions, or that may be asset based but are off balance sheet, the comparative power of the ratio will be weakened.

The Relationship between ROA and ROE

ROE and ROA, as suggested, are mathematically correlated through the equity multiplier or whatever other ratio measuring leverage or gearing.

HYPOTHETICAL ILLUSTRATION: ROA, ROE, AND LEVERAGE

Assume that ABC Bank's financial statements show the following:

- Net income of \$1 million on total assets of \$50 million
- Total shareholders equity of \$5 million
- No fixed assets or liquid assets

ROA (nonaveraged)¹⁷ is therefore \$1 million divided by \$50 million or 2 percent. ROE is \$1 million divided by \$5 million or 20 percent. Both are very respectable figures for a bank. The relationship between the two is dependent on the equity multiplier, which is equal to assets divided by equity or \$50 million divided by \$5 million: 10×. Note that sometimes leverage is expressed as a fraction rather than as a multiple, that is, $\text{assets/equity} = \$5 \text{ million}/\50 million or 10 percent—the relationship is still the same.

Suppose ABC Bank increases its leverage by expanding its loan book without increasing its capital, that all of its assets are in the form of loans, and that by doubling its loans outstanding it can double its income. If ABC Bank doubles its loans, its assets will increase to \$100 million from \$50 million and profit to \$2 million from \$1 million. Equity remains unchanged at \$5 million. The resulting increase in profit is proportional to the increase in assets, and

EXHIBIT 6.11 Dupont Analysis by Country: 12 Banks in Egypt 2006 to 2010

Egypt	2010	2009	2008	2007	2006
Net interest margins	2.23%	2.04%	1.51%	1.50%	0.85%
Total Income/Total Asset	8.29%	7.49%	8.46%	7.64%	7.52%
Other Income/Total Asset	1.88%	0.73%	1.65%	1.10%	1.45%
Cost Burden/Total Asset	5.34%	6.07%	6.60%	6.38%	6.36%
Preprovision RoA	2.38%	1.20%	1.81%	1.24%	1.17%
Provisioning burden	41.81%	62.82%	−36.85%	32.62%	−0.71%
Tax burden	59.69%	57.27%	82.02%	89.22%	84.80%
ROA	0.62%	0.37%	0.28%	0.45%	0.29%
ROE	11.56%	6.86%	5.49%	9.56%	6.82%
Leverage	18.53x	18.45x	19.59x	21.35x	23.13x

commonly what may be referred to as a Dupont analysis is merely a modified version showing leverage and key profitability ratios.

The deconstruction of ROE is referred to as Dupont analysis, named after the U.S. firm E.I. Dupont du Nemours and Company. It can be used for a single company, or for the comparison between several companies, or for an aggregate of companies, for example the 12 largest banks in Egypt as in Exhibit 6.11.

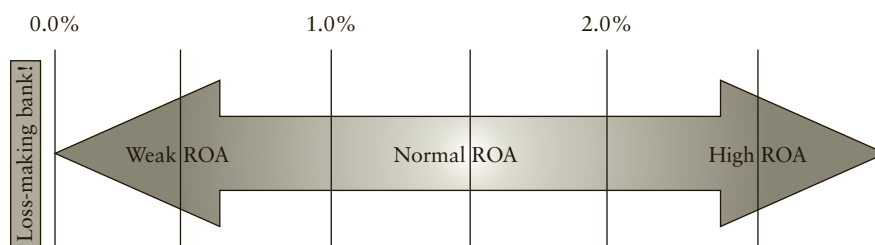
Advantages and Disadvantages of ROA

ROA's advantages are that it disregards the distorting effects of leverage and affords a simple easily calculated benchmark of profitability. It is probably the most popular overall profitability measure for bank credit analysis purposes. It is far from perfect, though.

The main disadvantage of ROA is that it does not take account of intrinsic risk levels associated with different types of assets. Also, as observed, it tends to penalize banks that have low levels of noninterest income. In addition, given the increasing exposure of banks to risks that do not have a linear relationship with total assets, the old-fashioned ROA is no longer of much relevance. At the very least, the denominator should display the bank's exposure to risk. This is, of course, discussed at length in this book.

ROA: Rule of Thumb

In most markets, healthy banks tend to generate ROAs between 1 and 2 percent. The mean will vary from country to country and be influenced by a host of factors, including the level of interest rates, the prudential regulatory regime, and the intensity of competition. In some markets, average ROA for commercial banks may be less than 1 percent, indicating that the banking sector possesses weak profitability. In other markets, average ROA can approach (and sometimes exceed 2 percent reflecting a highly profitable sector). Very low industry ROA tends to be associated with highly taxed banks (if the ROA is taken after tax) or those that are otherwise burdened by policy lending obligations. Very high ROA may be the result of oligopolistic behavior on the part of banks that is countenanced by government.

**EXHIBIT 6.12** Rule of Thumb: Spectrum of ROA Values**EXHIBIT 6.13** Rule of Thumb for ROA

Band of ROA	Remarks
<0.50%	Weak profitability; characteristic of state-owned banks or banks burdened by heavy policy lending obligations, or with heavy loan-loss provisioning burden.
$\geq 0.50\% < 1.00\%$	Mediocre profitability in most markets.
$\geq 1.00\% < 2.00\%$	Healthy level of profitability in most markets.
$\geq 2.00\% < 2.50\%$	Very strong profitability in most markets; watch for unusual business model or high-risk activities.
>2.50%	Unusually high ROA; scrutinize for high-risk activity.

Evaluating whether a bank's ROA is strong or weak therefore is best accomplished with reference to an industry or peer average. As a rule of thumb, however, the benchmarks in Exhibits 6.12 and 6.13 may prove helpful to novices, who should remember, though, that such benchmarks pertain to traditional banking with commercial loan credit risk as the major component of the bank's risk profile, but without any risk adjustment to the denominator.

The range of values exhibited in Exhibit 6.14 by the same group of Hong Kong banks observed in connection with ROE is wide but not atypical. Hang Seng Bank is again the most profitable with an ROA in excess of 1.6 percent, and many banks are in the range extending roughly from 1.0 to 1.5 percent.

Fine-Tuning ROA: Risk-Weighted Assets

As with ROE, it is quite common to alter basic ROA for analytical purposes. Concerning the ratio denominator, the advent of the Basel Accords has provided a new tool to tweak this ratio: risk-weighted assets.

Suffice it to say for the present¹⁸ that risk-weighted assets differ from ordinary total assets as reported on a bank's balance sheet in that they take into account the riskiness of the assets. As an example, under the original 1988 Basel Accord, because residential mortgage loans were viewed as less risky than unsecured industrial and commercial loans, they were assigned a risk weighting of 50 percent versus

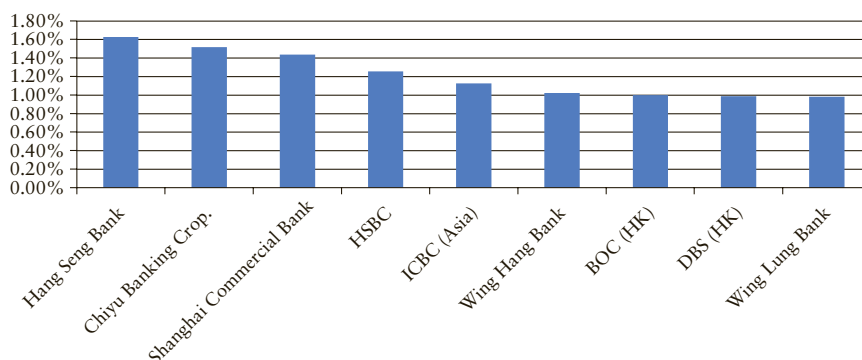


EXHIBIT 6.14 ROA, Selected Hong Kong Banks FY10

100 percent generally speaking for the latter. Further refinements of that approach will be discussed at length later in this book.

Return on Risk-Weighted Assets

Measuring returns (net income) against risk-weighted assets provides a more accurate indication of profitability as it takes into account the risk undertaken in generating them.¹⁹ Therefore, this more sophisticated variant of ROA, which may be called *return on risk-weighted assets* or RORWA, overcomes some of its limitations through the use of total or averaged risk weighted assets as the ratio's denominator.

$$\text{RORWA (percent)} = \frac{\text{Net income}}{\text{Total risk-weighted assets}} \times 100$$

Risk-weighted assets = Assets grouped into classes and multiplied by weightings as prescribed under the 1988 Basel Accord, and the Revised Accords called Basel II and Basel III when adopted in the relevant market.

Adjusted RORWA

Other variants of course are possible. An even more conservative approach—embraced by Basel II and Basel III—is to add assets that conventionally are treated as technically off balance sheet, notably *securitized assets*, to risk-weighted assets in the return-type ratio's denominator. This adjusted ratio can be seen in the box that follows. The larger denominator will result in lower ratio values.

$$\text{Adjusted RORWA} = \frac{\text{Net income}}{\text{Total risk-weighted assets} + \text{risk-weighted securitized assets}} \times 100$$

Adjusting for Asset Tangibility

Another possibility is to identify the portion of a bank's assets that are tangible in nature, although by reducing the denominator ratio values will increase. Whatever denominator is ultimately used, as with ROE it would generally be averaged over the current and previous financial period.

Core Profitability Indicators

The cash flow statement is of little aid in bank credit analysis. For this reason, to evaluate bank performance in relation to credit risk, the quantity and quality of an institution's sustainable earnings stream replace the movement of cash as the focus of review. It is therefore all the more important that the focus be on the portion of net income²⁰ that best reflects an institution's true earnings capacity. To accomplish this, other forms of earnings measures can be substituted as the ratio numerator.

Adjusting for the Interests of Others: Minority Interest and Preferred Dividends

As in relation to ROE, if a ratio's denominator can be adjusted to better fit the analytical objective, so can the numerator.²¹ One possible modification would be to make the same adjustments mentioned earlier in relation to *return on equity*; that is, to deduct from net income *minority interest* and *preferred dividend payouts*. If, however, the goal is to measure *core profitability*, then it may be preferable to move higher up the income statement to identify the source of the bank's sustainable earnings. Various alternatives are discussed in the following.

Adjusting for Credit Costs: Use of Pretax Profit or Net Operating Profit after Provisions

Another possibility is to use pretax profit as the ratio numerator. This has the effect of excluding the impact of any discrepancies in tax treatment, including those resulting from differences in tax liabilities or usable tax benefits.²²

Determining Recovery Capacity: Preprovision Income (PPI)

If the objective is to determine a bank's raw regenerative capacity—that is, its ability to earn its way out of distress—a better numerator might be PPI,²³ although of course the analyst would have to ensure that none of the stated income was overstated at the expense of the need for further provisions.

Retained Earnings to Average Assets and the Internal Growth Rate of Capital

A final possibility is to step down below the bottom line and measure retained earnings; that is, net income attributable to common shareholders after all dividend payouts, to identify the flow of earnings that is transmuted to internally generated capital. When average equity is used as the denominator in this ratio, it is termed the *internal capital generation ratio* and is predictably discussed with other capital indicators, even though it closely resembles other return-type profitability ratios.

A decision on the dividend payout ratio is very much in the hands of a bank's management, and is therefore subject to unpredictable fluctuations. When banks are under rehabilitation after a near collapse, however, regulators often forbid the payment of any dividend.

$$\frac{\text{Retained earnings to average assets (percent)}}{\text{assets (percent)}} = \frac{\text{Retained earnings}}{\text{Average assets}} \times 100$$

Retained earnings = Net income attributable to shareholders minus dividend payout

$$\frac{\text{Retained earnings to average equity (percent)}}{\text{(Internal capital generation ratio or internal growth rate of capital)}} = \frac{\text{Retained earnings}}{\text{Average shareholders' equity}} \times 100$$

Recurring Earnings Power

Bankscope, a leading bank financial database, uses a core profitability indicator it refers to as *recurring earnings power*, or REP. It defines this ratio as shown. Note the numerator adds back loan-loss provisions, which makes REP a simplistic gauge of net cash flow. It does not explicitly exclude nonoperating items.²⁴

$$\frac{\text{REP (percent)}}{\text{(as defined by Bankscope)}} = \frac{\text{Pretax profit} - \text{tax} + \text{loan-loss provision}}{\text{Average assets}} \times 100$$

Core Profitability Ratios Suggested by Rating Agencies

The rating agency Moody's has published reports specifically dealing with bank ratios and credit analysis. In respect of core profitability, it has two "favorite" ratios, one of which is a variant of RORWA, and another that Moody's denotes by the unwieldy abbreviation PPP-PfdDiv-consumNCO % BIS Assts+Seczd CC,²⁵ but that we would describe as adjusted core return on risk-weighted assets.

$$\frac{\text{Adjusted core return on risk-weighted assets (percent)}}{\text{(Moody's definition)}} = \frac{\text{PPI} - \text{preferred dividends} - \text{consumer net charge-offs}}{\text{Total risk-weighted assets} [+ \text{securitized credit card assets}]} \times 100$$

Moody's defines this ratio as PPI less preferred dividends and consumer net charge-offs as a percentage of risk-weighted assets plus securitized credit card assets.²⁶ Exhibit 6.15 shows illustrative indicators for a hypothetical institution.

EXHIBIT 6.15 Illustrative Profitability Indicators

Income Statement	US\$ mil	% Average Assets	% RWA	% Average NTAs
Interest income	680	4.89%	5.06%	5.04%
Interest expense	200	1.44%	1.49%	1.48%
Net Interest income	480	3.45%	3.57%	3.56%
NonInterest income	120	0.86%	0.89%	0.89%
Operating income	600	4.31%	4.46%	4.45%
Noninterest expense	380	2.73%	2.83%	2.82%
Preprovision income	220	1.58%	1.64%	1.63%
Loan-loss provisions	40	0.29%	0.30%	0.30%
NOPAP	180	1.29%	1.34%	1.33%
Pretax profit	200	1.44%	1.49%	1.48%
Taxes	60	0.43%	0.45%	0.44%
After-tax profit	140	1.01%	1.04%	1.04%
Preferred dividends	4	0.03%	0.03%	0.03%
Net income	136	0.98%	1.01%	1.01%
Minority interest	1	0.01%	0.01%	0.01%
Attributable income	135	0.97%	1.00%	1.00%
Common dividends	35	0.25%	0.26%	0.26%
Retained income	100	0.72%	0.74%	0.74%
Partial Balance Sheet	2004	Avg	2003	
Total assets	14,950	13,915	12,880	
Intangible assets	590	430	270	
Net Tangible assets (NTAs)	14,360	13,485	12,610	
Total risk assets	14,515	13,440	12,365	
Total equity	1,475	1,320	1,165	
Minority interests	165	163	160	
Preference shares	80	80	80	

RATIONALE FOR DEDUCTING CONSUMER NET CHARGE-OFFS FROM CORE EARNINGS

Commercial loan charge-offs are not charged against income in normal practice. Instead, they are deducted from the loan-loss reserve account on the balance sheet and represent the drawdown of loan-loss reserves, which had been previously deducted on the income statement. Consumer charge-offs are often charged to income. Why subtract consumer charge-offs from income and not do so in respect of commercial loans?

The answer is that when companies default on their lending obligations, banks are typically much more willing to *work out* the loans than with consumers; with regard to the latter, the amounts are simply too small to allocate staff. Consumer loans are usually sent straight to a collection agency and

written off. Commercial loans, in contrast, are less prone to immediate charge-off.

Another argument is that, for consumers, it is conceptually easier to assume that expected losses will take place in any event—which is why they are termed expected—based on recent history. The higher income associated with such loans should directly take care of those expected losses.

Naturally, if by identifying core income the objective is to measure the stream of income that the bank has can use to earn its way out of difficulty, what we are really doing is showing the stream of income that is really available to the bank to rebuild its balance sheet.

To be sure, this very rebuilding is accomplished, in large part, after the deduction of loan-loss provisions from the income statement to build loan-loss reserves on the balance sheet. The bank, however, may be delaying incurring provisioning costs to show a better bottom line, or it may be taking advantage of obviously poor business conditions to overprovision so that results will look better later. This would give a distorted view of the strength of the earnings stream potentially available for balance-sheet rebuilding.

As does Moody's, S&P publishes several ratios for bank analysis purposes. Among the key ratios used by rating agencies with regard to core profitability, those listed in Exhibit 6.16 are notable.

S&P defines core revenues as net interest income plus noninterest income adjusted for gains on sale of nonloan assets and long-term equity holdings,²⁷ to the extent that gains on sale of nonloan assets and long-term equity holdings are reported as operating items.

Fitch Ratings' methodology for evaluating banks is somewhat less explicit in defining ratios than Moody's and S&P, but a sample report shows the indicators deemed important, which are appended to the text. Operating profit as defined by Fitch, which is referred to in this book as net operating profit after provisions, to average risk-weighted assets seems to be the primary core profitability measure applied by the agency. Dominion Bond Ratings Services applies a similar ratio, income before provisions and taxes to risk-weighted assets, which is defined here as PPI to risk-weighted assets.

Margins and Spreads to Measure the Profitability of Specific Activities

Margin-type ratios allow the evaluation of specific categories of business activities, and the net interest margin (NIM) is probably the most widely used of these indicators for bank credit evaluation purposes. It measures the profitability of a bank's interest-earning operations.

EXHIBIT 6.16 S&P's Core Profitability Ratios

Ratio Name	Net Operating Income before LLP/Average Assets	Net Operating Income after LLP/Average Assets	Pretax Profits/Average Assets	Net Operating Income before LLP/Revenue (as defined by S&P)	Net Operating Income after LLP/Revenue (as defined by S&P)
Numerator	Noninterest income + net interest income – noninterest expense	Noninterest income + net interest income – noninterest expense-LLPs	Pretax income	Noninterest income + net interest income – noninterest expense	Noninterest income + net interest income – noninterest expense-LLPs
Denominator	Average assets	Average assets	Average assets	Net interest income + noninterest income adjusted for gains on sale of nonloan assets and long-term equity holdings	Net interest income + noninterest income adjusted for gains on sale of nonloan assets and long-term equity holdings

Data Source: S&P.

NIM Defined

The NIM is defined as net interest income divided by average earning assets.²⁸ Another way to think of the NIM is as the return (in net interest income) on (average interest) earning assets.

$$\text{NIM (percent)} = \frac{\text{Net interest income}}{\text{Average earning assets}} \times 100$$

Macro Influences on the NIM

Prevailing NIM will differ from market to market, from bank to bank, and from time to time. At the macro level, prevailing NIMs will depend upon variables including regulations governing bank activities and the level of competition. Again depending upon government policy, interest rates are subject to fluctuation as a consequence of economic forces operating at both a country and global level. All other things being equal, a high demand for credit relative to supply will increase interest rates; a low demand will depress rates. Since governmental policy and market factors will ordinarily affect all banks in largely the same way, a credit analyst will look for anomalies between a bank and its peers. A bank's individual strategy and management competence will enable it to perform more or less successfully than its competitors.

Whether an increase in interest rates has a positive impact on a bank's NIM depends upon whether deposits reprice faster than assets, or vice versa. This is illustrated in Exhibit 6.17.

While the overall level of interest rates should have a limited impact on the NIM, it remains obvious that high market interest rates would generally allow banks to obtain better spreads, as those spreads would still appear small in comparison to market interest rates. Conversely, low interest rates put a psychological cap to margins.

Micro Influences on the NIM

At the micro level, again several factors may affect a bank's NIM and explain changes in the profitability of a bank's spread business as well as why one bank may be able to charge premium rates while another is burdened by above-average funding costs. For example, a bank's NIM may rise as a consequence of the bank's strategy to expand its deposit base. The bank might open new branches allowing it to capture low-cost deposits. Similarly, short-term tactics may have an impact. For instance, an aggressive bank seeking to gain market share may cut lending rates or raise borrowing rates above that of its peers to expand its loan book or mobilize additional deposits. Changes in bank fundamentals and market perceptions may also play a part. This can occur when other banks become less willing to lend to an institution that is believed to have problems, raising the cost of its funding sourced through interbank borrowing.

Why Funding Costs Vary

There are a variety of reasons one bank might have the advantage of lower funding costs, and consequently higher spreads, assuming yields are the same as other banks.

EXHIBIT 6.17 Impact on NIM of Changes in Interest Rates

If when prevailing interest rates rise . . .	a bank is able to increase the interest rate it charges on loans at a more rapid rate than the rate it pays for funding, then . . .	it is asset-sensitive and . . .	such an increase in market interest rates will cause the bank's	gross interest income to increase faster than gross interest expense increase, resulting in	a rise in net interest income and therefore	a higher net interest margin . . .
	the interest rate a bank pays for funding increases at a more rapid rate than that which it can charge for loans, then . . .	it is liability-sensitive and . . .	such an increase in market interest rates will cause	gross interest expenses to increase faster than gross interest income, resulting in	a decline in net interest income and therefore	a lower net interest margin . . .
If when prevailing interest rates fall . . .	a bank must reduce the rate it charges for loans at a more rapid rate the interest rate it pays for funding declines then . . .	it is asset-sensitive and . . .	such a decrease in market interest rates will cause	gross interest income to decrease faster than gross interest expense declines, resulting in	a decline in net interest income and therefore	a lower net interest margin . . .
	the interest rate a bank pays for funding decreases at a more rapid rate than that which it can charge for loans, then	it is liability-sensitive and . . .	such a decrease in interest rates will cause	gross interest expenses to fall faster than gross interest income resulting in	a rise in net interest income and therefore	a higher net interest margin . . .

REASONS FOR CHANGES IN INTEREST RATES

There are many influences on the interest rates that banks can charge on loans and those they pay on deposits or other funding. A bank's spreads are affected both by external factors such as government policy and general business conditions, as well as by the institution's strategy. Outside influences on prevailing interest rates, and therefore on a bank's NIM, include:

- The level of prevailing business activity
- The propensity of consumers to save or spend
- Expectations concerning inflation
- The amount of competition in the financial sector
- Government policy

Government policy can also influence interest rates directly or indirectly through a variety of mechanisms, including reserve requirements (the percentage of funds required to be deposited in the central bank). Related to government policy is the existence of cartels, quasi-cartels, or other rate fixing either mandated or tolerated by government. While some banking markets are highly competitive, others are not. Insofar as government policy toward competition allows the formation of cartels, bank profitability will be increased. Regulation can also squeeze bank margins by placing ceilings on interest rates or by burdening banks with policy obligations, such as subsidized lending.

An institution with a wide branch network will normally have a lower cost of funds than one with a small branch network, because the wide branch network will typically include regions where there is less competition among banks for deposits, and interest paid on deposits will be correspondingly lower, absent regulation. Also, when interest rates change, the bank with a large customer base will tend to see less relative outflow of deposits than a smaller one, all other things being equal. Another reason funding costs may be lower is that the bank is less aggressive in pursuing new deposits and does not therefore need to pay higher rates to attract them. Customer loyalty and high-quality service may enable the bank to pay lower rates than its rivals. A third reason for low funding costs may be that the bank entered the capital markets at a propitious time and was able to fund its loans when interest rates were comparatively low. As interest rates rise, its funding costs will remain lower, at least for a while, than the costs of banks that were not so lucky in their timing.

Why Yields Vary

As with funding costs, there is a variety of reasons why one bank may show higher yields than another. One is the composition of the bank's loan portfolio. A bank with a conservative portfolio²⁹ will tend to earn less interest on its assets than a bank whose lending is concentrated on smaller companies and individuals. In contrast,

NET INTEREST INCOME AND THE NIM

In an earlier chapter, the importance of looking at net interest income in evaluating a bank's spread business was noted. Since interest income and interest expense tend to rise and fall together, focusing on net interest income permits the performance of this part of a bank's franchise to be separated from changes in economic conditions resulting in the rise or fall of prevailing interest rates. Although a bank's net interest income levels will vary as interest rates and economic conditions alter, the effect is much less pronounced than with gross interest income levels. In addition, the perception of the acceptability of the NIM—to the extent that bank depositors and/or borrowers can figure it out—changes with the level of interest rates.

To illustrate, suppose DEF Bank (see Exhibit 6.18) in Year 1 earns \$10 million in interest on earning assets of \$100 million, providing an average yield of 10 percent.

Imagine further that the bank funds these assets from \$100 million in deposits, on which it pays 3 percent interest, and interest expenses are \$3 million for the year. This leaves net interest income of \$7 million. (Let us ignore market risk and assume that interest-earning assets and interest-bearing liabilities reprice immediately.)

In Year 2, suppose that interest rates increase by 2 percent. Average lending rates are now 12 percent, while deposit rates are now 5 percent. The bank now earns \$12 million on the same assets, but pays \$5 million. Net interest income is still \$7 million. That gross interest income rose by \$2 million is essentially irrelevant, but the perception is that a margin of 7 percent out of a lending rate of 12 percent is more acceptable than a margin of 7 percent out of a lending rate of 10 percent.

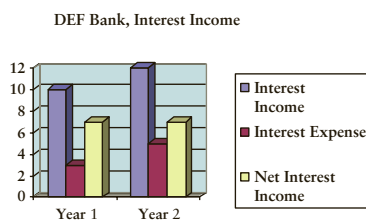
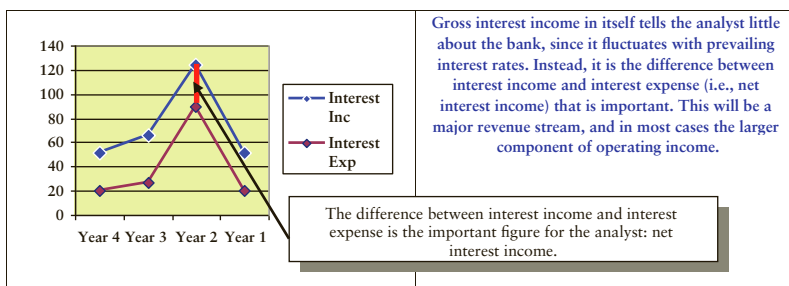


EXHIBIT 6.18 Net Interest Income and a Bank's Spread Business

a bank that concentrates on high-risk lending will tend to have greater yields, although elevated levels of nonperforming loans (NPLs) associated with riskier lending will ultimately tend to dampen net interest revenue. In such case, the interest from those loans deemed nonperforming will either be not forthcoming, or if there is a subsequent collection, it will ordinarily be recorded as a noninterest income item.³⁰ Note that although nonaccrual loans will not generate revenue and therefore will not contribute to the numerator of the NIM, the ratio as conventionally calculated disregards that situation and, for example, does not take account of loan-loss provisioning, which would reduce the denominator.

THE IMPORTANCE OF PROVISIONING AND PROFITABILITY

The level of provisioning that a bank takes in a given year will also have a significant impact on its bottom line. This is the fourth critical element affecting bank profitability. The amount of additional loan-loss reserves that a bank will set aside depends upon several factors. The regulatory environment, including the amount of discretion allowed to bank management, as well as any changes affecting loan-loss reserve requirements will be a strong influence. Other regulations, such as those governing the taxability of provisions or write-offs will also have an effect. Beyond these considerations, the level of NPLs the bank is experiencing, management's belief about future prospects for asset quality, and management's conservativeness will also play a part in determining the amount channeled into credit costs. Because ample provisioning is an important aspect of bank creditworthiness, the credit analyst, when reviewing bank profitability, should perform a profitability analysis using preprovision net profit figures as well as ordinary net profit figures. This will better reveal the bank's underlying profitability, and what proportion of profit has been siphoned off either to prepare for expected loan losses or to repair damage that has already occurred.

If a bank changes its business strategy to more aggressive lending, it may take 6 to 18 months before the high-risk loans display materially higher levels of nonperformance, possibly longer in a buoyant economy. Nevertheless, a bank that properly provisions against loss may still net higher yields from a higher-risk loan portfolio, despite greater loan losses. The final and perhaps most critical reason for higher average yields may be superior management. A bank that is able to target and market to customers who have not been well served by its competitors may have a comparative advantage over its rivals that enables it to earn more on its interest-generating assets. Similarly, where credit analytics are robust, management will be able to better price loans according to their risk, and be better able to spot and avoid problem credits.

Net Interest Spread

While in evaluating a bank's interest-generating activities, *net interest income* is the critical figure, and *NIM* the crucial ratio, the former merely represents the end

product of a bank's spread business and the latter its overall profitability. In itself, these indicators only tell us how profitable a bank's interest-generating activity is, but do not explain why one bank can lend more profitably than another, or why performance has changed over time.

It can therefore be useful to observe the relative changes between gross interest income and gross interest expense, and their associated ratios, average cost of funding, and average yield.

- If gross interest income increases by a greater percentage than gross interest expense, then net interest income will rise.
- Conversely, if gross interest expenses increase by a greater percentage than gross interest income, net interest income will fall.

Whether total interest income growth outpaces growth in interest expense, or vice versa, will depend upon changes in the bank's average cost of funding vis-à-vis its average gross yield. Funding costs and yield then are two additional elements affecting bank profitability. Ratios to measure these attributes can be computed as shown in Exhibit 6.19. Changes in the growth of each of the constituents of net interest income will be reflected in both net interest income and the NIM as well as in a third ratio, the net interest spread (NIS). The NIS closely resembles the NIM. It is often expressed as the difference between the average yield on earning assets and the average cost of funding those assets.

The average cost of funding is useful in discerning whether a bank enjoys a lower cost of funding than its peers, or than it has historically, or whether its cost of funding is comparatively high. In the same way, its average earnings yield is helpful to show whether a bank is able to achieve higher yields than its peers or than it has in the past. Finally, the NIS—again the difference between the average funding cost and the average yield—like the NIM shows the basic profitability of a bank's spread business.

The difference between the two is that the NIM shows the overall profitability of this earnings component irrespective of capital structure. The NIS, in contrast, looks only at interest-bearing liabilities and interest-earning assets, and therefore disregards earning assets funding that does not fall within the category of interest-bearing liabilities: noninterest bearing liabilities and equity capital. The NIS will generally be close to the NIM, but will differ to the extent that the denominator, average interest-bearing liabilities, diverges from average interest-earning assets.³¹

Exhibit 6.21 shows how the NIS, together with its constituents, and the NIM vary across the sample peer group. Note that the NIM is almost invariably higher than the NIS.

$$\text{Average cost of funding (\%)} = \frac{\text{Gross interest expense}}{\text{Average interest-bearing liabilities}} \times 100$$

$$\text{Average gross yield (\%)} = \frac{\text{Gross interest income}}{\text{Average (interest) earning assets}} \times 100$$

$$\text{Average cost of funding (\%)} - \text{Average gross yield} = \text{Net interest spread}$$

EXHIBIT 6.19 Average Cost of Funding, Average Gross Yield, and NIS

NIM VERSUS NIS

Assume that a Bank FGH has the balance sheet listed on the left-hand side of the table in Exhibit 6.20 and Bank IJK has the balance sheet on the right. Suppose that Bank FGH generates \$600 million in interest income on its \$8 billion in interest-earning assets and Bank IJK earns the same. Bank FGH pays \$120 million in interest on its \$7.2 billion in interest-bearing liabilities, while Bank IJK pays \$160 million on its \$8 billion. Net interest income for Bank FGH is \$480 million, while that for Bank IJK is \$440 million. We can compute average funding costs, yields, NIS, and NIM.

EXHIBIT 6.20 Illustration of the Difference between NIS and NIM

Simplified Balance Sheets							
Bank FGH				Bank IJK			
Cash and Non-Interest-Earning Assets	1.0	Non-Interest-Bearing Liabilities	1.5	Cash and Non-Interest-Earning Assets	1.0	Non-Interest-Bearing Funding	1.0
Securities, Loans and Other Interest-Earning Assets	8.0	Interest-Bearing Deposits and Other Funding	7.2	Securities, Loans, and Other Interest-Earning Assets	8.0	Interest-Bearing Deposits and Other Funding	8.0
Fixed Assets	1.0	Shareholders' Equity and Other Capital Funds	1.3	Fixed Assets	1.0	Shareholders' Equity and Other Capital Funds	1.0
		Avg Yield	Fndg Costs			NIS	NIM
Bank FGH		7.50%	1.67%			5.83%	6.00%
Bank IJK		7.50%	2.00%			5.50%	5.50%
							Diff
							−0.17%
							0.00%

Margin-Type Ratios for Other Purposes

Although the NIM is the most commonly used margin ratio, it is possible to use margin-type ratios to measure the profitability of particular segments of business activity such as credit card operations, as the table in Exhibit 6.22 illustrates.

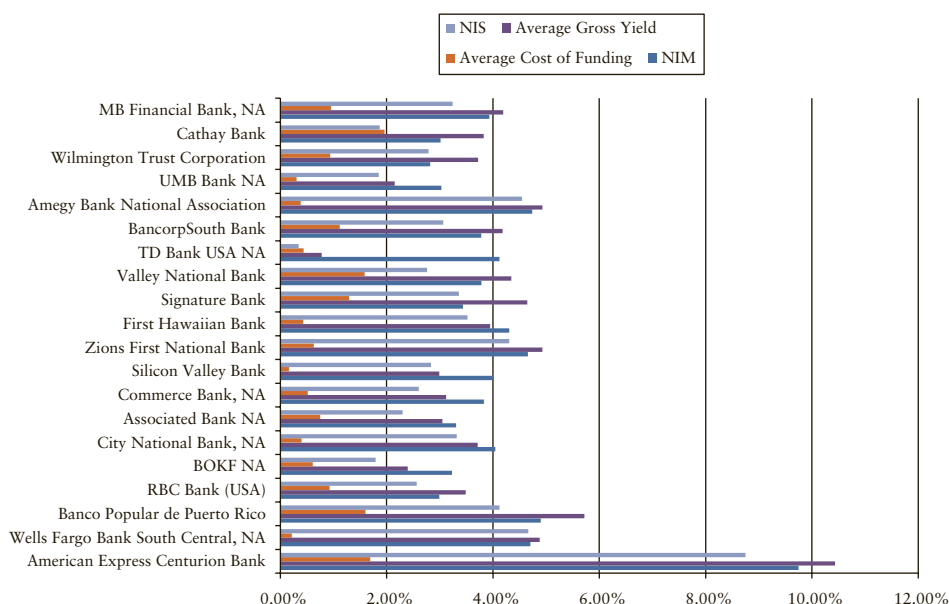


EXHIBIT 6.21 Comparative Funding Costs, Yields, and NISs for Selected Banks

Naturally, the costs associated, for example, with maintaining a network of branches are a factor in a bank's strategic decisions regarding the funding of its assets. A large branch network will provide the bank with low-interest funding, but will have associated costs that interest margins do not apprehend. This takes away some of the strength of interest margin comparisons.

EXHIBIT 6.22 Profitability of Credit Card Operations in the UK, around 2005 (figures for illustration only)

	Subprime Issuer	Prime Issuer
Net interest margin on cards	20.7%	6.4%
Credit card fee income	5.8%	2.3%
Charge-offs (i.e., credit costs)	-11.0%	-3.2%
Net credit card margin	15.5%	5.4%
Operating expense	-6.5%	-2.3%
Pretax return before marketing	9.0%	3.1%
Marketing expenses	-4.0%	-1.5%
Pretax credit card return	5.0%	1.6%
Ancillary product fee income	2.0%	1.0%
Pretax return on assets	7.0%	2.6%

Cost-Efficiency Ratios: Measuring Profit Destroyers

The final category of profitability ratios is the category that measures relative costs rather than revenue. Differences in the expense burden a bank carries can have a large impact in its profitability vis-à-vis its peers. It can also explain why one bank has superior performance and another below-average performance. Since we have already looked at the impact of funding expense—that is, interest expense—in the context of a bank's spread business, cost analysis tends to focus on noninterest expense, particularly operating expenses or overheads.

Cost Efficiency

The term *efficiency* is sometimes used to refer to a bank's ability to optimize income relative to costs.

- A bank with high efficiency—that is, it makes efficient use of the inputs to its noninterest expenses such as staff compensation represent—will have an efficiency indicator that is a low number.
- A bank with poor efficiency will exhibit a higher number.

A first step in evaluating overheads is to look at noninterest expenses as a whole to ascertain how noninterest expenses have changed over the previous fiscal period. Significant changes in percentage growth from previous periods, or substantial deviations from the industry and the bank's peer group should be noted.

- Have expenses increased or declined?
- Is the rate of change in line with changes in operating revenue?
- How does it compare with the bank's peers?

As with income, we can look at a bank's overall operating expense level, or break it down by categories. Recall that these are comprised mainly of compensation, occupancy, administrative, and marketing expenses. Where a large deviation from historical or peer norms in the bank's overall cost-base has been observed, the categories of expenses that have caused it should be identified. The basic question that arises is this: How have the constituent elements of each expense category caused the overall category value to change?

Cost-Efficiency Ratios

There are two fundamental indicators of cost efficiency. One considers overheads relative to profits; the other overheads relative to assets.

Cost-Income Ratio

The first is the cost-income ratio. This measures noninterest expenses (excluding loan-loss provisions) as a proportion of operating income. It is sometimes merely called the efficiency ratio. The formula is shown as follows.

$$\begin{aligned} \text{Cost-income ratio (percent)} &= \frac{\text{Noninterest expenses (excluding loan-loss provisions)}}{\text{Operating income}} \times 100 \\ \text{Operating income} &= \text{Net interest income} + \text{Noninterest income} \end{aligned}$$

Cost-Assets Ratio

The other is the cost-assets ratio, which measures noninterest expenses as a proportion of total (average) assets. The formula is shown as follows. Ideally, average assets should be used in the denominator so that the rise in expenses over the course

of a financial period is measured against the mean asset level and not the end-period figure, which will tend to dampen the ratio value.

$$\text{Cost-assets ratio (percent)} = \frac{\text{Noninterest expenses (excluding loan-loss provisions)}}{\text{Total (average) assets}} \times 100$$

Application

Significant changes in these ratios from previous periods, or substantial deviations from the industry and the bank's peer group should be noted. Where an anomaly, such as a very high rate of increase, is observed, the categories of expenses that have caused it should be identified. Both metrics are useful, but for most banks changes in assets fluctuate less than operating income levels. Consequently, the cost-assets ratio (sometimes called the cost margin) is both more stable and less easily manipulated, so it better reflects a bank's core efficiency. Since there is usually a close correlation between a bank's assets and its earnings, its use is also appropriate. Note that as the operating expenses among inefficient banks approach PPI, the cost-income ratio will peak below 100 percent unless the bank is making an operating loss. Operating costs to assets will vary along with the profitability of the bank's spread business and its relative contribution to PPI.

What Affects Bank Efficiency?

Average efficiency ratios vary significantly from bank to bank, and from market to market. Generally, a bank's overheads will vary with the type of business it engages in, the business environment in which it operates, its size, and management's ability to rein in expenses. Likewise, the type of franchise a bank holds will affect its cost ratios. Consumer banking that employs a large branch network will tend to have substantially higher costs per unit of income or assets than a wholesale banking operation that only requires a single or few offices. Finally, a larger bank may very well benefit from economies of scale, although if the increase in scale is the result of a merger the anticipated synergies may not always be forthcoming as rapidly as expected.

At the macro level, differences in the structure of the economies of each market, their relative level of economic and technological development, the strength of labor regulation, as well as geography will have an impact on the cost base of the industry. Certainly, administrative expenses such as commercial rents, utility charges, and compensation will vary from market to market depending upon supply and demand that may not correlate directly with revenues. Similarly, regulations and the degree to which outsourcing and technology are employed, among other numerous considerations, will have an impact on the institution's cost base. But this is compensated to a large extent by the level of economic development having an impact on transaction size.

To illustrate, a bank that operates where wages are relatively low may not be able to exploit that advantage because the cost of consummating a transaction does not rise directly with the volume of the transaction. The cost of arranging a \$1 million loan is not likely to be much more than the cost of arranging a \$100,000

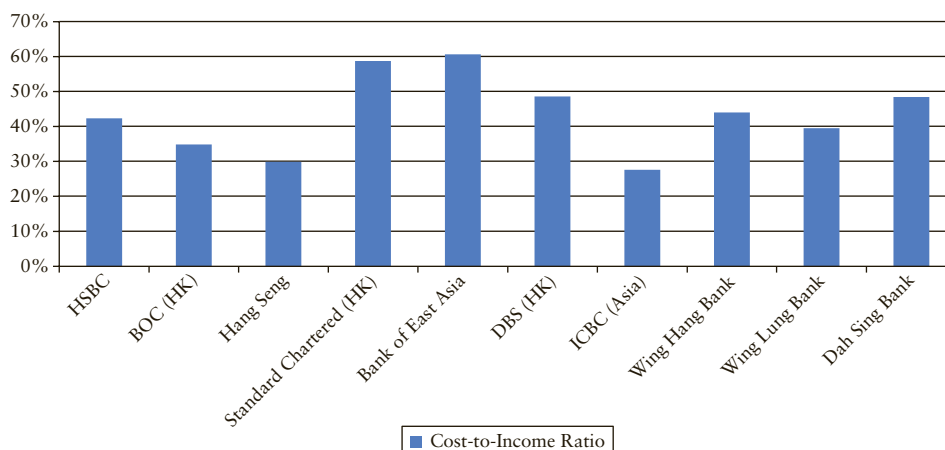


EXHIBIT 6.23 Cost-Income Ratio, Selected Hong Kong Banks, 2010

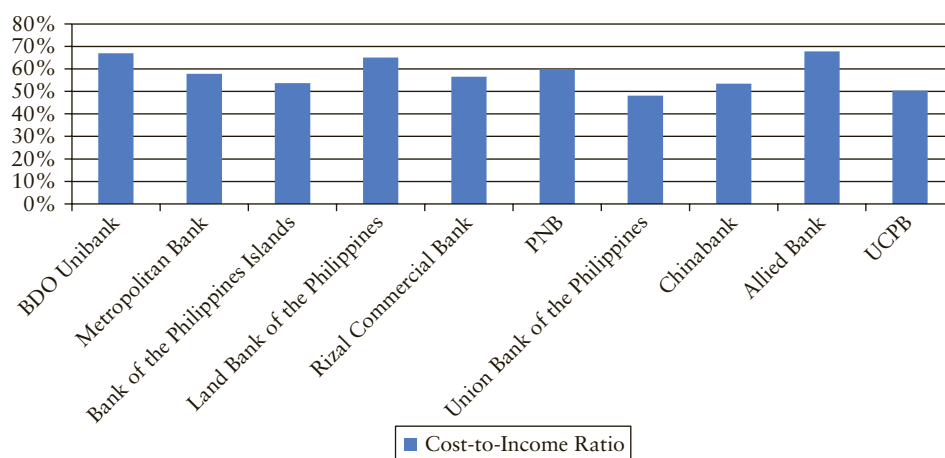


EXHIBIT 6.24 Cost-Income Ratio, Selected Philippine Banks, 2010

loan. For example, Hong Kong banks have very high efficiency levels (see Exhibit 6.23). With much wealth concentrated in a small geographic area, comparatively small branch networks, a regulatory regime that is friendly to companies, and a high level of technological development, administrative costs per transaction are relatively low.

In contrast, the Philippines, a widely dispersed island nation with a substantially lower per capita GDP, less-developed technological infrastructure, a widely dispersed geography, and labor laws that are more sympathetic to workers, overall efficiency is much lower. For these reasons, cross-border comparisons are often inapt (see Exhibit 6.24).

Is Better Efficiency Always Credit Positive?

As a general rule, better efficiency exhibited in a lower cost-income or cost-assets ratio is a positive consideration in a bank's credit assessment. As with most indicators, however, the rule does not necessarily hold true at the margins, and it is always a good idea to subject unduly low ratios to some further scrutiny. Why is the bank able to maintain operating costs at levels significantly below its peers? Assuming that the factors described immediately above are not present, it may be the result of better management and tighter rein on expenses. It may also be short-term gain at the expense of longer-term pain. A bank that fails to keep pace with market wages, that neglects to train staff order to keep up with regulatory and product changes, or that fails to invest in new technology may be either substantially raising its operational risks, or making itself vulnerable to initiatives on the part of its rivals. These operational risks may not be as visible as the difference in the amounts spent on the improvements, but in the longer term they may be just as real. It is, of course, difficult to factor such considerations on a quantitative basis, but it is nonetheless a matter that the analyst should not fail to consider.

An efficiency ratio that shows total assets as a denominator might be difficult to gauge against standard benchmarks since some of those assets might not carry operational costs as high as those carried by commercial banking. Currency or treasury traders, for example, cost a small fraction of the assets they build and manage. Therefore an out-of-range efficiency ratio may not necessarily be a good indicator of a bank's strength or lack of it.

Efficiency ratios that measure costs against assets and those that measure costs against income may give conflicting signals.

WHY THE COST-ASSETS RATIO AND THE COST-INCOME RATIO MAY CONFLICT

Assume that Hydrangea Bank's noninterest expenses actually remain constant over two succeeding periods, and that total average assets decline while operating income rises. This might occur as a result of the bank's enjoying unusually strong trading gains. Or perhaps its marketing of fees and commission type businesses is bearing fruit generating strong noninterest income while net interest income remains comparatively flat. Whatever the cause, the result will be that the bank's cost-assets ratio will rise at the same time its cost-income ratio declines (Scenario 1 in Exhibit 6.25). Likewise, it is possible, though perhaps less likely, that assets grew while operating income declined. Again, there is any number of reasons this could occur. The consequence would be the same divergence in the ratios but in opposite directions (Scenario 2). Where rate of changes in assets and operating income sandwich the rate of change in noninterest expenses, the same phenomena will occur, as shown in Scenarios 3 and 4.

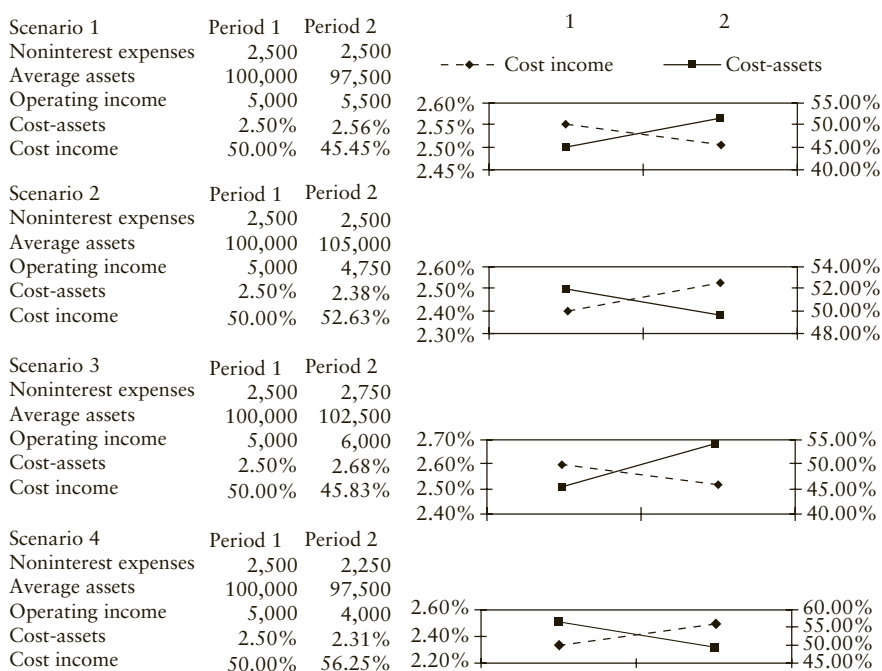


EXHIBIT 6.25 Conflict between the Cost-Income Ratio and the Cost-Asset Ratio

PROFITABILITY: AN ILLUSTRATION OF A PEER ANALYSIS

To illustrate the use of earnings and profitability ratios in bank credit analysis, consider the peer group referred to earlier in the chapter. The institutions are listed in Exhibit 6.26 in descending order of their net income growth over FY10, with their growth from FY07 to FY10. For the purposes of this example, only one indicator is presented. A full analysis would consider a larger number of metrics and explore the influences on bank profitability in more detail.

Comparing Earnings Growth

How can we analyze the earnings growth of a series of banks in the United States, such as those in Exhibit 6.26, in comparison with their peers? It is very difficult, and quite often irrelevant. Looking at published net income from 2007 to 2010, the figure most readily obtainable, it can be seen that substantial fluctuations took place with most banks, admittedly during an unprecedented crisis.

Interestingly, though, the average growth—that is in fact the growth of the grand total of the batch of banks—remained stable between 9 and 11 percent. Only one bank—UMB Bank—displayed modest positive or negative growth figures.

Substantial variations in earnings show either that a bank actually went through rough times or that it is generally prompt at recognizing losses and profits.

EXHIBIT 6.26 Comparing a Bank's Earnings Growth with That of Its Peers

Bank Name	Net Income US\$ mil 2010	Change	Net Income US\$ mil 2009	Change	Net Income US\$ mil 2008	Change	Net Income US\$ mil 2007
First National Bank of Omaha	192	121.2%	87	1.2%	86	243.5%	25
City National Bank, NA	146	106.6%	71	41.3%	121	40.6%	203
East West Bank	169	98.3%	85	317.9%	(39)	123.2%	169
Signature Bank	102	62.8%	63	44.5%	43	59.0%	27
U.S. AgBank, FCB	511	57.7%	324	7.2%	350	16.9%	421
American Express Centurion Bank	1,341	44.8%	926	2.6%	951	25.4%	1,275
First Citizens Bank & Trust Company	208	38.4%	150	18.1%	127	1.5%	125
Commerce Bank, NA	219	32.6%	165	6.2%	176	17.0%	212
Silicon Valley Bank	129	31.9%	98	20.5%	123	13.2%	142
Bank of Hawaii	184	27.1%	145	23.3%	189	0.4%	190
FirstMerit Bank NA	106	25.8%	85	30.0%	121	1.9%	123
Wells Fargo Bank South Central, NA	569	25.3%	454	22.1%	583	63.8%	356
Northern Trust Bank of Florida, NA	101	8.1%	94	33.3%	141	3.0%	137
Valley National Bank	145	6.8%	136	45.2%	93	41.7%	160
UMB Bank NA	70	2.8%	72	17.0%	86	49.2%	58
First Hawaiian Bank	213	7.8%	231	3.6%	223	7.5%	207
Bank of America Oregon, NA	345	28.2%	480	282.5%	125	96.3%	64

(Continued)

EXHIBIT 6.26 (Continued)

Bank Name	Net Income US\$ mil 2010	Change	Net Income US\$ mil 2009	Change	Net Income US\$ mil 2008	Change	Net Income US\$ mil 2007
Bank of America California, NA	327	36.8%	517	65.5%	312	444.3%	57
Bank of America, Rhode Island, NA	388	41.8%	667	10.7%	603	854.3%	63
BancorpSouth Bank	33	71.1%	113	13.2%	131	12.2%	149
	5,498	10.8%	4,962	9.2%	4,544	9.1%	4,164
Average Growth		10.8%		9.2%		9.1%	

Data Source: Bankscope.

In emerging markets, it is easier to hide fluctuations in earnings, and a bank's profitability is often seen by the public at large as the only sign of creditworthiness.

Inside the sample, it would be tempting to compare banks on the basis of earning growth, but this of course would be incorrect, unless the reasons for the differences in growth are known. As an example, a corresponding growth in total assets comes to mind.

Even a comparison focusing on a narrower item such as core earnings growth, for example, would not be too relevant.

Comparing Profitability

Having obtained an overview of the banks' earnings growth, profitability ratios can be compared, for example ROAA. The institutions³² are listed in Exhibit 6.27 in descending order of the ROAA generated by each in FY10. In the adjacent columns two other key profitability indicators are indicated: *net interest margin* and *recurring earnings power*.

It is apparent that ROAA ranged mostly—ignoring the two extreme readings—from about 0.7 percent to about 2.3 percent, the NIM—ignoring two unusually high ratios—from 2.7 percent to almost 5.2 percent, and the REP—ignoring the two extreme readings—from 1.20 percent to 6.4 percent.

As the reader will have gathered, a solid NIM usually translates into a strong ROA. When it is not the case, that is with, say, more than a 250 basis points difference, it might indicate that the bank has efficiency problems. With a difference below 150 basis points, either the bank is very efficient or it generates substantial noninterest income. Sometimes those factors combine to propel a bank out of the

EXHIBIT 6.27 Peer Comparison on ROAA, NIM, and REP 2010

Bank Name	Net Interest Margin % 2010	Return on Average Assets (ROAA) % 2010	Recurring Earning Power % 2010
American Express Centurion Bank	9.747	4.955	9.617
Bank of America Oregon	4.441	2.291	4.194
Bank of America, Rhode Island	4.419	1.773	3.792
First National Bank of Omaha	8.174	1.75	6.363
U.S. AgBank	2.731	1.691	1.926
Wells Fargo Bank South Central	4.7	1.572	4.061
First Hawaiian Bank	4.304	1.471	2.597
Bank of Hawaii	3.555	1.445	2.49
Bank of America California	4.799	1.4	4.523
First Citizens Bank	4.878	1.233	2.687
Commerce Bank	3.826	1.207	2.292
Valley National Bank	3.78	1.02	1.812
Signature Bank	3.432	0.981	2.275
Silicon Valley Bank	4.001	0.904	1.816
FirstMerit Bank	4.156	0.863	1.887
East West Bank	5.231	0.821	2.243
Northern Trust Bank of Florida	2.846	0.809	2.153
City National Bank	4.042	0.7	1.762
UMB Bank	3.025	0.666	1.2
BancorpSouth Bank	3.776	0.245	1.747

Data Source: Bankscope.

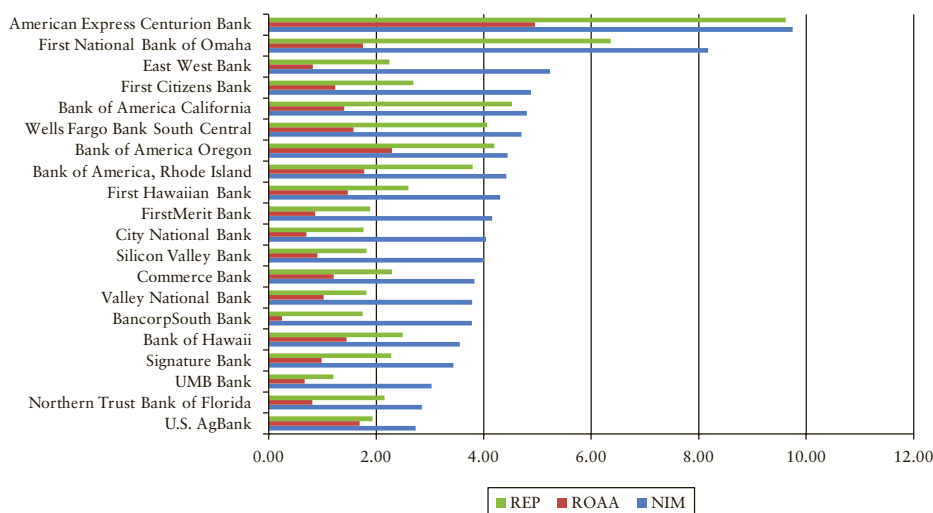
expected range. Other factors, such as those related to quality of earnings, might still alter the picture.

At the very least, it seems obvious that relying on those indicators in isolation would not necessarily provide a clear picture of the profitability of the institution. The analyst would have to study in much detail the notes attached to its financial statements.

Empirical Relationship among Profitability Ratios

At this point, it is worth repeating that the numerator and, more importantly, the denominator of most profitability ratios can be defined in a variety of ways. For the purpose of comparing several types of ratios, we use the simplest ones in each category, but obviously similar comments could be made about the more elaborate ratios.

It has been shown that there is a mathematical relationship among ROA, ROE, and leverage. Is there any evident correlation between ROA and these other profitability ratios such as REP, NIM, and the cost-income ratio? It can be expected because ROA, REP, and NIM have a similar form in that each compares some measure of return against some measure of assets. The cost-income ratio measures two income statement items, so the connection is more tenuous. The empirical results bear this out. See Exhibit 6.28.

**EXHIBIT 6.28** ROAA, NIM, and REP

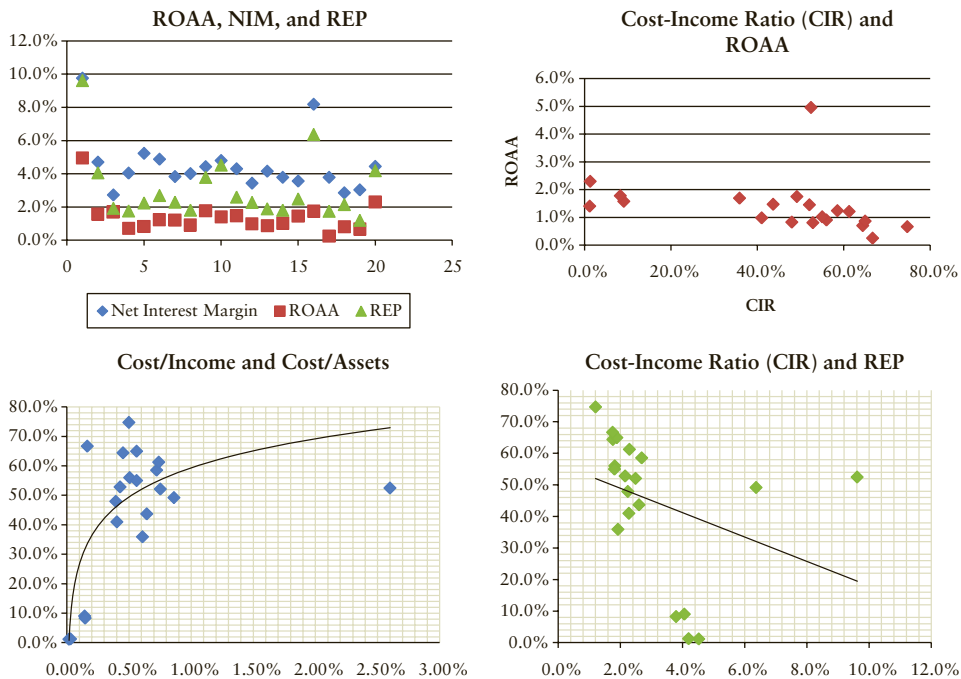
Data Source: Bankscope.

ROA, REP, and NIM

Of the banks in the group, there is an apparent correlation among the first three ratios, while the cost-income ratio appears to be all over the map, so to speak, in relation to return and margin-type indicators. If the data from the banks in Exhibit 6.28 is plotted on a scatter chart, this relationship becomes even more apparent, as can be seen in Exhibit 6.29. The correlation among ROA, REP, and NIM is explained by them all being similar in structure. Each compares an adjusted line item on the income statement—net income, PPI, and net interest income respectively—with a corresponding line item on the balance sheet, namely average assets or in the case of NIM, average earning assets.

While by and large similar, the differences among the ratio values generated can be explained by the numerators and denominators of REP and NIM, and ROA varying from bank to bank. One institution, for example, may have a high proportion of net interest income to operating income, so its NIM may diverge from its ROA. Likewise, one bank may have a low proportion of loan-loss provisions to PPI while another may have a higher proportion, causing REP to diverge from ROA. The differences among ratio values result from variations in the bank's asset and revenue mix. In practice, these divergences may tend to follow a statistically normal distribution.

The cost-income ratio, in contrast, looks at two income statement items, neither of which is directly linked to the numerators or denominators in the preceding ratios. Since the cost-income ratio measures the relationship between operating expenses and PPI, it only has a much looser relationship to ROA and NIM. The relationship between the cost-income ratio and REP is closer, as the bottom right diagram depicts, possibly because the denominator of the cost-income ratios is similar to the numerator of REP.

**EXHIBIT 6.29** Correlation among Profitability Indicators

Data Source: Bankscope.

PROFITABILITY RATIOS: NO MYSTERY

There is really no mystery to profitability ratios. They are all of the same cloth. The more sophisticated the ratio, the more refined the question it seeks to answer. But simplicity can be a virtue. In the end, for rough-and-ready ranking purposes, the more refined ratio may not convey significantly more information than simple return on average assets. The analyst needs to decide whether the time required to collate the data and compute the ratios across a universe of banks is worth the effort.

MACRO-LEVEL INFLUENCES ON BANK PROFITABILITY

In evaluating the earnings and profitability component of the CAMEL framework, the analyst needs to understand and to be able:

- To explain why the bank is more or less profitable than average.
- To explain why its profitability is improving or declining.
- To evaluate the impact of its performance on its creditworthiness.

The critical questions to be asked are:

What is causing the bank's earnings and profitability to increase, decrease, or remain constant?

What accounts for differences in profitability between the subject bank and its peers?

What causes present performance to diverge from historic performance?

To answer these questions, the analyst must look at the bank's earnings drivers and at the obstacles to earnings growth and profitability. An understanding of the bank's business is helpful in this respect, but much can be gleaned solely from its financial statements and key indicators. There are questions to consider in a general assessment of earnings and profitability.

EARNINGS AND PROFITABILITY

First considering earnings, to what extent have net profits changed in percentage terms compared with one or more previous periods? Points to consider are:

- How have net profits changed?
- By what percentage upward or downward?
- Does this represent a continuation of prior trends or a break with them?
- How does the bank compare with its peers in terms of critical profitability ratios?
- Is it materially above or below the mean?

What constitutes weak or strong growth will depend upon the market and the economic environment, including the stage of the credit cycle, but in normal conditions the following guidelines may prove helpful.

Losses are invariably unfavorable with a possible mitigating factor being that they have emerged because of adoption of stronger prudential rules or better disclosure. Negative growth is usually unfavorable, but can be expected in some circumstances such as after the collapse of an asset bubble.

Rules of Thumb Concerning Earnings Growth Rates

Growth rates:

- Less than 2 percent would in most situations be considered weak or mediocre.
- From 2 percent to 8 percent generally can be construed as fair to middling.
- From 8 percent to 15 percent are by and large considered healthy, and in the double-digit range normally strong.
- From 15 percent to 30 percent would ordinarily be viewed as high to very high growth, potentially raising questions toward the higher end.
- More than 30 percent are extremely high and may be suspect.

Consider ameliorative factors such as mergers, growth from a low base, and an asset bubble or an overheating emerging economy.

Profitability

How has the bank's ROA changed? Note that ROA will be adversely affected by high levels of loan-loss provisions.

Compare trends in pretax, preprovision ROA with after-tax, after-provision ROA to determine changes in core profitability. Are there any unusual items, either income or expense items, that explain a significant change in the bottom line?

Have high levels of loan-loss provisioning been put aside that have cut into profits? Has there been an acquisition of another institution (note: this will tend to increase profits, but effect on profitability will vary)? Have there been unusual sales of real property or subsidiaries that have inflated profits?

The financial statements, if they are doing their job, merely reflect an underlying economic reality. The deeper question to be answered is this: What is impelling these changes at the macroeconomic and micro levels?

To understand what is causing a bank to be more or less profitable than its peers, or to be experiencing increases or decreases in profitability relative to them, the analyst can take comfort that while there is an unlimited number of potential causes, these can be divided generally into two categories.

1. The first embraces those causes that are affecting all or most banks in the sector. These can be termed *sector-wide influences*.
2. The second encompasses those that mainly affect the specific bank being examined and not most or all of its peers. These are *bank-specific factors*.

Of course, reality is complex, and in any particular instance the interaction of bank-specific characteristics with sector characteristics may exacerbate or dampen the emergence of significant changes in earnings and profitability.

Sector-wide influences are those that affect the entire banking industry to some degree. Macrolevel factors can be further divided into those that are cyclical in character and those that are secular in nature.

- Cyclical changes are those that correlate with the economic or credit cycle, as will be discussed in Chapter 7.
- Secular changes represent long-term changes that span more than one business cycle and are usually impelled by technological innovation, the development of new markets or products, or demographic and social change. Regulatory reform can also have a major impact on bank profitability.

Related to secular factors are one-off changes that result from special circumstances such as the need to prepare for the so-called Y2K problem that caused banks'

operating expenses to rise in many markets in 1999. In practice, both types of sectoral changes are likely to intermingle and contribute to result in a significant change in earnings or effectively cancel each other out. Identifying the most salient influences can be quite difficult, although it becomes easier as one's knowledge of the local market increases.

Changes in Profitability: Cyclical and Secular

When the economy is booming, demand for credit is high, and banks are frequently able to earn fat margins (especially where savers have few alternatives to bank savings). Fee business is also likely to be strong. All other things being equal, banks will show more robust performance when the economy is expanding. In the same manner, when the credit cycle is turning and an asset bubble has burst,³³ there will probably be surplus liquid funds coupled with a dearth of lending opportunities, as companies retrench and banks seek to rebuild their balance sheets. Where banks engage mainly in unhedged floating-rate lending and loans are more interest-sensitive than funding costs, the central bank's reduction of prevailing interest rates could cause loans to reprice faster than deposits, decreasing spreads and bank profits industry-wide.³⁴

Secular factors, as stated, span more than one business cycle and represent long-term changes in a bank's overall operating environment. What might cause a change in profitability among banks in a given market? First consider possible reasons for a general improvement in bank earnings or, more accurately, a general improvement of one or more of their profitability ratios.

- One possibility is that banks have found new business opportunities enabling them to generate stronger earnings. The development of securitization, for example, has tended to push up bank profits in some markets.
- Another example is that amid a loose regulatory regime, banks have formed a cartel to maintain higher interest rates on lending, with the result that the profitability of the sector as a whole is lifted.

Unsurprisingly, a bank's operating environment can also deteriorate as a consequence of similar types of developments. Examples of adverse effects include the following.

- New government regulations or directives may make it difficult for banks to make profits. Regulatory changes increasing or restricting competition, requiring higher or lower levels of capital or loan loss provisions, or limiting or expanding the types of business banks can engage in and the imposition of policy lending obligations are all apt to have some impact on profits. Increasing mandatory reserve ratios, imposing restrictions on real estate-related lending, and similar measures may adversely affect the banking industry as a whole.
- Another phenomenon that is seen is the deleterious impact general financial deregulation can have on banks that were once protected from competition. These institutions, forced to vie with new entrants to obtain new business may find it hard to compete as margins decrease and drag down the profitability of the sector as a whole.

- Finally, long-term changes in business conditions may make it hard for banks to maintain thick spreads as companies in need of capital obtain access to additional financing options (disintermediation).

Naturally, the analyst should remain aware of the impact of the yield curve, on the one hand, and of the bank's risk profile, on the other. Those two factors may contribute a combined overall effect on profitability that may overshadow the influence of the above factors.

SUMMARY OF FACTORS AFFECTING THE PROFITABILITY OF A BANK'S INTEREST-GENERATING BUSINESS

Some of the reasons for changes in the profitability of a bank's interest-earning business have already been mentioned. Fundamental factors include the institution's asset mix, deposit mix, and asset quality.

Asset Mix

A bank's asset mix may explain above- or below-average yields. Other things being equal, a greater proportion of funds being placed into higher-risk assets, for example loans, will cause a bank's NIM and ROA to rise—but of course not necessarily the ratios that use risk-adjusted assets as a denominator. A greater proportion of funds being placed into low-yielding liquid assets will cause the NIM and ROA to fall. Similarly, making higher-yielding but riskier loans (or fixed income securities) will push the NIM and ROA upward, while a more conservative policy concentrating on lending to blue-chip companies at prime rates of interest will keep the NIM and ROA at lower levels.

The regulatory framework and environment may also have an impact on asset composition. Bank supervisors may encourage banks to maintain higher levels of liquidity by raising mandatory liquidity thresholds. Likewise, during harsh economic conditions, bank management may seek to hold a higher proportion of its assets in easily liquefiable assets and in relatively safe but lower-yielding loans.

Deposit Mix

A bank's mix of liabilities (and equity) may explain above- or below-average funding costs. An increased proportion of demand (checking) deposits means lower funding costs; in contrast, an increase in *time deposits*, and more particularly in *purchased funds* (interbank funds or brokered deposits) will tend to push up funding costs.

In contrast, a bank with a large branch network is also likely to have lower funding costs than a bank that maintains only a few branches. The wider the bank's branch network, the more probable it will encompass areas where there

is comparatively little competition for deposits. Normally, nonurban residents are less interest-rate-sensitive than depositors who live in large cities.

Asset Quality and Provisioning Levels

Finally, a bank's asset quality may help to explain comparatively higher or lower net yields. A higher level of NPLs (i.e., low asset quality) means less net interest income received. Moreover, the accounting treatment of NPLs is such that once a loan is deemed nonperforming, typically any income ultimately received will no longer be categorized as interest income. Conversely, a lower level of NPLs (i.e., a loan book with high asset quality) translates into a higher level of net interest income for a given portfolio of loan assets, all other criteria being equal.³⁵

MICRO-LEVEL INFLUENCES ON BANK PROFITABILITY

When the bank being evaluated registers substantial changes in earnings and profitability not seen in the group as a whole, or where it diverges significantly from the peer group, the possible causes need to be considered. Five fundamental possible reasons for such differences or divergence in bank profitability at the micro level are:

1. Operating costs are higher or lower than those of its peers.
2. Provisioning is higher or lower than that of its peers.
3. Funding costs are higher or lower than those of its peers.
4. Yields are higher or lower than those of its peers.
5. Noninterest income business is more or less robust than that of its peers.

Peer Comparison of Operating Costs

As with revenue, operating expenses should be monitored for changes both overall and in their relative composition—compensation, occupancy, and so on—as well as for material movements in concomitant ratio values. An initial inquiry ought to determine the following:

- Have noninterest expenses increased, decreased, or remained about the same?
- How have changes in various expense categories affected the bank's overall cost burden?
- How does cost control compare with past experience and peers? How do these changes compare with previous periods and with the record of competitors?
- How does the bank's operating efficiency, as reflected in the cost-income and cost-assets ratio, compare with past performance and peers?
- Operating expenses that are comparatively higher or lower than the mean may be the result of temporary circumstances or be more fundamental in character.

Temporary blips or dips in overheads may be the response to specific circumstances, while others may represent longer-term competitive advantages or disadvantages. An example of the first type might be a consequence of change in strategy that entails the establishment of new branches or a new marketing campaign. An example of the second type might be the economies of scale a large institution is able to enjoy, providing it with a competitive edge over its smaller rivals.

- A comparatively low cost base is almost always an advantage to a bank and credit positive, but extremely low overheads may either reflect a difference in the institution's business model that makes it not strictly comparable with its peers, its costs savings may be undercut by reduced revenues or higher funding costs, or it may be at the expense of the bank's long-term health. A simple illustration is the bank that maintains a branch network that is small relative to the bank's assets. This strategy may work well if management is targeting a niche market or if it is able to obtain low-cost funding through other channels, such as through the Internet. But if the small branch network means that the bank must rely mainly on purchased funds, resulting in higher funding costs and greater liquidity risk, then the cost efficiencies may be outweighed by these factors.

Possible causes of changes in a bank's cost base and divergence from the mean include the following:

- (*Lower*) Management is keeping a tight rein on expenses.
- (*Higher*) Management is too much of a spendthrift.
- (*Lower*) Management is hiring second-class staff or not hiring enough staff.
- (*Higher*) Management is too top heavy.
- (*Lower*) Management is not spending enough on staff training.
- (*Higher*) Management is investing in information technology or other capital-intensive initiatives.
- (*Lower*) Through merger or organic growth, the bank has realized economies of scale.
- (*Higher or lower*) The bank's franchise has special characteristics that cause it to have higher or lower costs than average (e.g., a far-flung branch network,³⁶ wholesale banking).
- (*Lower*) The bank has changed its strategy, such as closing retail branches to concentrate on wholesale banking.
- (*Higher*) The bank has changed its strategy, such as opening retail branches in a shift from wholesale to retail banking.

Peer Comparison of Provisioning Costs

Loan-loss provisions are an estimate of a bank's credit costs that are deducted from operating income even if the expected credit losses do not materialize. High rates of loan-loss provisioning are therefore a drag on net profits. From a credit perspective, it is often better to see a bank err on the side of prudent provisioning levels than on increasing the bottom line. But higher loan-loss provisioning charges are not always credit positive. If, for example, management deducts greater amounts of provisions

because its asset quality is poorer than that of its peers, or to make its bottom line look better in the next financial period, higher provisions may justifiably be viewed as credit neutral or even credit negative.

To evaluate the impact of provisioning charges, the analyst needs to consider why these charges are higher or lower than those of the bank's peers. The amount of loan-loss provisioning deducted from operating income can reflect a number of scenarios and needs to be examined closely to determine whether it is precautionary, reactive, or used for earnings management purposes, among other reasons.

THE SIGNIFICANCE OF LOAN-LOSS PROVISIONING

The critical questions, easier posed than answered, with regard to loan-loss provisioning, concern whether the level of loan-loss provisioning is appropriate to conditions. In brief, the analyst needs to determine the following:

- Has the bank underprovisioned? If so, what would profits and profitability be if the appropriate level of loan-loss reserves had been set aside?
- Has the bank overprovisioned? If so, what would profits and profitability be if the appropriate level of loan loss reserves had been set aside?

A high level of loan-loss provisioning relative to that of peers, depending upon circumstances, may signify any of the following:

- The bank is engaged in an intrinsically higher-risk lending business than are its peers (that is hopefully justified by higher returns).
- The bank is responding to an unexpected decline in asset quality, and is remedying the situation before it gets out of hand.
- The bank's asset quality is substantially worse than its peers, and it is attempting to rebuild its balance sheet to a more acceptable level.
- The bank is more prudent than its peers in setting aside high levels of precautionary provisions.
- The bank is attempting to reduce its taxable income (if located in a jurisdiction where loan-loss provisioning is taxable).

Various scenarios are illustrated in the table in Exhibit 6.30. On the left, three potential provisioning levels are shown: too high, just right, and too low. The six columns to the right show possible reasons for the three scenarios. Appropriate levels would by definition reflect the appropriate level of precautionary prudence or a response to an eroded asset quality position (the two columns on the left labeled "precautionary" and "reactive"). The remaining four columns to the right show common causes of overprovisioning and underprovisioning:

	Type	Precautionary	Reactive	Error	Tax driven	Earnings smoothing	Excess
Reason		Appropriate level of conservatism	Appropriate reaction to asset quality problem	Miscalculation—too much or too little	To reduce taxable income or driven by tax regulations	To build pool of LLRs or to boost bottom line	Excessive conservatism or risk appetite
Amount	Excessive			✓	✓	✓	✓
	Appropriate	✓	✓				
	Too little			✓	✓	✓	✓

EXHIBIT 6.30 The Significance of Loan-Loss Provisioning Levels: Multiple Scenarios

Source: Independent Company Research.

1. Error (e.g., miscalculation, inadequate risk management systems, incompetence)
2. Tax driven (e.g., regulations limit or encourage a fixed level of provisioning, other regulatory considerations)
3. Earnings smoothing (e.g., reducing loan-loss provisions to show better earnings results even though a higher level is called for)
4. Excess (e.g., excessive risk appetite, excessive conservatism).

Thankfully, in many countries, the regulatory authorities are now imposing rules on provisioning. It remains the case, though, that bank managers keep some discretion in provisioning decisions, especially in emerging markets where many tricks are available to them should they wish to conceal weaknesses in asset quality. Factors that will tend to cause loan-loss provisioning to be higher or lower, with a corresponding effect on the bottom line, include the following:

- *(Lower)* The bank has better asset quality than its peers do owing to excellent credit control, and therefore has low provisioning costs, resulting in stronger earnings.
- *(Higher)* The bank has poorer asset quality than its peers do as a consequence of inadequate credit control, and therefore has high provisioning costs. Alternatively, new management is attempting to clean up the bank's balance sheet by provisioning adequately and writing off loans quickly.
- *(Lower)* Because of a strongly conservative management, the bank is provisioning more than it needs to, notwithstanding that asset quality is at reasonable levels.
- *(Higher)* The bank has shifted its strategy to focus on higher-yield businesses, such as lending to small and medium-sized enterprises, and is provisioning in advance for expected higher credit costs that are priced into the new activity.
- *(Lower)* The bank has shifted its strategy to focus on more conservative lower-yield activities, such as holding a higher proportion of government securities that require lower loan-loss provisioning levels on account of their minimal default risk. Alternatively, legitimate concerns about liquidity in view of developing business conditions have impelled the bank to reduce its loan book and place more of its funds into liquid assets including government securities.

- *(Higher)* Management justifiably fears the burst of an asset bubble in the near future and sets aside higher-than-usual provisions to build up loan-loss reserves against an imminent collapse in asset quality.
- *(Lower)* The bank is not provisioning as much as it should, given the state of its loan book, for any number of reasons, including a desire to make earnings look better, a high risk appetite, and a desire to reduce tax liability.
- *(Higher)* There has been a regulatory directive narrowing the bank's discretion in provisioning. (Other banks are already in compliance.)
- *(Lower)* There has been a regulatory directive penalizing banks for *over-provisioning* to minimize their tax liability.
- *(Higher)* The bank is provisioning more than it should, given the state of its loan book, to make earnings look better in future reporting periods.³⁷

Peer Comparison of Funding Costs

As with operating expenses, an edge in funding costs may be inherent in the bank's credit profile and business model that gives it a competitive advantage over peers. For example, a highly rated well-established bank with a strong deposit network will ordinarily have lower funding costs than a bank that has a lower credit rating and depends to a large extent on purchased funds. Correspondingly, funding cost advantages or disadvantages may be quite short term in nature and only relative to changing conditions. For instance, a bank that relies heavily on the interbank market for funding and market conditions will usually be at a disadvantage to banks with strong deposit bases. But under certain conditions, such as when a turn in the economic cycle causes prevailing interest rates—and the bank's funding costs—to fall more rapidly than for its rivals, it may temporarily benefit.

Possible causes of higher or lower funding costs include the following:

- *(Higher)* The bank relies heavily on the interbank market (or other purchased funds) for funding, and therefore pays more than peer banks with wide branch networks.
- *(Lower)* The bank has a well-established branch network with loyal customers who are not especially sensitive to deposit rates. Alternatively, deposits are sourced from less interest-sensitive markets (e.g., rural markets).
- *(Higher)* The bank is paying more for funding as a result of unfortunately timed debt issues it made in the capital markets. Alternatively, expecting higher interest rates, the bank has locked in funding at a rate it considered low, but interest rates have fallen, rather than risen.
- *(Lower)* The bank is paying less for funding as a result of propitiously timed debt issues it made in the capital markets allowing it to obtain longer-term funding at exceptionally low rates. Alternatively, the bank has a strong treasury department that has enabled it to optimize funding costs relative to those of its peers.
- *(Higher)* The bank is paying more for funding due to a lower or deteriorating credit rating. Alternatively, the bank does not benefit from government support enjoyed by peers and therefore can pay lower amounts for funding (while paying higher rates for deposits).
- *(Lower)* The bank is paying less for funding due to a higher or improving credit rating. Alternatively, the bank has the benefit of government support, and

therefore can pay lower rates for deposits and funding since depositors are confident about its survival.

- *(Higher or lower)* There has been a change in the bank's prospect for external support in the event of distress.

Peer Comparison of Yields

Reasons for differences in yields parallel those responsible for variance or differences in funding expense. As a consequence of the bank's business model, for instance, it may reap greater yields by taking greater risks. Likewise, the causes may be of a temporary nature. Possible causes of material changes include:

- *(Higher)* The bank engages in higher-risk lending than its peers do, such as commercial real estate lending or industrial and commercial lending to small and medium-sized businesses. Or the bank engages mainly in unsecured lending to second-tier borrowers.
- *(Lower)* The bank is aggressively competing to expand market share to obtain a critical mass in the market.
- *(Higher)* The bank engages in—or has it shifted more assets into—a high-yield business such as consumer finance.
- *(Lower)* The bank engages in lending only to blue-chip companies and maintains a large pool of liquid assets. Or the bank has shifted a higher proportion of funds into more liquid assets or lower-risk lending.
- *(Lower)* The bank engages almost exclusively in collateralized lending.
- *(Higher)* As a result of good or improved credit management, the bank is doing better in minimizing problem loans and keeping extant loans performing.
- *(Higher)* The bank is using funds more efficiently, placing a higher proportion in risk assets.³⁸
- *(Higher)* The bank's treasury department has unfortunately locked in high interest rates on short- and medium-term securities as prevailing interest rates have fallen.
- *(Lower)* The bank's treasury department has locked in low interest rates on short- and medium-term securities as prevailing interest rates have risen.

Peer Comparison of Income

Noninterest operating income flows largely depend on the bank's franchise and strategy. Review of the revenues generated by specific business lines or categories of noninterest income is recommended. The next box provides an overview of noninterest income relative to earnings and profitability analysis.

Relevant causes of material changes or variance include the following:

- *(Higher or lower)* The bank's noninterest income stream is composed primarily of more volatile components, which have registered a high positive (negative) fluctuation.
- *(Higher or lower)* The composition of the bank's noninterest income stream has changed as the result of strategic decisions or other causes resulting in more (less) profitable components dominating.

- *(Higher or lower)* Government policy or regulatory changes have positively (adversely) affected the bank's noninterest income flows.
- *(Higher or lower)* The bank's noninterest revenue stream has increased as the result of an acquisition or decreased as the result of a divestment.
- *(Higher or lower)* Items that can be classified analytically as nonoperating have had a significant positive (negative) impact on operating income.

OVERVIEW OF EARNINGS

Net Interest Income Stream

From a top-level perspective, the following questions should be considered in relation to the flow of net interest revenue:³⁹

How does the rate of increase (decrease) in net interest income compare with past trends and peers?

Does it correlate to changes in the volume of loan assets, funding expense (average funding cost), and gross interest income (average yield)?

Is the bank's spread business comparatively profitable or unprofitable? If profitable, is it the result of high-risk lending? Does it compensate for unprofitable noninterest-earning business activities?

How do the NIM and related ratios—including the average cost of funding, yield on funds advanced, and NIS—compare with peer and historical values?

Can changes and divergence from the norm be explained in terms of changes in the bank's funding costs, yield, and asset-liability mix?

Overview of Noninterest Income Stream

The noninterest income component of operating income may be composed of both relatively low-risk fee and commission-type business and comparatively higher-risk trading activities. (Noninterest income categories such as securities and forex trading tend to be more volatile than fee and commission income.) By observing how earnings from noninterest-income-generating business activities contribute to operating income, trends in earnings may begin to become apparent.

Important to keep in mind is the fact that higher-risk trading activities usually involve larger transactions. Losses or profits from such activities tend to be substantial, and they may create a degree of volatility in profitability, something particularly visible in jurisdictions where banks publish quarterly results and do not—or are not allowed to—manage their profits.

From a top-level perspective, the following questions should be considered in relation to the sources of noninterest income:

- What changes can be observed in the bank's noninterest income level?
- Separating this revenue stream into its component parts—fees and commissions and trading income—how has each contributed to the overall change?

- Overall, how volatile is the noninterest revenue stream?
- Does the bank have adequate noninterest income to help diversify its revenue stream?
- Has the bank benefited from any exceptional gains recorded as noninterest income?
- Did the bank have unusually high or low noninterest income the previous year, resulting in a false decline or rise in noninterest income during the current period?

Impact of Nonoperating Items

Although accounting regimes are limiting the ability to obscure the impact of one-off gains or losses, they remain one of the most common ways in which substantial changes in net income are effected. A typical exceptional gain might be the sale of a subsidiary. A typical loss might be a write-off of an equity investment. Similarly, the acquisition of another bank through merger may have a major impact on reported net income as the two banks' operations are merged. Of course, assets will usually expand in turn, and overall ratios may or may not be significantly affected. Finally, there is the possibility that both secular changes and internal factors are combining to have a positive or negative effect. The subject bank may be less or more vulnerable to business cycle, regulatory, or policy changes. Note that items that may properly be included within operating income under local accounting rules nonetheless may for analytical purposes be excluded from it.

The causes of changes in earnings and profitability and differences from the norm are graphically illustrated, on a selective basis, in the diagram in Exhibit 6.31.

QUALITY OF EARNINGS

Earnings quality presents two aspects.

The first concerns intrinsic differences in the sustainability of the various components of a bank's earnings stream. These have already been discussed in the main part of the chapter.

The second concerns the use of aggressive accounting practices, sometimes coupled with opaque reporting and disclosure practices to make an enterprise's financial performance appear better than it is.

Financial Reporting and Earnings Quality

Looking at earnings quality generally, the techniques employed to make financial results look better can be boiled down into a few basic categories. First, the techniques can be divided into those that attempt to make profits look better now and those

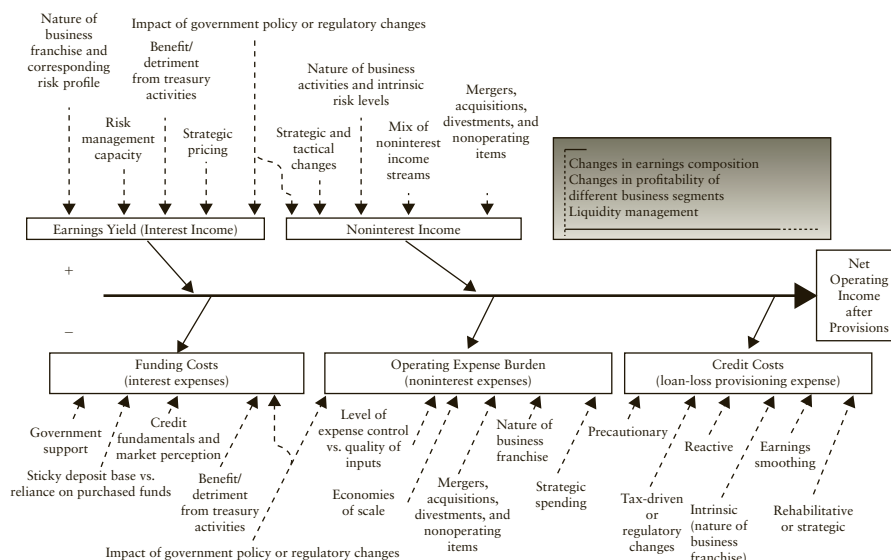


EXHIBIT 6.31 Causes of Changes in Earnings and Profitability

that attempt to make profits look better later. Often the second, somewhat counterintuitive tactic is used as part of an earnings-smoothing strategy designed to make the firm's earnings look highly stable. Steadily growing earnings are much more appealing to analysts and investors than those that are erratic. The latter suggest a more volatile and therefore a more risky investor or counterparty.

It is evident that there are only so many ways earnings can be boosted. One way is to report a company's accounts in such a manner that revenue appears stronger than more conservative accounting practices might justify. Another way is to minimize expenses by not reporting expense items to the extent that conservative practice would require. The same techniques obviously work in reverse to make results look better now either to appear better in future, or to show steadily increasing profits.

How Banks Cook Their Books

In a number of countries, there is a growing pressure on financial institutions to make full and frank disclosure of their financial health. Even in the most advanced economies, it remains possible for banks, with or without the help of lenient regulations, to hide some of the truth, whether on their liquidity, capital, asset quality, or earnings. In general, banks employ similar earnings management techniques as do nonfinancial firms. The structure of banks' financial statements and the way they are reported differs from that of corporations in some respects. Consequently, where earnings management by a corporate will often involve a firm's inventory account and sales figures, since a bank has no "sales" per se and little in the way of physical inventory, earnings management sometimes manifests in a different manner. The loan-loss reserve account is one of the most popular vehicles for bank earnings manipulation as discussed further below.

REPORTING OF FINANCIAL RESULTS BY BANKS DIFFERS FROM THAT EMPLOYED BY NONFINANCIAL FIRMS

The most important differences are as follows.

When the Product Is Money

Banks trade primarily in money, which is inherently fungible and therefore the most commodity-like commodity. For the most part, banks sell no physical product that has inherent value and that can be inventoried; instead, funds held in various forms are effectively a bank's inventory, and are indistinguishable in substance from the proceeds from debt or equity, which finances capital and inventory in a nonfinancial company. Consequently, cash flow analysis is largely irrelevant to bank credit analysis. Healthy, rapidly expanding banks may, and often do, have negative cash flow.

Banks Use Extremely High Leverage

Banks are highly geared. Their financial performance and condition are therefore highly vulnerable to changes in earnings constituents and to the impairment in the value of their earning assets (loans and securities). They are likewise very vulnerable to changes in economic conditions, including changes in interest rates, and to tax regulation. Since they are unlike nonfinancial firms, in that their assets are mainly financial in character and are funded primarily by deposit liabilities, the concept of debt takes on a special meaning to banks. Bank debt in the form of customer deposits and short-term borrowings cannot be directly compared to debt held by nonfinancial companies.

Banks Are Subject to Special Accounting Treatment

Owing to their quasi-policy role within a nation's economy, banks are highly regulated and often subject to special accounting requirements and reporting guidelines; that is, regulatory accounting principles.

Bank Balance Sheets Mirror Those of Nonfinancial Firms

Banks' balance sheets mirror the balance sheets of industrial companies. In other words, if Company A borrows \$1 million from Bank B, the loan of \$1 million is a liability to Company A but an asset to Bank B.

Boosting Revenue**Using Loan Classification**

Where discretion is permitted to banks in classifying problem loans, and this is usually the case, bank managers may employ a lax view of what constitutes a nonperforming asset. Particularly, in emerging markets where bank supervision and enforcement are weak, this technique may be exploited by managers who wish to

both hide asset quality problems and make net income seem higher than is actually justified.

The failure to classify an NPL properly both understates a bank's true credit costs and overstates its net income.

- It understates credit costs by inadequately provisioning against probable loan losses.
- Lower loan-loss provisioning is, of course, reflected in higher apparent profits. In addition, an improper classification shows as generating income loans that have in fact stopped performing. The interest, while fully counted as income, is not actually received since it is capitalized into the loan amount.

Finally, restructured loans may be reported in ways that understate actual cost. The economic cost of restructuring a loan is the present value of the cash flow of interest and principal that is given up by the bank and would otherwise have been received, based on the original interest rate and repayment terms. U.S. GAAP, for example, conforms to this principle by requiring that restructured loans be restated at net present value discounted at the original rate contracted. Some countries allow the calculation of such costs in ways that significantly diverge from this principle. Thailand, for instance, has no requirement as to revaluation of restructured loans. As mentioned, restructured loans may be classified as performing under quite lax criteria in some markets, which creates a picture of better asset quality than is actually the case under international standards. In addition, in many countries, some restructured loans are simply not classified as such: for example, when a fresh loan is disbursed—whether directly or indirectly—for the purpose of allowing the borrower to service his existing loans.

One-Off Items

A common technique by which banks manage earnings is through the use of a one-off item. Characterization of one-off income items as operating income will make an enterprise look considerably better in the present. To the extent that investors and analysts do not take note of an aggressive characterization, they will think that the bank is performing better than it actually is. Even if properly classified, nonoperating items will usually boost taxable income and therefore the reported bottom line. Note that if management's objective is to smooth income, the well-timed recognition of losses might be achieved.

The one-off technique can be employed in several ways.

CHARACTERIZATION OF ONE-OFF EXPENSE ITEMS AS NONOPERATING EXPENSES

Characterization of one-off expense items as nonoperating expenses will make an enterprise look better in the future and to some extent make it look better in the present, insofar as the one-off item is disregarded owing to its nonoperating status. Even where the item is properly classified for accounting purposes, an adjustment for analytical reasons may be appropriate.

EXHIBIT 6.32 Property Revaluation Rules

U.S. GAAP: APB 6: "Property, plant, and equipment should not be written up by an entity to reflect appraisal, market, or current values which are above cost to the entity."	IAS 16 allows revaluations but they must be updated at least every three to five years; asset revaluations directly increase equity, unless they reverse a devaluation counted as an expense.
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Asset Disposal

The easiest is simply to save unneeded assets for disposal during a cyclical downturn or when results are unexpectedly poor. This can be accomplished by, for instance, selling unwanted subsidiaries or property or securities holdings. The trick, of course, works best when those assets are not recorded in the books at their market value.

Property Revaluation

Not all accounting frameworks allow banks to boost their income through the revaluation of real property held by the bank. Applicable rules in two major regimes are shown in Exhibit 6.32.

A timely boost to income through property revaluation can come in handy to banks looking to improve their earnings results.

BOOSTING THE BOTTOM LINE THROUGH PROPERTY REVALUATION

In 1998, one Hong Kong bank, suffering from heavy provisioning charges, which reduced the bottom line, took an exceptional loss when its property revaluation reserves were insufficient to offset the downward property revaluation occasioned by a sharp decline in local property prices. (It charged revaluation loss first against these reserves, then against the profit and loss [P&L].) Wording in the notes to the accounts indicated that in future upward property revaluations would be recognized as income.

Other Nonoperating Items Used to Boost Income***Boosting Income through Bank-Owned Life Insurance***

Similar to pension income being passed through the earning statement, bank-owned life insurance (BOLI) is life insurance taken out by a bank on its key officers. BOLI may result in tax-free boosts to income that is nonoperating in nature and similar in quality to pension income.

Boosts to Income from Pension Gains

A related technique involves the diversion of gains on employee pension funds into bank profits. Applying this method, for example, overly optimistic return

assumptions are made to boost “pension income” to levels in excess of plan requirements. This pension income is effectively nonoperating in character, and should be classified as such by bank analysts. Note that demographics should, if anything, increasingly result in actuarial losses in pension funds.

BENEFITING FROM “EXCESSIVE” PENSION GAINS

In funding employee pension plans, companies make certain assumptions concerning returns on pension assets and administrative expenses. Items considered include:

- Service cost
- Interest cost
- Amortization of prior service costs
- Amortization of pension gains or losses
- Estimated return on pension assets
- Pension expense or income

These assumptions influence computed pension income. For example, in 2009, lackluster returns in the capital markets for one unnamed bank increased its return assumption to 11.0 percent from 9 percent, resulting in a 34 percent increase in pension income from \$46 million to \$62 million, more than doubling the contribution of pension income to pretax income (1.05 to 2.3 percent). Discretion in making certain assumptions allows firms to set these assumptions at levels that allow the creation of “excessive” gains. Management may seek to channel such “excess” pension income, which is typically unrealized, into the bank’s P&L account, thereby superficially improving results.

Aggressive Income Recognition

Rather than use one-off techniques, management might use aggressive accounting techniques to boost income or reduce expenses. There, of course, may be overlap, since very often one-off techniques involve aggressive accounting treatment.

Reclassification of Debt Securities to Boost Income

How gains and losses are reported on a bank’s securities holding affords the possibility of bumping up the institution’s bottom line. Mere reclassification of assets can boost income. This can be done by reclassifying certain securities holdings and does not even require that the bank actually realize the gain through the sale or transfer of the assets. Recent regulations in some countries now make it more difficult for banks to do so.

CHERRY PICKING ASSETS FOR RECLASSIFICATION

A bank transferred a substantial portion of its debt securities portfolio from the held-to-maturity category to the category of nontrading. Since held-to-maturity securities were valued at historical cost, but nontrading securities were valued at fair value, the mere transfer resulted in an unrealized gain of approximately US\$120 million, which boosted net income for the period. Note that booking mark-to-market gains or losses on trading securities, whether realized or not, to the income statement is not uncommon practice worldwide, although categories of securities do vary from country to country. Such trading gains, however, should not be confused with recurrent income, and any mass reclassification to take advantage of unrealized gains should be viewed with suspicion.

Use of Aggressive Gain-on-Sale Accounting to Boost Revenue

Banks have followed a technique pioneered by finance companies, recording the gain on the sale of assets (e.g., mortgage loans) in connection with a securitization. The effect is to enable the bank to record as income today revenues it will not in actuality receive until months and years in the future. In addition, depending upon disclosure rules, the method of calculating the gain-on-sale is not always transparent, enabling the bank to boost the numbers by applying overly optimistic assumptions.

Creative Expense Classification***Masking Loan-Loss Provisions***

It is fairly rare that a bank attempts to avoid putting loan-loss provisions through the P&L, but it does happen. An example is drawn from the FY99 financial statements of an unnamed Canadian bank. You can see that the note refers to C\$550 million in “provision for credit losses charge to retained earnings.” To its credit, the bank offered a seemingly plausible explanation for this unusual accounting treatment, implying that it was at the behest of regulators, and it did acknowledge that the treatment did not conform to GAAP. An adapted excerpt from the notes to the accounts is shown in Exhibit 6.33.

Creative Use of Merger-Related Charges

Banks that have grown through acquisitions frequently have used one-time write-offs at the time of purchase to avoid passing expenses, ostensibly associated with the cost of the merger, through the P&L. In 1997, for example, at least one U.S. bank took a one-time charge in advance against acquisition-related charges. This enabled the bank to reduce goodwill and associated amortization costs, while also avoiding recognizing the expenses resulting from integration in the P&L, thereby making results look better than they otherwise would have. Sometimes, temporarily worse results caused by one-off charges taken at the time of acquisition actually are sought to make postmerger results look better.

<i>As of October 31 (\$ millions)</i>	Specific provisions	Country risk provision	General provision	1999	1998	1997
Balance, beginning of year	\$ 1,270	\$ 571	\$ 600	\$ 2,441	\$ 2,078	\$ 2,327
Newly acquired subsidiaries— balance at date of acquisition	—	—	—	—	160	138
Write-offs	(653)	—	—	(653)	(565)	(587)
Recoveries	133	—	—	133	57	70
Provision for credit losses charged to income	485	—	150	635	595	35*
Provision for credit losses charged to retained earnings	—	—	550	550	—	—
Other, including foreign currency adjustment	1	(26)	—	(25)	116	95
Balance, end of year	\$ 1,236	\$ 545	\$ 1,300	\$ 3,081	\$ 2,441	\$ 2,078

EXHIBIT 6.33 Masking Loan Loss Provisions

Rule changes concerning goodwill now make the use of many such techniques—particularly those using so-called pooling accounting—infeasible in many markets. Mergers create fertile ground for accounting manipulation. The use of purchase accounting has not eliminated all such techniques, which include dampening the earnings of the acquired company during the premerger period to display better results on a consolidated basis. Another technique is to write down assets at one time to create reserves that can be later used to top up income. In a banking context, one approach might be to set aside large loan-loss provisions before an acquisition or a disposal is to depress current period income and to provide a potential boost to future period income, or to smooth earnings as discussed in the next section.

Smoothing Earnings and Improving Future Results

Recognizing that both equity and credit analysts dislike erratic earnings results, which imply a higher-risk business profile, managers may attempt to time the recognition of certain expense and income items to smooth earnings. Earnings smoothing means to alter the timing of recognition that might otherwise show more erratic, and apparently volatile earnings results, so as to give the impression that the enterprise is growing at a steady but robust pace. Sometimes, a boost may be required, while at other times results must temporarily be muted, preferably in such a way that the treatment can be reversed in a future reporting period.

A related tactic is to save up bad news so that it occurs in a single reporting period. Although it will be evident that the company has experienced tough times, by recognizing unusually high expenses or unusual low revenue generation all at once, subsequent periods may very well look much better than they otherwise would as the organization appears to climb back to its normal performance profile from a low base.

PRESCIENT OPINION FROM A RATING AGENCY

The bank's profitability is overstated in its accounts. The bank has taken provisions directly from retained earnings to loan-loss reserves on the balance sheet. . . . The amounts transferred equated to 46 percent and 44 percent of the after tax profits for 1994 and 1993, respectively. . . . In itself the transfer is not a major concern. . . . The concern is that this may reflect the bank's management style and that other parts of the accounts are not a true reflection of its position.

—Thomson BankWatch report on Siam City Bank,
September 14, 1995

One might correctly argue that smoothing earnings is of limited use to a bank that suffers from chronically low profitability, and that a bank cannot hide losses or weak profits forever. Yet, banks in some countries—very often state-owned banks—do hide losses forever or at least for quite a long time. They simply keep restructuring bad loans.

Smoothing Earnings through Loan-Loss Provisioning

Bank managers often attempt to smooth earnings through the loan-loss reserve account. By deducting higher-than-needed loan-loss provisions during good times when profits are above normal, loan-loss reserves may be built up to be returned to operating income later when they are ostensibly no longer needed.

Note that this strategy differs from the arguably prudent approach of building up loan-loss reserves during the expansionary phase of the credit cycle to tide the bank over through the expected deterioration in asset quality in the contractionary phase. If management is merely employing a precautionary approach to the loan-loss reserves, they will not ordinarily be returned to operating income during lean times. It is the excessive buildup of reserves “justifying” their later reduction and conversion back into income that distinguishes earnings management from prudent provisioning. Effectively, the bank is using the cover of loan-loss provisioning to shift current revenue to a later period to be tapped later to smooth reported earnings.

Smoothing Earnings through Merger-Related Items

Merger reserves mentioned above can also be used to smooth earnings. For example in 4Q99, Chase obtained a boost to reported earnings from the reversal of previously established restructuring reserves. It reversed \$127 million of restructuring reserves that were established in March 1998 as part of a \$510 million restructuring plan. Another method used in the merger context is *push-down accounting* by which the acquired company recognizes deal costs for its own acquisition, rather than the acquirer. This practice may have the effect of understating the company's stand-alone performance in the period of the acquisition, while for the acquirer, deal costs may be a recurring operating expense that is avoided through this technique.

NOTES

1. Walter Altherr, "Japanese Banks: False Dawn," Jardine Fleming Securities (Asia) Ltd. (November 1998).
2. Although it is true that not all deposit-taking institutions have mainly commercial aims, the vast majority do. A fair number of banks, it should be noted, are not solely profit-oriented. In some economic systems, government policies direct bank credit to chosen sectors, such as agriculture or particular industries. But even in the case of banks that play such a policy role and rely upon government support, the avoidance of excessive losses and the production of at least modest profits are desirable. Although external support provided by a government may underpin such an institution's creditworthiness, a reasonable level of profitability is always a positive factor in assessing such an institution's credit risk profile.
3. It is apparent that the tools used to measure risk-adjusted profitability have some relevance to the evaluation of earnings quality.
4. Although there are limitations on net income and other revenue-based metrics as indicators of an enterprise's ability to service its obligations, it is nevertheless generally true that the income statement remains the best tool we have to measure bank performance.
5. In reality, it is not always the case that higher returns are achieved only at the cost of higher risk. *Ceteris paribus*, this proposition is undoubtedly true, but in practice all other things are not equal, even though this may not be evident from outside observation. Superior management and business strategies may account for an institution's achieving higher profitability without additional risk.
6. Liquid assets such as interbank deposits or government securities generally earn a modest return, in line with their perceived risk, as compared with bank lending or less highly rated debt securities.
7. It is usually not necessary to understand a bank's limitations and strengths in every single respect, or even as well as one of its senior managers might. Although outside analysts may not possess all the information available to a senior manager, they often have the benefit of perspective, and over time can develop a sixth sense about when something does not quite ring true. This is an advantage that senior managers may not as readily enjoy from their limited vantage point within an institution.
8. Even on the reported income statement, there may be a line item for net income, or its equivalent, as well as net income attributable to shareholders.
9. In Chapter 4 we defined a bank's core earnings as its preprovision income (PPI).
10. Preferred dividends (preference dividends) are deducted to obtain that income attributable to common shareholders. Note that depending upon the accounting regime, preferred dividends might not be deducted even to obtain net income.
11. A bank or other enterprise is obliged to pay a fixed dividend to preferred shareholders absent unusual circumstances: for example, making a loss. Admittedly, preferred dividends represent a long-term funding expense, but nevertheless are in essence economically equivalent to interest expense.
12. Although it is unlikely that all line items will change at the same rate, with regard to profit items one is generally looking for moderate rates of growth. Unusually high or low rates of growth should draw our attention, and negative items, that is, losses, though comparatively infrequent in respect of net income, obviously raise a red flag.
13. Earlier, it was noted that common-size ratios and fundamental ratios overlap and differ more in purpose than in structure. A number of critical fundamental ratios are also common-size ratios, but not all common-size ratios are fundamental ratios.
14. Insofar as the NIM differs from operating income to average assets, it can be inferred how profitable the bank's non-interest-earning businesses are compared to its

non-interest-earning business. To illustrate (assuming no nonoperating items and no “other” items), if total average assets were 100, average earning assets 80, net interest income 4, noninterest income 2, pretax profit of 2.5, and net income 1.5, the bank’s net interest margin would be 5 percent ($4/80$), the ratio of noninterest income to nonearning assets 10 percent ($2/20$), and operating income to assets 6 percent ($6/100$). If return on assets is 1.5 percent, the difference over 4.5 percent might hypothetically be broken down as follows: attributable to tax, 1 percent ($1/100$); loan loss provisioning, 1 percent ($1/100$); and overheads, 2.5 percent ($2.5/100$). Note that the NIM is almost invariably greater than a bank’s ROA since NIM does not take account of operating expenses, loan-loss provisions, or taxation. A conceivable exception is possible when noninterest income is much greater than noninterest expenses, but the difference would have to be of such a magnitude as to make such a circumstance, at least for commercial banks, almost impossible in practice.

15. In other words, unless specified, ROE is used to mean return on average equity (ROAE), and return on assets (ROA) is used to mean return on average assets (ROAA), and so on.
16. Economic capital will be discussed in detail in Chapter 9.
17. In actuality, bank analysts normally use a similar ratio to ROA called *return on average assets*, sometimes abbreviated ROAA, but often used as shorthand for ROA, and similarly *return on average equity* (ROAE), but still commonly referred to as ROE. The structure of this ratio is exactly the same, except that the denominator is composed of the average of the current year’s assets and the previous year’s assets.
18. Weighing assets for risk will be discussed in detail in Chapter 9.
19. By measuring net income against risk-weighted assets rather than total nominal assets, RORWA takes into account the level of risk incurred in creating profits.
20. Net income, as ordinarily reported and published by banks, is simply the figure to which reference is most commonly made. It may be replaced, however, with measures of earnings drawn from the income statement to convey information more useful for certain purposes.
21. Recall that in Chapter 4 it was shown that there are a number of ways to define earnings and more than one “bottom line” can be drawn from the balance sheet.
22. Here the impact of nonoperating items that have affected net income might also be considered, and either pretax profit or net operating profit after provisions (NOPAP) employed as a critical measure of core earnings capacity, taking into account credit costs.
23. If the aim is to measure a bank’s strength to absorb credit costs, NOPAP, which has already deducted such credit costs, would not be the best measure. Instead, preprovision income (PPI) is a more appropriate numerator since it represents core earnings after operating expenses that can be tapped to rebuild the balance sheet.
24. As defined, REP is essentially equivalent to ROA where net income less loan-loss provisions is substituted for net income.
25. This stands for preprovision profit minus preferred dividends minus consumer net charge-offs to Bank of International Settlements risk-weighted assets plus securitized credit card assets.
26. This is an adjusted form of what we referred to in Chapter 4 as operating income.
27. Recall that loan-loss reserves are the bank’s estimate of its credit losses. If a bank overestimates its credit costs, the initial loan-loss provisions charged in the income statement may have exaggerated the bank’s actual credit costs. Likewise, if the bank underestimates its credit costs, the initial loan-loss provisions charged to the income statement may have been insufficient to reflect such credit costs. The best measure, therefore, is what the actual credit costs are for the fiscal period being examined. As a rule, net charge-offs are the best measure in regards to consumer lending. For reasons already mentioned, it may not be the best for commercial lending. So if we view PPI as a measure of the bank’s ability to earn its way out of problems, it is appropriate to first deduct consumer net

charge-offs to account for current credit costs (as well as preferred dividends), and thereby showed the amount now available for future balance-sheet rebuilding. It should be noted that in some countries, banks face huge hurdles in actually writing off any loans. In these markets, deducting net charge-offs would not be recommended.

28. The NIM is similar to the net interest spread, which is discussed subsequently.
29. A conservative portfolio is one in which most of the interest-earning assets in the form of loans to blue-chip—that is, highly rated—corporations or in government securities (or other liquid assets).
30. Note that this depends on properly recording interest income on NPLs as interest in suspense. In some less-developed markets, interest income may be accrued through the income statement for a considerable period despite being uncollectable.
31. The reason for any divergence will usually be because the bank is able to fund its earning assets partly through liabilities not counted as interest-bearing liabilities or partly through capital. Therefore, a strongly capitalized bank's NIM will tend to exceed the NIS. An alternative method of calculation uses average earning assets as the denominator for both ratios, which, by disregarding equity and non-interest-bearing liabilities, results in the NIS (being defined as average funding cost as a proportion of average earning assets minus average gross yield on interest-earning assets) being equal to the NIM.
32. The institutions on the list are the same as those appearing in Exhibit 6.5.
33. The business cycle is described in the next chapter.
34. More particularly, under central bank guidance, short-term interest rates are kept low while long-term interest rates remain elevated, enabling banks to profit by borrowing short and lending long. This can occur by the bank investing in long-term securities supported by short-term funding. Changes in interest rates can be effected by a central bank through a variety of means including changing the overnight rate at which the central bank will lend funds to commercial banks. Another method is to increase or decrease the percentage of funds that the bank must keep on reserve with the central bank. Policy measures of this sort are usually used to respond to cyclical economic changes, although they may be maintained for extended periods.
35. Note that asset quality as well as other factors may cause a bank to take a higher or lower level of provisions. This will have a direct impact on net profit and profitability, although it will not directly affect net interest income.
36. In this case, higher overheads may possibly be mitigated by lower funding costs.
37. Therefore, even if the bank's asset quality is good and the current level of loan-loss provisioning ample, it nevertheless provisions more than necessary so that it can reverse provisions the next year to smooth out earnings and make its shares more attractive to investors.
38. This attribute may be evaluated by measuring the proportion of total assets that are in the form of risk assets (earning assets).
39. Recall that high levels of nonperforming loans will adversely affect not only the bottom line (through loan-loss provisioning), but also net interest income (through nonaccrual of interest income).

Asset Quality

Evaluating the quality of a bank's assets is both the most important—and normally the most difficult—facet of bank analysis.

—Moody's Investors Services, *Global Credit Analysis*¹

Although the concept of asset quality is relevant to nonfinancial firms, particularly with regard to the valuation of inventory, it comes into its own in the realm of bank analysis. Because the vast bulk of a bank's assets are financial assets backed by a comparatively small proportion of capital, any substantial discrepancy between the value of those assets at the time of acquisition and their reported value becomes a significant analytical concern. As hinted earlier, an increasing proportion of risk assets or risk transactions nowadays affect what is generally termed as *asset quality* without being *earning assets* such as plain loans or securities.

Not all of a bank's customers will pay back the funds they have borrowed. Some will make repayments for a period of time and then default on the full payment of *interest* and *principal*. In other words, some loans that a bank makes will become *nonperforming*. Indeed, that a portion of a bank's loans will become *nonperforming loans*, or *NPLs*, is practically certain and an inherent risk and cost of banking.² Notwithstanding that earnings from non-interest-income-producing sources is becoming increasingly important to banks in many countries, in nearly all markets the majority of deposit-taking institutions still generate most of their operating revenue from advancing funds—that is, on their (interest) earning assets.³ Even in the United States, where disintermediation is further along than in most countries more than half of *operating income* will, on average, be in the form of *net interest income*. It is fair to say that problem loans will continue to remain a matter of concern for the vast majority of banks—and that *asset quality* will continue as a fundamental criterion of bank credit risk—for the foreseeable future.

This chapter is appropriately titled “Asset Quality” rather than “Loan Quality.” In the past 10 to 15 years, commercial banks—or whatever denomination describes retail and corporate banking as opposed, for example, to investment banking—have discovered the thrills of playing the markets. Ostensibly for the purpose of reducing or smoothing their risk profile through risk management techniques, many commercial banks have started taking proprietary market positions with the view of generating additional profits. Talking restrictively about loan quality no longer captures the whole spectrum of a bank's exposure to a string of instruments ranging from foreign exchange positions to options to financial derivatives, the list of which constantly expands.

In recent years, the increase in off-balance-sheet items in the accounts of even the most traditional commercial banks has taken away the notion that the analysis of asset quality could be confined to that of loan and interest-earning nonloan assets. Quite a number of off-balance-sheet entries do create downside—and in fact upside—risks that affect the creditworthiness of banks in very much the same way.

Asset quality beyond that of a bank's loan portfolio is usually assessed on parameters and market—or model—values that are not specific to that bank. By contrast, the quality of a loan portfolio remains very much dictated by the creditworthiness of the borrowers, the value of any associated loan collateral, and generally the terms of the loans. For that reason, considerable attention ought to be given to a bank's management of its loan portfolio, which occupies the bulk of this chapter.

As with all aspects of bank credit analysis, the asset quality of a bank must be measured against the institution's peers and its own past performance. Similarly, its evaluation can be divided into two parts. The first and most obvious is ratio analysis. The second looks at the less quantifiable aspects of a bank's financial assets, taking account of softer criteria such as the state of the credit cycle and the economic climate. The following questions are examples of the queries employed in this second part.

- Has the economy in the subject country been in an expansion phase for several years?
- Have banks on average been expanding their lending rapidly?
- Has total bank credit to GDP been swiftly rising?

An affirmative answer to these questions may make the analyst wary about bank asset quality. Correspondingly, if the banking industry in a particular market appears to be relatively healthy in terms of capital, liquidity, and generally low NPL ratios, and the economy is entering a recovery phase, concerns about expansion in bank credit will be lessened. At the micro level, the analyst also has to evaluate the extent to which any impairment of the bank's loan book is moderated by the bank's ability to generate profits and, more importantly, the degree to which loan losses are cushioned by loan-loss reserves and equity.

A NOTE ON TRANSPARENCY

One would think that data would be fairly easy to obtain, and in most developed markets that would be a correct presumption. In emerging markets, however, and even in some well-emerged markets, this is not always the case. To be sure, figures for total loans, loan-loss provisioning and loan-loss reserves are almost always readily available. It is data related to the composition of the loan portfolio itself, namely nonperforming and problem assets, that is more difficult to come by. Although the situation is much better than it was a decade ago, the analyst should be prepared for the occasional opaque financial statement where the true volume of nonperforming loans, to take the most conspicuous example, seems nowhere to be found.

Several terms are used to refer to the volume of the loan book. Among them are *total loans*, *gross loans*, and *net loans*. Sometimes they are used almost synonymously, although conventionally they have more specific meanings.

NOTE ON TERMINOLOGY: SIZE OF THE LOAN BOOK

- *Gross loans* refer to the loan principal plus unearned (interest) income.
- *Total loans* refer to the face value of the loan, that is, loan principal net of interest income.
- *Net loans* refer to total loans minus loan-loss provisions.⁴

ASSET QUALITY AND NPLS: AN INTRODUCTION

Asset quality refers to the *credit quality*⁵ of a bank's interest-earning *portfolio* and means the degree to which a bank's *earning assets*—its *financial assets* such as *loans* and *bond holdings*—are performing and are likely to continue to perform in accordance with their terms. In other words, asset quality denotes the level of *credit risk* associated with a bank's financial assets. Those assets that have become *non-performing*—where borrowers or issuers have ceased making interest payments—are no longer generating the return expected. These assets can be said to have become *impaired*, the impairment in value being commensurate with the probability of failure to produce the anticipated return. Ultimately, if payment is not forthcoming within a reasonable period of time, such assets must be construed as having little real value and be removed from the *balance sheet*, or written off,⁶ thereby reducing equity capital.

The Importance of Asset Quality

Asset quality is an extremely important component of a bank's credit profile. Some would argue that it is the most important, since poor asset quality has been shown to be the root cause of most bank failures, although the apparent cause may be insufficient liquidity or capital.

Good Asset Quality Allows a Bank to Prosper and Grow

A bank with satisfactory asset quality will ordinarily be able to maintain sufficient profits to generate the necessary capital for expansion of its franchise, while maintaining compliance with regulatory norms. However, as a bank's asset quality decreases, credit costs rise, and the institution's ability to generate capital internally (or attract external capital) is reduced.

More generally, a bank that exhibits good asset quality is probably well managed, and it is likely to exhibit safe levels of all the other quantitative ratios.

Robust asset quality is the preeminent indicator of a healthy bank. When loans and other financial assets produce interest income in accordance with the agreed terms, the anticipated return is generated, and the institution will ordinarily have sufficient income and cash to meet the immediate needs of depositors and creditors, maintain sufficient capital, and generate a profit for shareholders. In contrast, poor asset quality is probably the greatest fundamental cause of bank failure.⁷ Elevated levels of nonperforming assets result in the destruction of capital by diminishing the value of loans and securities relative to the liabilities that have funded them. Since

banks operate on comparatively thin margins and are highly leveraged, the impact of a relatively small volume of NPLs can entirely consume a bank's capital, rendering it insolvent.

Poor Asset Quality Erodes Profits and Stymies Growth

Loans that become nonperforming will have to have loan-loss reserves set aside against them, and may ultimately have to be written off. Provisioning against loan-loss reserves cuts into profits, impairing the bank's ability to create new capital, while writing off a loan directly reduces existing capital. Although banks are in the business of taking risks and expect some losses—taking account of these in pricing assets—when credit costs are too high, the institution's ability to meet its financial obligations may ultimately be endangered.

In this context, it would of course be relevant to remember that expected losses and unexpected losses are telling different stories.

Deteriorating asset quality can not only result in *insolvency*, it can also cause revenues to decline, credit costs to rise, and, in serious cases, liquidity to evaporate. *Illiquidity*—the inability to fulfill financial obligations as they arise—is the proximate cause of the majority of historical bank failures, and a bank's greatest worry. Liquidity problems generally—but not always—have as their root cause degraded asset quality, which destroys capital, drains cash, saps creditor and depositor confidence in the solvency of the institution, and thereby narrows the cushion of liquid reserves. A loss of confidence inevitably means a loss of funding and, absent support from the government or a highly creditworthy parent company, is in many circumstances lethal, resulting in a bank being forced to shut its doors.

Asset Quality Control as Effective Credit Risk Management

More broadly, the term *asset quality* also encompasses an assessment of management's capacity, in view of the risk control policies and systems it has put in place, to maintain credit risk exposure—and this should of course also apply to market risk—within a prudent range. This includes both maintaining optimal levels of asset quality in the first sense of the definition as well as other aspects of credit risk management.

Specifically, a bank's credit policy should work to minimize the following signals of a bank collapse waiting to happen, as well as less egregious problems:

- Excessive related-party lending
- Poorly documented loans that weaken the bank's rights as a creditor
- Excessive concentrations of loans to problem industries or regions experiencing economic decline

In addition to its impact on other financial criteria, a bank's relative asset quality also speaks to the competence of management in controlling the credit risk associated with a bank's financial assets. As observed, when a bank is poorly managed, it is likely that liquidity and solvency problems will occur. These problems in turn will ultimately cause a deterioration in the institution's overall creditworthiness vis-à-vis depositors and creditors. In other words, as conceptually illustrated in Exhibit 7.1, poor credit quality in a bank's loan book—which is merely one element of the CAMEL framework—may ultimately manifest in the weakening of the bank's credit profile.

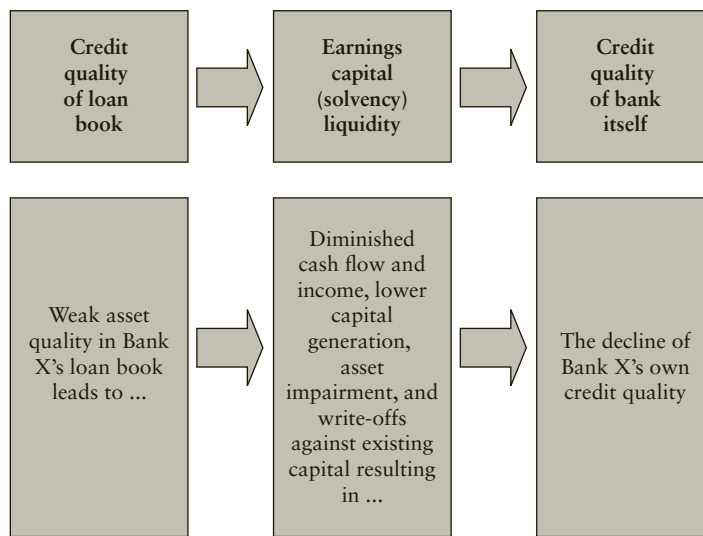


EXHIBIT 7.1 The Knock-On Effect of Asset Quality Problems

Hence, the ability to maintain prudent standards and effectively control credit risk is one of the most fundamental responsibilities of bank managers, and an inability to effectively perform this function is both damning in itself and implies deficiencies by management in other areas. Accordingly, because it has a direct relationship to *solvency, income, cash flow, and liquidity*, as well as to management's competence, a bank's asset quality is arguably the most important criteria in evaluating its financial condition.

The Measurement of Asset Quality

Asset quality is also one of the most difficult aspects of a bank's financial condition to measure. This is especially the case during periods of economic prosperity when even the dodgiest borrowers can make their loan payments without difficulty. In some markets, this difficulty is exacerbated by deficiencies in disclosure and significant differences in definitions and terminology. In addition, such definitional discrepancies and discretion in classifying problem loans are vulnerable to abuse. Empirical evidence shows what bank analysts have long known: that some financial institutions "use loan loss accounting to manage earnings and capital."⁸

Furthermore, aggressive or improper accounting for problem assets may very well substantially overstate income and solvency to manage the expectations of investors. As part of the process of bank credit analysis, it is therefore vital to understand how the reporting of loan losses affects bank profits and solvency, and to consider what a bank's loan-loss accounting has to say about the integrity of management.

Nonperforming Loans (NPLs)

There is no "universal standard" for determining when a loan becomes nonperforming. . . .

—Ernst & Young⁹

So long as a borrower continues to repay the principal and interest on a loan in a timely manner, it is termed a *normal* or *performing* loan. If the loan is not paid when due, it is referred to *overdue* or as *past-due*. After a certain period of time, most often 90 days, an *overdue loan* (*past-due loan*) may be deemed *nonaccruing* or *nonperforming*, subject to the definitions utilized by the bank or imposed by the regulatory authority that supervises it.¹⁰

Impaired Assets: Definitions and Terminology

Definitions of what constitutes a nonperforming loan (NPL) vary considerably throughout the world. The range of definitions is particularly broad in emerging markets. In addition, the governing criteria are in a state of flux as regulatory authorities seek to improve applicable standards. Lastly, in many jurisdictions, banks have considerable discretion in determining whether a loan is deemed to be nonperforming or not. In other words, both regulators and individual bank managements may adopt less or more stringent policies in designating past-due loans as nonperforming loans. Accordingly, what is determined to be a nonperforming loan often differs from bank to bank, and especially from country to country.¹¹ What is labeled a nonperforming loan in Country X at Bank J may be considered a performing loan in Country Y. Likewise, within Country X what is interpreted to be a nonperforming loan by Bank K may vary from the interpretation used at Bank L. To complicate matters further, nonperforming loans are commonly grouped according to specified characteristics. As with the general definition of a nonperforming loan, the eligibility criteria and labels applied to each of these various classifications of so-called problem loans (or problematic loans) also diverge considerably from one jurisdiction to another. The consequences of being assigned to a particular category may also differ.

As—or if and when—countries adopt recommendations under Basel II and Basel III, the definition of a nonperforming loan of asset may become more consistent across markets. It will remain that, accounting rules—and more importantly regulatory scrutiny—not being the same in every jurisdiction, the analyst will never be able blindly to rely on banks reporting their asset quality with reliable accuracy.

There is no universally accepted definition of a nonperforming loan that is applicable in all markets. Nevertheless, it is true that as a practical matter, a loan in which interest (or principal) is more than 90 days overdue is widely regarded by bank analysts as constituting an NPL. It is sometimes called the international standard, and indeed there does seem to be some convergence toward this measure globally. Given that the following definition is in wide use among bank analysts and for lack of a better measure, it is also adopted in this book.

NPL: A WIDELY ACCEPTED ANALYTICAL DEFINITION

An NPL (nonperforming loan) is a loan in which interest (or principal) is more than 90 days overdue. Depending on how it is defined, it may exclude special mention loans that have not met the 90-day aging criterion.

NPLs are themselves a subset of *problem loans*, or *classified loans*.¹² Loans that are problem loans but which remain performing (i.e., not past due) are generally termed *special mention* loans and are discussed further below. So-called *restructured loans* may fall under the rubric of *problem loans* although they are just as apt to be deemed *nonperforming assets* or *problem assets* as opposed to *nonperforming loans*. In parallel with the distinction made between gross loans and net loans, *gross NPLs* may also be distinguished from *net NPLs*.

- *Gross NPLs* is the broadest NPL figure, without deductions, although it may not include restructured loans, foreclosed assets, and technically performing special mention loans.
- *Net NPLs* means NPLs less applicable loan-loss provisions, or collectively total loan-loss reserves. Some banks may also deduct collateral to diminish nominal (i.e., gross NPLs) even farther.

NPAs and Foreclosed Assets

Another term, *nonperforming assets* or NPAs, is largely synonymous with problem loans, but also includes assets not strictly categorized as loans, as well as foreclosed assets obtained upon a borrower's default and restructured loans. In the United States, such foreclosed assets, which are almost always real estate collateral, are referred to as *other real estate owned*, or OREO. Perhaps the broadest term is *problem assets*, which includes NPAs and *special mention* loans.¹³

NOTE ON TERMINOLOGY: NPL AND RELATED TERMS

An **NPL (nonperforming loan)** is—generally—a loan in which interest (or principal) is more than 90 days overdue. Depending on how it is defined, it may exclude **special mention loans** that have not met the 90-day aging criterion.

A **classified loan** is one that is classified as other than **performing** or **normal**; that is, a classified loan is one that has been designated as **special mention**, **substandard**, **doubtful**, or **loss**.

Problem loans (problematic loans) and classified loans mean basically the same thing, unless the classification scheme is so lax that the analyst has deemed it necessary to adjust the classifications to better correspond with economic reality.

Gross NPLs is the more conservative NPL figure referring to nonperforming loans without deduction, for example, of loans that have been restructured. **Net NPLs** means nonperforming loans after deduction for defined items. It is the less conservative figure.

Nonperforming assets NPAs, include NPLs and other nonperforming assets (e.g., defaulted securities) together with restructured loans and foreclosed assets.

Problem assets include NPLs plus special mention loans.

Restructured loans are those where the bank and the borrower have negotiated or *worked out* new terms in order to enable the customer to resume

performance. Invariably, the terms of such loan workouts afford the borrower some concessions with regard to the timing, interest rate payable, or the total sum that must be repaid.

Rescheduled loans are similar to restructured loans but tend to connote loans with respect to which payment terms have been extended without any meaningful concessions on the part of the borrower. This is sometimes called **cosmetic restructuring** or **evergreening**.

Loan Classification

Loan classification systems have emerged because the binary division of all financial assets into those that are performing and those that are nonperforming is an inadequate measure of asset quality both for management and for financial reporting purposes. It is evident that there exist degrees of loan performance. Clearly, the fact that a loan is overdue does not mean that a bank will recover nothing from it. It is possible that the borrower may be able to make partial payments, or that the bank may be able to foreclose the loan and take possession of the collateral securing the obligation, which it can sell to mitigate its losses. In other words, as observed in earlier chapters, the bank's loss given default (LGD) with respect to the asset may fluctuate.¹⁴ Conversely, the fact that loan payments are still current is not necessarily predictive of continuing repayment in the future. The lender may be aware that owing to changing economic conditions, for instance, the borrower will very likely be unable to continue to make full interest payments. Loans that fall in this category, often called *special mention* loans remain technically performing but are identified as being at a significant risk of default. To estimate and take account of the probability and severity of loan losses for internal credit management purposes, regulators have promulgated, and the Basel Committee has suggested, classification systems with which banks must comply. These schemes provide benchmarks with which to grade loans in terms of the loss the bank is likely to suffer and to allocate loan-loss reserves to compensate for the relevant credit costs.

Types of Problem Loan Classification Schemes

Loan classification systems both define the various classes of problem loans and specify what actions a bank must take to reflect the credit costs and accompanying asset impairment associated with each class. These actions typically include the:

- *Cessation of accrual of interest income* on loans classified as nonperforming.
- *Deduction of loan-loss provisions* of a certain percentage of the outstanding amount of the loan corresponding to the loan classification assigned.
- *Write-off* (charge-off) of all or a portion of the loan against shareholders' equity.
- *Claw-back* of any net interest income previously accrued to account for its nonreceipt.

Loan classification systems may be based upon a *qualitative* standard, allowing a wide degree of discretion to the bank in classifying its loans, upon a *quantitative* standard, or upon a combination of the two.

Quantitative standards usually involve the use of aging criteria that classify problem loans on the basis of how long they have been nonperforming. Qualitative standards naturally allow a bank's management to use its own judgment concerning the probability and extent to which the loan will result in a loss to the bank.

Impact of Definitional Differences

Loan classification systems and their corresponding categories and definitions can be fairly complex. They also vary across and within markets. A diversity of terminology is also used to refer to nonperforming loans generally, as well as to specific subcategories of problem loans, which can create confusion as to what is meant by a particular label.¹⁵ Hence comparing the volume of nonperforming and classified loans, and the various associated asset quality ratios, is often fraught with difficulty. For the same reasons, it can be hazardous to take reported or headline figures, such as the average ratio of NPLs to total loans for banks in Country X, at face value.

In order to overcome the differences in definition and classification in place from one market to another and one institution to another, bank analysts frequently employ their own working definition of what constitutes a *nonperforming loan*. Usually, this definition is similar to best practices widely accepted in highly developed economies. The result is that reported NPLs often must be adjusted from that reported locally for analytical purposes. Consequently, the term *NPL* has both a general widely understood meaning and a more particular local technical meaning.

Classified Loan Categories

Classified loans are commonly grouped into a number of internal categories. Unless a bank groups its loans or assets in categories imposed by regulators, it would usually have four internal categories. These classifications, listed in order of increasing expected loan losses, are:

1. Special mention
2. Substandard
3. Doubtful
4. Loss loans

Each category encompasses loan assets deemed to have roughly comparable loss rates. Accordingly, from top to bottom, each attracts increasingly stringent provisioning requirements.

Special Mention Loans

The first of the four classifications is *special mention loans* defined in the box below. These are loans that are *not* more than 90 days past due, and therefore technically still performing (and outside the category of nonperforming loans), but that represent exposures to a borrower that has experienced deteriorating financial conditions. As a consequence, they are subject to a substantial risk of loss. This class of assets is the least worrisome of the larger universe of problematic loans, and not all jurisdictions recognize this category or require loan-loss provisioning in respect to it.

SPECIAL MENTION LOANS

Special mention loans are loans that are still performing, and therefore not technically NPLs, but with respect to which there is a material risk of loss. Special mention loans are, however, generally deemed to fall within the categories of classified loans, problem loans, and problem or nonperforming assets. They attract the lowest provisioning requirement of all classified loans (e.g., 5–10 percent).

Substandard Loans

Substandard loans are loans more than 90 days past due that display a well-defined weakness on the part of the borrower or in respect of collateral securing the loan, thereby giving rise to a substantial risk that full payment will not be forthcoming. This class of problematic loans is clearly within the nonperforming category. Substandard loans generally attract a loan-loss provisioning requirement, but the provisioning required is usually less than that associated with doubtful and loss loans.

SUBSTANDARD LOANS

Substandard loans are loans that are more than 90 days past due and in respect to which there is a substantial risk of loss. Substandard loans attract a moderate provisioning requirement (e.g., 25 percent).

Doubtful Loans and Loss Loans

Doubtful loans are loans past due more than 90 days, where full repayment is unlikely. Loss loans (i.e., *bad loans*)¹⁶ are loans due more than 90 days and considered to be uncollectible in full, notwithstanding the prospect of some partial recovery. Both doubtful loans and loss loans generally attract a loan-loss provisioning requirement equal to a large percentage of the nominal value of the asset, or in the case of loss loans 100 percent of the nominal value.

DOUBTFUL LOANS AND LOSS LOANS

Doubtful loans are loans that are more than 90 days past due and in respect to which collection in full is improbable. Doubtful loans attract a high provisioning requirement (e.g., 50 percent).

Loss loans are loans that are more than 90 days past due and deemed uncollectible. Loss loans usually attract a 100 percent provisioning requirement, and, depending on local rules, should be written off.

A basic classification system could use a combination of a single 90-day aging criterion—for example, any loan more than 90 days past due would be presumed to display substantial risk of loss and therefore would be classified as substandard at the very least—and a qualitative judgment concerning the risk of loss associated with a particular asset. Other classification systems may rely more heavily on bright-line classifications based on aging criteria.

Finally, observe that in most circumstances (depending upon the structure of the classification scheme and applicable local rules), a loan that is past due will be placed in the three categories in a sequential basis. That is, it will be first classified as *substandard*, then with the likelihood of recovery diminishing as *doubtful*, and lastly as a loss loan, as the probability of resumption of performance dims.¹⁷

Comparisons Are Difficult

What is a typical breakdown of a portfolio into the foregoing classifications? It varies, depending on the country and its culture, as well as on the type of loans most likely to be found in bank portfolios. Local rules have an enormous impact on the classification, its uses, and its misuses.

Since, for example, large syndicated loans are apt to be of a higher credit quality than the entire loan book of a single bank, banks with substantial syndicated loan portfolios usually look better. It certainly appears also that loss loans are written off more quickly in the United States and do not have an opportunity to build up to become a substantial portion of NPLs. In some countries, loans languish in a given category for years before moving on to another one.

In short, comparisons across borders, or even between different types of banks are difficult, if they have to be made on the strength of published figures along nonperforming loan classification.

Asset Quality and Credit Risk

As discussed earlier, asset quality of a commercial bank is essentially synonymous with credit quality. Both are concerned chiefly with credit risk, or rather with the ultimate impact of their credit risk on the profitability and capital adequacy of a bank. The difference is largely one of context. In bank credit analysis, a discussion of asset quality considers how the credit quality of borrowers, as well as counterparties and issuers, collectively affects the quality of the financial assets held by the institution being evaluated.¹⁸ This conceptual point is elaborated further in the section on the effects of classification that follows. In a parallel fashion, measurement of asset quality involves essentially the estimation of credit risk. Different scales, however, are customarily used in each case. Asset quality assessment is mainly concerned with the proportion of performing assets to nonperforming assets and the accompanying impact on the bank's financial condition. Bank credit assessment—the subject of this book as a whole—is concerned more with establishing probabilities of default and estimating loss severity with respect to an individual institution or a set of institutions under evaluation.

The concept of risk mitigation is to reduce the probability or consequences of a threat, in this case a threat to the quality of a bank's asset. If a bank manager succeeds in mitigating a specific threat, for example by covering a single bond exposure with the use of a credit default swap (CDS), it is plain that the accounts will record, at two different places, a couple of transactions, which, taken jointly, exhibit a low risk

to the creditworthiness of the bank, while each of them separately would point to a potential loss or a potential profit. Therefore it has become difficult to analyze each asset—or liability, or off-balance-sheet item for that matter—in isolation. The reader should keep that fact in mind at all times.

Effects of Classification

Loan classification operates at two levels: significance and impact.

Accounting Significance

On one level it is merely an arbitrary labeling exercise that divides financial assets according to identifiable characteristics that, it is hoped, bear some meaningful relationship to their expected loss. No matter how arbitrary, the operative loan classification may also trigger an accounting event that is likely to affect the bank's regulatory compliance obligations and nominal earnings, assets, and capital. Therefore classifying a loan as problematic will, among other things, directly affect the amount of provisioning a bank must allocate against its loan book. In turn, loan-loss provisioning (LLP) charges tend to erode reported net income and potentially diminish the value of the shareholders' stake in the institution.

Fair Value Accounting

Recent crises have rekindled the debate about fair value accounting. There is a major conceptual distinction to be made between likely loan losses flowing from known events and likely loan losses flowing from expected future events. The general approach has long been that loan-loss provisions should be put aside at the discretion of bank management, essentially on the basis of events affecting the borrower or the fate of the loan, rather than on the basis of some estimate of future loan losses.

The difference is substantial. Banks must take reasonable credit risks and charge interest accordingly, very much the way insurance companies charge a premium that rewards them for an estimate of future casualties over a portfolio of risks. Insurance companies amortize the premium over the risk period. Banks do not do the same thing. As long as a loan is performing, banks record interest income without setting aside that part of interest income—beyond funding and administrative costs—that might be needed if and when the loan ceases to perform.

Putting a fair value on a loan would basically reduce its accounting value today in exchange for a future stream of excess interest. Not doing so has the perverse result that banks record artificially higher profits in early periods in the life of a given loan, followed by artificially lower profits when the loan deteriorates. In peace times,¹⁹ the loan portfolio diversification and the law of large numbers would even out the positive and negative impacts. When war times²⁰ follow peace times, if loans were kept at historical value, banks would be confronted with loan losses at the very moment when the business environment is not conducive to putting aside loan-loss provisions.

The other side of the coin is that, leaving aside the administrative complexity of a loan fair value approach, management discretion would have a devastating impact. Without the fair value accounting of loans—that is, with only event-driven decisions about provisioning—bank owners and managers already have considerable leeway,

especially in emerging markets, in the management of earnings through the massaging of provisions.

The Headache of Accounting Standards

An additional issue creating headaches for analysts and bank accountants alike is the fact that each jurisdiction operates under its own set of accounting rules.

The International Accounting Standards (IAS) long were the norm in many countries. They were issued and suggested but not imposed under the umbrella of the International Accounting Standards Committee.

A new set of standards, the International Financial Reporting Standards (IFRS) have existed since 2001 under an independent body called the International Accounting Standards Board (IASB).

The United States has its own Financial Accounting Standards Board (FASB), which produces its own Generally Accepted Accounting Principles (GAAP), very much in the way other countries build their own GAAP with or without reference to external standards.

In the European Union (EU), standards are built from IAS and IFRS sources. The adoption of Basel II has meant a distinction between the need to cover unexpected losses with regulatory capital and the need to address incurred losses and expected losses with appropriate measurement techniques and the usual general or specific loan-loss provisions.

Within each jurisdiction, even where standards are clearly adopted, there are a number of variations around the way some items are to be accounted for either in the balance sheet or in the income statement, let alone around their impact on regulatory capital and liquidity requirements.

Exploring those often-changing regulations would be going well beyond the scope of this book. The bank credit analyst should remember that, whatever the set of rules under which a bank records entries related to the quality of its assets and the attending impact on its earnings—and, beyond, on its capital ratios—it is in general better to give limited credence to published figures. Granted, some banks in some countries are both well regulated and well managed, but, most of the time, as the saying goes, “what you see ain’t what you get.”

Economic Impact

No matter how arbitrary the classification scheme, all loans will generate a measurable after-the-fact loss ranging from between zero and 100 percent that the initial designation seeks to approximate in advance. Insofar as the classification system is consistent and bears some reasonable correspondence to economic reality, it will convey additional valuable information both to managers and investors about the bank’s risk profile.

Since, by nature, financial accounting merely represents an attempt, albeit often an inadequate one, to mirror economic reality, lax accounting requirements can only shield bank management temporarily from consequences of fiddling with its true asset quality position. In a best-case scenario, lenient classification standards—so-called regulatory forbearance—afford struggling banks a reprieve so that confidence can be maintained and financial health restored. Ultimately, the reality is that insofar as it is impaired, a nonperforming loan represents a destroyed asset. Its destruction obliterates, to a corresponding extent, the wealth of the bank’s shareholders, and this reality cannot be circumvented. For these reasons, the volume and

proportion of a bank's earning assets classified as nonperforming should be viewed both as an essential indicator of its financial condition, and, more skeptically, as a mere approximation of its true asset quality.

Needless to say, NPL levels are closely watched by investors and by equity and credit analysts. In the end, what is most important in terms of loan classification is not the label but the underlying characteristics of the loan book.

Transparency and Disclosure Issues

Lack of transparency, however, may make the task of estimating asset quality more difficult than it need be. Although bank managers generally recognize an obligation to reveal profits and capital, they may be reluctant to disclose in detail the volume and sources of the bank's dud loans unless required to do so by regulation. This reluctance arises from the institution's need to maintain public confidence in its soundness. For example, the Development Bank of Singapore expressed the sentiment quite diplomatically in its 1999 Annual Report:

The incidence of bad and doubtful loans is a particularly sensitive aspect of a bank's operations. If the losses are material, they can reduce the capital resources of the bank and affect its ability to grow and develop its business. If large losses are disclosed in the financial statements, it may lead to a loss of confidence in the bank's management and a reduction in its credit ratings. This will lead to an increase in the bank's cost of borrowing and make it more expensive to raise capital.

As the quote above makes clear, banks have rational reasons to limit disclosure of their true financial conditions especially with respect to their level of nonperforming loans when asset quality is in decline. Full disclosure can damage shareholder, investor, and depositor confidence, cause a downgrade in credit ratings and have a consequent adverse impact on the bank's business, especially with regard to the cost of funding. Moreover, as was discussed in the preceding paragraphs, definitions and classification systems for problem assets are subject to considerable divergence.

Why Is an Excessive Level of NPLs Dangerous?

The reasons for the importance of asset quality as a criterion of credit risk have been addressed above, but it may still be worth exploring in more depth why nonperforming loans or more generally nonperforming assets (NPAs) are so undesirable for banks. Simply put, loans are the assets out of which a bank produces the bulk of its profits. Because margins on bank loans are low—money is, after all, merely a commodity—and because banks are highly geared, the complete loss of a single bad loan can wipe out the profits generated from dozens of performing loans. More importantly, as alluded to earlier, a defaulted loan represents a demolished asset the destruction of which will ultimately have a direct impact on a bank's equity capital. When a significant portion of a bank's financial assets are wiped out, the effect is analogous to a manufacturing firm losing a factory or a significant portion of its inventory through fire or other calamity.

ASSET QUALITY VERSUS CREDIT QUALITY

While the terms *asset quality* and *credit quality* are often used synonymously, they have somewhat different connotations. *Asset quality* usually connotes the credit risk associated with an entire portfolio held by an entity. The term *credit quality*, however, tends to connote the creditworthiness of a particular entity being evaluated. The same principal attribute—credit risk—is being assessed, but it is approached in each case from a somewhat different perspective. Thus, with reference to a hypothetical Bank X, the term *asset quality* refers to the relative degree of credit risk and expected loss associated with the institution's asset portfolio. The term *credit quality*, however, implies the credit risk associated with a particular entity or investment. Credit quality is relevant to the twice-remote approach of looking at counterparties whose creditworthiness is relevant to the creditworthiness of the bank under scrutiny. In respect of Bank X, we could speak of its credit quality meaning the credit profile of the bank itself from the vantage point of outside counterparties, investors, and analysts. The same term might also be used by Bank X's management to refer to the credit risk associated with a particular financial asset held by Bank X. Thus, the credit quality of each asset within Bank X's portfolio has presumably already been evaluated by Bank X's credit analysts. The adequacy of this process itself forms part of overall evaluation of the credit quality of Bank X by an outside bank credit analyst. As part of that process, the credit analyst will examine Bank X's asset quality—including its loan review procedures, its earnings and profitability, its capital strength, and so on.

Compensating for the hole created in a bank's balance sheet when a material portion of its loan book goes bad—like a manufacturing facility destroyed by fire and without insurance—can place a heavy burden on an enterprise and, in severe cases, ultimately wipe out its capital. Undeniably, asset quality problems can trigger a vicious cycle that ultimately leads to bank failure unless support is forthcoming. The way this can happen is as follows.

Interest Income Stops Accruing and Provisioning Costs Rise

As loans are classified as nonperforming, the bank will be compelled to stop accruing interest on those assets. This means that gross interest income will fall, which will be accompanied by a corresponding decline in net interest income, since funding costs will remain unchanged. The bank will probably have to increase its loan-loss provisioning to compensate for the loans that will be written off and that will thereby reduce loan-loss reserves.

Net Income Falls

The increased provisioning costs will be charged against operating income and will fall through to the bottom line reducing net income attributable to shareholders.

Capital May Be Impaired

If charge-offs exceed existing loan-loss reserves and available profits from other sources, shareholders' equity will have to be written down. Since banks are required to maintain minimum levels of capital, it is probable that new funds will need to be found to bring the institution's capital back to regulatory thresholds. In addition, to bolster capital levels, it is not unlikely that the bank will need to reduce loan growth, or even shrink its loan portfolio.

Costs of Long-Term Finance Rise

However, as the market learns of the bank's problems, as it almost inevitably will, investors will be reluctant to inject new capital into the organization, since eroding asset quality is likely to herald lower profitability and reduced returns on investment.

Costs of Short-Term Finance Rise

For the same reason, depositors may withdraw funding, in whole or in part, or at least demand higher rates on deposits and other funds provided. In a worst case, a bank run might be triggered. Even if deposit insurance exists for retail depositors, commercial depositors may be loath to provide funding at market rates. The result will be higher funding costs, reduced liquidity, or both. Neither is likely to be viewed favorably by investors.

NPLS: DESTROYED ASSETS

Imagine, for instance, that an industrial company, Widgets, Ltd. purchases a machine, the Widgetomatic (an automated widget producer) at a cost of \$1,000,000 capable of producing a product, which based on reasonable estimates, can add \$75,000 to the firm's bottom line in a single year. The firm finances the cost of the machine by borrowing \$1 million from Green Bank at a 5 percent rate of interest, secured by the firm's other assets.

Now, suppose that this wonderful machine catches fire and is completely destroyed. Widgets, Ltd. has no insurance and no product liability claim or any other valid claim to obtain compensation of its losses.²¹ The asset, the Widgetomatic, has been irreparably burnt to a crisp, and will be unable to generate any revenue for the company. Instead, it has become a cost. Widgets, Ltd. must still repay the funds it borrowed to buy the machine, but is now obtaining no revenue whatsoever.

Loans, too, are assets that can be impaired or destroyed. A nonperforming loan is not unlike the incinerated Widgetomatic. Suppose that Green Bank unwisely lent \$1,000,000 to Company X, without undertaking proper due diligence. Company X's management agreed to pay an above-market rate of interest, that a key business manager at Green Bank, since out of a job, estimated would increase the institution's net income by \$75,000 a year for the duration of the loan. The loan, rather imprudently, was not secured.

As a result of financial problems, Company X collapsed one month after the loan was extended, having used the funds to pay performance bonuses to

its top executives. A more senior creditor has a claim on its paltry remaining assets. Under the laws in Country Z where both Green Bank and Company X are situated, Green Bank's claims are unsecured and rank *pari passu* with all other unsecured creditors. Given Company X's resources, there is no viable legal claim against the firm's assets. Nor is there any valid claim against the firm's owners or managers.

Green Bank is in an almost identical position to that of Widgets, Ltd. The company owned an apparently profitable asset that occupied the larger portion of a small factory, while the bank's supposed profitable asset was a mere five-page document with appropriate signatures affixed. Both assets are now essentially worthless and both enterprises have black holes on the asset side of their balance sheets. These destroyed, that is, impaired, assets must be written off, which under the accounting equation, will reduce shareholders' equity and correspondingly the capital of each enterprise by roughly \$1,000,000.

Margins Fall, thereby Increasing the Temptation to Engage in Riskier Business

With funding costs higher, the bank's margins on lending will be reduced. To maintain profits, the bank may seek to gain higher yields on higher-risk assets, such as loans to less creditworthy companies, or other more volatile business. Alternatively, the bank will suffer reduced earnings in the short term, exacerbating its predicament.

The Result Is Likely to Be Either a Higher Risk Profile and Ultimately Worsening Asset Quality, or Reduced Earnings

Both consequences will naturally increase the bank's overall *credit risk*, and make it more vulnerable to more loans becoming nonperforming.²² Alternatively, squeezed margins will reduce net interest income further, and the vicious cycle, illustrated in Exhibit 7.2, will continue unless and until the bank can clean up its balance sheet by writing off problem assets, rebuilding loan-loss reserves and capital, and improving credit policies.

To sum up, poor asset quality reflected in high levels of NPLs is likely to have the following adverse effects on a bank:

- Diminished earnings and profitability
- Reduced solvency and liquidity

Lower net profits and overall profitability are mainly the result of:

- Increased loan-loss provisioning charges to the income statement
- Higher funding costs and reduced margins
- Erosion of capital and consequent need to rebuild the balance sheet

Lower solvency and liquidity are respectively caused by:

- The *impairment of assets*, which in turn erodes capital
- *Reduced access to sources of funding* occasioned by the bank's higher risk profile

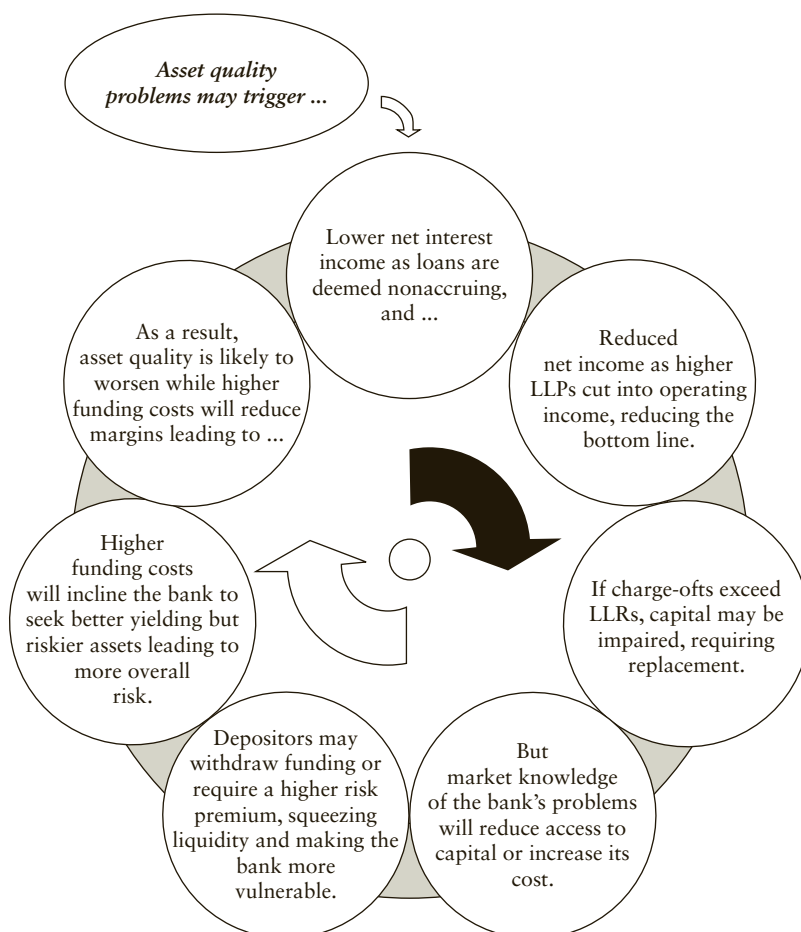


EXHIBIT 7.2 Asset Quality Problems Can Trigger a Vicious Cycle

If the bank is fortunate and favorable conditions prevail, it will be able to earn its way back to financial health. The process, however, may be slow and painful. If not, outside support may be required.²³

RESOLUTION OF NPLS

What happens to NPLs and why might they accumulate on the balance sheet and form a high proportion of all classified loans? Once a loan is classified as a nonperforming loan—that is, for example, as a substandard, doubtful, or loss loan—there are essentially four possible outcomes, succinctly described as write-off, foreclosure, work out, and bad bank.²⁴

Write-Off

A nonperforming loan can return to performing status, perhaps as a result of restructuring, or it can be written off. To write off a loan means to remove it from the

balance sheet.²⁵ When a loan is written off from the asset side of the balance sheet, an equivalent adjustment must be made from the liability side. Since funding obligations are not affected by loans gone bad, equity is reduced by the amount of the write-off. To rebuild equity, the bank must divert profit that might otherwise have been distributed to shareholders to retained earnings. Accounting rules may vary across jurisdictions, but usually the charge-offs and write-offs hit the balance sheet through the P&L. Sometimes banks are allowed to deduct those charge-offs and write-offs directly from equity so as avoid having to post losses.

Foreclosure

Another possibility is foreclosure. This is actually a variation on the first option, and includes the ultimate write-off of the loan, but also involves the recovery of any collateral used to secure the loan. A loan may be secured either with recourse or without recourse. *With recourse* means that the bank will be able to make further claims against the borrower to the extent that the funds realized from the collateral do not cover the borrower's extent obligation. *Without recourse* means that the outstanding debt is satisfied by the collateral, irrespective of whether its sale price will satisfy the amount of principal and interest outstanding. In addition, a creditor may be a senior secured creditor or a junior secured creditor. Senior creditors have first priority to the collateral and have the right to have their claims for repayment settled in full ahead of junior creditors.

Also, a distinction should be drawn in this connection between a borrower that has become insolvent and voluntarily or involuntarily (i.e., through the demand of creditors) entered bankruptcy proceedings and one that is nonbankrupt. Short of forcing a borrower into a bankruptcy, a bank may be able to recover collateral through summary judicial procedures, which allow a creditor to expeditiously take title to the collateral securing the loan. In a bankruptcy proceeding, recovery of claims tends to be a more drawn-out process. Such proceedings conventionally come under the authority of a special legal tribunal (e.g., a bankruptcy court), which may empower an administrator or trustee to settle claims.

Normally, as shown in Exhibit 7.3, a hierarchy of claims is followed whereby the most senior ranking creditors are paid first with remaining amounts realized from the sale of the debtor's assets flowing to each successive creditor level.²⁶ If assets are sufficient, the final remainder will come to rest in the hands of common shareholders. Hence, even where there is no collateral securing a loan and the bank therefore stands as an unsecured creditor, it can still make claims against a deadbeat or insolvent borrower, and very possibly may recover some funds owing to it.

Workout

Finally, a bank may negotiate with the borrower to *work out*, or restructure the loan.

Revised terms may be agreed that enable the borrower to make payment under more relaxed terms. Usually the bank will give up some benefit owing to it, while it is assured of some recovery without the cost of engaging in expensive legal procedures. Similarly, the borrower benefits from being able to remain in business and to avoid being forced into bankruptcy. The new agreement will represent a restructured loan.

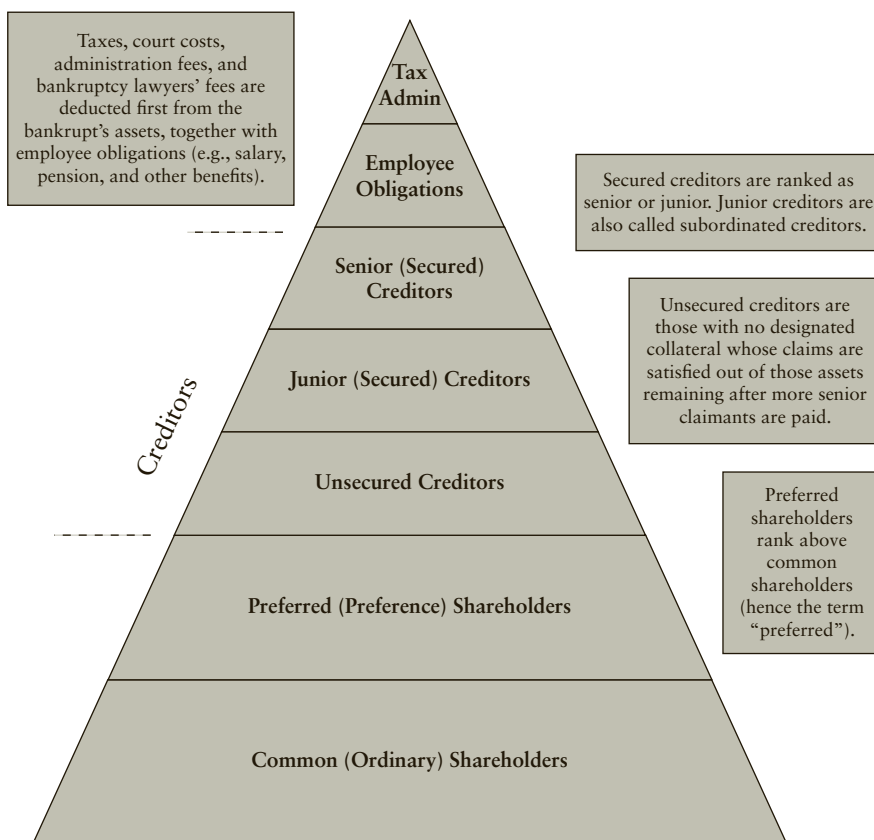


EXHIBIT 7.3 Hierarchy of Claims

Probably in most cases, loan restructuring is a *bona fide* effort on the part of the bank to preserve the benefits of the financial asset the loan represents, and to maintain its relationship with the customer.

But in jurisdictions where restructured loans are still²⁷ considered to be outside the definition of nonperforming loans and bank supervision is lax, banks may engage in hasty restructuring—in effect mere rescheduling—in an attempt to reduce nominal NPLs. So-called cosmetic restructuring (also called *evergreening*) may create a false impression, since the reality is likely to be that many of these loans will ultimately return to a nonperforming state. In the meantime, however, the bank's asset quality superficially appears better than indicators would show.

RESTRUCTURED LOANS

Restructuring means to adjust the terms of a loan so that it can functionally return to performing status, although formal treatment of restructured loans within classification schemes varies. The rationale for restructuring is that where a borrower's business is fundamentally sound, despite the customer experiencing temporary repayment problems (i.e., liquidity problems), it may

be in the bank's interest to attempt to put the borrower back on a sound footing by extending the term of the loan, providing some relief in respect of the interest rate, or otherwise relaxing the terms of the obligation. Restructuring should be based upon the assessment that the overall recovery to the bank through restructuring will exceed the value of recovery should it foreclose and write off the loan.

Depending upon regulatory and accounting guidelines, restructured loans may in certain cases be reclassified as performing, usually after they have continued to perform for some period of time. Some loans will nonetheless relapse to their previous state, again becoming nonperforming. The relapse rate therefore becomes a relevant indicator in evaluating asset quality of a bank with significant holdings of restructured loans. It should be emphasized that restructuring can be abused by banks seeking to make their asset quality appear better than it is. Where restructured loans escape classification as NPLs, restructuring may be used as a way to avoid high nominal NPL rates and write-offs against capital. This can be accomplished through so-called cosmetic restructuring or evergreening whereby loan terms are relaxed with regard to the viability of the borrower's business.

Bad Bank

In situations where the bank's exposure to nonperforming loans or generally to substantial asset losses makes it unlikely for it to survive, it is possible to segregate part of the weak assets and to sell them to a special-purpose company, usually called an asset resolution company or a *bad bank*.

The bad bank is funded either by the bank or banks whose weak assets the bad bank takes over, or it is funded with public money.

In Japan, the banking industry created the Cooperative Credit Purchasing Company, or CCPC, in 1992 to acquire from banks, with funding from the selling banks themselves, real estate held at collateral. The banks would only get back their funds upon successful sale of the assets, and only the actual proceeds. This allowed some of them to delay recognizing most of their losses for well over 15 years.

As another example, the UK government established the UK Asset Resolution in October 2010 for the sole purpose of holding weak assets such as those inherited from Northern Rock.

China, France, the United States, and other countries have made use of bad banks to take weak assets out of bank balance sheets.

ACCOUNTING FOR NPLS

Banks must establish an allowance for loan and lease losses because there is credit risk in their loan and lease portfolios. The allowance, which is a valuation reserve, exists to cover the loan losses that occur in the loan

portfolio of every bank. As such, adequate management of the allowance is an integral part [of] a bank's credit risk management process. . . . A bank that fails to maintain an adequate allowance is operating in an unsafe and unsound manner.

—U.S. Comptroller of the Currency, *Comptroller's Handbook*²⁸

The Impact of Accrual Accounting

Like nearly all companies of a significant size, banks normally use accrual accounting to measure profit and loss. Under the accrual principle, income is earned and expenses are incurred when certain conditions are met, and such financial inflows or outflows are matched insofar as they correspond with one another irrespective of the timing of cash flows. Interest income, being the main generator of a bank's operating revenue, is therefore recognized on an accrual basis.

Accrual of Interest Income

Interest payments that the bank expects to receive in the applicable reporting period on advances of principal made to borrowers is therefore accrued periodically in an "Accrued Interest Receivable" account. This account is intended to include the interest income earned by the bank—that is, that which it is entitled to receive in respect of the current period, but which it may only receive in subsequent reporting periods. In other words, interest income is recognized on the income statement, at least initially, irrespective of whether or not the borrower has actually made a periodic interest repayment in accordance with the terms of the loan agreement.

Where the borrower's payments are en route to the bank—the check being in the mail, so to speak—the use of an accrued interest receivable account is entirely appropriate and proper. When, however, a significant delay occurs or other circumstances arise that casts doubt on repayment being made in full on a timely basis, the justification for continuing to recognize the subject interest income as "earned" is undermined. If substantial doubt is cast upon the ultimate receipt of cash corresponding to the income to be recognized, some adjustment will be required. Depending upon the pertinent regulations and internal policies in place at the institution, both the payment in question, future payments, and the loan itself may need to be recharacterized. The loan may be classified into one of the several categories other than normal or performing that we identified above. This accounting event, as was suggested at the outset of this chapter, may have one or more consequences including:

- The increase of *loan-loss provisioning* against operating income
- The *charge-off (write-off)* of the loan or loans affected
- The *foreclosure* of assets held as collateral to secure the loans
- The *reduction of capital*

In addition to the classification of the past-due financial asset (which will be reflected on the balance sheet), the erstwhile income also needs to be recharacterized. Chiefly this is accomplished by ceasing the accrual of interest income, although loan-loss provisions may also be seen in part as adjusting for the increased probability that cash flows will not match expected income. The cessation of the accrual of interest

income, and its impact upon the reporting of interest income already accrued as well as that which might be recovered in future will inevitably affect the bank's financial statements and therefore be of concern to the analyst. We will examine each of these effects in the sections below.

When May Income Be Recognized?

Under *accrual-based accounting*, the concept of *income recognition* applies to a bank's interest income in the same manner as it does to its other *operating income*. Two underlying principles govern the recognition of income:

1. Is the presumed income quantifiable?
2. Is it probable that the presumed income will ultimately be accompanied by a corresponding inflow of cash?

Thus, to be recognizable putative income must be both measurable and likely to be received.

When Is Income Properly Accrued?

If eligible for recognition, the income *earned* but not yet received may be *accrued*. In the context of interest income, the concept of accrual is relevant in two circumstances:

1. First, when the period over which interest income is earned spans more than one reporting period.
2. Second, when the loan becomes overdue.

The first situation is a normal situation faced by both banks and other companies. The second, which is related to problematic loans and a potential accompanying adverse impact on asset quality is one of particular importance to banks. It is addressed next.

Interest income is normally accrued when an interest payment period spans more than one reporting period. The portion that has been earned corresponds to the period of time that has elapsed during which interest is earned but has not yet been received due to the timing of the periodic payments. There need be no question of a loan's status; merely the fact that interest has accumulated on a daily basis underlies the accrual of interest income.

Interest Income Accrual for Problematic Loans

Interest income will continue to accrue on a loan until the bank acts to recharacterize the receipt of the ostensible revenue as so unlikely to be received that the cessation of accrual is justified. Typically, this occurs when a loan is past due for some significant period of time, conventionally 90 days, although the actual period depends upon local accounting standards and prudential regulations. There is no simple or single answer to the question of when the accrual of interest should be terminated. Ultimately, the decision to stop or continue the accrual of interest requires a prediction about the future—namely the likelihood that the putative interest income will

eventually be received. In the end, this forecast necessarily involves a subjective judgment. Accrual of interest income is illustrated in Exhibit 7.4.

Variations in Treatment

Its very subjectivity makes the foregoing determination vulnerable to external considerations such as the apparent performance of the bank as a whole. Unless required by law, however, a bank’s managers or owners may be tempted to continue to characterize interest as accruing since it will make earnings look better. Consequently, regulators impose guidelines to clarify the conditions under which accrual of interest should be stopped and frequently set forth somewhat arbitrary rules to limit

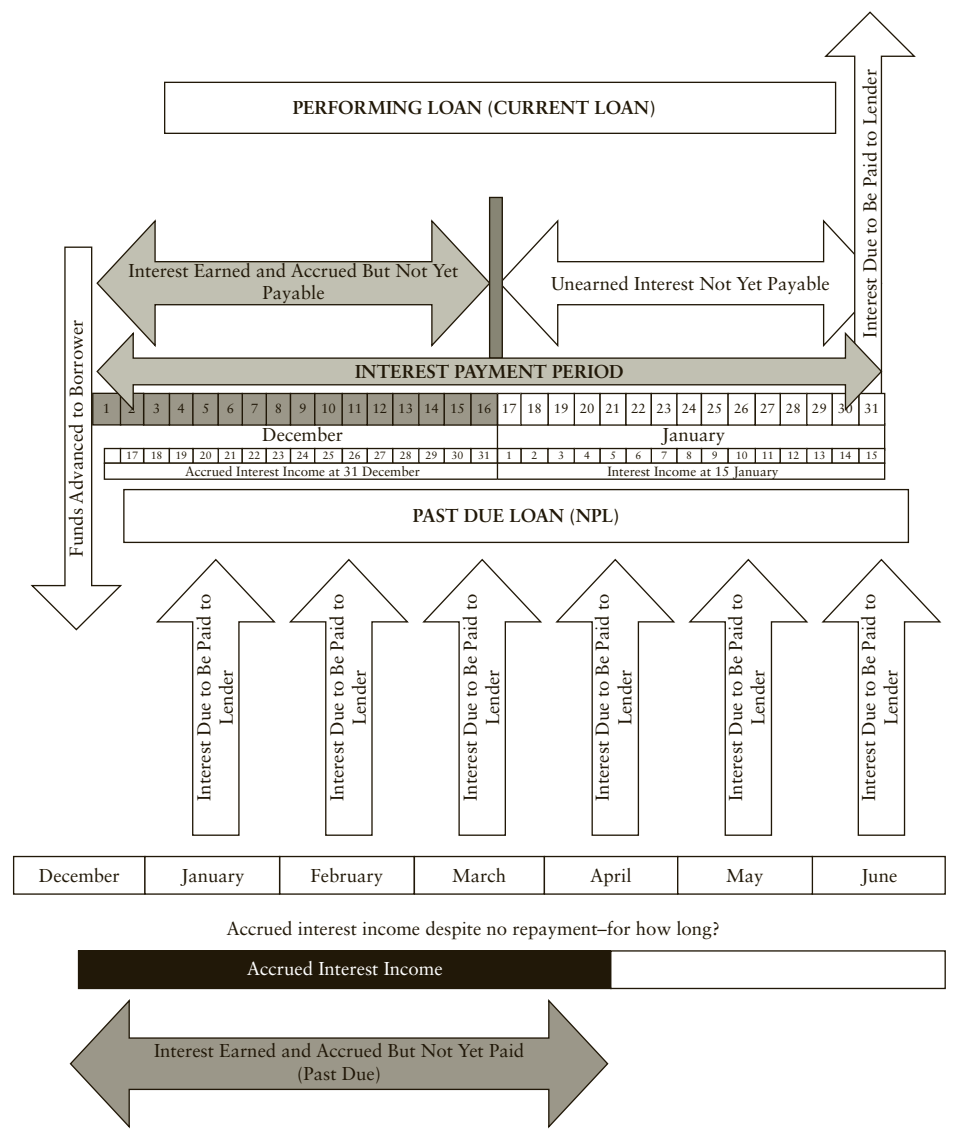


EXHIBIT 7.4 The Accrual of Interest Income: When Does it Cease?

the discretion of individual banks. The subjective nature of the decision to accrue interest income is also responsible for the fact that accounting and regulatory policies governing this determination—as with many aspects of bank accounting—vary considerably from one country to another. Likewise, within the discretion allowed them by regulators, individual banks may also differ in their treatment of this income item.

The analyst should remember that interest accrued but not received, if it is not recognized as a potential loss, is conceptually—and most often technically—capitalized. This means that any unpaid interest simply increases the amount lent to the borrower, and, if he is delinquent, it does not matter much that the bank's loss is on the principal or on the interest. In any event, interest when actually paid may very well have been funded by the loan itself. This shows the limitations of putting too much trust in a single indicator.

ACCRUING INTEREST INCOME

Assume that on December 15, Jambalaya Corporation borrowed \$100,000 from Riverside Bank on an interest-only basis at the rate of 0.5 percent per month (6 percent per annum), with payment due on the 15th of each month. Each monthly payment, needless to say, would equal \$500. Like many institutions, Riverside Bank's fiscal year ends on December 31. On that date it will have effectively earned 16 days' worth of interest equivalent to \$258.06 even though it has not yet received that income in cash. It expects that it will receive this amount together with the remainder of the monthly payment of \$241.94 on January 15. On December 31, Riverside Bank will have effectively earned 16 days' worth of interest despite not yet having received that income in cash. It expects that it will receive this amount together with remainder due on January 15.

But what if a borrower does not repay the loan when due and payments remain delinquent for several months? How should the interest that has been accruing be reported? Assume that Roseville Corporation borrowed \$100,000 from Riverside Bank on the same terms as Jambalaya Bank. Riverside Bank will accrue interest income in the same manner as above when the interest payment period spans its quarterly reporting periods. However, suppose in this instance that Roseville did not make payment on the loan when due on January 15, nor on February 15, nor on March 15. At what point should Riverside Bank cease accruing interest on the loan?

It is this question that makes the second situation of greater relevance to the bank analyst. If a borrower misses a single loan payment, either through a technical error or as a result of a very temporary condition, and thereafter resumes making payments in full on a timely basis, it may be inappropriate to cease accruing interest. But if a bank continues to accrue interest for a long time after the borrower has stopped making loan repayments, and there is little likelihood of payment resuming, continuing to accrue interest would not reflect economic reality. If the accrual of interest were stopped on a loan past due a single day, income would be understated. If the accrual of interest continued on a loan past due an entire year, income would be overstated.

In general, the criteria that are used to judge whether the accrual of interest on a loan should cease are similar to those used to determine whether a loan is *nonperforming*. Both *quantitative* criteria (e.g., how long the loan is past due) together with qualitative criteria (e.g., what is the bank's judgment concerning the borrower's ability and willingness to repay) are considered.²⁹ Be aware that in some markets a distinction is made between the two conditions, and accrual may be required to cease before a loan is formally deemed nonperforming. Similarly, between the time a loan is determined to be nonperforming and the time it is formally written off, accrued interest income may be required to be recorded in a special account. Separate from this issue but related to it is the matter of loan-loss provisioning. The cessation of interest accrual may trigger a provisioning requirement, either directly or indirectly. Loan-loss provisioning is discussed in the next section.

Impact of the Cessation of the Accrual of Interest

Regulations and policies vary not only with respect to when interest accrual should cease, but they also vary considerably as to the consequences of cessation. Where a loan is deemed nonperforming and to be written off, critical issues are whether interest income already accrued must be reversed, and the timing and mechanics of such a *clawback*.³⁰ Under best conservative accounting practice, a clawback is required. A related issue is whether the clawback should be made against current profits in the income statement or against equity capital in the balance sheet.

In some markets, when, under relevant regulations and bank policies, a loan is deemed to be nonperforming, income from the assets is no longer recognized or permitted to accrue, and the portion of the bank's accrued income receivable attributable to the loan is erased. Interest accrual will still be calculated off-balance-sheet, however, as memorandum interest. In the event, the bank recovers some portion of the loan, the recovery will be first allocated to this off-balance-sheet accrued interest and second to the repayment of principal. In other markets, the classification of a loan as nonperforming also stops the recognition of interest income, but interest continues to accrue on the balance sheet in a suspended interest or *interest in suspense* account. In still other markets, banks may choose from two options:

1. Stop income recognition, but record the unpaid accrued interest in a suspended interest income account.
2. Recognize the interest income but set aside loan-loss provisions against the entire amount of the interest income accrued and recognized.

Some countries, like the United States, have specific procedures for accrual interest clawback.

U.S. PROCEDURE CONCERNING CLAWBACK OF ACCRUED INTEREST

In the United States, the procedure concerning accrued interest income on delinquent loans is described in the *Allowance for Loan Loss Handbook*:³¹

When a loan is charged off, any accrued but uncollected interest on the bank's books from both current and prior periods should be charged against current earnings. The one exception is when specific provisions have been made to the allowance for uncollectible accrued interest. In that case, the accrued interest should be charged to the allowance. For discounted loans, the unearned portion of the loan balance should be charged against the unearned discount account.

Loan-Loss Provisions, Charge-Offs, and Loan-Loss Reserves

Since a bank's lending activities inevitably give rise to problem loans, accounting conventions have developed as a means of capturing this cost of doing business. One should be careful to understand and use the proper terms for the various accounting entries associated with impaired assets.

Loan-Loss Provisions versus Loan-Loss Reserves

Loan-loss provisions or LLPs refer to the charge made against operating income in the income statement to take account of identified probable loan losses during a given reporting period, and in some markets expected losses in the loan portfolio not yet identified. They are a flow item, newly allocated in each financial period for that purpose.

In contrast, loan-loss reserves or LLRs are a balance sheet item that measures cumulative provisions, minus write-offs. Loan-loss reserves therefore constitute a bank's cumulative pool of reserves to provide a cushion against total loan-losses. At the end of the reporting period, new loan-loss provisions are added to cumulative loan-loss reserves from the previous period from which net charge-offs have been subtracted. The resulting new figure reflects loan-loss reserves for the current fiscal year.

During a financial period, as the bank registers new *past-due loans*, it will usually allocate new *loan-loss provisions* (LLPs) that will be charged to operating income at the end of the period. In addition to these *specific* loan-loss provisions, the bank may also set aside *general* loan-loss provisions not specifically tied to identified problem loans.³² The amount of each category of provisions will be guided by the bank's own analysis as well as prudential guidelines and regulations concerning loan-loss provisioning requirements. Ordinarily, the bank will report on its income statement taxable earnings after deduction for loan-loss provisions, while on its balance sheet it will show its loans net of loan-loss reserves (LLRs) and any *unearned income*.³³ The new loan-loss provisions will be added to the existing pool of cumulative loan-loss reserves. Those NPLs deemed loss loans and written off will soak up an amount equivalent to the loss from the pool of loan-loss reserves. This amount will therefore be deducted from the pool of LLRs.

In some countries and in some circumstances, asset losses can be deducted directly from equity without being recorded as losses in the income statement. This is of course unacceptable. More importantly, this is something the analyst should

always check before reaching an opinion on asset quality or on the truthfulness of the accounts.

Net Loans versus Gross Loans

Another way to think about the loan-loss accounting process is that when a bank sets aside loan-loss provisions during a single reporting period, say of one year, it is in effect stating that it is withdrawing an equivalent sum from pretax profit on the income statement as an adjustment for the losses it expects it will incur as a result of a certain percentage of loans generating losses. On the balance sheet, loan-loss reserves are, in accounting terms, *contra-assets*. When they are deducted from total loans, the result is *net loans*. The deduction may be displayed on the face of the balance sheet, or in a footnote to the accounts.

Net loans, in other words, represent the volume of outstanding loans against which no loss is anticipated and no loan-loss provisions have been charged. It therefore approximates the true accounting value of a bank's loan book.

Charge-Offs, Recoveries, and Net Charge-Offs

Some borrowers will fail to repay their loans as scheduled, and these past-due loans may subsequently be classified by the bank as nonperforming. Typically, a large portion of these NPLs will be written off. Others may be restructured, and in some jurisdictions some may be kept on the bank's books as performing even though the reality is that they have ceased to generate interest payments. As the bank writes off those assets classified as loss loans, total loan-loss reserves (LLRs) will be reduced by the same amount.

It is often the case, however, that banks are able to recover some proportion of the potential loss on a nonperforming loan as a result of collection efforts, negotiation with the borrower, the sale of collateral, or the loan's returning to performing status. If some amount of loan-loss reserves has been specifically allocated to a given loan, the amount recovered as a result of these actions, will be designated as a recovery and added back to the pool of LLRs.³⁴

In addition, recoveries are significant because, in the long run, the difference between the total amount of loans written off, or charged off, and recoveries represents the bank's true credit costs. This remainder amount is a critical figure, which is termed the bank's net charge-offs, or NCOs. The computation of NCOs is shown below:

$$\begin{aligned}\text{Net charge-offs(NCOs)} &= \text{Total charge-offs} - \text{Recoveries} \\ \text{Charge-offs} &= \text{Write-offs}\end{aligned}$$

The process of adding LLPs to LLRs, deducting charge-offs, and adding back recoveries is further illustrated in Exhibit 7.5.

General Provisions versus Specific Provisions

The purpose of loan-loss reserves is to identify and provide for probable, but for the most part as yet unconfirmed loan losses. As stated by the U.S. Comptroller of the Currency:

Income statement for period (excerpt)		Balance sheet at beginning of period (excerpt)	
Preprovision income	275	Liquid assets	1,000
LLPs	125	Total loans	8,000
NOPAP	150	LLRs	300
Nonoperating items	0	Net loans	7,700
Pretax profit	150	Other assets	1,000

Note to balance sheet		Balance sheet at end of period (excerpt)	
Loan-loss reserves (LLRs) at beginning of period	300	Liquid assets	1,000
– Total charge-offs	– 30	Total loans	8,000
+ Recoveries	+ 5	LLRs	400
+ New loan-loss provisioning	125	Net loans	7,600
= LLRs at the end of period	400	Other assets	1,125

EXHIBIT 7.5 Loan-Loss Accounting: Camellia Bank (\$000s)

*The allowance [for loan losses] is a valuation reserve maintained to cover losses that are probable and estimable on the date of the evaluation. The allowance is not a cushion against possible future losses; that protection is provided by capital. . . .*³⁵

A further distinction is made in a number of countries between general loan-loss provisions and specific loan-loss provisions. In those markets that make this distinction, the two categories can be defined as follows:

1. *General provisions* are those that are set aside before the fact against expected losses within the portfolio. In some countries, however, statute or regulation may oblige banks to set aside a certain percentage of total loans, typically 1 to 2 percent, as so-called *statutory provisions*.
2. *Specific provisions* are those set aside against identified problem loans.

In markets where no distinction is made between general and specific loan-loss provisions, all provisions are taken against specifically identified problem assets. The existence of two classes of provisions is entirely dependent upon local rules and regulations. The distinction between general and specific provisions is not employed in the United States approach to provisioning, but it is quite common in many markets.

General Provisions: Statutory

The difference between general and specific provisions has to do with how bankers and, particularly, regulators think about credit risk. Statutory provisions effectively mandate a basic minimum level of provisioning. As tools for managing credit risk have improved, the case for statutorily imposed minimum provisioning levels has diminished. The rationale for statutory provisioning requirements is also undermined by the fact that they bear no clear relationship to asset risk. Nevertheless, in emerging markets where bank regulation and supervision are weak, an argument can be made that such a basic minimum reserve is justified.

From an analytical point of view, how loan-loss provisions are allocated is not as important as their total volume relative to problem assets. The object is to discern the extent of the gap between the nominal value of the loan book and its true value, the difference being attributable to deficiencies in asset quality and being in need of provisioning.

This said, some see statutory general provisions as being more of a capital indicator than an asset quality indicator,³⁶ since—if one disregards tax considerations—they are nothing short of an increase of equity that could have been achieved by retaining more of the earnings instead of taking away such provisions from total earnings through the P&L.

General Provisions: Nonstatutory

Under a regime providing for both general and specific provisions, where no statutory requirement exists the case for general provisions is that their availability encourages management to provide some additional cushion against loan losses based on historical experience. Customarily, management will project the portion of its loan book not yet identified as problematic assets that will become so in the ensuing reporting period based on previous loss data or reasonable projections taking into account relevant influences such as economic conditions and asset composition. Under this approach, the general loan-loss reserve corresponds to anticipated loan losses, but with no clear indication as to where the problems are going to come from.

In emerging market banks, management will in fact often make good use of general and even of specific provisions to smooth out the bank's income over a number of periods.

Specific Provisions

Specific provisions vary with each class of loans and are only properly applied to loans identified as problem assets. Loan classification schemes, which we discussed earlier, will establish the criteria for specific loss provisioning. Management will also look at the borrower's history and at his financial prospects, as well as at the value of any collateral associated with the loan. As a rule, the higher the probability of nonperformance, the higher the provision should be. Once a loan is identified as a loss, it will have a specific provision of 100 percent assigned to it. Ultimately, when the loan is charged off, the loss reserve account will be reduced by an equivalent amount.

WHAT CAUSES EXCESSIVE NPLS

[B]ank credit expansion sets in motion the business cycle in all its phases—the inflationary boom, marked by expansion of the money supply and by malinvestment; the crisis, which arrives when credit expansion ceases and malinvestments become evident; and the depression recovery, the necessary adjustment process by which the economy returns to the most efficient ways of satisfying consumer desires.

—Murray N. Rothbard, *America's Great Depression*³⁷

What about all the bad investments made during the boom? Well, that was so much wasted capital. But there is no obvious reason why bad investments made in the past require an actual slump of output in the present. Remember . . . economic slumps are not punishments for our sins, pains we are fated to suffer.

—Paul Krugman, *Return of Depression Economics*³⁸

Although some nonperforming loans are inevitable and accepted by analysts, investors and managers alike as a part of the cost of engaging in the banking business, extremely high levels of nonperforming loans are in almost all situations viewed unfavorably. Hence the causes of excessive NPLs are a matter of vital concern to bank analysts.

There are two fundamental causes of high levels of nonperforming loans and the poor asset quality they normally signify:

1. Inadequate credit management
2. A severe deterioration in economic conditions

In reality, both factors are usually present to some degree, and are often intertwined. As the quotes that lead this section suggest, economic conditions strongly affect a bank's business. The impact is in large part explained by the close linkage between the relative availability of credit on the one hand, and the level of economic activity and growth on the other. It is fair to say that banks both strongly influence the economy and are strongly influenced by it.

Of the two causes of elevated NPLs, however, inadequate credit management is the greater concern. While almost all bankers recognize that ebb and flow of economic activity mould the terrain in which they operate, achieving the proper balance of risk and reward is more easily sought than achieved. Predictably, in an economic slump most or all banks in a particular market will be adversely affected and display deteriorating asset quality. But a downturn will tend to separate the wheat from the chaff, exposing deficiencies in credit management that have been present all along. Banks with prudent management will in most cases rank among the survivors, while less well-run banks are brought to their knees.

An awareness of the linkage between economic activity and bank asset quality will therefore help the analyst better assess whether a particular institution is in a good position to endure anticipated events. Possessing a recognition of how changing

conditions are likely to affect the operations of the bank being evaluated will aid the analyst in gauging the severity of the challenge faced by management, its capacity to respond, and the adequacy of actions already initiated.

MACROECONOMIC INFLUENCES ON ASSET QUALITY—ECONOMIC, BUSINESS, AND CREDIT CYCLES

Macro-level influences on bank profitability were mentioned in Chapter 6, and the reader will have gathered that asset quality is highly influenced both internally by management and by external factors, including prevailing economic conditions. These external elements affect individual bank creditworthiness generally, and profitability and asset quality specifically.

It is evident that economic activity tends to oscillate, with periods of expansion followed by periods of contraction. The character and causes of such phases and the transitions between them have been a focus of study since the mid-nineteenth century, and a number of theories have attempted to explain these recurrent if not regular changes in the business climate that are termed business cycles or economic cycles. Despite having been a subject of scholarly inquiry for a century and a half, the business cycle³⁹ has yet to be pinned down and made predictable. The nature of the cycles—if they can be called that for their patterns are more like the weather than, for example, the mechanical cycle of the internal combustion engine—vary in character, duration, and magnitude.⁴⁰

Credit Cycles

Credit cycles have an enormous influence on asset quality, and they deserve considerable attention from the external analyst, much more than the space allocated to this topic in the present chapter would indicate.

The nature, severity, and duration of credit cycles are difficult to predict, and, thankfully, most of them do not end in a banking crisis. Credit cycles are well known to economists, and it is not the purpose of this book to explore them in detail. They are, however, extremely relevant to banking crises, especially systemic banking crises, and this book devotes an entire section to credit cycles in the context of our analysis of banking crises in Chapter 14.

If a lack of familiarity with credit cycles is an obstacle to a perfect understanding of their importance in the context of asset quality, the reader may wish to visit the relevant section in Chapter 14 prior to reading the rest of the present chapter.

THE IMPORTANCE OF THE CREDIT CYCLE

Understanding the position of the local banking industry in the relevant economic and business cycle facilitates credit risk assessment. While cycle phases are easy to see after the fact, the transition points are rarely obvious beforehand. Nevertheless, if a trend in a leading indicator can be spotted, it may give a clue concerning the timing of the transition to the next stage of the cycle. The clearest credit cycle indicators are those that define the cycle itself. In the

chapter on country risk, we discuss macroeconomic indicators in more depth. Because they are closely related to the banking industry, the most relevant metrics for asset quality include:

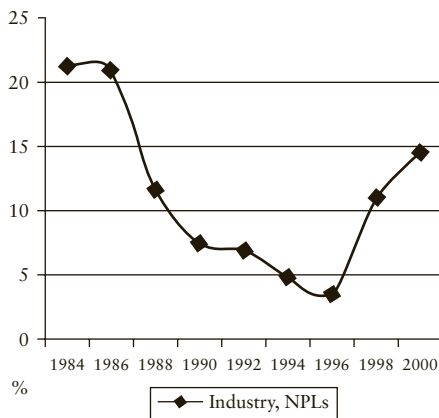
- Industry loan growth
- Industry problem assets, including NPLs
- Industry provisioning and coverage levels
- Related indicators that affect problem asset growth

The next subsection will show an example of a credit cycle over many years, spanning a period exempt of banking crises as well as a period during which a financial and banking crisis took place.

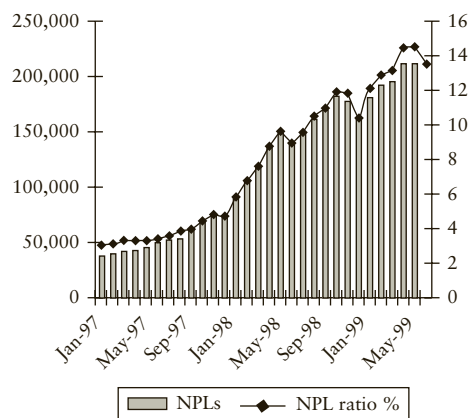
Another subsection will also illustrate a useful tool for analysts to gauge market stress through the number of bankruptcy petitions.

The Credit Cycle Illustrated

The Philippines offers a good example of both a long-term and short-term credit cycle, which is illustrated in Exhibit 7.6. The diagram on the left-hand side shows a long credit cycle in the Philippines, which began in the late 1980s as the country's economic and political situation began to stabilize following the overthrow of the dictator Ferdinand Marcos, and the banking and economic crisis that was still



The longer credit cycle can be seen in the above diagram, which traces the decline in nonperforming loans from the Philippine banking crisis of the early 1980s through recovery during the later 1980s and early 1990s, expansion during the mid-1990s and contraction from 1997 to 2000.



The chart above shows the rise of total nonperforming loans (NPLs) in the Philippines banking industry from the onset of the Asian financial crisis in mid-1997 through mid-1999, together with total NPLs. Note that it does not take account of other problem assets, namely restructured loans and foreclosed assets.

EXHIBIT 7.6 NPLs in the Philippines

Data Source: Bangko Sentral ng Pilipinas.

occurring. Banking industry NPLs fell from a peak of over 20 percent and declined rapidly as the country partook of the Asian boom when economic reforms were implemented under the Fidel Ramos administration in the mid-1990s. Bottoming out at below 3 percent, NPLs shot up again with the onset of the Asian crisis in mid-1997 as the diagram on the right-hand side of the illustration depicts.

In the aftermath of the crisis, prudential regulations promulgated by the new central bank put a lid on banking excesses, and NPLs growth eventually slowed in 2000. Much of the reduced growth, however, was the result of more loans classified as foreclosed and restructured. Nonperforming assets (NPAs), which broadly include both categories of assets continued to rise—from 1997 to 2000—at a steady rate, as Exhibit 7.7 shows. Note that although NPL growth slows by mid-1999, other problem assets grow at a rapid pace from early 1999.

The downturn in the global economy in 2001–2002 stalled economic and banking system recovery causing the NPL ratio to resume its upward climb, but to stabilize and recede in later years, as depicted in Exhibit 7.8.

The line graph from the Philippine central bank illustrates the peak in post-Asian crisis NPLs at over 18 percent and their subsequent decline during the ensuing recovery.

Hong Kong: Bankruptcy Writs

Sometimes less obvious indicators can provide a good indication of the change in the credit cycle. In Hong Kong, where the banking industry depends heavily on the property market, indicia such as unemployment and personal or small business bankruptcy can be regarded as a leading indicator for defaults on mortgage loans.

Another unusual indicator—the result perhaps of the fact that Hong Kong newspapers frequently publish such detailed information—is the number of bankruptcy petitions made by banks. A peak in such petitions usually signals a change in the credit cycle to the recovery phase.

Credit Mismanagement

In addition to the implementation of effective policies and procedures, even more qualitative criteria may be relevant to evaluative process. The analyst may wish to

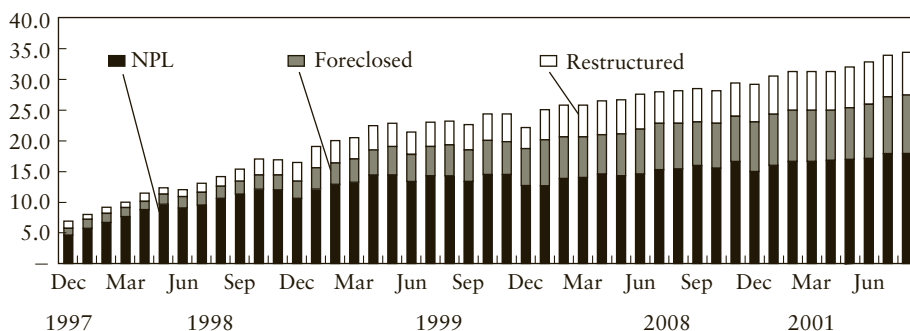


EXHIBIT 7.7 Asset Quality in the Philippines, 1997–2004

Source: Salomon Smith Barney, “Philippine Banks: Asset Quality in Pictures,” August 2001; Bangko Sentral ng Pilipinas (Philippine Central Bank), 2004 Annual Report.

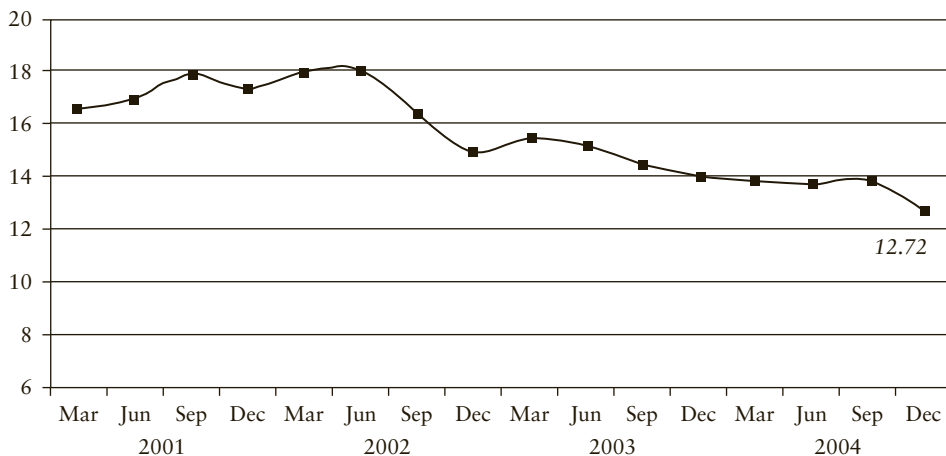


EXHIBIT 7.8 NPLs in the Philippines, 2001–2004

Source: Philippine Central Bank (BSP).

consider the strength of a bank's *credit culture*, an admittedly somewhat nebulous term, which embraces the seriousness with which a bank and its staff take the credit risk management process. With the consideration of this second aspect of asset quality, it is evident that its comprehensive evaluation goes beyond merely scrutinizing the credit quality of a bank's loan book. The characteristics of the bank's financial asset portfolio together with an assessment of the strength of its credit risk management systems are integral to the process.

According to a study by the U.S. Office of the Comptroller of the Currency (OCC), the consistent element in bank failures in the United States has been poor control of credit risk. Each of the following more specific deficiencies were found to be a potential precursor of poor asset quality.⁴¹

- Loan growth in excess of the bank's ability to manage
- Overconcentration of lending
- Lack of clear standards and excessively lax loan terms
- Inadequate controls over loan officers
- Lack of compliance with loan policies
- Inadequate systems for identifying loan problems
- Insufficient knowledge about customers' finances
- Lending outside the market with which the bank was familiar

It is precisely these items in respect of which the bank analyst must be on the lookout. Of the eight, those most amenable to quantitative review are the first (excessive loan growth), the second (overconcentration), and the last (lending outside the bank's home market, including foreign lending). Loan growth is discussed in the next section, while loan book composition is addressed in the latter part of the chapter, as are standards, controls, and the overall credit culture of the institution. These often can be readily identified from disclosed data. The last items are more qualitative in nature. If the opportunity exists, they may be explored in discussions with management or gleaned from conversations with those familiar with the institution.

DATA AND RATIO ANALYSIS

The problem with banking ratios is the lack of consistency in how they are calculated.

—Anthony Lok⁴²

Data and indicators used in the analysis of asset quality are not as straightforward as they could be in the analysis of profitability or even capital adequacy. For the quality of the bank's assets is often in the eyes of its management. Granted, many assets—nonloan assets, in particular—can be marked to market, but the problems lie with the assets whose continuing value can only be estimated, and such value may yet not translate into a quantifiable, visible entry into the accounts of the bank.

Key Data and Indicators

The most critical figures to obtain for the periods being examined are:

- Total loans (this will almost always be available in the annual report)
- Total nonperforming loans
- Total nonperforming assets (i.e., NPLs + foreclosed assets + restructured loans)
- Loan-loss provisioning (annual)
- Total loan-loss reserves (cumulative)
- Net charge-offs
- Preprovision income
- Total shareholders' equity

Figures concerning loan portfolio composition are also important, but they are discussed in a later part of this chapter.

With the foregoing data and the information normally supplied in the bank's audited financial statements, the following asset quality ratios can be easily calculated:

- Year-on-year *loan growth*
- Nonperforming loans to total loans (*NPL ratio*)
- Loan-loss reserves to nonperforming loans (*NPL coverage ratio*)
- Loan-loss reserves + equity to nonperforming loans (*dead bank ratio*)
- Nonperforming assets (NPAs) to total loans

Also helpful but not quite so critical would be the following ratios:

- Loan-loss provisions to preprovision income
- Net charge-offs to preprovision income

Loan Growth and Related Ratios

How rapidly a bank is growing can influence its asset quality. Excessive loan growth is a potential indicator of both current and future problems. Too rapid an expansion in lending suggests that a bank is expanding its loan portfolio beyond its credit

management capability and is a negative factor in making a credit assessment. Prospective difficulties are consequently implied since there is inevitably a lag between the time a bank makes a loan and the time the loan becomes delinquent; an increase in lending in one financial period may very well not be reflected in higher NPLs until a later period.

Moreover, unusually high rates of loan growth suggest that a bank is advancing funds faster than it can internally generate or otherwise acquire new capital, thereby raising potential solvency concerns. Given typical return on asset values for banks, extremely high rates of growth will almost always result in a weakening of capital strength, all other things being equal.

UNUSUALLY HIGH RATES OF LOAN GROWTH—A RED FLAG!

Rates of loan growth substantially above the market norm, particularly if over 20 to 25 percent per annum, are a warning of potential asset quality problems. Such rates suggest that the bank is almost certainly outpacing its ability both to generate capital internally and to effectively apply credit controls.

In regard to capital, assume, for example, that a bank is able to maintain a nonaveraged return on assets ratio of 2 percent—a high value in nearly all markets. Assume that the bank's business and exposure largely flow from taking asset risk. Assume further that its equity multiplier (asset-to-equity ratio) is 12.5 $\times$, a reasonably conservative value, which would result in a nonaveraged ROE of 25 percent—also a high value. Finally, assuming that the bank returns no dividends, it follows that it would at best be able to grow its capital by 25 percent per year. Therefore, loan growth beyond that level would result in a diminution of non-risk-adjusted capital ratios.

Additionally, high rates are a danger signal that functions as a potential precursor of a sharp rise in problem assets. There are several reasons for this. First, if a bank is growing its loan book at rates in excess of 15 to 20 percent, it is unlikely that it has acquired the relevant manpower and is scrutinizing loan applications with proper care. Second, there is on average a lag of at least several months between the creation of a pool of new loans and a portion of that pool becoming nonperforming. Finally, and a corollary to this fact, high rates of asset growth inherently tend to mask asset quality problems since the most commonly employed measure of asset quality, the NPL ratio, will normally decline in nominal terms as new loan issuance rises substantially.

For all the foregoing reasons, loan growth significantly higher than peer levels warrants further scrutiny and monitoring, and even if in line with peers, abnormally high rates of loan growth should be monitored carefully. This is especially so when there are no mitigating factors, such as the following:

- The bank is a recently formed institution expanding from a low base
- It has made a recent acquisition
- It has just received a capital injection in preparation for planned expansion

Measuring Loan Growth and Asset Growth

Loan growth is customarily measured on either an annual or annualized basis.⁴³ End-year figures are most commonly used, but year-on-year or other periodic figures may also be employed. The construction of the ratio is shown later. As an alternative to loan growth, earning asset growth or total asset growth is sometimes used to measure the rate of a bank's expansion.

Extracting total loans (or earning assets) from the balance sheet may occasionally require some parsing of the disclosed figures. The amount of total assets, however, is practically always available and has the additional virtue of being able to capture debt securities holdings, which in some circumstances may be significant to an appraisal of the bank's overall asset quality.⁴⁴

Loan Growth in Emerging Markets

In emerging markets, rates of loan growth are often higher than in industrialized countries. This occurs for several reasons.

- First, in these markets, the economy is often developing from a low base and financial intermediation is expanding rapidly.
- In addition, because the capital markets in such countries are less evolved, companies must rely heavily upon bank credit, thereby channeling most of the demand for new finance into demand for loans.
- Finally, new banks that are formed following deregulation will see huge loan growth at first, since they are escalating from a small base.

Expansion in lending volume in excess of 30 percent per annum among individual institutions, particularly new banks, is not uncommon in these situations.⁴⁵ Nevertheless, very high rates of growth still warrant close scrutiny notwithstanding that the entire sector may be advancing funds at a frenzied pace. For example, in Indonesia, the loan growth for the banking system as a whole reached an extraordinarily high 30 percent per annum in the two years before the market crashed in 1997–1998, while some individual banks posted loan growth rates as high as 50 percent.⁴⁶

PRESCIENT OPINIONS FROM A RATING AGENCY

A major area of concern is asset quality. Like most of Indonesia's banks, this bank has grown its portfolio in recent years at a punishing rate. As with portfolios everywhere, the true quality of a creditor does not become apparent to outside observers until well after the bank is aware that problems exist. The experience in Indonesia is that this delay is rather longer than one would find in many other jurisdictions. After posting years of rapid growth, one can only conclude that a major reckoning is at hand. . . . Nothing suggests that this bank will be able to escape this reality.

—Thomson BankWatch report on Bank Utama, December 6, 1996

Further, the rising bankruptcy levels are also likely to be indicative of an expansion in substandard loans as well, and . . . these problem asset totals are believed to be much higher than the “official” figures presented. At this stage it appears that despite the government’s efforts to require the banks to improve their reserves against problem assets, the bad debt problem which has long plagued the Korean banking sector will continue to persist, at least in the intermediate term. This is particularly troublesome when one considers that Korea’s economy has been generally quite healthy in recent periods. Should the economy show signs of easing, the probability of sharply higher levels of problem assets would likely be high.

—Thomson BankWatch, Banking System Update on Korea,
September 29, 1995

Rule of Thumb

How much growth is too much? The Indonesian experience illustrates one extreme. It is evident that expansion at this breakneck pace exceeded the industry’s ability to perform prudent credit assessments. Abnormally high rates of loan growth, as suggested, also imply that a bank is exceeding its ability to internally generate or otherwise acquire new capital. As a rough rule of thumb, loan growth rates over 20 to 25 percent per annum should give the analyst pause, and 15 percent would be a more conservative ceiling, especially in more mature markets.⁴⁷ Bearing in mind the current phase of the credit cycle and the degree of bank intermediation in the local economy, loan growth of between 5 and 15 percent per annum can generally be viewed favorably in a normal economy. Loan growth of 5 to 10 percent is respectable and characteristic of a mature economy experiencing stable economic conditions. Growth rates of 10 to 15 percent would not be out of line in a healthy developing economy. At the other end of the spectrum, loan growth of 0 to 5 percent would tend to indicate either a very large bank or a mature economy in a weaker phase of the credit cycle, while negative loan growth is characteristic of a bank or banking sector that is rebuilding its balance sheet following an economic shock.

NPL Growth as an Indicator

The most critical asset quality indicators are those that take account of nonperforming loans (NPLs). The rate at which problem loans are increasing or decreasing—the NPL growth rate over, say, one year—is one such measure.

Significance of NPL Growth

Sharply rising NPLs, particularly when they are disproportionately high to total loan growth, are generally taken as a danger sign by bank analysts. Likewise, falling NPLs are usually a positive sign. But this is not always true, nor are rising NPLs always credit negative. It is important not to accept nominal percentage changes at face value, but to understand the reasons for the change and any variance from the norm.

What is most critical is how the rate of growth in absolute NPLs compares with total loan growth. Absolute NPL growth of 6 percent per annum when the rate of loan growth is 2 percent means something entirely different from the same rate of NPL growth when lending is expanding at a rate of 18 percent per annum. In addition, while NPLs most obviously may be increasing or decreasing as a consequence of a growing or diminishing asset quality problem, there may be other causes. Most notably, instead of being attributable to variations in credit fundamentals, it is possible that particular changes or variance in problem loan levels may be partly or wholly explained by differences in how NPLs are defined.

ASSET QUALITY: FUNDAMENTALS VERSUS DEFINITIONS

Superficially, higher or lower NPLs indicate poorer or better asset quality. Consider, however, whether NPLs are higher (or lower) than expected owing to definitional differences or changes. In particular, consider whether the cause might be:

- A tightening or relaxation of the definition of nonperforming loans (tighter in the sense of more encompassing standards means more NPLs, all other things being equal)
- The inclusion or exclusion of restructured loans from classification as nonperforming loans (the bank may be restructuring loans to make its asset quality look better)
- Decreasing or increasing charge-offs to loan-loss reserves or to capital, thereby increasing or reducing the rate at which NPLs are removed from the balance sheet

A change in definition, unless it is a relaxation from a standard that is stricter than international best practice, can be viewed as a form of regulatory forbearance. The effect, if not the intention of regulatory forbearance, is typically to allow a bank's performance or financial condition to appear better than it would under prior regulations. Where definitional concerns result in substantial deviation from the international norm, the NPA-based indicators, discussed below, can be a useful supplement to NPL-based ratios.

Loans Worked Out or Restructured

Definitional issues in relation to NPLs can manifest in several ways. In addition to differences in aging criteria applicable to problem loan classifications, there may be discrepancies in the treatment of restructured loans as well as those that were previously nonperforming and that have been returned to normal status.⁴⁸

Impact of Write-Offs and Provisioning

Additionally, NPLs may decline as a result of being written down against provisions or written off against capital. Loans that have been removed from the balance sheet will naturally no longer appear on a bank's financial statements, and the volume of NPLs will thereby be reduced. Note that the ability of banks to write off loans is

restricted directly or indirectly in some markets. In such circumstances, write-offs are of less value as an asset quality indicator, and instead loan-loss reserve coverage becomes a more important metric, particularly when cross-border comparisons are made.⁴⁹ Finally, as with many financial ratios, movement off a low base is apt to be less meaningful than might first appear. Do not be led astray by large increases in the rate of NPL growth due to a rise off a low base.⁵⁰

NPL Ratio

Perhaps the single most important asset quality indicator is the ratio of problem loans to total loans, or the NPL ratio, the formula for which is shown below. This indicator, when available for individual banks, several peer banks, and preferably for the banking industry in the subject market as a whole, will provide an excellent picture of a bank's comparative asset quality. The usefulness of the ratio, however, depends upon consistent definitions of its numerator and denominator.⁵¹

Definition of the NPL Ratio

The *NPL ratio*⁵² shows nonperforming loans as a proportion of the bank's loan book and therefore reflects the extent to which—on a percentage basis—a bank has lent money to borrowers who are not paying it back. Since the problem loans are no longer generating interest payments to the bank, their value is much reduced.

$$\text{NPL ratio (\%)} = \frac{\text{Total nonperforming loans}}{\text{Total or gross loans}} \times 100$$

Significance of the NPL Ratio

An NPL ratio that is higher than the bank's peer group (and industry) is a superficial indicator of inferior asset quality. The same meaning is conveyed if its NPL ratio is increasing more rapidly or is decreasing less rapidly than its peers. Conversely, a bank that has an NPL ratio that is lower than its peers, that is, increasing less rapidly or decreasing more rapidly than its peers, suggests superior asset quality. However, as with loan growth and NPL growth, it is essential to probe beneath the surface to determine what may account for differences among banks and trends in the NPL ratio. Insofar as regulatory changes and definitional differences affect the volume of NPLs, they will in turn affect the NPL ratio (see formula above).

Banks usually have some degree—nowadays thankfully reduced by better regulatory constraints—of discretion in deciding whether a loan is nonperforming or not. Some take a relatively conservative approach to classifying loans, and as a result have higher nominal NPLs than institutions that are comparatively reluctant to grade their loans as nonperforming. Thus, equivalent levels of true asset quality can give rise to a range of NPL values. Consistency is therefore a matter of concern when comparing banks, since lax or inconsistent NPL definitions can skew results not only on a cross-border basis but also within markets.⁵³ At the end of the day, management will often find ways around regulations, and the analyst is advised not to take NPL figures and ratios at face value, especially in emerging markets.

SIGNIFICANCE OF THE NPL RATIO

When the NPL ratio of a bank diverges significantly from its peer group, from the industry average, or from the bank's own historical performance, it provides a potential indication concerning of the institution's asset quality. Higher than average or increasing NPLs, for example, suggest but are not a definitive indicator of mediocre or deteriorating asset quality. Other factors, however, may account for ratio values that appear better or worse than peers. In some situations variations in performance can be traced back to definitional differences, or to lax prudential regulations that allow a bank to avoid formally classifying a problem asset as an NPL. Moreover, in some markets, nominal NPL figures, even if ostensibly measured according to international standards, should be taken with a grain of salt. In short, a bank with a high NPL ratio does not necessarily have inferior asset quality, nor does one with a low ratio inevitably have better asset quality.

Rule of Thumb

What is an acceptable level of NPLs relative to total loans? It depends. The local market, economic conditions, and how strictly the bank classifies NPLs are all relevant. In a research report, dated 1999, for Nomura Securities analyst Anthony Lok observed:

Historically, problems have started to emerge once NPLs breach 5 percent in advanced economies, while full-blown systemic banking crises have occurred at NPL levels as low as 10%. In emerging economies, Latin American and Asian banking crises featured peak NPL levels of 15–35 percent.⁵⁴

As a rule of thumb, the ranges in Exhibit 7.9 may prove helpful to the novice analyst.

NPL Coverage

Although it is true that *ceteris paribus*⁵⁵ above-average NPL levels are credit-negative, in practice all other things are rarely equal. For instance, as just discussed,

EXHIBIT 7.9 NPL Ratio, Rules of Thumb

NPL Ratio	Developed Markets	Emerging Markets
< 1%	Excellent	Excellent
1–2%	Good	Excellent
2–5%	Fair	Good
5–10%	Moderate-to-severe problems	Fair
10–20%	Severe-to-extremely severe problems	Moderate-to-severe problems
over 20%		Extremely severe problems

how NPLs are defined will affect the NPL ratio. Therefore an elevated ratio of nonperforming loans, while superficially undesirable, does not necessarily mean that a bank has inferior asset quality.

What is most important is how NPLs, defined in a consistent manner, compare with the amount of reserves a bank has set aside to cover its bad loans—that is, the extent of its NPL coverage. Hence, the corresponding indicator—the NPL coverage ratio—can be said to be the most vital asset quality measure since it relates the volume of the buffer afforded by LLRs with the need for that buffer. In a sense, it is a risk-adjusted asset quality metric.

NPL Coverage versus NPL Ratio

Thus, NPL coverage (or NPL cover) measures the degree to which a bank's problem loans are cushioned against impairment by loan-loss reserves. The NPL coverage ratio can be defined as the total (cumulative) level of loan-loss reserves (not the annual provisioning) divided by the volume of nonperforming loans (see the following formula).

$$\text{NPL coverage ratio (\%)} = \frac{\text{Loan-loss reserves(LLRS)}}{\text{Total nonperforming loans(NPLs)}} \times 100$$

Put another way, the NPL coverage ratio indicates the extent to which the bank's loan-loss reserves will absorb the losses incurred on those loans designated as nonperforming. If the ratio value is high, it suggests that the bank has a large cushion against loan losses and coverage is strong; if low, the cushion is correspondingly smaller and weaker. So long as it maintains adequate provisions to cover the problematic assets, a bank with nominally elevated NPLs may nevertheless be viewed as having sound asset quality. Because it compares NPLs with loan-loss reserves, NPL coverage ratio conveys more information than does the NPL ratio alone. The NPL coverage ratio's disadvantages are the same as all NPL-based ratios; namely the definitional differences that exist from market to market and to a lesser extent from bank to bank.

How Much NPL Coverage

How much NPL coverage is adequate is a matter of judgment taking into account the bank's operating environment, the composition of its loan book, and other factors. The customary level of provisioning will vary from country to country depending on several factors, including:

- Applicable prudential regulations
- The profitability of the bank and banking system as a whole
- The stage of the economic cycle

The same definitional concerns that apply to NPL growth and the NPL ratio logically apply to NPL coverage. Ordinarily, NPL cover that exceeds 100 percent can be regarded as fundamentally adequate, since such a value signifies that loan-loss provisions are at least theoretically equal to expected and identified losses.⁵⁶

Coverage of between 50 and 100 percent can be regarded as borderline, meaning that there may possibly be mitigating factors such as strong capital ratios or a highly conservative asset portfolio suggesting that such a value might be minimally adequate. A figure of less than 50 percent can be deemed generally unacceptable.

As always in respect of such ratios, the analyst would have to scrutinize the denominator before drawing any conclusion. If regulations about recognition of bad loans are lenient, the ratios may tell a good story in comparative terms within a given market, but would not be relevant in absolute terms or in international comparisons.

NPL COVERAGE: OAK BANK VERSUS MAPLE BANK

Imagine two banks: Oak Bank and Maple Bank. Oak Bank maintains a very conservative loan book lending primarily to the most creditworthy large enterprises at very low spreads. Maple Bank's expertise is lending to the small and medium-sized enterprise sector (SME) in which it believes it enjoys a competitive advantage at higher spreads than Oak is able to earn from lending to the blue-chip corporate sector. Since Maple's customers are less creditworthy than Oak's, it can be expected that the rate of default on its loan book will be relatively higher. Perhaps this translates into an ambient NPL ratio of 1 percent for Oak Bank and 2 percent for Maple Bank. At first glance, Oak's asset quality looks better than Maple's. But if Maple's loans are priced at an appropriate level allowing for the provision of ample loan-loss reserves to cover the higher credit costs, then its asset quality may very well be better than that of a bank with a lower rate of NPLs, but which has set aside a negligible level of loan-loss reserves. If Maple's NPL coverage is 150 percent but Oak's is just 75 percent, the superficial hypothesis that Oak's asset quality is better than Maple's appears less valid.

In a number of emerging markets, however, many banks will not achieve the 100 percent coverage level, especially during depressed economic conditions. In such environments, depending upon peer figures, it might be said that an ordinary NPL coverage ratio of 50 percent can be regarded as minimally adequate if the bank is fundamentally profitable, reasonably well-capitalized, or is in the process of aggressively provisioning against NPLs occasioned by a recent downturn. In the same situation, a ratio of between 25 and 50 percent implies borderline asset quality, possibly mitigated by extremely strong fundamentals, while one having less than 25 percent is generally unacceptable and deserves careful inspection. The table in Exhibit 7.10 summarizes these rudimentary benchmarks that, it must be emphasized, should be applied with discretion.

Adjusted NPL Coverage

After current profits, *loan-loss reserves* are the first line of defense against a rash of *problem loans*. But if a bank has not provisioned adequately for NPLs, then it must dig into its capital. *Shareholders' equity* therefore provides the second and last line of defense against a rising tide of sour assets.⁵⁷ Accordingly, if it is unable to achieve an

EXHIBIT 7.10 NPL Coverage Ratio, Rules of Thumb

NPL Coverage	Emerging Markets	Developed Markets
> 200%	Excellent	Excellent
100–200%	Adequate	Adequate
50–100%	Barely adequate	Borderline
25%–50%	Borderline	
< 25%	Generally unacceptable	Generally unacceptable

NPL coverage ratio of 100 percent, at the very least the sum of the bank's *loan-loss reserves* and *equity capital* should exceed that figure. The ratio that expresses this relationship—defined as loan-loss reserves plus equity divided by NPLs—has been referred to as the *dead-bank ratio* and provides a broader measure of a bank's ability capacity to survive a crisis.⁵⁸

This adjusted NPL coverage ratio shows the broadest coverage for problem loans. If the figure registered is less than 100 percent, this means that if all NPLs were total losses (resulting in 0 percent recovery), then loan-loss reserves and equity would be exhausted before outstanding obligations to depositors and other creditors would be satisfied. In other words, the bank would be technically insolvent. Whatever the name given, this indicator, including as it does both loan-loss reserves and equity in the denominator, gives the analyst a reasonable idea of the bank's ability to survive a worst-case scenario in which NPLs rise to unanticipated levels. In addition to functioning as an asset quality indicator, it is also a measure of bank solvency and highlights the link between these two aspects of bank credit analysis.

Related Ratios

Other asset quality ratios that attempt to measure loan-loss reserves against variables other than NPLs or problem assets are much less useful to the analyst.⁵⁹ For example, loan-loss reserves to total loans merely compares cumulative loan-loss reserves to total loans and in itself does not convey much information, since the level of LLRs is not very meaningful without knowing the level of problem loans they are meant to provide against.⁶⁰

$$\text{Loan-loss reserves to total loans (\%)} = \frac{\text{Loan-loss reserves (LLRs)}}{\text{Total loans}} \times 100$$

NPA Ratio, Coverage, and Related Indicators

In addition to the preceding ratios, various other metrics can aid the analyst in refining the asset quality profile of a bank. The most important of these are ratios that consider a bank's nonperforming assets or NPAs. Nonperforming assets, as discussed, encompass not only nonperforming loans in the strictest sense, but also foreclosed assets and restructured loans, as well as loans that are not technically nonperforming but show signs of weakness (i.e., special mention loans). As a broad measure of problem assets, NPAs facilitate a fuller understanding of the ability of the bank to withstand rapid erosion of the quality of its loan book.

NPL COVER: A FURTHER ILLUSTRATION

The diagram in Exhibit 7.11 indicates the NPL and loan-loss reserves for three hypothetical banks in the imaginary country of Floralia. Each has a loan book of \$100 million. The average NPL ratio among banks in Floralia is 5 percent, and the average NPL coverage ratio is 0.8 or 80 percent.

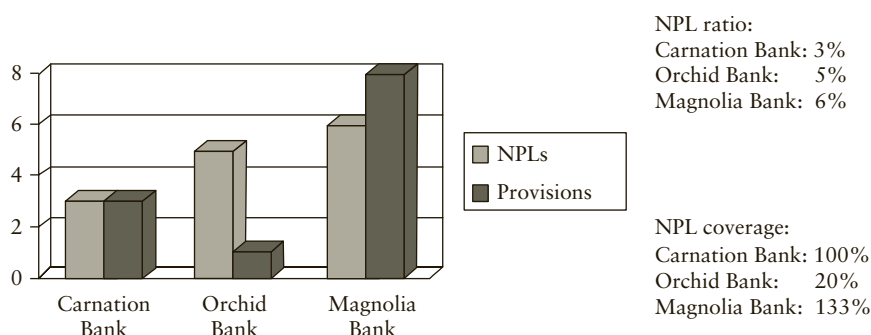


EXHIBIT 7.11 NPL Coverage versus NPLs

Carnation Bank, shown on the left, maintains a conservative loan book with just \$3 million in NPLs or 3 percent of its total portfolio, a low ratio for the country of Floralia although high by developed country standards. It has an equivalent amount of loan-loss reserves, providing NPL coverage of 100 percent.

Orchid Bank's NPLs are just 1 percentage point higher than Carnation's at 5 percent. Its level of loan-loss reserves is just \$1 million or a mere 20 percent of the outstanding NPLs.

Magnolia Bank, shown on the right, has the highest NPL level of the three at 6 percent of book. But its coverage ratio is even higher than Carnation's at 133 percent of NPLs.

Although Carnation certainly appears to have good asset quality and strong provisioning, it cannot definitively be said that its asset quality is necessarily better than that of Magnolia. To be sure, its NPL level is less than half that of Magnolia's, but Magnolia's NPL coverage seems to be more than adequate to cover any foreseeable loan losses. Orchid, since its NPL levels are moderate but its NPL coverage is low, would appear to be the worst of the three. More information naturally would be needed to come to a firm conclusion about which ranks better in asset quality. The point here is that the bank with the lowest NPLs does not necessarily have the best asset quality.

Adjusted NPL coverage ratio(%)(aka dead bank ratio)

$$= \frac{\text{Loan-loss reserves(LLRs)} + \text{Shareholders' equity}}{\text{Total nonperforming loans}} \times 100$$

Example 1: Durian Planters Bank has NPLs of \$350 million. Its loan-loss coverage amounts to \$200 million, and it has shareholders' equity of \$100 million. Together, the bank's \$300 million of equity and reserves is less than its

outstanding NPLs. With a dead bank ratio of 85.7 percent, DPB may be regarded as technically insolvent, although actual solvency would depend upon the expected recovery rate on the nonperforming assets among other factors.

Example 2: Rubber Tree Bank also has NPLs of \$350 million. Its loan-loss coverage also amounts to \$200 million; shareholders' equity is, however, double that of DPB at \$200 million. RTB's dead bank ratio is 100 percent, making it barely solvent.

The principal reason, however, that nonperforming assets rival nonperforming loans as a key input in asset quality metrics is that NPAs are less subject to variation among banks and markets. The volume of NPLs, as observed, will depend upon the manner in which banks classify problem assets. For example, regulations in one country may make it more difficult to dispose of NPLs than in another. Existing laws or the inefficiency of the legal system may cause foreclosure to be deferred for long periods of time, and prevent loan losses from being recognized and thereby written off in the interim.⁶¹ While the nonperforming loans may be fully provisioned against, the higher nominal NPL ratio implies that that asset quality is worse than is actually the case.

Conversely, restructured loans and other identifiable weaker-quality loans may escape formal classification as NPLs allowing NPL-based ratios to overstate asset quality. By substituting the broader measure of nonperforming assets for NPLs in each of the foregoing three ratios, the allocation of problem assets among various categories and related definitional concerns will have less impact on the analysis. The NPA-based indicators therefore facilitate the making of cross-border comparisons.

Definition of Nonperforming Assets

Nonperforming assets can be approximated by adding foreclosed assets and restructured loans to NPLs.⁶² Note that assets that are not loans but are substitutes for loans are included in the definition of *loans* in the context of the discussion of NPAs. NPAs are meant to be an approximation rather than a perfect calculation of actual problem assets.

$$\text{Nonperforming assets (NPAs)} = \text{Nonperforming loans} + \text{Special mention loans} + \text{Restructured loans} + \text{Foreclosed assets}$$

Example: Purpleheart Bank has the following breakdown of problem assets. Total NPAs or problem assets were \$1,200,000, while classified loans were \$800,000 and NPLs were \$500,000	Special mention	\$300,000
	Substandard	\$200,000
	Doubtful	\$180,000
	Loss	\$120,000
	Restructured	\$240,000
	Foreclosed assets	\$160,000
	NPAs	\$1,200,000

Although its use facilitates international asset quality comparisons, one disadvantage of employing NPAs as a metric input is that in some markets it can be difficult to obtain detailed figures concerning "restructured loans," and to a lesser extent concerning foreclosed assets. While NPL data is also not necessarily easily

obtainable, it is nevertheless one of the most popular indicators employed by regulators, analysts, and the financial press in discussing bank asset quality, and is frequently available when figures on restructured loans and foreclosed assets are not.

NPA Ratio

The NPA ratio parallels the NPL ratio in structure and has a similar function. It is designed to take in a broader class of problem assets than the NPL ratio; the denominator of the NPA ratio, as a matter of convention, remains total loans rather than total assets.

The NPA ratio can be divided into component ratios:

- The *NPL ratio*
- The *foreclosed asset ratio*
- The *restructured loan ratio*

$$\text{Nonperforming assets ratio (\%)} = \frac{\text{Nonperforming assets}}{\text{Total loan}} \times 100$$

A bank has, for instance, total assets of \$5 billion and a loan book of \$4 billion, while NPLs amount to \$320 million resulting in an NPL ratio of 8 percent. If it does not have restructured loans or foreclosed assets on its books, its NPA ratio is equal to its NPL ratio.

Another bank, in contrast, has problem assets in excess of just nonperforming loans, such as in Exhibit 7.12. Assuming total loans of \$160 million, NPLs of \$5 million would generate an NPL ratio of 3.125 percent, foreclosed assets of \$1.6 million would generate a foreclosed asset ratio of 13.33 percent, and restructured loans of \$2.4 million would generate a restructured assets ratio of 20 percent, for a total NPA ratio of 7.5 percent.

Foreclosed Assets Ratio

When a nonperforming loan is identified as a *loss loan* and written off the balance sheet, the volume of NPLs is in turn reduced. Any foreclosed collateral, however, remains on the bank's books until sold and represents a nonearning asset, although in some cases the asset may continue to earn an economic return in the form of rents. It is normally best practice for a bank to dispose of any acquired foreclosed assets quickly. This is because most banks simply do not have the expertise and are not

EXHIBIT 7.12 NPA Components

NPA Component	Amount	% of NPAs	As a Percentage of \$160 Million of Total Loans
Special mention	\$3 million	25%	1.875%
Substandard	\$2 million	16.67%	1.25%
Doubtful	\$1.8 million	15%	1.125%
Loss	\$1.2 million	10%	0.75%
Restructured	\$2.4 million	20%	1.5%
Foreclosed assets	\$1.6 million	13.33%	1%
NPAs	\$12 million	100%	7.5%

organized in a fashion that will enable them to optimize the return on foreclosed collateral. Frequently, such collateral is acquired when asset prices are falling, and the bank may not be in a position, either as a matter of economics or of law, to carry the property until market values recover and the property can be sold at a profit. In these circumstances, the real property owned by the bank will be a net burden to it as the costs of maintenance, taxes, and other overheads can easily exceed returns.⁶³ Foreclosed asset are therefore usually a drag on profitability.

Foreclosed assets are denominated in different ways and treated differently depending upon the market. In the United States, the term OREO is used to refer to other real estate owned, real estate being as in all countries one of the most common forms of property used to secure a loan. In the Philippines, for instance, the term ROPOA, real estate and other property owned or acquired, is a more all-encompassing term that is equivalent to foreclosed assets. The foreclosed assets ratio, depicted in the equation below may be considered together with the NPA ratio.

$$\text{Foreclosed assets ratio (\%)} = \frac{\text{Foreclosed assets}}{\text{Total loans}} \times 100$$

The foreclosed asset ratio provides a useful adjunct to the NPA ratio and to the NPL ratio by showing what proportion of a bank's loan book is taken up by foreclosed assets. The same can be said of the restructured loans ratio described in the next subsection.

Restructured Loans Ratio

Although data on restructured loans can be more difficult to obtain than data on NPLs and foreclosed assets, if available, it is a helpful ratio to monitor. The restructured loans to total loans ratio is shown below.

$$\text{Restructured loans ratio (\%)} = \frac{\text{Restructured loans}}{\text{Total loans}} \times 100$$

NPA Coverage

The NPA coverage ratio is defined as total loan-loss reserves divided by nonperforming assets. It parallels the NPL coverage ratio. NPAs are merely substituted for NPLs in the ratio's denominator. An adjusted NPL coverage ratio (not shown), which corresponds to its NPL counterpart, can also be applied.

$$\text{NPA coverage ratio (\%)} = \frac{\text{Loan-loss reserves (LLRs)}}{\text{Nonperforming assets (NPAs)}} \times 100$$

Ratio Distortions

The interrelationships of the various indicators described above can sometimes give rise to results that are superficially deceptive. Owing to the lag between loans being made and the recognition of nonperformance, problem loans will appear to be relatively low when a bank or banking industry is expanding rapidly, particularly when a substantial number of new banks have entered the market, perhaps as a result of financial deregulation. In a similar fashion, when a bank adopts more conservative policies such as cutting back on new lending or increasing its loan-loss provisions, its

asset quality will often begin to look worse in some respects. A contracting loan book will push up its NPL ratio in nominal terms, while increasing loan-loss provisioning in response to defaults will reveal problems previously hidden. An illustration appears in the following box.

RATIO DISTORTIONS

Consider the hypothetical Gardenia Bank in the Republic of Tropicalia, an imaginary emerging market. Gardenia Bank is growing extraordinarily rapidly together with many of the newly formed banks following deregulation by the Tropicalia government. During its first seven years the bank grows at an annual rate of between 15 and 40 percent, as seen in Exhibit 7.13.

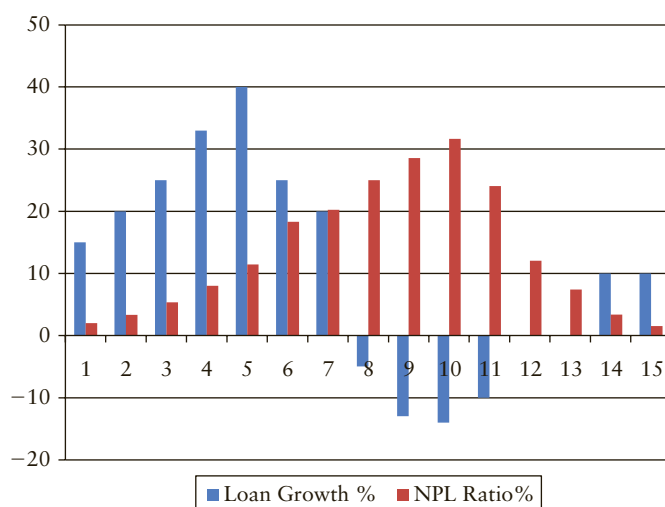


EXHIBIT 7.13 The Masking Effects of Explosive Loan Growth

Due to poor credit management, among other reasons, NPLs double each year over the same period. The bank's NPL ratio remains in the single digits for the first four years, not bad for an emerging market. By Year 5, however, the rapid expansion is beginning to take its toll as the NPL ratio reaches the double digits, doubling from 11 to 20 percent by Year 7.

New management takes over the bank that year and begins cleaning up the balance sheet. Loan growth is stopped, and the bank calls in existing loans. It also begins writing off bad loans and with the help of an injection of capital from new owners rebuilds its capital. But between Years 8 and 10, even as NPLs are stanchd and then reduced, the NPL ratio continues to rise as a function of the ratio's shrinking denominator.

Finally, in Year 11 following huge charge-offs taken over a period of four years, the NPL ratio is pushed down to a moderate level, and eventually the bank is restored to a modicum of financial health. The behavior of the various indicators during this period shows how they may easily be misinterpreted if viewed only in nominal terms.

Credit Cost Ratios

Several asset quality ratios seek to measure *credit costs*, which are echoed in the relationship between loan-loss provisioning charges (LLPs) and net charge-offs (NCOs) on the one hand and some other variable such as total loans or preprovision income on the other.

Net Charge-Offs to Loans

Probably the most useful of these credit cost ratios is net charge-offs to loans, illustrated in the following equations, because it relates the rate at which provisions are being made to the rate at which bad loans are being removed from the balance sheet.

$$\text{Net charge-offs (NCOs) to total loans} = \frac{\text{Net charge-offs}}{\text{Total Loans}}$$

$$\text{Net charge-offs (NCOs)} = \text{Total charge-offs} - \text{Recoveries}$$

Example: Utile Bank's total loans in Period 1 amount to \$200 million. Total charge-offs of problem loans were \$1.5 million, and total recoveries were \$500,000 resulting in net charge-offs of \$1 million. NCOs to total loans were therefore 1 percent.

Banks vary in the aggressiveness of their charge-off policies, which are to a large degree dependent upon the regulatory and tax regime in which they operate. Therefore, the value of the information provided depends greatly on the regulations in the applicable market governing write-offs. Regulation may encourage or discourage a rapid write-off of problem loans. For instance, in the United States at present, write-offs of bad loans are tax deductible while loan-loss provisioning is not. In other markets, it may be the reverse.⁶⁴

Although write-off policy varies considerably globally, the NCO-to-loans ratio still can be of benefit in comparing peer banks in a given market. But the usefulness of the ratio remains limited since even where aggressive charge-offs are encouraged, it does not indicate whether the write-offs are the result of forward-looking prudence or after-the-fact remedial action.⁶⁵

Recoveries to Write-Offs (%)

Recoveries pertain to loans that have been written off from which some partial collection is made or that return to performing status following write-off. As a rule, write-offs will precede recoveries. If a loan returns to performing status before being written off, it is usually not treated as a recovery.

The recovery to write-off ratio will therefore vary with:

- The aggressiveness of the bank's write-off policy
- The average loss-given-default (LGD) ratio of the bank's loan book

As just mentioned, write-offs will be affected mainly by accounting rules, bank prudential regulations, and the relative conservativeness of bank management.

A bank with a conservative reporting policy will write off loans aggressively, albeit within reasonable bounds and in compliance with local rules. A less conservative bank will tend to write off problem assets less aggressively in the hope that eventually the loan will return to full performing status prior to write-off.

LGD—the amount recovered on a defaulted loan—will vary according to a number of factors. These include:

- The industry of the borrower
- The seniority of the loan
- The extent to which it is secured by collateral
- The strength of creditors' rights in the country in which the loan was advanced

The meaning of the recoveries-to-write-off ratio, shown in the equation below, will correspondingly differ depending upon what combination of factors is pushing write-offs and recoveries up or down.

$$\text{Recoveries to write-offs ratio (\%)} = \frac{\text{Total recoveries}}{\text{Total charge-offs}}$$

Example: Larch Bank registered charge-offs in the most recent financial period of \$136 million. Recoveries for the same period were \$32 million. The recovery-to-write-off ratio was therefore 23.5 percent.

Loan-Loss Provisions to Total Loans and Related Ratios

Other credit cost ratios include LLPs to total loans and LLPs to net income (see equations below). Because they do not compare the level of provisioning with the magnitude of a bank's bad debt problem, neither of these indicators is especially useful although they do give some idea of how hard a bank is trying to raise its overall reserves.

$$\text{Provisioning to Total Loans} = \frac{\text{Loan-loss provisions (LLPs)}}{\text{Total loans}}$$

Example: In Year 1, Spruce Bank set aside loan-loss provisions (LLPs) of \$100 million against a loan book of \$10 billion. In Year 2, the bank increased its provisioning by 25 percent to \$125 million against a loan book that grew 20 percent to \$12 billion in the same period. Its LLP to total loan ratio was 1.00 percent in Year 1 and 1.04 percent in Year 2 since the 25 percent rise in provisioning was nearly matched by the increase in total lending.

$$\text{Loan-loss provisioning to net income} = \frac{\text{Total recoveries}}{\text{Total charge-offs}}$$

Example: In Year 1, Pine Bank set aside loan-loss provisions of \$100 million against net income of \$125 million, resulting in an LLP/NI ratio of 80

percent. In Year 2, the bank's earnings fell by 25 percent to \$93.8 million, while its LLPs fell 5 percent to \$95 million. Despite the decrease in provisioning, Pine Bank's LLP/NI ratio increased to 102 percent since the contraction in total loans outpaced the reduction in provisioning.

While possibly helpful in some circumstances these indicators are for the most part not worth the time it takes to compute them and are listed principally for reference purposes.⁶⁶ Without knowing the bank's proportion of NPLs or the regulations to which management is subject, it is impossible to discern whether a high level of provisioning evidences conservatism—that the bank is prudently building up its loan-loss reserves against some anticipated adverse event—or whether the adverse event has already occurred and the bank is scrambling to restore reserves that have been (or will soon be) lost due to write-offs.⁶⁷

Credit Costs and Profitability: The Capacity to Absorb Bad Loans

The impact of *credit costs* on *profitability* was discussed in Chapter 6, where it was noted that poor *asset quality* can adversely affect earnings. Recall that this occurs because a high level of problem loans results in reduced interest income and a higher burden in the form of provisioning requirements that eat into the bank's net profits. From the opposite perspective, bank profitability can be a factor in evaluating asset quality.⁶⁸

Holistic Nature of Asset Quality Evaluation

At its core, the evaluation of asset quality is holistic in nature. It is not enough to look at the extent to which a bank's loan book has been impaired. It is also necessary to consider a bank's capacity to absorb such losses as well as management's ability to prevent further asset quality degradation in the future.

Loan-loss reserves and shareholders equity are two bulwarks against bank failure resulting from eroded asset quality. A third is profitability. In theory at least, a relatively profitable bank will be better able to withstand a crumbling loan portfolio, earn its way out of problems, restore capital, and rebuild loan-loss reserves in the process. Hence, all other things being equal, a highly profitable bank can be seen as having better asset quality than one that is less profitable. As an illustration, a bank that focuses on high-risk, high-yield lending may have better asset quality than a bank with comparatively low-risk, low-yield earning assets, so long as the more aggressive bank properly estimates its credit costs and prices its loans accordingly so as to be able to set aside adequate reserves to compensate for expected losses.

NCOs to Preprovision Income and LLPs to Preprovision Income

The following ratios compare credit costs, as measured either by loan-loss provisioning or net charge-offs, to operating income streams in order to gauge the ability of the bank to absorb expected loan losses without consuming capital. They are therefore not so much measures of quality of the existing loan book, but instead function mainly as indicators of a bank's prospective capacity to respond to

problem assets. The ratio of net charge-offs to preprovision income is shown below; loan-loss provisions to preprovision income are also depicted below.

$$\text{NCOs to preprovision income (\%)} = \frac{\text{NCOs (Total charge-offs - Recoveries)}}{\text{PPI (Preprovision income)}} \times 100$$

Example: Yew Bank had charged off £100 million in bad loans in the last financial year but recovered £20 million resulting in net charge-offs (NCOs) of £80 million. Preprovision income was £240 million. Thus, its NCO-to-PPI ratio was 80/240 or 33.3 percent.

$$\text{Loan-loss provisions to net interest income (\%)} = \frac{\text{LLPs}}{\text{Net interest income}} \times 100$$

Example: Tulipwood Bank generated \$2,256 million in net interest income for the year, and set aside LLPs of \$423 million. Its LLP/NI ratio was 18.75 percent.

$$\text{Loan-loss provisions to preprovision income (\%)} = \frac{\text{LLPs}}{\text{PPI}} \times 100$$

Example: Hemlock Bank registered operating income of 1,800 million and preprovision income of 600 million at December 31, leaving 1,200 million in preprovision income. The bank set aside provisions of 480. Hence, its LLP-to-PPI ratio was 48/120 or 40 percent.

Loan-Loss Provisions to Net Interest Income and Net Charge-Offs to Net Interest Income

The numerators in the foregoing ratios can also be compared against other streams of income, such as net interest income, as shown below. The first depicts loan-loss provisions to net interest income while the second shows net charge-offs as a proportion of the same revenue stream. Like the previous indicators, they provide some indication of a bank's ability to absorb credit costs, but do not say much about the current state of its loan book.

$$\text{Loan-loss provisions to net interest income} = \frac{\text{LLPs}}{\text{Net interest income}}$$

Example: Tulipwood Bank generated \$2,256 million in net interest income for the year and set aside LLPs of \$423 million. Its LLP/NI ratio was 18.75 percent.

$$\text{Net charge-offs to net interest income} = \frac{\text{NCOs}}{\text{Net interest income}}$$

Preprovision Income to Net Loans

Another ratio that may be useful for comparison purposes in this regard is preprovision income (PPI) to (average) net loans (see below.) The higher the PPI-to-net-loans ratio, the more bad loans a bank can afford to write off in a given year. For example, a bank with a ratio of PPI to net loans of 4 percent can provide against or write off 4 percent of its loan book every year. In other words, it can, in theory, afford an NPL ratio of 4 percent before it has to dig into existing reserves to cover loan write-offs.

$$\text{Preprovision income to net loans} = \frac{\text{PPI}}{(\text{Average}) \text{ Net loans}}$$

Example: If Walnut Bank's preprovision income was \$108 million, and its average net loans \$2,790, the bank's PPI-to-net-loan ratio was therefore 3.87 percent meaning that it could absorb a loss of 3.87 percent of its loan book each year without experiencing a decline in coverage ratios.

NCOs to NPLs

The ratio of net charge-offs (NCOs) to net loans previously examined merely shows how fast a bank is actually shedding NPLs from its balance sheet but ignores the rate of NPL formation. For example, if a bank has a loan book of \$10 billion and annual NCOs of \$50 million, that \$50 million figure indicates that an equivalent amount of NPLs are being written off the balance sheet. The ratio does not, however, provide an indication as to what percentage of NPLs is being written off. It may be that NPLs are rising at \$100 million annually, in which case only half are being charged off and the remainder is piling up on the balance sheet. Or the bank may be writing off the NPLs at a faster pace than they are arising thereby reducing any accumulated NPLs.

The ratio of NCOs-to-NPLs shown in the following equation addresses the foregoing deficiency in the NCO-to-net-loan ratio by affording some indication of the relationship between the pool of NPLs and the time it will take to bring aggregate NPLs down to a base level, assuming no new NPL formation. For example, if total NPLs amount to \$100 million and NCOs for the year were \$20 million, then in five years all NPLs would be removed from the balance sheet, assuming no new NPLs form during the period. Thus, if NCOs are high relative to average total NPLs then they are less apt to accumulate on the balance sheet, and the bank will tend to report a steady or decreasing level of problem loans. If, however, NCOs are low relative to NPLs then NPLs are more apt to accumulate. This can build up to a large overhang against capital threatening solvency, unless correspondingly large levels of provisions are set aside against them.

$$\text{NCOs to NPLs (\%)} = \frac{\text{NCOs}}{\text{Average NPLs}} \times 100$$

Example: Walnut Bank's net charge-offs (NCOs) were \$28 million. Average NPLs were \$42.5 million. The bank's NCO-to-NPL ratio was therefore 65.9 percent.

Assuming sufficient financial disclosure, it can be helpful to compare the rate of new NPL formation with NCOs for the relevant period (see below). If NCOs exceed new NPLs, then the ratio will of NCOs to new NPLs will exceed 1, and the pool of aggregate NPLs will decline. If new NPLs exceed NCOs (and the ratio is below 1), then the pool of NPLs will swell.

$$\text{NCOs to NPLs (\%)} = \frac{\text{NCOs}}{\text{Average NPLs}} \times 100$$

Example: Balsa Bank registered \$110 million in NPLs at the end of Period 1 and \$100 million in NPLs at the end of Period 2 resulting in a net negative change of \$10 million. NCOs for the period were \$11 million. Therefore new NPLs were \$1 million. NCOs exceeded new NPLs.

Net new NPLs = NPLs at end-Period 2 minus NPLs at end-Period 1 plus NCOs

$$\text{NCOs/New NPLs} = \frac{\text{NCOs}}{\text{New NPLs}}$$

Example: Mansonia Bank's NPLs at the end of Year 1 were \$250 million, and at the end of Year 2 they were \$275 million. Net charge-offs (NCOs) during Year 2 were \$25 million. Since the bank's NPLs increased by \$25 million between the end of Year 1 and the end of Year 2, notwithstanding NCOs for the period of \$25 million, it can be assumed that new NPLs registered during the year were \$50 million. Since new NPLs exceeded NCOs, aggregate NPLs rose. The ratio of NCOs to new NPLs was 25:50 or 50 percent.

Example: Kapur Bank's NPLs at the end of Year 1 were \$250 million and at the end of Year 2 they were \$250 million. Net charge-offs (NCOs) during Year 2 were \$20 million. Since the bank's NPLs remained constant and NCOs were \$20, it can be assumed that new NPL formation between the end of Year 1 and the end of Year 2 was also \$20 million. NCOs equaled new NPLs and therefore aggregate NPLs remained constant. The ratio of NCOs to new NPLs was 20:20 or 100 percent.

Example: Melunak Bank's NPLs at the end of Year 1 were \$250 million, and at the end of Year 2 they were \$200 million. Net charge-offs (NCOs) during Year 2 were \$70 million. Since the bank's NPLs declined by \$50 million and NCOs were \$70 million, it can be assumed that new NPL formation between the end of Year 1 and the end of Year 2 was \$20 million. NCOs exceeded new NPLs, and therefore aggregate NPLs declined. The ratio of NCOs to new NPLs was 70:20 or 350 percent.

NCOs to LLPs

Depending upon applicable charge-off policies, the NCOs-to-loan-loss-provision ratio provides a clue to the extent to which a bank's loan-loss provisioning is preventative and the extent to which it is reactive. If LLPs far exceed NCOs, it suggests

that the bank is either building up or rebuilding loan-loss reserves. If LLPs are close to NCOs, it suggests that the bank is maintaining a level of provisions and simply replacing those reserves soaked up through the write-off of bad loans. If LLPs are substantially below NCOs, it suggests that the LLRs are being eroded more rapidly than they are being replaced and that the bank is playing catch up and responding to an asset quality crisis.⁶⁹

$$\text{NCOs to LLPs} = \frac{\text{NCOs (Net charge-offs)}}{\text{LLPs (Loan-loss provisions)}}$$

Quantitative Asset Quality Checklist

Definition and discretion. How are nonperforming loans (NPLs) defined? What is the difference between NPLs and overdue loans? What degree of discretion does the bank have in defining NPLs?

Amount of NPLs. What is the volume of NPLs and the ratio of NPLs to total loans? How do these compare with the previous period? What are projected NPLs for the current year?

Amount of classified loans. What is the volume of classified loans (substandard, doubtful, and loss) in absolute terms and their respective ratios to total loans?

Restructured loans. What is the volume of restructured loans in absolute terms and their ratio to total loans? What criteria, if any, must be satisfied before restructured loans are considered to be performing?

Foreclosed assets. What is the volume of foreclosed assets in absolute terms and in relation to total loans?

Security. What percentage of loans are secured? What percentage of problem loans?

Write-offs. What was the volume of loans written off in the last fiscal year? What were the net write-offs?

Loan-Loss Provisioning. What amount of provisioning was set aside in the last fiscal year? How does it compare with the previous year? Are there any regulations or regulatory requiring provisions or disincentives limiting provisions?

Loan-Loss Reserves and Coverage. What is the bank's total stock of (cumulative) loan-loss reserves? What is it in relation to total loans and problem loans (NPL coverage)?

LOAN BOOK COMPOSITION, CREDIT CULTURE, AND OTHER SOFT FACTORS

You can't take a sharp picture of a fuzzy object.

—Joseph Grundfest, former SEC commissioner

It should be evident that the evaluation of asset quality, like all aspects of bank analysis, includes both quantitative and qualitative elements. Having explored aspects of asset quality analysis, it is time to turn to the softer side of asset quality evaluation; that is, those facets of asset quality assessment that are somewhat less quantifiable than those were examined in the preceding section. These include a bank's:

- *Business and regulatory environment*
- *Credit management policies and procedures*
- *Composition and risk profile of the financial asset portfolio*⁷⁰

The business environment, discussed in connection with the economic and credit cycles, and the regulatory environment are addressed separately, which leaves loan portfolio composition and credit culture, policies, and procedures to be developed in the coming sections.

Review of Loan Portfolio Composition

Inferences about the quality of a bank's asset portfolio and prospective trends can be drawn from the composition of a bank's loan book. While ratio analysis is backward-looking, loan portfolio composition may suggest problems that have yet to be captured in asset quality indicators. Specifically, attributes of the portfolio including average loan maturity, industry of the borrower, and borrower type may suggest potential concerns.

Maturity/Tenor

By tenor, loans are typically classified as *short-term*, *medium-term*, or *long-term*, defined as follows:

- Short-term loans: Term of one year or less
- Medium-term loans: Term of one to five years
- Long-term loans: Term in excess of five years

A large portion of bank loans will ordinarily be *short-term*. This is partly a consequence of the bulk of bank of funding being short-term in nature. Banks rightly seek to avoid an excessive mismatch between their earning assets and funding liabilities, although some disparity is inevitable.⁷¹ In addition, short-term exposure is as a rule less risky than medium- or long-term exposure: there is simply less time for things to go wrong. From an analytical perspective, therefore, a proportion of medium- and long-term loans that is significantly greater than the bank's peers implies higher credit risk and calls for further investigation.

Borrower Type

Lending can be *wholesale* or *retail* in character. Wholesale lending means that the bank is lending in large denominations mainly to businesses, including financial institutions. Retail lending refers to lending made to individuals, households, and small or family-run businesses.⁷² Neither wholesale nor retail lending is necessarily more risky than the other. While both wholesale and retail lending each imply a certain type of borrower, the criteria can be further refined as to the types of consumers or firms within the broad categories.

Within the *wholesale lending* sector, a bank might focus on lending to larger (blue-chip) companies and the *middle market* within its geographic franchise, or alternatively concentrate on providing finance to small and medium-sized enterprises (SMEs). On average, larger companies are apt to be more creditworthy than SMEs, but yields will be lower on corporate lending than on middle-market lending. So a bank that concentrates on lending to mid-sized and small companies may be prone to a higher level of NPLs but may generate sufficiently ample yields to cover the higher risks entailed.

Finally, so-called *policy lending*, which can occur in either a wholesale or retail context, raises separate concerns. Policy lending—also called *directed lending*—means lending for noncommercial purposes in furtherance of government policy ends. Most policy lending is in effect subsidized lending, the subsidy being borne either—reluctantly—by the private owners of the bank or by the taxpayers through the government. Because subsidized loans are priced lower than the risks associated with them would warrant, they are potential agents of asset quality erosion. The relationship between policy lending and state ownership of industry is a delicate matter.

POLICY LENDING AND STATE-OWNED ENTERPRISES

Policy lending, the lending for noncommercial policy-driven purposes mandated by government, commonly is associated with an implicit or explicit promise that support will be provided by the government in exchange for the bank's engaging in this otherwise risky activity. The promises, however, may not be as sweeping or firm as creditors sometimes assume. When a local or provincial government is the primary backer of the enterprise, decisions at the national government level will take precedence over local authorities, and support may be withdrawn.

Lending to state-owned enterprises (SOEs) also can raise somewhat difficult issues of analysis since it is regularly linked to the problem of policy-lending SOEs, which may comprise a substantial portion of the economy in countries with a socialist legacy and are frequently poor credit risks. Often loss-making, almost invariably inefficient and reliant on state subsidies—of which policy lending is the primary one—SOEs have a tendency to “not service their debt obligations in a timely manner,” as Robert Morris Associates’ “A Guide to Analyzing Foreign Banks,” put it. In many countries, the banking regulators allow banks to treat assets to these companies as fully performing despite massive delays in payments. More importantly, poor performance in the nongovernmental sector of a bank's portfolio may be masked by the continuous accrual of interest on NPLs.

In evaluating the presence of high SOE lending in a bank's portfolio, special attention must be paid to the relationship between the bank and the government and particularly to the scope and depth of government support. Generally speaking, banks wholly owned by a national government and dedicated to a specific policy purpose are the safest bets for government backing. They are frequently regarded as being quasi-sovereign risks. Those depending upon support by subnational, local, or provincial entities tend to be justifiably regarded as greater credit risks.

Review of the Loan Portfolio by Sector

Another way of categorizing loans by borrower type is by dividing loans into functional groupings, five of which are described below. Default rates vary considerably within each category and from market to market, so generalizations are difficult to make. Probably the most salient consideration in evaluating the character of a bank's loan portfolio is whether there is adequate diversification, a point discussed further below. Bank loans may be further classified by a more detailed sectoral breakdown, and each country has its own customary groupings. These are often designated by the local bank regulator.

Commercial and Industrial Loans

Commercial and industrial loans are loans made either on a short-term basis to provide working capital to meet seasonal or specific needs or, less frequently, medium- and long-term loans used to fund capital expenditure or infrastructure projects (project finance).⁷³ *Trade finance*, which includes off-balance-sheet finance such as the provision of documentary *letters of credit*, while often seen as a separate category of lending, can be viewed as a subset of commercial and industrial loans.

Consumer Loans

This category, which is comprised of loans to individuals and families, spans a wide variety of types, including residential owner-occupied mortgage loans, credit card loans, personal loans, and auto loans.

Real Estate Loans

These include loans to finance all types of real estate, including the development of and investment in residential and commercial property, as well as for the purchase of houses, apartments, and other real property. Home mortgage loans are sometimes put into this category, but may also be classified as consumer loans.

Agricultural Loans

These include loans to farmers and ranchers to fund seasonal planting and livestock-raising operations, among other things. Such loans may be secured by the crops harvested.

Interbank and Financial Institution Loans

This category includes loans made to other banks, as well as to finance companies, securities houses, and other financial organizations.

Making Sense of Loan Composition: Overconcentration

Putting all one's eggs in a single basket is a proverbial recipe for disaster. Likewise, a bank that made all its loans to just one borrower would obviously be vulnerable to any vagaries in the financial health of that customer. As with all investment, diversification is a key to minimizing risk. Equally, disproportionate lending to a narrow range of borrowers or borrower types poses perils for banks, especially in view of their high intrinsic levels of leverage. Such overconcentration can manifest itself in a number of ways that will vary according to market conditions and by the governing regulatory framework.⁷⁴

Relevant questions in regards to overconcentration include the following:

- In what geographic or trade area does the bank do or intend to do business? Is it highly concentrated in one city or region?
- How fragmented is the local banking market?
- Does the bank have international exposure? To what countries? What percentage of lending is outside the bank's home market?
- What proportion of the bank's lending is in a foreign currency as opposed to being in the local currency?
- Does the bank have a high concentration of loans in any particular region or sector, especially those that are depressed or that are especially vulnerable to sharp cyclical or secular changes?
- Is the bank subject to policy lending obligations? Are those obligations greater or lower than peers?
- Who are the bank's top 10 borrowers? What percentage of the portfolio does each represent? Are any of these borrowers in distress or potential distress?

Identifying overconcentration in the loan portfolio must, of course, be accompanied by an evaluation of its significance. It is, however, hard to generalize about the credit implications of particular categories of lending without reference to the historical experience in that country and the expertise that a specific bank has developed. The types of loans a bank makes will depend upon the market (and submarkets within which it operates), its size, and its strategy. In Hong Kong, for example, because the property industry is so important to the local economy, a very high proportion of bank loans are made to fund home purchases and for other property-related purposes. In other markets, savings and thrift banks specialize in home mortgages, and commercial banks do comparatively little business in the field. This said, within particular markets, however, the analyst will ordinarily rapidly gain an understanding of what sort of lending tends to be problematic.

Some rules of thumb are suggested in Exhibit 7.14, with the caveat that there is no substitute for familiarity with the local market conditions and some clue as to the bank's level of experience and expertise in a particular type of lending.

Overconcentration by Industry

When too high a percentage of its loans are made to firms in any given industry, the bank becomes vulnerable to a downturn in that particular industry. Concerns are heightened when a significant proportion of loans are made to vulnerable sectors.

Overconcentration in a Vulnerable Industry

The higher the level of risk associated with the sector, the lower the threshold of danger. Consequently, overconcentration is especially dangerous when the industry is one in decline; highly cyclical in character; or prone to drawing excessive credit, creating an asset bubble—absent ameliorating factors. Hence, even seemingly low nominal percentages of lending to vulnerable sectors can raise a red flag. The property sector, particularly lending for the purposes of funding real estate development or speculation, is a prime example. Others include the oil and gas and computer chip sectors. Excessive lending to a mature industry that is in decline, or is about to be negatively affected by new regulations or deregulation, can raise similar concerns. Another danger arises when banks themselves are the target of deregulation.

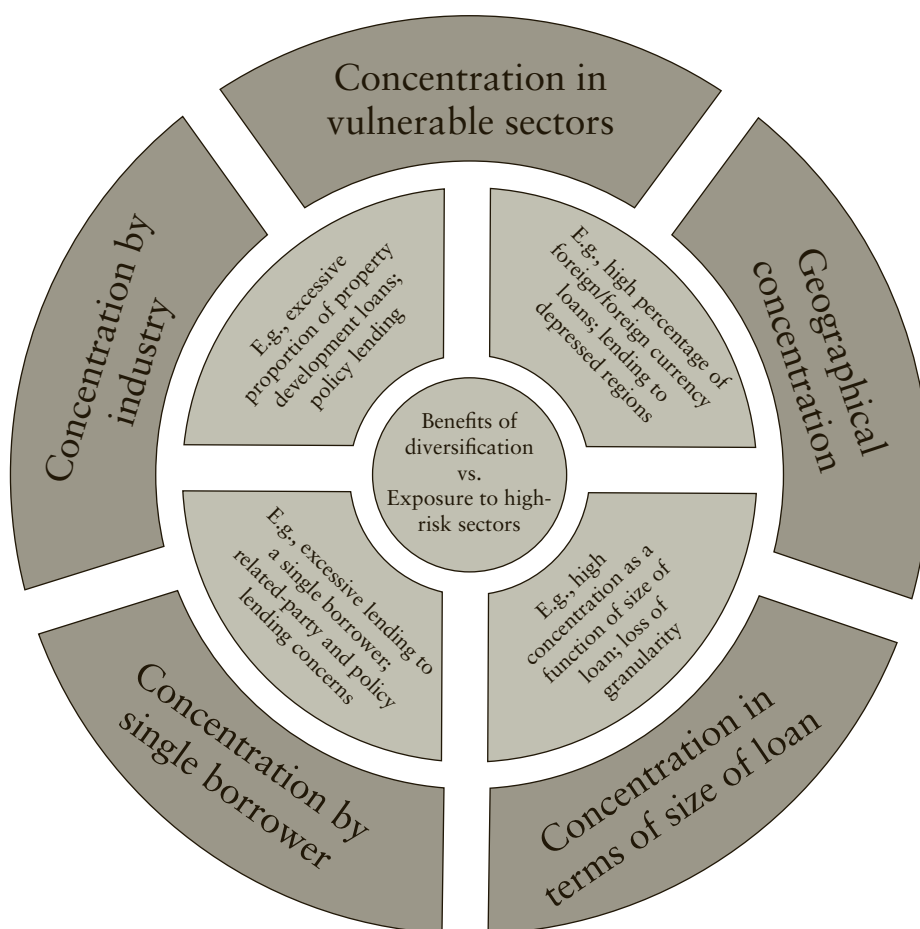


EXHIBIT 7.14 Types of Overconcentration

BANK DEREGULATION AND BANK RISK

Deregulation has been correlated with bank failures. Recently privatized government-owned banks are particularly vulnerable to financial deregulation.⁷⁵ Specifically, banking crises are more likely to occur within a few years after deregulation than before, as banks that were once protected by central bank support and artificially high profit margins are unable to cope in the more competitive environment that ensues. Typically, prior to privatization their primary business was lending to large but inefficient state-owned enterprises or regulated private corporations. Despite high credit costs relative to spreads, they survived by virtue of an implicit government guarantee of deposits and occasional cash injections.

Deregulation allows the entrance of new players into the financial market, including foreign players, who are often able to optimize their risk/return ratios

and outperform stodgier institutions. Bank staff brought up in an environment where loans were made to state-owned companies or other large enterprises with few questions asked are likely to find it difficult to make the transition to one where credit risks must be rigorously examined and the concomitant returns sufficient to justify the risks.

The credit culture in such banks is not likely to change immediately with privatization, especially if support continues. Yet if the culture is not reformed, such banks' lack of experience in making commercially viable loans may result in the bank making errors in a more competitive environment, errors that prove fatal.

Overconcentration by Geography

Too high a percentage of loans made in a particular region within its franchise area makes a bank vulnerable to an economic downturn or a cataclysm affecting that locale. Depending upon the size of the bank, the overconcentration may be in one urban or rural district, city, province, or even a country. When the region concerned is one in decline or in a cyclical depression, potential problems are aggravated.

The very structure of the banking system in some countries may affect diversification.

OVERBANKED MARKETS AND BANK DIVERSIFICATION

In some markets, such as Thailand, Canada, and Australia, a small number of banks dominate the entire country. In these circumstances, banks tend to be well-diversified. In other markets, the United States being the most noteworthy example, the banking system is highly fragmented, and many banks are associated with particular towns, cities, and regions. This type of industry structure tends to work against the diversification of a bank's loan book, especially geographic distribution.

Foreign Lending and Foreign Currency Lending

A distinction can be drawn between foreign lending and foreign currency lending, as they raise separate risk issues. Although foreign loans are often made in a foreign currency, foreign lending and foreign-currency lending are not the same. It is possible for a bank to lend funds overseas in the currency of its home region, in which case the loan would be a foreign one, but not a foreign-currency loan. In the same way, a bank may conceivably make a domestic loan in a foreign currency, as well as in the local currency.

Domestic foreign-currency lending can give rise to excessive aggregate foreign-currency borrowing that, in turn, can pose a severe threat to the health of individual banks and of a country's banking system as a whole. Loans made in a foreign currency create foreign-exchange risk. Loans made abroad give rise to country and sovereign risk, and may raise concerns about the bank's capacity to lend astutely in a foreign jurisdiction.

Overconcentration by Size

A small number of large loans is intrinsically more risky than a large number of small loans.

Overconcentration by Individual Borrower

Too a high a percentage of loans made to a single individual or company holds a bank hostage to the fortunes of that borrower. The analyst should beware of a hidden overconcentration achieved through cross-holdings and the use of nominee companies. Such veiled linkages may be difficult for the analyst to ferret out, however. Overconcentration is particularly hazardous when, in addition, the borrower is a related concern, meaning an insider or a party related to bank shareholders or management.

RELATED-PARTY OR INSIDER LENDING

Related-party lending means lending to individuals or entities that are related to the bank's shareholders or to its management. Related-party lending would occur for instance if a bank lent to a company owned by one of the bank's shareholders. There may be good economic reasons for providing credit to a related party, but in general such loans are suspect. During financial crises, bank failures are not uncommonly associated with excessive related-party lending. In many jurisdictions, bank regulators require related-party lending to be disclosed or limit such lending to a specified percentage of a bank's total portfolio. Unfortunately, related-party lending is often accomplished by subterfuge, and regulations proscribing it are thereby circumvented.

Overconcentration in Real Estate Lending

Real estate-related lending has been so often implicated in cases of bank distress, that it deserves some additional attention. As observed, the property market tends to be highly cyclical in nature. Long boom periods are usually followed by steep and rapid busts. Unfortunately, over the decades banks have demonstrated a fatal attraction to property-based lending. Not all such lending, however, is equally risky. At one extreme, loans to real estate developers are among the riskiest category of property-related lending. Such developers are subject to severe cash flow problems when the market turns. Lending to purchase existing real estate for investment purposes is less risky than real estate development. Finally, home mortgage lending to finance owner-occupied housing, which is often categorized as consumer lending, is likely to be comparatively low risk, shelter being a basic human need and homeowners with positive equity in their dwellings highly inclined to make their mortgage payments. The various categories⁷⁶ of real estate-related lending are illustrated in Exhibit 7.15.

EXHIBIT 7.15 The Real Estate Connection: Banking and Property Exposure

Type of Real Estate Exposure	Description	Risk Level	Remarks
Real estate development	Development of raw land or redevelopment of existing developed areas for sale or lease	High	Real estate development is subject to numerous risks including the developer's ability to service debt and complete the project, construction risks, location, marketing, and economic risks.
Real estate investment management	Leasing and management of commercial or residential real estate	Medium	No completion risk. Loan may be secured by real estate or by contract cash flow.
Commercial or residential mortgage lending (for investment purposes)	Lending for the purchase of commercial or residential structures for investment purposes	Low-medium	Main risk is inability to rent at a level to cover debt service or decline in market values.
Owner-occupied residential mortgage lending	Lending for owner-occupied single-family occupancy of a residential unit	Low	Every individual's need for housing and the limited costs savings of alternatives reduce defaults. May be classified as consumer lending.
Hidden real estate loan	Loan ostensibly for other purposes diverted into real estate development or speculation	High	Issues of loan security and risk management.
Collateral against non-real estate loans	Use of real estate as security	Low	In itself not real estate lending. Decline in value of property may impair value of security (collateral).

Appraisal of the Bank's Credit Culture, Policies, and Procedures

Evaluating the effectiveness of a bank's credit risk management encompasses both an examination of its formal policies procedures, as well as the informal credit culture that underlies them (discussed in the following subsection and the accompanying box). Formal procedures may very well appear prudent, but it will be hard for an analyst to answer the following questions and thereby discern how they are being applied in practice.

- What is the attitude of the bank's managers toward credit review?
- What incentives, institutional or pecuniary, may affect credit decisions?

Although an appraisal may be difficult without meeting bank management in person, much can be gleaned from a bank's annual report, and from third parties.

Role of the Analyst and Asset Quality Assessment

Learning about and evaluating a bank's credit policies and procedures is difficult without making a bank visit and meeting with management, since such information may not be fully disclosed in public reports and the rules without a culture of enforcement are of little value. Rating agencies have an edge in this regard since their independent status entitles them in many jurisdictions to inquire about nonpublic information and gain an understanding of actual credit risk practice at a particular institution.⁷⁷

Nonagency analysts, of course, must rely on public data, although a bank visit can be helpful especially in respect to unrated institutions or those in markets where disclosure is lacking. Often, for reasons of practicality the nonagency analyst will be unable to meet with management and will have to rely upon publicly available information and the bank's financial statements. Assuming a modicum of transparency, the analyst ought to be able to make a rough assessment of the bank's asset quality using only ratio analysis.

Depending upon the level and reliability of disclosure, as well as the regulatory environment, such assessments should be viewed as provisional. To some degree, the counterparty credit analyst may need to rely on rating agency reports for an evaluation of credit policies and procedures. Where such information is lacking, it may be wise for the analyst to err on the side of conservatism.

Taking account of the preceding qualifications concerning disclosure, bank management should be able to articulate its goals and objectives in definite terms.⁷⁸ In this context, management should be able to describe its current franchise, growth expectations,⁷⁹ and the type of lending it wishes to target and avoid. For instance, a bank may want to emphasize consumer credit card business and home mortgage loans, but avoid auto loans, which in some markets have elevated default rates. Likewise, in the SME sector, a bank may want to focus on working capital and trade finance, but avoid medium- and longer-term project finance.

Credit Culture

The term *credit culture* refers to bank management's overall commitment to effectively managing credit risk.

WHAT IS MEANT BY CREDIT CULTURE?

Credit culture, like the term *corporate governance*, is an implicitly normative term. It ordinarily refers to a set of shared beliefs as to the desirability of lending being done on the basis of prudent and profit-oriented criteria. It involves a combination of prudent rules and policies based on experience.

A strong credit culture usually includes the following characteristics:

- Integrated credit risk management, including a comprehensive approach to asset quality and credit risk management.
- An explicit approval system for granting loans, with clear delegation of authority and accountability.
- Diversification of risk by avoidance of overconcentration in particular segments, sectors, or firms.
- Separation of the loan marketing function from the credit function (coupled with the independence of the latter from the former).
- Incorporation of credit quality concerns into the staff performance review process.

A bank that lacks a credit culture, or has a weak credit culture, is likely to be characterized by a significant proportion of its lending being done for noncommercial or imprudent reasons resulting in poor asset quality. Policy banks arguably lack a credit culture because their *raison d'être* is noncommercial; that is, to achieve the government policy objective. Yet policy lending ostensibly has overt and clearly defined aims. More problematic is the commercial bank in which lending for noncommercial reasons is pervasive or where government influence occurs beneath the surface. The latter could manifest itself through a legislator's request that a company in this representative's district be favored with a loan.⁸⁰ Where bank officers owe their jobs to political patronage, refusal of such requests may have personal repercussions. Whether the result of explicit government policies or more nebulous political criteria, such lending is often associated with a high proportion of NPLs and correspondingly low asset quality.

Ideally, a bank should have developed a sophisticated credit culture that imbues its lending function, and that is expressed in both written credit policies and established operational credit approval and loan monitoring procedures. A bank with a strong credit culture thus will tend to have both well-defined implicit and explicit guidelines and standards that govern the bank's lending. The absence of a credit culture suggests that loans are made without rigorous concern for credit standards that take account of willingness and ability to repay the loan.

Such lack of concern may be the result of overaggressiveness on the part of bank's management, who become preoccupied with expansion and gaining market share at the expense of prudent credit control.

The credit culture of an individual bank can be hard to separate from the environment in which it operates. Every market in a sense has a prevailing credit culture. For instance, in many markets, particularly emerging markets, after relationship, collateral is more or less the sole criterion in the decision whether or not to lend. This is the so-called pawnshop mentality of which some emerging market banks have been accused. A related issue is the matter of collateral quality whereby the nominal value of collateral cannot be taken at face value owing to weaknesses in the legal infrastructure and creditors' rights.

PAWNSHOP MENTALITY AND COLLATERAL QUALITY

The term *pawnshop mentality* refers to an overreliance on collateral to approve a loan. Where collateral is properly valued, rights to foreclosure are enforceable, and a deep and liquid market for the assets exists, dependence on collateral in lieu of financial analysis may be justified.⁸¹

It depends upon the proper valuation of the collateral, its quality, and any accompanying legal risk. The collateral assets may be overvalued, so when an economic crisis occurs, collateral that once appeared to be sufficient may be worth a fraction of its nominal value. Or the prevailing legal infrastructure may be so weak, or fraught with corruption, that enforcing the security is difficult or infeasible. Laws governing bankruptcy or insolvency may be antiquated or unduly biased toward debtors.

Alternatively, the laws may exist on paper, but in practice are all but impossible to enforce in a timely manner. Indeed, the process may be so time consuming and expensive as to make a mockery of the time value of money. Finally, in certain countries, some individuals or the companies they own may be so powerful, as to in practical terms be immune from legal process.

Noncommercial reasons for granting a loan could include among other things family relationships,⁸² acquaintanceship, and social or political stature. At one end of the spectrum, a loan may be granted on the basis of the perceived standing of the borrower, or because of a personal relationship—for instance, because the bank's chairman plays golf with the director of the firm seeking the loan. In its milder manifestations, so-called name lending or relationship lending is not necessarily bad practice. Certainly, knowing the character of the borrower is relevant to the credit process.⁸³ But because this type of lending is typically undertaken without proper financial analysis of the loan applicant and procedural safeguards, it is subject to abuse. For instance, insufficient or poor quality collateral may be accepted. At the other extreme, the basis for the loan may be the reciprocal giving of favors—such as a loan on the one hand, and some business favoritism or political dispensation from a politically well-connected individual on the other—or by kickbacks to individuals. These practices in their more egregious forms are called *crony capitalism*.

Review of Credit Policies and Procedures

Credit policies and procedures concern a bank's overall aims in advancing funds, the corresponding guidelines adopted that govern its lending decisions procedures, and the rules and procedures that have been implemented to ensure effective execution of the governing principles and objectives. A systematic lending strategy and supporting procedures can enable a bank and its shareholders to meet earnings goals with acceptable levels of risk. Direct evaluation of such policies and procedures, however, is principally the province of the rating agency analyst as opposed to the counterparty credit analyst. The former are much more likely to have access to the type of information just described than the latter because the institution that is the target of analysis can be expected, with some justification, to regard much of this information as proprietary in nature.

Credit Policies

Among the matters that ought to be addressed by such policies are:

- Data and documentation required for loan or credit applications
- Lending and credit exposure limits and authority thresholds for officers, and various committee levels
- Limits on concentration whether by sector and geographical-area/lender as well as restrictions on related-party lending
- Documentation requirements, including expected loan covenants
- Eligible collateral and collateral appraisal methods
- Policies governing monitoring of borrowers in general and problem borrowers in particular
- Policies governing loan-loss provisioning and cumulative reserves, write-offs, collection, and workouts.

Lending policies will vary over time and place and from bank to bank. What is prudent in one country or region may not be in another. In mature markets, a wider range of lending activities may be viewed as reasonable in light of industry diversification.

Credit standards, as observed, may also change throughout the business cycle. During a period of robust expansion, credit officers may relax lending standards, believing a more liberal lending policy to be consistent with the reduced default rate that a period of prosperity brings. In contrast, following a collapse in asset prices, a bank may stop fresh lending altogether and enter into a period of consolidation.

A threshold query is whether the bank has a written credit policy. There is no guarantee that a written loan policy will be followed. But the existence of one at least suggests that the bank's managers have given some thought to better managing their credit risks. The bank's intended composition of its loan portfolio should be addressed in its credit strategies and policies. Illustrative questions in respect to lending objectives include the following:

- Is management content with the current portfolio composition?
- In what ways does the bank seek to alter its portfolio mix?

In addition to having articulated loan policies, a related line of inquiry concerns how the bank approaches the decision to grant a loan. Relevant points here include

whether or not the bank engages in name lending, extending credit on the basis of a borrower's perceived status in the business community.

Credit Procedures

A corollary to having clear policy aims and objectives in place is the need to have effective procedures and guidelines to properly manage credit risk at all stages of the loan cycle. A bank credit analyst, however, is not a bank examiner, and a detailed knowledge of the criteria a bank applies in granting and monitoring loans will not be sought in most circumstances, although the rating agency analyst at least will want to see evidence of sensible credit policies. The information provided here is therefore provided chiefly for reference purposes.

To avoid growth in problem credits, it is of course also essential that banks have adopted audit and enforcement measures to ensure that existing procedures are followed and that good judgment is exercised in making decisions about which loans to extend. Adequate procedures would normally include:

- Establishing appropriate lending criteria and policies and procedures for loan approval, as described above
- Securing the loan by obtaining collateral or security in the form of real or other property to which the bank can take title in the event of default
- Monitoring the loan portfolio and keeping abreast of problem loans, while setting aside adequate reserves to take account of foreseeable losses
- Taking appropriate steps when a problem appears with a particular credit
- Taking prompt measures to write off defaulted loans, pursue collection efforts, and to dispose of any property obtained through foreclosure to optimize overall recovery

In practice, not all banks give due attention to the preceding steps. The less developed the banking system, the less sophisticated the bank, the less likely that it will approach the management of credit risk in a methodical way.

Although access to information can sometimes be difficult, the qualitative aspect of asset quality analysis should not be given short shrift. Quantitative metrics such as the NPL ratio may have a superficial precision, but they are also lagging indicators. An awareness of the composition of the bank's loan portfolio, its credit approval policies, and its procedures together with the state of its credit culture will alert the analyst to prospective asset quality problems that have not yet materialized in numerical ratios. Bearing in mind that counterparty credit analysts will often find it infeasible to explore these matters in great detail, relevant lines of inquiry are summarized in the next subsection for quick reference.

Credit Allocation and Monitoring

Credit approval procedures, at the internal level, tend to be hierarchical in nature, the structure of the hierarchy representing a pyramid comprised of various loan committees each having authority over specified lending thresholds. Warning signs in this context include too much loan approving authority at the branch level, and a

mingling of the marketing and credit functions. The latter occurs, for example, where the bank officer originating the loan is also responsible for approving it. When marketing staff have a significant role in credit decisions, there will be an incentive to give the benefit of the doubt to the loan applicant. This is not apt to result in strong asset quality.

Credit review goes beyond the mere scrutinizing of the loan applicant before credit is extended. The bank is likely to also have in place policies and procedures for an ongoing and periodic review of its loans. The purpose of this monitoring is to recognize problem loans at an early stage and to take appropriate actions.

While late payments are conspicuous signs of loan problems, ideally the bank's ongoing credit review and monitoring will take account of developments that may adversely affect the customer's business and herald potential problem assets. Such a review would ordinarily seek to confirm that:

- The borrower's financial condition has not materially changed for the worse, and it remains able to repay the loan.
- The loan documentation is complete and the bank has perfected any relevant security interests in collateral.
- The loan is in compliance with laws and regulations affecting the bank.
- The loan conforms to the bank's loan policy and is likely to be profitable for the bank.

Customarily, credit monitoring policies mandate the review of loans at regular intervals. The degree of attention that can be given to particular loans will necessarily depend upon both the size of the bank and the average size of its loans. In general, the riskier the loan, the more frequently it should be reviewed. Depending upon the size of the bank and the average loan size, a bank may deem certain loans unworthy of regular review as the cost is not likely to be justified by the benefits. Similarly, some banks may undertake review of certain classes of loans only at random. Clearly, however, large loans should be subject to a regular review in any event.

QUALITATIVE REVIEW OF ASSET QUALITY: A SUMMARY CHECKLIST

Credit Culture and General Lending Policy

- How can the bank's credit culture best be characterized? Does management appear to be committed to prudent and rigorous credit risk management? Does it have adequate experienced staff for this purpose?
- Does the bank appear to have a coherent lending policy? Is the lending policy articulated in writing?
- What are the bank's lending targets to specific sectors and in terms of loan growth? Is the bank entering (or has it entered) any new sector?

Portfolio Composition

- What does the composition of the loan portfolio suggest about overall asset quality?
- Is there evidence of overconcentrations in the bank's loan portfolio, particularly in respect of a particular industry, borrower, region or problematic sector, or type of lending?
- Does the bank engage in substantial real estate lending, especially to property development or construction companies? Has the bank made a significant portion of its loans to declining or highly cyclical industries?
- Is a high percentage of loans made outside the bank's main franchise area, particularly in risky markets? What is the bank's proportion of international loans?
- What percentage of loans are effectively investment-grade quality, whose risk of default is extremely remote?
- Is the bank exposed to major problem credits?
- What sectors are responsible for the bulk of the bank's NPLs? Are there any sectors that the bank avoids for lending purposes?
- Does the bank engage in substantial foreign currency lending? If so, how is it funded?

Credit Policies and Procedures

- Does the bank appear to have coherent loan approval procedures? How are they monitored and enforced? Is money being lent for speculative purposes?
- Is there an effort to minimize overconcentration in the bank's portfolio, by loan size, borrower, and industry? Is there substantial related-party lending?
- Do credit officers appear to have sufficient experience or training in credit analysis?
- Do marketing staff members have credit assessment and approval responsibilities?
- Is there adequate ongoing monitoring of outstanding loans? How often are loans reviewed?

NOTES

1. David Stimpson, Editor, *Moody's Investor Service—Global Credit Analysis* (New York: IFR Books, 1991).
2. A bank that has negligible nonperforming assets may be pursuing an overly conservative lending policy. While credit analysts may find little to complain about in such an institution, equity analysts might opine that such a bank is not maximizing the value of its franchise.
3. Besides loans, other forms of *earning assets* include, of course, *debt securities* and *equipment leases*, as well as miscellaneous financial assets that produce predetermined returns. Although *investment securities*, leases, and other earning and nonearning assets are also subject to *default* risks, the loans that banks make are the focus of most asset

quality concerns. The reasons are as follows: first, in the vast majority of markets, loans to businesses and consumers still comprise the lion's share of such assets; and, second, most investment securities are publicly rated by one or more rating agencies. Assuming the ratings are at an investment grade level, the loss rate on a bank's securities portfolio will be comparatively low. Furthermore, owing to the external ratings, there will be more certainty concerning *expected loss* on the portfolio compared with the loan book, since loans as a rule are for the most part unrated by outside agencies.

4. Some banks also subtract collateral cover to arrive at *net loans*.
5. *Credit quality* means the relative absence of credit risk; high credit quality means low credit risk.
6. To *write off* means to declare the value of an asset to be effectively zero—a loss asset—and to remove it from the balance sheet. To *write down* an asset means to reduce its nominal or book value from its original level while letting it remain on the balance sheet. In this book, the terms *write off* and *charge off* are used synonymously.
7. It is, however, possible to have too much of a good thing. A commercial bank with zero credit costs may be constrained by overly conservative lending policies that restrict returns to a below-average level.
8. L. I. Hasan, "Determinants of the Loan-Loss Allowance: Some Cross-Country Comparisons" 33 (Helsinki: Bank of Finland discussion papers, December 2003), 7.
9. *Nonperforming Loan Report: Asia 2002* (Hong Kong: Ernst & Young, 2002), hereafter *NPL Report Asia 2002*.
10. The amount of time past due that a loan must be before being considered nonperforming will vary with the term of the loan and local rules. Usually, an ordinary loan with, for example, a maturity of three years and requiring monthly payments that is past due one or two days will not necessarily be considered nonperforming although it would be technically past due.
11. See *NPL Report Asia 2002*: "Determining the appropriate category for an NPL is a complex, subjective process. The same NPL might be categorized as 'doubtful' by one lender and 'substandard' by another."
12. The term *classified* is ordinarily used to refer to a loan classified as other than normal or performing. But a loan need not be *nonperforming* or even *past due* to be so *classified*. On the basis of economic changes or changes in the financial condition of a customer, a bank may classify a loan as *special mention* and thereby as problematic even if payments or principal are still current.
13. It should be noted that rigid distinctions between these terms are rarely maintained, and they may be used synonymously to some extent.
14. Recall that the concept of *loss given default* was mentioned in Chapter 1, in which expected loss was defined as the product of the probability of default, the exposure at default, and loss given default. But even if it is able to regain some fraction of the value of the loan, the bank is unlikely to come out ahead when a loan fails to perform. While the bank may be able to foreclose the loan and obtain title to the security, which typically will be in the form of real property, it is likely that the circumstances that led the bank's customer to default occurred in less than sanguine economic conditions. Property values may very well be depressed or falling, and collateral when sold may be insufficient to cover the outstanding principal and interest. Ultimately, the lending bank may very well end up recovering only a portion of the funds that it extended.
15. To illustrate, consider the definitions of the terms *gross loans* and *net loans*. *Net loans*, for instance, usually refers to *total loans less loan-loss reserves*. But in some markets banks may deduct collateral cover (i.e., $\text{net loans} = \text{total loans} - \text{loan-loss reserves} - \text{collateral}$) to arrive at a lower nominal net NPL figure. For the purposes of this book, we attempt to employ terminology in common use worldwide, and, while acknowledging that many variant terms are equally acceptable, to the extent possible we will gravitate to

our preferred appellations so as to avoid confusing the reader. Hence, when referring to various terms that may have several possible meanings, unless otherwise indicated we intend the so-called international standard definitions and labels as defined in the main text.

16. The term *bad loan* is an informal term frequently used loosely to refer to problem loans. Depending upon the context, the term may be intended to encompass all classified loans, or it may be used to denote just NPLs, doubtful and loss loans, or only loss loans.
17. It is rare for reclassification to occur in the other direction, particularly once a loan has been classified as a loss loan. Of course, as discussed below, some loans will return to performing status, but these assets are frequently grouped separately and may be subject to eligibility criteria based on continued performance for some period of time.
18. Of course, from the perspective of the borrowers, issuers, and counterparties, these bank financial assets are financial obligations and therefore financial liabilities.
19. In this context, *peace time* means a quiet, comfortable period during which loan quality is maintained.
20. In this context, *war time* means a deterioration of the micro- or macroeconomic conditions.
21. The assumption that there is no claim against the maker of the incinerated machinery is employed to emphasize the destruction of the asset and correspondingly capital. Of course, the bank is likely to have a claim against the borrower and—to stretch the analogy a bit—so might a manufacturing firm have legal recourse against the maker of the machine for product liability.
22. Alternatively, the institution may look to riskier noninterest activities such as proprietary trading to boost the bottom line. In all cases, the likelihood is that risks to the bank will increase.
23. We will discuss this situation in detail in Chapters 14 and 15 concerning bank crises and the rehabilitation of the distressed bank.
24. This assumes that a nonperforming loan is not reclassified as performing.
25. In some countries, even after a loan is commercially considered a loss, a bank would have to keep a nominal amount within its loan assets on the balance sheet to represent that lost loan during legal proceedings, which might take ages.
26. In insolvency proceedings, once satisfied privileged claims under local laws, and absent enforceable covenants to the contrary, each class of creditors will ordinarily rank equally with other class member—that is, on a *pari passu* basis. The Latin *pari passu* means “with equal step,” or that the treatment of all involved will be without partiality or favor.
27. Until regulations on such matters are reinforced either locally or under international pressure.
28. “Allowance for Loan and Lease Losses—Comptroller’s Handbook,” published and updated by the Comptroller of the Currency in the United States.
29. Things have improved since, but a report by Salomon Smith Barney published in 2001 compared the policies governing cessation of interest accrual in six key Asian markets. It found that in Thailand, for example, that the sole regulatory criterion was the “length of the past-due period.” There the borrower’s ability to repay was irrelevant. Contrasting with Thailand’s policy was that of Hong Kong, where the only consideration was “the bank’s subjective judgment of a debtor’s ability to repay” so long as the obligation was fully secured and overdue less than 12 months. (However, it was observed that as a practical matter the length of time the loan was overdue was taken into account in making this judgment.) In further contrast, in Singapore, after three months had elapsed, a bank was no longer allowed to continue to accrue interest income, unless fully provided for (through loan loss provisions). Finally, the bank was only permitted to recognize any interest income received on a cash basis.
30. Clawbacks are reversals of interest income already accrued.

31. Op cit.
32. In some jurisdictions, the bank will be expected (or permitted) to set aside provisions only against specifically identified problem loans (specific provisions). In others, an additional provisioning requirement for as-yet unidentified problem loans (general provisions) will be mandated or customary.
33. "Unearned income" in this context means interest income nominally generated by the loan that has been received or accrued but not yet earned. Examples of unearned income include payments of interest on loans subsequently deemed to be nonaccrual loans or in respect to any unearned portion of the discount on discounted loans (i.e., where the funds advanced are less than a 100 percent of the amount due at maturity). Accordingly, net loans can be defined as the outstanding principal amount of the sum originally advanced, less loan loss provisions representing the bank's anticipated loss on the portfolio less any income deemed to be unearned.
34. Likewise, if an erstwhile problem loan returns to current status and has reached maturity having performed in full provisions charged against it having remained unused will be returned to the pool of loan-loss reserves.
35. Op cit.
36. A. Lok, *Banking Guidebook: Nomura's Standardised Financial Statements and Ratios* (Hong Kong: Nomura Securities, Asia Pacific, November 1, 1999) (hereinafter *Banking Guidebook*): "We believe . . . that the general provision should be used . . . in assessing solvency [and not] to assess the adequacy of provisioning unless [it] is higher than the statutory minimum . . . since any bank that runs down its general provisions to cover NPLs is only deferring the . . . charge to later periods when it must build [it] back to the statutory minimum." As we shall see, loan-loss provisions were eligible as part of Tier 2 capital under the original Basel Accord, so long as the provisions in question were properly classified as general provisions not specifically identified with a particular problem loan.
37. Murray N. Rothbard, *America's Great Depression* (Richardson and Snyder, 1963), 20, cited in Dr. Jim Walker, "Asianomics: The Unbearable Knowledge of Impotence: From Austria to Asia" (Credit Lyonnais Securities Asia, May 11, 1998), 3.
38. Paul Krugman, *Return of Depression Economics* (New York: W.W. Norton, 2009), 71.
39. Business cycle theory was a key element of the Austrian school of economics as developed by economists such as Joseph Schumpeter and Ludwig von Mises, which held that business cycles were inevitable. The Austrian school was largely subsumed by the theories of John Maynard Keynes, who believed that the business cycle could be smoothed out if not eliminated by government intervention. See Dr. Jim Walker, "Asianomics: The Unbearable Knowledge of Impotence: From Austria to Asia," Credit Lyonnais Securities Asia, May 11, 1998. Further exploration of this somewhat arcane corner of macroeconomics need not be pursued in any detail here, as our approach is for the most part practical rather than theoretical. Nevertheless, the relevance of business and economic cycles to the banking industry is widely acknowledged and provides the background for a good deal of comment concerning the banking industry. For our purposes, description of archetypal cycles, or patterns, provides the analyst with a useful shorthand that facilitates a better understanding of the impact of macroeconomic change on the banking industry.
40. The very term *business cycle* is arguably misleading. Although business cycles are associated with periods of boom and bust (and bull and bear markets), the ups and downs in the business cycle as it affects a particular economy do not necessarily lead to financial crises. However, such crises occur frequently enough, and when they do they highlight in their aftermath the causes that engendered them. For an accessible discussion of the anatomy of a financial crisis from an historical economic perspective, the classic work written by Charles P. Kindleberger and Robert Z. Aliber, *Manias, Panics and Crashes: A History of Financial Crises*, 5th ed. (Hoboken, NJ: John Wiley & Sons, 2005), provides a

refreshing contrast to much of the literature on the subject, which is inclined to be on the one hand highly theoretical research heavily laden with econometric formulas, or on the other the screeds of doomsayers and gold bullion salesmen. We discuss banking crises and their impact on banks in more detail in Chapters 14 and 15.

41. This list is a paraphrase of items enumerated in a study by Joseph F. Apadoford, "Credit Quality: CEOs Set the Tone," *Magazine of Bank Administration* (June 1988) cited in George H. Hempel, Donald G. Simonson, and A. Coleman, *Bank Management: Text and Cases*, 4th ed. (New York: John Wiley & Sons, 1994) t, 371.
42. Op cit.
43. By *annualized* is meant the equivalent annual rate; for example, a quarter-on-quarter growth rate of 4 percent would imply an annual rate of 16 percent, as if that growth had prevailed throughout the year and disregarding any compounding effect.
44. Note, however, that total assets growth gives signals that are less telling than loan asset growth due to the spectacular increase in nonloan assets in many jurisdictions.
45. During boom periods, 50–60 percent year-on-year loan growth is not unprecedented, but such extreme rates of expansion often are followed by disastrous consequences.
46. See Philippe F. Delhaise, *Asia in Crisis: The Implosion of the Banking and Finance Systems* (Singapore: John Wiley & Sons, 1998), 125.
47. Twenty percent growth would be less worrisome if the industry average were 40 percent, but such a figure still suggests that an asset bubble is building and that the pace of lending and other asset quality indicators should be closely monitored. An exception arises where, for example, the high rates of loan growth are the result of a merger or acquisition and the financial statements of the entities are consolidated.
48. Depending upon the jurisdiction, NPLs may return to performing status and be permitted to be classified as such, provided certain criteria are met. Attention must be paid, however, to the relapse rate. This is the rate at which loans that have returned to performing status, often following restructuring, default and go back to being nonperforming.
49. While falling NPLs arising from aggressive write-offs are generally favorable, where the analyst is comparing banks on a cross-border basis, differences may lead to erroneous conclusions. In one country where aggressive write-offs are permitted, NPLs may fall sharply or never rise to conspicuously dangerous levels while in another the problem loans remain on the balance sheet and provisions are set aside to cover them. The net effect is essentially the same, notwithstanding that one bank has nominally low NPLs but high NPL coverage, while the other has nominally low NPLs but lower coverage.
50. Example: Hypothetical Ebony Bank had an NPL ratio in Year 1 of 0.1 percent. In Year 2, the bank's NPL ratio rose to 0.3 percent. The average among its peers is 1.9 percent. Despite the fact that Ebony Bank's NPLs tripled (rose by 200 percent), the bank remains relatively sound, and it cannot reasonably be said to have an asset quality problem.
51. It should, however, be kept in mind that both the volume of nonperforming loans and the nonperforming loan ratio are historical and lagging indicators. In view of the aging criteria applicable in most markets, loans having a maturity of two years or less, which make up the lion's share of most bank loan portfolios, will not manifest NPLs until at least three months have elapsed with perhaps the majority manifesting within 6 to 12 months of being advanced.
52. The ratio occasionally goes by other names, such as the impaired assets ratio.
53. Aside from disclosure and definitional issues, it is also possible that a significant difference or variance may be attributable to fundamentals that in some circumstances may mitigate its impact. At the micro level, banks differ from one another in their risk appetite and business franchises. For these reasons, this ratio should be supplemented with measures of charge-offs as well as indicators that define problem assets more broadly and

that consider problem asset coverage. Of course, as mentioned previously, the rate at which a bank writes off loans will also affect the NPL ratio.

54. *Banking Guidebook*, 8.
55. This Latin expression means “other things being equal.”
56. Depending upon the phase of the credit cycle, banks may have coverage ratios that are more than double or triple NPLs. In these circumstances, the level of coverage acts more as a way to rank banks that have otherwise similar asset quality ratios.
57. In a distress situation, both loan-loss reserves and capital function act as buffers against asset impairment, and therefore, equity coupled with NPLs, provides the broadest measure of a bank’s capacity to remain solvent amid a rising tide of problem loans.
58. To be sure, realistically, not all NPLs will result in 0 percent recovery, and so some degree of conservatism is built into the 100 percent benchmark. In addition, so long as the bank has access to liquidity, it can continue to operate as a going concern, notwithstanding a theoretical shortfall in its ability to cover problem loans. But, as we have noted, it is hard to predict when NPLs will peak, so it is probably fair to insist that banks that are to be considered the most creditworthy meet or exceed this standard.
59. The same can be said of similar ratios that use loan-loss provisions such as LLPs to total loans.
60. That is, without knowing to what extent LLRs cover problem assets, it is impossible to determine whether asset quality is superior or inferior to the norm. (This is why NPL coverage is much more useful.) Reasons for significant changes in the percentage upward or downward can be worth pursuing, however.
61. In contrast, where foreclosure or equivalent procedures allow a bank to take title to collateral relatively quickly, nonperforming loans can be held down or even reduced as foreclosed assets rise.
62. To be sure, this calculation merely approximates actual nonperforming assets. Certainly, some restructured loans may currently be performing, while foreclosed assets will in most cases afford some measure of recovery. Including all of these items as nonperforming assets may therefore seem unduly conservative. However, the cost of maintaining foreclosed property may not be reflected in the financial statements of the bank as a credit cost. Finally, since in many markets the prevailing tendency is to underreport problem assets this approximation of NPAs serves to compensate for that bias.
63. Consequently, it is not uncommon for a bank to end up recovering only a portion of the funds that it extended, even if at the time the funds were advanced the collateral was nominally sufficient to cover the outstanding loan.
64. In yet other markets, regulators cap the level of gross reserves such that additional provisioning beyond that level is no longer tax deductible. This is a conspicuous disincentive to additional provisioning.
65. Just as there are a number of determinants of loan-loss provisioning that vary in significance, the same can be said of loan write-offs. Low levels of write-offs may imply that there is a disaster waiting to happen, while high volumes of write-offs may be the result of the loan book having just been cleaned up. Finally, write-offs will lag behind new loan growth so the significance of increasing write-offs may be masked in the same way as rising NPLs.
66. Unexplained fluctuations or trends in loan-loss provisioning levels, however, may provide provocative lines of inquiry for the analyst. Nevertheless, when evaluating a bank’s asset quality, it is almost always better to concentrate on the key asset quality ratios such as the NPL and NPA ratios, NPL coverage, and NCOs to PPI (or LLPs to PPI) together with qualitative evaluation of a bank’s loan book, than to spend significant time calculating and perusing this category of supplementary indicators.
67. In other words, merely knowing how much a bank is provisioning in a given year relative to gross loans says very little about whether the amount of loan loss provisions is adequate

- to the volume of problem assets. Moreover, the ratio alone provides almost no indication, and it is impossible to discern whether the bank is acting preventively or remedially.
68. One should note, however, that absolute levels of profits or of loan losses do not always adequately capture the balance between risk and reward in a loan portfolio.
 69. These hypotheses assume that the bank is able to write off its problem loans expeditiously. However, as already observed, differences in regulation may affect the ability of banks to engage in preventative provisioning as well as to aggressively write off bad assets.
 70. Concededly, each of the preceding features is quantifiable to a degree, particularly the first and last items. GDP figures, unemployment rates, and industry loan growth, for example, can give some indication of state of the economy. Similarly, the percentage of a bank's advances that are unsecured, concentrated in high-risk sectors, or made to friends and relatives of the bank's directors and top managers may shed some light on the riskiness of its loan portfolio. The ability of a bank to manage its credit risk, however, is largely a matter of judgment, although some tangible measures, such as the extent to which credit policy is documented, may function as a sign of a more deep-rooted credit culture.
 71. Asset liability mismatch is discussed again in the context of funding and liquidity in later chapters.
 72. Note that *retail lending* is not synonymous with *consumer lending*. Retail lending encompasses consumer lending, but not the converse.
 73. *Project finance*, which often includes other types of financing besides bank finance, is usually medium to long term in nature, although bridge loans for project finance purposes may be short term.
 74. A critical point to consider is that the benefits of greater diversification may be outweighed by the hidden hazards an expansive lending policy may entail. A bank that lends in many different countries and to too many different types of borrowers may be taking on board fatally risky assets in its quest for high levels of nominal diversification.
 75. In a similar fashion, financial deregulation can pose a threat to hidebound banks that practice name and good old boy lending.
 76. Note that lending collateralized by property, although mentioned here, need not necessarily be categorized as real estate related. It is not uncommon for real estate to be used to secure other types of loans.
 77. In addition to having created effective credit approval procedures, a prudent bank will maintain pristine loan files, with full documentation. A detailed examination of this aspect of credit control is more an issue for the bank examiner than even for the rating agency analyst.
 78. For example, a bank may inform a rating agency analyst that its strategy is to develop a portfolio that is 40 percent large corporate, 40 percent middle market, and 20 percent consumer, in the process reducing its emphasis on corporate and middle-market lending.
 79. Earlier, the relevance of loan growth to asset quality was discussed. New banks often want to grow their loan books very rapidly, and this is facilitated by the low base from which they are expanding. More mature players often find it difficult to expand their loan base in a competitive market and may be content with modest growth. The degree of aggressiveness, of course, will have some bearing on the amount of attention that is paid to credit quality. At least, the agency analyst, if not the counterparty credit analyst, should be able to obtain some projection from management of the coming year's loan growth.
 80. This practice has been referred to as *behest lending*.
 81. Recall our warning that collateral can be used to circumvent the need for credit analysis.
 82. Lending to insiders or their families or business associates falls under the rubric of "related-party lending."
 83. At best, *name lending* and *related-party lending* can be defended on the grounds that they represent lending based upon a perceived understanding of the borrower's character and earnestness. More often, however, they are a poor substitute for systematic credit analysis, and at worst, mask cronyism and poor credit practices.

Management and Corporate Governance

I made a mistake in presuming that the self-interests of organizations, specifically banks and others, were such that they were best capable of protecting their own shareholders and their equity in the firms.

—Alan Greenspan, Chairman of the U.S. Federal Reserve 1987–2006,¹

I've met more masters of the Universe than I would like to, people who were grossly over-rewarded and didn't recognize that. . . . Let us be quite clear: there has been mismanagement of our banks.

—Lord Myners, City Minister²

[A] company's corporate governance can have a material impact on its credit quality, particularly where governance practices are weak. Academic research shows a statistical link [and] the experience of recent corporate scandals—in which governance mechanisms were in many cases not strong enough to check errant management behavior—also highlights the negative credit implications of exceptionally poor corporate governance.

—Fitch Ratings³

A bank's performance and financial condition, which are described by the various indicators defined in the preceding chapters, are a product of the environment in which the institution operates, of the decisions taken by its management, and of chance. Although the business and regulatory conditions are largely outside management's control, how the bank responds to its operating environment is within its province. Consequently, while luck will inevitably play a part in affecting a bank's relative success in achieving stated objectives, management remains the critical variable that determines a bank's success or failure. As a 2003 investment report observed:

*Banks are creatures of regulation—which govern balance sheet make up (through liquidity and reserve requirements; capital requirements; concentration limits) and . . . profitability (margins—through shape of yield curves, provisioning, capitalization). . . . [T]he operating environment—economic circumstances (economic development, cyclical positioning), geography and demographic profile are key determinants of profitability. . . . Given restrictions on most parameters of business, banks that consistently deliver better profitability do so mainly due to better management.*⁴

Indeed, management's influence is pervasive, its decisions affecting nearly all aspects of the institution's operations and resulting credit profile. As a result, to comprehensively evaluate a bank's credit risk profile, the quality of its management must be considered.⁵

Management within the CAMEL framework refers primarily to the competence of the bank's managers, a necessarily subjective judgment of their experience, expertise, and vision. Whether management or bank owners act dishonestly is another important matter. An evaluation of management may also include other qualitative concerns such as the bank's ownership and organizational structure, as well as the capability of staff. Although quantitative indicia may occasionally function as plausible proxies for more intrinsic attributes—for instance, funds spent on staff training and compensation and incentives may possibly correlate with staff competence—ultimately an assessment of management is very much a qualitative exercise. In addition, the governance of the institution is relevant to an assessment of management's abilities, as will be discussed in more detail in the second part of this chapter.

While there is a consensus concerning the importance of management as a factor affecting bank creditworthiness, and to the overall attributes desired in management—leadership, competence, integrity, and so on—there is much less agreement over how to go about its evaluation. Unlike with the other CAMEL elements, there are no key ratios almost universally employed and corresponding benchmarks, nor is there wide concurrence over how to integrate management assessment into the overall credit analysis process. In addition, the systematic appraisal of corporate governance matters is relatively new. Accordingly, the evaluation of a bank's management tends to be more individualized according to analyst and organizational requirements than the analysis of the institution's financial attributes. For this reason, in comparison to preceding chapters, this one is less prescriptive and more exploratory in character than the chapters dealing with the analysis of earnings, asset quality, capital, and liquidity. No single recipe is provided. Instead, the objective is to consider the impact of management—together with the criteria by which its quality and competence can be judged—and the quality of a bank's corporate governance on the institution's credit profile. To this end, the following section will examine management's role and provide an overview of management evaluation.

AN OVERVIEW OF MANAGEMENT APPRAISAL

Because management quality is an attribute that is not directly measurable, it is different in kind from capital, asset quality, earnings, and liquidity—and more difficult to evaluate. The difficulty has increased as the challenges confronting management have grown. In decades past—and perhaps still in a few locales—running a bank was not an especially demanding role. In the United States, for example, until banking deregulation began to accelerate in the 1970s, the strict regulatory apparatus put in place following the Great Depression of the 1930s afforded senior bankers a comparatively easy life. The so-called 3–6–3 rule applied: collect deposits at 3 percent, lend them at 6 percent, and be on the golf course by 3 o'clock in the afternoon. Respectability, prudence, good judgment, and the capacity to inspire confidence were as important as or more important than intellect, training, educational qualifications, and experience. The potential for spectacular returns was more limited, but so were risks.

With banks no longer protected in most markets, either by state acquiescence to industry cartels or by means of protectionist barriers to entry, competition has intensified. Innovation in the financial markets has flourished over the past few decades, with results both good and ill. In this rapidly changing industry landscape, management has been confronted with ever-greater challenges, and the concept of prudent management has become harder to define. At the same time, as the challenges with which management must cope have increased, astute management has become more critical both to a bank's success and to its survival. Choices faced by management require increasingly sophisticated analysis and the evaluation of multiple trade-offs.⁶

Moreover, these same choices, together with the concomitant activities undertaken by the bank, are influenced, to some extent predictably, by the internal constraints and incentives to which management is subject. The same constraints and incentives, that is, the corporate governance of the bank, together with the decisions that result, can have a positive or negative impact on both its owners and creditors. Consequently, a comprehensive evaluation of management undertaken as part of a bank credit analysis should also consider the quality of the institution's corporate governance. In the credit analysis of a viable bank, rather than being treated as a separate element to be evaluated, franchise value is usually reflected in the assessment of management, its track record, and its current financial condition, particularly in respect to profitability and liquidity, as well as the operating environment.

The Function and Importance of Management

In a commercial bank, as opposed to a bank set up to fulfill a specific social policy purpose, management's fundamental objective is to work to maximize the bank's return on its owners' capital in a manner that is sustainable over the long term. This is easier said than done, however, and in reality, at the same time that it is competing with its rivals in the marketplace, management confronts conflicting demands from its shareholders (who are often not a static group) as well as from other actors in the marketplace including depositors, other creditors, and employees, among others.

COMMERCIAL VERSUS NONCOMMERCIAL BANKS

The objectives of management depend, generally speaking, upon whether a bank is primarily a profit-making commercial institution, or whether it serves non-commercial policy objectives set by government or charter. The overarching objective of commercial enterprises is, of course, to maximize sustainable returns for their shareholders. Strong management is vital in this regard because, as suggested in the preceding introduction, it is the key variable that determines the enterprise's ability to survive, to prosper, and to optimize its profits. Although the route to maximizing shareholder value is not always obvious, it is widely accepted that a bank's sustainable level of profitability is closely linked with the competence of its management. Similarly, the quality and competence of a bank's managers has an undeniable impact on the financial condition, and corresponding credit quality, of the institution they oversee. Banks that are

fundamentally noncommercial in their orientation are typically wholly or majority owned or controlled by the government. As such, they frequently benefit from a greater or lesser degree of state support or subsidy. In such organizations, management, to be sure, plays an important role. State ownership, however, tends to overshadow other criteria in the analytical process because the risk profile of a government-owned bank will ordinarily be regarded as closely correlating with the sovereign risk of the state. In emerging markets, state influence is often imposed in less obvious ways, so as a control over the top management of major private-sector commercial banks.

Although the shareholders' interests are paramount, to pursue their interests effectively, management must see that the bank successfully meets the needs of the markets formed by its customers (for bank loans and other products), by its depositors (for the safety of funds deposited, an acceptable return on such funds, and ancillary services), and by its other creditors. In addition, to acquire and retain competent staff it must satisfy the labor market by providing adequate compensation and benefits. Additionally, of course, it must comply with government directives imposed by the bank regulator as well as other state agencies. Finally, to maintain (or obtain) a given credit rating, it must meet the implicit thresholds set by rating agencies.⁷ In this environment, a significant part of management's job is to balance the demands from each group while looking to the owners' overall best interests.⁸ The behavior of each cohort with respect to the bank is influenced by their objectives, whether monetary or nonmonetary. To some extent, the requirements of one can only be satisfied at the expense of another, although each is broadly cognizant of the fact that the relevant choices they face are subject to the pressure of competition and therefore necessitate trade-offs, such as between risk and return, or cost and convenience. To illustrate, in respect to leverage, equity investors as a whole are more tolerant of high leverage than debt holders and tend to be less averse to the increased risk it implies. The search for easy returns with a blatant disregard for safety was one—and sometimes the major—cause for bank failures in the 2008–2012 crisis. Holders of bank paper and counterparties, like depositors, seek to maximize yield while at the same time minimizing risk within predefined expectations. Generally, they tend to be more risk averse than shareholders, and are sensitive to any significant deterioration in asset quality, impairment of capital, or loss of liquidity that threaten a bank's continued survival.

Beyond the level of risk aversion of owners and creditors, there is a myriad of trade-offs to be made. Customers and depositors respectively want the cheapest rate on loans and the highest rate on deposits. Depositors normally place a premium on the safety and liquidity of their deposits, while borrowers ordinarily focus primarily on the cost of funds. Both generally desire low transaction costs and good customer service. Staff desire to maximize compensation, an objective that is potentially in conflict with the interests of shareholders and creditors insofar as higher wages reduce profits, and poorly designed incentives increase risk.⁹ Regulators' usual overarching goal is the safety and soundness of the banking system. In this light, bank supervisors often focus on curbing banks' exposure to excessive risk through the implementation of widely applicable prudential norms. This tendency is generally tempered, however, by

recognition that banks must maintain adequate profits to remain in good financial health and must thereby further that goal. In practice, reconciling the interests of each group in the overall interest of stakeholders' is facilitated by several factors.

One is that the composition of interest groups overlaps. For instance, borrowers are usually also depositors, and depositors are often borrowers. Bank managers themselves are also staff, and staff may be owners as well through share ownership. Another is that individual actors within each group have different preferences. Some investors may be more risk averse, others less so. Similarly, monetary considerations are not the sole drivers of behavior. An attractive mix of products and services, for instance, may include innovative offerings that fulfill unmet customer needs. Likewise, depositors may be drawn toward appealing nonmonetary benefits such as good customer service or convenience, notwithstanding a bank's not having the highest savings rate.¹⁰ Finally, stark trade-offs are eased in a business landscape whose contours are constantly shifting. By seizing opportunities and avoiding mistakes, a bank can achieve higher returns relative to risk than its rivals.

It is evident that a bank's survival depends, to a material degree, upon management's ability to generate sufficient revenue (i.e., maintain sufficient core profitability), hold down credit losses (i.e., maintain good asset quality), maintain adequate reserves as a cushion against both expected and unexpected losses (i.e., maintain capital adequacy), and upon management's capacity to ensure access to liquid funds (i.e., maintain ample liquidity). In turn, a bank's net profits and profitability depend upon management's ability to control operating costs and to expand revenues through the execution of effective business strategies. By changing the bank's franchise, its marketing approach, its funding method, by effectively exploiting product innovations—or by any combination of these or any number of other strategies—management may potentially succeed in increasing a bank's revenues.¹¹ Alternatively, it may fail to do so and set the bank on a path of decline. It is probably not going too far to ascribe a bank's financial results over any significant period of time to the competence of its management.¹²

Difficulties in the Evaluation of Management

It should be apparent from the foregoing discussion that the quality of a bank's management is a critical element of a credit risk analysis. There is a problem, however, in taking this element into account when analyzing a bank: a lack of quantitative indicators. The evaluation of management is a highly subjective exercise that from a credit perspective is particularly difficult. Past financial results are, of course, measurable. But to what extent a bank's track record reflects well or poorly on its management is hard to ascertain as cause and effect can be obscured by external factors. It is often hard to differentiate the extent to which a bank's current level of credit strength is attributable to benign operating conditions, and luck, as opposed to the astuteness of the management team. During a boom, it requires no great skill to generate healthy profits.¹³ Similarly, management's capacity to effectively respond to future operating conditions will always be a matter of speculation. No matter how stellar the record of a management team, however, its future performance cannot be predicted with certainty.

Despite the hazards of extrapolating from past performance, management's track record tends to be given substantial weight in evaluating management. Owing

to the preeminent importance of profitability in the equity investor's calculus, so long as the markets remain liquid, good share performance is accepted as *prima facie* evidence of management competence.¹⁴ For the credit analyst, however, the task is not so easy. A major reason is that, absent an actual event of default made known to the markets, credit quality is not objectively apparent in the same manner as, for example, share price. Indeed, strong short-term profits generated by a bank with a large appetite for risk can mask a declining credit quality.¹⁵ From a credit vantage point, the ideal management team is one that seeks to strike a happy medium between near-term performance and long-term financial strength that will ensure the bank's capacity to continue to function as a healthy going concern for the foreseeable future.¹⁶ It is most assuredly not one that pursues short-term profit growth at the expense of the institution's long-term financial strength. The problem is that if financial indicia comprising a bank's track record are excluded, there are no easily quantifiable standards for appraisal.¹⁷ So it is perhaps understandable that even credit analysts focus on the financial fundamentals that comprise management's track record—that is, financial performance and condition as registered over three to five years¹⁸—and that the separate appraisal of management tends to be allocated relatively little weight as compared with the weight assigned to capital, asset quality, earnings, liquidity, and other relevant factors.

One way to address the lack of quantitative indicators is through the use of a scorecard approach, illustrated in Exhibit 8.1.¹⁹ Scorecards are commonly used in an effort to quantify evaluations that are essentially qualitative in nature. Typically, they employ some sort of ranking or grading scheme whereby meeting a specific set of

EXHIBIT 8.1 Indicative Scorecard Approach to Management Competence

	Weight (%)	Score	Weighted Score
Management Assessment and Corporate Governance			
Management skills	15%	6	0.9
Management structure	10%	5	0.5
Ownership quality	10%	2	0.2
Risk management structure and policies	25%	4	1
Franchise	10%	5	0.5
External meddling risk	10%	2	0.2
Corporate governance	20%	3	0.6
Weighted Management/Corporate Governance Score			3.90

Risk Factor Score Definitions (from 7 to 1)

7	Outstanding
6	Very High
5	High
4	Average
3	Substandard
2	Well below average
1	Critically low

criteria places a given bank within a numerical range, between, say, 1 and 7 with respect to the attribute for which a numerical score is sought. Scorecards work best when each grade level is described by a number of specific criteria that must be met for the corresponding score to be obtained, as this facilitates the assignment of scores in a reasonably consistent manner.

While scorecards permit the quantification of qualitative judgments, they are subject to the same weaknesses as subjective evaluations of any kind. There is also a significant amount of potential model risk depending upon how the scoring system is constructed and weighted. Problems of consistency can arise where guidelines concerning scoring are ambiguous or analysts receive insufficient training in the use of the model. In the case of the evaluation of management quality, the process begins by identifying the principal criteria that describe the attribute. To accomplish this, management quality could be divided into multiple subattributes, each to be ranked according to grading guidelines.²⁰ Alternatively, the ranking of management quality as a single attribute can be made to depend upon the presence or absence of particular elements or subattributes, while a hybrid approach would combine elements of both where evaluation of the subcriteria is embedded within the overall evaluation of the primary attribute.²¹ Management competence *stricto sensu* is assumed to be comprised of two fundamental elements; namely the quality of effective strategic planning together with the resulting designated objectives, and the quality of leadership and capacity to achieve results as demonstrated by management's track record. The higher the score, the more each attribute has been displayed or has been credibly shown to exist within the management team.²² How the various attributes get selected and scored depend on the analyst's perception of their importance to gauge the creditworthiness of a specific bank or group of banks.

Lines of Inquiry

Whether preparing for a bank visit or seeking to profile management from afar, the analyst should review the bank's past performance and changes in top personnel during the previous several years, and should consider the challenges the bank faces in view of current and prospective operating conditions. In the process, an attempt should be made to ascertain and understand management's strategy.²³ Whether or not an on-site visit is undertaken, the objectives remain similar; for example, to understand the bank's business model, management's future goals and plans and the capacity of management to execute those plans effectively, and to check how the bank's past affects it in the present and going forward. A preliminary assessment can ordinarily be accomplished through the review of written material received from the bank, including its annual report, as well as through background material obtained from the bank's website. The annual report and the accompanying statements of the chairman and president can be particularly useful. Additional background data can be obtained from proprietary news databases such as Factiva, LexisNexis, and Bloomberg, as well as from other print and online sources. If a bank meeting is scheduled, this preparation can serve to provide a base from which lines of inquiry may be pursued during meetings with bank officers. By formulating questions in advance, the analyst will be in a position to make best use of the time available, and should be able to gain a better idea of how the bank's managers conduct the bank's affairs and plan for the future. Finally, in addition to meeting with the bank's senior management, the analyst,

where feasible, should visit with other relevant participants in the bank's marketplace, such as foreign banks, accountancy firms, brokers, and others, to understand these key players' views of the bank's management and operations.

Needless to say, the preliminary assessment, thanks to a thorough analysis of all data and documents available, should inspire a range of questions for the analyst both on the quantitative and on the qualitative sides, including critical lines of inquiry that would help prepare a profile of management. Whether questions are posed before or during the bank visit, at its conclusion the analyst will want to have answers to most of her questions before completing a written analysis.

The following paragraphs will give some guidance to the analyst.

Status

How has the bank grown to its present position? Has it grown internally or by acquisition? If by acquisition, how effectively have the acquired companies been incorporated into the organization? To what extent have new cost savings or efficiencies been realized through mergers? Are any mergers expected in future? What synergies are anticipated from recent or anticipated business combinations?

How old is the bank? Who founded it? Did it originally have a purpose other than commercial banking? Has its mission changed? Is the bank a comparatively new one, or has it survived and prospered through many business and economic cycles?

Ownership and Organizational Structure

Ownership

- What is the current ownership structure? For example, is the institution closely held and dominated by a single shareholder, or are a high proportion of shares owned by the public? Is it licensed as a domestic commercial bank on an equivalent basis to the other leading commercial banks in the country? If not, what is its status?
- If closely held, is it a family-owned and managed bank?
- If listed, is its stock widely traded?
- Is the bank a finance arm of a larger nonfinancial conglomerate, or is its owners' business primarily finance related?
- Is the bank owned in full or in part by the government? Relative to its peers, to what extent, if any, could it be said to be government controlled?
- Was it founded as a government bank? If so, when was it privatized?
- Are there any plans for privatization?
- To what extent is management constrained by government policy objectives? Does the bank have a mission other than commercial banking (e.g., agricultural development) that constrains its operations?
- Is the bank primarily owned by domestic shareholders? If there is significant foreign ownership, what relationship, if any, is there with a foreign bank or financial group? What restrictions, if any, are there upon its business activities in the country?

Organizational Structure (see also corporate governance)

- How does the bank's ownership structure affect management? How does the organizational structure affect its capacity to manage risk and to optimize profits? How hierarchical is the bank? To what extent can the bank be described as highly bureaucratic in character?

- Is there an organizational chart? How are reporting lines and authority mapped? To what degree do actual reporting lines differ from formal ones? If the bank is a subsidiary of a holding company, how much independence does it have vis-à-vis the parent?

Franchise, Market, and Competitive Position

- What is the bank's main franchise? How has it evolved over the years? Are significant changes expected or needed in the near to medium term?
- What market does the bank primarily serve? Is it a global bank, or mainly domestic in focus?
- What is the bank's competitive position generally and with respect to its market niche? Within its market, is it a first-tier, middle-tier, or third-tier player?
- Is the bank too big to fail? (That is, could it expect support from the state should it become distressed?) Does it have a particular market niche in which it is a dominant player?
- What is management's view of the bank's competitive position? Its strengths, weaknesses, opportunities, and threats?
- What does management see as the bank's key competitive advantages?
- What does management see as the bank's weaknesses or areas for improvement?
- What institutions does management perceive to be the bank's most direct competitors?
- How capable is the marketing team? What controls exist against mis-selling or other sources of legal liability or reputational risk?

Leadership, Decision-Making Processes, and Overall Management Competence

Management, Competence, and Style

- How competent is management? What is the level of experience of the CEO and other senior managers?
- Is the bank professionally managed as opposed to being managed by (family or group) owners? How closely are the bank's owners involved in management? What ownership stake, if any, do managers have in the bank?
- What is management's leadership style? Is leadership dependent upon a particular visionary *individual*, or is it effected through a governing committee or other means? Is there significant state interference in management?
- How are decisions made? Are decisions made in an *authoritarian* or *consensual* style? How are *controls* exercised? If authoritarian in nature, is there nonetheless an *esprit de corps*? Do managers give the appearance of having a high degree of *motivation*?

Planning Processes

- How is strategy formed? How well formulated is the bank's strategic planning process? Does the plan consider various possible scenarios?
- Is there a planning department, and, if so, what is its role within the organization? To what extent are outside consultants used in the planning process?
- Does senior management consult with directors concerning planning issues? What input do middle management, risk managers, and other staff have in planning or policy formulation?

Quality of Strategy

- Is there a business/management plan? Are management's objectives over the next three-to-five years, or more, clearly stated and realistic?
- What are the institution's best opportunities for future earnings growth? What challenges does it face?

Risk Appetite

- What is management's appetite for risk? How aggressive is the bank in competing for business?
- How keen is management to expand the bank either through acquisition or organic growth?

Track Record

- What is management's track record in respect to financial performance and in meeting strategic objectives? Is there a track record of strong profitability? Is there a history of adequate liquidity? Is there a history of conservative capital ratios?

Management Integrity and Compliance (see also Corporate Governance)

- What is the reputation of senior management? Does it have integrity? Does it inspire trust?
- Has the bank complied strictly with laws and regulations affecting it? Has it been subject to any sanction by relevant authorities? Is there any reason to think that corners are being cut, or that fraud or criminal activity might be taking place?
- Is there adequate disclosure and transparency in the bank's reporting? Is there evidence of managed earnings, window dressing of the accounts, or of any aggressive accounting techniques being employed?

Quality and Effectiveness of Risk Management and Related Systems and Controls

Information Systems

- How effective are the bank's internal monitoring and forecasting of economic and business conditions? To what extent are outside economic forecasts used?
- Has there been enough investment in information technology to meet the bank's compliance, risk management, and decision-making needs in light of its current and anticipated business activities?

Regulatory Compliance

- Are there comprehensive staff policy manuals? How often are they revised?
- What are the reporting requirements of managers and staff vis-à-vis securities trading and other areas where potential conflicts of interest or compliance issues exist?

Risk Management and Risk Controls

- How effective is the bank's risk management function? Does it conform with best practice standards? If not, what deficiencies exist? Are risk controls and risk reporting fit for purpose?
- Is the risk management function viewed as a critical one by top management? Are risk managers respected within the bank? Do they have enough authority?

- How frequently does the top management meet? Do risk managers participate? Are there effective channels for communication between senior management and senior risk managers? Between junior and senior management?
- Have there been recent changes in auditors or accounting methods, and if so why?

The Bank's "Company Culture" and Its Institutional Strength

Company Culture

- What is the bank's "company culture?"²⁴ Is there a strong credit culture?
- To what extent is the bank an innovator or a traditional institution? That is, is the institution's market known for doing things the way it has always been done, or does it pride itself on product and technological innovation?
- If an innovator, does it innovate out of strength or weakness; that is, is innovation prompted by success in other areas or because its existing franchise is threatened or shrinking and profitability is falling?
- If a traditional player, is the bank in a comfortable market niche, or is it seeking to expand the scope of its operations?
- Does the bank have the resources to support its innovation, or does its reach exceed its grasp?
- How smoothly have past acquisitions been integrated with respect to personnel, processes, and policies? To what extent do factions exist within the bank as a result of prior acquisitions or other causes?
- Does the bank tend to promote from within or does it seek new blood from outside the organization? Is there resistance to new approaches brought in from outside the organization? How is staff morale?
- How target oriented is the bank vis-à-vis sales and lending goals? Is sufficient attention being paid to risk management? What incentives influence the behavior of bank officers and staff?

Institutional Strength: Staff Morale, Training, Treatment, and Compensation²⁵

- How effectively does the bank operate as an institution largely dependent on the quality of its staff?
- How could staff morale be characterized? How are relations between rank-and-file bank staff and management? Between middle or junior management and senior management? Is rank-and-file bank staff unionized? Has there been a history of industrial action?
- How often and on what criteria are staff generally evaluated? What incentive compensation exists; are there stock options? Are salary levels including benefits comparable with peers? On what basis are bonuses or other incentive payments allocated?
- Is training provided to staff, other than on the job? What opportunities are there for promotion? What are the succession arrangements? Is the personnel department actively involved in recruitment? Is there strong investment in staff training and promotion of a unique organizational approach to banking?

The Bank Visit

A bank visit provides the analyst with the opportunity to meet senior bank managers, and is probably the best way to obtain a reasonably complete picture of management's

strategy and capabilities. It is difficult to make this appraisal at a distance, and for this reason a bank visit can be a real aid in evaluating management. The bank visit may be particularly useful where the bank is comparatively new or where data are otherwise limited. Additionally, if there are concerns of an especially subjective nature, a visit may be necessary to coming to a judgment about the bank's future prospects.²⁶ For these reasons, the work of the rating agency analyst and the investment analyst practically requires that each meet periodically with the management of the banks that he covers.

In contrast to bank credit analysts employed at the rating agencies or part of buy- or sell-side fixed-income teams, counterparty credit analysts ordinarily will not have the time or resources to engage in regular bank visits. These visits are likely to be less frequent or limited to institutions that are either being evaluated for the first time or in developing markets where assessment at a distance is especially problematic. The advent of the Internet has consequently been a boon to desk-bound analysts affording much more immediate access than was previously available. Many listed banks make available audio recordings and webcasts of their periodic meetings with bank analysts. This can provide an excellent opportunity to understand the bank's point of view and also to benefit from questions posed by other analysts.

Mention should be made of some of the potential disadvantages of making a bank visit. Aside from the cost and time required, there is also the risk of being swayed by spin over substance. A well-crafted presentation, personable managers radiating charm, or the ability of bank personnel to speak in the analyst's language should not, however, cloud her critical perspective. There also is a danger in agreeing—sometimes for want of another option—to let the bank put forward a specially trained representative to handle questions on behalf of the management.

EVALUATING MANAGEMENT DURING THE BANK VISIT

With the exception of evaluations made as part of a solicited rating by a major rating agency, which can last a day or more, a full and comprehensive appraisal of management will be difficult. Where the invitation to visit is extended as a courtesy rather than as part of a solicited rating, it is likely to be fairly brief, ranging from as little as a half-hour to perhaps two to three hours in the case of unsolicited ratings. In these circumstances, usually the best that can be expected is to form some basic understanding of management's strategy, the plausibility of their plans, and some initial judgment concerning the likelihood of the plans being carried out successfully—while in the process forming some impression of management's capability and integrity.

Although it is commonsensical, nonetheless it may be worth noting that during the bank visit the assessment of management is perhaps best performed indirectly. Naturally, direct questions concerning management competence or integrity may prove offensive. To soften the impact of the inquiry, questions should be directed toward gaining a better understanding of the bank's business. For instance, the analyst will probably want to ask management about recent

initiatives and the future direction of the bank's business. Likewise, instead of asking whether the bank is an innovator or a follower, the analyst might inquire as to what new products or initiatives the bank has introduced during the past year, or what its plans are with regard to a recent trend. The answers may very well reveal relevant information about the institution's strategy, its approach to doing business, its relative position in the industry and so on. When compared with those of its peers, the answers may prove even more illuminating. They will almost certainly suggest related questions to pose.

The deeper aim will likely also be furthered because it is likely that the answers proffered will reveal something about the competency level of its managers in the process. It should also be borne in mind that unless the analyst is conducting a due diligence investigation on the part of a rating agency, or in a similar independent capacity—and even such a status might still face a number of constraints—there may be an understandable reluctance on the part of bank management to be completely open about its business strategy.²⁷

Ideally, the analyst should go beyond a general discussion of the bank's business to gain an appreciation of the process by which decisions are made and the quality of the individuals comprising the bank's management team. These topics are naturally more sensitive than those relating to a general discussion of the bank's franchise, and again they may be better answered indirectly.²⁸ Much can be gleaned through small talk with the individuals concerned about their education, experience, and current role. Rather than ask individual managers about their own situation, questions can be put in the institutional context.

Finally, the nature of a bank's human resource management system, as well as of its bank's recruiting and training programs can also give some indication of the quality of its managers. As with any complex and risk-sensitive organization, management should have a clear organizational structure, clear job descriptions, and reporting lines that permit adequate controls, well-planned recruiting strategies, suitable training, regular performance evaluations, and salary review. In addition, learning what factors are considered to be most important in the performance review of bank staff and officers can suggest something about the qualities valued by the organization.

Nevertheless, familiarity can create comfort. Consequently, most rating agencies send a minimum of two analysts when performing a full rating of a bank. It is also possible that the analyst may also face the opposite situation—one in which management appears to be reluctant to provide information to the analyst, and will disclose comparatively little about operations or strategy. In some instances, this may be attributable to the local business culture, or even the domestic regulatory regime, which militates against frank disclosure of a bank's financial condition. It might also be noted in this context that unless the bank has solicited a rating from a ratings agency, and agreed to the terms of the condition of the ratings agreement, management is under no obligation to meet with a particular analyst. Nevertheless, banks that have correspondence or other business relationships with each other will normally cooperate and meet with visiting analysts.

Moreover, the analysis must consider policies that, in an effort on the part of government to reduce *moral hazard*, may be quite opaque. In sum, this aspect of the analytical process necessarily requires keen judgment, and is in consequence principally qualitative in character.

QUANTIFYING THE UNQUANTIFIABLE: THE EXAMPLE OF RATING MANAGEMENT CAPABILITY

In both bank and corporate credit analysis, management competence figures highly, and its evaluation perhaps epitomizes the attribute that cannot readily be measured in numerical terms. It is not that no quantitative indicators can be found.

We can measure, for instance, the number of senior officers having an MBA, their years' experience in the industry, and so forth. The question that arises, however, is this: How relevant are these indicators to the attribute being measured? Ultimately such objective indicators are perhaps only weak proxies for more incisive subjective evaluations. Bill Gates, after all, was a college dropout, yet he founded one of the world's most successful and profitable companies.

The quantitative constituent of bank credit analysis, as with credit analysis generally, is mainly accomplished through the examination of financial ratios. In contrast, qualitative analysis calls for judgment and insights not readily reducible to figures. To systematize the assessment process, however, it is often desirable to attempt to quantify the unquantifiable.

How, for example, can an evaluation of management competence be converted to a numerical value? To convert qualitative and subjective judgments into numbers, the commonplace scorecard approach probably provides a better method to quantify qualitative judgments.

Resembling the letter grades assigned in schools or the proverbial questionnaire query "rate this item on a scale of 1 to 10," a scorecard acts as a vehicle to convert an analyst's judgments into numbers. For any particular category of criteria, the figures are compared against absolute or relative benchmarks to assign a level of risk to the attribute being measured.

As with the preceding illustration, we might rank management on a scale of 1 to 10, with 10 being a brilliant management team with an excellent track record, and 1 for a management team with no comprehensible strategy and an abysmal track record, retaining its position only through political connections or through family control and nepotism.

Other, largely qualitative, criteria applicable particularly to banks but that also are relevant to some extent in corporate credit analysis include:

- The regulatory environment
- Business climate
- Company strategy
- Industry positioning

These, as well, are amenable to the scorecard approach.

CORPORATE GOVERNANCE

Corporate governance relates to the internal means by which corporations are operated and controlled . . . [and the] relationships between a company's management, its board, its shareholders, and other stakeholders. . . . [It] provides the structure through which the objectives of the company . . . and the means of attaining those objectives . . . are determined. Good corporate governance should provide proper incentives for the board and management to pursue objectives that are in the interests of the company and shareholders. . . .
—OECD²⁹

Corporate governance . . . encompasses the interactions between a company's management, its board of directors, and its financial stakeholders (e.g., shareholders and creditors). This approach emphasizes the financial dimension of corporate governance, and does not focus on nonfinancial stakeholders.
—S&P³⁰

As a result of high-profile corporate and financial scandals during the past two decades, the failure of U.S. corporate Enron in 2001 being a notable example, the issue of corporate governance has become a significant concern to the financial community. The financial crisis that started in 2008 has also revealed a number of severe corporate governance weaknesses in financial institutions. Shareholders, particularly institutional investors, debt investors, rating agencies, government regulators—not to speak of senior corporate and bank managers who are often the target of these concerns—have become increasingly aware of the dangers posed by inadequate oversight of management. The problems and conflicts embraced by the term *corporate governance* are by no means new, although the term encompasses matters that previously were often discussed as separate topics—the principal-agency problem, minority shareholders rights, and so on.³¹ As Adam Smith stated in the *Wealth of Nations* with reference to company directors:

*[Because they are] managers rather of other people's money than of their own, it cannot be well expected that [they] should watch over it with the same anxious vigilance with which the partners in a private co-partnership frequently watch over their own. Negligence and profusion, therefore, always prevail.*³²

So what exactly is meant by corporate governance, and why is it so often a problem?³³ Underlying almost all discussions of corporate governance, and what constitutes best practice, is what is sometimes referred to as the agency problem. This refers to the legal principal-agent relationship that is intrinsic to the structure of the modern corporation in which “management (the agent) makes decisions about the disposition of the company's assets on behalf of stakeholders (akin to the principals).”³⁴ Because the interests of the principal and the agent are not identical, the agent may be tempted to act in a manner that benefits the agent at the expense of the principal. Over time, various legal rules and principles have developed to regulate this relationship, particularly in regard to behavior of the agent that is deemed adverse to the principal.

The economic interests of shareholders and creditors are for the most part quite similar. Both wish to prevent managers from running the firm to their detriment. For this reason, although creditors do not stand in formal agency relationship with management, they might be said to stand in a quasi-principal–agent relationship.³⁵ Finally, there are also issues related to the existence of majority (controlling) shareholders.³⁶

Stakeholders

While the protection of shareholders rights and interests vis-à-vis the managers that serve them constitute the core of corporate governance and provides the impetus for the development of the field, the interests of other stakeholders in the firm have not been ignored. Corporate governance considers a wide range of potential stakeholder conflicts and embraces a wide variety of best practice norms depending upon who is defined as a stakeholder in the firm and consequently included within the circle of interests deemed worthy of protection.³⁷

In examining the impact of corporate governance from a credit perspective, the analyst will naturally almost exclusively focus on the interests of the key financial stakeholders—that is, shareholders and creditors—and their relevant rights, with a particular emphasis on the interests of creditors as opposed to shareholders.³⁸ Nevertheless, since much of the methodology for corporate governance evaluation has been developed from the vantage point of the traditional owner-manager conflict, an examination of the shareholders' perspective when discussing corporate governance is unavoidable.³⁹

Fortunately, however, there is significant overlap in the interests of shareholders and those of creditors. In many, if not most, instances where a corporate governance concern is raised, these interests are similar. In the banking context, for instance, both owners and creditors would look askance at a major loan made to a vice president's wife's company as a favor and not justifiable on purely commercial terms.⁴⁰ There are situations where the economic interests of shareholders and creditors differ.⁴¹

WHEN THE ECONOMIC INTERESTS OF SHAREHOLDERS AND CREDITORS DIVERGE

In consequence of their legal status as holders of a residual claim on a firm's assets, shareholders' returns can vary widely depending upon how the firm performs. If profits are high and exceed market expectations, shareholders may enjoy excellent returns on their investments, reflected in dividends or share price appreciation. Conversely, shareholders may be completely wiped out should the company fail, its liabilities exceeding its assets. In contrast to shareholders, creditors are limited to a fixed return, being able to at best receive what they are due under the relevant loan agreement or bond indenture. Because they do not benefit from a firm's high-risk actions that may generate commensurately high returns, they are naturally more risk averse than shareholders. Simply put, the economic interests of shareholders and creditors are, at times, in conflict. As Fitch Ratings succinctly put it in a special report entitled "Evaluating Corporate Governance: The Bondholders' Perspective":

*[B]ondholders prefer corporate actions that maximize the certainty of full and timely repayment on their obligations [while] equity holders prefer actions that maximize the present value of their investment.*⁴²

The credit analyst should therefore be on the lookout for management actions that benefit the shareholders at the expense of the creditors. In addition, actions that do not necessarily directly impinge upon the creditors' returns, but increase credit risk beyond that expected by them, are also of concern.

One example of such a conflict in economic interest concerns dividend policy. Shareholders directly benefit from higher dividends, while creditors are at least potentially harmed since funds channeled to shareholders as dividends are not available for future debt service and do not augment capital, strengthening solvency, as do retained earnings.

Another illustration occurs when the failure of the firm becomes an imminent possibility. If an enterprise is in dire straits, the shareholders, facing a total loss of their investment, may be inclined to prefer a highly risky course of action that at least offers the possibility, even if small, of some recovery of their investment, if not an ample return. Creditors, however, will favor the prudent course that preserves as many of the firm's assets as possible and increases the likelihood that they will recover most, if not all, of the principal they have extended.

It might seem that shareholders, who after all own the company and appoint management, have all the cards in their hands. But creditors hold several aces as well.

- First, bondholders ordinarily benefit from contractual provisions in the bond indenture that protect their rights and may offer some protection against risky behavior by management. Creditors may hold, or have rights to specified collateral, or benefit from additional covenants in the bond indenture, or in the case of a bank, the loan agreement. So-called negative covenants are intended to protect creditors by prohibiting the firm from taking actions that are detrimental to creditors' rights. Customarily, negative covenants limit a borrower's right to engage in business combinations, borrow beyond a certain amount, or engage in actions that cause specified financial ratios to go outside predetermined ranges.
- A second check on management actions that harm creditors is market discipline. The borrower that the debt markets perceive as acting against creditor interests will be punished by seeing the value of its debt securities decrease and the cost of any new borrowing increase.
- Finally, where the borrower (or counterparty) is a bank, its regulators provide an additional check by proscribing business activities or strategies that the supervisor deems excessively risky. While bank regulators are primarily interested in protecting ordinary customer deposits, their strictures also benefit other creditors by constraining excessively risky behavior by bank management.

Corporate Governance Defined

Corporate governance is a term that nominally refers to how a corporation is governed; that is, how decisions are made and carried out. In other words, corporate governance answers the following question: *In whose interests is a firm managed?* As was discussed earlier, the interests of the various groups that have a relationship with a commercial enterprise may, and often do, conflict.⁴³ Although this topic was raised in the context of the banking industry, the same fundamental underlying divergence in economic interests exists in respect to all companies. While the term *corporate governance* can be used purely descriptively, it frequently has a prescriptive connotation, postulating certain best practices that constitute a model framework at both the firm level and as minimum legal norms affecting all companies.

These best practices embrace fundamental principles of fairness intended to consider and reconcile the needs of the affected interest groups, often collectively referred to as *stakeholders*.⁴⁴ In general, they embrace a variety of structures and practices designed to ensure the effective stewardship of the firm by management on behalf of such stakeholders. Since the interests of each stakeholder group, or subgroup, vary according to their status—as majority shareholders, minority shareholders, bondholders, and so on—the ideal corporate governance regime will differ depending upon which groups it is designed to protect. The question of how to define stakeholders for corporate governance purposes is an important one.

DEFINING CORPORATE GOVERNANCE

Corporate governance has been defined as “the relationship among various participants in determining the direction and performance of corporations. . . .”⁴⁵

The groups that are universally identified as participants are shareholders, directors, and management. Although this definition is purely descriptive in nature, the term *corporate governance* typically has a strong normative connotation, with most commentary aimed at identifying *best practices*, rather than toward merely describing extant structures and behavior. Over what constitutes the proper norms of good governance, however, there is a variety of opinion, partly because the specific rules will depend to a large extent on which groups are deemed to be stakeholders for corporate governance purposes.

There are four basic definitions ranging from narrowly exclusive to broadly inclusive.

1. Under the most exclusive definition, *stakeholder* is seen as synonymous with *shareholder*, and good corporate governance norms are defined as those that enable shareholders collectively to receive the maximum return on their investment in an enterprise. Minority shareholders’ rights are a principal focus of concern.
2. A less exclusive definition takes in as relevant stakeholders creditors, including bondholders, counterparties, and depositors. Some rating agencies, for example, define financial stakeholders as being comprised of shareholders and creditors.

3. A more inclusive definition takes in as relevant stakeholders not only shareholders and creditors but also employees and the long-term health of the corporation.
4. An even more inclusive definition of stakeholder extends so far as to embrace the community as a whole. Commentators who adopt this expansive view see good corporate governance as being directed not merely toward maximizing shareholder value in the short or long term, but also toward extending benefits toward society as a whole.⁴⁶
5. A new, broader, definition of stakeholder has appeared in the wake of the financial crisis that started in 2008. Financial and regulatory reforms are being discussed or implemented with a view to mitigate systemic risk. Such systemic risk goes beyond the simple contagion effect. Globalization and the very advent of risk mitigation tools have increased correlation between portfolios, in turn making improbable events potentially devastating for entire financial systems. Most financial institutions nowadays thus have an indirect stake in the well being of other financial institutions.

In the commercial context, the financial stakeholder approach (the second of these five definitions) takes center stage, the focus being on identifying those best practices that will tend to maximize value to equity investors. Credit analysts, however, are primarily concerned with the interests of creditors and focus on those areas where the interests of shareholders and creditors may differ. One methodology published by an investment bank provides an illustration of this difference: The absence of bank or large creditor representatives on a company's board is regarded as a positive corporate best practice. Obviously, a company's bankers or bondholders might have a different view. Nevertheless, since bondholders do not possess an ownership stake in the issuer, their interest in good corporate governance is limited to actions that may impair an issuer's creditworthiness or otherwise increase their credit risk exposure.

Commercially oriented approaches to the development of corporate governance best practices, and, correspondingly to the evaluation of corporate governance, typically focus on safeguarding the interests of the owners of the firm with a particular emphasis on the protection of minority shareholders.⁴⁷ So enterprises that exhibit all of the following traits would usually be said to display good corporate governance.

- Management *runs the firm for the benefit of shareholders* (rather than for the benefit of management).⁴⁸
- *Transactions are undertaken on the basis of commercial criteria*⁴⁹ (instead of, for example, entering into a transaction because the other party is the CEO's brother-in-law).
- *Shareholders are provided with fair and accurate reporting* concerning the firm's financial results and its operations.

Since management cannot monitor itself, another body must act on behalf of stakeholders' interests. That body is the board of directors, which is charged with

supervising management and looking out for the long-term interest of the company and its owners. It should therefore be unsurprising that there is a broad consensus that an independent and competent board is a prerequisite to good corporate governance. Among the board's most crucial functions are monitoring and proscribing common types of management behavior that have historically produced results detrimental to the interests of the firm and its principal stakeholders. The specific abuses will vary according to the particular corporate governance approach and relevant circumstances, but with respect to commercial enterprises transactions undertaken for noncommercial criteria such as related-party transactions and the diversion of firm resources for managers' personal use almost always raise a red flag.⁵⁰ Consequently, treatment of related-party transactions and management remuneration are commonly used as discrete categories of criteria to be assessed as part of an overall corporate governance evaluation.

Finally, stakeholders cannot properly evaluate management's success or failure without access to reports concerning the firm's activities and performance during the preceding financial period. For this reason, timely, accurate, and objectively audited disclosure is a requisite of good corporate governance. While there are differing views on levels of emphasis and specific items, there seems to be a broad consensus that a good corporate governance regime requires:

- *An independent and effective board of directors*, having sufficient experience, knowledge, judgment, and integrity to safeguard stakeholders' interests.
- *Independent auditors*.
- *Adequate disclosure to enable those stakeholders to evaluate management performance*.

The impact of deficient corporate governance on credit quality is similar to that of poor management. The financial performance and condition of the enterprise is weakened, thereby reducing its creditworthiness, making it more vulnerable to insolvency and default. In respect to a bank, one or more of its key financial attributes—earnings, asset quality, capital, and liquidity—fall below levels that would have been achieved had good management and corporate governance practices been in place.

Evaluating Corporate Governance: An Overview

Most approaches to evaluating corporate governance do so by comparing actual governance practices at a particular organization or in a particular country with a best practices ideal. At the firm level, to the extent that the organization conforms to best practice, it is said to display good governance. To the extent that it diverges, it can be said to display poor governance. Of course, function must take precedence over form. Divergence in form may be acceptable if the desired governance function is still maintained. Although primarily a qualitative exercise, there are a number of objective binary or quantitative metrics that may be useful in assessing an enterprise's corporate governance regime.

Of these, the attribute of board independence and competence is perhaps the linchpin of corporate governance because it is the board that supervises management and acts as the guardian of shareholders' interests. As the body that acts on behalf of shareholders as a whole, it is the firm's governing body, although in practice much

authority—sometimes too much—is delegated to management. When independent and possessing adequate knowledge and skills to effectively oversee management, the board functions as a check upon the actions of management that can damage the interests of the company's financial stakeholders. Indeed, it is the board that ultimately has the authority to prohibit related-party transactions⁵¹ that have not been thoroughly vetted and to ensure broadly that management is generally acting in the interests of financial stakeholders. Yet at the same time, the board must maintain a good working relationship with management. An adversarial relationship might give rise to infighting that would be antithetical to the long-term prosperity of the company and its shareholders. Corporate governance assessment methodologies therefore typically devote significant attention to examining factors that are likely to affect board independence and competence, impinging on its ability to, for example, monitor related-party transactions and management compensation. In respect to board independence, the presence of any of the following categories of individuals on the board is generally viewed as potentially detrimental to the interests of most financial stakeholders.

- A current or former senior manager of the company or of an affiliate or subsidiary of the company
- A current or former board member of an affiliate or subsidiary of the company
- A majority or controlling shareholder, or where the shareholder is a company, a current or prior manager or board member of the company
- A significant supplier of goods or services to the company
- A person in a close family relationship or having significant business dealings with any of the above, including any individual having a relationship that is likely to impair his or her independence (e.g., an individual who holds an interlocking directorship in relation to another board member).

As for board effectiveness, its evaluation depends both on the skills and disposition of the individual board members as well as upon the structure of the board itself. It is evident that for the board to monitor management effectively on behalf of shareholders, it must have a sufficient understanding of the company's business, and a requisite degree of expertise, as well as timely and relevant reports from management. Assessing these qualities, like that of management competence, is largely a qualitative exercise and perhaps more difficult to perform than the evaluation of management itself. Structural issues take in criteria such as the size of the board, whether it is a single-tier or two-tier board, the process of electing board members, the length of their terms, the use of committees to perform specialized board functions, and the level of authority delegated to management.

The other key elements of corporate governance summarized at the end of the previous subsection—namely, the independence of auditors and the transparency and quality of corporate disclosure—are, as suggested, primarily a function of the independence and effectiveness of the board. Auditor independence is often rated indirectly by examining the policies and procedures governing the appointment of auditors and the degree to which auditors do nonaudit work for the enterprise, thereby giving them a financial incentive to curry favor with management. So not only will the identity and reputation of the auditor be relevant to this exercise, but also its relationship to the company, the relative amount of nonaudit work it performs, whether the auditor has changed in the recent past, and the reasons for the

change. As with the independence of the auditor, transparency and disclosure are normally the result of decisions taken by management but subject to board oversight. The level of disclosure and its apparent fairness and accuracy in terms of financial reporting, which can be evaluated from the data disclosed, is a factor to be considered in assessing the company's overall quality of corporate governance.

In addition, specific dubious behavior or activities frequently subject to managerial abuse may be subject to discrete evaluation. In addition to those already mentioned—that is, related-party transactions and unjustifiable remuneration or other cash payments made to or on behalf of managers—other potentially problematical activities include transactions undertaken for political purposes, insider trading, and takeover defenses that function to entrench existing management. Finally, a critical corporate governance factor that may influence the described behavior is the ownership structure of the company. For instance, a company that is partly state owned may be subject to political influences that adversely affect its private shareholders. Similarly, minority shareholders in a company that is dominated by a single family may be subject to unfair treatment.

Like credit ratings, corporate governance evaluation must take into account the environment in which the firm operates. To illustrate, shareholders' rights and particularly the ability to enforce those rights will be greatly influenced by the prevailing laws and regulations as well as by the strength of the legal and judicial infrastructure. To account for this, the corporate governance environment can be evaluated separately from corporate governance at the firm level, or this country-level assessment can be incorporated into a company analysis. As a rule, corporate governance is evaluated at the country level in a similar manner to a firm-level analysis; that is, relevant laws and regulations that set forth shareholders' and creditors' rights and that, more specifically, impose minimum corporate governance norms that are compared with a model code or framework.

Finally, these questions arise: (1) How much weight should be given corporate governance considerations in a bank credit analysis, and (2) more specifically, how much weight corporate governance should be assigned relative to basic management competence? There is no single answer to either question. However, it should be noted that historically the weighting usually given to management evaluation as compared to a bank's financial attributes is relatively low, perhaps 15 to 20 percent of the overall score, although these weightings are rarely explicitly stated or acknowledged. As for corporate governance vs. management quality, the former has only been formally considered as a ratings criterion in the past couple of decades, so it has to be said that in the past it has not been heavily weighted. Although more than half of this chapter is given over to a discussion of corporate governance, in most cases it is probably not weighted as heavily as management competence, and only comes to the fore in situations where the analyst has reason to believe there are serious problems. This may be due in part to logistical difficulties in obtaining the necessary data to be able to perform a corporate governance appraisal across a wide range of markets. Recent events, however, strongly suggest that corporate governance will be considered more closely in future.

Bank Corporate Governance

Among the key distinguishing characteristics of banks are their high leverage and their reliance on retail depositors for funding. High leverage magnifies the risks of

imprudent lending, while the special responsibility banks have toward their depositors underlines the need for caution in their operation. Insofar as misgovernance results in a bank failure, the adverse impact on the economy as a whole will often be greater than would the impact of the collapse of an ordinary nonfinancial company. These characteristics of the banking industry, which it might be noted also, provide a justification for intensive regulation of the sector, call for a higher standard of corporate governance in banks than in nonfinancial companies. Banks are not merely selling products and services, but they are also managing other people's money.

The monitoring of related-party transactions is of special concern in respect to the banking industry. Being in the business of lending money, banks are especially vulnerable to misgovernance, as the problem is very likely to manifest itself in funds being advanced for more risky purposes than would otherwise be the case. Another distinguishing feature of banks is that they tend to have more stakeholders than nonfinancial companies. Stakeholders encompass depositors, financial counterparties, and various classes of debt holders.

Approaches to Corporate Governance Focused on Creditors and Banks

The Basel Committee on Banking Supervision has considered the subject of corporate governance in a number of reports from the vantage point of bank regulators, whose desire to maintain the safety and soundness of the banking system is not dissimilar to the collective interests of creditors. In *Core Principles of Banking Supervision* (in the version dated 1997), the committee noted the importance of corporate governance and ensuring that banks put in place “a management structure with clear accountability, a board of directors with ability to provide an independent check on management, and independent audit and compliance functions.”⁵² Subsequently, the committee updated its principles in a document⁵³ that lists them in the following way:

The Basel Core Principles define 25 principles that are needed for a supervisory system to be effective. Those principles are broadly categorized into seven groups: Objectives, independence, powers, transparency and cooperation (principle 1); Licensing and structure (principles 2 to 5); Prudential regulation and requirements (principles 6 to 18); Methods of ongoing banking supervision (principles 19 to 21); Accounting and disclosure (principle 22); Corrective and remedial powers of supervisors (principle 23); Consolidated and cross-border banking supervision (principles 24 and 25).

In a more recent consultative document, dated December 2011,⁵⁴ it increased the number of its principles to 29 and considered what measures could be taken to improve bank corporate governance, in particular through greater public disclosure and transparency and with enhanced financial reporting and external audit. It also published *Enhancing Corporate Governance for Banking Organisations*⁵⁵ in which it defines the corporate governance objectives in the following terms:

From a banking industry perspective, corporate governance involves the manner in which the business and affairs of banks are governed by their boards of directors and senior management, which affects how they:

- Set corporate objectives;
- Operate the bank's business on a day-to-day basis;
- Meet the obligation of accountability to their shareholders and take into account the interests of other recognised stakeholders;
- Align corporate activities and behavior with the expectation that banks will operate in a safe and sound manner, and in compliance with applicable laws and regulations; and
- Protect the interests of depositors.

Key Categories of Best Practices

The list of best practices can be divided into three categories:

1. The independence of the board
2. Its accountability to shareholders
3. The independence of external auditors

Board Independence

Measures of board independence include (a) board size and (b) the percentage of directors that (1) are insiders; (2) have financial relationships with the bank; and (3) hold other directorships or other directorships that are interlocking.⁵⁶

Board Accountability

Board accountability criteria include (a) the use of staggered terms; (b) member attendance at meetings; (c) stock compensation to board members; (d) stock ownership of board members; (e) overextension; and (f) length of tenure. In addition the report identifies two additional board composition criteria—average age and gender—that are not included in the list of methodological criteria. Again, these are linked where possible to quantitative benchmarks.

Auditor Independence

Principles of auditor independence include the following: (a) auditors should not engage in nonaudit work for the company (b) auditors should be changed frequently, for example every five years; (c) the internal audit committee should be completely independent of the bank; and (d) one or more members of the audit committee should have audit-related experience.

Evaluating Corporate Governance: How to Quantify the Qualitative

Corporate governance methodologies have been developed by a number of organizations, including rating agencies, consulting firms, investment banks, and multi-lateral institutions. In the previous section, the key aspects of corporate governance and its evaluation were discussed, and several methodologies described.

This does not solve the problem of quantifying the qualitative judgment. The reader will be familiar with scoring models where the qualitative judgment translates into a set of scores on various aspects of the person or the organization under scrutiny.

Those are subscores that a model has yet to digest in order to produce a final score upon which a decision will be taken or a rating assigned.

Quantifying the qualitative judgment is best done, as discussed earlier in this chapter, by assigning subscores to various factors or attributes. The model will give weights to the various subscores before adding them up. Given the judgmental source of the opinion, the model will itself be built on judgment. As one can imagine, anyone can create her own model based on her own scales, and based on the importance given to each of the key aspects that are examined.

A further obstacle is conceptual. It can be said that, in general, not all ratings and scores assigned by rating agencies on quantifiable data are necessarily predictive, since they are largely based on past performance. They permit a degree of extrapolation, however, without which they would have no value.

Assessments on governance are probably even less predictive, or, at the very least it could safely be suggested that good governance should not give a bank a higher overall rating, but that bad governance should negatively weigh on the final rating.

Although the initial focus of governance scrutiny, in the 1990s, was on emerging markets in the wake of the Asian and Russian debt crises, the spate of corporate failures in the United States, such as Enron, Global Crossing, and WorldCom, seems to have been a factor in the roll-out of such assessments. The recent discovery that Greece was less than candid in the disclosure of its balance sheet shows how important it has become to give governance the place it deserves.

These assessments are entirely distinct from credit ratings, although they are designed to have a complementary role in providing additional information to investors and risk managers. They might be described in separate assessments of individual companies and the countries in which the firms are situated.

Corporate Governance Methodologies

In an effort to offer some further background concerning corporate governance assessment, this section surveys several more of these methodologies and compares them. Although the approaches discussed are primarily aimed at protecting shareholders' interests, many of the good corporate governance attributes identified will benefit creditors as well. For this reason, they are worth consideration by the bank credit analyst. A few of the methodologies take the creditor's perspective or consider the special characteristics of bank corporate governance.

SocGen: Rating Corporate Governance in Emerging Markets—A Country-Focused Approach

In a 67-page February 2000 report entitled "Standards of Corporate Governance: The Straight and Narrow," SG Global Equities, the brokerage arm of Société Générale, articulated a scoring system based on an assessment of five key corporate governance categories: shareholder rights, the enforcement of shareholders rights, disclosure, analyst access to management, and board independence. SocGen's analysis was directed solely at countries rather than at specific companies, the scope of coverage being the emerging Europe, the Middle East, and Africa. The methodology, however, would seem to be adaptable to measuring corporate governance in other circumstances and for various types of companies, including banks.

The scoring model seeks to measure various corporate governance practices against a best-practice standard, which is implicit in a perfect score. Within each category, various governance issues or practices are ranked from best to worst and

enumerated, while a score is assigned depending upon the “grade” of practice that is followed by the firm, or that is applicable in the market in which the firm does business. There were a total of 12 issues or practices identified, and the maximum score in each case is 3, making 36 the maximum score. The model therefore takes the form of a scorecard, to be filled out by the analyst. For example, under the broad category of shareholder rights, five possible answers are enumerated concerning the statement of best practice: “Directors’ dealings should be published on a regular basis.” Each of the possible answers has a score assigned corresponding to the degree to which the company achieves the posited standard. The five answers and their respective scores are:

1. Local regulations/authorities require this, and it is complied with, 3.
2. Local regulations/authorities require this, but compliance/enforcement is not consistent, 2.
3. Local regulations/authorities require this, but no meaningful compliance, 1.
4. No local regulations, and no compliance, 0.
5. Consistent evidence that this is an area of serious abuse, -1.

The possible answers for each of the 12 issues or practices follow a similar pattern. Since the approach is geared toward evaluating overall corporate governance within a particular market, the emphasis is on regulatory requirements and broad industry practice, as opposed to specific or discretionary policies that might be implemented by particular companies. Each of the 12 questions is given an equal weight, so that weighting of criteria is implicit in the number of questions in each category. Six of the questions concern shareholder rights, including the enforcement of shareholder rights, giving a weight of 50 percent to this area of governance. Three questions concern disclosure and two questions concern board independence..

As a broad-brush approach to evaluating corporate governance across markets, this approach has the virtue of simplicity and affords some discretion to the analyst in determining to what extent a particular best practice standard has been reached instead of compelling a binary “yes/no” evaluation. It encompasses the three aspects of corporate governance embraced universally: shareholder rights, board independence, and quality of disclosure. Although “quality of analyst and investor access” is made a discrete category, it could be more broadly thought of as part of “disclosure.” Likewise, “enforcement of shareholders’ rights” could be included within the category of shareholders rights generally, or as more an environmental factor than a company-specific factor. Although the model is ostensibly designed to be used to assess corporate governance in a particular market, some of the questions are rather company-specific for a country-oriented model; for example, A.1. “Does the company make it unduly difficult or expensive to cast votes?” Presumably, as the methodology is explicitly aimed at evaluating a country as a whole, this question should be interpreted as being posed in general terms: that is, “do companies in this market tend to make it unduly difficult or expensive to cast votes?” Alternatively, perhaps, each listed company of interest would be ranked individually on this point and the results averaged. Conflating company and country evaluations in the same questionnaire, however, might be criticized as addressing neither as comprehensively nor as precisely as might be accomplished were these done separately.

Deutsche Bank: The Four Pillars of Corporate Governance—Scoring the Firm

Deutsche Bank has addressed several investment reports to the subject of corporate governance, of which a report on corporate governance called “Global Corporate Governance: Valuing Corporate Governance in South Africa.”⁵⁷ The bank used a scoring system based on rankings of several categories of qualitative criteria. Four “corporate governance pillars” comprise 80 percent of the total score.

Corporate Governance Pillars (Company Level, 80 Percent Weighting)

These four pillars are board independence, shareholder treatment, information disclosure, and best practices.

Board Independence As with the approaches already discussed, board independence ranks as a critical criterion of governance quality. The salient questions are:

- Is the board independent?
- Is it effective?
- Are its interests aligned with shareholders?

Shareholder Treatment Are shareholders, particularly minority shareholders, treated fairly? Is the board performing its stewardship function effectively? Is the board’s interest aligned with the long-term interest of the shareholders?

Information Disclosure Is disclosure timely, transparent, and adequate to present a fair picture of the company’s financial condition?

Best Practices Is good corporate governance reflected in the company’s actions? Does the company acknowledge the importance of good corporate governance?

Corporate Governance Environment (Country Level, 10 Percent Weighting)

In addition to these four pillars, a 10 percent weighting is given to the corporate governance environment of the country in which the company is headquartered. Variables affecting the scoring in this category include the following.

Legal and Institutional Infrastructure This is based on applicable laws, regulations, and listing rules; effectiveness of enforcement; and institutional support (e.g., through shareholders’ associations, pension funds).

Level of Corruption This is as ranked by Transparency International.

Average Company Score Average company scores based upon the four pillars.

Discretionary Input (10 Percent)

Ten percent of the total score is left to the analyst’s discretion based upon his or her impressions of the subject company.

The scores having been computed, Deutsche Bank then ranks companies within its corporate governance universe into five tiers. Calculating the comparative performance of the companies within each of the tiers, the report concludes “it pays to

be good”; that the evidence shows that shareholders are willing to pay a premium for companies with perceived superior corporate governance.⁵⁸ Unlike the approach adopted by some other observers, country criteria explicitly affect the overall score assigned to a company, since country factors are given a 10 percent weighting. This does away with the need for separate country scoring mechanisms, and probably does not impose too great a burden on a company that is an otherwise perfect example of corporate governance best practice in an otherwise imperfect market.

The four company-specific categories—board independence, shareholder treatment (i.e., shareholder rights), information disclosure, and best practices—are reasonably given equal weighting, although the last of these categories is a bit of a catch-all that allows for a degree of subjective input. This coupled with an explicit 10 percent allowance for the analyst’s subjective input gives Deutsche Bank’s approach a considerably more qualitative character than other models. This perhaps reflects its intended global applicability and implicit use by experienced analysts who have a deep understanding of the companies and markets they cover. Depending upon one’s perspective, that could be viewed as an advantage or disadvantage. Of the bank and brokerage house models, Deutsche Bank’s approach is among the broadest in the criteria embraced in a single score.⁵⁹

CLSA: 57 Varieties of Corporate Governance Criteria

CLSA, a highly regarded securities brokerage firm focused on the Asia-Pacific region, has applied a corporate governance assessment methodology to Asian companies, many of which may be classified as emerging markets. In a 224-page report entitled “Saints & Sinners: Who’s Got Religion—Corporate Governance in Emerging Markets” and published April 2001, the brokerage outlined its approach, which is broadly based on the OECD principles of corporate governance. These principles consider five aspects of governance and prescribe five corresponding objectives of a good corporate governance framework—the rights of shareholders and stakeholders, disclosure, and board effectiveness and accountability. In summary, the principles declare that corporate governance frameworks should:

1. Protect shareholders’ rights.
2. Ensure the *equitable treatment of all shareholders*, including minority and foreign shareholders.
3. *Recognize stakeholders’ rights* as established by law and encourage active cooperation between corporations and stakeholders.
4. *Ensure that timely and accurate disclosure* is made on all material matters regarding the corporation.
5. *Ensure the board functions effectively* and is accountable to shareholders and the company as a whole.

Working from these principles, CLSA analysts identified 57 governance variables that were divided into seven categories and weightings:

1. Discipline (15 percent)
2. Transparency (15 percent)
3. Independence (15 percent), that is, board and committee independence

4. Accountability (15 percent), that is, accountability of board to shareholders
5. Responsibility (15 percent), that is, enforcement of rules
6. Fairness (15 percent), that is, fair treatment of minority shareholders
7. Social awareness (10 percent), that is, social responsibility with respect to child labor, environmental safeguards, and so on

A questionnaire was created in which each of the 57 variables was transformed into questions to be given yes/no answers. The first category, management discipline, considered the issue of management's focus on shareholder value rather than, for example, expanding market share and on empire building. Specific attributes in this category included the following questions—does management:

- Place priority on corporate governance?
- Have incentives to attain a higher share price?
- Focus on defined core businesses?
- Have an appropriate estimate of cost of equity?
- Have an appropriate estimate of cost of capital?
- Apply conservatism in issuing of equity or dilutive securities?
- Ensure that debt is manageable and used only for projects with adequate returns?
- Return excess cash to shareholders?
- Discuss corporate governance in its annual report?

The second category considers the accuracy, fullness, and timeliness of disclosure. Board independence and accountability toward shareholders are addressed in the third and fourth categories. A fifth category of corporate governance criteria is “responsibility,” meaning the enforcement of shareholder rights. A sixth category concerns fairness, meaning fair treatment of shareholders, particularly minority shareholders. Finally, social concerns constitute a seventh aspect of corporate governance.

CLSA's approach can be distinguished by the breadth of criteria employed and the inclusion of some standards not embraced by SG or Deutsche Bank, such as “social responsibility.” The yes/no questionnaire has the virtue of simplicity at the expense of precision. Answers to questions such as “Does management place priority on corporate governance?” obviously imply a significant degree of subjective judgment on the part of the analyst answering them. Similarly, the question “Does management have incentives to attain a higher share price?” begs the question of whether those incentives are likely to be effective or whether they may encourage the use of aggressive accounting techniques or management of earnings in an effort to raise the share price.

Survey Summary

The foregoing methodologies share similarities. All use, either implicitly or explicitly, a checklist of best practices against which a subject company is compared. All, with the exception of CLSA, generally limit themselves to financial stakeholder concerns. Board independence and effectiveness, shareholder rights, and disclosure are major focuses of each, but there are significant differences as to the weightings given each

aspect. Aside from weightings, there are two other areas of difference: (1) the relative emphasis placed upon quantitative input vis-à-vis qualitative input; and (2) the degree to which there is reliance upon a dialog between the analyst and the rated company in respect to corporate governance issue, which in turn corresponds to the degree of analyst time required.

The more qualitative approaches either directly or indirectly call for more subjective judgment. This is inevitable. There may be some deficiencies in a methodology that evaluates corporate governance on the basis of whether specific measurable criteria are satisfied. The danger is that the appearance of good corporate governance is achieved at the expense of substance. In addition to imprecise criteria that can lead to inconsistencies, qualitative approaches can also fail to capture governance deficiencies.⁶⁰

NOTES

1. Alan Greenspan, in testimony to Congress, October 2008, www.iht.com/articles/2008/10/23/business/24greenspan.php.
2. Lord Myners, January 2009, www.telegraph.co.uk/finance/newsbysector/banksandfinance/4327363/Gordon-Browns-City-Minister-criticises-banking-bonus-culture-and-mismanagement.html.
3. K. Olson et al., "Evaluating Corporate Governance: The Bondholders' Perspective," Fitch Ratings (Special Report), April 12, 2004, 2.
4. Sunil Garg et al., "Asia's Best Managed Banks" [investment report], Fox-Pitt Kelton, January 7, 2003.
5. Note that a bank's operating environment and risk management capacity are discussed elsewhere in this book. This chapter only considers management quality and corporate governance.
6. Applying the concept of risk-adjusted return, seemingly high-risk strategies have come to be viewed as not necessarily fatal to creditworthiness, provided that the relevant risks are managed effectively, and that returns are commensurate to the risks taken. Similarly, a superficially low-risk strategy will not necessarily bolster creditworthiness where returns are so disproportionately feeble as to make the strategy unsustainable over the longer term.
7. Rating agencies aim to function as neutral agents conveying information about bank risk and offering an opinion on the creditworthiness of individual institutions. As a consequence, their views reflect the interests of debt investors and counterparties, notwithstanding that the primary source of their revenue is the institutions they evaluate.
8. It is evident that shareholders' interests may vary considerably, particularly in a widely held publicly traded bank. While certainly more sophisticated investors recognize the importance of maintaining an adequate degree of capitalization and liquidity, others may have a high risk appetite and seek maximum short-term returns. Where such diversity exists, management should seek to meet the needs of an ideal shareholder, one who wishes to obtain the greatest sustainable return on investment; to maximize returns without putting the long-term survival of the bank at risk.
9. To be sure, investors recognize, however, that extreme parsimony is likely to dent staff morale and over time weaken performance. In a similar vein, bank employees will take account of considerations that are not purely monetary in nature, including working conditions, opportunities for career development, and job security.
10. As another illustration, the goals of the bank regulator may be somewhat in conflict with the policy objectives of other government agencies, which may favor a greater stock of

- housing, better working conditions for the bank's labor force, more credit provided to certain economic sectors, or other social benefits.
11. To illustrate, by spotting opportunities where a prized benefit can be provided at comparatively little cost, management can potentially retain or increase business without sacrificing rates or prices. Similarly, by using sophisticated risk management techniques, the bank may be able to price its loans more efficiently and enhance profits. Likewise, implementation of improved information technology infrastructure can open new channels of product delivery, personalize marketing, and reduce transaction costs.
 12. With regard to the profit or loss on any particular transaction or set of transactions, there is, of course, almost invariably an element of chance.
 13. The process is further complicated by the potential that strong presentation skills and the impression of credibility exuded by experienced senior bank officers may not be accompanied by the leadership and ability to execute a plan that translates into positive results.
 14. Of course, were this the sole factor, no separate evaluation of management would be necessary or required since the evaluation of the bank's financial fundamentals would also constitute an implicit evaluation of management.
 15. It should be noted, however, that the price movements of comparatively liquid debt securities do often convey some information concerning overall creditworthiness and in so doing reflect upon management competence.
 16. In technical terms, it might be said to be management that seeks an optimal combination of risk to return by maximizing risk-adjusted profits for its banking franchise as a whole. As a practical matter, however, it is extremely difficult to identify with any precision the middle ground between excessive prudence and unfettered short-term profit maximization.
 17. This said, mention should nonetheless be made of potential quantitative indicators that might be considered as proxies for the caliber of management. These include indicia such as number of years of experience, educational qualifications, professional experience, training expenditure per staff member, and compensation levels. While probably useful to a limited degree, these criteria hardly embrace the myriad of traits, including good judgment, integrity, and business acumen that make a management team first-rate. Parenthetically, while compensation levels may correlate to some extent with the quality of management, broadly defined, there is a danger in looking at the numbers alone. A bank that attempts to skimp on remuneration to its key managers is usually pennywise and pound-foolish. Because salary and bonus levels are primary markers of success in the financial industry, the unduly parsimonious bank will find that its best employees leave and mediocre ones remain. At the other end of the spectrum, excessive compensation can be detrimental to a bank's health, contributing to excessive risk taking—as seen in the years leading to the 2007 crisis—especially where the monetary rewards are linked to short-term profits, or are paid out irrespective of the bank's overall performance. With the exception of past financial results achieved, there are comparatively few, if any, quantitative indicators of much use in evaluating management. It is impossible to escape the need to make subjective judgments concerning management quality.
 18. Nonetheless, the stronger the track record exhibited and the greater the extent to which current management demonstrates a capacity to respond to changing conditions, the more reasonable it is for the analyst to acquire a greater confidence in the bank's capacity to achieve similar results in future. The same applies to the bank that has consistently displayed poor or mediocre results. In the same manner, a change of management or a change of strategy may support either a more or a less favorable assessment of the institution's future prospects.
 19. Scorecard approaches are commonly seen in credit risk models used as part of a bank's internal credit risk management framework.
 20. As an example of how management quality might be divided into a number of sub-criteria, various attributes of the bank and management might each be rated separately on

a scale of 1 (weak) to 7 (very strong) and aggregated to calculate an overall management quality score. These might include the bank's level of commercial orientation (vs. a policy or bureaucratic orientation); the experience and reputation of top management; the quality of strategic planning; management's capacity to execute plans and meet goals (based on track record); the strength of credit and organizational culture; the sophistication and effectiveness of risk management and operational controls; staff morale and management's commitment to staff training and professional development; and the strength of customer and depositor relationships.

21. Exhibit 8.1 provides a simplified illustration. It is assumed here that management competence is deemed to be one of two or three major attributes of management quality, one of which perhaps is the quality of corporate governance. They might just as easily have been ranked separately and weighted to calculate an overall management quality score. Likewise, the number of subcriteria might be increased, to include subattributes such as management integrity or management sophistication.
22. So to achieve an overall rank or score of 5 or 6 in respect to management quality, management, as an indicative matter, must have clearly articulated, worthy, and realistic medium- to long-term goals plus a plausible strategy to achieve them. Execution ability is reflected in a mixed track record coupling overall success with some missteps. Note that this illustration is purely for indicative purposes and has not been used as part of an actual bank credit analysis process.
23. Managers of controversial or well-known banks may be profiled in the local or international business press.
24. The term *company culture* takes account of the less tangible organizational attributes that may nonetheless provide clues to the bank's future performance. The term encompasses a cluster of attributes such as how much the institution embraces innovation.
25. These criteria overlap to a degree with those relating to corporate governance discussed in the latter part of this chapter.
26. Consider, for example, a situation where management of a newly privatized bank is making an effort to shed a tired and stodgy brand for a new and more innovative one. In this context, the analyst should be alert for deviations between image and reality, and consider this question: To what degree has the bank really changed? A newly privatized bank, for example, may have much the same workforce as it did when it was a turgid bureaucratic organization, and the same company culture may still prevail notwithstanding a change in ownership. More generally, the image the bank tries to present can also offer clues both as to management's philosophy and the prevailing corporate culture.
27. Such reluctance may arise out of a concern that the information revealed could, if made known to competitors, lessen the bank's competitive advantage. In respect to a publicly listed bank, there is generally more openness about these matters, in the interest of keeping shareholders and prospective investors informed. At the same time, however, regulatory or ethical norms may prohibit the sharing of such information in a manner that does not allow for its immediate dissemination to the investor community at large. One should be careful however, not to confuse language and presentation skills with banking skills.
28. Questions relating to the caliber, credentials, and compensation of bank staff, particularly at upper and middle management levels, are likely to be intrusive if made in a blunt or interrogative fashion. *Sensitivity* is the watchword. Although these subjects may be hard to address, they should not be ignored.
29. OECD definition, as quoted in K. Olson et al., "Evaluating Corporate Governance," Fitch Ratings (Special Report), April 12, 2004.
30. Standard & Poor's "Corporate Governance Services—Frequently Asked Questions," October 8, 2002.
31. See Fitch Ratings, op. cit.

32. Adam Smith, *An Inquiry Into the Nature and Causes of the Wealth of Nations*, Book V, Chapter 1: "Of the expenses of the sovereign or commonwealth" (1776).
33. Part of this section and the following section are adapted from an article by Jonathan Golin entitled "Commercial Approaches to Assessing Corporate Governance: A Survey of Methods," *Corporate Governance International* 6, no. 4 (December 2003).
34. *Ibid.*, 4.
35. Strictly speaking, in legal terms, management is not an agent of creditors in the same manner that it is in relationship to shareholders. A true principal-agent relationship exists between shareholders of a firm (its owners) and the managers they employ while the relationship between creditors and management merely resembles a principal-agency relationship; hence the term a quasi-principal-agent relationship.
36. Conflicts between majority or controlling shareholders and minority shareholders present another potential conflict of economic interest and a corresponding governance issue. Nor does it distinguish between different classes of creditors, who may also have interests in opposition to each other, such as between bank depositors and covered bondholders.
37. At the periphery, this merges with the interests of the world as a whole, and the subject of corporate governance at this level considers essentially the same issues as the field of social and ethical investing.
38. A divergence between the interests of creditors—that is, of bondholders, counterparties, depositors, and other creditors—and those of shareholders arises intrinsically because of the different legal positions of shareholders and creditors.
39. Most of the corporate governance evaluation methodologies examined in the following section focuses on the interests of shareholders.
40. Comparatively few reports have looked at the governance of banks as opposed to corporate governance generally, and even fewer have considered the point of view of creditors in contrast to that of shareholders. However, as these are of particular relevance to this book, several are discussed in some detail in the pages that follow. The creditors' perspective has been examined by Fitch Ratings in a 2004 report entitled "Evaluating Corporate Governance: The Bondholders' Perspective." In addition to considering the impact of corporate governance on bondholders generally, Fitch Ratings has also discussed bank corporate governance in a 2005 report focusing on European banks. J. Dow et al. "The Framework for Corporate Governance in Major European Banking Systems—Still Evolving, but Good Progress Made" (Special Report), Fitch Ratings, April 4, 2005.
41. This is not to minimize the importance of looking out for situations in which the interests of the shareholders and creditors groups may diverge, as such conflicts may escape notice in markets or in parts of a banking sector that are not closely followed by analysts.
42. Fitch Ratings, "Evaluating Corporate Governance," 4.
43. The reference here is not only to the conflict between the interests of management and those of shareholders, but also to other potential conflicts of economic interest between or among other shareholders and bondholders.
44. Who is a stakeholder for corporate governance questions is a fundamental initial query that affects how best practices are defined.
45. R. Monks and N. Minow, *Corporate Governance*, 2nd ed. (Blackwell, 2001).
46. This view of corporate governance is encompassed in the following quotation: "Corporate governance is concerned with holding the balance between economic and social goals and between individual and communal goals. The governance framework is there to encourage the efficient use of resources and equally to require accountability for the stewardship of those resources. The aim is to align as nearly as possible the interests of individuals, corporations, and society." Sir Adrian Cadbury in the Global Corporate Governance Forum, World Bank, 2000. An interesting list of definitions can be found at <http://corpgov.proxyexchange.org/library/corporate-governance-defined>.

47. Put in economic terms, a good corporate governance regime will guard against management behavior that reduces the value of the owners' investment in the firm, collectively and individually.
48. Protection of shareholders is ordinarily taken to include the protection of minority shareholders against abusive treatment by the majority or controlling shareholders of the firm.
49. That is, for profit.
50. Obviously, this is applicable to commercial institutions. A policy-oriented organization by definition would operate on a noncommercial basis.
51. Related-party transactions are dealings between a company and insiders, whether managers, directors, or other staff, as well as their respective families, friends, or associates. The danger of related-party transactions is that the transactions will be undertaken primarily for noncommercial reasons, and therefore that the company will either make a loss as a result or forgo more profitable opportunities as a consequence. Typically, the most hazardous related-party transactions in this regard are those entered into between management and the company; or between major shareholders and the company. It is, of course, possible that a related-party transaction does not constitute such a sweetheart deal, and in fact represents a great business opportunity.
52. Basel Committee on Banking Supervision, *Core Principles for Banking Supervision*, Basel, September 1997.
53. On page 8 of the 2006 version of Basle Committee on Banking Supervision, *Core Principles for Banking Supervision*, www.bis.org/publ/bcbs129.pdf.
54. Available at www.bis.org/publ/bcbs213.pdf.
55. On page 4 of www.bis.org/publ/bcbs122.pdf.
56. Interlocking directorships are defined as situations "where an insider director of company A sits on the board of company B, and an inside director of company B sits on the board of company A." *Core Principles for Banking Supervision*, Basel, September 1997.
57. Deutsche Bank Securities, Inc., "Global Corporate Governance: Valuing Corporate Governance in South Africa," August 19, 2002.
58. Deutsche Bank Securities, "Global Corporate Governance." "Our analysis," the report said, "suggests that South African companies in our top tiers outperformed the MSCI SA index and those in the bottom tiers over a one-, three-, and five-year period." Deutsche Bank's approach clearly separates the elements of corporate governance from those that are company-specific.
59. The description of the methodology in the report, however, does not include many specific criteria.
60. For example, CLSA includes as one criterion the existence of incentives on the part of management to improve the share price of the company. Such incentives may indeed exist, but their effectiveness needs to be evaluated as well. In contrast to stock grants, stock options may provide unduly strong incentives to push up a company's share price at the expense of earnings quality, as managers have no downside with options.

Capital

A well-managed bank needs no capital, whilst no amount of capital can save an ill-managed bank.

—Walter Bagehot¹

The capital of a bank should be a reality, not a fiction; and it should be owned by those who have money to lend and not by borrowers.

—Hugh Maculloch, U.S. Comptroller of the Currency, 1863

It is difficult to proceed far in bank analysis without coming to grips with the impact of capital on both the creditworthiness of banks and on perceptions of their creditworthiness. The idea that bank capital is the ultimate measure of bank creditworthiness is widespread.² Much obeisance is paid to the importance of banks maintaining robust levels of capital, while banks in turn tout their capital levels as a sign of financial soundness. Indicators such as the Basel *capital adequacy ratio* (CAR),³ which came to the fore after the advent of the first Basel Accord of 1988, have acquired an almost talismanic significance, which seems to have only increased in the wake of the controversies that preceded the adoption of the second iteration of the Basel Accord (Basel II) in 2005. The aftermath of the credit crisis of 2007–2012, where governments rushed to recapitalize banks at taxpayer expense, has contributed to the ubiquitous perception that capital is by far a bank's most vital financial attribute.

The belief that capital strength is the preeminent mark of a highly creditworthy bank is no recent phenomenon, as the quotations from Walter Bagehot and Hugh Maculloch that lead this chapter suggest, but instead has a long history. This is because the belief rests upon a core truth, which is that a bank that has liabilities that exceed its assets is insolvent and cannot long survive without support or the credible guarantee of support through the infusion of funds. Significantly, however, neither of the two referenced quotations unreservedly extols that belief. Instead, both implicitly ask us to consider the function of capital, and whether and under what conditions it may be the paramount measure of bank soundness. Bagehot uses hyperbole to suggest that high levels of capital pale against the importance of other factors, presumably those of prudent management, while Maculloch points to the dangers of taking capital indicators at face value. Both cautions are applicable more than a century later.

Contemporary comment echoes these nineteenth-century voices. Howard Palmer, author of *Bank Risk Analysis in Emerging Markets*,⁴ refers to the “myth of capital,” the notion that a bank’s capital position compensates for other deficiencies it might have—that the capital measure functions as some “supreme indicator of the market’s confidence.”⁵ He criticizes those who “reach for capital as a measure of bank performance before any other figure is considered.”⁶ In the same vein, the rating agency Moody’s Investors Services in a report entitled “Bank Rating Methodology: Analytical Framework for Banks in Developed Markets,” states that “capital inadequacy cannot truly be considered the cause of bank failure” and that “capital is important, but it is not a leading indicator of credit health. . . .”⁷

The Moody’s report also makes reference to “regulatory” capital. It is significant, and hints, if at all needed, at the existence of more than one category of capital and, by implication, to more than one category of capital to measure. Indeed, there are different types of capital, and their respective relevance to bank creditworthiness is one of the main themes of this chapter.

Of course, capital is important, and, moreover, under certain circumstances, it is arguably the most critical of all of a bank’s financial attributes. Most of the time, however, it is not the be-all and end-all of bank credit analysis. To evaluate capital strength and a bank’s overall credit profile, one must understand what capital is, its function, the various ways it can be defined and measured, and when capital can “become a fiction” as well as whether or not the fiction serves a useful purpose. With this knowledge, one can then ascertain under what circumstances capital becomes most relevant, and when other characteristics of the bank assume greater significance. To this end, this chapter looks at the function of capital in its various forms, measures of capital strength, the concept of capital quality, and the role of capital evaluation within the framework of bank credit analysis.

From a regulatory standpoint, the only acceptable form of capital is a bank’s own funds—under certain definitions discussed elsewhere in this chapter—and the regulator would ignore other components of a bank’s value, except when a decision to rehabilitate the bank is considered. Those other components exist, without which the market capitalization of a bank would just be equal to its equity, which is seldom the case. A bank as a going concern has an explicit forecast period value and a long-term continuing value that are taken into account formally in valuation estimates and informally in share prices. The regulator should, of course, be conservative and should disregard the contribution of anything that may disappear if the bank runs into financial difficulties.

THE FUNCTION AND IMPORTANCE OF CAPITAL

The primary function of capital is to support the bank’s operations, act as a cushion to absorb unanticipated losses and declines in asset values that could otherwise cause a bank to fail, and provide protection to uninsured depositors and debt holders in the event of liquidation.

—U.S. Federal Reserve⁸

Although there are differences in views concerning the weight that capital ratios should be accorded relative to other metrics—and concerning the best way to

measure capital—as indicated towards the conclusion of the preceding introduction, there is no denying that the level of capital backing a bank can count on is almost universally a significant indicator of its financial health. Certainly, capital is *a* vital sign of bank financial strength.⁹ But what makes it so?

Why Capital Is Important

The Functions of Capital

Consider first the purpose of capital. As noted in *Bank Director Basics*, a publication of the U.S. Federal Reserve Bank intended to educate nonexecutive bank directors, “[b]ank capital serves the same purposes as capital in any other business.” It is important to banks for the same reason it is important to any firm. By providing the wherewithal to establish the enterprise in the first place, and over time, supplying the funds to expand, add new business lines, and invest in technology, it facilitates the bank’s creation and its subsequent investment and growth. More significantly from our perspective, capital sustains firms through difficult business conditions by providing a buffer against unexpected losses, and thereby affords creditors some comfort that their debts will be repaid.¹⁰ Therefore, a level of capital that is more than sufficient to meet the organization’s needs bolsters market confidence in the institution.

Capital Becomes More Important When There Is Little of It

Although these functions of capital are operative in all businesses, they are especially critical to banks. This may be explained by a very peculiar attribute of banks that historically has served to underline the comparative importance of their capital levels: high leverage. Relatively speaking, banks simply do not have very much of it. In other words, compared to other types of firms, banks do not have much capital at risk in proportion to the remainder of their balance sheets. Another way to say the same thing is that banks employ exceptionally high leverage, or gearing, compared with other types of businesses. Irrespective of the industry, high leverage magnifies both gains and losses and thereby makes a firm vulnerable to sudden financial reverses. Banking organizations are not immune from this effect.

Leaving aside for the moment the reasons for banks’ universally to employ high levels of leverage compared to other types of commercial enterprises, it is somewhat paradoxical that there is a corresponding perception, nearly as universal, that banks are conservative businesses. The reality is that banks are constantly juggling with assets that are mostly realizable in the long term and with liabilities that are mostly callable in the short term. It requires an act of faith from depositors and creditors that this juggling will not come to an abrupt end.

The difference in the amount of gearing, or leverage, employed by banks and other financial companies relative to their nonfinancial counterparts is not small. In fact, it is so large as to be practically within a different order of magnitude. Bearing in mind that nonfinancial companies possessing no significant debt are not particularly rare, a rule of thumb is that a conservative level of debt for a nonfinancial company is not more than one part debt to one part equity. The corresponding capital-to-assets ratio for such a company would be 50 percent or more.

A less conservatively structured company might desire greater leverage and accordingly take on proportionately more debt. As a rough measure, such a firm

might employ as much as two parts of debt against one part of equity, thereby exhibiting a capital-to-assets ratio of about 33 percent and in fact most companies would find themselves in a range between 33 and 50 percent. A nonfinancial company taking on debt in excess of the preceding threshold would, by almost any standard, be viewed as a high-risk proposition.

In contrast, it would not be at all unusual for a bank to employ as little as one part capital to 20 parts debt corresponding to a capital-to-assets ratio of about 5 percent, while leverage as much as twice as great as that could well be viewed as acceptable—at least until recently. Accordingly a bank with a capital ratio of 10 percent would, under prevailing standards, be considered to be very prudently capitalized. Thus, nonfinancial companies structured with at least a minimal degree of conservatism would display capital-to-asset ratios ranging from 33 to 100 percent. The corresponding values for banks, which as rule are expected to be managed with some degree of prudence, might range from 3 to 10 percent. It is evident that because banks have proportionately so little capital at risk compared with nonfinancial companies, the importance of differences in the actual amounts they do employ is accentuated.

In recent years, the gearing or leverage of banks has increased considerably. A perverse outcome of the Basel regulations was to invite banks to search for the best return to the *regulatory risk* ratio for their activities. Whether the actual risk was higher than the regulatory risk is a consideration that started becoming increasingly disregarded. As a result, there has been a shift towards less *regulatory risky* transactions, among them subprime loans and sovereign debt—or should we write *subprime sovereign debt*?—which, on average, has allowed an increase in total assets against a stable level of equity, and therefore higher leverage. It is striking to see how European banks saw their leverage explode in recent times. For example, while Australian banks have kept their crude leverage (being the ratio of total assets to equity) in the vicinity of 13 to 18 in the past 10 years—which is already higher than in the United States—Dexia Bank at the end of 2010 was thought to be holding about EUR 10.7 billion in equity to sustain a EUR 556 billion balance sheet, for a crude leverage of 52 times.

Owing to high gearing, it only takes a comparatively small amount of losses for a bank's thin layer of supporting capital to erode to a degree that confidence in the institution begins to disappear as well. For this reason, a bank's capital ratios are closely scrutinized by a bank's creditors and counterparties.

THE IMPACT OF HIGH LEVERAGE IN BANKS

In view of the exceptionally high levels of leverage they employ, banks are potentially high-risk businesses, which, unless run in a consistently conservative manner, are vulnerable to failure. No wonder then that banks traditionally built imposing edifices to impress their current and prospective depositors. While a highly geared nonbank enterprise would be perceived immediately to be the high-risk venture that it is, banks must cultivate exactly the opposite image; not only have banks historically sought to impress with their offices, but they have also emphasized the relative size of their capital bases, thin though they may be by any universal standard.¹¹

Over the longer term, image naturally must be backed by substance. Although taken for granted except in times of crisis, the challenge involved in

persuading depositors and creditors to have confidence in a banking organization is not an insignificant one. To mitigate the bank's inherent vulnerabilities, the market expects banking institutions to be even more conservatively managed than the typical nonbank enterprise. A bank's shareholders generally view the enterprises in which they have taken an ownership stake as comparatively conservative investments in which a degree of return is sacrificed for a commensurate reduction in risk. To an even greater degree, those from whom banking institutions receive most of their funding—that is, their depositors—tend to be risk averse, traditionally receiving a comparatively low fixed rate of interest with the expectation that their deposits are both liquid and not at significant peril of loss. For these reasons, bank managers and their shareholders favored prudence over performance, and have sought prospective depositors, counterparties, and investors on that basis.

The continued funding of bank assets through deposit liabilities or other obligations depends almost entirely upon the confidence on the part of the very same depositors and bond holders that their funds are safe and available. To paraphrase Bagehot, where there is no confidence, there is no bank worthy of the name. So the risk that this trust might be lost poses an enormous danger for banks. Should a bank's assets become impaired—that is, should some of those to whom the bank has advanced loans miss payments on those loans or otherwise cause doubt to arise concerning whether or not payment will be made on time and in full—or, graver still, should a significant number of borrowers fail to fulfill their financial obligations to the bank—then the bank will either make less profit on those loans than expected or suffer a loss. The amount of the loss will depend, among other things, upon the scale of the default and any corresponding recovery from the borrowers in default.

If a loss is incurred, it will either come out of other profits (directly, or indirectly through loan-loss reserves); or, if profits are insufficient, the losses will be deducted from capital. Since a modicum of capital is required to maintain confidence and sustain operations, any capital lost must be replenished out of future profits or, absent a crisis severe enough to engender an offer of government assistance, from the very same risk-averse shareholders mentioned.

Capital as a Regulatory Standard

A bank's depositors and bondholders are not the only parties keeping a weather eye on its capital strength. Bank supervisors also watch capital ratios closely; that modicum of capital just mentioned is likely nowadays to represent an elaborately defined regulatory requirement. In addition to the discipline exercised by the market, banks must comply with minimum capital requirements imposed by regulators. These rules form a significant part of the regulator's arsenal, together with the other prudential regulations to which banks are subject, and related enforcement powers. Since the early 1990s, minimum bank *capital requirements have become more uniform and increasingly ubiquitous*, to the extent that another function of capital exists that is specifically applicable to banks: regulatory compliance.

Capital as an Organizing Principle

Finally, another function of capital within the banking sector derives from developments in financial theory, notably the concept of economic capital discussed later in this chapter. Briefly put, the relative level of bank capital has come to be seen as an organizing principle with which to measure a bank's capacity to respond to risk events. In this context, capital is not only a benchmark against which risk can be evaluated, but also the critical parameter within a framework that enables risk-adjusted performance to be measured, alternative business strategies to be compared, and investment and capital allocation decisions to be systematically made. This is the concept of economic capital that is discussed later in this chapter.

Is Capital Important at All?

Is capital important at all? As argued elsewhere in this book, there are situations where an explicit or implicit recourse exists against the shareholders of a bank, should capital be found insufficient. This would apply to state-owned banks, for example, when explicit guarantees are given or implicit support is expected, provided the state is creditworthy. It is the same principle that applies to military vehicles circulating without any third-party insurance, as they do in many countries, even where compulsory third-party insurance exists for the protection of all road users.

There is no doubt whatsoever that many state-owned banks in emerging markets—and since 2007 in some OECD countries—operate without an appropriate level of capital. The fact is carefully hidden through creative accounting while depositors and creditors, knowingly or unknowingly, maintain their confidence in those weak financial institutions in spite of the situation.

When Is Capital Important?

When is capital important? In view of the foregoing discussion, it should be apparent that capital is always important. From an analytical perspective, however, the importance of capital depends upon the situation. In brief, capital is most important when it is conspicuously weak or in decline, a condition that is most likely to occur during turbulent or worsening operating conditions.

As a practical matter, the relative importance of capital in the analytical process is subject to several influences. One is that capital is at least in part a residual item. Consequently, it is the result of the interaction of other financial elements. Change in the values of these other elements may therefore provide earlier indications of changes in capital ratios, and take on greater relative importance, notwithstanding that they will ultimately be captured in the values of the applicable capital indicator.

Additionally, as the succeeding section will make clear, capital can be calculated in different ways, and the usefulness of each depends upon whether the manner of calculation is fully disclosed and upon the purpose for which it is used.¹² As a simple example, consider regulatory capital, which is explained in more detail later. Regulatory capital, as defined in the original Basel Accord and in its revised forms (Basel II and Basel III) allows for some items to be considered as capital that will disappear upon the bank's insolvency. These items do not therefore provide a cushioning function against loss and so ought to be discounted when analyzing the bank. Quite clearly also, the continuing value of a listed bank as—often inaccurately—embedded in its

price-to-book ratio loses its relevance if the bank disappears as a going concern. Finally, and most importantly from the position of a practicing bank analyst, capital recedes in importance during ordinary operations under stable or benign operating conditions. Capital assumes its greatest importance when it is already evident that there is too little of it—that is, when it has been materially impaired—or when it is significantly decreasing. This is most likely to occur when stressful conditions plague an entire banking industry. So long as capital indicators have been within a desirable range and have been stable or increasing, capital remains in the background as a critical concern.

Under such favorable conditions, more attention should be paid on those financial attributes of a bank that are prone to adversely affect capital or to otherwise cause the bank problems. For example, strong sustainable earnings allow for the rapid internal generation of capital, while signs of poor or deteriorating asset quality, such as above-average loan growth, are often precursors of subsequent capital erosion. Similarly, should conditions deteriorate, a strong liquidity and funding profile, especially where the funding is from sticky sources (e.g., insured retail deposits), may allow a bank to compensate for weaker nominal capital indicator values, but liquidity and funding attributes deserve close scrutiny in such times. In contrast, the seemingly better-capitalized bank that relies almost exclusively upon wholesale funding sources will probably be at greater comparative risk than one with profile just mentioned. Lastly, the competence and prudence of management, although perhaps the hardest attributes to gauge, may—when adverse features or changes are detected—provide the best early warning of elevated default risk.

Summary of Functions

To sum up, the main functions of bank capital are to:

- Fund startup, investment, and expansion to take advantage of new business opportunities
- Absorb unexpected losses and thereby allow the bank to remain solvent and functioning, and in the event of bank failure to fulfill the institution's obligations to depositors and other creditors
- Inspire confidence in depositors and creditors, regulators, and analysts and thereby avoid the discipline to which aggressively risk-seeking banks may be subject
- Comply with regulatory requirements
- Function as a yardstick against which risk management and capital allocation decisions can be more easily made

CAPITAL CONFUSION

To comprehend the role of bank capital, it is critical to distinguish between the different definitions that may be applied. Do not confuse *shareholders' equity*—capital as commonly understood in nonbank contexts—with *bank regulatory capital*.

Also, bear in mind the following important points:

- Equity capital, or shareholders' equity, is simply total assets minus total liabilities, and represents the everyday accounting view of capital.
- Financial capital is the corporate finance view of the firm, and considers the structure of the composition of the financing element of the balance sheet in terms of long-term debt vs. shareholders' equity.
- Regulatory capital is an artificial construct only applicable—in the context of this book—to banks, which is defined by bank regulators. Under the Basel Accords, regulatory capital encompasses shareholders' equity plus other items including certain eligible medium- to long-term subordinated debt. It is divided into fundamentally two types, with Tier 1 capital being largely, but not wholly, equivalent to equity capital and Tier 2 capital being very roughly equivalent to qualifying subdebt.
- Analytical capital, or adjusted capital, is whatever the bank analyst defines it to be in line with the relevant analytical objectives. For a bank credit analyst, it usually represents a narrower definition than regulatory capital, which excludes low-quality capital from relevant metrics. Note that the decision of what to include or exclude may take account of the quality of the assets on the bank's balance sheet from which equity capital is derived.
- Economic capital is not capital in the conventional sense of the word—that is, representing funds contributed to, available to, or retained by an enterprise. Instead, economic capital is a measure of risk, in which the amount of risk is measured in units of capital required to absorb expected and unexpected losses to a certain confidence interval.

Why Are Banks So Highly Leveraged?

To return to the question put aside earlier, why are banks so highly leveraged?

The answer is that banks are highly leveraged because banking is a commodity business, trafficking in the most fungible of all commodities: cash. The business of banking is taking calculated risks, where the bank's comparative advantage derives in part from the information it obtains as it intermediates between savers and borrowers. While this comparative advantage is real, and a particular bank may develop a level of expertise in a specific banking or geographical franchise, so long as there are rivals for the same business margins will tend to be undercut.

From the prospective borrower's standpoint, the overriding consideration is understandably the cost of the loan. It is therefore not easy for banks to differentiate themselves in respect to this core business in any manner but on the rate offered. Consequently, so long as significant competition exists in a given market, and no cartel or government regulation acts to maintain a threshold level of interest rates, margins on supplying banking products to depositors and customers will ordinarily be extremely thin, generally much thinner than is characteristic of a manufacturing or service enterprise. For banks to make decent profits and compete for capital along with other industries, they need fairly high leverage to earn an adequate return on

equity, because they must compete not only with other banks for capital but with all manner of other enterprises.

Another reason banks are highly leveraged is that unlike nonfinancial enterprises, which are expected to be able to eventually require less debt as they mature, in the ordinary course of bank's business there is no expectation that it will ever clear its debts by repaying all of its obligations. The bulk of these liabilities are, of course, financial obligations to depositors and to other banks. Consequently, so long as the bank accepts deposits or seeks commercial funding, it will have outstanding debt. The bank is expected to roll over (or replace with new deposits) the largest part of its liabilities indefinitely, as long as it continues to operate. Instead of being a cause for concern, being in a constant state of debt is an essential part of a bank's business model.

Instead, cause for concern would exist where a bank over an extended period steadily reduced its outstanding obligations (and its leverage) beyond the range considered to be extremely conservative within the banking context. Such a move would suggest that the bank was unable to attract new deposits or funding, or that the bank was exiting the commercial banking business based largely on deposit-based funding. It would certainly call for some explanation, which, if the result of an inability to access funding at going rates, implies some underlying credit problems.

REASONS FOR HIGH BANK LEVERAGE

One reason banks are highly geared is because they are dealing in the ultimate commodity, cash. Another is that ongoing debt is an inextricable element of their business model.

WHAT IS CAPITAL?

Because capital serves different purposes and because the relative importance of those purposes may differ among customers, managers, owners and regulators, complete agreement is lacking on exactly how capital should be defined.

—Mona Gardner and Dixie Mills¹³

This chapter began by addressing the relative importance of capital in the evaluation of a bank's financial strength—or conversely, the degree of credit risk that exposure to it engenders—and surveying the various ends that bank capital serves, with an emphasis on the similarities and differences between bank capital and capital in nonbank enterprises. In this connection, it was mentioned that capital can be defined and measured in different ways. To further illuminate the role capital plays in underpinning a bank's creditworthiness, in the succeeding sections of this chapter, each of the principal types of capital is considered in turn and in greater depth.

As do many terms in finance, and as the quotations above suggest, *capital* has a variety of meanings that depend upon the circumstances in which it is used. The word's

chameleon-like character comes perhaps from the fact that it is applied to several concepts that are related but distinct. Much of the ambiguity surrounding capital is dispelled if we first consider three separate, albeit overlapping, views of capital, all of which have been alluded to in preceding chapters and in the first part of this chapter. Consider the following five meanings of capital:

1. Capital as *total finance*, including short-term funding and, particularly, *long term funding*, whether debt or equity.
2. Capital as the *owners' investment* in a company.
3. Capital *as defined by regulators* for regulatory purposes.¹⁴
4. Capital *as defined by bank analysts* that is consonant with their analytical objectives.
5. Capital *as defined by a bank's own management* for internal accounting, risk management, and performance-evaluation purposes.

Each of these categories or facets of capital is elaborated upon in the pages that follow. Observe that the first two types of capital share a commonality in that they are also applicable to entities other than banks. The last three are generally applicable only to enterprises that come within definitions of banks under the relevant national law.

Financial Capital: Capital as Total Finance

From a corporate finance perspective, *capital* refers to all the funds used to finance an enterprise. When considering the meaning of capital from this vantage point, the liability side of the balance sheet, together with owners' equity, represents the funds used to acquire or maintain ownership of a bank's assets, classified by source.¹⁵ The position of equity and debt on the balance sheet, together with their components, is illustrated in Exhibit 9.1. (Note that the components are not drawn to scale and also that our focus here is on the liability side of the balance sheet, the asset side being provided merely for reference purposes.) Examining Exhibit 9.1, we can see the entire balance sheet of a bank from the corporate finance perspective, divided into its component parts according to the character of its assets and liabilities. More particularly, the term *financial capital* can be used to refer to capital viewed from a corporate finance vantage point. It includes funds made available to a firm on a relatively long-term basis and is therefore generally defined to exclude short-term funding.

Financial capital therefore encompasses two distinct categories of funds:¹⁶

1. *Long-term debt (loan capital)*
2. *Owners' or shareholders' equity (equity capital)*

The term *loan capital* is not an oxymoron, but instead refers to medium- and long-term debt used to finance an enterprise.¹⁷ Loan capital may take the form of a loan obtained from a bank, parent company, or other entity, or, in the case of a more established enterprise, a bond issue. Loan capital, together with equity capital, comprises financial capital, which constitutes comparatively long-term finance.¹⁸ To finance expansion, for example, management may opt for varying combinations of debt, equity,

ASSETS					LIABILITIES		
Liquid Assets	Cash and cash equivalent	Financial Assets		Funds borrowed and other liabilities	Short-term or current debt	Debt (i.e., liabilities)	
↑	Short-term debt securities						
Less Liquid Assets	Medium- and long-term debt securities						
↓	Loans						
Illiquid Assets	Fixed Assets			Invested Capital	Med- & long-term debt ("loan capital")	Equity (owners' residual claim on assets)	
	Intangible Assets	Non-financial assets					

EXHIBIT 9.1 Loan Capital versus Accounting (equity) Capital

and hybrid capital to meet its needs. A bank, therefore, normally uses short-term funding to finance its own lending, and this short-term funding is deemed to be separate from its longer-term debt issues, while naturally, both sources are used for lending.

Owners' equity encompasses the owners' investment contribution to the firm in exchange for shares, which function as units of ownership. The value of such shares will vary over time depending upon the accumulation of profits or losses, and distribution or dividends. Note that the funds originally placed at risk are invested indefinitely, and ordinary or common shareholders as a rule have no right to the return of their invested funds (used to purchase their shares) or to any guaranteed dividend or other return. Unless there is a secondary market for such shares, only upon the winding up of the enterprise will its common shareholders be entitled to payment, and then only to that residual amount that remains after all outstanding liabilities have been fulfilled.

Equity Capital: Capital as the Owners' Investment (Accounting Capital)

In the everyday milieu of investment reported upon in the business media, the term *capital* is frequently used to refer purely to the owners' (e.g., the shareholders') interests in a firm's assets—that is, shareholders' equity or equity capital.¹⁹ This is also the fundamental view of capital from an accounting perspective. Consequently, unless further specification is provided, capital is usually equated with *owners' equity*, so it is defined as the difference between a firm's *assets* and its *liabilities*. Since equity capital is what remains when financial obligations are subtracted from assets, it is by definition distinct from debt. Obviously, this is a narrower definition than that of *financial capital* just described.

Shareholders' equity was discussed in Chapter 5 and will not be further explored here, except for a few remarks related to bank equity.

CAN DEBT BE CAPITAL?

For those who approach bank analysis from a financial accounting, rather than a corporate finance or economics, perspective, it may seem somewhat odd to call debt "capital," or to allow debt obligations to be characterized that way, since the debt holders certainly want their money back in the event of financial distress. But as is evident from the discussion in this chapter, this usage is in line with the corporate finance view of the firm, and there is a case to be made that certain forms of debt finance may in some circumstances increase a bank's financial strength.

Sources of Bank Equity Capital

As with nonfinancial companies, there are two primary sources of capital: (1) internal generation by earnings; and (2) external contributions, as through an initial public offering, or subsequent offerings, including a rights issue.

Internal Generation

Once operations have begun, earning assets, funded mainly by deposits and borrowings, generate revenue. This stream of revenue, net of

- Expenses (including interest expenses and operating expenses that are paid to depositors, creditors, and suppliers)
- Loan-loss provisioning, produces net profits.

Net profits are divided into two parts:

1. One, dividends represent the portion of net profits distributed to shareholders.
2. Two, retained earnings, which represent profits that are not distributed to shareholders but are ploughed back into the bank and so constitute the bank's internally generated capital. This is a very important component of a bank's equity as banks almost always expand and need fresh capital to sustain additional volumes of risk assets.

External Contribution

Founding shareholders invest in the banking corporation in exchange for ownership interests (certificates of issued shares) in the institution. This investment constitutes the bank's paid-in capital or initial equity capital, which is comprised of the:

- Par value of the shares
- Any surplus paid for the shares in excess of their par value

This equity capital goes primarily to set up operations by obtaining the necessary premises and equipment, hiring staff, and so on.

Regulatory Capital: Capital as Defined by Regulators

In nearly all jurisdictions, banks are highly regulated entities. Bank regulators have long taken an interest in bank capital levels and have also endeavored to define capital for regulatory purposes. Capital so defined—what may be termed *regulatory capital*—is applicable only to banking organizations. Consequently, although it may bear some resemblances to the types of capital discussed earlier in this section, it should be clearly distinguished from them.²⁰

Regulatory Capital Defined

What is regulatory capital? Broadly speaking, it is whatever regulators say it is. Since the advent of the Basel Accord in 1988, however, *regulatory capital* is generally synonymous with capital as defined under that agreement and its successor, the revised accord (Basel II) adopted in 2005, subject to jurisdictional differences.²¹

Applying the Basel definitions, regulatory capital generally encompasses shareholders' equity as well as certain other items enumerated below. In respect to shareholders' equity, it includes most of the elements just discussed (*paid-in capital, surplus retained earnings*), subject to some adjustments. Regulatory capital, however, also includes a number of other line items that are not deemed to be capital in the usual accounting sense of the term, including certain forms of eligible medium- and long-term debt. The particular components of regulatory capital under the first Basel Accord and the revised accords are discussed in further detail later in this chapter as well as in Chapter 12.

Rationale for Regulatory Capital

The rationale for this broader view of bank capital, taking account of certain items other than shareholders' equity, has two main bases.

First, in respect to *balance sheet items that have a similar function to shareholders' equity* in providing a buffer against probable loan losses, there may be some justification in deeming them eligible to qualify as bank capital. As an example of this rationale, one item with a capital-like function is the category of loan-loss reserves. Subject to limitations, loan-loss reserves may constitute part of bank capital under various bank regulatory regimes.²²

Second, certain items of regulatory capital are included to enhance *market discipline*.²³ For example, the issue of medium- and long-term subordinated debt by banks is thought to create an incentive for market participants to scrutinize the credit quality of these issues and thereby closely monitor bank credit quality.²⁴ Imprudent management, it is thought, will be disciplined by investors demanding a higher risk premium for holding the paper of the subject bank and thereby raising its cost of funding in the capital markets.²⁵

Fundamentally, the underlying justification for a broad definition of regulatory capital is that it serves to make banks sounder and correspondingly less vulnerable to failure.²⁶ Although it is recognized that bank capital is not the sole measure of this attribute, minimum capital adequacy requirements are seen as a fundamental set of tools in the regulator's arsenal. As one commentator has noted:

Capital inadequacy in relation to the risks being run is behind virtually every bank failure. . . . That said, it is important to keep things in

*perspective. A bank's primary risks are not taken care of solely by throwing more capital at them. . . .*²⁷

The risk of using a particular definition of regulatory capital as a central indicator of bank creditworthiness is that it remains, at heart, an artificial construct that is subject to manipulation for the sake of appearance. In fact, regulatory arbitrage has become the norm in recent years.²⁸ In referring to the original Basel Accord, one central banker expressed equally strong concerns “that regulatory capital . . . is becoming a safety and soundness irrelevancy, and simply a compliance requirement.”²⁹

Let us now turn to a fourth type of capital: analytical capital.

Analytical Capital: Capital as Defined by the Analyst

From an analytical perspective, regulatory capital has some deficiencies.

Most importantly, it may include some elements that are less equity-like than others. For example, as just mentioned, some portion of subordinated debt may be included in regulatory capital. While the rationale for its inclusion is plausible, regulatory capital nevertheless defines as capital a mix of items that, critically, vary in their capacity to absorb losses and cushion economic shocks.

As far as this variance exists, it can be said that the quality of the capital also varies. High-quality capital is capital with a strong capacity to absorb losses and cushion economic shocks. Low-quality capital is capital with a weak capacity to absorb losses and cushion economic shocks. In view of such distinctions in capital quality, analysts may wish to define capital in a manner that eliminates weak forms of capital and contains only those items with the best buffering potential—those that constitute high-quality capital.

This can be done by making some adjustments in capital as reported by the bank in its financial statements. In making these adjustments, the salient question to be answered is this: *What definition of capital makes sense from an analytical perspective?* Just because certain balance-sheet items are permitted to be defined as capital by regulators, and consequently by bank accountants, does not mean that analysts should accept this definition as holy writ. Although the analyst can make use of accepted accounting and regulatory definitions, they should not be applied blindly.

In evaluating capital quality, either for purposes of defining analytical capital or as part of the overall analysis of the bank, it is critical not just to look at the liability side of the balance sheet where capital normally resides. Since capital—or at least *equity capital*—is a residual item, it is only as good as the assets that contribute to define it. Consequently, the asset side of the balance sheet must also be closely scrutinized.

Remember that when a bank's assets, typically its loans, become impaired, and are charged off, equity diminishes. Even if an actual write-off has not yet occurred, where the assets from which capital is a putative remainder are in actuality unprovisioned or underprovisioned loss assets, then equity capital must, correspondingly, be adjusted downwards. Likewise, those assets whose value is likely to be overstated on the balance sheet may, from an analytical perspective, be adjusted to reflect a value closer to reality. Again, the effect will be to further diminish equity. As an example, *intangible assets* such as goodwill or deferred tax assets that provide no or negligible cushioning against economic or financial shocks may be properly deducted from analytical capital.

Worse yet, the bank owners might have indulged in connected-party lending, in which case the entire amount involved should be deducted from the posted capital to provide a true picture of the analytical capital.

Bear in mind that there is no single answer as to how capital should be defined (nor how much is enough). Perhaps the most conservative approach is to define as capital only those items that have a strong capacity to absorb loss, namely core equity capital, and in the process excluding items that only have value insofar as the bank remains a solvent and profitable entity.³⁰

MEASURING CAPITAL STRENGTH: TRADITIONAL RATIOS

[C]apital by itself is not an effective predictor of future financial difficulty. . . .

—DBRS³¹

This section surveys traditional measures of bank solvency, which is another way to refer to relative capital strength. These indicators are, for the most part, quite similar to the leverage and other solvency ratios used in corporate credit analysis.

Non-Risk-Weighted Measures

The most basic ratios for measuring capital strength are *equity to assets* and *equity to deposits*. The use of these ratios in the banking industry dates back at least to the early 1900s.

Equity-to-Deposits Ratio

In the United States, an early rule of thumb of sound bank capitalization was one part capital to each 10 parts of deposits. The constraints imposed by this rule became apparent as the American economy expanded in the 1950s and 1960s, and banks began to increasingly obtain funding from the money markets.

$$\text{Equity-to-deposits ratio} = \frac{\text{Total shareholders' equity}}{\text{Total deposits(or average deposits)}} \times 100$$

Example: Cedar Bay Bank has total deposits of \$16 billion, debt and other liabilities of \$2 billion and shareholders' equity of \$2 billion. Its equity-to-deposits ratio = $2/16 = 0.125$ or 12.5 percent.

Equity-to-Assets Ratio

Equity to total assets succeeded equity to deposits as the ratio of choice, and a ratio of 8 percent capital to total assets came to be favored as a benchmark within the U.S. Federal Reserve system. A variation of this ratio excludes preferred shares. A reverse of this indicator can also be used; it is called the equity multiplier or the leverage

ratio, which is simply a way to express leverage as a multiplier rather than as a fraction. An equity-to-assets ratio of 5 percent simply becomes a leverage ratio of 20 times. Reversing the numerator and denominator results in another traditional ratio: assets to equity.

$$\text{Equity-to-assets ratio} = \frac{\text{Total shareholders' equity}}{\text{Total assets(or average assets)}} \times 100$$

Example: Cedar Bay Bank has total assets of \$20 billion and shareholders' equity of \$2 billion. Its equity-to-assets ratio = $2/20 = 0.10$ or 10.0 percent. Its leverage (assets to equity) ratio is $20/2$ or 10×2 .

As suggested in the preceding discussion, capital may be defined differently for different purposes. The same, of course, applies to capital ratios, which are subject to a number of variations. Most of these variants are found in the corporate credit realm, or in connection with equity analysis, and are not commonly used in for bank credit analysis purposes.³² Many of the modifications concern the definition of capital, irrespective of whether it forms the numerator or denominator of the ratio. *Capital* may be defined simply as shareholders' equity, more broadly as regulatory capital or more narrowly by, for example, subtracting questionable items. Intangible assets that often have an attenuated relationship with economic reality, such as goodwill, are often deducted from capital for ratio-calculation purposes.³³

Rudimentary Risk-Adjusted Capital Ratios

Analysts still use the equity-to-assets and other non-risk-adjusted leverage ratios especially in emerging markets, where if nothing else, a bank will invariably reveal its total assets and shareholders' equity in its annual report.³⁴ They also remain of some use in making cross-comparisons within a bank peer group. The equity-to-assets ratios have serious limitations, however, but not for arithmetical reasons. Keeping in mind that the purpose of capital is to provide confidence that the bank can absorb losses and maintain a cushion to meet its obligations—meaning its liabilities—not counting equity, whose total is almost equal to its assets, an equity-to-assets ratio is as predictive as an equity-to-liabilities ratio. Instead, the problem is that the equity-to-assets ratios do not take account of the riskiness of the assets and the concomitant probability of loss.³⁵ An asset is an asset is an asset, no matter if it is a U.S. Treasury bill or a loan to the local neighborhood watering hole, and the risk that its value may fall is considerably higher than the risk—in fact, the opportunity—that a bank liability could fall.

Risk-adjusted ratios overcome this limitation,³⁶ to the extent that risk adjustment actually corresponds to different levels of economic risk. Instead of treating all assets equally, they take account of the riskiness of a bank's assets to measure the relative strength of its capitalization. A bank with only high-risk assets requires more capital to achieve the same numerical value than a bank with a mix of high-risk and low-risk assets. This is the principle behind the first Basel Accord. Before the Basel Accord, ratios based on total assets were commonly disclosed, and more rudimentary

risk-adjusted capital indicators compared equity capital to total loans or to total risk assets. In other words, pre-Basel, risk assets generally referred to loans or somewhat more broadly to all earning assets at risk of default.³⁷ So the equity-to-loans ratio, discussed next, can be viewed as a primitive form of risk-adjusted capital ratio.

Equity to Loans

The equity-to-loans ratio simply means a bank's total shareholders' equity divided by its total loans (see the equation that follows). The rationale for using equity capital to total loans is that since loans are the riskiest assets banks generally hold, it is the proportion of capital to these assets that is more relevant in evaluating a bank's solvency. Of course, nowadays banks carry huge risks outside the strict perimeter of traditional lending, but the equity-to-loans ratio remains useful.

$$\text{Equity-to-loans ratio(\%)} = \frac{\text{Total shareholders' equity}}{\text{Total (customer) loans}} \times 100$$

Example: Cedar Bay Bank in Year 1 has total assets of \$20 billion, loans outstanding amounting to \$16 billion, and shareholders' equity of \$2 billion. Its equity-to-loans ratio = $2/16 = 12.5$ percent vs. its equity-to-assets ratio of 10 percent.

Since total loans will always be smaller than total assets, equity-to-loans will result in a higher ratio value than equity-to-assets.

Equity to Risk Assets

In addition to its loans, a bank's other earning assets are, of course, generally subject to credit risk. Accordingly, a variant of the equity-to-loans ratio is another basic type of risk-adjusted indicator, which simply lumps all assets subject to credit risk, that is, effectively all earning assets, into the denominator of the ratio as shown in the equation that follows. Again, note that this definition of risk assets is not the same as risk-weighted assets as defined under the Basel accords.

$$\text{Equity-to-risk-assets ratio(\%)} = \frac{\text{Total shareholders' equity}}{\text{Risk assets}} \times 100$$

Example: Cedar Bay Bank has total assets of \$20 billion, loans outstanding amounting to \$16 billion and other risk assets of \$2 billion. Its equity-to-risk-assets ratio = $2/18 = 11.1$ percent vs. its equity-to-assets ratio of 10 percent and its equity-to-loans ratio of 12.5 percent.

Internal Growth Rate of Capital: A Measure of the Link between Profitability and Capital

The internal growth rate of capital, or IGRC, is closely linked to profitability, and refers to the rate at which a bank is able and willing to retain its earnings—in fact its

so-called retained income—and use the funds to increase its capital.³⁸ If a bank has an historic record of profitability and the ability to sustain similar rates of earnings growth going forward, it will be able to earn itself out of most potential problems. Only a liquidity crisis, most likely occasioned by a severe economic shock, or an extremely severe capital impairment would, as a rule, pose a threat. Otherwise, in most cases, over time, a bank will be able to rebuild its capital strength to satisfactory levels, again assuming reasonable profitability. Think about it this way. If, after dividend payout, a bank is able to generate a return on assets of 2.0 percent, it can effectively afford to write off at least 2.0 percent of its loan book each year, without impairing capital.³⁹

$$\text{Internal growth rate of capital (IGRC)(\%)} = \frac{\text{Retained income}}{\text{Average shareholders' equity}} \times 100$$

Example: Cedar Bay Bank in Year 1 has total assets of \$20 billion, loans outstanding amounting to \$16 billion, and average shareholders' equity of \$2 billion. Its net income last year was \$400 million, of which \$200 million was distributed to shareholders in dividends. Retained earnings are therefore equal to \$200 million, which divided by average shareholders' equity of \$2 billion generates an IGRC of 10 percent.

While the IGRC evinces a bank's ability to create capital, and is consequently an important supplement to balance-sheet indicators, it has some weaknesses as an indicator. Most importantly, it is subject to large fluctuations as earnings oscillate. Moreover, there is no assurance that high historical rates of internal capital growth ensure the future ability to replenish capital at the same rate. In addition, the IGRC varies directly according to the dividend payout policy, which the bank may vary at will over time. It can also complicate analysis in respect to bank holding companies. For example, low dividend payouts on the part of the bank subsidiary, while superficially positive at the subsidiary level, can increase double leverage at the holding company level, giving rise to internal stresses within the group and reducing the apparent benefit of a high IGRC.

Aside from concerns in respect to bank holding companies, when examining the IGRC, the analyst should consider whether:

- The bank has been internally generating capital at a healthy rate.
- There have been any infusions of new capital.
- New capital injections, if any, have come from a public offering of shares, or a contribution by existing shareholders through a rights offering, for example.
- Capital increases have been accomplished through accounting legerdemain—through a revaluation of reserves for instance.

Of course, even more critically, the analyst should make note of any decline in capital, and corresponding indicators, such as one resulting from write-offs of loan losses.

Holding Company Structure and Double Leverage

In some markets, most significantly the United States, major banking groups use a holding company ownership structure. A key benefit of such an approach is that it enables a bank to expand into other types of nonbank financial business, such as

insurance or securities brokerage, while at the same time keeping separate accounts for each business and not distorting the core bank's financial statements as a result.

One aspect of employing this structure is that it enables a bank's owners to increase financial leverage in a nonobvious way. Typically, a bank holding company, separate and distinct from the bank it owns, is formed and acquires total ownership interests in banking subsidiaries and sometimes in nonbank subsidiaries. Insofar as the holding company is able to borrow funds, whether in the money or capital markets and convert some of the borrowed proceeds into equity into the bank subsidiary, leverage on the part of the ultimate owners of the bank is effectively increased.

Where parent leverage over the bank subsidiary exceeds that of the bank subsidiary itself, double leverage is said to occur. As stated in the U.S. Federal Deposit Insurance Corporation's *Manual of Risk Management Examination Policies*:

The double leverage ratio is the equity of the subsidiary, or in the case of multiple subsidiaries the combined equity of all the subsidiaries, divided by the equity of the holding company.

So double leverage exists when shareholders' equity on the balance sheet of the parent (i.e., holding company) is less than that of the bank it owns.⁴⁰ The double leverage ratio is illustrated in this equation:

$$\text{Double leverage ratio} = \frac{\text{Shareholders' equity of bank subsidiary}}{\text{Shareholders' equity of bank holding company}} \times 100$$

High levels of double leverage are viewed as credit negative. The reason is well-stated in the above-referenced manual.

*Even when the holding company is financially sound, supervisory concerns may arise as the parent issues long term debt to fund equity capital in the subsidiaries. Although this capital raising activity, known as "double leveraging," does increase equity capital in the subsidiary, too much debt at the holding company level can generate pressure on the subsidiary to upstream additional dividends.*⁴¹

If a bank holding company uses a substantially higher amount of leverage than its bank subsidiary, its creditworthiness and corresponding credit rating may fall below that of the subsidiary. This has been the case with several bank holding companies in the United States. As a rule of thumb, double leverage in excess of 115 to 120 percent is regarded as a warning signal, calling for greater scrutiny into the banking organization's level and allocation of capital.

REGULATORY CAPITAL AND THE FIRST BASEL ACCORD

[In] the mid-1970s, American bankers began running into competition they could not understand. They encountered groups of banks that priced loans and extended credit in ways that made no sense. They were Japanese banks.

—R. Taggart Murphy⁴²

[The] most potent influence in boosting capital ratios has been the Basel frame-work itself. . . . Banks now have no choice but to maintain capital ratios possibly higher than economic need might otherwise suggest.

—Eddie Cade⁴³

Background to the 1988 Capital Accord

Beginning in the mid-1970s and accelerating throughout the early 1980s, bankers in the United States and Europe began to face new and unprecedented competition from Japanese banks.⁴⁴ While Japanese manufacturing companies were becoming known worldwide as formidable competitors during the same period for their ability to produce extremely high-quality products for exports at affordable prices, the competitiveness of Japanese banks was something of a puzzle.

It could not be efficiency. Anyone walking into a bank in Japan at the time could see that they were models of inefficiency—hugely overstaffed and lagging their Western counterparts in the adoption of technology. A simple retail-banking transaction such as a foreign remittance could take upward of a half-hour.⁴⁵ As bank analysis began to catch up with the increasing global reach and dominance of Japanese financial institutions, it became evident that the principal reason Japanese banks could undercut their foreign competitors was that they were undercapitalized. With the firm promise of government support, these behemoth financial institutions were leveraged to the hilt. As a result, they could achieve reasonable returns on equity with comparatively low returns on assets, making them formidable competitors, in international lending, to banks operating from other parts of the world.

Even before prudential concerns about Japanese banks—which, of course, is an angle different from the competition issue—began to mount, efforts were already under way to strengthen international banking supervision. In 1974, after the fright inspired by the collapse of Herstatt Bank, a small German bank active in the foreign exchange markets, the central bank governors of the so-called Group of Ten countries (G10) established the Basel Committee on Banking Supervision. In 1983, the Basel Concordat was formed by the G10 committee⁴⁶ to provide a framework for international banking supervision through national banking authorities. By the mid-1980s, a consensus had formed among the G10 nations, given impetus by the perceived threat posed by Japanese banks, that a level playing field concerning bank capital level was necessary.

The Basel Committee on Banking Supervision, sometimes referred to as the Basel Committee meets about four times a year. The committee has no formal supranational supervisory authority, and its conclusions, in themselves, have no legal force. Its mission instead is to formulate broad supervisory standards and guidelines, and to recommend statements of best practice with the hope that national regulators will implement such norms through their own legal and supervisory frameworks.

In December 1987, initial proposals were published and by June 1988, agreement was reached among the G10 to establish a minimum risk-weighted capital requirement for internationally active banks in the 12 countries.⁴⁷ The accord provided that by March 1993, the ratio of capital to risk-weighted assets must be a minimum of 8 percent. To gain the assent of the Japanese authorities, a concession was made that allowed Japanese banks to count 45 percent of their “hidden reserves”

as capital. At the time, the skyrocketing Tokyo stock market, of which bank shares made up about 25 percent of the capitalization, made this compromise attractive.⁴⁸

Formally known as the International Convergence of Capital Measurement and Capital Standards Systems, the accord was previously commonly referred to as the Basel Capital Accord⁴⁹ and is now referred to as Basel I (the official website uses Roman numerals, although Arabic numerals are sometimes used by commentators and analysts). The comprehensive revision to Basel I was released in mid-2004 and is informally referred to as Basel II.⁵⁰ The most recent revision is called Basel III.

Although the accord was not enforceable by an international body—it was remarkably successful, if popularity were the only gauge. Even if not fully binding even on the original 12 members, it was implemented largely according to its terms. Ultimately, more than 100 countries worldwide have used the accord (or a variant of it) as a benchmark for their own national banking supervision, even though most of them did not participate in negotiating the accord and did not formally approve it. The European Union (EU) contributed to its acceptance by subsequently making the framework embodied in the accord mandatory on all EU countries and required that member states enact national banking laws implementing it.

Regulatory Capital and the 1988 Accord

The Basel Accord gave an international imprimatur to the concept of regulatory capital. The heart of the accord and the innovation that made it so successful was the standardization it brought to capital measurement, which was epitomized in the establishment of an official risk-weighted CAR (capital adequacy ratio). Until the accord was implemented, not only was there no international consensus on how much capital was a reasonable minimum, there was no firm consensus on what constituted capital.

Before Basel, countries varied widely in how they defined capital for regulatory purposes. Reconciling the different approaches, even assuming that adequate disclosure was forthcoming was an analyst's nightmare. Whatever the deficiencies in the 1988 accord generally, and more specifically in the agreed definition of capital adequacy, by building a uniform definition and methodology for calculating regulatory capital adequacy and by establishing a benchmark under that methodology, the original agreement represented a great advance over what had existed before.

At the time of this writing, some countries have yet to adopt the subsequent Basel II rules, and it is therefore important to explore the first Basel rules, all the more so because many of its principles were transposed to Basel II.

The CAR and Its Calculation

The Rosetta stone of the Basel Accord that has ostensibly enabled regulators, investors, and creditors to divine whether an internationally active bank had sufficient capital strength, is the CAR. This ratio in a single number measured a bank's capital as defined against its assets to determine whether or not a bank met the threshold. The magic number above which a bank was deemed to have achieved satisfactory capital strength was set arbitrarily at 8 percent.⁵¹

The accord specified in precise detail how capital adequacy was to be calculated: what balance-sheet items were to be included in which category of capital and how

risk weightings applied. Some discretion, however, was left to national regulators in defining and implementing the scheme.⁵² Although the CAR was originally intended to apply to internationally active banks in countries within the G10, soon the minimum of 8 percent risk-weighted CAR became a near universal measure of bank solidity, even though that CAR did not mean the same thing in every country.

$$\text{Capital adequacy ratio (CAR)}(\%) = \frac{\text{Tier 1 capital} + \text{Tier 2 capital}}{\text{Risk-weighted assets}} \times 100$$

While sometimes complex in execution, the method of calculation was conceptually quite simple. The numerator, capital, was separated into core (Tier 1) capital and supplementary (Tier 2) capital (see the following equation). At least 50 percent of a bank's risk weighted capital had to be core (Tier 1) capital.⁵³ The denominator, assets, was divided into separate components based on defined risk weightings of 0 percent, 20 percent, 50 percent, and 100 percent. Risk weightings are summarized in Exhibit 9.2.

Definitions of Capital under the Basel Accord

By accepting a definition of capital broader than that of shareholders' equity, the Basel Accord acknowledged the concept of regulatory capital, but did so within a framework that recognized a distinction in the quality of capital. It divided regulatory capital into core capital or Tier 1 capital, on the one hand, and supplementary capital or Tier 2 capital, on the other.

The approach to risk-weighting assets has changed dramatically in the past 20 years, but under the first Basel Accord, as befits a regulation intended to establish uniform rules for bank capitalization, capital was defined with some precision, and the definition has remained conceptually unchanged in the new Basel accords.

Core (Tier 1) Capital

The strongest form of capital—the one that afforded the greatest comfort to creditors and is therefore of the highest quality—is shareholders' equity. When owners have placed their own funds at risk, there is, at least in theory, a strong incentive to see that such capital is managed prudently. Accordingly, common shareholders' equity, with some slight modifications, was identified as core capital, under the Basel scheme. Besides ordinary equity capital, Tier 1 capital also encompasses eligible preferred shares and certain other items that are discussed in greater detail next.

EXHIBIT 9.2 Basel I Weightings

Type of Asset	Risk Weight
Cash and OECD government securities	0%
Banks in OECD countries	20%
Residential real estate mortgages	50%
Ordinary corporate loans	100%

Source: Basel Accord.

Equity Capital

Tier 1 capital includes what we have previously described as equity capital, subject to certain adjustments. More specifically, Tier 1 capital includes first and foremost, ordinary or common shareholders' equity, composed of:

- Issued and fully paid ordinary shares/common stock
- Surplus (i.e., the difference between the par value of the shares and the amount paid for them by investors)
- Retained earnings (recall that retained earnings represent internally generated capital)
- Minority interest
- Disclosed (equity) reserves

Preferred Shareholdings

In addition, Tier 1 capital also includes:

- Eligible preferred or preference stock, specifically noncumulative perpetual preferred shares
- Certain hybrid securities, such as convertible bonds meeting regulator specified criteria

Perpetual preferred share issues eligible for inclusion in Tier 1 capital are sometimes referred to as *Tier 1 preferred*, or more loosely, and somewhat confusingly, as *Tier 1 bonds*.⁵⁴

Not eligible for Tier 1 status is cumulative preferred stock—that is, stock that gives the holder the right to receive a dividend deferred on account of insufficient profits at the time payment was due is excluded.⁵⁵

Tier 1 Exclusions

Excluded or subtracted from otherwise eligible Tier 1 capital are the following:

- Treasury shares (i.e., the bank's own shares held by the bank) as well as the bank's holdings of its own eligible Tier 1 bonds
- Interim net losses
- Goodwill

Supplementary (Tier 2) Capital

In contrast to Tier 1 capital, Tier 2, or supplementary, capital included certain forms of subordinated debt as well as various other balance sheet items. Subordinated debt, that is, debt junior to the claims of the bank's depositors and senior creditors, could comprise up to half of total capital so long as it met certain eligibility criteria.⁵⁶ Aside from subordinated debt, certain hybrid debt and reserve items were allowable as Tier 2 capital. Again, these are discussed in more detail in the succeeding section.

Notwithstanding that Tier 2 is generally comprised of debt obligations rather than equity capital, it was believed that they provided additional protection against bank failure.⁵⁷ To be sure, in the case of Tier 2 subordinated debt, it could be argued that these obligations bear a number of resemblances to shareholders' equity and that vis-à-vis depositors and senior debt holders, they are akin to equity capital. Another

argument in favor of Tier 2 capital is that these debt issues stimulate market discipline. Nevertheless, from a credit analyst's perspective, Tier 1 capital or shareholders' equity still remains the key measure of the bank's ultimate capital cushion.

Tier 2 capital was rather more complex than the definition of Tier 1 capital. It generally has been deemed to encompass the following, subject to specified eligibility criteria:

- Subordinated debt and certain other financial instruments, including
 - Subordinated term debt
 - Hybrid debt
- Reserves, including
 - General loan-loss reserves
 - Revaluation reserves
 - Undisclosed (hidden) reserves

Subordinated Term Debt and Hybrid Instruments

Subordinated term debt is, in most instances, the principal component of Tier 2 capital. To qualify, it must:

- Have a minimum maturity of no less than five years.
- Be subordinated only to senior debt obligations.
- Have appropriate amortization provisions as part of the agreement governing the issue.

The referenced amortization provisions require that in the last five years in which the issue is outstanding, the qualifying capital must be amortized (i.e., reduced) to reflect its increasingly current debt-like quality.⁵⁸

Lower Tier 2 Capital versus Upper Tier 2 Capital

Ordinary subordinated debt that meets the foregoing requirements is permitted to be counted as Tier 2 capital. Notwithstanding the general principle, total Tier 2 capital can be as much as 100 percent of total Tier 1 capital. Unless it has specific identified equity-like characteristics, it may only count as Tier 2 capital up to 50 percent of total Tier 1 capital. To gain the full measure of the Tier 2 allowance, the bank as the issuer of the putative Tier 2 instruments must be able to escape the obligation to make periodic fixed payments to the investors during extreme financial conditions.⁵⁹

The ability to escape such obligations renders the instrument more equity-like—just as there is no obligation to distribute a dividend to common shareholders—and clarifies whether Tier 2 capital is deemed to be classified as upper Tier 2 capital or as lower Tier 2 capital.⁶⁰ Upper Tier 2 capital is deemed to be the more equity-like portion of Tier 2 capital and therefore is not subject to the 50 percent limitation on issuance. Lower Tier 2 capital is more debt-like and is subject to a limitation on issuance of no more than 50 percent of Tier 1 capital.

Upper Tier 2 Capital

In contrast to lower Tier 2 capital, upper Tier 2 capital is subordinated debt that contains both the following equity-like eligibility features:

- The ability to defer interest payments at the option of the issuer under defined conditions so that the obligation is effectively not entirely fixed
- The ability of the issuer to impose losses—for the sake of the solvency of the issuing bank—on the upper Tier 2 debt holders through the write-down of principal or interest

Typical circumstances that would permit a reduction in interest payments to investors include the issuer's:

- Inability to pay dividends to shareholders
- Inability to meet regulatory capital requirements
- Recording of an accounting loss during the applicable year

Lower Tier 2 Capital

Lower Tier 2 refers to ordinary subordinated debt that, by reason of the absence of the foregoing features, is deemed to be more debt-like than equity-like. As a consequence, Lower Tier 2 is subject to the 50 percent limit on issuance.

Hybrid Debt

Hybrid debt instruments that do not qualify as Tier 1 capital—for example, long-term preferred shares in Canada, mandatory convertible debt instruments in the United States—may be classified as Tier 2 capital, provided they meet certain criteria.⁶¹

Reserves

Certain reserve items on the balance sheet may be eligible to be Tier 2 capital in specified circumstances.

Undisclosed (Hidden) Reserves Hidden reserves are equity reserves representing additional value not reflected directly on the bank's balance sheet. They typically arise through substantial appreciation in a bank's securities holdings. To qualify as Tier 2 capital, such reserves ordinarily will have been first registered on the profit and loss (P&L) and, most importantly, will have been accepted as legitimate by the relevant local supervisory authorities.⁶²

Revaluation Reserves Revaluation reserves, permitted in some countries, which result when an adjustment to the historic cost of assets produces an unrealized gain, are permitted to be classified as supplementary capital in some circumstances. Regulatory approval is key, with the revaluation required to have been "considered by the supervisory authority to be prudently valued, fully reflecting the possibility of price fluctuations and forced sale." Revaluation reserves typically arise through the substantial appreciation in a bank's real property holdings.⁶³

General Loan-Loss Reserves General loan-loss reserves, in contrast to specific loan-loss reserves ascribed to particular identified problem assets (e.g., loans), are included within Tier 2 capital up to a limit of 1.25 percent of weighted risk assets.⁶⁴

Additional Tier 1 and Tier 2 Exclusions

In addition to the deductions previously mentioned in connection with Tier 1, when calculating a bank's capital adequacy, some additional items must generally be subtracted from the whole of regulatory capital.⁶⁵ An example is holdings of another bank's capital exceeding 10 percent of the capital base of the holding bank.⁶⁶

Tier 3 Capital

In addition to Tier 1 and Tier 2 capital, there is also a somewhat obscure category of capital known as Tier 3 capital. It came into being following the Basel Committee's 1996 revision to the first accord, "Amendment to the Capital Accord to Incorporate Market Risk."⁶⁷ An arguably weak form of capital, Tier 3 capital is a form of short-term subordinated debt designed to support market risk exposure in trading and treasury activity by financial institutions.⁶⁸ It does not count as regulatory capital for the purpose of supporting credit or counterparty risk.

Tier 3 instruments must be unsecured, fully paid up, subordinated to senior creditors (Tier 3 paper ranks equally with upper Tier 2 paper), have a minimum maturity of two years (being not repayable before maturity without the agreement of bank regulators), and be subject to a deferral of payment of interest or principal, at the discretion of the regulator, if a bank has not maintained its minimum capital requirements.⁶⁹ Tier 3 capital is generally permitted in Europe and in the United States, albeit not in all jurisdictions, but not elsewhere. Overall, the market for this type of debt has been limited to date.

Summary

Generally speaking, Tier 1 capital is of a high quality, being composed largely of shareholder funds placed at risk (shareholders' equity), with deductions both from the asset and liability side of the balance sheet of its weakest components, such as goodwill (on the asset side) and certain equity reserves (on the liability side), such as revaluation reserves. Other less strong components of Tier 1 capital include preferred shares that meet Tier 1 eligibility standards and the equity engendered by deferred tax assets. Tier 2 capital on the whole is less equity-like than Tier 1 capital, and is largely synonymous with certain eligible subordinated debt. Exhibit 9.3 lists the various forms of regulatory capital in relation to the liability side of the balance sheet as a whole.

Risk-Weighted Assets

On-Balance-Sheet Assets

Under the first Basel Accord, both on-balance-sheet assets and off-balance-sheet exposure were weighted according to imputed levels of risk. As seen above, the risk weighting buckets were kept to a small number. The benefits of simplicity and political agreement came at the expense of economic reality. The risk categories were so broad and the selection criteria, which reflected a simplistic rationale, so unrefined as to belie their intended purpose. Membership in the OECD, a group that encompassed an enormous diversity of nations, was deemed to correlate with low

EXHIBIT 9.3 Tiers of Regulatory Capital (in Relationship to the Liability Side of the Balance Sheet)

	Type of Liability	Characteristics	Tier 1, 2, or 3 or None and Remarks	Caps and Restrictions
Not Capital	Customer Deposits	Not capital; protected by capital.	Funding, not capital	No capital-related restrictions.
	Senior debt (includes commercial, i.e., interbank depositors)	Not capital; protected by capital.	Funding, not capital	No capital-related restrictions.
Tier 3	Tier 3 capital	Tier 3 capital Maturity $\geq$ 2 years. Subject to regulatory deferral but cumulative, and principal write down by issuer permitted. Only counts as capital for trading book purposes, not against credit risk.	Tier 3 capital	Applicable only to trading book. Does not count as part of capital to meet 8 percent regulatory threshold.
	Other subordinated debt qualifying as Tier 2 capital	Lower Tier 2 capital Maturity $\geq$ 5 years. Cumulative, nondeferrable, no principal write-downs by issuer but must be amortized for regulatory capital calculation purposes during last 5 years.		Lower Tier 2 capital may not exceed 50 percent of Tier 1 capital for ratio calculation purposes.
Tier 2	Junior subordinated debt (perpetual/step up perpetual)	Upper Tier 2 Capital Cumulative, but deferrable at issuers' option and may be written down by issuer in defined circumstances.	Tier 2 capital	Total Tier 2 capital not to exceed Tier 1 capital for ratio calculation purposes.
	Other allowed Tier 2 items	Certain defined equity reserves, e.g., evaluation reserves, hidden reserves, etc., and general loan-loss reserves, which, technically speaking, are a contra-asset (that reduces the book value of balance sheet assets).		<i>Goodwill and specific LLRs, among other items, are disallowed.</i>

(Continued)

EXHIBIT 9.3 (Continued)

	Type of Liability	Characteristics	Tier 1, 2, or 3 or None and Remarks	Caps and Restrictions
Tier 1	Other allowed Tier 1 items	E.g., deferred tax assets (limitations vary)		Tier 1 capital must comprise at least 50 percent of total regulatory capital (i.e., Tier 2 capital may not exceed the amount of Tier 1 capital).
	Hybrid Tier 1	Certain convertible bonds and specified instruments (may comprise up to 15 percent of core Tier 1 capital)		
	Eligible Preference Shares (Preferred Tier 1)	Noncumulative perpetual preferred shares (also known as "Tier 1 bonds")	Tier 1 capital	
	Common Shares	<i>Ordinarily, the major component of</i>		← EQUITY CAPITAL
	Minority Interest	<i>Tier 1 capital. Including minority interest</i>		
	Retained Earnings	Internally generated capital		

risk vis-à-vis nonmember countries.⁷⁰ Similarly, a single risk weight for all corporates was equally faulty in conception. Consider, for example, the proverbial loan to a hypothetical neighborhood business—call it Joe's Bar—and a loan to GE, a major U.S. corporation.⁷¹ Under the first Basel accord, both were weighted equally, at 100 percent, notwithstanding an obviously wide divergence in their risk profiles.

Off-Balance-Sheet Commitments

Off-balance-sheet commitments, which until the 1980s had often escaped regulatory scrutiny, were captured under the Basel rules. Such contingent liabilities were effectively treated as risk assets under the accord. These risks, which had been causing increasing anxiety among bank regulations, were assigned a capital charge, and certain off-balance-sheet commitments sometimes required as much capital as ordinary loan assets. The risk weighting of off-balance-sheet items is a lengthy and complex matter which is best explored by accessing the various texts of the Basel Accords,⁷² as this would divert this book away from its main topics.

Basel-Related Risk Ratios and Supplementary Capital Indicators

Risk-weighted ratios are intended to take into account the risk profile of a bank's assets in measuring its capital strength. The usefulness of the ratio will naturally vary to the extent that the risk-weightings of assets reflect economic reality and to the extent that the capital measured is actually available to absorb unexpected economic loss. Since the first Basel Accord was implemented, the CAR and its component ratios, particularly the Tier 1 ratio, have come to assume more importance in bank analysis. Basel II and Basel III brought considerable changes to the CAR, but mostly in respect of its denominator. In addition to the CAR, other risk-weighted ratios, which may be used for comparative purposes when CAR numbers are not available, or as an alternative to the CAR, are shown in the section that follows.

CAR

Although a minimum of an 8 percent CAR ratio is required for banks that are subject to Basel rules, actual ratio values vary considerably from country to country. CAR values, however, are only reflective of true capital strength insofar as the data they measure provide an accurate reflection of the depth of a bank's capital cushion. As is discussed below, there were significant deficiencies in the 1988 accord, which detracted significantly from the utility of the CARs as a measure of bank soundness. Basel II, though a great improvement over the original accord, still has material weaknesses, which Basel III may be able to address.

Nevertheless, all other things being equal, a high CAR is generally viewed favorably by bank credit analysts. As a rough rule of thumb, ratios in the 10 to 12 percent range are normally considered acceptable, those in the 8 to 10 percent range will tend to be seen as adequate although possibly marginal in some markets, depending upon peer averages and the local operating environment. In contrast, ratios in the 12 to 15 percent range would be ordinarily viewed as quite strong, although again it very much depends upon the market. At the turn of the century, for example, a CAR of 14 percent for a Hong Kong bank would not have been significantly above the industry average.

In fact, because of the limitations of the first Basel Accord, many analysts preferred paying more attention to a bank's Tier 1 ratio than to its total numerator (being Tier 1 + Tier 2), as Tier 1 had the virtue of representing, generally speaking, a higher quality of capital.

$$\text{Tier 1 ratio}(\%) = \frac{\text{Tier 1 capital}}{\text{Risk-weighted assets}} \times 100$$

Other Similar Ratios

Some other ratios are variants of the Tier 1 ratio and may involve as the denominator, for example, the tangible common equity, being shareholders' equity less all intangible assets, excluding mortgage servicing rights⁷³ for conforming mortgages or would broaden the denominator to include certain off-balance-sheet items—namely off-balance-sheet securitized assets—that would not figure as prominently in risk-weighted assets under the original Basel Accord, as amended.

Evaluating Capital Strength: When Is Capital Important and How to Measure It?

At the outset of this chapter, the suggestion was made that capital is overrated as a measure of a bank's financial strength. It is not that capital is unimportant. Instead, because it is so important, it is one of the attributes of a bank most prone to manipulation. Not only are there a number of ways to define capital generally, even within the Basel framework, but also banks still have considerable discretion in how their ultimate capital ratios are calculated. Moreover, as suggested earlier, capital is not a leading indicator; instead, it is influenced by changes in profitability and in credit costs. By the time danger signs are apparent in reported capital ratios, it is probably too late to take remedial action.⁷⁴ Although capital is not generally as useful an indicator for analytical purposes as might be expected, there is a time at which it comes into its own as perhaps the most critical indicator. During a banking crisis, when banks are in potential distress and in need of state support, their capital ratios are of preeminent importance. All other things being equal, the less capital a bank possesses, the greater the cost of its recapitalization and of its ultimate rehabilitation.

Assuming a bank is not in distress, how should capital be evaluated? As with the other CAMEL financial attributes, key capital ratios should be evaluated historically and in comparison with the bank's peers. Three different types of indicators should be considered: traditional leverage indicators, Basel-style ratios and other adjusted solvency metrics, and finally the capacity of the bank to generate capital internally should be considered. In each case, as with other financial attributes, the analyst should be on the lookout for banks that consistently show significantly below-average ratio values or that exhibit a clear adverse trend. Not all indicators carry equal weight from a credit perspective. A bank that has a somewhat higher CAR than another bank, but achieved the better ratio value through a greater proportion of Tier 2 subordinated debt issuance, should not necessarily be considered better capitalized than a bank with a lower CAR but with a higher Tier 1 ratio. This

illustrates the point that a review of a bank's capital position should also take account of the quality of bank capital, as well as its nominal volume. As another example, consider the impact of intangible assets such as goodwill on the balance sheet. Although deducted from capital under the Basel rules, goodwill will contribute to nominal assets and therefore to nominal equity. Its capacity to cushion loss is, however, essentially nonexistent. In addition, capital ratios should be considered in light of the stability of its operating environment and the bank's risk appetite.

A final point to bear in mind is that insolvency alone will not necessarily trigger a bank failure. Although insolvency is ordinarily at the root of and frequently accompanies bank failure, the collapse itself is almost invariably triggered by a shortfall of liquidity rather than mere technical insolvency. Technical insolvency means that the true value of a bank's assets is exceeded by its liabilities, but nevertheless, owing to ample liquidity (generally provided by the central bank), the institution is able to continue to keep its doors open. As a matter of fact, a fairly sizable number of Asian banks, in the 1980s and 1990s, were technically insolvent, but continued functioning normally with state backing. Some still do, in spite of the economic emergence of Asia. As would be expected, unless required to do so by the regulator, banks that are technically insolvent tend not to broadcast their true state of affairs, particularly in countries where appearances and the need to save face count for much. Instead, as capital has acquired such perceived importance, banks are given an incentive to inflate capital indicators and to hide the volume of the NPLs in their loan book. This accounts for much of the obfuscation and lack of disclosure in respect of NPLs, and for the skepticism with which capital ratios are justly viewed.

THE BASEL II AND BASEL III ACCORDS AND THE CONCEPT OF ECONOMIC CAPITAL

The aim is for capital requirements to be better correlated to actual economic capital needs, as calculated by each bank according to its own model.

—Morgan Stanley, *Bank Capital A–Z*

Problems with the Old (1988) Capital Accord

The first Basel Accord brought some undeniable benefits, as noted in the previous section of this chapter. They included a degree of uniformity in the approach taken by regulators worldwide to set minimum prudential capital levels. Equally, they established a forum in which bank regulators from all over the world could share their experience, views, and knowledge.

It was not long before the deficiencies of the scheme soon became evident. The primary criticisms of the 1988 accord, as it was implemented, was that it created distortions in the way banks allocated their funds, causing them to take on credit exposure in ways that would not otherwise be economically justified. The risk weightings, although acknowledged to be an improvement over the hodgepodge that the great diversity in national regulatory approaches constituted, were highly arbitrary and did not accurately reflect the true risks involved in lending decisions.

PROS AND CONS OF THE 1988 BASEL ACCORD

Benefits of the 1988 Accord

- Created a standardized approach to measuring capital adequacy and a relatively level playing field, at least compared to what had previously existed, particularly among the OECD countries that were originally parties to the accord.
- Although risk buckets were crude, precise definitions as to what qualified as core and secondary capital represented an improvement on the mish-mash of definitions that had previously existed.
- Established a forum for dialog among bank regulators from around the world.

Key Problems with the 1988 Accord

- Took a meat-ax approach to risk weightings: too few gradations and distinctions not in line with reality (e.g., low risk weightings accorded banks).
- Made no distinction in creditworthiness of corporates: CCC requires no more capital than an AAA borrower. Short-term lending to banks imputed to be 1/5 as risky as lending to nonfinancial corporations.
- Made unfair distinction between OECD and non-OECD countries.
- Encouraged regulatory arbitrage; perverse incentives would seem actually to encourage riskier lending.

Rather than focusing solely on the commercial reasons for a given lending decision—and, concomitantly, on the weighing of the actual economic risks and returns attributable to a particular transaction—it became more important for banks to consider the impact the transaction would have on its regulatory capital allocation. Because of the divergence from regulatory capital from economically risk-weighted capital, an incentive was created for financial market participants to engage in regulatory arbitrage.

Regulatory arbitrage, as the second quote that leads this chapter suggests, occurs when the principal motivation for undertaking a particular transaction is its regulatory benefit rather than any commercial benefit.⁷⁵ In other words, the crude manner in which risk weightings were determined under Basel I resulted in banks making loans that on a true economic risk-adjusted basis would not have been attractive. Instead, the scheme provided an incentive for banks to take additional risk and obtain higher yields for the same level of regulatory benefit.⁷⁶

To express this perverse outcome in credit rating terms, under the first Basel framework, a loan to a CCC-rated company, or even to an unrated company, required no more capital to be allocated as percentage of the assets lent than a loan to an AAA-rated company. The consequence was to encourage certain types of transactions at the expense of others, namely high-risk, high-yield loans whose risk weighting made them more attractive from a capital allocation perspective than lower-risk, lower-yield loans.

The distorting effect of the original accord was soon recognized by commentators, as well as the committee itself. In a speech given at a conference of banking supervisors in September 2000, William McDonough, president of the Federal Reserve Bank of New York and chairman of the Basel Committee on Banking Supervision acknowledged that “in recent years . . . the [existing] Accord has exhibited serious short-comings,” adding that:

[o]ne significant weakness is that the Accord’s broad brush structure may provide banks with an unintended incentive to take on higher risk exposures without requiring them to hold a commensurate amount of capital.⁷⁷

Since the accord in effect caps leverage—taking into account the risk weighting of assets and excluding the use of Tier 2 capital (subordinated debt), the 8 percent CAR is roughly equivalent to a leverage ratio of 12.5—banks were enticed to achieve requisite levels of return on equity by looking for higher yields (for a given risk weighting). In other words, because the most creditworthy corporations were lumped in the same basket as the least creditworthy, banks were given an inducement to maximize their return on capital by making higher-yield loans to less creditworthy customers.

In addition, to reduce capital requirements (while simultaneously seeking to improve ROE), banks were encouraged to take on exposures that were risk weighted below 100 percent at the expense of corporate lending. This gave banks an additional incentive to securitize both mortgage loans and other financial assets, particularly those that were more conservative and ordinarily lower yielding.⁷⁸ The paradoxical consequence of Basel I is that its design actually undermined the objectives of soundness and security it sought to achieve.

One additional example serves to illustrate the point well. Since exposures to banks generally attracted low-risk weightings, especially for short-term loans, banks were encouraged to engage in interbank lending with other financial institutions. Of course, the banks that borrowed interbank funds often lent to domestic corporations. In Thailand, such short-term foreign-currency funding contributed to the onset of the Asian crisis. Another, similar result of the Basel risk approach is, of course, the banking house of cards created in the first decade of this century.

Basel II

Basel II will be discussed further in Chapter 12 on risk management, together with Basel III, since those rules address both capital and liquidity issues and are gaining a growing relevance to risk management techniques. Basel II is explored here, while Basel III is touched upon, in the sole context of capital adequacy.

As the deficiencies of the first Basel accord became increasingly apparent through the mid-1990s, support grew for a major revision that would remove the many incentives to regulatory arbitrage inherent in Basel I’s rather simplistic structure. This reform was undertaken between 1998 and 2004 by the Basel Committee on Banking Supervision, and involved the production of numerous working papers, consultation with a myriad of interested parties, and three major iterations of the consultation documents. In mid-2004, the final version of the revised accord was released and substantially the same form was signed the following year.⁷⁹ The agreement was signed in 2005 and was to be implemented over the next several years as various

governmental and regulatory bodies started considering how to implement its provisions in local markets.

As of the end of March 2012, most major jurisdictions (among them the European Union, Canada, Australia, Japan, India, Mexico, Brazil, and Hong Kong) had fully implemented Basel II, while some of them had partly done so (among them the United States, China, Indonesia, and Russia).⁸⁰

The onset of Basel II brings with it a more sophisticated approach to estimating credit risk. As delineated in its governing document, the revised accord did away with the one-size-fits-all approach to risk-weighting assets that Basel I employed to calculate *regulatory capital*.⁸¹ In its place, Basel II uses a more sophisticated rating-based risk-weighted methodology to measure capital, with the explicit goal of discouraging regulatory arbitrage.⁸² In essence, it enables finer distinctions to be drawn between the more and less risky assets held by a bank than was previously the case. By allowing more accurate gradations in the risk weightings that are ultimately assigned to bank assets, correspondingly greater refinement can be achieved in regulatory capital allocation, and the gap between regulatory capital and economic capital can be narrowed.

While the definition of regulatory capital is little changed under Basel II, the method of weighting the riskiness of bank assets and other credit exposures has undergone a dramatic overhaul. In the process, calculating regulatory capital requirements has become much more complex. Nevertheless, the CAR concept established under Basel I remains intact as the primary measure of capital strength under the revised accord. Equally significant, but in contrast to Basel I, Basel II looks not just at credit risk and market risk in setting capital requirements, but also at operational risk.

Indeed, Basel II incorporates a comprehensive enterprise risk management methodology into the regulatory framework established for purposes of setting minimum capital requirements. For this reason, a detailed discussion of Basel II is deferred to a later chapter of this book dealing with risk management. However, before moving on from our survey of bank capital, it is worthwhile to examine the view of bank capital that underpins the revised accord. This view is embodied in the concept of economic capital that is the subject of the remainder of this chapter.

Economic versus Regulatory Capital

In view of the preceding discussion, it should not be surprising that the concept of economic capital and the reforms instituted by Basel II are closely connected. If the problem with the first Basel Accord was the regulatory arbitrage it fostered as a consequence of the divergence between economic and regulatory capital, the purpose therefore of Basel II is to narrow the very same gap between *regulatory capital* and *economic capital* in order to deter regulatory arbitrage. If regulatory capital were completely aligned with economic capital, regulatory arbitrage would be nearly impossible.⁸³ To reduce the gap between economic and regulatory capital, the revised accord adopts a more fine-tuned approach to risk weighting assets, which was outlined in the previous section of this chapter.

While the focus of Basel II is on the calculation of *regulatory capital* adequate to withstand unexpected credit risk, market risk, and operational risk confronted by banks, the accord also implicitly recognizes that banks, based on their own experience and objectives and on their own strategic and business reasons, will inevitably

calculate the requisite *economic capital* in somewhat different ways. Despite these perhaps unavoidable gaps between economic and regulatory capital, Basel II considerably reduces the gulf between the two that prevailed under Basel I, and is thereby likely to go far in accomplishing its objective of reducing regulatory arbitrage. In addition, Basel II imposes an obligation on banks to calculate economic capital. However, it does not, for the most part, specify in detail how economic capital is to be measured, other than that it should be accomplished in accordance with “sound practice” guidelines as set forth in paragraph 528 of the revised accord, in a somewhat different manner from regulatory capital.

Calculating Economic Capital

Economic capital can be calculated by establishing expected loss (EL), which is the multiplication of exposure at default (EAD), probability of default (PD), and loss-given-default (LGD), and then estimating unexpected loss (UL) to the predetermined confidence interval, then multiplying the result by a factor to achieve the desired confidence interval and then further adjusting the result by a factor to take account of the impact of portfolio diversification.⁸⁴

The diagram in Exhibit 9.4 illustrates economic capital in graphical form. On the vertical axis, the diagram shows the probability of loss. The horizontal axis shows the potential amount of loss that could be experienced by the bank. Determination of economic capital requirements is dependent upon management's risk preferences. These risk preferences can be expressed as the risk of insolvency within a given time period that management is willing to tolerate. The greater the confidence level, the more remote is the possibility of insolvency. Confidence levels are usually expressed in percentage terms and typically cover a one-year period.⁸⁵ The confidence level

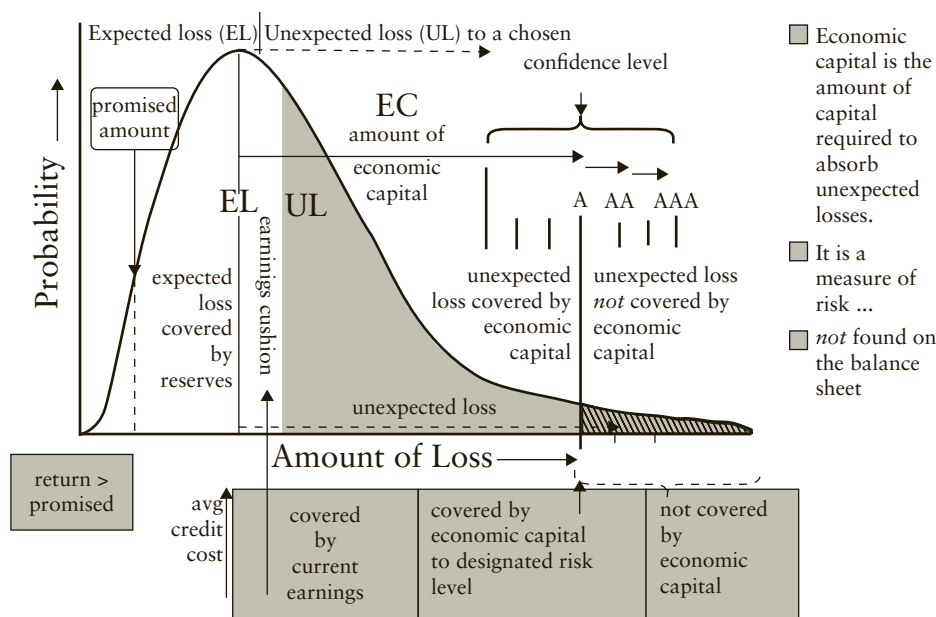


EXHIBIT 9.4 Economic Capital

EXHIBIT 9.5 Credit Rating and Implied Confidence Level

Rating	One-Year Default Rate	Implied Confidence Level	Equivalent Probability (approximately)
AAA	0.01	99.997	1 in 30,000
AA	0.03	99.95	1 in 2,000
A	0.07	99.93	1 in 1,400
BBB	0.22	99.78	1 in 300

Source: Rating agencies.

required can be mapped to the desired credit rating as shown in Exhibit 9.5. Most banks employing economic capital models choose a confidence level between 99.96 and 99.98 percent, which corresponds to the expected default rate associated with a double-A credit rating.

Basel III

While Basel II was an attempt to revise the simplistic Basel accord of 1988, it did so essentially by refining the way assets or other risk exposures should be weighted so as to create the risk-weighted denominator.

Basel III goes one step further. Not only does it further refine the numerator of the capital adequacy ratio, it also creates rules about a liquidity buffer. Further discussions around Basel III will be found in Chapter 12.

NOTES

1. Quoted in Eddie Cade, *Managing Banking Risks*, 1st ed. (Woodhead Publishing Ltd, in association with The Chartered Institute of Bankers, England, 1997), 25.
2. See Sam Theodore, Christopher T. Mahoney, and Gregory W. Bauer, "Bank Rating Methodology: Analytical Framework for Banks in Developed Markets" (New York: Moody's Investors Service, April 1999), 35.
3. Although the often-used CAR is sometimes referred to as the "BIS ratio," this usage is inaccurate. In fact, many countries use the term CAR to define their own version of the regulatory capital ratio, so it may not even be correct to restrict the term to its Basel version.
4. Howard Palmer, *Bank Risk Analysis in Emerging Markets* (London: Euromoney Publications, 1998), 101.
5. Ibid.
6. Ibid.
7. The report is dated September 2006.
8. U.S. Federal Reserve, *Commercial Bank Examination Manual*, Sec. 3020 (Washington, DC: 1995).
9. Obviously, a key point of this chapter is that contrary to apparent public opinion, capital is not always *the* vital sign of a bank's financial health and that in routine bank credit analysis, it is rarely the preeminent financial indicator.
10. The cushioning function—sometimes referred to as "loss-absorbing ability"—and additional functions of capital are discussed later in the chapter. Note that as with other financial concepts or characteristics explored in this book, it is helpful to consider capital first in a general sense and then in the more specific context related to banks and to credit analysis.

11. While it may be obvious that a bank in dire straits has become technically insolvent, this will rarely if ever be openly admitted. Even when the facts belie the desired impression, as a practical matter banks are compelled to maintain a façade of discretion to ensure their continued survival.
12. Incidentally, because capital is watched so closely, it is a value that is among those most apt to be massaged to improve appearances.
13. Mona Gardner and Dixie Mills, *Managing Financial Institutions: An Asset Liability Approach* (Chicago: Dryden Press, 1994), 481.
14. Regulatory capital is, generally speaking, applicable only to banks and to certain other types of financial companies, such as insurance companies. Regulatory capital as defined for banks is distinct from that applicable to insurance firms.
15. In other words, collectively, all of a firm's liabilities and equity fund its assets. While short-term debt in the abstract constitutes a source of funding for both financial and nonfinancial companies, it is typically viewed as separate from sources of longer-term finance. Capital in the corporate finance context—or what we have termed here as *financial capital*—is therefore used primarily to refer to funds advanced for an extended period, whether in the form of a permanent ownership interest in the enterprise (equity capital) or in the form of medium- to long-term debt (loan capital). So in the corporate context, current liabilities such as trade credit, while technically part of the funding that supports a firm's assets, are excluded from loan capital.
16. Of course, ordinary equity capital, also known as *shareholders' equity* or *owners' equity*, is not a fixed financial obligation, but instead represents the owners' residual stake in the enterprise. The distinction between the two elements of financial capital—capital as the owners' investment and capital as long-term funding—is muddled by the existence of preferred (or preference) shareholders, who straddle the roles of creditors and investors, although arguably they are more like the former than the latter. The instruments they hold (i.e., preferred stock or preference shares) exhibit both ownership and creditor characteristics. As a result, these securities come under the rubric of *hybrid capital*, a category of financial instrument that shares both debt and equity characteristics. Another form of hybrid capital is *convertible debt* (i.e., debt obligations convertible into shareholders' equity under certain circumstances).
17. *Debt*, of course, means funds obtained as proceeds under a financial obligation that earn a predetermined rate of interest instead of dividends, without any residual stake in the enterprise, and that are required to be repaid within a specified period.
18. There are no precise standardized definitions for either *loan capital* or *financial capital*, the latter term being generally synonymous with *capital employed*. In that sense financial capital can be defined as funds available to acquire real capital, such as the machinery used to produce goods, and so on. Real capital is indistinguishable from assets such as property, plant and equipment, and inventory. In the context of examining definitions, mention should be made of *working capital*. Though not often heard when discussing the composition of bank balance sheets, the term is frequently used in reference to nonfinancial companies. It refers to the excess of current assets over current liabilities, and is perhaps more a measure of liquidity than of capital in the sense used here. The comparison of current assets with current liabilities is akin to gap analysis, which compares various maturities of bank funding with the corresponding maturities of the same institution's financial assets. Gap analysis is a tool used in asset-liability management that has only tangential relevance to the present discussion.
19. Other synonyms for owners' equity include *net worth*, and *share capital*. Recall that the claims of shareholders are subordinated to the claims of creditors and depositors, indeed to all creditors of the firm, and consequently merely hold a residual right to any assets that may remain after the satisfaction of creditors' claims upon liquidation. Given that shareholders must absorb all losses before gaining access to any residual assets, there may

very well be nothing left for them after outstanding obligations to creditors have been fulfilled.

20. *Bank regulatory capital* is specifically defined for public policy and supervisory purposes, and is not to be confused with capital as the term is understood in normal investment and accounting contexts. Although there is substantial overlap between ordinary capital and bank regulatory capital, it may be better to think of the latter as a separate and artificial construct.
21. Before the establishment of the 1988 Basel Accord, definitions of *regulatory capital* varied considerably from country to country, but in the years since there has been a considerable convergence. *Regulatory capital* has been defined as the “minimum amount of equity capital or other funds that the institution must maintain to comply with regulatory requirements.” KPMG International, *Basel II—A Closer Look: Managing Economic Capital* (Zug: KPMG International, 2003.)
22. As mentioned in the earlier discussion of the balance sheet and asset quality, loan-loss reserves are characterized in accounting terms as a contra-account, more particularly contra-asset, and represent a deduction from a particular category of assets, in particular a deduction from the asset category of loans. Often, they will be disclosed in a note to the accounts, namely those notes referencing the asset side of the balance sheet. Note that although they are referred to as “reserves,” loan-loss reserves or provisions are wholly different from equity reserves, which fall within the category of shareholders’ equity. Note also that a frequent appearance of contra-account items is off-balance-sheet, for example, when the account item is an off-balance sheet asset entry and a contra-account item of the same amount appears as an off-balance-sheet liability entry.
23. Shareholders’ equity, while the consummate cushion against asset impairment, likewise encourages market discipline since shareholders will be loath to lose funds they have placed at risk and will therefore have an incentive to monitor the institution’s financial condition.
24. Subordinated debt issuance could possibly contribute to a bank’s creditworthiness in other ways. Although such debt must be repaid eventually just as depositors must be repaid, the long tenor means that there is little risk that immediate payment will be required and hence, there may well be some buffering effect against asset impairment. Perhaps more importantly, by providing additional longer-term funding, and correspondingly reducing dependence on volatile short-term commercial funding, qualifying subordinated debt may strengthen a bank’s overall credit profile. Finally, of course, the funds contributed by subordinated debt holders signified by the bank’s financial obligation to them remain, in theory at least, available to more senior creditors.
25. Concomitantly, like the canary in the coal mine, a loss of value in subordinated debt instruments may conceivably provide an early warning signal should the bank’s credit situation deteriorate.
26. It could also be argued that one reason that regulatory capital is more inclusive than shareholders’ equity is public relations. The broader the definition of capital, the higher nominal capital ratios, and, at least to the unsophisticated, the less thinly capitalized these highly leveraged institutions appear. From this vantage point, the broad definition of regulatory capital by banks is characterized pejoratively as a form of window dressing. A rosier interpretation is that a broad definition compensates for the shortcomings of book value accounting, which limits the valuation of some assets to their historical cost. However, as mark-to-market accounting has become more widespread, this argument becomes less persuasive.
27. Eddie Cade, *Managing Banking Risks*, 26 ff.
28. The phenomenon of regulatory arbitrage, in essence, constitutes the manipulation of regulations with the objective of effecting compliance and displaying better-appearing

numbers while at the same time maximizing yield irrespective of the higher risks entailed. It is the compliance with the letter of a law while simultaneously taking actions that contravene its objective.

29. Attributed to Larry Meyer, a U.S. Federal Reserve governor, "Sweeter Basle," *The Economist*, January 18, 2001.
30. The adjustment of financial statements for analytical purposes is, of course, not limited to capital. In regard to the income statement, for instance, the concept of depreciation is well accepted by accountants and regulators. Yet corporate credit and investment analysts often ignore depreciation and adjust net income to correspond more closely to cash flow. Consequently, earnings before interest, tax, depreciation, and amortization become as or more important than net income as a prime indicator of a company's ability to service debt and therefore of its creditworthiness. Note that earnings before interest, taxes, depreciation, and amortization is often abbreviated as EBITDA.
31. Alan F. Reid, William Schwartz, Roger Lister, et al. *Lifting the Lid on Ratio Analysis* (New York: Dominion Bond Rating Service, 2005). But see Cade, *Managing Banking Risks*, 25: "[b]ecause the primary problem is always something else (e.g. credit risk . . .) there is a fallacious tendency to deny the relevance of capital levels in banking failure. . . . [O]ther risk categories may occasion the unexpected losses, but the level of [capital] determines whether the bank continues in business or goes under."
32. Variant ratios replace shareholders' equity calculated at book value with "market capitalization" (the product of the number of shares outstanding and the current price per share). Since market capitalization will often exceed book value during ordinary circumstances, but will surely decline during turbulent conditions, this is a much less conservative or useful indicator for credit analysis purposes. The other side of the ratio may substitute total assets with debt, whether defined as all financial obligations without limitation, or limited to certain types of obligations such as long-term debt. For bank credit analysis purposes, total assets generally works better as the denominator, since banks are expected to have a high level of outstanding liabilities. Differentiation among the different types of financial obligations is most often seen in funding and liquidity indicators, not in capital ratios.
33. Moody's and other rating agencies use these ratios, for example.
34. After all, provided the analyst knows the basis upon which a ratio—however basic—is expressed, that ratio can at least be used for comparison purposes. Furthermore, even basic ratios built on truthful information can be more predictive than elaborate ratios built on questionable sources.
35. This is another way of saying that they are not *risk-weighted* capital adequacy measures. The advantage of risk-weighted ratios is explained in the following section.
36. To elaborate, the equity-to-assets ratio does not distinguish in any way among the type of the assets being compared with capital, or the comparative credit risks associated with them. In a banking context, a financial asset represented by a low-risk government security held by the bank affects the equity-to-assets ratio in exactly the same fashion as does a high-risk loan to a customer. That is, the riskiness of the assets held is not taken into account by the ratio. So if two banks both have an equity-to-assets ratio of 10 percent and an equal proportion of nonearning assets, but the earning assets of the first are all in the form of high-risk loans, while those of the second are all in low-risk loans, the ratio will show no difference between them. In other words, a very high-risk bank and a very low-risk bank can easily have identical equity-to-asset ratios. This is a serious limitation of non-risk-adjusted indicators.
37. That is, assets at risk. The term includes loans and other obligations at risk of default, and is, broadly speaking, synonymous with earning assets.
38. Retained income (also called undivided profits) is what is left of the after-tax income after the dividend payout, if any.

39. Actually, return on loans, or earning assets, would be the corresponding measure, but return on assets is in more common use and is also a bit more conservative. To illustrate, if loans comprise 80 percent of hypothetical Furnace Creek Bank's (FCB) total assets such that in a given year they represented \$8 billion out of \$10 billion in assets, and 1 percent of the bank's loan book went bad on average, it would cost FCB \$80 million per year to provision fully against those loans (or to compensate for the loss of equity when they were written off, assuming no provisioning). On \$10 billion in total assets, an ROA of 1 percent is equivalent to a net after-tax profit of \$100 million, well in excess of each year's loan losses, while a 1 percent return on earning assets would generate profit equivalent to the loan losses. In actuality, a bank that performed like this would be in pretty dire straits, but the point is that high profitability contributes to a bank's capital strength, and is more important in some respects than its static capital position on the balance sheet.
40. By the same token, where the ratio value is equal to 100 percent or less, there is no double leverage.
41. www.fdic.gov/regulations/safety/manual/Section4-3.html.
42. R. Taggart Murphy, *The Real Price of Japanese Money* (London: Weidenfeld and Nicolson, 1996).
43. Cade, *Managing Banking Risks*, 29.
44. Japanese banks could price their loans at very competitive levels and offer them in substantial volumes.
45. Both authors of this book resided in Tokyo a few years apart in the 1980s, and they can personally attest to this inefficiency. Even more routine matters, such as deposits transactions, could take inordinate amounts of time. Our recollection was that it was a several step process. Take a number. Fill out the paperwork. Sit down, wait to be called. Initiate the transaction. Sit down and wait to be called. Complete the transaction. From a customer service standpoint, it had its advantages, affording one the opportunity to relax comfortably while waiting rather than having to stand in a long line, which was the practice in New York.
46. In fact, with Luxembourg and Switzerland, 12 countries comprise the group that took the decision: (1) Belgium, (2) Canada, (3) France, (4) Germany, (5) Italy, (6) Japan, (7) Luxembourg, (8) the Netherlands, (9) Sweden, (10) Switzerland, (11) the UK, and (12) the United States. Five French banks were also viewed as operating with unfairly low levels of capitalization. The accord's "principal motivation was to persuade major Japanese and French banks to increase their equity capital, thereby achieving a more 'level playing field.' . . ." Fitch IBCA, untitled paper concerning frequently asked questions about the Basel Accord, 2000 ["FAQ"].
47. Basel Committee on Banking Supervision, "International Convergence of Capital Measurement and Capital Standards" [The 1988 Basel Accord], June 1988. Two fundamental objectives lie at the heart of the Committee's work . . . firstly . . . to strengthen the soundness and stability of the international banking system; and secondly that the framework should be . . . fair and have a high degree of consistency in its application to banks in different countries with a view to diminishing a source of competitive inequality among international banks." Section 3.
48. The subsequent market crash in 1990 caused every Japanese bank to fall below the interim BIS guidelines by September of that year. Murphy, *Real Price*, 190.
49. The 8 percent ratio was long known as the "Cooke ratio" after a Bank of England manager who was among those promoting the accord and was a member of the early committee.
50. At the committee's website, the German spelling for the Swiss city has replaced the English spelling, and is now the most widespread. Although the committee's secretariat is provided by the Bank for International Settlements (BIS) in Basel, Switzerland, it is separate and distinct from the BIS. So for example the term "BIS II" is technically

inaccurate, since the accord is an initiative of the committee not of the BIS in its own capacity. See www.bis.org/bcbs/aboutbcbs.htm.

51. Why 8 percent? The number is essentially an arbitrary one around which a political consensus formed. As noted earlier, the 8 percent figure had been used as a key capital threshold in the United States before the formation of the Basel Accord, and this may very well have influenced its choice as a global measure. While, as observed, the scheme represented a major advance over the confusion that had reigned before, form soon triumphed over substance, sometimes with paradoxical results. After the crash of Japan's stockmarket in 1990, much of the capital generated by the hidden reserves with which the country's banks had expected to meet the 8 percent bar disappeared as prices fell.
52. In fact, legal systems and accounting rules differed, often substantially, from one jurisdiction to the next. Particularly affected were the rules surrounding the definition of what went into Tier 1 capital. More recent differences among jurisdictions are still making comparisons difficult in the EU, as for example the way assets are being marked to market according to fuzzy notions such as whether an asset is available for sale or is intended to be held to maturity.
53. A transitional scheme provided flexibility on this and other items through the end of 1992.
54. Note that some regulators permit redemption 10 years after the preferred share issue without impairing Tier 1 eligibility.
55. Conventionally, preferred shareholders are entitled to receive dividends unless profits are insufficient. The terms of the right will be specified in the original documentation authorizing the shares.
56. Subordinated notes or bonds are fixed-income securities, the holders of which have claims against the bank's assets that are subordinated to, or follow, the satisfaction of all depositor and senior debt holder claims; in some cases, such instruments may be convertible to common shares at the option of the holder.
57. The rationale for this approach has already been discussed. Putting aside for a moment the potential benefits in buttressing the capital strength of banks, there was also a cosmetic benefit. By allowing these additional balance-sheet items to be counted as capital, banks, of course, appeared to have more capital at their disposal than they had under the traditional measures. At the same time, the concept of risk weighting of assets reduced the bank's apparent assets, since certain types of securities and loans were deemed to have lower weightings as a percentage of their actual balance sheet weight when calculating capital adequacy. The effect was to increase the value of the resulting ratio in comparison to the traditional equity-to-assets ratio, making banks on average look stronger as a result.
58. Consequently, as a matter of practice, many subordinated debt issues designed for issue by banks have call features allowing the bank the opportunity to repurchase the debt from investors during the last five years of the issue's term. Further deductions may be required by national authorities in respect of cross-holdings of bank capital by other banks.
59. Note that although this asset class is termed upper Tier 2, it is actually more like equity than lower Tier 2. The tiers of capital are apparently envisioned as numbered top-down rather than bottom-up.
60. "The Special-Corporate Study, Asian Bank Subordinated Debt—A Primer," JP Morgan Securities Asia, January 24, 2000 (analyst: Stephen Long), was helpful in the preparation of the following material concerning the distinctions between lower Tier 2 capital and upper Tier 2 capital.
61. The criteria generally involve their being able to support losses without triggering liquidation.
62. The rationale for including hidden reserves was that they were similar to retained earnings. The accord noted, however, that "many countries do not recognize undisclosed

- reserves” and because of this fact and because of their nontransparent quality, such reserves were excluded from core capital.
63. Where “very substantial amounts of equities” are held by banks at historic cost but which if sold could produce a gain, they are permitted to be included in supplementary capital subject to a 55 percent discount. This section of the rules encompasses the exception sought by the Japanese banks, referred to earlier.
 64. The limitation was added by a November 1991 amendment to the accord.
 65. Total regulatory capital is equal to the sum of Tier 1 and Tier 2 capital.
 66. Other mandatory deductions may include intragroup lending representing a long-term investment in an affiliate or subsidiary; guarantees of another bank’s obligations; and investments in life insurance companies.
 67. Contemporaneously, it came into force in the EU in 1996 after the promulgation of the EU Capital Adequacy Directive. For a thorough discussion of Tier 3 capital, see Financial Institutions Special Report: *Tier 3 Capital—Well-Named and Unloved* (London: Fitch IBCA, July 2000).
 68. Market risk is the risk arising from changing market prices or rates, including interest rates and exchange rates (identified more specifically as currency risk or foreign exchange risk).
 69. The last restriction is referred to as a *regulatory lock in* clause. While any deferral ordered by the regulator is cumulative, no additional interest payment as a result of the deferral is required. In addition, Tier 3 capital generally may not be written down to maintain bank solvency, nor is there usually an amortization requirement. There are, however, several quantitative restrictions on the amount of Tier 3 capital that may be employed; for example, Tier 2 + Tier 3 capital may not be more than 100 percent of Tier 1 capital. See Jackie Ineke, Olivia Guillard, and Carlo Mareels, *Bank Capital A–Z*, Fixed Income Research *Bank Capital A–Z* (London: Morgan Stanley, 2003).
 70. Founding members of the OECD included the following nations: Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the UK, and the United States. Later-joining members included Japan (1964), Finland (1969), Australia (1971), New Zealand (1973), Mexico (1994), the Czech Republic (1995), Hungary (1996), South Korea (1996), Poland (1996), the Slovak Republic (2000), Chile (2010), Estonia (2010), Israel (2010), and Slovenia (2010).
 71. Note the low weighting on loans to banks within the OECD, and to short-term lending to banks generally.
 72. See the BIS website www.bis.org.
 73. Associated with mortgage securitization transactions.
 74. Notwithstanding the expectation that in the aftermath of the current global crisis, capital requirements are likely to be tightened, it remains uncertain how much leeway will remain in the ability of management to influence its reported capital ratios. The problem may not be entirely solvable since there is always a trade-off in regulation between formal precision and the potential for divergence from economic reality.
 75. A bank can increase the risk—and therefore the return—of its transactions by shifting them to riskier ones within the same regulatory risk definition, and this would not penalize the bank’s regulatory capital.
 76. To reprise an earlier illustration, a bank in an OECD country lending to a Greek bank for one year, would only have to weight that exposure at 20 percent, while a bank’s loan to a highly creditworthy company, such as General Electric (GE) in the United States, would be weighted 100 percent. That is, the first Basel accord deemed a hypothetical loan to GE to be five times more risky than a loan to a Greek bank. In fact, a loan to any random small business—to a Joe’s Bar or Sid’s Used Car—would, like GE, also be assigned a risk weighting of 100 percent.

77. The Basel Committee in its own papers has recognized the limitations of the Basel I framework, noting that “a bank’s capital ratio, calculated using the current Accord, may not always be a good indicator of its financial condition. The current risk weighting of assets results, at best, in a crude measure of economic risk, primarily because degrees of credit risk exposure are not sufficiently calibrated as to adequately differentiate between borrowers’ differing default risks.” A New Capital Adequacy Framework: Consultative paper issued by the Basel Committee on Banking Supervision, July 1999 [New CAF], Section 6.
78. Overall, the original Basel Accord did much to spur the growth not only of securitizations, but also of derivatives and structured transactions generally during the 1990s, and as such was arguably a contributing cause to the global credit crisis of 2007–2012.
79. The June 2004 final proposal differed from previous versions in the following key areas: (1) simplification of the calculation of minimum capital requirements in relation to loan-loss provisioning and expect loss estimation; (2) amendments to the calculation of risk weights based on asset types for the Internal Ratings Based (IRB) approach to credit risk; (3) changes to the calculations for securitized transactions and clarification of the Credit Conversion Factors (CCF) for eligible liquidity facilities; and (4) amendments to the operational risk measurement methodologies available to banks, among other items. See KPMG’s *Basel Briefing* 8, October, 2004. Basel Committee on Banking Supervision, *International Convergence of Capital Measurement and Capital Standards: A Revised Framework*, June 2004.
80. The BIS (www.bis.org) publishes progress reports from time to time.
81. *Regulatory capital* has been defined as the “minimum amount of equity capital or other funds that the institution must maintain to comply with regulatory requirements.” KPMG, *Basel II—A Closer Look: Managing Economic Capital*, 2003. *Economic capital* is not explicitly defined in Basel II. It has been referred to as capital “defined in a manner that the organization considers reasonable for its purposes.” www.riskglossary.com/articles/capital_allocation.htm.
82. *Regulatory arbitrage* refers to transactions solely or mainly aimed at reducing regulatory capital needs without altering the economic impact on the financial institution.
83. From a slightly different vantage point, economic capital can also be understood as capital perfectly adjusted for risk and quality that represents a bank’s perfectly calculated economic capacity to absorb economic shocks to a specified confidence interval.
84. While the bank credit analyst should be aware of the concept of economic capital and how to use it as part of a bank review, the intricacies of its calculation are largely the province of internal teams within the bank or of providers of intellectual or technical advice to banks. So our discussion of that topic is limited. Note that there is no specific linkage between economic capital values required and the regulatory capital requirements under Basel II.
85. When percentages are converted into fractions, they are perhaps more readily understood. A 99.96 percent confidence level simply means that the probability that the bank’s unexpected losses will not exceed a certain threshold (covered by economic capital) in a 12-month period is 9,996/10,000. Conversely, the probability of the bank becoming insolvent during the next 12 months is 4/10,000.

Liquidity

[L]iquidity . . . is at the core of banking. Credit institutions typically transform short-term, liquid liabilities into long-term, illiquid assets. In so doing, banks allow customers to smooth out their consumption and investment patterns. . . . In providing this important economic function, banks protect their customers against liquidity problems, but—at the same time—become exposed to such risks themselves. In an extreme case . . . runs, even on sound banks [may occur] when customers withdraw their deposits on a massive scale.

—European Central Bank¹

Lack of access to funding sources and weak liquidity management are typical factors that lead to bank failures.

—Dominion Bond Rating Services²

Banks are machines for taking liquidity risk [borrowing] short-term deposits and [making] long-term loans. . . . [T]he credit squeeze has proved that liquidity has been forgotten in the quest to regulate banks' capital.

—*Financial Times*, Editorial³

Liquidity is arguably the most difficult attribute for a bank analyst to explore. Not only are there myriads of possible ratios, each one offering its own reading of a bank's liquidity profile, often at odds with that provided by the other results, but a bank's liquidity changes by the minute.

If seasoned analysts and competent regulators can barely agree on near-universal grounds to pass judgment, retail depositors—and even sophisticated investors—often remain in the dark until it is too late. Worse yet, even when it is not too late, any panic contributes to shorten the life of a financial institution.

In a pivotal scene in Frank Capra's classic 1946 film, *It's a Wonderful Life*, a bank run is brewing in a small American town.⁴ The protagonist, George Bailey, played by James Stewart, has just married his childhood sweetheart, and the couple is headed for their honeymoon. En route to the train station, George sees a crowd advancing toward the Bailey Building & Loan Association, a modest single-branch savings institution that he reluctantly took over on his father's death. Jumping out of the taxi, George rushes into the

building just ahead of the panicked depositors. He soon learns that Uncle Billy, the institution's bumbling accountant, has misplaced \$8,000 in cash. No funds remain to pay the anxious depositors, who now stand before George demanding their money back.⁵

With the Bailey Building & Loan Association on the brink of collapse, George makes a last-ditch effort to save the savings and loan.⁶ Imploring his customers to believe in the bank, he tells them, "You're thinking about this place all wrong, as if I have the money back in the safe." Then, in a sentence that encapsulates the essence of fractional reserve banking, he continues: "[Y]our money's in Joe's house and in Mrs. Maitland's house and a hundred others. . . ." His impassioned appeal, together with the US\$2,000 that would have paid for the honeymooners' sojourn, saves the day. George proceeds to negotiate partial withdrawals with most of his depositors, regains their confidence, pacifies the bank examiner, and keeps the institutions' doors open until the end of banking hours and, the film implies, happily ever after. While neither a good illustration of a well-managed bank nor robust regulation, the scene is not a bad dramatization of the risks banks face, and the vital importance of liquidity to a bank's survival.⁷ Not only at the Bailey Building & Loan Association, but also for almost any bank, the depositors' money is indeed not in the vault, but mainly advanced to borrowers as loans. There would be nowhere near sufficient cash to pay all depositors at once, or any substantial portion of them, should they elect to withdraw their funds simultaneously.

The American humorist and writer, Mark Twain, is reported to have declared, "A banker is a fellow who lends you his umbrella when the sun is shining, but wants it back the minute it begins to rain." While they may be fair-weather friends, banks themselves are not immune from the very same phenomenon.⁸ A bank too will find it difficult to obtain an umbrella in the form of new cash inflows from depositors should they lose confidence in an institution's soundness—at least in the absence of full and effective deposit insurance.⁹ Instead, they will withdraw their funds. Depositors, of course, are lenders to the bank itself, and it is they who provide the lion's share of bank funding. For most banks—and from a global vantage point, for the great majority—retail depositors constitute, in the aggregate, the largest single source of bank funding. Likewise, debt investors and bank counterparties are less likely to advance funds, while equity investors will be inclined not to buy or to sell shares.

In the not very distant past—that is, just a few years ago—the bank run, and the dangers it poses, seemed to be either (1) a quaint relic associated with nineteenth-century financial panics or with the Great Depression of the 1930s, or (2) a phenomenon only found in contemporary times in emerging markets. Recent events, however, have made plain what has always been so: Bank runs remain an ever-present threat, even if obscured in normal operating conditions. In today's financial world, bank runs tend to occur without a telltale crowd outside the institution's doors. Instead, as confidence in a bank's solvency ebbs, funds are quietly withdrawn through telecommunications. But this is not always the case. In 2008, just such a queue formed outside the doors of the branches of one of Britain's largest banking institutions, the first bank run of its kind in many years.

BANK CRISES IN BRIEF¹⁰

Classically, the bank crisis erupts according to a recognizable pattern. Initially, a shock in the form of some newsworthy event triggers an abrupt change in sentiment that results in sudden movements in prices or interest rates. Amid the ensuing uncertainty, the ability of banks to borrow in the market will often be adversely affected, with the most vulnerable banks worst hit. Assuming that it is still possible to obtain funding, the cost of funds to the more vulnerable banks will rise sharply, and confidence in their soundness may erode, thereby aggravating their funding problems. During such tumultuous periods, depositors and investors may become reluctant to entrust their savings to domestic banks in the absence of effective deposit insurance or a credible government guarantee. Under severe market conditions, or in countries susceptible to instability, signs of potentially acute social, political, or economic turmoil may spark deposit flight, while ordinary savers may believe their cash to be safer placed under the mattress than in accounts with the local banks.¹¹

The run on Northern Rock was the result of a confluence of factors. At bottom was the typical loss of confidence in the bank among depositors, counterparties, and investors. As the market for mortgage-backed securities (MBS), among other funding instruments, crumbled in mid-2008, Northern Rock became increasingly vulnerable to a funding shortfall. Among its peer group, it had the smallest customer depositor base in relative terms, while it was the most reliant on the money and capital markets for funds. In addition, limitations on deposit insurance coverage in the UK—including a minimum uncovered amount—caused depositors to fear the partial loss of funds placed with the bank, or at the very least the loss of immediate access to their funds, and this was a critical factor. Northern Rock was bombarded with a wave of online withdrawals, and when the bank's website jammed, anxious depositors naturally lined up in front of the bank's doors to withdraw cash in person.

As it turned out, Northern Rock was merely the first in a sequence of major bank failures in Europe and in the United States. By late 2008, it was apparent that a fully fledged banking crisis was underway. Bank crises will be examined in detail later in this book.

WHAT IS LIQUIDITY AND WHY IS IT IMPORTANT?

This illiquidity problem arises even if the banks are solvent and profitable. If depositors come to doubt the soundness of banks, they will run on them. The banks must start unloading their assets to get the cash to satisfy depositors. If many banks do this, the simultaneous dumping of assets into

the market depresses asset prices with the ironic result that banks can become insolvent as a result of the run—thereby justifying the public's decision to run in the first place.

—G. Thomas Woodward¹²

Liquidity refers to the ability of a bank, or any other enterprise, to meet its obligations as they fall due.¹³ Satisfactory liquidity means the ability to access *sufficient* cash for that purpose through a number of sources, which will be discussed in subsequent sections.

Conversely, liquidity risk for a bank is the risk of being *unable to honor deposit withdrawals* or *to make repayments* of other liabilities at maturity (or of current obligations).

The liquidity gap analysis is the most basic and the most efficient tool that banks use to measure potential liquidity problems. The approach is to estimate the volume of assets that will mature within a preset period—called a time-band—and to compare it to the volume of liabilities that will mature within the same period. The difference is the liquidity gap for that period, positive if assets exceed liabilities or negative if liabilities exceed assets. The analysis moves forward over time, making the gap analysis an ever-moving tool.

Naturally, the selection of the various time-bands is important since shorter periods would result in a greater volatility of the results.

The accuracy of the gap estimates is much higher for the short term than it is for the long term, and at the same time, the need to manage the gaps is more pressing over the immediate future periods. As a result, immediate future time-bands should be shorter than distant future time-bands. For example, a bank would look at its gaps over every day and every week of the next month or two, while it would, for example, regroup years 4 and 5 ahead in a single time-band.

As the reader would have gathered from earlier chapters, the likely maturity of quite a number of assets and liabilities simply cannot be measured accurately. Some estimates must be used—for example, on the liability side, the fact that, say, 82 percent of the funds held in all savings accounts of a given bank in normal circumstances would remain in those accounts for over one year.

It is evident that a bank's capacity to access its various sources of liquid funds affects its overall liquidity profile. Since a bank's overall liquidity includes its ability to obtain new funds (the last source of cash usually listed)—and correspondingly *liquidity risk* encompasses the risks associated with a bank's potential inability to obtain new funds—funding and liquidity are usually discussed and analyzed together. More broadly, liquidity risk includes not only the risk of *being unable obtain funds needed at an affordable price* and “within a reasonable time period to meet obligations as they become due . . .”; it also includes the risk of *incurring onerously high funding costs* on an ongoing basis,¹⁴ a situation that may lead to insolvency. Because the bulk of most banks' funding ordinarily comes from customer deposits, the assessment of its deposit base is a key step in appraising a bank's liquidity position. All other things being equal, if a relatively high percentage of a bank's funding is in the form of customer deposits, it is likely to possess a comparatively stable and reliable funding base. Where, however, a high proportion of total funding comes from commercial sources, as opposed to customer deposits, it will be

exposed to greater funding risk. Generally speaking, the greater the reliance on customer deposit funding, the stronger the bank's funding and liquidity position; the greater the reliance on commercial sources of funding, the weaker the bank's funding and liquidity position.¹⁵

The Importance of Liquidity

A bank that, for whatever reason, cannot obtain needed liquid funds will quickly face severe problems. Its failure to honor deposit withdrawals or to make repayments on other obligations will cause its depositors and creditors to quickly lose any remaining confidence in its soundness, although typically confidence has already eroded by this stage; otherwise cash would not be in such short supply. Without deposit insurance, government guarantees or some other means to restore that confidence, the institution's collapse is almost inevitable. Even if the bank was in reasonable financial health before facing a cash crunch, should a bank run begin it will feed on itself and turn a false perception into reality. As the quote that leads this section suggests, even a good bank with a clean balance sheet and sound management, and that is making solid returns will ultimately collapse if access to liquid funds dwindles or dries up. Dominion Bond Rating Service put it well: "[W]ithout adequate cash resources to meet short-term liquidity requirements, a bank will find it impossible to continue its operations even if its capital or solvency remains acceptable."¹⁶

Indeed, a lack of liquidity is the proximate cause of most, if not nearly all, bank failures and in this respect its importance can hardly be overstated.¹⁷ Correspondingly, sufficient liquidity can more than compensate for other substantive deficiencies in a bank's financial condition. Whether it is able to draw cash from operational cash flows, liquid asset reserves, the money market—or, *in extremis*, from a strong parent company or the state—access to liquid funds can allow an otherwise fragile bank to continue operating, indefinitely if necessary where such access is unrestricted.

The very nature of fractional reserve banking—that is, the conventional business of banking in which a depository institution lends all but a fraction of its funding, (the remainder of which is held in reserve)—means that banks are innately vulnerable to a potential need for liquid funds greater than that which they can have at hand. In addition, banks, as a practical matter, must borrow short and lend long. Most bank funding is subject to immediate withdrawal, while the great majority of the loans it advances will be for comparatively longer terms and less easily called in on short notice—notwithstanding terms in the loan agreement that may give the bank a right to call the loan upon relatively short notice. Since, on the other side of the balance sheet, a high proportion of bank funding may be withdrawn on no, or very short, notice, a bank's cash requirements are particularly hard to predict. Banks' essentially inherent inability to repay all depositors should a high proportion withdraw their funds at once, and the difficulty in predicting future cash requirements constitute the banking industry's Achilles heel. It explains why maintaining, and cultivating the confidence of depositors and investors alike—not to speak of counterparties, regulators, rating agencies, and other market participants—is so critical to the industry's survival. In this sense, banking is very much a confidence game.

LIQUIDITY PROBLEMS: CAUSE OR EFFECT?

In 1997, Finance One, Thailand's largest finance company, collapsed, an event that was widely perceived as the spark that set off the regional Asian crisis of 1997–1898. The company's owner later attributed its demise to mere liquidity problems, declaring: "FinOne did not die from financial failure, it died from [a lack of] liquidity. It was a plain and simple run on deposits."

But this, of course, begs the question of why this particular finance company experienced the run. Do depositors run on banks at random? Perhaps. It is more likely, however, that those institutions with some comparative weakness become the targets of anxious depositors. In the case of the Northern Rock bank run in 2007, the fact that it depended more than any other bank in its peer group on commercial funding as opposed to customer deposits, certainly made it a conspicuous first target of a loss in institutional and then public confidence.

The demise of Fortis Bank in Belgium in 2008 was triggered by a lack of liquidity in the markets, at the very time when the bank had foolishly gambled that the markets would carry it over a temporary overreliance on interbank funding for the cash acquisition of a large chunk of ABN Amro Bank at what in hindsight looked like a hefty price. The bank would likely have survived the pricing mistake, if not for the liquidity angle.

Although there may be circumstances where external factors—or pure bad luck—cause even a reasonably solvent, liquid, and well-run financial institution to founder, in most instances bank distress has its root in deficient bank management. To be sure, minor vulnerabilities may be exaggerated through rumor, but more often than not, the old proverb applies: Where there's smoke, there's fire. Management deficiencies may range from inadequate liquidity management to a more pervasive general mismanagement that has allowed a bank's capital position to become impaired through inadequate credit risk controls, excessive dividend payouts, or mispriced loans, among other things. Characteristically, inexperienced or incompetent management expands lending too aggressively without maintaining adequate credit assessment capability. The result is deterioration in asset quality to a greater degree than the peer group average and a consequent impairment of capital. Loan and capital write-offs, and reduced profitability, erode market confidence, setting the stage for a potential run on the bank.

Inevitably, less than salutary economic conditions—and the rumor mill—will aggravate the impact of managerial mistakes already made. It is more usual for the weaker institution, the risk profile of which is conspicuously worse than its peer group, to become the target of depositors' fears. Weakness may first manifest in other fundamentals, but ultimately the common characteristic of collapsing banks is an inability to meet funding requirements triggered by (1) the market's loss of confidence in the ability of the particular bank and its management or (2) a generalized loss of confidence pervading the market as whole. Confidence is intrinsically mercurial and its evaluation elusive. It is fair, however, to infer that depositors and

investors exposed to bank credit risk will draw confidence from an institution that exhibits healthy profitability and a strong balance sheet. Large institutions thought to be too big to fail will benefit from the perception that government support is inevitable. Likewise, a reasonable expectation will often arise that institutions wholly or majority owned by the state will receive comparable support. In addition, perceptions concerning the operating environment in which the bank conducts its business, the quality of the regulatory regime, and economic expectations will influence depositor and investor behavior.

In sum, liquidity problems often constitute the late-stage symptoms of a more fundamental ailment, typically exacerbated by operating conditions. Irrespective of the cause, evaluation of a bank's liquidity position comprises a crucial—if often difficult—part of the process of gauging its overall credit risk.

Uses and Sources of Liquid Funds

Having defined liquidity and established its significance, it may be useful—as a prelude to discussing the evaluation of liquidity and key indicators—to survey in somewhat greater detail a bank's uses and sources of funds. Its access to each of these sources, relative to the present and anticipated gaps between current assets and current obligations, forms the basis for the evaluation of its liquidity risk.

Drains on Liquidity

In routine operations, banks, like any other business, require cash to meet everyday operating expenses. These include mundane items such as

- Rent to the lessor
- Compensation to staff
- Payments to suppliers
- Tax payments to the government

In addition to the preceding cash requirements, to which all enterprises are subject, banks have other substantial cash requirements. These include:

- *Interest paid* to depositors, interbank lenders, and holders of debt securities issued by the bank
- *Depositor withdrawals* and principal repayment in respect to borrowings (e.g., debt securities)
- *Margin calls* on securities in connection with proprietary trading operations (e.g., for repo finance purposes)
- *Advances* in the form of loans as well as other extensions of credit¹⁸

These uses of liquidity are illustrated in condensed form in Exhibit 10.1.

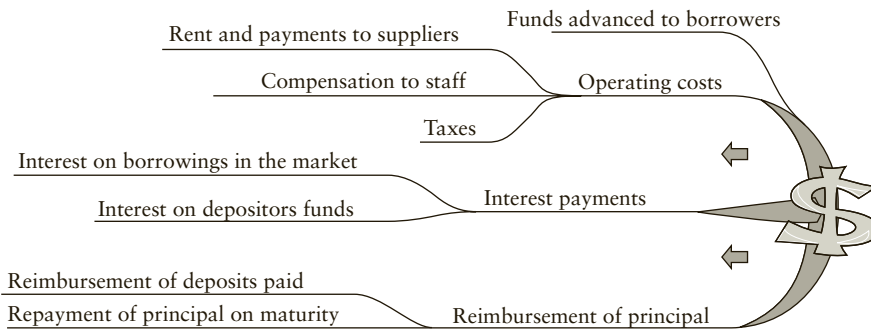


EXHIBIT 10.1 Uses of Liquidity

The diagram divides liquid fund uses into four groups: (1) the advance of liquid funds to borrowers for lending purposes; (2) ordinary operating expenses; (3) funding costs (interest payments); and (4) reimbursement of principal payments to depositors (depositor withdrawals) and other creditors.

Although, as suggested, a bank's cash needs are difficult to predict, nonetheless under normal conditions its management is likely to have a reasonably good idea of anticipated cash outflow and cash inflow. Naturally, however, some degree of variation from the norm can be expected. On any given day, depositors may withdraw more funds than expected, as may customers having various open lines of credit.

Sources of Liquidity

Liquidity can be sourced from any of the following:

- *Cash flow from operations*
- Existing *cash reserves*
- The *conversion of marketable noncash assets into cash* through sale
- *Net incoming deposits* (i.e., the excess of new deposits over deposit withdrawals—a supply not generally available to nonbanks)
- *New cash infusions* from other funding sources, that is, investors or lenders

A bank can and will tap cash flow from operations to meet current obligations. Operational cash inflows include the repayment of loan principal by borrowers, interest payments from borrowers' fees paid by customers, and the institution's trading gains. These incomings will meet much of the institution's ordinary liquidity needs.

Cash reserves provide another layer of liquidity. Some cash will be kept in the vault, but in ordinary banking practice such cash holdings will be kept to minimal levels, both to reduce incentives for crime and because they do not generate a return—not to mention the obvious staff cost associated with the handling of physical currency. Most of the institution's other liquid assets take the form of accounts with the central bank or to be held as negotiable instruments or marketable securities. Collectively, coin, currency, and those assets just described that are essentially equivalent to cash constitute in the aggregate the bank's liquid or *quasi-liquid* assets.¹⁹ The bank's other assets will invariably be less liquid in character,

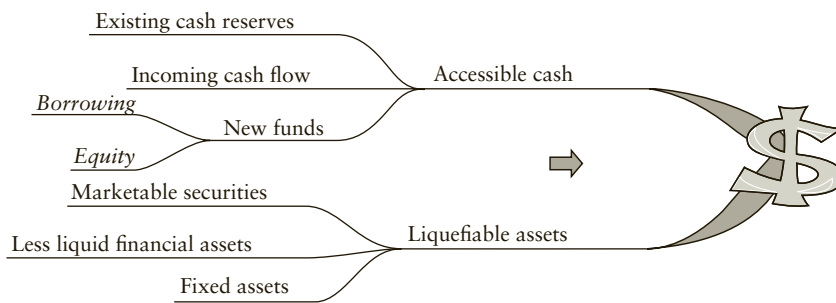


EXHIBIT 10.2 Sources of Bank Liquidity

being loans to borrowers, or to a much smaller extent, nonmarketable securities or fixed (or intangible) assets.²⁰ In extreme circumstances, the more liquid of such assets can be sold to generate cash.

In addition to operating cash flows, cash reserves, and assets convertible to cash, a bank can look to funding inflows as a third layer of liquidity. An excess of new deposits over deposit withdrawals provides additional funding of a kind not generally available to nonbanks. Insofar as net cash inflows from deposits together with operational cash flow and liquid asset reserves do not meet the bank's liquidity needs, it might seek additional cash either by, on the one hand, borrowing from creditors or depositors, or on the other, by obtaining an injection of new capital from shareholders, old or new, from debt investors and other creditors, as well as from equity investors.²¹ A last layer of liquidity is government funding. With the exception of this last category, these sources of liquid funds are depicted in the diagram in Exhibit 10.2.

Liquidity Management

Within the ambit of normal swings in the business cycle, liquidity problems at individual banks will tend to be caused by unexpectedly low cash inflows or by unexpectedly high cash outflows, as opposed to arising from some deficiency in the bank's business model or its strategy, or from extreme economic conditions. Among the most common causes for reduced cash inflows are declining yields on floating rate assets, increased default rates, and maturity date changes. Increased cash outflows may arise from increasing funding costs; rising deposit withdrawals; or the acceleration of obligations issued by the bank, owing for example to a rating downgrade.

The concept of managing liquidity is simple to comprehend but can be complex in its implementation. To anticipate and respond to potential cash shortfalls, prudent management will aim to implement policies that make certain that the institution has ample liquidity available to meet its operational needs for cash, as well its compliance requirements in terms of liquidity.²² The basic approach is to attempt to gauge prospective cash needs and to prepare for them in a cost-effective manner that does not unduly burden the bank's overall performance.²³ Internally, various techniques including gap analysis and cash flow analysis are used in an attempt to ensure that an adequate supply of cash is available to meet current and upcoming obligations. Many of the methods are also used in asset-liability management, which primarily aims to manage market risk but also considers that aspect of liquidity risk insofar as it arises from a mismatch of asset and liability maturities.²⁴

THE RELATIONSHIP BETWEEN MARKET RISK AND LIQUIDITY RISK

Liquidity risk is closely related to other categories of risk that banks confront, particularly market risk. Unless its asset liability management is wanting, a bank is unlikely to face a liquidity crisis in isolation. Instead, other risks, and losses arising from them, are likely to be contributing causes to any shortfall of liquidity to which a bank becomes vulnerable. For instance, market risk may contribute to liquidity risk. A bank's trading activities create the risk of loss due to adverse movements in rates and prices of currencies. If a bank borrows against the securities held in its trading portfolio, a decline in the value of financial collateral resulting from changes in prices or interest rates in respect to trading positions can result in unexpected margin calls, thereby increasing the demand for liquid funds. This last situation is one of the many problems that Dexia Bank encountered in 2011.

Another example of the relationship between market risk and liquidity risk is the manifestation of what is termed *refinancing risk*,²⁵ which arises from a mismatch between the maturities of liabilities and assets. Refinancing risk is the risk that to continue to fund a given asset, funds must be borrowed at a rate that is significantly higher than the original funding cost, reducing consequent profits. The higher cost of funding not only reduces spreads, it also makes the cost of purchasing additional liquidity comparatively more expensive. To manage market risk, banks use both simple gap analyses and much more sophisticated techniques.

To describe one such technique, it is common practice for a bank to set so-called stress loss exposure limits for each trading desk. Stress exposures are decided by measuring the potential losses that might be incurred on *open positions* should rates, prices, and volatilities change within specified ranges. These are set to reflect potential market conditions within a range of probabilities. Market risk in this context is usually quantified by the *value-at-risk methodology*, which measures the probability of trading loss on both capital and earnings at a given confidence level. A daily value-at-risk of US\$10 million with a 99 percent *confidence level* means that in one day out of 100 can the bank be expected to lose more than US\$10 million.

While these models seem to work most of the time,²⁶ it has become apparent since the 1998 demise of Long Term Capital Management, a prominent hedge fund, that they have serious limitations. A key criticism²⁷ is that such models, as conventionally employed, systematically underestimate risk, owing to an over-reliance on recent historical data and a failure to effectively account for scenarios involving a collapse in market liquidity.

Liquidity Management Plan

In view of the importance of liquidity and the rapidity with which it can evaporate, a critical objective of the bank's board of directors and senior officers is to manage the bank's liquidity to ensure that such a scenario does not occur. More specifically,

a well-managed bank will have actively managed its liquidity and put into place a liquidity management plan. Ideally, a major element of a liquidity management plan will be a contingency funding plan. Indeed, this could be said to be the core of liquidity management insofar as other aspects of liquidity planning overlap to a great degree with asset-liability management, treasury operations, and strategic planning.

The purpose of this plan is to put emergency funding in place in the event that projections go awry. Of course, projections are less likely to hold during periods of high market volatility. Occasionally, market conditions are so extreme as to trigger extremely serious system-wide liquidity problems in the banking industry.²⁸ If this occurs, a contingency funding plan may provide only temporary relief. Maintaining liquidity during episodes of financial turbulence is considerably more difficult than under normal market conditions, and in some circumstances it may be impossible. Nevertheless, putting in place a contingency funding program when market conditions are relatively benign, assuming such funding is available at a reasonable cost, is one of the most important steps a bank can take to ensure survival.²⁹

The United States Federal Deposit Insurance Corporation has identified a number of characteristics of an effective liquidity management plan, which are condensed in the following.³⁰ According to the agency, such a plan should:

- Assess risk arising as a consequence of potential unusual and worst-case liquidity scenarios (e.g., a rating downgrade).
- Identify key indicators that function as warning signals of an imminent liquidity problem.
- Evaluate potential alternative sources of funding, including backup lines of credit, and create plans for accessing such funding sequentially in the event of an emergency.
- Establish emergency monitoring procedures, including the monitoring of available assets during the crisis period to expedite access to funds.
- Evaluate options for contingent funding and implement the contingent funding options chosen.

These criteria may serve as a checklist for analytical purposes.

Contingent Funding Facilities (Bank Facilities)

As mentioned, to avoid a shortfall of liquidity, a bank may arrange an emergency borrowing or backup facility with another financial institution. The contingent funding facility is a form of funding that grew out of a bank's relationship with its correspondent banks. Traditionally, should a bank wish to arrange borrowing facilities, it would turn first to these banks with which it had already established a relationship.³¹ The overall reliability and assuredness of such facilities will differ depending on the type of arrangement, and the credit profile of the institution providing the facility. Be aware, as well, that it is common for the relevant facility document also to include financial covenants that the bank must adhere to. If the bank fails to meet any of these covenants, then again the funding facility may be withdrawn quickly, and the bank's liquidity position may be dangerously weak. These considerations will, in turn, affect the liquidity profile of the bank benefiting from them.³²

Among the safer arrangements is to have in place a contractual lending commitment from another bank that is legally binding. In times of need, such as a severe liquidity squeeze brought on by a sharp change in market conditions, this type of committed facility can be very important. Such a backup liquidity arrangement could be structured as a committed line allocated for a specific issuance program of, for example, commercial paper provided or guaranteed by a highly rated bank for an extended period. Alternatively, a bank may have a similar type of contractual lending commitment, again legally binding, but with an escape clause that will come into play in certain circumstances. So-called material adverse change clauses enable the institution providing the facility to withdraw it if there is any significant negative change in markets affecting the bank to which it is offered or in that bank's financial condition.

A bank may well have commercial paper backup lines and revolving commitment vehicles, but these will be of little value if material adverse change clauses are incorporated into the governing legal agreements, as is frequently the case. Should turbulence persist and any support for the bank's liquidity position be withdrawn, its funding profile would suddenly deteriorate. This, of course, would further constrict the already tightening liquidity position of the bank in question. It might be noted that not all banks require committed backup facilities to provide comfort in times of stress. Highly rated, strong, and large institutions with a significant franchise and distribution network will obviously have less need for such commitments. Highly robust top-tier banks will ordinarily continue to be able to source funds from the interbank market. Under extreme market conditions, however, even top-tier banks may find it difficult to source funds in the interbank market.

A bank may be more vulnerable to changes in market sentiment because it has a weaker market reputation, a less stable deposit or funding base, or a lower credit rating, among other possible reasons. A bank's response to a potential liquidity shortfall will therefore depend upon its level of vulnerability, which in turn will be strongly influenced by the extent to which it has ready access to external funding, whether through a contingent funding line or otherwise, to government support, or to some other source of new funds. In general, the more confidence-sensitive the bank, the more precisely defined its alternative liquidity should be. Alternative sources of liquidity may come if not from the government, then potentially from affiliate banks both within and outside the country. In any case, a significant criterion in evaluating both liquidity and the bank's creditworthiness generally is the probability of a given bank receiving liquidity support in a crisis, and the relative strength of that support.

In the unfortunate circumstance that external support is not available, the path most conserving of liquidity would be for the bank to limit lending to conserve cash.³³ It may shut down lending operations entirely, or provide only short-term advances to existing customers known to be very high-quality credits, while requiring strong collateral in all possible circumstances. Funds will be kept in cash or high-grade marketable government securities.

The Cost of Liquidity and the Trade-Off with Bank Profitability

Liquidity has a cost, which is a function of market conditions and the risk profile of the bank. If liquidity needs are met through holdings of high quality short-term assets, generally the cost is the income sacrificed by not holding

*longer term and/or lower quality assets. If funding needs are not met through liquid asset holdings, a bank may be required to incur additional liabilities, possibly under adverse market conditions at an undesirable cost.*³⁴

—FDIC

The cost of liquidity comprises both (1) the opportunity cost of profits foregone through the sacrifice of yield for liquidity, and (2) the added cost of arranging funding and arranging backup credit lines. It is evident that there is an ineluctable trade-off between performance and liquidity. At one extreme, the perfectly liquid bank that maintains most of its assets in the form of cash or government securities would likely not generate sufficient margins to compete effectively with its peers.³⁵ At the other extreme, a bank that puts all of its available funds into high-yielding but risky and illiquid loans might generate extraordinarily high profits, but its probability of experiencing a liquidity shortfall would be relatively much greater. Confronted with the dilemma, prudent bank management will seek to find some happy medium between a high degree of safety that results in lackluster performance, on the one hand, and high profits that are coupled with severe liquidity risk, on the other.

In practice, the range of liquidity strategies is narrower. To maximize safety, the conservative bank will be inclined to hold a large proportion of high-quality liquid and short-term or callable assets, and fund those assets through stable customer deposits, and to avoid the use of *purchased funds* to finance new customer advances. A bank that functions as such a *liquidity warehouse*, however, will be limited to low yields since the safest and most liquid assets will generate relatively weak returns, and attracting adequate deposits may require the bank to expend more on marketing, customer service, to maintain branch networks, and possibly to pay higher rates on deposits. In contrast, to maximize profitability, the aggressive bank will tend to minimize its holdings of liquid assets, and increase its gearing through the use of purchased funds obtained from other banks and commercial creditors, or by issuing short-term debt instruments. Note that while money-market funding may superficially seem to be expensive, it can actually be cheaper than customer deposit funding when the full costs of the latter are considered. As rating agency DBRS put it in a recent report:

*While core deposits often have lower interest costs, their overall cost may be higher than wholesale funding, when the costs of maintaining a network of branches and providing full banking services are considered.*³⁶

To increase earnings relative to assets and equity, the bank will also have an incentive to fund higher-yielding assets, and correspondingly less liquid assets, such as loans to small and medium-sized enterprises, as well as advancing funds to customers in the subprime market. To increase returns further, the bank may, in addition, provide borrowers with more favorable terms, such as longer maturities or fixed interest rates, to the detriment of its own liquidity.

Within limits, an aggressive profit-oriented strategy will work so long as confidence in the bank remains strong and there is no doubt about its liquidity. At some point, however, this approach becomes vulnerable to overreach. So while the generally more profitable banking strategy of reducing liquidity will have its advantages during benign market conditions, it can prove disastrous should perceptions concerning the bank's soundness change.

ELEMENTS OF BANK LIQUIDITY ANALYSIS

Liquidity is traditionally defined as the ability of credit institutions to fund increases in assets and meet obligations as they become due.

—European Central Bank, citing the Basel Committee on Banking Supervision³⁷

Liquidity to meet obligations cannot always be perfectly predicted from balance sheet analysis. Market perception can be important.

—Thomson BankWatch³⁸

In the preceding section, it was shown that a bank has four primary sources of cash which it can potentially draw on to:

1. Pay for funding (i.e., make interest payments).
2. Reimburse principal borrowed.
3. Settle bills and other operating expenses, whether ordinary or unexpected.
4. Continue to advance loans.³⁹

They are:

1. Current and imminently expected *operational cash flow*
2. Existing *liquid assets* and *quasi-liquid assets* (as well as the bank's stock of less marketable albeit potentially liquefiable assets that can be sold for cash)
3. *Net deposit inflows*
4. Untapped capacity to borrow in the money (or capital) market or obtain cash infusions from new or existing shareholders

It was also mentioned that another possibility was liquidity support from the government.

Analysis of the first source comes under the rubric of cash flow analysis and is mainly evaluated internally for management purposes rather than by outside bank credit analysts. It also has much less of a buffering function than the remaining categories. For these reasons, it is only briefly discussed in this chapter.⁴⁰ The second source encompasses two aspects of the same source of liquidity: *asset liquidity*, discussed in the immediately following subsection.⁴¹ Asset liquidity, sometimes termed *market liquidity*, refers to a bank's capacity to tap existing cash reserves or to sell financial assets in exchange for cash in a timely manner. The third and fourth sources together comprise a bank's *funding liquidity*, sometimes termed *cash liquidity*, which refers to a bank's ability to raise cash through net deposit inflows or other borrowings (as well as through cash contributions to equity). For purposes of analysis, liquidity can therefore be divided in two fundamental categories: (1) asset liquidity and (2) funding liquidity (see Exhibit 10.3).

Asset Liquidity

A term loan is nothing but an illiquid junk bond.

—Joseph Grundfest, former SEC commissioner⁴²

EXHIBIT 10.3 Types of Liquidity

Types of Liquidity	Sources of Cash	Financial Statements
Operational cash flow	<i>Cash flow from operations</i>	Cash flow statement
Asset liquidity	<i>Cash reserves</i> <i>Liquefiable assets</i>	Assets
Funding liquidity	Access to funding (cash injections)	Liabilities

Asset liquidity refers to a bank's ability to access cash reserves or to convert various forms of financial or other assets into cash on short notice and without significant loss.⁴³ In the event of an unexpected need for a large amount of cash, whatever funds the bank has on hand, together with whatever financial assets it can quickly convert to cash, provide a pool of liquid funds out of which it can satisfy any current obligations. An ample pool of high-quality liquid assets therefore acts as a critical buffer for banks in the event of sudden rise in cash outflows or should there be some interruption in expected cash inflows.

In analyzing asset liquidity, the relevant questions that the analyst must consider are these: How much cash does the bank have on hand to meet current obligations relatively speaking and to what extent can cash be procured on short notice by accessing existing stocks of cash or by quickly selling existing assets. Consequently, the proportion of cash and other liquid assets to nonliquid assets is an important criterion of whether the bank is maintaining an adequate level of liquid assets.⁴⁴

Characteristics of Liquid Assets

Readily marketable high-quality liquid assets include, as a rule, widely traded government securities and central bank balances. Highly marketable financial assets, such as widely traded debt securities issued by subnational governments or by large and frequent corporate issuers, are also apt to be viewed as relatively liquid, albeit not so liquid as national government securities. Generally speaking, loans and securities typically held to maturity and not frequently traded are viewed as relatively illiquid.

In general, to be defined as liquid assets for bank analysis purposes, the relevant instruments should:

- Have *nil or very low credit risk* (i.e., be investment-grade securities)
- Be *highly marketable* through liquid secondary markets

All other things being equal, the shorter the maturity of the instrument, the more liquid it is likely to be. Whether the monetary authorities, such as the central bank, are prepared in the course of their normal money-market operations to advance funds against particular instruments—for example, repurchase (repo) finance—is another attribute of liquid assets. If certain assets are classified as eligible collateral for repo finance purposes, that eligibility contributes to their liquidity.

A Liquidity Hierarchy: Ranking Bank Assets by Liquidity

Since the topic of relative asset liquidity was examined previously in the discussion of the balance sheet, it is only touched upon lightly here. Conventionally, assets will rank in liquidity more or less in the following manner:

1. Cash
2. Funds due from the central bank
3. Funds due from other banks
4. Highly marketable government securities
5. Other marketable securities (e.g., subnational government securities, investment-grade commercial debt securities, widely traded exchange-listed equity securities—all assuming a liquid market exists)
6. Marketable loans and advance (e.g., eligible for trading in deep and liquid secondary mortgage market)
7. Less-marketable securities
8. Nonmarketable loans
9. Real estate holdings
10. Strategic investments (equity)
11. Fixed assets
12. Other essentially illiquid assets

The most liquid of all assets is, of course, cash, while fixed assets and some intangible assets are among the least liquid.

Liquidity Quality

Applied to a bank's assets, the concept of liquidity quality refers to the degree to which a gap exists between an asset's nominal liquidity and its actual liquidity.

Actual liquidity may indeed differ from *nominal liquidity*. In certain markets, securities of a type that would be characterized as *liquid assets*—government securities, for example—may not really be liquid. The usual reasons for a lack of liquidity are the absence of a secondary market and limits on the negotiability of the securities. In addition, securities that are liquid during ordinary circumstances may be illiquid during periods of market stress. Where there is reason to doubt the actual liquidity of nominally liquid assets, they are—for analytical purposes—properly excluded from the category.

Just as some apparently liquid assets are not in fact liquid, some apparently illiquid assets may in fact have a higher degree of liquidity than appears at first sight. For example, in some countries, certain types of loans may in reality be marketable. Characteristically, this occurs where a secondary market has been established in loans that meet specific standardized criteria, or that are eligible to be placed into pools and converted into securities through the process of securitization.⁴⁵ So the analyst should consider whether and to what extent a bank's loan book is composed of assets that are marketable for these or other reasons.⁴⁶

The first step is to divide a bank's assets into various categories of liquidity, as discussed in a previous paragraph, and then to apply various ratios to assess the sufficiency of the liquid assets immediately available to the bank. This would determine whether the bank holds an adequate level of liquid assets.

Yet that quantitative assessment will be tempered by the qualitative assessment of how truly liquid those assets might be.

An answer to the following questions would be needed:

- Are the bank's nominally liquid assets, such as marketable debt securities, actually liquid? That is, can they be readily sold for cash with minimal transaction costs?
- In particular, are government securities easily marketable, and is there a deep market for them? Conversely, are there restrictions on the negotiability of government securities holdings, or a thin secondary market that is apt to dry up under harsh conditions?

Not all government securities and ordinarily highly marketable commercial debt securities are equally liquid. In some emerging markets, for example, domestic government securities will not be liquid, even under normal conditions, because there is no secondary market, or there are restrictions on the bank's ability to sell the securities. Moreover, even if securities are liquid in normal circumstances, the market for many securities may dry up at exactly the time it becomes necessary to liquidate them. Or they may only be saleable at a deep discount.

Restriction on the Negotiability of Government Securities

With respect to the first scenario, the negotiability of government securities, especially in the Middle East and South Asia, may not be what it seems. Consider the case of Kuwait's long-term government debt bonds. After the first Gulf War in 1990, in an effort to rehabilitate its banking sector, the Central Bank of Kuwait purchased Kuwaiti banks' nonperforming assets and issued to them long-term government debt bonds (GDBs). By end-1992, the central bank had issued \$16.6 billion of GDBs, which represented 52 percent of the banking industry's aggregate balance sheet.⁴⁷ Although the GDBs were government securities of a quite highly rated sovereign, and it would seem appropriate to classify them as liquid assets, they were illiquid because there was no secondary market in which they could be traded. Consequently, a calculation of the liquid assets ratio for analytical purposes should exclude these GDBs.⁴⁸

Similarly, during the European debt crisis of 2011–2012, while theoretically Greek state bonds could be traded, their market price was hugely affected by uncertainties to the point where any sale was akin to a fire sale and the marked-to-market value applied to those assets was technically⁴⁹ reducing the level of quasi-liquid assets of the bond holders.

Loss of Liquidity

As an illustration of the second scenario, consider the South Korean banking industry going into the 1997–1999 Asian crisis. Before the crisis, Korean banks' headline liquidity ratios appeared to be conservative. For instance, most banks displayed prudent loans-to-deposits ratios of 65 percent and less. This was in large part because a substantial proportion of the banks' credit exposure to the *chaebols*, the large conglomerates that accounted for an outsized proportion of South Korea's output and borrowing, was classified as marketable securities. Since some lending to the

conglomerates took the form of debt security holdings instead of loans, nominal liquidity levels appeared satisfactory. When the market for these securities evaporated during the Asian financial crisis, however, Korean banks' liquidity position proved to be much shallower than it had seemed since the financial difficulties experienced by the *chaebols* were reflected in an evaporation of liquidity with respect to their debt issues. In both of the preceding sets of examples, actual liquidity and nominal liquidity differed substantially. In view of these scenarios, the analyst needs to first determine whether the nominally liquid assets under review are, in fact, truly liquid under normal conditions, and second to make a judgment whether they are likely to be liquid under more extreme conditions.

Funding Liquidity

Lack of access to funding sources and weak liquidity management are typical factors that lead to bank failures.

—Dominion Bond Rating Services⁵⁰

Asset liquidity is an important measure of an institution's overall liquidity. But it is only a partial measure. An insufficiency of liquid assets does not in itself make a bank illiquid. Instead, the salient question is: Where a bank shows signs of weak asset liquidity, does it, in compensation, have strong funding sources, or in case of need, the capacity to obtain cash in another way? This type of liquidity, what can be called *funding liquidity*, refers to a bank's capacity to access new funding to refinance maturing liabilities or to meet other myriad obligations.

A bank's overall liquidity position is based not only on the relative liquidity of its assets, but also on the composition of its funding liabilities.

The existence of a stable deposit base and an active presence in the relevant interbank market may afford a bank a liquidity buffer well beyond that which it can obtain from its liquid asset holdings alone. So regardless of the volume of its liquid assets, a bank that is justifiably viewed as a sound credit, or is guaranteed by the government, will ordinarily be able to access funding freely and at relatively little cost.

An understanding of a bank's sources of funding and ability to obtain funds in the money markets is critical to a comprehensive evaluation of the institution's liquidity position. The principal objective of evaluating a bank's funding profile is to assess its ability to draw upon existing or new sources of funding, as opposed to its capacity to tap existing cash reserves or liquefy current assets into cash.

From where does the bank obtain funding to finance its earning assets? Recall from Exhibit 10.2 that a bank can draw on both internal cash flows and external sources of funding. Putting aside ordinary cash flow from operations, there are essentially two outside sources from which a bank can draw funds: equity investments and borrowings.

Equity finance, however, is a source of long-term funds that is expensive and time consuming to arrange. The sale of equity is simply not appropriate to funding loan advances and other ongoing liquidity needs, except indirectly to the extent that the bank can leverage its additional equity to borrow.

Most of a bank's financial assets are funded by borrowings—that is, debt—as was discussed in connection with the high levels of leverage characteristic of banks.

Borrowings can be divided into two types: (1) borrowings from the money and capital markets and (2) borrowings from depositors, that is, deposits. (It may also be possible for a bank to borrow funds from the government. This is usually done through the central bank. The circumstances under which the government will lend funds to banks vary by country.) These funding sources vary in their stability and concomitantly in their liquidity.

A key distinction can be made between customer deposits and money-market funding, also referred to as commercial funding. Generally speaking, commercial funding obtained through the money market (also referred to as *purchased funds*)⁵¹ is viewed as less stable than funding from customer deposits. In other words, retail customer deposits are regarded as the more stable source of funding. More stable funding in turn increases liquidity, since it means that it is far less likely that the bank will have to scramble for liquid funds to meet current liabilities. This said, banks that maintain access to the money markets as active participants in these markets and thereby have more diversified sources of funding tend to be viewed more favorably from a credit perspective than banks that rely solely upon customer deposits.

Deposit Funding

In contrast to nonbank financial institutions, commercial banks and other similar institutions benefit from the ability to fund themselves through deposit taking. A preliminary question to answer is this: Are deposits sourced from individuals or businesses? Deposits made by businesses, particularly those made by large commercial enterprises on a transactional basis (wholesale deposits) tend to be a more volatile source of funding, more akin to money-market funding than retail deposit funding. As an illustration, large banks in South Africa have traditionally relied heavily on wholesale deposits. While these are usually rolled over, in times of crisis they are prone to withdrawal, thereby weakening these banks' overall funding profiles.

In contrast to wholesale depositors, retail depositors tend to be less interest-rate-sensitive on the whole than commercial depositors. They tend to make their decision whether to maintain a deposit account with a particular institution on a host of factors including customer service and convenience, as well as the interest rate paid on deposits. In the retail investor's eyes, the inconvenience involved in changing banks for the sake of earning a slightly higher return will often outweigh any marginal interest rate benefit obtained. In addition, personal relationships and habit may underpin the loyalty of retail depositors to a particular institution.

This comparative stability, or stickiness, of retail deposits gives many consumer-oriented banks a funding advantage over their counterparts that must rely to a greater degree on funds accessed from the money markets (i.e., commercial funds or purchased funds).⁵² To evaluate the stickiness of bank deposits, the analyst should examine the composition of a bank's deposits. Not all banks benefit equally, however. Large retail banks with extensive branch networks may have an advantage over their wholesale and smaller retail counterparts. Generally speaking, large retail banks tend to enjoy cheaper funding costs attributable to their deep customer deposit bases, although as previously noted when the costs of maintaining a branch network are accounted for the cost advantage may diminish considerably or even disappear.

Another variable to consider is whether deposits are largely *at call*, able to be withdrawn on demand, or time deposits, where withdrawal cannot be made before

maturity without penalty. Most customer deposits fall into the former category, notwithstanding that they tend to be rolled over indefinitely. For purposes of funding stability, and to match liabilities with assets better, it is usually beneficial to have some longer-term deposits.

Another matter to consider in relation to deposit funding is concentration risk. The composition of deposits should be examined to ascertain, if possible, whether dominance of large depositors whose withdrawal of funds could conceivably cause liquidity problems is acute. This could occur as a consequence of privatization. It is typical for a state-owned bank to rely heavily on deposits from the government and its agencies. Should the bank be privatized, these deposits may be withdrawn.

A related issue concerns deposits made by related parties. Significant related-party deposits may be either good or bad depending on the reasons behind the funds, as well as the strength of the related party. Positively, related-party deposits are likely to be a fairly stable funding source because the related party is unlikely to withdraw funds, assuming it has no pressing need for cash, especially if the bank is suffering liquidity problems. Negatively, such deposits may represent a large proportion of deposit funding, and consequently pose concentration risk, especially if the bank lacks the ability to quickly replace the funding should it be withdrawn.

A final point to note in connection with deposit funding is that many countries have deposit insurance schemes in place. These guarantee the nominal value and liquidity of deposits up to a certain size, and tend to bolster an institution's liquidity. Their primary purpose, however, is to protect depositors and to reduce financial system fragility. The effectiveness of such schemes depends on several factors, including depositors' confidence that they will suffer minimal if any losses in the event of bank failure and that access to their deposits will be maintained, or at worst restored quickly under such circumstances.⁵³

Commercial Funding

The main alternative to customer deposits as a source of short-term bank funding is the money market. The money market includes the interbank market, which encompasses interbank borrowing and lending, and other short-term funding available from commercial sources, such as commercial paper. While nominally deposits, both wholesale deposits and interbank deposits are much more akin to commercial funding than to deposit funding. Funds generated from this type of borrowing go by a variety of terms, but are referred to here as *commercial funding*.⁵⁴ As a general rule, commercial funding comprises the following sources of funds:

- Interbank funding
- Repo finance and securities lending
- Commercial paper and other short-term debt issued by banks
- Wholesale or brokered deposits

A bank's capacity to borrow in the market is largely a function of (1) the maturity and depth of the money markets, (2) the bank's own credit standing, and (3) the bank's presence in the market and track record with investors. Highly creditworthy banks will, of course, have easier access to commercial funding than those rated as greater credit risks. Where local money markets, including an

interbank market, are not well developed, banks may have to rely heavily, if not exclusively, on funding from customer deposits.

Naturally, prevailing market conditions and investor appetite have a huge impact on the capacity of banks generally and on any given bank to obtain new funds. When markets are stable and sentiment is positive, a bank that meets minimal financial strength thresholds is unlikely to have substantial funding difficulties. The real test comes when, due to either internal or external challenges, market sentiment changes and the bank's liquidity position can be squeezed. This has occurred many times in the past, including during the revision of this book. Banks that rely heavily on foreign funding or international credit lines can be particularly at risk, with smaller and weaker institutions being most exposed. When sentiment changes, counterparty banks quickly become more risk averse and may abruptly cut credit lines. If lines are cut at an unfortunate time, the bank may be placed in difficult straits.

For instance, if a bank has significant syndication facilities from foreign banks, and these facilities come due at a time when markets are negative on the country and funding has been squeezed, the institution may find itself in an untenable situation, especially if its financial health is cast into doubt, whether by rumor or by verifiable evidence of a deteriorating balance sheet. At best, depositors and other creditors may be reluctant to advance additional funds without some further risk premium being paid. The result will be that the bank's funding liquidity, and therefore its overall liquidity position, will be weakened. One relevant question to pose to management, should the opportunity present itself, is: How long could your bank survive without having to access the money markets?

Interbank Borrowing

The interbank market in most countries is an informal market among local banks for overnight or short-term funds. It is among the most sensitive of commercial funding sources and therefore characteristically volatile. Generally speaking, it can be a comparatively expensive funding source. Banks that rely too heavily on interbank funding are vulnerable to liquidity problems when market conditions sour.

THE TURKISH BANKING CRISIS OF 1994

The 1994 crisis in Turkey exposed many Turkish banks' overreliance on international credit lines and bank syndicated borrowings from overseas. When the country's sovereign risk rating was downgraded in the early part of 1994, sentiment soon changed as the economic crisis took hold. International banks cut credit lines, squeezing many domestic banks' liquidity positions, which then led to rumors regarding the solvency of some banks.

This eroded confidence in the banking sector and customer deposits were soon withdrawn in large quantities to the point where the government stepped in and placed a blanket guarantee on all deposits to restore the system's liquidity. Turkish banks realized from the crisis the advantage of possessing a good base of domestic deposit funding because it is the most stable source of funds for banks.

Repo Finance

Another type of commercial funding is through repos, which are a form of collateralized off-balance-sheet finance that is effectively secured by government fixed-income securities. Repo finance and reverse repo finance—borrowing and lending using repurchase agreements—have become an important source of funding for banks in many countries. Repos generally have maturities of less than one week, but may be rolled over. Under a repo agreement, one counterparty agrees to sell to another a fixed-income security, usually government treasury bills or bonds, with a guarantee to repurchase it later at a predetermined rate and price. In effect, repo finance is a form of secured lending. Repos create both liquidity and interest rate risk due to the mismatch in maturities between the repos and the longer maturities of banks' interest-earning assets, as well as pricing differences. In a situation where there is a significant fall in the appetite of repo finance providers, rates will rise accordingly. For example, market interest rates may be 4 percent, but the overnight repo rate may be 5 percent. Those banks relying on repos as a principal source of funds could face a funding shortage, particularly in the short term, and the resulting loss from the materialization of market risk could be significant.

Bank Commercial Paper

Like corporate issuers, banks may issue commercial paper. These are short-term obligations generally having a maturity of less than one year. Other short-term debt issues or obligations that do not qualify as Tier 2 capital can be regarded for analytical purposes as a principal source of funding.

Wholesale or Brokered Deposits

While the difference between customer deposit-based funding and money-market funding is clear in the main, the two categories are blurred at the borders. While smaller businesses may bank with a single institution, large corporates are likely to have several banking relationships and will place funds on deposit with various

COMMERCIAL FUNDING (PURCHASED FUNDS)⁵⁵

There are, as discussed, two principal sources of short-term bank funding:

1. Customer deposits (made by individual consumers and small and medium-sized businesses)
2. Funding other than customer deposits

As a category, customer deposits, also called *core deposits*, generally encompass the following types of deposits, where placed by retail consumers and small- and medium-sized businesses:

- Short-, medium-, and long-term deposits⁵⁶
- Certificates of deposits
- Cash margin (collateral) placed by a bank's customers

Funding not meeting this definition includes deposits made by large commercial enterprises or on a purely transactional basis and related forms of funding. These include:

- Interbank deposits (funds placed with a bank overnight or for other relatively short periods by other banks)
- *Purchased funds* or *brokered deposits*, including large-denomination certificates of deposit
- Short-term funding obtained from the cash proceeds received from issuing securities, such as commercial paper, in the market

The markets in which these instruments trade are the money markets. Money markets are notoriously sensitive to current business and economic conditions, changes in government policy, and current events. A bank that depends heavily on purchased funding can easily find itself facing liquidity problems should market conditions turn.

It is apparent that overuse of such funding increases a bank's liquidity risk and in consequence its credit risk. In addition, it is generally, but not always, more expensive than core deposit funding and nearly always more volatile. The use of purchased funds is not, however, wholly adverse from a bank credit perspective. Mixing this funding with funding from customer deposits does afford a bank the benefit of diversification of funding sources, which can have a positive impact on a bank's credit profile.

At this stage, it is appropriate to mention that the notion of *purchased funds* could be helpful to make a distinction, in interbank markets, between interbank funds received by a bank without the bank seeking them, on the one hand, and interbank funds actively sought by a bank—which could then be akin to purchased funds. While the funds passively received might prove volatile and therefore be seen as providing less stable funds, it is also true that if a bank finds itself frequently in need of interbank funds, those funds might be seen as purchased funds and may be an indication that the bank is putting too much reliance on that kind of funding source.

Although no hard and fast rules can be stated, the following rules of thumb may prove helpful:

- Where a high proportion of a bank's total funding base comes from commercial sources, as opposed to retail customer deposits, it will usually have, and be perceived to have, a comparatively less stable and more expensive funding profile.
- Where a high proportion of a bank's total funding base comes from retail customer deposits, as opposed to commercial funding, it will usually be perceived to have a comparatively more stable and cheaper funding profile.
- A bank may lend aggressively and fund itself conservatively, or lend conservatively and fund itself aggressively, but it should not do both: that is, fund itself aggressively and lend aggressively.
- An optimally conservative customer deposit/total funding ratio is about 80 percent.

banks according to the terms being offered and the company's particular needs at the time. There is no bright boundary between a deposit made by a business and wholesale funding. The latter term implies that the deposit has been made on a transactional basis and purely according to commercial criteria. Similarly, a large deposit is arranged through a third-party broker, so it would be more appropriately characterized as commercial funding instead of as customer deposit funding.

Medium- and Long-Term Debt

The presence of medium- and long-term debt on a bank's balance sheet gives the institution a significant advantage in respect to its liquidity profile. These debt obligations usually take the form of subordinated debt securities. Subordinated debt finance was discussed in the preceding chapter in the context of its role in strengthening a bank's regulatory capital position. It also provides the bank with a core of very stable funding that reduces the typical asset liability maturity mismatch that is endemic to banking. Consequently, it is able to manage its projected balance sheet better and, at least partly, to match its medium- and long-term assets with corresponding liabilities. This affords the bank greater flexibility and diversity than an institution that must rely solely on equity finance and short-term deposit and money-market funding. So the appearance of significant medium- and long-term debt on a bank's balance sheet usually elicits rating advantages for the bank in the eyes of the analyst.

It is particularly an advantage in emerging markets because banks in these operating environments often find it difficult to obtain medium-to-long-term funds by other means, as other financial institutions are not willing to take on longer-term risk in markets of high volatility. It tends to be only the very highly regarded banks or highly rated institutions that are able to gain access to such finance. As was mentioned in respect to loan syndications, however, it is important to be aware of the dates when repayments are due to ascertain whether the bank can comfortably meet its obligations. Especially in times of market volatility and negative sentiment, a bank may have difficulty rolling over its medium- and long-term debt, thereby putting pressure on its liquidity position.

Securitization and Liquidity

Loans are normally regarded as fairly illiquid from a bank credit analysis perspective, but this is not always the case. In some markets, a strong legal infrastructure and standardized terms enable these otherwise nonmarketable financial instruments to be sold. This, of course, increases both the liquidity of the loans themselves, and the overall liquidity profile of the bank relative to those banks that are unable to dispose of loans in this manner. The regular sale of *whole loans*, however, is much less important than another financial technique that has done much to increase banks' overall liquidity; that is, securitization. Until the recent credit crisis struck in 2008, banks were increasingly securitizing assets, such as mortgages, to diversify funding, reduce capital needs, and improve their return on equity. Suddenly mortgage loans came to possess a significant amount of liquidity.

The discussion that follows has lost much of its relevance following the implosion of securitization markets in 2008 as part of the global financial crisis. Nevertheless, it may prove useful in understanding the background to the crisis.

What Is Securitization?

Securitization is the process in which a financial institution that owns a loan portfolio converts it into debt securities issued to investors. To effect this transformation, the bank pools the mortgage loans or other financial assets it wishes to sell and transfers them to a bankruptcy-remote special purpose vehicle (SPV). The SPV, in turn, pays the bank for the pooled loans from funds obtained through the bundling of these underlying assets into securities. Typically the resulting securities are issued in two or more tranches, each tranche having specific risk and return characteristics, based on the investors' claim to the cash flows from the interest payments and principal repayments on the underlying assets.

Traditionally, it was more expensive for a bank to securitize assets against obtaining funds from the wholesale market. The extra cost can be 40 basis points and above, and banks tended to securitize only if they could obtain capital relief (i.e., reduce the requirement in the balance sheet of a corresponding level of capital to be held against the value of, say, the mortgages funded through the securitization issue). If the bank needs to increase the yield of the issue to attract investors to the securitization issue to a greater level than the equity relief, then the securitization deal is not cost-effective funding.

Nevertheless, securitization does have compensating funding, liquidity, and asset-liability management advantages that make it attractive, notwithstanding some increased funding cost. Securitization can improve a bank's funding profile as well as its asset liquidity. As assets are moved off balance sheet, capital is freed up, while at the same time, default risk and other risks arising from holding the underlying assets are transferred to investors. The matter of asset concentration is also of some relevance.

Advantages of Securitization

Securitization increased the liquidity of some assets, notably mortgage loans, by creating a secondary market for them. Even if a bank retained its loans on balance sheet, rather than converting them into marketable securities, the fact that securitization existed as an option made such loan assets more liquid. Additionally, securitization also provided an extra funding source for the bank apart from commercial paper and deposits. By selling a portion of its loan portfolio to the SPV and indirectly onward to investors, the bank receives immediate cash proceeds that it could use to fund new advances to borrowers. Overall, the liquidity profile of the bank could be improved. Asset-liability management is also improved through the reduction of the volume of medium- and long-term financial assets held on the balance sheet. Credit risk and market risk, including prepayment risk, associated with the underlying assets are correspondingly reduced as these are transferred to investors in the securities issued by the SPV. Finally, capital requirements are reduced, since the loans to which capital had to be allocated have been removed from the balance sheet.⁵⁷

Disadvantages of Securitization

The foregoing advantages hold so long as there remains market appetite for these securitized assets. Banks, however, could and did become overdependent on the securitization as a source of funding. In the event of a collapse in demand for securitized financial products, such banks would confront a severe loss in liquidity. The chief disadvantage of securitization, though, was its long-term impact on the mortgage supply chain. By fragmenting the various roles once performed by one institution,

accountability for managing the risks of the underlying assets disappeared. The result was to create securities, the actual risk of which well exceeded their rated risk. The consequent huge losses borne by investors and in turn by the banks that originated and, in part, held such securities played a major role in contributing to the severity of the credit crisis in 2008 and 2009.

SECURITIZATION IN EMERGING MARKETS⁵⁸

In emerging markets, with a few exceptions, the state of development of domestic capital and financial markets, limited the use of securitization techniques. It nevertheless offered some advantages that made it attractive in such countries, notably access to longer-term funds for banks. In emerging markets, it can be difficult for banks to gain access to funding with a tenor of more than one year. So securitization of future receivables became attractive as a financing tool in non-investment-grade countries beginning in the late 1980s. The underlying assets are usually hard-currency-generating export receivables, credit card receivables, and remittance receivables.

Typically, in a future financial flow transaction, the interest and principal repayments under a debt instrument are paid out of, and secured by, the cash flows from transactions such as credit card receivables. The important factor in these transactions is that they capture hard-currency cash flows offshore and pay the debt service on the bonds before transferring excess funds back to the originator.⁵⁹

Securitizing future receivables is very advantageous for banks in countries whose sovereign rating is below investment grade. A good bank with a low domestic credit risk seeking funds on the international market will otherwise be constrained by its country's sovereign rating, and hence its own rating will be capped at the *sovereign ceiling*, thereby raising the costs and restricting the availability of funding.⁶⁰ Through these structures, a bank can achieve a higher rating than the sovereign rating by structuring the issue so as partly to overcome and reduce the related risks.

Central Banks and Lenders of Last Resort

Although it is usually the more vulnerable institutions that are the target of sudden demands for withdrawals, even the most financially sound banks can potentially be subject to a liquidity shortfall or be hit by a run on deposits. A generally healthy bank might face difficulties because of negative news about the bank or the banking system. Perhaps the timing of the news is unfortunate, occurring perhaps just as the bank is about to roll over some existing funding facilities. Or, as is more often the case, the bank in question may not be in such robust health. Although it is not in terminal decline, perhaps it faces difficulties because poor asset quality has depressed cash flows.

For obvious reasons, governments and central banks cannot tolerate significant disruption to their banking systems.⁶¹ Consequently, central banks may be effectively

compelled to extent their traditional role of lender of last resort to that of its extreme meaning of the expression. This simply means that the state functions, usually through its central bank, as the final source of liquidity to a bank. Temporary facilities from a central bank are usually secured against sound assets. Alternatively, the central bank will discount credits of a solvent bank.

These facilities are normally extended to a basically fit but somewhat vulnerable institution that faces a liquidity shortfall. A central banker's decision—as lender of last resort—whether to support a distressed bank often depends upon the answers to four sets of questions:

1. How large a facility is needed? For how long? What are the costs and risks associated with providing the facility?
2. Is the bank solvent? Is it a fundamentally sound bank facing a difficult situation owing to a confluence of circumstances?
3. Is management reasonably competent? Can it do better with the assistance of regulators?
4. Would the failure of the bank cause unacceptably high disruption to the national financial system?

The original Basel Communiqué⁶² commented on the subject of lenders of last resort. It describes the intention of intervention by lenders of last resort as providing financial support to banks under temporary liquidity pressure, not to cover the losses of badly managed banks at the expense of the entire national community.

The External Evaluation of Liquidity

The adequacy of a bank's liquidity will vary. In the same bank, at different times, similar liquidity positions may be adequate or inadequate depending on anticipated or unexpected funding needs. Likewise, a liquidity position adequate for one bank may be inadequate for another. Determining a bank's liquidity adequacy requires an analysis of the current liquidity position, present and anticipated asset quality, present and future earnings capacity, historical funding requirements, anticipated future funding needs, and options for reducing funding needs or obtaining additional funds.

—Federal Deposit Insurance Corporation⁶³

Depending on both the financial condition of the bank and its operating environment, a bank's liquidity problems may be patently acute—that is, a liquidity shortfall is imminent or has occurred—or merely latent. By the time a bank's liquidity problems have become acute, it is probably too late for analysis. The market will likely already be aware of the bank's distress, or rumors will be swirling. Therefore, it is more likely that the analyst will be called upon to evaluate banks that may have material weaknesses in their liquidity position, but where any acute problems have not materialized.

For the external analyst, such an evaluation is a challenging task compared with an assessment of other CAMEL financial attributes; namely earnings, asset quality, and capital. There are two basic reasons for the difficulty. First, as has been shown, banks' liquidity needs are somewhat unusual, particularly when compared with those of corporates. Unlike nonfinancial companies, a bank normally enjoys a regular

inflow of cash from deposits. But they are more vulnerable to sudden and uncontrollable demands for cash. An unexpectedly high volume of deposit withdrawals or drawdowns on credit lines, for example, can quickly result in a potential liquidity shortfall. Consequently, in contrast to earnings, asset quality, and capital problems, which tend to manifest in a more measured manner, a lack of liquidity may not be at all evident until it makes its appearance suddenly and with full effect.

Second, and related to the first reason, in evaluating a bank's liquidity position, the credit analyst is almost always handicapped by data limitations. A bank's expected cash flow is important to its overall liquidity, but mainly for internal management purposes. Data on cash flow are likely to be too limited to figure importantly in external credit analysis. At a minimum, weekly cash flow projections are required to effectively identify forthcoming liquidity requirements. This level of periodicity and the detail required are simply unavailable to those—other than regulators—evaluating a bank from outside the organization. So, as observed, a bank's cash flow statement is of little use for external analytical purposes.⁶⁴

Given that the outside analyst will rarely have access to sufficient data to draw a complete picture, the best that can be hoped for is a rough indication of the degree to which a bank is vulnerable to liquidity problems. This can be accomplished by comparing an institution's most recent and historical liquidity positions with its peers, taking account of the most critical influences on bank liquidity and applying key indicators.

Beyond the difficulty of accessing reliable data on a timely basis, a quantitative analysis of liquidity is confronted with the difficulty of selecting which ones of many possible indicators are likely to provide the best snapshot of a specific bank's liquidity at a specific moment.

PRESCIENT OPINIONS FROM A RATING AGENCY

This reliance on overseas funds may offer immediate cost savings, but may impact the bank's liquidity in times of financial stress. In early 1995, there was a substantial withdrawal of capital from Thailand. The Bank of Thailand protected the currency and assisted the liquidity of the market. Problems may arise again and if they are more severe, the liquidity of some members of the banking industry will be tested.

—Thomson BankWatch report on Bank of Asia, July 15, 1995

Generally speaking, external observers have long been tolerant of the low liquidity ratios, drawing comfort from Bank of Thailand's long established practice of offering protection to institutions in trouble. Recent events surrounding the Bangkok Bank of Commerce have eroded confidence in the Bank of Thailand's capacity to identify and address potential problems in the banking industry.

—Thomson BankWatch report on Siam City Bank, July 17, 1996

A simple table with the broad definitions of main asset and liability categories is useful to explore (see Exhibit 10.4).

EXHIBIT 10.4 The Liquidity of Some Categories of Assets and Liabilities

Item		Common Definition	Remarks
Liquid assets		Cash and cash equivalent, including the highest-grade most marketable government securities, ordinarily having a maturity of less than one year.	Most liquid assets
Quasi-liquid assets		Broader definition of liquid assets including all marketable securities having a reasonable degree of liquidity.	Generally liquid assets.
Risk-weighted assets		Risk assets weighted as per the Basel accord (as revised).	More refined version of risk assets.
Long-term assets		Net loans, debt securities having a remaining maturity > one year, equity securities, foreclosed assets, minus acceptances of other banks.	Least liquid assets, generally speaking.
Total deposits		Total customer deposits + Interbank deposits (and other large commercial deposits)	All deposits
Customer deposits (Core deposits)		Total customer deposits (not exceeding \$100,000) (i.e., effectively retail consumer and small- to medium-sized business deposits, excluding brokered deposits).	Most stable deposits
Non-core funding (Commercial Funding)	Noncore deposits	Total deposits – core deposits; i.e., time deposits with balances of \$100,000 or more + brokered deposits + foreign deposits + deposits made by banks	Less stable deposits
	Other commercial Funding	Interbank borrowings (and central bank borrowings) + repo finance + commercial paper + subordinated debt + mandatory convertible notes + other borrowings (including mortgage indebtedness and obligations under capitalized leases)	+ debt funding
Customer deposits and short-term funding		Total customer deposits + Interbank deposits (and other large commercial deposits) + other short-term funding (e.g., commercial paper, repo finance)	Same as customer deposits but includes other short-term funding.

(Continued)

EXHIBIT 10.4 *(Continued)*

Item	Common Definition	Remarks
Total deposits & borrowings	Total customer deposits + Interbank deposits (and other large commercial deposits) + other short-term funding (e.g., commercial paper, repo finance) + other funding (e.g., medium-term debt) but excluding hybrid capital and subordinated Tier 2 debt	Same as above but includes longer-term noncapital funding.
Stable funds	Total customer deposits + other deposits + medium/long term liabilities + free capital funds	Same as above but includes debt capital funding and free capital.
Free capital	Total capital (i.e., Tier 1 + Tier 2 capital) – fixed assets – investments in subsidiaries	Portion of equity and debt capital funding available for noncapital purposes.
Noncapital debt funding	Debt funding minus hybrid capital and subordinated Tier 2 debt.	Debt funding other than capital.

Measuring Asset Liquidity

This section surveys the quantitative liquidity indicators that may be available to the external analyst.

There are three basic types of fundamental quantitative indicators.

1. The first type evaluates a bank's asset liquidity by comparing its holdings of relatively liquid assets with relatively less liquid assets.
2. The second type measures funding stability by comparing a bank's relatively stable funding liabilities to less stable funding liabilities.
3. Finally, a third type of fundamental ratio compares the various categories of a bank's assets with the various categories of its liabilities to gauge asset liability mismatch.

The categories of ratios, and the definitions of the terms used in individual ratios, are discussed in more detail in Exhibit 10.5.

There are only a few indicators used to measure asset liquidity, and most, like the quasi-liquid assets ratio, are some variant on the liquid asset ratio defined in the equation shown. Both the liquid assets ratio and the quasi-liquid assets ratio are fairly basic indicators that can be easily generated from data provided from banks' annual reports or quarterly financial statements.

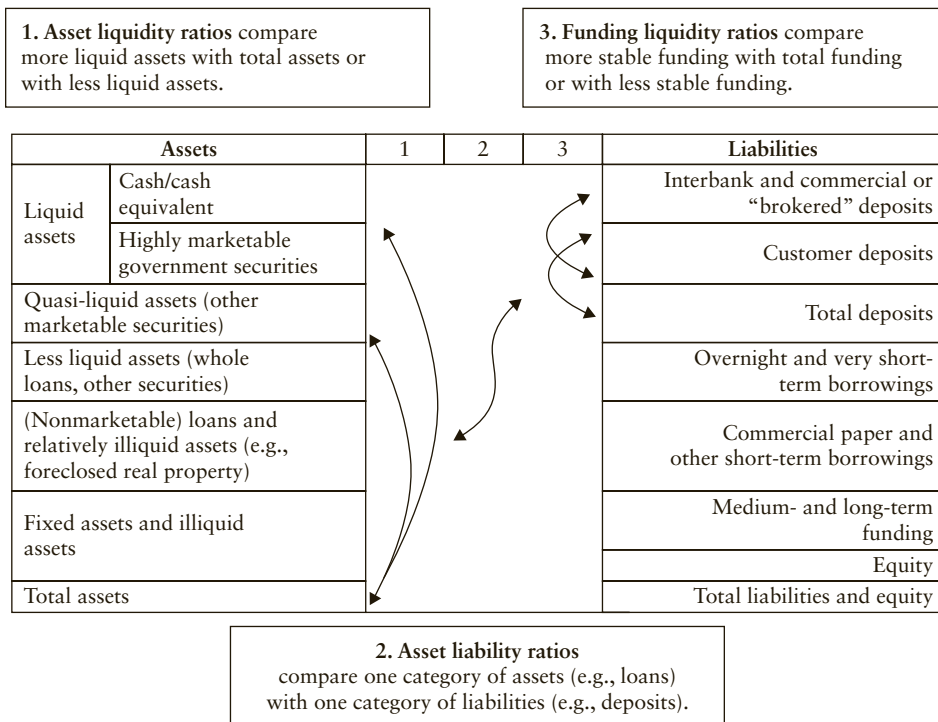


EXHIBIT 10.5 Liquidity Ratio Components

Liquid Assets Ratio

The liquid assets ratio is the basic measure of asset liquidity. It measures the percentage of total assets that consist either of cash or assets that ostensibly can be sold in a time of crisis at a minimal cost to the bank. Normally, liquid assets encompass funds having a maturity of one week or less, such as cash, deposits with the central bank, treasury bills, government securities, and interbank loans (i.e., funds deposited with other banks or advanced through the local interbank market).

$$\text{Liquid assets ratio (percent)} = \frac{\text{Total liquid assets}}{\text{Total assets}} \times 100$$

The usefulness of the liquid assets ratio depends upon how liquid assets are defined (including the breadth or narrowness of the definition), and perhaps more importantly, to what extent the definition takes account of actual liquidity of assets as opposed to their nominal liquidity. For this reason, it should be used with care when comparing banks on a cross-border basis, or even within the same country where there is wide variation in how banks report the various assets on their balance sheets.

SORTING OUT THE RATIOS

In common with nearly all the financial ratios commonly employed for bank analysis purposes, a multitude of variant indicators can be devised depending upon how the numerator and denominator are defined.⁶⁵ Definitions vary by institution and analyst. In using these tools, it is not a question of the more, the better. Most bank analysts focus on a few key indicators, and use others to consider points relevant to a particular evaluation. With regard to liquidity and funding, there are fundamentally three types of ratios:

1. *Asset liquidity ratios* compare one category of assets against another category or against a benchmark such as assets as a whole, typically gauging the relative volume of liquid assets to illiquid assets. The principal ratio is the liquid assets ratio, which is liquid assets divided by total assets. Other asset liquidity ratios, such as pledged government securities to total government securities, follow a similar pattern.⁶⁶

2. *Asset-liability ratios* compare one category of assets against one category of funding liabilities or vice versa. Examples include loans to total deposits (which reflects the relationship between illiquid loans and relatively stable deposit funding) and the interbank ratio (which shows the extent to which the subject bank is a net borrower or lender in the interbank market)

3. *Funding liquidity ratios* compare one category of funding liabilities against another category of funding liabilities or against a benchmark such as total assets or total liabilities. Examples include customer deposits to total deposits (the proportion of total deposits that are placed by retail and small business customers relative to total deposits), a category that encompasses

large commercial depositors and, conceivably, borrowing from other banks through the interbank market in the form of deposits. Likewise, the ratio of brokered deposits to total deposits shows the relationship between commercial deposits arranged on a transactional basis through a broker versus total deposits, the lion's share of which would in most cases be customer deposits.

Regulatory Liquidity Ratios

In addition to establishing minimum prudential thresholds with respect to capital ratios and other financial variables, regulators frequently require that banks maintain certain minimum liquidity ratios.⁶⁷ These ratios may be defined differently from the liquid assets ratio. For example, a liquid assets ratio used for regulatory purposes might be calculated to be the bank's average daily proportion of liquid assets, as defined by the regulator, to total assets calculated on a rolling monthly basis. More importantly, outside analysts will frequently not have access to the data necessary to compute these ratios. Out of necessity and to allow comparability across jurisdictions, bank credit analysts normally employ the simpler definition of liquid assets described in the previous subsection or the broader definition of quasi-liquid assets.

REGULATORY LIQUIDITY THRESHOLDS

Most regulators require that banks maintain eligible liquid assets as a minimum proportion of all liabilities maturing within a short time horizon, for example seven or eight days. Regulatory definitions, however, may differ from those generally calculated from publicly disclosed financial data.

The requirement for liquidity reserves under common sense prudential rules and the requirement for actual liquidity reserves under local prudential rules are not the same thing. The former is of relevance to the analyst and will soon be the subject of regulation under Basel III. The latter is a local practice whereby banks actually deposit, usually with the central bank, a certain amount of funds. Such practice both allows the central bank to act on the multiplier and, especially in some emerging markets, gives government access to cheap and discreet funding that might not be counted as fiscal debt.

Quasi-Liquid Assets Ratio

The quasi-liquid-assets ratio is a sometimes used variant of the liquid assets ratio that places quasi-liquid assets in the numerator instead of mere liquid assets. It is therefore a somewhat broader measure of asset liquidity with no precise definition. A practical definition of *quasi-liquid assets* would be those financial assets that are generally marketable, but do not meet the strictest liquidity criteria.⁶⁸ Because there is,

however, no uniform agreement on what constitutes quasi-liquid assets, this indicator can be adapted by the analyst to overcome limitations in financial reporting among banks and across jurisdictions.

QUASI-LIQUID ASSETS

The term *quasi-liquid assets* refers to marketable securities or, more often, collectively to the sum of liquid assets and marketable securities. The reason for the use of the term is to distinguish between those assets that meet the strictest liquidity criteria and those that are somewhat less liquid.

$$\text{QLAR} = \text{Quasi-liquid asset ratio (percent)} = \frac{\text{Cash} + \text{Near cash} + \text{Interbank assets} + \text{Government securities} + \text{Marketable securities}}{\text{Total assets}} \times 100$$

Net Loans to Total Assets

A somewhat cruder measure of liquidity, but useful where information concerning asset liquidity is lacking, is to simply weigh net loans⁶⁹ against total assets to see what proportion of a bank's assets are in the form of these less-liquid earning assets. Indeed, the loan-to-assets ratio might be defined as the illiquid assets ratio, since it compares relatively illiquid loan assets to total assets.

$$\text{Net loans to Total assets (percent)} = \frac{\text{Net loans}}{\text{Total assets}} \times 100$$

Pledged Government Securities to Total Government Securities

When government securities are pledged as collateral, they cannot be freely sold and thereby used to bolster a bank's liquidity. This ratio answers the relevant, but not always obvious, question mentioned earlier: How liquid are those assets categorized as liquid assets by the bank? More specifically, this ratio asks the question: What proportion of the government securities on the bank's books have been pledged as some form of collateral, and are therefore not immediately available for conversion into cash for other purposes? In other words: How much of the stated amount of government securities is actually available in case of need?

$$\text{Pledged government securities to Total government securities (percent)} = \frac{\text{Pledged government securities}}{\text{Total government securities}} \times 100$$

Unfortunately, data to calculate this ratio is not widely available to the external analyst, not to mention that it changes very frequently. When it is available, however, this otherwise little used indicator is a welcome addition to the analyst's toolbox. A variant of this ratio is *pledged securities to total securities*.

Measuring Funding Liquidity

This category of liquidity indicators examines the composition of a bank's funding and compares its various elements to provide a measure of funding stability. Before looking at the ratios, recall that short-term bank funding can be divided between deposits and money-market funding, or to be more precise, between customer deposits and other funding (i.e., commercial funding).

Customer Deposits to Total Deposits

The customer-deposits-to-total-deposits ratio (customer deposits divided by total deposits) is a measure of the strength and stability of the bank's customer deposit base. Equally, it indicates the extent to which the bank is dependent on generally more volatile interbank or commercial sources for funding. Multibranch retail banks will conventionally have high customer-deposits-to-total deposits ratios, while specialist institutions, such as development banks with small branch networks, may have very low ratios. In compensation for this weakness, however, the latter category may have special access to certain forms of high-quality funding, such as medium-to-long-term facilities—for example, bonds or debentures.

$$\text{Customer deposits to Total deposits (percent)} = \frac{\text{Total customer deposits}}{\text{Total deposits}} \times 100$$

Brokered Deposits to Total Deposits

A variant indicator of the customer-loans-to-deposits ratio is the brokered-deposits-to-total deposits ratio. In employing this ratio, a distinction may or may not be made between brokered deposits and other types of purchased funds.

$$\text{Brokered deposits to Total deposits (percent)} = \frac{\text{Brokered deposits}}{\text{Total deposits}} \times 100$$

Other possible variants of this ratio pattern would be to substitute interbank deposits, purchased deposits, commercial deposits, or noncustomer deposits, defined as appropriate, for brokered deposits. (Note that some of these terms are essentially synonymous.) Similarly, instead of total deposits, total funding could be used as the denominator. So the following are potential variant indicators:

- Interbank deposits (i.e., interbank borrowing) to total deposits
- Interbank deposits to total funding
- Purchased funds to total funding

The Notion of Stable Deposits

As will be discussed in a later chapter, Basel III is introducing requirements for the maintenance by banks of sufficient stable deposits to meet potential short-term cash outflows.

Banks have long used such an approach, with very simple tools giving a predictable, if intuitive, level of deposits, which, balancing lost ones with fresh ones, gave a good approximation of potential shortages.

Asset-Liability Mismatch

So far, we have looked at ratios comparing assets with assets and liabilities with liabilities. A bank's ability to maintain liquidity under difficult conditions, however, is appropriately measured by comparing a bank's core illiquid assets, that is, mainly loans, with its core, that is, stable funding. In general, by measuring the mismatch between the two, the bank's vulnerability to liquidity problems is revealed. This section therefore surveys ratios that provide such comparison, which is a fundamental determinant of liquidity risk.

Net Loans to Total Deposits and Net Loans to Customer Deposits

The loans-to-deposits ratio and its more refined cousin, the loans-to-customer-deposits ratio, are perhaps the most traditional and widely used measures of overall liquidity. Defined in the two equations that follow, each reflects the mismatch between relatively illiquid loans on the asset side of the balance sheet and relatively stable deposit funding on the liability side.⁷⁰ The net-loans-to-customer-deposits ratio is generally perceived as the better indicator of relative funding stability.

The higher the ratio, the more the bank is depending on noncore deposits and wholesale funding sources to fund its loan book rather than relying on more stable sources of funding. Conversely, a lower ratio is more credit positive in that it suggests that the bank is funding its loan portfolio primarily from core deposits. Ideally, total loans should not exceed core deposits.

$$\text{Net loans to Total deposits (percent)} = \frac{\text{Net loans}}{\text{Total customer deposits} + \text{Interbank deposits}} \times 100$$

$$\text{Net loans to Customer deposits (percent)} = \frac{\text{Net loans}}{\text{Total customer deposits}} \times 100$$

Prevailing loans-to-deposits ratios, however, vary with the market and economic conditions. Ratios will tend to rise during times of prosperity as lending expands and to decline in economic downturns. Also, in developed markets the ratio may be higher on average, even exceeding 100 percent because of the widespread use of other sources of funding, such as repo finance or securitizations. In less sophisticated financial markets, however, strong banks will usually maintain more conservative loans-to-deposits ratios, generally in the 70 to 90 percent range, with even more cautious institutions holding to the 50 to 70 percent range. Finally, observe that, as normally computed, loans-to-deposits ratios do not make adjustments for the fact that in some markets a bank may be able to sell loans or to pledge loans as security to borrow funds. Insofar as these circumstances exist, the analyst should be aware that the risks addressed by these indicators will be ameliorated to a corresponding degree.

The Interbank Ratio (Interbank Assets to Interbank Liabilities)

Another important ratio of this type is the interbank ratio. By dividing a bank's interbank assets (loans to or deposits placed with other banks) by its interbank liabilities (borrowings or deposits from other banks), this ratio answers the question: What is the

bank's net position in relation to the interbank market? Is it a net lender (good) or a net borrower (not so good)?

Interbank assets are generally defined as instruments in which another bank is the counterparty and where the obligation flows from that counterparty back to the bank. These assets generally include deposits due from banks: certificates of deposits and similar instruments, including overnight deposits placed with other banks and time deposits placed with other banks. Interbank liabilities normally include deposits received from other banks, call loans from other banks, vostro accounts,⁷¹ overdrafts, funds borrowed from banks, repos, drafts, and remittances outstanding.

A related indicator, which answers the same question, is the bank's net interbank asset position, which is simply equal to interbank loans (due from banks) minus interbank borrowings (due to banks). Put another way, a bank's net interbank asset position is equal to total deposits with banks minus total interbank liabilities. If the resulting figure is a positive number, the bank is a net lender in the interbank market. If the net interbank asset position is negative (i.e., if interbank liabilities are greater than interbank assets), the bank is a net borrower in the interbank market. The former scenario is credit positive, the latter, credit negative.

$$\text{Interbank ratio (percent)} = \frac{\text{Interbank assets to Interbank liabilities (percent)}}{\frac{\text{Total deposits placed with other banks (Interbank assets)}}{\text{Total deposits received from other banks (Interbank liabilities)}}} \times 100$$

$$\text{Net interbank asset position} = \text{Total interbank assets} - \text{Total interbank liabilities}$$

The interbank ratio provides the same information but in a common-size format that gives an indication of relative size of the bank's net interbank position. If interbank assets divided by interbank liabilities yield a number greater than 1 (i.e., more than 100 percent), the bank is a net lender, while the bank is a net borrower if that number is less than 1.

RULE OF THUMB: BETTER TO BE A LENDER THAN A BORROWER

A bank that relies heavily on "purchased funds" will not only be paying more for funding than its peers with extensive branch networks, but it will also be more vulnerable to sudden withdrawals and to a liquidity crunch. Since in difficult times bank lenders are more likely to cut lines than customers are to withdraw their deposits, a large net taker of funds is likely to be subject to higher liquidity risk.

In most situations, it is generally preferable, from a credit perspective, that a bank be a net lender in the interbank market than a net borrower. So a ratio of 100 percent or more is credit positive; less than 100 percent credit negative, all other things being equal. As mentioned elsewhere, from a profitability angle, depending on the true cost of gathering and maintaining retail deposits, being a net borrower is not

necessarily a negative factor. Analysts should be aware, generally, that year-end figures must be treated with circumspection, as they may not reflect the typical position. A variant ratio is interbank borrowings to total liabilities. Additional variant ratios are described in the following.

Loans to Deposits and Short-Term Funding

A modification of the loans-to-deposits ratio, this ratio incorporates additional types of short-term funding frequently found in more developed markets to better reflect the bank's actual funding sources.

$$\text{Net loans to (short-term) Funding (percent)} = \frac{\text{Net loans}}{\text{Total customer deposits} + \text{Short-term funding}} \times 100$$

Net Loans to Total Deposits and Noncapital Funding

Net loans to total deposits and borrowings uses moderately broad definitions of deposits and funding in the ratio's denominator, incorporating all debt other than long- and medium-term capital finance.

$$\text{Net loans to Total deposits and Noncapital funding (percent)} = \frac{\text{Net loans}}{\text{Total deposits} + \text{Short-term funding} + \text{Other funding (excluding Subordinated debt and Hybrid capital)}} \times 100$$

Net Loans to Stable Funds

The net-loans-to-stable funds ratio is similar to net loans to customer deposits, but adds to the denominator of the ratio other stable funding including medium- and long-term debt, as well as free capital. This indicator considers the fact that customer deposits are not necessarily the only type of bank funding that is comparatively stable, and takes account of the extent to which a bank's own unencumbered capital and other forms of less volatile funding contribute to its funding stability.

$$\text{Net loans to Stable funds (percent)} = \frac{\text{Net loans}}{\text{Customer deposits} + \text{Official deposits} + \text{Medium-/Long-term liabilities} + \text{Free capital funds}} \times 100$$

Stable funds are defined as total customer deposits plus official deposits plus medium-term/long-term liabilities plus free capital funds. Note that:

- Official deposits are deposits with, or borrowings from, central governments or central banks.
- Medium-term/long-term liabilities include loans and notes with more than one year to maturity.
- Free capital funds are defined as total equity capital minus financial and non-financial subsidiaries minus fixed assets.

Core Deposits to Total Assets

Another ratio sometimes seen in bank liquidity analysis ratio roundups is the core-deposits-to-total-assets ratio. It simply measures core deposits, that is, customer deposits, against a bank's total assets and provides information similar to the customer-loans-to-deposits ratio, albeit with less precision since it merely compares core funding with total assets rather than with more vulnerable assets. So it resembles a common-size ratio and could just as easily be grouped with funding liquidity ratios.

$$\text{Customer deposits to Total assets (percent)} = \frac{\text{Total customer deposits}}{\text{Total assets}} \times 100$$

Purchased Funds to Earning Assets

Another indicator that can be used to measure reliance on purchased funds is the ratio of purchased funds to earning assets, but the net-noncore-funding-dependence ratio, discussed later in the section, is a much better indicator.

$$\text{Purchased funds to Earning assets (percent)} = \frac{\text{Purchased funds}}{\text{Earning assets}} \times 100$$

Illiquid Assets to Purchased Funds

An alternative ratio, constructed somewhat differently, is the illiquid assets-to-purchased funds ratio. Illiquid assets are primarily composed of loans, but would also include fixed assets and other nonliquid assets on the balance sheet.

$$\frac{\text{Illiquid assets (or Loans) to}}{\text{Purchased funds (percent)}} = \frac{\text{Illiquid assets (or Loans)}}{\text{Purchased funds}} \times 100$$

Short-Term Assets to Short-Term Liabilities

The ratio, short-term assets to short-term liabilities, is similar to the current ratio used in corporate credit analysis, and is sometimes referred by that name. The ratio, shown in the box below, attempts to answer the question: What proportion of a bank's depositors can demand immediate repayment of their deposits without causing a liquidity shortfall for the institution?

$$\text{Short-term assets to Short-term liabilities (percent)} = \frac{\text{Short-term assets}}{\text{Short-term liabilities}} \times 100$$

For bank analysis purposes, short-term assets will generally include cash and cash equivalent, interbank loans (i.e., deposits with other banks), and other short-term investments. Short-term liabilities will be made up of customer deposits and interbank and other short-term borrowings. In emerging markets where, in case of difficulties, access to funding is sometimes very limited, a current ratio below 25 percent sends a danger signal. Above 50 or 60 percent, the ratio translates into much reduced interest margins, which may be too high a cost to pay for maintaining a comfortable surplus of short-term liquidity.

Note that minor variations in the definitions of the numerator and denominator may be responsible for the ratio being given different names, as shown in the two equations that follow. For instance, it is sometimes referred to as the liquid-assets-to-customer-and-short-term funding ratio (short-term assets being essentially equivalent to liquid assets and short-term liabilities being essentially equivalent to short-term funding) or as the deposit runoff ratio.⁷² Likewise, it is effectively interchangeable with the short-term-assets-to-deposits-and-noncapital-funding ratio shown.

$$\text{Deposit runoff ratio (percent)} \sim \frac{\text{Liquid assets}}{\text{Customer and Short-term funding}} \simeq \frac{\text{Short-term (} \sim \text{liquid) assets}}{\text{Short-term liabilities (} \sim \text{Customer deposits + Money market funding)}} \times 100$$

$$\text{Short-term assets to Deposits and Noncapital funding (percent)} = \frac{\text{Short-term assets}}{\text{Deposits and Noncapital funding}} \times 100$$

Net Short-Term Liabilities to Total Assets

This ratio, among several liquidity indicators favored by Dominion Bond Rating Services, weighs a bank's dependence on short-term funding relative to short-term assets, by comparing its excess of short-term funding over short-term assets against total assets. In so doing, it provides a more precise measure of the degree of mismatch between a bank's current assets and its current liabilities.⁷³ A negative ratio is credit positive. The greater the ratio above zero, the more vulnerable the bank is to funding risk.⁷⁴

$$\text{Net short-term liabilities to Total assets (percent)} = \frac{\text{Short-term liabilities} - \text{Short-term assets}}{\text{Total assets}} \times 100$$

Wholesale Funding Reliance Ratio

Another ratio preferred by DBRS is what it calls the wholesale-funding-reliance ratio. This indicator shows the proportion of funding that is essentially noncore—that is, other than customer deposit funding in relationship to total funding. It resembles a similar-sounding indicator called the net-noncore-funding-dependence ratio, but is defined in a slightly different way.

$$\text{Wholesale funding reliance ratio (percent)} = \frac{\text{Interbank borrowings (and Central bank borrowings) + Repo finance + Noncore deposits + Subordinated debt + Mandatory convertible notes + Other borrowings}}{\text{Core deposits + Wholesale funding}} \times 100$$

Net-Noncore-Funding-Dependence Ratio

The net-noncore-funding-dependence ratio used by the U.S. Federal Reserve Bank is a fairly sophisticated measure of the relationship between a bank's noncore funding and its illiquid assets. It resembles the net-short-term-liabilities-to-total-assets ratio,

the principal difference between the two being that the latter's denominator is total assets rather than long-term assets. Using long-term assets instead of total assets means that the degree of mismatch between liabilities and assets is emphasized.

In the context of the funding dependence ratios, noncore liabilities (or funding) comprise time deposits with balances of \$100,000 or more, foreign deposits, interbank borrowings, repo finance, commercial paper, other borrowings (including mortgage indebtedness and obligations under capitalized leases), and brokered deposits.

The definition of short-term investments is the sum of interest-bearing bank balances (including interbank loans), reverse repo finance, debt securities with a remaining maturity up to one year and, in addition, acceptances of other banks.

The definition of long-term assets is the sum of net loans and debt securities with a maturity longer than one year, equity securities and foreclosed assets, from which acceptances of other banks are deducted.

$$\text{Net noncore funding dependence ratio (percent)} = \frac{\text{Noncore liabilities} - \text{Short-term investments}}{\text{Long-term assets}} \times 100$$

Net-Short-Term-Noncore-Funding-Dependence Ratio

A companion ratio to net-*noncore*-funding-dependence ratio is the *short-term-noncore*-funding-dependence ratio. This indicator focuses on a bank's reliance on short-term commercial funding, as opposed to all commercial funding, by including only short-term noncore funding in the numerator. Compared with the previous indicator, this ratio attempts to further isolate and measure the most extreme mismatches.

$$\text{Net short-term noncore funding dependence (percent)} = \frac{\text{Noncore short-term funding} - \text{Short-term investments}}{\text{Long-term assets}} \times 100$$

Foreign-Currency Assets to Foreign-Currency Liabilities

Theoretically, a liquidity indicator that concentrates on the degree to which foreign-currency funding is matched or not to foreign-currency assets is more relevant to market risk management than it is to liquidity management. The ratio of foreign-currency assets to foreign-currency liabilities provides an indication as to whether a bank holds a position, positive or negative, in other currencies. In fact, it should be computed separately for each foreign currency. If the risk is not hedged, the bank may be exposed to foreign exchange risk.

If assets and liabilities in a given foreign currency are roughly of equal value, the bank does not have a position, but it does not follow that it does not have liquidity gaps in that currency. The relevance of this ratio to the matter of liquidity is only that it can give some signal as to a funding problem beyond the market risk problem. The significance of the ratio, however, is severely limited by the fact that it only catches items recorded on the balance sheet and ignores off-balance-sheet transactions that a bank would enter into in the normal course of its risk management decisions.

$$\frac{\text{Foreign-currency assets to}}{\text{Foreign-currency liabilities (percent)}} = \frac{\text{Foreign-currency assets}}{\text{Foreign-currency liabilities}} \times 100$$

Foreign-Currency Loans to Foreign-Currency Funding

The foreign-currency-loans-to-foreign-currency-funding ratio is a more relevant ratio, provided it is used separately for each foreign currency. It uses the same basic data as the previous ratio, but uses the loans-to-deposits ratio to target the portion of total assets—that is, its loan book—that is among the least liquid, and is therefore most difficult to shed should market conditions change.

$$\frac{\text{Foreign-currency loans to}}{\text{Foreign-currency funding (percent)}} = \frac{\text{Foreign-currency loans}}{\text{Foreign-currency borrowings} + \text{Foreign-currency deposits}} \times 100$$

It bears repeating that, as discussed in respect of the above two ratios, a bank has to manage its assets and liabilities not only in its local currency but also in any single foreign currency in which it makes transactions. That prudential rule applies both to liquidity and to currency risk angles.

Other Issues in Evaluating Bank Liquidity

Institutional Size and Liquidity

The size of a financial institution frequently has a bearing on its liquidity and funding options. All things being equal, a big bank is likely to obtain a higher credit rating than a smaller bank and possess access to market funding, such as the interbank and fixed-income securities. Generally, smaller banks often rely more heavily on a narrow funding base, and are therefore more vulnerable to liquidity risk.

The large bank's advantages can be attributed both to economies of scale, and the perception that the most dominant institutions in a given market are either inherently more creditworthy than smaller ones, or more apt to be beneficiaries of government support in the event that they experience distress.⁷⁵ Moreover, a large bank will usually have at its disposal more funding options and consequently will probably have a more diversified funding base than its smaller rivals. Additionally, having in all likelihood a more sophisticated treasury operation, it may very well be able to use hedging instruments more efficiently.

Such an institution will often have greater access to the money and capital markets generally, as well as to international markets, and may be able to attract more stable medium- and long-term funding. If retail oriented, it can be expected to have a pervasive distribution network, enabling it to more easily scoop up retail deposit funding than its smaller competitors. Finally, under the too big to fail principle, during a period of financial distress, the larger bank is more likely to receive state support than less-critical institutions.

The Impact of Other CAMEL Criteria

As will all of a bank's financial attributes, its liquidity position will be affected by management strategy as it manifests in the character of the other CAMEL financial elements. We have already seen that profitability, capital, and asset quality are closely connected

with other CAMEL attributes, and this is no less true of liquidity. Earlier it was observed that liquidity risk is closely linked to market risk, specifically in the form of interest rate risk—including refinancing and reinvestment risk—and foreign currency risk.

High levels of liquidity are likely to affect profitability negatively because very liquid assets are likely to be low yielding. In addition, poor asset quality will have an unfavorable impact on liquidity, not to speak of the detrimental effect on profitability and capital. Similarly, a highly solvent bank—that is, one that has low leverage—will tend to be more liquid, all other things being equal. In addition, a well-capitalized bank is less likely to be a victim of a flight to quality in the event of financial turmoil.

Liquidity and Reserve Requirements

Almost invariably, domestic bank regulations dictate reserve requirements and other liquidity guidelines that influence the overall adequacy of liquidity in a market. Minimum reserve requirements differ from country to country, and of course are subject to change.

As discussed earlier, the minimum reserve requirements serve a different purpose, but clearly they contribute to the creation of a liquidity buffer that the banks might otherwise neglect.

The analyst needs to be aware of this to see whether minimum requirements are above, below, or at generally accepted levels. Some examples of various regulatory liquidity requirements that have been imposed in the past⁷⁶ are outlined in Exhibit 10.6.

EXHIBIT 10.6 Sample Liquidity Requirements Circa 1999

Country	Selected Liquidity Requirements
Saudi Arabia	Banks must maintain liquid assets equal to at least 20 percent of their deposit liabilities. Banks must also maintain statutory deposits with the Saudi Arabian Monetary Authority amounting to 7 percent of their average demand and 2 percent of their average savings and time deposit liabilities.
Philippines	Banks are required to set aside a certain portion of their peso deposit liabilities as reserves in the form of liquid assets. At least 25 percent of the liquidity reserve must be kept in the form of deposits with the central bank of the Philippines. Banks can keep up to 75 percent in cash in vault and/or eligible government or foreign securities. Banks also have to maintain a 10 percent reserve requirement on their peso-denominated common trust funds and other similar managed funds. The reserve requirements are as follows: Regular Reserves: 9 percent for deposits and deposit substitutes for commercial banks. Liquidity Reserves: 3 percent on government and quasi-government deposits.
Czech Republic	Banks are required to keep a minimum cash reserve balance with the central bank equal to or greater than 2 percent of primary deposits. Other minimum liquidity ratios are also utilized.
Turkey	The reserve requirement ratio is set at 6 percent. Two percent of Turkish lira deposits must be held as free deposits with the central bank. The liquidity requirement may be met on the average of daily data for the reserve requirement period, rather than on a continuous basis.

Source: Central Banks.

Basel III will introduce new constraints in relation to liquidity, but such constraints may, depending on the local circumstances, not be sufficient to ensure safe levels of liquidity. It would be dangerous to assume that, provided a bank is Basel III compliant, its liquidity levels are acceptable, as this would be as foolish as assuming that a bank that is compliant with Basel II or Basel III necessarily exhibits adequate capital ratios.

Window-Dressing Liquidity

It is important to be aware that certain window-dressing practices may change the apparent liquidity position of the bank at year-end. These include:

- Overnight back-to-back transactions with banks to boost liquidity ratios.
- Transferring overdrafts or loans to or from associated companies over the end of the accounting period; since overdrafts or loans between associated companies and third parties are not consolidated in group financial statements, this may change the apparent liquidity of the group.
- Repaying a loan shortly before the end of an accounting period and reborrowing it shortly afterward, with a view to avoiding its disclosure.
- The temporary rearrangement of financing transactions, so that at the end of the accounting period they are dealt with as off-balance-sheet items.

An example worth exploring is the trick used by Lehman Brothers to massage its end-of-period liquidity and capital adequacy through repos.⁷⁷ In short, Lehman used its London affiliates to handle repos on its behalf in such a way that, unlike traditional repos, the transaction was seen as a sale rather than as short-term financing. Lehman used London instead of the United States because it could get only English lawyers to state that such transactions could be booked as sales. Technically, a sale would reduce the bank's assets and therefore reduce its leverage and increase its liquidity. Shortly before its collapse, Lehman Brothers was able, using that trick, to remove about US\$50 billion from its assets and to appear less leveraged.⁷⁸

Assessing Bank Holding Company Liquidity

Bank holding companies were discussed in the previous chapter in relation to capital and leverage. To review, bank holding companies are corporate entities that own banks—the main bank owned by the bank holding company is usually referred to as the lead bank—and frequently also own nonbank subsidiaries.⁷⁹ The bank holding company structure affects liquidity as much as or more than it affects leverage, and the discussion of the analysis of bank holding companies is therefore resumed here.

What Is a Bank Holding Company?

Because the bank holding company's primary asset is merely its equity stake in its lead bank, it must rely mainly on dividends *upstreamed* to the parent holding company from the subsidiary lead bank for fulfillment of the parent's financial obligations. These dividends are dependent upon the continued profitability of the lead bank subsidiary. Prudential regulations may constrain the bank holding company's access to its lead bank's resources, despite its nominal authority over the lead bank. In the United States, for example, regulators have the power to stop payments

when deemed necessary to maintain the soundness of the bank subsidiary. In addition, bank subsidiaries in the United States may not lend on an unsecured basis to their parents. Finally, only banks and not bank holding companies have access on a routine basis to funding from the central bank's (i.e., the Federal Reserve Bank's) discount window. Consequently, the bank holding company, although it owns its lead bank, holds a weaker position from both a liquidity and an overall credit perspective in comparison to its subsidiary. For the same reason, evaluating the liquidity profile of a bank holding company requires a somewhat different procedure from the one that is applicable to a bank.⁸⁰

Moody's Approach to Evaluating Bank Holding Companies

The approach of the rating agency Moody's Investor Services to the analysis of bank holding companies illustrates this approach. In Moody's view, the typically weaker creditworthiness of a bank holding company relative to its subsidiary bank or banks and their lack of access to central bank liquidity call for an alternative approach to analyzing their liquidity. The agency believes that the general creditworthiness of a holding company, its banks, and its nonbank subsidiaries—together with the reputation of these entities and management's attitude towards liquidity risk—are all relevant to a comprehensive evaluation of bank holding company liquidity. Moody's employs what it calls a *plausible stress scenario* for this purpose. Applying this methodology, the bank holding company's liquidity requirements are calculated by adding the principal and interest due within one year on the bank holding company's obligations, and deducting any cash and cash equivalent on the bank holding company's own balance sheet (excluding its subsidiaries). To the resulting sum, parent-only noninterest expenses, and the parent's own revenues, excluding any intragroup dividends from subsidiaries and securities gains, are deducted. In addition, the liquidity of any marketable securities on the bank holding company's balance sheet is evaluated and discounted as appropriate.

The agency uses two critical ratios to evaluate bank holding company liquidity as a part of its overall analysis. They are what it calls the comprehensive ratio and the adjusted comprehensive ratio. The comprehensive ratio is equal to the bank holding company's total liquid assets minus any debt maturity within one year plus the bank holding company's net income exclusive of all subsidiary dividends, with the resulting sum being divided by the group's consolidated net income. The significance of the ratio is that it "captures the potential cash demands that a bank holding company faces within a year if it loses access to the funding market."⁸¹

A positive ratio suggests that conversion of the bank holding company's liquid assets into cash would be adequate to fulfill the bank holding company's cash demands for a one-year period. Correspondingly, a negative ratio suggests that the sale of such assets would be insufficient to meet the bank holding company's current (i.e., one-year) liquidity demands. Where the bank holding company has a negative position that is more than 100 percent of the group's consolidated net income, the implication is that even the upstreaming of all dividends to the bank holding company would still leave a shortfall in liquidity. The advantage of this ratio is that the data used to compute it is publicly available. The main disadvantage is that it does not take account of the liquidity of the subsidiaries, which become relevant in those circumstances where the bank holding company has acted as a guarantor of its subsidiaries' debts.

$$\text{Comprehensive ratio (percent)} = \frac{\text{Bank holding company's total liquid assets} + \text{Bank holding company's current debt maturity} + \text{Bank holding company's own net income (excl. Subsidiary dividends)}}{\text{Consolidated net income (for the group)}} \times 100$$

Note that the bank holding company's own income is defined by Moody's to comprise:

- Its net interest income
- Its fees
- Its noninterest income other than dividends and securities gains less noninterest expense (excluding special charges)

Adjusted Comprehensive Ratio

The adjusted comprehensive ratio is identical to the comprehensive ratio but for one change. A 50 percent haircut (i.e., discount) is applied against the value of those securities that come under the category of "Other debt and equity securities" in the bank holding company's public filings. The rationale for the 50 percent discount on the value of these securities is to take account of a potential decline in their value in case of adverse market conditions. This ratio is therefore simply a more conservative variant of the comprehensive ratio.

$$\text{Adjusted comprehensive ratio (percent)} = \frac{\text{Bank holding company's adjusted liquid assets} + \text{Bank holding company's current debt maturity} + \text{Bank holding company's net income (excluding Subsidiary dividends)}}{\text{Consolidated net income (for the group)}} \times 100$$

$$\text{Bank holding company's adjusted liquid assets} = \frac{\text{Total liquid assets} - (\text{Other debt and Equity securities} \times 50 \text{ percent})}{1}$$

Backup Facilities

Moody's believes that backup facilities should ordinarily be excluded in evaluating bank holding company liquidity. Its rationale is that "if a BHC drew down a liquidity facility from a third-party bank, it may signal to the market that it is facing funding difficulties throughout the total organization." While it might save the day in the short term, it would likely have a longer-term negative impact on the group as a whole, with trust, custody, funds management, and treasury operations being particularly vulnerable to the reputational damage.⁸²

The DBRS Approach

Canadian rating agency DBRS emphasizes the importance of assessing both bank and bank holding company liquidity while maintaining a distinction between the two. The relationship between the parent company and its lead bank subsidiary is viewed as a reciprocal one in that the bank holding company will issue debt to *downstream capital* into its subsidiary banks and will also commonly keep a large portion of its liquid assets in the form of deposits with its lead bank. In turn, "management expects to be able to draw on these deposits to meet parent company obligations."

DBRS sees four potential scenarios in the relationship between a bank holding company's liquidity and that of its lead bank, as illustrated in the diagram below: They are a strong parent–strong subsidiary, a strong parent–weak subsidiary, a weak parent–strong subsidiary, and a weak parent–weak subsidiary. Of the four possible scenarios, a strong parent–strong subsidiary is obviously positive and a weak parent–weak subsidiary obviously negative. In DBRS's view, however, neither of the two alternative scenarios is necessarily wholly adverse, although it believes that of the two, the weak parent–strong subsidiary is preferable since weak bank holding company liquidity is more easily remedied than weak lead bank liquidity.

Liquidity Coverage Ratio

DBRS favors what it terms the liquidity coverage ratio as a measure of the strength of a bank holding company's liquidity position. The numerator is similar to Moody's comprehensive ratio, but the denominator rather than being consolidated net income is comprised of short-term liabilities. This indicator is defined as "eligible funding" to "short-term payment obligations," and correspondingly seeks to measure the extent to which current liabilities are covered by sources of liquid funds.

Definition of Eligible Funding Eligible funding includes:

- Cash and balances due
- Marketable securities
- Interbank lending and reverse repos
- Interest income
- Management fees from subsidiaries
- Dividends from nonbank subsidiaries
- Other operating income

It excludes dividends from bank subsidiaries.

Definition of Short-Term Payment Obligations Short-term payment obligations include both fixed and contingent liabilities. The former would generally encompass:

- Commercial paper and similar obligations
- Interest payment obligations, for example, to preferred shareholders
- Short-term nonborrowing liabilities
- Term borrowing maturing within one year
- Noninterest expenses
- Dividend payments to common shareholders
- Contingent liabilities such as guarantees to bank holding company subsidiaries

$$\text{Liquidity coverage ratio (percent)} = \frac{\text{Liquid assets} + \text{Short-term borrowings} + \text{Incoming cash flow}}{\text{Short-term payment obligations}} \times 100$$

DBRS prefers that the ratio exceed 100 percent, meaning that "eligible funding sources to equal or exceed obligations."⁸³ In other words, it is optimal for a parent

company to have a liquidity coverage ratio of more than 100 percent. A weaker ratio would be compensated to a degree by strong lead bank liquidity. Note that all items refer to the parent company only; that is, the bank holding company, and not to the group.

Bank Liquidity: Emerging Markets versus Developed Markets

Exhibit 10.7 shows bank liquidity ratios across seven emerging markets circa 1999. Note the considerable variations in indicator values.

Liquid asset ratios ranged from 16.6 to 51.1 percent, and 20.0 to 61.7 percent for the broader quasi-liquid asset ratio. Loans to customer deposits ranged from 26.5 to 97.9 percent, and loans to stable funds from 26.7 to 89.8 percent.

Interestingly in the context of the European debt crisis, Greece's banking sector displayed the highest average liquidity ratios, 10 years before the crisis. The banking sector in Greece had low net-loans-to-customer deposits and net-loans-to-stable-funds ratios, while its liquid-assets-based ratios were high. These liquidity ratios largely reflected the fairly weak lending environment in the past caused by poor economic fundamentals, which saw high market interest rates. As a result, a high proportion of Greek banks' asset base comprised domestic government securities while their loan portfolios were relatively modest. The situation in 2010 was similar, which partly explains why the Greek debt crisis was also a crippling one for Greek banks.

In contrast, weaker liquidity ratios were seen for South Africa. For the big four banks—ABSA, Standard Bank of South Africa, FirstRand, and Nedcor—their loan portfolios comprised a high percentage of total assets. The relatively tight liquidity is a factor of the market in which the banking sector is very well developed and sophisticated. Individual lending, such as home mortgages, is well developed. South African banks also rely heavily on wholesale deposits. This reflects the sophisticated characteristic of the corporate sector with the large corporates placing their funds only with the main clearing banks. Generally, South African banks' liquidity ratios are more in line with those of developed markets.

Saudi Arabian banks possessed solid liquidity ratios, aided by well-capitalized balance sheets. Banks included in the Bahraini sector are the offshore-licensed investment banks. Loan-based liquidity ratios are moderate due to the fact that

EXHIBIT 10.7 Bank Liquidity Ratios across Seven Emerging Markets

Average Sector Liquidity Ratios at End 1999	Indonesia	Greece	Saudi Arabia	Bahrain	Malaysia	Thailand	South Africa
Net loans/ customer deposits	26.50	54.11	74.27	85.70	85.71	86.33	97.85
Net loans/stable funds	26.66	47.17	60.81	32.17	74.04	81.52	89.82
Liquid assets ratio	16.61	51.10	40.54	30.46	26.55	18.38	17.60
Quasi-liquid assets ratio	50.88	55.98	47.52	61.65	37.40	21.09	19.95

Source: Bank financial statements.

lending is not a focus of these institutions. In addition, the offshore banks in Bahrain cannot tap the local deposit market and rely quite heavily on interbank funding, particularly from Gulf-based institutions. Their assets profile included significant portfolios of marketable securities and unquoted investments. The reason for the large jump between the liquid assets ratio (30.46 percent) and the quasi-liquid assets ratio (61.65 percent) is due to the large portfolios of *marketable securities*.

Malaysia's banking sector reflected overall sound liquidity ratios, but the sector averages did overlook some noted variances. For example, Public Bank's management has traditionally been very conservative, and the institution has always maintained very high liquidity with ratios being superior to the market. Through this practice, the bank recorded a much lower level of bad debt compared to the market.

Indonesia's low loan-based liquidity ratios was due to the massive restructuring—after the Asian crisis—resulting in large volumes of nonperforming loans having been removed from the balance sheets and replaced with government paper (which was, in actual fact, illiquid). Thailand's liquidity ratios, although improved from earlier times, were still relatively weak owing to an overhang of bad loans still on banks' books. In the years leading up to the Asian crisis of 1997, which began in Thailand, the country's banks registered exceptionally high loans-to-deposits ratios, in some cases as high as 125 percent, and abysmally low quasi-liquid asset ratios in the vicinity of only about 8 to 10 percent.⁸⁴ Tellingly, banks that registered such high loans-to-deposits ratio values suffered severe pressure when the crisis hit and international funding was withdrawn. They were all subsequently taken over by the government to be restructured.⁸⁵

As can be seen in Exhibit 10.8, liquidity ratios in four developed markets—Australia, Canada, Germany, and the UK—around the same time were much tighter

EXHIBIT 10.8 Liquidity Ratios among Selected Banks in Developed Markets, 1998

	Liquid Assets/ Total Assets	Gross Loans/ Customer Deposits	Gross Loans/ Total Assets
Australia			
ANZ Group	14.14	77.24	151.04
Commonwealth Bank of Australia	12.47	79.97	137.65
National Australia Bank	12.84	78.58	150.19
Westpac Banking Corporation	8.90	79.67	155.28
Canada			
Canadian Imperial Bank of Commerce	8.90	77.24	156.20
Royal Bank of Canada	12.44	75.94	126.17
Toronto-Dominion Bank	12.57	77.96	136.17
Germany			
Bayerische HVB	27.36	67.91	210.56
Deutsche Bank	43.30	46.58	118.71
United Kingdom			
Abbey National	36.79	52.38	132.79
Barclays	42.37	44.20	91.97
Lloyds TSB Group	22.44	57.72	107.53

Source: Bank financial statements.

than in emerging markets. This reflected a greater diversity of funding sources and their perceived high credit standing at the time. Liquid asset ratios were on average lower than in the emerging markets just shown (ranging from 8.9 percent to 42.4 percent, while loans-to-deposits ratios were held within a conventionally conservative range of 44.2 percent to 80.0 percent).

Liquidity Warning Signals

The analyst should be on the lookout for several warning signals of liquidity problems. Not all warning signals, of course, give way to a full-fledged liquidity crisis, but where such red flags are present, further scrutiny is warranted. The most salient indicators of emerging liquidity weakness can be found in information provided by the U.S. Federal Deposit Insurance Corporation, in particular in its manual for bank examiners. While the data available to which the bank examiner has access may not be available to the rating agency or counterparty analyst, the list is nevertheless instructive.

SERIOUS WARNING SIGNALS FOR U.S. FEDERAL DEPOSIT INSURANCE CORPORATION BANK EXAMINERS

General Warnings

- Volume of turndowns in the brokered markets unusually large, forcing the institution to deal directly with fewer willing counterparties.
- Rating-sensitive providers, such as money managers and public entities, abandoning the bank.
- The institution receiving requests from depositors for early withdrawal of their funds, or the bank has to repurchase its paper in the market.
- Transaction sizes decreasing, and some counterparties even unwilling to enter into short-dated transactions.
- An increasing spread paid on deposits relative to local competitors, or national or regional composites.

Warnings Related to Purchased Funds

- Ineffective management or the absence of appropriate expertise.
- Newly chartered institution with few relationship deposits and an aggressive growth strategy.
- Inadequate internal audit coverage.
- Inadequate information systems or controls.
- Identified or suspected fraud.
- High on- or off-balance-sheet growth rates.
- Use of rate-sensitive funds not in keeping with the bank's strategy.
- Inadequate consideration of risk, with management focused exclusively on rates.
- Significant funding shifts from traditional funding sources.

- The absence of adequate policy limitations on these kinds of funding sources.
- High delinquency rate or deterioration in other asset-quality indicators.
- Deterioration in the general financial condition of the institution.
- Other conditions or circumstances warranting the need for administrative action.

Source: Risk Management Manual of Examination Policies, Section 6.1—Liquidity and Funds Management.

NOTES

1. European Central Bank, “Developments in Banks’ Liquidity Profile and Management,” Frankfurt am Main, Germany, 2002, 7.
2. F. Reid, William Schwartz, and Roger Lister et al., *Lifting the Lid on Ratio Analysis for U.S. Banks* (New York: Dominion Bond Rating Service, July 2005).
3. Editorial: “Liquidity Reform: Credit Squeeze Means a Rethink of Banks’ Access to Cash,” *Financial Times*, February 14, 2008.
4. Information about the film can be found at http://en.wikipedia.org/wiki/It's_a_Wonderful_Life.
5. Building and loan societies, more commonly referred to as savings and loans, or S&Ls, were a type of financial intermediary in the United States that were developed to finance residential home purchases. Until 1913, nationally chartered commercial banks were generally prohibited from mortgage lending and were highly restricted in such lending until 1927. State-chartered S&Ls filled the void. (Another quite similar structure, mutual ownership, was also used.) Thousands of state-chartered S&Ls collapsed in the United States during the 1930s, and the industry again experienced a crisis in the 1980s after deregulation. See Mayer, *The Bankers: The Next Generation* (Penguin, 1996), 361–365.
6. In fact, a savings and loan association (S&L) is a relatively simple financial institution that essentially takes savings deposits and is usually restricted in the types and volumes of loans it can extend.
7. The concept of liquidity has been alluded to in previous chapters and is explained in the discussion that follows.
8. Banks are not unique in this respect. Insurers may react in a similar manner, refusing to insure against suppliers’ nonpayment when economic conditions worsen. The same dynamic occurs in other situations where there is an exposure to credit risk.
9. Deposit insurance protects bank depositors against the loss of their deposits should a bank fail. Such programs are usually state- or industry-sponsored, and coverage is usually automatic with respect to member institutions. The effectiveness of deposit insurance depends upon how the programs are structured and how they operate in practice, especially in respect to coverage limits and the speediness with which depositors regain access to their funds. The topic is discussed briefly later.
10. Bank crises are studied in detail in Chapters 14 and 15.
11. Flight to quality does indeed happen in times of market turmoil, and depositors seek safe havens in well-regarded and highly rated banks. The Asian crisis saw such much deposit flight from weaker banks to banks perceived as stronger (in most cases to strong foreign-owned banks, which were deluged with new depositors). During the 2008–2012 crisis, in particular from the end of 2011, the European Central Bank started becoming the only safe haven for interbank surpluses, and that included corporate funds routed for that

purpose through relatively safe banks controlled by corporates (as for example Siemens Bank in Munich, whose interbank assets—largely rumored to be parked with the ECB—jumped to EUR 4.48 billion in September 2011 from a mere EUR 10 million in September 2010).

12. G. Thomas Woodward, *Money and the Federal Reserve System: Myth and Reality*, Congressional Research Service Report for Congress, No. 96-672 E (1996).
13. See FDIC's manual: "Liquidity represents the ability to fund assets and meet obligations as they become due," Federal Deposit Insurance Corporation, Section 6.1—Liquidity and Funds Management, www.fdic.gov/regulations/safety/manual/section6-1.html#introduction. In relation to a bank, liquidity means the capacity to readily access cash by (1) converting marketable assets, such as government securities, into cash with minimal transactional costs, and by (2) obtaining new funds from investors or lenders.
14. Ibid. That is, liquidity risk takes in the risk that the costs of maintaining adequate liquidity will be too high. (This particular risk is a form of funding risk.) A bank that can fund its assets only at significantly above-market rates will, to maintain adequate profitability, tend to lapse into a chronically weak liquidity position. Over time, it will become increasingly vulnerable to a shortfall of cash in the event of an unexpected contingency. Unless the bank can extricate itself from such a situation, it will face a precarious future.
15. More stable funding increases liquidity, since it means that it is far less likely that the bank will have to scramble for liquid funds to meet current liabilities.
16. DBRS, *Lifting the Lid on Ratio Analysis*.
17. A guide published by a number of years ago by Robert Morris Associates, now the Risk Management Association, included the following still cogent observation: "Illiquidity, rather than poor asset quality, is the immediate cause of most bank failures." Robert Morris Associates, *A Guide to Analyzing Foreign Banks* (Philadelphia: Robert Morris Associates, 1988), 28. Aside from its reserves of liquid assets, illiquidity invariably flows from a lack of confidence on the part of depositors, counterparties, and investors. The lack of confidence may be utterly unjustified. Usually, however, it has its roots in some substantive weakness in the bank that often is exacerbated by external circumstances. Thus, although a lack of liquidity is the usual proximate cause of bank failure, and may conceivably have no justifiable substantive cause, in the vast majority of circumstances a combination of internal and external factors—poor asset quality, fragile solvency, management incompetence, and market conditions, for example—will play a part in a bank's demise. A lack of liquidity is only the final symptom before expiration.
18. Although nonfinancial companies may need cash for similar purposes on occasion, they will not be subject to the same type of liquidity requirements to nearly the same degree. Separately, note that should liquidity become problematic, a bank ordinarily will be able to curtail lending, except for those funds with respect to which an irrevocable commitment has been made.
19. The term *quasi-liquid assets* is a broad but somewhat imprecise category intended to include both those liquid assets strictly defined as cash or close cash equivalents, together with marketable securities deemed to be reasonably liquid. Highly liquid government securities are often defined to be liquid assets, and where this is the case, quasi-liquid assets could be expected to cover as well those marketable and highly liquid nongovernment securities and those marketable and reasonably liquid government securities not so defined.
20. Of course, some assets that do not meet this definition of liquid assets (or quasi-liquid assets where this terminology is used) may be, in reality, potentially liquefiable to a greater or lesser degree. Marketable securities, for instance, are easily liquefiable; loans are much less so. For analytical purposes, however, these less-liquid assets are frequently lumped together and labeled "illiquid."

21. Borrowing may be effected, in other words, by accepting new deposits, borrowing from other banks through the interbank market, or issuing new debt securities in the money or capital markets. Similarly, new cash contributions can be sought from shareholders in exchange for shares. Note that generally speaking, the issue of medium- and long-term debt and the issue of equity not used to generate liquidity but to improve a bank's capital position. Typically, they take too long to arrange and are too costly to use for funding purposes. Nevertheless, they do generate cash proceeds and as such are an additional source of liquidity even if this is not their main function.
22. Such prudential requirements vary across jurisdictions. They have received much attention in recent years.
23. Remember that there is a trade-off between liquidity and profitability.
24. Asset-liability management is touched upon briefly in the third section of this chapter, and is discussed further in Chapter 12 dealing with risk management. Note that the aforementioned techniques require highly detailed and timely financial data, which may not be disclosed in the financial statements available to external analysts.
25. Refinancing risk is one aspect of market risk. It exists when changing interest rates make it more expensive to roll over borrowings at maturity. It is the converse of reinvestment risk, which is the risk that changing interest rates will result in a lower yield on investments that must be rolled over at maturity.
26. See Chapter 12 on risk management.
27. For more on such angles, the reader should explore Nassim N. Taleb's excellent book, *The Black Swan: The Impact of the Highly Improbable*, rev. ed. (London: Penguin Books, 2010).
28. Where interest rates or exchange rates are volatile, when inflation rates are high or where there is great uncertainty about future economic or political conditions, severe liquidity problems are more apt to occur. Countries with weak or unstable economies are especially vulnerable to these types of conditions, but, as the global financial crisis of 2007–2012 has demonstrated, highly developed countries are by no means immune.
29. Under normal conditions, the cost of liquidity is likely to be less if it is perceived to be unneeded. Waiting until a bank's financial condition begins to deteriorate can prove counterproductive. Where a bank is seen as desperate for liquid funds, and therefore at imminent risk of distress, the willingness to provide funding will inexorably recede and the price of obtaining it will rise.
30. Federal Deposit Insurance Corporation, "Risk Management Manual of Examination Policies," Section 6.1, www.fdic.gov/regulations/safety/manual/section6-1.html#management.
31. Although the move in international banking today is toward transactional banking, in which the correspondent bank is looking for reciprocal business that the other bank can generate, in this context banking relationships remain important.
32. In taking account of the relative benefit of a liquidity backup facility, relevant analytical criteria therefore include the fundamental creditworthiness of both the bank providing and receiving the facility, considering their respective importance domestically or internationally; their respective dependence on confidence-sensitive funds (e.g., commercial paper), and the overall country status or sovereign rating applicable to each. The safest source of alternative liquidity is a committed line allocated for a specific issuance program such as commercial paper provided by a highly rated bank for a longer period and without clauses.
33. Credit crunches are usually prompted by a shortage of capital or by a lack of liquidity or by both simultaneously.
34. Federal Deposit Insurance Corporation, "Risk Management Manual of Examination Policies."
35. A deposit-taking institution that engaged in no lending, but instead placed its depositors' funds in highly marketable government securities would resemble a very conservative money market or bond fund more than a bank, albeit one with too much overhead to generate a decent profit.

36. Federal Deposit Insurance Corporation, "Risk Management Manual of Examination Policies,"
37. European Central Bank, *Developments In Banks Liquidity Profile And Management* (Frankfurt am Main, 2002), citing the *Sound Practices for Managing Liquidity in Banking Organisations*, Basel Committee on Banking Supervision, February 2000.
38. "Introduction to Bank Credit Analysis," Thomson Financial BankWatch (PowerPoint presentation prepared by Jane Cates), 1999.
39. While these are the principal uses to which a bank's cash will be put, the list is not meant to be exclusive.
40. Cash flow analysis is discussed in the context of a discussion of asset-liability management in Chapter 12.
41. Asset liquidity takes account of two categories of assets—(a) cash on hand and (b) cash proceeds obtained by liquefying some portion of a bank's existing assets—relative to other assets. Categories (a) and (b) are distinguished according to whether, on the one hand, the subject assets are (a) already in the form of cash or a form deemed sufficiently equivalent to cash, or are (b) of a less liquid character.
42. Quoted by M. Mayer in *Financial Market Regulation: Legislation and Implications*, ed. John A. Tatom (New York: Springer, 2011), 34.
43. During periods of financial turmoil, it often becomes difficult to dispose of financial assets without reducing the price of the assets to fire sale prices. The discount, that is the haircut the seller of such assets must provide, represents the cost of obtaining liquid funds on short notice.
44. As with all indicators, those in this category are evaluated relative to the bank's past financials, its peers, and the character of the institution's business activities.
45. In 2008, the origination of securitized products such as MBS dwindled as a result of the global financial crisis. At the time of writing, when and to what extent the market for securitized products will fully return remains uncertain.
46. Even if loans are in fact illiquid, the degree of illiquidity will vary with maturity, among other relevant criteria. An illiquid loan that has a maturity of three months can be said to be more liquid than an illiquid loan that has a maturity of three years. Therefore, another question to consider is the average maturity of the loans held by the bank being analyzed.
47. By end-1999, this percentage had fallen to 15 percent.
48. Nevertheless, in a crisis situation, it would not be unreasonable to believe that the Central Bank of Kuwait would provide liquidity in respect to the GDBs through discount and swap facilities.
49. Lenient rules about disregarding paper losses on bonds held to maturity do nothing to prevent a loss of liquidity.
50. DBRS, *Lifting the Lid*.
51. The notion of purchased funds will be discussed in detail later in the chapter.
52. Wholesale or investment banks, in contrast, do not have a significant customer deposit base, and their funding profile is generally regarded as higher risk and possessing greater volatility. Investment banks in particular concentrate on wholesale and bank funding, and usually their banking licenses prohibit them from taking retail deposits. Consequently, their funding tends to be less stable, and their overall liquidity and funding profile is generally inferior to that of commercial or retail banks. If market sentiment turns against investment banks, they can quickly lose access to the unsecured debt market and thus need to rely on the market as a secured borrower. Consequently, investment banks often carry lower credit ratings than commercial banks. To defend against the generally higher risk of loss of access to ordinary funding, even more so than a commercial bank, an investment bank must have in place a contingency plan for times of funding stress, and be fully aware of its realizable assets and how quickly they can be liquidated to meet maturing liabilities. After the collapse of Lehman Brothers and the sale of Merrill Lynch to Bank of America in 2008, the viability of the investment banking

- model came into question as the leading investment banks sought to shore up their funding by becoming commercial bank holding companies affording them access to deposit funding.
53. Deposit insurance schemes, however, may have adverse side effects by contributing to moral hazard. This is the risk that because insured depositors are no longer concerned with the overall quality and financial soundness of any given bank, bank management can take risks without fear of chastening in the marketplace.
 54. Other similar terms include *purchased funds*, *brokered funds*, *commercial funds*, *wholesale funding*, and *hot money*. These terms are used relatively interchangeably.
 55. See further comments later on a different meaning of purchased funds in interbank markets.
 56. Foreign currency deposits may or may not be categorized as customer deposits depending on the market and analytical preferences.
 57. The advance of new loans requires that the capital just deallocated be reallocated to the new loans. Note that not all securitizations are fully nonrecourse vis-à-vis the originating bank. Some include early amortization clauses designed to provide extra protection to investors against the possibility that the performance of the underlying assets (e.g., the loans comprising the securitized securities) do not meet designated criteria. Should the early amortization clause be triggered, the bank may be liable to repay principal to bondholders earlier than expected and will thereby face unanticipated liquidity demands that must be funded. Early amortization may have additional knock-on effects harming the bank's reputation in the marketplace and making it more difficult and more expensive to obtain funding through additional securitizations in the future. In addition, poorer than expected performance of the underlying assets can unexpectedly reduce the bank's cash flow.
 58. Before the current credit crisis, securitization was an effective tool mainly in a number of developed markets, including the United States, Australia, and the UK. The chief disadvantage of securitization, however, has been its role in fragmenting the mortgage supply chain and reducing accountability at each step in the process.
 59. See Dresdner Kleinwort Benson, "Global Fixed Income Credit Research Note," April 2000.
 60. The capping of the bank's rating at the sovereign level—the sovereign ceiling—reflects the notion that a bank can never be more creditworthy than the country in which it operates.
 61. Regulators may choose to allow a relatively small insolvent bank to fail, with shareholders and depositors suffering the losses. For larger banks, the reasons for intervention may include political considerations but are usually impelled by a desire to avoid a banking systemic collapse.
 62. The BIS was set up in 1930 to foster international financial stability and greater international central bank cooperation. The work of the Basel Committee on Banking Supervision (secretariat provided by the BIS) focuses on handling specific supervisory problems and enhancing standards of supervision, notably in relation to solvency.
 63. Deposit Insurance Corporation, "Risk Management Manual of Examination Policies."
 64. Cash flow is important, however, in analyzing liquidity for internal risk management purposes.
 65. Be aware that some analysts and organizations prefer to reverse the position of the line item in the numerator to the denominator and vice versa. For example, in respect to the loan-to-deposit ratio, an alternative construction would be to divide deposits by loans to create a deposits-to-loans ratio. In this case, of course, the relationships are reversed, and a high ratio becomes credit positive and a low ratio credit negative.
 66. In the case of pledged government securities to total government securities, the ratio reflects the proportion of government securities, which are ordinarily liquid assets that are pledged and therefore illiquid.

67. See discussion of prudential regulation in Chapter 13.
68. In addition to cash and cash equivalent, quasi-liquid assets include those assets generally regarded as marketable and therefore liquid, but may, for instance, not meet regulatory liquidity requirements. Quasi-liquid assets broaden the scope of liquid assets to embrace marketable investment-grade corporate debt securities, and listed and widely traded equity securities.
69. Net loans are loans net of unearned income and of any specific reserve for loan losses. Collateral, if any is held against the loans, is disregarded because taking it into account would imply having information on the liquidity, or the lack of it, of the collateral.
70. This ratio may be employed using either total loans or net loans in the numerator and total deposits or, preferably customer deposits, in the denominator. Sometimes “core customer deposits” is used as the numerator. Recall that net loans is equal to total loans, minus loan loss reserves. Moody’s defines core deposits as total domestic deposits excluding deposits over \$100,000.
71. A vostro account is, with respect to a bank, a demand deposit account maintained with another bank in a foreign country. Vostro accounts are mainly used to arrange foreign currency transfers between the two banks.
72. This notion of deposits running off under stress scenarios are at the heart of the Basel III liquidity standards.
73. The modifier “current” in this context means assets and liabilities maturing within one year.
74. See DBRS, *Lifting the Lid*.
75. As discussed elsewhere in this book, funding for troubled banks facing liquidity problems may be provided from a government or central bank (as lender of last resort) to prevent a banking system collapse. Under the too big to fail principle, banks whose collapse could trigger a systemic breakdown will tend to have priority in the receipt of state assistance. A bigger bank can also be expected to have a larger number of depositors resulting in more political damage to the government should it fail.
76. These regulations were in force circa 1999, and are provided for illustrative purposes only.
77. A repurchase agreement (repo), while in practice a buy and resell transaction, is essentially the pledge—usually in the short term—of some collateral, for example government securities, against financing, with the same or similar collateral returned at maturity.
78. See the Report by A.R. Valukas, Examiner, Unites States Bankruptcy Court, Southern District of New York, Chapter 11 Case No. 08-13555 (JMP), dated March 11, 2010.
79. Normally when engaging in the credit evaluation of a particular bank, it is best practice to analyze the consolidated entity rather than a subsidiary. It is therefore important to be aware of critical difference between a bank and a bank holding company insofar as it affects the analytical process.
80. Rating agency Moody’s Investor Services has published several methodological reports on the liquidity and credit analysis of bank holding companies, and this part of the chapter is drawn in large part from those reports. In addition, this section is informed by materials published by Dominion Bond Rating Services, and other rating and government agencies.
81. In the United States, the data necessary to compute the ratio can be obtained from the bank holding companies’ public regulatory (Y9) filings.
82. Moody’s allows for possible exceptions in cases where the bank holding company’s bank subsidiaries are funded largely by retail deposits and where the bank holding company is relatively small and its use of a backup facility would not invite intense market scrutiny. Moody’s also asserts that the ability of its subsidiaries to tap “excess dividend capacity” and upstream it to the bank holding company is of limited value from a liquidity perspective. It would be better in the agency’s view if dividends were regularly upstreamed

subject to regulatory blockage, rather than the bank holding company relying on an out of the ordinary upstreaming of dividends in the event of difficulty. Nonbank subsidiaries, however, may have more flexibility in upstreaming dividends and therefore such “excess dividend capacity” is likely in Moody’s view to have more merit in respect to those entities.

83. DBRS, *Lifting the Lid*.

84. See Philippe Delhaise, *Asia in Crisis: The Implosion of the Banking and Finance Systems* (Singapore: John Wiley & Sons, 1998), 93–96.

85. This is considered high for an emerging market bank due to greater market volatility and less stable overall funding profile.

Country and Sovereign Risk

Country risk is the exposure to a loss in . . . cross-border lending caused by events in a particular country which are, at least to some extent, under the control of the government. Country risk is therefore a broader concept than sovereign risk, which is restricted to the risk of lending to the government of a sovereign nation.

—P. J. Nagy¹

While the statement “a bank can never be better than the country within which it is located” is not necessarily true in all cases (for example, if a bank has sizeable assets and liabilities beyond the home country’s borders . . .), in general, it is a pretty good rule to follow.

—Robert Morris Associates²

Well before globalization increased the volume of cross-border commercial and financial transactions, banks were exposed to risk associated with the creditworthiness of other countries, either directly or through transactions entered into with banks located in other countries.

Whether the denomination attached to that risk—with country risk and sovereign risk often used interchangeably—it is clear that many banks are exposed to the creditworthiness of sovereigns, as well as to the creditworthiness of financial institutions operating in foreign countries, themselves affected both by the creditworthiness of their government (usually associated with the notion of sovereign risk) and by the risks specific to financial and commercial banking operations in their country (usually associated with the notion of country risk).

The economic weakness of a country has a direct impact on both its sovereign and country risks. Other notions such as policy risk or systemic risk are extremely relevant, more so than they were at the time when sovereign or country risks were the sole province of macroeconomists.

OVERVIEW

Beyond the traditional sovereign angle, no matter how robust an individual bank or banks, all financial institutions are subject to the political, legal, and economic milieu in which they operate.

This chapter will look at all relevant angles, but as many of them are actually beyond the scope of the book, for example public finance, they will at times receive limited coverage. The reader who is not yet familiar with those concepts should refer to other sources.

The Operating Environment

No matter how robust an individual bank or banks, all financial institutions are subject to the political, legal, and economic milieu in which they operate. So the assessment of the creditworthiness of banks in a particular country requires an understanding of the risks associated with operating in that country. Such risks will be influenced by, among other things:

- The structure of the local financial system
- The laws and regulations affecting the banking industry
- The strength and stability of the economy and government that oversees those rules
- The level of public confidence in the system

Periodically, some business magazines publish a list of the top 100, 200, or 500 banks in the region they cover, or if the world is their beat, they come up with a global ranking. These lists often purport to reveal which among those enumerated are the best banks. Like *Fortune* magazine's famous Fortune 500 list, it is likely that rankings of this kind are popular with readers. While, depending upon how tabulated, they can be of interest, rarely are they of much value, at least from a credit perspective. The problem is that determining what makes the best bank is an elusive quest. Usually the periodicals' editors sort the banks by assets, equity, profits, or some composite ratio. Such metrics taken out of context convey little useful information. If the criterion is *return on equity* or another profitability ratio, we have seen that banks can easily enhance profitability at the expense of safety. If capital ratios are the governing metric, it has been shown that the soundness of the best-capitalized banks depends upon their risk profile, while they will rarely be among the most profitable. If the controlling indicator is total assets, size tells us only a little about creditworthiness, and less about profitability, not to mention the impact of off-balance sheet items.

A greater difficulty with such rankings is that they rarely take into account the environment in which the bank operates. Banks cannot be analyzed in a vacuum. Some of the most lucrative and most well-capitalized banks in the world operate in developing countries with highly volatile economies subject to difficult operating conditions. Periods of rapid growth may be followed by sharp market corrections or downturns in business. While their admirable financial ratios should not be overlooked, neither should the challenging terrain in which they operate be ignored.

Economic conditions are not the only variable. A country's political, legal, and regulatory framework must be taken into account. The banks may very well be profitable because a banking cartel keeps deposit rates low and lending rates high. Capital may have been robust at the end of the previous financial year, but bank supervision may leave much to be desired, and deposit insurance may be nonexistent, subjecting weaker banks to deposit flight on the strength of rumors. In such

circumstances, high solvency ratios may be an effective requirement to maintain depositor confidence and to survive in the market. Just as an off-road vehicle needs high ground clearance and strong springs to absorb bumps and potholes, so too banks in emerging markets frequently need a greater cushion of capital than they would in a more developed, less volatile economy, and more earnings power to drive themselves out of difficult situations. Additionally, local accounting rules may allow the bank to show more impressive figures than an equally strong and profitable bank across the border. Rankings that rely solely on numerical ratios without adjusting for variations in the local operating environment and applicable accounting regime are, for this reason, mostly worthless.

Systemic Risk

A comprehensive credit analysis requires an evaluation of the impact of a bank's operating environment, which in turn requires assessments both of the sovereign risk associated with the market in which the bank is headquartered—also called country risk—and the systemic risk associated with the financial or banking system of which the bank is a part. To do so, the bank analyst must take account of the following facets of a banking organization's operating environment:

1. *Political character and policies.* What political dynamics drive government policy and underpin or undermine political stability generally and banks' performance and financial condition more specifically?
2. *State of the economy.* How strong, stable, diversified, and well-managed is the national and local economy, and to what extent is it subject to a boom-bust cycle or to a prolonged period of weak economic growth?
3. *Regulatory regime.* How do applicable laws and regulations, as well as relevant local customs and industry practices, affect the financial industry?
4. *Financial system.* How is the financial sector organized; how healthy is it, what is the degree of public confidence in the industry?

The first two of these items are primary elements of sovereign risk analysis, while items three and four are critical to analyzing the strength of the financial system in which the bank operates, that is, its systemic risk—also referred to as banking system fragility.³

The evaluation of sovereign risk focuses on an assessment of a country's political risk and economic risk. The evaluation of the fragility of a banking system focuses on the organization of the banking industry and the quality of the regulatory regime, including the government's capacity to bail out or support banks.

Sovereign (country) risk and systemic risk have an impact on each other. That is, the level of sovereign risk in a given country, arising from political or macroeconomic conditions (that are shaped by a government's economic policies), will affect the strength (or fragility) of the banking system. Similarly, the degree of systemic risk implicit in that system will affect the sovereign risk of the country as a whole. It can therefore be expected that the political environment, the legal and regulatory framework in which banks operate, and the shape and character of the financial system are invariably linked. A country's general legal and regulatory framework, which conventionally is included within sovereign risk analysis, will influence the nature and effectiveness of the relevant regulatory regime that specifically governs banks.

The Impact of the Legal and Regulatory Framework

It is evident that the capacity of banks to generate adequate earnings and maintain financial fitness is strongly affected by the regulatory environment in which they operate. Both government policies and macroeconomic conditions are influenced by a country's political environment and the corresponding objectives of its leadership. That is, the laws and regulations governing banks are ordinarily reflective of the prevailing political consensus, and as a consequence, a comprehensive bank credit analysis requires their consideration.⁴

At the broadest level, a strong legal and regulatory infrastructure, including the protection of property rights generally and creditors' rights particularly, are usually supportive of a healthy banking system. Insofar as the state establishes and enforces the rules under which financial markets can function relatively effectively—by, for example, allowing the enforcement of rights and other forms of legal redress—it facilitates the conduct of a banking business.

In the same manner, the heavy-handed use of law and regulations for policy purposes can adversely affect the health of the financial system. To the extent, for example, that government uses commercial banks as instruments of public policy by closely directing their activities to benefit certain social groups or economic sectors, it may impair the financial health of particular institutions or the industry as a whole.

SYSTEMIC RISK: Q&A

What Is a "Financial System"?

Very broadly speaking, a country's financial system encompasses its capital and money markets (and other financial markets), the legal and regulatory framework governing those markets, its institutions, and the rights and obligations in respect to financial instruments, together with the customs, techniques, and business practices that affect the delivery of services by such institutions and the sale, purchase, and trading of such instruments.⁵

What Is Systemic Risk?

Systemic risk broadly means a risk that affects all financial institutions in the same way, but it is properly defined as the risk that a country's financial system as a whole will be affected or even will fail—that is, in which the delivery of financial services such as lending, deposit-taking, and payments, is no longer carried out by the majority of significant financial institutions—and the risk of economic loss from such a calamity. As commonly used, systemic risk can be thought of as the risk of a domino effect, in which the failure of a relatively small part of a system triggers a series of events that result in a total system malfunction. According to George Kaufman of Loyola University and the Federal Reserve Bank of Chicago:

*Systemic risk refers to the risk or probability of breakdowns (losses) in an entire system [accompanied by] co-movements (correlation) among most or all [its] parts. Thus, systemic risk in banking is evidenced by a high correlation and clustering of bank failures in a country, a number of countries, or globally. . . .*⁶

Because of the fear of systemic risk, the threat of the collapse of a small but not insignificant financial institution will usually be taken quite seriously by bank regulators.

Is There Any Link Between Systemic Risk and Contagion?

Systemic risk is related to the concept of contagion, which can be understood as the tendency for a crisis episode affecting a financial system in one country to spread to another,⁷ regardless of the second country's fundamentals. Contagion, therefore, can be viewed as a form of cross-border systemic risk.

What Is the Relationship between Sovereign Risk and Systemic Risk?

It is evident that sovereign risk affects systemic risk. But, interestingly, systemic risk also affects sovereign risk. As Virginia Manzer of rating agency Standard & Poor's put it:

*Experiences of countries with fragile financial sectors have shown that banking system soundness is essential to a country's macroeconomic stability, effective demand management, and sustained economic growth.*⁸

Sovereign risk can be thought of as equivalent to overall country risk. As such, it encompasses systemic risk. Systemic risk can be thought of as equivalent to the fragility of the banking system. In short, sovereign risk encompasses and thereby affects systemic risk. Systemic risk contributes to sovereign risk. Both affect bank credit risk.

What Is Sovereign Risk?

In the context of bank credit evaluation, sovereign risk encompasses two meanings. First, it refers to the risk of sovereign default, including the default of national government-guaranteed obligations, and the direct or a concomitant impact of such a default on one or more banks in the country. Second, it refers to the risks that (a) state policies, manifested in laws, directives or regulations, or government incapacity, may impair or constrain the ability of a subject bank within a given jurisdiction to perform its obligations in accordance with their terms; and that (b) state policies or prevailing economic and business conditions (irrespective of such policies) may adversely affect the business environment in which the bank operates, thereby indirectly affecting its credit quality.

Country Risk versus Sovereign Risk

The concept of sovereign risk has been distinguished from that of *country risk*. Although the term *sovereign risk* may be used merely to refer to the risk of loss resulting from the default or decline in the credit quality of a government,⁹ it is also frequently used to refer to the broader risk that government *action* or *inaction* may impair the ability of other organizations or institutions to operate profitably. As mentioned at the beginning of this chapter, country risk encompasses the broadest spectrum of risks related to government action or omission.¹⁰

In practice, however, this real distinction is commonly ignored, and the terms *sovereign risk* and *country risk* are used interchangeably. One reason for this is that essentially the same analytical criteria apply to the evaluation of both. In almost all cases, the ultimate assessments made will be similar, although the conclusions may differ depending upon the specific area of concern.¹¹ In other words, an assessment made by a sovereign analyst concerning the risk of sovereign default will generally have similar underpinnings and may result in comparable conclusions to those made by a country risk analyst concerning the risk that government actions or adverse economic conditions will result in losses in respect of the obligations of enterprises under within that government's jurisdiction.¹²

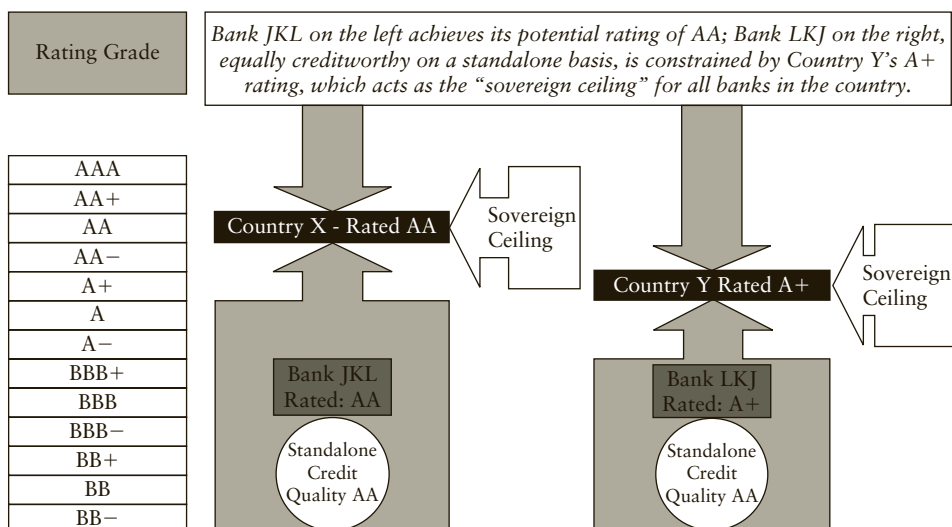
Sovereign Analysis as Part of Bank Credit Analysis

In view of the importance of a bank's operating environment to its financial health, a bank credit analysis should take account of this environment. To do this, the analyst will ordinarily either refer to a sovereign or country risk analysis or undertake such an analysis directly.

In either case, the aim is to identify and assess those economic strengths and risks confronted by a bank arising out of its operations in a particular market.¹³ Although more often than not bank credit analysts often will leave the evaluation of the impact of macroeconomic perils and the likelihood of a government defaulting upon its obligations to sovereign analysts, extracting such findings as required, a basic understanding and awareness of sovereign risk concerns is important to the effective analysis of banks, whether they are government owned or not.¹⁴

Other reasons why bank analysts should have more than a passing familiarity with the process of sovereign risk assessment include: (1) sovereign ratings function as a benchmark for the local bond market and for ratings assigned to banks within the sovereign jurisdiction; and (2) internally, sovereign or country ratings are used to establish overall limits with regard to exposure in a given national market.

More specifically, external *sovereign risk ratings* are used not only to evaluate the creditworthiness of debt instruments issued by a national government, thereby establishing a benchmark yield curve for the local bond market, but also to serve as a benchmark for the creditworthiness of firms, including banks, under the jurisdiction of the government. Since the government has the ultimate authority over private entities within its borders, in theory no private company, including a bank, can be rated higher than the sovereign. This concept is referred to as the *sovereign ceiling*.¹⁵ The rationale for a sovereign ceiling is that government has the power to block the performance of commercial transactions. See Exhibit 11.1. For instance, in an extreme situation, a government might impose a moratorium on foreign debt repayments. Exceptions are sometimes made to the sovereign ceiling, however.

**EXHIBIT 11.1** The Sovereign Ceiling

Since sovereign ratings are typically used to establish both country limits and country ratings that determine whether and to what degree a bank will take on credit exposure, *sovereign risk analysis* is as a practical matter important to risk managers. *Country ratings* internally assigned by banks may also determine the level of pre-established limits and, if relevant, provisioning the bank will have to set aside against exposures in those countries. Those provisions, imposed by internal policies or external regulators, will have an impact on the profitability of particular transactions.

THE SOVEREIGN CEILING

A rigid sovereign ceiling means that companies within a particular jurisdiction can be rated no higher than the rating assigned to the government.

Sovereign Risk Analysis versus Bank Credit Analysis: Differences and Similarities

Although there are similarities between sovereign risk analysis and bank credit analysis, there is an important difference between the two. Unlike the holders of obligations issued by financial institutions or corporates, creditors have no legal redress against a sovereign (i.e., a national government) without its consent. In other words, their ability to coerce repayment of a debt obligation is limited. For the same reason, no global bankruptcy court for sovereign borrowers exists that possesses the authority to compel a country to enter insolvency proceedings involuntarily or subject itself to such a court's judgment. In consequence, the sovereign's willingness to pay assumes a higher profile relative to its ability to pay than is usually the case

with ordinary commercial entities.¹⁶ As S&P analysts David T. Beers and Marie Cavanaugh observed in a report entitled *Sovereign Credit Criteria: A Primer*, “a government can (and sometimes does) default selectively on its obligations, even when it possesses the financial capacity for timely debt service.”¹⁷

The issue of *willingness to pay* is more important in respect to government default than it is with regard to the default of privately owned companies, owing to creditors’ power to coerce repayment from nonstate entities through the relevant national legal system. Sovereign risk analysis takes account of this issue by considering *political risk*, which implicitly encompasses willingness to pay. An illustration of political risk of this kind would be a situation where there has been a radical change in the political leadership of a country—or there exists the imminent prospect of such change—that suggests that the state’s creditworthiness has altered (or will alter) for the worse. More concretely, a similar conclusion could be drawn from actions undertaken by a country’s political leadership that effectively compel private firms to default, or increase the risk of default by private sector borrowers (through, for instance, the imposition of a general moratorium on repayment of foreign debt). In addition to making judgments of this type, which are frequently subjective in nature, an analysis of a government’s *capacity to repay*¹⁸ must be undertaken. Such an assessment, as might be expected, relies to a much greater extent on quantitative criteria such as foreign currency reserves, balance of trade, and other indicators discussed in later pages. Exhibit 11.2 shows the basic types of macroeconomic attributes and indicia that are relevant to this aspect of sovereign risk analysis.

EXHIBIT 11.2 Sovereign Economic Capacity versus Sovereign Economic Risk

Evaluate Overall Economic Capacity	Identify Potential Economic Risks
Economic fundamentals—secular	Monitor key economic data and indicators.
Economic fundamentals—cyclical	Consider relevant qualitative criteria.
Government economic policy	Ascertain the economic outlook.
Overall resiliency—that is, ability to withstand economic shocks	Take note of potential red flags.

ANALOGIES TO CAMEL

Although the terminology can vary a great deal, there are a number of similarities that might be noted between the credit analysis of a bank and that of countries. Although there is no comparable acronym to CAMEL, it does appear that analogs to the elements of the acronym can be found in sovereign risk analysis. This comparison is not intended to suggest more similarity between the two fields than actually exists, and is merely offered as a potentially interesting parallel.

Consider the “M” or “management” component of the CAMEL model. The appraisal of a nation’s political leadership and economic management of the country is in some respects similar to the evaluation of a financial institution’s management in bank credit analysis.¹⁹ In sovereign risk analysis, a great deal of attention is paid to a country’s *external debt position* (i.e., *external position*), or the context of its ability to service its external financial obligations. This is not entirely dissimilar to the consideration of capital adequacy in relation to the proportion of debt and to *liquidity* in terms of a bank’s ability to generate liquid funds to service the debt. Similarly, in sovereign risk analysis looks at *exports* and *currency reserves* as providing the foreign exchange to meet external obligations, and at the sufficiency of government *revenues* versus government *expenses*. Again, parallels might be drawn with an enterprise’s earnings (or cash flow) and liquid assets. The parallels between sovereign risk analysis and the CAMEL model are summarized in Exhibit 11.3.²⁰

EXHIBIT 11.3 Sovereign Analogs to the CAMEL Framework

CAMEL	Analogous Sovereign Risk Factors
Capital/Solvency	External position/balance of payments/relative debt
Asset Quality	Natural resources/infrastructure/human resources
Management	Political leadership, governance, and economic management
Earnings	Net exports (trade balance)
Liquidity	Foreign currency reserves

Sovereign Risk Analysis: An Overview

As with all forms of credit analysis, *sovereign risk analysis*²¹ is both a qualitative and a quantitative process. *Qualitative* aspects of the analysis concern mainly political and policy criteria, as well as what might be described as general criteria, while *quantitative* methods are mainly applicable to economic attributes as reflected in economic indicators. What appear to be objective measures, however, are, as always, subject to qualitative judgments such as those relating to the appropriateness of the indicator being employed, its definition, and its method of calculation. Correspondingly, in evaluating qualitative criteria, such as the state of a country’s physical infrastructure, it may be possible to use some proxy quantitative metrics such as road mileage per square mile, or port capacity in tons per year.²² Alternatively, *scorecard* approaches, which quantify qualitative judgments, may be applied.

To begin the process of sovereign analysis, the usual starting point is to build a profile of the country that will provide a context for the risk analysis that follows.

This profile will take into account the country's geographic position, natural resources, people, history, structure of government, and state of development, among other things. Some questions relevant to this process are enumerated in the first box below.

From this starting point, *sovereign risk analysis* proceeds to consider the following principal categories of criteria with respect to any given jurisdiction:

- *Political dynamics*, and the policies that flow from them, taking account of related background factors, the stability of the government and the strength and competence of the political leadership and civil institutions.
- *Legal and regulatory factors*, which are largely animated by policy and seek to implement policy objectives.
- *The structure of the economy and its performance*, including the country's level of resource utilization, its labor market, and level of development.
- *The condition of its domestic public finances (fiscal policy)*; the *effectiveness and prudence of monetary policy*; and its *external position* vis-à-vis foreign countries, their investors, and creditors.

COUNTRY BACKGROUND PROFILE: LINES OF INQUIRY

Relevant criteria to be considered in building a background profile of the country include its:

- Geographical situation and natural resources
- People and their history
- Physical infrastructure
- Position in terms of international relations
- History and current state of economic, political, and technological development
- Relevant recent developments

More specifically, the following questions might be posed as a starting point for the analytical inquiry:

Geographical Situation and Natural Resources

- How large is the country in terms of area and population?
- What are the advantages or disadvantages of its size and geographical position?
- Does it have access to the sea?
- What is the country's natural resource endowment? To what extent has it been used?
- What is the education level of the population? The quality of educational facilities? Literacy rate? Access to education? Skill levels relative to economic needs?

Population, Demographics, Health, and Education

- What are the country's infant mortality rate, life expectancy, and fertility rates? How do they compare with those of other countries?
- What is the *state of the country's* health system? What challenges does it face?
- What is the *age* distribution? Urban vs. rural population?
- What is the *ethnic, religious, and language composition* of the population?
- To what extent do *ethnic conflicts* or other comparable social conflicts or tensions exist?

Scale of Economy, Standard of Living, and Other Basic Data

- What is the size of the economy? Gross domestic product (GDP), gross national product (GNP)?
- What is the standard of living? GDP per capita?
- With what other countries might it be grouped for comparative purposes?
- What are the key attributes of the economy, for example, is it diversified or not; manufacturing vs. service, export driven or domestically focused, mature vs. emerging, and so on?
- What is the currency? Is it widely negotiable?

Political System

- What is the distribution of income, wealth, and land?
- What are the fundamental attributes of the political (e.g., constitutional monarchy, democratic republic) and economic system (e.g., democratic market economy, socialist market economy)?
- What is the *structure and type of government*? What are the *major institutions* and branches of government?
- Who is the present *leadership*?
- What *party* is in control? Does it share power with other parties? What parties constitute a viable opposition?
- What is the election schedule or other mechanism for *leadership succession*?
- *Local governance*. How are local regions and communities governed? How is government authority delegated or divided? What degree of autonomy do local governments have in imposing taxes, budgeting, and so on?

The Political Environment and Political Risk Analysis

The analysis of a market's political system and the social conditions that underlie it is one of the more difficult aspects of *sovereign analysis*. Its importance cannot really be minimized since the *political environment* influences the *structure* and *strength* of a country's economy, and similarly has an effect upon both fiscal policy and monetary policy. It also has a substantial impact upon the laws and regulations that enable the

implementation of both economic and other policies, which, in turn, directly affect overall sovereign risk.

Overview of Political Risk Analysis

The *stability* of a country's political institutions will often influence the ability of a commercial bank to meet its obligations. A situation where the political leadership champions populist or egalitarian policies may handicap banks in their effort to generate returns for shareholders. In the same manner, a country racked by civil war or disorder, or one where laws go unenforced, or are enforced arbitrarily, may not be conducive to the conduct of a banking business. From a *qualitative* perspective, political risk analysis is largely an exercise in understanding the dynamics underlying a country's governance, its policies, and their execution, with the aim of forecasting where current trends are likely to lead. Such analysis will therefore, in large part, be concerned with detecting signs of latent instability. Additionally, the quality and competence of the government and its institutions, as well as its prevailing values and ideology, are important criteria to consider when evaluating political risk. In a world of fixed exchange rates, reviewed and adjusted by governments, analyzing the psychology of powerful politicians—as was done about General De Gaulle, the French president, for example—was an important component of a proper sovereign analysis. A similar situation arose in 2012 when the fate of the euro—and perhaps of the world's economic stability—was in the hands of just a few politicians in Greece.

Political stability inherently affects political risk, and unsurprisingly instability is generally adverse for credit risk purposes because of the uncertainty it creates. In some cases, however, such as the fall of the Soviet bloc in the early 1990s and the emergence of transition economies in eastern and central Europe, dramatic political change may ultimately improve the overall credit risk environment.

Criteria affecting political stability include:

- The nature of the government
- Its legitimacy and accountability
- The degree of popular support it enjoys

Related factors include the government's ability to maintain a rule of law, and a reasonable level of internal and external security. A critical determinant of stability is the degree to which there exist latent or patent conflicts, either internal or external, among both the leadership and the populace. Such clashes may be propelled by conflicts of a political, religious, ethnic, or class nature. Until resolved satisfactorily, conflicts of this nature may persist for many years. Finally, macroeconomic conditions, such as the standard of living and gap between rich and poor (implicit in the question concerning ownership of land, natural resources, and wealth in the preceding table, may have an impact upon political stability.

The *quality and competence of the government and its institutions* is another characteristic that may affect political risk. Quality and competence will be reflected in, among other things: (1) the strength, capability, independence, and integrity of key branches or agencies of government, including the judiciary and the central bank; (2) the absence of corruption; (3) the capacity of the government to coordinate

effective policy initiatives, including its track record in implementing policies; and (4) in the prudence, efficacy, and commercial-friendliness of its policies.

As part of the process of analyzing a country's political outlook, the analyst is likely to pose questions pertaining to potential political risks, which may be sensitive in some emerging markets—and sometimes in advanced countries as well.

CATEGORIES OF POLITICAL RISK CRITERIA AND RELATED QUESTIONS

Among those characteristics relevant to political risk analysis, the most important categories of criteria are political stability and institutional capacity.²³

Type of Government

- Is the government democratic or nondemocratic?
- If democratic, what is the nature of the democracy? For example, is it a constitutional monarchy, pluralistic democracy, or two-party system?
- If nondemocratic, how could it be characterized? Totalitarian dictatorship, benevolent monarchy? Is the political environment conducive to widespread wealth creation?

Legitimacy

- How is government power obtained and maintained? Is the government perceived to have legitimacy?
- What forms the basis for the government's legitimacy? Hereditary monarchy? Adherence to a religion? Adherence to an ideology? Democracy?
- What is the level of government integrity? How widespread is corruption and cronyism?

Leadership

- What is the character of the present leadership? What are its views? What is the agenda of the leadership and the governing party?
- How politically astute is the leadership?
- To what extent do other parties share power or represent a viable opposition, and how do their agendas differ from the governing party? What political alliances exist? How durable are they?
- What priority is given to economic policy?

Parties, Interest Groups, and Policy Coherence

- What is the degree of consensus among political parties on economic issues? In the event that another party or coalition should gain control, how are economic policies likely to change? Are there competing agendas, political infighting?
- How coherent are various elements of the government in respect to economic policy issues in particular?
- To what degree do local and national governments agree or disagree on matters of key economic policy?

- Do local and regional governments have sufficient autonomy to undermine national policies?
- Who influences the government? What interest groups have power within the country? Examples of interest groups include farmers, trade unions, political parties, religious organizations, secular organizations, clans, or oligarchies. How entrenched are these groups?

Stability and Popular Support

- How stable is the government? Is there continuity in its policies? Overall, what degree of popular support does it enjoy? Is there growing discontent with the government or its policies?
- If an opposing party gains control, what would be the likely impact on government policies and the country's overall creditworthiness?
- What are the prevailing attitudes of the populace toward the government? For example, how much freedom of action does the government have in implementing new policies, such as an austerity program?

Security and Control

- Is there a growing internal opposition, particularly one that is resorting to violence?
- Is there a threat of a revolution or a *coup d'état*?
- Is there a threat of war from neighboring or other countries? If there is a war or insurrection in progress, what is the likely outcome?
- Does a government maintain control of the entire country? How secure is the government's control?
- Are there competing groups that control several of its regions? If so, how likely are they to gain control?
- If competing political factions or groups gain power, what policy actions might they take, and would these have an adverse impact on the country's creditworthiness?
- How are political alliances likely to affect government policy?
- In the event of unexpected financial turbulence, could the government draw on support from alliances or external institutions?

Institutional Quality, Integrity, and Independence

- How experienced and capable are government officials?
- What is the government's track record in performing on its obligations?
- Has the government previously issued a debt moratorium or otherwise prevented resident companies or individuals from performing their financial obligations?
- How capably has it sought to reform laws and regulations and rehabilitate distressed institutions? In previous financial crisis situations, how did the government perform?
- Is the bureaucracy politicized or relatively nonpolitical? Are bureaucratic norms and benefits sufficient to deter corruption?
- Is there significant corruption? How widespread is corruption and cronyism?

- How much independence do various institutions have relative to the government? Is the central bank independent?
- How strong is country's legal framework?
- Is the legal and regulatory framework conducive to business and entrepreneurship?
- Do laws and regulations generally facilitate business and commerce?
- How strong are creditors' rights? Can they be effectively enforced?
- Is the legal system perceived as efficient and fair?
- Is the judiciary independent?

Political Risk Modeling

Since political risk, like the analysis of management within the CAMEL framework, is largely qualitative in character, there are few, if any, standardized quantitative indicators that apply to it. While some efforts have been made to quantify the many factors that affect a country's political risk profile, the process employed merely transforms qualitative judgments to quantitative scores through a scorecard approach. Using this technique, the various qualitative criteria identified as important contributors to political risk are weighted, ordinarily by analysts and experts with in-depth knowledge of the country being analyzed. In addition to the global rating agencies, key providers of political risk analysis services include those in Exhibit 11.4.²⁴

The factors designated by another political risk consultancy, PRS (www.prsgroup.com, not listed in the preceding table) are illustrative. Predictably, there are variations in approaches taken by different political risk analysts, the criteria utilized, and the weightings assigned to each, but those listed in the table below comprise a typical framework methodology. PRS takes account of each of the following 12 country attributes, identified in Exhibit 11.5, and assigns each a weighting reflected in the possible number of points that may be assigned to that category.²⁵

Stability

This category considers the capacity of the country's government to execute its proposed agenda, and remain in power. It also includes the unity of the government, its backing by the legislative body, and the degree of its support among the populace.

Socioeconomic Conditions

Taking account of the rate of unemployment, poverty level, and the degree of consumer confidence, this category reflects pressures that could hold back government initiatives or trigger popular opposition to policies.

Investment Conditions

This category encompasses disincentives to investment arising from, among other things, the risks occasioned by the loss of contractual rights due to nonenforcement or biased enforcement, inability to repatriate profits, delays in payment, and expropriation.

EXHIBIT 11.4 Selected Providers of Political Risk Analytics

Provider	Remarks
BERI S.A. (Business Environment Risk Intelligence) www.beri.com	U.S.-based consultancy established in 1966 provides political risk analysis services to multinational banks and corporations. According to its website, the firm's Business Risk Service (one of a number provided) publishes "qualitative analyses and forecasts" using established numerical assessments concerning "operating conditions, political risk, and . . . external accounts" through its "Political Risk Index, Operations Risk Index, and Remittance & Repatriation Factor ratings." The consultancy provides a number of other services including, for example, its QLM-FE, or Qualitative Risk Measure in Foreign Lending—Financial Ethics Index, which attempts to measure criteria having an impact on credit risk but which "cannot be accessed through regularly published statistics."
Control Risks Group www.controlrisks.com	Control Risks bills itself as "an independent, specialist risk consultancy" that provides a range of "advice and services that enable companies, governments and international organizations to accelerate opportunities and manage strategic and operational risks." In 2005, it launched a new product, RiskMap that seeks to "forecast political and security developments in the year ahead for more than 150 countries."
Economist Intelligence Unit www.eiu.com	An affiliate of the Economist Group, The Economist Intelligence Unit or EIU has been providing country risk and analyses and forecasts for about 60 years, and presently covers more than 200 countries, while offering economic projections for the world's 150 largest markets.
Political and Economic Risk Consultancy (PERC) www.asiarisk.com	A Hong Kong-based research consultancy founded by former banker Robert Broadfoot, PERC focuses on major Asian markets, including China, Hong Kong, Indonesia, Malaysia, Philippines, Singapore, South Korea, Taiwan, and Thailand. The consultancy offers three primary publication services: a biweekly newsletter entitled Asian Intelligence; its Country Risk Monitoring Service, which profiles individual countries and attempts to explain "how and why business risks are changing in the country concerned and what companies should be watching for in the near and medium-term that could affect the business environment"; and finally, the consultancy's annual Business Environment Report reflects the perceptions of "1,000 senior expatriates living in Asia" concerning "subjective variables such as corruption, labor quality, intellectual property rights risks and other systemic shortcomings."

EXHIBIT 11.5 Criteria for Modeling Purposes

Item Number	Category	Maximum Score
1	Stability of the government	12
2	Socioeconomic conditions	12
3	Investment conditions	12
4	External conflict	12
5	Internal conflict	12
6	Religious tensions	6
7	Ethnic tensions	6
8	Military in politics	6
9	Law and order	6
10	Corruption	6
11	Democratic accountability	6
12	Bureaucracy quality	4
Total		100

Source: Based on the PRS Group's *International Country Risk Guide*.

External Conflict

This category of criteria considers the risk, or severity, of war or other armed external conflict, as well as the risk, or severity, of nonviolent forms of pressure from other countries or external bodies, such as trade sanctions, boycotts, and similar measures.

Internal Conflict

Internal conflict refers to the likelihood of a country becoming immersed in a military or similar conflict within its borders, and to what degree. Internal conflicts range from civil war at one extreme to disruptive strikes, mass marches, and protests at the other. Subcategories of internal conflict include a *coup d'état*, terrorism, or other forms of politically motivated civil disorder.

Religious and Ethnic Tensions

This category refers to the existence and severity of a conflict, or the potential conflict, between one or more religious or ethnic factions within a country that could give rise to internal conflict, as described immediately above.

Military in Politics

The presence of the military—unelected and unanswerable—having a significant role in the country's political system may be viewed as potentially contributing to weak government stability and democratic accountability.

Law and Order

The category of law and order is distinguished from politically motivated disorder and is composed of three subcategories:

1. The robustness and fairness of the country's legal infrastructure
2. The strength of law enforcement
3. The degree to which the populace is perceived to be law-abiding

Democratic Accountability

This category seeks to measure how much a country's government is responsible and accountable to its citizenry.

Corruption

This generally functions as a disincentive to foreign investment and contributes to dissatisfaction among the populace. Since corruption often functions as a form of de facto tax that is often unpredictable in its impact, it heightens political risk. Forms of corruption include bribery, extortion, a biased judiciary, nepotism and political patronage, and the emergence of black markets.

Quality of the Bureaucracy

Also referred to as institutional strength, this category of political risk criteria reflects the competence and integrity of the civil service and other government bureaucracies.

An Introduction to Macroeconomic Analysis

While an evaluation of the political aspects of the country pertain mainly to the government's willingness to repay obligations (or to the existence of conditions that facilitate their repayment), the remaining criteria discussed in the sections that follow largely concern the state of a country's economy and, in the case of a sovereign borrower, its capacity to repay.

A country's overall economic output is a fundamental criterion of *sovereign risk*, as it is an important determinant of a nation's ability to fulfill its financial obligations. The principal measure of output is GDP, or *gross domestic product*, which is the lodestar of macroeconomic analysis, and which is linked to quite a few of the most important metrics.²⁶ Although specific aspects of a country's economic situation need to be considered to form a thorough understanding of its debt-servicing capacity, an important first step is to obtain a perspective on the relevant economy's comparative size and health, as well as its momentum, which can be expressed in its growth rate. This provides the backdrop against which the importance of particular problems or challenges may be better understood.

Although economies tend to grow, and the ideal economy is one that grows at a steady and sustainable pace, in practice trends rarely follow a straight line.²⁷ Instead, the reality of economic expansion is that its velocity fluctuates, often sharply, over time in a wave-like pattern.²⁸ These waves represent economic cycles. A country's position in the *economic cycle*—which is the very same phenomenon as the *business cycle*—also has an impact on its sovereign risk profile. Consequently, an understanding of the dynamics of macroeconomic change is critical to sovereign risk analysis.²⁹

The strength of a country's economy may also undergo elemental changes as a result of a fundamental shift in conditions. Examples of catalysts for such a shift include major technological innovation, a permanent change in market demand owing to a change in fashion, and the exhaustion of a natural resource. The state of the economy is a crucial input in evaluating that strength.

Another aspect to be considered in forming a picture of the general state of a country's economy is understanding what it produces, which sectors are driving economic growth, and the degree of economic *diversification* it enjoys. A nation with an economy not too dependent upon any specific one sector—one that enjoys good

prospects—will tend to have a better sovereign risk profile than one that has tied its fortunes to a single industry. To the extent it is dependent on one source of income—tourism or a single agricultural commodity, for instance—such an economy will be vulnerable to a broad economic downturn if, for example, civil disorder in the region causes foreign tourism to collapse as occurred during the first Gulf War, or prices for the commodity fall sharply on world markets.

In the same way that bank credit analysis itself relies on quantitative metrics to measure an institution's financial health, so too does macroeconomic analysis also utilize such indicators to gauge the economic health of the country being reviewed. If the past is prologue, the general level of performance of the economy of a country is likely to provide some indication of its future. Much of this chapter is given over to a discussion of these indicators, their definition, and interpretation.

Upon gaining a broad understanding of a country's economy, its level of development and its economic performance, the next step is to examine how government actions, or inaction, are affecting that performance. Specifically, this entails reviewing the government's *fiscal policy* and its *monetary policy*, as well as its *trade policy*, which affects the country's *balance of payments* and its *external position*.

Fiscal policy has been described as “the use of the government budget to affect an economy,”³⁰ and its assessment requires that we look at how a government collects its revenues, spends the funds it collects, and borrows or lends the difference. Here, quantitative criteria, such as the extent of the government *budget deficit* or surplus, are important, but qualitative criteria also play an important part.³¹

Monetary policy, in contrast, is concerned primarily with keeping *interest rates* and *price inflation* within tolerable levels, and describes how a government seeks to influence these economic attributes by taking various actions to affect the *money supply*; that is, the volume of money circulating in the country, which is roughly the same as the supply of credit.

A country's *trade policy* and financing needs as reflected in its *external position* refer to its overall creditor or debt relationship with its trading partners, which is tied to its *trade balance*, a function of its level of exports relative to imports. The level of debt a nation has in relationship to other sovereign states is, for the sovereign analyst, a crucial consideration in its credit evaluation.

About Economic Indicators

To find out how well an economy is doing, we consider the various indicators that describe a country's economic condition, beginning with general economic indicators. Highly developed countries publish a large number of economic indicators on a frequent basis, which give market watchers possible signals of changes ahead. Close scrutiny of such indicators is an everyday matter for central bankers, sovereign risk analysts, investment analysts, and equity and debt traders, who are very much concerned with even relatively small fluctuations.

For bank credit evaluation purposes, however, intense monitoring of the many metrics available is less critical, but the analyst should nevertheless be aware of the types of indicators that are released and what they mean. It is worthwhile to follow them on a periodical, if not a daily basis, and certainly, when there are highly newsworthy changes in these indicators, the analyst should take note. The table in Exhibit 11.6 shows some of the principal economic indicators associated with the various facets of sovereign risk analysis.

EXHIBIT 11.6 Categories of Macroeconomic Indicators and Selected Key Metrics

General (Economic Welfare) GDP growth and trends; unemployment levels and expectations	Fiscal Position Fiscal deficit/GDP; budgetary balance
Monetary Conditions Money supply—changes in volume and composition; intermediation ratio; inflation rate; interest rates; yield curve	External (Trade) Position Current account status; net external debt/GDP; investment and savings rates; net external debt/exports; export cover; real exchange rate change

Categories of Economic Indicators

Economic indicators can be grouped in several ways. For example, indicators may be classified topically into employment-related indicators, production and activity indicators, market and price indicators, and so on. Rating agency Moody’s, which uses economic indicators primarily for sovereign risk analysis purposes, groups these metrics into the following four categories corresponding to a market’s general economic condition, and the fiscal, trade, and monetary components of macroeconomics:

1. Overall economic structure and performance
2. Government finance (i.e., fiscal indicators)
3. External payments and debt (i.e., external position indicators)
4. Monetary and liquidity indicators

Indicators can be distinguished between those that mirror the economic cycle—*procyclical* and *countercyclical indicators*—and those that do not. In addition, they can be differentiated according to their predictive power vis-à-vis the business cycle: that is, between those that are deemed to be leading, lagging, or coincident indicators. These are discussed later in this chapter.

ECONOMIC INDICATORS: A FUNCTIONAL CLASSIFICATION

Fundamental economic indicators include:

- The GDP, which can be used to measure the size of an economy, the overall standard of living, and the rate of economic growth.
- The *balance of payments* (BOP), which shows the country’s *external position* in relation to other countries.

Constituent indicators function as components of fundamental indicators. Automobile sales, for example, form part of a nation’s overall consumption, which is a basic component of GDP. Likewise, the merchandise trade balance is part of the balance-of-payments equation.

Indirect indicators are measures that, though not constituents of fundamental indicators, have an indirect relationship with them. The *unemployment rate* would be an example.

Specialized indicators include, for example, the number of personal computers per capita, average levels of education, and so on.

Note that the second and third groups (constituent and indirect indicators) often function as cyclical indicators, either signaling, describing, or confirming and predicting changes in the economic cycle.

Adjustments to Indicators

Quantitative indicators may be expressed in either *nominal* or adjusted terms as shown in Exhibit 11.7. Adjustments may be made for inflation or for seasonal effects, for example. As opposed to nominal rates, therefore, *real* interest rates are interest rates adjusted for inflation. When adjusted for inflation,³² an item such as GDP is said to be expressed in “real” terms; that is, *real* GDP. In reference to price, “real” corresponds to “*constant*”; hence figures expressed in *constant prices* linked to a base year are synonymous with inflation-adjusted prices, while those expressed in *current* prices include the impact of inflation and are synonymous with nominal prices. The adjustment factor is referred to as the *price deflator*.³³ Seasonal adjustments are effected through the use of year-on-year indicators instead of, for example, using sequential monthly or quarterly data. Thus, since in many Western countries, the December holiday season is responsible for a significant portion of retail sales, by comparing sales between December 2011 and December 2012, rather than between November and December of 2012, adjusts for the fact that December invariably registers higher retail sales than November. Exhibit 11.7 gives some typical adjustment that can be made.

When indicators must be compared transnationally, potential problems arise as a result of the fact that in relation to any given currency chosen as a unit of measurement the local currency will likely be either undervalued or overvalued in terms of its purchasing power. For instance, if one U.S. dollar purchases 500 grams of coffee in the United States, but the same 500 grams of coffee can be purchased for the local equivalent of 50 cents in Brazil or the local equivalent of \$2 in Japan, the difference in the cost of living will have a propensity to distort the meaning of indicators expressed in a particular currency.³⁴ Hence GDP per capita, which is often used as a proxy for a country’s standard of living (or level of development), may be adjusted for purchasing-price parity in order to reflect differences in costs of goods

EXHIBIT 11.7 Indicator Adjustments

Type of Adjustment	Nature of Adjustment
Real, constant	For effects of inflation
Seasonal	For effects of seasonal variation
Purchasing power parity	For under- or over-valuation of currencies

and services. As an illustration, China's nominal GDP per capita in 2011 was \$5,414 but taking in consideration the purchasing power parity (PPP) resulted in an adjusted figure of \$8,382.³⁵

GDP and Similar Ratios

GDP . . . paints the most comprehensive picture of an economy—the single measure that encapsulates what we want to know about how activity is and how much the economy can deliver.

—Morgan Stanley European Economics Team

GDP is regarded as the preeminent macroeconomic indicator because it is an all-encompassing measure of total economic activity within an economy; that is, it attempts to account for all such activity that takes place within a country's borders. It is the market value of all goods and services, measured in a given currency produced within a country (or region),³⁶ on an unduplicated basis (e.g., transactions involving second-hand goods are not counted) within a specified time period, typically one year. GDP is a particularly valuable indicator for sovereign risk assessment because it reflects the current condition of a country's economy and therefore when plotted over time mirrors the business cycle and the economic growth trend.³⁷

DEFINITION OF GDP

GDP is the aggregate unduplicated market value of all the goods and services produced within a sovereign state or subnational region during a given period, conventionally one year, during a specified period. It is the key measure of the overall size of the economy, as it indicates total domestically generated output.³⁸

GDP versus GNP

Another similar measure of the economy is *GNP* or *gross national product*, which is also now known as *GNI* for *gross national income*. As this last term suggests, like income to a firm, GDP and GNP are both *flow* indicators, measuring production, which is roughly the same as the aggregate of income flowing to individuals and entities within the country.

The difference between the GDP and GNP is that GDP takes account of *all* economic activity *within a country's borders*, regardless of whether those who own the assets that generate the activity are nationals of the country or not, while GNP takes account of all production generated by assets owned by a country's nationals, *irrespective of where the nationals or assets are located*. In other words, GDP is equal to GNP *minus*:

- *Income received by foreign capital*—such as direct investment in other countries or from holdings of securities issued in other countries—or foreign labor, *plus*
- *Income paid to foreigners* for their contribution of capital or labor supplied domestically (such as foreign direct investment, ownership of domestically issued securities).

Over time, GDP has become accepted as the more useful indicator for measuring the *size* of an economy and *economic growth rates*, particularly since the United States changed its leading economic metric from GNP to GDP in 1991. However, GNP is considered a better measure of the standard of living of a country's citizens as reflected in income per capita since it takes account of the revenue attributable to a nation's people regardless of their place of residence.

For the great majority of countries, GDP and GNP are within a few percent of each other, so as a rough gauge of total economic activity either is satisfactory. Although ordinarily the discrepancy between the two is not great, significant divergence exists in those countries that have a great deal of foreign investments abroad or holdings of foreign securities, or that export a significant portion of their labor or capital.³⁹ The oil-rich countries around the Persian Gulf generally fall into the first group, while labor-exporting countries fall into the second.

GDP AND ITS CALCULATION

A country's GDP is a key measure of its overall economic activity. GDP measures the total market value of goods and services produced within a country during a specific time period. The formula for GDP is:

$$\text{GDP} = C + I + G + \text{NX}$$

where

C = personal consumption expenditure (comprised of durable and nondurable goods and services purchased by consumers)

I = gross private domestic investment (comprised of residential and nonresidential fixed investment + change in business inventories + depreciation)

G = government expenditures and gross investment (including defense and nondefense)

NX = net export of goods and services (exports less imports), that is, *Exports – Imports* (or EX–IM). The entire equation can therefore be expressed as:

$$\text{GDP} = C + I + G + (\text{EX} - \text{IM})$$

Real GDP means GDP adjusted for the impact of price inflation.

Macroeconomics teaches that, for a given period, total final spending in an economy will be equal to the market value of all goods and services produced, which in turn will be equal to the total income generated. In other words, gross domestic production equals gross domestic income.

As a consequence of this equality, there are two ways to calculate GDP: the expenditure approach and the income approach. The expenditure approach tallies total spending on goods and services during the fiscal period, including:

- Spending by households (personal consumption)
- Expenditures by businesses (gross private investment)
- Government spending
- Net exports (exports minus imports)

The income approach, instead of calculating spending, adds up all income, including income to the government in the form of taxes. The relevant items are

- Salaries
- Dividends (including proprietor's income, interest income, and rents)
- Corporate profits
- Indirect business taxes (taxes minus net income earned overseas)
- Net income of foreigners
- Depreciation

What Can GDP, GNP, and Related Ratios Tell Us?

Indicators based on GDP and GNP can tell different things about an economy:

- *Total GDP* measures the size of an economy.
- *GDP per capita* (or preferably *GNP per capita*) is—usually—a basic indicator of the state of a country's development.
- The rate of change in GDP, or *GDP growth*, expressed in percentage terms indicates how rapidly an economy is growing or contracting.

This last indicator is particularly important to bank analysts, since it mirrors the current phase of the business cycle and the credit cycle and therefore directly influences a bank's activity and earnings.

Total GDP

Total GDP functions to measure either the relative size of a country's economy or an economy's growth trend, especially when displayed over comparatively long periods of time. On a comparative basis for a given time period, total GDP shows the relative size of one economy to another, as well as changes in their position over the medium term.

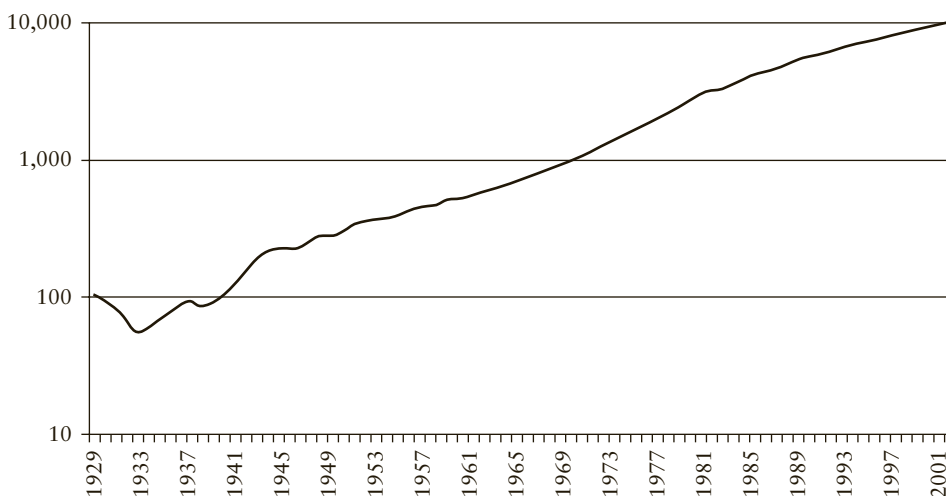


EXHIBIT 11.8 Total U.S. GDP in \$ Billion, Adjusted for Inflation (Logarithmic Scale)

Data Source: Federal Reserve Bank.

Over a period of decades, total GDP mirrors the longer-term trend in the economy, which in most markets is almost invariably upwards. Exhibit 11.8, which registers total GDP for the U.S. economy over a period of more than 70 years, illustrates this phenomenon. Observe how using a logarithmic scale to show total GDP smoothes out moderate business cycle fluctuations of a comparatively short duration. The exception is the Great Depression of the 1930s, which, by reason of its length and magnitude registers a conspicuous dip on the graph of the economic contraction it embodied before steady upward growth resumes in the post-World War II era.

GNP or GDP per Capita

GNP or GDP per capita is a widely used indicator for a country's average standard of living, and it is one of the metrics widely used to group countries in respect of their level of development. It is simply equal to a country's GNP or GDP divided by its population.

As a measure of a country's standard of living and economic development, GDP per capita has some significant deficiencies.

First, and this is true of all economic statistics, it is only as good as the data input, and this is supplied by each national government respectively. Some countries may have better data collection processes and therefore be more accurate in their calculations.

Second, GDP only includes the official economy. It generally does not include barter transactions, black market, or illegal trade, which in some nations—even wealthy nations such as Italy—may form a significant part of economic activity. The same is true with respect to production that does not enter the marketplace, such as canned goods produced for home use, or cooking services performed by homemakers. Hence GDP may be criticized for understating production.

Finally, GDP does not reflect the quality of goods and services produced, nor does it differentiate between goods and services that benefit peoples' lives and those that do not. Because it only measures production, the indicator also ignores intangible quality of life values such as job satisfaction, leisure time, intellectual and cultural life, and external costs such as resource depletion or environmental pollution. Ultimately, GDP is simply a summation of the market value of all goods and services produced by a society. Its main virtue is that it provides a relatively consistent measure of economic activity on a cross-border basis.

DEFICIENCIES OF GDP PER CAPITA AS A MEASURE OF A COUNTRY'S STANDARD OF LIVING

GDP per capita

- Is subject to quality of data input.
- Only includes the official economy.
- Does not differentiate among the quality of goods produced.
- Ignores nonmeasurable quality of life criteria.

GDP Growth

GDP growth is a particularly important indicator for bank analysis because it reflects both the general strength of an economy and its position in the credit cycle. GDP growth reported in nominal terms is reported as a simple delta ratio.

While *nominal GDP* is GDP reported at the price prevailing at the end of the period (*current prices*), *real GDP* is a measure of GDP against a given base year (*constant prices*, i.e., base year prices).⁴⁰ The use of real GDP growth as an indicator is preferable to nominal GDP growth since it adjusts for inflation. The calculation of GDP is not the same in every country, and this affects the validity of cross-border comparisons, but this topic will not be discussed here. If, however, inflation rates are low and the time span being examined is relatively short, both serve essentially the same purpose. In certain instances, a fall in nominal GDP can be an indicator of severe economic distress.⁴¹

Conventionally, GDP growth is expressed on a year-on-year basis, which adjusts for seasonal variations. A moderate to moderately high rate of GDP growth is almost universally perceived as credit-positive in the banking context. As with most comparisons of this nature, what is a low rate of growth, a moderate rate, or a high rate will depend on global and regional averages, particularly among similarly situated economies. Mature economies usually display comparatively lower GDP growth rates than emerging markets, and a growth rate that would be viewed as quite strong in the former might be seen as merely satisfactory in the latter. Accordingly, under normal recent conditions, a rate of 4 percent GDP growth in Sweden would likely be considered very strong, while the same rate of growth in China would be perceived as fairly lackluster.

RULE OF THUMB: GDP GROWTH RATES

Mature, highly developed economies such as Germany's rarely grow at annual rates of much more than 3 to 4 percent, and 2 to 3 percent is more typical average rate of increase. A growth rate of less than 1 percent, however, normally is a sign of a weak or faltering economy, and an economic contraction signals the onset of a recession. As a rule of thumb, a two percentage point or more decrease in the GDP growth rate (on an annualized basis) should be regarded as a warning signal.

While more is generally better when it comes to GDP, too rapid growth is not desirable. In mature economies, GDP growth rates of more than 4 to 5 percent are usually unsustainable and suggest that the economy is operating at or near full capacity. In this case, it will be in danger of overheating creating the risk of triggering high rates of inflation.

In emerging markets, GDP growth rates not only tend to be higher on average but will also generally exhibit greater fluctuations. Growth rates in the range of 5 to 6 percent are generally considered quite robust. Although very rapidly expanding markets, such as Japan in the 1960s or China in the 1990s, have registered growth rates as high as 10 percent or more, these have rarely been sustained for more than a year or two.

Growth that is too low, or worse negative, is undesirable since a country that is incapable of expanding its economy at a sufficiently brisk pace will be unable to absorb an expanding labor force or to offer the populace an improving standard of living. Over time, this situation can potentially give rise to social problems and, if frustrations boil over, political unrest. Whatever the reason, a sharp decline in GDP growth functions as a red flag for the sovereign analyst and the bank analyst alike, as it frequently heralds more difficult economic conditions ahead, and tougher times for the banking industry. As a general rule, more than a two percentage point annualized drop in GDP should be regarded as a warning signal.

In a similar manner, rapidly rising living standards on average can mask a widening gap between the rich and poor caused by structural changes in the economy. As with a low growth scenario, where high overall growth rates are the result of extremely rapid improvements in income and wealth for some, while others experience a decline in their standard of living or even stark poverty, deepening social conflict and political volatility can also result. Naturally, with respect to the causes of social and political instability, economic growth is but one of a number of factors that typically play a part, although it is not uncommonly a critical one.

Aside from this risk, extremely high growth rates generate other difficulties. At rates of growth of more than 4 or 5 percent for a mature economy and somewhat more for an emerging economy, overheating is prone to occur. *Overheating* simply means that the economy is operating at full or near full capacity. By definition, operating at full capacity means that labor is in short supply, which will tend to push up wages and salaries. For the same reason, an economy running at full capacity will

have a similar impact on the other factors of production, putting upwards pressure on the cost of raw materials, services, and land. If the macroeconomy is left by government officials to overheat, they risk triggering a sharp rise in the rate of inflation, potentially leading to the creation of an asset bubble. Should high rates of inflation take hold, it may be difficult to bring them down again. Monetary authorities will typically respond by tightening monetary policy; that is, taking steps to push up interest rates in order to dampen growth. But such measures frequently come with adverse side effects. It is preferable, therefore, to nip inflation in the bud before market expectation of continued inflation takes hold.

Although both nominal and real GDP figures almost always show apparently steady growth over the long term, percentage changes in GDP growth from quarter to quarter reflect changes in the business and economic cycle. Hence, in addition to functioning as a growth speedometer, changes in GDP growth rates mirror the business cycle. Indeed, as one commentator aptly put it:

To a large extent, GDP evolution “is” the [business] cycle. It is in fact with respect to GDP dynamics that other variables are defined as pro-cyclical (if they follow a similar path) or anti-cyclical. A recession is often defined as a two-quarter absolute reduction of GDP.⁴²

The terms *business cycle* and *economic cycle* are basically synonyms, while the *credit cycle* is merely the *economic cycle* viewed from a slightly different vantage point. Simply put, the credit cycle is simply the economic cycle as applied to banks. It encompasses both changes in a bank’s financial condition and strategy that result from the expansions and contractions of the macroeconomy.

The impact of the credit cycles on banks is described in Chapter 14 dealing with banking and financial crises. The duration of economic cycles is hard to predict, but they generally last for one to three years, although in extreme cases an economic downturn may continue for five years or more. From a credit perspective, we are usually concerned with the next 12 to 18 months, so understanding where an economy is in its business cycle is important to an evaluation of the banking system and individual banks. Sovereign risk analysis provides the tools to describe and to some extent predict the economic cycle.

Other GDP-Related Indicators

In addition to its usefulness as an indicator of a country’s standard of living and how its economy compares with those of other nations and as an indicator of economic trends and cycles, GDP is also useful because it gives us a common denominator against which to compare other economic metrics.

For instance national savings can be measured as *savings to GDP*; fiscal accounts as *public sector debt to GDP*; and bank lending (bank intermediation) as *bank lending to GDP*. Further, the composition of a country’s economy can be broken down by source, such as what percentage of GDP comes from agriculture, what percentage from manufacturing, and so on. From a credit perspective, countries with a high foreign debt burden will benefit more from sectors that contribute to export growth, or generate foreign exchange, than those that are domestically oriented.

While it is apparent that GDP (together with its sibling GNP) functions as a chief indicator of macroeconomic activity, less obvious are some of the linkages between

GDP and other economic indicators. First, quite a few macroeconomic indicators are in a sense derived from GDP. Since GDP (or GNP) represents a sum of all economic activity, each constituent of it generates other indicators. As an illustration, automobile sales make up a significant part of consumption. If interim auto sales figures are declining on a seasonally adjusted basis, it implies that overall consumption (the “C” in GDP) is as well. In the same way, if business investment is increasing, it suggests that overall investment (the “I” in GDP) is as well. Other indicators, such as the unemployment rate, function as a proxy for changes in the components of GDP: the greater the proportion of the population at work, the higher are both production and consumption.

Second, GDP is directly linked with the balance of payments discussed later in the chapter. In brief, if we examine the formula given,

$$\text{GDP} = C + I + G + \text{EX} - \text{IM}$$

we can see that its final part, exports minus imports, is identical to the definition of the current account on the balance of payments. When imports exceed exports, a *current account deficit* results. By definition, a current account deficit requires a *capital account surplus*, meaning that there must be net inflow of capital. A net inflow of capital suggests that a country’s savings are less than its spending on investment. So, a country’s savings and investment rates, which represent changes in the proportional mix of C, I, and G, will also have an impact on the balance of payments.

Finally, macroeconomics teaches that GDP is linked to the money supply and monetary policy under the quantity theory of money. (Monetary indicators are discussed in a later section of the chapter.) This theory posits the “equation of exchange” as shown in the equation that follows. The equation states that nominal GDP, which can be expressed as market price, P, multiplied by output, Y, is equal to the amount of the money supply, M, multiplied by the rate at which the money, V, circulates through the economy. V stands for the velocity of the money supply.⁴³ Since V and Y are relatively fixed, an increase in the money supply will, generally speaking, cause prices to rise.⁴⁴ (The relationship between investment and saving, the current account and the capital account, and the money supply and inflation are discussed in more depth later in this chapter.)

$$\text{Equation of exchange: } M \times V = Y \times P$$

Leading, Lagging, and Coincident Indicators

Economic indicators can be used to predict changes in the economy.

Measures of economic activity that tend to anticipate changes in the direction of an economy; that is, expansion or contraction are called *leading indicators* (Exhibit 11.9).

The opposite of leading indicators are *lagging indicators*, which lag behind changes in the economy by a number of months. These may be used by sovereign risk analysts for both to confirm a change in the economic cycle and to predict its duration and severity.

Finally, *coincident indicators* are those that describe the current state of the economy.

EXHIBIT 11.9 Leading, Lagging, and Coincident Indicators Illustrated

Leading		Lagging Indicators		Coincident		
Average hours worked per worker per wk.	Average new unemployment claims/week	Unemployment rate	Average duration of unemployment in weeks	Total employment		
Total manufacturers' new orders		Change in manufacturing labor cost per unit of output	Consumer price index (CPI) of services	Total sales		
Manufacturers' new orders, capital goods	Manufacturers' new orders, capital goods	Consumer credit to personal income	Commercial and industrial lending	Retail sales	Wholesale sales	
New residential building permits issued	Average time from order to delivery			Personal income	Index of industrial production	
Consumer confidence index	Business confidence index	Inventories to sales ratio, inflation adjusted (usually confined to wholesale goods)				
Stock market	Yield curve					
Inflation-adjusted Money supply(M2)						
Composite Index of Leading Indicators		Composite Index of Lagging Indicators		Composite Index of Coincident Indicators		

Source: Federal Reserve Bank.

SELECTED LEADING INDICATORS, LAGGING INDICATORS, AND COINCIDENT INDICATORS

Leading Indicators

- The average *hours worked per week* by workers in manufacturing. This is a leading indicator because employers usually adjust work hours before hiring more workers or laying off existing workers.
- Average weekly *new claims for unemployment* insurance. When workers are laid off (made redundant), they will usually file for unemployment benefits. A rise in new claims suggests slowing demand, and vice versa.
- *Building permits*, new residential housing. An increase in residential building permits heralds a rise in construction activity, which is often one of the first types of production to recover from an economic downturn.
- *Stockmarket index*. Large changes in equities prices often signal broader economic changes as investors and traders seek to take profits before a looming recession or get in on the ground floor of a period of economic expansion.
- *Consumer confidence index*. A poll of consumer expectations about the direction of the economy, this metric mirrors changing perceptions among consumers, which may very well be followed by changes in their behavior, such as by saving more and consuming less in expectation of potential job loss, for example. The business confidence index is a comparable leading indicator.
- *Manufacturers' new orders*. New orders received by manufacturers, which are usually impelled by rising demand, are a forerunner of increased production.
- *Money supply* (adjusted for inflation). Called M2 in the United States (meaning demand deposits, savings deposits, small-denomination time deposits, money-market mutual funds, and cash and currency), a declining money supply in real terms is a sign that bank lending may be dropping. A cutback in credit will tend to discourage investment and concomitant growth in production and economic expansion.
- *Yield curve*. The difference between the rates of long-term government bonds and short-term rates is often called the yield curve. A change the shape of the yield curve is widely regarded as predictor of changes in the economy. A normal yield curve is moderately upward sloping. When the yield curve inverts, so that short rates are higher than long rates, it is seen as predictive of an economic downturn.

Lagging Indicators

- The *unemployment* rate and average *duration of unemployment*. As discussed in the text, unemployment rates are lagging indicators because employers tend to delay hiring or reducing staff until changes in business conditions clearly warrant it. The duration of unemployment shows the

relative weakness of demand for labor. Duration falls during an expansion and increases after a recession has commenced.

- *Inventories to sales.* This ratio shows how quickly firms are able to dispose of their stock of goods. During a downturn, sales slow, and inventories rise, often beyond projections. The inventory-to-sales ratio tends to peak about halfway through a recession and tends to decline at the start of a recovery, so its fall is one of the first hints that an expansionary period is about to begin.
- *Consumer credit to personal income.* Consumers usually avoid increasing their debt levels until they feel secure in their employment and the compensation rises that are associated with a period of prosperity. So a material increase in this ratio from a trough that may occur a year or more following a recession is good evidence that a period of economic expansion is well under way.
- *Total commercial and industrial loans outstanding* (inflation adjusted). Total business loans (including commercial paper issues outstanding) tend to peak a year or more after the beginning of a contraction, since a fall in revenues associated with the slowdown, will temporarily boost credit demand. This includes loans and commercial paper, adjusted for inflation.
- *Average prime rate.* Usually tracks the interbank rate (in the United States the federal funds rate), which will normally be held lower by the central bank until well after a recovery commences.
- *Cost of labor and cost of services.* Service cost increases tend to rise at the outset of a recession and return to stability as a recovery resumes.

Coincident Indicators

- *Total employment* (of nonagricultural workers, including part-time and temporary employees) does not distinguish between permanent and temporary employees. This indicator is viewed as reflecting the current condition of the economy.
- *Personal income* excluding transfer payments. Aggregate inflation-adjusted income mirrors aggregate spending.
- *Index of industrial production.* This measure of total physical output of manufacturing, energy and resource extraction industries is viewed as descriptive of the current state of the economy.
- *Retail and wholesale sales.* Collectively, inflation-adjusted retail and hotel sales correspond to the economy's total spending, and consequently its total income and production.

The Foregoing Indicators Include

- General economic indicators, such as the unemployment rate.
- Monetary indicators such as interest rates and the money supply.
- Market indicators such as stock market indices.

Source: Federal Reserve Bank.

General economic indicators are discussed in the section that follows. Monetary and market indicators will be discussed towards the end of this chapter.

The Consumer and Business Confidence Indices—Illustrative Leading Indicators

Two important categories of leading indicators are the applicable business and consumer confidence indices for a particular country, particularly the latter. Both suffer from the same conceptual weakness that political events, such as the outcome of elections, might unduly influence them.

Among other sources, the European Commission publishes excellent data that can be downloaded, for example consumer confidence or construction data for any EU country. A comparison between such data and a country's actual business cycle—for example as reflected in change in GDP—with the appropriate time adjustment, usually demonstrates the predictive value of such indicators.

Similarly, Business Confidence indices published in the United States—or, in fact in many countries—reflect the mood of business owners and are often predictive of important trends, for example, in job creation.

Change in new manufacturing orders, that is, factory orders, is another key leading indicator, which anticipates greater or reduced investment in productive capacity. Durable goods orders represent new orders for manufactured goods placed with domestic producers, and they are a subset of manufacturing new orders.⁴⁵

The Unemployment Rate—An Illustrative Lagging Indicator

[The] strikingly close relationship between the economy's growth rate and the rate of change in the unemployment rate . . . is one of the few things economists are willing to call a "law" . . . with a straight face.

—Paul Krugman⁴⁶

The unemployment rate is the percentage of the eligible workforce that is unemployed, and constitutes both a lagging indicator and a countercyclical indicator. It is countercyclical in the sense that the indicator goes up during an economic downturn (as the proportion of the population unemployed increases) and falls during periods of rising prosperity as a greater percentage are in the work force.⁴⁷ A comparison of unemployment rates across countries is difficult as every country has its own calculation method.

UNEMPLOYMENT RATE

This is the ratio of those unemployed and actively seeking work or those made temporarily redundant and waiting to return to work⁴⁸ to the population of those eligible to work.

$$\text{Unemployment rate} = \frac{\text{Unemployed persons actively looking for work}}{\text{Adult civilians of working age}} \times 100$$

As Exhibit 11.10 illustrates, changes in employment are closely connected to changes in both production and consumption and therefore in GDP. An above-average or sharply increasing unemployment rate is indicative of a cyclical downturn and lackluster economic conditions. In general, the unemployment rate is viewed as a lagging indicator since most employers will reduce worker hours first before making employees redundant. In addition, regulatory restrictions may limit the ability of employers to terminate employment quickly. The indicator also functions as a fairly smooth measure (i.e., comparatively nonvolatile) of the economic cycle and is subject to only mild seasonal variations in most countries.

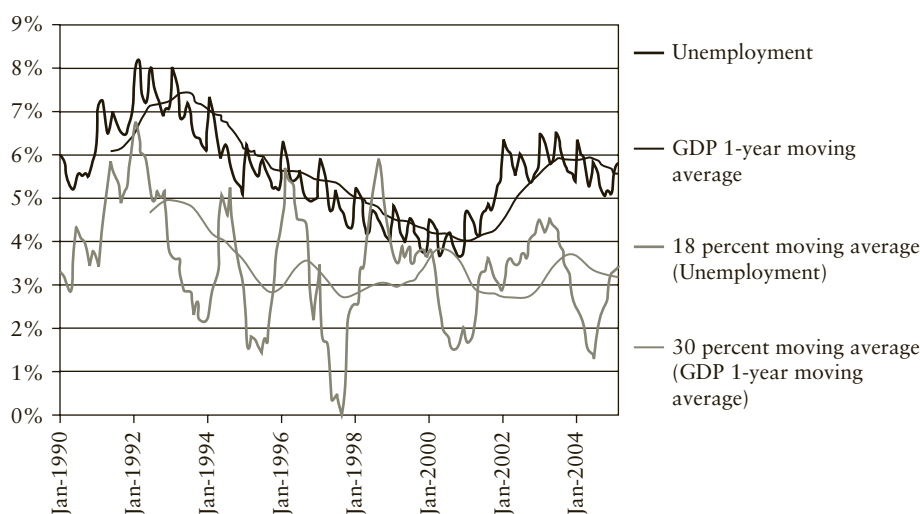


EXHIBIT 11.10 U.S. Unemployment Rate (percent) and Real GDP growth

Data Source: Econstats.

UNEMPLOYMENT IN MACROECONOMIC ANALYSIS

Although from a social welfare perspective full employment is desirable, from the vantage point of the market and monetary policy makers, the unemployment rate like GDP growth, should be neither too high nor too low. The optimal number depends on the market: a rate between 4 and 6 percent is typically viewed as ideal.⁴⁹ The reason is that policymakers tend to prefer to see a relatively low unemployment rate, but not one that is so low that wages are pushed up substantially, perhaps triggering inflation and overheating the economy. Most economists accept that there is a trade-off between price stability and high rates of employment. As one economist writing in a U.S. Federal Reserve Bank publication argued:

Consistent attempts to [achieve levels of unemployment of 4 percent by expanding] the economy beyond its potential for production will

*result in higher inflation while ultimately failing to produce lower average unemployment.*⁵⁰

Still, a prolonged period of above-average unemployment is likely to remain a negative indicator with regard both to sovereign risk and bank credit. From a sovereign risk perspective, it signals a fragile economy and the continuation of weak demand, because those who are without a job are not likely to be big consumers of new products and services. From a bank credit perspective, it suggests asset quality problems at banks since the primary cause of default on consumer loans such as credit cards, residential mortgage loans, and auto loans is lack of income from employment. Likewise a sharp increase in unemployment should be construed as a danger sign. As a rule of thumb, more than a two percentage point rise in the unemployment rate (on an annualized basis) should be viewed as a red flag from a sovereign and bank credit analytical perspective.

- *Cyclical unemployment*: unemployment associated with changes in the economic cycle, that is, a recession.
- *Structural unemployment*: unemployment resulting from secular changes in the economy where those unemployed do not have the skills for the employment on offer (e.g., out of work coal miners in a country where the economy is changing from heavy industry to one that is light industry and service based).
- *Frictional unemployment*: unemployment resulting from imperfections in the job market due to a lack of information about employment availability or appropriate candidates and the time or transaction costs involved in consummating a new employment arrangement.
- *The natural rate of unemployment*: the highest rate of unemployment consistent with stable prices.
- *NAIRU* (nonaccelerating-inflation rate of unemployment): the natural rate of unemployment.
- *Full employment*: the term has two meanings: (1) an effectively nil rate of unemployment, allowing for frictional unemployment, often used in this ordinary sense in political discourse; (2) the natural rate of unemployment.

Related to the unemployment rate are other employment indicators that are used to project changes in production, consumption, and other macroeconomic variables. Among the most important is the number of new unemployment benefits claims, which heralds changes in the unemployment rate.⁵¹ Associated employment metrics such as total employment, average labor costs, and the like may also form part of the sovereign risk analyst's calculus.⁵²

Additionally, changes in employment indicators in particular industries may also suggest the condition of a particular sector, such as heavy industry, relative to the economy as a whole, and may be relevant if a bank specializes in a particular type of

lending or has high concentrations of loans in certain sectors.⁵³ Finally, related to unemployment is the rate of consumer default echoed in new personal bankruptcy or insolvency filings made within a given period.⁵⁴

From a bank credit perspective, the unemployment rate is particularly useful both as a way to gauge where a nation's economy is in the business and credit cycle and as a signal of declining asset quality, especially in respect to consumer and retail lending. Falling unemployment rates are associated with economic expansion. If unemployment is increasing, however, it usually means that the economy is well into a cyclical contraction. At this stage, banks are likely to have cut back on lending and consumer default rates, which are directly correlated with and parallel to unemployment rates, are likely to be rising.

It should be noted that there are some definitional and other issues of which to be aware when comparing unemployment rates. In some markets, the unemployment rate may be understated since the true level may raise politically sensitive issues. In addition, the ratio does not capture underemployment, which occurs when people work at less than their full capacity or skill levels, and the ratio is also subject to variations in definition (e.g., who is classified as an eligible worker? does employment include part-time employment?). Finally, those who are not actively seeking employment, but who would work were the opportunity presented, are excluded from the ratio's calculation.

Retail Sales and Auto Sales—Illustrative Coincident Indicators

Personal income in the GDP formula represents total income received by individuals, and total expenditure represents the total spending by individuals. *Personal outlays* means *consumer purchases* of all goods and services, and both terms are essentially synonymous with the terms *retail sales* and *consumer spending*. Both personal income and retail sales are regarded as coincident indicators.

Auto sales are a subcategory of retail sales, and represent total sales of domestically produced automobiles and light-duty trucks. Seasonally adjusted, auto sales are generally viewed as a good indicator of rising or falling consumer demand. Another coincident indicator is the index of industrial production, which represents the total physical output of the nation's manufacturers, energy producers, and resource extractors.

FISCAL, MONETARY, AND TRADE POLICIES

Although the ratios and indicators discussed so far describe the state of an economy, alone they convey little information about what the relevant government has been doing to manage it.⁵⁵ This section aims to fill that gap by addressing the topic of economic management. Specifically, this section looks at three interrelated aspects of economic dynamics and their management:

1. Government spending and revenue collection, or fiscal policy
2. Interest rates, the rate of price inflation and the money supply, and government management of these economic attributes, or monetary policy
3. The country's trade and debt position in relation to other nations, sometimes referred to as its external position together with the government's management of its trading relationships, foreign borrowing and currency—abbreviated here as trade policy

Implicit in each of the three bulleted descriptions above is a set of indicators that measure economic performance with respect to that facet of the macroeconomy.⁵⁶ For example, fiscal policy takes account of government spending levels, revenue collection and concomitant tax rates, and the relationship between revenue and spending. It also takes account of the level of domestic borrowing. The principal indicators associated with each of the three aspects of macroeconomics is shown in Exhibit 11.11. They are discussed in greater detail in succeeding pages.

In addition, each of the foregoing aspects of economic management affects the other, as shown graphically in Exhibit 11.12 and as described at various points throughout the chapter. To illustrate, changes in monetary policy will tend to have an impact on the money supply and interest rates. Such changes will often have a knock-on effect influencing exchange rates, as well as the competitiveness of exports and the affordability of imports, thereby altering a country's trade balance. The same changes in the money supply and interest rates may also affect fiscal policy by making it more or less expensive for the government to finance its spending through borrowing.

Fiscal Policy: The Condition of Public Finances

Fiscal policy concerns how a government influences the economy through its collection of taxes, spending of those funds collected, and borrowing or lending the difference. This is very complex generally, but also very difficult to gauge. Indicators are not easily observable, and in any event, would not be very predictive⁵⁷ unless associated with other indicators. We shall limit our examination of fiscal policy analysis to a cursory review.

The evaluation of a government's fiscal policy primarily focuses upon the extent and nature of its (fiscal) deficit. Naturally, whether the government incurs a deficit or surplus depends upon its spending relative to revenues generated from taxation. The

EXHIBIT 11.11 Selected Fiscal, Monetary, and Trade Indicators

Fiscal policy is concerned with government spending, revenue, and borrowing.	Monetary policy is concerned with the money supply, interest rates, and prices.	Trade policy is concerned with exports, imports, and borrowing and lending to foreigners.
Fiscal Indicators	Monetary indicators	Trade indicators
Budget balance (whether surplus or deficit) to GDP; gov't spending to GDP (current vs. capital); cumulative gov't borrowing to GDP; gov't spending to revenue; gov't debt to revenue.	Money supply; availability of credit (intermediation ratio, aggregate lending growth, and LDP); interest rates (official discount rate, interbank rate, real interest rates, yield curve shape); inflation rate (CPI, PPI, various asset price indices).	Trade balance (whether surplus or deficit), services balance, net investment and transfers (investment, FDIs, etc.), terms of trade (comparative advantages); external debt position (gross and net), official reserve account, debt service capacity (debt service ratio, import cover ratio).

Source: Independent Company Research.

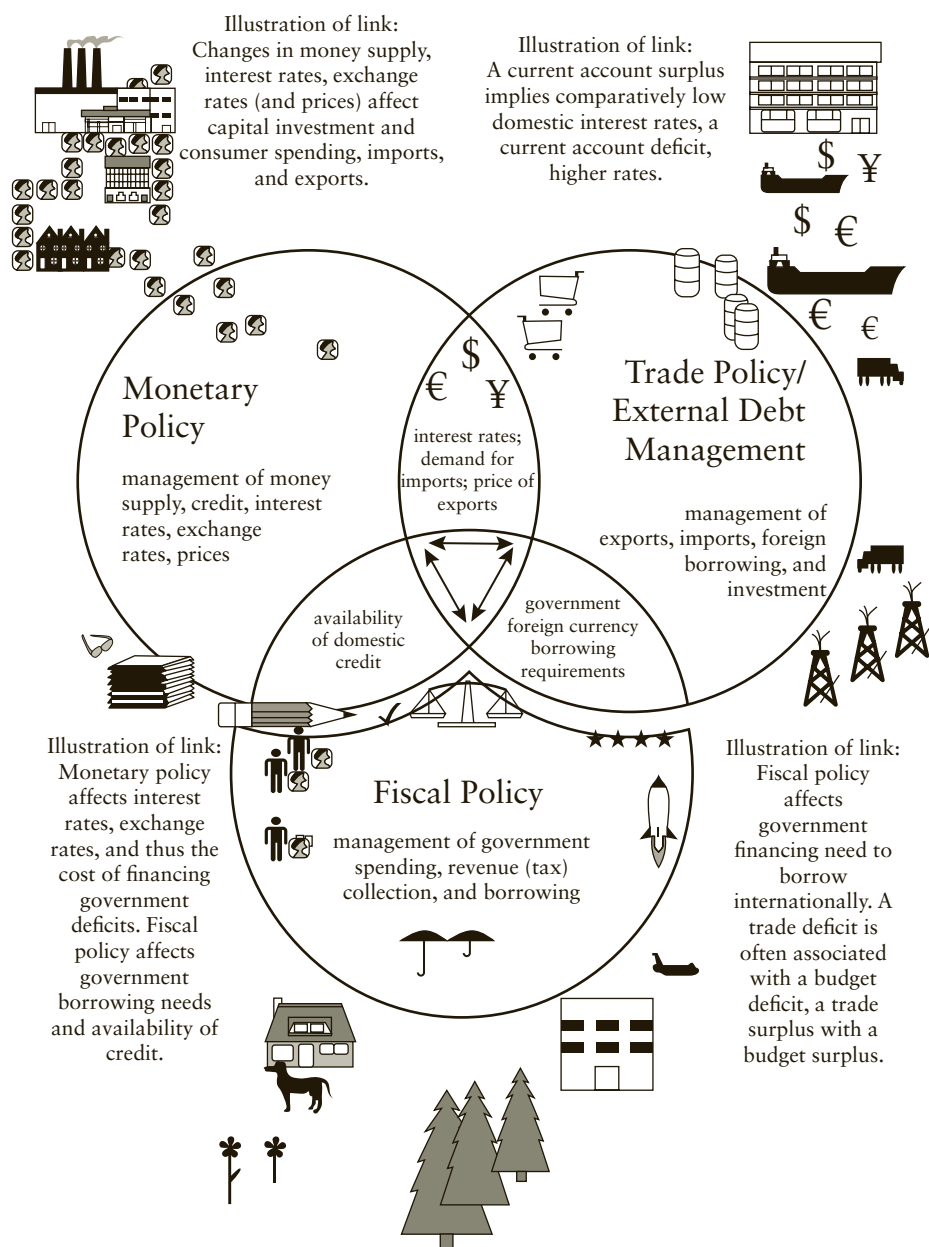


EXHIBIT 11.12 Fiscal Policy, Monetary Policy, and Trade Policy

volume of spending and revenues, in turn, will be affected by the country's particular circumstances, such as whether there is an ongoing war with another nation or the prospect of war, the government's policies as influenced by its political ideology or values, or strategic or tactical political considerations.

In addition, a country's fiscal balance will be affected by the government's competence in spending wisely and efficiently in order to get the most political and social benefit per unit of currency. Similarly, it will be affected by the tax rates set by

government and the corresponding volume of revenue collected. Again, such rates and manner of administration will be affected by ideology (e.g., does the government seek to create an egalitarian society with few classes seen as rich or poor?), policies (e.g., a pro business policy that limits tax rates so as not to hinder economic growth), political factors (e.g., is a particular interest group so strong politically that it can shift the tax burden to others?) and their perceived fairness.

Scope of Fiscal Policy Analysis

When examining a country's public finances, the scope of the analysis is ordinarily on their current state, how it got that way, and prospects for improvement or risks of deterioration. How the government manages its revenue, mainly from taxes, and its expenditures (public expenditure) is reflected in a number of fiscal indicators. These include (1) the budget balance, whether surplus or deficit, as a percentage of GDP; and (2) government revenue (and spending) as a percentage of GDP.

Absent mitigating circumstances, such as a natural disaster, a war initiated by another nation, or the impact of a regional financial crisis, responsibility for the present conditions lies squarely with the government and the officials that have been in charge of national economic management. Where public finances are in a good state, the implication is that the government has managed them capably; where not so good, it may be inferred that the country's economic managers could do better. Certainly, unforeseen internal or external calamities can place a strain on even the best-managed economies, while sometimes government decision makers are simply lucky.

Measures of State Spending, Revenue, and the Difference between the Two

In short, analysis of a government's fiscal position takes in the evaluation of its revenue collection ability, its prudence in spending, and its capacity to effectively finance the difference. Usually spending drives taxation, rather than the other way around, so spending is a good place to start.

Spending: Current and Capital Spending

As already mentioned, state spending to GDP is a primary measure of the cost side of the fiscal equation. As with firms, spending is frequently differentiated between spending on capital investment (e.g., the construction of roads, bridges, hospitals, and schools) and current spending (e.g., salaries of civil servants, the cost of heating those schools and hospitals). Hence, measures of relative government spending include both total government expenditure as a percentage of GDP and current spending to GDP. Needless to mention, governments often play with the spending descriptions when it suits them to do so for tactical or electoral purposes.

CURRENT SPENDING VERSUS CAPITAL SPENDING

Current spending (also called noncapital expenditure) means government payments for the consumption of goods and services. A more technical term for current spending is *government final consumption expenditure*, which is defined by the OECD as consisting "of expenditure, including imputed

expenditure, incurred by general government on both individual consumption goods and services and collective consumption services.”

Capital spending (also called capital expenditure) means government payments for public infrastructure and similar long-term investments.

Source: OECD⁵⁸

Until recently, countries used to exhibit a wide range of government spending, as a percentage of GDP, from under 10 percent for an emerging market like Tanzania to perhaps 35 percent for a social welfare state like Sweden, with the UK and the United States in the middle of the range. In recent years, the situation in some countries has taken a dramatic turn for the worse, some of it as a result of wars, some as a result of a costly rescue of the banking system, and many for ideological or political reasons, and not only in developing countries. France has not had a balanced budget since 1974, and it has now reached the implausible level of 55 percent of GDP being spent by the state, while an astonishing 75 percent of parents recently polled wished for their children to find a job with the state.

Revenue

In some countries, the government’s ability to collect tax revenues from resident individuals and companies is weak, and accordingly it may find it difficult to generate sufficient revenue to pay for needed public services.

Note that most government revenue comes from taxes, although in some countries the government may wholly own or partly own companies and banks, or even have a monopoly in certain sectors.

Budget Surplus or Deficit

The government may either run a *budget surplus*, where tax revenues exceed its expenditures, or more commonly a *budget deficit*, where expenditures exceed tax revenues. The budget or surplus is commonly measured against GDP. The relative budget deficit (or surplus) can be measured as a percentage of GDP, and it is an essential fiscal indicator.⁵⁹

$$\text{Budgeted surplus or deficit as \% of GDP} = \frac{\text{Government Revenue} - \text{Government Spending}}{\text{GDP}} \times 100$$

In any given period, should a budget deficit arise, it must be funded by borrowing. Governments typically borrow funds by issuing debt securities in domestic or foreign currencies that are purchased by domestic or foreign investors.⁶⁰

DEBT IN SOVEREIGN ANALYSIS

Government spending relative to revenues determines financing requirements. When revenues do not meet expenses, a deficit arises. Government deficits require financing, which is ordinarily accomplished through the issuance of

government debt such as treasury bills or bonds. Sovereign analysts therefore pay attention to the relative size of the fiscal deficit (or surplus) and government borrowing as a percentage of GDP.

Note that a country's fiscal debt and its external debt, as discussed later, are two distinct, albeit related, concepts. In general, all financial obligations within a country can be differentiated as follows:

- Obligations to the country's residents vs. obligations to nonresidents
- Public sector obligations (official obligations) vs. private sector obligations
- Domestic currency obligations vs. foreign currency obligations
- Current, recorded obligations vs. accrued and/or unrecorded obligations (such as future pension or medical care and liabilities)

In this portion of the chapter dealing with fiscal policy and the government's budget balance, the focus is upon domestic public sector obligations, and there is no need to differentiate between domestic and foreign obligations. In the context of a discussion of trade policy, the relevant distinction is between domestic and foreign obligation irrespective of whether the relevant obligors are public sector entities or private ones.

The government's cumulative deficit is referred to as its *public debt* or *national debt*. Cumulative debt increases if a government continues to spend beyond its means, being unable to cover the increase by boosting revenue. A jurisdiction's public sector debt is the *aggregate debt* of its central government, local government, and government-owned corporations (other than banks), and therefore encompasses all borrowings of such entities.⁶¹ The indicator *public sector debt to GDP* is a useful ratio to gauge the size of the budget deficit for comparative purposes. It measures a government's cumulative borrowing as a percentage of GDP.⁶² Also referred to as the (*national*) *government debt to GDP* or the *debt ratio*,⁶³ the ratio shows the relative cumulative public sector debt burden placed upon an economy.

$$\text{Public sector debt to GDP}^{64} = \frac{\text{Public sector debt}}{\text{Total GDP}} \times 100$$

Note that the public sector debt ratio, which measures government borrowing as a percentage of GDP, is also referred to simply as the debt ratio. Be aware that it differs from the *total-debt-to-GDP ratio*, which includes *private sector* as well as *public sector debt*.

Other fiscal indicators related to the preceding ones include those shown below. These are often computed with respect to the national (general) government, as opposed to all public sector debt.

$$\text{Cumulative government debt to revenue}(x) = \frac{\text{Government debt}}{\text{Government revenue}}$$

Government debt service to government revenue(%)

$$= \frac{\text{Government interest payments}}{\text{Government revenue}} \times 100$$

Tight or Loose Fiscal Policy

In response to economic circumstances and its fiscal condition, a government may run a *tight fiscal policy* or a *loose fiscal policy*. A *tight fiscal policy* means that the government is seeking to reduce its deficit by spending less than it earns in taxes or from other sources. Alternatively, it may mean that it is sharply reducing spending, even if it has not yet been able to cover existing spending with tax revenue. A tight fiscal policy is normally accomplished by cutting back on costs deemed to be not absolutely necessary and therefore unaffordable. Typically this is accomplished by reducing public services, capping civil service wages (or shedding staff), and raising taxes. At the extreme, a tight fiscal policy constitutes all, or a significant part, of what may be termed an *austerity program*. Such programs, although they may be necessary or well intended, are rarely popular with the country's residents, and may result in the government's loss of popular support.

In contrast to a tight fiscal policy, a *loose fiscal policy* means that the government is either spending more than it earns in taxes thereby increasing the deficit, or spending is rapidly increasing. Changing economic conditions can transform what appeared to be a reasonably loose fiscal policy into one perceived to be profligate. An economic downturn, for example, will tend to cause tax revenues to fall, increasing the imbalance between costs and revenue. Notwithstanding this, mainstream economic thought, which embodies the views of the noted economist John Maynard Keynes, suggests that a loose fiscal policy be implemented—that is, a fiscal stimulus—during a downturn to stimulate consumer demand.

TIGHT FISCAL POLICY VERSUS LOOSE FISCAL POLICY

A tight policy means that the government is spending less than it earns in taxes, or strongly reducing its spending. In contrast, a loose fiscal policy means that it is spending more than it earns in taxes (or rapidly increasing its spending).

There are a number of variations on the preceding primary ratios, as each can be defined in several ways depending upon which public sector elements are included or excluded. For example, the indicators can be computed based upon the *current* and *cumulative* balances calculated on a *consolidated* basis. Similarly, by excluding public sector entities, such as government-owned export insurance companies, or by excluding local and regional government balances or national pension obligations computed values will differ, and in fact will often differ substantially.⁶⁵ Finally, of course, whether the various line items are gross or net, and include extraordinary items and contingent liabilities, alternative indicators will affect the computed value of the indicator. Exhibit 11.13 illustrates some of the variations that can occur

EXHIBIT 11.13 Composition of Public Sector Debt

Public Sector			
Gross public sector debt. Gross debt of general government + Government-guaranteed private sector debt.			
General Government			PSEs
General government debt (gross.) All national, regional, and local government debt + debt on-lent to government-owned or private sector enterprises. General government debt (net). General government debt less government deposits, official reserves, government loan portfolio, and pension fund assets. General government contingent liabilities. All government-guaranteed debt and contingent liabilities.			
Central Government		Regional and Local Governments	Public Sector Entities (government-owned nonfinancial enterprises)
Central Government Debt Excluding Pension Fund	Pension Fund		
Gross debt (or surplus)	Gross assets	Gross debt (or surplus)	Gross debt (or surplus)
Net debt (or surplus)	Net assets (liabilities)	Net debt (or surplus)	Net debt (or surplus)
Net central government debt (excl. pension fund)	Pension fund net assets	Net regional/local government debt (or surplus) excl. pension fund (not shown)	Net PSE debt (or surplus) excl. pension fund (not shown)
Contingent liabilities i.e., government guaranteed obligations		Contingent liabilities regional/local	Contingent liabilities PSE
Net Central Government Debt		Net Regional and Local Government Debt	Net PSE Debt

Source of Terminology and Definitions: Standard & Poor's.

depending upon which components of the public sector are included. Exhibit 11.14 depicts the structure of a government budget while showing whether revenues exceed expenses or as is more often the case, expenses exceed revenues.

Quality of Economic Management

How effective is government's management of public finances or fiscal policy? If the economy has gone off the rails, the ability of the government and its economic managers to correct the situation and restore the nation to economic health becomes a more critical concern. The quality of fiscal economic management depends upon the extent of any structural imbalances (i.e., how severe is any shortfall in revenue

EXHIBIT 11.14 Government Budget Surplus or Deficit and Corresponding Financing

Public Sector			
Gross public sector budget balance – deficit (or surplus)			
General Government			PSEs
Government “revenues”			PSE “revenues”
Operating revenues (recurrent, from taxes, fees, interest payments, etc.)		Capital revenues (not requiring repayment)	
Government expenditures			PSE expenditures
General government operating expenditures (recurrent spending on salaries, interest payments, supplies, etc.)		General government capital expenditures (raises the value of physical assets)	
Pension expenditures			Other outgoings
Net lending (e.g., to state enterprises)			
Extraordinary expenditures (revenues)			Extraordinary items
Budget balance—surplus (deficit)			Surplus (deficit)
Central Government		Regional and Local Governments	Public Sector Entities
Central Government Debt Excluding Pension Fund	Pension Fund		
Revenues		Revenues	Revenues
Expenditures		Expenditures	Expenses
		Budget balance— surplus (deficit)	Profit (Loss)
Budget balance—surplus (deficit)			

Source of Terminology and Definitions: Standard & Poor's.

vis-à-vis expenses?) and primarily upon the government's overall competence, particularly in its capacity to correct any significant imbalances without incurring significant undesirable side effects. Questions to consider in assessing this capability include the following:

- Does the government's current situation reflect capable economic management?
- To what extent is its current situation the result of factors outside the government's control?
- How capable are those charged with managing the country's economy?
- How capable is the government in responding to crisis conditions? What is their performance track record?
- How plausible is the government's plan? Is it likely to meet with success?
- Will it be able to persuade the populace to go along with its plan?

IMPACT OF FISCAL POLICY ON THE BANKING SECTOR

Budget deficits can have a deleterious impact on the banking sector. First, insofar as they contribute to a higher perception of risk in the wider global market, internationally active banks based in the country affected will tend to have less access to foreign debt and capital markets or find them more costly than they otherwise would be. Purely at the domestic level, high budget deficits suggests that the government debt issues will crowd out private-sector issuance. This means that less of available savings will be channeled towards privately owned enterprises, making borrowing in those markets more expensive both for companies able to tap them directly and for banks that lend to companies unable to do so. Generally, this will make bank lending to such firms more costly, although banks may gain to the extent that disintermediation is discouraged.⁶⁶

Monetary Policy: Price Stability and Market Indicators

Monetary policy concerns how a government regulates or seeks to influence the following monetary variables: *interest rates*,⁶⁷ *inflation rates*, *money supply*, *availability of credit*, and *currency*.⁶⁸ These items are closely interrelated. Each affects the behavior of consumers, producers, and investors, and the behavior of each in turn affects the others.⁶⁹ For example, investors' expectations concerning corresponding market rates will materially affect their own willingness to advance capital. Their appetite for risk determines the risk premium they will demand over the risk-free rate, and accordingly influence prevailing interest rates and fixed-income prices. In turn, domestic interest rates affect the rate of inflation, the exchange rate, and the availability of credit—the last of which is closely intertwined with the money supply.

What are the objectives of monetary policy? While governments may have different priorities at different times, the fundamental objectives of monetary policy tend to be fairly straightforward, even if their attainment is not. They are stable prices, moderate interest rates, and an adequate supply of credit. The larger goal of policymakers is to achieve and maintain a healthy and sustainable level of economic growth.⁷⁰ Although governments must manage monetary policy to some extent, either actively or passively, opinions on the degree to which monetary policy can be used to increase economic growth over the longer term is one of those issues about which economists have significant differences of opinion.

Whatever those differences, there is little argument that monetary policy is linked to fiscal policy. As hinted earlier, monetary policy can only be distinct from fiscal policy where a country is sufficiently financially developed that the government has the capacity to borrow from investors to finance its debts. Otherwise, the only alternative available to a government that spends more than it collects in taxes is to print more money. Printing money to finance deficits, however, has the predictable negative effect of triggering inflation, which if not soon constrained, can spiral out of control.

As suggested in the preceding discussion, where the government is able to borrow in the money and capital markets, large fiscal deficits incurred by government can force up interest rates hindering economic growth and also potentially leading to inflationary pressures. In mentioning the danger of the government crowding out

private borrowers, it should also be emphasized that a modicum of government borrowing actually benefits the markets and may be a necessary condition to establish a threshold degree of depth and liquidity. Government borrowing serves to support by implicitly creating a benchmark yield curve that, by aiding the pricing of private debt, facilitates its issuance. Government borrowing also establishes a critical mass of market participants that fosters their further development.

In addition to its impact on fiscal policy, monetary policy also has an impact on trade policy since domestic interest rates influence exchange rates, which are perhaps the critical variable in determining whether exports rise or fall (and whether imports fall or rise). For this reason, an additional indirect monetary policy objective of most governments is to maintain a stable currency at an exchange rate that facilitates a competitive export position while maintaining a reasonable level of purchasing power in international markets.⁷¹

Relevant questions to pose when evaluating the effectiveness of monetary policy management primarily relate to the government's effectiveness in pursuing the preceding objectives. Correspondingly, the principal question to keep in mind when attempting to assess the quality of the government's monetary policy is this: How effectively has the government been able to manage the money supply and interest rates and, in the process hold down inflation to an appropriately low rate? Bear in mind that interest rates are simply the price of borrowing money, inflation reflects how much money is required to purchase the same item at different times, and the availability of credit is strongly affected by the quantity of money circulating throughout the economy. It should also be noted that deflation is as dangerous, and perhaps more dangerous, than moderate inflation. Consequently, central bankers normally target an inflation rate in the low single digits as opposed to zero inflation or a falling price scenario.

Inflation

The matter of inflation is, as the reader will have noticed, permeating many sections of this chapter. Monetary inflation (the increase in money supply) and price inflation (the actual upward changes to the price of a set of goods or services) do not share the same meaning, although monetary inflation usually⁷² prompts price inflation.⁷³ It is only for convenience that most people use the word *inflation* when *price inflation* is what they actually mean. We shall do the same.

All measures of *inflation* are broadly comparable and are constructed similarly. Each measures inflation by comparing the current price of a shopping list or basket of items with the cost of the same goods or services at a certain time. This functions as the base period from which the rate of change is calculated. The *GDP deflator* mentioned earlier is a form of inflation indicator. Two others that are widely used are the *consumer price index* (CPI) and the *producer price index* (PPI).

The consumer price index measures the rate of inflation for consumer goods and services. It is based on a basket of goods and services designed to represent those consumed by a typical household. The producer price index (PPI) is similar to the CPI, but does not measure services and also encompasses a much different shopping basket from the CPI index, focusing on commodities and capital investment items. As an indicator of inflation, it functions in the same way as the CPI, although changes in

the price of commodities and of goods at the consumer level tend to be more rapid than at the consumer level.

In addition to the CPI and the PPI, the employment cost index, which is a broad measure of the cost of labor that takes account of all employee compensation expenses including benefits, is another important indicator of inflation. Other types of price indicators, which may be used to signal inflation or other monetary effects, are the prices of real assets. Among those that sovereign risk analysts consider are:

- Commodity prices, such as the price of crude oil per barrel or gold
- Equity prices, such as the S&P 500 index or Tokyo's Nikkei 225 Index
- Real estate prices, such as the average price of a house in a particular market

THE IMPACT OF INFLATION

Moderate inflation, in the range of 1 to 3 percent, is generally not regarded as a problem, and is often seen as concomitant with levels of healthy and sustainable growth. A higher rate of inflation is, however, viewed as something to be avoided since, particularly when unanticipated, it helps some borrowers at the expense of creditors, and in doing so increases uncertainty and risk levels in the money and capital markets, which can impair overall economic growth.

Inflation comes in several varieties. There is hyperinflation, such as occurred in Germany in the 1920s or in Zimbabwe in the 2000s, which resulted in economic chaos and paralysis. Stagflation occurred following the oil shocks of the 1970s. A sudden rise in oil prices decreased purchasing power resulting in reduced aggregate demand, pushing the economy into a tailspin. Stagflation sees unemployment and inflation rising simultaneously. Although inflation is often associated with an overheating economy, stagflation couples inflation with stagnant economic growth; hence the name.

Deflation, which occurred in the United States in the 1930s and less severely in Japan in the 1990s, is feared at this writing in a number of countries, as it means that prices are generally falling across a wide range of products and services.⁷⁴ While it might appear desirable, deflation is dangerous because it causes consumers to put off purchasing in anticipation of lower prices. It is apt to result in increases in real interest rates, and can trigger a vicious cycle of falling demand and production—a liquidity trap.⁷⁵

Since nominal interest rates seldom go below zero, central banks are more limited in their ability to control deflation than inflation.

The banking system is especially vulnerable to deflation because funds borrowed under normal circumstances become prohibitively costly for borrowers to service encouraging defaults and collateral declines in value. The result is asset impairment for the institutions affected, and possibly insolvency.⁷⁶

The Money Supply and Its Management

An evaluation of monetary conditions and monetary policy begins with the money supply. The *money supply* can be defined in different ways. As is shown in the box entitled “M1, M2, and Other Definitions of the Money Supply,” it is more than the amount of coins minted and currency printed by the government still in circulation.

M1, M2, AND OTHER DEFINITIONS OF THE MONEY SUPPLY IN THE UNITED STATES

Budget deficits can have a deleterious impact on the banking sector. First, insofar as they contribute to a higher perception of risk in the wider global market, internationally active banks based in the country affected will tend to have less access to foreign debt and capital markets or find them more costly than they otherwise would be. Purely at the domestic level, high budget deficits suggest that the government debt issues will crowd out private-sector issuance. This means that less of available savings will be channeled towards privately owned enterprises, making borrowing in those markets more expensive both for companies able to tap them directly and for banks that lend to companies unable to do so. Generally, this will make bank lending to such firms more costly, although banks may gain to the extent that disintermediation is discouraged.⁷⁷

M1. Measure of the U.S. money stock that consists of

1. Currency outside the U.S. Treasury, Federal Reserve banks, and the vaults of depository institutions
2. Travelers checks of nonbank issuers
3. Demand deposits at all commercial banks other than those due to depository institutions, the U.S. government, and foreign banks and official institutions, less cash items in the process of collection and Federal Reserve float
4. Other checkable deposits (OCDs), consisting of negotiable order of withdrawal (NOW) and automatic transfer service (ATS) accounts at depository institutions, credit union share draft accounts, and demand deposits at thrift institutions

M2. Measure of the U.S. money stock that consists of M1 plus:

5. Savings deposits (including money-market deposit accounts)
6. Small-denomination time deposits (time deposits, including retail RPs, in amounts of less than \$100,000)
7. Balances in retail money market mutual funds

Excludes individual retirement account (IRA) and Keogh balances (a tax-deferred retirement account for individual or small enterprises) at depository institutions and money-market funds.

M3. Measure of the U.S. money stock that consists of M2 plus:

8. Large-denomination time deposits (in amounts of \$100,000 or more)

9. Balances in institutional money funds, RP liabilities (overnight and term) issued by all depository institutions
10. Eurodollars (overnight and term) held by U.S. residents at foreign branches of U.S. banks worldwide and at all banking offices in the United Kingdom and Canada

Excludes amounts held by depository institutions, the U.S. government, money funds, and foreign banks and official institutions.

In the United States, the measure M2 is a frequently used definition of the money supply, which essentially includes all cash and currency, and bank and money market accounts other than large time deposits and marketable securities, which fall into other money supply definitions.

The Money Multiplier and the Reserve Ratio

The money supply is also affected by, among other things, the volume of bank lending as Exhibit 11.15 graphically depicts.

In actuality, banks are engines of money creation, and the regulation of the banking system is one of the government's primary levers in controlling a nation's money supply. While there may be no such thing as a free lunch, it is nevertheless correct to say that banks create money out of nothing. The reason banks are able to create money in this manner is because funds on account with banks, being used in the same manner as currency, are properly counted as part of the money supply.⁷⁸ The amount of money created depends upon the *reserve ratio* imposed upon banks by the government.⁷⁹ This remarkable fact goes far in explaining the importance of depository institutions to the state of the economy in which they operate. Hence, banks are not merely subject to the effect of changes in the money supply, but as *de facto* conveyors of government monetary policy, they also very much affect it.

One of the principal purposes of the reserve ratio is to regulate the process of money creation by banks and to make sure it does not get out of hand.⁸⁰ To manipulate the money supply, the government may alter this ratio from time to time. As it is raised or lowered, it becomes either less or more costly for banks to lend their funds, depending upon the amount of funds required to be deposited with the regulator. (These funds are a cost to the bank but earn little or no interest.) Accordingly, they are encouraged to increase or decrease the amount of funds they advance to borrowers.⁸¹

THE RESERVE RATIO AND ITS IMPACT

Governments almost invariably require that banks keep a fraction of deposits received as a reserve with the central bank (or as vault cash), which may not be

lent against. The *reserve ratio*—sometimes called the minimum *operational ratio*—is generally defined as the amount of *statutory reserves* (i.e., deposits not permitted to be lent) divided by total deposits sufficient to meet the percentage threshold set by the government.

The reserve ratio:

$$(\text{Statutory}) \text{ Reserve Ratio (or Statutory reserve requirement)} = \frac{\text{Bank reserves}}{\text{Total deposits}}$$

The reserve ratio establishes the level of deposits that must be kept in reserve relative to advances. For example, if the minimum reserve ratio is 10 percent and a bank's deposits equal \$100 million, then the bank must keep \$10 million on account with the central bank, and may only lend \$90 million. Those funds in excess of its required reserves are the bank's excess reserves. Ten percent is not an unrealistic amount for the reserve ratio.⁸²

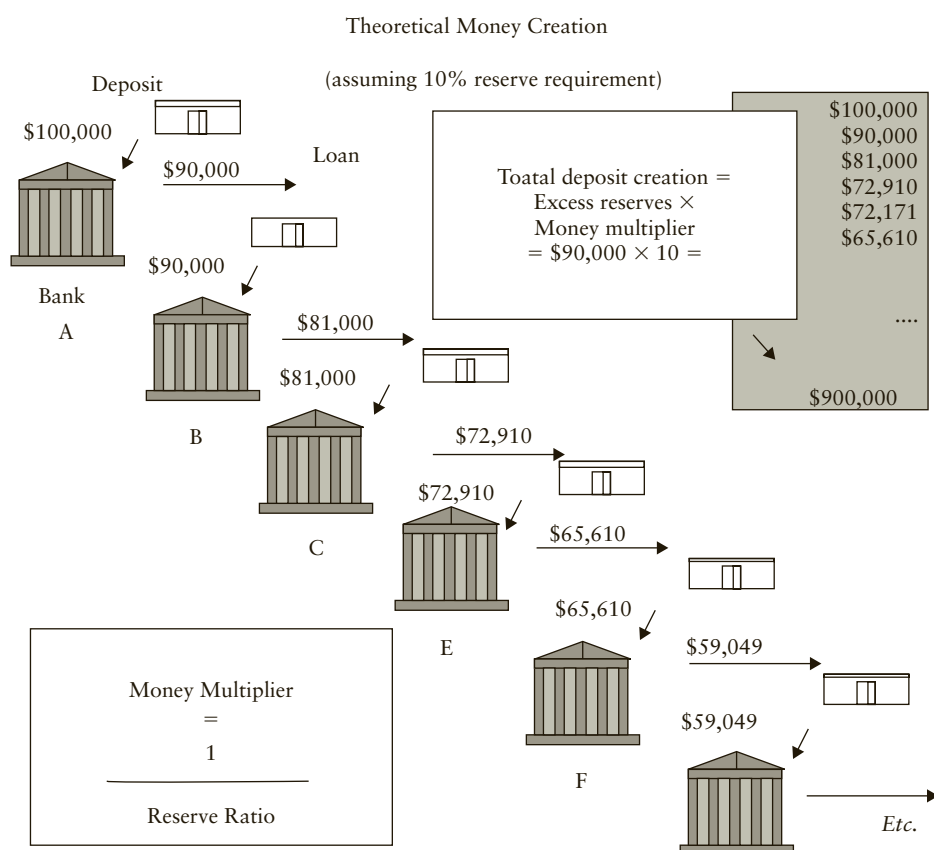


EXHIBIT 11.15 The Money Multiplier Effect

The Money Supply: Its Impact on the Economy and Its Management

Although the majority of, if not most, economists and central bankers do not agree entirely with all tenets of monetarism, discussed in the box below, one of its key principles is generally accepted. That principle is that *excessive money growth will trigger price inflation*. Therefore, sovereign analysts keep a look out for excessive money supply growth, which both triggers and functions as forerunner of price inflation. Exhibit 11.16 provides an illustration of the growth and relationship among these elements over a period of 30 years.⁸³

As a general macroeconomic principle, expansion of the money supply tends to bring about a rise in demand for goods and services. If the money supply expands too quickly, the increase in demand for the same number of available items can lead to price inflation. Inflation constitutes one of the predominant worries of most central bankers, who are wary of increasing the money supply too much for fear of sparking

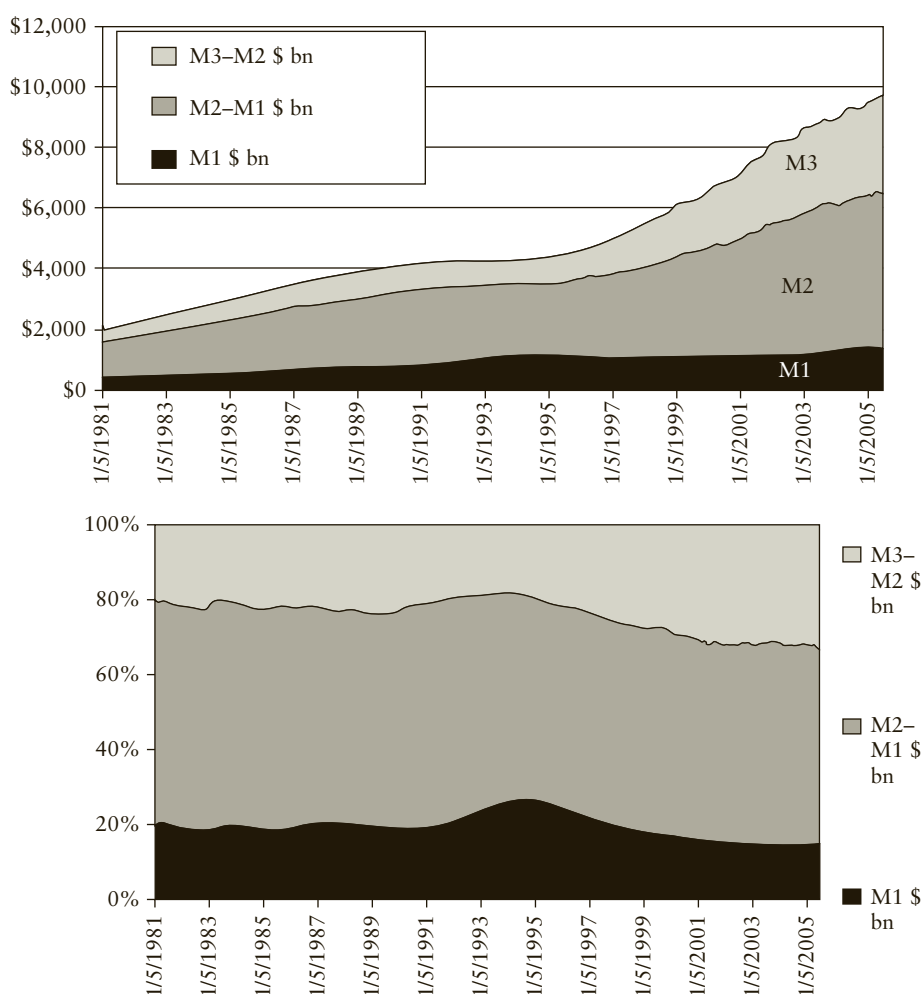


EXHIBIT 11.16 Growth of the Money Supply in the United States, 1981–2005

an inflationary spiral. In an inflationary spiral, inflation leads to the expectation of further inflation, which impels consumers to spend more (as tomorrow the price may be higher) but also for workers and producers to demand more in return for their labor and for their goods and services. Thus, inflation feeds upon itself.

MONETARISM

Monetarists believe that the central banks should focus on expanding the money supply at a rate that is steady and disclosed. Accordingly, they judge that the use of monetary policy instruments to attempt to stimulate or dampen the macroeconomy is basically counterproductive. Economists and policy makers have generally accepted one fundamental tenet of *monetarism*, which is that the money supply drives inflation. In the words of the famed monetarist Milton Friedman, “Inflation is always and everywhere a monetary phenomenon.” More specifically, “sustained money growth in excess of the growth of output produces inflation; to end inflation or produce deflation, money growth must fall below the growth of output.”⁸⁴

Hence it is widely agreed that a rapid loosening of monetary policy to allow the money supply to increase sharply is apt to trigger price inflation. Much less widely accepted is the monetarists’ dictum that the discretionary use of monetary and fiscal policy tools should be abandoned.⁸⁵ Mainstream economists, by and large, believe these tools have their uses. The consensus view is that, in the short term, tightening monetary policy will dampen economic growth (causing GDP and inflation to fall moderately, and causing real and nominal interest rates to increase somewhat); and that loosening monetary policy will stimulate economic growth (causing comparable increases in GDP and nominal interest rates and corresponding decreases in real and nominal interest rates) and intermediate between savers and borrowers by taking deposits from those with surplus funds (depositors) and lending those funds to entities in need of funds.

A bout of inflation is classically set off by a loose monetary policy that results in a growing money supply swollen by easy credit and aggressive bank lending. Such lax policies can be accompanied by the economy overheating, which means that it is operating at full capacity such that labor in particular is in short supply thereby pushing up wages and prices generally. When they perceive that inflation is about to accelerate, the usual response of central bankers is to reduce the rate at which the money supply is growing, if not to contract it. A number of tools, which are discussed further later in this chapter, are available that will normally aid in accomplishing this objective.

Perhaps needless to say, a central bankers’ other worry is the opposite phenomenon: that is, an economy at the brink of entering a recession, or even worse, a deflationary spiral. Severe recessions characteristically occur following an asset bust and are not infrequently accompanied by a financial crisis. In such circumstances, if the money supply does not expand rapidly enough, an economic slowdown of a

modest magnitude can become a more severe and prolonged recession. In extreme cases, the result can be price deflation.⁸⁶ Reduced demand brought about by, among other factors, a change of sentiment and substantial asset losses (e.g., equity securities, real property), pushes prices down, and causes production to fall. Resulting rising unemployment reduces demand further, thereby creating a potentially vicious cycle. In a deflationary spiral, expectations of continued falling prices causes demand to erode even more. Consequently, if the economy slows sharply and it appears to be entering a recession, central bankers will seek to take the opposite steps from those just mentioned in connection with stopping inflation. Whatever approach the central bank takes, monetary policy can almost always be characterized as becoming either tighter or looser, in other words restrictive or expansionary.

Monetary Policy Management Tools

To attempt to fend off a round of inflation or a recession by tightening or loosening monetary policy, there are three principal tools available to either slow or accelerate money supply growth. The use of the reserve ratio has already been discussed. The other two methods are (1) to alter the quantity of money in circulation by engaging in open market operations and (2) to encourage bank lending by lowering the rate at which banks are permitted to borrow from the central bank or otherwise relaxing terms and conditions. These methods are described in Exhibit 11.17 and further elaborated upon as part of the discussion on interest rates later in this section.

EXHIBIT 11.17 Monetary Policy Tools

Instrument	How It Works
The central bank can: <i>Alter the Reserve Requirement</i> Raise or lower the percentage of certain funding that banks may not lend.	By lowering or raising the reserve requirement, the central bank will both lower or raise the money multiplier and lower or raise the effective cost of funding to banks and encourage or discourage lending.
<i>Alter the Discount Rate</i> Raise or lower the cost of funding obtained by banks directly from the central bank.	By lowering or raising the discount rate, the central bank strongly influences the cost of short-term funding, which is passed on to borrowers and will lower or raise the cost of investment thereby encouraging or discouraging growth.
<i>Engage in Open Market Operations</i> Buy or sell government securities to or from banks.	By buying government securities from a bank, the central bank effectively injects new reserves into banks, which increases their supply of funds and causes interest rates to fall on the interbank market; alternatively, the central bank can sell government securities to banks, causing their reserves to decline and raising the cost of funding on the interbank market.

IMPLEMENTING A TIGHTER OR LOOSER MONETARY POLICY

A tight monetary policy is applied when the central bank wants to slow down the economy—to dampen inflation, for example. To do so, it may therefore do one or more of the following: (1) *raise the reserve ratio* limiting the supply of money and making it more costly for banks to lend; (2) *increase the discount rate* at which banks can borrow short-term funds from the central bank; or (3) *engage in open market operations*, as just discussed, selling government securities. All three will tend to cause interest rates to rise, reducing funds advanced, slowing spending, and dampening the economy.

In contrast, if economic growth is too sluggish and inflation is not a problem, the central bank will take the opposite actions, thereby relaxing its monetary policy. With the objective of lowering interest rates and expanding the money supply, it will (1) lower the reserve ratio, (2) decrease the discount rate, or (3) use open market operations to buy up more government securities from commercial banks.

Open market operations are a form of indirect intervention in the debt securities markets that allows the Federal Reserve Bank to influence interest rates in the direction it wishes. The Federal Reserve's website succinctly explains the process:

Suppose the Fed wants the funds rate to fall. To do this, it buys government securities from a bank. The Fed then pays for the securities by increasing that bank's reserves. As a result, the bank now has more reserves than it wants [which it] can lend . . . to another bank in the federal funds (interbank) market. Thus, the Fed's open market purchase increases the supply of reserves to the banking system, and the federal funds rate falls. When the Fed wants the funds rate to rise, it does the reverse, that is, it sells government securities. The Fed receives payment in reserves from banks, which lowers the supply of reserves in the banking system, and the funds rate rises.⁸⁷

Although a loosening of monetary policy will lower nominal interest rates and may temporarily lower real interest rates,⁸⁸ prolonged inflation will over time push up nominal high interest rates keeping real interest rates relatively constant. Inflation reduces the real return to lenders, and in an inflationary period, the nominal price of borrowing money ultimately goes up along with everything else. Consequently, market expectations will influence the effect of monetary policy changes. Over the long term—that is, where the policy actions have been anticipated by the market—loose monetary policy will contribute to inflation and high nominal interest rates, but the impact on real GDP, employment, or interest rates will be less than where the policy change has not been anticipated.

The Money Supply as an Indicator

As an indicator, the usefulness of the money supply depends upon the extent to which it registers the volume of liquid funds in circulation. In the United States, the money supply measures do not include funds invested in stock and bond funds, which may function as a form of relatively liquid assets.

According to Fed Chairman Alan Greenspan, “The historical relationships between money and income, and between money and the price level have largely broken down, depriving the aggregates of much of their usefulness as guides to policy.”⁸⁹ Nevertheless, the money supply and its components remain traditional indicators of inflation and continue to be used in sovereign risk analysis. The succeeding section examines other inflation indicators as well as interest rate and yield curve metrics.

Monetary Indicators

In addition to the money supply, other monetary indicators concern the same items with which monetary policy is concerned since they are inevitably linked directly or indirectly with the quantity of money in the economy. Aside from the reserve ratio, for instance, monetary policy indicators include measures of the availability of credit,⁹⁰ interest rates, and prices. Sovereign risk analysts are interested primarily in how current rates and prices compare to historic rates and to those in comparable countries, and are on the lookout for significant shifts in both.

Interest rates and prices are dealt with later as discrete categories. Interest rate indicators include the interbank rate, the prime rate, as well as various market-based rates. The official interest rate set by the central bank functions not only as policy tool but also as a sovereign risk indicator. A significant number is the real interest rate (the nominal interest rate minus the inflation rate).⁹¹ Finally, the yield curve constitutes a form of interest rate indicator. It reflects investors’ expectations about the future direction of the economy and is widely believed to have predictive value.

Prices are reflected in various indices, including inflation rate indices. The major inflation indicators are the consumer price index (CPI) and the producer price index (PPI). Some of the monetary policy tools discussed above also function as monetary indicators.

The Availability of Credit

Several ratios mirror the growth in the supply of credit available to prospective borrowers. These include the bank intermediation ratio, growth in credit (lending) and the loan-to-deposit ratio. The intermediation ratio is better at showing secular trends, while aggregate loan growth⁹² and the aggregate loan-to-deposit ratio provide a better indication on a cyclical basis of whether bank credit is expanding or contracting.

The Bank Intermediation Ratio The *bank intermediation ratio*—domestic (bank) credit to GDP (or bank assets to GDP)—was mentioned earlier as a measure of the extent to which bank finance has evolved within a market.⁹³ In a developing country, the banking sector typically grows rapidly from a low base as funds come out from under the mattresses and are deposited in bank accounts. This will cause a corresponding expansion in bank credit, as well as enlarge the money supply.

During a period when bank intermediation is rapidly increasing, it is not uncommon for banks to rapidly expand lending without paying sufficient attention to credit risk. At the same time, the country's banking inspection and supervision institutions may be struggling to cope with a rapidly growing banking sector. A fast rise in the intermediation ratio can portend the creation of an asset bubble, a corresponding potential decline in intrinsic asset quality, and a spike in banking system NPLs. Slow or even negative growth in this ratio once a threshold figure is reached is not necessarily unfavorable since it may be the result of a natural disintermediation, reflecting the lower relative importance of bank finance in consequence of the evolution of a country's money and capital markets.

Aggregate (Banking System) Loan Growth Aggregate loan growth constitutes the average change in loans and advances across the entire banking industry. As a general rule, lending increases year on year in an expansionary period, and declines during a recession. Loan growth rates in excess of 10 percent in a mature economy such as the United States can be taken as a potential indicator of the formation of an asset bubble that, when pricked, will be followed by a sharp downturn.

In addition to figures showing total credit growth, examining the growth pattern of specific types of lending can be useful. When an asset bubble forms in an elongated credit cycle, high rates of loan growth to the property sector or stock market, through margin lending, are frequently contributing factors. Exhibit 11.18 displays the high rates of mortgage loan growth in Hong Kong before the 1997–1998 financial crisis, and the sharp decline thereafter.⁹⁴

Aggregate Loan-to-Deposit Ratio Finally, the loan-to-deposit ratio for the banking system as a whole is a good indicator of the extent to which the local banking system is “loaned out”; that is, where banks have aggressively lent to the extent that good

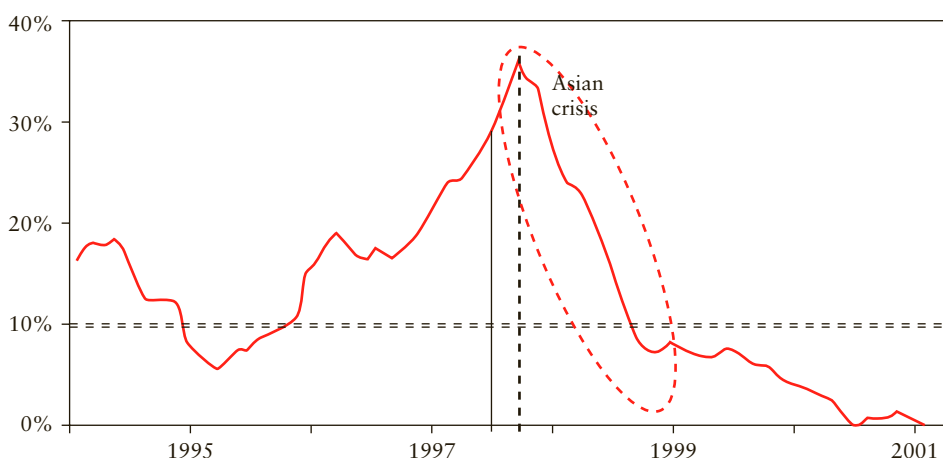


EXHIBIT 11.18 Hong Kong Mortgage Lending Year-on-Year Growth, 1994–2001

Source: Hong Kong government.

credit risks have become hard to find.⁹⁵ Generally speaking, ratios in excess of 100 percent should be considered elevated, if not excessive. Again, in Hong Kong, we can see the dramatic change in the loan-deposit ratio of the region's banking system during the boom and bust years of the 1990s.

BANK LENDING STANDARDS AS AN ECONOMIC INDICATOR

One empirical study found that changes in banks' lending standards had a higher predictive value than interest rates for projecting not just loan growth, which would be expected, but also real GDP and investment in inventory. All other things being equal, if banks are tightening their lending standards more than usual, the study found that "lower levels of loans and slower rates of output can be expected with a high degree of confidence."⁹⁶ The explanation for the stronger informational value of credit standards concerning prospective lending than interest rates is lenders are inclined to ration loans via changes in standards more than through changes in rates.

Interest Rates

Within an economy, interest rates, their prevailing level, and changes in those rates are important for a host of reasons. First, the level of interest rates affects the debt repayment ability of borrowers. When interest rates rise, companies' debt service burden generally increases, and the predictable consequence is often a decline in the profitability and the cash flow of the affected firms.⁹⁷ When interest rates fall, the opposite occurs. Interest rates therefore affect overall credit risk. Interest rates changes can have a positive or negative impact on banks and their risk profiles.

The impact depends in large part on the asset liability profile of the bank and upon significant unhedged trading positions it has taken. By extension, of course, interest rate changes can affect the financial health of the banking system as a whole. Finally, interest rate levels also affect the relative cost and availability of credit and consequently affect the growth of the money supply. Prevailing interest rates therefore directly affect a country's economic vitality and are, therefore, both economic indicators and metaphorical levers that can be adjusted to manage the economy. For this reason, the government, usually through its central bank, is often an important arbiter of interest rates. Adjustments are primarily made through the central bank's authority to set the rate at which it is willing to make short-term funding available to banking institutions.

Interest rate levels—more specifically, real interest rate levels—represent the time value of money, adjusted for credit risk and related factors.⁹⁸ The real interest rate is the nominal interest rate adjusted for the impact of inflation. To determine the real interest rate, the rate of inflation is simply subtracted from the nominal interest rate.

REAL INTEREST RATES

The applicable real interest rate is equal to the nominal interest rate minus the rate of inflation. The nominal interest rate and the real interest rate in a given market can vary dramatically.

Nominal interest rates in Hong Kong in the early 1990s were nominally moderate, but because of the high rate of inflation prevailing at the time, the real interest rate actually turned negative. Because Hong Kong employed a currency board system that pegged the local currency to the U.S. dollar, the (British) colonial government was proscribed from pursuing a monetary policy independent from that of the United States. Hence Hong Kong interest rates had to stay roughly in line with U.S. interest rates, notwithstanding that the economic situation in the colony was entirely different from that of the United States. At the time, the U.S. Federal Reserve Bank sought both to stave off a recession and to keep banks' interest rates costs low to enhance their profitability following the S&L crisis and the collapse of the junk bond market at the start of the decade. Although low interest rates suited U.S. financial conditions, such a loose monetary policy was inappropriate for Hong Kong given the levels of prevailing inflation and the economic and property boom the colony was then experiencing.

Real interest rates in Hong Kong ultimately turned negative, and unsurprisingly returns on investing in such an environment became very attractive. Although inflation gradually trended down, concerns about asset prices and inflation ultimately pushed both nominal and real interest rates up. The consequence was the creation of an asset bubble, which burst following the 1997–1998 Asian crisis. The same currency board regime that had prevented Hong Kong authorities from holding interest rates up forced a severe adjustment to the local economy when the asset prices collapsed and credit severely contracted. Real interest rates shot up as inflation turned into deflation.

Ireland at the end of the 1990s provides another example of the gap between nominal and real interest rates. Negative real interest rates also stoked a property boom in the Celtic Tiger and had a similar cause. The European Monetary Union forced then—and still does now—comparatively uniform interest rates upon countries that had widely differing economic situations.

In theory, real interest rates should not vary too much, and they typically hover around 3 percent.⁹⁹ Under some unusual circumstances, real interest rates in the real world can fall to zero or even below. This does not mean that the time value of money has disappeared. The phenomenon can occur, for example, where the central bank or other responsible agency has set the rate at which banks can borrow from it at an artificially low level, or where inflation outpaces expectations. The situation in Hong Kong in the early 1990s, discussed in the box earlier, provides an illustration. Likewise, real interest rates can rise to higher levels where, for instance, creditors become more risk averse owing to unstable market conditions.

Interest rate indicators, which can be drawn from yields on various types of financial instruments, are a form of market price indicator. In that sense, they resemble the CPI, PPI, and other inflation indices discussed in the preceding subsection. Interest rate indicators include the relevant local *prime rate*; *interbank rate* such as the federal funds rate in the United States, LIBOR in the UK; *official rate*; and the yield curve comprised of *yields on government securities* such as the 3-month U.S. Treasury bill and the 30-year U.S. government bonds.

The prime rate is the rate charged by banks for loans to their most creditworthy customers. The Hong Kong prime rate depicted in Exhibit 11.19 provides an example over the long term. The diagram shows the local prime rate charged by Hong Kong banks to lend to their best customers. The rise in the early 1990s was a response to inflationary conditions, but still lagged behind the rate of inflation. The rise in mid-1997 was primarily in response to the uncertainty and increased credit risk brought on by the Asian crisis.

The interbank rate is the rate banks charge each other usually for very short term or overnight borrowings, either directly or mediated through the central bank. LIBOR, which stands for the London Interbank Offered Rate, is the interbank rate among banks in London. The chart in Exhibit 11.20 shows the £LIBOR rate for the month of September 2007, an extremely volatile period for the market in the wake of the collapse of the Northern Rock building society.

Related to the interbank rate is a rate that, for the lack of a better term, may be referred to as the official rate. An example is the Bank of England official rate, which refers to the rate at which the central bank is willing to lend funds in pounds sterling to banks. It may represent a quasi-interbank rate as the so-called federal funds rate does in the United States. In the United States, the *federal funds rate* functions both as the official rate and as the interbank rate. Exhibit 11.21 shows this rate over a nearly 40-year time span.

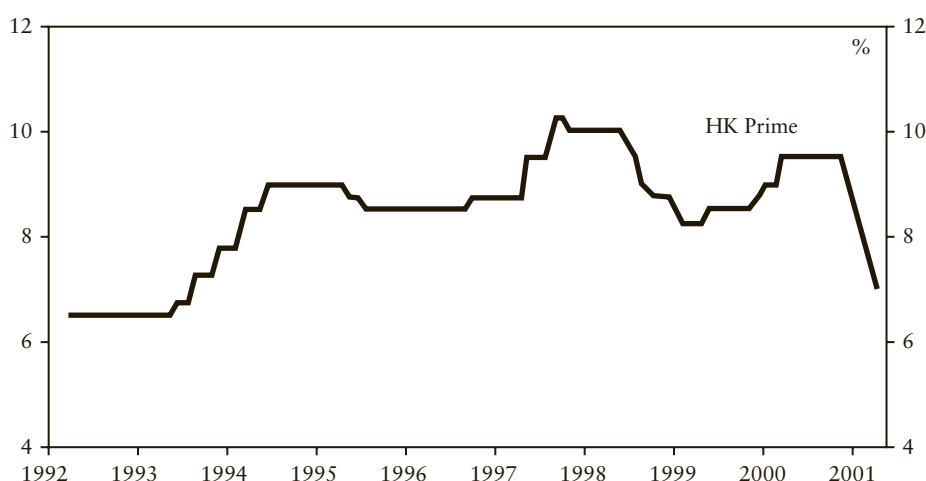


EXHIBIT 11.19 Prime Rate Charged by Banks in Hong Kong between 1992 and 2002

Source: Hong Kong Monetary Authority.

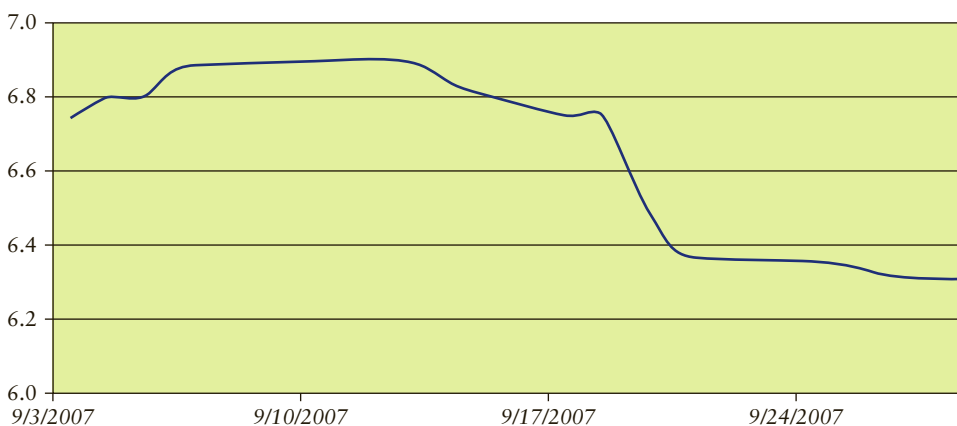


EXHIBIT 11.20 Pound Sterling LIBOR in September 2007

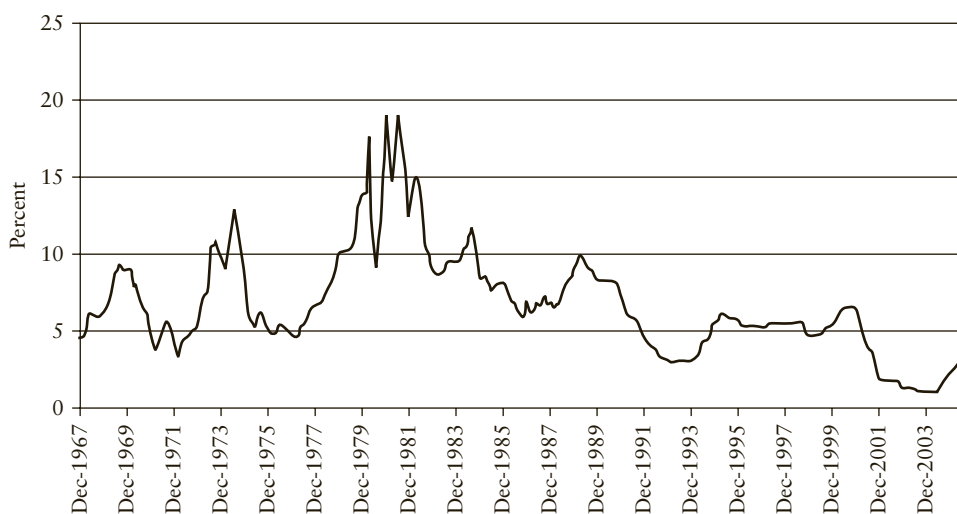


EXHIBIT 11.21 U.S. Federal Funds Rate (Interbank Rate), 1967–2005

The Yield Curve

The yield curve is a very important indicator, and it has an impact on bank earnings, liquidity, and asset quality. It is a major driver of risk management and risk mitigation strategies. For the sake of efficiency, yield curves are discussed in detail in this chapter rather than in each of the other relevant chapters.

Readers will be familiar with yield curves, and our purpose here is to show their impact on bank or banking system creditworthiness.

A yield curve shows interest rates in relation to another variable such as maturity, but also to credit quality. Classically, it is a graphic depiction of the interest rates for fixed-income securities (of the same credit quality) having varying maturities. Usually government debt issues are used to identify a benchmark yield curve that assists in pricing securities of lesser credit quality. The diagram in Exhibit 11.22

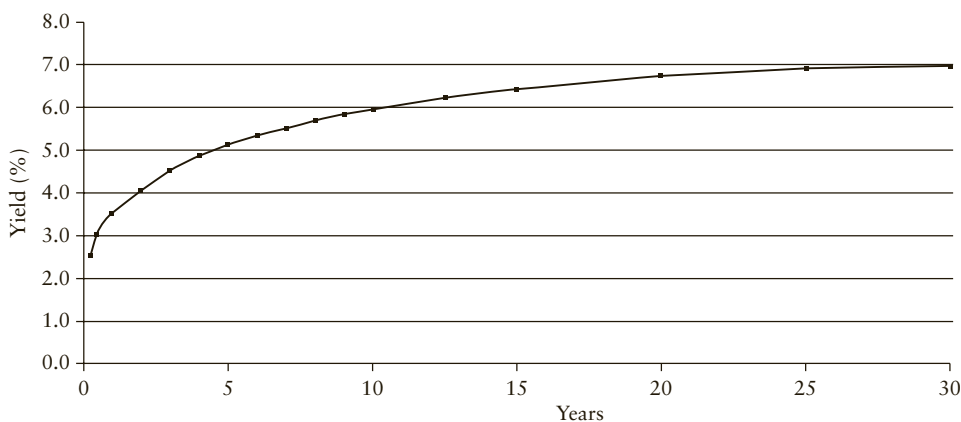


EXHIBIT 11.22 Idealized Benchmark Yield Curve

shows an idealized benchmark yield curve for maturities ranging from 3 months to 30 years. Observe that in normal circumstances it has any of the various shapes described further on.

Shape of the Yield Curve When interest rates in an economy change, rates for short-term funding and long-term funding do not necessarily change to an equal degree. As the shape of the yield curve changes, it therefore not only shows the overall relationship between yields and maturities, but also how this relationship has altered. What makes the yield curve interesting and useful is that it often signals changes in the economic cycle. The shape of the (benchmark) yield curve—whether it is mildly upward sloping, steeply upward sloping, downward sloping (“negative,” “inverted”), or flat/humped—functions as an economic indicator.

The Usual Yield Curve¹⁰⁰ The usual yield curve is a gently upward sloping one, which corresponds to a situation where short-term interest rates are somewhat lower than long-term rates. The gentle upward slope accords with the concept of the time value of money as well as the ordinary operative assumption that longer-tenor obligations bear higher credit risks. A yield curve with this shape signifies that investors in fixed-income obligations anticipate continued moderate levels of growth—that is, normal conditions—over the near- to medium-term, and that they have no particular concerns about the short-term or medium-long-term business climate. More specifically, they have no especial worries about inflation, nor do they expect dramatic changes in the economic environment. Hence, the curve expresses the ordinary expectation on the part of investors that funds put at risk for longer periods of time should generate a progressively higher rate of return.

The U.S. situation in the early 1980s is interesting. In December 1984, the U.S. economy recovered from a recession that ended in November 1982 and was expanding at a healthy pace, with annualized quarterly GDP growth in the 2 to 5 percent range. In this balmy economic environment, a classic normal yield curve emerged.

The Inverted Yield Curve Like a rapidly falling barometer, an inverted yield curve is an almost sure signal warning of an imminent economic contraction. In early 1981,

amid signs of stalling economic growth, the U.S. Federal Reserve Bank began loosening monetary policy by cutting short-term interest rates in an effort to stimulate the economy. Bond investors began to worry that this was their last chance to buy medium- and long-term bonds that were still paying historically high interest rates before the bottom dropped out of the market. For this reason, they were more than willing to forego their usual preference of higher returns for longer-term exposure for an opportunity to lock in these once-in-a-lifetime double-digit interest rates. Consequently, bond investors by August 1981 had already bid up the price of long-term securities and bid down yields (by bidding up the price, they were effectively willing to accept a lower interest rate in return) to the point that long-term bonds yields were lower than those of short-term securities. Retrospectively, August 1981 was already one month into a 16-month recession, the start of which was retrospectively dated to July 1981. By the first quarter of 1982, annualized GDP growth was deeply into negative territory, 30-year bond yields slid over the following five years from 14 to 7 percent while short-term rates, which at 15 percent had exceeded long-term rates, steadily plunged to between 5 and 6 percent.

The Steep Yield Curve A steep yield curve occurs when, for example, the spread between the yield on 30-year treasury bonds and on 3-month Treasury bills widens to unusual levels. In a normal yield curve, the spread is ordinarily about 3 percent. When it widens more than that, it means the bond investors want higher returns for putting their funds at risk for a lengthy period. Why might they have that attitude? It could be that they believe the economy is going to substantially improve in the foreseeable future. When the economy recovers, interest rates typically rise, as opposed to the fall in interest rates that frequently accompanies an economic downturn. In a rising interest rate environment, fixed-income securities tend to lose value. Hence, investors expect interest rates to go up, and want a higher return for longer-term securities to compensate them the loss in value of the securities that rising interest rates will bring. Short-term investors are not so concerned because over three months to a year, their downside risk is comparatively much lower, and they can benefit by purchasing higher-yielding securities when the opportunity arises.

The Flat or Humped Yield Curve In order for an inverted yield curve (discussed below) to form, it must first change from an upward sloping curve into a flat or slightly humped curve. Because the inverted yield curve is so often a reliable harbinger of a recession, bond investors are quick to exploit the trading opportunity it represents.

Not all flat or humped yield curves break into inverted formations, however, so this pattern is not as reliable a bellwether as the inverted yield curve. Nevertheless, its flat or humped counterpart does tend to be associated with a slowdown in the economy. In 1989, a humped pattern occurred when 30-year bond yields fell below 3-year bond yields for a 5-month period. Instead of transforming into an inverted shape, it took on a relatively normal appearance by early 1990.

Other Ways to Depict the Yield Curve In addition to their familiar shapes, yield curves can also be depicted by showing, over a long period, the spread between a long-term and a short-term interest rate, for example between the 1-year T-bill and the 30-year U.S. government long bond. This is called a “term spread.”

Naturally, a number of external factors may affect such depiction. Commentators seldom agree on the significance of the term spread trends or on their predictive value.

The Impact of the Yield Curve on Bank Earnings *A steepening yield curve.* In addition to being associated with a favorable economic outlook, a comparatively steep yield curve affords banks a better than usual opportunity to generate spread-related earnings based on effective asset liability management. By funding high-yielding long-term assets with cheap short-term funds, a bank is apt to be able to enjoy fatter spreads, particularly where it benefits from a sticky core deposit base.

A flattening yield curve. Where the yield curve is flattening, however, such opportunities diminish and increase interest rate risk, especially where a bank is dependent upon commercial funding or is locked into long-term instruments that are falling in value. The degree of risk is contingent on the degree of exposure that the bank has in various types of financial assets.

An inverting yield curve. Not only is an inverted yield curve a harbinger of an economic downturn, but the higher cost of short-term funding coupled with the outlook for lower interest rates ahead puts banks in a comparatively precarious situation. Short-term funding costs are relatively high, which discourage lending, while long-term rates may have a considerable distance to decline if a downturn materializes.

The diagram in Exhibit 11.23 illustrates the impact of the yield curve on bank margins. The y axis on the left is bank average net interest margins; the axis on the right depicts the yield spread between 3-month U.S. Treasury bills and 10-year notes. As we saw earlier, when the spread between long-term and short-term debt is increasing, the yield curve is steepening. In the United States, this occurred between 1990 and 1992 as the market anticipated better economic conditions ahead. Between 1992 and 2000, the yield curve resumed a more normal shape, and then steepened and flattened in three successive cycles before inverting in anticipation of the 2001 recession. A strong steepening was followed by mild flattening, which had repeated once by 2004. Observe how the one-year trailing net interest margin rose during periods of a steepening yield curve, and fell during a period when the yield curve flattened, suggesting that the majority of banks do better during conditions of curve steepening.

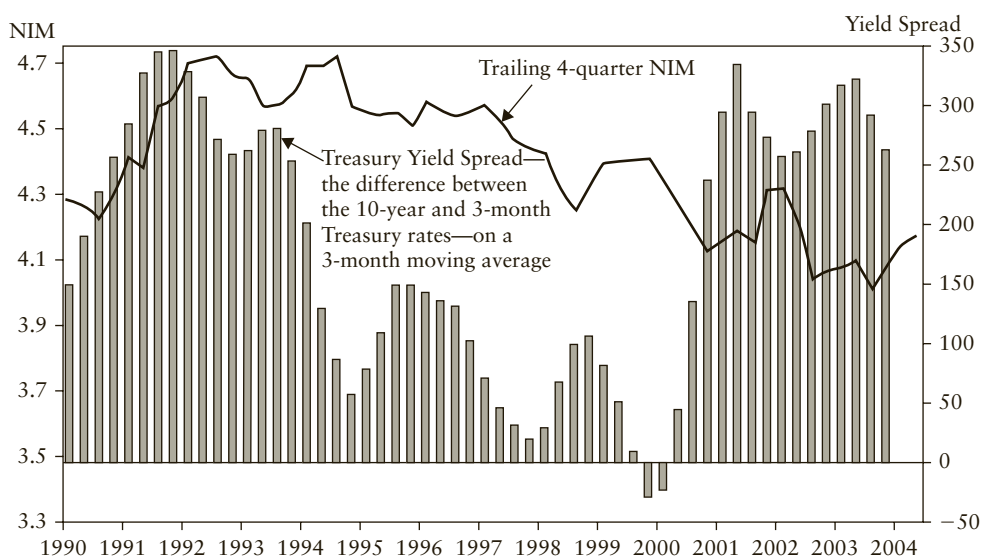


EXHIBIT 11.23 The Yield Curve and Bank Net Interest Margin

Source: U.S. Federal Reserve Bank and FDIC (NIM averages exclude “specialty banks”).

As mentioned earlier, as banks borrow short and lend long, even the usual yield curve makes their business intrinsically profitable since it widens net interest margins. A steep yield curve further helps them.

YIELD CURVE RISK

Yield curve risk is a form of market risk, specifically the risk that given a change in prevailing interest rates (yields), prices of fixed-income securities with a certain maturity (e.g., short-term securities) will shift more or less than the average change in interest rates. In other words, it is the risk that yield curves will steepen, flatten, or invert, rather than shift in parallel.

Bank Stock Prices as Cyclical Indicators

Bank equity analysts and investors in bank shares keep an eye on bank stock prices to make a profitable investment. But changes in securities prices may have some value both as economic indicators and as harbingers of a decline in a specific bank's financial condition or a banking system in its entirety.¹⁰¹ Banks are generally viewed as a *cyclical stock*, and empirical research confirms that returns on bank equity securities are "strongly asymmetric."¹⁰² There appears to be a consensus that success in investing in bank stocks depends in large part upon an analysis of macroeconomic conditions.¹⁰³ If that is the case, it would seem that bank stocks in turn might have some value as economic indicators in their own right.

A number of bank equity analysts have concluded that the prices and price-to-earnings ratios of bank stocks reach a zenith well in advance of an economic downturn, and that the severity and length of a recession and the correction in bank stock prices are correlated. These analysts observe historical patterns in which bank stocks ordinarily performed better than the market as a whole in the early stages of a recovery, especially during periods displaying moderate GDP growth, stable prices and steady or falling short-term interest rates.

The European debt crisis has offered interesting opportunities to study the behavior of bank stock prices under the stress of sovereign debt uncertainties.¹⁰⁴

Stock prices as cyclical indicators are substantially less relevant to banks in emerging markets where creative accounting might make it difficult for investors to take decisions, since published accounts and stock prices might not reflect the true state of affairs of a bank.

HOW THE ECONOMY AFFECTS BANK STOCKS

Trends in bank stock prices, as just discussed, may function as precursors of changes in the direction of the economy. But how does the economy in turn

affect bank stock prices? This, as might be expected, is a concern of bank equity analysts. In a report¹⁰⁵ focusing on the U.S. stock market, the following macroeconomic indicators were identified as signals of expected price changes in banking sector stocks.

Predicting an Upward Trend in Bank Stock Prices

Steep yield curve (see discussion in the preceding text)

Moderate GDP growth, i.e., 1 to 3 percent in a mature market

High or increasing purchasing manager's index

Expanding loan growth, with optimal levels being about 1 to 2 percent above historical GDP growth levels

Declining unemployment

Predicting a Downward Trend in Bank Stock Prices

Below average or above average GDP growth rates; i.e., < 1 percent or over 3 percent in a mature economy

Widening high-yield credit spreads

Declining loan growth

Rising unemployment

Low or declining purchasing manager's index

Trade Policy: Exports, Imports, and the External Debt Position

A critical component of sovereign analysis is an evaluation of a country's *trade balance* and its *external debt position*, together with its corresponding policies concerning exports, imports, foreign debt, and investment. A third, less often discussed variable, is the volatility of the trade balance or terms of trade, which embraces the frequency and degree of variation in its trade position. Consider a country that earns foreign currency primarily from exporting a single crop. Its trade position will be at the mercy of market fluctuation in the price of that crop.

In recent times, commodity prices were on a generally upward trend, a result of a combination of factors that include the substantial growth of China and other newly industrialized or emerging countries. Side effects also existed: for example, the rise of corn prices as a result of a quest by the United States to produce biofuels in a context of an overdependence on foreign oil. The volatility of commodity prices increased and made it difficult for some countries—such as Chile, which is a major exporter of copper—to manage its trade and the resulting impact on public finance.

The two attributes—the trade balance and the external debt position—are related. A country's trade balance, also referred to as its *balance of trade payments* is the difference between its exports and its imports. Under *balance of payments accounting*, a *trade deficit* (*current account deficit*) occurs when *imports exceed exports* and the balance must be funded through capital inflows, which take the form of proceeds from debt issuance to foreign investors (or foreign direct investment).

A chronic *trade deficit* will therefore tend to result in an elevated level of accumulated foreign debt, and a country in this situation will be referred to as a net debtor nation. In the same way, a *trade surplus* (*current account surplus*) occurs when *exports exceed imports* and such a surplus funds other nations' trade deficits. Instead of being a *net debtor*, a country in this situation will be a *net creditor* nation, lending or investing its surplus in foreign countries.

Note that a country's *balance of payments*, although it may be affected by the government's domestic *budget balance* (a result of its fiscal policy), is a wholly distinct concept, notwithstanding that there is a relationship between the two. Although the terms *surplus* and *deficit* are applied to both, the two accounts are best thought of as largely separate.

A country's *terms of trade* will have a major impact on both its *external debt obligations* and its capacity to meet them. The stronger a country's *net exports* and other sources of hard currency, the better able it will be to fulfill its external financial obligations. The evaluation of such capacity is, of course, the primary objective of sovereign risk analysis. Questions to consider in regard to evaluating terms of trade include the following:

- What comparative advantages does the country have in terms of exports?
- What is the country's export mix? Is it conducive to economic growth? For instance, is demand for its exports likely to grow?
- Are exports growing? Are imports growing? Are the rates of growth sustainable? What trends can be identified?
- Must the country rely on certain imports, such as petroleum, that make it vulnerable to a sudden change in prices?
- Is it dependent on a single export crop or any single commodity for the bulk of its foreign exchange earnings?
- How locked is the country to its current export/import mix?
- What sources of hard currency are available to the country? Does it have petroleum or other natural resources for which there is strong demand?
- To what extent have the country's natural resources been exploited? Are these resources sustainable? If not, how long are they likely to last?
- Does the country have good prospects of tapping unexploited resources? If they have yet to be fully exploited, what barriers exist to their exploitation?
- Is the country an exporter of manufactured goods? Does it have potential to be? What barriers exist to its developing a manufacturing industry?
- To what extent is the country dependent upon exports for hard currency? How strong is its own tourism a major contributor to foreign currency earnings?
- Is tourism growing, or does it have the potential to grow? What obstacles or vulnerabilities characterize the country's tourism resources?
- What is the country's capital balance (i.e., external capital position)?
- If imports exceed exports, what is the degree of the imbalance? Is it long-standing or of short and likely temporary duration?
- Is foreign capital financing productive investment, or is it funding conspicuous consumption? Are the country's imports primarily for investment purposes (e.g., to build physical infrastructure, purchase machines for factories) or for consumption purposes?

- To what extent is the country the victim of protectionism? To what extent is the country protecting noncompetitive industries?

The Structure of the Balance of Payments

The *balance of payments* (sometimes abbreviated BOP) is a way to account for transactions between entities within one country and other countries over a given period. It also provides a framework for understanding the relationship between the import and export of goods, services, and capital and its external debt position.

Current Account and Capital Account

The *current account* records international trade in goods and services and earnings on investments. The *capital account* records capital transfers, including loans to and borrowings from foreign countries, outbound and inbound foreign investment, and the purchase and sale of certain nonfinancial assets. Under today's global floating exchange rate regime—which for most countries started in 1973—the *official reserve account* performs a relatively insignificant adjustment function and therefore can be ignored for the purposes of this discussion. Putting aside this official *reserve account*, it is apparent that the balance of payments is equal to the balance between the current account on one side and the capital account on the other. In other words, the current account and the capital account are essentially the mirrors of each other. Any current account imbalance must be compensated for by a corresponding capital account imbalance and vice versa.

Balance of Payments

There is no such thing as a deficit or a surplus of the balance of payments, since the expression simply relays the fact that, in financial terms, what goes out of an economic entity must go in. There are of course fluctuations in the flows, in and out, related to current account transactions, on the one hand, and capital account transactions, on the other hand, with any imbalance—usually a modest adjustment—met by changes in the official reserves.

KEY POINTS CONCERNING THE BALANCE OF PAYMENTS

A deficit or surplus in a country's current account will result in a respective and compensating surplus or deficit in its capital account, and vice versa.

The reverse quasi-equivalence between the *current account* and the *capital account* means that any difference between a country's exports and imports (resulting in either a *current account surplus* or *current account deficit*) will be equal to the difference between capital outflows and capital inflows (creating a corresponding *capital account deficit* or *current account surplus*) to compensate for the difference, and vice versa a *capital account deficit* must exist to fill the gap when domestic investment *exceeds* domestic savings, a condition that will arise when a country is generating less savings (from both domestically generated income and from foreign trade) than it is investing domestically.

A *capital account surplus* must exist to fill the gap when domestic savings *exceed* domestic investment, which will occur when the country is generating more savings (from both domestically generated income and from foreign trade) than it can invest domestically. If a nation's domestic investment exceeds its domestic savings, then the difference must be financed by foreign investors; if a nation's domestic savings exceeds its domestic investment, then the difference must be invested in foreign countries either in the form of debt or equity (and aggregate foreign outbound foreign investment will exceed aggregate inbound foreign investment).

The current account balance is essentially equal to domestic savings less domestic investment.

The BOP is equal to the *current account* minus the *capital account* plus the *official reserve account*, each of which is further divided into subaccounts.¹⁰⁶

Despite the superficial resemblance in terminology, the balance of payments bears no other functional relationship to a company's balance sheet. For instance, unlike a company's balance sheet, it is not a cumulative measure. Instead, like GDP and a company's income statement, the balance of payments is a flow measure covering a designated time period, most commonly one year. Each of the foregoing main accounts is divided further into subaccounts—the merchandise account, the services account, the investment income account, and the transfers account—as shown in Exhibit 11.24.¹⁰⁷

EXHIBIT 11.24 The Structure of the Balance of Payments Account (including hypothetical data)

Structure of the Balance of Payments Account	Formula	Hypothetical Data
(1) CURRENT ACCOUNT		
(a) <u>Merchandise Account</u>		
Merchandise Exports	A	650
Merchandise Imports	B	850
Balance of Merchandise Trade – Surplus (Deficit)	$C = A - B$	<u>(200)</u>
(b) <u>Services Account</u>		
Service Exports	E	250
Service Imports	F	160
Balance of Services – Surplus (Deficit)	$G = C - D$	<u>90</u>
Balance of Goods and Services – Surplus (Deficit)—the “trade balance”	$H = C + G$	<u>(110)</u>
(c) <u>Net investment income</u>		
	I	(20)
(d) <u>Net transfers</u>		
	J	(30)
Current account balance – Surplus (Deficit)	$K = H + I + J$	<u>(160)</u>

Structure of the Balance of Payments Account	Formula	Hypothetical Data
(2) CAPITAL ACCOUNT		
(a) Private capital inflow	L	420
(b) Private capital outflow	M	290
Capital account balance – Surplus (Deficit)	$N = L - M$	130
<u>CURRENT & CAPITAL ACCOUNT</u>	<u>$O = K + N$</u>	<u>(30)</u>
<u>BALANCE – SURPLUS (DEFICIT)</u>		
(3) RESERVE ACCOUNT TRANSACTIONS	$P = -O$	30

The four subaccounts have the following functions:

1. The *merchandise account*¹⁰⁸ records total exports and imports of goods within the specified time period.
2. The *services account* records exports and imports of services.
3. The *investment income (net investment income) account* records income generated from foreign investments.
4. The *transfers account* records one-way transfers such as for foreign aid or workers' remittances.

To summarize, the *merchandise trade balance* is equal to the difference between goods exported and goods imported, while the *services balance* shows the same in respect to services. Transfers are one-way (nontransactional) transfers such as foreign aid or workers' remittances, while net investment income means net rents, interest, dividends, and profits from overseas.

Current account =	Merchandise trade balance		+	Services balance		+	Net investment and transfers	
	Exports of goods	Imports of goods		Exports of services	Imports of services		Net investment income	Net transfers

The equivalence between the current account and the capital account in the trade balance means that the difference between a country's exports and imports also measures the difference between capital outflows and capital inflows and vice versa. Unless restrictive of capital (investment) flows, however, government policies tend to have a more immediate impact on domestic economic conditions and therefore on the demand and salability of exports and imports than on capital flows. As a result, until the 1980s, the current account reflected in consumption of imports and the sale of exports more often drove the capital account than the other way round, and, consequently, market participants (and sovereign analysts) were more apt to focus on the

current account rather than on the capital account.¹⁰⁹ Globalization has since then changed the rules of the game, with the capital account playing an increasing role. Currency fluctuations will affect balance-of-payment values and, in fact, must be taken into account when evaluating them.

IMPACT OF CURRENCY FLUCTUATIONS ON THE BALANCE OF PAYMENTS

Since it is an account measurable in monetary terms, the currency selected to measure the current account and the capital account will have an effect on the calculated figures and may lead to distortions if large currency fluctuations occur between the currency used for measurement purposes and local currencies. Often, the balance of payments will be calculated in a reserve currency¹¹⁰ such as U.S. dollars or the euro. Consequently, if there is a substantial movement in the value of the local currency relative to the designated base currency, the results may be subject to misinterpretation.¹¹¹

Aside from maintaining a constant stable equilibrium, which is relatively rare in practice, it is common for countries to run a *trade deficit (current account deficit)*, or a *trade surplus (current account surplus)* for a number of years in succession, and each condition may be chronic or temporary in character, as shown in the matrix in Exhibit 11.25. Some countries remain in chronic current account deficit or surplus for decades, while others move from surplus to deficit or vice versa from time to time as economic policies and conditions change. South Africa's economy in the 1990s provides an illustration of a conspicuous shift from trade surplus to trade deficit over a 10-year period.

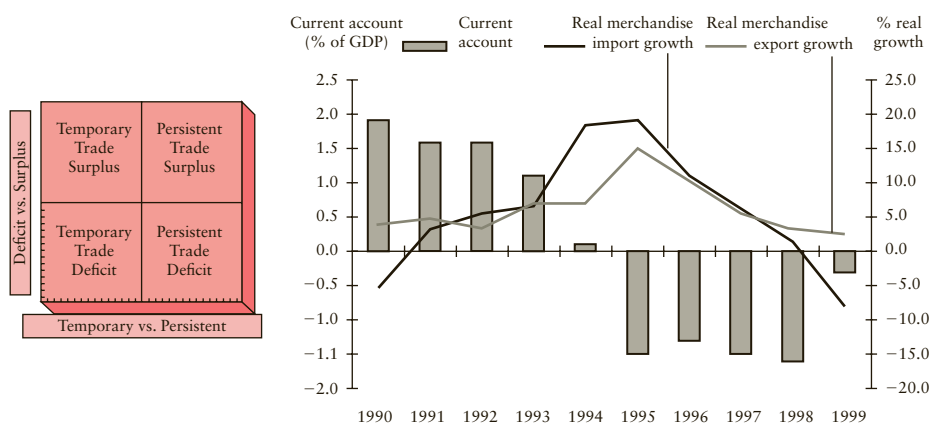


EXHIBIT 11.25 The Balance of Payments: Four Scenarios aside from Equilibrium and the Changing Trade Balance in South Africa in the 1990s

Source: Government publications.

Impact of a Current Account Deficit

When the current account balance is in deficit—that is, when imports exceed exports—the country will require an inflow of foreign capital.¹¹² A *deteriorating balance of payments* simply means that a country is importing more goods and services than it is exporting and thereby experiencing a rising *current account deficit*. This tends to be problematic since it means that a country is unable to fund its imports from exports and must increase its foreign borrowings or attract foreign direct investment (FDI). Hence, the magnitude of the current account deficit functions as an indicator of sovereign risk. To make it more amenable to comparison, however, it is helpful to compare the level of the deficit to the size of the economy. Hence, a useful indicator of the scale of the current account deficit is to express it as a percentage of GDP.

Oil-rich countries would usually exhibit large current account surpluses relative to GDP. Countries with strong export-driven economies, like China, Korea, Japan, and Brazil, register substantial current account surpluses, but on a percentage basis the figures are low since these economies are so large.

In contrast, a number of the developing countries in Africa and Latin America exhibit high current account deficits in proportion to the size of their economies.

In contrast to a capital account deficit, a *current account deficit* means that the country is investing more domestically than it is saving domestically and that the remainder must be financed by *foreign savings*. If, for instance, more capital is being invested in factories, real estate development, and physical infrastructure than is being saved domestically, then the current account will be in deficit, and the funds to finance the investment will have to come from elsewhere. Such a country is likely to be consuming a high proportion of its income, with much of that spending being for imports. A capital account surplus will be required to balance the national accounts.

**CURRENT ACCOUNT SURPLUS NATIONS (NET CREDITOR NATIONS)
VERSUS CURRENT ACCOUNT DEFICIT NATIONS (NET
DEBTOR NATIONS)**

Any deficit in the current account must be balanced with a surplus in the capital account and vice versa. A country experiencing a current account surplus is one in which domestic savings exceed the amount required for domestic investment. This occurs when savings from domestically generated income, contributed to by earnings from foreign trade, result in a pool of domestic savings greater than domestic investment needs.

Countries such as Japan, which historically as a result of its strong export industries have maintained chronic current account surpluses, therefore must simultaneously experience a net *capital account deficit* or outflow. That is, they (or rather both government and private enterprise together) must collectively lend *more* than they borrow. The consequence is that such countries become *net creditor nations* rather than net borrower nations. Since the supply of capital available domestically in Japan is plentiful relative to investment,

domestic real interest rates tend to be low. In the same way, given a lack of investment and lending opportunities at home, domestic capital in Japan—in search of higher yields—will tend to flow to debt or equity investment in other countries, the purchase of real estate and foreign debt securities, for example.

The United States has long been investing less than it has been saving while at the same time a high proportion of GDP has been taken up by consumption, including consumption of imports. Imports of clothes made in China, cars manufactured in Japan, electronic goods made in South Korea, and many other items have exceeded corresponding U.S. exports. As a result, the country's current account deficit has been steadily rising. Since the U.S. companies are investing more than U.S. residents are saving, the difference has to be funded by foreigners, whether in the form of Asian central banks holding U.S. government securities or in foreign direct investment in U.S. land and businesses. For the same reasons, the situation in major exporting countries is the mirror image of that in the United States. China, Japan, and Korea save more than they spend, and the difference represents the capital that partly funds U.S. spending.

Evaluating a Country's External Position

Absent FDI to compensate for chronic current account deficits may lead to elevated cumulative levels of foreign debt. Elevated levels of foreign debt require that a substantial proportion of the country's earnings go to pay debt service on past borrowing. Over time, excessive foreign borrowing will force a readjustment in the terms of trade, such as through a devaluation of the country's currency,¹¹³ making debt service even more burdensome. Onerous debt service obligations can have a crippling effect on the development of the domestic economy. For this reason, a rising current account deficit is traditionally viewed as an adverse signal, especially when it is financing consumption rather than productive investment.

In contrast, when a current account deficit is attributable to a buildup of a developing country's production capacity that will plausibly lead to higher export earnings in future, then a current account deficit will be viewed less negatively.¹¹⁴

A country's external debt embraces both *private sector debt* and *public sector debt* owed to nonresidents (i.e., mostly foreigners). Its external debt burden is an important element of sovereign analysis, but one should remember that it is the net balance that is relevant. Luxembourg and Hong Kong, for example, have large external debts, but those debts are mostly private sector debts balanced with substantial external assets.

Total Debt Burden

External debt indicators include gross external debt and net external debt (to which assets and liabilities associated to financial derivatives must be added), as well as a variety of ratios that relate debt and debt service levels to GDP, to exports, and to foreign currency reserves.¹¹⁵ In developed countries, which use a reserve currency or

a highly stable and convertible one, an alternative measure to external debt is the country's net international investment position. According to Moody's, however, calculation of this ratio is "notoriously difficult" and the ratio is not widely used.¹¹⁶

TOTAL (NET) EXTERNAL DEBT

Net external debt is equal to the debt incurred by the government and domestic private borrowers minus corresponding assets. Often, the figure employed by sovereign risk analysts is foreign currency debt as opposed to total external debt because traditionally this was deemed to be the important figure and also because it is easier to source.

As with the current account, a more useful measure for comparative purposes is to measure external debt in relation to the size of the relevant economy, that is, as a percentage of GDP. A high level of net exports provides the wherewithal to service external debt, so in addition to taking account of external debt relative to GDP another indicator—that of exports to external debt—should also be considered.

Measuring Debt Service Capacity

Foreign currency obligations must usually be paid for in a reserve currency,¹¹⁷ that is a currency that is accepted worldwide for the payment of debts. Countries are able to gather hard currencies with which to fulfill international obligations in one of several ways:

- Earnings through exports
- Tourism and other inbound sources of foreign currency
- Lines of credit to foreign currency
- Foreign currency reserves

The purpose of looking at foreign currency earnings (e.g., export earnings) and foreign currency reserves is therefore to ascertain whether the country's sources of hard currency are adequate to meet its immediate and medium-term needs. Relevant questions are:

- Is the country's currency a reserve currency? If not, what sources of foreign currency earnings does it have?
- Does the country have other sources of foreign currency funding on which it can draw?
- What is the level of the country's foreign currency reserves? Does it have foreign currency reserves adequate to its needs?

Debt Service Ratio The debt service ratio measures, for a given year, a country's total debt interest and principal repayments in relation to its annual export earnings

or revenues. Since it takes in a number of debt attributes such as interest rates and some attributes such as term structure on the one hand, and export earnings on the other, it is a prime indicator of the country's overall relative debt burden. A disadvantage of this ratio, however, is that it does not reflect problems associated with a high proportion of short-term debt to total debt stock. One should remember that short-term debt is subject to nonrenewal risk,¹¹⁸ and hence the debt service ratio is not a good indicator of potential problems associated with that risk.¹¹⁹

The debt service coverage ratio is the inverse ratio of annual debt service divided by annual export earnings.

Import Coverage Ratio as a Measure of the Strength of Foreign Currency Reserves As a sovereign risk indicator, import coverage or import cover is a proxy measure of the level of a country's liquid foreign currency reserves¹²⁰ expressed as a multiple in months of its current imports.

It reflects a country's access to liquid foreign currency reserves relative to the country's current external transactions requiring foreign currency. As such, it is an indicator of possible economic distress in the short term. As a rule of thumb, import cover of less than two months is viewed as a serious warning signal. Exhibit 11.26 shows the relationship between import cover and foreign reserves.

As foreign reserves increased, so did import cover, rising from relatively low levels for most of the 1990s to more comfortable levels by the end of the decade.

Foreign Currency Reserves Countries whose local currency is not a reserve currency, however, must maintain foreign currency reserves. Invariably under the control of a country's central bank, they provide an emergency pool to guard against the sudden withdrawal of the foreign credit needed to pay for imports. The need for foreign currency reserves is greatest when the local currency is volatile, nonconvertible, or not widely accepted. These reserves may be in the form of gold or silver bullion or, as is increasingly the case, in deposits in various reserve currencies as well as government bonds or other forms of very high-grade debt securities denominated in such currencies.

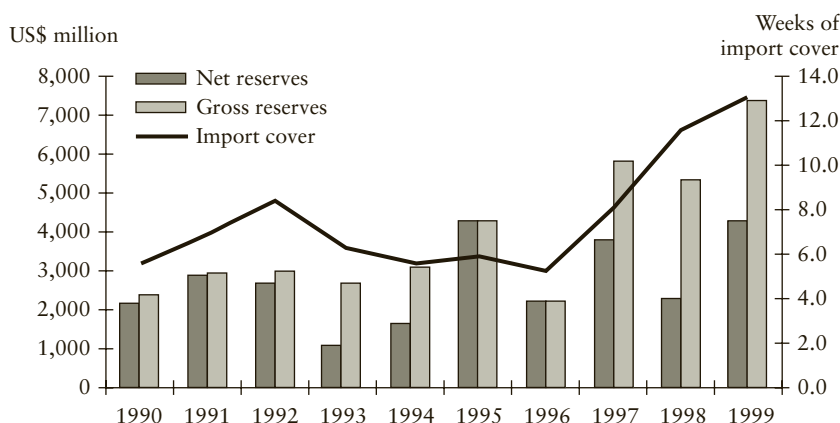


EXHIBIT 11.26 Import Cover in South Africa in the 1990s

Source: Government publications.

Typically, a country's foreign exchange reserves are measured in U.S. dollars at the current exchange rates prevailing at the end of each year. Comprehensive data concerning a country's foreign currency reserves may not always be available, although estimates usually can be found. Reported reserve levels can provide a false picture of actual reserve capacity in certain cases, such as where the central bank has sold foreign exchange forward.

The headline figures are misleading because the foreign reserves might be affected by a number of transactions that do not necessarily translate into improvement or deterioration of the nation's account. A single dollar lent by the World Bank to a poor country immediately increases that country's foreign reserves. Nearly 90 percent of China's reserves in the middle 1990s—a small fraction of its reserves in 2012—were borrowed money.¹²¹ A lot could be said about the flawed perception of the real meaning of such indicators during the 2011–2012 debt crisis in Europe.

M2 Compared to Foreign Currency Reserves Another measure of the protection afforded by foreign currency reserves is obtained by dividing the size of the money supply by such reserves. Recall that M2 comprises most bank deposits plus cash and cash equivalent. The resultant ratio provides an indicator of the potential stress resulting from capital flight.

Putting aside the reserve currency issue, evaluation of a country's debt service capacity brings us back where we started: to the current account and an assessment of the strength of exports relative to imports, as well as related vulnerability criteria. By looking at trends in exports and imports, as well as export potential, we can gain an understanding of what scenarios are likely to develop of the near-to-medium-term and how these may affect the ability of a country to service its external debt burden. The indicators below are relevant to this process. Changes in exports and imports¹²² may be expressed in nominal terms (that is, in respect of a nonlocal base currency such as US\$ or euros) or in real (that is, local) terms.

TRENDS IN THE EVALUATION OF A COUNTRY'S EXTERNAL DEBT BURDEN

Lines of Inquiry

- What is country's comparative external burden? (See e.g., net external debt.)
- What is the proportion of external debt incurred by private enterprises vs. the state sector and by financial institutions vs. nonfinancial companies?
- How is this allocated by various categories of creditors (i.e., multilateral institutions, foreign aid, bank debt, securities issues, supplier finance)?
- Does it generate sufficient foreign currency earnings from trade to support its foreign currency debt? (See various debt service ratios in this section.)
- What controls does the national government place on borrowings by its agencies and by subnational governments or state enterprises?
- How does the national government monitor such debt?
- As a whole, how sensible is the government's debt strategy?

External debt matters most in developing countries that do not have highly evolved local currency capital markets capable of providing the requisite finance for economic growth. Other contributing factors that may increase the risks of elevated external debt levels are a weak or historically unstable local currency and high rates of inflation, which are damaging both to domestic and nonresident investors in local currency debt. In contrast, external debt burdens are of comparatively less concern in developed countries where the capital markets are sophisticated and highly liquid, and that use either a reserve currency or one that is regarded as highly stable.

As with other forms of credit analysis, the larger the borrower's debt burden, the more likely that it will face difficulties servicing the debt. Traditionally, the critical external debt figure was foreign currency debt, and a major focus of sovereign risk analysis was on a country's central bank to be able to supply the foreign exchange necessary for domestic borrowers to repay foreign currency obligations on a timely basis. The capital account has become more relevant, since many emerging countries have opened their doors to foreign investment, and since generally capital flows more freely among most advanced countries. Foreign investors are more willing to accept currency risk by holding assets denominated in a local currency than in the past. This is in part due to a quest for higher returns and greater portfolio diversification, as well as in some instances the increased ability to hedge risk through interest-rate swaps. In addition, deregulation of capital flows by the authorities in various countries has widened the field in which such local currency investments may be made. The consequence, however, has been to heighten systemic risk since foreign investors are now more sensitive to changes in domestic economic conditions as a result of their greater exposure to currency risk.

Nowadays, adverse changes in both local macroeconomic indicators, as well as balance of payments indicators, are more prone to trigger capital flight than may have happened in the past when capital controls restricted the free movement of capital and most foreign investors held obligations denominated in a reserve currency. The freer movement of capital, while bringing with it many benefits, has also given rise to some negative side effects in the form of a greater propensity to currency crises and contagion than was experienced during the preceding era of fixed exchange rates and restrictive capital controls.¹²³ Hence gross external debt including both foreign currency obligations and local currency obligations to nonresidents is increasingly used as a supplementary sovereign risk indicator.

Other Indicators

Other indicators relevant to gauging a country's debt burden include those relating to specific attributes of the external obligations owed by a country's government, and its resident firms and individuals. These include the following:

- Amortization schedule and currency composition of gross debt
- Proportion of debt at floating interest rates

- Average terms of such debt as a spread over LIBOR
- CDS prices
- Record of recent foreign currency fixed-interest-rate bond issues, their coupon, price, maturity, and launch spread over the equivalent bond of the currency's government
- Maturities: short term vs. long term

Savings and Investment Rates

The topic of savings and investment rates falls mainly within a discussion of GDP and not within terms of trade, but is addressed here since it indirectly affects the latter and is frequently used to differentiate economies.¹²⁴ Under the GDP equation, income not used for consumption represents savings available for investment. *Investment* refers to all investment in capital goods by the private and public sector and thereby takes account of all expenditures on equipment and the construction of housing and other physical infrastructure, including, for example, factories, roads, ports, and telecommunications facilities. Obviously, such investment is critical to economic growth just as investment facilitates the earnings capacity of a single firm.

DEFINITIONS OF SAVINGS AND INVESTMENT

Gross savings may be defined as income not spent on consumption (GNP minus private and public consumption).

Gross domestic savings can be defined as “the difference between GDP and total consumption,” where total consumption includes consumption both by households and by the general government.

Gross national savings are equal to gross domestic savings, adjusted for net income received by the country's residents from capital invested abroad, and net transfers from other countries.

Both *gross national savings* and *gross domestic savings* are used as indicators in themselves and in ratios, but the national figure tends to be favored for ratio use.

As with GDP and GNP values, the corresponding indicators are usually fairly close to each other but can, in some countries, vary significantly.

Gross domestic investment encompasses gross fixed investment—all expenditures on capital goods (capital expenditure), including plant, equipment, and housing—as well as changes in inventories that are excluded from gross fixed investment.

Investment can be divided into *investment in fixed assets* and *investment in inventories*, and is typically expressed in gross terms, that is, gross domestic investment. Investment is also referred to as *gross capital formation*, which is the favored term for the World Bank, for example. The bank defines it as “outlays on additions to the fixed assets of the economy plus net changes in the level of inventories.”¹²⁵

For ratio purposes, *gross domestic investment* is used for the most part rather than gross national investment, although some countries report only the latter.

A country's investment rate can be defined as the percentage of total output that is directed toward investment as opposed to consumption. Conversely, a country's *savings rate* measures the extent to which its residents channel national income towards savings or towards consumption.¹²⁶ But what if investment exceeds savings? As suggested, this can only happen if capital in the form of debt or equity comes into a country from elsewhere. Insofar as the investment rate differs from the savings rate, it has an impact on the country's balance of payments position. Should investment exceed national (or domestic) savings, then an investment gap arises that requires a corresponding inflow of foreign capital. Conversely, a country with a current account surplus will generally have a high proportion of investment relative to consumption and funds available to lend or invest overseas.

High savings rates are associated with strong GDP growth, as Exhibit 11.27 generally illustrates. In developing countries in particular, a high national savings rate is normally viewed as beneficial, unless circumstances warrant a contrary view. Some of these countries achieved savings rates greater than 30 percent of GDP.

The East Asian miracle, which preceded the Asian crisis of 1997, was explained in large part by the high savings rates of the so-called Asian Tiger economies, as can be seen in Exhibit 11.28, with only the Philippines behind the rest of the pack.

Savings rates are strongly influenced by government policy, and some would add, by cultural considerations. Among the forces that affect savings rates are:

- Taxes
- Real interest rates on deposits or other investment vehicles
- Confidence in the formal financial system
- The existence of social welfare programs
- Government programs or policies that affirmatively encourage or compel citizens to save (or to consume)

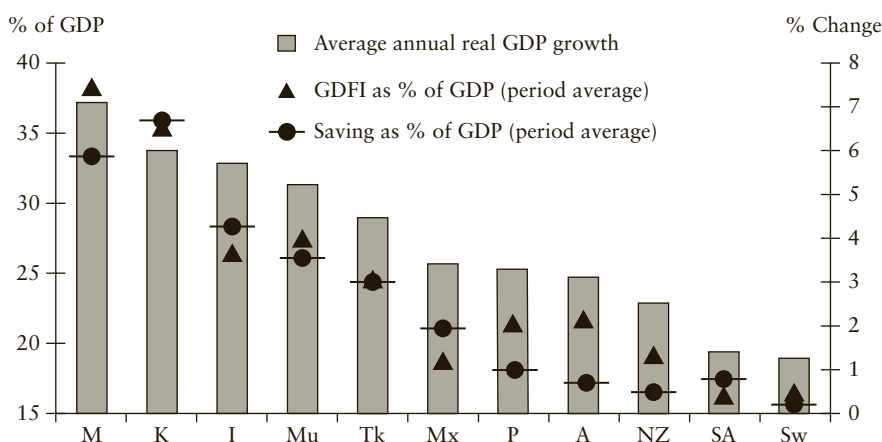


EXHIBIT 11.27 Savings Rates and GDP Growth Comparisons

Source: South African budget, 2000. Countries shown from left to right are: Malaysia, South Korea, Indonesia, Mauritius, Turkey, Mexico, Paraguay, Australia, New Zealand, South Africa, and Sweden.

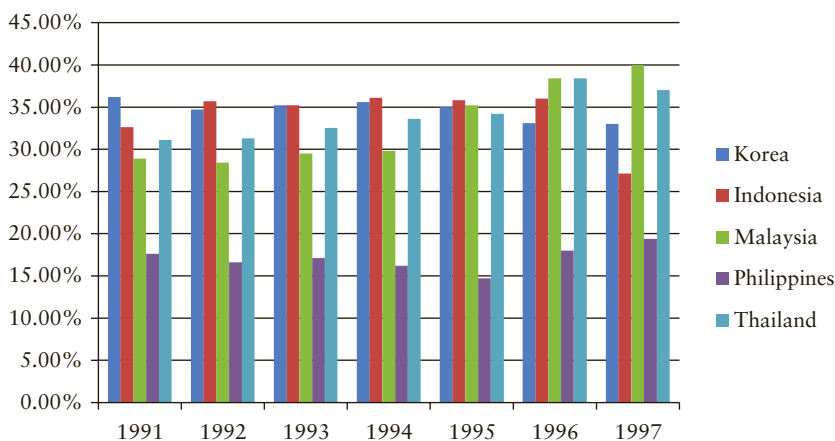


EXHIBIT 11.28 Domestic Savings Rate in Asia Prior to the 1997 Crisis

Source: Government publications.

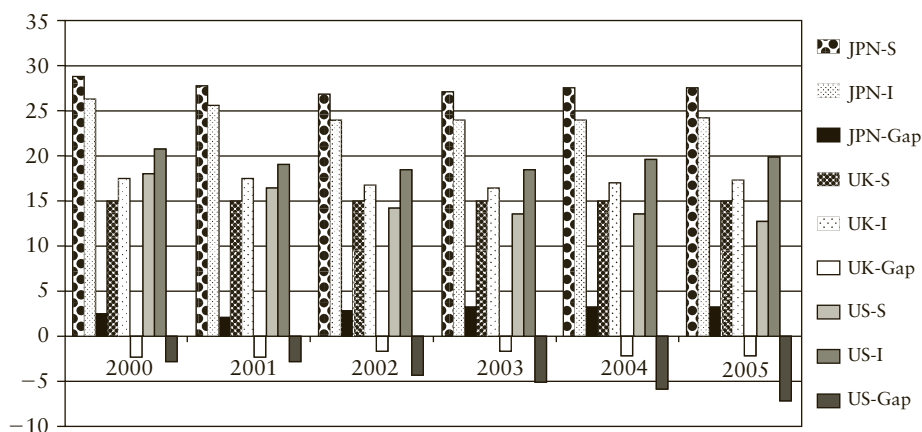


EXHIBIT 11.29 Investment Gap: Savings Less Investment in the UK, the United States, and Japan, 2000–2005

Source: Government publications.

Like high savings rates, high investment rates are generally viewed favorably. An unduly low investment rate, for instance under 10 percent, would suggest overconsumption at the expense of investment and that manufacturers are merely replacing machinery and other durable goods when they wear out and that would not bode well for economic growth. At the extreme, however, high investment rates can raise questions about the sustainability of economic growth and potential overcapacity. Also, if not funded domestically through adequate national savings, they will cause the current account deficit to widen.

The diagram in Exhibit 11.29 shows the gap between the savings rate and investment rate in the United States, UK, and Japan between 2000 and 2005. Observe that during the period displayed, savings in the United States did not cover

investment, resulting in a rising *investment gap*. Japan, in contrast, exhibits the opposite situation with its savings rate well in excess of its investment rate.

Where an investment gap exists, long-term capital in the form of foreign direct investment (FDI) is usually regarded as credit-positive, particularly in emerging markets, as compared with debt. There are several reasons for this. First, if the country's investment exceeds savings, FDI can fill the investment gap without raising external debt levels. Second, FDI usually brings with it the transfer of technology that can enable the country to develop its own industries and expand its exports. Disadvantages of FDI include the outflow in royalties, license fees, and repatriated profits that will have an unfavorable impact on the country's current account, as well as associated intergroup advances within the investing firm that will increase external debt levels. In addition, foreign ownership and control of domestic firms, whether private sector firms or privatized state-owned companies, can be politically sensitive, especially in strategic industries, and may be regarded as a threat to national sovereignty. Also, foreign investment in real property may contribute to a speculative asset bubble and also raise political concerns. Another concern relates to the quality of investment. Investment in manufacturing facilities might be regarded as being of higher quality than speculative investment in property. In the same manner, within a given sector, the more productive the investment, the better it is in economic terms.

CAPITAL PRODUCTIVITY

Not all investments are created equal. One investment may be highly productive, another one an utter waste. Measuring the amount of capital investment says little about the amount of output, and indirectly revenue, generated by the same input of funds. In one country, a great deal of investment might go to the building of lavish palaces for the ruler and his family, while in another the same amount of funds might be directed towards the construction of factories, roads, and ports. The country whose ruler directed capital towards the construction of palaces is not as likely to have the same output per unit of investment as one where investments have been made more judiciously. Another indicator called the *incremental capital-output ratio* (ICOR) is a measure of output generated by a given quantity of investment.

For comparative purposes, FDI is usually measured relative to GDP. That ratio is always comparatively low in highly developed countries such as the United States and Japan.

NOTES

1. P. J. Nagy, *Country Risk: How to Assess, Quantify, and Monitor It* (London: Euro-money Publications, 1984).

2. Robert Morris Associates, *A Guide to Analyzing Foreign Banks* (Philadelphia: Robert Morris Associates, 1988), 11.
3. The assessment of a country's legal infrastructure conventionally forms part of political risk analysis. The more particular reference here is to those laws and regulations that specifically concern, or target, banks and related entities. The overall legal and regulatory framework—its effectiveness together with the substance of the laws and regulations themselves—as well as those laws and regulations specifically governing the banking industry will affect banking system fragility, and in turn the creditworthiness of particular institutions. It is through both the broader framework and more specific regulatory regime that government policies affecting banks are implemented, and as such a link exists between the political risk component of sovereign risk and the regulatory component of systemic risk. The political aspects of sovereign risk analysis, along with economic risk, are discussed in this chapter. Fiscal, monetary, and trade policy are also discussed in more detail later in this chapter.
4. For example, successful populist movements may result in the enactment of laws severely restricting creditors' rights, to the detriment of the banking industry. In the same manner, a market-oriented government is likely to have as its goal the maintenance of a strong and healthy banking system, and to promulgate laws and regulations accordingly.
5. See generally Peter S. Rose and Milton H. Marquis, "*Money and Capital Markets: Financial Institutions and Instruments in a Global Marketplace*," 10th ed. (McGraw-Hill/Irwin, 2007). In this chapter, we apply the term *financial system* in a similar fashion, with a particular emphasis on the structure of the financial-services industry, while distinguishing between the legal and regulatory framework governing commercial transactions generally, and those rules having a direct or significant impact on banks and related-financial institutions.
6. George G. Kaufman, "Banking and Currency Crises and Systemic Risk: A Taxonomy and Review," presented at the Conference on Regulation and Stability in the Banking Sector at the Bank of the Netherlands, Amsterdam, Netherlands (November 4–5, 1999), 19.
7. Barry Eichengreen and Andrew Rose, "The Empirics of Currency and Banking Crises," <http://haas.berkeley.edu/~arose>.
8. Virginia L. Manzer, "Financial System Stress And Sovereign Credit Risk," *Standard & Poor's*, December 1997, p. 1.
9. For the rating agency Moody's Investor Service, the salient determinant of sovereign risk is whether the borrower has access to foreign exchange with which to satisfy external obligations, and this may depend on both the ability and the willingness of the central bank to make it available. A lack of foreign currency could arise out of economic fundamentals, or be imposed by the government (often to preserve what little foreign currency remains for the purchase of vital goods and services). Risk of this nature is quite similar to *transfer risk*, which is the risk of being unable to transfer funds transnationally as a result of the imposition of regulatory (or other) obstacles. Without the necessary foreign currency needed to fulfill financial obligations incurred in that currency, default would inexorably follow.
10. As commonly used, the term *sovereign risk* embraces two meanings that parallel the differences in the leading quotations. It may refer either narrowly to the risk of sovereign default—that is a government defaulting on its own obligations—or more broadly to country risk. The definitions below illustrate the same bifurcated meaning: "[Sovereign risk is] the risk to a lender that the government of a sovereign state may default on its financial obligations"; or "Also, the risk to a lender that unfavorable changes in the borrower's overall currency exchange position might imperil the payment of a loan"; or "Also the risk to a lender that unfavorable political events in the country of the debtor might imperil repayment" as suggested by Trade and Export Finance Online, www.tefo.org/trade-finance-glossary/s.html. Included in this wider definition is the risk of loss

affecting a *direct investment* in a particular country arising from government actions or omissions having a deleterious impact upon the investment. When direct investment is international in nature, it is often referred to as *foreign direct investment*.

11. Hence, the term *sovereign risk analysis* or *assessment* as used here embraces both the risk of national government default and risk of negative operational impacts on banks attributable to state actions or omissions, to economic conditions, or to other relevant causes.
12. This not always the case, and situations may arise where a government continues to fulfill its sovereign obligations to foreign investors and counterparties while imposing harsh restrictions on foreign-owned entities operating locally. Moody's provides an illustration of a hypothetical circumstance where "a nation undergoing international liquidity crisis may choose to cut off foreign currency transfers of foreign bank branches even though it may continue to honor its own debt servicing." Moody's *Credit Risk Analysis*, 160.
13. Economic risk can be defined as the risk that a bank's financial strength will be adversely affected by change in macroeconomic conditions resulting in impairment in its creditworthiness.
14. Sovereign risk analysts, whether employed by external rating agencies such as Moody's or S&P, or internally within banks or other financial institutions, are concerned with evaluating the credit risk associated with particular countries, the credit impact of enterprises doing business in those countries, as well as the securities that their governments issue. In this sense, sovereign analysts are a specialized type of credit analyst. As such, in regard to state obligations they will consider a country's capacity and willingness to repay its creditors. Depending upon the type of the review undertaken, bank credit analysts were, at least until the European debt crisis started, less likely to be interested in the creditworthiness of state obligations, and more interested in any potential unfavorable impact of state policies as implemented through laws and directives upon a given bank. (An exception would be a sovereign analysis of government bonds on the part of a state-owned bank guaranteed by state.) Of particular concern to the bank analyst would be the impact of a deteriorating business or regulatory climate on the banking system.
15. It could be argued, however, that the sovereign ceiling is not impenetrable.
16. This difference is muted in countries that have weak or ineffectual legal systems. In such countries, the quality of the legal system may be such that it may as a practical matter be difficult or impossible to obtain redress, whether as a result of weak creditors' rights, an inefficient, dilatory, or corrupt legal system, or the inability to enforce a favorable court judgment.
17. David T. Beers and Marie Cavanaugh, "Sovereign Credit Ratings: A Primer," Standard & Poor's Rating Service, December 1998, 21.
18. Or capacity to support entities under its jurisdictions to perform their financial obligations.
19. For example, how effectively does the government manage monetary policy? How effectively does the government manage its fiscal policy and external debt position? What is the government's willingness to service its foreign debt obligations? Note that both political risk analysis and an assessment of management at the individual bank level are extremely important yet essentially qualitative in nature.
20. Of course, the parallels are not exact. At this point, it may be stretching the analogy a bit, but the structure of an economy could be said to be analogous to asset quality. What is the quality of the resources with which the country has been endowed?
21. This section examines sovereign risk analysis with a particular emphasis on first, political risk analysis and, second, macroeconomic analysis. Systemic and legal and regulatory considerations will be discussed later.
22. Such measures are proxies because they do not capture all the factors relevant to the utility of the attribute described as implicit in the following questions. Is the port linked

- to the rest of the country by a good network of road and rail? Is it staffed by a comparatively efficient and corruption-free workforce? Are the roads well-engineered and well-maintained, or potholed and difficult to traverse?
23. Of the two categories, perhaps the most important criteria in political risk analysis, at least in the short term, are those indicative of the probability of threats to stability.
 24. There are a number of specialized providers of country risk analytical services offering information on the Internet.
 25. Such a scoring approach is useful, but of course its quality will always depend on the way the qualitative opinion is calibrated.
 26. As a result GDP is probably the most important headline acronym in economic comments.
 27. Note, however, that a long-term chart (e.g., covering a period of decades) showing total GDP graphed using a logarithmic scale will resemble a straight line so long as the cyclical fluctuations are not too great or long-lasting.
 28. Recall that Chapter 8, which dealt with asset quality, discussed the relevance of the *business cycle* to a bank's asset quality and its overall financial condition.
 29. Indeed, it would not be too much of an overstatement to say that sovereign risk analysis is essentially a macroeconomic exercise with a bit of political prognostication thrown in.
 30. D. N. Weil, "Fiscal Policy," *Concise Encyclopedia of Economics*, (The Library of Economics and Liberty, 2005), www.econlib.org/index.html.
 31. In the same way that evaluating the quality and competence of a company's management is largely a matter of judgment, the appraisal of the merits of the government's economic management, for instance, is a particularly subjective and qualitative exercise. To a lesser degree perhaps, evaluating a country's competitive advantages and diversification of its economy, referred to in the section above, also involve a degree of subjectivity.
 32. Inflation plays an important role in the reliability of indicators, but the measure of inflation itself is often incorrect or biased.
 33. When figures such as GDP are compared against a reserve currency such as the U.S. dollar or the euro, as in an international comparison of GDP, inflation is usually adjusted for through the exchange rate conversion.
 34. *The Economist* magazine uses its so-called Big Mac Index to gauge the overvaluation or undervaluation of currencies relative to the cost charged by McDonalds Corporation for the hamburger entrée in various countries.
 35. As per IMF's estimates.
 36. For example, one can measure the GDP of California (\$1.4 Trillion "Gross State Product" in 2010) or Hong Kong (GDP \$224 billion in 2010), both subnational units of sovereign nations.
 37. Naturally, for such time-based comparisons, to adjust for the effects of price inflation, GDP is often expressed in constant rather than in current prices.
 38. See, for example, canadianeconomy.gc.ca/english/economy/gdp2.html and AmosWEB, www.amosweb.com, where GDP is defined as "the total market value of all goods and services produced within the political boundaries of an economy during a given period of time, usually one year."
 39. The assets referred to in the phenomenon of capital flight, where holders of financial assets move such assets overseas in response to a fear of devaluation or other inauspicious circumstances, thus most of the time still contribute to GNP but not to GDP.
 40. The GDP deflator, which is used to convert nominal GDP to real GDP, is simply one measure of inflation. It resembles the consumer price index and producer price index, discussed later in this chapter, but is different in that it is intended to take account of changes in the mean price level of all goods and services that comprise GDP, and unlike the CPI the composition of the basket of goods and services measured may vary from year to year. Usually, the inflation rates it generates are close to the inflation rate calculated by

the CPI, but sometimes the rate can vary by a percentage point or more. To illustrate the calculation of real GDP, assume that in the fictional country of Floralia nominal GDP for the base year of 2005 was F\$120 billion and was F\$160 billion in 2010. If the GDP deflator in the base year was 100 and in 2010 was 108, the country's real GDP in 2010 would be equal to F\$148.15 or $160 \times 100/108$. The percentage change for the entire period would be 23.5 percent for the entire five-year period or an average of roughly 4.7 percent per year.

41. This can occur in a deflationary environment, where declining GDP implies the potential for a *liquidity trap*, also known as a *deflationary spiral*. This may occur because in a falling price environment consumers as well as firms have an incentive to defer purchases anticipating goods and services to become less expensive in nominal terms in the future. The resulting decline in consumption and production, however, can create a vicious cycle pushing real interest rates dangerously high. This topic is discussed again later in the section concerning monetary policy.
42. Valentino Piana, "Gross Domestic Product," Economic Web Institute, 2001, www.economicwebinstitute.org/glossary/gdp.htm#business. See also U.S. Congressional Budget Office, Glossary of Budgetary and Economic Terms, www.cbo.gov/showdoc.cfm?index=3280&sequence=0: "As a rule of thumb, though not an official measure, recessions are identified by a decline in real gross domestic product for at least two consecutive quarters."
43. *Velocity* means "average number of times that the dollar turns over in a given year on the purchase of final goods and services," Chapter 17, "Lecture Notes—Monetarism, Introduction to Macroeconomics," www.econweb.com/MacroWelcome/monetarism/notes.html ("Monetarism").
44. The theoretical foundations of monetarism, the primary counterpart to the Keynesian model of macroeconomics, are largely based on the assumption that the velocity of money, V , is stable.
45. Another important leading indicator is change in housing starts, which is generally synonymous with the issue of new building permits. Most construction projects require a building permit, so applications for new permits tend to be a precursor of new construction activity.
46. Paul Krugman, "How Fast Can the U.S. Economy Grow?" *Harvard Business Review* (July/August 1997).
47. That is, the rate falls as demand for goods and services and demand for labor rises and more workers are needed to staff stores and factories, and rises as demand for goods and services and labor declines.
48. See, for example, the U.S. Congressional Budget Office glossary where the unemployment rate for the United States is defined as the "The number of jobless people who are available for work and are actively seeking jobs, expressed as a percentage of the labor force," www.cbo.gov/showdoc.cfm. Related indicators are based on unemployment benefits claims.
49. "The belief that 4 percent unemployment and stable prices are inconsistent is shaped by the widely accepted "natural rate hypothesis." It argues that the economy's average equilibrium unemployment rate, often called the natural rate . . . is independent of monetary policy. Most current estimates . . . place in the range of 6 to 6 ½ percent. . . . Consistent attempts to [to achieve levels of employment of 4 percent by expanding] the economy beyond its potential for production will result in higher inflation while ultimately failing to produce lower average unemployment." Carl Walsh, "A Primer on Monetary Policy, Part I: Goals and Instruments," FRBSF [Federal Reserve Bank of San Francisco] Weekly Letter, 94-27, August 5, 1994.
50. Ibid.
51. For example, one such indicator is the percentage change in the periodic figure for new claims for benefits made by those who have been made redundant.

52. One should remain aware of the fact that a vast range of methodologies is used across countries to measure unemployment.
53. The analyst would have to explore historical data on the correlation between unemployment and business cycles in specific industries or sectors.
54. The reader may wish to refer again to our discussion of bankruptcy in Chapter 7 on asset quality.
55. It is very difficult for politicians to grasp the complexities of economics, but admittedly economists themselves often have conflicting opinions about what would be appropriate government economic policies.
56. Since the previous sections focused on general economic indicators, the specific indicators associated with each of these, the foregoing three areas were not explored. Instead, they are addressed in this chapter in connection with each area in turn.
57. The situation of Greece in 2011 and 2012 was an interesting example.
58. OECD Glossary of Statistical Terms, <http://stats.oecd.org/glossary/detail.asp?ID=1135>. The OECD uses certain defined terms to classify government spending. *Final expenditure* means the sum of “*final consumption expenditure*”—i.e., spending for goods or services—and “*gross fixed capital formation*,” or investment in fixed assets. *Gross fixed capital formation* is similar to *capital expenditure* (i.e., “the value of purchases of fixed assets . . . that are used repeatedly in production processes for more than a year”) except that *capital expenditure* does not deduct disposals of fixed assets while gross fixed capital formation does and also takes account of improvements. (If capital expenditure is added to final consumption expenditure, the result is national expenditure rather than government final expenditure.) *Gross fixed capital formation* is distinguished from *gross capital formation* in that that latter takes account of changes in *inventories*. *Imputed expenditure* means spending that does not take place in monetary terms and so can only be estimated.
59. As an example, the European countries committed under the 1992 Maastricht Treaty to keep their fiscal deficit below 3 percent per year, a promise many ended up not keeping.
60. Debt obligations of the government, or the public sector, represent a part, often a significant part of a country’s external debt, which is discussed later in this chapter. Foreign debt effectively becomes part of a country’s external or trade position.
61. *United Kingdom National Statistics Online, Guide to Public Sector Accounts*, www.statistics.gov.uk. In some circumstances, public sector debt may be defined as debt issued by the national government but not by local government units or public law entities.
62. The public debt is not the fiscal deficit. Under the Maastricht Treaty, the maximum public debt was set at 60 percent of GDP.
63. This ratio may also be referred to as Government Net Debt as a percentage of GDP.
64. Alternative terms for this ratio are *national government debt to GDP* and *general government debt to GDP*.
65. A detailed discussion of the significance of intra-account imbalances is beyond the scope of this book, however. Instead, the intent is merely to provide an overview of the key issues and indicators with which sovereign risk analysis is concerned.
66. The impact on long-run economic growth is thought to be negative because the private sector can be expected to invest the proceeds from loans (and equity) more efficiently than the government.
67. It is critical to distinguish between nominal interest rates and real interest rates; that is, the nominal interest rate less than the rate of inflation. Note that, in a deflationary environment, nominally low interest rates can still mean high real interest rates. Real interest rates are discussed later in the chapter. Interest rates that are volatile, or a monetary policy that is erratic, are both adverse from a sovereign risk perspective.
68. See “Macroeconomic and International Policy Terms,” compiled by Mary A. Marchant and William M. Snell (University of Kentucky, Online Publications, 2005), www.ca.uky.edu/agc/pubs/aec/aec75/aec75.htm.

69. When interest rates go up, bond prices go down, and vice versa. Likewise, inflation causes the real value of fixed income securities to fall, unless they are inflation adjusted.
70. To this end, other goals, such as the achievement of “full employment,” may be added, which may or may not be inconsistent with the referenced primary aim.
71. “The natural role of monetary policy in the economy is to maintain price stability. . . . Monetary policy can affect real activity only in the shorter term. . . . But ultimately it can only influence the price level in the economy.” European Central Bank (ECB), Objective of monetary Policy, www.ecb.europa.eu (2012).
72. This obviously did not happen during quantitative easing episodes in the United States, the UK, and Europe during the financial crisis that started in 2007.
73. This is the quantity of money theory, a tenet of monetarist theories. It works best in balanced circumstances.
74. Note that deflation should not be confused with disinflation, the latter meaning a decreasing rate of inflation.
75. Although occasionally deflation can be benign when it is not broad-based and occurs in response to improved productivity or when it constitutes a brief adjustment, usually falling price levels are to be feared rather than welcomed because usually deflation is the result of either a sharp reduction in the money supply (or the supply of credit) or a severe recession.
76. Past episodes of deflation do not bode well for the banking sector. During the Great Depression in the United States in the 1930s, the price level fell by 22.5 percent, and thousands of banks were closed. In the late 1990s, Japan suffered a prolonged albeit not as severe a price deflation. Its GDP deflator declined by 5.3 percent between 1995 and 2001, and this was a major contributing factor in the deterioration in the financial condition of the country’s financial institutions.
77. The impact on long-run economic growth is thought to be negative because the private sector can be expected to invest the proceeds from loans (and equity) more efficiently than the government.
78. In other words, when a bank lends funds, the majority of loan proceeds are never converted into currency at all but are simply added to the recipient’s bank account through a bookkeeping entry.
79. In theory, total deposit creation is equal to the money multiplier, which is the reciprocal of the reserve requirement times excess reserves. That is, if the reserve requirement is 10 percent (1/10), then the money multiplier is 10x, and conceivably \$1,000 in deposits could create an additional \$9,000, which together with the reserved deposit is equal to \$10,000. In practice, money creation does not work as efficiently as this, but just the same, it is an important contributor to the money supply, and the reserve requirement is a basic instrument of monetary policy.
80. Other tools to adjust the money supply are discussed below. They include (1) the central bank’s provision of short-term funding to banks and (2) a form of intervention in the debt securities markets known as open market operations. In the UK and in the United States, the use of this method to increase the money supply is known as *quantitative easing*.
81. In the United States, for example, the Monetary Control Act of 1980 allows the Fed to set the reserve requirement between 8 percent and 14 percent on accounts that allow payment through checks or other means from 0 percent to 9 percent on time deposits that are not held by individuals. Other countries have similar enabling laws that give comparable power to their banking authorities.
82. The reserve requirement imposed by central banks may be much higher or much lower, and it may fluctuate, sometimes frequently. In May 2012, for example, the People’s Bank of China, which is China’s central bank, adjusted its ratios downward by 0.5 to 20 percent for large banks and to 16.5 percent banks for smaller banks.

83. Observe the decrease in M1 as a component of the money supply. One reason is that U. S. banks have adopted the practice of using *sweeps*, which move funds from current accounts that are subject to a reserve requirement into savings accounts that are not.
84. Allan H. Meltzer, "Monetarism is Alive and Well," Carnegie Mellon University, January 1, 1991, 2.
85. Monetarists lost some ground in the wake of the financial crises that started in 2007, but this is more to do with politics than with fundamentals.
86. Normally, when money supply contracts, interest rates will rise, and inflation will go down because of lower aggregate demand. In some circumstances, however, the usual response will not occur. Instead, cutting the money supply to increase interest rates may, perversely, cause inflation to go up. It is as if the central banker steps on the brakes but inflation continues to accelerate. This can occur in countries—in particular in Asia—that have oligopolistic industry structures coupled with underdeveloped capital markets that force firms to rely almost entirely on bank finance. In such an environment, the firm's higher interest expenses are then passed on to the consumers.
87. Federal Reserve Bank of San Francisco, "About the Fed: What Are the Tools of U.S. Monetary Policy?" (2005), www.frbsf.org/publications/federalreserve/monetary/tools.html.
88. The real interest rate is the nominal interest rate minus inflation.
89. Fedpoint, "The Money Supply," U.S. Federal Reserve Bank of New York, (January 2003), www.ny.frb.org/aboutthefed/fedpoint/fed49.html.
90. Broadly speaking, the availability of credit takes in other criteria besides the money supply and the reserve ration, including intermediation, loan growth, and prudential standards. Of course, these items together with prices, interest rates, and inflation affect the money supply indirectly, if not directly.
91. Note that real interest rates may be based on various fundamental interest rate indicators.
92. This ratio may be referred to as overall domestic credit growth or as banking system loan growth.
93. Note that enclave-type financial centers such as Hong Kong, Singapore, and Mauritius may register unusually high intermediation ratios that reflect the concentration of banking institutions rather than necessarily a heavy reliance on bank finance.
94. During the period of rapid loan growth, property prices were rising sharply, and the economy was growing at a rapid pace although not nearly so quickly as the expansion of credit. Although Hong Kong is a somewhat unusual market in the fact that the property sector is a major part of the economy, and in the constraints on independence the monetary authorities face under a currency board system, it does provide a clear illustration of how the expansion of credit can fuel a property bubble.
95. The LDR may be expressed in terms of customer lending (excluding interbank lending) or in terms of total bank lending. (It is also sometimes calculated to include credit provided by nonbank financial intermediaries (NBFIs) as well.)
96. Cara Lown and Donald P. Morgan, "The Credit Cycle and the Business Cycle: New Findings Using the Loan Officer Opinion Survey," June 8, 2004, 22.
97. Moreover, the decline in profits reduces internal capital generation and will thereby tend to increase gearing.
98. Interest rates are also directly and indirectly affected by government policy actions, as well as by anticipated government actions (or nonaction).
99. Naturally, that comment would not hold in high inflation situations.
100. The material on the yield curve, particularly in respect of the historical examples, is drawn in large part from Fidelity Investments, "The Historical Yield Curve," <http://fixedincome.fidelity.com/fi/FIHistoricalYield?refpr=obrfr12>. There are many other good sources.

101. The value of bank stock prices as an indicator of banking system distress is addressed elsewhere. Here we will consider the use of bank equity prices merely as an economic signal.
102. Lieven Baele, Rudi Vander Vennet, and Astrid Van Landschoot, "Bank Characteristics and Cyclical Variations in Bank Stock Returns," Abstract, Ghent University, July 2003.
103. Of course, such an assumption might not hold in unusual circumstances.
104. See for example Chiara Angeloni and Guntram B. Wolff, "Are Banks Affected by Their Holdings of Government Debt?" (Brussels: Bruegel Working Paper 2012/07, March 2012).
105. Andrew B. Collins, Steven M. Truong, and R. Neal Kohl, "Large Commercial Banks: The 60-Second Bank Stock Primer," US Bancorp Piper Jaffray, April 2003.
106. The reserve account is also called the financial account. Federal Reserve Bank of New York, Fedpoint, www.ny.frb.org/aboutthefed/fedpoint/fed40.html.
107. Like the capital account and the balance of payments as a whole, the current account can be viewed as resembling a corporate cash flow statement in that it records cross-border movements of funds between a country's resident and nonresidents. The current account registers, over a given time period, exports and imports of goods and services, unilateral transfers as well as investment income generated by foreign assets and liabilities. By convention, monetary inflows are positive and monetary outflows negative. Hence, when all transactions are tallied, if incomings from other countries exceed outgoings to foreign countries, the current account balance is said to be positive; if funds paid are in excess of funds received, the current account balance is said to be negative.
108. At one time, the merchandise trade balance and the trade balance were synonymous, but the term *trade balance* now refers to the balance of both goods and services. See John Calverley, "Pocket Guide to Economics for the Global Investor," American Express Bank, 1995, 85.
109. The current account therefore is not just a fundamental measure of foreign trade but also the key indicator of a country's net trade position, which is another term for the "trade balance," discussed earlier. Further, as a result of its equivalence with the capital account, it indirectly represents the difference between a nation's savings and investment. In other words, by measuring the gap between exports and imports we can simultaneously measure the amount, if any, that a country must borrow from abroad, as well as the difference between savings and investment. This can be better understood by referring to the calculation of GDP.
110. A reserve currency is a well-recognized, liquid, and relatively stable currency frequently held by central banks as international reserves for purposes of the settlement of transnational claims arising from international trade and for intervention in the foreign exchange markets. At present, the U.S. dollar, the euro, and the Japanese yen are the primary reserve currencies, followed by the Swiss franc and British pound sterling.
111. Currency fluctuations tend to be of greatest concern in reference to developing countries, which are vulnerable to currency crises and attendant capital flight. Also, within a monetary union, such as the European Monetary Union or EMU, a member state's balance of payments is of little significance so long as the union remains intact.
112. See discussion later in the chapter concerning the connection between a current account deficit and foreign capital requirements.
113. This is not open to individual members of the euro zone, such as Greece.
114. Particularly in reference to emerging markets, a distinction is frequently made between trade deficits driven by excessive consumer spending and those that are driven by productive investment resulting in the import of commodities or capital goods. Although the nominal figures could be the same, if the country is borrowing to meet strong demand for productive investment, this is viewed less negatively, if not favorably. Such a scenario is often regarded as a necessary stage in the economic development of

- emerging markets, and consequently is regarded as likely to be temporary rather than chronic in character.
115. Ordinarily, external debt will be measured in units of a reserve currency, whether U.S. dollars, euros, or Japanese yen.
 116. This indicator is calculated by estimating the market value of the nation's foreign assets and subtracting that figure from the corresponding estimated market value of its liabilities, dividing by GDP and expressing the result as a percentage.
 117. Historically, payment that was ultimately transferable directly into gold or silver was the most trusted means of making international payments.
 118. Nonrenewal risk was a major risk for some euro zone countries in 2012.
 119. This is yet another example of the complexity of gauging a bank's or a country's liquidity.
 120. Definitions of the numerator vary. Another definition is "holdings of foreign exchange plus gold held in the additional reserves (valued at 75 percent of current market price). . . ." Kenneth F'Anson, Andrew Fight, and Patrick Vandenbroucke, *Bank and Country Risk Analysis*, vol. 4, *Country Risk Modeling* (Euromoney-DC Gardner Workbooks, 1999), 8.
 121. See more comments on foreign reserves in Delhaise, "Asia in Crisis," 196 to 198.
 122. It is important to be aware that exports do not just refer to goods but also services, which are becoming increasingly important in the international economy. Services encompass a wide range of activities including transportation, tourism, arts and popular culture, financial services, legal, medical, and other professional services, as well as services provided in the IT industry, by call centers, and so forth.
 123. A warning signal of systemic risk according to Moody's is a situation where domestic investors hold a large portion of the country's foreign-currency bonds as a hedge against a weak local currency.
 124. Recall that both consumption and investment contribute to a market's total GDP or gross domestic product, which is equal to its gross domestic income. Recall also that the formula for GDP provided earlier in the chapter, states that GDP is equal to the sum of personal consumption, private investment, and government spending, plus or minus the applicable trade surplus (or deficit).
 125. World Bank, "National Accounts: Output and Expenditure," www.worldbank.org/data/working/def7.html.
 126. In the national accounts, savings are defined as total income, that is, the aggregate of individual income, company income, and government tax revenue, less consumption (i.e., current expenditure), which includes all household spending (including spending on durable items such as automobiles and computers), government spending, and current company expenses (i.e., spending other than capital expenditure). Investments primarily encompass noncurrent spending by companies.

Risk Management, Basel Accords, and Ratings

Let no loans be made that are not secured beyond a reasonable contingency. Do nothing to foster and encourage speculation. Give facilities only to legitimate and prudent transactions. Make your discounts on as short a time as the business of your customers will permit, and insist upon the payment of all paper at maturity, no matter whether you need the money or not. Never renew a note or bill merely because you may not know where to place the money with equal advantage if the paper is not paid. In no other way can you properly control your discount line, or make it at all times reliable. . . . Pursue a straightforward, upright, legitimate banking business. Never be tempted by the prospect of large returns to do anything but what may be properly done. . . . “Splendid Financiers” in banking are generally either humbugs or rascals.

—Hugh Macculloch, U.S. Comptroller of the Currency, 1863

Risk management is not merely about reducing risks (although that is in many cases a necessity), but essentially about taking risks in an intelligent manner. Banking can be no more riskless than life itself.

—Eddie Cade¹

More and more the bank is a “risk machine.” It takes risks, it transforms them, and it embeds them in banking products and services. . . . They take risks more consciously, they anticipate adverse changes, they protect themselves from unexpected events, and they gain the expertise to price risks. . . .

—Joel Bessis²

Prior to a more elaborate discussion³ of Basel II and Basel III, as well as of the role of rating agencies, we would like to explore what is nowadays generally labeled *risk management* but has always been part and parcel of bank management.

It has been suggested previously that any particular category of risk, if realized to a sufficient degree, can potentially lead to a given bank’s default upon its obligations to investors, lenders, depositors, and counterparties, and thus to its demise. To these

entities, the risk of such default is, of course, a form of credit risk to which they accept exposure upon their entering into a transaction with that bank. Credit risk is also the principal peril to which the subject bank—and indeed all banks—become vulnerable when advancing loans, acquiring debt securities, or providing guarantees on behalf of customers—to mention only three common types of transactions in which banks engage.

Reference has been made in earlier chapters to a number of types of risk—other than credit risk—to which a bank may become vulnerable in the course of doing business. Banks typically take on exposure to liquidity risk, market risk, and operational risk, all of which can severely impair an institution's financial condition and potentially trigger its collapse should the associated worst-case loss be realized.⁴

Consequently, to comprehensively evaluate the credit risk a bank presents to those contemplating a credit transaction with it, all the principal risks to which the institution is exposed should be considered, for each category of risk will affect the bank's overall creditworthiness vis-à-vis its creditors and counterparties.

Actually, the CAMEL framework corresponds in part with the categories of risk just mentioned. Asset quality reflects credit risk exposure and management, liquidity risk relates to liquidity and liquidity management, while market risk has an impact on liquidity and funding as well as the earnings and profitability of the bank's spread business and its trading and treasury operations. Operational risk can arise from nearly any of a bank's activities, while solvency risk corresponds to capital adequacy and represents the institution's wherewithal to absorb losses should it be struck by a perfect storm worst-case risk scenario.

If the consideration of the magnitude of these risks is necessary in a comprehensive credit risk analysis, so too is a bank's capacity to manage these risks. In having the responsibility to oversee the risks the bank accepts, management's actions or omissions may potentially strengthen or weaken the institution's overall financial position. Risk management capacity embraces not just its risk management methods, but also the status of the risk management team within the institution. The salient question in this regard is this: To what extent are the views of risk managers taken into account by top management? Even more broadly, consideration may be given to the existence and strength of what may be termed a risk management culture akin to the notion of a credit culture discussed earlier in this book.

Within the CAMEL framework, the analyst's review of a bank's risk management function forms part of the criterion of management quality/competence represented in the letter "M" in the acronym.⁵ Management's decisions concerning risk—for it is ultimately senior management that receives the advice of its risk management team and has the final say—are among the most important it can make in determining a bank's future. Although the ultimate volume of losses (or profits) as a result of any given risk accepted is subject to chance, the decision as to whether to undertake a particular risk exposure is for the most part solely the decision of management. With respect to any given risk, a number of potential responses are ordinarily available. Any one or more of the following may constitute a potential course of action.

If the exposure has not yet been accepted, it can be avoided by rejecting the proposed transaction that would generate it. If already accepted, the risk can be shed through the termination of the relevant agreement or relationship, or by transfer without recourse to another party. Perhaps most importantly, short of eliminating

the risk, it can be mitigated by actions to reduce exposure or reduce the impact of any realization of the risk. Such mitigation can be effected through diversification, the purchase of insurance, or hedging mechanisms. In connection with credit risk in particular, additional collateral may be demanded or creditor protections sought. Of course, if the risk is deemed to be low relative to returns, there is the possibility of increasing the risk exposure.⁶

Although normally, any prospective transaction that engenders significant risk can be rejected, it bears repeating that the elimination of all significant risk is neither practical nor desirable. A bank with no appetite for risk cannot function, for risk is inherent in banking. Arguably, it is a bank's willingness to accept some calculated risk that through experience provides it with a competitive advantage, thereby enabling the institution to provide services for which there is a demand and to earn a profit. Yet too much risk can lead to its downfall. Put in rather simple terms, the function of risk management is to find an appropriate middle ground between too little risk and too much risk.

The task, however, is actually much more complex. To find the happy medium between too little and too much, a bank must identify, classify, and estimate the risks it confronts in view of corresponding historic and prospective returns. That is to say, management's aim should be to maximize risk-adjusted profits and in the process to optimize risk exposures relative to returns. As part of this process, a vital role performed by management is to balance at a strategic level the various risks to which a bank is subject relative to the returns generated by the corresponding businesses.

At the same time, management should ensure that the bank will be able to weather potential unexpected worst-case scenarios, except for those where the probability of occurrence is so remote that it is not reasonable or cost-effective to provide additional capital reserves or undertake additional credit risk mitigation. Insofar as it effectively performs the foregoing roles, top management fulfills its ultimate responsibility and objective to its stakeholders, which is to make certain that the bank continues to survive and prosper.

It is worth reiterating a critical distinction between management's view of bank risk and that of the bank credit analyst's. From management's vantage point, the primary subject of this book—credit risk—is merely one of several major categories of risks affecting a bank's overall performance and financial condition, and accordingly one of several that must be evaluated and managed.⁷ From the credit analyst's perspective, however, all the risks faced by a bank, together with the way in which they are managed, are variables affecting its overall credit risk. That is, to the credit analyst, all such bank risks can be reduced, in their measure or in their concept, to credit risk—to which counterparties, investors, and depositors become subject should they enter into a transaction with that bank. Exhibit 12.1 contrasts the credit analyst's perspective on risk management with that of the enterprise risk manager.

In view of the preceding discussion, it is evident that to make a judgment as to how a bank's risk management capacity contributes to, or detracts from, the strength of its overall credit risk profile, the analyst needs to be aware of the following: (1) the nature of the risks banks confront generally and common practices for managing them, and (2) the character of the particular risk environment in which the subject institution operates. Both attributes will, in turn, be influenced by various country/systemic risk factors, the bank's franchise, and management's risk appetite, among

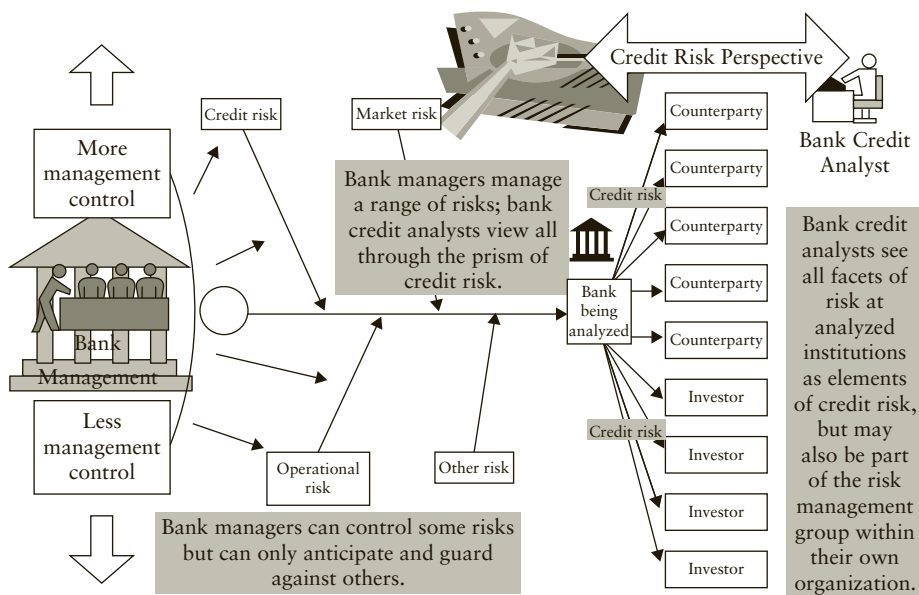


EXHIBIT 12.1 Bank Manager's View of Risk versus Credit Analyst's View

other factors. Additionally, understanding how banks typically manage their risks will aid the analyst in gauging the capacity of a specific institution to manage the risks it faces.⁸

To appreciate the potential perils posed by a bank's particular risk environment, country risk and related banking industry data for the subject market will be useful. Additionally, a bank's CAMEL financial attributes—earnings and profitability, asset quality, capital, and liquidity—will be helpful in ascertaining its risk environment.⁹ For instance, as alluded to above, the assessment of asset quality will implicitly take account of the relative level of credit risk to which a given bank is exposed in respect to its customers and counterparties. Regarding market risk, the magnitude of such risk—for instance, to what extent it may be adversely affected by sharp changes in market prices or interest rates—will vary depending upon several variables, including the institution's product mix; the proportion of domestic vs. international funding and operations; and its capacity to hedge market risk in the money and capital markets. In addition to its impact on trading profits, market risk at a structural level will have a bearing on spreads and upon overall earnings and profitability. It may also affect liquidity.¹⁰

Finally, to assess the quality and capacity of a bank's risk management function, relevant considerations include the bank's risk environment, management's risk appetite, the role and authority of the risk management team, and how its operations compare with industry best practices. Parenthetically, the more sophisticated the financial markets in which the institution operates, the wider the range of risk management methods that are likely to be available to it. Ideally, more choices should allow for more effective risk mitigation, but the risks implicit in the techniques themselves can also increase overall risk levels.¹¹

Put in a simplistic way, the bank's credit risk assessed by the bank credit risk analyst is a twice-remote risk. Where the analyst's ultimate aim is to evaluate the credit risk associated with a given bank, the same bank's ability to manage all the risks to which it is vulnerable—including but not limited to credit risk—will affect its overall credit risk profile. To facilitate an evaluation of these factors, the analyst should be generally aware of the nature of the main categories of risk banks face and have some understanding of basic risk management processes.¹² This chapter therefore represents an attempt to provide a broad-brush survey of risk management in order to provide some criteria against which a particular risk management regime can be assessed as to its quality and its effectiveness.

NOTE TO THE READER

Before proceeding further, it should be emphasized that since contemporary bank risk management is both a broad and—in many respects—a highly complex topic, the subject can be touched upon only superficially in a single chapter such as this. Although the fundamental concepts employed in risk management are often readily understandable, the devil lies in the details. Definitional issues and the advanced mathematics employed—not to speak of the often thorny implementation issues involved—can present a challenge for the nonspecialist. Therefore, the purpose of this chapter is merely an attempt to provide an introduction to the field for a limited purpose. That purpose is to aid the bank credit analyst in understanding the risks banks themselves generally confront, together with the best practices for their management, all in the context of evaluating the creditworthiness of the bank as a whole.

RISK AND THE IMPORTANCE OF RISK MANAGEMENT

*Risk is the fundamental element that influences financial behavior. . . . Much of the structure of the financial system . . . serves the function of the efficient distribution of risk.*¹³

—Robert C. Merton¹⁴

Risk in banking refers to the potential of a financial loss arising from banking activities and related transactions, including losses arising from the holding of financial assets. The term *risk management*, which has come into common parlance only in the past two decades or so, has been defined as “the complete set of policies and procedures which organisations have in place to manage, monitor and control their exposure to risk.”¹⁵ The ultimate risk to a bank is, of course, its own collapse, and the most fundamental goal of bank risk management is to prevent such a collapse. This is accomplished through the effective identification and assessment of risk, coupled with measures to reduce risks relative to returns and to ensure that

adequate safeguards are in place to protect against unexpected worst-case loss scenarios.

DEFINITION OF RISK

Risk is the potential or likelihood of loss on an investment or on a financial asset within its life or within a given time period.

The above definition—the most common one—refers, in the field of finance, to adverse risk. A more attractive definition, at least from a conceptual standpoint, is associated with the variability of the potential return of an investment. The “risk” is then seen, in a probabilistic way, as the likelihood of a gain as well as that of a loss.

As suggested in the introduction and implicit in the quotations above, the business of banking is inseparable from the acceptance of risk. Since a bank must assume risk to generate sufficient revenue to survive, the management of risk has always been a part of banking, notwithstanding that the term *risk management* was not generally used to describe this function. Until recent decades, however, the focus of such de facto risk management in banks was on credit risk. Prior to the adoption of advanced quantitative methods beginning in the late 1980s and early 1990s, a traditional approach to risk management had become ubiquitous throughout the banking industry. Historically, bankers mainly worried about credit risk, but from the 1970s onward market risk was becoming a growing concern.

By the time the more quantitative models began to transform the de facto risk management functions into a new specialized field, the management of credit risk was generally accomplished at the operational level through the use of limit systems and hierarchical approval protocols. Market risk was addressed in a more eclectic manner, largely under the rubric of asset-liability management, discussed briefly later in the chapter, together with methods of sensitivity analysis and valuation applied to trading operations.

In respect to credit risk, all but perhaps the smallest or least sophisticated banks had well-established policies and procedures subjecting any significant credit transactions to either a formal approval process or to delegated authority. Large transactions were reviewed by a credit committee, which at the highest level included the most senior officers of the bank, while more routine loan applications were merely subject to the approval of a credit officer. Such credit policies and processes, including internal loan grading systems, constituted a primary element of credit risk management.

Credit risk management was discussed in part in connection with asset quality in Chapter 7. Conventionally, it employed the use of limit systems that limited credit exposure to approved “credits” and required a chain of approval requirements, including the requirement that loans over a certain threshold be approved by committee. More generally, the management of credit risk is accomplished by evaluating prospective credit exposures through credit analysis, monitoring ongoing credit

exposures, setting credit limits, and using credit risk mitigants as appropriate, and shedding or renegotiating outstanding credit risks when necessary. In this chapter, credit risk management is examined in the wider context of enterprise risk management, with a particular emphasis on how measurement of credit risk is linked to overall enterprise risk management and capital allocation.

Credit policies and processes remain important, but in the past 20 years have been subsumed within the larger risk management function, which has been transformed during that period through the application of quantitative methods and developments in financial theory.

The origins of the transformation can be traced back to the 1970s. Financial innovation sparked by the deregulation of financial markets during that decade made risks other than credit risk increasingly important. Deregulation alone—freeing interest rates and foreign exchange rates to be set by market forces, for example—had a significant effect. But at the same time, market acceptance of increasingly sophisticated financial products such as mortgage-backed securities and interest-rate swaps, as well as other derivatives, played a role in increasing the prominence of risks other than credit risk in the banking industry. Notably, market risk became a major concern during this period. Liquidity risk and operational risk also increased, however, while credit risk analysis in many cases became more complex.

As a consequence, even before it was common for banks to have a separate risk management department charged with monitoring and managing banking risks on an enterprise-wide basis, the basic components existed, albeit varying in their state of development. It just took some time before the various strands of risk management were addressed in an integrated fashion.

Ultimately, the evolution of the financial markets coupled with consolidation and expanding scale in the banking industry and regulatory changes demanded a more systemized and comprehensive approach to risk management supported by robust theoretical foundations.

Purposes of Risk Management

Some of the purposes of risk management were mentioned in the introduction to this chapter. It bears repeating that the purpose of risk management is not to eliminate risk. Risk is intrinsic to banking, and empirical evidence supports what financial theory holds: that there is no profit without risk. A book on the subject of risk management produced by two leading investment banks put it well:

*Managers . . . who insist on zero risk tolerance condemn their firms to zero excess returns. . . . [A]lthough risk management cannot prevent losses, it can ensure . . . awareness of and comfort with the level of risk [undertaken].*¹⁶

If the purpose of risk management is not to eradicate risk, what is its objective?

Primary Purpose

Perhaps the overarching aim of modern risk management is to optimize the risk/return calculus of risk by maximizing risk-adjusted profits, while maintaining the

solvency of the bank. Another way of putting the first part of this proposition is that it is impossible to effectively manage risk without taking account of the returns that a prospective risk will generate. In the same fashion, transactions should not be rejected without considering the returns that will be foregone relative to the risks they engender. Hence, the aim is not merely to reduce risk irrespective of other considerations, but to optimize the relationship between risk and return.

The second part of the proposition, however, provides an important *caveat*: Worst-case risk scenarios sufficient to bring down the bank should be limited to all but the most remote possibility. To this end, risk managers seek to hold risk levels to prudent levels in line with the bank's strategy and risk appetite, while accounting for calamitous risk (i.e., unexpected worst-case scenarios) to a specified remote possibility. Since a bank's capital is its ultimate buffer against unexpected worst-case scenario losses, risk management is affected by and affects capital requirements. In sum, the central objective of modern enterprise risk management is to ensure that a bank's operations conform to its stated risk appetite by establishing model-justified limits on transactions and adequate reserves of capital.¹⁷

PURPOSES OF RISK MANAGEMENT

- Identification, measurement, and communication of risks to senior management
- Regulatory compliance under Basel II and Basel III
- Facilitation of pricing and strategic decisions
- *Key purpose*: Optimization of risk-adjusted returns subject to the establishment of model-justified limits and an adequate capital buffer to maintain the solvency of the bank in accord with the risk appetite of management and shareholders

Other Purposes

In addition to this all-encompassing purpose, a very practical aim of risk management is to identify risks that might otherwise remain hidden or viewed as insignificant, and to communicate information about such risks and the relative magnitude of risks to which a bank is exposed to its senior management. It could even be argued that one objective of risk management is to educate staff concerning risk generally with the goal of improving the bank's risk culture.

Yet another objective of risk management is regulatory compliance. With the arrival of the Basel II era, an integrated risk management function has to all intents and purposes become a supervisory requirement for many banks.

Finally, risk management has come to be viewed as an essential element of overall bank management. By providing data vital to strategic decision making and product pricing, among other things, that can potentially equip a bank with a competitive advantage over its rivals.

MARKET DEREGULATION AND THE RISE OF ENTERPRISE RISK MANAGEMENT

The origins of the modern approach to risk management can be traced to the 1970s. But to fully comprehend the changes in the financial services industry that catalyzed its emergence, it is helpful to go back somewhat earlier. In the United States, where the evolution of credit risk management into comprehensive enterprise risk management was particularly conspicuous, the story began at the conclusion of a long period of rigid control of the financial system. The Great Depression that followed the stock market crash of 1929 had brought about numerous financial reforms designed to prevent past abuses in the financial services industry. Among these were the separation of commercial banking from investment banking and the imposition of strict regulations governing both.¹⁸ Another important facet of the regulatory framework, a fixed foreign exchange regime, was put in place through international agreement following World War II to replace the prewar gold standard. Under the Bretton Woods framework, foreign exchange rates were effectively pegged against the U.S. dollar, while the U.S. government stood ready to exchange dollars for gold at a fixed rate.

While the banking industry during the 1950s and 1960s was relatively stable, it was also characterized by a lack of competition. Unsurprisingly, there was little innovation in financial markets and products. Instead, banking became a cozy business where the so-called 3-6-3 rule prevailed—borrow money from your depositors at 3 percent, lend it to others at 6 percent, and be at the golf course at 3 p.m. The Bretton Woods scheme of fixed currency rates, which had been abandoned in 1971, gave way to a fully floating foreign exchange market by 1976 creating the potential for currency risk at a new order of magnitude. Another seminal event was the establishment of the Chicago Board of Trade options exchange in 1973, which created the opportunity for financial traders to buy and sell standardized contracts for stock options, a form of derivatives. The same year saw the publication of a landmark paper by financial theorists Fischer Black and Myron Scholes that was to prove an invaluable tool for pricing the new instruments. Deregulation of the banking industry began gradually, but by the early 1980s, it was a clearly established trend in the United States.

Deregulation had a profound impact upon the banking sector. As the rules were loosened, the industry became more competitive, and bankers had an incentive to take bigger risks in the quest for greater returns. Some institutions were, arguably, victims of deregulation. The country's savings and loan industry, illustrated both phenomena. The industry as a whole was unprepared for the losses wrought by interest rate risk. Some S&Ls were taken over by aggressive players who took excessive risks with unfortunate results. Other S&Ls, many small community institutions, were swamped by losses engendered by interest rate risk. In the late 1980s, the industry had to be bailed out by the federal government. During the 1970s and 1980s, the growth—and increasing depth and sophistication—of the money and capital markets,

enabled large corporates to obtain finance directly and bypass commercial banks. The same developments allowed banks a wider variety of wholesale funding sources, while the growing popularity of new instruments such as interest-rate and currency swaps, as well as mortgage-backed securities created through securitization, marked the start of a period of dramatic innovation in financial products.

By the late 1980s, swaps and securitizations had become well-established financial products, the later continuing to expand and mutate into ever-more-complex variants until the credit crisis of 2007. These new instruments heightened risks for market participants who dealt with them. Market risk mushroomed, but credit risk, liquidity risk, and operational risk also increased. The complexity of many new instruments made them hard to evaluate from a credit perspective, and sometimes obscured increases in operational risk. Debacles involving rogue traders, such as that of Nick Leeson, who brought down Barings Bank, concentrated bankers' minds on the heightened importance of risk management.¹⁹

In the meantime, pioneering approaches to measuring and managing risk developed in academia and at banks themselves, such as Bankers Trust and J.P. Morgan in New York, provided the methodological underpinning for a statistically based approach to risk management. The parallel explosion of computing power accessible to banking staff afforded the number-crunching capacity to enable the new tools to be rapidly implemented. By the mid-1990s, the foundations were set for the integration of the field of risk management on an enterprise-wide basis, rooted in fundamental concepts of statistics and probability.²⁰

Within a decade, the industry had come to recognize the importance of enterprise risk management, together with the concept of economic capital that underpinned its application in the financial sector. By this time, major internationally active banks had substantial numbers of staff assigned to risk management and to implementing the new Basel accord (Basel II), which itself incorporated many of the assumptions and methods of the prevailing risk management methodology. Yet at the same time, as events were soon to reveal, risk was being systematically underpriced caused either by latent flaws in the models or in how they were being implemented.

Integrating Risk Management

The key to integrating the management of the various risks to which a bank may be subject lay in the further refinement of the broad definition of risk stated earlier to one that facilitates a consistent approach to its calculation across the bank. The lodestar that heralded a new era of risk management was the redefinition of *risk* to mean “risk in excess of expectations.” Outside the context of finance, the concept of risk as uncertainty is not necessarily an intuitive one. The risk of serious bodily harm as a consequence of an automobile collision while driving remains a serious risk with

respect to which all drivers maintain some awareness irrespective of the level of uncertainty associated with the outcome. But in finance, where risks are largely transactional and outcomes can be adjusted through transfers of funds or through transfers of risk, uncertainty has become the ubiquitous critical variable making risk and volatility synonymous.

Investors like risk as long as they can price it correctly. The same applies to bankers. Expected risk, or to be more precise the expected or average loss incurred from a given portfolio of assets attributable to a certain category of risk, neither was nor is the kind of risk that keeps bankers awake at night. As Eddie Cade, author of *Managing Banking Risks*, observed, under the approach to risk just described, “the expected outcome, no matter how dire, is not a risk”²¹ but rather the cost of doing business. Instead, as Cade explained, the risk that was of concern was risk that could result in above-average losses to an uncertain degree. Risk therefore came to be defined “as exposure to uncertainty of outcome . . . where the scope of uncertainty is reflected in the volatility . . . of potential outcomes.” Note that while some fluctuations from the mean might be better than expectations, these are ignored for risk management purposes, which is concerned with protecting against adverse risks. Thus, it is those outcomes that are significantly worse than expectations that are the concern of risk managers.

By redefining—or put another way, refining the definition of—risk to mean the variance of the outcome from the mean (i.e., volatility), the problem of managing risk was suddenly made amenable to a host of powerful quantitative and statistical tools. Not only did the equation of risk and volatility greatly facilitate the modeling and calculation of risk, the logic of applying the concept of statistical variance to risk was, as suggested above, consonant with the needs of bank management. It enabled risks to be reduced to numbers that were simple to understand, and facilitated the establishment of capital reserves sufficient to provide a cushion against worst-case unexpected losses.

Concomitantly, when used to establish a risk management system based upon the concept of unexpected loss, it allowed senior management to take some comfort in the adequacy of their institution’s buffer against catastrophic shocks. As a consequence, the concepts of statistical distributions, variance, and the concept of the standard deviation became central to the emerging field of risk management.²²

The further result was to transform bank risk management from a bunch of discrete, relatively unsophisticated, analytical techniques coupled with a collection of heterogeneous policies and procedures, to a highly quantitative systematic approach to risk with strong theoretical underpinnings. Renowned financial theorist Robert C. Merton was not overstating the case when he wrote in the foreword to a leading text on risk management that “mathematical models of modern finance practice contain some of the most complex applications of probability and optimization theory.”²³

Changes in the Organization of Risk Management

In a liberalized regulatory environment, banks out of necessity had to become more sensitive to the impact of increased risk on both returns and the institution’s own credit profile. Concepts of risk-adjusted return emerged during the late 1980s as the idea of the *middle office* developed. The middle office stood between the back office, which was primarily concerned with settlement and management accounting, and the

front office, which was the source of revenues generated by account officers and traders. During this period, multinational banks were typically managed on a regional basis with bank staff in each locale reporting to the country manager, who reported to a regional manager, who in turn reported to the head office.

The financial scandals of the 1990s and other developments mentioned in this chapter focus bank managers and regulators' minds on the need for integrated firm-wide risk management. For example, the often not readily apparent risks associated with derivatives trading created a new demand for systems to manage them, and for risk managers who were able to employ these systems effectively. The unidirectional hierarchical structure depicted above was not, however, conducive to such an integrated system of enterprise risk management. As a result, a dedicated risk management department is now commonplace and more refined management structures such as matrix structures are pervasive in the financial industry. See Exhibit 12.2 below.

The Process of Risk Management

As a separate function within the bank, the process of monitoring, assessing, and managing risk on a firm wide basis can be divided into three principal subactivities, namely:

1. The identification of risks and the formulation of risk management policies and procedures
2. The modeling, analysis, and monitoring of risk to establish limits on risk taking
3. Compliance and enforcement, including the auditing of risk management models, policies, and procedures

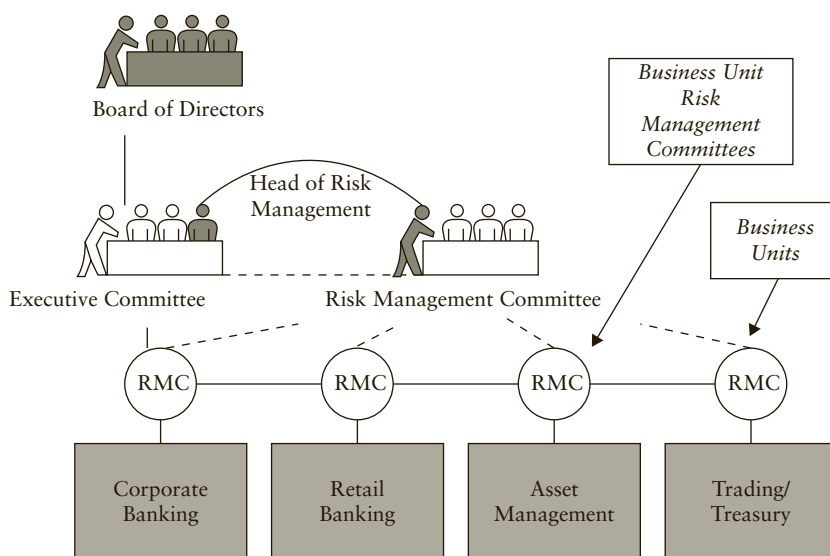


EXHIBIT 12.2 The Risk Management Group in Relation to Senior Management

Risk identification and the setting of policies and procedures is the first step. The process of managing risk is an iterative one since banks will customarily have some set of risk policies and procedures in place initially, whether developed internally or compelled by external regulation. As the risk management function evolves, these policies will naturally be modified in light of new developments and to meet changing needs and conditions.

Once the various risks the bank faces have been ascertained, they can then be monitored, measured, modeled, and analyzed. The second step in the process therefore includes the monitoring of risks on a daily basis and the collection of relevant data. The data collected by the risk management department is aggregated and distilled into reports, which in due course make their way to senior management and to the board of directors.²⁴ So-called management information systems (MIS) are the vehicles through which such reporting is effected. In what might be called the risk management engine, overall mechanisms take data and ratings as inputs and then model, analyze, and monitor them. As a next step, they establish limits as outputs.

Finally, risk auditing involves ensuring that the risk management system is operationally effective and that data integrity is maintained. Risk audit also encompasses the back testing of the bank's risk assessment models on the one hand and compliance responsibilities on the other. Risk managers will also follow up on violations of bank policy and take steps to maintain or re-establish that compliance.

RISK MANAGERS OR RISK DIPLOMATS?

Although the group which performs all the foregoing functions has as its nominal function *risk management*, this term is something of a misnomer in that its authority is rarely absolute. Instead, its chief role is to monitor and evaluate risk and report its recommendations to senior management, while in turn executing those policies implemented by the bank's top-ranking officials. As with credit officers, risk managers must often play a somewhat diplomatic role negotiating the inevitable conflicts between risk and return in an attempt to optimize both of them.

CATEGORIES OF BANK RISK

[For] the quantitative management of risks, generic concepts are useless. . . . The different types of risk have to be carefully defined, and such definitions provide a first basis for measuring risks and implementing risk management.

—Joel Bessis²⁵

A bank confronts various types of risks in the course of engaging in its business. Integral to the business of lending in particular is credit risk, but banks also confront many other types of risk.

In fact, there exist a myriad of identifiable risks, as almost any operational or financial attribute can be framed in terms of risk. Identified risks range from relatively benign behavioral risk to operational risk to serious liquidity risk. We do not intend to list them all, but they usually fit into a few major categories, which we shall explore.

Risk Classification

Risks can be categorized in a number of different ways for different purposes, and the distinctions frequently blur at the edges. Differences in classification depend on point of view and result from the fact that risk categories can be defined in different ways, often overlapping at the periphery, as well as from the multiplicity of inter-relationships they embody. Individual commentators and organizations have applied a variety of categories and labels to classify bank risks.²⁶ Broadly speaking, however, bank risk can most simply be divided into three categories: credit risk, market risk, and operational risk. This is the classification system employed by Basel II and is now probably the most common approach to categorizing bank risks.

Three Fundamental Categories of Bank Risk

These three basic categories of risk (see Exhibit 12.3) address the following types of potential losses:

- Market risk* refers to the risk of loss arising from changes in market prices or rates, including interest rates, foreign exchange rates, and other market-determined value measures that are adverse to positions taken by the bank.
- Credit risk* refers to the risk of loss from an obligor’s default, taking account of the severity of the loss, and thereby encompassing the risk of changes in credit quality reflected in a corresponding decline in the value of the associated assets or in an adverse change in the rating assigned to the obligor or to the relevant obligation.
- Operational risk* refers to the risk of loss arising from failings within the bank in its operational systems (such as lapses in internal procedures or technological breakdown), whether caused by human error, malfeasance, or mechanical breakdown, or from failings without it, such as adverse legal rulings, state actions, natural disasters, or criminal/terrorist activity.

Liquidity risk was sometimes considered as a minor issue, on the grounds that lenders of last resort were looking after their charges, until the 2007 crisis demonstrated how important it really was. It is now given much attention under Basel III. Credit risk, discussed in Chapter 1 and referred to in succeeding chapters, is further elaborated in this chapter, but in view of previous discussion is given relatively less attention than would otherwise be the case. Market risk and operational risk have not yet been addressed in any detail. Hence, a brief overview of each is provided in the pages that follow with further elaboration provided in the remainder of the chapter.

Credit Risk Default risk, risk of decline in credit quality	Market Risk Interest rate risk Currency risk Pricing risk	Operational Risk Business risk Technology risk Human risk
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EXHIBIT 12.3 Three Basic Categories of Risk

Other Key Risk Categories

The main three risks, as well as operational, internal risk, and event risks, are discussed separately next.

Not all categories of risk, however, fit neatly into these three main groups, including notably liquidity risk and solvency risk.

Liquidity Risk Liquidity risk, addressed in Chapter 10, refers to the risk that, irrespective of its solvency, an entity will not be able to pay its obligations as they fall due. As was suggested in that chapter, this may occur because (1) it has insufficient liquid assets to meet its obligations; or (2) it is unable to access funding to meet its obligations. Liquidity risk can also mean the risk that an entity will be unable to consummate a transaction as a result of insufficient market supply or demand.

While recognized as a distinct risk category, liquidity risk is very closely linked to market risk, as market conditions can have a dramatic impact particularly on liquidity. Moreover, market liquidity is a benefit for which investors are willing to pay more or, conversely, accept a lower return. In addition, liquidity is affected by credit risk in that the higher the credit quality of an asset, the more liquid it usually is. In a similar fashion, if a bank's asset quality—that is, the credit quality of its loan/securities portfolio—is weak or deteriorating, liquidity in the form of wholesale funding tends to evaporate as the institution's condition becomes known to (or suspected by) market participants, who perceive the bank as a whole as a greater credit risk.²⁷ Nevertheless, if required to be shoehorned within the tripartite classification system illustrated in Exhibit 12.3, liquidity risk is usually deemed to be a subcategory of market risk. Liquidity risk, however, is extremely hard to measure and to model, and for that reason is not well accounted for in most market risk models. Since liquidity risk management was discussed in Chapter 10, it is not further addressed in the following sections of this chapter.

LIQUIDITY RISK

Liquidity risk refers to the risk of loss arising from a counterparty's having insufficient liquid funds to meet its current obligations, whether due to a lack of sufficient funds on hand, to the inability to convert assets to liquid funds, or to funding problems, as well as to losses occasioned by insufficient market liquidity.

Solvency Risk Solvency risk, which is really the risk of insolvency, fits even less neatly into the tripartite classification of major banking risks than liquidity risk. In actuality, solvency risk, though described in terms of insufficient capital, is the risk of bank failure.²⁸ In contrast to liquidity risk, which is arguably a subset of market risk, solvency risk captures all categories of risk to which a bank is exposed.²⁹ Hence, the level of solvency risk will be driven by the bank's aggregate risks in relation to its capital. In the same way that—from an external vantage point—all the risks borne by an obligor determine its credit risk, from an internal vantage point, all the risks borne by a bank—relative to its capital—determine its solvency risk.

SOLVENCY RISK

Solvency risk is the risk that a bank may become insolvent, meaning that its liabilities exceed its assets or that under the applicable regulatory regime it has insufficient capital to continue to remain in operation. More broadly, it means the risk of bank failure from the perspective of regulators, bank management, or shareholders. To the external bank analyst, as well as to counterparties, investors, and other obliges, solvency risk is largely synonymous with overall credit risk.

Other Classification Criteria

In addition to breaking down risks into functional categories as just described, some other distinctions in risk classification are worth noting. The various risks to which banks are exposed can be distinguished on the basis of the degree of influence a bank has over them. For instance, a bank has a fair amount of control over its internal operational processes, such as the implementation of strong risk management policies and procedures within the bank, than it does over external operational risks, such as political revolutions, widespread civil disorder, headline acts of terrorism, or natural disasters, which are likely to be completely beyond management's ability to prevent, although their mitigation is a possibility.³⁰

Related to the control is the criterion of directness. Is the risk one that presents a cause followed by an effect, or is the chain of events longer and more complex, obscuring causation and thereby making potential losses more difficult to predict? Perhaps the most damaging occurrence, from an organizational perspective, is the loss of a bank's reputation. To a bank, the loss of the bank's good name is generally synonymous with a loss of depositor and customer confidence in the institution that may be, as a practical matter, impossible to repair.

A last useful distinction is whether the risk is solely adverse or whether there are some potential positive outcomes as well. Classically, credit risk is a one-way downside risk. At best, the bank only gets what it is due on a loan to a borrower; at worst, the borrower defaults.³¹ Conversely, market risk is generally viewed as two-way since there is the possibility of both gains and losses both in absolute and in relative terms.

Market Risk: The Risk of Loss through Adverse Market Movements

Market risk is . . . the exposure to the sensitivity of value of a . . . portfolio to changes in market parameters [including] foreign exchange rates, interest rates, equity market indices, implied volatilities and commodity prices.

—“The Practice of Risk Management”³²

Market risk is the risk of loss to a bank arising from changes in rates or prices, including interest rates and foreign exchange rates. More specifically, it is the risk that a change in interest rates, foreign exchange rates, or prices—such as the prices of real property, commodities, equities, or other securities—will adversely affect a

bank's financial performance or position.³³ In modern banking, market risk is practically as ubiquitous as credit risk, affecting all banks to the extent that they take positions³⁴ in a market economy where interest rates rise and fall or insofar as they engage in international transactions.

Generally speaking, interest rate risk is more important to banks than foreign-exchange risk (or other less common forms of market risk) since nearly all institutions are subject to it to a significant degree. Moreover, changes in interest rates directly affect the profitability of lending, which traditionally provides the lion's share of bank earnings,³⁵ and also may affect the bank's funding liquidity.³⁶ The impact on profits can be dramatic depending upon the nature and terms of its funding arrangements. For example, if a bank has many fixed-rate loans outstanding that are generating a comparatively low rate of interest and such loans are funded by comparatively short-term deposits and interbank borrowing, it will suffer should prevailing interest rates rise significantly.³⁷ For these reasons, interest-rate risk tends to receive the most attention of all market risks.

MARKET RISK

Market risk is the risk that changes in prices or rates will adversely affect a bank's financial performance or position.

The pervasiveness of interest rate risk arises from its being intrinsic to the structure of the typical bank's balance sheet. Most institutions borrow short from depositors and in the money markets and lend long, relatively speaking, giving rise to potential changes in the spread between the cost of funds and the yield on those funds. It follows that the level of such interest rate risk to which a bank is subject depends upon the structure of the balance sheet, particularly in respect to the interest-rate sensitivity of the corresponding assets and liabilities.³⁸

For this reason, the interest rate risk arising from the bank's balance sheet as a whole is sometimes referred to as structural interest rate risk. In the context of such risk, interest rate sensitivity concerns the relative vulnerability of a bank's net interest income stream to the changes in assets and liabilities resulting from interest rate changes. More specifically, an asset-sensitive bank is one for which its assets reprice faster than its liabilities; a liability-sensitive bank is one for which its liabilities reprice faster than its assets.

A classic illustration of the potentially catastrophic impact of interest rate changes on a class of financial institutions was that of American savings and loans institutions (S&Ls) in the late 1970s and early 1980s. For reasons that were partly historical and partly regulatory, the vast majority of these institutions were at the time borrowing on a short-term basis at variable interest rates but lending long at fixed rates. Interest rates rose significantly in the early 1980s as the U.S. Federal Reserve Bank experimented with monetarism in a bid to end the era of stagflation (inflation without corresponding economic growth). The spike in rates had a strongly adverse impact on the S&Ls. The spread between the institutions' yield on earning

ASSET-SENSITIVE BANKS VERSUS LIABILITY-SENSITIVE BANKS

Asset-sensitive banks hold assets that tend to reprice faster than their corresponding liabilities; in other words, loans reprice faster than funding costs. In a rising interest rate environment, asset-sensitive banks benefit since their interest revenues tend to rise faster than their funding costs. This means that increases in market interest rates can be passed onto borrowers quickly while depositors are apt to remain satisfied with relatively smaller increases in deposit rates. Usually such banks are highly creditworthy institutions with a high proportion of sticky deposit funding sourced from a well-established branch network. On the asset side of their balance sheets, short-term and floating rate instruments predominate.

Liability-sensitive banks are those whose liabilities tend to reprice faster than their assets; in other words, loans reprice faster than funding costs. Usually, such banks are relying heavily on commercial funding or other short-term floating rate sources, while a high proportion of their assets are either fixed-rate or are comprised of loans to borrowers that do not have a wide variety of funding options and are therefore unlikely to quickly find cheaper sources of funds. This type of bank benefits from falling interest rates since funding costs fall faster than interest revenues, which tend to remain at elevated levels for the reasons just given.

To an extent, banks will frequently attempt to make some adjustments in their asset/liability positions depending upon the interest rate environment. For example, in a falling interest rate environment, a bank will want funding costs to reprice faster than its revenues so that lower interest rates need not be passed on quickly to borrowers. Hence, it may move to more emphasis on commercial funding, or it will enter into sets of transactions to adjust its asset/liability profile.

assets and their cost of funds turned negative as rates paid on short-term deposits quickly outstripped their long-term, fixed-rate mortgage portfolio yields. High interest rates also reduced loan demand dampening operating income. The combination of these factors ultimately led to the collapse of numerous S&Ls and to what effectively amounted to a bailout of the entire industry in the late 1980s and early 1990s.

Other market risks related to interest rate risk include reinvestment risk, rollover risk, prepayment risk, and inflation risk. Reinvestment risk is the risk that, owing to changes in prevailing interest rates, the bank will, upon the maturity of an existing investment, be unable to obtain a new investment offering a return equal to or higher than that of the maturing investment.

In the same manner, rollover risk refers to the risk that the bank will, upon its repayment of an existing funding obligation, be unable to roll over funding at an interest rate equal to or lower than that of the maturing funding instrument.

Prepayment risk, which is commonly associated with long-term residential mortgage loans and instruments securitized from them, is the risk that borrowers will

prepay the mortgage prior to maturity thereby depriving a lending bank of a viable earning asset. Prepayment risk is by and large a form of interest rate risk akin to reinvestment risk. It tends to occur when borrowers are able to refinance their mortgaged property at a lower interest rate than the one originally obtained.

Finally, inflation risk is another form of market risk that is indirectly a form of interest rate risk. Except when holding inflation-linked bonds that provide for an increase in the coupon rate in accord with changes in the inflation rate, bond investors ordinarily lose out during periods of sustained high inflation, since rising costs reduce the buying power of their interest income. Since the real return of all investments is equal to the coupon rate minus the inflation rate, inflation can result in bonds generating a negative return in real terms.

Credit Risk: The Core Bank Risk

The mere prospect of a future default . . . on its customer obligations can effectively destroy those businesses.

—Robert C. Merton³⁹

Credit risk primarily comprises the risk of loss arising from a default of an obligation to the party subject to the loss, and takes in both the probability of default associated with the issuer and the severity of the loss associated with a given obligation in the event such default.⁴⁰ While default is binary in nature—an entity is either in default or it is not—the probability of default is a continuum ranging from no possibility of default (0 percent) to a defined default event (100 percent).⁴¹ Some credit risk exists so long as there is the possibility of default, however remote.⁴²

It follows that credit risk encompasses the risk of a decline in credit quality, referred to as credit spread risk in the context of marketable debt securities. If the obligor is rated, a significant decline in credit quality will almost inevitably result in a ratings downgrade creating what is termed *rating migration risk*.⁴³ In a sense, the two forms of risk are the same since debt ratings are intended to correspond with levels of relative credit quality and concomitant credit risk.

All other things being equal, both credit spread risk and rating migration risk, if realized, will cause marketable loans made to borrowers—or debt securities held by the bank—to decline in value as perceived credit risk rises. The demand for a higher risk premium, that is, a higher return relative to the risk-free rate, on the part of investors' accounts for the lower price they will be willing to pay for an instrument, the credit quality of which has declined.⁴⁴ Both a substantial increase in an issuer's credit spreads (versus the benchmark risk-free rate) and a ratings downgrade may trigger problems for a bank in accessing wholesale funding.

Credit risk can be divided into the risk arising from the default of an issuer or counterparty, on the one hand (issuer/counterparty risk), and the default risk associated with a particular issue or obligation (issue-specific risk), on the other. Ratings of securities issued by the same issuer can differ because of differences in their respective terms and conditions, including seniority, structure, and presence or absence of particular covenants.

Credit risk can be further differentiated according to whether it refers to the credit risk of a single transaction or to that of an entire portfolio. In the latter case, concentration

CREDIT RISK, CREDIT SPREAD RISK, AND RATING MIGRATION RISK

Credit risk is the risk of loss from an obligor's default. Credit risk takes in the amount of exposure at default and the severity of the loss over a given time period. Broadly speaking, it encompasses the risk of changes in credit quality (corresponding to an increased risk of default or of default severity) reflected in a corresponding decline in the value of the associated obligation (an asset from the point of view of the holder) or in an adverse change in the rating assigned to the obligor or to the relevant obligation.

Credit spread risk is the risk that, owing to a decline in an issuer's credit quality or in the credit quality of the subject obligation, the interest rate premium required by investors in its obligations will increase relative to the risk-free rate thereby generally reducing the market value of such obligations.

Rating migration risk is the risk of loss arising from a change in the prevailing credit ratings assigned to an obligor or an obligation, the ratings being a proxy for credit quality.

risk and asset correlation risks ordinarily must be considered when calculating the magnitude of credit risk expressed in terms of expected loss. Concentration risk, which is one aspect of asset correlation risk, occurs when a high proportion of the portfolio assets are obligations of a single obligor, or of multiple obligors that share basic characteristics. Obviously, where, for example, a bank's loan book includes a large percentage of loans to the same borrower, should that entity become insolvent, the bank will be very badly affected.

In respect to multiple obligors, relevant characteristics that generate concentration risk include the industry and the geography of the obligors. Where, for instance, a large proportion of a bank's loan book is comprised of loans to the shipping industry, industry concentration risk exists, and a downturn in the shipping industry can be expected to have an adverse impact on most, if not all, of the obligors in that subsector. Similarly, where most of a bank's borrowers are located in a specific region of the country in which the bank operates, a regional downturn or secular changes negatively affecting the region will again tend to hurt all the borrowers in that region. Their difficulties will in turn cause difficulties for the bank as these borrowers default upon their loans.

Industry concentration risk will result in a disproportionate adverse impact on a bank that has a disproportionate number of loans to the affected industry. While the potential harm presented to the bank in such a circumstance may appear self-evident, it is the correlation among firms in the same industry that explains the effect. Sometimes, the correlation among obligors may not be quite so obvious, and identifying the linkages can be problematic. But an effort must nevertheless be made to do so in order to properly calculate the expected loss of a credit portfolio.⁴⁵

Some forms of credit risk are categorized as separate risks as a result of their distinctive features. Chief among these are settlement risk (discussed in further detail later), transfer risk, and sovereign risk. Settlement risk is a subcategory of credit risk that arises in the course of the settlement of a transaction, that is, through the exchange of quantity of assets (including cash) for another, which is normally a matter of days at most. Sovereign risk is the risk of default by sovereign nations or subnational entities. Transfer risk concerns the risk of default arising through the cross-border default owing to the imposition of sovereign restrictions upon such transfers among other potential causes. Accordingly, it is closely related to sovereign risk.

As sovereign nations are, as a rule, not subject to bankruptcy proceedings, loss recovery is potentially more problematic than in respect to ordinary corporations and banks, there being no international bankruptcy court to which nations are subject. The former, of course, are subject to the jurisdiction of bankruptcy courts at the national or subnational level. Sovereign risk encompasses both the risk of sovereign default, and the political, economic and systemic risk criteria that are predictive of such a default. These sovereign risk factors may have a direct impact on operational risk as well as upon the other major categories of bank risk.

Settlement Risk

A somewhat specialized type of credit risk, alluded to earlier, is settlement risk. Settlement risk is a form of credit risk that arises during the process of consummating a transaction through the exchange and delivery of assets (including cash and securities) to each party as previously agreed. The risk ordinarily arises as the result of a gap occurring (e.g., through differences in time zones) between the time payment is made and delivery is received, creating the possibility that one counterparty may not complete its part of the transaction. For instance, a particular counterparty could, in the interim, become insolvent or be blocked through government action from making payment. Note that by definition presettlement risk is credit risk other than that arising from settlement; in other words, what is conventionally thought of as constituting credit risk.

As with credit risk generally, the longer the time gap, the greater the settlement risk. Real-time settlement, if and when implemented, offers a way to minimize if not eliminate settlement risk. Because settlement risk tends to be associated with much lower expected losses than other forms of credit risk, it is frequently categorized separately for risk management purposes vis-à-vis counterparties. As a result, the exposure limits for settlement risk tend to be considerably higher than for ordinary counterparty default risk.

SETTLEMENT RISK

Settlement risk is a subset of credit risk that usually⁴⁶ refers to the risk of loss to either party to a transaction where settlement—that is, the exchange of funds and relevant instruments pursuant to the transaction—occurs in different time zones such that one bank that will make payment is subject to the risk that the other bank will fail to do so.

Calculating Credit Risk

For risk management purposes, credit risk is commonly measured by expected loss (EL). Recall that the concept of expected loss introduced in Chapter 1 referred to a bank’s average credit costs. Since by definition EL is expected, it is properly viewed as a cost of doing business that management should take into account.

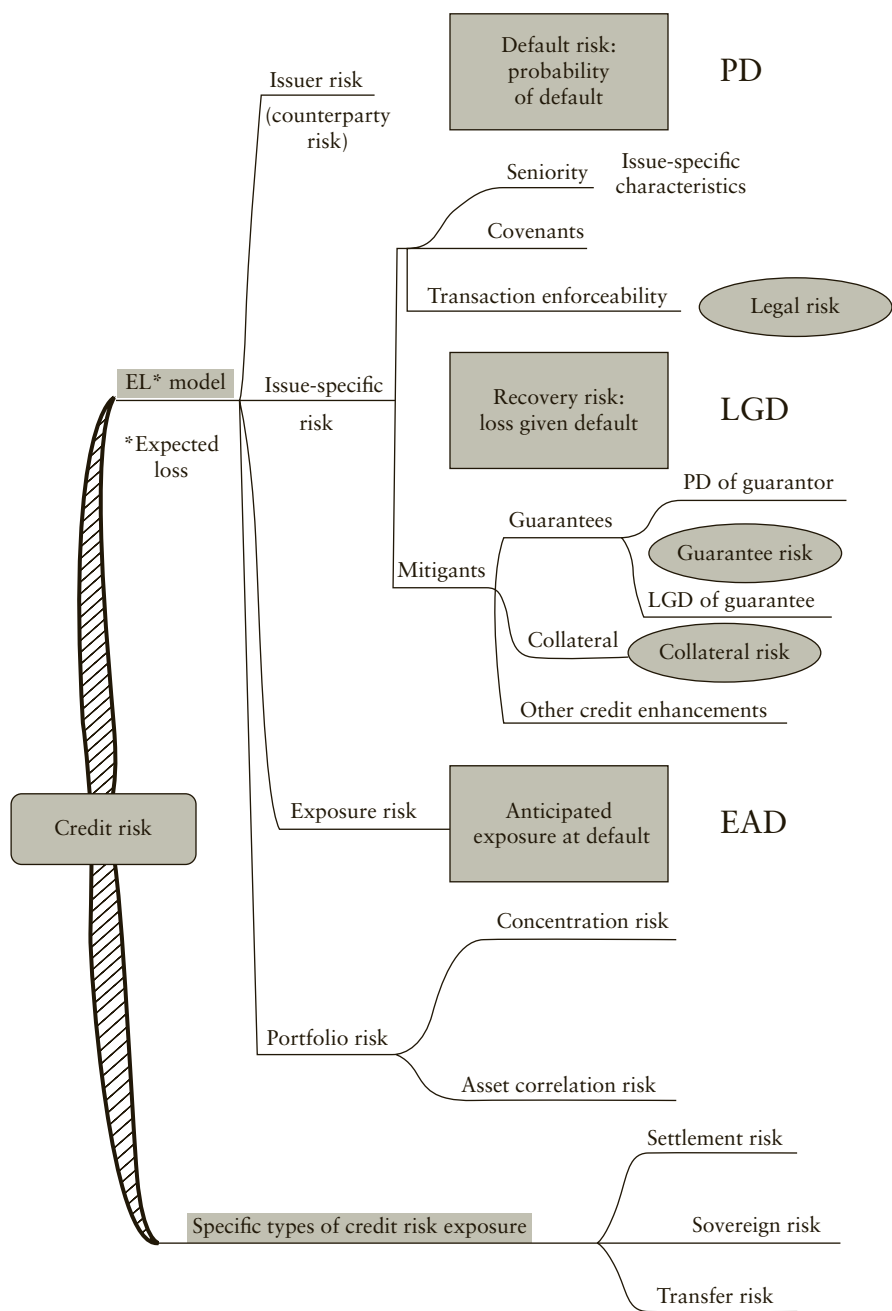


EXHIBIT 12.4 Taxonomy of Credit Risk

It can be either calculated retrospectively, or estimated prospectively, by multiplying the product of the probability of default or PD (sometimes referred to as expected default frequency or as risk quality) of the obligor, by the loss given default or LGD (sometimes called the loss severity) of the obligation, and by the exposure at default or EAD, which is the amount at risk at the time of default. Note that PD corresponds to issuer (i.e., obligor) risk, while LGD corresponds to issue-specific risk. The former is dependent upon the creditworthiness of the obligor, while the latter will naturally depend upon the particular terms of the obligation, including the seniority of investors and the specific investor protections provided by the governing agreement.⁴⁷

Although default is a binary criterion and EAD is a calculated amount, PD and LGD can range from 0 to 100 percent. Consequently, expected loss can range from 0 to 100 percent of EAD. To take account of concentration risk and correlations among various types of assets within a portfolio (asset correlation risk), EL is subject to adjustment factors. Finally, EL is calculated with reference to a specific time period. The longer the time period, the greater the expected loss and the greater the credit risk, all other things being equal. Exhibit 12.4 illustrates the various types of credit risk and the concept of expected loss.

As illustrated, the level of credit risk can be measured in respect to a given time period by estimating or retrospectively calculating the probability of default (PD), loss given default (LGD), and the amount of exposure at default (EAD). The product of $PD \times LGD \times EAD$ is the expected loss (EL), which somehow represents the average credit cost associated with the exposure. On a portfolio basis, EL may be adjusted for identified correlations among the assets that comprise the portfolio.

Operational Risk

Operational risk . . . has in the past received rather less attention than other aspects of risk.

—“The Practice of Risk Management”⁴⁸

Of the three major categories of bank risk—credit risk, market risk, and operational risk—the last is perhaps the hardest to define because, in a sense, it constitutes a catch-all grouping for significant risks that cannot be placed in the first two classes. A good general definition of operational risk was enunciated by the Basel Committee on Banking Supervision. It defined operational risk as “the risk of direct or indirect loss resulting from inadequate or failed internal processes, people and systems or from external events.” As with all risks a bank faces, these risks can have a knock-on impact on other risk categories potentially impairing a bank’s overall creditworthiness.

OPERATIONAL RISK

Operational is “the risk of direct or indirect loss resulting from inadequate or failed internal processes, people and systems or from external events.”⁴⁹

EXHIBIT 12.5 Functional Categories of Operational Risk

Classification	Loss-Generating Activity
Internal Fraud	Misappropriation of assets, tax evasion, intentional mismarking of positions, [bribery]
External Fraud	Theft of information, hacking damage, third-party theft, and forgery
Employment Practices and Workplace Safety	Discrimination, workers compensation, employee health and safety
Clients, Products, and Business Practice	Market manipulation, antitrust, improper trade, product defects, fiduciary breaches, account churning
Damage to Physical Assets	Natural disasters, terrorism, vandalism
Business Disruption and Systems Failures	Utility disruptions, software failures, hardware failures
Execution, Delivery, and Process Management	Data entry errors, accounting errors, failed mandatory reporting, negligent loss of client assets

Source: Basel Committee on Banking Supervision.

Operational risk can be classified in a number of different ways. In a paper published by the Basel Committee, operational risk was divided into seven classes by grouping some of the more conspicuous sources of such risk by key identifying characteristics. These can be seen in Exhibit 12.5.

Other useful distinctions in classifying operational risk include:

- The source of the risk (internal to the bank or external)
- The principal cause (human, technological, or natural phenomenon)
- The nature of the resultant loss (e.g., physical damage, fines or litigation costs and damages, loss of reputation, loss of business or loss of capacity to do business)

Because operational risk incorporates risks throughout the bank that are not identified as market risk or credit risk, responsibility for monitoring them and implementing appropriate procedures and risk mitigation will naturally be devolved in large part to the departments within the enterprise that have primary responsibility for them. Therefore, the IT department will be closely involved with establishing backup policies and safeguards against data loss, the HR department will have a similar role in vetting staff and training them concerning compliance requirements, and the legal department will do the same with regard to documentation requirements and keeping staff informed concerning the impact of new legislation.

For purposes of describing the range of operational risks, they are divided here into those that are primarily internal in origin and those that are external. Internal risks are further divided into two subcategories: first, what may be termed human factor risk; and, second, systems risk.⁵⁰

Internal Risks

Human factor risk refers to the risk of loss arising through either (a) human error or (b) malfeasance (including fraud and criminal activity). The two can be distinguished

largely with regard to intent. Risks caused by human error take in acts or omissions that are inadvertent or the result of incompetence, as opposed to being motivated by an individual's desire to personally profit from engaging in activity that is illegal or contrary to the rules or interests of the bank. They are conventionally addressed through the imposition of detailed procedures and controls aimed at reducing their occurrence and the severity of any losses. In practice, human factor risks often emerge from problematic transactions that are largely attributable to human error (transaction risk) or errant humans (operations control risk).⁵¹

Loss can arise from an intended transaction that is defective or an unintended transaction entered into against bank policy. Respective examples might include the failure to complete the documentation necessary to make a bank's claim on collateral securing loan legally enforceable and the violation of internal rules by a rogue trader. Hence, errors in documentation, booking, settlement, and delivery would fall into the first group. Unauthorized trading, embezzlement, and fraud would fall into the second.

Systems risk concerns the possibility of a breakdown in a bank's critical systems, including telecommunication, information technology, payments processing, or similar components of the organizational infrastructure.⁵² An illustration of this class of risk would be the loss of business that might occur should a bank's website need to be shut down for an extended period of time to repair poor programming or recover from intentional interference from inside or outside.

Event Risk (External Risk)

In addition to the primarily internal risks just discussed, operational risk includes risks associated with external events, particularly those of an extraordinary character. This category is sometimes termed *event risk*, *business risk*, or *business event risk*, and embraces momentous occurrences of a political, legal, or regulatory character, as well as catastrophic natural events that are apt to negatively affect the bank's financial performance or condition.

More specifically, event risk covers events in the following subcategories of operational risk:

- Legal risk, including legal or regulatory changes that may adversely affect a bank financially, the risk that the bank may be unable to comply with legal/regulatory requirements, as well as the risk of unfavorable court judgments or administratively imposed fines or penalties⁵³
- Business events that are not closely intertwined with changes in law and regulation, such as a ratings downgrade of the bank itself, the introduction of new technologies that put the bank at a disadvantage relative to key competitors, or loss of reputation owing to a scandal at the bank (reputation risk)⁵⁴
- Systemic risk, arising from events that reflect an overall failure of the banking system, or the closure or suspension of financial markets
- Political risk, including events of war, revolution, civil disorder, and terrorism—including more broadly events that signify changes in overall sovereign or country risk (which encompass political and systemic risk)
- Natural disasters, including damaging hurricanes, tsunamis, tornadoes, volcanic eruptions, earthquakes, and calamities of a comparable magnitude in respect to their impact on the bank (e.g., the loss of key bank officers in a plane crash)

Risk Map

Taking account of the preceding discussion and in spite of the difficulty inherent to such a classification, the division of risks shown in Exhibit 12.6 may be useful.

The three basic categories of risk—credit risk, market, and operational risk—form the core of the territory represented, but for practical reasons solvency risk and liquidity risk are added. This creates a risk classification framework that fairly

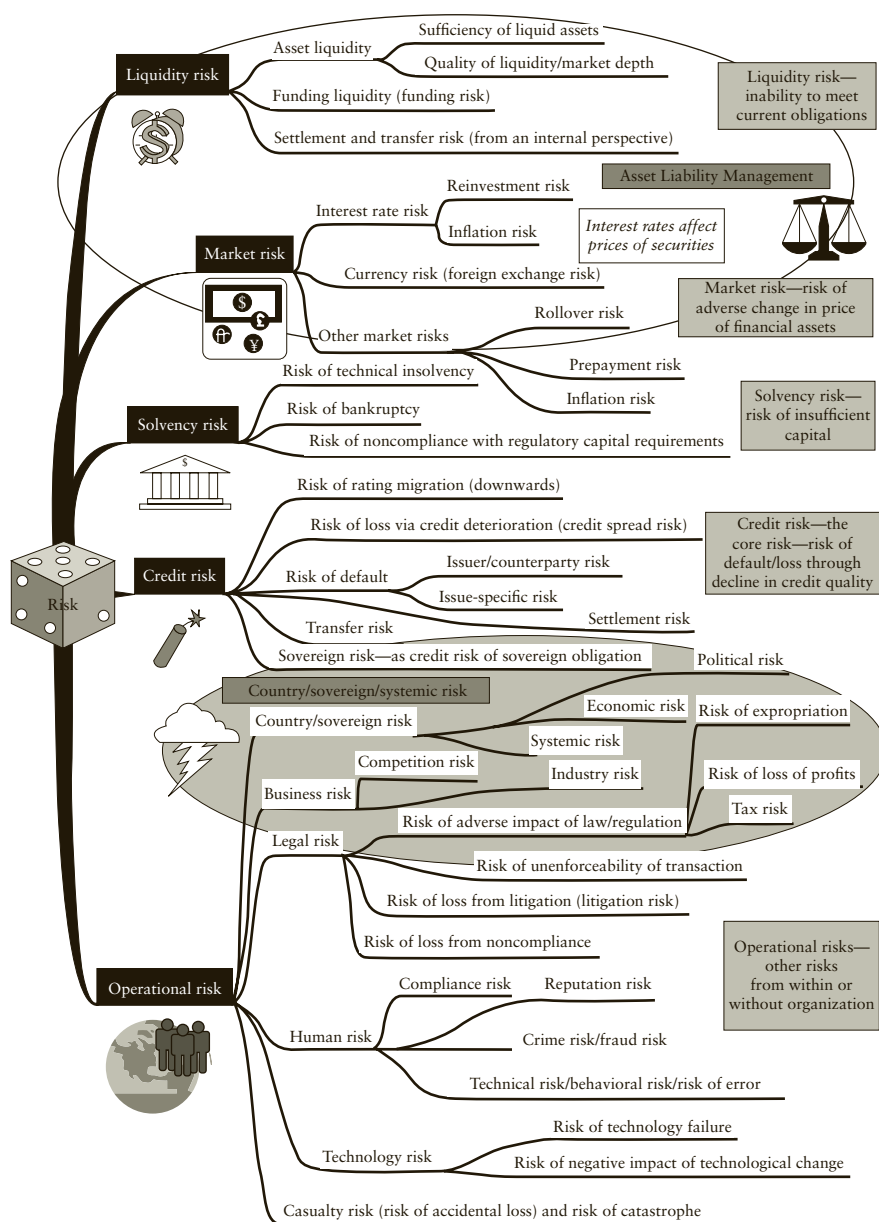


EXHIBIT 12.6 A Risk Map: Selected Classifications

corresponds to the CAMEL model. The evaluation of capital adequacy corresponds to solvency risk, asset quality to credit risk, earnings and profitability (roughly) to market risk, and liquidity and funding to liquidity risk.⁵⁵

RISK MANAGEMENT METHODS

[Value-at-risk is] . . . a number invented by purveyors of panaceas for pecuniary peril intended to mislead senior management and regulators into false confidence that market risk is adequately understood and controlled.

—Barry Schachter⁵⁶

Impelled by a desire to maintain a bank's solvency, risk management represents an attempt to anticipate prospective losses and—depending upon estimates of their variability relative to returns and potential magnitude—take steps to limit or reduce a bank's exposure to such losses. This is a more difficult task than it sounds, since it constitutes an effort to predict (and sometimes to alter) future outcomes arising from the full spectrum of a bank's operations, where each outcome is frequently subject to multiple influences thereby making causation difficult to ascertain. Unlike gambling where the odds of winning (or more likely losing) at the roulette wheel can be calculated with precision, in the real world of finance, computing expected losses and unexpected losses across the range of risks to which a bank is subject is in most instances an almost impossibly challenging task.

The volume and the heterogeneity of exposures and the myriad of dynamics affecting outcomes make it an undertaking of almost unimaginable complexity. Hence, in an effort to maintain institutional solvency, risk management in banking began as an art that relied heavily upon experience, judgment, and a very strong dose of conservatism.

Initially, risk management was largely synonymous with the management of credit risk, although banks have long implemented basic measures to prevent or mitigate operational losses through limits, controls, and standard operating procedures. As was previously touched upon, the development of the financial markets, product innovation, and the growth of bank participation in these markets and products during the later decades of the last century resulted in the need for greater management of market risk. In contrast to credit risk, critical market risk inputs, namely prices, interest rates, currency rates, and related market data, are more readily accessible and arguably more easily measured than their credit risk counterparts. Default and recovery rates have traditionally been proprietary, and character and historical data are comparatively limited.

As a consequence, the move to a more systematic approach to risk management began in earnest with market risk management. The development of asset liability management approaches represented a first move to a more evolved and methodical model of risk evaluation. The emergence of concepts such as value at risk (VaR) coupled with the computing power required to apply them in the real world, set the stage for the wholesale application of statistical and quantitative methods, first to market risk management and subsequently to the management of credit risk and operational risk as well.⁵⁷ This section provides a brief overview of risk management methods, focusing first on the area that has been at the forefront of change: market risk management.

Market Risk Management

Market risk is most apparent in connection with assets that trade in the financial markets, such as fixed-income and equity securities, and in wholesale funding liabilities. But loans and deposits are also affected by market risk, as suggested in the earlier mention of prepayment risk. Interest rate risk, and the management of such risk, is frequently divided according to the part of the bank's operations affected: namely, the banking book, and the trading book.

The *banking book* refers to the institution's overall funding and lending positions as just illustrated above, while the trading book refers to its comparatively short-term holdings of debt securities; that is, those that are actively traded by the bank's trading desk.⁵⁸ Broadly speaking, the management of market risk arising from the banking book is addressed through asset-liability management techniques, such as gap analysis. Analysis of the market risks associated with trading positions is more complex than banking book interest rate risk and is typically addressed through the application of the highly quantitative value-at-risk (VaR) methodology discussed later in this section.⁵⁹

BANKING BOOK VERSUS TRADING BOOK

It is necessary to distinguish between interest rate risk arising from the structure of a bank's balance sheet in respect to its general banking operations, including most importantly its lending operations, and the funding and allocation of the assets in its trading book. Generally speaking, the banking book is defined as all assets and liabilities that are intended to be held to maturity in order to generate an income over time. These activities traditionally have come within the purview of the bank's asset-liability committee.

In contrast to the banking book, balance sheet items that are utilized primarily for trading purposes including debt securities, repo finance, and similar instruments—the trading book—are managed in a somewhat different manner. The risk management of the trading book is characterized by the use of mark-to-market accounting and risk quantification methodologies, and, since the mid-1990s, VaR methodologies. Risk management of both usually involves the establishment of formal risk exposure limits and appropriate risk monitoring and reporting.

Note that the distinction also affects the way latent profit or losses are accounted for.

Asset Liability Management (ALM)

Banks are by nature transforming money, in that it is near impossible for them to match the maturities of their assets and liabilities. The pursuit of advantage through asset/liability mismatches and interest rate differentials is at the heart of banking and of the quest for greater profitability. Unexpected or abrupt changes in interest rates,

however, can have an adverse impact upon a bank that has taken positions, or has received them through fresh customer or counterparty transactions, later found to be incorrect, or without adequate hedging. The bank's funding costs may rise, resulting in a squeeze in margins. Consequently, there has long been recognition of the importance of identifying and managing interest rate risk in respect to the banking book. Not only may profits be affected by changes in interest rates, but liquidity may constrain the institution's ability to access necessary funding. In a worst-case scenario, this could threaten the survival of the bank. For the latter reason, asset liability management (ALM) was introduced in Chapter 10 concerning liquidity.⁶⁰

In addition to its use in the context of liquidity risk management, ALM constitutes the traditional form of market risk management with respect to the banking book, and remains an indispensable part of modern banking operations. It encompasses a broad range of bank activities, including the identification of imbalances between earning assets and funding liabilities, on the one hand, and the formulation and implementation of strategies to mitigate or ameliorate identified risks, on the other. Responsibility for these matters lies with a bank's asset liability committee or ALCO.

First, ALM requires that bank management actively monitor and manage the bank's asset and liability position, together with the size and extent of its interest rate (as well as its interest rate in foreign currency) exposure, in order to identify the magnitude of the interest rate risk to which the institution is exposed.⁶¹ This is accomplished through the application of gap analysis and sensitivity analysis techniques. Second, ALM embodies a variety of methods that may be implemented to reduce the severity of any negative outcome such risk is likely to generate, or even mitigate it entirely.⁶²

Among other additional techniques, risk mitigation incorporates the following fundamental elements:

- Setting policies regarding acceptable interest-rate risk taking account of liquidity thresholds, limits on debt, and minimum capital levels
- Cultivating and maintaining access to a range of funding sources
- Employing hedges to mitigate mismatches of various kinds

THE ROLE OF THE ASSET-LIABILITY MANAGEMENT COMMITTEE

In a bank, the asset and liability committee (ALCO) is responsible for implementing asset and liability management. ALCO's main objective is to plan and control financial flows and to structure and develop the bank's balance sheet in order to achieve target levels for profitability while providing for adequate liquidity, capital, and risk management. As part of this process, the ALCO will normally restrict aggregate risk exposure in relation to the bank's capital. A conservative aggregate risk exposure limit of 5 percent of capital and reserves is considered prudent. In addition, the ALCO will periodically set, as required, applicable interest rate levels and terms for both assets and liabilities and make

decisions regarding permitted maturities and pricing with respect to various types of assets and liabilities. Depending on the operating conditions faced by the bank, the ALCO will meet on a daily, weekly, or monthly basis as required.

In managing the bank's assets and liabilities, a bank's ALCO considers issues such as interest rates and yields, the size of the loan and investment portfolios, loan maturities, time deposits and investments, foreign currency positions, exchange rates, and inflation rates together with other macroeconomic factors. The ALCO ideally will strive to ensure that acceptable levels of financial risk, excluding credit and operational risk, are identified and optimally managed, while still aiming to achieve the financial objectives of the bank.

Other important objectives include cash flow forecasting and strategic planning, maintaining a strong presence in selected target markets, holding an adequate pool of high-quality marketable assets, and maintaining diversification in funding sources. Liquidity risk management forms an integral part of balance sheet management. Although primarily responsible for the banking book, the ALCO will consider the risk and correlation within the bank's investment and trading portfolios, as well as in relation to the banking book, allocating funds in the process.

Key techniques used in ALM are gap analysis, balance sheet stress testing, and earnings-at-risk modeling. Gap analysis measures the volumes of assets and liabilities subject to repricing within a given period, ideally on a monthly or three-monthly basis. Accordingly, assets and liabilities are classified according to their contractual repricing characteristics. In addition, the cumulative difference between rate-sensitive assets and rate-sensitive liabilities is calculated, say up to five years and over, to pinpoint any possible large mismatches.

Banks use balance sheet stress testing and net interest income simulations for a variety of possible interest rate scenarios, against which the effect of interest rate movements can be measured. In the process, the impact of a 1, 2, and 5 percent change in short-term interest rates on earnings is measured. Depending upon the relationship between assets and liabilities, the impact may be disproportional to the change in interest rates. For example, a bank's exposure to an adverse 1 percent movement in interest rates could conceivably have a 2 percent overall impact on net interest income.

The next step is to calculate the projected change in bank profits and equity over a forecast period based on the most likely high and low interest rates scenarios for the period. The bank will then implement strategies for the mitigation of interest rate risk. Such strategies range from the frequent repricing of loans and deposits to the use of complex hedging techniques. Unless cost-effective hedging strategies are available through interest rate swaps, or other hedging mechanisms, a good rule of thumb is that loans over one year in tenor should be extended only at a floating rate. Similarly, if a bank has a fairly large mortgage book, all of this lending should be done at variable rates, or at the very least, only a small proportion done at longer-term fixed rates.⁶³

GAP ANALYSIS

By examining the composition of a bank's balance sheet in terms of its interest-sensitive assets and liabilities, it is possible to profile its interest-rate sensitivity through the division of such instruments in accord with their maturity or repricing characteristics. To perform an interest-rate sensitivity analysis, the first step therefore is to divide assets and liabilities into maturity categories (sometimes called buckets). See the example in Exhibit 12.7, which provides an illustration of such a profile, where the buckets are divided into five repricing categories designated as follows, with instruments having a maturity of:

1. Less than 3 months
2. From 3 months to 6 months
3. From 6 months to 12 months
4. From 1 year to 5 years
5. Over 5 years

A separate category is provided for non-interest-sensitive assets.

Plus Ultra Bank—Interest Rate Risk Analysis	<3 months	3–6 months	6 months–1 year	1 year–5 years	>5 years	Non-interest Sensitive	Total			
Shareholders' funds						9000	9000	1. Add up liabilities by maturity to arrive at Total Liabilities in each maturity category.		
L/T Subordinated debt					500		500			
Deposit, current, and other accounts							0			
Deposits by banks	5000						5000			
Deposits by customers	48000	1500	2500	13000	1000		66000	2. Do the same for assets to arrive at Total Assets for each maturity category.		
NCDs and repos	5000	1500	500	500			7500			
Other liabilities						7000	7000			
Total Liabilities	58000	3000	3000	13500	1500	16000	95000			
Cash and S/T funds						7000	7000	3. Subtract Total Assets from Total Liabilities in each maturity category, adjusting for off-balance-sheet gaps.		
Other S/T securities	5000	200	100				5300			
Government securities	1000	800	200	1000	1000		4000			
Loans to banks	4000						4000			
Loans to customers	49000	3500	3200	11000	4000		70700	4. Add up gaps for each maturity category to reach cumulative gap.		
Other assets						1000	1000			
Customers' indebtedness						500	500			
Associates/investments						1500	1500			
Fixed property						1000	1000	5. Divide the cumulative gap by total assets to obtain cumulative gap as % of total assets.		
Total Assets	59000	4500	3500	12000	5000	11000	95000			
<i>Off-balance-sheet items</i>	<i>3000</i>	<i>–100</i>	<i>–100</i>	<i>–2800</i>						
Interest rate sensitivity gap	4000	1400	400	–4300	3500	–5000	0			
								5. Divide the cumulative gap by total assets to obtain cumulative gap as % of total assets.		
Cumulative gap	4000	5400	5800	1500	5000	0	0			
Percentage of total assets	4.2%	5.7%	6.1%	1.6%	5.3%	0.0%	0.0%			

EXHIBIT 12.7 Gap Analysis

The additional steps are as follows:

Step 2: Add up liabilities by maturity to arrive at Total Liabilities in each maturity category.

- Step 3:** Do the same for assets to arrive at Total Assets for each maturity category.
- Step 4:** Subtract Total Assets from Total Liabilities in each maturity category, adjusting for off-balance-sheet gaps. Gaps for off-balance-sheet items are calculated in the same way as balance sheet items.
- Step 5:** Add gaps for each maturity category to reach cumulative gap.
- Step 6:** Divide the cumulative gap by total assets to obtain cumulative gap as a percentage of total assets.

Maturity mismatches are part and parcel of banking—often because of the short-term nature of deposits and long-term nature of loans. Nevertheless, banks should avoid excessive maturity mismatch as a high negative position may present problems if the funding environment becomes more difficult. Particularly important when examining maturities is the need to look at the maturity structures of long-term debt, if any. Long-term debt should be spread out so that the obligations do not mature during the same financial period to facilitate retirement of the debt. Should large amounts be coming due within the year, the analyst should pay extra attention to the bank's liquidity position to ascertain whether she is comfortable with the institution's capacity to meet obligations coming due.

Maturity mismatch analysis is similar to interest-rate sensitivity analysis, with the emphasis being on the maturity mismatch instead of the degree of asset or liability sensitivity. The object is to discern whether there is any large mismatch whereby the bank is funding long-term assets with short-term liabilities. Again, maturing assets and liabilities are broken down based on maturities or repricing intervals, but are not limited to those that are interest-rate-sensitive. The cumulative position is also calculated to determine whether the bank is carrying a positive (assets in excess of liabilities) or negative (liabilities in excess of assets) gap. The extent of the gap position is measured against the total assets of the balance sheet. A negative gap greater than 10 percent of the asset base starts to be on the high side, and the position needs to be examined more closely to determine if there are likely to be any problems.⁶⁴

While this is conceptually different from the matter of liquidity gaps, an important consideration for the bank's funding position is the degree to which foreign currency liabilities match foreign currency assets. Put simply, does the bank possess a well-matched currency position? It is preferable—and, in most countries, mandatory—for a bank to run a closely matched foreign currency position.⁶⁵ If the bank is running an unmatched foreign currency position, it is exposing itself to currency risk, which, if the movement is adverse, can add significantly to the bank's cost of funds through a large foreign currency translation loss.

In high inflation economies, short foreign currency positions tend to be very common, and many banks in emerging markets strategically run short foreign

exchange positions as part of their overall asset-liability policy.⁶⁶ Such a strategy tends to be very lucrative (usually in these economies a lot of bank funding goes to high-yielding domestic government securities) providing that the rate of devaluation of the local currency is below the inflation rate.⁶⁷ This is particularly the case where domestic interest rates are elevated, making funding locally through either deposits or the interbank market expensive compared with low interest rates in hard currency economies. The yield difference between borrowing in the high interest rate domestic market and borrowing in the low interest rate hard currency market is highly tempting and can boost substantially the bank's interest rate spread. The bank will borrow at, say, 7 percent in U.S. dollars and on lend in local currency, say at 25 percent. This will be very profitable providing the local currency does not devalue significantly against the U.S. dollar.⁶⁸

Yet this carry trade could prove deadly whenever wild currency fluctuations overly disrupt the conditions under which the trade was profitable.

UNMATCHED CURRENCY POSITIONS

An unmatched position exposes the bank to foreign currency risk—either loss or gain. There are two ways a bank can have other than a matched currency position:

1. Foreign currency liabilities exceed assets: short position. A short position benefits if the foreign currency loses strength and value falls vis-à-vis the local currency.⁶⁹
2. Foreign currency assets exceed foreign currency liability: long position. A long position benefits if foreign currency gains strength and value rises vis-à-vis the local currency.

Currency positions generate profits or losses of a trading nature. Such profits or losses are independent from profits or losses arising from maturity mismatches, although banks may—and do—manage both situations jointly, with the view of creating a net profit.

Losses on one front may create profits on another front, and only the net profit—or loss—is, in fact, relevant.

Prior to the Asian crisis of 1997–1998, some countries, such as Indonesia and, most importantly, Thailand had high foreign currency exposure through borrowing overseas, and when the Indonesian rupiah and Thai baht fell sharply against the dollar, many local banks suffered heavy losses. To truly reflect the real funding cost, when banks strategically run short foreign currency positions, it is recommended that the analyst include any translation loss from the foreign currency position as part of normal interest expense.

ILLUSTRATION: ASSESSING FOREIGN CURRENCY EXPOSURE

Bank Gamma is located in a volatile, high-interest-rate, high-inflation-rate country, has an interest on average earning assets ratio of 22.47 percent and a funding cost ratio of 10.11 percent, culminating in an interest differential ratio of 12.36 percent, which is substantial.

However, the bank consistently runs a short foreign exchange position as part of its funding strategy in order to obtain cheap foreign funding, and it directs the funds to government paper and/or loan customers where the domestic interest rates are substantially higher.

The foreign exchange translation loss (the domestic currency is depreciating consistently against the U.S. dollar) amounts to US\$127 million. This was—as intended—more than compensated by an increase of US\$160 million in interest income, being the difference between actual interest received in local currency and actual interest paid in foreign currency on the short position.

The bank made an extra US\$33 million through the arrangement, which, as mentioned elsewhere, is frowned upon by regulators but often tolerated in emerging markets.

There seems to be no consensus on how to account for this. The bank will post a higher interest income at the expense of a negative foreign exchange trading income, in this example to the tune of US\$127 million.⁷⁰ Accounting adjustments are always possible, of course.

Note that the fictional illustration above assumes that the bank itself takes the foreign currency exposure. It borrows in U.S. dollars, sells the U.S. dollars for the local currency, lends the local currency, receives interest in local currency, recovers the principal in local currency, sells the local currency for U.S. dollars and pays back principal and interest in U.S. dollars.

When the bank does not take the foreign currency exposure, it borrows in U.S. dollars and lends the proceeds in U.S. dollars, which it recovers later, with interest, and pays it back. In that situation, the local borrower carries the foreign exchange risk, while it also enjoys some of the interest differential (between U.S. dollar rates and local rates). In our example, the bank would typically charge him, say, 16 percent on the U.S. dollar loan, keeping for itself a good margin in spite of the fact that it no longer, in theory, carries a foreign exchange risk.

The relevance to the risk analysis is that, should the borrower default, the borrower's foreign currency risk becomes the bank's currency risk, as the bank will still have to pay the foreign currency loan to its creditors. This played a role in the Asian crisis of 1997, as well as in the Argentina crisis of 2001, and it may yet play a role in Hungary in 2013.

Value-at-Risk

A bank's trading activities within the financial markets create the risk of loss due to adverse movements in rates and prices of currencies, interest rates in the domestic and international capital and money markets, equities, and commodities, as well as option volatility. This risk of loss owing to changes in market rates and prices is of a

different character from the structural market risk discussed in the context of the banking book. Value at risk (VaR) is the core methodology that started being used in the early 1990s to evaluate this type of market risk.⁷¹

What Is VaR? The term *value at risk* or VaR refers both to a risk assessment methodology and to a term defined—actually a numerical value calculated—using that methodology.⁷² As a methodology, VaR has been described as a “category of risk metrics that describe probabilistically the market risk of a trading portfolio.”⁷³ VaR methodologies emerged out of efforts to estimate the maximum losses apt to be caused by market movements; that is, to answer the question: “What is the worst-case loss scenario the bank could reasonably have to face over a given time period?” As such, VaR methodologies represent an attempt to apply common statistical concepts to the analysis of market risk.⁷⁴

As mentioned earlier, this can be accomplished by defining risk as the degree of uncertainty concerning losses; that is, volatility. VaR is ostensibly able to compute the probability of worst-case scenarios, thereby enabling risk managers to limit such scenarios to an acceptably remote possibility.

Within the VaR methodology, value at risk “is an estimate, with a given degree of confidence, of how much one can lose from one’s portfolio over a given time period.”⁷⁵ A daily value-at-risk of US\$10 million with a 99 percent confidence level means that, in one day out of 100, the bank can be expected to lose more than US\$10 million. This is the financial industry equivalent of deciding to make preparations for the once-in-a-100-year flood, but not for the once-in-a-1,000-year flood. Like disaster planning, the governing assumption⁷⁶ of financial risk management is that while it is sensible to prepare for periodically recurring calamities, it is not practical to conduct a business preparing for catastrophic events, the occurrence of which is judged to be so extremely remote that it is not cost-effective to prepare for them.⁷⁷

WHY VAR BECAME SO POPULAR SO QUICKLY

Although the statistical and mathematical concepts upon which VaR is based can be traced back many years, in its contemporary form the methodology had its origins in the early 1990s. First developed at J.P. Morgan, it only took a few years before VaR became a buzzword in the emerging and rapidly growing field of risk management and a ubiquitous tool to manage market risk at leading financial institutions.

The rapid spread of VaR, in particular, can be attributed to several factors.

First, J.P. Morgan did not keep the methodology proprietary, but effectively gave it away.

Second, notwithstanding that the mathematical computations required by the model to generate useful results are complex, and the volume of data required is high, the underlying concepts were fairly easy to grasp. (Propitiously, the requisite computing power had become widely available within the prior 10 years.)

Third, VaR models by and large generated results that were simple and easy to comprehend—namely at the most fundamental level a single number that expressed the maximum loss that could, in theory, be incurred by a financial institution in respect to the defined portfolio.

Fourth, the flexibility of the VaR model made it applicable to essentially any class of assets that comprised a portfolio, and with adjustments it could take into account most, if not all, of the critical variables such as asset correlation.

Finally, from the late 1990, securities regulators began to require that financial institutions active in trading quantify market risks for disclosure purposes.

For all the foregoing reasons, VaR was the obvious tool to accomplish this task, and the implicit regulatory imprimatur it received cemented its status as the most prominent tool for the calculation of financial risks.

Why VaR Works, to the Extent That It Does. The reason that VaR works, to the extent that it does, is that price volatility closely resembles, and can be measured by, the statistical concept of dispersion, meaning the degree to which outcomes are spread around the point of central tendency (e.g., the mean). This is fortunate because it means that the fluctuation of prices and other financial attributes can be analyzed and predicted using statistical methods. In essence, VAR-type methodologies, at least in their simpler forms, utilize well-established statistical techniques to estimate the probability of certain unknown future outcomes, given two known criteria: a portfolio's current value and the distribution of past and present price and rate movements. Given enough historical data to map a distribution of possible outcomes, the calculation of worst-case scenarios for a defined market portfolio to a prescribed confidence level is a problem to which VaR can provide a solution.

The concept of volatility and how VaR works can be illustrated simply by looking at the movement of the price of an equity security at frequent intervals, for example at the end of each trading day over a period of 200 days. (VaR is typically computed for one-day periods.) If the change in the price of the stock, expressed as a percentage increase or decrease from the previous data point is plotted as a histogram, the result will look a lot like a statistically normal distribution curve as shown in Exhibit 12.8.⁷⁸

The horizontal axis shows the percentage price change, which is on the right, was positive, and if on the left was negative. The vertical axis shows the number of times the price movement fell into a defined range of percentage change (e.g., 0.0 to +0.5 percent) from the previous price point. Volatility represents the degree to which the price movements deviate from the mean, which is the average or expected return.⁷⁹

Assuming a normal distribution, 68 percent of all such price movements will be within one standard deviation of the mean, 95 percent within two standard deviations and 99.7 percent within three standard deviations of it. Each of these percentage figures reflects a confidence interval, meaning that for the period described the specified percentage of events will not exceed a loss (or gain) represented by the

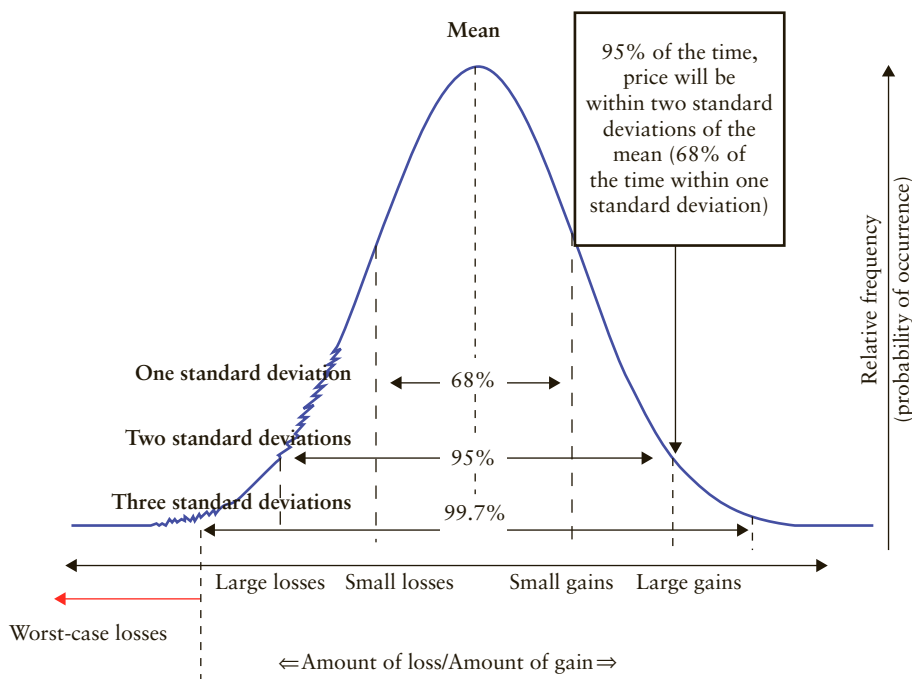


EXHIBIT 12.8 A Distribution of Prices over a Short Time Period

vertical lines to the left and right of the mean. Of course, the problematic data points are those to the far left, for they represent large unexpected market losses.

The same concept can be employed to analyze the volatility of the bank's total trading portfolio, or its constituent parts, and to facilitate the comparison of the riskiness of various trading operations within the bank. By applying the statistical characteristics of a normal distribution, to consider VaR in its simplest form, and by considering data from a bank's trading portfolio together with the relevant historical market data, volatility can, in theory, be predicted and managed. Through the calculation of potential maximum losses from market risk (worst-case scenarios) to certain confidence levels (often to either a 99.7 or 95 percent confidence levels) and the setting of *stress loss* exposure limits for each trading desk, management is afforded critical tools to manage its trading risk. Significantly, for a given time period and to a defined level of probability, it will be able to sum up in a single number the worst-case-loss scenario across a bank's entire trading operation.⁸⁰

VAR VARIANTS

Value-at-risk is not a single unified methodology, but one that comprises several different methods. When used to measure market risk, VAR models seek to measure the probability of a worst-case scenario occurring over a short time frame.

There are basically three approaches to computing value at risk:

1. The historical method
2. The variance-covariance method
3. The Monte Carlo simulation approach

In brief, the first method assumes that history will repeat itself. The second assumes a normal distribution of market returns and statistically extrapolates from historical data to encompass potential scenarios that have not as yet come to pass. Finally, the third method uses computer-generated random trials to create a myriad of potential scenarios. Although not strictly speaking based on historical data, the underlying assumptions and results tend not to differ greatly from methods 1 and 2.

Extending VaR to Other Risks VaR is the market risk analog of economic capital. Indeed, signaling its link to the VAR approach, the economic capital model is sometimes referred to as *credit VaR* and economic capital itself as *capital at risk*.⁸¹ Recall that the concept of economic capital, introduced in Chapter 9, and shown again below in Exhibit 12.9, is a measure of risk expressed in terms of the amount of capital that would be required to cover maximum unexpected losses to a certain confidence interval. It illustrates what is actually a VaR concept applied to credit risk—or rather, in the spirit of the most recent theories, to all risks that would eat away at shareholder’s equity—what is termed *economic capital* is essentially the same concept, albeit expanded, as *value at risk*. The major difference is that what has been termed value at risk is used in connection with market risk, while *economic capital* is the term used in respect to credit risk and enterprise risks generally. Note that the distribution is skewed to the right, which is here used to show loss rather than gain, and is a reminder that only bad outcomes are relevant to the analysis, since credit loss is essentially a one-way downside risk.

Hence, VaR can be applied as well to market risk, credit risk, and operational risk and to bank risks in the aggregate, thereby allowing for the estimation of the bank’s total economic capital requirements.⁸² The rise in the popularity of VaR can be attributed in large part to its amenability to the use of quantitative methods and statistical analysis as well as its capacity to function as a unifying vehicle bringing credit, market, operational risk, and risk management with the same paradigm.

Criticisms of VaR VaR has come under heavy criticism as a risk management tool, particularly since the outbreak of the present credit crisis. As the quotation that led this section of the chapter suggests, the methodology has been said to encourage complacency. A similar criticism was made in 1999 by Nassim Taleb, part-time trader and professor who asserted that the unstated rationale of VaR is to provide “the alibi bankers will give shareholders (and the bailing-out taxpayer) to show documented due diligence and will express that their blow-up came from *truly unforeseeable circumstances and events with low probability*—not from taking large

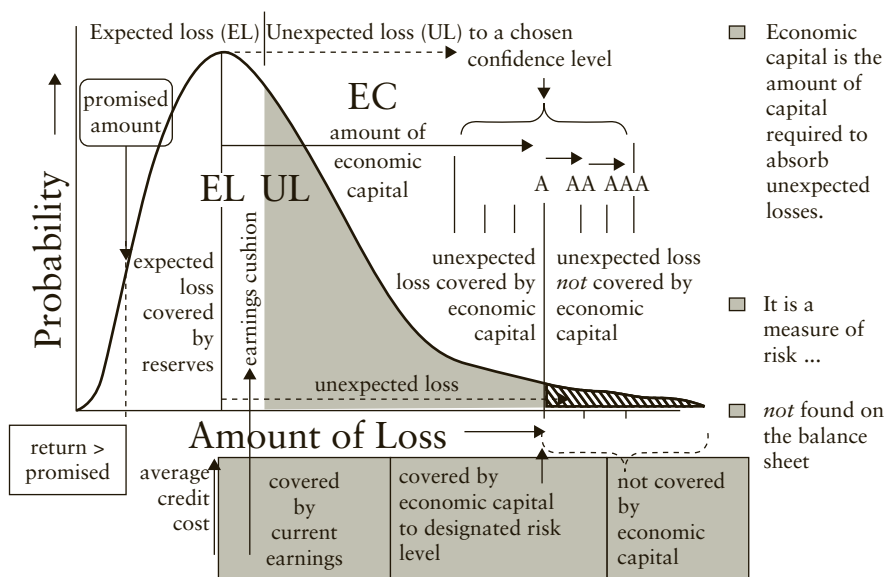


EXHIBIT 12.9 Economic Capital—Analog of Value at Risk

risks they did not understand.”⁸³ Nassir Nicholas Taleb is the renowned author of *The Black Swan*, a brilliant, if iconoclastic, study of the improbable.⁸⁴ These and similar critiques were voiced well before the present crisis, but the criticisms have since, with justification, become stronger.

CRITICISMS OF VAR IN LIGHT OF THE CURRENT CRISIS

The story that I have to tell is marked all the way through by a persistent tension between those who assert that the best decisions are based on quantification and numbers, determined by the patterns of the past, and those who base their decisions on more subjective degrees of belief about the uncertain future. This is a controversy that has never been resolved.

From the introduction to *Against the Gods: The Remarkable Story of Risk*, by Peter L. Bernstein⁸⁵ (quote leading the article by Joseph Nocera entitled “Risk Mismanagement—What Led to the Financial Meltdown”)

An article in the *New York Times Magazine* by financial writer Joseph Nocera entitled “Risk Mismanagement—What Led to the Financial Meltdown,”⁸⁶ examines the role of VaR models in potentially contributing to the current global credit crisis. The article begins with a frequently heard anecdote about VaR model outputs at Goldman Sachs in December 2006 just prior to the

onset of the crisis that Mr. Nocera seeks to verify. To provide the context for the tale, he interweaves an explanation of the origins and development of VaR, which ultimately leads to the larger issue with which he is concerned; namely did VaR contribute to the crisis? The story goes that the Goldman's VaR models were providing indications "that something was wrong." From its CFO, David Viniar, Mr. Nocera learned that in December the firm's mortgage business had been in the red for 10 days running. Viniar said that "by the 10th day we thought that we should sit down and talk about it." A meeting of senior staff on various trading desks and a number of risk managers scrutinized a report that included all the firm's current trading positions, and the corresponding VaR outputs. In the end, they discussed how the MBS market "felt." According to Viniar, the consensus was that "it felt like it was going to get worse before it got better . . . so we made a decision: let's get closer to home." As a consequence, MBS positions were reduced and hedged, so that the firm was able to escape much of the grief experienced by its Wall Street peers the following summer.

This anecdote, which is both an implicit criticism of VaR's ability to provide accurate outputs during unusual market conditions while at the same time suggesting that this fact itself provided a useful warning signal, ultimately appears to confirm the importance of relying upon expert judgment and intuition. It sets the stage for an exchange of views between VaR's more notable advocates and critics, suggesting along the way that VaR has many acknowledged weaknesses. In one interview, one of VaR's sharpest critics admits that if given a choice of one indicator upon which to make a decision, he would choose VaR. Although the article does not clearly resolve the issue it raises, it suggests that while VaR has some usefulness, the numbers it generates should be taken with a pinch of salt and an ample dose of experience-based judgment.

As it is based on historical data incorporating normal market conditions, VaR works well under normal market conditions. There is no denying, however, that VaR methodologies make a number of assumptions that experience, particularly recent experience, has demonstrated limit their utility, particularly during unusual market conditions.⁸⁷ The most critical flaw is the assumption that the future will generally resemble the past. It usually does, but not always. Moreover, the period covered by the historical data may be too short to cover all significant market permutations. During unusual market conditions, such as those when liquidity is substantially below historical levels or when a few players can distort the market, the model is prone to error. Like a flotilla of large fast boats maneuvering in a small harbor, the actions of the dominant participants in particular markets, together with those of their imitators, may affect the market in quite unforeseen and unpredictable ways, generating unusual eddies and whirlpools that throw the vessels off course and onto the rocks.

VAR AND THE FALL OF LTCM

Long Term Capital Management (LTCM), founded by Wall Street legend John Merriwether in 1993, was one of the world's most prominent hedge funds until it collapsed in late 1998. Despite its having among its board members, two Nobel Prize winning financial theoreticians (Nobel prize winners Myron Scholes and Robert C. Merton), employing top Wall Street traders and utilizing the most state-of-the-art VaR models to guide its investment decisions, the firm foundered in the financial turmoil that swept international markets during the third quarter of 1998.

The combination of two market crises in Asia and in Russia resulted in poor market liquidity that so distorted normal market scenarios and correlations that the models upon which the hedge fund relied provided faulty guidance. The values of bonds upon which the highly leveraged fund placed its bets, which under more normal conditions would predictably have converged, instead diverged causing it massive losses. Despite the highly sophisticated value-at-risk modeling tools LTCM employed, it had to be unofficially rescued by the U.S. Federal Reserve bank acting in concert with major investment banks.

Advocates of VaR generally acknowledge its limitations. They typically argue, however, that it has been misused or that it is the best tool available to model market risk, even if it only works well during normal market conditions. Despite the criticisms that VAR encourages an overreliance on quantitative measures of risk at the expense of human judgment, the qualitative alternative, expert judgment based on many years of experience, is by definition difficult to obtain and therefore in comparatively short supply. As for other quantitative methods, there are no good alternatives to VaR for the purpose of risk modeling. For these reasons, their view is that it does not seem likely that VaR will soon be abandoned as the principal quantitative approach for risk management.

In the wake of the most recent banking crisis, however, the use of VaR is being rethought. Although it is likely to remain an important tool in the risk manager's toolbox, it can also be expected that it will be utilized more judiciously in the future, with a greater recognition of its intrinsic limitations.

Later in this chapter, Richard Lumley⁸⁸ contributes comments where he argues for a more judgment-based approach to risk management.

Equity-at-Risk

There are, of course, many other ways to gauge the strength of a bank's balance sheet in the face of adverse developments. As an example, some researchers suggest the equity-at-risk approach.

The approach restricts capital to the sole core equity—not dissimilar to the Tier I concept—without consideration for other types of quasi-equity and subordinated arrangements.

The equity at risk is measured as the equity that would be lost under stress—whose extent should of course be part of the definition—and the regulatory minimum is the equity remaining after such loss.

Any depletion of the remaining equity below the regulatory minimum would trigger regulatory intervention.

The difficulty with such an approach is the definition of stress. Yet, recent events—the 2007–2012 crisis—show that adjusting the definition of stress is a tool that regulators can use, and in fact have used, as seen in the hurried stress tests put in place in Europe and in the United States in 2011. In short, traditional capital adequacy ratios are rather static in nature when compared to the dynamism of a “variable geometry” stress-based ratio. It would impact a bank’s equity management and create difficulties for weaker institutions, but the creditworthiness of the better banks would appear more stable and more robust.

Note that some proponents of another type of equity-at-risk approach for banks suggest the computation of the net present value of all assets and liabilities, including those resulting from off-balance-sheet items, the result being the net value of the institution. The net value would vary with the same several factors under stress, as well as with the interest rates that would be used in the computation. Regulators—and shareholders alike—would then be able to react to weaknesses in net present values under various scenarios. A similar approach is encountered in insurance ratings.

Credit Risk Management

The subject of this book is the assessment of banks themselves, as counterparties, rather than the more restrictive evaluation of the banks’ internal credit management function, which is only one of several tools in that assessment, and the purpose of this section.

Credit risk arises both through lending and longer-term investments (the banking book) as well as through those assets held merely for treasury or other short-term dealing purposes (the trading book).

Credit risk analysis is often performed as part of a bank’s credit risk management function. Risk analysis, however, comprises just one facet of a larger role that includes the quantification of credit risk with the aid of credit risk models, the implementation of a universe of policies, procedures, and protocols comprising a bank’s limit setting and credit approval processes, credit portfolio analysis, and techniques to actively manage credit risk through pricing decisions, risk mitigation and transfer techniques, and recovery and workout activity. Like market risk management, credit risk management is a vast subject, the surface of which can only touched upon here.

Analysis and Quantification

A significant portion of the work of managing credit risk is concerned with the analysis and quantification of such risk, and evaluation of the existing or prospective obligor being the first step in this process. Evaluation of the credit risk associated with an obligor is normally effected with the aid of internal credit risk models that generate internal ratings to represent gradations in credit quality. These models, which will vary with the type of obligor, provide data that is used to estimate the probability of obligor default.

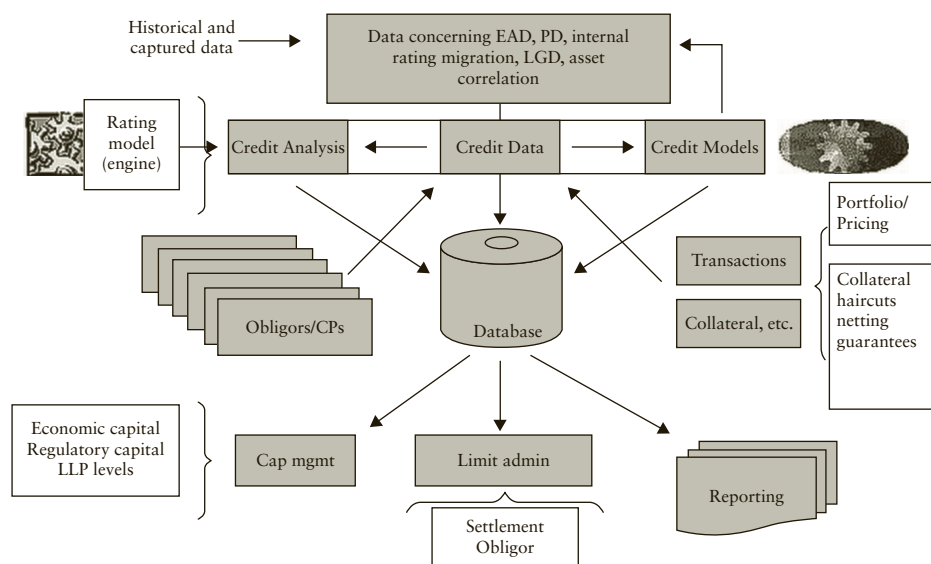


EXHIBIT 12.10 Credit Risk Management Infrastructure: Interaction of Data, Functions, and Models

As was observed in Chapter 2, credit risk evaluation can be subdivided by the nature of the obligor: bank/counterparty risk, corporate credit, retail, sovereign, and structured finance. Retail exposure will be quantified mainly through automated dynamic credit scoring models, while each major type of product to which exposure is undertaken will have a corresponding credit risk model applied. Generally, the more complex the product, the less automated the credit assessment process.

The overall amount of credit risk generated by a particular transaction will be affected not only by the obligor, but also by type of financial product generating the credit risk exposure. It will also be affected by decisions taken at various stages of the credit approval process including decisions to require certain covenants in the agreement governing the obligation or to require collateral or other credit risk mitigants. While the probability of default is associated with the obligor, the nature of the product or facility, together with the applicable terms and conditions, are key criteria in establishing loss-given-default.

Finally, the impact of correlation on the larger portfolio of assets of which the subject transaction represents a part needs to be taken into account. These estimates of PD, LGD, and EAD, together with appropriate portfolio adjustments, will be channeled through the bank's credit risk management systems with the ultimate objective of calculating its economic capital requirements. The diagram in Exhibit 12.10 depicts this process in a simplified form and also suggests how the various aspects of credit risk management interact to affect controls and capital allocation.

Control Infrastructure: Limits and Approvals

Fundamental to credit risk management is the control infrastructure utilized to govern approvals of loans and the many other types of transactions into which a bank might enter. In the discussion of asset quality assessment in Chapter 7, this aspect of credit risk management was discussed briefly. Major components include

the setting of limits with regard to customers and counterparties, and the policies, processes, and procedures governing the approval of transactions.

Limits may restrict overall lending by country (and sometimes by region or industry), by the type of product or facility contemplated, and by whether the exposure takes the form of funds advanced or arises from the settlement of the transaction. Among typical bank products that generate credit risk exposures are overdraft facilities, revolving lines of credit, trade finance, financial leases, repo finance and securities lending, interbank lending, and the acquisition of debt securities of various types, to name but a few.

Approval policies and procedures comprise the other major component of a bank's credit control infrastructure. Often quite detailed, these internal regulations set forth the terms and conditions under which approvals with respect to particular products will be granted, including minimum and maximum maturities, relative amounts, mandatory covenants, and whether or not collateral, guarantees, or other risk mitigants are required. The approval process is conventionally hierarchical in structure whereby the larger the proposed advance or facility, the more reviews it will undergo, and the more senior the management level that must approve it. Their primary objective is to reduce unnecessary credit risk by avoiding unattractive credit exposures.

Credit risk management also incorporates a range of techniques and products that can be used to mitigate credit risk or reduce exposures to acceptable levels. They include traditional credit risk mitigants such as loan guarantees, standby letters of credit, joint and several liability, and, perhaps most commonly, collateral. In recent decades, products such as mortgage insurance, the sale or securitization of loans and credit default swaps are among those that have been used to shed risk.

At the strategic level, management's decisions concerning what business activities and market niches in which to focus do reflect a bank's risk appetite as well as the judgment of management as to where its competitive advantages lie. Pricing decisions do not affect risk per se, but do affect the risk/reward calculus that underpins effective risk management.

Credit Risk of Portfolios and Risk Mitigation

To measure credit risk across the bank loan book and trading book, and with respect to particular portfolios, various techniques analogous to market VaR techniques have come to be utilized, primarily over the past decade or so. These techniques allow for the accounting of concentration and asset correlation risk, and also are integral to the enterprise risk management function and the allocation of capital. In addition, a variety of methods has been utilized at the portfolio level to reduce, transfer, optimize, or increase credit risk.

Operational Risk Management

Of the three major categories of risk, management of operational risk is the least developed. Operational risk management tends to focus on activities that are vulnerable to loss as a consequence of error, malfeasance, or systems failure, including:

- Payment processing
- Securities trading and clearing

- Data entry; documentation
- Collateral monitoring

At most banks operational risk management is apt to be found in a plethora of standard operating procedures, guidelines, and policies—what might be referred to collectively as *controls*. Such controls attempt to reduce operational risks among the numerous activities undertaken by the bank in the course of doing business.

Two simple examples provide an illustration of this approach.

1. *Principle:* To avoid embezzlement of bank funds and related forms of self-dealing, authority for making payments should be divided such that no one individual has sole authority to disburse funds.
Control: Two signatures are required on all significant payments.
2. *Principle:* A record should be made of all trading transactions entered into on behalf of the bank to prevent disputes as to whether a particular transaction was consummated and, if so, under what terms it was agreed.
Control: Record all telephone calls made and received by traders.

As these examples show, operational risk management will be concerned with both (1) identifying operational risks; and (2) formulating, revising implementing and enforcing controls to mitigate such risk. In this sense, operational risk analysis and operational risk management are closely linked.

Another way to address operational risk is through the use of insurance. Finally, operational risk management becomes more problematic when functions are outsourced to third parties.

Operational risk differs from the other major categories of risk with which banks must deal in that they are highly variegated. For this reason, they are not easily systematized, although the treatment of operational risks under Basel II constitutes an attempt to do so, and more sophisticated institutions are applying VaR methods to operational risk management.⁸⁹ Quantification of operational risks remains challenging, however, and establishing correlations generally difficult as there may be no apparent connection between one operational risk and another.

Owing to the heterogeneity of operational risk, and to the fact that it can arise in nearly any corner of the bank, responsibility for formulating and implementing operational risk management will, however, necessarily be devolved to a great extent to the institution's various functional groups. For example, the IT department will be responsible for formulating data backup strategies and off-site storage of critical data, and HR for keeping staff informed of bank-wide rules and regulations and avoiding unnecessary litigation. Likewise, as with the groups responsible for market risk and credit risk, there are apt to be operational risks arising that are best addressed within each group.

It follows that to accomplish their objectives, those responsible for overseeing operational risk management will need to coordinate extensively with the heads of the bank's various departments and groups in connection with the identification and measurement of such risks, reviewing controls already established, identifying areas for improvement, and from time to time implementing new controls.

THE COLLAPSE OF BARINGS: MARKET RISK OR OPERATIONAL RISK?

The collapse of Barings Bank in early 1995 highlighted the importance of operational risk and its management.⁹⁰ Barings' failure was triggered by the actions of rogue trader Nick Leeson operating in the bank's Singapore office. Owing to loss positions amounting to over £800 million, Leeson broke the bank resulting in its subsequent sale to ING Bank of the Netherlands for the nominal sum of 1 GBP.

Although the losses that caused the bank's collapse by speculative unhedged derivatives trades were expressions of market risk, the true problem revealed by the scandal was a failure of adequate operational risk management. Leeson's unauthorized trading was made possible by his having had effective control over both the front and back offices thereby enabling him to hide his improper trades and the losses he was actually making. Ultimately, senior management bore responsibility for the collapse as they had failed to ensure that there were adequate controls over the trading operations and risk exposures. Worse, it later became apparent that senior management lacked sufficient understanding of the bank's critical businesses, notably that of its options and derivatives trading in Singapore.

The Management of Solvency Risk

Solvency risk, as mentioned previously, is the risk that a bank will have insufficient capital to meet its obligations; that is, to remain a viable institution.

In fact, solvency risk can be understood as comprising a bank's aggregate risk—to which its entire internal risk management process is directed—relative to capital.

It was also observed that, from an external vantage point, all bank risks lead to credit risk, while from an internal vantage point, they all lead to solvency risk. In this sense, solvency risk is the internal counterpart to a bank's credit risk profile. See Exhibit 12.11.

The concept of solvency is perhaps better captured by the phrase “capital adequacy,” which, of course, prompts the rejoinder, “adequate for what?” Managing solvency risk, like determining irrespective of regulatory requirements how much capital is adequate, cannot be accomplished without considering the nature and magnitude of risks to which the bank is exposed.

Consequently, the management of solvency risk must encompass not only capital management, discussed in part in Chapter 9 on capital adequacy, but also the risk exposure and the risk management of the bank as a whole, which has been the subject of this chapter.⁹¹ Thus, if the key aim of risk management is the managed bank's continued survival, and if solvency risk is equivalent to risk of bank failure—which in the modern risk management framework can be defined as the bank having insufficient economic capital to absorb unexpected risks—then risk management and capital management are two sides of the same problem.

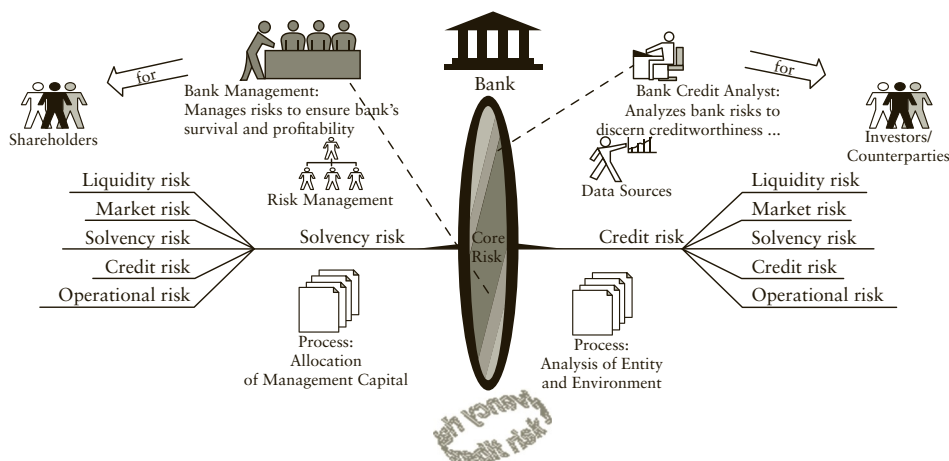


EXHIBIT 12.11 A Bank's Solvency Risk Mirrors the Credit Risk It Presents as an Obligor

Value-at-risk type methodologies, despite their flaws in practice, have become the accepted means to measure and monitor not only market risk, but also the risks of the banking enterprise as a whole. In the same manner, VaR techniques enable volatility in market outcomes to be measured in terms of value at risk, so too can other forms of risk be evaluated applying essentially the same methodology.

Analogously to value at risk, economic capital, which as mentioned is sometimes called capital at risk, becomes the master measure of enterprise risk. Although the foregoing paragraphs imply that economic capital is merely a measure of risk, as was stated in Chapter 9, it can also be seen as an expression of capital requirements for a given risk profile. Economic capital implicitly embodies the following question: To a given degree of confidence, what is the worst case overall that the bank could experience within a given time period and accordingly how much capital would be required to ensure the bank's solvency notwithstanding such a loss?

It is therefore perhaps unsurprising that Basel II, which is ostensibly concerned with setting regulatory capital requirements, employs the same conceptual view of risk that is embodied in the economic capital framework. In view of this connection, an understanding of VaR and economic capital helps in understanding Basel II, for the new capital accord can be viewed as the expression of a comprehensive bank risk management framework along such lines.

BASEL II AND BASEL III

We have touched upon the Basel accords a number of times in this book, and we devoted an entire section on Basel II in the context of capital adequacy in Chapter 9.

Perhaps surprisingly to the reader, discussions about Basel are, to a large degree, irrelevant to the purpose of this book.

The Basel accords, as stated already, are not mandatory. The countries that adopt them do so voluntarily, or, more precisely, under the pressure of international markets.

In the context of a book looking at bank credit analysis, it would be conceivable to consider Basel compliance—or at least regulatory compliance in respect of capital adequacy and liquidity, if other regulations were imposed outside of the Basel framework—as a given, or as a black box. The assumption should be that, regardless of the stringency—or the lack of it—of the local regulators, regulations exist and the banks are abiding by them, failing which they would not operate. If the authorities are comfortable with their banks, so should we. Well, to a degree only.

What Basel has achieved—very slowly—over the years is essentially to refine the denominator of capital adequacy ratios (essentially the risk-weighted exposure) so that the risk profile of a bank is better captured. It even clarified and improved the definition of the numerator (essentially the capital). Basel stopped short of deciding whether the minimum ratio levels were safe.

It is the view of the authors that such minimum levels are, if anything, too low. Alas, raising them would reduce shareholder returns to levels where banks would not be able to attract the fresh equity that their natural expansion—in particular in vibrant emerging markets—and, unfortunately, their occasional losses require. This is not going to happen.

It also seems that it makes little sense to attempt to impose the same minimum levels in every country in the world. Quite obviously, banks in countries where regulators are inquisitive and powerful should enjoy lower minimum capital levels, while banks in countries where creative accounting reigns, with or without the authorities' approval, should face higher capital requirements.

We invite readers to look at the Basel accords as they would look at any local rule: A bank that is compliant under the rules is not necessarily a good credit. The reader should try to figure out independently to what extent the bank is creditworthy.

For those reasons, for purposes of this book our interest is primarily in how it broadly affects capital adequacy and liquidity requirements and upon its relationship to risk management.

At the end, what is important to the analyst is the reality behind the figures and ratios. Whether management knows what it is doing, whether it has built the proper tools, and whether the bank's structure and chain of command operate smoothly are major considerations.

Basel III is not in place yet and, in many countries, even Basel II is not yet implemented. Basel II will as a result remain, for some time, a natural basis for capital rules.

The 1988 Basel rules, as well as Basel II, were discussed earlier in the context of capital adequacy, as those rules were predominantly geared to setting uniform capital rules for international banking. We shall now devote the rest of this chapter to Basel II and Basel III under angles that increasingly pertain to bank risk management in general.

Basel II

As discussed in Chapter 9, after a number of years of negotiation, and considerable controversy,⁹² agreement was finally reached upon the revised Basel capital accord, commonly referred to as Basel II, and on June 26, 2004, the new accord was officially published.⁹³

The Basel II accord is now to be superseded by Basel III, which intends to increase regulatory capital and to introduce liquidity requirements.

The Basel II framework is structured with three so-called pillars approach to the management of bank risk:⁹⁴

1. Minimum capital requirements
2. Supervisory review of capital adequacy
3. Market discipline

Bank solvency (i.e., capital adequacy) requirements, which were practically the sole focus of Basel I at its outset, are covered in Pillar 1, which now encompasses not just credit risk but also market risk and operational risk.⁹⁵ Pillars 2 and 3 take account of the role of outside actors in constraining excessive risk taking by banks. See Exhibit 12.12.

Pillar 1 retains the same 8 percent capital adequacy requirement set forth in Basel I, but makes significant changes in its manner of calculation. Capital definitions remain largely unchanged, but the method of risk-weighting assets has been transformed. As discussed in Chapter 9, one of the primary objectives of Basel II was to reduce the gap between regulatory and economic capital in order to diminish regulatory arbitrage and the risks it engendered and for which it failed to account.⁹⁶ To accomplish this goal, the revised accord adopts a ratings-based approach to replace Basel I's fixed and somewhat arbitrary risk-weightings.

The new accord provides that the credit risk of bank asset portfolios be weighted for risk using credit ratings. Three different approaches are available to banks, subject to the approval of their national regulators (see Exhibit 12.13).

The first approach, the revised standard approach (RSA), can be thought of as a more refined version of the Basel I approach, one which assigns risk-weightings, based on external ratings; that is, those provided by regulator-approved outside credit rating agencies such as Moody's, S&P, Fitch, and several regional and national rating agencies around the world.

EXHIBIT 12.12 The Three Pillars of Basel II

Pillar	1 Capital	2 Supervision	3 Market Discipline
Scope	Minimum capital requirements	Supervisory review of capital adequacy	Effective use of market discipline
Major change	Rating-based risk-weightings either external (standardized) or internal (FIRB or AIRB); consideration of operational risk	Greater regulator involvement in supervising capital requirements (e.g., by overseeing IRB systems)	More emphasis on disclosure and reporting requirements

Source: Independent Company Research from Basel Committee material.

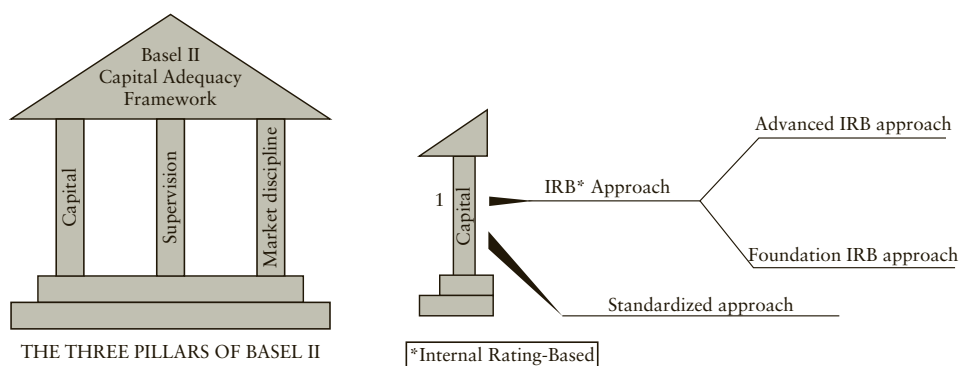


EXHIBIT 12.13 The Three Pillars and Three Approaches of the Basel II Framework

Source: Independent Company Research from Basel Committee material.

While an external ratings-based approach alone would have represented progress towards this end, Basel II went much further in an attempt to more closely align regulatory capital with economic capital and in so doing to incorporate state of the art approaches to managing bank risk at the enterprise level in the very framework of Basel II.

The other two approaches available under Basel II are internal rating based. In other words, they utilize a bank's internally generated ratings (instead of external ratings) to weight their assets to adjust for credit risk. The second approach or the foundational internal rating based approach (FIRBA), permits banks to calculate probability of obligor default (PD), but delegates the setting of benchmark loss-given-default (LGD) values and exposure at default (EAD) to the local bank regulator.⁹⁷ The third and most complex approach to meeting the capital adequacy requirement is the advanced internal rating based approach (AIRBA), which leaves essentially all the credit risk calculations to the bank itself.

Another noteworthy difference in Basel II from Basel I is that it takes account of operational risk. Market risk, which was addressed in subsequent revisions to the first accord, is treated in a more integrated fashion in its successor. Although it is a framework for calculating regulatory capital, not economic capital as such, many of the same concepts and methods of calculation as are used in VaR and the economic capital model are also employed in Basel II, especially in respect of the two IRB approaches to risk-weighting assets.

Leaving the mechanics of the calculation aside, the essence of Pillar I, and of Basel II as a whole, is the aggregation of risk in each of the three major risk categories—credit risk, market risk, and operational risk—to calculate a minimal capital requirement. In the revised standard approach, prescribed risk weightings (taking account of predetermined supervisory adjustments) are assigned to all assets, resulting in an aggregate minimum regulatory capital requirement. In the IRB approaches, more of the variables are left to supervisor-approved internal models than to preset proxy values, but the ultimate result, too, is a minimum capital requirement that attempts to account for the three categories of risk that a bank confronts.

A few more general points can be made about the Basel II model and its relation to VaR and economic capital. First, there is recognition that economic capital as defined by a bank for its own risk management purposes and regulatory capital will diverge. Although precisely measurable in an ideal world, in practice economic capital as calculated can only approximate true economic capital. In contrast, regulatory capital, to be fit for its policy purpose, must be precisely measurable. Hence, it is reasonable that regulatory capital be defined in a manner that facilitates its calculation, and that it should not vary with a bank's preferred risk appetite. In this light it is understandable that regulatory capital is set arbitrarily at a fixed level of, say, 8 percent, irrespective of a bank's risk appetite, which, of course, will affect its economic capital requirement. For a similar reason, regulatory capital does not attempt to take into account all operational risks as strategic and general business risks are excluded.

Postcrisis Risk Management: A View from the Trenches

The global banking crisis that is still ongoing at the time of the writing of this book has brought about a rethink of existing approaches to risk management and has already impelled regulatory reforms in an attempt to remedy its flaws. Richard Lumley, who has worked both in credit and market risk management and is currently senior vice president (SVP)—International Credit Risk with Northern Trust in London, questions whether these efforts are likely to be effective.

A CRITIQUE OF BASEL II STYLE RISK MANAGEMENT BY RICHARD LUMLEY (2009)

The Committee believes that the revised Framework will promote the adoption of stronger risk management practices by the banking industry, and views this as one of its major benefits.

— Basel Committee on Banking Supervision (2006)

However, when looking at Basel's new approach, the question that must be asked is: Given the cataclysm of the Global Financial Crisis, what is the purpose of tinkering at the edge of existing (discredited) regulation? This is not rearranging deckchairs on the Titanic; this is rearranging cushions on deckchairs on the Titanic while taking orders for dinner.

—Dr. Patrick McConnell, Basel 2.0

To widen the market and to narrow the competition is always the interest of the dealers.

—Adam Smith, *Wealth of Nations*

It has been asserted that modern day risk management was a major contributor to the 2008 credit crunch, which as of this writing is still continuing. Unsurprisingly, regulators are now in the process of seeking to comprehend how state-of-the-art risk management systems got it wrong, and address identified weaknesses. But, as is evident from history, unwanted side effects are always a possibility when undertaking regulatory reforms. The question, which therefore must be posed, is this: Will the steps being taken to improve risk management at banks provide better protection from future unknown events? Unless the reforms address certain fundamental issues in the banking sector, the answer will be in the negative. Worse, the reforms could make things worse, hampering positive changes, whilst increasing the risks of future systemic failures triggered by the deficiencies of a handful of poorly managed complex institutions.

Great expectations accompanied the advent of the new Basel Accord (Basel II), since its formal endorsement by the Basel Committee on Banking Supervision in 2004 and publication the following year, particularly given that the flaws in the previous accord had become so conspicuous. The Basel II framework is based on statistical risk management concepts, which since its launch have come to shape much of the thinking among regulators about risk management. In some countries, the new accord has effectively been made legally binding upon regulators, while in others it has been informally adopted in large part at the supervisory level. Although it has not been in effect a long time, having only come into force in the EU in 2007, the new accord represents the culmination of a vast amount of research, preparation and consultations that had taken place over the prior decade. Although Basel II cannot be blamed for the current crisis, the view of risk management that it represents was a contributing factor.

In the past two decades, modern risk management has evolved through a belief in and adoption of statistical methods and methodologies. Whilst risk management at financial institutions is typically broken down into the core categories of credit, market and operational risk, the first two of these have been by far the most critical. Prior to Basel II, banks had embedded risk management practices with respect to credit risk and market risk within their institutional processes and practices. Although perhaps sometimes weak in clearly communicating their overall appetite and risk tolerance, internally as well as externally, key individuals within the corresponding operational groups were involved in the formulation of risk management systems and procedures. Risk awareness and appetite developed through an iterative learning process that came to form part of the bank's institutional memory. Crucially, unlike a number of today's problem banks, top management also came from a risk taking background.

With the onset of the drive towards implementing Basel II, which was anticipated from the beginning of the current decade, its approach to risk management, based upon VaR and economic capital methodologies, came to the fore. Chief risk officers, brought in to implement the new accord and possessing the budgets to do so, became the new overseers of risk management.

Within the bank, concepts once mainly the province of theoreticians and specialists, such as probability of default and loss given default, became the language of the many.

In the process, the more qualitative judgment-based approach that had evolved organically over many years was displaced. Individuals who considered risk taking to be an “art” rather than a “science” were marginalised. Experienced-based judgments and intuitions that could not be justified numerically came to be given less credence. In effect, that which could not be quantified was ignored. Paradoxically, in seeking more quantitative and “accurate” risk measurement to feed the new statistically based models and allow the assessment of transactions using them, institutions built up even greater concentrations of risk than they might otherwise have done. It was, of course, the flawed assumptions underlying these models, particularly those in respect to correlation, which allowed this situation to occur.

Modern day risk management with its attention to the establishment of VAR (market risk), internal ratings, default measurements (credit risk) and risk transfer have deflected management attention from understanding real risk drivers and gave a false impression of the levels of capital required to sustain the business. The creation of what may be termed “ivory tower” risk management groups within financial institutions, staffed by specialists with little front-line experience in credit or market risk, made it harder to see the “wood for the trees.” The high degree of specialization in modern day risk professionals concerned with a narrow range of risks under their remit and blinkered to wider issues contributed to the erosion of sound risk management. Huge amounts of time could be spent on determining the most preferred modelling approach to a complex problem when it was apparent to those with risk management experience at the transaction level that the underlying parameters and model building blocks were already hopelessly flawed through either a lack of data or predictability.

In addition, at the operational level, systems deficiencies meant that some non-vanilla transactions could be subject to ambiguity concerning which approval procedures were appropriate. Given financial incentives, the front office was naturally keen to identify the easiest approval route and often prevailed. At the same time as the foregoing changes were occurring in regulatory and risk management practices, there was a significant growth in securitisation and an accompanying move towards the origination and distribution of risk. This commoditization of risk, although permitted and implicitly encouraged by the governing models, allowed situations to occur that seemed to be divorced from economic reality. Often the banks owned or controlled the entities that held the repackaged risk, but were setting relatively little capital aside. The justification was that their risk exposures were slight since nominally they only retained so-called super senior tranches of risk or stand-by facilities. In light of the events that have occurred over the past two years, it is clear in retrospect that risks were systematically understated and underpriced, and as a consequence insufficient capital was set aside to cover them. Consequently, the model-based approach to risk management

called for under Basel II turned out to be a contributing cause of the current banking crisis.

The emergence of the current crisis caught regulators and politicians, not to speak of most market participants, by surprise. Panic replaced complacency and at the height of the turmoil, moves were made to support major banking groups by a number of national regulators. Although understandable as an antidote to the risk of a complete systemic breakdown, these initiatives have had some adverse consequences. Permitting the failure of weaker and less well-managed institutions would have cleared the field for more nimble, smaller institutions to step in and feast on the opportunities. Whilst of course it would have taken time for new entrants and sound existing participants to grow, this could arguably have been a better solution than allowing lumbering incumbent failed institutions to restructure, while shielding creditors and shareholders from the greater losses they would have otherwise incurred. To be sure, the equity markets and a large swathe of creditors would have felt more pain, the banking system would have suffered some damage, but the removal of vested interests and opening the door to new entrants would have permitted real change—in effect, the revolution of a failed approach.

Given the incremental efforts that have been made so far, regulatory reforms seem to be leading towards the following:

- Increased regulation, rules and capital to undertake business, increasing the costs and resources required to stay in business
- Increased demand for risk management ‘experts’, i.e., specialists, who by virtue of their narrow risk management focus, increase the complexity of managing a business, thereby allowing a greater potential for gaps to develop in the risk management framework
- Risk management becoming rules based causing transactions to be sanctioned because they follow the rules and fit into the process rather than because they are a prudent risk for the bank

Instead of achieving the objectives sought by Basel II’s framers, greater reliance on the regulatory framework may very well lead to the deterioration of a bank’s own risk management capability. Attention by senior management is in danger of being diverted towards meeting these new regulatory requirements. In complying with them, a false sense of security will likely be created, based upon the belief that meeting these new hurdles will cover all reasonable risks. Moreover, problems with legacy portfolios, the constant stream of bad news as well as the need to scramble for capital has distracted senior management away from developing new business. Adding more regulatory oversight and reporting requirements will only increase the costs and stress being faced by the institutions affected and their managers.

Building upon a flawed risk management and regulatory methodologies and permitting so much support to failed institutions will not drive forward the changes required in modern day financial markets. Like the driver just narrowly escaped from his high performance sports car that has slid off a

pot-holed gravel mountain track and exploded below in a ball of fire, the answer may be to change to a more suitable vehicle for the terrain and road and drive a bit more slowly. Instead, the current approaches to regulatory reform resemble an attempt to modify a sports car to traverse the rocky road of reality by increasing ground clearance, installing four-wheel drive, oversize tires, adding a roll cage, attaching a fire extinguisher, and so forth.

To continue with the allegory, changing the driver's attitudes and habits is, of course, another challenge. So is changing the culture and attitude of failed financial institutions and moving an institution to rebuilding their franchises and reputation. Ultimately, however, this is probably more critical than additional legislation and rules. It is essential that banks return to a culture of risk awareness by making risk taking and the formation of risk-judging capacity an integral and strategic part of their operations.

Basel 2.5

In the wake of the financial crisis, a wider set of rules—beyond Basel II but not Basel III yet, hence its name—was put in place at the end of December 2011 in Europe and in many large countries, with the notable exception of the United States where, for conceptual reasons, another set of rules will be implemented to achieve a similar result.

Those halfway rules specifically target the trading portfolio of banks, as opposed to their banking books. The original distinction, among other things, between the two sets of assets, was that the trading book was supposed to require capital, liquidity, and accounting rules different from the rules applied to assets in the banking books, which would be kept to maturity.

The crisis that started in 2007 showed that, in situations of stress, market liquidity may dry up and counterparties may go bust, taking away the liquidity or position protection of myriads of transactions. This problem is of course compounded by the fact that, in such situations, the netting of positions may lose its conceptual capacity to reduce capital needs.

Basel 2.5 introduces additional capital requirements for the trading book, in particular for credit-related items. Securitized assets—certainly a main source of difficulties in the crisis—attract a scrutiny they never received. The trading book is to be divided into five buckets, for example, a stressed VaR bucket, each of them measured for risk and the relevant capital needed.

Importantly, those new rules impose capital requirements that are independent from and additional to those imposed under Basel II, and they do not make any allowance for risk offsets or even double counting between the two sets of rules.

Details can be found in the “Guidelines for computing capital for incremental risk in the trading book,” published in July 2009, as well as in the updated “Revisions to the Basel II market risk framework,” published in February 2011, both available from BIS publications on www.bis.org. The Basel Committee also published “Interpretive issues with respect to the revisions to the market risk framework” in July and in November 2011.

From the bank analyst's standpoint, Basel 2.5 is a welcome development, if only because it forces the larger banks either to increase capital or to deleverage. Most importantly, it reinforces the banks' reliance on risk management tools that, had they been available to them 10 years ago, would have spared them some misery.

Basel 2.5 is not without its problems, though. To start with, there has long been some grey area between the banking book and the trading book, with banks gaming the system, whenever possible, by parking their transactions where it suited them best.

The complexity of Basel 2.5—and of course Basel III—will be a burden for many banks, not to mention a source of disconnect between rocket scientists who handle the figures and bank managers who do not understand them.

A number—but not all—of Basel Committee member countries had already implemented Basel 2.5 by the middle of 2012.⁹⁸

Basel III

Like the first two Basel accords, Basel III has very complex rules, which cannot be explored in detail in our book. The reader should refer to BIS publications on www.bis.org for all the rules and their updates.

We shall nevertheless attempt to give a general view of the new rules.

Basel III has two major areas of requirements: those on capital (or solvency) and those on liquidity.

Solvency Requirements

Common Equity

The minimum common equity requirement, which was 2 percent in earlier accords, becomes 4.5 percent. This is not, in fact, a major constraint since many countries have long enacted local regulations requiring much higher levels of Tier 1 capital. Such local emphasis on common equity was sometimes due to the fact that, in many countries, financial markets are not sufficiently mature to offer substitutes to common equity or quasi-equity. Even in mature markets, though, many of the world's large banks actually have sufficient⁹⁹ Tier 1 capital to meet that new common equity requirement.

A capital conservation buffer of 2.5 percent of common equity is added, which, if not maintained, would trigger constraints on a bank's dividend policy.

In addition, a countercyclical buffer—based on local macroeconomic circumstances—of up to 2.5 percent of common equity is introduced.

Very importantly, most large banks would already meet the common equity requirements of 10.5 percent (being 4.5 percent + 2.5 percent + 2.5 percent) if they used—and abided by—the Regulatory Capital rules under Basel II.

Basel III, however, is also changing the way the numerator and the denominator are calculated.

It changes the definition of regulatory capital by taking away the eligibility of certain items, such as

- Intangibles
- Deferred tax assets
- Some types of financial instruments

It also changes the way risk-weighted assets (RWA) are calculated, especially in respect to trading activities and as a result of a requirement to mark to market certain loans.

On top of the above, the Financial Stability Board¹⁰⁰ recommended an increase of between 1 and 2.5 percent of equity ratio for systematically important financial institutions (SIFI), a list of which, comprising about 30 banks, FSB members prepared in 2011. The rationale was that such banks would be more sensitive to systemic events and that their demise would create ripple effects in financial systems.

Of note is that, independently, the European Commission decided that a minimum solvency ratio of 9 percent (to supersede the first 4.5 percent + 2.5 percent of Basel III) would be imposed on EU banks and that banks would have to comply in 2012, rather than at the end of 2018.

Basel III improves on Basel II in many other ways, which will not be explored here in detail.

For example, Basel III looks at the matter of wrong-way risk where the exposure increases when the counterparty's creditworthiness weakens. Basel, of course, looked at counterparty default, but did not address the credit valuation adjustment risk (CVA), which under Basel III prompts a capital charge for potential mark-to-market losses associated with the deterioration of the creditworthiness of a counterparty.

Leverage

STRESS TESTS

The idea of submitting banks to stress tests is not new, but it took the global financial crisis of 2007–2012 to bring the major Western economies to test their major banks under stressed conditions with the view of determining whether they would need fresh capital under those unusual conditions.

The Federal Reserve launched its Comprehensive Capital Analysis and Review (CCAR),¹⁰¹ which resulted or contributed to result in the major bank holding companies in the United States increasing their Tier 1 ratios, on a weighted average basis, from 5.4 percent in Q1 2009 to 10.4 percent in Q4 2011. Part of the increase was provided by a restrictive common dividend policy.

The end of 2011 stress tests—the third round of such tests—showed 15 out of 19 major U.S. banks having enough capital to withstand a financial shock associated with “a peak unemployment rate of 13 percent, a 21 percent drop in housing prices, and severe market shocks, as well as economic slowdowns in Europe and Asia.”¹⁰²

The European Banking Authority (EBA) ran stress tests on 90 banks in 21 countries in 2011. They showed that 20 banks—as at the end of 2010—would, under stress, fall below the 5 percent threshold of Core Tier 1 ratio, but that only 8 of them would have failed had the tests been taken at the end of capital raising actions undertaken between January and April 2011, with 16 banks remaining with a Core Tier 1 ratio of between 5 and 6 percent.¹⁰³

The EBA tests were criticized for disregarding some of the bond holdings in the books of the tested banks. To be fair, it would be impossible, in the midst

of a crisis such as the European sovereign debt crisis, to value such assets correctly. The problem, unfortunately, also spread to the numerator of the ratios, with Dexia Bank, for example, which collapsed a few weeks later, posting an acceptable Tier 1 capital ratio only as a result of regulations that allowed it to disregard the impact of some of its losses on its true equity. Irish banks in 2010 also looked in good shape just a few months before calling for help.

The old notion of leverage was for long the main gauge of capital adequacy. Banks used to have to keep the ratio between total assets and capital, or sometimes between total deposits and capital, within certain limits. In those times, a bank's overall risk was concentrated on its loans, which made the bulk of total assets, with deposits providing most of the funding. Typically, a bank's leverage would seldom exceed 20 times.¹⁰⁴ We touched on that in Chapter 9.

Leverage limits are back in fashion. Banks will have to bundle their loans and other on-balance-sheet assets, as well as all commitments, regardless of any risk weighting. The total shall not exceed 33.33 times the regulatory capital or, in an arguable misuse of the definition of leverage found in many texts, including those on Basel III, the Tier 1 leverage ratio should be at least 3 percent.

Quite obviously, Basel III addresses the various situations where some netting should apply, for example in terms of loan provisions (on the balance sheet), or where the notional economic exposure should be considered (off-balance-sheet).

SYSTEMATICALLY IMPORTANT FINANCIAL INSTITUTIONS (SIFI)

The G20 countries decided in November 2011 that some banks—29 banks around the world—were systematically important and needed special scrutiny. The SIFI approach is not distant from the too-big-to-fail concept, but with some differences.

The too-big-to-fail bank is just a bank whose failure would impose sufficient socioeconomic and financial stress upon its country to justify attempting to salvage it at (almost) any cost. Such banks' absolute size is not a predominant factor.

A systematically important bank is a bank whose failure is feared not just because it is too big—all such banks are—but because its failure could inflict contagious damage to other financial institutions and drag them down in a spiral of failures. Such contagion knows no borders.

In the United States, the Dodd-Frank Wall Street Reform and Consumer Protection Act specifically requires the Federal Reserve to undertake stress tests on systematically important financial institutions.

Liquidity Requirements

Basel III introduces two notions, one in terms of the liquidity of the assets, and the other one in terms of long-term funding. It also monitors several other indicators.

Liquid Coverage Ratio (LCR)¹⁰⁵

High-quality liquid assets are so called if they can easily be converted to cash at little or no loss. They must at all times exceed the total net cash outflows of the next 30 calendar days. That relationship is expressed by the LCR.

High-quality liquid assets are defined as cash, deposits at central banks, and sovereign debt of countries that are weighted at 0 percent (under the existing Basel II Standardized Approach rules). In addition, provided they do not reach a ceiling equivalent to 40 percent of the total high-quality liquid assets, 85 percent of the sovereign debt of countries weighted at 20 percent (under the existing Basel II Standardized Approach rules) and 85 percent of certain high-quality bonds are also counted as high-quality liquid assets.

The net cash outflows are equal to outflows less inflows—but net cash outflows cannot be smaller than 25 percent of outflows—along the following lines:

Major *cash outflows* items are:

- Retail deposits run-off (5 percent; or 10 percent if they are not considered as stable)
- Unsecured wholesale customers run-off (5 to 25 percent)¹⁰⁶
- Unsecured funding by nonfinancial corporate, sovereigns, central banks, and public sector entities run-off (75 percent)
- Unsecured funding from other legal entity customers run-off (100 percent)
- Secured funding backed with high-quality liquid assets run-off (from 0 to 15 percent, sometimes higher)
- Drawdowns on committed credit or liquidity facilities (from 5 to 100 percent, depending on the type of facility and the recipient)

Major *cash inflows* items are:

- Reverse repos and securities borrowing (from 0 to 100 percent, depending on the type of asset and the use of the collateral)
- Expected loan repayments (50 percent)
- Other wholesale inflows (100 percent from financial institutions or in respect of maturing securities, 50 percent for others)

Net Stable Funding Ratio (NSFR)¹⁰⁷

The assets and activities over a one-year horizon must be funded with at least a minimum amount of stable liabilities. The NSFR ratio is the ratio between what is available and what is required and, of course, it must be higher than 1.

Major *stable funding* items¹⁰⁸ are:

- Capital and preferred stock (of over a one-year maturity)
- Any liability of over a one-year maturity

- A portion of retail deposits maturing within one year or expected to remain over one year (between 50 and 100 percent)
- 50 percent of deposits from nonfinancial corporate and public entities

Major *relevant asset categories* are:

- 5 percent of the securities over one year; that is, 0 percent risk-weighted (under existing Basel rules)
- 20 percent of the securities over one year; that is, 20 percent risk-weighted (under existing Basel rules)
- 50 percent of gold and of certain equity securities and bonds
- 65 percent of residential mortgage with a residual maturity of over one year
- 85 percent of retail loans and loans to small business customers
- 5 percent of the undrawn portion of committed credit and liquidity facilities

Maturity Mismatch and Funding Concentration

In addition to the LCR and NSFR, Basel III requires some reporting in terms of contractual maturity mismatch, which banks in most countries already provide, albeit without reference to the Basel III classification.

Concentration of funding is monitored, both from significant counterparties and from significant instruments, and the mismatch is also studied for all significant currencies.

Available Unencumbered Assets

Assets that could be used as collateral for borrowing or that are eligible for secured financing with central banks should be reported as well. This notion is not too distant from the notion of marketable securities in the traditional quasi-liquid asset ratio analysis.

Others

The LCR should be reported and monitored in all significant currencies.

While the LCR should be reported at least monthly and the NSFR at least quarterly, banks should be equipped to provide weekly or even daily reporting in stressed situations.

Some Lateral Thinking about the Basel Accords

This entire book is built around the ideas that, in bank financial analysis, it is near impossible to put together a robust set of control ratios, that compliance with local or international regulations is not predictive of creditworthiness, and that what you see is not always what you get.

Robustness

In most countries, import duties on alcohol are a source of headache for finance ministries. Should it be based on value, on volume, on weight, on the degree of alcohol per volume, or on a combination of those criteria?

Some countries opted for a simple classification where, say, whiskey and champagne duties would be charged on a combination of value and gross weight

(of the beverage itself, its bottle, and its packaging.). As a result, importers attempted to reduce the weight of the glass bottle. This might explain why, suddenly, whiskey bottles became thinner. Champagne importers were not so lucky since inside pressure would not permit a significant thinning of bottles of Champagne.

Similarly, the actual cost of each ounce of meat on a T-bone steak is more difficult to measure than it is on a filet.

In banking, it is not too difficult to find ratios that tell something significant and consistent about a bank's profitability—although of course the choice of a correct denominator, for example, is increasingly difficult to make, as was discussed in Chapter 6.

Gauging capital adequacy is a challenge, as discussed in Chapter 9. There are several possible approaches both for numerators and for denominators of capital adequacy ratios. What Basel I and, especially, Basel II achieved was to narrow the possible approaches to only a few internationally consistent ones.

Bank liquidity is a minefield. It is a moving target on slippery ground. Countless ways exist to look at it. Basel III is—among other achievements—bravely attempting to offer rules and parameters.

The Worth of Compliance

If you are compliant, you have proved nothing, but if you are not compliant, you *have* proved something!

To a large extent, the Basel accords—which, remember, were created to establish a level playing field for international lending—achieved some degree of consistency across the globe. Most jurisdictions, though, had a bank regulatory framework well before the Basel accords existed. Banks have long had to comply with various sets of rules, mostly local rules.

The notion of economic capital—discussed in this book—is often misunderstood as meaning that a certain set of assets and liabilities requires a certain level of capital. This is too simplistic. Other factors make a difference: the country or countries where the bank operates, the strength of its shareholders, the skills of its management, the vigilance of the regulators, the local accounting rules, the local skill at cooking the books, and many more.

Whatever the Basel accords or local government regulations might indicate, each bank will have its own minimum economic capital, always higher than the regulatory capital. How much higher is determined by the strength of the rules, by those other factors, and by the environment in which the bank operates.

If a bank does not comply with capital rules or with liquidity rules, it has proved something. A bank that does comply has, in our view, proved nothing.

Water will always run along the path of steepest slope. Bank managers, bank traders, and bank accountants will always try to maximize returns, growth, or bonuses within the constraints of the rules. This is human nature. Creative accounting will always exist.

Therefore, once again, in the analysis of bank financials, be prepared to accept that what you see is not always what you get.

As a result, the bank credit analyst should have the entomologist's look for the Basel accords, for their implementation, and for a bank's compliance under those and

other rules. At the end, though, analysts will make up their own minds about the creditworthiness of a bank.

Accordingly, the Basel tools are useful, but less relevant than the many tools suggested in this book for a proper analysis of bank risk.

We did, of course, cover the Basel accords in several chapters, but the reader who has a particular interest for them will find a bounty of information on the Internet, including on the BIS website.

RATINGS

Sed quis custodiet ipsos custodies?

—Latin aphorism¹⁰⁹ meaning “But who will guard the guards?”

Ratings are here to stay, but there is no denial that the way banks and investors use them has changed in the recent past and will change further in coming years. The complexity of a number of risk items has gone beyond the skills of the traditional analyst, let alone those of loan officers, and yet, at the same time, for regulatory reasons, many banks will soon have to rely on third-party opinions for capital adequacy compliance purposes. Rating definitions have always been vague and ordinal, leaving some space for in-house analysts and credit managers to pepper their conclusions with proprietary insights and on-the-ground experience, including so-called judgment-based input.

Under the cardinal approach, though, the rating must provide some measurable quantity—such as a probability of default—rather than a ranking for creditworthiness, which would be ordinal. Traditional ratings provide some of both approaches since the rated items get ranked by creditworthiness, and, at the same time, the rating category system provides an approximate, yet sometimes very precise, mapping between the rating category and a quantity such as the probability of default.

Nowadays, ratings are increasingly supposed to use the cardinal approach and to provide a probability of default with a high degree of confidence, along relatively rigid processes that take away the sometimes subjective input that used to give traditional ratings their value. This is bound to bring more financial disasters. If scoring models tell us all we want to know, and if we blindly build risk decisions and capital rules on them, then mistakes in the models or a simple shift of the background they were built against will precipitate markets into further gregarious fads leading to further international crises.

Origins and Development of the Rating Industry

The ratings industry had its origins in the mid-nineteenth century when companies arose in the United States to provide merchants with credit information on customers.

Towards the end of the century, the concept of merchant credit information agencies was adapted to the securities markets. Instead of serving merchants seeking information on the creditworthiness of purchasers or sellers of goods, rating agencies provided a service for investors in bonds, offering views on the creditworthiness of

debt issuers. *Poor's Manual*, which contained opinions on both equity and debt securities, was published in 1890. In 1909, John Moody published a book that evaluated the creditworthiness of railroad companies and the bonds that they issued, while distilling those evaluations into ratings symbols. The rating symbols provided investors summary risk assessments designed to assist them in making decisions concerning which bonds to buy or sell and at what price, a function the ratings continue to provide.¹¹⁰ Moody's Investor Services (Moody's), incorporated in 1914, and Poor's Publishing Company, soon found a market in analyzing and evaluating investments in fixed-income securities, although Poor's (later Standard & Poor's) also covered equity securities. Both expanded as the capital markets evolved in the United States in the early twentieth century, and were joined by competitors, including Standard Statistics Company in 1922, and Fitch Publishing Company in 1924. Rating agencies originally charged subscribers a fee for obtaining rating information in contrast to the general present practice of charging for the rating itself, and thereby provided "unsolicited" rather than "solicited" (i.e., paid) ratings.

By the mid-twentieth century, corporate debt ratings were so well-established that institutional investors came to expect a rating from one of these two major agencies as a prerequisite to purchasing a fixed security from organizations other than the federal government. In a sense, a rating assigned by a recognized agency came to be viewed as a passport into the bond and commercial paper markets. Ratings also became a means to distinguish between so-called investment-grade and non-investment-grade securities for many institutional investors. The significance of this distinction between investment-grade and speculative-grade bonds (popularly known as *junk bonds* or more recently as *high-yield bonds*) became more important as securities laws in the United States, which were strengthened during the 1930s, gradually came to give investment ratings a regulatory imprimatur in certain instances. For example, the U.S. Banking Act of 1935 prohibited nationally regulated banks from purchasing securities other than investment grade. The imprimatur was strengthened in the United States in the 1970s through the creation of the designation Nationally Recognized Statistical Ratings Organization (NRSRO). The distinction came to be an important one in many instances. For example, when an NRSRO assigns an investment-grade rating to an offering of nonconvertible debt or preferred stock, the issuer benefits from streamlined registration requirements. Ironically, rating agencies themselves have not universally welcomed the incorporation of ratings into regulation, fearing a loss of independence and the erosion of credibility that political pressures could bring to bear.¹¹¹ A belief that regulation would foster *rating shopping* has also helped to fan this reaction. Some of the controversy over the revisions proposed in 1999 to the 1988 Basel Capital Accord was the result of these concerns.

Beginning in the mid-1970s, the business model of the debt rating industry, which by then consisted of two major agencies previously mentioned, Moody's and Standard & Poor's (S&P),¹¹² as well as their large competitors Duff & Phelps and Fitch, had begun to shift from one that was subscription-fee-based to one that charged issuers for the privilege of obtaining a rating. The change took some time to implement, and unsolicited ratings continued to be provided to a greater or lesser degree.

But by transferring the cost of a rating to the issuer and reframing the cost of a rating as part of the cost of borrowing in the capital markets—and a comparatively

low one at that compared with underwriting, legal, and accounting fees—rating agencies were able to hold down subscription costs and distribute ratings more freely to investors. Rating agencies, moreover, by this time had become so ensconced (at least in the United States) in shaping investor perception that they were usually able to justify their charge by asserting that a rating in most cases tended to reduce the cost of borrowing for the issuer, as compared to an unrated issue. The argument was made that even a modestly rated issue would fare more favorably than an unrated one, as the latter would be perceived by investors as intrinsically more risky. Although there is an apparent conflict of interest raised in a circumstance where the issuer pays for the rating fee, rating agencies have argued plausibly that they are constrained from showing bias toward those who pay the piper both by professional standards and because their business rests upon their credibility and reputation with investors. As Chester Murray, Moody's managing director for Europe said, "If we started inflating ratings to get business, it would become readily apparent to investors and our overall credibility as a rating agency serving the global credit markets would be at stake."¹¹³

Contemporaneous with this evolution in the industry, a number of specialist rating agencies that concentrated on particular industries, most notably the banking and insurance industries, were established. These included BankWatch in 1975, which was initially established as a division of the boutique investment banking firm Keefe, Bruyette and Woods and was to become the largest bank rating firm, and IBCA Ltd in 1978. Notable banking failures in the United States, such as that of Continental Illinois and others, provided the impetus for the creation of these specialist agencies. They were given further nurturance from the increasing concern about bank risk that came about in the 1980s as a result of a number of causes: the Latin American debt crisis, anxiety about the use of off-balance-sheet finance, deregulation, and the latent savings and loan crisis, and finally the increasing prominence of Japanese financial institutions in world markets. The major agencies did not cover banks extensively during this period, unless they were major issuers of debt securities, and coverage remained scant into the 1990s, as most banks relied primarily on their depositors and the interbank market, rather than the capital markets, for funding. Thus, at a time when credit information on international banks was limited, the specialist agencies provided a needed service to counterparties on a subscription basis.

These specialist agencies differed from the major agencies both in coverage, in the type of rating they provided, and in the needs they fulfilled. Unlike the major agencies that became integral players in the capital markets, the specialist agencies were oriented towards counterparty and correspondent banking transactions, including trade finance. The major agencies were highly focused on the capital markets. In contrast, the specialist ratings tended to be used by credit officers at banks and corporations in one of their key tasks: establishing credit lines and limits for financial institution exposure, a function that took on increasing importance with the spread of derivatives for hedging and trading purposes. Their coverage was generally broader than the universe of financial institutions seeking to tap the capital markets for funding.

Moreover, there was relatively little incentive—at least, at first—for banks to pay to be rated as the business models of the major agencies generally required, so the specialist agencies initially relied on a subscription model of revenue to subsidize a

wider range of coverage than that provided by the major rating agencies. This, of course, was the same approach the major agencies had taken in their early days. Ultimately, the specialist agencies also attempted to move to a paid rating revenue model, particularly in developed markets, but continued to rely heavily on subscription revenue. In the case of paid ratings, they functioned in a similar manner to the major agencies, although usually instead of rating particular securities, they often focused on appraising the overall creditworthiness of financial institutions in respect to its outstanding obligations. Later, individual securities were rated, but often as generic classes of securities rather than as individual issues.

The late 1980s and 1990s were characterized by global expansion within the ratings industry as equity and fixed-income investment in emerging markets flourished. Sovereign ratings, which assessed the creditworthiness of governments, came into more widespread use. All agencies broadened coverage to meet customer demand with major players setting up overseas offices, while the secondary and specialist agencies grew primarily through acquisition of, and joint ventures with, regional rating agencies located in emerging markets such as Asia, Latin America, the Middle East, and eastern Europe. A related phenomenon was consolidation within the industry. Although the ratings industry based in the United States was essentially an oligopoly, in the latter part of the 1990s, consolidation occurred as both the general and specialist agencies recognized the virtues of size. Also motivating consolidation in the specialist rating industry was the consolidation of the banking industry itself, which went through an unprecedented wave of mergers and acquisitions during the decade.

This phenomenon of consolidation in the rating industry was epitomized with the mergers of Fitch with IBCA, with Duff and Phelps and Thomson BankWatch in rapid succession between 1997 and 2000, creating the third of the current Big Three rating agencies.

A Paradigm Shift

Internal or external ratings are used by banks for three major purposes:

1. In support of commercial decisions
2. As a basis for the pricing of a transaction
3. As a tool to measure economic capital and compliance capital

Commercial Decisions

Banks take commercial decisions—whether on credit, treasury, foreign exchange, investment, or other types of transactions—every day. This is done under the umbrella of a preestablished strategy that balances the bank's risk appetite—looking for a reasonable risk against a reasonable return—with its risk tolerance—beyond which no potential return is worth taking the risk—tempered by side consideration such as diversification needs or a wish to expand geographically or into new markets.

The strategy must be consistent with a thorough evaluation of the risks and rewards of a wide range of opportunities. To guide credit origination teams, in particular, the measure of credit risk is crucial, and that includes not only probability of default, but also exposure at default and loss given default—which are affected by

the strength of any collateral—including often-disregarded recovery costs and opportunity costs.

External ratings have long provided banks with all they needed for their commercial decisions—of course, provided there was a rating available on the risk item—while the judgmental input of seasoned credit managers or credit officers added a layer of protection.

Internal ratings or risk categories were put in place to make it possible to map external ratings—including conflicting rating opinions from several sources—into internal risk categories. They also provided a structure to shelter the many internal rating opinions that bank officers had to attach to myriads of transactions that were not covered by rating agencies.

Pricing

If a transaction is within the risk tolerance of a bank, it remains important to price it correctly. The notions of expected loss and unexpected loss are discussed elsewhere, but in short the bank has to price its transaction in such a way that, all told, within reason, it will likely not upset the balance sheet and destroy shareholder value, after allowance is given to funding and administration costs.

If ratings tell the truth, a mapping with the pricing of the transaction is easily seen, but there are a number of other components such as a correct estimate of the administration costs, the impact of competition, or the required return on equity. Those are beyond the scope of this discussion.

Compliance Capital

This is where Basel II and Basel III have brought a major change in the way ratings are used. As explained earlier, the compliance capital must be measured at all times, along very complex rules.

Among those rules figures prominently the use of external assessments provided by acceptable third parties such as recognized rating agencies.

External Credit Assessment Institutions (ECAI)

ECAI are institutions—such as rating agencies—that are recognized for the quality of their assessment of third-party risk. That recognition is usually local, and therefore the ratings are only as credible as the local regulators who granted the ECAI status.

The EU inscribed the concept in its regulatory framework in January 2006,¹¹⁴ whose sole purpose is “to provide a basis for capital requirement calculations in the Standardised Approach and the Securitisation Ratings Based Approaches” under the EU’s Capital Requirements Directive (CRD), which is built around the Basel Accords’ recommendations.

The Basel guidelines include criteria that local regulators should consider when granting ECAI status. Large rating agencies have obtained that recognition—or its local variety—in most of the countries they cover. In addition, many regional or local rating agencies obtained the status domestically or in a limited number of countries.

A local recognition is just local, and, very naturally, reciprocal recognition is sparingly granted.

In the United States, a recognition as a Nationally Recognized Statistical Rating Organization (NRSRO) is a recognition as an ECAI. Similar situations are found in many countries.

Basel II introduced the reference to the IOSCO Code of Conduct Fundamentals for Credit Rating Agencies to determine ECAI eligibility.

Relevance to Compliance Capital

Under the Basel so-called standardized approach, banks must use credit assessments provided by eligible and recognized ECAIs to determine the risk weights of their credit exposures.

Basel III went further than Basel II in requiring that procedures, methodologies, assumptions, and the key elements underlining credit assessments, among other things, be made publicly available.

Such requirements took away some of the judgment-based input, except where rating agencies were able and willing to quantify qualitative opinions in a consistent and reproducible way.

While a bank will obviously keep the freedom to enter into a transaction that its managers feel more comfortable with than an external rating would suggest, the impact on the bank's compliance capital will be determined by the external rating.

This has rendered external ratings increasingly important, in spite of the—unjustly—negative image politicians painted of rating agencies in the wake of the crisis that started in 2007.

Rating vs. Scoring

Rating agencies are justified in claiming that they should not use only scoring tools to reach conclusions on creditworthiness. After all, if a well-built scoring system existed, anybody could feed a selection of data into a model and draw the appropriate conclusions. Rating agencies would disappear.

Credit scoring is in fact widely used by a number of financial institutions, essentially to measure large numbers of small risks with a statistical tool that catches most of the components of creditworthiness, with the view of selecting risks and of pricing them with a good degree of accuracy at the lowest possible cost. Female drivers are safer than male drivers. A family of four with two breadwinners is a better risk for a mortgage loan than a single unemployed piano bar player.

The trick in scoring is to catch and use the right risk components, which is something difficult in normal circumstances and perilous under the stress of financial storms.

Scoring systems are not accurate in the prediction of individual loan losses, but they are helpful in handling large numbers, as their predictive power increases with larger pools. They also allow the risk taker to concentrate on sorting out good from bad applicants or files in borderline situations.

Can such a tool be used in bank counterparty risk analysis? It can, in two narrow ways: to create composite scores and to quantify qualitative opinions.

Composite Scores in a Risk Scoring Model

A typical process would be for a rating agency to establish a methodology profile of the type of issuer or the type of issue under scrutiny and to determine both a weight and a subscore card for each risk factor, both quantitative and qualitative, along predetermined dimensions.

The matrices produce a composite rating score that is the starting point for a judgment-based adjustment. If the process is dynamic—in that it gauges the risk against past findings¹¹⁵—it will also detect sudden changes and flag them for the benefit of the analysts.

Quantifying Qualitative Opinions

The current trend in regulatory circles is for any rating opinion to be quantified,¹¹⁶ regardless of whether the components of the opinion are actually quantifiable. The skills of the asset liability team are not immediately quantifiable, while earnings would always be.

This is where the scoring card is most useful, in that it allows the quantification of the unquantifiable—the qualitative opinion—however subjective or changing the criteria could be. The analyst rates this or that quality on a scale of 1 to 7, for example.

Rating Modeling

The ultimate scoring process—respectfully called a rating model—is one that attempts to mimic the analyst's line of reasoning with the view of producing a mechanical rating that will predict the most correct probability of default over a long period, of perhaps 5 or 10 years.

Model building implies a complex process that back-tests the results on past data and compares them to the actual event. This creates substantial difficulties in the handling of issues or issuers that, over the years, have left the original cohort, as information about their fate may not be available.

In the bank rating context, this is particularly vexing as many of the banks that merge or get taken over may otherwise have failed. As we know, very few ailing banks actually disappear.

In the corporate world, companies actually fail and disappear, and the robustness of the models relies on the research undertaken to track companies that disappear from a cohort. For example, if the original cohort is made of all listed companies in Mexico above a certain market capitalization, it is plain that, over 10 years, the list would have been modified, making the back testing a delicate matter.

An additional obstacle is that good models are evolving. For them to remain accurate, they must be regularly fine-tuned to their micro and macro environments. A good model created today may not be very predictive if applied to 10-year-old data, and if it is predictive when applied to 10-year-old data, then it may not be predictive as from today.

It remains that only rating models should be able to provide the huge quantity of rating opinions required by Basel II and Basel III. The number of human analysts needed to cope with the task would be substantial.

NOTES

1. Eddie Cade, *Managing Banking Risks: Reducing Uncertainty to Improve Bank Performance* (London: Global Professional Publishing, 1999).
2. J. Bessis, *Risk Management in Banking* (New York: John Wiley & Sons, 1998).
3. The authors wish to thank Mr. Richard Lumley, SVP—International Credit Risk with Northern Trust in London, for his contribution of an essay on the future of risk management, to be found in this section, as well for his thoughtful comments on various parts of the manuscript for the first edition of this book in 2001. In addition, the authors acknowledge the contribution of Mr. Darren Stubing to the book's first edition, particularly in respect to the material concerning asset-liability management, parts of which have been retained in this edition.
4. Each of these major risk categories can be further divided into numerous subcategories.
5. The significance and rating of management competence and quality was the subject of an earlier chapter.
6. See Cade, *Managing Banking Risks*, 5.
7. In other words, the credit analyst is only interested in these matters insofar as they affect credit risk.
8. This said, the roles of the credit analyst and the enterprise risk manager are distinct, and the credit analyst's interest in the risk management process will for the most part be limited to that necessary to form an understanding concerning a particular institution's risk management challenges in relation to its risk management capability.
9. Parenthetically, not only does the quality of a bank's risk management affect the credit risk posed by the bank to counterparties, the evaluation of the quality of each of the subcategories of risk management, such as credit risk management and liquidity risk management, are closely linked to the assessment of the corresponding financial attributes. Thus, as was addressed in Chapter 7 dealing with asset quality, a bank's management of its lending function and the quality of the financial assets it creates are inextricably linked with credit risk.
10. Liquidity risk is closely linked to market risk in that market developments may have a severe adverse impact on its liquidity position, not to speak of the effect on earnings and on asset quality.
11. Note that the risk environment and risk management capacity affect each other. For instance, in banks operating under the Basel framework, the risk management process may very well influence the levels of capital it seeks to maintain, as well as the allocation of capital internally.
12. Moreover, many credit analysts employed by financial institutions are part of their own institution's risk management process and consequently should have some knowledge of the work that risk managers undertake.
13. Foreword to Michel Crouhy, Dan Galai, and Robert Mark, *Risk Management* (New York: McGraw-Hill, 2000), xiii.
14. See also Robert C. Merton, "Commentary: Finance Theory and Future Trends: The Shift to Integration," *Risk* (July 1999): 48–50.
15. "The Practice of Risk Management," co-authored by Goldman Sachs and SBC Warburg Dillon. Although the term *risk management* dates back to at least the 1950s, it only came into widespread use in the financial industry in the early 1990s.
16. Goldman Sachs, "Practice of Risk Management," 60.
17. Risk management may also limit activities undertaken to a threshold level of risk irrespective of potential returns, risk adjusted or not.
18. For example, investment banking and commercial banking were segregated.
19. Other scandals during the 1990s that highlighted the importance of risk management included the bankruptcy of Orange County, California, at the hands of Robert Citron,

- and traders Toshihide Iguchi at Daiwa Securities and Yasuo Hamanaka at Sumitomo Corporation, and the collapse of the super-hedge-fund, Long Term Capital Management.
20. Until the mid-1990s, risk was measured at the level of the business unit rather than being aggregated across the bank, and risk management reflected was heterogeneous in character, embracing a collection of discrete approaches to specific concerns. See Philippe Jorion, *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed. (New York: McGraw-Hill, 2007), 411.
 21. Cade, *Managing Banking Risks*, 3.
 22. This is not to say, however, that the equating of volatility and risk did not result in some problems.
 23. Merton, "Finance Theory and Future Trends," 1.
 24. Depending upon the jurisdiction and various circumstances, the conclusions of such reports may in some form be made available to regulators and to shareholders.
 25. Bessis, *Risk Management in Banking*, 1.
 26. To provide just a single example, text on bank risk classified market risk as a separate category from both interest-rate risk and foreign-exchange risk, market risk being defined to be limited to trading risk. Bessis, *Risk Management in Banking*, 8–11. Another separates operational risk and legal risk. Crouhy et al., *Risk Management*, 37.
 27. As with all bank risks, liquidity risk when viewed from the external credit analyst's perspective contributes to credit risk. In the same manner that weak asset quality impairs a bank's solvency and thereby increases the credit risk it presents to counterparties, so too does weakness in the bank's liquidity position reduce its overall credit quality.
 28. Insolvency to a bank technically means the circumstance where it has insufficient capital to cover its obligations and will be forced to close its doors. As such, it is synonymous with bank failure. This is notwithstanding the fact that a technically insolvent bank can continue to survive so long as it has liquidity whether provided by the state or obtained elsewhere. For this reason, the term *solvency risk* is not ideal. It remains, however, the most common term used to describe the risk of bank failure from the perspective the bank's own managers and shareholders.
 29. Solvency risk has been described as a secondary risk that "by definition . . . is disqualified as a cause of bank failure." Cade, *Managing Banking Risks*, 19.
 30. Needless to say, such catastrophic events can result in great harm not only financially, but also to property and to human life. Damage can include the loss of vital business records, major facilities, and even of key personnel, impairing the capacity of the bank to continue to operate. In this context, it may be worth mentioning the distinction between financial risk and nonfinancial risk, whereby credit risk, liquidity risk, and market risk would be considered financial in character, while political risk (risks occasioned by political changes, including war or revolution), strategic risk (concerning general top-level business decisions made by senior management), and many aspects of operational risk would be viewed as nonfinancial in character. Since all significant risks with which a bank deals can, if realized, result in a negative financial impact upon it, the usefulness of this distinction is questionable.
 31. Where, however, portfolios are involved and credit risk can be translated into prices, the possibility of lower than expected credit losses and gains on securities holdings becomes possible.
 32. Goldman Sachs, "The Practice of Risk Management."
 33. The word *changes* in the definition is intended to comprise not only changes in nominal rates or prices but also the volatility of rates and prices and correlations among various instruments. In the case of interest rate risk, changes take in the shape of the yield curve as well. Note that there are, as might be expected, a number of definitions of market

risk. Another definition is “the risk that changes in financial market prices and rates will reduce the value of a security or a portfolio.” Similarly, there are quite a few additional subcategories of market risk other than interest-rate and currency risk, including among others yield curve risk and equity price risk. Crouhy, Galai, and Robert Mark, *Risk Management*, 177.

34. Banks take positions by virtue of being the initial counterparty of myriads of transactions, not all of which the bank is able, or wishes to cover through an identical, reverse transaction.
35. At least for banks whose traditional “warehouse banking” activities remain their core business.
36. Market risk and liquidity risk can be linked. If, for example, interest rates spike upward even temporarily and the bank is dependent on short-term funding to support long term lending, it may experience a shortfall of interest revenue—and cash—relative to its funding costs. The linkage can also be seen clearly in situations where market in a particular security is thin. If a bank must accept a deep discount in order to liquefy a particular financial asset, such a liquidity risk can also be seen as a form of market risk.
37. This is so unless it has adequate hedging mechanisms in place that, at a cost, may substantially mitigate the intrinsic interest rate risk.
38. Structural interest rate risk will vary with the nature of the change in prevailing interest rates and the degree of mismatch in the repricing term characteristics of those assets and liabilities. Potentially, these variables may interact in a manner that adversely affects the institution’s financial performance and condition.
39. Foreword to Michel Crouhy, Dan Galai, and Robert Mark, *Risk Management*, xiv.
40. As has been discussed, traditionally credit risk was largely viewed as tantamount to default risk, i.e., the probability of default, but the costs of default were not necessarily taken into account in a systematic manner. Nor did the assessment of credit risk generally consider the diminution in value resulting from events short of default that evinced a decline in credit quality. When banks mainly held loans to maturity, this was a sensible view of credit risk. With the development of the field of enterprise risk management, however, a more sophisticated view of credit risk emerged, one that took account, before the fact, of a bank’s expected amount at risk at the time of default, the severity of the default (exposure at default or EAD), and which considered the significance of the decline in an obligor’s credit quality reflected in a diminution in the market value of that obligation.
41. Note that credit risk by definition exists in advance of a potential event of default, so its calculation is naturally based on the hypothetical assumption of a default occurring. The estimated probability of the default occurring will, by definition, always be a prediction that can only be tested after the fact (i.e., back tested). The accuracy of estimated probability of default will vary among other things with the quantity and quality of the historical default data available and with the precision of the models employed.
42. While a zero percent probability of default is impossible in the real world, nevertheless government domestic obligations in the national currency are, by convention, generally deemed risk free. Assumed to be risk free, they are frequently utilized as a risk-free benchmark against which other obligations are compared. Events that might negate this assumption have long been held as conceivable, but admittedly so remote as to be justifiably ignored. The events that unfolded in Europe in 2011 and 2012 proved that assumption unjustified in some situations, but one would have to remember that, for instance, the Greek government was in no position to print its domestic currency, the euro.
43. Rating migration risk of the bank itself from a risk management perspective almost certainly falls into the category of operational risk as it mirrors the overall creditworthiness of the bank, notwithstanding that it is a proxy for solvency risk. Where the

downgrade was occasioned by a sovereign downgrade and therefore not due to a relative decline of the credit quality of the bank in relation to its peers, it would still be operational risk albeit perhaps better considered with sovereign risk within that grouping. Where, however, the downgrade affects not the bank itself but the bank's largest borrower ratings downgrade is a proxy for credit risk and should be considered together with that category of risk.

44. Note that in practice it may be difficult to ascertain to what extent price movements are the result of credit concerns as opposed to more general market movements attributable to fundamentals or technical market factors. Consequently, in this situation, credit risk and market risk are closely related.
45. For instance, a global bank may have significant credit exposure to obligors in a certain Latin American nation as a consequence of operations in that country, and for similar reasons a comparable level of exposure to obligors in a particular Asian country. Superficially, it might not be expected that an economic crisis in one of the two countries would have much impact on the ostensibly healthy economy of the other. As both are included in the class of so-called emerging markets, in actuality investors and counterparties may perceive a link and demand higher risk premia in non-crisis-affected market resulting in a significant unfavorable knock-on effect upon the obligors in that country. The correlation in such a case is less obvious than with regard to obligors in the same industry, but no less real.
46. Settlement risk is, strictly speaking, the risk that a counterparty does not deliver at time of settlement, not necessarily on account of time zone differences.
47. Note that the formula assumes that a default on one obligation automatically triggers a default on all obligations. There are exceptions. In practice, it is conceivable that an issuer might default on some obligations and not on others, in which case the issuer could be said to be in default while an obligation of the issuer remains nondefaulted.
48. Goldman Sachs, "The Practice of Risk Management."
49. This is the Basel II definition of operational risk. Note that some commentators limit operational risk to risks internal to the bank, and classify external risks as business risks or as event risks.
50. Notwithstanding this broad distinction, some risks that are technically external in character—notably criminal fraud perpetrated by outsiders against the bank—are more appropriately considered together with other internal risks since banks have an obvious responsibility to take measures to guard against such traditional and obvious risks.
51. That is, the transactions in question were either subject to inadvertent errors in execution or have been entered into in violation of bank rules.
52. The former are sometimes called technical risks, not to be confused with the latter technological risks.
53. Some legal risks may be purely internal in nature (e.g., did the bank comply with existing laws?), while in other cases the same category of risk may encompass unpredictable external events such as a court decision that unexpectedly goes against the bank or a change in law that adversely affects its particular business franchise.
54. As with many of the categories of risk discussed in this chapter, there is no bright line dividing one type from another, and some ostensibly external events may have an internal element that contributes to the risk (is a bank robbery a business event or perhaps in some case—where an insider was involved—partly the consequence of a failure of controls?), or even have some link to some other of the three major risk categories.
55. Since credit risk and liquidity risk have been discussed already to some extent, the emphasis of this chapter in respect to specific risk categories will be on market risk and

- operational risk. Liquidity risk was discussed in depth in Chapter 10, solvency risk in Chapter 9, and credit risk in Chapter 7, among other places in the text.
56. Barry Schachter, "An Irreverent Guide to Value at Risk," *Financial Engineering News* 1, no. 1 (August 1997), reprinted in *Risks and Rewards*, March 1998, 17–18, www.gloriamundi.org/introduction.asp.
 57. As Jorion has stated, "in comparison [with the measurement of credit risk], the measurement of market risk using VaR is a simple affair." Jorion, *Value at Risk*, 433.
 58. Since both aspects of banking operations have historically been subject to some degree of interest rate risk, excepting those periods when governments have imposed strict control over borrowing and lending rates, managing the risk associated with interest rate movements is well—established as a cornerstone of prudent bank operations.
 59. "Market risk is primarily measured with Value at Risk." Jorion, *Value at Risk*, 262.
 60. In the context of asset liability management, liquidity risk is affected by, among other things, anticipated withdrawals, the composition of the loan book in terms of maturities, the maturities of funding instruments such term deposits, and the expected additional demand for new funds, all in light of mandatory liquidity ratios and other regulatory constraints.
 61. In especially volatile operating environments where extreme interest rate fluctuations are not uncommon, a prudent bank will need to reprice interest rates on new domestic currency-denominated assets very frequently, perhaps even daily.
 62. As with risk management generally, the overall objective is to optimize mismatches of assets and liabilities in order to maximize risk-adjusted returns.
 63. As suggested in the preceding discussion of asset sensitivity, in an environment characterized by rising interest rates, a bank is likely to endeavor to be as asset-sensitive as possible so that its assets reprice faster than its liabilities enabling the bank to capitalize from widening spreads. In a falling interest rate environment, of course, the converse is true.
 64. Naturally, such thresholds would depend on the nature of the bank's activities, as some of them may preclude reaching conservative ranges.
 65. It is worth remembering that such foreign exchange positions—and this is also true for maturity mismatches—must be analyzed on the strength of both on-balance-sheet and off-balance-sheet items.
 66. Banks in developed countries may run currency positions from time to time, but interest rate differentials are usually not wide enough to make this a part of an overall funding strategy. Moreover, hedging strategies are far more sophisticated, which will reduce the risk.
 67. High-yielding domestic government securities are usually a function of large fiscal deficits and hence high borrowing requirements.
 68. Even taking into account the loss on the foreign currency position through the devaluation of the local currency, the yield differential will—or rather the bank hopes it will—more than make up for it.
 69. If a bank is short in the foreign currency, it loses if the foreign currency gains strength.
 70. If nothing else, this creates distortions in the apparent profitability of otherwise separate activities, not to mention difficulties in strategic decisions or in matters as mundane as staff bonuses.
 71. Prior to the advent of value at risk, discussed in the succeeding subsection, interest rate sensitivity analysis formed the mainstay of analyzing the market risk of the trading book. The specific techniques employed focused on the impact on portfolio value of parallel shifts in the yield curve as well as in the change of shape of the yield curve. These techniques, however, did not address the probability of a particular level of loss arising from market risk or take into account benefits from portfolio diversification. Nor were they useful in measuring the volatility of particular securities or portfolios. Because of

- the limitations of these techniques, they were supplanted in large part by value-at-risk methodologies. The value-at-risk concept itself later lost some ground.
72. More properly, it is a class of methodologies that includes variations on the simplest approach that is described here. Although the statistical and mathematical methodologies upon which VaR is based date back hundreds of years, VaR in its contemporary form (and described as such) had its origins among a team of quants working at JPMorgan in the early 1990s.
 73. See www.riskglossary.com.
 74. Note that although the term VaR is most commonly used with reference to market risk, the underlying methodology can be—and often has been—used, with some adjustments, to estimate credit risk and operational risk, as well as the aggregate risk to which a bank is subject. The concept, however, is perhaps most easily described with reference to market risk, which because its two-way character shows loss characteristics more closely resembling the bell-shaped normal loss distributions than does credit risk.
 75. Paul Wilmott, *Paul Wilmott Introduces Quantitative Finance* (Chichester: John Wiley & Sons, 2001), 356.
 76. A good source of inspiration about such matters is Nassim Nicholas Taleb, *The Black Swan: The Impact of the Highly Improbable*, rev. ed. (New York: Penguin Books, 2010).
 77. See Goldman Sachs, “The Practice of Risk Management,” 83.
 78. In the illustration the horizontal axis represents the defined ranges of positive or negative changes in price, e.g., 0 percent to +0.50 percent, and the vertical axis corresponds to the number of data points for a defined range of change. Over a short time frame, the most likely net outcome is no significant change in the stock’s price. Note that since the diagram is actually comprised of data points it does not show that the most frequent outcome would be zero, but that it would approach zero since the highest concentration of outcomes would tend to be clustered around this figure. The larger the price movement, the less likely it is to occur. This example is adapted from Wilmott, *Wilmott Introduces Quantitative Finance*, 357.
 79. Over such a brief time horizon, it is reasonable to assume that the mean change is zero. Since stock prices tend to rise over time, over a longer period of time, the mean change in price would likely be a positive figure.
 80. Stress exposures are decided by measuring the potential losses that would be incurred on open positions, should rates, prices, and volatilities change according to specified ranges, which are set to correspond to potential market conditions.
 81. Economic capital as a single number is analogous to VaR as a single number.
 82. In actuality, the model was first widely applied in the market risk context, which was viewed as simpler to model, before being extended to credit risk.
 83. Nassim Taleb, “Against Value-at-Risk: Nassim Taleb Replies to Philippe Jorion,” 1999, expanded in 2008 in “The Fourth Quadrant: A Map of the Limits of Statistics,” an EDGE Original Essay.
 84. Taleb, *Black Swan*.
 85. Peter L. Bernstein, *Against the Gods: The Remarkable Story of Risk* (New York: John Wiley & Sons, 1998).
 86. *New York Times*, January 2, 2009.
 87. See, for example, JPMorgan’s unexpected trading losses in 2012.
 88. Richard Lumley is SVP—International Credit Risk with Northern Trust in London. See note 1.
 89. See Cade, *Managing Banking Risks*, 192.
 90. The failure of Barings was followed by a number of similar scandals involving errant traders, including Jérôme Kerviel at Société Générale, who single-handedly prompted a loss of close to EUR 5 billion in January 2008. Continued financial volatility in the last

half of the 1990s and the 2000s supported the growth of the risk management function within financial institutions. The advent of Basel II required all major banks to pay serious attention to risk management.

91. That is, volatility in credit losses and operational risk losses (i.e., unexpected loss) bear the same relationship to economic capital requirements that market volatility bears to value at risk. Note that economic capital requirements will vary with a bank's risk appetite, which is reflected in the confidence levels chosen. Such confidence levels can be mapped to the probabilities associated with a composite agency credit rating.
92. The controversies are too numerous to discuss here. Suffice it to say that some provisions in earlier iterations of Basel II that were deemed impracticable or that would result in onerous capital requirements were relaxed to a degree. There particularly concerned certain risk weightings, and the treatment of loan loss provisions and collateral.
93. The full text of Basel II, formally referred to as "International Convergence of Capital Measurements and Capital Standards," can be found at www.bis.org/publ/bcbs107.pdf. Implementation of the new accord was to take place over time beginning in June 2007.
94. Mirroring the three-pillar approach, the revised capital accord is itself divided into three distinct sections.
95. Although the original version of Basel I did not address market risk, later versions did. Basel II, however, for the first time required that operational risk be considered in a systematic fashion.
96. Recall that Basel I set risk weightings based on rigid criteria having only a tenuous relationship to actual credit risk.
97. The foundation IRB approach is intended to be a transitional one for banks, ahead of the advanced IRB approach. The advanced IRB approach allows banks to set all parameters with their own internal data and analysis.
98. See the "Report to G20 Leaders on Basel III Implementation," July 2012, BIS.
99. At least as per their published financials.
100. The FSB was established in April 2009 by the G20 group of countries to coordinate the regulatory and supervisory work of national authorities and international standard setting bodies. It is hosted by the BIS.
101. The authors apologize to the reader for the Federal Reserve's use of an acronym similar to that of the Capital Adequacy Ratio in the same context.
102. See the Federal Reserve's press release dated March 12, 2012.
103. See EBA's press release dated July 15, 2011, and the 2011 EU-Wide Stress Test Aggregate Report of the same date.
104. Just consider that, with a capital ratio of 8 percent, with all assets weighted at 100 percent and with no substantial side risk, a bank would have a leverage of 12.5 times.
105. The reader should check the details on www.bis.org.
106. For customers with an operational relationship with the bank.
107. The reader should check the details on www.bis.org.
108. All of them unencumbered.
109. Juvenal, *Satire* 6.346–348.
110. Companies and banks also use counterparty risk ratings to help establish risk exposure limits and for risk management purposes, as discussed later in the section.
111. See, for example, "Ratings Agencies as a Catalyst for Growth in Capital Markets: Moody's Special Comment," June 1997.
112. Poor's started rating securities in 1916. It merged with Standard in 1941 and was acquired by McGraw-Hill in 1961.
113. Quoted in Katherine Morton, "Time to Face the Image Problem," *Risk Publications*, April 2000, www.riskpublications.com/credit/apr00/features/credcoverstorv.htm. The *Financial Times*, in a special report dated May 8, 1998, cited in defense of the rating

agencies, a 1994 Federal Reserve Bank study that found that the discipline provided by concerns about their reputation kept agencies from inflating the ratings of their paying customers, and that a subsequent study “found little evidence that issuers shopped around for more favorable ratings.”

114. See “Guidelines on the recognition of External Credit Assessment Institutions,” 2006, Committee of European Banking Supervisors (CEBS).
115. This is not always done as it can lead to a situation where a past mistake gets compounded.
116. Meaning in this context that they are expressed as a (range of) figure(s) or a (range of) ratio(s), or a symbol associated with them.

The Banking Regulatory Regime

Recognize that banking . . . is a disaster-prone 19th century technology. The [banking] problem is unsolved now, nor will it ever be, I'm afraid, given the moral hazards posed by the absurdly high leverage ratios in banking, by deposit insurance, by the doctrine of "too big to fail," and by the increasing likelihood that the IMF—at least until it runs out of money—always stands ready in the wings to bail out bad banks and bad creditors generally.

—Prof. Merton Miller¹

None of the risks to which banking is prone can be avoided entirely. As was observed earlier, they can only be minimized, mitigated, or optimized—in other words, managed.²

Left to their own devices banks may behave in ways that increase their already considerable risks. Were the concomitant costs borne solely by the bank's owners, as is the case with ordinary commercial enterprises, such high-risk behavior would be of little concern to policy makers. But, as must already be evident, banks differ in material respects from their corporate counterparts, and imprudent risk taking on the part of banks poses severe risks to the economy as a whole. Imprudently managed banks are prone to conflict with and undermine government's economic policy goals, and the failure of one or more major institutions can catalyze a crisis that paralyzes the entire financial system.³

Governments have therefore learned (and sometimes forgotten again) the benefits of placing strict constraints on bank operations. In so doing, they can counterbalance banks' collective inclination to take on excessive risk. Policy makers most commonly accomplish this by establishing an intensive regulatory and supervisory regime to oversee financial institutions. Those that take deposits are often more intensively regulated since ordinary depositors are not expected to be financially sophisticated. As is the case with risk management, the purpose of government regulation is not to eliminate risk, but to compel banks to manage risks in a reasonably prudent manner in order to foster a sound financial system.⁴ If, as suggested earlier, banking is a confidence game in that only access to a quantity of liquid funds and public confidence separates a healthy bank from one that must close its doors, then the trust of market participants and depositors in the institutions that comprise the industry is essential. Upon what foundation such confidence should be based is an interesting question.⁵

Certainly, the financial condition of a given bank, as reflected in fundamental solvency, liquidity, and profitability indicators, coupled with the quality of management, is an important, if not the most important, source. Just as the financial strength of an individual bank is affected by the vitality of the banking sector as a whole, including the regulatory regime to which it is subject, so too is confidence, in part based on the strength of the industry as a whole. Similarly, whether a banking system is strong or fragile depends upon a number of variables, of which the legal and regulatory framework is one. Other criteria, notably the state of the economy, and the related field of sovereign risk analysis, have already been addressed in previous chapters.⁶ This chapter is concerned with assessing the strength and resilience of a country's financial system, with a strong emphasis on the quality of the regulatory regime.⁷

The proposition that the legal and regulatory framework governing banks has a significant influence on the risks banks take, on the strength of individual banks, and on confidence justifiably placed in the industry as a whole, is therefore perfectly sensible. To take a simple example, if creditors' rights are unenforceable, then a bank as lender will face both the higher probability of default and a greater loss severity. Likewise, because banks form a relatively small community of firms, which frequently deal with each other as counterparties, if individual institutions are permitted to engage in high-risk activities, the risks faced by the industry as a whole rise.⁸ If one bank defaults on a transaction, it may force other banks with which it is dealing to default on additional transactions, with unpredictable effects.

Conversely, where the legal system is efficient, where banks operate under firm but reasonable regulation, and where the public reasonably believes that banks are being properly supervised, confidence in the banking system is likely to be stronger and less easily undermined. For these reasons, a single bank cannot be fully evaluated without reference to the industry of which it is a part and applicable laws and regulations to which it is subject.

Consequently, one of the aims of this chapter is to survey the causes of banking system fragility.⁹ More specifically, the chapter looks at how the intrinsic vulnerability of the banking system provides the impetus for government oversight of the industry, and aims to survey those attributes that contribute to, or mitigate, banking system fragility. It attempts to offer a basic outline for the purpose of, first and foremost, understanding and evaluating a banking system in the context of the evaluation of an individual bank or peer group, and, secondarily, for purposes of illustration, the preparation of a written banking system review.¹⁰

The following aspects of a bank's financial *and regulatory environment* are considered. They are the quality or efficacy of

1. The legal infrastructure generally, and that of the bank regulatory regime specifically
2. The substantive laws and regulations within that regime, including the scope and effectiveness of prudential regulation
3. Bank supervision and enforcement

Concerning the *structure and viability of the banking system*, key criteria include

4. The degree of consolidation of the banking industry, as well as its structural features
5. The overall financial health of the institutions that comprise it, both individually and in the aggregate

6. The nature of the local business environment, together with the prevailing credit culture
7. The financial health of banks' customer base
8. The degree of foreign participation, together with its openness to new entrants

Each of these attributes may affect the level of ambient risk within a country's financial system, and each therefore may have a direct or indirect impact on a bank's financial condition and its creditworthiness.

In parallel with the facets of systemic risk evaluation just mentioned, a section looks into the rationale for the intensive regulation of banks by government, the types of approaches employed, and substantive rules that an analyst can ordinarily expect to find in a bank regulatory regime. A further section examines the features, both qualitative and quantitative, that describe a banking industry—such as size, level of consolidation, and common ownership arrangement—that provide additional criteria for fragility assessment.

Before concluding this introductory section, a note of caution is in order. Although the chapter enumerates a plethora of criteria, and lists numerous questions to consider in preparing a banking system assessment, banking system analysis should not be reduced to an exercise in ticking the boxes, and it is not the intention to provide a checklist for this purpose.¹¹ Instead, the assessment of systemic risk is very much a holistic endeavor, and should be approached as such. As has been noted:

*Unfortunately . . . there is no evidence: that any universal set of best practices is appropriate for promoting well-functioning banks; . . . or that detailed regulations and supervisory practices should be combined to produce an extensive checklist of best practices in which more checks are better than fewer.*¹²

In this light, what follows should be taken as an attempt to provide a bird's-eye view of systemic risk analysis, and to offer a variety of criteria that might be considered in preparing a banking system report. It is not intended to set forth a particular methodology, but instead to be a general survey of the key elements out of which such a methodology might be constructed.

THE BANKING SYSTEM REVIEW

Analysis of banking system fragility and corresponding systemic risk often takes the form of a written profile of the banking system. Such a report goes under an assortment of names, among them: banking system report, banking system update, and banking system review. The term *financial* may be substituted for *banking*, *industry* or *sector* for *system*, but ordinarily the content remains roughly similar. Typical format for a banking system report can be found in those produced by the major rating agencies.

Reports may range anywhere from 1 to 2 pages for internal purposes to 5 to 200 pages or more when prepared by rating agency or investment bank

analysts. The latter typically include abbreviated individual bank reports together with the relevant systemic analysis. How much detail to go into in profiling a banking system depends very much on the analyst's role, and on the purpose of the review. Rating agency analysts provide detailed profiles of the banking sectors for which they are responsible, frequently on an annual or biannual basis, with shorter updates prepared as required in the interim. Sell-side analysts working at investment banks may likewise write up system overview reports with a comparable level of frequency. Counterparty credit analysts, however, although likely users of such reports, probably have fewer occasions to perform a comprehensive banking system review. It may become necessary to research and write one from time to time, especially when a country's banking system is undergoing changes or has problematic aspects. This chapter may be helpful in understanding the criteria to consider in preparing such a report.

OVERVIEW

[T]here are advantages to examining an array of supervisory and regulatory policies simultaneously to identify those that enjoy a strong, independent relationship with financial development and stability.

—J. R. Barth, Gerald Caprio Jr., and Ross Levine¹³

It has been observed that banks are highly regulated entities, and, on this basis alone, the nature of the legal and regulatory environment in which they operate is an important consideration in their credit evaluation. The rationale for their close supervision is more fully explored in this chapter. Together with sovereign risk, the structure and condition of the banking industry, including the character of the applicable legal/regulatory regime, is an essential element of a bank's operating environment. The evaluation of a regulatory regime should take account of

- *Government policies* affecting banks (and regulators), particularly those constraining bank activities such as those directing lending to specific sectors or enterprises and tax policies
- *Institutional structure* of the regulatory and supervisory apparatus, including its scope of authority and degree of independence from political influence
- *Overall approach to bank regulation and supervision* reflected in that structure and in the behavior of the relevant institutions
- *The competence and integrity* of their leadership and staff

Beyond these broad considerations, additional specific parameters relevant to assessing the quality of a jurisdiction's bank regulatory regime include:

- *Structural regulations* governing the financial services industry: that is, those rules governing the establishment and activities of specific types of financial institutions

- *Extent and significance of state ownership of banks* and of state support to banks in distress
- *Key prudential regulations*, notably those dealing with asset quality, solvency, and liquidity, as well as those covering accounting, management, governance, and disclosure
- *Nature and impact of social policy directives*, particularly those concerning policy lending, that affect the banking sector
- *Authority of bank regulators and supervisors* to enforce applicable rules, along with their capacity and authority to respond appropriately to problem banks and distressed banks
- *Quality of the legal system*, including its fairness, efficiency, and creditor-friendliness, coupled with its capacity to enforce decisions made by regulators or courts

THE RATIONALE FOR REGULATION

In almost all countries around the world, banking institutions are subject to close supervision and monitoring by governments. Why, however, is such regulation so pervasive? One reason, it might be ventured, is that banking is an inherently risky business. The risks with which the banks must grapple—among them liquidity risk, credit risk, market risk, systemic risk, and operational risk—are in large part intrinsic to modern banking and do make the industry vulnerable to distress.¹⁴ As an illustration, because banks' stock in trade is the ultimate commodity—money—they must employ very high leverage to compete for investment capital. High leverage, however, magnifies the adverse impact of small missteps, or to put it another way, the correspondingly low levels of equity capital afford a thin cushion against asset impairment.

But the riskiness of banking in itself is not sufficient to justify the level of regulatory scrutiny to which the sector is subject. Other equally risky commercial endeavors do not shoulder the same regulatory load. Indeed, as we discussed in Chapter 1, looking at the nominal survival rate of banks compared with ordinary commercial enterprises, the former, superficially at least, do not appear to face great peril. Hence, while it may be a factor, risk alone does not supply the essential rationale for close regulation. If it is not risk, what then is the critical feature of banking that so distinguishes the industry from other lines of business that a special regulatory regime is needed? What makes banks different from their nonbank commercial counterparts?

Are Banks Special?

In 1982, the president of the Federal Reserve Bank of Minneapolis, E. Gerald Corrigan, published an essay entitled, "Are Banks Special?" in which he outlined the three characteristics that make banks essential to a healthy economy, and therefore worthy of protective regulation. In brief, he identified three distinctive features of the banking industry; namely the role of banks in (1) supporting a national payment system; (2) providing backup liquidity to nonbanks; and their (3) functioning as a "transmission belt for monetary policy."¹⁵

To elaborate upon the first point, the main bulwark of contemporary payment systems is the existence of demand accounts. An omnipresent feature of contemporary banking, such accounts make it possible to transfer funds quickly to third parties through the use of checks, direct transfers, or cash payments made from immediate withdrawals. Demand accounts, however, come at a price. Recall that in essence, banking involves accepting deposits in small, irregular amounts at erratic intervals, and lending out the same funds in larger blocks for fixed terms. The inevitable result is some gap in tenor between a bank's assets (loans) and its liabilities (deposits), and the inescapable potential of a shortfall in liquidity.¹⁶ The ubiquity of demand accounts magnifies banks' intrinsic structural asset liability mismatch, exacerbating the already difficult challenge banks face in matching assets and liabilities. As was suggested in previous chapters, the consequence is to heighten both their liquidity and market risk.

Second, banks function as providers of credit to companies and individuals, and in doing so foster economic growth. Perhaps more important, however, is the liquidity banks make available to both nonfinancial companies and nonbank financial institutions through short-term credit lines. These backup sources of liquidity help firms of all stripes to weather both seasonal fluctuations in sales and normal variance in cash flow, thereby facilitating economic stability.

Finally, as discussed in Chapter 11, owing to banks' ability to expand the money supply, among other functions, they provide a vehicle through which government can implement monetary policy. In doing so (and by acting in the other capacities mentioned), banks are cast in a quasi-public role, one that is integral to a cultivation and maintenance of a robust modern economy. It is their central position within the economy that most of all sets banks apart from nonbank enterprises, and provides the principal justification for their special regulatory treatment.

In addition to the rationales already provided, there exist other reasons for bank regulation. According to Barth, Caprio, and Levine, authors of *Bank Regulation and Supervision, What Works Best*, the "classic treatment of regulation holds that market power, externalities, and informational asymmetries create a constructive role for the strong helping hand of government to help offset market failures, and enhance social welfare."¹⁷ According to these scholars, each of the three attributes—market power, informational asymmetry, and externalities—is relevant to the banking industry.

Many banks enjoy an oligopolistic position, and gain market power as result. In many markets, a small number of banks enjoy that oligopolistic position, if not an effective monopoly within the communities they serve.¹⁸ Close regulation of their operations compensates for an absence of effective market discipline that would presumably occur in a more competitive market. Informational asymmetry simply means that by reason of cost or availability, one party to a transaction has access to more relevant information about the transaction than the other.¹⁹ Such asymmetries can be found in various aspects of banking. For example, a bank's experience in lending and deal flow gives it an informational advantage over potential new entrants into the market.²⁰ Informational asymmetries can support the contention that banks need close supervision owing to the special risks they face, as well as the assertion that their close supervision is necessary to help to protect others who need protection from banks, which leads to the next regulatory rationale: economic externalities.

The term *externalities* refers to situations where not all the costs of a transaction are borne by the parties to the transaction, and are therefore not captured in the

market pricing mechanism. A good example of externalities occurs in the area of environmental degradation, where (assuming there is no regulation), a polluting chemical manufacturer does not bear all the costs of the befouled water, earth, and air it creates. These, instead, are borne in large part by those who live near the polluter and downstream from it. The analogy in the business environment and the link with the banking industry comes from the havoc wrought by the manifestation of systemic risk. When bank imprudence leads to economic collapse, capital flight, and widespread bankruptcies, not to speak of unemployment and poverty, clearly others who are innocent of recklessness or malfeasance bear costs that should be borne by the culpable banking institutions and their management. Just as the chemical industry is subject to environmental regulation to protect the public from the harmful effects of toxic pollutants, so too should the banking industry be carefully monitored to protect others from the knock-on effects of a systemic and accompanying economic meltdown.

For all the foregoing reasons and others, governments have an important interest in maintaining the health of their country's banks.²¹ This is even more so in emerging markets, where capital and money markets are, in most cases, poorly developed. Those having authority over banks will consequently, in the interest of national economic development, attempt to support the strength of the banking system in any number of ways. These include, among others:

- Placing restrictions on the business activities in which a bank may engage (structural regulations)
- Implementing prudential regulations restricting a bank's operations, including setting threshold levels of financial strength and imposing minimum disclosure standards (prudential regulations)
- Making available or requiring deposit insurance
- Acting as a lender of last resort to a troubled bank

The last two items take the form not of restrictive regulations but instead of affirmative programs or actions intended to inject liquidity into the banking system. It might be said that deposit insurance schemes are designed to prevent deposit runs from starting, while lender-of-last-resort facilities are intended to prevent runs from spreading, but in practice, both programs probably reinforce each other. Finally, a critical justification—implied by the immediately preceding rationale—is the potential cost of a collapse, even of a partial collapse, of the banking system.

The Dangers of a Banking Crisis

During the Great Depression of the 1930s, the failure of thousands of banks in the United States and the concomitant loss of millions of dollars of depositors' savings and wealth were both cause and consequence of economic paralysis. During a severe banking crisis like the Great Depression—one that could be characterized as approaching a total banking system failure—the ability of a significant number of banks to effectively perform these vital economic functions breaks down, resulting in a partial paralysis of a country's economy.²² In a worst-case scenario, a country's banking system as a whole may collapse. The risk of this happening is what is termed *systemic risk*.²³ Though very infrequent, a banking system failure of sufficient

magnitude to trigger an all-embracing systemic collapse will bring with it economic damage of such severity that governments are, as practical matter, obliged to guard against its occurrence. Like the banking crisis of that period, similar banking crises—some large, some small—have occurred throughout the modern era.

A related and equally compelling motivation for a separate and tighter regulatory regime is the cost of repairing the banking system following a crisis. Less severe banking crises than the worst-case scenario posited above are correspondingly less rare events, but even a run-of-the-mill crisis will place an enormous burden on the public purse. Leaving the system unrepaired is not a viable option. Winston Churchill famously said that “democracy is the worst form of government except all the others that have been tried.” The same can be said of the banking industry in reference to systems of finance. Despite its many deficiencies, the alternatives are limited and appear to be no better than the system that widely prevails today.

Moreover, experience teaches that problems at a single bank are in many cases endemic. As a result, the closure of one institution can easily erode confidence in the banking system as a whole, potentially triggering runs against other institutions. In a conventional fractional reserve banking system, as was discussed in Chapter 10, the rumor that motivates a bank run can easily become a self-fulfilling prophecy, bringing down even a comparatively healthy institution. Although the identifiable cause of a banking crisis may be some external event that catalyzes the panic, invariably irresponsible, or even reckless, decision making by bank managers is at least partly to blame. The dangers that imprudent bank management poses to a country’s economy, to its government’s treasury and to the social welfare of its populace, provide an even stronger justification for intensive regulation of the banking industry.

Consequently, it is desirable and in the end almost certainly less costly to nip such potential problems in the bud. Governments seek to do so by establishing minimum standards of prudent bank management. With this aim, a regulatory regime is set up, and staff is assigned to examine and supervise the banks under their charge.

BANKING SYSTEM FAILURES AND THEIR IMPACT

It is rare for a banking system to completely break down. Banking crises of varying degrees of severity are less rare, but with few exceptions they do not result in a paralysis of the entire financial system. As a general rule, a banking system failure will be preceded by a banking crisis of some magnitude. It is, however, possible to conceive a catastrophic external event triggering a systemic breakdown. Such a scenario would include not only a paralysis of bank lending, but also bring about the stoppage of the national payment system, causing the reversion to a cash economy, or even a barter economy. The modern economy as we know it would cease to exist.

Not all banking crises lead to a far-reaching systemic failure, although they do invariably increase systemic risk. Almost by definition, a banking crisis is associated with some malfunctioning—or imminent malfunctioning—of the financial system. It is the prospect of a systemic impact that transforms a single distressed bank into a potential banking crisis. It is all a matter of degree. To use a metaphor, in a banking crisis it is as if a convoy of cars are caught in a

severe storm. Visibility is poor, and one of the more vulnerable vehicles stalls in the middle of the road. It is hit by another, and a chain of collisions ensues. As the intensity of the storm increases, visibility worsens, and the road turns into a ribbon of slippery tarmac interspersed with pothole ponds. More cars and trucks drop out of the convoy, victims of flooding, slippery surfaces, and more collisions, if not with other vehicles hitting boulders washed down from an adjacent cliff. In a mild banking crisis, most of the vehicles carry on albeit more slowly and cautiously than before the storm. In a worst-case scenario, the convoy is no longer moving. Some vehicles have crashed, while others sit by the roadside awaiting assistance.

Even a fairly mild banking crisis can prevent banks from adequately performing their credit-providing function, thereby damaging the economy. Equally, periods of even comparatively moderate financial turmoil can impose substantial costs on government and by extension upon taxpayers. Costs include the provision of liquidity to distressed institutions, the bailout of depositors, and the injection of new capital into the banking system. For these reasons, not only are publicly owned banks contingent liabilities of the state, but ultimately so are private sector banks.

The magnitude of potential costs reinforces the maxim that the credit assessment of a country's banking sector forms an integral part of sovereign credit analysis.²⁴ The 1997–1998 Asian crisis is a case in point and is now recognized as having been, at its core, a banking crisis.²⁵ While regional macroeconomic fundamentals remained largely intact, profligate banks, often acting in alignment with government policy, triggered the consequent financial turmoil. The crisis that began in 2008 and that is still in progress as this book is being written drives home the point. Can banking crises or banking system failures be predicted? Banking system failures, by definition, are catastrophic events. Not only do they occur infrequently, but, while they frequently share common features, each crisis has its unique imprint. This alone makes them hard to forecast. Perhaps more significantly, the participants though close to the market are often blind to the warning signs of crisis, which after the fact stand clear for all to see. Nevertheless, they have attracted the interest of scholars, and significant research has been undertaken in an attempt to discern the symptoms of a nascent crisis. This research and its findings are discussed in more detail later in the book.

Moral Hazard: The Perverse Incentive

Moral hazard has been defined as the notion “that if you compensate people when bad things happen to them, they might get careless.”²⁶ It can occur in any situation where such compensation for loss can be expected, including those scenarios where there is a formal agreement between an insurer or guarantor, on the one hand, and a beneficiary on the other. For example, a business owner that purchases fire insurance may be less likely to take reasonable precautions, such as the installation of smoke alarms and a sprinkler system in a warehouse, if he believes he will suffer no loss if a fire breaks out and the building burns to the ground.²⁷ Consequently, the insurance

company may require that the firm install such equipment and take other reasonable precautions as a condition to obtaining insurance. Additionally, it may refuse to compensate 100 percent of the loss, and only offer partial coverage instead.²⁸

Although moral hazard classically arises in the context of an insurance relationship, a similar dynamic may occur in any situation where risk is delinked from return. Thus, a broader definition of moral hazard might, perhaps, extend to situations where the potential benefits of imprudent behavior far outstrip the likely costs. In the banking context, the term *moral hazard* is used primarily to refer to scenarios in which one party benefits from an explicit or implicit guarantee. A common example is the situation where a reasonable expectation exists that the government will supply a distressed institution with liquidity or an injection of capital, or would otherwise ensure its continued survival. If bank management acting in line with the interest of shareholders reasonably expects the institution they oversee to be bailed out should it run into difficulty, they can be expected to take greater risks than they would should the owners bear the risk directly.

While the reason for such a bailout may be the avoidance of systemic risk and not to save the bank's owners, shareholders and management may still come out ahead thanks to realized gains or previous remuneration. Of course, if the government's policy is to bail out distressed banks without making its owners suffer any loss, moral hazard increases. Creditors, too, benefit from such largesse, and may even act as if the government is providing them a guarantee as well. In the same manner, explicit deposit insurance also creates moral hazard since it absolves depositors from any responsibility to vet the institution with which they entrust their funds. The benefits of retail deposit insurance, however, are usually judged to outweigh its costs, including those arising from moral hazard.

Judging that such promises of support will only encourage risky behavior, governments and regulators avoid often offering explicit guarantees in an effort to reduce moral hazard. Nonetheless, contemplation of the calamitous consequences that may unfold from failing to act frequently compels state authorities to rescue a failing institution, providing it additional funds directly or guaranteeing its obligations. Political influence may permit management, investors and creditors to escape the losses they could expect to incur under a laissez-faire regime. In view of this experience, even if there is no explicit state commitment of support, a bank's creditors and owners may plausibly reckon that a particular institution is too big to fail, and that the authorities will ultimately come to its rescue. The same phenomenon can occur in the international milieu. Foreign banks that make risky loans overseas may anticipate a bailout from multilateral agencies such as the IMF and the World Bank. This scenario, too, creates moral hazard, allowing such creditors to profit from high-risk transactions without necessarily having to shoulder the losses.

Ways in which moral hazard arises in the banking industry are shown in Exhibit 13.1.

The too-big-to-fail doctrine, that is, that a dominant bank will inevitably be bailed out by the government, is relevant to the creation of moral hazard since it may imply a *de facto* guarantee by the government even if the government does not provide an explicit guarantee and disavows any intention to rescue distressed banks.

Note that withdrawing deposit insurance coverage is only practicable before the bank becomes distressed as at that stage the withdrawal of insurance would penalize depositors more than the bank's owners or managers, and further would very likely

EXHIBIT 13.1 Moral Hazard in the Banking Industry

	Cause	Description	Incentive	Mitigants
Classic Moral Hazard	Deposit Insurance	Government insures depositors funds if a bank fails.	Depositors no longer have an incentive to scrutinize the bank's financial condition and risk appetite. Will look for highest deposit rates irrespective of risk, since they have nothing to lose. Bank need not worry about loss of funding or bank runs; deposit insurance subsidizes the bank, and it is encouraged to take higher risks than it would otherwise do.	Limits on deposit insurance; deposit insurer may have right to examine the bank, set conditions of insurance, call in regulator for prompt corrective action, or withdraw coverage where imprudent banking practices are identified.
Quasi-Moral Hazard	Guarantees (explicit or implicit)	Government guarantees to bail out banks if they get into trouble.	Banks owners (and indirectly managers and creditors) are encouraged to take higher risks for higher returns, since the owners do not need to worry about the loss of their stake in the bank, and a bailout often cover prospective losses to creditors.	No bailouts of banks, or if there is a bailout, shareholders and nonretail depositors and creditors bear losses.
		International agency indirectly bails out failed banks and their creditors.	Banks owners and creditors are encouraged to take higher risks in anticipation of outside assistance.	No bailouts of banks or creditor; both must bear loss.
	Little at Stake	Bank owners employ leverage well above industry rates (no implicit guarantee or bailout).	Banks owners (and managers) are encouraged to take higher risks for higher returns, since the amount they have at stake is negligible relative to returns. Same situation may occur when capital is impaired.	Capital regulation: Bank owners must have substantial funds at risk. Market discipline: Investors and uninsured depositors will flee undercapitalized banks.

trigger a funding crisis for the institution. Withdrawal before the bank becomes distressed, however, would result in the loss of the imprimatur that deposit insurance provides, and the threat of disqualification would function as an incentive for more prudent banking.

THE REGULATORY REGIME

Implicit in the rationales for banking regulation are its primary objectives. Broadly speaking, governments seek to promote sound banking systems in order to:

- Enable banks to carry on playing their usual beneficent economic role
- Prevent a bank crisis and its associated social and economic costs
- In the event of a banking crisis, to reduce the risks and severity of a systemic failure

In addition, the state may wish to use the banking system to promote ancillary objectives such as consumer protection or housing for low-income families.

Categories of Bank Regulation

While the business of banking is largely similar all over the world, regulatory structures differ considerably. Nonetheless, many common features may be found. Bank regulations generally fall into one (or more) of three categories as follows and as illustrated in Exhibit 13.2. *Structural regulations* govern entry into the banking business and the types of business activity in which banks may engage. *Prudential*

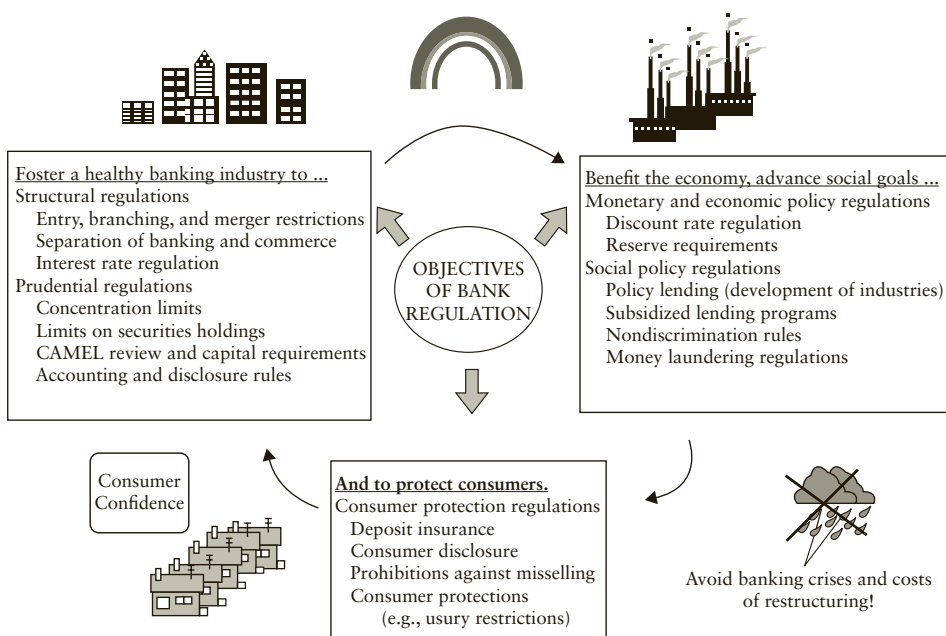


EXHIBIT 13.2 Goals and Means of Bank Regulation

regulations constrain a bank's operating freedom in those business activities that are permitted. *Policy regulations* seek to encourage or direct banks to fulfill particular social or economic policy goals.²⁹

Structural Regulations

Structural regulation concerns the organization of the banking industry. This category of rules sets the basic terms under which banks may conduct business. This category of bank regulations answers questions such as:

- What kind of business activities is a bank permitted to perform? (e.g., may a bank own and operate a cement manufacturing firm?)
- What business activities are prohibited?
- What distinguishes banks from other corporate entities?
- Who may establish a bank? Who may own a bank? (e.g., may a steel manufacturing firm own a bank?)
- What qualifications, if any, are necessary to establish or operate a bank?

Overall, structural regulations mold the shape of the banking industry present in a country.

Prudential Regulations

Prudential regulations concern the conduct of permitted banking activities. Their purpose is to ensure that banks operate according to minimum standards of prudence; hence their name. Characteristically, they require the bank to meet certain performance thresholds, or they impose restrictions on various aspects of a bank's operations. Questions such as the following are matters for prudential regulation.

- How much capital must the bank maintain?
- How much capital is required against off-balance-sheet assets?
- How much loan-loss provisioning must the bank charge against problem loans?
- What percentage of a bank's assets should be liquid or quasi-liquid?

The main distinction between structural regulations and policy regulations is between rules that proscribe certain activities and those that limit the degree to which a bank may engage in activities generally permitted.³⁰ Prudential regulations either examine financial benchmarks or deal with ownership, management, governance, reporting, disclosure, and related matters.

Policy Regulations

Policy regulations attempt to channel bank behavior in order to achieve social goals beyond that of a safe and sound banking system. Although all regulation is policy-driven, policy in this context refers to substantive social welfare policies separate from general economic and monetary policies. An example would be a policy to help the agricultural industry that required banks to allocate 5 percent of their portfolio into agricultural loans.³¹ Policy regulation deal with questions such as:

- Should banks be required to lend on a nondiscriminatory basis?
- Should banks lend to designated economic sectors, such as agriculture to undeveloped regions in the country, on other than commercial terms—that is, where the returns are not commensurate with the risks incurred?
- Should banks be restricted in what interest rates they charge consumers?
- Should banks lend to companies that employ a large number of voters, even if those companies are not competitive with foreign firms?

Although the preceding classification provides a useful way of organizing the topic, the categories are not mutually exclusive, and the distinctions between each are not particularly important. It is evident that a single regulatory feature may address multiple objectives and therefore fall into more than one category. Structural regulations, for example, are likely to have an underlying policy purpose similar to that of prudential regulations. Hence, restrictions on the ownership of banks by individuals or industrial groups, classified as structural, can also be seen as an attempt to prevent related-party lending, which would ordinarily be included within the category of prudential regulations. Likewise, deposit insurance schemes can be viewed both as social policy-oriented, being designed to protect the consumer, and as prudential regulations intended to enhance bank liquidity and banking system stability.

Scope of Authority and Enforcement

That a government has promulgated well-written and comprehensive regulations is probably less important overall to the cultivation of a sound banking system than the institutional capacity of the responsible regulatory agency. While virtually all schemes of banking supervision include regulations of the types described, the devil is in the details.

What is crucial is how a bank regulatory regime operates in practice, and not merely what it looks like on paper. Regulatory bodies vary considerably in their authority, scope of responsibility, and enforcement powers. In some jurisdictions, regulators may take a more proactive approach seeking to prevent problems from developing in advance; in others, they will be more reactive, allowing banks more freedom of operation, but stepping in quickly and with a heavy hand if things go awry.

Probably the majority of bank regulations impinge on banks' freedom of action in ways that firms in other industries would likely find oppressive. Yet some regulations actually function to make banks more profitable. Regulations establishing interest rate limits on deposits, for example, will often act to protect the local banking industry's profitability. Likewise, restrictions on new entrants into the market, whether domestic or foreign, will also tend to stifle competition, thereby benefiting existing players.

Most bank regulatory banking schemes reveal a tension between restricting institutions' freedom in the pursuit of maintaining the "safety and soundness" of the financial system and allowing them autonomy to transact opportunistically in the market subject to market constraints. Accordingly, there is a spectrum of approaches applied in practice.

At one end is the command approach, in which the bank supervisory body exerts authority over the banks within its jurisdiction, compelling comprehensive disclosure

to the regulator and requiring banks to meet its standards. The bank supervisory authority sees itself as the guardian of the public interest, but colludes with members of the industry in revealing little about their respective operations, in the interest of preventing the public from jumping to conclusions that might impair confidence in the system as a whole. Under such a regime, however, bank regulators may implicitly or explicitly promise to assist banks in trouble, thereby reducing systemic risk. The role of market discipline in the regulatory process is minimal.

At the other end of the spectrum is the market-discipline-based approach, which has increasingly come into favor, particularly since it has received the Basel Committee's imprimatur in the revised accords. Under this approach, regulatory actions are combined with an emphasis on transparency and comprehensive public disclosure. The emphasis is on creating effective incentives and disincentives to shape bank behavior, as opposed to relying solely on enforced formal compliance with regulator-imposed rules. As was noted by researchers Barth, Caprio, and Levine in "Bank Regulation and Supervision: Lessons from a New Database":

*[B]ank regulatory . . . practices that (1) . . . enforce accurate information disclosure, (2) empower private-sector [bank ownership], and (3) foster [private sector monitoring] work best to promote efficient bank intermediation and stability. . . .*³²

The theory is that by requiring banks to disclose comprehensive financial information, the power of the market assists regulators by rewarding "good" banks with lower funding costs and punishing "bad" banks on the basis of their disclosed financial figures. This, it is believed, will encourage banks to establish appropriate levels of risk management. Although bank shareholders still have a perverse incentive to maintain maximal leverage, market discipline assists bank regulators in prompting banks to act more prudently than they might otherwise do.³³ Utilizing the market in this manner necessitates a high level of transparency on the part of banks, which must be monitored for compliance by the regulator. It also requires fairly wide and deep financial markets. Banks that issue debt securities will be most affected by market discipline. Whether the regulatory regime leans more on the command and control³⁴ approach or market discipline, the areas of regulation will typically address many of the items discussed in the following subsection.

Regulating the Structure of the Banking System

Structural regulations, as mentioned above, comprise rules that govern entry into the banking industry, limit the number and location of branches, restrict the category of businesses that banks may conduct, and place constraints on ownership, as well as laws and directives that affect the fundamental character of the banking system.³⁵

Although the broad purpose of structural regulations is to encourage the development and maintenance of a sound banking system, sometimes they seek to do so by reducing competition with the aim of protecting bank profits, by restricting the number of new banking licenses, thereby shielding existing players from new competition.

An example would be a regulatory regime that tolerates or even gives its stamp of approval to a banking industry cartel that fixes interest rates on behalf of its

members. The fatter margins they enjoy as a result might be rationalized on the ground that by lifting bank profits it is fostering a stronger system. In actuality, interest-rate fixing constitutes a form of subsidy that may benefit banks at the expense of borrowers and retail customers. Although anticompetitive regulation detracts from economic efficiency, it tends to allow the protected banks to maintain oversized margins,³⁶ which provides an additional buffer against distress.³⁷ This, in turn, bolsters their creditworthiness allowing such an approach to be reasonably construed as credit-positive, at least in the short term so long as the protections remain in place. Over the longer term, however, a protectionist policy is apt to be credit-negative as such defensive practices tend to dull the commercial skills of bankers that have been shielded too long from competition. Loan portfolios of banks long insulated from competition are often poorly diversified, a feature that makes them more vulnerable to an economic downturn.

Such practices, however, may bode ill for individual banks and for the banking system as a whole when the deregulation or liberalization of the banking industry is imminent. When barriers to entry are removed, the banks previously protected from competition must face it. Increased competition comes both from new entrants, both domestic and foreign, as well as from the more nimble and aggressive among the existing players. Although ostensibly a positive development, history shows that the implementation of deregulation is often highly destabilizing in the short term. Remarkably such regulatory reform initiatives are frequently followed by banking crises.³⁸ Institutions that were previously state-owned are particularly vulnerable. When the cloak of government support is withdrawn, the newly privatized banks are often revealed to be quite weak on a stand-alone basis. Consequently, deregulation may be reasonably perceived to be credit-negative for a banking system and a substantial portion of the banks that comprise it. The analyst should therefore be very careful when assessing the fragility of banking systems—and the creditworthiness of banks—that have recently been deregulated or where widespread deregulation and privatization is imminent.

Activity Restrictions

Restrictions on a bank's business activities, as suggested, overlap with more basic structural regulations. Generally, the former encompass prohibitions engaging in (1) nonfinancial intermediary business activities of any kind (e.g., real estate investment); (2) designated financial intermediary activities that regulators deem appropriate to separate from commercial banking (e.g., insurance or securities brokerage); or (3) specific financial intermediary activities, which, though often undertaken by commercial banks, are assigned to other specialist institutions under the applicable regulatory framework.³⁹

One rationale for prohibiting banks from entering into the insurance, real estate, or securities business is that conflicts of interest between the various divisions of a "universal bank" could harm depositors and ultimately undermine bank soundness. Another justification is the belief that banks, as heavily leveraged enterprises, should stick to their knitting and not engage in potentially high-risk activities outside their usual domain. Finally, should banks engage in nonbanking activities, it might be argued that the special character of banks that provides the basis for intensive bank regulation is compromised.

The trend among policy makers and regulators over the past several decades has been towards greater consolidation of the banking industry. Rigid restrictions on the scope of businesses in which banks can participate within the financial sector have in large part fallen into disfavor.⁴⁰ Nevertheless, good arguments have continued to be made with respect to the merits of a more segmented approach to the banking industry. Notably, a significant minority view in the United States argued for the separation of commercial banking from investment banking.⁴¹ This said, there does appear to be a wide consensus, however, that banks should be forbidden from engaging in nonfinancial business.

The crisis that started in 2007 brought back some questions about whether bank depositors, creditors, and taxpayers should carry the burden of losses arising from nonwarehouse banking activities. To the extent that banks play a crucial economic role in providing personal, commercial, or industrial credit, that burden may in the end have to be shared, but should this remain the case where banks indulge in what some people call gambling activities? The debate is open.

Entry Barriers

Barriers to entry can take many forms. Bank regulators recurrently limit entry into the banking industry by prospective domestic entrants as well as foreign ones. Aside from outright barriers, onerous and time-consuming licensing requirements can constitute *de facto* bars to entry.⁴² The nominal rationale for creating barriers to entry, whether overt or covert, is usually a prudential one: namely, that only qualified and well-capitalized enterprises should be permitted to set up as depository institutions.

Other reasons include the following:

- Entry barriers enable protected banks to earn ample (read: excess) profits that contribute to their financial strength.
- Entry barriers allow existing banks time to adjust to new entrants. (Regulators may believe that domestic institutions will be unable to match the skills of new entrants with predictable adverse outcomes.)
- Entry barriers are necessary for existing banks to achieve the economies of scale that will enable them to compete effectively in a regional, or even in the global, arena.

The foregoing arguments may be in support of a broader policy view that a consolidated banking industry comprised of a comparatively small number of large banks is stronger than a fragmented sector with a large number of small institutions. According to this view, banks need to have a critical mass in order to effectively function, and that an overbanked sector leads to invidious competition that weakens the system as a whole.

Ownership Restrictions

Ownership restrictions on banks come in several variants. They may limit who may own banks and to what extent, or what types of companies banks may own or be owned by. In addition to outright restrictions, regulators may require state consent to investment in bank shares by an individual or group, foreign or domestic, above a certain percentage.

Prudential objectives are the primary justification for ownership restrictions and account for the use of broad prohibitions against banks investing in stocks, or limitations on such investment to a proportion of bank capital, with typical limits ranging from 10 percent (Korea) to 40 percent (Singapore). Regulations of the same general nature may prohibit or limit a bank's investment in nonfinancial nonbank business, in nonbank financial businesses or in other banks. Similar considerations motivate regulators to prohibit banks from investing in real estate, to limit such investments to a proportion of bank capital (e.g., 25 percent of capital), or to restrict the time period of applicable investments.⁴³

It is not uncommon for ownership restrictions to target nondomestic investors. In this context, the rationale is not prudential, but instead nationalistic; that is, to prevent the domination of a local banking system by large foreign banks. Particularly for a small nation, the fear that large and efficient foreign financial institutions may drive local players out of existence is not without some justification. Hence, regulators may impose foreign ownership restrictions that, in their most usual form, restrict foreign shareholders to a minority position.

The primary categories of structural regulations are illustrated in Exhibit 13.3.

Another category of ownership restrictions includes rules that are designed to remove or mitigate conflicts between the interests of owners and those of the bank's depositors and creditors. For example, to discourage banks from channeling depositors' funds to loans to insiders, regulations may curb the percentage of shares that may be held by any individual, corporation, or corporate group. Alternatively,

EXHIBIT 13.3 Summary of Structural Regulations

Restrictions on:	Comments and Examples
Permissible activities	Limitations on banks engaging in nonfinancial businesses or in designated financial intermediary business (e.g., securities brokerage, insurance, or investment banking), and limitations on activity by class of financial institution. Restrictions may be imposed directly as well as through entry (licensing) or ownership restrictions.
Entry into the industry	Caps on bank licenses; minimum capital requirements (see prudential regulation); vetting of prospective owners and managers; limitations on foreign bank entry; and suspensions on the establishment of new banks. Barriers to entry may be direct or imposed through ownership restrictions or capital requirements (the latter primarily a form of prudential regulation).
Ownership	Prohibitions or restrictions on ownership of banks, including limitations on their ownership by nonfinancial firms (such as where a bank becomes part of larger industrial group); restrictions on ownership by individuals or family groups; limitations on the concentration of ownership; and restrictions on foreign ownership.
Competition through other means	Limitations on branching; acquisition or merger; the freedom of banks to set lending and deposit rates (this may also be construed as a policy regulation); and tolerance of regulated monopolies or cartels.

regulators may limit the percentage of voting shares that may be held by such individuals or entities. In Asia, some jurisdictions have no limits on such shareholding (e.g., Hong Kong, Philippines), while others have imposed restrictions on individual shareholdings (e.g., 20 percent, Singapore) and on corporate shareholdings (e.g., 10 percent, Thailand). In addition, percentages above a certain threshold may be subject to regulatory approval. A last and distinct group of ownership-related regulation concerns the role the government itself plays as an owner of banks within the financial system. As government ownership of banks is closely related to policy lending, it always needs careful scrutiny.

Prudential Regulation of Banking: Financial Standards

Prudential regulations consist of the substantive requirements governing bank operations. These are rules intended to discourage banks from taking undue risks and to encourage them to manage their operations prudently and maintain adequate financial strength. This type of regulation includes, for example, reserve requirements; capital requirements; and other operational restrictions, such as those governing lending.

Subcategories of lending restrictions include limits on related-party lending, residential mortgage lending, and single-borrower lending. Since prudential rules are too numerous to address in detail, this chapter will merely provide a broad-brush sketch of the more important types commonly found in bank regulatory regimes.⁴⁴

Capital Requirements

The most pervasive and long-standing prudential regulations concern capital requirements. Basel II and Basel III are in essence capital requirements. Since capital and its regulation were discussed in Chapter 9, the topics are here referred to only in brief. Recall that the purpose of bank capital is principally to provide a cushion against unexpected loss. This explains why almost all bank regulatory regimes require banks to maintain minimum capital levels. Their main objective is to ensure that banks maintain a critical mass of capital in order that there be a strong probability that they will remain viable.

Capital requirements, however, can also be used for other ends. For example, an absolute capital threshold can be used to establish a conventional entry barrier to discourage new entrants to the banking industry. In a similar fashion, minimum capital requirements can also be used as incentives to encourage bank mergers. By raising capital requirements on a staggered basis industry consolidation among banks is thereby encouraged. This use of capital requirements functions as a kind of retroactive entry barrier.

Accordingly there are two main types of capital requirements:

1. Minimum capital ratios (i.e., relative capital thresholds), which specify that designated capital ratios of the bank be greater than a certain percentage amount. This first type is designed to impose minimum levels of solvency on licensed banks, which is accomplished using both traditional regulatory capital ratios and modern risk-weighted capital ratios.⁴⁵
2. Minimum absolute capital thresholds, which specify that the bank's capital be greater than a certain absolute amount. This second type applies almost

exclusively to the establishment of a new bank. Additionally, statutory reserve requirements, discussed earlier in connection with monetary policy management, are akin to regulatory capital requirements.⁴⁶

Each country has its own banking regulations, and it should be remembered that they do not even have to follow the rules under the Basel Accords. Therefore, rating the regulatory regimes around the world is an interesting exercise, which has a bearing on the overall perception of bank counterparty risk of a given country. One of the most visible sets of regulations is about capital ratios, but a country might ostensibly be stringent on that point while being lax on other, less prominent regulations.

Asset Quality Requirements

The maintenance of asset quality is perhaps the most critical area of bank operations subject to prudential regulation. Impaired assets quickly eat into capital and, given the high leverage banks employ, can quickly render an institution insolvent. By restraining banks from taking on excessive risk in their loan portfolios, and by attempting to ensure that they properly take account of credit costs through adequate loan-loss provisioning, most of the difficulties banks might otherwise be likely to experience can be avoided.⁴⁷

Regulation of asset quality can be divided into two groups:

1. Substantive rules designed to prevent asset quality from weakening (i.e., to prevent the bank from making mistakes in extending loans).
2. Disclosure rules that deal with loan-loss accounting and reporting, including loan-loss provisioning and write-offs.⁴⁸

The first category is comprised mainly of restrictions on lending and minimum collateral requirements. This set of rules seeks to prevent banks from making too many high-risk loans or taking on too much concentration risk through excessive lending to any single entity or in any single sector. Requirements that banks mitigate risks through adequate collateral backing also come within this rule group. The second group of asset-quality rules set the required levels of transparency in loan-loss reporting.

The same, of course, generally applies to any risk taking, of the nature of a credit risk, that does not strictly fall into the on-balance-sheet assets category.

The principal types of asset-quality regulations, and their regulatory rationales, are summarized in Exhibit 13.4.

The reader should note that Basel II and Basel III have attempted to classify, for capital-adequacy-compliance purposes, the various ways asset quality is measured. Countries that follow Basel II or Basel III are likely to adjust their national regulatory requirements accordingly.

Restrictions on Intergroup, Related Party, and Insider Borrowing

When a bank permits a significant portion of depositors' funds to be used to finance enterprises with which the bank's owners or employees are connected, the risks to depositors are heightened. In such circumstances, there is a danger that funds are

EXHIBIT 13.4 Asset Quality Regulations

	Type of Restriction	Purpose
Group 1	Restrictions on intergroup/insider borrowing	To deter the creation of poor-quality loan assets; avoid capital impairment
	Single-borrower limits	To reduce concentration risk
	Limitations on real estate lending	To avoid (excessive) high-risk loan asset creation
	Limitations on margin or share lending	
	Limits on foreign currency exposure	To limit FX risk
Group 2	Collateral requirements and restrictions	To mitigate credit risk from prospective asset impairment
	Definition and classification of nonperforming loans	To accurately report credit costs and loan losses
	Interest clawbacks and interest accruals	To more accurately account for credit costs
	Write off policies	
	Mandatory provisioning requirements	To provide an adequate cushion against asset impairment; mitigate credit risk
	Tax deductibility of provisions or write-offs	To encourage provisioning

being advanced on noncommercial terms, owing to the relationship of one or more of the parties with the bank. Hence, the existence of rules preventing significant lending to individuals or entities closely connected to a bank's shareholders, managers, or staff provides an important safeguard against the subversion of prudent credit standards.

PRESCIENT OPINION FROM A RATING AGENCY

It must be stated that it is possible for any bank in Indonesia to report figures in compliance with regulations while actual exposure to intergroup lending or volatile economic sectors, such as real estate, is considerably higher. This can be accomplished through indirect means. . . .

—Thomson BankWatch, Report on several Indonesian banks,
January 1996

Normally, regulations of this type limit the degree to which defined related parties can borrow from the bank, and also impose arm's-length dealing standards. Typical directives restrict aggregate related-party advances to a percentage of bank equity (e.g., 5 to 25 percent), enforce collateral requirements, or prohibit such loans outright. In considering the benefits of such regulations, attention must be paid to the definition of related parties and the level of enforcement. Note that as mentioned

earlier, when loans are granted, directly or indirectly, to a bank's owners, the effect is to render their capital contribution fictitious.

Single Borrower Limit

In some jurisdictions, regulators may set a ceiling on the amount a bank may lend to any single borrower separately from restrictions on related-party lending. The purpose of such restrictions is to prevent an overconcentration of commercial lending to an individual, entity, or group. Usually, these restrictions are expressed as a percentage of total capital, for instance, 25 percent of paid up capital (as in Hong Kong).⁴⁹

Limitations on Real Estate Lending

Because financial history teaches that banks often run aground on the shoals of excessive real estate lending, bank regulators may impose limits or guidelines on property-related lending. As this type of lending is closely associated with asset bubbles and the banking crises that often flow from them, regulators may attempt to constrain lending to property developers. What is defined as property-related lending varies, but it ordinarily excludes mortgage lending for owner-occupied housing, a form of lending that historically was much less risky than lending for property development or investment.⁵⁰ Amid the current credit crisis that began in 2007 in the United States, this general rule has come into question for reasons to be discussed in a later chapter. Restrictions on real estate lending may be defined as a percentage of deposits or of total loans (e.g., 20 percent of total loans, Philippines), or restrictions may take the form of collateral requirements.⁵¹

Limitations on Margin or Share Lending

In some countries, an attractive business for banks is share or margin lending. This is the practice of lending to brokers or investors to finance investment or speculation in stocks. In share lending, loans are secured by equity securities, in a similar manner to the way that property investment is secured by a mortgage on property. Margin lending can refer to unsecured lending—beyond the margin—for the purchase of shares. Quite a number of jurisdictions forbid margin or share lending by banks, Thailand being one example. Others constrain it to certain types of financial institutions, limit the percentage lent to a proportion of total loans or place an absolute cap on this type of lending.

Restrictions on Foreign Currency Exposure

To reduce the local banks' exposure to foreign exchange risk, as well as the dangers arising from an external position reflecting high levels of foreign debt, it is not uncommon for regulators to restrict foreign currency risk. A limit in the range of 10 to 15 percent as a proportion of Tier 1 capital (shareholders' equity) is typical. Regulators may also curb foreign currency lending and borrowing, or impose reserve requirements on foreign currency borrowing to constrain short-term foreign currency inflows.

Collateral Requirements and Restrictions

To reduce local banks' credit exposure, regulators may require designated types of loans to be collateralized to a specified minimum degree. Real estate mortgage loans

are often subject to such restrictions or guidelines. The requirement may be expressed as a loan-to-value ratio (LTV) or as collateral-to-loan ratio. The loan may for example not exceed 70 percent of the appraised value of the property. Restrictions may also be mandated on the type of collateral deemed acceptable. To illustrate, a bank may be prohibited from accepting its own shares or shares of a subsidiary as collateral. Alternatively, collateral may be limited to certain types of real and personal property.

Definition of Nonperforming Loans and Loan Classification Schemes

In general, regulators require banks to classify loans according to the probability that they will default and become uncollectible. While some discretion may be allowed to the bank, regulators typically identify minimum thresholds based upon the amount of time that principal or interest remain unpaid or have been delayed.⁵² Most commonly, regulators classify loans into four categories, excluding loans that continue to perform.

Depending upon the classification scheme used, a loan does not have to be overdue to be “classified.” So-called special mention loans may include loans where the borrower is experiencing financial difficulties, but is nonetheless fully current in its payments. Such loans are still performing, thus not technically NPLs. Not all regulators, however, employ the special mention classification. The customary international standard to define a nonperforming loan is any credit that is past due more than 90 days.⁵³ In many countries, especially emerging markets, looser definitions prevail, which, of course, tend to make the asset quality of the banks of that country appear better. A six-month threshold instead of a three-month threshold has been common in developing countries, although the trend is towards a general acceptance of the three-month definition. The fact that a loan is nonperforming does not mean that a loan is a bad loan, although in everyday parlance the terms may be used synonymously. A bad loan really means a loan that is in the significantly impaired category; that is, *doubtful* or *loss*.

Cessation of Interest Accrual and Clawbacks

Regulations governing interest accrual determine under what circumstances a bank may no longer accrue interest. Cessation of interest accrual refers to the time when the loan becomes overdue, after which banks are required to stop accruing interest. When interest accrual ceases, interest income must be reserved, it is no longer reflected on the income statement, and it cannot contribute to profits.

The international standard concerning the cessation of interest accrual is three months (90 days), the same as the definition of a nonperforming loan. In some countries, however, loans can be overdue six months or more before the bank is required to cease interest accrual. This has the effect of inflating bank income, albeit temporarily, as long as the interest income is later clawed back. In the case of a bullet or balloon payment (or a quarterly, semiannual or annual payment), a payment overdue as little as one day can compel interest to cease accruing.

Interest clawbacks refers to regulatory accounting rules (i.e., regulations governing bank accounting) that require a bank to reverse the recognition of accrued interest in the income statement once a loan has been classified as nonperforming. International best practice requires clawbacks of interest income accrued but not received at the time a loan is deemed nonperforming (which is usually at the 90-day

threshold). In some countries, however, there is no clawback obligation, or long grace periods exist. Both have the effect of inflating bank income.⁵⁴

Write-Off Policies

When a loan is written off, actual credit costs are realized. In effect, it is a moment of truth when the amount of asset, or a figure closely approximating it, becomes known. Writing off loans is encouraged and can be accomplished quickly in some countries, while in others the regulatory or legal framework functions to discourage or delay write-offs.⁵⁵ As a general rule, policies that facilitate, or better require, quick write-offs are preferable, as otherwise, problem loans may remain on a bank's books for long periods, making assessment difficult. When law or regulation make it hard for a bank to write off impaired assets, actual credit costs and their negative impact on reserves and equity is obscured.⁵⁶ The amount of time after which a write-off is mandatory also varies. In quite a number of emerging markets, regulations have often seemed to discourage write-offs.

Mandatory Provisioning

Regulations sometimes provide for minimum general provisioning, applicable to performing loans, and specific provisioning applicable to classified loans. The requirements concerning performing loans differ, but a general provisioning requirement of from 0.5 to 2 percent is not unusual.

Tax Deductibility of Provisions or Write-Offs

The extent to which provisions or write-offs are tax deductible will affect banks' willingness to take them. In the United States and the UK, loan-loss provisions are not tax deductible, but write-offs are. In other markets, provisions may be deductible up to a percentage limit, but not write-offs.

Liquidity Requirements

A lack of liquidity, as discussed, is the proximate cause of most bank failures. Therefore, requiring banks to maintain a minimum level of liquidity is an important objective of bank regulation. Liquidity management, however, is among the more difficult areas to regulate effectively, partly because liquidity needs can change so rapidly. The issues have already been discussed to some extent in Chapter 10, so their treatment here is intentionally brief.

Liquidity requirements tend to fall into one of the following types:

- Minimum liquid asset ratios or other liquidity ratios
- Restrictions on maturity mismatching of assets and liabilities
- Restrictions on mismatching of foreign currency assets and liabilities

The underlying purpose of minimum liquidity requirements is to make certain that the bank has an adequate pool of liquid assets, or access to cash through funding lines, to meet its reasonably anticipated needs. Hence, bank regulators in many countries require banks to maintain a minimum level of liquidity in order to satisfy current obligations. This can be accomplished through a variety of approaches, ranging from the setting of simple ratio thresholds to more sophisticated

regulatory limits that use asset liability techniques to set minimum requirements. A simple type of liquidity regulation requires banks to maintain a certain proportion of their assets in liquid form, either in cash, or in a form readily convertible to cash, such as government bonds or treasury bills. Compliance is effected through regular reporting of each institution's liquidity ratios, as defined by the regulator, that specifically measure designated liquid or current assets in proportion to total assets.

Since liquidity is strongly affected by the relationship between both sides of the balance sheet, the use of liquidity ratios alone may obscure the extent to which earning assets and funding liabilities may be mismatched, whether in respect of (a) maturity or (b) currency.⁵⁷ A very rudimentary method of gauging such mismatch is through the bank's loan-to-deposit ratio. This ratio weighs a bank's nonliquid assets (i.e., loans) relative to the bulk of its potentially loanable funds. By limiting the bank's LDR, some degree of liquidity might be preserved.

Restrictions on loan-to-deposit ratios, when utilized, vary, but 100 to 120 percent is a common threshold. More sophisticated regulatory approaches adapt asset-liability management techniques to monitor mismatches between assets and liabilities, and to mandate limits upon the level of mismatch permitted.

Mismatching in respect to the currencies of the bank's assets and liabilities is of particular concern to regulators since it can create fertile conditions for a currency crisis. A bank may borrow in U.S. dollars and then make loans in the local currency of the country (ordinarily on a floating rate basis). Since wider-than-normal currency fluctuations can potentially cause the borrower's debt service payments to vary enormously (and the bank's funding costs to do the same), it is apparent that liquidity risk, in this context, is closely tied to foreign exchange risk. Should the local currency decline dramatically in value, the payments required of borrowers may rise to intolerable levels resulting in defaults and leaving the lending bank with a shortfall in cash to meet its funding obligations. Regulators must therefore encourage banks to carefully manage their risk in order to avoid having a shortfall of foreign currency funds when those obligations become due. Regulators can go further by limiting foreign exchange transactions with a view to reducing foreign exchange mismatches that would result from a bank having a net positive or negative position in a foreign currency. For example, the amount of any overnight foreign exchange position may be limited to a percentage of the bank's equity capital (e.g., 15 to 25 percent). Note, however, that banks may maintain matched foreign currency positions, but still be vulnerable to foreign exchange risk where they on-lend funds to local customers in foreign currency. While technically the risk is with the borrowers, if a major currency devaluation occurs and borrowers are unable to afford the new exchange rate, their foreign exchange position becomes the bank's position, and their problem becomes the bank's problem.

Impact of Deposit Insurance Programs and Other Government Policies

Another type of regulation that can facilitate bank liquidity is the existence of a deposit insurance program. Whether implicit or explicit, such a program will reduce the likelihood of a bank run or smaller blips in demand for liquid funds. Yet, as mentioned earlier, deposit insurance or deposit schemes in general raise moral hazard issues as well.⁵⁸

DEPOSIT INSURANCE AND LIQUIDITY

Turkey implemented a full 100 percent deposit guarantee in the midst of its banking crisis in 1994 to halt the spread of deposit runs in the sector. While effective at the time, the blanket scheme has caused a number of problems over the years that can be attributed to the moral hazard the program brought about. As is often the case, the scheme, while implemented in the public interest, resulted in some unintended consequences. Weaker and poorly run banks were able to survive on the back of the guarantee scheme by offering above-market rates to depositors. Rather than being used to supply credit to the private sector, and thereby promote economic development, the deposited funds were often invested in Turkish government paper where high real yields prevailed. In a few cases, insured deposits were used to fund related-party loans, with the expected deleterious impact on the banking system. Finally, the full deposit guarantee indirectly had an adverse impact on the well run and financially sound banks by pushing up funding costs for all banks. In summary, as this example shows, deposit insurance programs can end up functioning as a subsidy to weak banks, and as a quasi-tax on stronger institutions.

Prudential Regulation: Management, Disclosure, and Transparency

An effective regulatory regime includes periodic bank examinations to evaluate the institutions under its charge. The CAMEL model, outlined earlier, is widely employed for this purpose. Concerning evaluation of management and related matters represented by the “M” in the acronym, bank examiners and supervisors will have occasion to evaluate the competence and quality of bank management and governance.⁵⁹ An effective bank regulatory agency is one that is not hesitant to appraise the quality of management in a fair and impartial manner. In most cases, bank regulatory authorities should also be willing to adopt a consultative approach towards bank management, having the capacity to advise management on improving its practices where appropriate. Where gross incompetence is detected, or there is material evidence of fraud, a strong regulatory agency should have the authority to take appropriate corrective action. Needless to say, regulators ought to take an aggressive stance towards fraudulent or criminal activity, such as money laundering.

A related area of bank prudential regulation concerns the level and quality of disclosure of financial data, which is critical to bank analysis and therefore to imposing effective market discipline on banks. If financial reports are late, incomplete, or inaccurate, it may be difficult if not impossible to perform an analysis that represents more than simply a pro forma exercise. Accounting presentation or local specificities might also affect the validity of financial reports. Naturally, a lack of independence of auditors, especially in emerging markets, is a negative signal, while limited disclosure remains a major factor in many countries.

Timeliness and Frequency

Failure to disclose financial results on a timely basis tends to make credit analysis a problematic exercise. Under the best of circumstances, reported annual results are

ordinarily two to three months out of date at the moment they are published. Longer delays mean the evaluation may be based upon the bank's financial situation six months or more past. Unfortunately, in some countries (banks in South Asia are among the most egregious offenders), reports published a year to two years after the end of the financial year are not unheard of. Plainly, such reports have little predictive value.⁶⁰

Related to timeliness is the issue of frequency of reporting. Quality, of course, is also a major issue. Only annual results may be available, and interim results, if forthcoming, may be highly abridged, possibly providing only limited balance sheet data and no information concerning income and expenses. Events having a material impact on the bank's financial condition, such as taking a charge for writing off losses, might not be announced at all, but instead left to be discovered following publication of the bank's annual report.

Another problem is that crucial data such as the level of nonperforming loans may not be disclosed. Moreover, data reported may be inaccurate as a result of a high degree of discretion permitted to management, or in more extreme cases, as the result of incompetence or even fraud. Taking into account the risk inherent in uncertainty, generally, the more frequent and comprehensive the financial reporting, the better and the more highly the analyst is likely to rate the sector and banking system as a whole.

Because public confidence is so important to banks, however, bank managers are frequently loath to disclose any more than the minimum financial and performance data. It therefore ordinarily falls upon bank regulators to set those minimum standards. From a bank analyst's perspective, quarterly or even monthly reporting would be near ideal. In the majority of countries, however, banks need only publish comprehensive financial reports annually, although some lesser degree of interim disclosure may also be mandatory.

Accounting Presentation

Consistency in accounting presentation aids comparison of the financial conditions of banks, and bank analysis as a whole. The world is basically divided into those countries that use International Accounting Standards (IAS), Generally Accepted Accounting Principles (GAAP) based on U.S. GAAP, or local accounting standards that are historically derived. Although IAS and GAAP are not necessarily superior to local accounting standards, international investors or counterparties are unlikely to be familiar with their idiosyncrasies. Consequently, these widely used standards, or rule systems that are not at too great a variance from them, do facilitate the analytical process.

This said, it may be noted that local accounting standards are sometimes characterized by reporting methods that obscure rather than reveal banks' true financial condition. They may be either unduly rigid or formalistic at one extreme, or allow for too much discretion on the part of bank management at the other. Problems include the use of peculiar definitions regarding the impairment and loss of financial assets such as loans. Nonperforming loans may be defined in a vague manner that makes it difficult for the analyst to fathom the true asset quality of the bank, while masking the bank's probable real level of loan losses. More often than not, in comparison with IAS or GAAP, they tend to reduce both the quantity and the quality of the information disclosed.

Related to the issue of accounting standards is the matter of the uniform reporting of financial data. Consistent presentation of financial data over a period of fiscal years with the same year-end facilitates comparisons among banks. When consistency is not present, comparisons over time and across markets become problematic. Similarly, it is troublesome to compare banks that end their fiscal year in June and issue full financial reports then, with those that end their fiscal years in December.

Consolidated Reporting

Another accounting standard issue concerns consolidation. Normally, it is best to analyze banks on a consolidated basis. Banks that report on a consolidated basis, including unconsolidated financial statements where appropriate, are less able to hide problem accounts in subsidiaries. Some Japanese banks have done this in the past, reporting financial and nonfinancial affiliates on an unconsolidated basis, thereby obscuring the magnitude of their financial difficulties. An exception may arise when the bank itself is but a minority subsidiary or an affiliate in a much larger group. In such circumstances, it would be appropriate to analyze the bank on an unconsolidated basis, although the creditworthiness of the larger group would be highly relevant in terms of possible shareholder support.

Undisclosed Financial Items

Depending upon the jurisdiction, it may not be unusual for one or more banks to omit disclosure of items critical to effective credit analysis, such as information pertaining to asset quality, especially nonperforming loans. In addition, a lack of data concerning risk-weighted capital and off-balance-sheet activity or the absence of comprehensive breakdown of income, expense, asset, and liability items can present obstacles to effective analysis.

Principal accounting, reporting, and disclosure issues are summarized in Exhibit 13.5.

EXHIBIT 13.5 Accounting, Reporting, and Disclosure

General Problem	Specific Issues
Timeliness of reporting, depth of reporting	■ Delays in financial reporting of from six months to two years beyond the reports' due dates. ■ Lack of interim financial updates or abbreviated interim reports. Failure to report information about material events at the time they occur.
Accounting idiosyncrasies	■ Nonstandard loss and impairment definitions for financial assets. Nonhomogeneous accounting standards. ■ No consolidation of the financial results of related companies. Capitalization of ordinary and recurring expenses.
Undisclosed relevant financial data	■ The recognition of future cash flows as current income. ■ Failure to disclose off-balance-sheet derivatives activity, NPLs, etc.
Limits on independence of auditors and freedom of speech	■ Lack of independence of auditors. Constraints on freedom of speech to openly discuss the financial problems of an issuer.

Policy Regulation: State Interference in the Banking Sector

All bank regulation has a policy purpose. At its core, the primary policy objective of bank regulation could be said to cultivate and maintain a safe and sound banking system that is able continue to perform a beneficial role in facilitating and maintaining economic growth. Conversely, all governments desire to minimize systemic risk so as to avert the damaging losses and high cost to the public that inevitably accompany banking crises. Since public confidence is essential to the effective operation of the banking system, the protection of depositors can be seen as an integral part of fundamental government policy towards banks.

Policy Regulation Defined

What is meant in this section by *policy regulation* are directives that go beyond what is necessary to preserve the health of the banking system, and that are not generally applicable to all commercial enterprises (e.g., health and safety regulations). Policies of this kind include mainly those that seek to protect or assist consumers or to protect or assist business and industry. Social policy laws and regulations include, for example, those that attempt to limit racial, ethnic, or gender discrimination in lending, or seek through affirmative action to help a designated group; and those that protect consumers by compelling banks to disclose information concerning contemplated transactions with the bank, that is, truth in lending-type laws; or restricting the interest rates that banks can charge consumers on loans; those that promote development in designated sectors by requiring banks to channel credit to selected industries or firms deemed by the government to require assistance, or provide incentives such as guarantees, for certain types of credit; and those that reduce crime by mandating that banks monitor their customers for potential money laundering or other related illicit activity.

Few governments are entirely without some such policy regulations specifically directed at the banking sector, and there is no reason that banks should be immune from broader socially—or politically—oriented policy initiatives. When such directives come from state authority, any disadvantage one bank might suffer by implementing particular socially desirable policies when its competitors do not, is eliminated.

Profitability Depletive Policies versus Profitability Enhancement Policies

Some policy directives, however, are more intrusive than others and pose a comparatively greater burden on bank earnings. The strength of the justification for imposing such costs also varies. While some policies, such as those that seek to make housing more widely available, have a clearly understandable social benefit, others may effectively function as subsidies invidiously discriminating in favor of particular industries. Whatever the ends sought, government policies that drain substantial profits from a country's banking institutions are inimical to a strong banking system. Internally generated bank capital is nothing more than retained profit (i.e., net profits excluding dividend payout), and strong earnings enhance a bank's profitability. It is evident that a bank's earnings are, in the long run, the ultimate foundation of its financial strength and shield against bank failure and the systemic risk that accompanies it. Accordingly, there exists an inherent conflict between the imposition of burdensome regulations on a bank for social policy purposes, and the adverse effect such regulations will have on the fundamental policy purpose of having sound banking system and the economic benefits that accrue from it.

The impact of policy regulation on banks individually and collectively depends upon the character and intensity of such regulation. Broadly speaking, policy regulations can be divided into profit-depletive regulations and profit-enhancing regulations. Profit-depletive policies are likely to include policy lending requirements (i.e., requirements or guidelines that a bank lend to certain companies or sectors determined by the government), excessive reserve requirements, obviously, high tax rates relative to other enterprises. *Policy lending* refers to lending that is mandated by government to advance some designated substantive policy ends. It is among the most common types of policy regulation and is also among the most perilous to bank creditworthiness.

THE PITFALLS OF POLICY LENDING

The extent to which bank lending is forcibly directed by the state towards certain sectors or institutions varies considerably around the world. Banks that engage in policy lending, also called *directed lending*, are most of the times state-owned. Some may have been established for the specific purpose of assisting a certain sector. For example, in many countries, there exists a national agricultural bank that was originally founded to provide credit to farmers. Often, however, privately-owned commercial banks have been subject to policy lending mandates. This was the case in South Korea and Japan where, for decades, banks were effectively commanded—through so-called administrative guidance—to lend to certain companies or sectors favored by the government. During most of the post-World War II era, banks in both countries provided loans at low margins to the country's leading industrial enterprises. Policy lending can even be said to exist in the United States, where, for example, various laws have compelled banks to provide loans in low-income districts from which they take deposits.

Policy lending is also commonly found in *socialist market economies* such as China and Vietnam, where banks, especially those owned by the government, have been obliged to provide loans to state-owned enterprises. The problem arises from the fact that many SOEs are loss making and will never be able to repay the loans. In India and Sri Lanka, policy lending mandates have been used to oblige banks to extend loans to farmers. In both of the foregoing examples, the loans are in fact anticipated to remain uncollected or uncollectible. They are therefore loans in name only, and in reality constitute disguised direct subsidies. In both circumstances—supporting SOEs or providing “credit” to farmers—the banking industry functions as a convenient vehicle through which subsidies can be channeled. While not all policy loans are destined to become nonperforming, they are subject to above-average credit costs. This is to be expected since, almost by definition, the borrowers targeted by policy lending mandates cannot obtain credit at an affordable rate, if at all, from commercial institutions. As a result, policy loans almost invariably represent a subsidy to the borrower and a cost to the lending bank.

As mentioned, in some countries, specialized institutions are chartered as policy banks, having been established to fund agricultural, community, or

industrial development, or to benefit a particular region or social group. A major part of, if not the whole, loan portfolio, of these institutions is devoted to policy lending. These are nearly always state-owned or state-guaranteed institutions, albeit sometimes the link is with a provincial government (or other subnational entity) or a quasi-state entity. On account of their ownership or direct control by the state, and the support it provides either explicitly or implicitly, such policy banks tend to benefit from a funding advantage in relation to private sector banks, in attracting deposits by virtue of their link with the government. For the same reason, they may also be more easily able to raise long-term financing that permits them to on-lend for long-term development projects.

Although profit making is not policy banks' chief goal, the bank's charter might allow it to generate some level of profit. In the same vein, the bank may have been established with a view that its profit-making activities will be used to subsidize its policy lending and development activities. Policy banks' state imprimatur and access to less costly funding potentially enables them to enter markets that commercial banks have avoided. A bank with a special role in agricultural loans may benefit from its wide branch network in rural areas, which often makes it the only bank in town in the country's more remote areas. It could also leverage its large rural branch network on behalf of multilateral institutions such as the World Bank or the African Development Bank, providing these larger institutions with greater distribution capability than they would otherwise have.

What then does it mean from a bank credit standpoint, and at a systemic level, when a bank is compelled to engage in policy lending? At the level of individual bank analysis, the short answer is that it is, generally speaking, credit negative when imposed upon non-state-owned commercial banks. Policy lending implies, as mentioned, above-average credit costs owing to the generally below-average asset quality of the corresponding loans. Since the lending is not undertaken on commercial terms, consequently the credit risk tends to be underpriced leaving little spread, if any, for operating profit. When undertaken by a state-owned bank, however, policy lending is only mildly credit negative or even neutral in effect, since as a rule state-owned banks, owing to the implicit or explicit promise of support, represent, to a degree, sovereign risk. In this context, it should be mentioned that even private sector banks stand a good chance of some state support in the event of distress. Of course, it is only fair that a bank that, at the behest of the government, foregoes opportunities to make higher profits in order to offer loans at below-market rates should receive some recompense for the foregone profits and state-induced losses that contributed to its distress.

A more comprehensive answer, and one particularly applicable to gauging the effect of policy lending on banking system risk, is that the impact on credit risk really depends upon the pervasiveness and severity of the policy lending mandates. Any quid pro quo intended to compensate for the affected banks' higher risk, reduced profits, or potential losses that policy lending generally entails should be considered. If, as is commonly the case, this recompense takes the form of prospective state support, then account should be taken of the government's capacity to supply it. This is especially so when the government

in question is a local or provincial government that may not have the capacity or the political will or autonomy to provide the requisite degree of support. An illustration of such a situation occurred in China in the late 1990s when the national government allowed some financial institutions supported only by local or provincial governments to collapse.

Finally, it should be noted that from an analytical point of view, identifying policy lending can be a challenge. It is easy to recognize when a bank is wholly state-owned and its charter explicitly directs the institution to lend to a certain sector. It is more difficult to discern when banks that have partial or no state ownership are compelled to lend to borrowers deemed worthy by the government on an ad hoc basis. In such case, the level of support that can be expected by the government is usually even more ambiguous than when policy lending is effected through a systematic program or quantitative mandate.

Governments may also implement profitability enhancement policies. The purpose may be to subsidize and therefore strengthen the banking sector through government support. They may also serve the purpose of benefiting politically favored groups. Policies of this sort may be explicit subsidies, or indirect such as where the government approves of, or tolerates, a cartel in which banks collude to maintain artificially high lending rates and artificially low deposit rates.⁶¹ High borrowing costs act as drag on growth; the cartel is in effect imposing a private tax on the economy. Banks may benefit from cartels in the short term, but over the long term they have a deleterious effect on the industry. They inhibit the development of a credit culture, competitive skills, and innovation. When support is withdrawn, the banks will find themselves ill-equipped to compete with new entrants into the domestic market or with foreign counterparts domestically or regionally.

In the implementation of policy regulation, incentives and subsidies are generally preferable to regulation by decree, such as a rule requiring that 20 percent of all lending must be directed to small businesses. The use of market mechanisms to implement policy goals offers banks more flexibility and has a less adverse impact on profits. While better for the banking industry, subsidies and incentives may cost the government more than simply requiring banks to allocate credit by fiat. Of course, the potential failure of the banking system owing to weak profitability and the consequent deficiency in capital constitutes a contingent liability on the state that may not be recognized in the risk-reward calculus.

Finally, it should be noted that regulations that are nominally prudential in character may also function as hidden policy directives. As an illustration, rules imposing excessive reserve requirements on banks could be used as means for the government to obtain low cost or financing at the industry's expense (statutory reserves typically pay little or no interest) rather than for the purpose of protecting depositors and creditors or managing monetary policy.

Consumer Protection and Other Social Policy Regulation

A special category of social policy regulation concerns consumer protection. Unlike policy lending, which provides subsidized credit to designated deserving recipients,

consumer protection regulation is aimed at preventing banks from engaging in sharp practices at the expense of consumers. Most regulations of this type that affects banks do not offer consumers subsidies, but instead they set standards for financial institutions to abide by in their dealings with consumers, and they grant the latter, or agencies acting on their behalf, the rights to enforce such standards. Consumer-oriented bank regulation might, for example limit the interest rates that can be charged for loans or other financial products; limit fees for overdrafts; proscribe unethical collection activities; and, perhaps most significantly, require banks to provide adequate disclosure concerning financial products and potential conflicts of interest. Since consumer protection regulation affects most or all banks in the industry and because they do not normally have a large impact on bank profits, they are not for the most part significant influences on systemic risk. It is only when particular institutions become subject to enforcement actions by the regulator that consumer protection regulations may become relevant to the analysis of a particular bank.⁶²

Finally, in light of their promise to restore funds to a bank's depositors should the institution fail, insurance schemes by their own terms come within the ambit of the consumer protection, notwithstanding that they are also intended to enhance bank liquidity and banking system stability. Although typically designed as discrete programs, with a regulatory framework of their own, operationally they are closely connected with the bank regulatory regime. As deposit insurance programs may strongly influence banking system stability, they are discussed further below.

Deposit Insurance

A deposit insurance scheme insures retail depositors that their deposits will be repaid in the event of a bank failure. Most such programs limit cover to a fixed amount, for example £35,000. Under some programs, deposits are insured in full; under others, a deductible amount equal to a relatively small percentage of the total is applicable, in most cases up to a certain limit. Although a government-run deposit insurance scheme is almost always established first and foremost to protect consumers—that is, ordinary depositors—and is justified on that basis, such a program also tends to boost confidence in the banking system and contribute to banking system stability.

The reason is that so long as there exists confidence in the insurance coverage, and a belief that payouts will be relatively prompt in case of a bank failure, depositors' incentive to join a bank run is greatly diminished. Banks therefore need not maintain as high liquidity and capital ratios as otherwise might be required both to bolster depositor confidence and to have sufficient liquid funds to resist a run. Consequently, bank runs are discouraged and bank failures reduced. In addition, insofar as deposit insurance reduces the risk premium demanded by borrowers, it provides an indirect subsidy to banks by making their cost of funding cheaper. Under most schemes, however, banks pay for the coverage although it may be administered by a government agency, thereby reducing the amount of subsidy. Deposit insurance programs have also become increasingly popular since first implemented on a wide scale in the 1930s.

Where no program exists, bank runs are more likely to occur. For this reason, when needed, a number of countries introduce blanket guarantees of depositors' funds in an attempt to restore stability to the sector. Guarantees were put in place in many countries in the wake of the 2007–2012 crisis.⁶³ Even where a deposit insurance program does exist, it may be prone to delays in repayment, and the objective of

maintaining depositor confidence may not be met.⁶⁴ This was the situation in the Northern Rock bank run of 2007. Under the applicable deposit insurance program, depositors would not only be subject to delays in repayment, they would also incur a loss of 5 percent of their deposits in the event that the bank failed completely.

Critics of deposit insurance programs assert that they increase moral hazard resulting in bankers taking greater risks than they would otherwise (the U.S. Savings & Loan crisis of the 1980s is often pointed to as an example) and that they diminish market discipline by reducing the incentive of market participants to monitor bank strength. If neither banks nor their depositors worry too much about depositors being repaid, then bank managers are apt to take greater chances with depositors' money than they might otherwise do.⁶⁵

The findings of empirical research are not all on one side, however. Two scholars of emerging market bank crises concluded after running "10,000 regressions," that the "evidence that deposit insurance, by weakening market discipline, heightens crisis risk in emerging markets is of questionable robustness." They contended that by solving the problem of bank runs deposit insurance reduced risks as much as it encouraged crises by increasing moral hazard.⁶⁶ A perhaps less contentious proposition is that the proliferation of deposit insurance has increased the severity and cost of banking crises that do occur by promoting extended two-phase crises in which during the first, or silent, phase the absence of depositor scrutiny that insurance allows enables bad loans to build up to extraordinarily high levels.⁶⁷

Another angle worth considering is that deposit insurance schemes distort competition since they unduly benefit the weaker and less creditworthy banks while, all along, the cost of the deposit insurance is disproportionally shouldered by the safe banks or by the average taxpayer.

DEPOSIT INSURANCE

A formal national deposit insurance program was established in the United States in 1934. Worldwide growth of such schemes was gradual. By 1964, six other countries adopted deposit insurance programs; by 1984, there were about 20 such systems in place, and by 2000, about 70.⁶⁸ As of September 2005, there were about 80 deposit insurance programs in operation, and in the same month Malaysia announced the initiation of one.⁶⁹ Those governments that did not have deposit insurance systems have not always left ordinary depositors in the lurch. For example, Thailand did not have one in place during and after the Asian crisis. Nonetheless the government promised to make retail depositors whole in respect to banks that have failed. Following the Northern Rock bank run in the UK in 2007, the government took a similar step providing a guarantee to depositors to compensate for a deposit insurance scheme perceived by the public as deficient. Europe started its schemes in 1994, covering modest amounts that most countries had to increase in 2008 or 2009 in the wake of the financial crisis. Some of those systems would not have the financial ability to meet casualties in a situation where a large portion of the banking sector would fail, and as a result, some governments added their unlimited guarantee—for what such guarantee would be worth, of course.

Coverage limits on deposit insurance schemes as well as regulations vary globally. In Canada, the deposits in most financial institutions are covered up to C\$100,000, but not all types of accounts, and only those in Canadian dollars.⁷⁰ In Brazil, the Fundo Garantidor de Creditos⁷¹ insures up to Reais 70,000 only (worth about US\$34,000), but it covers each affiliated bank separately. Deposit insurance programs may be privately funded and operated, or run by the government, voluntary or mandatory. Premiums are ordinarily paid for by the insured banks, calculated as a percentage of total deposits. A number of countries have no deposit insurance program, while in others, especially in emerging markets, the government implicitly or explicitly guarantees bank deposits.

Concededly, there is a possibility that deposit insurance programs, and particularly those that are unduly generous, may have long-term damaging effects. Irrespective of the debate among scholars, governments often reckon that the costs associated with the potential liability of having to pay out on claims is exceeded by the benefits that accrue to the banking industry and to the economy as a whole. Since it is also the bank regulator, the government is in a position to reduce risks by imposing strict regulations and acting quickly to deal with troubled banks. If the insurance scheme is designed in a prudent manner with sensible account and claim limits accompanied by rigorous checks on the banks by regulators, then much of the harmful impact of deposit insurance schemes can be precluded. An understanding of the features of a deposit insurance program can help in determining whether it contributes to moral hazard.

The mainstream view, therefore, is that the benefits of deposit insurance outweigh its risks. Deposit insurance programs are generally viewed to reduce systemic risk and to promote financial system and economic stability. For these reasons, most bank credit analysts will tend to view the existence of deposit insurance as a positive indicator in evaluating a banking system. In analyzing the impact of the existence or nonexistence of a deposit insurance program, some relevant questions include the following:

- Is there a deposit insurance scheme in place? If not, does the government implicitly or explicitly guarantee bank deposits?
- If there is a formal deposit insurance scheme, is it well-designed?
- Who runs the program? A government agency or corporation, a private entity or public-private partnership?
- Is the program funded by the government, the banks, or both? Is the program mandatory? Is it adequately funded?
- What is the ratio of funds to bank assets?
- Is there a limit on payouts? If so, what is it? Is there a limit on a per person rather than on a per account basis?
- Does the deposit insurance agency have a role in bank supervision?
- In the event of imminent or actual bank failure, does the agency have the legal authority or right to gain access to bank assets to mitigate its losses?

- Does the agency have the capacity to take legal action against bank directors or officers?
- How long has the agency been in operation? What is the agency's track record in respect to paying out claims?
- Are claims paid in a timely manner thereby furthering the objectives of the program?

THE STRUCTURE AND STRENGTH OF THE REGULATORY APPARATUS

Until now, our focus has been on substantive regulations, and the impact of such regulations upon a banking system. A full assessment of a regulatory framework will also consider the strength of the institutions charged with regulating the banking industry, including their capacity to enforce their directives. The purpose of this section is to identify relevant criteria to be considered assessing the quality of bank regulation, supervision, and enforcement.

Structure of the Regulatory Regime

Separate from the structure of the financial system as a whole, the structure of the regulatory agencies that oversee it differs from one country to another. Certain common patterns may be distinguished. One is between those markets in which there is a single bank regulator, and those in which responsibility is shared among a number of agencies. The more traditional approach was for each subsector of the financial services industry to be handled by a separate regulator, with banking ordinarily coming under the authority of the central bank.

The United States is an example of a country where there is a high degree of decentralization and specialization in bank supervision. Although the central bank, the Federal Reserve Bank, has primary responsibility at the national level for banking regulation, a number of different institutions both at the state and federal levels have substantial authority over the depository institutions that come within their domain. At the national level, the Securities and Exchange Commission has primary authority over the securities markets, but this structure is replicated in a number of states.⁷²

Over the past several decades, the trend in financial system regulation has been towards creating an integrated albeit specialized supervisory agency leaving the central bank with responsibility for monetary policy. For example, just between 1985 and 2004, 29 countries took a similar path creating integrated financial supervisory agencies.⁷³ The emergence of the trend is attributable to greater integration in the industry "as the dividing lines between differing kinds of financial institutions become increasingly fuzzy."⁷⁴ The underlying rationale is that as boundaries between traditional categories of financial services blur and individual institutions engage in an array of businesses that transcend these boundaries, it is sensible to have one agency overseeing the entire industry.

In the UK, for example, bank supervision was in 1997 transferred to a unified financial supervisor, the Financial Services Authority, which took over this role from the Bank of England, with the latter continuing to have a separate role in monetary policy. Not only is the FSA sole bank regulator, but it is the sole regulator of all financial institutions operating in the UK. Prior to this transfer, the UK followed the

more traditional approach in which the central bank had primary responsibility for monetary policy and bank supervision.

Despite this trend towards an integrated regulatory and supervisory financial services agency, there is a range of views as to whether a unified regulator is better than the more traditional approach. One view is that specialized agencies will have greater expertise in the industry they regulate. Another is that it depends on how the integrated agency is organized, how it coordinates with the central bank, what level of authority it has, and so on. Still another view argues that which is best depends upon the state of financial development of the country. From this last perspective, it is thought that an integrated financial services authority may make more sense in more developed countries with an established culture of regulatory independence. It may work less well, however, in developing markets, where a greater risk of both banking and currency crises makes the traditional approach a better alternative. It is also asserted that in such markets the established role of the central bank will tend to shore up the independence of supervisors, ensure sufficient funding, and provide a center of professional competence.⁷⁵

Supervision and Enforcement

No matter how the banking regulatory apparatus is structured, supervision and enforcement are usually accomplished through similar measures. These include:

- Compulsory periodic reporting by the bank to the regulator
- Ongoing monitoring and review of disclosed information
- Regular on-site bank examinations
- Consultative discussion between the regulator and the bank
- Supervisory orders or penalties imposed upon the bank
- The enforcement of the regulator's orders administratively or through the judicial process by calling upon the police power of the state to impose sanctions

Beyond informal consultative guidance, supervisory orders typically require that a bank make changes in operations or pay fines as penalties for noncompliance. Illustrative changes in operations that a regulator might demand include mandating that a bank limit its lending in certain sectors or improve its risk management procedures as specified by the regulator. Similarly, required changes may incorporate meeting specified capital or liquidity thresholds. Where management has been noncompliant or the bank is in distress, sanctions may be imposed. These may range from fines to compelling changes in management or ownership (including nationalization), or even closing the bank. In situations where fraudulent or criminal activity is found, similar actions may be taken and criminal prosecution undertaken.

Competence, Independence, and Authority

Just like evaluating management quality, evaluating the quality of bank regulation is a challenging and largely subjective process. First, the defects in bank supervision, by their nature, will tend not to be willingly revealed, or may not be apparent until a crisis strikes. Second, as the examples below suggest, surface appearances may belie

significant deficiencies in the regulatory agencies' capacity to function effectively. A regulatory scheme that looks good on paper may not shine in practice.

While acknowledging the difficulty in appraising the agency or agencies charged with responsibility for supervising a country's banking system, it remains an important element of a comprehensive assessment of the systemic risk in a given country. Some critical issues to bear in mind when attempting to appraise the competence and effectiveness of a regulator are addressed next.

A significant consideration is the competence of the regulatory agency, which naturally rests on the competence of the staff members who comprise it. Competence is also affected by the resources available to the agency. In some jurisdictions, bank supervisory agencies are inadequately staffed and overworked, and enforcement is lax. In others, inadequate resources, including insufficient training and access to IT resources, may hinder bank examiners in doing their job effectively. Poorly paid examiners may suffer from low morale or, in more extreme cases, be vulnerable to corruption.

Lack of resources will tend to impair the independence of the agency, an attribute that in sufficient measure is critical to its success. Independence, of course, also rests upon the regulator's charter and upon the clarity of its delegated authority to set regulations and issue supervisory orders. Similarly, its grant of authority, as well as the political, legal, and judicial framework within which it operates, are critical to its success and will influence its ability to enforce its rules and orders.

In some countries, however, regulatory agencies do not possess the preceding attributes to a lesser or greater degree. For example, the bank supervisory agency may not have the resources to hire adequate numbers of professional staff to competently examine the banks under its charge. Likewise, it may lack the authority to take appropriate actions to remedy the problems it confronts vis-à-vis problem banks. Alternatively, the regulator may in theory have the authority to impose sanctions on errant banks, but cannot effectively wield that authority in practice. For example, this may occur because the enforcement machinery, whether effected through the courts or administrative commissions, is creaky and unable to act on a timely basis. Alternatively, it may occur because the owners of the errant banks have the political clout to deflect sanctions that the agency attempts to impose, thereby rendering the agency dependent upon other nodes of authority within the government.

Yet another potential problem that can potentially compromise an agency's authority and independence in some countries is the vulnerability of bank examiners to personal lawsuits or other retribution from bank owners who believe that they are being unfairly targeted.⁷⁶ Such litigation can have a chilling effect on the willingness of bank examiners to thoroughly analyze the banks they are supervising.

As suggested, there are three principal criteria that ought to be taken into account when judging the quality of regulation: the regulator's competence, independence, and, authority. Competence, however, is quite difficult to evaluate on a systematic basis. The outside analyst will rarely be privy to the reports issued by the bank examiner, and, except perhaps after a banking crisis has occurred, there will be little firm evidence upon which to base a judgment. Nevertheless, it may be possible to form an impression based upon informal discussions and reading between the lines of reports on regulatory agency actions. Authority and independence, however, are more amenable to a fact-based evaluation and are therefore elaborated upon below.

Authority

The authority granted to regulators varies considerably among countries. The naïve analyst might assume that regulators, being the authorities, must have authority. However, the authority of the regulator over the banks it supervises is rarely absolute, and in some countries may, as a matter of practice, be more constrained than one might imagine. In their paper for the World Bank, *Bank Regulation and Supervision: What Works Best*, J. R. Barth, Gerald Caprio Jr., and Ross Levine survey bank regulations around the world. Interestingly, views are mixed concerning the benefits of the bank regulator having strong powers. The authors of the aforementioned study came to the conclusion that “greater supervisory power [was] positively associated with government corruption,” although it was beneficial when the political openness was “particularly advanced.”⁷⁷

Among the survey questions are the following, which are paraphrased from the original text:⁷⁸

In regard to external auditors:

- Do regulators have the right to meet with external auditors to discuss the bank’s financial statements without the consent of bank management?
- Must external auditors report directly to the regulator any material evidence that bank directors or officers or senior managers are involved in illicit activities?
- Can the regulator initiate legal proceedings against external auditors for negligence?

In regard to the authority of the regulator:

- Does the regulator have the power to compel the bank to change its internal organizational structure?
- Can the regulator compel management to set aside additional provisions to cover actual or potential losses?
- Is the regulator empowered to suspend the director’s decision to issue dividends and/or bonuses and/or management fees?
- Does the regulator have the authority to declare a bank insolvent?
- In the event it is necessary to protect depositors, does the regulator have the authority to supersede shareholders’ rights, to replace management, and/or to replace directors?
- Under law or regulation, are there specific provisions that mandate automatic intervention, such as “prompt corrective action,” should a specified deterioration in bank solvency occur?
- Does the regulator have the power to initiate civil or criminal legal proceedings against the bank’s directors and officers?
- What level of discretion does the regulator have in forbearing from applying certain prudential regulations?

Independence

For bank regulation to be effective, the regulatory agency must be able to make decisions and act without undue interference from outside or special interests, including politicians, other government bodies, and those organizations and

individuals subject to regulation.⁷⁹ At the same time, the regulatory agency's power should not be absolute, but instead be subject to some sort of judicial or administrative oversight in order to safeguard against abuses of power. Ideally, the enabling laws and regulations that define the scope of the agency's authority and accountability will strike a proper balance in this regard.⁸⁰

Relevant questions to answer as part of an evaluation of regulatory independence include the following:

- What degree of independence is granted to the regulatory agency under law? To whom does it report?
- Who appoints the head of the agency? May she be removed arbitrarily (at will)?
- Are there institutional protections buttressing the agency's independence?
- Does the agency have budgetary authority?
- Are its actions subject to deference by the courts and other institutions?
- Are supervisors compensated adequately?
- What is the staff turnover rate? Do bank supervisors who leave the agency often seek or obtain employment in the banking industry?
- Are supervisors subject to litigation for their role as supervisors?

THE QUALITY OF THE LEGAL SYSTEM AND CREDITORS' RIGHTS

In any given jurisdiction, the existence of an effective legal system—one that is comparatively fair and consistent in application and that allows relatively speedy and cost-effective legal redress—stands as a fundamental precondition to the efficient and profitable operation of banks. Particularly critical to banking system health are the existence of insolvency and bankruptcy laws and related mechanisms for protecting creditors' rights. If those that borrow from a bank can renege on their debts with impunity, notwithstanding their ability to pay, then a financial institution can quickly be brought to its knees, and the consequences to the financial services industry as a whole can be grave.

Of course, when economic conditions are rosy, it is in the interest of all parties to play by the rules. When conditions turn sour, however, the very economic survival of a debtor may be at stake, and the debtor may be inclined to skirt its legal obligations to repay a loan, if the conditions permit. It is evident that without an effective legal system, the debtor's problems may rapidly become those of its bank. (Of course, if the obligor lacks the capacity to repay its debt, the same thing will happen.) If enough of the bank's customers take the same course, the bank may collapse, with knock-on effects throughout the financial system as a whole.

Accordingly, from the vantage point of the bank credit analyst, a country's financial system should be underpinned by a supportive legal framework that upholds creditors' rights, that is bolstered by a common belief in the sanctity of contracts, and that enables their effective enforcement. Such a framework—one that incorporates both suitable substantive rules and efficient enforcement mechanisms—is more likely to foster a healthy banking system and facilitate effective bank supervision.

Conversely, it is understandably difficult for a banking system to achieve first-rate levels of strength and stability amid a weak legal framework, as it will be unable

to equip banks with the creditors' rights that define their ability to earn their keep. As has been pointed out:

*If the legal system makes it difficult to seize or to transfer the collateral behind delinquent loans, or for debtors to pledge collateral for bank loans, or to adjudicate cases of corporate or individual bankruptcy, then both banks' credit losses and the cost of borrowing for firms will be (abnormally) high.*⁸¹

If, in a severe economic downturn, the legal infrastructure does not provide appropriate support to creditors' interests, it will not only aggravate the situation, possibly contributing to a banking crisis, but will also make recovery more difficult. More specifically, the existence of an insolvency law that can be utilized to protect creditors' rights in a timely manner is critical to the ability of a bank to enforce its claims under a relevant loan or security agreement. In a number of emerging markets, insolvency laws, though they may exist, are not infrequently archaic, little used, or difficult to enforce in practice. A dearth of judges with expertise in insolvency procedure, as well as a lack of specialized bankruptcy courts, may limit the ability of a bank to enjoy the benefit of legal process. A related barrier may be the policies and procedures embodied in extant law. Procedures may be cumbersome or biased in favor of debtors.

Like efficient and modern bankruptcy laws, simple and time-sensitive foreclosure procedures buttress the financial strength of banks. Procedures that permit foreclosure within three to six months are the international norm. In some markets, however, a borrower may have the right to redeem foreclosed assets for up to a year or more, or foreclosure procedures may be unduly cumbersome. While a judgment may be forthcoming, it may be difficult for a creditor to seize and dispose of the subject collateral.

During the Asian crisis—in September 1998—investment bank Warburg Dillon Read, observing that international best practice afforded creditors the ability to foreclose on loan security within three to six months on average, estimated the following average times for such proceedings in several Asian countries. The amount of time required varied anywhere from approximately three months in Singapore to as long as 5 to 10 years in Thailand. It also observed at the time that “the main issue is not whether an insolvency law exists, but whether it is practiced and enforceable.” Bankruptcy reforms in Thailand, previously noted, appear to have since reduced the time required considerably.

As was briefly mentioned in Chapter 1, a number of attempts have been made to measure the quality of country's legal system using various qualitative and quantitative criteria. The databases and indexes that have been developed have been primarily intended for economic research purposes, and for use in economic development organizations to gauge the climate for foreign direct investment. They may nevertheless be useful for credit assessment—more so perhaps for internal credit risk modeling and rating purposes than for day-to-day credit analysis—and are mentioned only in passing here.

An excellent source of information is the Worldwide Governance Indicators (WGI) project supported by the World Bank and provided by the Brookings Institution, the World Bank Development Research Group, and the World Bank Institute.⁸²

The WGI reports draw from a number of sources and they aggregate the results in six dimensions: voice and accountability, political stability and absence of violence, government effectiveness, regulatory quality, rule of law, and control of corruption.

The scope of the WGI goes beyond the strict “rule of law” angle, but the other components should, in particular in emerging markets, play an indirect role in the determination of the quality of the legal system in the context of the credit risk analysis.

The WGI goes back to 1996 and it is currently updated every year.

There are a variety of similar indices used to produce a composite index for credit risk modeling purposes.

GAUGING BANKING SYSTEM FRAGILITY

Banks will never be wholly safe—nor should they be.

—The Economist⁸³

As many indicators relevant to a bank’s strength were examined in some detail in earlier chapters, and also, in some cases, were referred to the discussion of prudential regulation earlier in this chapter, it is not necessary to repeat the discussion concerning their use and significance here.

In the following sections, we shall concentrate on the following banking system attributes: size; ownership characteristics; local credit culture and level of consumer and business confidence generally, and confidence in the banking system specifically; recent developments and anticipated government initiatives with reference to the sector over the near-to-medium term; and the state of local money and capital markets development.

Profiling the Banking System

The regulation of a banking system is, as suggested, a difficult exercise. If it is accepted that the regulators should adjust their work to fit the needs of the market they oversee, it is relevant to examine banking systems in general. Putting aside the items and other features mentioned in previous sections, the remaining attributes of banking systems that are amenable to quantification can be summarized in a table such as that shown in Exhibit 13.6.

The table in Exhibit 13.6, adapted from a J.P. Morgan report on the banking systems of central and eastern Europe, includes the following key criteria and divides them into six categories that are useful to describe a banking system, not only in the context of regulatory tools, but also as a basic shopping list for the important factor an external analyst should consider. The six categories include:

1. Size (relative to GDP)⁸⁴
2. Market structure
3. Foreign and state ownership
4. Concentration
5. Market penetration
6. Retail development

EXHIBIT 13.6 Metrics to Describe a Banking System: The Polish Banking System (1999 to 2001)

	1999	2000	2001
Size of the financial system relative to GDP	(Degree of intermediation)		
Banking assets/GDP (%)	57	60	63
Deposits/GDP (%)	37	38	40
Market structure	(Degree of consolidation)		
No. of banks	77	74	71
No. employees	149,638	144,237	138,824
No. of branches	10,222	11,245	10,509
Concentration	(Degree of consolidation)		
Market share top three banks in assets (%)	44	44	44
Market share top five banks in assets (%)	48	47	51
Banking penetration	(Related ratios)		
Banks per million of population	2.0	1.9	1.8
Banks per million of bankable population	2.9	2.8	2.7
Branch per population (per thousand)	3.8	3.4	3.7
Income reach per bank (\$ billion)	2.01	2.13	2.48
Fee income/GDP (%)	0.65	0.75	0.79
Fee income per capita (€)	25	33	36
Fee income per bankable population (€)	36	49	52
Noninterest income per bankable population (€)	175	237	263
Banking revenues per capita (€)	124	165	187
Bank revenues per bankable individual (€)	183	241	271
Banks revenues/GDP	3.28	3.73	3.71
Deposit per population (€)	1,408	1,702	2,044
Retail development			
Household debt/GDP (%)	5.5	6.6	7.1
Mortgages/GDP (%)	1	1.4	1.9
Consumer credit/gross wages (%)	4.3	5.1	5.2
Ownership structure			
Foreign ownership	47	70	69
State ownership	24	23	23

Source: Adapted from JPMorganChase & Co. and NBP (the Polish central bank) data.

These principal parameters are addressed in the remainder of this section albeit in a slightly different sequence from that in Exhibit 13.6.

Size

The size of a banking system is an important parameter in its evaluation. As with individual banks, the larger the industry, the more resilient it is likely to be to economic shocks. Size is usually measured by the total banking assets of the industry, aggregated from the balance sheets of its constituent institutions. (Size relative to GDP is an indicator of bank intermediation.) Variant indicators include total risk assets, total deposits, and total equity. The size of a banking system generally corresponds with the size of a country's economy and its level of economic development.

In the 2000s, though, and until the 2007–2012 crisis, the growth of bank balance sheets—especially in Europe—considerably weakened the traditional link between

the size of an economy and the size of its domestic banking system. Comparisons between countries over the years reveal substantial changes in the relative sizes of the banking sector, but it can be safely argued that those changes are no longer a good indicator of the degree to which a country's banking system has developed relative to the size of its economy. This used typically to be reflected by several *intermediation ratios*, one of which was introduced earlier in the context of economic analysis.⁸⁵ Recall that the ratio mentioned was calculated by dividing domestic bank assets by GDP.⁸⁶ It functions as a rough indicator (nowadays very rough) of economic development in a country, subject to adjustment for idiosyncratic local factors. Intermediation can also be measured through comparable metrics, such as deposits to GDP. Intermediation indicators generally reflect the extent to which formal financial intermediation has replaced informal finance networks.

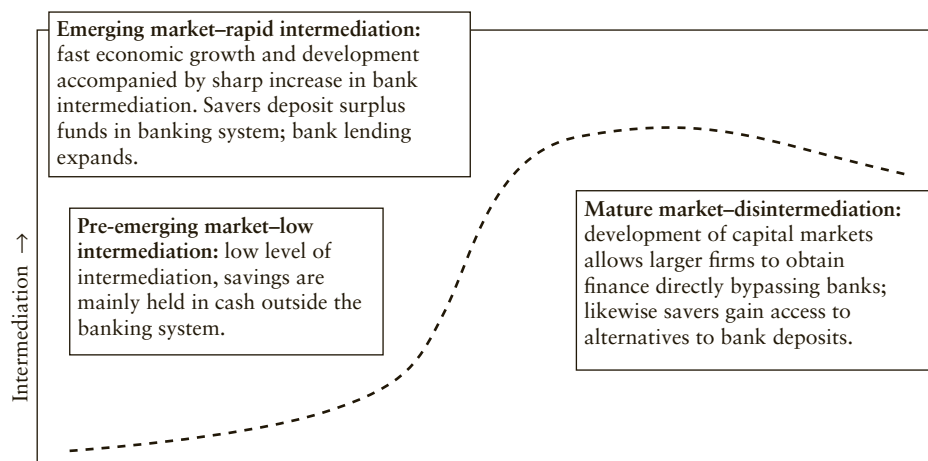
Less developed countries, such as Sierra Leone, Guatemala, and Moldova, characteristically exhibit relatively low levels of intermediation. The ratio for each country circa 2005 was in fact less than 50 percent, a fairly meager ratio value. As economies grow, intermediation expands, and bank debt as a percentage of GDP rises. At that time, the intermediation levels for four Southeast Asian countries, all of which might be considered to be emerging markets, ranged from under 70 percent in the Philippines and Indonesia to between 110 and 125 percent for Thailand and in excess of 150 percent for Malaysia.⁸⁷

As economies grew and became more diverse, intermediation tended to expand rapidly. As always, ratio values appear to peak in highly-developed countries in which the banking industry remains the predominant source of corporate finance. For example, the extraordinarily high respective intermediation ratios of 313 and 164 percent for Japan and Germany corresponded to the extensive development of the banking system, and the heavy reliance on bank finance in both countries.

At some stage, additional economic development was accompanied by developments in the capital markets that resulted in the emergence of alternatives to bank finance. Notably, the maturation of a strong domestic fixed-income market, in which institutional investors such as pension funds and insurance firms participate, gave companies the capacity to borrow directly from investors instead of using banks as intermediaries. The result was disintermediation in respect to banks, which, unsurprisingly, causes intermediation ratios to stabilize or decline. For these reasons, the intermediation ratio for the United States was only in the 60 to 70 percent range, a fairly low nominal level for an industrialized country. A prototypical pattern of intermediation followed by disintermediation is shown in Exhibit 13.7.

The American banking system in its traditional lending capacity plays a less important role in providing finance vis-à-vis the capital markets than it does in Japan and most European countries. In addition, significant amounts of bank assets are taken off-balance sheet through securitization and through the use of derivative instruments. As well, foreign institutions provide a significant amount of credit within the U.S. economy.⁸⁸

While risk management strategies prompted the growth of securitization and generally the use of derivative instruments, they also brought about, particularly in Europe, an increase in interbank and similar transactions. As a result, total assets remaining on bank balance sheets often grew without much relation to the actual growth of economic activity. This is the reason why intermediation indicators are nowadays less telling.

**EXHIBIT 13.7** Intermediation and Disintermediation**Concentration**

Bulgaria's economy in 2011 was just about 1.5 percent the size of Germany's. Bolivia's economy was about 1 percent the size of Brazil's economy. California's economy was about the same size as India's economy. Talking about the relative size of banking systems is probably a bit futile. Nevertheless, looking at bank concentration is more interesting. In some countries, one or two banks completely dominate the market, while in others the sector is much more fragmented. Even in highly decentralized systems, however, there are usually a few outsized institutions that comprise the sector's top tier. Perhaps the most common situation is to have a few top tier banks, 5 to 10 second-tier banks, and the remainder of the commercial banking sector comprised of smaller institutions. Because the condition of the largest banks within a given banking sector greatly influences the strength of the sector as a whole, it is important to be aware of the identity and financial condition of the largest banks in the sector.

Concentration—sometimes called *consolidation*—refers to the extent to which banking services are provided by a relatively small number of large institutions, or through a larger number of smaller institutions. A banking system in which there are a large number of small institutions is sometimes said to be *overbanked*. The implication is that a large number of small institutions will generate an intensely competitive environment resulting in a substantial number of marginal players. In such an environment, the argument goes, a substantial number of banks, unable to tap economies of scale, will struggle to earn mediocre profits and generate adequate capital. Some will be motivated to take excessive risks thereby potentially placing the industry as a whole in peril, while others will suffer from chronic weakness and long-term capital erosion. All but the largest institutions will find it hard to compete with foreign entrants or on a cross-border basis. The need to supervise a large number of banks also makes regulation more problematic. For these reasons, a good number of regulators and policy makers takes the view that a banking system benefits from a relatively small number of strong institutions, albeit not so few that there is insufficient competition. The rationale, in short, is that it is better to have banks with the critical mass that will enable them to exploit economies of scale and achieve a modicum of stability.

The level of concentration in a banking system can be measured by the degree to which most banking assets are concentrated in the largest, for instance, top 3, 5, or 10 banks, or how many of the largest banks have aggregated total assets exceeding a certain percentage of the total assets of the domestic banking sector, for example 70 percent. Such indicators vary widely around the world, although there is an evident global trend towards greater consolidation in the banking industry, which tends to be associated with higher efficiency and more stable financial institutions. In the majority of countries, the percentage of assets in the three largest banks is between 35 and 70 percent, and, in many, fewer than 10 banks hold the vast majority of all banking system assets.

There are a number of significant exceptions, however, to these averages. For historical reasons and because of its federal structure, the United States has an extremely large number of banking institutions—over 8,000—and Germany, which like the United States has a federal government structure, has over 2,500. Despite their large populations and the somewhat dissimilar role banks play in each country, both have among the highest number of banks per capita in the world, and are also among the least concentrated systems.

Bank concentration issues recently took a fresh dimension when the international community, through the Financial Stability Board (FSB), established in 2009, decided to single out about 30 *systematically important financial institutions* (roughly half from Europe and half from the United States and Asia) for special scrutiny, with the view of preventing their failure, on the basis that they were too big to fail, or more precisely that a failure would have systemic repercussions. Quite obviously, their absolute size was a relevant factor in their selection, and there is some link between absolute size and concentration.

A jurisdiction having the population of large metropolis occupying a comparatively small geographic region, Hong Kong is another example of a banking system that long had a large number of banks relative to its size. The reasons are largely historical, being rooted in its history as a British colony and later as a trade center, a stepping-stone to Mainland China, and as a financial center. Foreign participation is substantial, and there is also a significant number of small family-owned or controlled institutions. Nevertheless, two banking groups control the bulk of banking system assets: the Bank of China, based on the mainland, and HSBC, which also owns Hang Seng Bank. The largest independent Hong Kong-based bank, Bank of East Asia, only had a market share of 3 percent of the total.

Types of Banks and Financial Institutions

An understanding of the particular categories of financial institutions operating in a given financial system enables the analyst to be aware of the degree of competition in the sector and the constraints under which commercial banks operate. While commercial banks exist in almost every country, there is a number of variant types of depository institutions as well as nonbank financial institutions that often exist alongside traditional commercial banks. Some are idiosyncratic to a particular country, such as German *Landesbanken* or Spanish *Cajas*, while others are of a more or less standard type. *Savings and loans* in the United States resemble *UK building societies* in many respects, for example, and such housing finance-oriented institutions have counterparts in a number of countries around the world.

Sometimes a particular type of financial institution may have a strong competitive advantage owing to special dispensations carved into law, to favorable regulatory treatment, or to some other reason making it more difficult for other types of banks to compete. For instance, a particular type of financial institutions may be loosely regulated. While the loosely regulated institutions may gain a short-term competitive advantage, such a situation may potentially trigger problems in the banking system as a whole. In Thailand, for instance, weak finance companies were early victims of the collapse of the bubble during the 1997 Asian crisis and set the stage for the difficulties that commercial banks subsequently experienced.

In general, banking organizations can be classified into one of the following seven types.⁸⁹

1. *Commercial banks* (private-sector and state-owned) are depository institutions that provide loans to companies and consumers, while offering a range of ancillary services.
2. *Investment banks* underwrite securities, provide merger and acquisition advice, and invest in private equity transactions among other activities. They are generally not funded through customer deposits but instead through the money and capital markets.
3. *Merchant banks* are traditionally more trade-oriented than investment banks. They are involved in trade finance (in particular in commodities), project finance, and syndicated loans. They underwrite securities, and they have private placement activities.
4. *Housing finance institutions* specialize in providing mortgage finance to individuals and households. They include depository institutions, such as U.S. savings and loans and UK building societies, as well as nondepository institutions.
5. *Universal banks* are commercial banks that offer a wide range of other financial services, including not only those offered by other types of banks such as investment banking, but also services traditionally offered by nonbank financial institutions such as securities brokerage, asset management firms, and insurers. (When insurance is provided by a bank, it is often referred to as *bancassurance*, a word coined in Europe when banks and insurance companies started merging.)
6. *Specialist banking institutions* are institutions established with a specific, often somewhat narrow, specialist purpose such as providing funding to cooperative banks, promoting export finance, and similar, usually government designated functions.
7. *Development banks* are policy-based institutions intended to finance various types of government-approved development institutions.

Money and Capital Markets Development and Other Factors

If I [could] turn back the clock and have a wish [list] . . . high in its ranking would be a well-functioning Thai baht bond market.

—The Governor of the Bank of Thailand

The state of development of a country's money and capital markets can have a significant impact on the strength of a banking system. These markets provide new

sources of funding for banks, and may provide both new lines of business and vehicles to hedge risk. At the same time, they also herald increased competition in the provision of finance and funding sources. Where the development of money and capital markets is embryonic, banks shoulder substantially all the burden of financing development, potentially straining their capacity. Where money and capital markets development is far advanced, banks may be compelled to radically change their business models to survive.

Generally speaking, the existence of active domestic capital markets is a positive credit signal, but much depends on how developed and stable those markets are, and what linkages exist between banks and the securities markets. Relevant questions include:

- Is there an active domestic equity market? A domestic money market? A domestic bond market?
- Are local institutional investors (e.g., insurance companies, pension funds) active in these markets?
- How deep and liquid are the local markets? How well managed are they?
- Is there a framework of rules or regulations that foster equitable treatment of market participants?
- How strong is the legal and regulatory infrastructure governing domestic securities markets?
- What are the local corporate governance and disclosure norms?
- Are there local rating agencies active in the market?

As money and capital markets develop, they tend to reinforce the banking system by providing new sources of funding and capital. If poorly supervised, however, they may have a detrimental impact. For example, a securities market in which unethical or fraudulent activity is rampant will be more apt to weaken rather than strengthen the financial system as a whole. Similarly, where banks engage in, or facilitate, securities lending, they may promote speculative activity that heightens systemic risk. In addition, as alluded to earlier, at a certain stage of economic development, disintermediation is likely to increase, presenting banks with a challenge to their existing business models.

It goes beyond the scope of this book to analyze money markets and capital markets. The relevance to the credit analysis of banks is that it has been plausibly asserted that countries in which the growth of the banking industry has outpaced the development of the money and capital markets become more vulnerable to banking crises. A robust and liquid local bond market is increasingly seen as essential to provide a developing economy with the resilience to absorb shocks. In contrast, where the entire burden of providing finance to companies rests upon the banking sector, the economy of the relevant market is more vulnerable to a collapse.

Credit Culture and Depositor Confidence

It has been observed that all business operates on trust. Hence, the existence and state of the local credit culture, though intangible, contributes to a robust banking system. The term *credit culture* is used here to refer to prevailing local attitudes toward

financial obligations that underpin and inform a more formal system of legally defined creditors' rights. The term also refers to bankers' attitudes towards credit risk.⁹⁰ Both attributes affect the health of a banking system. Phrased as questions, they are:

Do borrowers take their financial obligations seriously and recognize the consequences of default?

Do bankers take proper credit considerations into account when making loans, or are relationships the prime criterion?

The historical role of banking and commerce within the country will affect these norms. Countries making a transition from a command economy, where credit was dispensed as a subsidy, to a commercial economy tend to experience challenges in this regard.⁹¹ The level of depositor confidence in banks is another critical factor that affects the strength of a banking system. In emerging markets especially, banks may be distrusted, perhaps due to previous failures, and savers may as a result keep their nest eggs under the mattress. As noted previously, the existence of reliable deposit insurance can mitigate such concerns.⁹²

Ownership Characteristics

Another characteristic of banking systems concerns the ownership of individual banks and, especially, the degree to which the state itself is a player in the banking industry. In connection with a banking system review, the concern is not only with the way in which banks are managed, effectively or not, but also with the level of support likely to be provided by their owners in the event of financial problems.⁹³

There are a variety of possible ownership arrangements based on the following distinctions: privately owned vs. state-owned, national government vs. subnational-government-owned, publicly listed vs. nonlisted, shareholder-owned vs. depositor-owned, foreign-owned vs. domestically owned, part of a group vs. independent, part of a financial group vs. part of a nonfinancial group.

The type of institution and the form of ownership are sometimes related. For instance, in the UK, traditionally building societies were mutually owned, that is owned by their depositors. During the 1990s, a number of building societies transformed themselves into more conventional shareholder-owned banks.

Family and Group Ownership

In developing markets particularly, a large portion of the banking sector may be in the hands of a family group, a closely held company, or a noncommercial enterprise. Family ownership can be a source of strength for the bank as it may entail robust backing for an enterprise in which the owners have considerable equity at risk. Perhaps more often, however, it is seen as credit-negative. The reason is the risk of governance problems. As an illustration, a bank's family owners might effectively divert depositors' funds into the financing of the family's other investments, thereby treating the depositors' funds as their own personal piggybank. For this reason, family ownership, in which a close shareholding group controls the majority of the shares of a banking institution, is proscribed in some countries.

Another potentially problematic ownership scenario is the financial institution owned or controlled by a larger nonfinancial corporate group. This situation can give

rise to a clash between the interests of the shareholders and those of the depositors in the group-controlled bank, as well as between the interests of the former and the bank's creditors. The impact of a conflict of this nature can be exacerbated where the regulatory environment is lax, or where the larger group is politically well connected. In mitigation of the risks in these cases, there may be implicit government backing for bank depositors, and possibly creditors as well. Nevertheless, such noncommercial dynamics are rightly viewed as generally credit negative since they usually increase risk and uncertainty. In the end, the significance of either family or industrial group ownership of commercial banks depends upon the particular business and regulatory environment as well as upon the quality of governance of the subject institutions. In this regard, it should be noted that ownership or control may be effected through nominee companies and obscured from public view and as a result, especially where the banks are not listed on the stock exchange, information may be difficult for bank credit analysts to obtain.

Foreign Ownership

Another material attribute of banking systems is the extent to which foreign-owned institutions form a significant part of the industry. Foreign participation in a country's banking system can be a sign not just of open markets, but an overall positive influence due to the technology and training they can bring to the sector. Concerning the percentage of bank assets owned by foreign banks, there is a wide range of participation from a negligible percentage in countries such as Mexico, Finland, and Japan to more than 90 percent in markets such as Luxembourg, New Zealand, or Gambia. Those jurisdictions where more than half of banking system assets are held by foreign banks tend to be relatively small countries (Paraguay), jurisdictions that have membership in or a close affiliation to a larger political unit (Macau to China, Guernsey to the UK, Estonia to the Nordic countries) or themselves function as global financial centers (Hong Kong, the United Kingdom).

The form of foreign ownership may be significant. A foreign branch of a bank based in another country may be restricted from engaging in a range of banking activities open to domestic institutions. Where a foreign bank acquires a domestic bank, or establishes a separate domestic subsidiary, it is more likely that it will be able to operate under essentially the same terms as local players.

It is important for the analyst to remember that the branch of a foreign bank—let alone a subsidiary—is at the mercy of changes in local regulations.⁹⁴ A branch in a foreign country is an extension of the foreign bank, and its banking activities are built on the parent bank's capital adequacy, something that is not open to a subsidiary, which as a legal entity in the foreign country is constrained by the level of its own local equity. If local circumstances prevent the branch from paying out funds to depositors or creditors, they can exercise a recourse against the parent bank. This is why, occasionally, some banks *ring fence* their branches in respect of some transactions, thereby isolating the parent bank from being exposed to the vagaries of local political decisions.

State Ownership

Perhaps surprisingly to those whose experience of banks has been in countries where the banking sector is, or has been, largely in the hands of the private sector,

governments around the world are major owners of banks. Although a number of nations have essentially no government-owned banks at all, including Canada, Denmark, South Africa, the UK, and the United States, to name several examples—ignoring of course the nationalization or quasi-nationalization of banks during the 2007–2012 crisis—government ownership of banks is quite common worldwide, particularly in developing countries.

In a number of countries, the national government owns more than 35 percent of the country's 10 largest banks. State ownership of banks in relation to the entire banking sector usually is less than one third of total local banking assets, but even in the 10-to-35-percent range the government's presence in the sector can have a major influence on the industry. Moreover, state-owned banks are often among the largest banks in the country, and thereby may play a part disproportionate to their industry market share. Among the most developed countries in the world, Germany was notable for the major role played by government-owned banks, which held over 39 percent of all banking system assets as late as 1999.

Even where the state does not possess an ownership interest in a bank, it may have a large influence on banking operations through formal regulation and supervision, or through more informal "administrative guidance" whereby it effectively directs banks to undertake particular activities. It is also not uncommon for other government agencies or enterprises, such as the postal service, to function as a financial institution and to perform many functions similar to that of banks, such as taking deposits. Also, in some countries, Spain for example, banking institutions owned in significant part by regional governments may have a significant impact on the banking industry. The relevance and evaluation of state ownership and support are discussed further in the next section.

STATE OWNERSHIP AND STATE SUPPORT OF BANKS

While there may be an occasional exception, governments are usually bad bankers.

—Morris Goldstein⁹⁵

Although the global trend is towards privatization, state ownership of the banking sector is substantial in many markets around the world, including Germany, Brazil, India, and China. State ownership may take a number of forms, and the choice of structure may have different origins and motivations. These include:

- Direct and 100 percent national government ownership of banks
- Majority national government ownership
- Minority government ownership

Where not all shares are owned by the government, they may be in the hands of private investors, or they may be effectively held by it through state agencies such as

state pension funds, sub-national government units, party organizations, or other “safe” institutional hands.

Rationale for State Ownership

While the original reasons for state ownership of the banking system are diverse, the principal rationale for state ownership within a market economy is to compensate for market failures. Commercial banks may justifiably be unwilling to provide credit to sectors of a country’s economy that the state deems to be in need of assistance. Examples include the provision of the credit to farmers, fishermen, and those in economically depressed areas that the commercial banking sector is unable to reach. The risks of lending in these sectors relative to the potential return and the high transaction costs relative to the volume of funds advanced make them commercially unviable.

Many state-owned banks were originally founded with a strong developmental mission. If not to finance development that could not be funded commercially, they were typically set up to facilitate foreign trade, particularly exports, provide long-term finance to industry, or merely add critical mass to a nascent banking system. Over time, however, as their development objectives have been reached, and the economies of their countries have become more diverse, many state-owned banks have evolved into institutions that are otherwise almost indistinguishable in their operations from private sector commercial banks. On the whole, state ownership of banks appears to be most popular in emerging markets.⁹⁶

The chief criticism of state ownership is that it detracts from economic efficiency at best, and at worst uses banks inappropriately for aims that would be better accomplished through direct subsidies. While perhaps justified initially, the tendency of organizations to perpetuate themselves has led to large government banks that have outlived their original useful purpose, and merely replicate functions that could be better provided through the private sector. Other criticisms are that state ownership of banks:

- Encourages corruption and cronyism resulting in credit being channeled to politically connected enterprises
- That in so doing it erodes incentives to develop keen commercial banking skills
- That it crowds out private sector banks, which might function better as engines of economic expansion

Empirical research suggests that countries where government ownership of the banking sector is high do lag in economic growth and experience slower rates of development in their financial systems.⁹⁷

Note that the provision of short-term credit to banks in the ordinary course of business is not generally viewed as government support, insofar as it is provided on a commercial basis and subject to collateral requirements. The government earns a return on its funds, while the banks benefit through a stronger funding profile. Such arrangements to make funds available to banks through the country’s central bank outside of distress situations are quite common and can be viewed as a characteristic feature of modern banking.

Significance of State Ownership

From a credit analysis perspective, the significance of state ownership is primarily the implicit or explicit expectation of a degree of support in the event that the bank runs into problems. Where a bank is wholly owned by the state, there is, generally speaking, a strong implication that the government will provide financial support should the institution experience financial distress. Hence, a wholly national government-owned bank is almost always assessable as sovereign risk. This is, of course, credit-positive for an individual bank, but the effect upon the banking system as a whole is likely to be less so as discussed in the next subsection.

Banks that are majority national government owned usually fit into this category as well, although less than whole ownership may be a step in an ongoing privatization plan. The terms of such divestment schemes are usually announced in advance and closely scrutinized by market participants. They may include provisions calling for government guarantees of the newly privatized entity for some transitional period, with the government remaining a significant minority stake in the institution to give confidence to the market.

Insofar as state-owned banks play a major role in the sector, the impact on the banking system will vary, however, depending upon the condition of government finances, economic conditions, and the competence with which such state-owned or state-controlled banks are managed. The belief that the government will support such banks will very likely afford them a funding advantage in relation to private banks, unless a strong deposit insurance scheme covers depositors in both types of institutions. This can have a correspondingly deleterious impact upon private sector banks within the country.

Distressed Banks and State Support

Aside from the question of state ownership is the matter of state support, which may be extended to both state-owned and privately owned banks alike, although it is usually considerably more likely to be forthcoming in the case of the former. In assessing the creditworthiness of a bank or banking system, the promise of government support is an extremely weighty factor. As with state-owned banks, such a promise converts the risk of a bank or bank system failure into a quasi-sovereign risk. When the promise of support exists, the relative importance of the stand-alone performance of a bank or banking system is diminished, provided of course that the state is creditworthy.

Although state support provides reassurance in assessing bank creditworthiness, it is not without costs. Indeed, it is apt to have a long-term negative impact on the banking system. First, the huge cost of fixing failed or failing banks is passed on to the country's taxpayers. Second, the impact of widespread bank distress and failures on the national economy can be expected to be extremely damaging. Finally, and perhaps of equal significance, the expectation of a bailout in the event of bank failure contributes, as we have observed, to moral hazard. Hence, the promise of support in the event of a crisis creates the perverse incentive that encourages excessive risk taking and makes a banking crisis more likely to occur. In an attempt to diminish moral hazard and avoid these adverse effects, regulators may employ "constructive ambiguity" to keep bankers in the dark as to whether they will receive assistance or not.

NATURE OF STATE SUPPORT

The nature of the support provided to a bank, whether state-owned or not, depends upon the condition of the bank and the banking system as a whole. From a credit analysis perspective, the promise of support in itself provides a bank with a number of major benefits. First, it gives the bank access to cheaper funding since the obligation is viewed as being essentially equivalent to sovereign risk. Second, it will tend to forestall a liquidity crisis since the promise of support will tend to give confidence to creditors, investors, and counterparties.

Aside from the normal short-term credit it provides to banks through the central bank, the state may in an emergency, relax its normal criteria and function as a lender of last resort, providing ample funds to a distressed institution with the objective of keeping it liquid and preventing its failure. Once a bank's critical liquidity needs have been relieved, the government may play a key role in rehabilitating the bank that it has assisted.

The amount of such support required will depend upon the severity of the bank's condition. For some banks, access to liquidity alone may be enough for it to quickly rebound to health. In more serious cases—for example, where a bank has been rendered totally insolvent—more intensive care may be necessary. Taking the sick bank under its wing, rehabilitation can mean the government's taking over bad loans that the bank has accumulated, injecting capital, and shopping the bank around to private investors. Further, state guarantees may be offered to prospective bidders as a sweetener. In extreme situations, one or more banks may be nationalized in an effort to provide the failed institution with a considerably cleaner balance sheet (i.e., fewer bad loans and more capital) so that it can be privatized later. Bank recapitalization and rehabilitation are discussed in more detail later in the book.

Be aware that state support is not necessarily a black or white proposition. While government-owned banks are likely to be rescued especially when owned by the national government—or perhaps by a local subsovereign—privately owned banks may be allowed to fail. Governments may come to the aid of some privately owned banks (e.g., those too big or too small to fail) and not to others. The level of support that may be forthcoming to privately owned banks is often very hard to ascertain in advance, and it is to some degree the same in respect of partly government-owned banks and those owned by state enterprises or by provincial or municipal governments.

While it is almost taken for granted that banks too big to fail would be supported, the notion of a bank too small to fail—and therefore worthy of support—relies on the idea that the cost involved to salvage it most of the time would exceed the social and economic cost of letting it collapse. Such social and economic costs are usually higher in emerging markets, but even in the United States very weak small banks often end up being swallowed by a stronger competitor.

In respect to the estimation of the probability and extent of support, the most problematic government-owned banks to analyze are those that are wholly or mainly owned by local governments or their agencies. The greater the distance from the national sovereign core, which may be defined as incorporating key national government institutions such as the treasury, central bank, and executive branch, the less likely that strong support will be provided. Provincial and local governments may wish to support local financial institutions that they wholly or partially own, but may be limited in their economic ability or legal capacity to support those institutions. Of course, in some instances, willingness to provide such support may also be lacking.

Consequently, whether such subnationally owned banks qualify as sovereign risk depends greatly on the particular division of power and authority between the national government and subnational units. In some countries, such as the United States, which has a decentralized federal structure, the credit rating of individual states (e.g., California) may be entirely separate from that of the national (federal) government, with there being little or no prospect that the national government will bail out California (and its creditors) in case of a default. In the same way, many U.S. states retain a high degree of autonomy in terms of setting taxes and in budgetary authority. In other countries, however, a highly centralized system of government is employed that allows minimal fiscal autonomy to local government units or that oversees their financial obligations. In such circumstances, local governments can be viewed as arms of the national government, at least from a credit perspective. The Netherlands is an example of such a country where municipalities and their agencies are effectively Netherlands's sovereign risk.

Gauging the Probability and Extent of State Support

For the analyst evaluating a particular bank, the question of state support is an important and unavoidable criterion that must be borne in mind. How can the probability and degree of state support for problem institutions be predicted? There is no easy answer, but the following questions and guidelines may be helpful.

- Has the government expressly promised assistance to troubled banks or their depositors?
- Is there an implicit or informal promise relayed through regulators or bank officials?
- What is the history of government support for problem institutions?
- Is there an element of government ownership in the subject bank(s)?
- What would be the likely political and economic of the failure of the subject bank or banks?

In summary, the probability that support will be forthcoming and the quality of that support can only be determined by weighing the particular circumstances and announced policies of the government carefully. Some rating agency reports and specialized ratings may attempt to provide guidance on this matter as well.

Exhibit 13.8 sums up some of the key questions to be considered in evaluating a banking system that were raised or alluded to in this part of the chapter.

EXHIBIT 13.8 Relevant Points to Consider in Understanding a Banking Industry and Evaluating Systemic Risk

Recent developments and outlook

- What, if any, are the notable characteristics of the banking system?
- How has the industry evolved in the past 10 years?
- What challenges has it faced?
- What recent developments have occurred?
- What is the outlook for the near-to-medium term?

Industry dimensions

- How large is the banking sector?
- How does it compare in size with neighboring countries?
- How many banks are there?
- What are the different types of financial institutions in the country, what role do they play, and what is their relative importance?
- To what extent has the industry been consolidating?

Regulation

- Who regulates and supervises the banking system?
- Is it a unified financial system regulator, or are banks regulated by the central bank or a specialist agency?
- What is the overall perception of the quality of regulation?
- What financial reforms, if any, have been undertaken?
- Are there any plans to initiate new financial system reforms?

Ownership

- What percentage of the financial sector is state-owned?
 - What role do government-owned financial institutions play in the sector?
 - Are there any plans for privatization?
 - What is the role of foreign banks?
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NOTES

1. Speaking before the Asia Society in Hong Kong, January 19, 1998.
2. Expecting a bank entirely to eliminate risks to depositors and creditors would be tantamount to forcing it to keep shareholders funds equivalent to its risk exposure or perhaps to its risk-weighted assets.
3. Of course, the behavior of wayward banks also conflicts with the government's objective of maintaining a sound banking system in order to facilitate the achievement of its economic policy goals.
4. There are various methods to accomplish this, from setting down arbitrary benchmarks that banks must meet to closely monitoring those banks identified as weak, and requiring them to make adjustments to their activities. These methods are discussed in a later section.
5. Given the importance of continued depositor confidence to banks' survival, it is not inaccurate to say that banking is a form of confidence game.
6. Sovereign risk, the subject of the preceding chapter, is occupied with a broad examination of political and economic risk factors with the goal of appraising a country's capacity and willingness to repay its financial obligations and evaluating the impact of related risks on the ability of local banks (and other enterprises) to meet their corresponding obligations. The health of a country's banking system is integral to the health of its economy as a whole. In particular, a banking system failure can have a devastating impact on an economy. Thus, not only does such sovereign risk influence systemic risk (i.e., the risk of systemic failure), but systemic risk influences sovereign risk.

7. Economic and political factors also affect systemic risk, but in view of their consideration in the sovereign context, and in view of the previous discussion in that connection, are given short shrift in this chapter.
8. It would not be unusual for a given institution to be involved in multiple transactions with peer banks at any one time. Such transactions could include interbank borrowing and lending, as well as the clearing of payments, providing of guarantees, and the buying and selling of financial instruments of various kinds to each other, including foreign exchange.
9. Note that when applied to a particular market, consideration of the corresponding criteria provides the basis upon which its systemic risk can be evaluated.
10. Recall that the operating environment risk variable in the analysis of an individual bank is based upon the level of sovereign risk and of systemic risk associated with the locale. This chapter, together with the chapter on sovereign risk, is meant to function as a survey of the operational environment component of the analytical calculus. For illustrative purposes, a discussion concerning the preparation of a banking system review is included, and, for the sake of brevity, references may be made to the impact of the various regulatory and systemic attributes on the banking system as a whole rather than on their respective consequential impacts on individual banks. These consequential impacts are understood and assumed.
11. E.g., adding a point if the regulator has promulgated appropriate restrictions on related-party lending and deducting a point for the absence of single-borrower limits.
12. James R. Barth, Gerard Caprio, Jr., and Ross Levine, "Bank Regulation and Supervision, What Works Best," *Journal of Financial Intermediation* 13 (2004), 206.
13. Barth et al., "Bank Regulation and Supervision," 2004, 207.
14. The full list of risks the banks are exposed to is, of course, longer.
15. E. Gerald Corrigan, "Are Banks Special?" 1982 annual report essay, Federal Reserve Bank of Minneapolis. In 2000, Mr. Corrigan, by then a managing director at Goldman Sachs, reviewed his 1982 essay and restated his support for its general conclusion: that banking should be treated separately from other forms of commerce. See E. Gerald Corrigan: "Are Banks Special? A Revisitation," Special Issue 2000, Federal Reserve Bank of Minneapolis. Note that the three points discussed paraphrase the points made in that essay and build upon them. The actual essay can be found online at www.minneapolisfed.org/publications_papers/pub_display.cfm?id=3527.
16. Put another way, this inherent mismatch between a bank's assets and its liabilities gives rise to both market risk and liquidity risk.
17. James R. Barth, Gerald Caprio, Jr. and Ross Levine, "Bank Regulation and Supervision, What Works Best," University of Minnesota, Carlson School of Management, World Bank Policy Research Working Paper No. 2725, August 2001, 4, citing Arthur C. Pigou, *The Economics of Welfare*, 4th ed. (London: MacMillan & Co., 1938). As Barth, Caprio, and Levine observe, there is a range of views as to whether regulation of banks enhances welfare, diminishes welfare ("governments implement regulations in a *grabbing-hand manner* that supports political constituencies" [emphasis in original]) or is neutral towards it ("stronger official regulation and supervision, while not necessarily or intentionally contributing to corruption, will not boost in performance and stability").
18. Of course, some market power that banks hold arises from their regulation. Whether or not banking tends towards a natural oligopoly or monopoly, there is no question that licensing restricts the new entrants into the banking market. In some countries, there is a moratorium on net banking licenses, frequently based on the rationale that a more consolidated banking system is a stronger banking system. In others, obtaining a new license can be a costly endeavor with no guarantee of success. As with public utilities, the quid pro quo for a grant of monopoly status is regulation in the public interest.
19. In addition to the illustration provided, the dynamic applies to the bank (as borrower) vis-à-vis its depositors (as lenders). The depositors see the bank's imposing edifice, the

dignified ambiance of the banking hall, and assume it will act as a prudent fiduciary of the depositor's funds. The reality, of course, may be somewhat different. Unless the depositors are in a position to function as credit rating analysts or, better yet, as auditors or as bank examiners, there will be a significant informational asymmetry between the two parties.

20. Couched as expertise in risk assessment, the same informational asymmetry contributes to a bank's capacity to earn a return on the funds it advances. Another example of an informational asymmetry occurs when a prospective borrower applies to a bank for a loan. The borrower presumably knows more about its own intentions concerning the loan proceeds than the bank.
21. For example, it can be argued that the fairness principle further validates a special regulatory regime for banks. In other words, the government should undertake to provide a basic level of protection to ordinary consumers who deposit funds with enterprises that hold themselves out to be banks from the loss of such funds through reckless business practices. A basic level of protection would include the imposition of a licensing scheme for banks, and the imposition of minimal standards of prudence on those institutions that receive a license.
22. Although financial turmoil may predate and thereby trigger a full-fledged banking crisis, once triggered, the worst damage follows. With banks no longer able to perform their vital quasi-public roles, the economy enters into a tailspin.
23. Systemic risk was already mentioned in Chapter 11.
24. In fact, the combination of a banking crisis and a financial crisis is a frequent occurrence. It is called a twin crisis. Twin crises will be discussed in Chapter 14.
25. This is Delhaise's argument in his book *Asia in Crisis: The Implosion of the Banking and Finance Systems* (Singapore: John Wiley & Sons, 1998), 223–229.
26. Tim Harford, *The Undercover Economist* (London: Little, Brown Book Group, 2006 & 2012), 130.
27. In other words, moral hazard occurs when the company owner who has fire insurance scrimps on fire safety to save costs or overloads electrical circuits by running machinery at a higher level than its rated capacity in order to increase production and earnings. If the factory burns down—not to worry—insurance will cover it.
28. It is not unusual for insurer to require the insured to bear a portion of the total loss in the form of a deductible amount.
29. These typically include those mentioned above—i.e., consumer protection or housing for low-income families—as well as objectives such as providing credit to farmers, encouraging urban or regional regeneration, encouraging exports, and promoting certain industries.
30. Thus, a restriction on engaging in real estate investment would be a structural regulation since it limits the scope of bank's activities; however, caps on real estate lending would constitute a prudential regulation since real estate lending is generally permitted, with the latter restriction merely limiting the volume or proportion of such lending or prescribing minimal standards.
31. Such a policy could be implemented in many different ways. For instance, the state might form a development bank, the mission of which is to promote agriculture. Or, as suggested, it might direct privately owned commercial banks to allocate credit in particular ways. In addition, because economic risk, sovereign risk, and systemic risk are so closely related, objectives may overlap. Examples would be a policy against inflation that might be accomplished by tightening monetary policy (e.g., by lifting interest rates or increasing required bank reserves) or a policy to encourage widespread home ownership. Both might in fact strengthen the banking system even while promoting more general economic or social goals.
32. Barth et al., "Bank Regulation and Supervision," 2004.

33. In a paper for the World Bank entitled “Bank Regulation and Supervision: What Works Best,” August 2001, economists James R. Barth, Gerald Caprio, Jr., and Ross Levine survey bank regulations around the world. They distinguish between two fundamental views of bank regulation that parallel our own and that pervade discussion of the topic: (1) the view that government regulation of banks is a generally beneficent one that lends a helping hand to compensate for the market failures described earlier; and (2) the less sanguine view that such regulation functions as government’s “grabbing hand” that tends to cause more or as much harm as it does good, the implication being that market mechanisms including private sector monitoring are preferable to regulation by government fiat. They observe that “[o]ver the years, economists have advocated greater reliance on the private sector and expressed misgivings with official supervision of banks,” noting that “[m]any supervisory agencies encourage private monitoring of banks. . . .” They go on to point out that criticisms of what they refer to as “private sector monitoring,” which includes rating agencies and auditors, as well as market participants, seem to have most applicability where capital markets, accounting standards and legal systems are “poorly developed.”
34. “[T]he policy approach common to many areas of environmental regulation [where those regulated] are given highly specific regulations, often including specific technologies to adopt [and] need to satisfy rigid regulatory requirements often based on ad-hoc judgments of regulators.” R. Wigle, “Environmental Policy Instruments,” info.wlu.ca/~wwsbe/faculty/rwigle/ec238/ref/p-instr.shtml.
35. This catch-all category could comprise various sorts of rules affecting the industry as a whole, such as the imposition of ceilings or floors on lending and deposit rates, or the laws or absence of laws permitting banking cartels.
36. See, for example, the case of the Philippines in Delhaise, *Asia in Crisis*, 168.
37. Protection mechanisms inevitably generate distortions and inefficiencies in the financial industry. In some countries, certain financial institutions are provided with special advantages that can sap the financial strength of competing institutions. In Japan, for example, the post–World War II banking system afforded the postal savings bank a number of inherent advantages over commercial banks in attracting deposits. This held down the profitability of commercial banks, and contributed to the ultimately dire consequences that many of Japan’s commercial banks confronted in the 1990s. Germany’s private banking sector, some would argue, has also been disadvantaged by direct and implicit subsidies granted to state-sector banks.
38. See, for example, *Banks and the Finnish Credit Cycle 1986–1995*, Bank of Finland Studies, E:7 (Helsinki: Bank of Finland, 1997), 25 ff.
39. For example, in Japan, although reform is underway, the banking industry was divided into several categories of depository institutions, each of which had a specified scope of activity in which banks of that class could engage. These groups included ordinary commercial banks, which included so-called major “city” banks and regional banks; long-term credit banks; trust banks; rural and *shinkin* cooperatives; and government financial institutions, including the Postal Savings Bureau and several government development banks.
40. This trend has been reversed by the current financial crisis.
41. The barrier between the two, following years of erosion, was finally dissolved in 1999.
42. Restrictions on branching are another form of entry barrier. In the United States, until recently, banks in rural areas were protected from competition from branch banks.
43. Yet another kind of ownership restriction exists in Hong Kong, where it is mandated that foreclosed property owned by the bank be disposed of within a certain period time.
44. In this, and the other parts of this section, the financial attributes of capital, asset quality, and liquidity that were discussed in earlier chapters are again addressed from a systemic risk perspective. In view of the prior discussion, treatment here is relatively cursory, and, unavoidably, there is some overlap with the earlier material.

45. Recall that capital, as defined by the regulator is termed regulatory capital. Regulatory capital may conceptually be calculated on a non-risk-weighted basis, as was the case traditionally, or in risk-weighted terms, as is the case in the first Basel Accord and in Basel II and Basel III. Note that minimum reserve requirements, which specify that a bank keep a certain amount of funds with the central bank, function effectively as a regulatory capital requirement, although they are normally categorized as such. These are discussed in Chapter 1 in connection with monetary policy management.
46. Banks have routinely been required by law or regulation to maintain a minimum level of reserves, normally expressed as a percentage of bank deposits, with the central bank. They go by the names mentioned as well as being referred to as statutory reserves, obligatory reserves against deposits, or similar such terms. Completely separate from and not to be confused with loan-loss reserves, the conventional rationale for these compulsory reserves was to provide an emergency liquidity or capital buffer. Other measures are now employed to mandate adequate bank liquidity and capital, so that statutory reserves now function more as an instrument of monetary policy rather than to back up bank liquidity. When the statutory reserve is increased, liquidity is absorbed by the central bank since the funds placed on deposit are not available to lend. In addition, a bank's funding costs are effectively increased because the funds placed in statutory reserve earn little or no interest. Since the deposits placed in reserve are unavailable to lend, the average cost of funds to the bank is raised. Statutory reserve requirements depend upon the monetary policy of the relevant country.
47. Once asset quality has started to erode, prompt measures must be undertaken to prevent further damage and the collapse of the bank as a whole.
48. Again, since much of this material was discussed earlier, the treatment here is relatively cursory, and some repetition is unavoidable. Note, however, that the perspective in this section is somewhat different from Chapter 7's, in that asset quality parameters are being considered from the vantage point of systemic risk.
49. In South Korea, following the Asian crisis, broad restrictions were imposed on lending to the country's largest conglomerates (*chaebols*) and to certain sectors in South Korea. The objective was to reduce concentration risk in bank loan portfolios and to promote a more diversified economy.
50. Note that in some markets such regulations are not infrequently circumvented in practice, and funds lent ostensibly for industrial purposes have been known to find their way into speculative property investment schemes.
51. Restrictions on lending to borrowers in a designated sector such as real estate, or business segment, such as margin lending, usually have a similar rationale. Especially in less developed countries, the analyst should be on the look-out for subterfuges being employed on a widespread basis to avoid such restrictions.
52. Broadly speaking, these agency criteria are used to define nonperforming loans, although the term *nonperforming loan* may itself be an explicitly defined term.
53. As per the Advisory Expert Group on National Accounts, "A loan is nonperforming when payments of interest and/or principal are past due by 90 days or more, or interest payments equal to 90 days or more have been capitalized, refinanced, or delayed by agreement, or payments are less than 90 days overdue, but there are other good reasons—such as a debtor filing for bankruptcy—to doubt that payments will be made in full." A. Bloem and R. Freeman, "The Treatment of Nonperforming Loans," IMF, 2005.
54. See Chapter 7 for more explanation on how interest accrual, interest in suspense, and interest clawbacks work.
55. Prior to the Asian crisis, for example, the official policy in one country was that write-offs were not required for 21 months until after the loan became overdue, or until litigation was concluded.

56. In some countries as a matter of policy, write-offs may be discouraged, either directly or indirectly. The approval of bank regulators may be required, for instance. In other cases, the requisite legal mechanisms for effecting write-offs may be cumbersome, or tax policy may discourage them.
57. As has been emphasized, fractional-reserve banking has an inherent liquidity risk in that virtually all banks have more illiquid assets such as loans than liquid ones, while a large proportion of the bank's liabilities will almost certainly be demand deposits or other current or short-term liabilities.
58. A related issue, although largely a matter of discretion rather than explicit regulation is the willingness of the central bank or monetary authority to function as a lender of last resort.
59. Ownership and investment restrictions, previously discussed in connection with the structure of the banking system may have several prudential or social policy objectives. For example, limiting foreign ownership of banks is usually motivated by a desire to maintain national control over banks as instruments of monetary policy and to protect local banks. Limits on individual shareholding tend to be impelled by a belief that a family-run bank, for instance, might channel depositors' funds into family-related enterprises. Note that, for reasons of space, however, prudential regulations setting standards of governance are omitted from this chapter.
60. Attitudes do appear to be changing, especially among the countries that were hard hit by the Asia crisis, as the value of disclosure is impressed upon banks and regulators.
61. As suggested earlier, while such policies certainly may benefit bank creditworthiness in the short-to-medium term by enhancing the financial strength of institutions affected, in the long run they are frequently negative both for the economy and the banks involved.
62. Another class of laws affecting the banking industry is those designed to prevent insider securities trading, money-laundering, and other criminal activity. These rarely affect analysis of the banking system and are therefore outside the scope of this chapter.
63. Somewhat counter-intuitively, however, two researchers concluded that deposit insurance was associated with an increased risk of a banking crisis. Asil Demircug-Kunt and Enrica Detragiache, "Does Deposit Insurance Increase Banking System Stability? An Empirical Investigation," *Journal of Monetary Economics* 49, no. 7 (2002).
64. "Even insured depositors may respond to bank risk if there is some uncertainty or costs involved with recovering deposits in the case of bank failure." Kaoru Hosono, Iwaki Hiroko, and Kotaro Tsuru, "Bank Regulation and Market Discipline around the World," RIETI Discussion Paper Series 04-E-031 (October 2004).
65. "Deposit insurance removes bank risk from depositors, eliminating depositors' incentives to monitor bank behavior and to limit the banks' risk taking through risk-based pricing or rationing. Therefore, provided the premium paid by the bank for deposit insurance does not adequately reflect bank risk, bank value can be increased by increasing the riskiness of the bank's portfolio." Vesa Vihriälä, *Banks and the Finnish Credit Cycle 1986–1995*, Bank of Finland Studies, E:7 (Helsinki: Bank of Finland, 1997), 21. Other arguments advanced against deposit insurance are that it penalizes better-capitalized and better-managed banks and is a subsidy to weaker institutions. It has been also asserted that deposit insurance discourages financial-sector reform and consolidation, since it enables banks that would otherwise be taken over to continue to operate. K. Irving, J. Emmanuel, and A. Scarff, "Focus: The Deposit Insurance Debate," *Asia-Pacific Banknotes* (Merrill Lynch, October 29, 2001), 3.
66. B. Eichengreen and C. Arteta, "Banking Crises in Emerging Markets: Presumptions and Evidence" UC Berkeley, Center for International and Development Economics Research (August 2000), 31.

67. D. O. Beim, "What Triggers a Systemic Banking Crisis?" (New York: Columbia University Working Paper, June 19, 2001), 2.
68. The actual number of schemes is difficult to establish, as there are many implicit or temporary deposit insurance systems in place.
69. C. S. Tan, "Deposit Insurance Launched," *Malaysia Star*, September 2, 2005, <http://biz.thestar.com.my/news/story.asp?file=/2005/9/2/business/11929670&sec=business>.
70. Canadian Deposit Insurance Corporation, www.cdic.ca.
71. Fundo Garantidor de Creditos, www.fgc.org.br.
72. Because of its federal structure and large size, the United States represents a rather extreme variant of this more traditional approach, in respect to a significant policy role in regards to banking policy.
73. Martin Cihak and Richard Podpiera, "Is One Wathdog Better Than Three? International Experience with Integrated Financial Sector Supervision," IMF Working Paper, March 2006, 4.
74. C. Goodhart, "The Organisational Structure of Banking Supervision," Financial Stability Institute, IMF, 2000, 7.
75. Ibid.
76. That is, they either do not benefit from legal immunity when acting in their capacity as agents of the state, or claimants may subject individual examiners to litigation rather than being obliged to direct the complaint against the agency as an institution.
77. Barth et al., "Bank Regulation and Supervision."
78. Paraphrased from Barth et al., "Bank Regulation and Supervision."
79. R. K. Abrams and M. W. Taylor, "Assessing the Case for Unified Financial Supervision," in Douglas Arner and Jan-Juy Lin, eds., *Financial Regulation: A Guide to Structural Reform* (Hong Kong: Sweet & Maxwell Asia, 2003), 41–42. While, as noted, bank regulatory policy may, as a matter of law, be obliged to incorporate social policy objectives, the intrusion referred to here concerns attempts to go beyond this agenda in an attempt to tell the agency how to do its job or to influence it to favor or disfavor particular regulatory actions.
80. On the one hand, regulators ought to have some degree of insulation from political intervention. This can be attained to a degree through legal protections against arbitrary dismissal of regulators, budgetary independence, and a modicum of deference by the courts to the expertise of the agency. On the other hand, there must be checks against misuse of the agency's regulatory powers and against excessively severe (or lax) regulation that is not in furtherance of the authority's objective. See Abrams and Taylor, "Assessing the Case for Unified Financial Supervision." As with the review of the quality of regulation generally, evaluation of these items' regulatory regime, is necessarily difficult and almost invariably highly subjective. It is, nevertheless, a critical part of this process, since an agency that is subject to political influences, and even corruption, will likely be ineffective in accomplishing the mission it was intended to perform.
81. Morris Goldstein and Philip Turner, "Banking Crises in Emerging Economies: Origins and Policy Options," BIS Economic Papers—No 46 (1996), 24.
82. <http://info.worldbank.org/governance/wgi/index.asp>.
83. "Handle with Care," *The Economist*, October 3, 1998.
84. In this table, intermediation ratios, including deposits to GDP, are used to measure banking system size rather than absolute figures.
85. As was observed, bank lending functions to expand the money supply. Therefore, where a sharp rise in the intermediation ratio is the result of cyclical rather than secular changes, it suggests the possibility of an asset bubble and a corresponding potential decline in intrinsic asset quality.
86. Recall that in economic analysis, the chief purposes of the bank intermediation ratio are to gauge the availability of credit and the volume of the money supply as part of a

macroeconomic analysis. Note, also, that there is more than one way to define bank assets. Values will naturally vary depending upon how bank assets are defined and measured. Relevant questions to consider are the following: Does the definition of banking assets include only on-balance-sheet assets? Or risk-weighted assets? How are banking institutions defined?

87. Interestingly, over the period shown the ratios in all four countries appeared to be declining, perhaps as a result of increasing disintermediation.
88. Another point to bear in mind is that financial centers such as Hong Kong or Mauritius may exhibit extraordinarily elevated ratios. This reflects a high concentration of banking institutions in relatively small jurisdictions, rather than necessarily an undue reliance on bank finance. A potential alternative measure of intermediation would be the proportion of banking assets to total financial assets. According to one measure, the percentage of bank assets to total financial assets was less than 20 percent of financial assets in the United States; in contrast, the ratio is near 40 percent for Japan, just over 50 percent for the UK, and over 60 percent for Germany.
89. Yet a major rating agency—which will not be named—wrote in a report published on April 18, 1997, barely 10 weeks before the collapse of dozens of Thai finance companies triggered the Asian panic: “Despite the much talked-about oversupply of housing, we found little evidence, as at February, of significant problems in the finance companies’ real estate development loan portfolios. While this form of lending is clearly more risky, on the whole finance companies appeared to take reasonable precautions, such as requiring an appropriate level of pre-sales before making a commitment. Most real estate lending is to low rise, stand-alone residential estate developments, which are inherently less risky than condominium developments.” The document ended with “We do not expect a repeat of the mid-1980s banking crisis, when a rash of defaults occurred across the sector.” Quoted in Delhaise, *Asia in Crisis*, 213.
90. The credit culture, as we discussed in Chapter 1, will be influenced by the quality of a country’s legal system, the strength of creditors’ rights, limits on banking activities, the nature of the regulatory regime, and the extent to which its directives can be readily enforced. All of these, in turn, are important considerations in their own right.
91. Nations that were once part of the former Soviet Union have seen excesses and numerous bank failures arising from dramatic change in the role of banks. In the same way, countries that practice various forms of state-directed capitalism (e.g., Japan) or that remain nominally socialist (i.e., Marxist-Leninist) are prone to problems arising from the shifting function of credit within their economies.
92. Note that while deposit insurance tends to increase depositor confidence, it also increases moral hazard, allowing banks to engage in risky lending without having to worry about an outflow of retail deposits.
93. State support for banks was discussed earlier. Ownership was also briefly discussed in an earlier chapter in the context of management and corporate governance.
94. Foreign banks in Argentina got a taste of that problem during the 2001 crisis.
95. Goldstein, KDIC presentation.
96. See Gerard Caprio, “Banking On Crises: Expensive Lessons from Recent Financial Crises,” World Bank Policy Research Working Paper (1999).
97. See R. La Porta, Florencio Lopez-De-Silanes, and Andrei Shleifer, “Government Ownership of Banks,” KSG Working Paper No. 01-016 (Cambridge: Harvard Institute of Economic Research Paper No. 1890), <http://post.economics.harvard.edu/hier/2000papers/2000list.html> (2000). La Porta and colleagues found that state ownership of banks is common throughout the world, particularly so in less developed countries, and that a high percentage of state ownership of the banking sector is associated with slower growth and financial development.

Crises: Banking, Financial, Twin, Economic, Debt, Sovereign, and Policy Crises

When bubbles explode, banks fail.

—Richard Duncan¹

What made the Great Depression great [was that an economic downturn was] allowed to disrupt the operation of financial intermediaries. . . . [I]t is . . . the instability of banking systems that distinguishes economic crises from ordinary recessions. . . .

—Eichengreen and Arteta²

The financial sector is the economy's plumbing system. A company's failure . . . is like a broken sink, but a failure in the financial sector threatens the entire water supply.

—Charles Morris³

The financial world is confronted almost permanently with financial crises prompted by numerous causes, some of them acting in concert, and it is nearly impossible to catalogue them in a satisfactory way. Take your pick. Banking crises masquerade as sovereign crises; foreign currency problems generate banking, financial, and twin crises; while economic or policy crises spark sovereign crises.

Worse yet, there is always a lack of consensus on the causes of crises, let alone—when there is an agreement on the source of the crisis—on where on a spectrum of relevance the identified causes produce their effects.

Aristotle taught us that four causes can bring a natural effect or event. The material cause explains the effect on the basis of what material the effect is made of. The efficient cause pertains to the way the material is processed to create the effect. The formal cause is about how the process is conceived. The final cause relates to the purpose of the effect.

David Hume's theory of causality pointed mostly at perception, where the perception of the existence of a cause being always followed by an effect established causality. He clarified that we cannot always satisfy ourselves that such connection is established, but that, realistically, we have to accept the causality.

More recently, probabilistic causation attempted to link causes and effects through probabilistic tools, on the grounds that it is possible to measure the impact of a cause on the probabilities of an effect. A number of tools used in financial risk management rely on such an approach, which makes sense in a universe where the perceived or measurable links between cause and effect are difficult to establish.

Confronted with multiple causes and multiple effects, we are modestly attempting to guide the reader through the maze of links between microeconomic and macroeconomic data or situations, on the one hand, and the creditworthiness of banks, in isolation or in a system, on the other hand.

Our definitions of the various events—often, but not necessarily—called crises may at times differ from those offered by other commentators. Yet, we need some level of categorization if our analysis is to be of any use to the bank credit analyst.

The reader will, by now, have realized that the creditworthiness of a bank largely depends on the skills and wisdom of its management, shareholders, and regulators, but that a bank is permanently at risk of suffering from exogenous situations and events.

The first part of this chapter will explore crises affecting a single bank, or of a banking system. We will then examine the all-important topic of the financial or currency crisis, followed by the twin crisis, which is the near-conjunction of a banking crisis and a financial crisis. We shall also examine economic, sovereign, debt, and policy crises for the impact they produce on the creditworthiness of banks.

The second part of this chapter will explore early warning systems (EWS), the modeling of bank crises for predictive purposes. In this context, it looks at the fruits of the empirical testing of the various potential causes of bank crises as identified in the first part of the chapter.

A banking crisis should be distinguished from the isolated collapse of a single bank. The first situation represents a breakdown in the capacity of the financial system as a whole to provide ordinary banking services, including, most notably, credit to businesses and consumers in the form of loans or similar facilities. The demise of an individual institution, even a prestigious one, does not necessarily herald the onset of such a crisis. Nor is the failure of a single institution likely to have much impact on the wider economy, despite the painful losses that may be suffered by the bank's shareholders, its management, its commercial creditors, and—depending upon the deposit insurance scheme in place—its retail depositors. Therefore, as a rule, banks not perceived to be critical within the financial systems in which they operate can close their doors—or, as is more likely, will have them ordered shut by the bank regulator—without triggering unwelcome side effects.

Indeed, during ordinary times when markets are comparatively stable, regulators take over noncritical local and provincial banking institutions quite regularly without much notice being taken, particularly in countries such as the United States, where thousands of small community banks still exist. When, however, the problem bank is a major financial institution of critical importance to the banking system as a whole, and therefore one that is perceived as too big to fail—to apply the usual label—it can be expected that the market will take notice of news of its troubles. Where the bank's difficulties are grave enough to potentially result in its collapse, a market upset can occur and a period of market turmoil may ensue. Banks in danger of failing are often said to be “distressed” or “in distress.”⁴

A conspicuous example of a collapse of a well-known bank that did not prefigure the collapse of the country's banking system was the demise of Barings Bank in 1995.⁵

Instead of triggering a systemic crisis, the legendary merchant bank sank quietly, without waves of panic engulfing the UK's banking system. The proximate cause of its collapse was the wayward behavior of a rogue trader in its Singapore office, who, through deception and fraud, ran up losses that exceeded the bank's equity capital and rendered it insolvent.⁶ While Barings was well known in financial circles, and had a long history, by the mid-1990s it was no longer an essential cog in the country's banking system. The Bank of England therefore declined to arrange a bailout, and the once venerable institution was sold to ING Bank in the Netherlands for the paltry sum of £1.

Compared to Barings' downfall, the implosion of Northern Rock in 2007 was no isolated occurrence without further reverberation. Instead, it was a warning of much worse to come. Among the top 10 deposit-taking institutions in the UK, the Rock was the domestic bank most vulnerable to a liquidity crunch, having long relied on securitization and wholesale sources to fund its huge mortgage-loan book. A former building society, this one-time regional institution had grown enormously during the preceding decade into one of Britain's largest mortgage lenders. The breakdown of the securitization model that followed the subprime meltdown in the United States in 2007, accompanied by the seizing up of the London interbank market in August of the same year, rendered Northern Rock's position untenable.⁷ The bank's takeover by the government did not mark the end of the matter, but instead heralded a series of events that rocked even larger UK banks not long afterward. Indeed the circumstances were difficult, and the liquidity crunch was an external factor, but the liquidity problem had been in existence for at least a couple of years when Northern Rock failed, which makes its collapse a borderline case of self-inflicted problems, and that bank's crisis probably still belongs to the single-bank-failure category.

The same comment can be made about the demise of Fortis Bank in 2008, which was prompted by the amateurish decision to buy a chunk of Dutch bank ABN AMRO for an inflated price and—a mortal sin under the interbank illiquidity circumstances—with borrowed money. Naturally, the then-largest bank in Belgium could hardly fail on account of a single cause, and there were most likely other weaknesses that the bank had built up over time.

Banks do fail in isolation or by themselves, but they usually do not disappear. Almost everywhere, banks tend to be salvaged or merged with other banks. In 2011, in the United States, 89 banks failed and were quietly merged with their acquirer. Even banks that fail in the wake of extraordinary events such as the 2007–2012 crisis usually do not disappear.

As demonstrated, it is difficult to draw a line between a single bank's failure and the systemic problems that may fell what is sometimes a large chunk of the banks of a given country.

WHY LOOK AT BANKING CRISES?

Banking crises are important to consider as part of the process of bank credit analysis for several reasons. First, a banking crisis defines one end of a spectrum in respect to the maximum level of stress a bank is likely to confront. It represents the worst-case scenario that should always be borne in mind as a possibility,

however remote, when preparing a credit analysis. Second, as a practical matter, most bank failures occur during bank crises. Consequently, it is wise for the analyst to be aware of the typical causes of a crisis, potential signs that conditions are changing, and the crisis' likely impact.⁸ Third, it is helpful to be aware of the process by which insolvent banks are typically resolved by regulators. Usually, banks are resolved in a different manner from nonfinancial entities. If the analyst is working on behalf of investors or counterparties with exposure to a distressed bank, it is necessary to keep informed concerning the restructuring process, which is likely to be adapted to conditions existing at the time instead of after the same template that was used in the past or in another country.

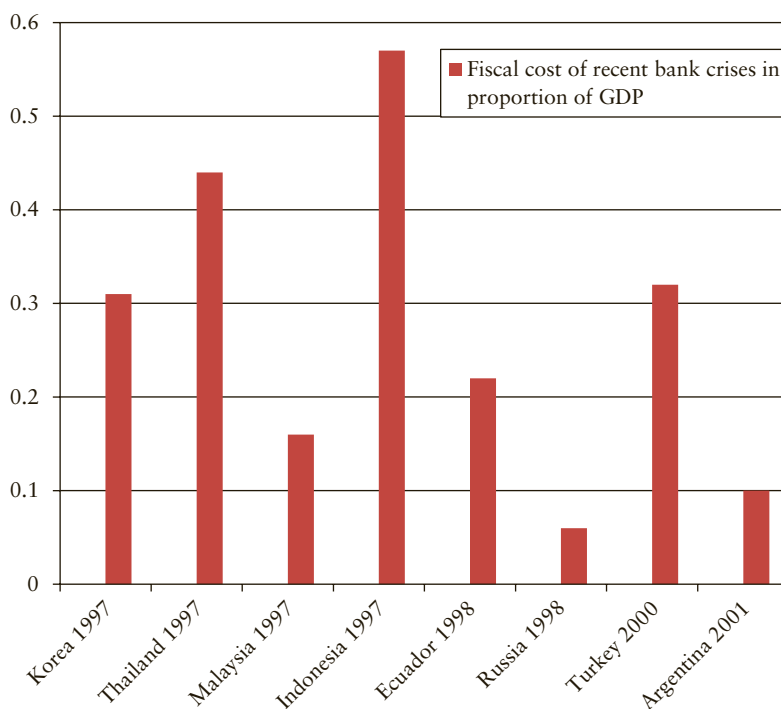
The question of what causes banking crises is a complex one that has been approached from a variety of perspectives, and while there is a degree of consensus, no definitive answer has been reached. Over the past two decades, a large volume of empirical research concerning both the causes of banking crises and, in addition, the possibility of predicting such episodes has been produced. The research has been undertaken primarily by staff researchers at the IMF, at the World Bank, and at various central banks as well as by academic scholars.⁹ Much of the interest in engaging in these studies has been motivated in large part by the increasing frequency of bank crises since the 1970s and the huge economic cost they have wrought, especially in developing markets. Rebuilding a country's financial system after such events is almost always a very costly enterprise.

This chapter looks at the question of what constitutes a banking crisis, the causes of such episodes of financial instability, how such events differ from other forms of financial crises, as well as why banking crises seem to be occurring more frequently over recent decades. As can be seen in Exhibit 14.1, the cost of recent crises in proportion of GDP is very high. It can be argued that not all of those costs were specifically prompted by banking crises as opposed to the attending financial crises, but crises are indeed very costly.

We shall attempt to define a banking crisis by describing its essential features. These include both its impact on the financial system and how it manifests. The discussion then naturally proceeds to a consideration of what might be the potential leading indicators of a banking crisis before discussing the question why banking crises have become so frequent over the past 30 years or so as compared with the preceding era. The macro and broad policy and institutional factors that differentiate mass bank distress from the isolated bank failure are also discussed. The micro and macro sources of bank crises are explored, as well as currency crises and their relationship to bank crises.

AN INTRODUCTION TO BANKING AND FINANCIAL CRISES

The standard way that countries—particularly developing countries—got into trouble in the past was that their governments did the wrong thing by

**EXHIBIT 14.1** Fiscal Cost of Recent Crises

Source: IMF.

running bad fiscal and monetary policy. [These days] economic crises are not caused primarily by bad fiscal policy or bad monetary policy [but are] financial in nature rather than stemming from poor macroeconomic policies.

—I. Macfarlane, Governor of the Reserve Bank of Australia¹⁰

There is no shortage of theoretical models of the causes and consequences of banking and financial crises.

—Barry Eichengreen and Andrew Rose¹¹

The contrast between cases as different as Barings, Northern Rock, and Fortis Bank shows that there is a threshold somewhere, with a critical distinction between the failure of a single problematic bank and a banking crisis. The former results mainly from deficiencies within a single organization during comparatively normal market conditions. A banking crisis is something more.

Defining a Banking Crisis

[T]here is no unambiguous definition of banking crisis.

—A. Gaytán and A. Christian¹²

What is a banking crisis?¹³ Described by a variety of terms—banking panic, credit crunch, and systemic crisis—it is sometimes encompassed within the more

generic phrase “a financial crisis,” which is best reserved for the description of a country or sovereign crisis, as will be discussed later.

All these labels refer to what is fundamentally the same phenomenon, although it may take a variety of different forms.¹⁴ That phenomenon is the impairment, and potentially the ultimate collapse, of the capacity of a country’s banking system to fulfill its critical economic role. That role is the supply of essential banking functions to facilitate commerce and economic growth, including, most critically, the supply of credit to businesses and consumers.¹⁵ A *banking crisis*, as the term is used here, is an episode of financial instability arising contemporaneously with the severe distress or failure of one or more of the country’s significant banks.

Implicitly, a banking crisis adversely affects the capacity of the banking system to function in its usual manner, so it constitutes the core manifestation of systemic risk, or at least some definition of systemic risk.¹⁶ Terms such as *banking system fragility*, or conversely *financial system stability*, are used to estimate the likelihood that a banking system, or the financial system as a whole, will suffer damage as a result of a crisis. Various banking system fragility indices—alternatively described as financial strength indicators or as aggregate macro prudential indicators—have been constructed in an attempt to estimate the likelihood of a crisis occurring. This topic is discussed in more detail later in the chapter. Again, it is the functional impairment of the banking system, or the threat of such damage, that differentiates a banking crisis from a single problem bank and from a spate of coincidental and isolated instances of bank failure. While the two phenomena are often related, a single financial institution in distress does not necessarily imply the existence of a banking crisis; instead it may or may not turn out to be a precursor of a banking crisis yet to unfold.¹⁷

Banking Crises, Liquidity, and Capital Adequacy

In general, a banking system (like an individual bank confronting failure) will not suffer impairment in its capacity to function so long as it possesses ample liquidity and/or the confidence of depositors and creditors that either its owners or its regulators or the country’s government will provide the capital to absorb the economic losses that will otherwise permit a systemic crisis to arise. Conversely, a distressed banking system that is no longer able to provide a full level of service to market participants probably no longer has adequate liquidity and capital nor the expectation of adequate liquidity and capital fully to absorb potential losses that were once unexpected but that have subsequently become foreseeable.¹⁸ In other words, it faces a real risk of illiquidity or insolvency,¹⁹ or both.

Both liquidity and solvency are needed,²⁰ but one should remember that the proximate cause of bank failure is often a lack of liquidity—which may be the result of the market perception of its insolvency—rather than a shortage of capital—which in any event can easily be hidden. When operating conditions are relatively stable, capital adequacy is not necessarily the most important financial attribute in assessing a bank’s financial strength. The sufficiency of bank capital adequacy, however, does come to the fore when higher-than-expected loan losses occur, unusually volatile market conditions prevail, and the insolvency of a significant bank or a number of such institutions becomes an imminent possibility, if not an actual occurrence. Solvency is so closely associated with bank crises that weak capital adequacy among a substantial portion of the major institutions that comprise the core of a country’s

banking system practically defines the existence of banking crisis. Indeed, a banking crisis has been so defined by one of the principal researchers on the subject as an episode, the outcome of which is that “the net worth of the banking system has been almost or entirely eliminated.”²¹ By this definition, where a number of institutions that together represent a substantial portion of the assets of a given banking system either (1) possess insufficient capital to function effectively (for which capital ratios well below the regulatory minimum are a proxy), or (2) are already insolvent but remain still functioning (usually as a result of state support), an ongoing or impending banking crisis can be said to exist.²²

It is evident that capital adequacy is undoubtedly of critical importance to institutions confronting a crisis situation, and equally to their creditors and depositors, because as a bank approaches insolvency and its collapse becomes a real possibility, the thickness of its cushion against losses determines its ability to survive. In these circumstances, at least for private sector banks, capital adequacy is arguably the single most important indicator of bank creditworthiness. It might reasonably be hypothesized that the metric is also the best predictor both of a bank’s and of a banking system’s short-term vulnerability to collapse during normal market conditions. In an ideal world, perhaps, it might be. For reasons discussed later, however, capital-focused definitions of a banking crisis are mainly useful after a crisis has fully unfolded, when capital weakness both at the systemic and individual bank levels becomes a serious concern.²³ At this stage, it becomes particularly relevant for purposes of ascertaining the depth of the problems facing a bank or a banking system and the cost of rehabilitation.

Solvency ratios, in fact, are generally not the best leading indicators of a banking crisis. The reasons are several. First, capital ratios are not affected until after a bank has experienced losses. As leading indicators as to the likelihood and severity of those losses, other indicators such as those pertaining to asset quality (e.g., the proportion of nonperforming loans (NPLs) to total loans), are apt to provide an earlier warning of distress than capital ratios—at least in countries where potential losses are promptly recognized. All other things being equal, the latter indicators will only fall when the bank has written off the corresponding loans, and capital ratios will therefore tend to lag behind asset quality and profitability indicators in reflecting changes in a bank’s financial position. This is not to deny, however, that high capital ratios to some degree may justifiably provide some comfort to those with credit exposure to the institution.

In addition to the volume of capital, another relevant consideration concerns the quality of a bank’s capital. When insolvency becomes an imminent possibility, the quality of capital becomes a matter of the highest importance, particularly to those at risk of loss from the bank’s default. The highest quality of capital—the best buffer against such loss—is comprised of owners’ funds at risk (i.e., shareholders’ equity), which is akin to Tier 1 capital without the add-ons allowed under the Basel Accords. Lower in quality is regulatory capital that consists of the bank’s own financial obligations to others, such as Tier 2 subordinated debt, since it does not provide much of a buffer as a bank approaches insolvency but instead becomes an additional obligation, albeit one relative to which most ordinary counterparties will stand in a senior position.

An additional limitation of capital indicators is the difficulty in measuring capital on a dynamic basis, particularly during periods of financial turmoil.²⁴ In times of

stress, if asset quality is the main factor affecting the viability of the bank—for example, when the bank is exposed to real estate risk or to sovereign bond risk in an environment where prices are falling substantially—there is a technical difficulty in gauging the potential loss the bank is facing. Marking to market such assets in troubled times is difficult, and asset values can move down—or thankfully up—by leaps that dwarf the bank's equity. Banks are usually worth more than their net equity, but such fluctuations could see banks technically—if not publicly—go bankrupt one week, recover the next, and fail again the following week.

By definition, equity capital is a residual item, obtained by subtracting obligations from assets; thus it can only be identified and measured with reference to the value of a bank's assets and its obligations. But how are both the value of a bank's assets and the cost of its liabilities to be calculated in a quickly changing market environment? A bank's liabilities may be defined in fixed terms (e.g., a no-interest current deposit amounting to \$9,321.37), and the bank is liable to repay them at their corresponding face value. This is not always true, since the bank could conceivably buy back at a profit its own debt in times of distress, but generally only the economic value of its asset holdings is subject to daily market fluctuation. Where a liquid market exists for a particular class of assets, value will correspond closely to market price. Under volatile market conditions, however, the value of a bank's assets will fluctuate more wildly. So too will its true level of capital, irrespective of that amount reported in its most recent financial statement.

During periods of market turbulence (like those associated with crisis conditions), however, the value of comparatively illiquid loans or securities (i.e., instruments that are not regularly traded in financial markets), as well as the value of ordinarily liquid assets that turn illiquid under such conditions, can become wildly uncertain.²⁵ The paradox is that the moment when you really need to know the amount of capital a bank possesses may very well be the time when it is most difficult to measure.²⁶ In this light, it is perhaps unsurprising that there may be considerable uncertainty on the part of outsiders, and even within the institutions themselves, concerning the true capital position of a country's banks day to day immediately before and during the early stages of a banking crisis.²⁷

Finally, reported capital levels are easily manipulated. Capital, as conventionally reported on the bank's balance sheet, is but one part of a snapshot of a bank's financial condition as it appeared at the end of a given day. There is no requirement that it be a typical day. So it is apparent that with adequate preparation capital levels can be tweaked to some degree, perhaps to a greater extent than the flow indicators on the profit and loss. Management needs only make some tactical changes in the few weeks before the reporting date to make the results look better for external analysts. Bear in mind also in this regard that regulatory regimes vary in the discretion they afford institutions under their charge to determine loan classifications and by implication expected losses.²⁸ As a crisis approaches, only the strongest banks have an incentive to provide full disclosure. For the others, the incentive is to avoid, at least within the limits of law and regulation, giving the slightest hint of a potential inability to meet regulatory capital minimums. The upshot is that the quantitative data on solvency that emerges from the smoke and fog that surround the early stages of a banking crisis is apt to be of limited value to the analyst.²⁹

The wariness of senior management in this regard is not misplaced. Any indication that a particular bank is suffering from weakness in the adequacy of its capital

will tend to cause liquidity problems or exacerbate any that are already evident. Skittish counterparties will decline requests for funding, and investors will dump the banks' securities hoping to beat others to the exits, while depositors will withdraw deposits unless credible deposit insurance exists. Conversely, the existence of liquidity problems or the receiving of support from the central bank may be taken to imply a capital deficiency as well. For this reason, regulators tend to avoid publicly disclosing the existence of a new commitment, or a change in commitment, to supply liquidity. To disclose a commitment in the midst of the uncertainty that characterizes the precrisis environment could be taken to mean that not only are the beneficiary banks in trouble, but also that there is significant risk of a banking crisis developing. In view of the signal it potentially sends, the central bank and bank regulators generally avoid revealing support for institutions facing difficulty for fear that the disclosure of such information could spark the very conflagration they seek to avoid.

DEFINING THE BANKING CRISIS: A RANGE OF VIEWS

A banking crisis occurs when one or more banks collectively holding a substantial portion of the banking assets within a country experiences a decline in asset quality that impairs their capital to such an extent, that a significant number effectively become insolvent, thereby limiting their capacity to function. Because the changes in economic and business conditions that invariably accompany banking crises change so rapidly, there is a considerable uncertainty on the part of banks themselves concerning their true capital position. Capital is what remains after liabilities are deducted from assets net of losses incurred during the relevant reporting period. But during a crisis, not only is the value of the bank's assets in the form of loan and securities holdings hard to ascertain, but also the estimation of losses arising from delinquent financial obligations is perhaps even more difficult.

Without support from the state as the lender of last resort, the weakest banks are prone to suffer a sudden loss of funding and liquidity. When this occurs, they will be unable to meet their obligations to repay their depositors or other creditors, and will be forced to cease operations. Since banks often have significant credit exposures with each other through the interbank market or in respect to settlement or other obligations, the failure of multiple banks in a given banking sector is apt to trigger panic in the domestic financial markets and may very well result in real losses to the remaining banks. Commercial depositors or holders of securities issued by the failed banks will experience losses that may force them to default on other obligations.

Definitions necessarily depend upon purposes. For the purpose of understanding the significance of the banking crisis, what is needed is a definition that conveys the multifaceted causes and effects of such an occurrence. For the purpose of developing an early warning system to forecast a banking crisis or other type of financial crisis, a somewhat different type of definition is needed, as is discussed in this chapter. To this end, and in light of the preceding

description, a fairly comprehensive general-purpose definition of a (systemic) banking crisis can be formulated as follows.

The term *banking crisis* refers to a series of events that typically begins with one or more banks in a given market suffering from the external perception of a situation of liquidity weakness or of severe loan losses that erodes capital to below the level at which market participants start losing confidence. They lose confidence in the affected bank or banks, and ultimately, as the crisis deepens, in the soundness of the banking system as a whole; unless state support is provided to facilitate their recapitalization, the loss of confidence in turn causes the affected banks to fail and the supply of bank funding to evaporate forcing sharp sector-wide cutbacks in lending that accelerate the economic downturn that is by then well under way, resulting in more borrower delinquencies, more NPLs, even less capital, less lending, and so on in a vicious cycle until a substantial proportion of the banks in the sector are effectively insolvent, and the banking system is paralyzed.³⁰

For the sake of comparison, some other definitions of banking crises from various research reports are provided here:

[I]n a systemic banking crisis, a country's corporate and financial sectors experience a large number of defaults and financial institutions and corporations face great difficulties repaying contracts on time. As a result, nonperforming loans increase sharply and all or most of the aggregate banking system capital is exhausted. This situation may be accompanied by depressed asset prices (such as equity and real estate prices) on the heels of run-ups before the crisis, sharp increases in real interest rates, and a slowdown or reversal in capital flows. In some cases, the crisis is triggered by depositor runs on banks, though in most cases it is a general realization that systemically important financial institutions are in distress.

—Luc Laeven and Fabian Valencia³¹

[A] banking crisis [is] a period in which significant segments of the banking system become illiquid or insolvent [as] evidence[d by] large scale bank failures, the adoption of emergency measures by the government (deposit freezes, nationalizations, deposit guarantees, bank recapitalization plans), whether there were reports of significant depositor runs, the level of nonperforming loans (at the peak of the crisis), and the costs of the bailout.

—A. Demirgüç-Kunt, E. Detragiache, and P. Gupta³²

[A banking crisis is] . . . the occurrence of either “(1) bank runs that lead to the closure, merging or take over by the public sector of one or more financial institutions . . . or (2) . . . the closure, merging, takeover, or large-scale government assistance of an important financial institution.

—A. Gaytán and A. Christian, “A Review of the Literature on Early Warning Systems for Banking Crises” (October 2002)³³

Systemic financial crises are an extreme form of financial collapse that results in wealth destruction that cripples the financial system and magnifies economic decline.

—Charles W. Calomiris, Daniela Klingebiel, and Luc Laeven³⁴

In this context, it is worth drawing attention to the obvious point that bank crises vary in their severity. Gerard Caprio, one of the leading researchers in the field refers, for example, to weak or borderline crises as opposed to severe or systemic crises. The essential difference would seem to be that in borderline crises the damage to the banking system as a whole is not substantial and repairs can be undertaken without the need for massive state intervention. The severity of a crisis can only be known, for the most part, after the fact. So, unless there is good reason to think otherwise, out of conservatism, the bank analyst should treat any impending crisis as a potentially severe one.

Causes of Banking Crises

Much analysis has been done on the root causes of the financial crisis that has swept Asia. Our view is simple; too much money was lent to the wrong people. A combination of financial liberalization, inflows of cheap money from overseas and ill-conceived infrastructure projects caused bankers to be swept up with the times and go on a lending spree with little regard for risk.

—Robert Zielinski³⁵

The existing empirical literature on banking crises has not produced agreement on their causes.

—Barry Eichengreen and Carlos Arteta³⁶

Almost always and everywhere banking crises are a complex interactive mix of economic, financial and structural weaknesses.

—David Llewellyn³⁷

Notable bank crises in the 1990s, including the Asian Crisis of 1997–1999,³⁸ provided an impetus for new research efforts not only into the causes of banking and currency crises, but also into their prediction and prevention.³⁹ The greater damage banking crises cause in emerging markets vis-à-vis those in developed countries (mirrored in the larger share of GDP they consume), has provided additional motivation for research into financial crises generally, while making enquiry into their causes in emerging markets practically a subspecialty within the field. Finally, dissatisfaction among investors, economists, and other market participants with conventional market and economic indicators, such as interest rate spreads and credit ratings issued by the major agencies, has been another factor driving research.⁴⁰

As the global financial crisis that started in 2007 has made plain, the problem of financial system instability is still with us and remains as great of a concern as ever. In this light, there is much to be gleaned from the large body of research concerning

not only the causes, and resolution, of financial crises, but also concerning their prediction and potential prevention. Although some of the work is highly technical or theoretical in nature, much of it easily accessible to the lay reader and can afford practical insights applicable to bank credit analysis.⁴¹ The remainder of the present section looks primarily at the question: “What causes bank crises?” while question “How can they be predicted?” is examined in the next section.⁴² Their resolution and prevention will be discussed later. It should be mentioned that in view of the volume of the literature on the subject, a comprehensive survey and evaluation of all the theories and hypotheses concerning the potential causes and combinations of causes of banking crises is beyond the scope of this book. Instead, the material is highlighted selectively, with the hope that readers having the interest will be inspired to delve deeper into the material on their own.⁴³

Symptoms of Distress: Major Indicators

Despite the distinction made between isolated instances of bank failure and a banking crisis, the symptoms displayed by banks weathering a banking crisis are generally the same as an individual problem bank experiencing distress during comparatively stable business conditions.⁴⁴ They can be summed up as follows:

- *Poor asset quality*, reflected in high or rising level of NPLs and other indicators
- *Shortage of liquidity*, resulting from a loss of confidence and a corresponding loss of access to funding implied in an increase in the loans-to-deposits ratio and comparable liquidity and funding indicators
- *Declining real capital adequacy* ultimately approaching a state of insolvency that is reflected in decreases in the equity-to-assets ratio, the Tier 1 capital ratio, and other comparable indicators
- *Deteriorating earnings and profitability*, owing to higher credit costs and a fall-off in lending revenue and mirrored in year-on-year declines in the rate of earnings growth, return on assets, net interest margin, and other indicators

In brief, they represent the panoply of CAMEL ailments that were discussed in Chapters 6 through 10.⁴⁵

Interestingly, for the analyst, this similarity in symptoms makes the prospective identification of a banking crisis particularly difficult since it cannot be known in advance whether the first failure of a significant bank in a long while (1) marks the beginning of what will be a series of bank failures resulting in a fully-fledged banking crisis, or (2) is merely an isolated occurrence attributable to idiosyncratic characteristics specific to the failed bank. Again, the symptoms exhibited in a single bank are insufficient to tell us if a crisis is underway. In most cases, one or more systemic elements must be present as well. The challenge of recognizing a banking crisis in the making is substantial.

Banking Crises and Systemic Weaknesses

In a severe banking crisis, even the strongest banks, judging by the criteria discussed earlier in this book, will struggle to survive. Recollect the distinctions between a banking crisis and one or more isolated individual bank failures discussed so far:

- The scale of the losses (reflected in system-wide NPLs to total loans) is much greater.
- There is a significant reduction of the capacity of the banking system to provide the usual range of banking services, including, most critically, credit and payment services.
- There is a material risk of the situation deteriorating into a total collapse of the financial system.

For a full-blown banking crisis to emerge, there must be, in addition to idiosyncratic causes specific to a particular bank, significant macroeconomic or pervasive institutional weaknesses present as well.⁴⁶ In a crisis, such systemic causes substantially outweigh in impact the idiosyncratic causes—to which isolated bank failures can be materially attributed. That is, in an isolated bank failure, idiosyncratic causes outweigh in importance any systemic factors that may be present, while in a banking crisis, systemic causes outweigh in importance any idiosyncratic causes that may be present.

Idiosyncratic causes usually come under the heading of poor bank management and pertain to a bank's internal operations and hence to the same matters as systemic causes of the micro type, discussed here.⁴⁷ Depending upon the type of deficiency, they may manifest in a pattern of weak financial fundamentals (e.g., in the form of bad loans or poor solvency). Microeconomic financial fundamentals are discussed briefly in this section but need not be examined in great depth as their significance in enhancing or undermining bank creditworthiness has already been explored in detail in the context of the CAMEL framework.

Systemic causes, of course, refer to problems that face the affected banking system as a whole, and typically arise out of government policies or regulations or relate to economic conditions affected by government policy or exogenous events.⁴⁸ Hence, systemic causes could include, for instance, weak prudential regulations, ineffective supervision and enforcement or shortcomings in a government's economic policy management, (e.g., maintaining a loose monetary policy for too long, setting off an inflationary trend that leads to an asset bubble).

The contributing systemic weaknesses or vulnerabilities, which constitute the underlying causes of the crisis, can be generally grouped into those (1) of the micro type, concerning internal operations and pertaining to management, governance or institutional failures such as ineffective or lack of enforcement; and (2) those of the macro type, concerning the operating environment and pertaining to regulatory deficiencies at the policy level, weaknesses in macroeconomic policy, management or in macroeconomic conditions, or external events. In a banking crisis, there is typically a mix of both categories of contributing factors. As one of the most prominent researchers on the subject put it:

*[T]he origins of crises are both internal and external. To focus myopically on one side only misses the essential point that systemic crises have both macro and micro origins. Almost always and everywhere banking crises are a complex interactive mix of economic, financial and structural weaknesses.*⁴⁹

In the same way that micro and macro factors influence the credit strength of an individual bank, usually both micro and macro factors are critical to the failure of a bank as well as to the eruption of credit crisis.

Micro-Level Factors (Weaknesses in Internal Operations)

As suggested, for purposes of considering endemic internal weaknesses, financial fundamentals are best construed as reflections of more underlying problems. That is, internal weaknesses tend to be eventually revealed in poor numbers, but the weaknesses are not the numbers themselves but the practices—whether acts of commission or acts of omission—that were more direct or proximate causes of the weakness. For instance, high NPLs indicate poor asset quality, which in turn is indicative of correspondingly high levels of loan losses that have destroyed capital, and very possibly have rendered the affected bank or banks insolvent.

Where the poor numbers come to affect the system as a whole, the cause is likely to be systemic, as for example where average NPLs for a banking system reach 15 percent. Concerning an individual bank, however, it may be unclear prospectively (in contrast with the usual retrospective clarity) whether the financial attribute in question has mainly idiosyncratic causes, or whether systemic weaknesses are playing a significant part.

Consider, for example, a bank that has high NPLs in the 15 percent range. Why did the bank suffer loan losses? Some suggestive questions for further inquiry include the following:

- Was it solely the responsibility of management and the deficient underwriting practices they neglected to improve, or were these practices endemic throughout the banking industry and condoned by the regulator?
- Did management have incentives to be more concerned with short-term results than with the asset quality of the bank two years ahead? In this regard, to what extent did shareholder pressure to generate a return on equity equal or superior to other banks' by taking comparable or higher risks play a major part in its decision to tolerate deficient underwriting practices?
- Was the bank awash in cheap funding? If so, what type of business was most attractive in such an environment?

Internal weaknesses commonly associated with bank failure include, among other practices, the following:

- Aggressive lending practices (e.g., purposefully lending to high-risk borrowers, allowing weak underwriting practices) or noncommercial (i.e., politically driven, policy or related party or intragroup) lending
- Ineffective risk management, insufficient attention paid to the recommendations of risk managers, or the absence of an internal credit culture (including insufficient attention paid to risk management issues or a dearth of credit analytical skills or even effective risk management practices that are undermined or circumvented by senior managers)
- Inadequate internal controls and supervision
- Poor disclosure and insufficient transparency.

Unless prevalent among many or most institutions in the system and reasonably attributable to external causes, nearly all of the internal shortcomings of a bank can be attributed to deficient management and weak or incompetent corporate governance.⁵⁰

Naturally, some factors will be outside the control of management, the directors, and the bank's owners. For instance, certain regulations or state intervention may require the bank to make lending decisions that management is well aware are not in the interest of the bank from a financial strength standpoint.

PRESCIENT OPINION FROM A RATING AGENCY

The traditionally tight management of the banks by the government has had its benefits in the past, yet as deregulation continues, many of the underlying problems facing the banks have surfaced and it is clear that these will not be resolved easily. Perhaps one of the most important is the lack of a strong credit culture. . . . Unfortunately, the government-imposed policy lending of the past has left many of the banks (particularly the old, nationwide commercial banks) with heavy bad debt problems (although documenting the magnitude of these problems is very difficult given the absence of meaningful disclosure). Resolution of these asset quality issues is one of the key hurdles facing the banks in the current environment. Other challenging issues include such structural problems as the relatively weak capital positions of most banks, bureaucratic management systems, overcapacity, and rising labor costs.

—Thomson BankWatch, Banking System
Report on Korea, December 1994

Irrespective of the culpability of individual bank managers and owners, when weaknesses of the type enumerated above spread uncontrolled throughout the financial sector and impair the soundness of the banking system contributing to a crisis, government should bear some responsibility. In allocating responsibility, illustrative queries include the following:

- Were regulators complicit in failing to rein in aggressive lending practices?
- What about policymakers? Were they happy to look the other way so long as the government's tax revenues from the financial industry allowed for generous public spending?
- Did the government's and the central bank's economic management policies collectively fuel the asset bubble—the collapse of which brought on the bank's misfortunes—and benefit from public support during the prosperous boom years; or did they attempt to curb the asset bubble?
- Did the government intervene in the banking system in a manner that was deleterious to banks' financial health (e.g., compel banks to engage in policy lending in favored economic sectors)?
- Was corruption or cronyism a significant factor affecting banks' business decisions?

Macro-Level Factors (External Weaknesses)

External weaknesses are those that are outside the control of a bank or its immediate stakeholders and effectively form part of its operating environment. Unsurprisingly,

some of the same reasons that explain the increased frequency of crises over the past several decades can be viewed as broad systemic causes that may be relevant to a particular bank's crisis. Such macro causes include financial liberalization and deregulation, particularly too rapid liberalization and deregulation; the proliferation of new and often complex financial products that are difficult for market participants to understand; the increased dependence on quantitative risk management models that are flawed or misapplied; and the rapid ebb and flow of huge pools of institutional capital throughout the world. In regard to any given banking crisis, the mix of systemic causes will vary with the circumstances, of which the following encompass most of those that are not embodied in events.

Excluding event risk (exogenous catastrophic events or shocks) arising from war, terrorism, or natural disasters, as well as economic and market developments beyond the control of the relevant national government, macro-type weaknesses generally concern government policy and regulation in respect to banks and can be divided into the following categories:⁵¹

- Unsound macroeconomic or regulatory policies, including deficiencies in the prudential regulation of banks or of the corresponding legal framework
- Weak management or execution of policies
- Heavy-handed intervention by government into bank operations on social policy grounds to the detriment of the financial health of the banking industry⁵²

External factors may be somewhat more critical in emerging markets and less developed countries, where growth levels are liable to be high, as compared with mature markets, and where often the local economy is easily buffeted by storms originating overseas in large developed economies. Another reason is that financial markets in developing countries are often less deep and liquid than those in developed countries, with one result being that banks tend to bear the brunt of financing economic growth. It is to be expected that the capacity of such banking systems to cope with and adapt to the high rates of growth of emerging markets will be strained. The same holds true of supervision of the banking system and management of the economy. Here, too, the capacity of government institutions to keep up with the demands placed upon them may be stressed.⁵³

Triggers

Critical exogenous events, such as the collapse of a major financial institution, a stock market crash, a natural disaster, war, or civil disorder, often occur contemporaneously with the onset of a banking or currency crisis and function as triggers, catalyzing other systemic factors into an explosive combination or intensifying a crisis already underway. If one or more of such vulnerabilities is present, and a crisis is ignited, the sequence of events that ensues often follows a similar pattern.

- Those who would have previously accepted credit exposure to banks lose confidence, making them much less willing to do so.
- Banks see sources of funding become costly or dry up entirely. Liquidity is thereby weakened, making the funding of new loans problematic, while margins are squeezed.

- Existing asset quality problems reveal themselves as lending declines. Bad loans turn sour and become a larger percentage of a contracting loan book.
- Profits take a hit from new loan-loss provisioning requirements generated as a result of the souring loans. As these requirements become increasingly difficult to meet from dwindling earnings, bank capital is impaired.

Once the crisis begins to affect the real economy, as it almost inevitably must, a vicious cycle is initiated making the restoration of confidence and bank capital, earnings, asset quality, and liquidity increasingly difficult.⁵⁴

Currency or Financial Crises and Their Relationship to Banking Crises

*[A financial crisis] is a situation where it becomes publicly known that the central bank has run out (or is about to run out) of reserves and could shortly become unable to service the country's foreign-currency debt obligations.*⁵⁵

—Alexandre Lamfalussy⁵⁶

Distinct from a banking crisis, but often associated with, or even also defined as a financial crisis, is a currency crisis, which describes the precipitous decline in country's domestic currency relative to a reserve currency, such as the U.S. dollar or the euro. A currency crisis can also be said to occur when a speculative attack intended to force a devaluation costs the affected country's central bank a substantial sum to avert. The currency crisis often turns into a financial crisis when the country is no longer in a position to defend its currency. The essence of a currency crisis is either a large devaluation of a country's currency, or the threat of such a devaluation that results in substantial expenditure of funds by the central bank. As is a banking crisis, a currency crisis is invariably accompanied by a rapid shift in market sentiment. The manifestation of a full-blown currency crisis will almost inevitably be followed by a destruction of wealth that puts the brakes on the real economy, and it may trigger a banking crisis if one is not already in progress.⁵⁷

There are quite a number of theories explaining the mechanisms behind financial crises. Some of them include contagion (usually but not always purely regional), some include sudden stops (a small loss of confidence results in a deterioration of the current balance leading to further losses of confidence). Many models include a credit boom preceding the loss of confidence.

Naturally, speculative attacks on currencies are a frequent occurrence. Whether it is possible or expected that the country will nationalize the foreign debt exposure of nonstate agents, and whether the country can afford to do so, are also factors.

Typically, however, while a currency crisis can be triggered by a speculative attack, one can also occur because of a general collapse in market confidence.⁵⁸ An adverse change in a country's economic fundamentals will often set the stage for a sharp readjustment in the exchange rate of the local currency.⁵⁹ As an example, the stability of the Thai baht between 1991 and 1996 was followed by a collapse of 45.81 percent of its value in 1997.

The actual crisis will be set in motion "either by a sharp, substantial, and disorderly decline in the exchange rate in one country, frequently, although not always,

from levels set by a fixed . . . exchange rate standard, a crawling peg standard, or a currency board.”⁶⁰

Potential warning signs of a currency crisis include the following.

- Export growth has been declining.
- The local currency shows signs of being overvalued.
- A high rate of import growth is registered, or an increasing proportion of consumer imports.
- A substantial mismatch exists between foreign currency liabilities and foreign currency assets.

Although a currency crisis will tend to make the country’s exports more competitive and the country itself more affordable as a tourist destination, in the near term the plunge in the value of the domestic currency is apt to result in a ballooning in the country’s foreign-currency debt when measured in local terms. Capital flight is also a common phenomenon as those with means seek to preserve the value of their domestic currency assets before devaluation further erodes their value. Neither bodes well for a quick economic recovery.⁶¹

FINANCIAL CRISIS CLASSIFICATION: DEFINITIONS AND TYPES

Some researchers have attempted to divide the two fundamental types of financial crises into various subcategories, or, alternatively, posit a larger number of fundamental types.

The following distinctions are often seen:

- Emerging-market (developing country) banking crises vs. mature-market (developed country) banking crises
- Liquidity-type banking crises (also referred to as bank panics or as self-fulfilling crises) that are spread through contagion and are akin to bank runs vs. fundamental-type crises (also referred to as solvency-type bank crises) where weak bank fundamentals and insolvent banks are a fundamental feature of the crisis-affected banks⁶²
- (Mild) borderline banking crises vs. (serious, full-scale, systemic) banking crises (where the existence of a crisis is defined by reference to additional distinguishing characteristics)⁶³
- Discrete banking crises vs. twin banking crises (i.e., banking crises accompanied by currency crises)
- Bank crises attributable primarily to internal (i.e., endogenous, micro-economic, bank specific) factors⁶⁴ vs. bank crises attributable primarily to external (i.e., exogenous, macroeconomic, systemic) factors⁶⁵

The foregoing efforts to draw distinctions among various categories of bank distress are not necessarily purely academic exercises since they represent an attempt to elucidate the various facets of a complex phenomenon that a banking

crisis constitutes. Nonetheless, these additional subcategories are not essential to the purposes of this chapter and are mentioned here only in passing.

Last, it perhaps should also be mentioned that attempts have been made to define different phases of a banking crisis.⁶⁶ These distinctions are considered in the next chapter in connection with a discussion of bank repair and recapitalization.

Twin Crises

Though a banking crisis and a currency or financial crisis are different phenomena—one being the threat or actual collapse of a country's banks and the other the threat or actual collapse of its currency—the two types of events are related. The occurrence of one can trigger the other, or intensify its severity. For example, where a country's banks have lent large amounts to borrowers using foreign currency funding, they will invariably feel the consequence of a plunge in the local currency. Likewise, a banking crisis will have a negative effect on the market's perception of the country and its currency, thereby putting pressure on its rate of exchange. It is therefore not unusual for banking and currency crises to occur in close time proximity to each other, particularly in emerging markets or in countries whose currency is not a reserve currency or is otherwise vulnerable to speculative attack.

Various researchers have sought to elucidate correlations between financial and banking crises, not just in terms of effects but also in respect of causes. Kaminski and Reinhart seem to have coined the expression twin crises in a seminal work on the topic.⁶⁷

In a presentation entitled "The Anatomy of Financial Crises," Carmen Reinhart showed that there were two primary theories concerning causation: one held that currency crises tend to cause banking crises; the second the reverse.⁶⁸

The first theory holds that a speculative attack on the currency is likely to trigger bank runs. High interest rates imposed to defend against the attack will have a negative impact on the local business environment, making finance more expensive, reducing demand for property and big-ticket consumer goods. Should an asset bubble have built up, a tighter monetary policy may cause it to burst and trigger a banking crisis, which invariably is followed by cutback in bank lending. As the central bank tries to maintain its exchange rate by selling foreign currency and hiking interest rates still further, the already deteriorating conditions are further aggravated resulting in a shortage of funding and liquidity throughout the economy. Companies facing falling demand and a lack of credit default on their obligations and enter bankruptcy. Meanwhile, the country's declining exchange rate catches out those who had borrowed in foreign currency on an unhedged basis. While banks themselves may be fully hedged, as their customers suffer, "the borrowers' exchange rate risk becomes the bank's credit risk."⁶⁹ The solvency of the affected banks is threatened as a result, while the more marginal institutions become particularly vulnerable to bank runs.

The second theory asserts that the high cost of bailing out distressed financial institutions by a central bank erodes its ability to defend against speculative attacks and maintain existing exchange rates, particularly if they are fixed or partially fixed.

As problems in the banking sector become evident, market sentiment is affected. A flight to quality or capital flight may occur as investors move funds into foreign-currency deposits locally or back to safer havens offshore. This flow of funds from domestic-currency to foreign-currency deposits reduces demand for the former and increases demand for the latter, putting further pressure on the exchange rate.

Concerning empirical support as to whether banking crises triggered currency crises or vice versa, one scholar found no apparent link between currency and banking in the 1970s when financial markets were highly regulated and many countries had extensive capital controls. However, in the 1980s and 1990s, after financial sector liberalization and the easing of capital account restrictions, it appeared that banking crises most often began before currency crises.⁷⁰ It is not surprising that the increasing deregulation in financial markets through the 1980s and 1990s revealed linkages between the two types of crises that had previously remained obscured. There is evidence to suggest that “twin” crises, that is, where banking and currency crises occur roughly contemporaneously with each other, are more likely to occur in developing countries than in developed ones.⁷¹

As an example, one of the authors of this book explained in his book on the 1997 Asian Crisis that the temptation to borrow cheap dollars was too much to resist:

*Undeniably, more foreign funds were made available to Thailand than were actually needed or could safely be swallowed. . . . The banks were paying relatively little for their US dollars borrowings, simply because too many dollars were chasing too few opportunities. . . . Unfortunately, the foreign funds so gathered were directed at the wrong investments, owing to the artificial “positive carry” created by the fixed exchange rate. A large chunk of excess foreign lending and investment went to property: Bangkok is lined with half-finished buildings, while many of those that were built in recent years lay empty. . . .*⁷²

It would seem that, given the right combination of circumstances, a currency crisis can trigger a banking crisis and vice versa, and that once catalyzed, the two crises feed off each other in a “mutually reinforcing” manner.⁷³ Whatever the actual causation mechanism, there seems to be good evidence that the two phenomena are related. Consequently, currency vulnerability should be considered as a potential warning signal of a banking crisis. Should a currency crisis ultimately develop after a banking crisis, it will tend to amplify the banking crisis, activating a vicious spiral.

There is a wide range of contradicting opinions as to which type of circumstances is most likely eventually to generate a twin crisis. In 1996, Bernanke, Gertler, and Gilchrist⁷⁴ developed a theory about financial accelerators: Any difficulty faced by the financial sector translates into investment and production problems for the economy as a whole. Borrowers become weaker, risk premiums increase, and their access to financing gets compromised. This, in turn, weakens them further in a vicious circle (it is a *procyclical* phenomenon). Such a phenomenon induces a flight to quality for the financial sector.

As suggested by Gertler, Gilchrist, and Natalucci in 2003,⁷⁵ the financial accelerator theory should be extended to the behavior of governments and foreign

investors or lenders. It is especially so in small open economies and in markets where domestic companies can borrow in foreign currencies (either locally or abroad), as the example of Thailand demonstrates.

A deteriorating perception of the risk by foreign lenders may affect the stability of the country's currency. In turn, this may affect the government's monetary policy, usually in the direction of higher interest rates, which compounds the procyclical phenomenon. Korea in 1997 and 1998 was a typical example.

In many emerging markets, banks are the major sources of financing, simply because capital markets are, for all sorts of reasons, poorly developed. In the midst of a financial crisis, the flow of credit could also be constrained by regulatory or prudential ratios affecting the banks. There are generally two types of such constraints: on liquidity and on capital adequacy. If either of those constraints results in a limitation to the capacity of banks to lend, the country may face a deadly credit crunch. In a credit crunch, even strong, legitimate borrowers are denied full access to credit.

As Bliss and Kaufman⁷⁶ rightly suggest, the strength of the banking sector is therefore a primary concern. One might argue that the government has the option to reduce prudential constraints in order for the banks to accommodate the asset misallocation required by the market. There are two reasons why such an approach is dangerous. One, in a relatively open economy, international market forces will sanction banks that post deteriorating ratios. Two, the fragility of the banks might increase to the point where a bank crisis erupts.

Kaminski and Reinhart⁷⁷ developed an interesting model looking at twin crises. The model covers 20 countries over 25 years (from 1970 to 1995) that had relatively small, open, economies, with an exchange rate that was fixed or in a narrow band, or under a crawling peg. The countries suffered 26 banking crises and 76 financial crises. The model uses warning flags and the probabilities of a crisis associated with them.

Kaminski and Reinhart offer several thoughtful conclusions:

- Bank crises usually take place before financial crises but bank crises reach their highest point after financial crises.
- Financial liberalization (in particular anything that translates into excessive credit growth) plays a role in twin crises, and particularly in bank crises.
- Twin crises are more devastating because of cross-feeding between the two types of crises.
- Economic fundamentals are deteriorating in periods preceding twin crises.

Not surprisingly, Chang and Velasco⁷⁸ claim that bank crises take place before financial crises whenever the central bank commits to help domestic banks meet foreign obligations under a fixed currency regime.

The other way around, Mishkin⁷⁹ uses Mexico's example in 1994 to show how a financial crisis can precipitate a bank crisis through both a lack of confidence and the mechanical effect of the foreign exchange positions of the banks (or their borrowers).

Many observers, among them Goldfajn and Valdes,⁸⁰ agree that the two types of crises are in fact so intimately linked that they always feed each other.

ASIAN CRISIS OF 1997–1999: CURRENCY CRISIS OR BANKING CRISIS (OR BOTH)?

The Asian crisis of 1997–1999 arguably began primarily as a currency crisis, with an attack upon and the subsequent devaluation of the Thai baht. In Thailand, however, problems in the financial sector had already become apparent before the currency crisis, and a full-fledged banking crisis might have ultimately developed in any event. Certainly, the attack against the local currency added fuel to the fire and led in short order to a full-fledged banking crisis in Thailand. Through contagion, the process replicated itself in other Asian countries. In the countries hardest hit, banking crises were accompanied by currency devaluations.

More fundamentally, the Asian banking systems exhibited various levels of strengths (or rather weaknesses) as follows:

- All banking systems would have, at one point, met with serious difficulties, but none of them (except in Thailand) was in short-term danger, in that counterparty risk on financial institutions would have been high on a 180-day horizon.
- Korean banks were in deep trouble, but not to the point of creating a financial crisis, thanks to a wealthy economy.
- Banks in Hong Kong and the Philippines were filthy rich, thanks to wide interest margins (and in the case of Hong Kong, as well as Singapore, to high productivity ratios).
- Taiwan banks were in trouble, but protected by a rich government. Indonesian banks were doing relatively well (with serious potential flaws, though).
- In India and China, the state ownership of the banks, as well as exchange controls, protected the weak banking systems from any trouble.

The Asian banking systems simply could not cope with Asia's extraordinary growth rate, a situation compounded by the fact that bank intermediation—as opposed to other avenues—remained the largest source of financing.

Various suggestions are offered by researchers to limit the twin crises:

- To approach cautiously the elimination of exchange controls when they exist
- To limit foreign currency borrowings from abroad by domestic banks
- To keep an eye on the current account deficit
- To reduce the economy's reliance on bank credit
- To reduce bank failures in the midst of a crisis
- To enforce strong controls on bank capital adequacy
- To implement risk management controls in banks and in large firms
- To gather fiscal surpluses (a war chest) meant to finance potential bank bailouts

Having surveyed the causes of banking crises, the stage has been set to examine in greater detail both these causes and the leading indicators that register their presence. It is a short step from understanding how banking crises arise to seeing the advantages in using that understanding to attempt to forecast episodes of bank distress. Such techniques for predicting bank crises would obviously be of immense benefit to regulators, depositors, investors, and counterparties, among others, assuming, of course, that forecasts could be generated with a reasonable degree of accuracy.

Credit Cycles, Symptoms and Features of a Prototypical Banking Crisis

Panics do not destroy capital; they merely reveal the extent to which it has been previously destroyed by its betrayal into hopelessly unproductive works.

—John Stuart Mill⁸¹

In its classic stylized description, a banking crisis is often, but not always, preceded by a protracted period of economic expansion.

Although the nature, severity, and duration of credit cycles remain difficult to predict, recognizable phases in the sequence have been identified. Usually a business or economic cycle is divided into four stages:

1. Recovery and expansion
2. Growth and prosperity
3. Contraction
4. Recession

The term *credit cycle* refers to the recurring changes in a bank's operations that appear to correspond closely to the economic changes. The key parameters of the credit cycle are asset growth (mainly loan assets relative to other variables) and asset quality as reflected in:

- The proportion of NPLs and their coverage by loan loss reserves and equity
- The risk profile of banks' loan portfolios
- The tightness or laxity of banks' lending policies

What differentiates the extreme boom–bust-and-banking crisis cycle from the textbook procession of the ordinary business cycle is the emergence of problems in the banking system, including the real possibility of a complete breakdown of the banking system. If this happens, timely recovery from the preceding economic contraction may become practically impossible, unless the state intervenes.

To put the features of financial distress described above into the context of a banking crisis, the dynamics of the boom and bust phases are described in some detail over the next several pages, after which the potential warning signs of a potential banking crisis are discussed.

We could start our exploration at any point of the economic cycle, but we shall begin with the most enticing stage.

Recovery and Expansion

During the recovery phase of the economic cycle, improving business conditions lead firms to expand. New workers are hired to respond to increasing orders, and plans are made to enlarge production capacity. Banks, flush with deposits and sensing the more salutary business climate, resume lending, which had slowed during the contraction that preceded a reawakening of demand. The growth of credit and the accompanying new investment help fund economic expansion. Additional employment of staff expanding productive facilities leads to more consumer spending, which leads to more orders for retail and wholesale goods and services, which leads to further growth in a virtuous cycle.

Amid a burgeoning economy and emerging prosperity, companies require yet more financing to fund further expansion. Broadening investment opportunities attract capital looking to benefit from them, resulting in a more general rise in the price of assets. Economic growth and price appreciation creates new capital that facilitates further development and creates an environment in which banks can lend profitably to a wider scope of firms with apparent safety. Through the money multiplier effect, the upsurge in bank credit results in a swelling money supply, which eventually triggers price inflation.

During the recovery phase of the credit cycle, asset quality is generally good-to-excellent across the banking sector, loan-loss provisioning burdens are low to moderate, and banks have a relatively high proportion of liquid assets to loans. In fact, during periods of strong economic growth, nonperforming loans as a percentage of total loans will tend to fall or remain essentially flat, and banks will be inclined to relax their credit standards. The reasons are as follows:

- In the early stage of recovery, demand for credit relative to supply will likely be strong. Hence, banks will at least initially be able to be selective in choosing borrowers, and will be in a position to charge spreads sufficient to adequately provision against expected nonperforming loans. Chastened by the most recent downturn in the business cycle, credit managers will be loath to be too lax in approving marginal loan applications.
- As the recovery gains steam, however, borrowers will, for the most part, be doing good business and generating sufficient cash flow to service their debts. The acceleration in economic growth will be accompanied by a boost in business confidence resulting in stiff competition among providers of finance for the best lending opportunities, which will prompt banks to neglect safe practices.
- The sharp rise in lending will mute NPLs since rapid loan growth will result in problem loans growing slowly in relative terms. In other words, high rates of loan growth will dampen the ratio of NPLs to total loans.⁸² Moreover, as there is normally some delay before new loans go bad—at the very least borrowers will probably be able to service the debt initially out of the cash infused as a result of the loan⁸³—the absolute volume of nonperforming loans will tend to remain artificially constrained for some period of time.

Contraction

Businesses slow production as lower consumer spending shrinks demand for their goods and services. In consequence, some firms collapse while others lay off workers in a bid to stanch the flow of red ink. As they do so, demand for bank lending wanes,

while bad loans swell as investments envisioned with rose-colored glasses fail to make projected returns.

Meanwhile, increasing numbers of unemployed consumers, like their producer counterparts, no longer have the funds to pay off loans assumed in more buoyant times. With aggregate demand falling, borrowers do less business, and generate less cash flow from which to service their debts. Some have severe difficulty keeping current with their interest payments to the banks from which they borrowed.

Nonperforming loans rapidly rise and banks tighten their credit standards for reasons that parallel the opposite behaviors during the expansionary phase.

The Boom-Bust Arc

As shown above, the expansionary phase of a business cycle does not necessarily end with a crisis and may simply lead to a contraction that will, sooner or later, lead to another expansionary phase.

If, however, the expansionary phase ends up in a crisis, the usual business cycle is more prolonged or displays a sharper rate of credit and economic growth and subsequent contraction or both. Prototypically, a gradual increase in prosperity is followed by a period of rising expectations that culminates in the creation of an asset bubble. In the final boom phase, asset prices continue to rise, seemingly unendingly, while investors enter a euphoric state humming the mantra “This time it’s different.”

The change from boom to bust follows no set pattern, although it seems that frequently some event or shock triggers the rapid change in sentiment. The catalyst, for example, might be a news report concerning problems at a single bank, perhaps a somewhat marginal institution, or the release of an unexpectedly high or low quarterly indicator. It could be disappointing results at a key bank, or unexpected events affecting other sectors to which banks have some exposure. Similarly, the impact may be immediate, causing investors and depositors anxiety and unleashing the characteristic panic, or, alternatively, the transition into an economic downturn may be gradual and cumulative. Whichever scenario occurs, it will not be long before the disagreeable effects of the crisis start to be felt by the banks in the form of loan losses, reduced cash flow and earnings, and diminished capital, and then throughout the economy.

Once the crisis is underway, banks, companies, consumers, and ultimately the government attempt to cope with the new market climate, and tackle the damage to the banking system and the misallocation of capital that has caused it.

There could be a flight to quality resulting in a shift of deposits to those banks perceived as the most creditworthy, in particular foreign banks. Inevitably, depositors would favor the more prudently run banks as a safe haven for their funds. Meanwhile, banks that accept deposits and investors themselves would seek refuge in holdings of comparatively low-risk government securities.

Prevailing interest rates, though they may be nominally low, may still be high in real terms as inflation diminishes. Depressed demand may even cause price deflation to take hold. Less money circulating through the economy would keep a downward pressure on prices.

Banks, facing shortfalls in liquidity and capital, invariably cut back on the supply of credit. Moves to restrict lending, however, while an understandable consequence of banks’ desire to remain liquid and clean up their balance sheets, are apt to have an adverse impact on the economy and may set off a chain reaction that combines with

falling demand to drive the economy into a downturn. The government meanwhile will urge banks to resume lending to kick-start the economy while warning them against resuming their old profligate ways. Banks are caught in a double bind, albeit one in large part of their own making.

Boom

As explained, the boom that precedes the bust classically begins after some period of steady recovery as the economy emerges from the last contraction. As years pass, memories of previous cyclical downturns fade and seem ever less relevant to present circumstances. Driven perhaps by the thinning margins achievable in their traditional franchises, banks look for new ways to boost profits. The more pioneering among them find unexploited, or long-neglected, banking franchises that can generate above-average margins, perhaps by marketing newly available financial products or gaining a comparative advantage by using the latest technologies.

Not infrequently an innovation or change in the real economy creates new investment opportunities that impel an expansionary cycle. Among the many possibilities, the innovation might be an emerging technology, a new trend, a natural resource discovery, rising demand for an existing agricultural product, or demographic change. Whatever it is, banks will seek to generate new business from the new economic activity that flows from the developing industry. This could, for instance, include arranging financing to the boom industry, providing credit to the industry's suppliers, or servicing through private banking products the beneficiaries of the new wealth that radiates from the expanding sector.⁸⁴

Whether a new franchise is found or an existing one is revitalized, it initially represents an actual opportunity for profit and growth. Should the innovation prove successful, whether it was developed by the bank or exploited by it, its peers will move as a herd into what has become a proven business niche. Irrespective of the original impetus that got the boom rolling, as the expansion gains momentum, its distinguishing characteristic is explosive credit growth. In addition to pent-up demand from the past economic contraction, the needs of existing industries, and that being generated by the bank's own marketing efforts of its new product or to the new sector, will see year-on-year growth in advances swell at a prodigious rate.

Initially, credit proceeds will be directed primarily toward productive investment. But as the boom approaches its peak, much will be diverted into speculation or consumption. Investors sitting on unrealized gains feel richer and are more willing to spend on all manner of goods and services increasing aggregate demand, which banks may foolishly support.⁸⁵ Although the bulk of the flow of capital unleashed initially will pour into the most rapidly expanding industries, other parts of the economy will benefit as well. More often than not, the wealth created by the favored industry eventually finds its way into the equity and property markets. As underlying markets heat up, margin lending on share purchases, the sale of investment products, and mortgage lending may become highly attractive businesses in their own right.

Overall, the expansion in credit will characteristically be preceded or accompanied by one or more of the following external changes:

- The relaxation of lending standards
- The lowering of regulatory standards

- The loosening of domestic monetary policy or the liberalization of the domestic market to allow greater inflows of foreign capital—so-called hot money

Such changes will naturally boost loan growth, further feeding the boom. In their quest for revenue growth and under pressure from shareholders, even skeptical managers in conservative banks may feel they have little choice but to go along to get along.⁸⁶

As a result, underwriting standards are loosened even more, and credit becomes still cheaper for borrowers in real terms and becomes increasingly available. Both equity and property will therefore tend to register a strong and steady rise in prices. Almost unwittingly, banks may cross a threshold beyond which they are not generating sufficient returns to cover their credit costs, although this may not be perceived prospectively. Meanwhile, benefiting from underpriced credit, borrowers continue to borrow, and funds keep flowing into purchases of equity securities and property, as well as consumption. More optimistic assumptions are now accepted as plausible, and business confidence levels remain high. New players are drawn into the market, including among them those who had previously avoided such endeavors. The traditional principles of valuation are thrown out the window, and the new entrants eschew recognized methods of analysis for tips and recommendations, not infrequently based on dubious metrics.

Precrisis Situation

Proverbially, hindsight is better than foresight, and perhaps nowhere is this more evident than after the emergence of a banking crisis. As a country's banks descend into a state of profound distress, the problems that triggered the downward spiral consistently appear almost self-evident. Almost always, they can be traced back to the loose lending practices that prevailed during the euphoric period that preceded the bust—at least in a boom-bust banking crisis. What was obscured at the time becomes crystal-clear after the fact.

It could be said that there is nothing intrinsically wrong with a boom except its excesses. But then excess practically defines the phenomenon, and distinguishes a boom from a period of sustainable growth and prosperity. Even as an asset bubble continues to build and the crisis to come remains in a latent state, hints of trouble will already be surfacing. They can be found in signs that expectations have far overshot the constraints of economic reality and that the asset quality of banks is vulnerable to rapid deterioration. Indications that a bubble is still building, evidenced in relative price ratios reaching or exceeding historic highs, is probably the best harbinger of increased vulnerability to a banking crisis. Nevertheless, this is easier said than done, because predicting the timing of a crisis is almost impossible to get right, and it is even harder to know when the bust has bottomed out. Moreover, we should insist, not every bubble leads to severe problems in the banking sector.

Those who predict a correction too early are derided as inveterate naysayers, and their contrarian views tend to be ridiculed or at least taken with a grain of salt. The most trenchant commentator may be given the sobriquet Doctor Doom or something similar, and complacency continues to reign. Even when the first signs of trouble appear, the predominant stance of market professionals may be one of denial. Even where fundamentals could be better, there may be very good rational explanations concerning why a banking crisis is unlikely and would be an irrational response to

the situation at hand—even if it is an understandable response in a particular narrow segment of the market or in another country. Banks continue to finance transactions that depend upon increasingly rosy expectations of profitability. These late-stage transactions can succeed only as long as prices keep rising at a rapid clip and consumer spending keeps pace. Hence, current positive trends are extrapolated well into the future, while the possibility of major reverses is discounted.

So long as the boom maintains momentum, a self-perpetuating dynamic continues to operate until some unknown threshold is reached. At that point, asset prices have deviated so far from historical norms—as represented for example in the price-earnings ratio of a bundle of company stocks or in average house prices relative to average disposable income—that a correction becomes inevitable.

The boom by now, at least in the eyes of the most experienced and clear-eyed participants and observers, has become a speculative bubble. Among the more hard-headed disbelievers, the *greater fool theory* starts to take hold. No longer believing that valuations are defensible but envying the profits made by both the sophisticated and the simple-minded, they buy only with the intention of unloading the asset to a greater fool once an incremental gain is achieved.⁸⁷

In the case of equities and property, either the volume of continued earnings required to support the price of the securities goes beyond any realistic possibility of achievement, or the incomes of prospective house purchasers can no longer support asking prices. In either case, a decline in demand becomes inevitable, confidence in ever-rising prices evaporates and the process of correction begins. With no prospect of capital gains through price appreciation, a gradual decline quickly snowballs into a severe fall.

Precrisis: What to Watch For

Characteristic signals of a swollen asset bubble include the following.

- Asset prices in relative terms, such as house prices in comparison to average annual income or stock prices compared to company earnings, will be near or at historical highs. The analyst should keep a lookout for the “irrational exuberance” and extreme sort of excesses typically found toward the peak of an asset bubble.
- Elevated levels or sharp increases in bank intermediation will be reflected in relatively lofty values in the key ratios, such as the following: total-bank-lending-to-GDP; customer-loans-to-deposits, short-term-foreign-funding-to-total-domestic-bank-deposits; and rates of loan growth. At the macro level, however, there may be some evident deterioration in the country’s external position.
- Rising incomes and asset prices coupled with high loan volumes are likely to mask any problems; instead, aggregate bank financials are likely to look good—perhaps too good. Signs of weakening asset quality, capital, and liquidity may be there but not conspicuous. Note that banks that rely heavily on the money markets for funding will be the most vulnerable in a downturn. Toward the top of a bubble, banks often show superficially strong earnings albeit with profitability and margins exhibiting some recent flattening.
- Finally, any recent deregulatory policy initiatives that may have encouraged looser lending policies should be taken into account.

Bust

Ultimately, the bubble bursts, marking the change from boom to bust.

There are archetypal scenarios but no uniform pattern. The three scenarios enumerated below are among the most common.

1. Early and sharp decline. The shift to the crisis may occur quickly in a dramatic fashion, as some event presents an enormous shock to the market—the failure of a major bank, for example.
2. After a period of uncertainty that prevails for several weeks or months, a panic arises, and prices fall precipitously.
3. The contraction phase sets in gradually without a period of panic and an accompanying plunge in prices; problems in the banking industry surface gradually as the recession grows deeper.

As scenario three illustrates, a banking crisis need not coincide with a precipitous fall in asset prices, such as a stock market crash. It is, however, not unusual for a banking crisis to begin with such a period of panic (as scenarios one and two illustrate) that may last for days or weeks, during which key financial markets are characterized by either a steep plunge in prices or an equally sharp widening of spreads. When markets experience such sharp corrections, the onset of the crisis is clearly marked.

Nevertheless, such an event is not a mandatory prelude to a serious banking crisis. When the banking system reaches a threshold level of vulnerability—and the crisis of 2007–2012 demonstrated that banks are vulnerable to many more factors than the traditional bubbles, for example to liquidity issues—a crisis could conceivably surface at any time during the phase of economic contraction. In the same manner, absent such vulnerability, a banking crisis is not inevitable as the business cycle turns after a prolonged expansionary phase. This said, banking is a cyclical industry, and member institutions face much tougher times when the economy is contracting than they do during a period of rising prosperity.

The strong returns of the peak boom period weaken during the transition period.⁸⁸ Less vigorous price rises dampen expectations resulting in flattening or declining loan growth. This is accompanied by higher NPLs and NPL ratios as the expanding denominator no longer can mask growth in the percentage of duff loans. At the same time, rising NPLs will compel the bank to set aside higher levels of loan-loss provisioning to compensate for the decline in asset quality arising from growth in delinquencies. That is, as the bank's credit costs outpace growth in its operating income, such costs will consume an increasing chunk of pretax income.⁸⁹ These charges will increasingly eat into bank earnings.

Since lower growth in lending corresponds to lower growth in operating income, the impact of the bite will be exacerbated. Moreover, since revenue previously generated from newly classified NPLs will no longer accrue and flow through to the bottom line, net income will take an additional hit, resulting not only in lower net operating profit (after provisions) but also in less net income relative to assets (weaker profitability). A prolonged period of poor financial performance can be expected to result until the bank's problem loans are resolved.

A Brief History of the Modern Banking Crisis

Since the late 1970s bank insolvencies have become increasingly common. Where these failures are systemic, they can drain a country's financial, institutional, and policy resources resulting in large losses, misallocated resources, and slower growth.

—Gerard Caprio and Daniela Klingebiel⁹⁰

Systematic banking . . . crises have in fact occurred so frequently, in so many countries, and at such great public cost, that they must be considered a global public policy issue of the first magnitude. They are a sobering reminder that the modern world is far from stable financially.

—Daniel O. Behm⁹¹

At least since banking systems first began to flourish in Europe and North America in the eighteenth and nineteenth centuries, banking crises, or bank panics as they were once commonly called, have been a periodic event in financial history. As reflected in accounts such as Charles Kindleberger's classic text of 1978,⁹² fascination among the public and scholars with such phenomena, and efforts to understand their causes, are nearly at least as old. It was the Great Depression of the 1930s, however, that focused great economic minds on the causes and on the resolution of these phenomena. During the period of extraordinary financial stability that prevailed between 1945 and 1971 under the Bretton Woods scheme, concern about such episodes seems to have receded somewhat, only to well up again after its collapse, particularly from the early 1980s onward.⁹³

Until the events of the crisis that started in mid-2007, outright implosions of high-profile banks in leading industrial countries that resulted in substantial losses to depositors or creditors were, in living memory, comparatively rare.⁹⁴ Notably, the period from the end of World War II until the mid-1970s was one of comparative financial stability in many markets. When Chicago-based Continental Illinois Bank collapsed in 1984, it was a momentous event within the U.S. financial community, and constituted the largest bank failure ever in the United States, a record that held until the demise of Washington Mutual in 2008.⁹⁵ The postwar era of relative calmness in the financial markets coincided with the period during which the fixed exchange rate regime established under the Bretton Woods agreement remained in force (1945–1971). Throughout this period, banks in a good many countries were subject to fairly tight regulations and supervision; within the often strict regimes that prevailed, the financial services market was divided up into separate franchises among various categories of institutions and cross-border capital flows in many countries, especially, in developing countries, were restricted. In addition, the range of financial products of most banks remained comparatively limited and generally uncomplicated, and the corresponding risks were on the whole not difficult to comprehend.

When the Bretton Woods foreign currency regime was unilaterally ended by the United States over a period of several years beginning in 1969, it catalyzed the onset of a movement toward financial liberalization—that is, opening up borders to the free flow of capital, which was accompanied by a move to deregulate domestic financial systems—and the period of relative stability in financial markets ended. Over the succeeding four decades after the breakup of the treaty, the frequency of

banking crises rose substantially. According to an IMF study published in 1996, “A review of the experiences since 1980 of 181 countries reveal[ed] that 133 ha[d] experienced significant banking sector problems at some stage during the past fifteen years.”⁹⁶ In 2008, Luc Laeven and Fabian Valencia wrote a well-documented IMF working paper with a database of systemic bank crises.⁹⁷

Banking crises were of course experienced not only in countries where the banks and the regulatory regimes that supervised them were objectively fragile. Those regulators had understandable difficulties in coping with conditions of rapid economic growth and a corresponding expansion in financial intermediation. The crises also erupted in the most highly developed countries possessing ostensibly the most sophisticated markets and regulators. Although relatively infrequent in any given domestic economy, when viewed from a global vantage point, banking crises have practically become a regular occurrence and more frequently so than during the Bretton Woods era.⁹⁸ Is this a coincidence? If not, why has it occurred?

FINANCIAL INSTABILITY: THE EXCEPTION OR THE RULE?

The past 40 years have seen an increase in both the frequency and severity of financial crises, encompassing banking crises as well as currency and debt crises, as compared to the period spanned by the Bretton Woods regime. Because of the huge economic and social costs such episodes of financial instability impose, the problem of their high incidence has come to be regarded as one of the most serious public policy issues confronting central bankers and other relevant policy makers. In view of their relative rarity during the Bretton Woods era, it is worth posing the following question: “Are frequent banking crises really that exceptional in the broader historical context?” More specifically, in respect to the frequency and severity of financial crises, have the past 40 years been that exceptional compared to the era that preceded Bretton Woods? In other words, was the preternatural stability of the Bretton Woods era the anomaly? Which experience was the exception and which was the rule?

In a paper entitled “Is Our Current International Economic Environment Unusually Crisis Prone?” Michael Bordo and Barry Eichengreen sought to answer these questions. They show that banking crises were actually quite frequent during the period between the World War I and World War II (1918–1938), and moderately frequent during the classical gold standard era that prevailed from the late nineteenth century through the outbreak of World War I.⁹⁹ Although they found that the severity of banking crises during these historical periods was comparable to those in the recent era, they observed that during an age when “the ideology of *laissez faire* and hard money reigned supreme” crises did tend to be shorter in duration since there were few competing policy objectives to interfere with the adjustments compelled by the gold standard. Twin crises were unusual but extremely severe when they did occur.

During the interwar years, the gold exchange standard that replaced the pure gold standard allowed the use of currency reserves in addition to gold, and

relied on a commitment to convertibility. According to Bordo and Eichengreen, this regime “was notoriously crisis-prone” and featured severe banking, currency, and twin crises. They attribute its weakness to a lack of confidence in the scheme and to the increasing financial fragility that prevailed during the period. The Bretton Woods regime implemented after World War II brought about a dramatic change in financial stability. As Bordo and Eichengreen state:

The Bretton Woods System, established in reaction to the problems of the interwar period, placed limits on capital mobility . . . [and] governments created elaborate systems of regulation to reduce risk-taking in the domestic financial sector [buttressed by a] financial safety net in the form of deposit insurance and lenders of last resort. . . . [T]he result was virtually no banking crises for the better part of four decades.¹⁰⁰

The authors concluded that in respect to financial crisis frequency and intensity during the past century and a half, the Bretton Woods era has been the exception.

Financial Deregulation and Liberalization

By introducing a greater degree of currency risk into the global economy, the change from a fixed exchange rate system to a floating rate system was in itself significant. The rescission of the Bretton Woods fixed-exchange-rate system, however, also coincided with the emergence of a change in prevailing thinking concerning appropriate financial regulation policy. Reflecting a more general disillusionment with rigid regulatory structures that hindered competition, innovation, and economic growth, a school of thought that favored a more market-oriented economy was gaining strength at this time.¹⁰¹ The aim was to unleash market forces to revitalize an economy that was stagnating during the 1970s. In the United States, the new approach to policy represented a swing of the pendulum away from the mixed (command and market) economy that had come into during the Great Depression. In the context of the financial sector, the view that the banking industry was too tightly regulated began to take root. By the end of the decade, financial deregulation and liberalization were about to become a major new trend, not just in the United States but globally.¹⁰²

The election of Ronald Reagan as president of the United States in 1980 marked a watershed in regulatory policy.¹⁰³ In the United States, banking reform meant primarily *deregulation*, which had the aim of replacing the regulatory regime put in place during President Franklin Roosevelt’s New Deal in the 1930s with a more flexible market-oriented regulatory regime. Internationally, new thinking manifested in an effort to encourage *financial liberalization*. This meant persuading developing nations to remove capital controls that many had established to insulate themselves from foreign investment, which they understandably often perceived as colonialism in a new guise, and from the vagaries of the financial markets. The goal in the global context was to allow the free flow of capital across national borders, which it was

thought would encourage trade, investment, and economic development, and would contribute to greater economic growth worldwide. Although the movement to deregulate financial markets and liberalize the global flow of capital was probably the strongest in the United States and the UK, few countries were immune from the new economic policy ethos. The theoretical and literal bankruptcy of Marxist-style socialism that had become apparent by the end of the 1980s gave the movement further impetus. By the end of the decade, the movement had become the favored policy approach in most of the nations of the world that aspired to strong economic growth. Its tenets concerning the benefits of financial deregulation and liberalization figuratively were encapsulated in a phrase: the Washington Consensus.

As the new reforms were implemented, their long-touted benefits in fostering economic growth did appear to bear some fruit. Banking system stability, however, did not appear to be among the advantages of the new policy regime.¹⁰⁴ Instead, the opposite phenomenon was observed. By the early 1990s, it seemed apparent that when one financial regulatory system was suddenly replaced with another, the result too often was a currency crisis, a banking crisis, or both a few years later. This occurred not only in developing countries, but also in developed ones. For example, the U.S. S&L crisis was in significant part the result of deregulation of the U.S. financial system, to which dramatic changes in U.S. monetary policy contributed.¹⁰⁵ Financial instability, it appears, was an unintended consequence of bank deregulation and financial liberalization. The correlation was so evident that deregulation and financial liberalization came to be seen as a good leading indicator for financial crises occurring a few years later. The reasons this occurs were well expressed by Christian Larraín in a paper concerning bank restructuring published by the World Bank.

*Financial institutions in recently liberalized financial systems often lack the experience to manage these risks, and, in the face of stronger competition, institutions will tend to be pushed toward riskier investments.*¹⁰⁶

There is now significant agreement among both scholars and practitioners that increased instability in financial systems around the globe was frequently the consequence of the adoption of financial liberalization and deregulation initiatives in too rapid a manner and without adequate preparations.¹⁰⁷ Indeed research findings suggest that there is a strong link between the two, and that liberalization and deregulation have left behind a more turbulent financial industry environment where crises erupted with increasing frequency.

Ironically, though, it is now clear that the unwavering determination of the international community to submit banks to increasingly stringent capital adequacy rules exacerbated if not triggered the debt crisis in Europe. Those rules were needed—and they are arguably insufficient—but banks swiftly moved from reasonable if not excessive compliance to an attitude of sticking to the rules while attempting to game the system.

Hot Money

Contributing to greater financial instability was the rediscovery of emerging markets in the 1980s and early 1990s by global institutional investors. The increasing importance of institutional investment from the 1970s onward came about for

several reasons, of which the deregulation of financial markets was only one. Changes in the pension industry, increasing national wealth, and an aging population in the developed countries, the revival of mutual funds in the United States and elsewhere, an increasing recognition of the benefits of portfolio diversification as supported by financial theory, high rates of growth in some newly industrializing countries, and other factors combined to create and drive a huge pool of institutional investment capital to search of yield and safety through diversification by asset class, industry, geography, and other factors. Significant portions of these assets, which during the Bretton Woods era would likely have remained domestic in character, were now available for global investment.

Beginning in the 1960s, even before the abrogation of Bretton Woods, and particularly in the 1970s as Japan's miracle economy reached a fever pitch, interest in what came to be called emerging markets gained momentum. In the 1980s, banks recycled petrodollars in Latin America and elsewhere, which culminated in that decade's debt crisis and the invention of so-called Brady Bonds used to resolve the obligations incurred. Among institutional fund managers, interest in these developing countries continued to grow. (The term *emerging markets* actually came into extensive use during that decade.) By the mid 1990s, so-called hot money was like an enormous tidal wave of capital sloshing around the world. Unlike a tsunami, however, this wave's damage was not so much when it came in, but when it receded just as suddenly as it had arrived.

As this vast accumulation of capital flowed into and out of these markets as investor appetites and perceptions changed, its destabilizing attributes became more evident. The number of significant national banking crises climbed during the 1980s and into the next decade. These were followed, of course, by a major regional financial crisis—the Asian crisis of 1997–1999. As suggested in an earlier discussion, what began as a currency crisis after the devaluation of the Thai baht, was soon transformed in Thailand into a domestic banking crisis and then into a regional financial crisis. Through contagion, as well as the inherent instability of banking systems generally, it was not long before countries whose economic profiles differed substantially from Thailand's were also in the midst of their own currency and banking crises.¹⁰⁸

The devastating crisis that started in 2007 is just the latest of a series of examples of the impact of contagion on banking and financial systems.

AN EXAMPLE OF CONTAGION: THE ASIAN CRISIS IN THE 1990S

Just after the devaluation of the Thai baht at the beginning of July 1997, the head of the capital markets group of a major bank in Hong Kong issued a report that sought to play down the significance of the event. In his report, he questioned whether the term *devaluation* was really applicable to the Thai central bank's surprise action. Characterizing the bank's allowing the baht to float as a "good move" that was inevitable at some point, the analyst declared that the change of policy was an ineluctable outcome given the underlying volatility that was present in all Asian currency markets. Making passing reference to the U.S. savings and loan (S&L) crisis of the late 1980s, France's

more recent difficulties with Credit Lyonnais, and Japan's ongoing banking-sector problems, he noted wryly that "[w]eak supervision and poor lending practices are not hallmarks of banks in emerging economies alone." Consequently, the analyst stated that he remained cautiously optimistic about Thailand's macro fundamentals, unequivocally stating that "at present, there are no serious sovereign risk considerations." Not only was he sanguine about Thailand's outlook, he was generally upbeat about the prospects for other countries in the region, observing that "each Asian economy must be considered and analyzed separately despite superficial similarities."

The views of that analyst, which sought to minimize the threat of what was to become a major regional financial crisis, were by no means unique. Instead, they reflected the conventional wisdom in Asian financial centers in the weeks after the Thai baht devaluation. Some of the rating agencies, for their part, were equally hopeful, based on Asia's recent decade or more of success in fostering strong export-led growth, financial reforms, and registering apparently strong macroeconomic fundamentals. Not long before the devaluation, IBCA found no serious problems with the finance companies, soon to spark the disaster, while Standard & Poor's held a telephone conference explaining to listeners why Thailand was not like Mexico and therefore not vulnerable to a replication of the 1995 peso crisis.

So, with a few exceptions, the suddenness and magnitude of the Asian crisis did surprise the experts, even those based in the region. On the basis of conventional macroeconomic thinking, a regional crisis seemed to be unjustified. While there were some difficulties in the banking systems of the four countries most affected by the crisis, the macroeconomic fundamentals were on the whole quite favorable, at least by the standards then used to gauge the propensity to a balance of payments crisis. Some readjustments were, of course, expected. But these were expected to occur at different times country by country; the tempest in Thailand would dissipate in due course and similar flare-ups might occur from time to time elsewhere. What was astonishing to those on the ground was the scale and speed of the contagion that transformed what appeared to be a purely national imbroglio into a regional calamity.

A more sympathetic interpretation of events was that the devaluation of the Thai baht was a sign of more fundamental problems not just in Thailand and served as a wakeup call to those who remained complacently enamored of the East Asian economic miracle. As a result, from the other side of the world the anxiety from which the contagion arose was more palpable; at a distance of 8,000 miles or more, it was easier to conflate the problems in one country with other seemingly similar East Asian countries. Whether justified or not, like a bank run, the self-fulfilling nature of such a contagion-driven crisis meant that it was soon in each individual or individual entity's best interest to get off the ship before it foundered. So the crisis spread rapidly through the region, with damaging effect.

Recent Factors Affecting the Financial Systems

The past 15 years have been rich in fresh—and sometimes dramatic—developments. Politicians and traditional bank managers must acquire skills in economic and financial management they never had the opportunity to grow up with. The proverbial rocket scientist now provides them with mathematical solutions they do not grasp.

Financial Innovation and the Failure of Risk Management

A substantial contributing factor to the increasing frequency of crises may be found in the procession of new financial products that has presented banks with a wider range of significant risks. Like icebergs in a foggy sea, the hazards these innovative products pose are often hidden and surprisingly large. To manage the risks generated by these complex products, as well as those generated by the broad array of business activities in which banks engaged, new methods of calculating and hedging against them were developed. The emergence in the late 1980s of a comprehensive and integrated approach to enterprise risk management and sufficient cheap computing power to drive it seemed to herald an evolutionary breakthrough in the capacity to measure and mitigate risk across the entire banking and financial system.¹⁰⁹

These new methods and technologies to manage risk—what has been called the new paradigm in risk management—were accompanied by regulatory initiatives that reinforced their acceptance. Most notably, discussions concerning the revised Basel Accords (Basel II and now Basel III), which began in the mid-1990s, ought to harness the power of these mathematical and statistical techniques that had quickly become ubiquitous among the world's leading financial institutions. Indeed, the models and accompanying theory that made use of these quantitative methods provided the conceptual underpinning for much of the Basel II and Basel III frameworks.

The widespread acceptance and use of the new sophisticated risk management methodologies, backed by the ongoing changes in bank regulation just mentioned, seemed to open the door to a new era of stability in the financial services industry. In this new financial world of risk-sensitive bankers, like-minded regulators, and enlightened central bankers well schooled in the science of economic policy management, it was thought that the risk and frequency of financial crises would be dampened, if not largely eliminated. Some even imagined that the business and credit cycle had been tamed and that severe banking crises had been relegated to history—that in brief the new paradigm of risk management effectively took the risk out of banking.¹¹⁰

Instead, looser regulation and the rapid growth of increasingly complex financial instruments coupled with the misapplication of risk management models and, on occasion, an astonishing disregard by senior management for the risks they were incurring led to a general underpricing of risk. This was especially the case in those countries with highly evolved capital markets. The unfortunate result was the global financial crisis of 2007. Needless to say, this crisis, which is still evolving, has caused the merits of the new paradigm to be called into question.

With the benefit of hindsight, it appears that the stronger risk management framework that has emerged in recent years was flawed in both the manner of its implementation and its interpretation. More critically, it was outpaced by aggressive risk taking. The global financial crisis has dashed the most optimistic hopes of an end to financial crises, much as the rise of Islamic extremism and Western countries'

difficulties in dealing with it have put the lie to the argument that the failure of Marxism heralded an end to history.¹¹¹ It can be expected that adjustments will be made to reflect the lessons learned from this most recent episode of financial turbulence. Nevertheless, it is prudent to expect that the risk of severe banking crises will continue well into the future and that their causes, prevention, and mitigation will remain a concern for some considerable time.

Domestic and International Financial Architectures

Traditionally, it is the domestic financial architecture of each country that inevitably has a major influence on local bank operations. Public policy decisions relevant to the banking industry are reflected in the various extant statutes, regulations, directives, and judicial decisions as well as in the established practices and procedures governing banks and their dealings with counterparties, depositors, and others, including the government. In its entirety, this body of rules, together with the various incentives and disincentives embedded in it, forms the major part of what can be referred to as a country's financial architecture.

A measure of bank intermediation—similar but distinct from the ratio of bank intermediation to GDP—is found in the proportion of total financial intermediation that the banking system provides, as opposed to corporate equity or fixed income. Interestingly, the relative role of bank intermediation in Asia in 1995 was overblown, as Exhibit 14.2 is showing, and this fact largely contributed to the 1997 crisis.¹¹²

Changing conditions, of course, will bring about new policy decisions and rules that reshape the architecture. Accordingly, it is not so much a static edifice but instead it is analogous to a dynamic cityscape, constantly under construction and changing shape in response to new political, economic, social, and policy developments.

Although there is no definite end to the causal links, there is one more level that can be thought of as a figurative last stop in the causes of banking crises that remains to some degree within human control. It is what is sometimes termed the *international financial architecture*, and it refers to the system that structures and informs the

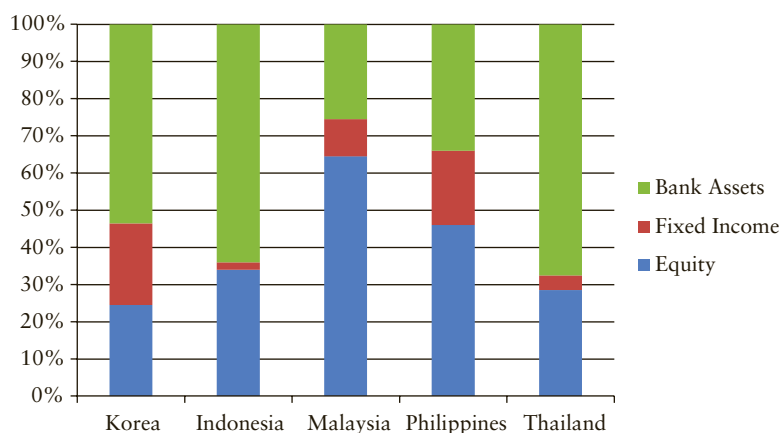


EXHIBIT 14.2 Financial Intermediation in Asia in 1995

movement of capital at the global level. Analogous to a country's domestic financial architecture,¹¹³ the corresponding international financial architecture is relevant to a study of banking crises. Of the few systems that have been used in the past two centuries or so of global commerce, none has been found to be perfect. Roughly speaking, the gold standard prevailed until World War II. For nearly three decades after the war, Bretton Woods governed the system. Since Bretton Woods, the U.S. dollar has taken over from the gold standard and functioned as the world's reserve currency, notwithstanding that it is no longer backed by gold. As both the gold standard and Bretton Woods had their own weaknesses, so too does dollar hegemony. One that has been recognized as potentially contributing to instability in the financial system at the global level is an imbalance in savings and investment vis-à-vis the developed world and the developing world. This topic is discussed further next.

A provocative view of the relationship between the role of currency, specifically the U.S. dollar, and banking crises has been put forward by bank analyst Richard Duncan in his book, *The Dollar Crisis: Causes Consequences Cures*.¹¹⁴ Duncan sets out the hypothesis that the financial instability of the post-Bretton Woods era, reflected in the increased frequency of banking crises, is almost wholly attributable to the "dollar standard" that succeeded the gold standard and the fixed currency regime that prevailed under the Bretton Woods agreement until its collapse in 1971. Although his main point is larger than that addressed in the preceding discussion, his views are relevant to an examination of the causes of banking crises.¹¹⁵ Duncan's basic argument is that due to the dollar's de facto role as a reserve currency, there are no effective constraints on U.S. current account deficits. As a consequence, the global money supply has expanded to inordinate levels over the past two decades, as is evident in the high levels of foreign currency reserves held largely in U.S. dollars by America's major trading partners, including Japan, Korea, and particularly China. Such money supply growth, Duncan contends, has undermined other countries' capacity to conduct an independent economic policy, and has resulted in an inexorable rise in bank intermediation. His approach is encapsulated in his view that "the linkages between capital inflows, accelerating credit expansion, the development of asset price bubbles, followed by systemic banking crises, are so obvious that the burden of proof should be on anyone who argues otherwise."¹¹⁶

Leaving aside Duncan's assertion that the U.S. dollar is at the root of global financial instability, there is certainly a good deal of support for the view that bank credit expansion tends to create asset bubbles that subsequently metamorphose into bank crises. It echoes views expressed by Jim Walker, who was widely credited as one of the few economists to foresee the Asian crisis. Championing the theories of the Austrian School of Economics¹¹⁷ in the midst of the Asian financial turmoil, Walker asserted that undue expansion in bank credit was at the heart of subsequent banking crises.

Based on the literature and experience, it seems evident that there is no single cause, or set of causes, that invariably triggers a banking crisis. Insofar as this book propounds a view on the subject, it is just that. Instead, there are numerous causes and combinations of causes—some evident, some obscure, some more important, some less, some closer to the event, some with roots in the past—that contribute to the emergence of a given crisis. Similarly, some factors will be present in some crises but not in others. An inquiry into the causes of banking crises inevitably raises questions about the nature of causation generally, but this is definitely out of the scope of this book.

EARLY-WARNING SYSTEMS OF FINANCIAL CRISES

There is a general consensus among economists that early-warning systems, no matter how sophisticated, will not be able to forecast crises with a high degree of accuracy. Even economists who construct such models are aware of this, and see these models as no more than useful supplements to more informed country analyses.

—Abdul Abiad¹¹⁸

[I]t is possible to construct simple composite indicators of banking crises that can be useful in assessing the risk of future financial distress with a reasonable degree of confidence . . . [but] such indicators of financial imbalances should at best be used as one element in a more thorough assessment of vulnerabilities, and never as substitutes for it.

—Claudio Borio and Philip Lowe, “Assessing the Risk of Banking Crises”¹¹⁹

The rise in financial instability over the better part of the past four decades (and particularly in the past two), as observed earlier in this chapter, has inspired some scholars and researchers to attempt to identify the principal causes of currency and banking crises. As a result, a large volume of scholarship concerning the causes and prediction of bank crises exists, much of it of recent vintage.

Going further, some researchers—many of whom have been associated with multilateral institutions such as the World Bank and IMF, or regulatory agencies—have sought to develop models that seek to predict these unfortunate events.¹²⁰ The fruits of their efforts are often termed early-warning systems. Attempts to develop such systems date back at least the 1970s. Interest in early warning systems was given a boost, however, by the various episodes of financial turmoil that occurred during the next three decades, culminating with the current crisis. The Asian crisis of the end of the 1990s was a serious hint at potential collateral damage from financial imbalances. As the authors of one survey of early warning models, James Bell and Darren Pain, observed:

*The serious disruption to financial markets and banking systems in south-east Asia in the late 1990s provided a timely reminder of the economic and social costs that can be associated with periods of financial, and in particular banking, instability. It has also stimulated a significant body of research on potential early warning indicators of banking crises.*¹²¹

Typically, the process of constructing a prototype for an early warning system begins with the identification of criteria that are thought to have some correlation with the events that the model seeks to predict suggesting a possible causal relationship. Various statistical and quantitative methods are then used to ascertain the degree of correlation in an effort to evaluate the predictive quality of the variables, often by using an historical dataset for testing purposes.

Early warning systems designed to predict banking crises mainly employ macroeconomic indicators, aggregated or averaged microeconomic indicators (e.g., average bank financial ratios), and related institutional data pertaining to the institution's size,

type of ownership, franchise, and related items. Market price data may also be used. While qualitative inputs are sometimes made use of, typically by using a scorecard, most models rely primarily upon quantitative data. Of course, quantitative inputs frequently serve as proxies for qualitative criteria, and the models are sometimes designed with the intention that qualitative reviews will function as a check upon model results.¹²²

To put the development and use of early warning systems in context, it is worth noting that providing a warning of conditions posing an elevated risk of bank failure is at the heart of bank credit analysis. Although most such analysis is targeted toward individual banks, the analyst cannot ignore the probability of a systemic bank failure when preparing a particular bank credit evaluation since most bank collapses occur during a crisis. Consequently, when weak CAMEL aggregates and relevant macro-economic indicators suggest an increased systemic risk for an entire banking system, it should be so noted by the analyst. This said, it is probably fair to say that the incorporation of an explicit and systematic approach to forecasting the probability of banking crisis as a core part of the conventional analytical process has traditionally been the exception rather than the rule.¹²³

The potential benefits of being able to predict banking crises with some degree of accuracy, however, make the concept of an early warning system an entrancing possibility that deserves a closer look. It is understandable that regulators would have a keen interest in such systems. In fact, the notion of the devastating effect of correlation is given much respect in the various attempts being made to prevent the contagion of distress. There is also, of course, a commercial interest in any methodology that would potentially enable market participants to better manage their own risks or their investment selections and perhaps gain a comparative advantage as a result. This section therefore surveys the range of approaches, largely focusing on a single example of an early warning system methodology for illustrative purposes.¹²⁴

Early Warning Systems as Predictive Models

An early warning system, as the preceding introduction suggests, is a predictive model designed to forecast with some reasonable degree of accuracy the probability of the onset of a financial crisis within a given time horizon. These systems have been devised with the goal of predicting either a currency crisis, banking crisis, or both, as defined by the underlying objectives and corresponding methodology. In addition, early warning systems may attempt to forecast, to some extent, the timing, duration, and severity of the expected period of the specified type of crisis.¹²⁵

It should be noted that the term *early warning system* has also been used to refer to similar systems designed to refer to the prediction of individual bank failures.¹²⁶ Indeed, as described in 2003 by Jagtiani, Kolari, Lemieux, and Hwan Shin in an article entitled “Early warning models for bank supervision: Simpler could be better,” early warning systems appear to have been first adopted by bank regulators for this very purpose:

Over the past two decades, various off-site monitoring systems have been created to identify developing financial problems at banking institutions. . . . Supervisors use the output from these monitoring or early warning system (EWS) models to determine which organizations need increased supervisory scrutiny, identify specific areas of concern, accelerate on-site examinations

*of institutions showing financial deterioration, and allocate more experienced or more specialized examiners to institutions with financial problems.*¹²⁷

Although the term is still used in this manner, as generally applied in this book, however, “early warning system” unless otherwise indicated is intended to refer to an empirically tested model that attempts to forecast financial crises, that is, a period of severe financial instability, as opposed to predicting individual bank distress or failure.¹²⁸ To avoid confusion, the latter are classified here as probability-of-default models, a term that is largely interchangeable with default prediction models, expected loss models, or rating models, depending upon the context. Related to early warning systems are financial fragility indicators (banking system fragility) and macro prudential indicators, a term that usually refers to any metrics believed to have a potential link with vulnerability to a crisis.

Although there is some similarity between the probability of bank default (PD) models and early warning systems, owing to their focus on the potential for individual bank failures irrespective of whether a crisis occurs, PD models typically rely more heavily on microeconomic and market pricing data than do early warning systems. Crisis prediction models, in contrast, rely mainly on macroeconomic indicators or microeconomic data that have been aggregated on an industry-wide basis, and thereby transformed into macro data. Again to avoid confusion, it should be noted that some early warning systems use mainly microeconomic inputs, such as average capital adequacy ratios (CARs) for instance, and therefore closely resemble some PD models.

Measures of Financial System Fragility

Methodologies to measure banking system fragility have been historically used by rating agencies, central banks, and others to gauge the vulnerability of such systems to instability. Financial system strength, of course, is merely the same attribute viewed from the opposite perspective. Like early warning systems, financial fragility assessment methodologies employ both quantitative data inputs and qualitative judgments to grade the potential for a systemic crisis arising. In terms of complexity, financial fragility assessment methodologies might be placed midway between ad hoc assessments made by analysts from time to time and the more sophisticated quantitative constructions and empirical testing involved in the development of early warning systems.¹²⁹ Since financial system fragility is another way of expressing a banking system’s vulnerability to systemic risk, and since systemic risk is in essence the risk of a banking crisis, all three approaches—financial fragility methodologies, analyst-driven approaches, and early warning systems—aim to evaluate what is essentially the same risk.

Most of the banking system fragility indicators used by practitioners draw upon the research conducted by the likes of Morris Goldstein, Philip Turner, Patrick Honahan, Graciela Kaminsky, and Carmen Reinhart, all of whom had sought to identify critical macro and micro indicators that might offer advance warning of an impending crisis and went on to become among the leading lights in the field of bank crisis research and early warning systems.

Using their core research findings, it has been possible to establish a number of tools that bank credit analysts can use to supplement their findings on the credit-worthiness of individual banks, by adding systemic or external factors.

The same tools are now increasingly used in the construction of models that seek to predict default or crises without having to consider every single factor that may have a close or a distant impact on the result.

A reasonable balance between complexity and accuracy must be found. At the end, the models share some flaws as follows:

- They have to assume that the same causes produce the same effects over time and in changing social, financial, and political environments.
- They neglect factors that other models incorporate, meaning that, unless there is redundancy, some of them are flawed.
- They do not always demonstrate plausible causation. Can marriage be counted as a predictor of divorce simply because all divorces seem to be preceded by a marriage?

Models end up attempting to demonstrate that there is an apparent causal link between the observed factors and the risk, whether individual or systemic. Back testing scores high on a regulator's list, but good models are those that adjust to changing circumstances, and banking and finance are changing faster than ever. Back testing an updated model carries little meaning. Nevertheless, models are useful, and they need ingredients or inputs.

The concept of early warning system, which uses the same ingredients or inputs without the kind of accuracy demanded by regulators for, say, compliance under Basel II or Basel III, works quite well as an elaborate set of red flags. We shall examine some of them in the context of crises. Red flags or their equivalent were discussed in our earlier chapters around the CAMEL analysis of single banks.

Survey of Early Warning Systems

Most, if not all, early warning systems are based on the monitoring of a set of indicators to predict crisis-fertile conditions. They differ according to the statistical and quantitative methods employed, the definitions of crisis events, the number and type of indicators, and the specific criteria chosen. The universe of early warning systems can be navigated more easily by understanding the basic categories into which each model falls. In parallel with the division in the literature concerning the primary causes of banking crises, which usually focused either macro or micro factors, early warning systems tend to fall into the same two basic categories. That is, early warning systems can be classified into those that focus (1) on microeconomic criteria and (2) those that focus on macroeconomic criteria.

An excellent survey of indicators of financial crises can be found in an article¹³⁰ written by specialists Graciela Kaminsky and Carmen Reinhart with Saul Lizondo.

The survey of early warning systems by James Bell and Darren Pain, published in 2000 by the Bank of England,¹³¹ one of the first such surveys, provides a good overview of the state of the art in early warning systems at the start of the millennium. In the article, Bell and Pain pose the threshold question about whether systemic banking crises are predictable.¹³²

Supporting unpredictability is the myriad of potential causes of a banking crisis, and by implication the complex interactions among such criteria and the possibility that banking crises arise as a result of pure or random contagion. These scholars

observe, however, that empirical research indicates that pure contagion is quite rare. If most crises are not random in nature, they are, by implication, possibly predictable to some extent.

Based on the research literature, Bell and Pain suggest that both micro- and macrocriteria can be useful in predicting bank failures and bank crises. In respect to microcriteria, the link between financial fundamentals and vulnerability to insolvency is inferred; with regard to macrocriteria, they note that logical linkages between such criteria and bank behavior—for example, low real interest rates providing an incentive for banks to expand lending to excessive levels (associated with deteriorating asset quality)—suggest that they could be helpful in predicting bank crises. This, of course, is particularly ominous considering what brought to us the crisis that started in 2007.

It is not uncommon to combine micro and macro indicators in the same model. This is understandable since, as has been suggested, macro factors seem to play a part in allowing a banking or currency crisis to take hold, or in catalyzing a crisis when bank fundamentals are weak. One study showed that “the introduction of macroeconomic variables significantly improves the explanatory power of models based on micro prudential indicators only.”¹³³ It might be hypothesized that micro models require additional macro input to be reasonably effective in forecasting systemic banking crises. As the authors of a research paper for the Central Bank of Chile put it, “[P]eer group analysis can detect outlier banks, but fails to detect systemic problems . . . [their use] for predicting [such problems] is limited, but the extensive analysis they allow are a natural complement of an EWS [Early Warning System].”¹³⁴

A BASIC EARLY WARNING SYSTEM FOR BANKING CRISIS PREDICTION

A fairly uncomplicated example of an early warning system focusing on banking crises is one developed by researchers Claudio Borio and Philip Lowe. Using the existing literature on the causes of bank crises as a guide to potential indicators, Borio and Lowe came to the conclusion that a fairly “small set of variables” should be adequate to cover the principal conditions necessary for a breakdown in the banking system to occur.¹³⁵ As proxies for the phenomena of excessive lending, the buildup of an asset bubble, and external imbalances, they settled on three key indicators—namely:

1. The bank intermediation ratio (private sector credit/GDP) as an excessive lending proxy
2. Equity prices adjusted for inflation as an asset bubble proxy
3. The real exchange rate as an external imbalance proxy

Should any of several combinations of these three variables diverge substantially from their normal range, this signaled a warning of ensuing financial

distress. Borio and Lowe found the following principal combinations to be the most predictive of a crisis occurring:

- Excessive lending and an asset bubble
- Excessive lending plus external imbalances
- Or all three indicators being in the red zone

Despite the simplicity of the model, the researchers asserted the model to be predictive of past crises based on nearly 40 years of historical data.

In terms of the quantitative methods applied, Bell and Pain divide early warning systems into those that use the *signalling* approach and those that used the *qualitative response model*.

The signalling approach compares indicator values in periods of comparative tranquility with corresponding values in identified periods of crisis. Indicators are selected on the basis of whether changes in their behavior between normal times and periods of crisis provide a reliable signal of a crisis. An indicator is said to signal an impending crisis if that indicator crosses some designated threshold value. Either an actual crisis subsequently occurs within a designated time horizon, or it is deemed a false alarm. Therefore, the threshold must be chosen so that it must strike a balance between having many false alarms (type II errors) and the risk of missing many actual crises (type I errors). This is called minimizing the noise-to-signal ratio.

The qualitative response approach uses statistical regression techniques to identify selected critical indicators. Based on the pattern, direction, and extent of changes in their values, a higher or lower probability of a bank crisis occurring is indicated.

Subsequent contributions to the debate—many of them very thoughtful—were provided by a number of researchers: among them, contributions from Hanschel and Monnin in 2005;¹³⁶ from King, Nuxoll, and Yeager in 2006;¹³⁷ from Hendricks, Kambhu, and Mosser in 2007;¹³⁸ and from Reinhart and Rogoff in 2009.¹³⁹

As can be expected, there cannot be a wide consensus on which specific set of variables is the most predictive. Even under the assumption that reaching such consensus would make sense, the reality is that the appropriate set of variables—not to mention the appropriate weight given to each of them—would shift over time as macro and micro conditions evolve.

Micro Models

Micro models mainly use financial fundamentals in the form of ratios of the type employed in the CAMEL methodology, as discussed earlier in the book, to gauge credit risk, market risk, liquidity risk, and solvency risk. Additionally, these models may use other relevant individualized or aggregate data relating to bank operations, ownership, and bank securities prices. A number of studies have shown that CAMEL indicators do appear to have significant predictive value with respect to individual bank failure. For example, a Federal Reserve Bank of New York research paper written by A. Estrella, S. Park, and S. Peristiani found that “capital ratios . . . scaled

by total assets, risk-weighted assets and gross revenues . . . [were] very significant predictors of failure.”¹⁴⁰ Likewise, a 2005 study written by M. Arena and published by the Bank of Canada found that, for crisis-hit East Asia and Latin America banks, institution-level “fundamentals not only significantly affect the likelihood of bank failure, but also explain a high proportion of the likelihood of failure for failed banks (between 50 and 60 percent).” The study acknowledged, however, that bank ratios alone were insufficient to account for cross-country differences in crisis outcomes and that macroeconomic inputs were important for this purpose.¹⁴¹

Not all researchers have achieved good results with micro criteria. For instance, Goldstein, Kaminsky, and Reinhart, in their early warning system model,¹⁴² determined that the use of micro indicators, such as the CAR, net profits to operating income, and operating costs to assets is not particularly predictive. It may be conjectured that the choice of micro indicators is critical in this regard, and there are plausible reasons why these ratios were not effective for model purposes.

AN ILLUSTRATIVE MICRO MODEL

An illustrative micro model used for research purposes was developed by González-Hermosillo in 1999. It is of interest both as an example of a micro model, and because it provides some empirical confirmation of the validity of the CAMEL model. The González-Hermosillo approach¹⁴³ employed proxy indicators to account for credit risk, market risk, solvency risk, and liquidity risk while also taking account of macroeconomic criteria, including the bank intermediation ratio, and policy variables, such as government support, that potentially generate moral hazard.

The research found that, prior to failure, nonperforming loans typically increase while equity capital ratios (i.e., solvency) decline quickly. In addition, the research found that the higher the percentage of a bank’s assets held in liquid assets, the lower the probability of failure and somewhat less definitively the higher the percentage of property-related lending, the higher the probability of failure. Other criteria displayed mixed results, reflecting the fact that there could be alternative explanations for higher or lower values. For example, high loan margins could imply either appropriate loan pricing or overly aggressive risk taking.

Macro Models

Macro-based models are at the core of the most common type of early warning systems, which is perhaps unsurprising in light of the foregoing discussion (i.e., that macro factors appear to constitute the essential additional elements that will allow a severe systemic crisis to develop in a banking system where the financial condition of a significant number of institutions has deteriorated). The macro-based approach is explored in the following subsection.

Design Issues in Macro Models

In addition to the choice of statistical methods to be used, developers of macro models must address similar issues in the process of constructing, refining, and testing the model. Among these are:

- What objectively defines a banking crisis?
- What time horizon is to be used, and how far ahead is the model designed to provide predictive results?
- What data sample should be used?
- To what extent is the selection of indicators constrained by data availability?
- What indicators are the best candidates for optimal predictive power?
- Can the use of these indicators be explained by logical causal relationships?
- Should emerging markets be treated separately from developed markets?
- To what extent, if any, should financial institutions other than banks be considered in model development?
- In models that use the signaling approach, how should the balance between Type I and Type II errors be optimized?
- How is the model intended to be used, and how will that affect its design?

DEFINITIONS OF A BANKING CRISIS FOR EARLY WARNING SYSTEM PURPOSES

For the purpose of developing a model for an early warning system, a more forward-looking definition that relies mainly on objective or quantitative criteria is required. What are called for in particular are definitional criteria, the data for which is readily accessible. The definition below, used in one early warning model, provides an illustration. In this definition, a banking crisis exists if one or more of the following tests are met:

- The average NPL ratio for the banking sector is greater than 10 percent.
- The cost of rescuing the affected banks would amount to a least 2 percent of GDP (the cost being measured as the capital destroyed by impaired assets).
- The putative crisis was characterized by pervasive bank runs, or the government rescue.

Note that of the three basic criteria, the first two can be formulated in numerical terms, while the last is subjective in nature. Additional guidance is provided, however, to assist in the determination as to whether it is met in a given instance. Government rescue is further defined to include one or more of the following types of response measures:

- The issuance of government blanket guarantees or similar emergency actions¹⁴⁴
- Large-scale nationalization of banks
- Deposit freezes (bank holidays)

A second definition deems a banking crisis to be any episode that includes one or more of the following:

- Large-scale bank runs
- Multiple bank closures
- Crisis-induced bank mergers or takeovers
- Crisis-induced nationalization banks¹⁴⁵

Which Macro Indicators Are Most Effective?

One of the more interesting aspects of the research into early warning systems, as earlier suggested, is the question of which indicators are most effective at predicting bank crises. Based on their survey of a number of early warning systems, Bell and Pain found the following criteria that function as proxies for the following attributes to be the most predictive:

- Overborrowing, that is, excessive lending, easy or mispriced credit
- A loose monetary policy reflected in a ballooning money supply or an unusually high volume of money circulating in the economy
- A downturn in economic growth relative to past performance or lower than expectations, including possibly a slowdown in industrial production

What may be inferred to be more immediate signals of an imminent banking crisis include:

- A plunge in the value of the domestic currency on world markets
- Increasing or high real domestic interest rates
- Decline in borrower creditworthiness accompanied by a rise in borrower defaults
- Sharp declines in the prices of assets including real property or securities

Finally, the evidence suggests that a key criterion that is potentially predictive of a banking crisis is a recent program of financial liberalization or deregulation.¹⁴⁶

An Early Warning System Methodology Illustrated

With too many extant early warning systems to survey properly in this limited space, readers may perhaps be better served by a sample of the results that can be obtained, or hoped for, and by the reference to the various sources of information.

The rest of this chapter will, as an illustration, explore some interesting findings.

The approach developed by Morris Goldstein, together with Graciela Kaminsky and Carmen Reinhart, and elaborated by the three in *Assessing Financial Vulnerability: An Early Warning System for Emerging Markets*, published by the Institute for International Economics in 2000,¹⁴⁷ as well as in a number of academic papers, is among the most accessible and comprehensive among the body of research reviewed. It is an early warning system of the macro type,¹⁴⁸ but many of its indicators touch on micro data, which, obviously, are relevant to a direct analysis of individual banks.

Its objective, according to the authors is to identify *empirical regularities* that would allow both public officials and private market participants to identify vulnerability to financial crises in advance of their occurrence.¹⁴⁹ Model development was based on the following guidelines:

- A wide range of crises would be examined to minimize sampling error and to maximize the number of identifiable potential causes (116 crises in 25 countries spanning a period of 25 years were examined).
- Equal attention would be given to currency and banking crises.
- Monthly data would be used insofar as possible.
- A variety of statistical techniques would be applied.
- Out-of-sample data would be used for testing.
- The inherent limitations in the model would be recognized.

In view of the set of major potential causes of financial crises listed in Exhibit 14.3 and generally discussed in more detail in this section, the next task was to identify appropriate quantitative indicators. Ultimately, 24 indicators were empirically tested. Based on the statistical testing undertaken, the researchers found that there was a wide range in the predictive power of the indicators selected for testing. While there are evident recurrent patterns preceding crises, accuracy was modest: There was one false alarm on average per two to four true signals. Based on the indicators utilized, emerging market banking crises were more difficult to predict than currency crises.

Although the model is nominally addressed toward emerging economies, it would appear to have a broader range of applicability. Though the criteria it employs are probably more conspicuously evident in developing countries, for the most part developed economies are not immune from the conditions or events they encompass.¹⁵⁰ The crisis that started in 2007 definitely corroborated the idea that analysts should observe advanced economies through the same lens.

Early warning systems, as noted, generally begin with an identification of potential crisis causes from which proxy indicators are selected for empirical testing. In one of Goldstein's other papers, he identifies 10 of the "leading suspects" of bank crises. Some have been discussed or alluded to previously but, for ease of access and use as a summary, they are presented, in Exhibit 14.3, in Goldstein's easy format,¹⁵¹ and then discussed generally, in the light of the findings of Goldstein and other researchers.

EXHIBIT 14.3 Causes of Banking Crises (Goldstein)

1	Highly volatile operating environment
2	Booms and burst bubbles
3	Large currency mismatches
4	Inadequate preparation for financial liberalization
5	Directed lending and high levels of government ownership of the banking system
6	Weaknesses in the accounting, disclosure, and legal framework
7	Distorted incentives
8	Lack of good internal and operational controls at banks
9	Exchange rate regime problems
10	Small size of banking system

Source: M. Goldstein.¹⁵²

A Highly Volatile Operating Environment

The first category of circumstances conducive to crisis formation is a highly volatile operating environment. Symptoms of such volatility include wide fluctuations in interest rate spreads, exchange rates, and other key economic indicators.¹⁵³ In a volatile operating environment, these will tend to swing more widely and dramatically than in more stable economies. Consequently, there is a greater level of market risk for both banks and their customers, which translates into a higher credit risk for the creditor counterparties. During a sharp economic contraction, a substantial proportion of a bank's borrowers will be hard pressed to repay their loans, while its customers involved in exporting or importing goods will suffer harsher terms of trade from their overseas counterparts than will vendors in more economically stable countries. Severe conditions may be made worse by poor economic and fiscal management by government authorities. Only the better-capitalized and more liquid banks will be able to weather such conditions without difficulty.

Booms and Burst Bubbles

Those mechanisms were discussed earlier in this chapter and will not be repeated.

Some economists believe boom-bust cycles are endemic to market economies. Whether or not this is the case, there are specific factors that seem to contribute to their occurrence, including too rapid financial liberalization (i.e., opening the economy to a free flow of foreign capital) and too rapid deregulation of the banking system; instability of the local currency (or the inability of the government to effectively maintain currency stability); and a loose monetary policy. Emerging markets, which have seen dramatic increases in capital inflows and investment from a low base, seem to be particularly susceptible to the formation of asset bubbles, although they obviously are not infrequent occurrences in developed markets.

A rapid influx of capital, if occurring in tandem with the deregulation of the banking system (listed here as a separate factor) and a sharp rise in intermediation, can contribute to causing a dramatic expansion in bank credit. Where too much money is chasing too few investments, prices are pushed ever upward.

When a flood of capital lifts all boats, it is not easy, especially for inexperienced bank officers, to differentiate strong credit risks from weak ones.¹⁵⁴ Similarly, bank regulators may be unable to respond effectively to the changed conditions, and in some cases may lack sufficient actual authority to rein in excessive risk taking of the banks under their charge. Therefore, in addition to weaknesses in economic policy, responsibility for the crisis can be attributed to inadequate risk management both on the part of the regulator at the national level and on the part of individual banks. Like those who oversee them, bankers often do not have the experience to recognize a crisis in the making.

Large Currency Mismatches

For several reasons, emerging markets are frequently plagued by high domestic interest rates. When the gap between domestic borrowing rates and foreign currency borrowing rates widens, there is an incentive for those companies and banks that have access to foreign currency funding to borrow overseas—or even through local banks—at a lower cost. The price that borrowers pay, whether direct corporate borrowers, indirect corporate borrowers, or the banks, is the currency risk arising

from a possible devaluation of the domestic currency. If a devaluation occurs in such circumstances, asset quality problems are very often aggravated because devaluation results in financing costs increasing in local currency terms.¹⁵⁵ In Mexico, before the 1995 tequila crisis, Goldstein observes, “60 percent of the liabilities of large and medium sized Mexican companies were denominated in foreign currencies—versus only ten percent of sales.”

The currency mismatch was a particularly important factor in respect to the Asian crisis of 1997, and many commentators found this indicator remarkably predictive in retrospect.¹⁵⁶ Thailand, for example, registered a huge jump in foreign currency liabilities in the mid 1990s, where short-term borrowing constituted about 70 percent of total funding before the crisis. As Delhaise remarked, “Undeniably, more foreign funds were made available to Thailand than were actually needed or could safely be swallowed.”¹⁵⁷ A long period of relative stability in the Thai baht encouraged complacency, and the government had even inadvertently encouraged such foreign currency borrowing through tax incentives¹⁵⁸ until it made a belated attempt to reduce it 1996. Other crises that affected Asian countries displayed the same phenomena, albeit to a lesser degree.¹⁵⁹

Inadequate Preparation for Financial Liberalization

Perhaps surprisingly to some, another condition that often precedes banking and financial crises is financial deregulation.¹⁶⁰ As perceived benefits of open economies and unrestricted capital flows became widespread over the previous two to three decades, changes in perception were followed by a wave of deregulation that has since spread throughout the world. Although intended to foster economic growth, financial deregulation paradoxically often acted to increase risk levels and trigger banking crises, and thereby impose significant economic costs.

In Indonesia, as noted by Philippe Delhaise, “[u]ntil 1995 when the authorities started feeling embarrassed by the number of applications [for banking licenses], there were no specific rules on how to obtain that license, except perhaps asset size, but being well connected certainly helped.”¹⁶¹

Empirical evidence supports the hypothesis that “financial liberalization often precedes banking crises,”¹⁶² but most likely it is inadequate preparation for deregulation, rather than deregulation in and of itself, that brought about bank crises.

New private sector banks were allowed to enter the financial industry in many countries, and in certain cases, government-owned institutions, long accustomed to allocate credit for policy reasons, were forced to compete on commercial terms. The result was “rapid rates of credit expansion,” more volatile interest rates, and generally greater market and credit risk than bank supervisors unfamiliar with the new environment were able to cope with.¹⁶³

Directed Lending and High Levels of Government Ownership of the Banking System

In many countries, national and local governments may own banking institutions, in whole or in part. The phenomenon is widespread and includes in Europe countries such as Germany, where the private banking sector is comparatively small, Brazil in Latin America, and numerous Asian nations, including India, as well as other emerging markets. The rationale was that governments were better at allocating capital than the private sector, or that it fitted the economic model—not necessarily socialist, but sometimes cultural as in Korea.¹⁶⁴ State ownership of banks, however,

has tended to result in a great deal of overall economic inefficiency through the misdirection of capital. In the mid-1990s, as Delhaise pointed out, “[i]n Taiwan, Indonesia, China, and Korea, many of the major banks [were] controlled by the state, while some of the major names in Malaysia, the Philippines, and Thailand also [fell] into that category.”¹⁶⁵

Moreover, even where significant state ownership of banks exists, often the remainder of the banking sector is in the hands of a narrow political and economic elite and the national government may cultivate an environment in which cronyism flourishes, if not outright corruption.

Policy lending continues to be used—mainly in developing countries—to channel bank credit to particular sectors or entities, often at subsidized below-market rates. This ultimately heightens the likelihood of a bank crisis.

Weaknesses in the Accounting, Disclosure, and Legal Framework

A lack of timely, current, and comprehensive financial reporting is another factor that contributes to a crisis-prone environment. Lax accounting standards tend to mask asset quality problems and make bank solvency and creditworthiness appear better than it is in fact. This was discussed earlier in the book.

Finally, weakness in the legal system has often made it difficult for banks to obtain or enforce judgments against delinquent borrowers, or to foreclose on pledged collateral. Therefore, the impact of weak asset quality was more severe than it would be in countries having more developed legal frameworks.

Distorted Incentives

This includes moral hazard as well as regulatory arbitrage, in particular where insufficient capital encourages the banks to make high-stakes gambles with depositors’ funds. A weak regulatory system, constrained by a lack of independence, is a further distorted incentive.

Lack of Good Internal and Operational Controls at Banks

Deficiencies of this sort have been implicated in a number of bank failures and, where present throughout a banking system, can increase the risk of a crisis. The lack of a robust risk management controls is a critical factor.

Exchange Rate Regime Problems

Countries that have fixed exchange rates are more inherently vulnerable to currency crises since a fixed exchange rate amid a sea of floating-rate regimes inevitably becomes subject to challenge. There are different types of fixed exchange rates, and some are highly susceptible to attack, while others tend to be somewhat more resilient. Because fixed-rate schemes do have some advantages, there are trade-offs to be made in determining which system is best in particular circumstances.

Small Size of Banking Systems

Small banking systems intrinsically share the same problems as small banks, namely a vulnerability to high levels on concentration risk, constraints on their ability to diversify, and an obvious incapacity to achieve economies of scale. Although it is not

always the case, small banking systems may be more vulnerable to political pressures from the local business elite, and, for the same reason, regulators may not enjoy sufficient independence to be able to compensate for this dynamic.

Which Indicators Worked Best in the Goldstein-Kaminsky-Reinhart Model?

As to which indicators were most predictive, the best indicators of an imminent banking crisis were found to be:

- An overvalued exchange rate
- Financial liberalization
- Changes in total output, that is, GDP
- Changes in stock prices before the crisis
- Changes in export levels
- Changes in real interest rates

Indicators having a moderate predictive ability included:

- Real interest rate differential
- M2/reserves
- “Excess” M1 balances
- Domestic credit/GDP
- Reserves
- Terms of trade

Finally, domestic-foreign interest differentials, credit ratings, lending/deposit interest rates, imports, and bank deposits were found to be generally ineffective indicators. The authors attributed this to the lack of emphasis on institutional factors in the models developed.

Forecasting Currency Crises Specifically

While this is less immediately relevant to the bank credit analyst, unless she deals with emerging markets—or the likes of Greece in 2012—an early warning system for currency crises proposed by Abdul Abiad in a 2003 working paper¹⁶⁶ provides a useful counterpoint to the model just discussed. Having surveyed the extant research, Abiad enumerated 22 potential attributes (Exhibit 14.4), classifying them into the following three categories:

1. General macroeconomic indicators, such as GDP growth
2. Capital flows indicators, such as the percentage of portfolio capital flows to total capital flows
3. Financial fragility indicators (i.e., indicators related to the strength of the banking system), such as the loans-to-deposits ratio

The attributes identified by Abiad parallel in some respects those enumerated in the preceding section concerning banking crises, reflecting some similarity in their respective origins. A particularly useful aspect of Abiad’s approach is his assignment of a proxy indicator to each of his 22 criteria.

EXHIBIT 14.4 Potential Banking and Currency Crisis Causes or Triggers Linked to Quantitative Indicators—Early Warning Indicators

Category	Concept		Measure
Macroeconomic indicators	External imbalance/ real overvaluation	1	Deviations of real exchange rate from trend
		2	Current account balance/GDP
		3	Export growth rate
	Inadequacy of reserve cover	4	M2/Reserves, level
		5	M2/Reserves, growth rate
		6	Reserves growth rate
	Overexpansion of credit	7	Growth rate of real domestic credit, deflated by nominal GDP
	Slowdown in the real economy	8	Industrial Production, growth rate
		9	Real GDP, growth rate (interpolated from quarterly GDP)
	Asset price boom/bust	10	Stock market performance, growth rate
Capital flows indicators	Monetary tightening	11	Real interest rate
	Possible cause of reversal of flows	12	LIBOR
	Lending boom	13	Bank assets/GDP, growth rate
	Short-term debt	14	Short-term debt to reserves
	Composition of capital flows	15	Cumulative non-FDI flows/GDP
		16	Portfolio flows, share in stock of total capital flows
Financial fragility indicators	Capital adequacy	17	Bank reserves/Total bank assets
	Bailing out by the central bank	18	CB credit to banks/Total bank liabilities
	Confidence in banks	19	Bank deposits/M2, level
		20	Bank deposits/M2, growth rate
	Ability of banks to mobilize	21	Loans/deposits, level
	Deposits	22	Loans/deposits, growth rate

Source: Abdul Abiad, "Early-Warning Systems: A Survey and a Regime-Switching Approach," IMF Working Paper WP/03/32, February 2003.

NOTES

1. R. Duncan, *The Dollar Crisis: Causes, Consequences, Cures* (Hoboken, NJ: John Wiley & Sons, 2002 and 2005), 129.
2. B. Eichengreen and C. Arteta, "Banking Crises in Emerging Markets: Presumptions and Evidence," Center for International and Development Economics Research (CIDER) Working Papers C00-115, University of California at Berkeley, 2000, 1.
3. Charles R. Morris, *Money, Greed & Risk: Why Financial Crashes and Crises Happen* (New York: Times Business/Random House, 1999).

4. The term *distress* or *distressed* is a common term to refer to a bank exhibiting a weakened financial condition, particularly in respect to its liquidity or solvency, such that it can no longer function optimally and is at constant risk of failure. See an interesting database of bank insolvencies in G. Caprio and D. Klingebiel, "Bank Insolvencies: Cross-Country Experience," World Bank Policy Research Paper, 1996. Solvency, or rather its lack, is the core attribute of the distressed state and weak capitalization is usually the root cause of its liquidity problems. Often, distressed banks will be already relying upon liquidity supplied or promised by the government's lender of last resort. Not infrequently, other metaphorical terms are used to describe institutions of this type, that is, banks that remain open but are dependent on the state support and cannot function at full capacity. *Sick banks* and *zombie banks* are among the terms applied. Note that some scholars define the term *distressed* with greater exactitude. A banking system has been said to be in distress "when a significant fraction of the banking sector is insolvent but remains open."
5. Another example was Bank of Credit and Commerce International, an institution that at its peak was among the world's 10 largest privately owned banks with head offices in Luxembourg and hundreds of international branches operating in 78 countries. Founded by a Pakistani financier in 1972, the bank grew rapidly over the next decade. In the late 1980s and early 1990s, Luxembourg asked for assistance from other countries in regulating the bank, and after investigations between 1987 and 1991, BCCI was revealed to be engaged in criminal activity, including money laundering, on a massive scale. The Bank of England shut down the bank's operations in the UK in 1991. Liquidation proceedings and litigation continued for over decade, culminating in the first lawsuit ever to be filed against the Bank of England. The bank's liquidators had made a claim on behalf of thousands of ordinary depositors who lost funds when BCCI collapsed, alleging that the UK's central bank had been "willfully negligent" in its supervision of the errant institution. The lawsuit was finally dropped in November 1997. See BBC News, "BCCI Liquidators Drop 1£bn Case," November 2, 2005, <http://news.bbc.co.uk/1/hi/business/4399336.stm>.
6. The phrase *proximate cause* is a legal term that is convenient to use to refer to the element of causation closest to the referenced effect, that is, a banking crisis. As used in this chapter, it may be contrasted with the terms *fundamental cause*, *underlying cause*, or similar phrases that are used in this chapter, to refer to the deeper sources of a crisis or that contribute to its manifestation. To illustrate, in reference to the collapse of Barings Bank, while the proximate cause of the bank's demise were the actions of trader Nick Leeson, the more fundamental cause of its collapse was the failure of its owners and managers to implement the basic internal controls that would have prevented the unauthorized trading to go undetected for three years.
7. The loss of liquidity in the London market was triggered by a pervasive uncertainty over which banks had undisclosed exposure to subprime mortgage-backed securities.
8. Of course, as the introduction to this chapter sought to illustrate, isolated instances of individual failure can occur even in benign conditions and do not constitute a crisis. The possibility that failure is the result of causes predominantly internal to the bank should always be borne in mind.
9. A single search using the keyword phrase "banking crises" of the IDEAS (Internet Documents in Economics Access Service), which uses the RePec (Research Papers in Economics) database of economics research brought up 1,158 documents. These services can be reached at <http://repec.org/>. Researchers James Bell and Darren Pain note the importance of the work at the World Bank and at the IMF in respect to developing such early warning systems of banking crises.
10. I. Macfarlane, "The Changing Nature of Economic Crises," Talk by the Governor of the Reserve Bank of Australia to the Australian Business Economists 13th Annual Forecasting Conference dinner, Sydney, Australia, December 4, 1997.
11. B. Eichengreen and A. Rose, "The Empirics of Currency and Banking Crises," Haas School of Business at the University of California, Berkeley, 1998.

12. A. Gaytán and A. Christian, "A Review of the Literature on Early Warning Systems for Banking Crises," Working Paper (Documento De Trabajo) no. 183 (Huérfanos Chile: Central Bank of Chile, October 2002), note 3, 3.
13. As with many financial terms, the definition is not standardized and will depend upon the purpose for which it has been constructed. While a researcher developing a specific financial model to predict banking crises may require a quantitative definition to function within an algorithm, such definitions are apt to be rather arbitrary as to what they include and what they exclude. For the purposes of this book, the referenced definition as supplemented by indicative benchmarks should be satisfactory for the present purposes.
14. Distinct from different manifestations of banking crises are other types of financial crises, including most notably *currency crises*. Approaches to classification of financial crises differ, and there is no uniform approach. While often related to a banking crisis in that it may contribute to a banking crisis or potentially result from one, the currency crisis—also known as a foreign exchange crisis, a current account crisis, or as a balance of payments crisis—can be identified and distinguished by the devaluation of the local currency that is its core component. A third type of crisis, the debt crisis, is widely recognized, but is outside the scope of this chapter. See, for example, Jan P.A.M. Jacobs, Gerard H. Kuper, and Lestano, "Currency Crises in Asia: A Multivariate Logit Approach," Department of Economics, University of Groningen, 2004, 2. Some further divide the basic categories into further subcategories.
15. Other critical functions supplied by the banking system, such as its acting as a transmission belt for monetary policy, were discussed earlier.
16. Recall that the concept of systemic risk has been defined as "the likelihood of a sudden, usually unexpected, collapse of confidence in a significant portion of the banking or financial system with potentially large real economic effects." P. Bartholomew, L. Mote, and G. Whalen, "The Definition of Systemic Risk," U.S. Office of the Comptroller of the Currency, Department of Economic and Policy Analysis, Bank Research Division, Washington, DC, 1995, 9.
17. There could be exceptions. An individual bank failure alone would not imply a banking crisis unless it happened to be by far the dominant institution in the country such that its failure would trigger severe losses at the other remaining significant banks and paralyze the entire system in the process.
18. This factor has been regarded as so closely linked to the impairment in functions that a banking crisis has also been defined as one in which all or substantially all of the banking capital in a country is wiped out.
19. Note that *insolvency* and *bankruptcy* do not mean exactly the same thing. While both point to a situation where liabilities exceed available assets, *insolvency* refers to the impossibility to discharge financial obligations, as opposed to the legal concept of bankruptcy. A bankrupt bank is obviously insolvent, but an insolvent bank might still escape bankruptcy, if only through creative accounting.
20. In early 2012, several Spanish banks had plenty of funding available, even though their solvency was in doubt.
21. G. Caprio and D. Klingebiel, "Bank Insolvencies," 6. Incidentally, this goes far to explain why capital has come to be so highly regarded as an indicator of bank creditworthiness, although as was noted earlier, it is not necessarily the best such measure for the working banking analyst during ordinary conditions.
22. In other words, a banking crisis exists when the equity capital in the banking system is well below regulatory minimums and is approaching or equal to zero. Note that it is often the case that actual capital levels will not be apparent until after events have unfolded. The capital deficit, however, will ultimately be evident from bank balance sheets, once the full extent of bank losses has been recognized.

23. As was noted more generally in the discussion of capital in Chapter 9, capital ratios are less helpful when viewing a potential banking crisis prospectively.
24. Daily monitoring of capital could be used to generate inputs for an early warning system at the regulatory level, but would probably not be feasible for use by outside analysts.
25. For example, a government security purchased for \$9,891 that pays \$10,000 in six months has a market value that is comparatively easy to estimate, assuming the bank is located in a country with an active local debt market. In contrast, as an essentially illiquid asset, a loan of \$10,000 that pays interest at 6 percent per annum in two semiannual installments is not, despite the fact that its value can be modeled in the abstract without too much difficulty. The valuation of both the relatively liquid asset and the essentially illiquid one is complicated further when the bank must raise liquid funds quickly or when the liquidation of the bank itself is contemplated; the need to sell an asset quickly means that a large discount may have to be offered to the prospective buyer. The fire sale effect, of course, diminishes the amount of capital cushion available to absorb losses. In view of these difficulties under such circumstances, bank managers may simply be unable to report their capital positions daily with a high degree of accuracy.
26. Ideally, the price of an instrument will reflect its economic value, taking into account demand for the asset class, supply, and other relevant criteria such as credit risk, a variable that the assigned credit rating is intended to evaluate. Because in most countries the secondary loan market is limited, loans are an example of an asset class in respect to which value is comparatively difficult to measure. The value of such assets may instead only be able to approximated, on a net present value basis for instance, taking into account current and historical default rates, expected changes in the economy (e.g., a recession translates into a higher rate of default), and the capacity of the bank to recover any collateral in a timely manner in the event of delinquency and the value of that collateral in the market once acquired. For instruments where a market exists, these factors will be taken into account; where they do not exist, they will need to be approximated.
27. In partial defense of banks during a period of market turbulence—at least with respect to financial reporting albeit not with regard to responsibility for incurring losses in the first place—senior managers may not, in fact, know precisely how much capital they really have at a given time, notwithstanding the information provided by their internal information reporting systems, owing to fluctuations in the value of volatile asset holdings or the opaqueness of the risks they pose.
28. As discussed in Chapter 13, under some regulatory regimes strict aging criteria are applied, while others banks have considerable discretion with regard to loan classification.
29. A more detailed discussion of the predictive quality of some of the ratios discussed will be provided later.
30. Other critical functions supplied by the banking system, such as its acting as a transmission belt for monetary policy, were discussed earlier. One key indicator of the supply of credit would be rates of aggregate loan growth, which are influenced both by the capacity and willingness of banks to lend as well as by the demand for credit and the availability of creditworthy loan applicants. Note also that in the classic banking crisis problem, banks' aggregate capital has become so severely impaired that it is unlikely they will be able to remain open without continued state support, let alone return to financial health in a timely manner. At a minimum, state support means the continued supply of liquidity, that is, the promise of liquidity when needed or guarantees to depositors and creditors or both. There is no good solution, because the provision of support creates its own problems, while if the distressed banks are allowed to collapse, there is significant risk that the remaining banks will also break, and that the survival of the whole financial system will be imperiled.
31. Laeven and Valencia, "Systemic Banking Crises: A New Database," IMF Working Paper, 2008.
32. A. Demirgüç-Kunt, E. Detragiache, and P. Gupta, "Inside the Crisis: An Empirical Analysis of Banking Systems in Distress," January 2004 draft.

33. Ibid., 8.
34. Charles W. Calomiris, Daniela Klingebiel, and Luc Laeven, "A Taxonomy of Financial Crisis Resolution Mechanisms: Cross-Country Experience," World Bank Policy Research Working Paper 3379, August 2004, 1.
35. Robert Zielinski, banking analyst, "Restructuring Asia's Banks: The Process and Institutions," Lehman Brothers, September 29, 1998.
36. B. Eichengreen and C. Arteta, "Banking Crises in Emerging Markets," Abstract.
37. David T. Llewellyn, "Some Lessons for Regulation from Recent Bank Crises," Loughborough University, Loughborough, UK. Paper presented at the Second International Conference on The New Architecture of International Monetary System, Florence, October 15, 1999.
38. Philippe Delhaise, co-author of this book, published a book on the Asian crisis of 1997 whose title clearly refers to the weaknesses of the banking and finance systems: *Asia in Crisis—The Implosion of the Banking and Finance Systems* (New York: John Wiley & Sons, 1998).
39. See Eichengreen and Arteta, "Banking Crises in Emerging Markets," 3: "the causes of banking crises, in emerging markets and generally, has become a key question for policy makers, prompting the growth of a large empirical literature." Martin Mayer refers to six notable financial shocks of the 1990s, namely, the one affecting the United States in 1990–1991, the failure of the European ERM in 1992, the unexpected interest rate hikes by the Federal Reserve in 1994, the Mexican peso crisis of 1995, the Asian crisis of 1997–1998, and the Russian defaults of 1998. See M. Mayer, "Risk Reduction in the New Financial Architecture: Realities, Fallacies, and Proposals," Levy Institute Working Paper No. 268 (April 1999), <http://ssrn.com/abstract=165550>. This enumeration, of course, was by no means comprehensive.
40. See C. Reinhart, M. Goldstein, and G. Kaminsky, "Assessing Financial Vulnerability: An Early Warning System for Emerging Markets," Institute of International Economics, 2000, 6–8.
41. This is particularly so in respect to those research efforts aimed at constructing models to predict financial crises (early warning systems), which are discussed later in this chapter, as well as the general question: How can banking crises be prevented?
42. Rather than espousing any particular theory of causation, this chapter looks at the broad categories of potential crisis causes—which perhaps might be better called contributing factors. As will be seen, the categories of contributing factors responsible for banking crises share a number of similarities with the major reasons for their increasing frequency noted just above.
43. In light of the large volume of research on this subject, much of it arcane, the work of others has been relied upon to some extent to sort through the mass of findings.
44. That is, any given bank in distress, whether owing to a banking crisis or to circumstances particular to that institution, will typically show symptoms that are indistinguishable from those exhibited by a bank facing collapse during a more volatile period, for example, as part of a wave of similar incidents during a banking crisis. This similarity in symptoms makes the identification of a banking crisis difficult prospectively. It cannot be known in advance whether the first bank failure is the start of what ultimately proves to be a series leading to a crisis or is merely an isolated implosion resulting from deficient management, possibly coupled with a run of bad luck. The problem of recognizing a banking crisis in the making will be returned to in Chapter 15.
45. Parenthetically, management likewise will have some degree of responsibility in both cases, although generally more so in the first scenario for reasons that should soon become clear.
46. See, for example, Owen Evans, Alfredo M. Leone, Mahinder Gill, Paul Hilbers, Winfrid Blaschke, Russell Krueger, Marina Moretti, Jun Nagayasu, Mark O'Brien, Joy ten

- Berge, and DeLisle Worrell, "Macro Prudential Indicators of Financial System Soundness," Occasional Paper (192) (Washington, DC: International Monetary Fund, 2000). To be sure, the isolated bank is much less likely to fail when operating conditions are buoyant. Instead, failure is likely to occur when stress increases, even if the stress is far short of that which would occur in crisis conditions.
47. The principal distinction is that when idiosyncratic in character the deficiencies are not likely to be widespread throughout the banking system and therefore will probably not be attributable to regulatory action or inaction or to specific policies.
 48. Note that the issue arises concerning whether the adverse outcome is attributable to the exogenous event or to the government's response to it. As an illustration, consider the failure of a national government through its central bank to maintain sufficient foreign reserves to ward off speculative attacks on the domestic currency, thereby making it vulnerable to devaluation.
 49. Llewellyn, "Some Lessons for Regulation from Recent Bank Crises," 4–5. See also Caprio and Klingebiel, "Bank Insolvency" ("caused by a mixture of bad luck, bad policies—both microeconomic (regulatory) and macroeconomic—and bad banking").
 50. Owners and to a lesser extent staff may also have some impact, depending, among other things, upon the level of influence they can exert. In a closely held bank, owners will be able to exert a high level of control over management and may themselves be senior managers.
 51. As was discussed, banks by virtue of their quasi public policy role are highly regulated entities in virtually all countries, and subject to a regime of regulation and supervision unlike no other business sector. In this sense, ostensibly external factors may impinge heavily upon internal operational decisions. Conversely, it is not unusual for the banking industry to have some influence over the government policy toward banks and the laws and regulations that affect the industry.
 52. This last category can be broken down further into practices that (a) are motivated by genuine good faith convictions about the role banks should play not only as key components of the financial system, but also as transmitters of government social policy; and (b) are motivated by personal gain. More specifically, the two subcategories would respectively include (a) intervention into bank decisions to effect social policies through policy lending (also called directed lending or connected lending); as well as the (b) tolerance or active promotion of cronyism (e.g., where banks make loans at the behest of government officials for personal or partisan reasons) or corrupt practices (e.g., where a government official in exchange for favors or the expectation or promise of future favors uses the power of the state to invidiously discriminate against certain institutions to the detriment of the health of the banking industry or that of the public interest as a whole).
 53. A list of such categories that was prepared in the context of early warning systems both for banking and currency crises is provided in a later section.
 54. See generally David O. Beim, "What Triggers a Systemic Banking Crisis?" Columbia University Working Paper, June 19, 2001, 5. Beim concludes that a bank crisis may be triggered by the withdrawal of funding by one or more of more agents: (1) depositors; (2) the local government, as guarantor; (3) private external lenders; and (4) intergovernmental financial institutions.
 55. A. Lamfalussy, *Financial Crises in Emerging Markets* (New Haven, CT: Yale University Press, 2000).
 56. Alexandre Lamfalussy was economic advisor to the Bank for International Settlements (BIS) from 1976 to 1983. He was also, from 1994 to 1997, the first president of the European Monetary Institute, which was dissolved in 1998 when the European Central Bank was created.
 57. The term *real economy* is used in contradistinction to the financial economy, that is, the change in market conditions in respect to equity equities, bonds, and other traded instruments.

58. The following definitions of a currency crisis are illustrative: (1) "[A] crisis that occurs when participants in an exchange market come to perceive that an attempt to maintain a pegged exchange rate is about to fail, causing speculation against the peg that hastens the failure and forces a devaluation." "Deardorff's Glossary of International Economics," www-personal.umich.edu/~alandear/glossary/c.html2; and (2) "[The occurrence of] a speculative attack on the exchange value of a currency [which either] results in a sharp depreciation in the value of the currency or forces [a government] to expend large volumes of international currency reserves and sharply increase interest rates to defend the prevailing exchange rate." <http://enbv.narod.ru/text/Econom/ib/str/261.html>. See also A. Kibritcioglu, "Monitoring Banking Sector Fragility," *Arab Bank Review* 5, no. 2 (October 2003). Note that a currency crisis is sometimes referred to as a balance of payments crisis reflecting the underlying imbalance between exports and imports that generally creates the underlying pressure for a devaluation of a currency.
59. "More recent research suggests that the initial adverse shocks may not only originate from poor domestic macroeconomic policies that make maintenance of the existing exchange rate difficult . . . but also from . . . policies that may be appropriate for achieving domestic goals, e.g., employment and price stability, but are inconsistent with maintaining the existing exchange rate." G. Kaufman, "Banking and Currency Crises and Systemic Risk: A Taxonomy and Review," Working Paper Series WP-99-12, Federal Reserve Bank of Chicago, 1999, 24.
60. *Ibid.*, 5.
61. The foregoing is, of course, a generalized description. Researchers have drawn many distinctions in developing typologies of bank and currency crises. Also, it has been shown that the impact of twin crises is not, however, larger than the sum of the parts. Michael M. Hutchison and Ilan Neuberger, Abstract, "How Bad Are Twins? Output Costs of Currency and Banking Crises," EPRU Working Paper Series 02-09, Economic Policy Research Unit (EPRU), University of Copenhagen, Department of Economics (2002): "twin crises are 'bad' only in that they entail output losses associated with both currency and banking crises, not because there are additional feedback or interactive effects further damaging the economy."
62. Matias Fontenla, "All Banking Crises Are Not Created Equal," Chapter 3 of "Financial Crises in Developing Countries," University of Texas at Austin, Dissertation (2003). For empirical support concerning the mechanism of a self-fulfilling bank crisis, see M. Blejer, Ernesto V. Feldman, and A. Feltenstein, "Exogenous Shocks, Deposit Runs and Bank Soundness: A Macroeconomic Framework," Working Paper (97/91) (Washington, DC: International Monetary Fund, July 1997). "In a model where all banks are initially solvent, an exogenous shock affects confidence, causing a flight from deposits into domestic and foreign currency. Real interest rates increase unexpectedly, affecting firms and raising the share of the banks' nonperforming assets. This increase causes genuine solvency problems and accelerates the bank run."
63. Examples of such distinguishing characteristics could be the occurrence of bank runs, bank collapses, "massive government intervention." See, for example, Carl-Johan Lindgren, Gillian Garcia, and Matthew I. Saal, "Bank Soundness and Macroeconomic Policy," (Washington, DC: International Monetary Fund, 1996), cited in G. Kaufman, "Banking and Currency Crises and Systemic Risk: A Taxonomy and Review," Working Paper Series WP-99-12, Federal Reserve Bank of Chicago, 1999. (See also E. Davis, "Toward a Typology for Systemic Financial Instability," Paper developed from lectures, November 12, 2003, www.ephilipdavis.com.)
64. Patrick Honohan, "Banking System Failures in Developing and Transition Countries: Diagnosis and Prediction," BIS WP No. 39, 1997. Honohan also distinguishes crises that are caused by state intervention into the banking system through government ownership of banks.

65. It could be argued that a crisis with causes that are mainly microeconomic in origin does not amount to a bank crisis as such, there being no systemic factors present that threaten the system as a whole, other than potentially the possibility of a bank run like self-fulfilling crisis taking hold.
66. David O. Beim, "What Triggers a Systemic Banking Crisis?" 2. And Edward Kane, "Capital Movements, Banking Insolvency and Silent Runs in the Asian Financial Crisis," *Pacific Basin Finance Journal* 8 (1999): 153–175, cited in Beim.
67. G. L. Kaminsky and C. Reinhart, "The Twin Crises: The Causes of Banking and Balance-of-Payments Problems," Board of Governors of the Federal Reserve System, International Finance Discussion Papers, no. 544 (March 1996).
68. See C. Reinhart, "The Anatomy of Financial Crises," seminar paper for the Seminar on Capital Flow Volatility, Contagion and Crises Management at the University of Maryland (1999). Although the research into the causes of currency crises and banking crises originally represented two separate strands of inquiry, over time they have converged as the interrelationship between the two, particularly in emerging markets, has become clearer.
69. Ibid.
70. Ibid.
71. If only because, as most scholars agree, a weak macroeconomic situation, as often found in emerging markets, is often a factor behind systemic banking problems.
72. Delhaise, *Asia in Crisis*, 85–88.
73. See Kaufman, "Banking and Currency Crises and Systemic Risk," 19.
74. B. Bernanke, M. Gertler, and S. Gilchrist, "The Financial Accelerator and the Flight to Quality," *Review of Economics and Statistics* 78, no. 1 (1996).
75. M. Gertler, S. Gilchrist, and F. Natalucci, "External Constraints on Monetary Policy and the Financial Accelerator," National Bureau of Economic Research Working Paper no. 10128 (2003).
76. R.R. Bliss and G.G. Kaufman, "Bank Procyclicality, Credit Crunches, and Asymmetric Monetary Policy Effects: A Unifying Model," Federal Reserve Bank of Chicago Research Department (2003).
77. G.L. Kaminsky and C.M. Reinhart, "The Twin Crises: The Causes of Banking and Balance-of-Payments Problems," *American Economic Review* 89, no. 3 (1999): 473–500.
78. R. Chang and A. Velasco, "Financial Fragility and the Exchange Rate Regime," Economic Research Reports RR#98-05, CV STARR Center for Applied Economics, New York University (1998).
79. F.S. Mishkin, "Understanding Financial Crises: A Developing Country Perspective," National Bureau of Economic Research Working Paper, no. 5600 (1996).
80. I. Goldfajn and R.O. Valdes, "Capital Flows and the Twin Crises: The Role of Liquidity," International Monetary Fund Working Paper 97/87 (1997).
81. In an article titled "Credit Cycles and the Origin of Commercial Panics," read before the Manchester Statistical Society, December 11, 1867.
82. This point is discussed in more detail later in this chapter.
83. This is often, mistakenly, disregarded by analysts.
84. Financial innovations can contribute to a boom. Consider the impact of long-term fixed-rate mortgage products, high loan-to-value mortgages, aggressive mortgage products, home equity loans, discount brokerages, and any number of other products that may increase the flow of new capital into an economy.
85. This is the so-called wealth effect.
86. For their part, shareholders—looking for short-term earnings growth and unsophisticated about banking risks—may not be able to understand why the bank they hold shares in cannot keep pace with its peers.

87. Still others begin to see high potential returns in *short selling* the more vulnerable of the high-flying equities. By short selling, they commit to selling the security to prospective buyers for a certain price at some future date in the expectation of being able to purchase it at a lower price in the interim.
88. The transition period as used here refers to the period that begins when market prices peak, whether for shares or property near their peak, and ends with the widespread recognition that a banking crisis, or at least a financial crisis of some sort, is in progress. It is generally characterized by increasing uncertainty until a shift in perception occurs among market participants and then among the public at large.
89. There are two main reasons. First, as a matter of good banking practice, management ought to seek to maintain at least the same relative buffer against anticipated losses as conditions become more volatile. If it does not, unless the bank has not set aside a large cache of reserves to draw upon to absorb the loss caused by the sharp rise in delinquencies, coverage of NPLs will show a comparative decline. Second, even if management does not wish to voluntarily replenish its pool of loan-loss reserves, amounts charged off loan-loss reserves, prudential regulatory requirements concerning mandatory minimum loan-loss provisioning may compel it to do so, diverting more operating income away from the bottom line.
90. G. Caprio and D. Klingebiel, "Bank Insolvency: Bad Luck, Bad Policy, or Bad Banking?," Annual World Bank Conference on Development Economics, 1996, 1997, The International Bank for Reconstruction and Development/The World Bank.
91. David O. Beim, "What Triggers a Systemic Banking Crisis?"
92. Charles P. Kindleberger, *Manias, Panics, and Crashes: A History of Financial Crises* (New York: John Wiley & Sons, 1978).
93. See the discussion of the formation of the Washington consensus in a later section of this chapter.
94. See, for example, G. Hoggarth, J. Reidhill, and P. Sinclair, "On the Resolution of Banking Crises: Theory and Evidence," Working Paper no. 229, Bank of England, London (2004).
95. Yet, even before this bank failed, market confidence had already been shaken by other events, including some outside the financial sector.
96. C. Lindgren, G. Garcia, and M. Saal, "Bank Soundness and Macroeconomic Policy," International Monetary Fund, Washington, DC (1996), quoted by M. Hanazaki and A. Horiuchi, "A Vacuum of Governance in Japanese Bank Management" revised version of a paper presented at the NBER (National Bureau of Economic Research) Japan Project Meeting held April 17–18, 1998, in Cambridge, MA.
97. Laeven and Valencia, "Systemic Banking Crises: A New Database," IMF Working Paper, 2008.
98. In a study on bank crises in 2000, Eichengreen and Arteta counted just one banking crisis in the period from 1945 to 1971 out of 21 significant industrial and emerging markets reviewed, observing that the 25 years "after World War II was one of tight financial regulation and control which left little scope for banking crises." B. Eichengreen and C. Arteta, *Banking Crises in Emerging Markets: Presumptions and Evidence* (August 2000), citing Bordo and Eichengreen, "Is Our Current International Economic Environment Unusually Crisis Prone?" (1999). Taking note of the large amount of support for the view that financial crises are an increased risk factor for banking crises, Eichengreen observed that from 1975 to 1997 the IMF had tallied 54 banking crises and that the World Bank had enumerated even more. *Ibid.*, 20. According to one of the pre-eminent researchers on bank financial crises, there have been at least "112 episodes of systemic banking crises in 93 countries" between the late 1970s and 2001. M. Goldstein, "Implications of Early Warning Models of Crisis," paper prepared for KDIC International Financial Symposium on "The Financial Crisis and Beyond," Seoul, Korea, Institute for

- International Economics, Washington, DC, December 11, 2001 (hereinafter “EW Implications”). See also Gerard Caprio and Daniela Klingebiel, 2003, “Episodes of Systemic and Borderline Financial Crises,” working paper and database, World Bank website, periodically updated, which reported 117 episodes; R. G. Rajan, “Has Financial Development Made the World Riskier?” National Bureau of Economic Research, WP 11728 (2005).
99. Currency crises, however, were rare during the gold standard era.
 100. Michael Bordo and Barry Eichengreen, “Is Our Current International Economic Environment Unusually Crisis Prone?” article prepared for the Reserve Bank of Australia Conference on Private Capital Flows, Sydney, August 9–10, 1999 and for a chapter in *Capital Flows and the International Financial System*, (Sydney: Reserve Bank of Australia, 1999).
 101. The mixed economy, which coupled free market capitalist principles with a strong dose of state control and regulation, presented itself as the happy medium between laissez-faire capitalism and the command-economy that had taken hold in the more doctrinaire socialist and communist countries, the latter generally strongly influenced by the example of the Soviet Union.
 102. A number of trends contributed to the shift in the governing paradigm from that which Franklin Roosevelt’s New Deal embodied to that which was embraced during Ronald Reagan’s administration.
 103. The election of the Conservative Party in the UK under the leadership of Margaret Thatcher in 1979 brought about policy changes to some extent analogous to those made during the Reagan Administration.
 104. Concerning the increase in the rate of banking crises, see, for example, Gerard Caprio, “Banking on Crises: Expensive Lessons from Recent Financial Crises,” World Bank Policy Research Working Paper (1999).
 105. The policies advocated by the Chicago School economist Milton Friedman, whose monetarist and free-market fundamentalist views came to prominence during this period, when adopted dramatically changed the operating environment for S&Ls. They removed the strictures that had safeguarded the industry from the imprudent and even abusive practices on the part of management and owners that followed deregulation, while under the Fed’s experiment with monetarism. Interest rates shot up to double-digit levels. While under the burden of long-term fixed rate mortgages at nominally low interest rates, the staid S&L industry suddenly faced double-digit interest rates and was forced to compete for funding with the newly deregulated money market checking accounts. Suffering from low or even negative spreads, and a dwindling capacity to attract new deposits, many S&Ls failed or became vulnerable to takeovers by owners and managers willing to offer high deposit rates guaranteed by the federal government and use the proceeds to make high-risk gambles with depositors’ savings. The result was the S&L crisis of the 1980s. It would be presumptuous, however, to lay the blame on Friedman and the Chicago School theories for the excesses of deregulation.
 106. Christian Larrain, “Best Practices for Efficient Banking Restructuring,” World Bank, 1999.
 107. While causation does not necessarily follow from correlation, there are good reasons to attribute the greater frequency of bank crises since the 1970s, in large part, to the quickening pace of financial liberalization and bank deregulation as well as to the accompanying financial innovation they both encouraged.
 108. As noted earlier, notwithstanding that Thailand’s macroeconomic fundamentals were not bad by the standards employed at the time to assess sovereign risk, a single national crisis would not in itself had been so surprising. It was the unprecedented scope and magnitude of the crisis that was the big surprise.
 109. The changes in the regulatory environment, particularly in the last decade of the previous century onward, engendered a new profession—risk management.
 110. Given that the current crisis (the global financial crisis of 2007–2012) began when the new regime had been barely put in place in some countries and was not yet implemented

in others, it cannot be said definitively that Basel II might not have had a salutary impact had it been operational for a longer period.

111. *The End of History* was the title of popular political treatise written by Francis Fukuyama. The end of history was a metaphor for the emergence of market-oriented democracy as the superior form of government, and a figurative end to contention concerning this proposition. The rise of Islamic fundamentalism and the failure of Western policy initiatives to spread democracy in the Muslim world, coupled with the failure of market economies to perform as expected in some countries may have tarnished the credibility of Fukuyama's hypothesis.
112. See Delhaise, *Asia in Crisis*, Chapter 1, "The Asia Panic," for more comments on that topic.
113. Public policy decisions relevant to the banking industry are reflected in the various extant statutes, regulations, directives, and judicial decisions as well as in the established practices and procedures governing banks and their dealings with counterparties, depositors, and others, including the government. In its entirety, this body of rules, together with the various incentives and disincentives embedded in it, forms the major part of what can be referred to as a country's financial architecture. The domestic financial architecture of each country will inevitably have a major influence on local bank operations. Changing conditions, of course, will bring about new policy decisions and rules that reshape the architecture. Accordingly, it is not so much a static edifice but instead is analogous to a dynamic cityscape, constantly under construction and changing shape in response to new political, economic, social, and policy developments.
114. R. Duncan, *The Dollar Crisis*.
115. The theory Duncan propounds is less concerned with the various aspects of the relationship of banking crises and currency crises than it is in the role of the U.S. dollar as a reserve currency in increasing financial instability.
116. Ibid. Duncan is not alone in his view.
117. The Austrian School of Economics favors deduction over induction in the interpretation of economic development, but its application to business cycles is often considered as weak by many economists.
118. Abdul Abiad, "Early-Warning Systems: A Survey and a Regime-Switching Approach," 2003, IMF Working Paper WP/03/32.
119. Claudio Borio and Philip Lowe, "Assessing the Risk of Banking Crises," BIS Working Paper No 114, December 2002.
120. Because of the linkage between the two types of crises, many of the research studies consider the prediction of both.
121. J. Bell and D. Pain, "Leading Indicator Models of Banking Crises: A Critical Review," Bank of England, *Financial Stability Review*, no. 9 (December 2000): 113.
122. While the causes of the crises enumerated earlier possess much explanatory value, it is apparent that with some exceptions they do not easily lend themselves to the construction of a predictive model.
123. At the rating agency level, where CAMEL ratios for the banks in a particular country imply a weakness in their collective financial strength, the usual analytical response would be to warn of the increased systemic risk and recommend that a lower rating be assigned for the sector and as appropriate for its constituent institutions in accordance with their comparative creditworthiness, taking into account the country or sovereign risk rating assigned and the prospect of state support.
124. The abundance of research on the topic of financial crisis causes and early warning systems, particularly after the onset of the Asian crisis in 1997, includes several surveys of the existing literature. A number of these have been relied upon in the preparation of this chapter.

125. The term has also been used to refer to similar systems designed to refer to the prediction of individual bank failures. Such models in this book are termed probability-of-default models.
126. Similar models may attempt to predict the occurrence of some state such as undercapitalization that implies the greater likelihood bank failure.
127. Julapa Jagtiani, James Kolari, Catherine Lemieux, and Hwan Shin, "Early Warning Models for Bank Supervision: Simpler Could Be Better," *Federal Reserve Bank of Chicago* 3Q/2003, "Economic Perspectives," 2003. According to the article, the U.S. Federal Deposit Insurance Corporation started the development of early warning systems in the mid-1980s.
128. As a model to evaluate the risk of a bank's default is termed a credit risk model, so too might these models be thought of as systemic risk models. The process of model development resembles that of other risk evaluation models.
129. That is, the term *early warning system* tends to connote a methodology that has been empirically tested. To facilitate testing and also to avoid bias, the bulk of the model inputs is ordinarily quantitative in character. This is not necessarily the case with respect to the metrics constructed for the purpose of gauging financial system fragility, which may merely be composite ratios that have been constructed based upon a reasonable notion of what criteria in combination imply greater or lesser systemic risk or ratings that incorporate qualitative judgments. Moreover, while a banking system fragility rating may merely represent an attempt to show whether conditions are more or less conducive to the occurrence of a banking crisis without purporting to forecast such an event, an early warning system seeks to predict whether or not a crisis will occur within a given time horizon, although predicting crisis timing within a predetermined range is usually beyond its design criteria. A banking system fragility index may nevertheless add significant value to the analytical process.
130. G. Kaminsky, S. Lizondo, and C. Reinhart, "Leading Indicators of Currency Crises," IMF Staff Papers 45, no. 1 (March 1998).
131. J. Bell and D. Pain, "Leading Indicator Models of Banking Crises."
132. Note that in Bell and Pain's schema, EWSs that use microeconomic criteria encompass models intended to predict individual bank failure. How models of individual bank failure are extended to predict bank crises is not specified in detail. Micro-based models to predict bank crises would seem to be extensions of models used to predict individual bank failures that indicate a high potential for crises when a significant proportion of individual banks show vulnerability to failure.
133. O. Evans et al., "Macroprudential Indicators," 17.
134. Gaytán and Christian, "A Review of the Literature on Early Warning Systems for Banking Crises," 24.
135. Borio and Lowe, "Assessing the Risk of Banking Crises," 46.
136. E. Hanschel and P. Monnin, "Measuring and Forecasting Stress in the Banking Sector: Evidence from Switzerland," BIS Working Paper, No. 22, Basel 2005.
137. Th. B. King, D.A. Nuxoll, T. J. Yeager, "Are the Causes of Bank Distress Changing? Can Researchers Keep Up?" *Federal Reserve Bank of St. Louis Review* 88, no. 1 (2006): 57–80.
138. D. Hendricks, J. Kambhu, P. Mosser (eds.), "Systemic Risk and the Financial System," *Federal Reserve Bank of New York Economic Policy Review* 13, no. 2 (2007): 65–80.
139. C. M. Reinhart, K.S. Rogoff, *This Time Is Different, Eight Centuries of Financial Folly* (Princeton and Oxford: Oxford University Press, 2009).
140. A. Estrella, S. Park, and S. Peristiani, "Capital Ratios and Credit Ratings as Predictors of Bank Failures," *Federal Reserve Bank of New York*, Abstract (September 1999). See also C. H. Shen and M. F. Hsieh, "Prediction of Bank Failures Using Combined Micro and Macro Data" (Taipei, Taiwan: National Chengchi University, June 2004), which

- found some predictive value in the following fundamental indicators: ROA, noninterest expenses to total assets and to a lesser degree equity to total assets.
141. M. Arena, "Bank Failures and Bank Fundamentals: A Comparative Analysis of Latin America and East Asia during the Nineties using Bank-Level Data," Working Paper (2005–19) (Ottawa, Ontario: Bank of Canada, 2005), 17–18.
 142. Morris Goldstein, Graciela Kaminsky, and Carmen Reinhart, "Assessing Financial Vulnerability: An Early Warning System for Emerging Markets" (Washington, DC: Institute for International Economics, 2000).
 143. B. Gonzalez-Hermosillo, "Determinants of Ex-Ante Banking System Distress: A Macro-Micro Empirical Exploration of Some Recent Episodes," IMF WP/99/33 (1999).
 144. A. Demirgüç-Kunt, "The Determinants of Banking Crises in Developing and Developed Countries," IMF Staff Papers 45, no. 1 (1998): 81–109. Washington, DC: International Monetary Fund.
 145. As an alternative to a purely event-based approach, some researchers have proposed a definition of a banking crisis based on a "money market pressure index" reflecting the "banking sector's aggregate demand for central bank reserves," asserting that the defining characteristic of a bank crisis is a "sharp increase in the banking sector's aggregate demand for central bank reserves" as a result of loss of liquidity in the industry. The index is based on "the ratio of total reserves held by the banking system to total deposits" and changes in the short-term real interest rate. The rationale is that "in a period of high tension in the money market, this ratio increases either because the central bank makes additional reserves available to the banking system, or because depositors withdraw their funds from the banks." J. von Hagen and T. Ho, "Money Market Pressure and the Determinants of Banking Crises Center for European Integration Studies" (Bonn, Germany: Zentrum für Europäische Integrationsforschung, Universität Bonn, July 2004), 4.
 146. See Bell and Pain, "Leading Indicator Models of Banking Crises."
 147. C. Reinhart, M. Goldstein, and G. Kaminsky, "Assessing Financial Vulnerability: An Early Warning System for Emerging Markets," Institute for International Economics, 2000.
 148. The statistical and related quantitative aspects of the methodology will not be discussed.
 149. Ibid., 1.
 150. Goldstein acknowledged that at least with respect to some of the criteria, they are not necessarily relevant only in emerging markets.
 151. Note that these criteria are not mutually exclusive with those found in most such lists.
 152. M. Goldstein, "The Asian Financial Crisis: Causes, Cures, and Systemic Implications," Institute for International Economics, 1998.
 153. For example, private capital flows, growth rates.
 154. See, for example, Gabriel Jiménez and Jesús Saurina, "Credit Cycles, Credit Risk, and Prudential Regulation," Banco de España, January 2005, which found "strong empirical support of a positive, although quite lagged, relationship between rapid credit growth and loan losses," Abstract, 1.
 155. If the bank has borrowed in foreign currency and lent in local currency, then the devaluation will remain a market risk problem for the bank instead of being transformed into borrower credit risk.
 156. D. Rodrik and A. Velasco, "Short-Term Capital Flows," NBER Working Paper No. 7364 (1999), found that countries with a ratio of short-term external debt to reserves that was higher than unity roughly tripled their probability of getting into a crisis. Furman, Stiglitz, Bosworth and Radelet concluded that ". . . the ability of this variable [i.e., short-term debt to reserves] to predict the crisis of 1997 is remarkable." J. Furman, J. E. Stiglitz, P. Bosworth, and S. Radelet, "Economic Crises: Evidence and Insights from East Asia, *Brookings Papers on Economic Activity* 1998, no. 2 (1998): 51.
 157. Delhaise, *Asia in Crisis*, 85.

158. The government scheme was intended to encourage nonresidents to borrow from nonresidents but it was diverted to residents borrowing from nonresidents. See Delhaise, *Asia in Crisis*, 83–84.
159. Indonesia, Korea, and the Philippines also exhibited large currency mismatches that came back to haunt them as contagion spread throughout the region. Indonesia, the worst-hit country in Asia by the crisis, entered a vicious cycle of a falling currency creating further debt distress and a further devaluation. This happened in spite of the fact that “[Indonesia] had been better than its rivals in managing its currency over the past few years.” Delhaise, *Asia in Crisis*, 135.
160. Eichengreen and Arteta observe that between 1945 and 1971, when the Bretton Woods scheme collapsed, “tight financial regulation and control” prevented major bank crises from developing, with one exception. Eichengreen and Arteta, “Banking Crises in Emerging Markets: Presumptions and Evidence,” 2.
161. Delhaise, *Asia in Crisis*, 124.
162. G. L. Kaminsky and C. Reinhart, “The Twin Crises: The Causes of Banking and Balance-of-Payments Problems,” Board of Governors of the Federal Reserve System, International Finance Discussion Papers no. 544 (March 1996).
163. In addition, moral hazard contributed to the explosive mix. Implicit guarantees of government support of troubled banks that had been made in a highly regulated, comparatively low-risk banking environment continued post-liberalization.
164. While it benefits certain groups and individuals in the short run and may have some positive economic effect when the banking sector is in an embryonic state, the impact of policy lending is usually, albeit not always, negative.
165. Delhaise, *Asia in Crisis*, 23.
166. Abdul Abiad, “Early-Warning Systems.”

The Resolution of Banking Crises

[B]ank restructuring [is a] . . . package of macroeconomic, microeconomic, institutional, and regulatory measures taken in order to correct incentives and to restore problem banking systems to sustainable financial solvency and profitability

—World Bank¹

A banking crisis is an event. Bank restructuring is a process.

—Andrew Sheng²

As a banking crisis begins to unfold, there is not necessarily any single occurrence that makes it plain to all that whatever has occurred, or is in progress, is a banking crisis. As the quotation at the top of the page suggests, crises are not self-identifying. Indeed, some time may pass before even the most astute observers recognize that a crisis is underway. Looking back, from a vantage point in the present, there may be any number of vivid events that seem in retrospect to have marked yet one more step into the abyss. Leading into the current crisis that started in 2007, for instance, there were a number of such moments. The murmurs of a subprime crisis in early 2007; the evaporation of interbank liquidity in London in late summer of that year; the bank run at Northern Rock in 2007, the bailout of Bear Stearns in 2008, and the collapse of Lehman Brothers in October of that year, an event that highlighted a month of upheaval in the financial industry. As each juncture was passed, the sense that something extraordinary was occurring increased. By the time the way stations had all been traversed, there was no doubt that a banking crisis of enormous magnitude was in full swing.

Despite the fact that the future is always uncertain, there is a human tendency to assume that tomorrow will not be so much different from today. If tomorrow does turn out to be extraordinary, the expectation is that normal life will resume the day after.³ Consequently, as a crisis develops, each event—which in retrospect leads inexorably to the cataclysm—seems, in the moment, to be only a temporary aberration that will soon be displaced by more usual patterns. This explains why there are likely to be many premature declarations that the worst is over.⁴ Unless impelled by a severe shock resulting from severe underlying (and possibly unseen) imbalances, a divergence from the norm is ordinarily self-correcting; and, if not, a return to equilibrium ordinarily can be easily achieved through modest outside intervention.

In 1998, for example, the Russian debt crisis created a global liquidity crunch in certain asset classes. Although the reverberations from the event brought down the mega-hedge fund Long-Term Capital Management and threatened larger harm, LTCM was wound down in an orderly resolution through a collaborative effort among major banks guided by the U.S. Federal Reserve Bank. Except as a reminder that even genius can fail, it had little lasting impact on the U.S. banking system as a whole.⁵

While a banking crisis, on the one hand, and a single institution in distress, on the other, represent two distinct phenomena that produce qualitatively different systemic outcomes, they are related in that each involves the collapse or rescue from collapse of at least a single bank.⁶ It is evident that the least financially robust institutions always tend to fail first; or, as one researcher has put it, “bank failures result from the interaction of vulnerability and systemic shocks, where the weakest banks are the ones most likely to fail.” But the symptoms associated with an individual bank’s failure are the same whether or not a crisis looms. Whether the bank in question collapses quietly or whether its failure rocks the entire banking system to its foundations, it will exhibit those characteristics common to nearly all problem institutions:

- First, an unexpectedly high level of defaults by borrowers will cause NPLs to rise to such an extent that its capital will correspondingly fall towards zero, that is, a state of insolvency.
- Second, profitability will weaken to the point that the bank is unable to effectively generate capital internally.
- Third, the two foregoing processes will call into question its capacity to attract deposits and commercial funding so as to make its liquidity position suspect.

In light of the frequency with which banks in distress, irrespective of the origins of the distress, display such symptoms means that determining into which category a particular problem bank fits is practically impossible to accomplish in the moment. Are the problems at the bank purely of internal origin and idiosyncratic to that institution, or are they systemic? Is the bank the canary in the coal mine that, in this case, is not only warning of catastrophe but is also potentially itself the trigger of the mine’s collapse? The answers will be quite evident retrospectively but almost difficult to divine prospectively.

The preceding chapter focused on the causes and prediction of financial crises. This chapter examines their resolution. The aim of the chapter therefore is to provide a synopsis of the restructuring process and the key issues it raises. In the first section, an overview of the process is provided, into which some views concerning how bank restructuring is best effected have been incorporated. In the second section, a case study shows how bank restructuring was effected in one country in the not-too-distant past.

Note that, as used in the banking context, the term resolution has two meanings. It is used both to refer:

1. To the process of clearing problem assets (i.e., nonperforming assets including the delinquent obligations of borrowers and foreclosed assets) from bank balance sheets

2. To distressed or insolvent banks themselves

In this latter context, resolution refers to the winding up of distressed banks either in a straight liquidation or as part of a larger restructuring process in which the failed bank is split into two parts—the so-called good bank–bad bank approach.

Under the good bank–bad bank approach, which is one of a number of ways to deal with one or more failed banks, the bank is split into two separate entities. One part of the bank becomes the bad bank, which holds all the problem assets. The bad bank enters a process of liquidation. Other parts of the bank become the good bank, which is the surviving entity. The good bank is recapitalized, sometimes as a step prior to the acquisition by another bank. The overall process of resolving a bank crisis by rehabilitating distressed banks and undertaking reforms to prevent a future crisis (or at least lessen its impact) is called *bank restructuring*, or in situations where more than a few banks are in distress, *systematic bank restructuring*.

Bank restructuring is a term that is applied both to the rehabilitation of a single distressed bank and to the rebuilding of an entire banking system should a critical number of the major banks in a country become so undercapitalized that the functioning of the system is impaired. In other words, it can be used in a narrow sense to mean the reorganization of a single institution, or more broadly to refer to rehabilitation of the entire banking industry. In this broader sense, which is the sense used in this chapter, it is commonly applied to the comprehensive process of rehabilitation—from the onset of severe distress until the bank or banks have been restored to financial health.⁷

It is such a restoration that is the immediate objective of bank restructuring, with the larger goal being economic recovery, followed by healthy and sustainable economic growth. In other words, bank restructuring is not—or, at least, should not be—undertaken for the sake of the banks themselves. Instead, its rationale is to enable the banking system as a whole to resume its normal functioning as a source of credit and finance as well as other critical financial services needed by producers and consumers.

The underlying impetus for bank restructuring is the realization that the banking system can no longer effectively perform the functions expected of it—that is, to supply credit and thereby foster economic growth. The reason it cannot do so is simply a lack of capital. That is, there is no longer sufficient capital in the system to constitute a credible buffer against potential economic losses. In consequence, the requisite confidence in the system can no longer be maintained to supply the ongoing liquidity that banks require to operate.

Bank restructuring therefore represents the solution to the problem created by banks that do not have enough capital to function normally. Unsurprisingly, the core element of any restructuring plan is a program to recapitalize (or to facilitate the recapitalization of) vulnerable or distressed banks with the aim of restoring them to a sound state of solvency. It also generally includes, at a minimum, the following elements:

- The provision of emergency liquidity to distressed banks within the banking system as an interim measure
- The recapitalization of banks having weak solvency
- The removal of their bad assets

- The undertaking of reforms (e.g., making changes in the prudential regulation of banks) to improve risk management and reduce overall systemic risk

Finally, since confidence is critical to bank operations, a successful restructuring program will also restore public confidence in the banking system. Supplementary elements include corporate restructuring, which means providing assistance to corporate borrowers to work out delinquent obligations with banks for the purpose of mitigating bank loan losses and facilitate their and the economy's recovery. The major restructuring methods are encapsulated in Exhibit 15.1.

Although bank restructuring—in the context used in this chapter, that is, a comprehensive and systematic scheme to rehabilitate a banking system—is commonly undertaken following a crisis, the latter is not a prerequisite to the former. Bank restructuring can be initiated while market and economic conditions remain relatively benign, and that may be the preferable course of action. Political pressures, however, often obstruct efforts to undertake fundamental reforms of banking system until a crisis changes the political landscape.⁸

RECOGNIZING THE CRISIS

The initial hurdle to overcome in any bank restructuring, whether individual or systemic, is the lack of incentives for both bank managers and supervisors to recognize and resolve insolvent institutions.

—Marjory Waxman⁹

Uncertainty is another fundamental characteristic of a crisis, the civilian equivalent of the “fog of war.” Unhelpfully . . . very few crises self-identify in advance, notwithstanding the experts who, in hindsight, had clearly seen it coming.”

—Kevin Lynch¹⁰

It is natural that in retrospect the causes of a crisis emerge with greater clarity. Similarly, as time passes, the inadequacy of both existing safeguards and, particularly, the initial responses to events on the part of regulators become more evident. With 20-20 hindsight, it is easy to forget that the responses of regulators or central bankers to events as they occurred may have seemed the height of prudence at the time, even if later developments proved otherwise.¹¹

The Fog of Crisis

A number of factors may contribute to a delay in perceiving the magnitude of a looming banking crisis, including a lack of transparency in financial reporting or accounting standards that obscure the reality of bank solvency draining away by the day, as well as the political perils internal and external of crying wolf.¹² Not only are responsible officials expected to remain calm and display confidence in the banking system, they also are predisposed to act cautiously until the need for decisive action

EXHIBIT 15.1 What Exactly Is Being Structured? How Is Restructuring Being Implemented?

Type of Restructuring	Target of Restructuring	Process Phase	Overall Aim	Objectives and Examples of Typical Modifications
Operational restructuring	Bank operations	Recapitalization	Improve the bank's overall performance (e.g., enhance sustainable level of profitability)	Changes in senior management to improve the bank's credit culture Improvements in credit controls and credit evaluation process Staff redundancies and sale of excess branches to improve efficiency Additional cost-cutting measures to improve efficiency Use of technology to improve bank's operational efficiency Sale of peripheral business lines or subsidiaries to raise cash, improve efficiency, or enhance business focus
Capital restructuring ^a	Capital structure (and balance sheet)	Recapitalization	Improve bank's capital adequacy (i.e., solvency) and overall financial condition	Fully recognize loan losses to accurately calculate the bank's existing capital position and amount of capital required to restore full solvency (NB backdoor capitalization via AMC) In exchange for ordinary or preference shares, inject cash, liquid government obligations, or other assets appropriate to the circumstances In exchange for a subordinated note qualifying as Tier 2 capital, supply the bank with funds in accord with regulatory strictures Wipe out existing shareholder equity to compensate for losses Nationalize the bank to prepare the bank for subsequent privatization
Financial restructuring ^b	Financial assets (and balance sheet)	Problem asset disposal	Resolve problem NPLs through negotiations ("workouts") with the borrower gain partial recovery and rid the	Reschedule the borrower's delinquent obligation (e.g., by extending its maturity) Reduce the interest rate on the outstanding delinquent loan or other obligation Reduce the principal amount or eliminate penalty charges (i.e., debt forgiveness) Receive an equity stake in the borrower in exchange for all or a portion of the outstanding obligation (debt for equity swap)

(Continued)

EXHIBIT 15.1 (Continued)

Type of Restructuring	Target of Restructuring	Process Phase	Overall Aim	Objectives and Examples of Typical Modifications
			bank's balance sheet of cash-draining illiquid assets	Negotiate with workouts with the bank's creditors (i.e., the creditors of the distressed bank) to reduce principal payments (improves bank's capital position) Use the good bank–bad bank approach to enable bad bank to focus on shedding NPLs Purchase NPLs from the bank via AMC to improve efficiency of recovery process
Regulatory restructuring	Bank regulatory structure	Reform	Improve prudential regulation and bank risk management at the national level	Strengthen prudential regulation to improve bank risk management, and if necessary increase agency authority Impose stronger reporting and disclosure requirements on banks Set higher corporate governance standards to rein in management abuses Improve the legal infrastructure especially pertaining to creditors' rights and insolvency proceedings to make recovery easier Reorganize the bank or financial regulatory structure to operate more effectively
Corporate restructuring	Delinquent corporate obligors	Corporate restructuring	Facilitate negotiations of workouts between corporate borrowers and banks	Mediate workouts between corporate borrowers and banks Assist with resolution of complex transactions involving multiple lenders Help corporate borrowers recover obligations from their own delinquent customers or other obligors

^aRecapitalization will necessarily change the bank's capital structure through the infusion of new assets (e.g., cash) and changes in equity and liabilities depending upon the form the capital infusion takes.

^bFinancial restructuring can refer to the restructuring of financial assets such as loans on the bank's asset side of the balance sheet (financial asset restructuring, financial instrument restructuring), or it can refer more broadly to the process of "cleaning up" the bank's balance sheet. In this latter sense, it would take in capital restructuring (shown separately on this table) as well.

has become unmistakable. The onset of a crisis tends to be a period of great uncertainty to which the intrinsic instability of the banking industry contributes. Because of the vulnerability of all fractional-reserve banking systems to a loss of confidence, an element of uncertainty invariably exists as to whether the level of aggregate capital and the overall financial strength is sufficient to prevent both individual bank failures and a systemic breakdown.¹³

Before the first wave of announced losses, bank closures, and accompanying panic breaks, uncertainty reaches a peak. Then, as the next problem bank teeters on the edge, the news media fills with a cacophony of conflicting signals—jeremiads warning of impending catastrophe, the voices of moderation, the paeans to the status quo, and the calming official rhetoric that is intended to maintain public confidence. The last of these comes from the bank regulators and other responsible officials, who, of course, are just doing their jobs in an attempt to bolster public confidence. When markets are jittery, to display anxiety or the expectation of a calamity is only apt to make things worse. There is always a possibility that so long as confidence is maintained, disaster can be deferred and perhaps even avoided.

The uncertainty characteristically present before a latent crisis becomes a patent one in an acute manifestation of what has been called the “information problem” in finance.¹⁴ During the period immediately before and after the events that mark the beginning of widespread acknowledgement that a crisis is occurring, the flow of information becomes especially critical to the operation of financial markets. But it is at this very time that obtaining information becomes more difficult owing to the “suddenness of changed perceptions” that results from striking shifts in asset prices.¹⁵ After the climactic events have occurred, the shift in perception begins to move the market participants, regulators included, to ask themselves, *what was the cause of what just happened, and what is to be done about it?* For some market participants, the need to answer the second question may be unavoidable. With the risk of loss rising daily, their decision will directly affect the outcome they confront. In a flight to quality, they may decide to cut their losses, and put their funds into safe havens. The more daring may seek to profit from the pending plunge in prices. For regulators and policymakers, inclined to act with caution but with no personal stake in the outcome, it may take longer to realize that it is no longer business as usual.

Denial precedes acceptance, and a critical first step in the restructuring process is to recognize that a crisis is in progress and that time is of the essence. In this context, it has been observed that the instinctive response of the authorities to a crisis at the early stage of a banking crisis is to offer too little rather than too much assistance.¹⁶ This inclination is understandable, in that regulators and central bankers are rightly concerned not only about the immediate impact of their actions, but also about the future effect, especially in respect to setting a bad precedent and thereby creating moral hazard.¹⁷ To act too quickly and give too generously risks giving the impression that the government will always rush to rescue a bank that gets itself into trouble. Yet, at the same time, regulators understand that the banking system is subject to intrinsically unstable events and that runs on banks, whether they take place outside the organization’s front doors or in the money market, are self-fulfilling.¹⁸ Delay therefore can be, and waiting too long risks making a bad situation worse.¹⁹

Once the situation is officially acknowledged and labeled as a crisis, the dialog changes. Former optimists now condition their optimism on the government coming

to the aid of banks, while government officials now must explain why previously unimaginable amounts of public funds are required to rescue the banking system. As filtered through the news media, the novelty of the event can contribute to the impression that it is a virtually unprecedented occurrence without parallel in recent times. Given that few people predicted it, it is inferred that it could hardly have been prevented. The fact that in any given country banking crises occur rarely or make an appearance more than once every generation contributes to the sense that the current episode was almost practically impossible to foresee or guard against. Should crises develop in rapid succession in a number of countries, the perception that the event is without parallel in modern history will become even stronger. Like the boom that preceded it, many will think “this time it really is different.” Struck by the apparent uniqueness of the event, instead of looking to remedies that have worked successfully in the past, they may advocate remedies on the basis of organizational self interest or upon what seems to be politically palatable.

To be sure, each crisis has its own defining shape, its own fingerprint so to speak, that is the result of internal and external weaknesses interacting in a complex and dynamic manner.²⁰ As conditions change, so too do banking crises. But as one scholar has suggested, it is just such differences—perhaps one might call them mutations—that permit banking crises to elude collective memory and recur over time:

*It is because the details vary that crises can occur. “This time it will be different” is the response given by boosters to words of caution as a bank or banking system moves into risky territory; and indeed it usually is, though not in a good way.*²¹

There is no one-size-fits-all solution to all banking crises and, equally none of the various options are without their trade-offs. Trade-offs acceptable in one country may not be acceptable in another. Choices concerning the best method to repair a national banking system may be influenced by the specific causes of the bank losses that have come to infect the system, and by the manner in which the crisis evolved until the decision point.²² Nevertheless, when the longer view is taken, the similarities among financial crises tend to overshadow the differences. In looking at many different bank crises, the same core elements are consistently present: a sharp decline in capital owing to a sharp rise in the NPLs and other problem assets relative to their total assets. Consequently, while the restructuring plan may differ in a number of respects, a good program will always incorporate measures to restore capital, to shed problem assets, and to enable banks to maintain healthy solvency levels while preventing the large-scale recurrence of loan losses.

Learning from the Past

Whether you look at the U.S. savings and loan crisis of the 1980s, the Norwegian and Swedish crises of the early 1990s, the Mexican crises of the early 1980s and mid 1990s, or the current . . . crisis, analysts will tell you that they could have predicted the eventual collapse of a particular banking system because of its underlying systemic problems; yet somehow

no one recognized these inherent problems until they were in the midst of a banking crisis.

—Marjory Waxman²³

Just as there are frequently seen patterns of causation in the origins of banking crises, so too do the successful solutions exhibit many similarities. Much can therefore be learned from examining how bank crises have been dealt with in the past.²⁴

How to Rebuild a Banking Industry

In a paper entitled “Experiences of Resolution of Banking Crises,”²⁵ Stijn Claessens, an economist at the World Bank, distilled the lessons gleaned from attempts to restructure banking systems in a number of countries around the globe over the preceding 60 years. The approaches outlined in his paper remain highly relevant, and this chapter draws on his work as well as on the work of others who have delved into this subject. It is apparent that the technology for restructuring banking systems was already well developed by the early 1990s, and that the processes and techniques, some of which have been employed at least as early as the 1930s, still comprise the larger part of today’s best practices.²⁶

In *Bank Restructuring: Lessons from the 1980s*, also published by the World Bank,²⁷ Andrew Sheng spelled out what he viewed as the principles and elements of bank restructuring. Although in his view the causes of bank crises were too “complex and . . . country-specific . . . to generalize a solution” for every case, Sheng identified common principles and elements that characterized their successful resolution.²⁸ The principles,²⁹ as seen in Exhibit 15.2, underpinned the formulation of a successful program to fix a country’s broken banking system, and the implementation of its four basic stages (elements). These stages, which are described next, are found in nearly all comprehensive restructuring programs.³⁰

Diagnosis refers to the need to reach an understanding concerning the nature of the crisis and its underlying causes, as well as its likely magnitude and duration. The purpose of the diagnosis phase is to determine the gravity of the crisis and what actions, if any, should the government take in response. Diagnosis generally begins with the recognition that there is an out of the ordinary problem affecting the banking sector that needs to be examined more intensively.³¹

Damage control means the taking of urgent measures to stop losses from escalating, and is relevant to the onset of the crisis as well as to its immediate aftermath. It is discussed later in this chapter under the heading “First response.”

Loss allocation denotes the political and policy decisions as to who bears the burden of replacing lost capital. It refers to what might broadly be called the planning and rehabilitation phase of bank restructuring when distressed banks are recapitalized, bad assets are resolved, and associated restructuring measures are undertaken. This phase normally takes place after emergency measures have been implemented and a comprehensive bank restructuring plan has been prepared. This is discussed in “Planning to rebuild” and “Recapitalization and Restructuring.”

Rebuilding the incentive structure comprises the implementation of regulatory and corporate governance reforms to encourage the development of a strong credit culture to facilitate the sustained success of the recapitalization and restructuring scheme and to avoid repeating the mistakes of the past.

EXHIBIT 15.2 Bank Restructuring: Lessons from the 1980s (Andrew Sheng)

Principle	Corresponding Element
1. Financial stability rests on the government's ability to maintain a stable currency.	<i>Diagnosis.</i> What are the causes of the crisis? What type of assets are the sources of the losses incurred by the banks?
2. Banks fail because of losses in the real sector, compounded by poor risk management and fraud.	
3. Liberalization programs often fail to take into account the wealth effects of relative price changes, and inadequate supervision creates further losses.	
4. Bank losses ultimately become quasi-fiscal deficits.	<i>Damage control.</i> The implementation of measures to prevent losses from increasing.
5. Failure recognition is important because a banking crisis is a solvency problem, not a liquidity issue.	
6. Stopping the flow of future losses is critical.	
7. The method of loss allocation determines the success of the restructuring program.	<i>Loss allocation.</i> Who bears the cost? Determining how bank losses will be allocated, where new capital will come from, which institutions will receive it, and under what conditions.
8. Success depends on sufficient real sector resources to rebound efficiently and safely, on the budget's ability to tax winners and wind down losers, in order to pay off losses, and on adequate financial sector reforms to intermediate without disturbing monetary stability.	
9. Rebuilding a safe and profitable banking system requires good policies, reliable management, and a strong institutional framework.	
10. Time and timing are of the essence.	<i>Rebuilding the incentive structure.</i> Creation of institutional frameworks that prevent imprudent activities.

Source: Andrew Sheng, "Bank Restructuring: Lessons from the 1980s," World Bank publication.³²

Views from the Front Lines the Last Time Around

Andrew Sheng's book came out in 1996, the year before the onset of the Asian financial crisis, and seems to have been influential in shaping the thinking of those who grappled with the question of how to repair the region's damaged banking systems.³³ As they sought to educate their investor clients in the depths of the crisis concerning future prospects, bank analysts covering the region's equity securities also came to grips with the problem policymakers faced, commented on the efforts made, and proposed their own solutions. Their prescriptions add some color and flavor to a topic that is sometimes missing from the academic literature.

In light of the foregoing prescriptions, it should be apparent that when the problem of widespread bank insolvency is considered, there is considerable agreement concerning how to solve it. Subsequent to the Asian crisis, the volume of papers and other publications that look at the question of how to rebuild a banking system has substantially increased. A number of them drew on Sheng's findings.

Several investment bankers produced suggestions, for example Stephen Frost, then at UBS in Singapore. In his January 1998 report, six months after the Thai baht devaluation, serious problems were evident in Thailand, Indonesia, and Korea, and Frost set out a dozen essential steps to repair the banking systems in the principal crisis affected countries, most of which are condensed into six groups as follows.³⁴

Restore confidence in the banking system by guaranteeing the funds of ordinary retail depositors in full and treating commercial depositors and creditors, whether foreign or domestic, equitably. (Let large depositors suffer some loss, however, to prevent moral hazard.)

Create incentives to encourage banks to resolve NPLs (e.g., tax incentives, institutions to purchase NPLs from banks).

Remove obstacles to the write-off and recovery of problem loans, and to banks enforcing their rights as creditors whether through the foreclosing on loan collateral, the initiation of bankruptcy proceedings, or otherwise, while streamlining foreclosure and bankruptcy proceedings (and shortening the time required to effect the foreclosure and liquidation of collateral).

Assess bank losses to ascertain which banks should be recapitalized and which should not; then take action: shutting down and liquidating the nonviable banks and forcing them to merge with stronger institutions, while making capital available, or facilitating the recapitalization (e.g., through government guarantees) of the fundamentally viable banks.

Further assist capital raising and accumulation, and the profitability of the banks in the process of rehabilitation through various measures, including allowing them to shed staff more easily; dampening interest rates to keep bank funding costs low; and permitting foreign banks to acquire local banks.

Seek to improve risk management both internally within banks and within the regulatory infrastructure through improvements in prudential regulation and enhanced enforcement (e.g., single-borrower limits), while compelling banks to improve their financial disclosure and become more transparent.

In addition to the foregoing prescription, Frost evaluated the magnitude of banking crises in three Asian countries (Thailand, Korea, and Indonesia) and assessed the response of the authorities to the crisis to date in each of several categories of remedial actions. In a summary table, he grouped the criteria against which he conducted his assessment into the following categories:

- Key structural features of the banking industry
- Overall precrisis sector-wide profitability expressed as ROA
- The source of the crisis
- The NPL ratio
 - Precrisis
 - Current estimates (classified by sector where possible)
 - Estimated peak NPLs
- Guarantees provided to depositors, if any

- Current loan-loss reserves to total loans and equity to total loans
- The cost of recapitalizing the banking system in total, and then in relation to
 - GDP
 - Government expenditure
 - Financial system equity
 - Multiple of equity raised annually in the capital markets
- The number and identity of any banks closed to date and percentage of total system assets represented
- Bank rescues, if any
- Announcements concerning banks to be recapitalized, if any
- Foreign ownership, including limits on foreign ownership
- Whether a program to resolve bad debts had been set up
- Changes in bankruptcy laws, if any

FIRST RESPONSE

Policy responses to crises naturally divide into immediate reactions in the heat of a crisis, and long-term responses after the end of the crisis. . . .

—Charles W. Calomiris, Daniela Klingebiel, and Luc Laeven³⁵

To bring to mind the sort of circumstances in which a banking crisis frequently arises, and in which regulators must take action, imagine a hypothetical country where the economy has been going strong for a number of years. The stock market is doing well, and housing prices, which have been rising steadily for more than a decade, are so high that the state of the property market amounts to an almost constant refrain to everyday life. At some stage, of course, boom turns to bust, and over the next few days, weeks, or months—as the bubble deflates—borrowers default in increasing numbers. The resulting flood of NPLs quietly but quickly begins to erode bank profits, reserves, and finally the thin layer of capital upon which the entire banking system stands, imperiling the survival of the affected institutions and the capacity of the financial system to function. The weaker banks are worst affected, the strongest banks seem to be little affected, and the majority are somewhere in between. One of the weakest fails, but the banking system is still regarded as fundamentally strong.

Before long, news reports mention another bank and then another that may be in difficulty, but this is fervently denied by their respective managements. On a road show, a senior manager at one of the institutions that has been the target of stories is asked by fund managers about the news reports and about the bank's solvency. The manager assures those present that the bank is quite strong and that they should have no concern. But the rumors persist, and some begin to suspect that the problems of weak capital are not confined to one or two isolated institutions.

Which banks are in trouble? Which are not? No one knows, or no one is saying. The last interim financial statements were published just a month ago and the last annual reports six months before that. Results were off from their peak, but on the whole exhibited fairly robust numbers. Amid uncertainty in the market as to which banks are sound, funding first becomes more costly and then becomes less available.

A panic arises, making all the banks vulnerable to a shortage of liquidity and imperiling the survival of the affected institutions and capacity of the financial system to function. Not long afterwards, the bank, the manager of which attempted to reassure his questioners, is taken over by regulators, and awareness that a severe banking crisis is underway crystallizes.

SUPPLY LIQUIDITY AND STOP THE BLEEDING

In the early stages of any crisis . . . international experiences offers a clear guide . . . don't provide liquidity on an on-going basis . . . until you are satisfied that oversight is more than adequate, don't close a financial institution in the middle of a systemic crisis until there is a credible system wide policy on resolution; don't announce a blanket deposit guarantee if depositors are merely running to quality within the system; and don't . . . [undertake major initiatives] . . . except in the context of a coherent and workable plan.
—Stijns Claessens³⁶

In connection with a serious financial crisis, it is important first and foremost to maintain the banking system's liquidity. It is a matter of preventing large segments of the banking system from failing on account of acute financing problems.

—Urban Backstrom, former Governor, Swedish Riksbank³⁷

It was mentioned in the preceding chapter that liquidity concerns predominate in the early days of a crisis. Although the classic bank run in which crowds gather outside the bank's doors is now rare, the same event effectively playing out in the money market is not at all unusual. While such panic is often underpinned by concerns about the solvency of local banks, it can also be set off by external shocks, as when a bank panic in another market is transmitted across national borders, a process referred to as *contagion*. Whether triggered by concern about the fundamentals of the domestic banking system, or by some seeming similarity between two far-flung markets, liquidity crises are driven in the short term by uncertainty and the fear it inspires. Once ignited, a self-perpetuating dynamic takes hold, making it rational to join the run. The result is that just at the moment when the need for liquid funds to meet current obligations rises, access to liquidity in the money market may disappear.

When liquidity evaporates, banks find themselves suddenly in immediate peril. Unless they can manage to borrow new funds, or obtain cash by liquefying assets, the banks with insufficient cash to meet their current obligations will fail. Should conditions be ripe, an entire banking system can be brought down in short order. In recognition of this danger inherent to fractional reserve banking, most governments throughout the world assign the role of responding to intermittent irregularities in the supply of liquidity obtained from the interbank and money markets to the country's central bank. We all now know from the current financial crisis that even a central bank may find itself helpless in the face of an interbank market grinding to a halt.

The central bank, functioning as a lender of last resort, provides the requisite liquidity to the banks no longer able to access adequate funding in the money market or from deposits. Assuming that the market has confidence in the central bank's wherewithal and commitment to see the crisis through, the liquidity crunch will vanish as quickly as it arose.³⁸ If, when liquidity is supplied, whatever fundamental concerns that exist about solvency are not sufficiently great to justify a full-blown banking crisis, conditions will soon return to normal. If not, then the liquidity crunch becomes a prelude to the second stage of a developing banking crisis.

In contrast to the first phase of a crisis, the second stage of a banking crisis is impelled not by uncertainty, but by knowledge. The uncertainty is greatly reduced but the revealed reality concerning the solvency of the banks within the system may be as bad or be worse than imagined. That is, once the government has indicated it will support the banking system with liquidity, banks are more willing to acknowledge their losses, and the principal concern becomes the underlying solvency of the banking system, and how to rebuild capital to adequate levels. Since solvency issues cannot be resolved simply by turning on the tap, the second stage of a banking crisis usually continues over an extended period.

Emergency Liquidity

By making liquid funds liberally available to banks—that at least upon preliminary assessments appear not to be beyond redemption—the central bank buys time for the country's regulators and policymakers, and prevents the premature collapse of institutions that are still salvageable. Equally important, it helps to preserve a modicum of confidence in the banking system as a whole, or at least to prevent the complete collapse of confidence. To the extent that confidence is lost, efforts to preserve the banking system will be undermined, and the magnitude of the problem will expand. Once emergency liquidity has been provided, the authorities should act decisively to ensure that a modicum of confidence is restored in the banking system and prevent loan losses from escalating further. If, despite the infusion of liquidity, confidence in the country's banks remains weak or erodes further, regulators will face a pressing challenge to prevent a further loss of confidence.

One step that might be taken, should an effective deposit insurance program not exist, is the implementation of a blanket deposit guarantee policy. Many countries resorted to that policy in 2009 and 2010. Such an all-encompassing guarantee is frequently limited to retail depositors, but sometimes wholesale depositors and commercial counterparties also benefit. While it creates some risk of future moral hazard, a blanket guarantee will normally serve to restore depositors' faith that their deposits are not at risk while, at the same time, relieving the pressure on banks' liquidity. The exception would be a situation where the government is not trusted, or where legitimate doubt exists concerning the capacity of government to honor its commitment. Another step the government could seek would be to educate the public concerning the need for banking sector reform and evoke confidence through its actions, making the rehabilitation of the financial system a priority. Naturally, in doing so, a fine balance must be drawn so that the populace becomes aware of the importance of the maintaining the banking system without confidence being further depressed.

Curb the Rise of Asset Losses

Although the proximate cause of bank failure is almost invariably a dearth of liquid funds, the vulnerability of banks to liquidity problems increases when their capital buffer becomes too thin. Liquidity crunches rarely come out of the blue. The market anxiety that drives them reflects a lack of confidence in banks' capacity to absorb the losses they have already made or can be expected to make in the near future owing to souring asset quality. As more borrowers default on their obligations, NPLs and provisioning costs rise, existing loan-loss reserves are consumed, and capital inexorably declines. It is apparent that the more fundamental cause of bank distress is the weak asset quality of their loan portfolios.

In the early days of a crisis, apart from the liquidity problems banks face, their loan losses are apt to still be climbing at this stage. As a result, there remains a substantial risk of bank insolvencies as capital is eroded. Should loan losses rise to the level of shareholders' equity, the bank is deemed insolvent.³⁹ Once the point of insolvency is reached, however, a bank's financial position may still continue to get worse, with shareholders' equity moving into negative territory, and creating a debt overhang. For this reason, as the value of shareholders' equity slides towards zero and the bank approaches insolvency, it is critical that it be watched very closely by regulators. Otherwise, it conceivably could incur losses that would consume its capital several times over, including, of course, any unsecured funding that the state would likely be making available in the interim. Losses of such magnitude not only complicate the rescue of the bank, making it potentially impracticable, but also increase the cost to the government of repairing the banking system.

The danger of losses mushrooming as solvency levels decline is aggravated by the change in the incentive structure that occurs as capital falls to ever more unsustainable levels. Stijn Claessens observed that the rapid obliteration of bank capital during a crisis causes a dramatic shift in the perceptions and the risk preferences of its shareholders' and its managers.

Managerial and shareholder incentives suddenly shift for a financial institution when it becomes insolvent: managers have no incentive to run the institution on a viable basis and their actions often speedily drain away resources—including liquidity support from the central bank.”⁴⁰

When capital falls well below regulatory minimums, management (and owners) can react in a number of ways. They may:

- Gamble
- Engage in business as usual
- Rein in lending insofar as possible, by limiting new lending to all but the most creditworthy customers and applicants, and conceivably reducing the bank's total financial assets

Gambling, in this context, means betting on a high-risk, high-return transaction (this stance is sometimes referred to as “gambling on resurrection”)⁴¹ or short-term

strategy to save the bank and the managers' jobs. A strategy that will pay off well enough in the remote chance that it succeeds, but that will otherwise probably cause large losses on top of those already sustained.

Business as usual means continuing to operate in the same manner that got the bank into trouble in the first case; that is, rolling over NPLs and credit lines thereby advancing funds without a reasonable prospect of timely repayment. This may be disguised through cosmetic restructuring, where the new advances are ostensibly provided on the basis that the existing loans have been restructured.⁴²

Reining in lending, the third approach is an attempt to rebuild capital. Assuming all capital has not been exhausted, merely shrinking the asset base will create some improvement in capital ratios. The degree to which the bank will need to cut back depends of course upon its financial condition, the severity of the crisis, preexisting commitments, and management's risk appetite.

Of the three approaches listed, the last is probably the only acceptable basis for continuing to extend liquidity until recapitalization and restructuring initiatives can be commenced, after which more normal lending can again resume. Continuing to lend other than on this basis—that is, as part of gambling strategy or business as usual approach—will likely lead to even greater losses down the road, and will only make the task of repairing the banking system more difficult at its end. It is therefore critical that regulators should take whatever measures necessary to prevent loan and other asset losses from increasing distressed institutions from continuing to extend credit to delinquent borrowers.

A Problem with Authority

An important question to be answered at this early stage in the restructuring process is whether the regulatory body responsible for supervising banks has adequate authority to respond effectively to a crisis. It is easy to assume that all bank regulators have comparable powers, but this is definitely not the case. In some countries, the powers of the regulatory agency are actually quite limited. In evaluating the strength of the regulator, the critical question to pose is this: Does the agency have the power to compel institutions under its charge to take particular types of remedial steps consonant with urgency of the situation?

As Hoggarth, Reidhill, and Sinclair observed:

*[T]he supervisory agency may lack the power to write down capital, force a merger or close an institution; and if it does, may face prosecution by creditors and owners for damages. In other countries the [agency] . . . can replace managers and the board of directors, close a bank, inject capital and nationalise [it].*⁴³

To answer the foregoing question at the operational level, more specific queries include an assessment of whether the regulator has the power, without the voluntary consent of shareholders or management (but subject to appropriate guidelines), to:

1. Impose restrictions on banking operations generally
2. Compel the bank to cease lending or to lend only up to a certain maximum amount
3. Impose restrictions on specific banks identified as distressed

4. Compel the write-off or write-down of existing loans
5. Pursue the recovery of the collateral intended to secure such loans
6. Put in place new credit and credit risk management policies
7. Shut down an insolvent bank
8. Force its sale or merger
9. Transfer the assets of the bank in whole or in part to another entity
10. Compel the write down of problem loans and their sale to another entity
11. Replace a bank's senior management
12. Compel the sale of subsidiaries or divisions
13. Take over the bank and nationalize it
14. Compel staff redundancies

In some countries, as suggested, such powers are available to the regulatory agency; in others, the agency may possess few or none of them. In some, the powers are formally delegated to the agency, constitute an accepted part of the regulatory regime, and are readily enforced; in others, bank regulators may possess the relevant powers on paper for all of these issues but may be unable to exercise such authority in practice; while, in still other countries, even formal authority may be lacking.

If existing regulatory bodies do not have the authority to respond effectively in a crisis, legislation or appropriate decrees should be enacted to grant them emergency powers to enable appropriate intercession in bank operations. It may also be advisable to strengthen the authority of the regulatory agency on a permanent basis to enable it to more effectively enforce prudential rules.

Planning to Rebuild

In principle, the goals of financial restructuring are to reestablish the creditor-debtor relationships upon which the economy depends for an efficient allocation of capital, and to accomplish that objective at minimal cost.

—Charles W. Calomiris, Daniela Klingebiel, and Luc Laeven⁴⁴

Systemic bank restructuring aims to improve bank performance—that is, restore solvency and profitability, improve the banking system's capacity to provide financial intermediation between savers and borrowers, and restore public confidence.

—C. Dziobek and C. Pazarbasioglu⁴⁵

Banking crises have a long history. The history of systematic policy-driven efforts to repair crisis-hit banking systems is much shorter. Nevertheless, in view of the greater frequency of financial crises during the past four decades, governments throughout the world have collectively built up a wealth of experience in addressing the question: What is the best way to rehabilitate our banking system? In its broad outline, there is a fair amount of consensus at the conceptual level concerning what must be done to resolve a banking crisis.⁴⁶ This is particularly the case examining the views of analysts, the World Bank, IMF, and similar multilateral institutions, and much of the scholarly literature.

This should not be surprising, for failed or failing banks tend to display very similar signs of distress—such as poor or deteriorating asset quality, insufficient capital, weak profitability, and so on—even if there is no single cause or consistent combination of causes that have brought on their condition. Of these, a lack of capital is the most critical. As a number of commentators have emphasized, a bank restructuring program should be tailored to the local operating environment. In practice, this means that institutional and, particularly, political constraints will shape both the design and the implementation of the program. It is particularly at this level that constraints on government action become very real. Bank restructuring schemes can be highly controversial at the grass-roots level, where, for understandable reasons, the notion of bank bailouts may not be popular.⁴⁷ There are good reasons for this, but there are no easy solutions, as ultimately the government has little choice but to pick up the bulk of the tab. Nevertheless, there is a balance to be struck in respect to the allocation of losses and costs, as well as the allocation of prospective benefits, and this should be done in a manner to create reasonably equitable and optimal outcomes among all the relevant stakeholders, including the government on behalf of its taxpayers.⁴⁸

The Plan to Rebuild: Aims, Overview, and Features

Until now, the focus has been on the initial stages of the crisis—the uncertainty that prevails and the need to provide emergency liquidity and to take steps to maintain confidence in the banking system, as well as the importance of halting further lending to delinquent borrowers while continuing to fund viable enterprises. In the early stages of a crisis, in addition to taking emergency measures, those responsible for overseeing the banking system should be working to assess the scope of the banking system's losses and the corresponding costs of recapitalizing. In coordination with senior policymakers, the critical task facing the government will be to understand what is needed, determine what can be supplied, and to implement the decisions taken through a banking system restructuring plan.

Objectives of the Restructuring Plan: Restore the Soundness of the Banking System and Confidence in It

The objectives of a bank restructuring scheme, and the basic elements of the process, are comparatively easy to understand and are reducible to a small number of items. As has been noted, a healthy banking system is an essential element in fostering economic growth. It underpins economic growth by supplying credit to viable and growing businesses and consumers—that is, efficiently allocating capital to those who can make best use of it—and by transmitting government monetary policy into the real economy and by facilitating the circulation of funds to pay for goods and services. The primary goal of a systemic bank restructuring scheme is therefore to restore the banking system's capacity to function as engine of economic growth. Ideally, this goal should be accomplished at a minimal cost to taxpayers and in a manner that allows for a rapid economic recovery but does not sow the seeds for another banking crisis 5, 10, or 20 years' time. In light of a bank restructuring program's overarching aims, the objectives of a restructuring plan are twofold.

First, the solvency of the banking system must be reestablished along with any ancillary measures needed to restore the system's functional capacity. Second, not

only must these substantive changes be made, but also measures must be taken to similarly restore the public's confidence in the banking system's soundness and integrity.

The Distinctive Features of a Bank Restructuring Program

The devil truly is in the details, however, owing to the special challenges posed by such a project and its somewhat unusual combination of features, which ordinarily include enormous costs and the political contention such costliness engenders; the establishment of a special regime to implement the program; a degree of unavoidable complexity that almost inevitably involves a number of different regulatory bodies; the need to tailor the program to the local environment; and the shortage of time available.

Cost and contentiousness. Banking system restructurings are among the most expensive event-based public expenditures. As discussed in the previous chapter, all-in cost of banking crises is not infrequently more than 10 percent of GDP, although not all the costs are reflected in the injection of new funds. In view of the large sums involved, and the complexity of the problem, both the economic capacity and political will to effect good public policy may be lacking.

Special regime. Nominally an individual bank restructuring is the financial services equivalent of corporate bankruptcy proceeding. A systemic bank restructuring, however, has no real analog in the corporate world. Relative to the funds and energy expended on restoring the banking system, the rather paltry effort by the U.S. government in 2008–2009 to save General Motors and provide assistance to the other big three carmakers in the once-dominant American car and truck manufacturing industry only serves to illustrate the stark difference in approach to a nonfinancial industry relative to the banking sector.

The usual approach is for insolvent banks and corporates to be dealt with under separate legal regimes, and banks do not commonly come under the jurisdiction of the same courts that handle corporate bankruptcies. Instead, initially at least, insolvent banks tend to come under the authority of the bank regulator. One rationale is that banks are already very intensively regulated, and have in place a ready-made oversight infrastructure for the regulators to supervise them. The regulators and supervisors presumably also have the necessary expertise to deal with an insolvent or materially undercapitalized institution, and therefore it is sensible that they should have at least first responsibility for deciding how to deal with distressed banks.

Bank nationalization, which is often considered a potential approach to resolving distressed banks in a banking crisis can in this sense be thought of as a sort of oversized bankruptcy hearing. Instead of being handled by a court and an appointed administrator, in view of the costs involved and the deep public interest in the outcome, bankruptcies of large banks during a crisis are in part dealt with through the political process rather than through the judicial process. Naturally, along with the bank's shareholders, creditors, and other stakeholders, one of the parties at the table is the government, acting both in its public policy capacity and on behalf of taxpayers who will in the end likely foot a large share of the bill for recapitalizing the banking system. Especially at the system-wide level, bank insolvency is not simply a legal or administrative matter.

Complexity. In addition to the enormous burden it typically places on the public treasury, and in turn upon the taxpayer, a noteworthy characteristic of systemic bank restructuring is its complexity, arising partly from the need to coordinate the process across a number of government departments. A scheme of this sort can be characterized as a cross-disciplinary cross-agency endeavor, one that requires expertise in banking and bank supervision, macroeconomics, and economic policy management, as well as law and accounting. Participants in the bank restructuring program will normally include the central bank and agency responsibility for bank supervision, the deposit insurance agency, the treasury ministry, and those responsible for economic planning and formulating new legislation in other departments, as well as senior members of the specialist institutions established to dispose of bank assets and assist in the crisis resolution.

Country-specific criteria. It seems plain that the severity of the crisis will play a part in determining the remedy. For example, persuading private investors to invest funds in insolvent banks in the aftermath of a very severe crisis without some level of support from the government is likely to be a challenge.

In addition, there are bound to be differences in the approach taken to bank restructuring from one jurisdiction to another. Laws and procedures governing insolvency and bankruptcy proceedings, and the institutional arrangements under which bank supervision is conducted, vary considerably around the world. Hence, generalizations about the bank restructuring process will not necessarily be applicable in any particular jurisdiction. While there exists no uniform template, there is nonetheless a fair amount of consensus on the principal features that should be included within a bank-restructuring plan.⁴⁹

Finally, it is worth stressing that the reason the task of repairing the banking system is taken on by the state is because there are no other practical options. In the midst of a crisis, there will likely be little, if any, interest on the part of private investors in putting new funds at risk in banking enterprises that are already underwater, certainly not without some government guarantees that represent a contingent cost to the state.⁵⁰ It should be recognized that when bank restructuring is undertaken during a severe crisis, the options are especially limited. Banks that during stable conditions would be regarded as too far gone to resuscitate may be treated differently in the depths of an extreme banking crisis. Although some triage will probably be necessary, the normal rules governing treatment of distressed banks tend to be put aside when the banking system is hanging by a thread.

Assess the Damage

While the first actions just discussed (providing emergency liquidity support and curbing distressed bank operations that threaten to significantly increase loan losses) are still being taken, the next task is to assess the likely severity of the crisis, including the estimated total cost and the steps necessary to restore the banking system to full functioning. Both the system as a whole and particular institutions must be examined. Indeed, it will be difficult to obtain an accurate picture of the system as a whole without taking a close look at specific banks.

A critical question that needs to be answered quickly is whether the government will be able to resolve the crisis without a substantial expenditure of state funds, or whether it will need to take responsibility for recapitalizing a significant number of

banks.⁵¹ In view of the assessment reached concerning the causes, scope, and magnitude of the banking crisis, together with the cost of its rehabilitations—in particular, the cost of recapitalizing the system to acceptable levels—appropriate steps can then be taken to develop a comprehensive plan to address the crisis. The plan should address in detail the manner in which the recapitalization, restructuring, and reform of the banking system is intended to be accomplished.

Often, the government will find it appropriate to establish temporary agencies and institutions to implement necessary measures.⁵² Finally, as mentioned, it may be necessary to establish new laws and regulations setting out the government's authority to intervene to compel changes in bank operations, management, and ownership for the purpose of implementing agreed-upon remedies.

Estimate the Loss

Replacing lost capital, however, is frequently a very costly endeavor, the cost being a function of the volume of capital destroyed, and not one that governments undertake lightly. Moreover, bank restructuring cannot be solved solely in reference to logic and objective rational criteria. Instead, it involves difficult, and often politically contentious, policy decisions as to how the rebuilding of the sector can best be accomplished. These include deciding which banks should be resuscitated and which should be allowed to fail; which should be transferred intact to other banks, and which should be liquidated; which should be effectively nationalized, and which should be allowed to continue operating solely in private hands despite some capital impairment. In this context, it should be borne in mind that recapitalization is not undertaken for the benefit of the banks' shareholders, its creditors, or other stakeholders, but because it is an economic necessity.

Where the state is to be the main provider of new capital, it is usually not so simple a matter as the government simply handing over cash to the banks. If the state were not involved, and a laissez-faire approach prevailed, shareholders and the bank's creditors would shoulder the brunt of the losses, while the bank's management and staff, its depositors, and suppliers would also suffer. Certainly, where the government is involved, the calculus must change, and the allocation of losses among the various stakeholders becomes a political decision.

How to allocate bank losses is perhaps the most difficult decision in the entire restructuring process. This is because it will affect the entire design and structure of the program and because it is likely to be the most politically charged since the affected groups have much to lose. Their interests are described below in reverse order. Instead, the government will almost certainly wish to set conditions and requirements with regard to the funds it supplies, including changes in ownership, management, and operations, allocating losses among the various parties concerned. These include the bank's shareholders, management, staff, depositors, borrowers in default, and the government as a proxy for the taxpayers.

Winners and Losers

Concerning which banks to save, all other things being equal, deeply insolvent banks should not be supported. There are several reasons for this course of action. One is that the banks in deepest trouble were likely the worst managed, and will therefore probably be the most difficult to turn around. Second, the sheer cost of recapitalizing a bank that is insolvent several times over would not be worth the expense. It would

be better to attend to those banks that still retain some equity or at worst are only marginally insolvent. Finally, funds will almost certainly be limited, so it is logical to focus on those banks that have the best chance of learning their lesson and will be able to maintain their financial health in future.

This said, practical considerations will undoubtedly prevail, and rightly so. If the bank is deemed too big to fail—that is, its perceived importance to the banking system and economy is so great that its failure would have grave and long-lasting repercussions—then government support is probably necessary. Once the decision has been made to resuscitate one or more distressed banks, what is critical is that a solution be found expeditiously. If the institutions are allowed to linger too long on intravenous liquidity support, it becomes increasingly difficult to return them to full health.

Components of a Bank Restructuring Program

A bank restructuring program incorporates three fundamental components:

1. Identifying the scope of bank losses and how much capital is needed
2. Helping the banks obtain needed capital while disposing of bad assets
3. Making appropriate systemic reforms with a view to preventing the next crisis

A supplementary component, which may be necessary during some banking crises, is the restructuring of parts of the corporate sector. In so-called corporate restructuring, institutional support or the creation of incentives is provided by the government to encourage delinquent borrowers and banks to work together to work out problem loans. Ideally, in this way impediments to resolving outstanding obligations can be removed, unlocking capital in the process.

Each of these items is discussed later in this chapter.⁵³

Sharing the Losses and the Costs

Banking crises are difficult in part because of the large sums of money at stake and the strong interests of each category of stakeholders in the outcome of a bank's resolution. In respect to any one or more distressed banks, key stakeholders include the

- *Government* on behalf of the taxpayers (as well as in its leadership role)
- *Senior management* of the distressed bank
- *Shareholders* (i.e., the bank's owners), *creditors* (such as bondholders and counterparties), and *depositors*
- *Current customers and borrowers*, especially those that are delinquent
- *Bank's staff*

Of course, these are not the only stakeholders. Also possibly relevant to some aspects of the restructuring or resolution process are the interests of

- *Other banks in the system*, including potentially acquiring banks
- The distressed bank's *suppliers*
- Its *headquarters community*

In addition, although they may not bear any loss directly, it should be recognized that different parts of government may have policy or organizational/political interests at stake in the resolution process, and that such interests may be at stake in one or more distressed banks, and that these interests may sometimes conflict and affect implementation. Relevant government-related interest groups that may need to be considered in practice include:

- The *ministry of finance* or treasury, including the agency responsible for tax collection
- The *deposit insurance agency*
- The government's *senior leadership*
- *Other ministries* responsible for industries that benefit from bank lending
- The *interests of politicians* (and their constituents); and of those responsible for bank restructuring or resolution within the government, including
 - The bank regulator/supervisor
 - The central bank (if a separate entity)
 - Law enforcement agencies and the judiciary and special agencies established for bank restructuring and resolution
- Finally, the potentially relevant views and responses of the *mass media*, *rating agencies*, and *other market participants*

Government/Taxpayers

In nearly all bank restructurings, taxpayers will ultimately have to bear a significant portion of recapitalizing problem institutions. Although it is ultimately in the public interest that banks be recapitalized, the public may be justifiably wary of public moneys being given to failed institutions; so, too, may relevant agencies within the government such as the ministry of finance or the treasury.

It must be conceded that there is a certain unfairness to the situation, particularly with respect to most ordinary people who did not directly benefit from bank profligacy. Nevertheless, the public at large may not fully apprehend the consequences of letting the banking system collapse and many will not understand why it is necessary to bail out the banks. Without government efforts to educate the public, convincing them that apparent subsidies to the banking system are necessary to the cause of long-term economic growth and prosperity may be a tough sell.

Borrowers in Default

Where major banks are insolvent, almost invariably the banks will have been lending to companies that are both major players in the economy and in default on their loans. These delinquent companies and their shareholders, not to mention their workers, may very well resist restructuring initiatives. Where the legal framework is insufficiently robust, they may be able to delay foreclosure or other measures to induce payment. There are a number of possible reasons for their possible resistance. First, particularly if they are state-owned enterprises, they may have become dependent upon borrowing from banks to carry on their daily operations, and will not wish to see a halt to the endless supply of credit. Second, where bank restructuring is deemed necessary, they may suspect that corporate restructuring will not be far behind, which may make the redundancies of staff and management alike a more likely eventuality.⁵⁴

Depositors

Bank depositors and other creditors of insolvent banks are naturally mainly concerned that moneys owed to them be repaid. As discussed in the earlier section on deposit insurance, there is much to be said for insuring the deposits of small deposit holders in order to restore faith in the banking system, notwithstanding the moral hazard danger that may ensue. Should ordinary depositors be permitted to lose significant sums to failed banks, the impact on both confidence and the stability of the banking system could be profound. Moreover, on the grounds of fairness alone, governments will rightly seek to protect the ordinary depositor up to a reasonable limit. Very large depositors and commercial depositors, however, might be expected to share in the loss if bank capital is insufficient to satisfy all claims.

Management and Staff

Depending upon the bank, management may be either closely linked to the owners or instead function as professional salaried staff. Staff may potentially pay their share through redundancies and reduced real wages. To improve efficiency, staff may need to be made redundant, suffer lower compensation, or be replaced by newer potentially more competent or less costly new hires. To instill a stronger credit culture, changes in management may be necessary. As with each interest group named, such changes are likely to be resisted, particularly in countries where trade unions are strong. Management itself may have an interest in resisting any scrutiny of past decisions, or the potential disappearance of their fat salaries and social status.

Shareholders (Owners)

Ordinarily, the owners of the bank, like those of any company, are responsible for its capitalization. It is shareholders who supply capital initially by putting their funds at risk. If the company does well, they benefit greatly through dividends and stock price appreciation, especially when the shares are listed on an exchange. If it fails, they lose their investment.

Again, as with any such enterprise, where the bank's capital is eroded, shareholders would ordinarily bear the loss. Having benefited from the returns on their risk-taking in the past, they should also bear a substantial proportion of the cost of restructuring, by seeing their ownership interest wiped out if the bank is insolvent or reduced to the degree that capital has been impaired. Shareholders of insolvent banks obviously will not relish losing their capital, having their stake in the bank diluted, or being deprived of the advantages that accrue from bank ownership. In countries where a comparatively small number of business groups and clans dominate both political and economic life, such individuals may be able to organize considerable resistance to giving up ownership of institutions with which they have been closely identified. Note that, where the bank remains a going concern, the market value of their shares will almost certainly have declined substantially, something that would achieve what owners resisted.

To what extent will the shareholders' interest in the bank at the time of its recapitalization be diluted upon the injection of government funds? It is not uncommon for governments to demand shares in exchange for recapitalization. This is entirely reasonable so long as the government does not intend to continue indefinitely in the banking business. In the present era, most governments do not want to get into the banking system. Instead, the rationale for the government taking

ownership of a bank—a process frequently referred to as nationalization—is to prepare it for privatization two or three years down the road. This sort of nationalization, as seen for example in the UK in 2007 and 2008, is entirely different from the sort of expropriations made on behalf of the “people” or the “workers” that were not uncommon during the last century. The latter were meant to be permanent, state ownership of the means of production being regarded as a good in its own right rather than as a means to an end.

BANKRUPT BANKS: A SPECIAL TREATMENT?

When it comes to bankruptcy, in many countries banks really are special. For reasons developed both in earlier chapters and in the present one, governments usually wish to handle the insolvency of a bank in a different manner from that of a nonfinancial enterprise. Without recapping the reasons for special treatment for banks, suffice to say that the winding-up of a bank tends to have considerably greater ramifications than that of an ordinary corporation. Consequently, while the laws of some countries do subject banks and corporate to the same insolvency regime, most establish a special insolvency framework for banks.

Banks and corporate not only often differ in respect of the bodies that have authority over them, the substance of the legal framework governing banks can be expected to differ from that covering corporates. While corporate bankruptcy generally comes under the purview of the courts, bank insolvency proceedings often are placed within the charge of an administrative body, which is solely responsible for resolving the case. Significant differences exist in respect of other aspects of the law that governs banks. The bank regulator may be given the exclusive power to initiate proceedings, or at least have a right to participate in them. This contrasts with corporate bankruptcies where proceedings may be initiated voluntarily by the debtor company or incidentally by any legitimate creditor.

In view of the variance in bankruptcy laws around the world, no definitive guide can be presented here. It is fair to say that, at any given time, a bank will fall into one of three categories depending upon its financial condition as assessed by the regulator.

1. The bank remains sufficiently solvent during a crisis and retains some degree of freedom and independence.
2. The bank has voluntarily sought and received support subject to its acceptance of certain conditions.
3. The bank has failed to meet minimum solvency or other related requirements and, as a consequence, it has come under state control with its fate dependent upon the decision of the applicable administrative body.

Banks that fall into categories 1 and 2 will usually be immune from liquidation. They may be obliged, however, to comply with emergency directives

promulgated by the regulator. Likewise, they may be strongly encouraged, for example, to accede to a merger with a stronger bank or to accept other administrative guidance or restructuring initiatives preferred by the regulatory agency.

Banks in category 3 might be referred to as under *official administration*. A bank under official administration becomes vulnerable to liquidation owing to its insolvency, although it will not necessarily suffer that fate. The administrators in charge might elect—if they do not nationalize the bank outright—to recapitalize it or to employ other restructuring techniques with which it has little choice but to comply. These may include discussion with prospective buyers, tendering the bank for bids, negotiating the sale of the bank, and providing guarantees if required, valuing and selling the bank, repaying depositors, making staff redundant, and so on.

RECAPITALIZATION AND RESTRUCTURING

Managing a banking crisis is one of the most difficult tasks to confront a policymaker. Often measures must be decided quickly [and] almost inevitably, decisions will be guided by imperfect information. This is an intrinsic problem because the very business of banking is built on the possession of information not available to others. Moreover, the various actors may well have an incentive to distort the facts. Because banks lie at the centre of modern economies, policies can have far-reaching implications, political as well as economic.

— John Hawkins and Philip Turner⁵⁵

A crisis of significant magnitude implies that a significant portion of bank capital within the banking system has already been wiped out as a result of loans or other financing assets having soured on a wide scale. Simple arithmetic would show the surviving capital funds to be insufficient to sustain the volume of bank lending the country's economy needs to operate. The country would likely enter a period of sustained decline and stagnation. In the absence of a healthy functioning banking system, able to provide seasonal, short-term, and medium-term credit and fuel economic recovery, intervals of contracting output or tepid growth could be anticipated for a considerable time to come. For banks to resume their function as suppliers of credit and other financial services, the capital destroyed through loan losses must be replenished.

While the potential causes of crises are manifold, the options for dealing with one or more banks that are insolvent—or that no longer meet minimum regulatory standards of solvency—are limited. They can be even more restricted when fiscal, institutional, and political constraints are taken into account.⁵⁶ In short, either distressed banks get the fresh capital they need, or they are liquidated like any other corporation.

The first option—recapitalizing the bank in full—is a good choice for a single critical distressed bank or a larger number of banks, subject to the following limitations:

- The first and principal constraint is that recapitalization is very expensive and that the resources of the state are likely to be limited. Consequently, there is an understandable inclination on the part of policymakers and regulators to minimize the cost to the treasury and to involve the private sector in the endeavor. Direct state support therefore usually only forms a part of the solution to a banking crisis.
- A second potential concern is that a wholesale bailout of the sector might increase moral hazard. As suggested before, if bank managers get the impression that the government will bail them out should things go awry, they will be inclined to take greater risks than would otherwise be the case, with the result that over the longer term more bank failures would impose a greater burden on the public purse.
- Finally, capital alone does not address operational deficiencies. Merely injecting funds will help in the present but will not address those underlying problems that were responsible for the crisis in the first place. Without dealing with those problems, there is the risk of creating an even larger problem in the future.

The second option—full liquidation of the bank—is commonly used by regulators to close smaller marginal institutions that are not essential to the operation of the financial system. Such banks are not usually those institutions deemed too big to fail, nor are they normally at the heart of a banking crisis. Liquidating even one large critical bank presents some obvious problems. The liquidation of major institutions means that the remaining banks will have to take up the slack or that a new bank would need to be established to replace the old. The first outcome points towards oligarchy and the establishment of even larger banks. While in certain instances, some consolidation in the banking industry may be welcome, there is the risk of even larger and more damaging crises in the future. The second amounts to building what already exists.

In view of the disadvantages with both of the extreme options, in most cases it will be necessary to look at the range of options in between the two poles. It is from there that the most appropriate mix of measures to implement an effective bank restructuring program is likely to be drawn.

Private Sector–Based Solutions

It is naturally preferable that the burden of recapitalizing the banking sector be shouldered, as much as possible, by the private sector. The reasons are several. First, the cost of the state recapitalizing the system without private capital falls primarily on the taxpayer, who is probably least responsible for the creation of the crisis. Second, in market economies, governments only take over banks for a temporary period. Hence, ultimately, private capital will support the banking system. Therefore, if it can be obtained at a reasonable cost at the start of the process,

the benefits of delaying until the bank's balance sheet is further cleaned up may be outweighed.

In a crisis of any severity, however, private investors are unlikely to be willing to take over problem banks without some state intervention.⁵⁷ The reason is that private investors will rarely be willing to advance new capital to existing institutions without some assurances regarding limitation on their loss exposure. If the crisis is a mild one, however, or one where the regulatory authorities have been astute enough to act preemptively and stanch the flow of red ink early, it is conceivable that it will be possible for the authorities to resolve the banking system's problems by providing limited support; that is, short of full-scale recapitalization.

Among the most common options for support are:

- Encouraging mergers (e.g., by acting as co-investor, providing guarantees to the private sector investor concerning existing loans, losses, etc.)
- Applying regulatory forbearance⁵⁸ (e.g., allowing temporary exemptions to minimum capital requirements)
- Arranging earnings support (e.g., tax concession granted to identified banks, as also discussed below)
- Providing a modicum of short-term direct or standby financial support

In the most severe crises, major intervention by the state may be required to facilitate the recapitalization of the banking system. As part of the process, to use resources in as an efficient manner as possible, the authorities may decide to recapitalize only the most viable of the problem banks while shutting down and liquidating the others.

Encouraging Mergers

A popular approach to emerging distress at identified institutions is to encourage bank mergers. By "merging" a weak bank with a stronger peer, the hope is that the more robust institution will in effect recapitalize its frailer counterpart. In actuality, of course, such transactions are not mergers in the sense of a merger of equals, but the acquisition of the weak bank by the strong bank, typically at a superficially bargain price.

There are a number of motivations for the regulator to encourage mergers and acquisitions, including those that correspond to the advantages listed immediately below. Perhaps most importantly, they may be a way to effect the recapitalization of a distress bank on the cheap. Also, the regulator may genuinely hold the view that a fewer number of well-capitalized banks makes for a more robust and healthy banking industry than one comprising numerous smaller institutions.

Advantages of so-called strong bank-weak bank mergers typically include the following:

- Public funds are not necessarily required to effect the recapitalization.
- Cost efficiencies are bound to be realized by combining the two operations, and other synergies may be found as well.
- The top management of the weak bank who were responsible for its demise are removed.
- The better technologies and practices of the strong bank are made available to the remaining staff from the weak bank.

As is often the case with mergers, there are typically some aspects that are a win-win for both parties, the significant parties in this context being primarily the owners of the strong bank and the government. The strong bank may be willing or even eager to acquire its peer in order to expand its branch network comparatively inexpensively or to acquire a ready-made entrée into a desirable franchise, or even to reduce potential competitive pressure. Through more efficient management techniques and the use of technology, the bank may also be able to shed staff thereby leading to operating efficiencies.

There are, however, several disadvantages.

- The principal disadvantage is that the problems at the weak bank may overwhelm the strong bank and infect the latter resulting in the demise of two banks instead of one. In the end, the merger route may prove more costly than an early direct intervention.
- Another disadvantage, already alluded to, is that the government may not have the legal authority to force a merger. The strong bank may resist the merger, and the government may be compelled to provide subsidies, for instance, guaranteeing some of the weak bank's assets, to effect it.⁵⁹
- Thirdly, depending upon the organizational culture at both institutions, integrating the two banks may prove more difficult than anticipated. Many mergers involve teething problems as differences in systems, technologies and staff expectations are worked out.
- Fourthly, and related to the immediately preceding item, in countries with a strong tradition of labor rights, trade unions may vehemently oppose staff redundancies or more mild changes in work practices.
- Finally, the subsidies or concessions needed to consummate the merger and make it work in practice could ultimately comprise a significant portion of the cost of recapitalizing and restructuring the weak bank in the first place.

These disadvantages reflect potential costs or impediment that may not be immediately apparent and may undermine the perceived benefits of the merger, including anticipated cost efficiencies that would otherwise arise from the consolidation.

Regulatory Forbearance and Earnings Support

The lack of an effective prudential regulatory framework can contribute to a banking crisis. International best practice embraces an assortment of prudential regulations that, by establishing minimum standards or prohibiting problematic banking practices, are intended to reduce unnecessary risks to the bank. These include minimum capital adequacy requirements, the prohibition of related-party loans (with limited exceptions), and mandatory loan-loss provisioning requirement in respect to NPLs defined according to international standards, not to mention more recent liquidity requirements.⁶⁰

Imposing a stronger prudential regime or starting to enforce an existing regime on banks that have slid into chronic disability may not be practicable, however, in the short term. Such banks may not even be able to meet precrisis standards. High minimal capital adequacy thresholds required of banks that have no more capital, or

hefty minimum provisioning requirements applied to banks that have no profits, would be pointless—unless the objective is to compel their liquidation.

For this reason, the authorities may decide, especially after a crisis, to loosen rather than to tighten bank prudential regulations. Regulatory forbearance refers to such a relaxation of rules that normally govern bank operations.⁶¹ The term is defined in one paper on the subject of resolving problem banks as follows:

*When the regulator suspects that banks are financially weak, one option is to simply allow them to continue operating in the hope that their performance will improve in the future. This policy is often called regulatory forbearance.*⁶²

The rationales for the practice vary, but, as suggested, one justification is to give problem institutions some breathing space that enables them to gradually adjust to a new and stricter regime. Accordingly, regulatory forbearance is usually intended as a temporary or transitional step, with regulators aiming to work with bank management teams and owners to put in place new stronger prudential standards over time. A number of countries, from those in Latin America in the 1980s to those in eastern and central Europe in the early and mid-1990s, to those in South East Asia in late 1990s, have employed regulatory forbearance tactics. According to a 2002 report by the OECD, 11 of 18 of its member countries had used the technique in the process of bank restructuring.⁶³

Although appropriate for a limited time in limited circumstances, regulatory forbearance should probably be avoided unless no other options are feasible. Situations where it might conceivably be utilized with some justification include:

- During a firmly defined grace period before which more stringent regulations come into effect
- During an emergency situation to help support bank liquidity and profitability
- Where it is accompanied by other measures to enhance risk management and capital accumulation
- Where there is neither widespread recognition that problems exist in the banking sector nor sufficient political support for more potent measures

Certain types of regulatory concessions, it must be acknowledged, do provide a tangible economic benefit that may assist distressed banks in returning to financial health. Reduced tax rates on income would be one such example. For the most part, however, regulatory forbearance is akin to putting a plaster on a wound when major surgery is what is really required. Moreover, some regulatory forbearance practices amount to mere window dressing that makes a bank look better on paper, but does nothing to fundamentally improve its financial condition.

By not addressing banks' underlying problems, most regulatory forbearance measures only make eventual future intervention more difficult and costly than it would otherwise be. Finally, it has been suggested that regulatory forbearance may encourage bank management to take greater risks than they otherwise would, a phenomenon sometimes referred to as "gambling for resurrection."⁶⁴

REGULATORY FORBEARANCE

As Stijn Claessens put it, “[F]orbearance can be a risky and costly policy . . . but . . . regulatory changes need to balance the need for fundamental reforms with realism and political support.”⁶⁵

Regulatory forbearance refers to the relaxation of ordinary prudential rules or other rules governing bank operations.⁶⁶ Examples of relaxed regulatory requirements include lowering reserve requirements and minimum balance sheet liquidity, reducing capital requirements temporarily, and reducing loan-loss provisioning requirements. During the Asian crisis, an example of regulatory forbearance included the decision by the Bank of Thailand to reduce banks’ Tier 1 capital requirement from 6 to 4.25 percent.⁶⁷

There are a number of possible rationales for waiving ordinary regulatory requirements. One possibility, for example, is that by waiving rules that require banks to charge a certain proportion of a classified loan as loan-loss provisioning, the forbearance represents an attempt to allow banks to better earn their way out of their problems. (Of course, the true credit costs of the bank remain unchanged.) A second possibility is that the waiver, for example of taxes, constitutes a real economic benefit to the banks and therefore does further the ostensible objective of the first item. A third possibility is that by waiving such rules, they make banks’ results look better, thereby potentially making it easier for them to obtain funding or attract capital. (This, of course, is the more likely impact of the first item, even if on paper the bank is accumulating capital at a quicker pace.) A fourth possibility is that other more desirable approaches are not practical and regulatory forbearance is better than nothing.

Insofar as the use of regulatory forbearance does provide a real economic benefit, it highlights an important trade-off in the restructuring process between a focus on bank solvency, sound management, and risk reduction on the one hand, and an emphasis on economic growth, strong bank profitability, and corresponding internal capital generation on the other. It is therefore probably best used in a limited and temporary manner and, more importantly, as an adjunct to the implementation of structural risk-oriented reforms, recapitalization, and bad asset disposal rather than as a primary response to bank distress.

The heavy use of regulatory forbearance tactics at the expense of fundamental reforms tends to connote either a lack of political will to undertake reforms or the resources. Since regulatory forbearance is technical in character and does not necessarily involve unambiguous bailouts to the banking sector, it can be undertaken quietly. Arguably, if used as the key response to bank distress, it amounts to bank restructuring on the cheap: an attempt to preserve confidence in the banking system and to keep its fundamental problems concealed, with the hope that strong bank earnings and a robust economy will enable banks to earn their way back to financial health.

Somewhat related to the use of regulatory forbearance is what might be called earning support, and some of the techniques used to boost bank earnings may also fall within the category of regulatory forbearance. There are a number of ways in which a government can attempt to bestow the benefits of earnings support upon its banks.

One is for the government to provide support to banks in distress through the use of economic policy management. For example, the central bank might implement an aggressively loose monetary policy lowering domestic interest rates in order to help widen margins for banks and fuel economic demand. Thicker margins should lift bank earnings and thereby enable the institution to generate capital internally at a more rapid rate, while in the process simultaneously attracting new capital from external sources. Other potential earnings support measures that might be undertaken by the regulatory authorities include:

- Fattening bank margins by fiat (e.g., as through the imposition of limits on deposit rates while allowing interest rates on lending to increase to more elevated levels)
- Providing tax concessions or relief from loan-loss provisioning requirements⁶⁸
- Postponing dividends on the government's preferred shares it received in exchange for its recapitalization
- Waiving dividend payments on the preferred shares the government has taken in exchange for recapitalizing the bank

State-Funded Solutions

It is evident that recapitalization of the banking system is the crucial phase of the entire bank restructuring process. As to the source of new capital, it is generally preferable that, where feasible, private capital be utilized to rehabilitate distressed banks. The more severe the banking crisis, however, the more likely it is that private investors will be unable or unwilling to provide the needed capital to revive the banking system, at least for the time being.

Obtaining capital from private sector sources under such circumstances is likely to be very difficult, as was observed during the S&L crisis in the United States during the late 1980s. Therefore, where a crisis situation prevails, the brunt of the cost and risk of recapitalization will likely fall upon the state. The European debt crisis of 2011 and 2012 is a striking example.⁶⁹

STATE SUPPORT

State support is often used to refer solely to the provision of funds to banks, for instance in regard to the supply of cash by the government to banks for liquidity purposes during periods of market turbulence or to recapitalize them following a crisis. In order of decreasing levels of intervention, actions that come within the category of state support are:

- Recapitalization (whether the funds are injected solely by the state or alongside private sector investment)

- Longer-term funding short of a full scale recapitalization
- Direct or indirect subsidies of various kinds

In broader terms, however, state support may embrace a wide range of actions that, while they do not represent a direct transfer of funds, do come at a cost. The cost may be that of foregone revenue (e.g., tax concessions), contingent liabilities (e.g., through guarantees provided to the private buyer of a former distressed bank), or the ancillary costs involved in setting up the restructuring program together with the key institutions that will oversee it (e.g., seconding staff from state institutions, and making politically controversial decisions that risk losing public support for the program).

As observed earlier, the term *restructuring* refers not only to the entire process of fixing the banking system, but also takes in a number of more specific actions that form part of this comprehensive effort. These include operational restructuring, capital restructuring, financial (asset) restructuring, regulatory restructuring, and corporate restructuring. Recapitalization necessarily effects some changes in the capital structure of the bank (capital restructuring) and usually implies some changes in its operation (operational restructuring).

Changes in Ownership

Preferably, a bank due to be recapitalized with public funds will be transferred to state ownership to effect the restructuring of its operations, including the replacement of senior management with a new team. The reasons for taking this action, as mentioned, are to:

- Create a clean slate, so that new management can reform the credit culture of the failed bank
- Reduce moral hazard by holding existing shareholders accountable for the policies they approved

In theory, removing the existing shareholders in these circumstances should be no problem. Since existing shareholders are responsible—up to value of their share of equity—for the losses of a troubled bank, in the event it is rendered fully insolvent, their stake and hence ownership rights to the bank should be wiped out. In practice, however, this is not always the case.

To be sure, some governments will not give existing shareholders a choice in the matter, taking over the bank immediately upon its being deemed insolvent for regulatory purposes. In Latin American countries, generally speaking, state-sponsored restructuring initiatives have dictated that the receipt of official aid required a complete change in bank ownership.

In various other countries, however, it may not be politically feasible to remove the existing shareholders entirely, despite the letter of the law. The extent of ownership

retained in some cases may be substantial, while in others a face-saving arrangement is reached whereby the existing shareholders keep only a nominal stake in the restructured entity.

Other restructuring strategies include adopting a transitional period in which bank shareholders are given additional time to raise capital.⁷⁰ Note that a change of ownership, even if it occurs, may not be enough, however, to transform the bank's credit culture. If that is a worry, then the recapitalization plan might be set up so that new injections of capital are contingent upon the bank meeting certain targets at various intervals.⁷¹

Changes in Operations

Changes in ownership and management may need to be accompanied by changes in operating procedures. At a minimum, as mentioned above, the bank must stop lending to delinquent borrowers, or at least begin to wean them from an incessant supply of credit. Other operational changes will naturally vary with the particular circumstances. If none is in place, for instance, a system of controls and internal audits should be implemented at the earliest stage and an accurate assessment made of the extent of the bank's financial problems. Applying accounting standards that conform to international best practice, the status of the bank's loan book should be clearly determined, as well as the amount of capital necessary to bring the bank up to a benchmark capital adequacy level. At the more routine operational level, the bank may be obliged by the authorities to create a credit committee and promulgate a written credit policy. Similarly, training of bank staff in modern bank operations may be undertaken or subsidized by the government.

ASSET DISPOSAL AND REGULATORY REFORM

Without regulatory intervention, weak banks have the incentives to hold on to the non-performing loans (NPLs) and gamble for the small chance of recovering these.

—M. Tanaka and G. Hoggarth⁷²

If you owe a bank a hundred pounds, you have a problem. If you owe it one million, the bank has a problem.

—Adage attributed to J. M. Keynes

One of the main decisions in bank restructuring programs is how to manage the bad loans and other problem assets held by banks. Leaving NPLs on a bank's books after recapitalization is likely to undermine the objectives for which the new capital was supplied; namely, the return of the bank to normal operations and profitability as well as the recovery of the economy. A large standing pool of NPLs, however, is liable to prove a hindrance to the recovery of both banks and the economy as a whole. It will tend to distort the bank's financials and the performance measures derived from them, while allowing uncertainty to remain concerning the bank's actual credit costs. Although the bank may be able to shed its bad loans gradually

over a number of years, taking this tack risks dissipating the new capital injected. Whether or not fully provisioned against, a mass of moldering assets on the bank's balance sheet is bound to draw management's attention away from their main business: generating profits through the provision of banking services. For example, property acquired through foreclosure needs to be sold, a task that can be difficult in a weak market.

Ultimately, the NPLs and other problem assets need to be resolved and removed from the banks' balance sheets. Where significant sums are at stake, efforts ought to be taken to recover from delinquent borrowers at least some of the amounts outstanding on their nonperforming loans. This may require dunning and extended negotiations with the borrower, as well as involvement in legal actions, including bankruptcy proceedings, which may be time-consuming and, as a consequence, costly to undertake. Antiquated laws or ineffective regulations may impede both the collection of amounts payable and the ability of the bank to write off the loans. Adding to the distraction is the fact that the set of skills required to work out problem loans, pursue legal interests through the courts, maintain collateral foreclosed, and finally sell such collateral together with marketable or nonmarketable financial instruments tends to be of a distinctly different character from those that are vital to running a successful bank.

For these reasons, most systemic bank restructuring schemes create a special institutional regime for speeding up problem asset disposal. The most common approaches employed are described below. One of the features that is common to most such schemes is the conviction that it is important to separate the pool of NPLs and other problem assets held by the bank at the time restructuring begins into a separate entity or business unit. As hinted, the rationale is that, in view of the skills required to handle the resolution of problem assets and foreclosed collateral, it is preferable that this project be handled by a dedicated team of specialists and not by senior management of the originating bank. The reality is, of course, that such separation provides an opportunity further to blur the picture in terms of the ultimate credit costs involved.

Before moving on to describe the various institutional devices for disposing of problem assets, it is worth stating that the time required for such disposal can be expected to take two to three years at a minimum. The actual time needed will depend upon the efficiency of the legal system, the marketability and the carrying costs of the assets or the corresponding collateral, and the probability that optimal recovery will require at least some recovery in market prices. For all these reasons, systemic bank restructuring programs usually include a scheme intended to assist banks in their efforts to resolve problem assets and to remove them from their balance sheets.

Problem Asset Disposal: A Prelude

The last two of the potential regulatory reforms noted above concern the removal of bad loans and other problem assets held by banks, including so-called toxic assets, from their balance sheets. The removal of these assets and associated collateral from bank balance sheets, which is the subject of the following section, is generally viewed as a crucial element of a successful bank restructuring program. In brief, disposal of problem assets is normally achieved through the sale or transfer of such assets to a

separate entity, which takes on the task of working out the obligations through negotiations with the borrower or alternatively by writing off or writing down such assets to reflect the losses incurred.

In looking at problem assets, both the bank and the restructuring agency will closely examine the collateral recovered. The collateral will, or should, boost the amount recoverable thereby reducing net overall losses. However, in examining collateral it is important to take into account the legal system within the country. This may indeed be weak, as in the case of many emerging markets, with the bank or agency having difficulty pursuing both the defaulter and any collateral. The Asian crisis of 1997 highlighted the weak legal systems in countries such as Indonesia and Thailand. Where the ability of the creditor bank to enforce its rights is weak, the nonpayment mentality (i.e., “can’t pay, won’t pay”) is aggravated. Moreover, in many cases, the value of the collateral will be below the value booked.

Additionally, if the crisis is sector-wide, with many banks affected by NPLs, the value of collateral may fall due to a selling oversupply. Also, a large part of collateral may be connected to property and real estate. The value of such real estate is likely to fall sharply during a prolonged economic downturn thereby magnifying the situation of the net NPLs. Often, spiraling property prices and rapidly increasing loan facilities to the property sector have been a major feature of economic boom times that, often, ultimately leads to a nasty correction for banks. This was the case in Thailand, which saw spiraling property prices throughout the 1990s, causing many assets to be valued excessively. The subprime crisis of 2007 in the United States and the Spanish real estate bubble of 2011 are other examples.

WHAT ARE TOXIC ASSETS?

These are typically complex and often risky instruments that for a number of reasons have become illiquid or have lost their underlying value. In the recent credit crisis, certain tranches of various asset-backed securities and collateralized debt obligations that have exhibited precipitous declines in price could be categorized as toxic assets. In addition to their complexity and high transaction costs, possible reasons for their illiquidity include the following:

- A lack of appetite in the market for the asset class into which they fall or a collapse of the market in which they were previously bought and sold, and a corresponding uncertainty as to their fair price
- A drop in the credit rating assigned to such instruments from investment grade to non-investment grade or the absence of a current rating and corresponding uncertainty concerning the risks such obligations represent

Additionally, the following attributes may contribute to their illiquidity:

- The failure of embedded credit or liquidity enhancements that previously supported the marketability of such instruments
- A loss of confidence in the models that previously supported their pricing

Components of Asset Disposal Schemes

Schemes of this type normally address three points:

1. The removal of existing obstacles to problem asset disposal through legal and regulatory reforms
2. The separation of the bank's problem assets from its regular banking operations through dedicated institutional arrangements
3. The creation of disposal mechanisms that further the objectives of the overall restructuring process

Reform of Law and Regulation to Enhance Creditors' Rights and Expedite Asset Disposal and Recovery

Since existing laws and regulations may obstruct or delay the capacity of banks to rid themselves of problem assets, a systemic restructuring plan should seek to eliminate or reduce such impediments. Illustrative reforms might encompass the following measures:

- Clarifying the compliance obligations of banks with respect to NPLs (e.g., establishing, if they do not already exist, clear definitions of loss-loan categories and accompanying provisioning requirements)
- Tightening loan-loss accounting and reporting standards
- Improving the efficiency of the legal system with respect to creditors; rights and foreclosure of collateral and other security (as noted earlier)

Separate Problem Loan Disposal from Ordinary Banking Operations and Disposal Mechanisms

Of equal or greater importance than legal and regulatory reforms, however, are the institutional arrangements that are established to facilitate problem asset disposal. There are fundamentally two types of arrangements employed: the good bank–bad bank approach and the use of a government-owned or sponsored institution. Usually, these entities are temporary in nature, the intention being that they will be dissolved upon the removal of all precrisis problem assets from the affect banks' balance sheets.

Select Disposal Mechanisms

Arrangements for the disposal of problem assets can range from the use of existing markets to sell bank collateral or other assets to the use of government-sponsored public auctions. Where problem assets are transferred to a government or government-sponsored entity, the design of asset disposal arrangements may be integrated into the structuring of the entity. In any case, it becomes a two-stage process. The first stage is the transfer from the bank to the government entity on terms specified by the government; the second stage encompasses the workout and ultimate final recovery with respect to the asset.

The Principal Institutional Approaches

There are two principal institutional structures to resolve problem assets resulting from a banking crisis. The purpose, as noted, is to maximize returns and within a limited period of time to shed all pre-recapitalization problem assets from the recapitalized bank's balance sheet.

Good Bank–Bad Bank Approach

Upon recapitalization, the distressed bank establishes with regulatory approval a separate “bank” into which it transfers some or all of its bad assets. The bad bank’s sole task is to recover as much as possible from those assets in a timely manner. It takes no deposits and engages in no lending or other conventional banking activity. This is accomplished through negotiated workouts with the borrowers, voluntary or involuntary insolvency and bankruptcy proceedings, and through repossession and sale of collateral securing the nonperforming assets. When the disposal of assets is complete, the bad bank is wound up, and any remaining assets are liquidated.

As the bad bank actually buys the assets from the distressed bank, it has to fund those acquisitions. Funding is usually provided by the state, but there are situations where the distressed bank itself provided the funding.

Separate Internal Recovery and Resolution Department

The concept is the same as the good bank–bad bank approach, but instead of establishing a separate legal entity, the “bad bank,” a separate internal recovery and resolution department is set up within the recapitalized bank. Its role and function, however, are identical to that of the bad bank.

The AMC Concept

Asset management corporations (AMCs) are, in effect, state-sponsored “bad banks” that are established for a limited period of time for the purpose of assisting a country’s banks with the task of shedding their bad loans. They have become increasingly popular as a vehicle for resolving problem loans, and, in view of their importance and somewhat greater complexity, they are discussed separately in the following section.

Asset Management Corporations

An AMC is a special-purpose agency, often a quasi-governmental entity, which is set up for a limited period of time for the purpose of assisting recapitalized banks in resolving their pre-recapitalization problem assets. This is generally achieved through the AMC’s purchase of the affected banks’ bad loans and accompanying collateral from banks in exchange for cash or government paper and the AMC’s subsequent disposal of the assets over time. As described below, AMCs potentially have a great deal of flexibility as to how they handle the asset disposal. In practice, the degree of flexibility will depend upon their charter and the powers granted to them upon their formation.

HOW AN AMC WORKS

An AMC or asset management corporation is a state or state-sponsored agency that is established for a limited period of time for the purpose of facilitating the removal of problem assets from bank balance sheets. This is accomplished by paying the banks from which the comparatively illiquid assets are acquired, and reselling them to third parties. That is, the AMC purchases a problem asset

or pool of problem assets from each distressed bank according to specified terms and conditions and adds the purchased asset(s) to its own portfolio for subsequent resolution. In the process, the asset is transferred from the bank's balance sheet to the AMC's balance sheet.

While a number of variants exist, the classic structure of an AMC is that of a national entity responsible for the resolution of precrisis problem assets throughout the banking system. When chartered, a national AMC is often granted special powers to enable it to accomplish its mission more effectively. For example, the AMC may be empowered to manage and recover bad loans, reschedule facilities where appropriate, and negotiate on the settlement of bad debt from large corporations. National AMCs may also benefit from its capacity to tap funding under the wing of the central government and from the latter's imprimatur, as well as from the greater economies of scale its national scope affords. Consequently, AMCs set up along such lines are well positioned to be able to resolve problem assets more rapidly and with lower carrying costs than would an individual bank. For the same reasons, they are likely to be able to recover a greater percentage of the funds owed by the original obligor than would a smaller entity.

Pricing of assets acquired by the AMC will necessarily vary with the type of asset, its marketability, and the apparent obstacles to the obligor's full payment of the underlying obligation. As would be expected, a collateralized loan would tend to bring a higher price than an uncollateralized loan. This said, pricing terms may be based on a formula, such as a percentage of book value or in accord with some other ratio or algorithm. Alternatively, the purchase terms may be subject to adjustment depending upon the amount the AMC is able to recover on the asset, either through the sale of the collateral or through a negotiated workout with the obligor. It is common for AMCs to acquire distressed assets from banks on terms that are relatively generous to the sellers. This provides an incentive to the banks to get rid of their NPLs and also amounts to a form of covert recapitalization.

If operated effectively, the AMC can use its market power to recoup any modest premiums it has offered by, for example, selling some earlier acquired collateral at higher prices as the market ultimately recovers.

To aid sales and attract potential buyers of the assets, an AMC may guarantee buyers of impaired assets against losses (but at less than 100 percent and only for a certain period of time).

As an illustration of how AMCs operate in practice, Danamodal, Malaysia's AMC set up in the wake of the Asian crisis, employed just such a price adjustment mechanism. When Danamodal was able to sell the acquired assets at a price higher than it had paid the bank, it agreed to return to the bank 80 percent of the profit obtained. To fund its purchases, the AMC may use government-guaranteed bonds intending to sell the acquired assets before the government bonds mature.

To illustrate this flexibility, potential approaches to asset resolution and restructuring used by AMC's include the following:

- The AMC overpays for the banks' problem assets thereby effecting a backdoor recapitalization of the bank.
- The AMC, instead of paying the selling bank in cash, conditions the transaction upon the buyer's accepting, as payment for the problem assets, non-negotiable government securities that are redeemable after a certain time period. The effect is to limit leverage.
- The AMC underpays for the banks' problem assets but promises that, where recovery on the original obligation exceeds a certain threshold, it will share the profits with the selling bank according to terms agreed beforehand.

Note that, in some AMC arrangements, most famously those put in place by the Japanese banks in the 1990s, the AMC keeps a partial recourse on the distressed banks, whereby the banks would participate in further unexpected losses on the transferred assets.

Notwithstanding the foregoing description of a model AMC, they vary considerably in their fundamental attributes including their mission, ownership, source of funding, intended lifespan, and powers, among other characteristics. For instance, at one extreme, the AMC may be established to dispose of bad assets from bank balance sheets quickly, in order to ready the troubled institution for sale to private investors. At the other, the AMC's role is essentially to warehouse the bad assets with the hope that simply removing them from the banks' balance sheets until better economic conditions arrive will restore the distressed institutions to financial health. Somewhere in the middle is the rehabilitative AMC, which aims to return the banking system back to financial health while optimizing returns on disposed assets. See various AMC approaches in recent years in Exhibit 15.3.

Likewise, there is a range of possibilities with respect to the role of the national government in sponsoring or funding AMC's as opposed to AMC's that involve the private sector in ownership and operations. Note that where the AMC's are pure private sector entities formed by individual selling banks it is simply the good bank-bad bank approach by another name (such AMC's being, of course, the "bad banks"). Finally, AMC's may differ in the extent to which they take control of managing problem assets. One approach is for the AMC to take the assets over completely, while another is for the banks to actually continue to manage the assets without having to maintain them on balance sheet. Each of the preceding approaches

EXHIBIT 15.3 AMC Approaches in Recent Years

Rapid Liquidation/ Fire-Sale Approach	Rehabilitation and Optimization of Returns/Middle Approach	Long-Term Warehousing/ Gradualist Approach
USA's Resolution Trust Corporation	Sweden's Securum	Japan's CCPC
Thailand's FRA	Malaysia's Danaharta	Mexico's FOBAPROA
Korea's KAMCO		

has its advantages and disadvantages. Experience would suggest, however, that the centralized approach may be more suited to situations where there are a large number of banks facing loan problems, where there exists a massive amount of bad debt across the sector, and where coordination among various government agencies will necessarily play a greater part.

Regulatory Reform

The lack of an effective prudential regulatory framework can contribute to a banking crisis. Therefore, beyond merely injecting new capital or otherwise facilitating its infusion, it would be imprudent not to also attempt to identify and remedy the systemic problems that led to the crisis in the first place. Not infrequently, prudential regulations are inadequate and have permitted aggressive and risky banking practices to get out of hand.⁷³

Sometimes, however, the problem is not so much the absence of regulations, but a deficiency in enforcement. The solution in such cases may be to strengthen the authority of the regulatory body. This might be effected through new legislation that fortifies the powers of the agency as well as by funding commitments that provide for the hiring of sufficient trained staff to properly enforce its regulations and directives.

Unless clearly needed as emergency measures, regulatory reforms should follow a thorough review of banking regulations, as well as banking and other business practices that contributed to the crisis. Laws and regulations enacted in haste are prone to having unexpected consequences or be vulnerable to regulatory arbitrage. After review and consultation, appropriate changes in regulation and supervision should be considered. Among the reforms often needed are:

- Strengthening banking regulations, particularly prudential regulations and reporting requirements
- Ensuring that the bank regulatory agency and related entities have sufficient authority to effectively enforce the rules
- Reorganizing the agency to improve its effectiveness
- Removing barriers to full provisioning against loan losses
- Improving the efficiency of the judicial system with respect to creditor rights and asset disposal

Further Notes on Corporate Restructuring

Depending on prevailing operating conditions, a final component of a systemic bank restructuring program may involve corporate restructuring. In many countries around the world that have strong legal systems and effective corporate bankruptcy regimes, a separate corporate restructuring program would serve little purpose. In those countries where bankruptcy laws are weak and the process is unwieldy, including many emerging markets and developing countries corporate restructuring might be considered.⁷⁴

In the aftermath of a severe banking crisis, it is delinquent corporate borrowers, as a consequence of their inability (or unwillingness) to repay the money they borrowed, who are often most directly responsible for the bank's losses. Recall that a bank's and the banking system's nonperforming assets are—most of the time—the corporate sector's defaulted liabilities (i.e., delinquent obligations). To liquefy capital

frozen in banks' nonperforming assets and restore that liquidity to the financial system as a whole under the bank restructuring approach already discussed, it is apparent that there are several options.

An attractive course of action for the bank, or for the AMC that took over its delinquent assets and now stands in the bank's shoes, if a workout cannot be negotiated with a delinquent borrower, is to take legal action to potentially force the delinquent corporations into bankruptcy and thereby compel recovery of a portion of the value of those assets.

Yet, as with banks that find themselves in difficulty, the causes of corporate distress similarly vary. The company's problems may be severe, and it may be unable to service its debt having become insolvent. In some cases, however, otherwise good corporates, like good banks caught up in a crisis, may find themselves in difficulties due to weakened demand on the back of the weak economy and higher loan interest repayments as domestic interest rates rise. Or borrowers fall somewhere in between those two positions.⁷⁵

Corporate restructuring might then be the solution. It embodies a variety of approaches, and the appropriate response depends upon the delinquent borrower's condition. If appropriate, drastic reorganization, in which existing management is replaced and portions of the company divested, or even liquidated, may be viewed as the best course of action in order to free up the remaining capital that has become frozen in these unproductive investments or ventures by permitting some recovery to creditors.

At one end of the spectrum, existing management is left in place and terms of outstanding loans are renegotiated. At the other end, new management is installed, and subsidiaries and divisions are sold off or liquidated. The Asian crisis demonstrated the fact that the legal infrastructure in a number of countries was ineffective to deal with restructuring needs. Again, in such circumstances, recognizing the problem, a government may opt to bail out corporations that arguably should be restructured. To address some of the obstacles, several countries during the Asian crisis adopted written objectives and principles to guide corporate debt restructuring sometimes—as Thailand did—setting up special agencies with sweeping authority to resolve restructuring issues that were formalized by law or regulation.

THAILAND CORPORATE DEBT RESTRUCTURING PRINCIPLES

To further the long-term viability of the debtor, the plan should achieve a business, rather than just a financial, restructuring.

If the debtor's management is providing full and accurate information and participating in all creditor committee meetings, creditors "should stand still" for a defined (60 days) and extendable period.

Restructuring should not be used to hide NPLs.

Debt forgiveness should only be used as a last resort and only in exchange for stocks and warrants.

A lead creditor institution (and within it, a specified individual) must be appointed early in the restructuring process to coordinate according to defined objectives and fixed deadlines.

In major multicreditor cases, a steering committee, which is of a manageable size while representative of all creditors, should be appointed.

Decisions should be made on information that has been independently verified.

Creditors' existing collateral rights must continue.

New credit extended on reasonable terms to help the debtor continue operations must receive priority status.

Lenders should seek to lower their risk (e.g., through improved loan collateral), rather than to increase returns (e.g., by raising interest rates).

Any creditor that sells his debt claim should ensure the buyer does not impede the restructuring process.

Creditors should take account of the impact of any action on other creditors and on potentially viable debtors.

Source: Drawn from "A Framework for Corporate Debt Restructuring in Thailand," published by the Board of Trade of Thailand, the Federation of Thai Industries, the Thai Bankers' Association, the Association of Finance Companies, and the Foreign Banks' Association. Extracted from BIS Policy Papers No. 6, August 1999.

The need to salvage banks is often intricately linked to the need to salvage corporations. As with bank distress, the need for corporate restructuring may be acute or chronic. The acute need for corporate restructuring typically arises as a result of a financial crisis or panic. The chronic situation arises in countries such as South Korea, Japan, and China where a large part of the corporate sector is either state-dominated or operated on noncommercial terms. A contributing factor may be so-called crony capitalism, a term that encompasses a multitude of sins, but often refers to business transactions undertaken to achieve political ends, for example as a favor that benefits a privileged individual or group.

BANK RESTRUCTURING IN MALAYSIA DURING THE 1990S ASIAN FINANCIAL CRISIS: A PRACTICAL EXAMPLE

The purpose of this section is to discuss in somewhat greater detail the process of bank restructuring and recapitalization as implemented in Malaysia during the Asian crisis of 1997–1998, including the actual strategies, institutions, and operations that were employed. The purpose is to gain a more practical understanding as to how governments react to banking crises.

The Asian crisis officially began in Thailand on July 2, 1997, with the devaluation of the Thai baht, and it rapidly spread throughout the region. Most badly affected were Thailand itself, Indonesia, which saw an economic crisis trigger a political one with the ultimate deposing of its long-serving dictator, President Suharto, and South Korea, which also saw severe political repercussions from the turmoil. Malaysia was arguably less severely affected than these countries, but

suffered a greater impact than those that were not in the direct path of the storm—Singapore, Hong Kong (returned to Chinese sovereignty on June 30, 1997), Taiwan, and the Philippines among them. Still other countries in the region were only affected peripherally due to the relatively closed nature of their economies at the time. China, Vietnam, and India fall into this last category. Japan, of course, had been dealing with its own extended period of economic malaise since the crash of the Tokyo stock market at the end of the 1980s and the bursting of the real estate bubble.

In hindsight, there is a fair degree of consensus that the Asian crisis was in large part a banking crisis.⁷⁶ The East Asian economic miracle was essentially financed by bank lending. Although local capital markets have evolved to varying degrees of maturity in each of the countries just mentioned, there has on the whole been a comparatively great reliance on bank lending the region. While there were a number of contributing factors, arguably those countries whose currencies traded relatively freely and whose banking systems were most overstretched suffered the worst. One reason that Hong Kong, Singapore, and to a lesser extent the Philippines survived comparatively unscathed was due in part to the strength of their banking systems at the time the crisis hit.⁷⁷

State of the Malaysian Banking System Entering the Crisis

Compared to its counterparts in Thailand, Indonesia, and Korea, the Malaysian banking sector entered the 1997–1998 Asian financial crisis in a relatively solid position. A banking crisis in Malaysia in the 1980s had led to a complete overhaul of banking legislation and supervision such that, prior to the crisis, the country had adopted fully 22 out of the 25 BIS Core Principles of Banking Supervision. Nonperforming loans, which had soared following the earlier period of financial instability, steadily fell as regulators instituted reforms. Malaysia's economy expanded rapidly in the emerging markets boom of the early 1990s. As of June 1997, the risk-weighted capital ratio of the banking system was 12 percent, and nonperforming loans were below 4 percent on a gross basis, and 2.2 percent on a net basis.

Nor in mid-1997 did Malaysia appear particularly vulnerable from a macroeconomic standpoint. Compared with its neighbors, Malaysia had a smaller proportion of short-term external debt to GDP and its foreign exchange reserves appeared to be ample, seemingly insulating it to a degree from pressure on its currency or an exit of foreign capital. With more developed local capital markets than Thailand or Indonesia, investment could be more easily financed from local sources. This is not to say that there were no weaknesses. Consistently high rates of credit expansion had brought Malaysia's domestic debt-GDP ratio to 170 percent. By mid-1997, its private sector debt levels were above those of Thailand and Korea on a relative basis.

Though comparatively robust in relation to some other countries in the region, Malaysia's banking system was not overly strong, in spite of its posted ratios. Although its banks had engaged in little foreign currency borrowing, they were vulnerable in other ways. Margin lending on shares was widespread, more so than in neighboring countries, and such loans were vulnerable to turning into NPLs should the soaring stock market crash. The Kuala Lumpur Composite Index peaked at 1271 in February 1997, and thereafter plunged. At its peak, the market capitalization of the country's stock exchange reached an extraordinary 310 percent of GDP, versus

116 percent in the United States and 29 percent in Korea, a level that was the highest in the world.

The boom in the stock market created a wealth effect that encouraged excessive rates of bank lending, a significant proportion of which was advanced to politically well-connected conglomerates and to the real estate sector. Financial institutions floated on a sea of optimism. Also reflective of overvaluation of assets was a money supply that was also relatively large on a proportional basis compared to these countries. Finally, the banking industry included a number of vulnerable weaker players.

Watching events across the country's border in mid-1997, the Malaysian authorities preemptively sought to strengthen the banking sector to avoid the banking crisis that had emerged in Thailand, its northern neighbor. Malaysian regulators had charge of a banking system not apparently in imminent risk of collapse, and objectively stronger than those in nearby Thailand and Indonesia, but, as observed, one that did have some problems. Almost simultaneously with the July 1997 devaluation of the Thai baht, NPL figures began to trend upward from their lowest point in more than a decade, reflecting deteriorating asset quality. In response, the country's regulators took steps to address the deterioration in banks' loan portfolios. As this initiative undertaken when the NPL ratio was still below 10 percent and the banking system was remained apparently adequately capitalized, it probably held down the later costs of bank restructuring.

Nevertheless, the currency crisis that hit Malaysia was a serious economic shock. A comparatively small Asian nation in population, the country's economy was characterized by a high degree of export dependency, not to mention a solid dependence on Japanese investment flows. Malaysia's currency, the ringgit, was not official pegged, but it had been comparatively stable in the range of 2.5 to 3.0 ringgit to the U.S. dollar for most of the mid-1990s. Any dramatic changes in exchange rates were bound to have a disruptive impact. Since the devaluation of the baht, time and again pressure on the ringgit forced it out of its normal trading range. By September 1998, the Malaysian government finally conceded that efforts to prevent a collapse of the ringgit were too costly and a losing battle. That month it devalued the currency and fixed the exchange rate at 3.8 ringgit to the U.S. dollar. To bolster a fixed rate of exchange, currency controls were instituted.

A National Approach to Restructuring and Recapitalization

Given the importance of the banking sector, and its role in the economy, stable domestic financial markets were perceived to be a crucial element in the economic recovery process. With the stability of the currency in question, contagion spreading throughout the region and the flow of foreign funding in retreat, the banking sector, now facing a rising tide of loan losses that were eating away at capital, was revealed to be vulnerable. Hence a systemic restructuring program had to be established.

Program Objectives

The immediate need, of course, was to contain the crisis. This called for the provision of emergency liquidity to be followed by the creation of incentives to restart bank lending. The key aims were to halt the economic downturn and to foster an economic recovery. Note that there is an inherent trade-off between the two objectives.

Concentrating on cleaning up the balance would tend to defer a resumption of rapid lending growth until the bank's financial condition was almost fully restored. Focusing on jumpstarting the economy would call for lending to be stepped up as soon as possible. Certainly, there is some happy medium between the two.⁷⁸

The longer-term aim was to restore the financial condition of the banking system to a healthy state, capable of supporting strong economic growth and of absorbing economic shocks of a level comparable to the immediate crisis. To this end, the Malaysian restructuring plan set forth as key targets the:

- Resolution and disposal of problem assets
- Recapitalization of those institutions in need of new capital
- Further development of the local debt markets
- Consolidation of the banking sector into a smaller number of more robust institutions

By and large, the program was successful. Emergency liquidity was provided, bank mergers were encouraged and consummated, new capital was provided to selected banks, and NPLs were steadily reduced over the next two years. After Malaysia began its restructuring program in 4Q98, the aggregate NPL ratio started to flatten out. Between its peak in November 1998 and November 2000, the banking system NPL ratio had fallen from 14.9 percent to 9.8 percent (when measured on a three-month basis), according to official figures.⁷⁹

Core Institutions Involved

At the heart of Malaysia's systemic restructuring program were three specially formed agencies that were responsible for overseeing the restructuring process. They included:

1. A national asset management company, Danaharta
2. A special-purpose bank recapitalization agency, Danamodal
3. To facilitate corporate debt restructuring, the Corporate Debt Restructuring Committee (CDRC)⁸⁰

Danaharta

Danaharta was formed in September 1998 as Malaysia's national asset management company, and was intended to have a life of from 5 to 10 years. The AMC's task was problem asset disposal; that is, to resolve the banks' NPLs by removing them from the banking system and disposing of them, while selling the acquired assets (i.e., collateral) at the maximum possible value. Because secured loans generally afforded greater prospects of recovery than unsecured loans, the former were given priority when the agency acquired assets. For secured NPLs, the underlying collateral was valued by Danaharta's panel of independent valuers, while unsecured NPLs were acquired at a flat 10 percent of the principal outstanding of the NPLs. Also, because roughly three-quarters of structural NPLs were comprised of advances in excess of M\$5 million, the agency normally limited its acquisitions to these large problem loans. In this context, because in some instances obstacles in recovering funds from delinquent borrowers was partly attributable to the inability of such entities to recover funds from third-party obligors, Danaharta was empowered in certain cases to acquire problem assets from nonfinancial enterprises.

Special enabling laws permitted Danaharta to step into the shoes of the original lender upon the acquisition of nonperforming assets (thereby allowing it to assume the original lender's rights with respect to the obligation) and reinforced its rights as a creditor to recover delinquent obligation. More specifically, Danaharta was empowered to:

- Deal directly with distressed borrowers
- Negotiate the restructuring of the loan or relevant facility
- Assist the borrower by providing additional advances and, if appropriate, convert all or a portion of the outstanding debt to equity
- Take over management of the borrower

In contrast to the approach taken in some other countries, Danaharta adopted a workout strategy and avoided both the fire sale approach to disposing of problem loans as well as the warehousing approach.

Notwithstanding the agency's powers, banks generally had no obligation—unlike banks in some other countries—to transfer their bad assets to Danaharta. Strong incentives, however, were provided for them to do so. Banks that did not transfer problem loans to the agency were obliged to write down the loans, which would have an adverse impact on their capital position. Alternatively, banks could sell their NPLs to Danaharta, taking a haircut in the process and receiving zero coupon government-guaranteed bonds in exchange, which they could amortize or write down over up to five years.⁸¹

Thus, instead of having to show a relatively high proportion of illiquid NPLs or foreclosed assets (written down to some fraction of the assets' nominal value) on their balance sheets, they could hold government securities that were both liquid and interest earning. Another sweetener offered by Danaharta was a profit-sharing arrangement. Should the agency be able to sell its bad assets or collateral acquired from a bank at a price in excess of the price paid to it, Danaharta promised to return 80 percent of any such excess in recovery values while keeping 20 percent for itself. These incentives were very effective in persuading banks to shed their problem assets.

Danaharta was consequently able to successfully complete its mission by extracting a large volume of NPLs, both viable and nonviable, and foreclosed assets from the banking system. In the case of viable assets, Danaharta operated just like a bank, negotiating with borrowers to work out the loans. Workouts could take the form of mere rescheduling or the arrangement of a merger or acquisition by a white knight, in all cases subject to the review of an independent adviser and regulatory approval. During the workout period, a 12-month moratorium over claims against the borrower could be allowed. In the case of nonviable assets, collateral was foreclosed, or the business was sold through an open tender sale. Collateral acquired from the borrower was sold through various means including auction.

Danaharta, in the view of most analysts and commentators, effectively fulfilled its mission to help restructure or liquidate many of the NPLs in Malaysia's banking system. By the end of December 1999, Danaharta had acquired and was managing problem assets with loan rights amounting to 45.5 billion Malaysian ringgit (RM)—then equal to roughly US\$12 billion, a not insignificant figure for a country of that size—from the financial system.⁸² The book value of the loans removed from the banking system amounted to RM34 billion, representing approximately 42 percent

of NPLs in the banking system. The removal of these bad loans from the system reduced the NPL level to under 11 percent for the first time in July 1998.⁸³ Overall, Danaharta was generally considered a success having resolved 83 percent of NPLs on its books with an overall average recovery rate of 66 percent. The agency closed down at the end of 2005 within the time frame originally planned.

Danamodal

Created at the same time as Danaharta, Danamodal was set up to address and facilitate the recapitalization of distressed banks. To be more precise, its mission was both to recapitalize the banking sector and to assist in its consolidation, creating sustainably profitable institutions in the process. As occurred in Malaysia's previous financial crisis during the 1980s, bank owners were expected to share in losses. The agency was contemplated to have a limited life, recapitalizing the banking system within two years, and then over time disposing of its interests in the country's banks recovering all or part of the funds expended in recapitalization.

Assuming a target risk weighted capital adequacy of 9 percent—1 percent above the Basel minimum—the total cost of recapitalizing the country's banking system was estimated to be RM16 billion, or in the vicinity of US\$5 billion.⁸⁴ Bank Negara Malaysia, the country's central bank, contributed seed capital and in October 1998, Danamodal obtained further funding by instituting a debt issuance program. The funds raised were then provided to banks in need of recapitalization in exchange for irredeemable noncumulative, exchangeable, convertible preference shares, non-redeemable cumulative preferred shares, subordinated bonds, or other comparable securities. The intention at the outset was that the agency would ultimately widen its shareholder base by making equity available to a variety of investors.⁸⁵

Fourteen financial institutions were identified as in need of recapitalization, which proceeded steadily.

Corporate Restructuring via the CDRC

The government's first priority as the financial crisis ensued was to ensure the stability of the financial system, and as a result the activities of Danaharta and Danamodal were given priority. Subsequently, however, the Corporate Debt Restructuring Committee (CDRC) was established. Its purpose was to restructure corporate debt and work together with Danaharta in cases of large defaulted or problem obligations to domestic banks, targeting borrowers that had viable business and large debt obligations to more than one lender.⁸⁶

Significantly, the CDRC did not have the power to compel workouts. Its role, instead, was to act as a mediator and independent consultant in resolving outstanding debt obligations, and therefore it depended upon the cooperation of creditors and debtors.⁸⁷ The committee, comprised of bankers with workout experience, only accepted cases by application and limited its operations to the largest and most complex cases. To be eligible for assistance from the CDRC, the outstanding obligation had to involve five or more creditors and more than M\$100 million in outstanding obligations.

Initially, based on information taken from the bank and the corporate borrower, an investigation was conducted and a report prepared offering the CDRC's recommendation of a restructuring plan. This report served as a departure point for negotiations between or among the parties.

Various workout techniques were employed by the organization including:

- Divestment of assets
- Creditor haircuts
- New financing
- Debt equity conversions

In a few instances, nationalization coupled with government guarantees was utilized with a view to readying the banks receiving assistance for privatization.

By July 2001, roughly two years into its existence, about half of the cases pending before the CDRC had been restructured. Although progress had been made, flaws were observed in the structure of the program that allowed negotiations to drag on unnecessarily. Although the CDRC was perceived as a fair broker, it was seen as somewhat lacking in authority to push through compromises in regard to some of the more difficult cases. Therefore, in August 2001, in an effort to increase the pace of corporate restructuring process, a new chairman was appointed, and a number of the CDRC's policies were changed to expedite completion of workouts.

Key changes included:

1. Restrictions on banks removing loans from NPL status during the restructuring process
2. The imposition of strict deadlines at each stage of the restructuring process
3. A requirement that participating banks must send senior debt recovery staff to workout meetings on pain of being excluded from future discussions
4. Fuller disclosure requirements on the part of borrowers

These changes represented an attempt to give the CDRC more leverage in the bargain process by neutralizing some of the delaying tactics that various parties had been employing.

To a similar effect, the CDRC began disclosing basic terms of the workout to the press, and releasing the names of companies that were under review. Where the parties had been dragging their feet and resisting settlement, this name-and-shame approach increased the pressure on them to reach agreement. Within 8 months, only 12 cases remained, of which 2 comprised about 70 percent of all outstanding assets. Both Danamodal and the Corporate Debt Restructuring Committee were shut down in 2004, their missions having been completed.

Dynamics of Malaysia's Systemic Restructuring Program

The three agencies worked together to coordinate systemic bank and corporate restructuring. Danaharta, the AMC, effectively had the lead role of the three, acquiring—against bonds—problem assets from distressed banks at a discount. Through Danaharta's removing the relatively illiquid problem assets from the balance sheets of the country's distressed bank banks and providing liquid earning assets in return, the banks' financial condition was substantively improved, and they could focus on lending and attracting new business rather than expending energies on resolving problem assets. In addition, unless the banks' problem assets had already been fully written down or provisioned against, these transactions tended to result in more accurate reporting concerning their level of solvency.

With the AMC helping to clean up the asset side of the banks' balance sheets, Danamodal was available to assist with the liability side. The agency was responsible for recapitalizing banks whose capital had been impaired to unacceptably low levels. Set up with initial capital from the central bank and funded in a similar fashion to Danaharta through the issuance of government-backed debt securities, Danamodal provided capital to the banks in exchange for ordinary or preference shares. Finally, the CDRC functioned as a mediator between banks and corporate borrowers to help with the workout of large delinquent obligations, particularly multiparty obligations such as syndicated loans.

Other Aspects of Systemic Bank Restructuring

The facets of the Malaysian plan just discussed—namely recapitalization, restructuring, and asset disposal—form the core of any systemic restructuring scheme. As previously mentioned, however, a comprehensive program should consider the causes of the crisis and ways in which the banking system ought to be reformed in order to avoid the same type of crisis from happening again. One aspect of such reforms concerned prudential regulations governing the country's banks. The other concerned consolidation of the banking system.

In respect to changes in prudential regulation, one somewhat surprising facet of the restructuring program was the significant use of regulatory forbearance, discussed earlier in this chapter. Although some regulations were tightened, the relaxation of others received the lion's share of attention in the financial markets. The objective was to enable banks to more easily return to profitability and also to encourage them to lend more vigorously to stimulate the economy.

Policy changes included:

- Loosening the definition of NPLs from the international standard of three to six months
- Reducing specific provisioning requirements on substandard loans
- Allowing restructured NPLs to be classified as performing if not in default for a continuous six months rather than the 12 months previously
- Relaxing limits on lending for purchasing shares and unit trusts
- Ordering banks to set up in-house loan administration units going forward

Consolidation of Malaysia's Banking System

Malaysia's regulators and policymakers took the view, long propounded by the central bank, that in light of its comparatively small size, the country would benefit from a banking system that was comprised of a smaller number of larger and more robust institutions. Larger universal banks, it was thought, could more effectively compete with other banks in the region as well as eventually in the local market should it eventually be more fully opened to foreign competition. Moreover, a smaller number of universal banks could be more effectively supervised. During the periods of strong economic growth in the late 1980s and early 1990s, the central bank's call for mergers was largely ignored. (Only two market-oriented mergers were completed subsequent to the mid-1980s banking crisis.)

In the aftermath of the regional crisis, however, the time was ripe for a major reform of the banking industry. An added impetus was the desire to make Malaysian banks large and efficient enough to withstand the onslaught of foreign competition

anticipated with the relaxation of entry barriers planned in 2003, when financial markets were due to be liberalized under a World Trade Organization pact. In early September 1999, the Malaysian Finance Ministry together with the central bank ordered the country's 22 commercial banks, along with a somewhat larger number of ancillary financial institutions,⁸⁸ to merge into six groups led by the following nominated anchor banks: Maybank, Multi-Purpose Bank, Bumiputra Commerce Bank, Perwira Affin Bank, Public Bank, and Southern Bank.

The plan, however, was met with significant protests as several of the larger and well-managed banks were not nominated as anchor banks. On the back of some unanticipated difficulties perceived in the original plan and heavy lobbying by the affected interests calling for a larger number of such nodular institutions as well as greater flexibility in the program's implementation, the government relented. In February 2000, following an extended period of negotiation, the central bank announced a new plan allowing for four more anchor banks (RHB, Arab Malaysian, EON, and Hong Leong), and giving affected institutions more time and flexibility to arrange their own mergers. By the spring of 2002, the consolidation process was completed, leaving Malaysia's banking system with 10 major players.

In sum, excepting the somewhat heavy dose of regulatory forbearance, the Malaysian experience with bank restructuring in the late 1990s provides an interesting example of how to rehabilitate a financial system.

One could argue that Malaysia benefited from lucky circumstances in the few years following the Asia crisis, in spite of its bold decision—which infuriated the markets—to restrict the outflow of foreign investment that had happily piled up in the country during the boom years. Observers would also argue that the Asian culture provided a helpful support to the government's plan, and that the industrial and commercial sectors of Malaysia, or even some parts of its civil service were not as sophisticated as its financial sector, which might have been prompt at saving its skin at any cost. Others would agree that the opinion of Malaysia's elite as to the original cause of the Asian crisis was grossly flawed. It remains that Malaysia very skillfully used rehabilitation and restructuring tools to salvage and transform its banking sector for the benefit of the whole country.

NOTES

1. World Bank, cited by C. Larraín, "Best Practices for Efficient Banking Restructuring," World Bank, 2002.
2. A. Sheng, "Bank Restructuring: Lessons from the 1980s," World Bank, 1996, 1, cited in Roy I. Ramos, "A Tale of Four Bank Restructurings," Goldman Sachs, January 7, 1999.
3. Eventually, of course, the extraordinary situation, if it lasts for a prolonged period of time, becomes the new normality.
4. The view that the worst is over springs from the observation that mild deviations from the norm are self-correcting. When the deviation is mild, they normally are. The perspective is illustrated in the following excerpt from economist Robert Solow's review of Richard Posner's book, *A Failure of Capitalism: The Crisis of '08 and the Descent into Depression* (Cambridge: Harvard University Press, 2009), that summarizes Posner's argument: "A modern capitalist economy . . . can probably adapt to minor shocks . . . with just a little help from monetary policy and mostly automatic fiscal stabilizers. . . . [Therefore] [i]t is easy to be lulled into the comfortable belief that the [financial] system can take care

of itself if only do-gooders will leave it alone. But that same financial system has intrinsic characteristics that can make it self-destructively unstable when it meets a large shock.” R. Solow, “How to Understand the Disaster,” *New York Review of Books* 56, no. 8, May 14, 2009. The “large shock” is the large deviation from the norm that is prone to trigger a severe banking crisis. Concededly, to compensate, warnings of even worse disasters to come sell more newspapers and magazines, and will also appear in abundance as a crisis develops.

5. Roger Lowenstein’s book, *When Genius Failed*, tells the story of the pantheon of Wall Street trading gurus who came together to create LTCM and how despite the brainpower behind the fund, it was overwhelmed by unforeseen events. R. Lowenstein, *When Genius Failed: The Rise and Fall of Long Term Capital Management* (New York: Random House, 2001).
6. Obviously, there is no set number of failures or rescues that can occur before a banking crisis can be said to have been triggered, and in any event, it can only be known after the fact. It only takes one such event to potentially set off a systemic crisis; conversely, depending upon the size of the system, the level of concentration within the sector in respect to assets and deposits and the specific causes of each failure or rescue, it is conceivable that multiple bank failures could occur without such a crisis occurring.
7. Notwithstanding the wide use of the term *bank restructuring* to refer to the rehabilitation of an entire banking system, it should be acknowledged that it is the view of some that the latter should properly be termed *systemic bank restructuring*. See M. Waxman, “A Legal Framework for Systemic Bank Restructuring,” World Bank, “Bank and Financial Restructuring,” June 1998, 1.
8. See C. Dziobek and C. Pazarbasioglu, “Lessons from Systematic Bank Restructuring,” International Monetary Fund, 2. (As an IMF report observed, “often a banking crisis occurs before countries can muster the political will to undertake essential reforms.”)
9. M. Waxman, “A Legal Framework for Systemic Bank Restructuring.”
10. K. Lynch, “Managing through the Economic Crisis: The Civilian Equivalent of the Fog of War,” *Policy Options* (Montreal: Institute for Research on Public Policy, May 2009), www.irpp.org/po/archive/may09/lynch.pdf.
11. Occasionally, central bankers, regulators and even analysts concede their mistakes. In mid-2009, with the worst of a banking crisis evidently past, the Chairman of U.S. Federal Reserve acknowledged that the central bank did not act swiftly enough to curb irresponsible mortgage lending. Don Lee, “Ben Bernanke Says Fed Didn’t Act Quickly Enough to Stop Reckless Mortgage Lending,” *Los Angeles Times*, web edition, July 29, 2009.
12. Waxman, “A Legal Framework,” 2.
13. Even if an economic capital model is employed and the model states that capital is sufficient for all but a once in a thousand year event, there is always the possibility that the model is wrong.
14. S. Claessens, “Systemic Bank and Corporate Restructuring: Experiences and Lessons for East Asia,” World Bank (1998).
15. *Ibid.*, 22.
16. *Ibid.*, 8.
17. See, for example, Anthony M. Santomero and Paul Hoffman, “Problem Bank Resolution: Evaluating the Options,” The Wharton Financial Institutions Center 98-05-B (1998), 11.
18. Before the scope and severity of a burgeoning financial crisis have become evident, the authorities responsible for overseeing a banking system may have been aware of fundamental problems in the banking system. Moreover, they may have taken some steps to address them, or sought to institute further reforms, while aware of the dangers a failure to reform posed in the medium to long term without necessarily anticipating that one was imminent. Undertaking fundamental reforms when the banking industry is doing

- well—and firms, their shareholders, and staff are making a lot of money—is no easy task. Sometimes it takes a crisis and the urgency it brings to implement long-needed changes. In Stijn Claessens words, “financial crises concentrate minds.” *Ibid.*, 5.
19. Claessens note that government inaction will discourage potential private investors from advancing new capital to undercapitalized banks. In a crisis situation, they will expect some assurances from the state concerning limits on loan losses. As banks have a tendency to hide their problems, such an expectation is not an unreasonable one. *Ibid.*, 5.
 20. This is because, as was observed, the causes of banking crises are both internal and external, and interact in complex and dynamic manner. Recall from our previous chapter the categories of contributing factors: internal management problems and corporate governance weaknesses; external governance or institutional failures, including regulatory deficiencies; as well as macroeconomic frailty. See generally David T. Llewellyn, “Some Lessons for Regulation from Recent Bank Crises,” Institute for Development Policy and Management, University of Manchester (1999), 5.
 21. Patrick Honohan, “Banks in Trouble: Same Old Story or Something New?,” speaking notes for presentation to 5th Euroframe Conference, Trinity College, Dublin, June 6, 2008.
 22. This is not to suggest, however, that differences in the particular causes of a banking crisis and the manner in which it manifests will necessarily mandate major differences in remedies. Naturally, in preparing a plan to fix the banking system, the specific characteristics of the crisis, as well those of the country in which it took place and political realities in that country, will undoubtedly shape the response. Conceivably, those who argue that this time is different and that the new approaches are required may be correct. But, if so, the appropriate rejoinder is to ascertain whether or not the identified differences materially change what needs to be done to fix the crisis, and if so, in what way?
 23. Waxman, “A Legal Framework for Systemic Bank Restructuring.”
 24. One must, of course, acknowledge that it is possible that the crisis at hand—referring here to the events of 2007–2012 or whenever they occur in future—is different in kind from all previous crises.
 25. See S. Claessens, “Systemic Bank and Corporate Restructuring,” 1998.
 26. The template for a successful systemic restructuring can be traced back to the experiences of the United States in 1933, Japan in 1946, Argentina in 1980–1982 and, more recently, in Estonia in 1992. In all cases, the approach was similar and embraced the following core stages, which begin after the intense onset of the crisis has passed and insolvent banks have been closed. It includes the following steps: (1) Prepare a comprehensive restructuring plan and implement it decisively; (2) arrange for the injection of capital into marginally solvent institutions; (3) facilitate equitable loss-sharing arrangements vis-à-vis shareholders, depositors, bondholders, etc.; (4) separate distressed assets from new assets to expedite their resolution markets; (5) keep interest rates on deposits sufficiently high and withdrawal restrictions sufficiently lax so as to prevent a flight to coin and currency; (6) by taking firm actions that show that the system is solvent and the state is committed to its full restoration (i.e., steps 1–4), regain the confidence of depositors and creditors.
 27. See Andrew Sheng (ed.), “Bank Restructuring: Lessons from the 1980s,” *The World Bank* (1996).
 28. While it would certainly appear that the response to the crisis needs to fit local conditions, the reader must determine to what extent the response chosen was due to political expediency, or a genuine adaptation to local needs. Views appear to differ as to whether the resolution approach chosen depends on the cause of the crisis. It does appear that the approach will appropriately vary with the depth of the crisis. See A. Santomero and Paul Hoffman, “Problem Bank Resolution: Evaluating the Options,” *The Wharton School, University of Pennsylvania, Financial Institutions Center*, 1998, 6.
 29. In brief, of the 10 principles, the first 3 principles emphasize the importance of the macroeconomic environment, conspicuously the role of a stable domestic currency and a

strong regulatory regime, to bank stability. Principle 4 reflects the critical point that bank losses of failed banks in the end represent a fiscal burden upon the state, since it has no practical choice other than to recapitalize the banking system. Principle 5 stands for the proposition that even if the immediate cause of bank failures is a lack of liquidity, the liquidity shortage is driven by fundamental economic losses caused by loan or other financial asset defaults, which in turn have eroded bank solvency. Principle 6 follows from Principle 5; if delinquent loans are the primary source of bank distress, then further lending of this sort must be halted immediately to prevent further damage to the system. Principles 7 through 9 address the formulation of the restructuring scheme. The key issue to be resolved is how the losses incurred are to be allocated, or conversely who gets what as the scheme is implemented. It is also critical that the banking system postrestructuring is a healthy and profitable one, able to repay or generate a reasonable return on the capital injected during the restructuring. Finally, recapitalization should be accompanied by reforms to ensure that “a strong institutional framework” has been created. The last principle stands for the proposition that delaying the restructuring process will merely postpone the day of reckoning and make the program more costly and more difficult.

30. Although the various analysts and researchers who have examined the problem of bank restructuring have used different terminologies and a smaller or larger number of categories to describe the components of a restructuring scheme, they generally refer to the very similar activities and techniques that seem to divide naturally into a small number of phases.
31. This subject has been implicitly addressed in the previous chapter in connection with crisis causation and will be elaborated upon from time to time in this chapter, particularly with reference to the suitability of various restructuring strategies and techniques. To recap, the causes of a banking crisis are invariably multiple, involving any combination of factors. While illiquidity is almost always the proximate cause of bank failure, the cause of illiquidity is usually the market’s perception that loan losses have left banks’ balance sheets too thin to absorb the next shock. In this sense, insolvency or imminent insolvency precedes illiquidity. Bank losses in turn have a cause. The cause may be bad banking, or as is usual in a systemic crisis, there is some wider systemic cause. To some extent, the propensity to default, which is typically exacerbated by management—and risk management deficiencies—reflects genuine weaknesses in the real economy and its management by government. Government policies that have been linked to banking crises include, as mentioned earlier, government attempts to deregulate the financial system too quickly, and, as Andrew Sheng has noted, its inability to maintain a stable currency. Wide swings in the value of the domestic currency tend to have a deleterious effect on the financial sector by, for example, promoting the rapid flow and ebb of foreign capital. Rapid inflows of such foreign funds tend to overstimulate the economy and adversely affect the behavior of banks and corporate borrowers, encouraging excessive risk taking on the part of both. Note that following the initial diagnosis, it is naturally sensible that it be reassessed from time to time as the crisis advances.
32. This material is not derived directly from Andrew Sheng’s book.
33. Sheng’s views have been cited and elaborated upon by others, including a number of bank analysts, referenced in the pages that follow. Hence, although Sheng’s elements and principles are used to provide a framework for this overview of bank restructuring, they are not intended as a summary of Sheng’s thinking.
34. Stephen M. Frost, “What Is to Be Done?” *Asian Banking Focus*, no. 5 UBS Global Research Singapore, January 9, 1998, 14. Actually, the number of steps Frost outlined was a baker’s dozen (i.e., 13) but they have been condensed here to 12.
35. Charles W. Calomiris, Daniela Klingebiel, and Luc Laeven, “A Taxonomy of Financial Crisis Resolution Mechanisms: Cross-Country Experience,” World Bank Policy Research Working Paper 3379 (2004).

36. Ibid.
37. Urban Bäckström, "What Lessons Can Be Learned from Recent Financial Crises? The Swedish Experience," presentation at the Federal Reserve of Kansas City's Symposium, "Maintaining Financial Stability in a Global Economy," 1997.
38. The formulation of the phrase is attributed to Sir Francis Baring, who in a treatise on the Bank of England published in 1797 referred to the bank as "the dernier resort" from which all banks could obtain liquidity in times of crisis." T. Humphrey, "Lender Of Last Resort: The Concept In History," drawn from Thomas M. Humphrey and Robert E. Keleher, "The Lender of Last Resort: A Historical Perspective," *Cato Journal* 4, no. 1 (Spring/Summer 1984). Although there appears to be some question as to how consistently the rule, as formulated by Henry Thornton and Walter Bagehot, was followed by the Bank of England in the late nineteenth century, it appears that over time it was adopted as a general principle there and at a number of other central banks as the financial industry evolved in the late nineteenth and early twentieth centuries. According to Thomas Humphrey, the rule "stressed (1) protecting the aggregate money stock, not individual institutions, (2) letting insolvent institutions fail, (3) accommodating sound institutions only, (4) charging penalty rates, (5) requiring good collateral, and (6) pre-announcing these conditions well in advance of any crisis so that the market would know exactly what to expect."
39. In other words, as shareholders' equity can be seen as the excess of the value of a bank's total assets (mainly loans, but also securities and fixed assets) over its liabilities, if it falls owing to an increasing portion of the loan portfolio being deemed a loss (whether or not the loss has yet been financial reported as such), then a loss equal to, or larger than the amount of equity renders the bank insolvent.
40. Stijn Claessens, "Systemic Bank and Corporate Restructuring," Claessens points to examples (Venezuela, 1994; and Thailand, 1997) where unconditional liquidity support was money down the drain "into black holes of insolvency." Claessens further noted that in Indonesia the central bank's prolonged support for failing banks ultimately resulted in additional advances to delinquent creditors that predictably turned to losses exceeding the banks' equity by several multiples.
41. See, for example, M. Tanaka and G. Hoggarth, "Resolving Banking Crises—An Analysis of Policy Options," Bank of England Working Paper No. 293 (2006). Note that the term *gambling on resurrection* is sometimes used to refer to any strategy that depends upon a remote chance of success even if it does not involve actively placing high-risk bets but instead just muddling along doing business as usual.
42. While the motivation of management in the first case is that their big win will pay for all the previous losses; in the second it may be merely the desperate hope that there will be a sharp and strong rebound from the current downward trend, an attempt to hold on to their jobs as long as possible, or resignation coupled with inertia.
43. Glenn Hoggarth, Jack Reidhill, and Peter Sinclair, "On the Resolution of Banking Crises: Theory and Evidence," Bank of England Working Paper no. 229 (2004), 10.
44. Calomiris, Klingebiel, and Laeven, "A Taxonomy of Financial Crisis Resolution Mechanisms."
45. Dziobek and Pazarbasioglu, "Lessons from Systematic Bank Restructuring"; see note 8.
46. Of course, predictably, views diverge with regard to implementation depending upon policy preferences. There are a number of trade-offs implicit in making the decisions that shape the implementation process, and differences will naturally arise with regard to allocation of losses, the attention and degree of emphasis to be placed upon each program component, the order in which actions should be taken, and so on. The list of preliminary questions in the previous subsection is suggestive.
47. A divide often stands between those who argue that if banks must be bailed out, that is, recapitalized, such a costly project should be undertaken in a manner that does not

reward those who were responsible for the debacle. Such critics say that bailing out undercapitalized banks, including their shareholders and management, primarily at the expense of the taxpayer amounts to “socialism for the rich.” This, it is asserted, is a form of corporate welfare; whereby the bank’s profits are kept private, but its losses when failure becomes imminent are socialized, the tab being picked up by the state. As policy issues of a political character are largely outside the scope of this book, let the reader decide whether a restructuring program that leaves management of a failed financial institution in place and advances huge sums of state funds to it without obtaining ordinary publicly traded bank shares in exchange (thereby an opportunity to benefit from investment) represents the best use of public funds.

48. Although such losses are attributable for the most part to the bank’s management, and to a lesser extent its shareholders (and conceivably to sins of commission or omission by the regulator), the potential shoulders upon which this burden might rest, to a greater or lesser proportion, also include the distressed bank’s nonmanagerial staff, retail depositors, the deposit insurance agency (assuming one exists), commercial depositors and counterparties, institutional investors, retail investors, and the public (i.e., taxpayers) represented by the government.
49. As a number of commentators have emphasized, a bank restructuring program should be tailored to the local operating environment. In practice, this means that institutional and, particularly, political constraints will shape both the design and the implementation of the program.
50. Allowing investors to set up new banks without resolving the problems of the existing banks would in most instances be a recipe for disaster. A tangle of unresolved obligations would act as a drag on the economy and could very well create a major impediment to the ability of the new banks to become sustainably profitable. The government might need to take many of the same steps that would normally form a part of a restructuring effort such as facilitating the workout and resolution of existing delinquencies on the part of borrowers. Moreover, the liquidation of an existing banking system could shatter confidence for years to come, particularly if the new banks began to have problems. Given the risks, apparent savings in setting up a new banking system out of whole cloth might in the end be less than imagined.
51. This step is naturally related to the earlier admonition to diagnose the causes of the crisis.
52. This will be discussed in a later part of this chapter. Gaining approval and funding for the plan is apt to be politically challenging, but this is an issue outside the scope of this book.
53. This component is ordinarily referred to as *corporate restructuring*. It should be noted that the phases should not be thought of as being undertaken sequentially. Instead, there would normally be some overlaps so that phases run either in a staggered manner or practically simultaneously.
54. Of course, one likely reason that these companies have not been able to perform their obligations to the bank—thereby leading to the bank’s insolvency—is that they themselves are highly inefficient, if not loss-making enterprises, in need of new management and reorganization.
55. J. Hawkins and Ph. Turner, “Bank Restructuring in Practice,” BIS Policy Papers, no. 6, August 1999.
56. As noted, there is rarely a single cause to a banking crisis, but instead a combination of causes having varying proximity to the point at which the crisis might be said to begin.
57. A possible exception to this rule could occur in those emerging markets that have long protected the domestic banking market, or forced competition on invidious terms. With the state in desperate need of capital for its banks, foreign banks may see a golden opportunity to enter the market before the door slams shut again, and therefore may be willing to require fewer concessions from the government than they otherwise would. The exception, however, in a sense proves the rule because even if the foreign institution is

willing to make concessions on loan-loss guarantees to some extent, it is likely to demand some assurances concerning, for instance, the foreign bank's continuing presence in the market and whether it will be treated in a nondiscriminatory manner in comparison to domestic banks. Discussions of this nature are likely to be intensive. While state capital might not be required—at least to the same extent—some political capital may be earned in respect to enacting changes in the law, allowing the foreign bank to initiate staff redundancies. And so forth.

58. Forms of supplementary state support such as earnings support and regulatory forbearance can be construed as forms of subsidies.
59. Other types of incentives include guarantees of income, special tax breaks, and indemnification against legal actions against the weak bank.
60. Other measures commonly employed in bank restructuring—including the provision of state funding to banks, easing monetary policy to encourage bank profitability, helping banks with capital raising, or placing government deposits with banks—have on occasion been characterized as regulatory forbearance. Given that *regulatory forbearance* refers specifically to regulation, to apply the term to affirmative assistance, as opposed to restrictions on bank operations, may be stretching the definition too far. Irrespective of semantics, what is important to watch for is whether or not, in exchange for the benefits granted to them, appropriate reciprocal commitments or concessions are obtained from the banks.
61. Morgan Stanley Dean Witter analysts Wan Ismail and Rafael Bello define the term *regulatory forbearance* to mean the application of “greater tolerance when helping banks meet compliance standards during the recovery process.” Wan Ismail and Rafael Bello, “The Bank Rehabilitation Process: Asia and Mexico Compared,” Morgan Stanley Dean Witter, September 4, 1998, 5.
62. M. Tanaka and G. Hoggarth, “Resolving Banking Crises: An Analysis of Policy Options,” Bank of England Working Paper no. 293, Bank of England, International Finance Division, London, 2006.
63. Ibid.
64. M. Tanaka and G. Hoggarth, “Resolving Banking Crises.”
65. S. Claessens, “Experiences of Resolution of Banking Crises,” BIS Policy Papers, 1999.
66. See Claessens, “Experiences of Resolution of Banking Crises.” *Regulatory forbearance* as used here is intended to encompass the waiver of rules in a wide range of supervisory contexts. Some commentators define *regulatory forbearance* more narrowly and recognize other forms of forbearance such as accounting forbearance (waivers or exemptions from usual accounting and reporting requirements) and tax forbearance (relief from taxes to which banks would otherwise be liable).
67. Another example occurred in Malaysia. To aid the banking system in times of difficulties, the regulatory body may actually relax the definition of NPLs, which has the effect of reducing provisioning requirements while making the country's banks look relatively good compared with other countries that make use of a stricter definition. This was the case in Malaysia, which moved from a three-month definition to six-month. This gave banks breathing space, allowing them to record lower levels of bad loans and hence reduced the need for higher provision charges, thereby nominally boosting capital and profitability. This kind of situation is obviously a red flag for the external analyst.
68. This might also be construed as regulatory forbearance.
69. Clearly, such a situation is bound to transfer the problem to the state's fiscal balance, which in turn may become a further burden for banks in a deadly vicious circle.
70. This was one of the policies the Thai restructuring agency used in the aftermath of the Asian crisis.
71. Again, in Thailand, for example, stricter loan-loss provisioning requirements were to be phased in over a three-year period in order to give banks time to adjust to the new order.

At each target point within the period additional capital is required. In addition, the Thai government offered to inject capital, but provided that banks would meet certain conditions, including tighter loan loss provisioning at the outset. Consequently, through this requirement, existing shareholders would see their capital dwindle before receiving any government funding.

72. M. Tanaka and G. Hoggarth, "Resolving Banking Crises—An Analysis of Policy Options," Bank of England Working Paper No. 293 (2006).
73. International best practice with respect to prudential regulation of banks embraces an array of restrictions and affirmative requirements. To mention a few, these include minimum capital adequacy requirements, the prohibition of related-party loans (with limited exceptions), and requirement of full loan-loss provisioning against anticipated nonperforming loans, defined according to international standards.
74. In other words, developed countries often have an effective legal and regulatory regime in place that effects restructuring in its normal operation. Emerging markets, however, often lack the legal and regulatory infrastructure, such as effective foreclosure and bankruptcy laws, to effect restructuring on a wide scale. Without reform or a special regime to facilitate corporate restructuring, capital that could be redirected towards productive purposes stagnates.
75. Of course, even if the immediate cause of their troubles is an economic downturn, some delinquent corporates may be in trouble because they overleveraged themselves and channeled capital into unprofitable ventures or investments. Note that in some cases, just as the government might bail out a bank because it is too big to fail, it may take the view that it may be more beneficial to allow an industrial company to continue operating.
76. This is the very argument developed by Philippe Delhaise, co-author of this book, in his book on the Asian crisis: *Asia in Crisis: The Implosion of the Banking and Finance Systems* (Singapore: John Wiley & Sons, 1998).
77. As of July 1997, banks in all three countries displayed robust financial indicators, and regulation was reasonably good in all three, although the Philippines lagged behind, particularly in terms of ongoing bank supervision and examination. Furthermore, banks in the Philippines, in spite of abysmal productivity levels, were enjoying unusually high interest margins that had made them capital rich.
78. Rapidly expanding loan growth before a bank is restored to a basic threshold of financial health can recreate the same problems that brought it down the first time. Equally, however, if the economy dives into a sharp and prolonged recession, getting the banks back to optimal levels of profitability will be much more difficult.
79. Following the dot-com crash in 2000, weaker demand for electronics hardware, which was one of Malaysia's strong export areas, triggered another economic contraction. Weak economic growth persisted for several years, during which time NPLs rose as high as 11.8 percent in October 2001, before resuming a steady decline to their level of about 7 percent in 2005.
80. This agency was commonly referred to as the CDRC.
81. Collateralized loans were acquired at 95 percent of the appraised value of their collateral, with only land and stock accepted as collateral. Noncollateralized loans were acquired at 10 percent of their outstanding value.
82. Of these, RM35.7 billion (a bit more than three-fourths of the total) comprised the loan rights acquired from the banking system.
83. Danaharta embarked on a secondary wave of NPL acquisition in late 1999. However, as the majority of bad loans had already been removed from the system, the amount of NPLs to be acquired was minimal. The secondary phase focused on facilities including unsecured loans granted to public companies and loans from banking institutions with net NPL ratios in excess of 10 percent. Danaharta also entered the loan and asset management stage of the restructuring process. Danaharta conducted two restricted open

tenders to dispose of foreign currency loans and papers in August 1999 and December 1999, involving 43 accounts worth US\$394 million. The first tender offering saw a recovery rate of 55 percent with the second at 71 percent. In December 1999, Danaharta conducted its first open tender exercise involving foreclosed properties. A total of 44 properties were included in the tender with indicative value of RM123 million; 24 bids were successful for a total consideration of RM17.8 million. The unsold properties were transferred to an asset subsidiary of Danaharta at the minimum bid price.

84. Ultimately, only about half the contemplated RM16 billion was required, most of which was recovered by the agency.
85. To protect its investments, Danamodal conditioned financing on the use of a call option that enables it to acquire the remaining shares at nominal value.
86. Corporate restructuring in Malaysia was not as sweeping as in South Korea, where the overleveraging of the large conglomerates that dominated the economy became a significant political issue.
87. In practice, banks suffered minimal haircuts for the most part, and this encouraged their participation.
88. At the time when the consolidation program was initiated, Malaysia possessed 58 significant domestic financial institutions in the country, comprised of 21 commercial banks, 12 merchant banks, and 25 financial companies collectively operating nearly 3,000 branches in a country with a population of 22 million.

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A graduate of the University of California at Berkeley and Harvard Law School, Jonathan Golin has more than 15 years of experience in the financial sector. He worked at the *Economist* Intelligence Unit and as a credit analyst at Tokai Asia in Hong Kong before joining Thomson BankWatch Asia, a specialist rating agency, in 1997. After BankWatch was absorbed by Fitch Ratings in 2000, he wrote the first edition of *The Bank Credit Analysis Handbook*. He subsequently relocated to London, where he established his own research and bank credit analysis training consultancy. Clients included Rabobank, Genworth Financial, Euromoney Training, the African Development Bank, and the Malaysian Deposit Insurance Corporation. More recently, Jonathan was employed by the United Kingdom's main bank regulator, the Financial Services Authority (FSA), as a risk specialist.

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Delhaise has been living and working throughout Asia and the Middle East since the early 1970s. After 14 years in commercial and corporate banking, he established in Hong Kong and Singapore a group of companies servicing the financial industry. Among these companies was Capital Information Services Ltd, whose rating division was purchased in 1994 by Thomson Financial BankWatch, then the world's largest bank credit rating company, later to merge into Fitch Ratings. Under contract as president of BankWatch's Asia Division between 1994 and 1999, Delhaise was responsible for all ratings and research coverage of financial institutions across the

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